LATEST 2020-21 PRACTICE PAPERS Included

As per the CBSE **Revised Curriculum**
For the Academic Year **2020-21**



mtG

CLASS
12

CBSE CHAMPION

10 yrs. 2020-11 SOLVED PAPERS
With TOPICWISE GRAPHICAL ANALYSIS

CHAPTERWISE-TOPICWISE

MATHEMATICS

- Comprehensive Theory
- Delhi, All India, Foreign & Compartment Papers
- Answers as per CBSE Marking Scheme
- 10 Practice Papers Based on Latest Pattern

SCORE MORE 100% MTG GUARANTEE

MtG

CBSE CHAMPION

10 yrs. 2020-11 SOLVED PAPERS with TOPICWISE GRAPHICAL ANALYSIS

CHAPTERWISE-TOPICWISE

MATHEMATICS

CLASS 12

MtG Learning Media

MTG Learning Media (P) Ltd.

New Delhi | Gurugram

Price : ₹ 300

Revised Edition : 2020

Published by : MTG Learning Media (P) Ltd., New Delhi

Corporate Office : Plot 99, Sector 44 Institutional Area, Gurugram, Haryana-122 003
Phone : 0124 - 6601200. Web : mtg.in Email : info@mtg.in

Registered Office : 406, Taj Apt., Ring Road, Near Safdarjung Hospital, New Delhi-110029

Information contained in this book has been obtained by mtg, from sources believed to be reliable. Every effort has been made to avoid errors or omissions in this book. In spite of this, some errors might have crept in. Any mistakes, error or discrepancy noted may be brought to our notice which shall be taken care of in the next edition. It is notified that neither the publishers nor the author or seller will be responsible for any damage or loss of action to anyone, of any kind, in any manner, therefrom.

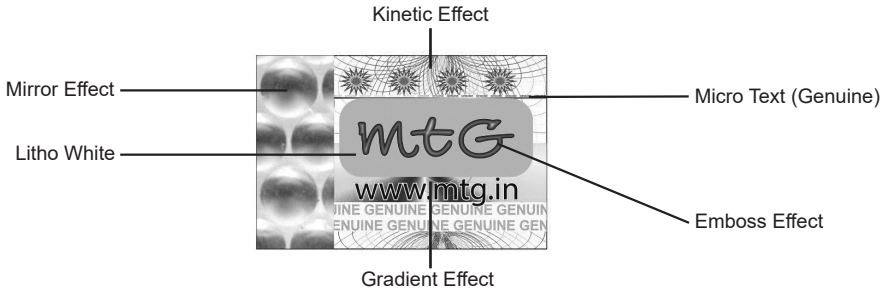
***© MTG Learning Media (P) Ltd.** Copyright reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior permission of the Publisher.*

All disputes subject to Delhi jurisdiction only.

Visit **www.mtg.in** for buying books online.

EVERY GENUINE COPY OF THE BOOK HAS A HOLOGRAM STICKER ON THE COVER

Move the book slightly to see the special features of the hologram



PREFACE

We feel pleased and delighted in presenting the revised edition of the book "**CBSE Champion Chapterwise-Topicwise Mathematics**" as per the CBSE curriculum for the academic year 2020-21. Special efforts have been put to produce this book in order to equip students with practice material including previous 10 years CBSE Board Examination questions. It will give them comprehensive knowledge of subject according to the latest syllabus and pattern of CBSE Board Examination. The book will be helpful in imparting students a clear and vivid understanding of the subject.

Salient features

Comprehensive and Lucid Theory : Well explained theory with tables for quick recap.

Topicwise Graphical Analysis : Graphical analysis of previous 10 years CBSE Board papers' questions (VSA, SA, LA 1 and LA 2) provided to let students figure out which chapter and which topic is to be revised hard and how much is the weightage of that topic.

Chapterwise-Topicwise Questions and Answers : Theory is followed by chapterwise-topicwise questions pulled from previous 10 years CBSE DELHI, ALL INDIA, FOREIGN and COMPARTMENT papers. Answers are given according to the CBSE marking scheme.

Strictly Based on NCERT Pattern : Questions are segregated strictly according to NCERT syllabus. Topicwise questions are arranged in descending chronological (2020-2011) order so that latest questions come first in practice and revision.

Formulae Highlight : Formulae have been highlighted for their reinforcement.

Key Concepts Highlight : Key concepts have been highlighted for their reinforcement.

Practice Papers with Objective Type Questions : 10 Practice Papers strictly based on the latest design and blue print issued by CBSE Board are also incorporated. Various types of questions like MCQs, Fill in the blanks, VSA, SA and LA types are included.

Solved CBSE Sample Paper : Solved CBSE sample paper is included with the marking scheme.

We are sure that the value addition done in this book will prove helpful to students in achieving success in board examinations. Every possible effort has been made to make this book error free. Useful suggestions by our readers for the rectification and improvement of the book content would be gracefully acknowledged and incorporated in further editions.

Readers are welcome to send their suggestions at **editor@mtg.in.**

All The Best

MTG Editorial Board

CONTENTS

1.	Relations and Functions	.....	**1**
2.	Inverse Trigonometric Functions	.....	**13**
3.	Matrices	.....	**29**
4.	Determinants	.....	**44**
5.	Continuity and Differentiability	.....	**85**
6.	Application of Derivatives	.....	**111**
7.	Integrals	.....	**144**
8.	Application of Integrals	.....	**187**
9.	Differential Equations	.....	**202**
10.	Vector Algebra	.....	**227**
11.	Three Dimensional Geometry	.....	**249**
12.	Linear Programming	.....	**283**
13.	Probability	.....	**303**
●	10 Practice Papers	.....	**328**
●	CBSE Sample Paper	.....	**377**

SYLLABUS*

Units		No. of Periods	Marks
I.	Relations and Functions	30	08
II.	Algebra	50	10
III.	Calculus	80	35
IV.	Vectors and Three - Dimensional Geometry	30	14
V.	Linear Programming	20	05
VI.	Probability	30	08
	Total	**240**	**80**
	Internal Assessment	**–**	**20**

Unit-I: Relations and Functions

1. Relations and Functions **[15 Periods]**

Types of relations: reflexive, symmetric, transitive and equivalence relations. One to one and onto functions, composite functions, inverse of a function.

2. Inverse Trigonometric Functions **[15 Periods]**

Definition, range, domain, principal value branch. Graphs of inverse trigonometric functions. Elementary properties of inverse trigonometric functions.

Unit-II: Algebra

1. Matrices **[25 Periods]**

Concept, notation, order, equality, types of matrices, zero and identity matrix, transpose of a matrix, symmetric and skew symmetric matrices. Operation on matrices: Addition and multiplication and multiplication with a scalar. Simple properties of addition, multiplication and scalar multiplication. Non-commutativity of multiplication of matrices and existence of non-zero matrices whose product is the zero matrix (restrict to square matrices of order 2).Concept of elementary row and column operations. Invertible matrices and proof of the uniqueness of inverse, if it exists; (Here all matrices will have real entries).

2. Determinants **[25 Periods]**

Determinant of a square matrix (up to 3×3 matrices), properties of determinants, minors, co-factors and applications of determinants in finding the area of a triangle. Adjoint and inverse of a square matrix. Consistency, inconsistency and number of solutions of system of linear equations by examples, solving system of linear equations in two or three variables (having unique solution) using inverse of a matrix.

Unit-III: Calculus

1. Continuity and Differentiability **[20 Periods]**

Continuity and differentiability, derivative of composite functions, chain rule, derivatives of inverse trigonometric functions, derivative of implicit functions. Concept of exponential and logarithmic functions.

Derivatives of logarithmic and exponential functions. Logarithmic differentiation, derivative of functions expressed in parametric forms. Second order derivatives. Rolle's and Lagrange's Mean Value Theorems (without proof) and their geometric interpretation.

**For latest details visit www.cbse.nic.in*

2. **Application of Derivatives** **[10 Periods]**

Applications of derivatives: rate of change of bodies, increasing/decreasing functions, tangents and normals, use of derivatives in approximation, maxima and minima (first derivative test motivated geometrically and second derivative test given as a provable tool). Simple problems (that illustrate basic principles and understanding of the subject as well as real-life situations).

3. Integrals **[20 Periods]**

Integration as inverse process of differentiation. Integration of a variety of functions by substitution, by partial fractions and by parts, Evaluation of simple integrals of the following types and problems based on them.

$$\int \frac{dx}{x^2 \pm a^2}, \int \frac{dx}{\sqrt{x^2 \pm a^2}}, \int \frac{dx}{\sqrt{a^2 - x^2}}, \int \frac{dx}{ax^2 + bx + c}, \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$\int \frac{px + q}{ax^2 + bx + c} dx, \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx, \int \sqrt{a^2 \pm x^2} dx, \int \sqrt{x^2 - a^2} dx$$

$$\int \sqrt{ax^2 + bx + c}\, dx, \int (px + q)\sqrt{ax^2 + bx + c}\ dx$$

Definite integrals as a limit of a sum, Fundamental Theorem of Calculus (without proof). Basic properties of definite integrals and evaluation of definite integrals.

4. Application of Integrals **[15 Periods]**

Applications in finding the area under simple curves, especially lines, circles/parabolas/ellipses (in standard form only), Area between any of the two above said curves (the region should be clearly identifiable).

5. Differential Equations **[15 Periods]**

Definition, order and degree, general and particular solutions of a differential equation. Formation of differential equation whose general solution is given. Solution of differential equations by method of separation of variables, solutions of homogeneous differential equations of first order and first degree. Solutions of linear differential equation of the type:

$\frac{dy}{dx} + py = q$, where p and q are functions of x or constants.

$\frac{dx}{dy} + px = q$, where p and q are functions of y or constants.

Unit-IV:Vectors and Three-Dimensional Geometry

1. Vectors **[15 Periods]**

Vectors and scalars, magnitude and direction of a vector. Direction cosines and direction ratios of a vector. Types of vectors (equal, unit, zero, parallel and collinear vectors), position vector of a point, negative of a vector, components of a vector, addition of vectors, multiplication of a vector by a scalar, position vector of a point dividing a line segment in a given ratio. Definition, Geometrical Interpretation, properties and application of scalar (dot) product of vectors, vector (cross) product of vectors, scalar triple product of vectors.

2. Three-Dimensional Geometry **[15 Periods]**

Direction cosines and direction ratios of a line joining two points. Cartesian equation and vector equation of a line, coplanar and skew lines, shortest distance between two lines. Cartesian and vector equation of a plane. Angle between (i) two lines, (ii) two planes, (iii) a line and a plane. Distance of a point from a plane.

Unit-V: Linear Programming

1. Linear Programming **[20 Periods]**

Introduction, related terminology such as constraints, objective function, optimization, different types of linear programming (L.P.) problems, mathematical formulation of L.P. problems, graphical method of solution for problems in two variables, feasible and infeasible regions (bounded or unbounded), feasible and infeasible solutions, optimal feasible solutions (up to three non-trivial constraints).

Unit-VI: Probability

1. Probability **[30 Periods]**

Conditional probability, multiplication theorem on probability, independent events, total probability, Bayes' theorem, Random variable and its probability distribution, mean and variance of random variable.

MATHEMATICS QUESTION PAPER DESIGN CLASS - XII

(2020-21)

Time : 3 hours **Max. Marks :** 80

S. No.	Typology of Questions	Total Marks	% Weightage
1.	**Remembering :** Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers. **Understanding :** Demonstrate understanding of facts and ideas by organizing, comparing, translating, interpreting, giving descriptions, and stating main ideas	44	55
2.	**Applying :** Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	20	25
3.	**Analysing :** Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations **Evaluating :** Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria. **Creating :** Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions	16	20
	TOTAL	**80**	**100**

Note:

1. No chapter wise weightage. Care to be taken to cover all the chapters
2. Suitable internal variations may be made for generating various templates keeping the overall weightage to different form of questions and typology of questions same.

Choice(s):

1. There will be no overall choice in the question paper.
2. However, 33% internal choices will be given.

MATHEMATICS

LATEST 2020-21 PRACTICE PAPERS

As per the **CBSE Revised Curriculum** for the **Academic Year 2020-21** issued on 7th July 2020

REVISED SYLLABUS*

CLASS XII

Time : 3 Hours **Max. Marks :** 80

Unit	Title	No. of Periods	Marks
I	Relations and Functions	17	08
II	Algebra	35	10
III	Calculus	57	35
IV	Vectors and Three - Dimensional Geometry	26	14
V	Linear Programming	13	05
VI	Probability	20	08
	Total	**168**	**80**
	Internal Assessment		**20**

UNIT- I: RELATIONS AND FUNCTIONS

1. Relations and Functions (9 Periods)

Types of relations: reflexive, symmetric, transitive and equivalence relations. One to one and onto functions.

2. Inverse Trigonometric Functions (8 Periods)

Definition, range, domain, principal value branch.

UNIT-II : ALGEBRA

1. Matrices (17 Periods)

Concept, notation, order, equality, types of matrices, zero and identity matrix, transpose of a matrix, symmetric and skew symmetric matrices. Operation on matrices: Addition and multiplication and multiplication with a scalar. Simple properties of addition, multiplication and scalar multiplication. Non-commutativity of multiplication of matrices, Invertible matrices; (Here all matrices will have real entries).

2. Determinants (18 Periods)

Determinant of a square matrix (up to 3 x 3 matrices), minors, co-factors and applications of determinants in finding the area of a triangle. Adjoint and inverse of a square matrix. Solving system of linear equations in two or three variables (having unique solution) using inverse of a matrix.

UNIT-III : CALCULUS

1. Continuity and Differentiability (16 Periods)

Continuity and differentiability, derivative of composite functions, chain rule, derivative of inverse trigonometric functions, derivative of implicit functions. Concept of exponential and logarithmic functions.

Derivatives of logarithmic and exponential functions. Logarithmic differentiation, derivative of functions expressed in parametric forms. Second order derivatives.

2. Applications of Derivatives (7 Periods)

Applications of derivatives: increasing/decreasing functions, tangents and normals, maxima and minima (first derivative test motivated geometrically and second derivative test given as a provable tool). Simple problems (that illustrate basic principles and understanding of the subject as well as real-life situations).

3. Integrals (15 Periods)

Integration as inverse process of differentiation. Integration of a variety of functions by substitution, by partial fractions and by parts, Evaluation of simple integrals of the following types and problems based on them.

$$\int \frac{dx}{x^2 \pm a^2}, \int \frac{dx}{\sqrt{x^2 \pm a^2}}, \int \frac{dx}{\sqrt{a^2 - x^2}}, \int \frac{dx}{ax^2 + bx + c}, \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$\int \frac{px + q}{ax^2 + bx + c} dx, \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx, \int \sqrt{a^2 \pm x^2} dx, \int \sqrt{x^2 - a^2} dx$$

Fundamental Theorem of Calculus (without proof).Basic properties of definite integrals and evaluation of definite integrals.

4. Applications of the Integrals (9 Periods)

Applications in finding the area under simple curves, especially lines, parabolas; area of circles /ellipses (in standard form only) (the region should be clearly identifiable).

5. Differential Equations (10 Periods)

Definition, order and degree, general and particular solutions of a differential equation. Solution of differential equations by method of separation of variables, solutions of homogeneous differential equations of first order and first degree of the type: $\frac{dy}{dx} = f(y / x)$. Solutions of linear differential equation of

**For latest information refer to www.cbse.nic.in*

the type : $\frac{dy}{dx} + py = q$, where p and q are functions of x or constant.

UNIT-IV : VECTORS AND THREE-DIMENSIONAL GEOMETRY

1. Vectors (13 Periods)

Vectors and scalars, magnitude and direction of a vector. Direction cosines and direction ratios of a vector. Types of vectors (equal, unit, zero, parallel and collinear vectors), position vector of a point, negative of a vector, components of a vector, addition of vectors, multiplication of a vector by a scalar, position vector of a point dividing a line segment in a given ratio. Definition, Geometrical Interpretation, properties and application of scalar (dot) product of vectors, vector (cross) product of vectors.

2. Three - dimensional Geometry (13 Periods)

Direction cosines and direction ratios of a line joining two points. Cartesian equation and vector equation of a line, coplanar and skew lines, shortest distance between two lines. Cartesian and vector equation of a plane. Distance of a point from a plane.

UNIT-V : LINEAR PROGRAMMING

1. Linear Programming (13 Periods)

Introduction, related terminology such as constraints, objective function, optimization, different types of linear programming (L.P.) problems. Graphical method of solution for problems in two variables, feasible and infeasible regions (bounded), feasible and infeasible solutions, optimal feasible solutions (up to three non-trivial constraints).

UNIT-VI : PROBABILITY

1. Probability (20 Periods)

Conditional probability, multiplication theorem on probability, independent events, total probability, Bayes' theorem, Random variable and its probability distribution.

MATHEMATICS (CODE NO. - 041)
Question Paper Design Class - XII
(2020-21)

Time : 3 hours **Max. Marks :** 80

S. No.	Typology of Questions	Total Marks	% Weightage
1.	**Remembering :** Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers. **Understanding :** Demonstrate understanding of facts and ideas by organizing, comparing, translating, interpreting, giving descriptions, and stating main ideas.	44	55
2.	**Applying :** Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	20	25
3.	**Analysing :** Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations. **Evaluating :** Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria. **Creating :** Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions.	16	20
	Total	**80**	**100**

1. *No chapter wise weightage. Care to be taken to cover all the chapters.*
2. *Suitable internal variations may be made for generating various templates keeping the overall weightage to different form of questions and typology of questions same.*

Choice(s):

There will be no overall choice in the question paper.
However, 33% internal choices will be given in all the sections.

INTERNAL ASSESSMENT	20 Marks
Periodic Tests (Best 2 out of 3 tests conducted)	10 Marks
Mathematics Activities	10 Marks

Note: For activities NCERT Lab Manual may be referred.

◈◈◈

EXAM BITES

This Pdf Is
Downloaded From
www.exambites.in

Visit www.exambites.in for
More Premium Stuffs,Latest
Books,Test Papers,Lectures etc.


jeeneetadda


jeeneetadda_official


jeeneetadda

VISIT NOW !!

PRACTICE PAPER 1
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) This question paper comprises four Sections A, B, C and D. This question paper carries 36 questions. All questions are compulsory.

(ii) Section A : Questions no. 1 to 20 comprises of 20 questions of 1 mark each.

(iii) Section B : Questions no. 21 to 26 comprises of 6 questions of 2 marks each.

(iv) Section C : Questions no. 27 to 32 comprises of 6 questions of 4 marks each.

(v) Section D : Questions no. 33 to 36 comprises of 4 questions of 6 marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 3 questions of one mark, 2 questions of two marks, 2 questions of four marks and 2 questions of six marks. Only one of the choices in such questions have to be attempted.

(vii) In addition to this, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

(Q.1-Q.10) are multiple choice questions. Select the correct option.

1. If the set A contains 5 elements and the set B contains 6 elements, then the number of one-one and onto mappings from A to B is

(a) 720 (b) 120

(c) 0 (d) None of these

2. If $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40}\begin{bmatrix} 5 & 10 & -5 \\ -5 & -2 & 13 \\ 10 & -4 & 6 \end{bmatrix}\begin{bmatrix} 5 \\ 0 \\ 5 \end{bmatrix}$, then the value of $x + y + z$ is

(a) 3 (b) 0 (c) 2 (d) 1

3. The area of a triangle with vertices (–3, 0), (3, 0) and (0, k) is 9 sq. units, the value of k will be

(a) 9 (b) 3 (c) –9 (d) 6

4. If $y = e^{x+e^{x+e^{x+\ldots \text{to}\infty}}}$, $\frac{dy}{dx}$ is equal to

(a) $\frac{y^2}{1-y}$ (b) $\frac{y^2}{y-1}$ (c) $\frac{y}{1-y}$ (d) $\frac{-y}{1-y}$

5. The normal at the point (1, 1) on the curve $2y + x^2 = 3$ is

(a) $x + y = 0$ (b) $x - y = 0$

(c) $x + y + 1 = 0$ (d) $-x - y = 0$

6. The anti derivative of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ equals

(a) $\frac{1}{3}x^{1/3} + 2x^{1/2} + C$ (b) $\frac{2}{3}x^{2/3} + \frac{1}{2}x^2 + C$

(c) $\frac{2}{3}x^{3/2} + 2x^{1/2} + C$ (d) $\frac{3}{2}x^{3/2} + \frac{1}{2}x^{1/2} + C$

7. If $f(x) = \int_0^x t \sin t\, dt$, then $f'(x)$ is

(a) $\cos x + x\sin x$ (b) $x\sin x$

(c) $x\cos x$ (d) $\sin x + x\cos x$

8. If $\left(\frac{1}{2}, \frac{1}{3}, n\right)$ are the direction cosines of a line, then the value of n is

(a) $\frac{\sqrt{23}}{6}$ (b) $\frac{23}{6}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

9. If $|\vec{a}| = 4$ and $-3 \le \lambda \le 2$, then the range of $|\lambda\vec{a}|$ is

(a) [0, 8] (b) [–12, 8]

(c) [0, 12] (d) [8, 12]

10. Let $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$. then $P(A'|B)$ is equal to

(a) $\frac{6}{13}$ (b) $\frac{4}{13}$ (c) $\frac{4}{9}$ (d) $\frac{5}{9}$

(Q.11-Q.15) Fill in the blanks.

11. _____ matrix is both symmetric and skew-symmetric matrix.

OR

If A is a symmetric matrix, then A^3 is a _____ matrix.

12. If $A=\begin{bmatrix}3 & 4\\ 1 & 2\end{bmatrix}$, find the value of $3|A|$.

13. Let A and B be two events. If $P(A|B) = P(A)$, then A is _____ of B.

OR

If A and B are independent, then P (exactly one of A, B occurs) = $P(A)\,P(B') +$ _____ .

14. In a LPP, the objective function is always _____ .

15. The unit vector normal to the plane $x + 2y + 3z - 6 = 0$ is _____ .

(Q.16-Q.20) Answer the following questions.

16. Find the principal value of $\sec^{-1}(-2)$.

OR

Find the set of values of $\sec^{-1}\left(\frac{1}{2}\right)$.

17. Write the value of $\int 2^x\,dx$.

18. Write the value of the following integral

$$\int_{-\pi/2}^{\pi/2} \sin^5 x\,dx.$$

19. What is the degree of the given differential equation $5x\left(\frac{dy}{dx}\right)^2 - \frac{d^2y}{dx^2} - 6y = \log x$?

20. The cartesian equation of a line is $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$. Write its vector form.

SECTION - B

21. Find the equation of the plane with intercept 3 on the y-axis and parallel to ZOX plane.

OR

Find the intercepts cut off by the plane $2x + y - z = 5$.

22. Find the domain of $\cos^{-1}(2x - 1)$.

23. Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P(A|B) = \frac{2}{5}$.

24. Is the function defined by

$$f(x)=\begin{cases}x+5, & \text{if } x \le 1\\ x-5, & \text{if } x > 1\end{cases}$$ a continuous function?

OR

Is the function defined by $f(x) = x^2 - \sin x + 5$ continuous at $x = \pi$?

25. Show that $|\vec{a}|\vec{b}+|\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b}-|\vec{b}|\vec{a}$, for any two non-zero vectors $\vec{a}$ and $\vec{b}$.

26. Prove that the logarithmic function is strictly increasing on $(0, \infty)$.

SECTION - C

27. Let $f: X \to Y$ be a function, define a relation R on X given by $R = \{(a, b) : f(a) = f(b)\}$. Show that R is an equivalence relation on X.

28. Show that the function defined as follows, is continuous at $x = 2$, but not differentiable.

$$f(x)=\begin{cases}3x-2, & 0<x<1\\ 2x^2-x, & 1<x\le 2\\ 5x-4, & x>2\end{cases}$$

OR

If $y=\cos^{-1}\left[\frac{3x+4\sqrt{1-x^2}}{5}\right]$, find $\frac{dy}{dx}$.

29. Evaluate : $\int e^x\left(\frac{\sin 4x-4}{1-\cos 4x}\right)dx$

30. Show that the following differential equation is homogeneous and then solve it.

$$y\,dx + x\log\left(\frac{y}{x}\right)dy - 2x\,dy = 0.$$

31. There are two bags, bag I and bag II. Bag I contains 4 white and 3 red balls while another bag II contains 3 white and 7 red balls. One ball is drawn at random from one of the bags and it is found to be white. Find the probability that it was drawn from bag I.

www.examglues.in

OR

A girl throws a die. If she gets a 5 or 6, she tosses a coin three times and notes the number of heads. If she gets 1, 2, 3 or 4, she tosses a coin two times and notes the number of heads obtained. If she obtained exactly two heads, what is the probability that she threw 1, 2, 3 or 4 with the die?

32. Maximise $Z = 8x + 9y$ subject to the constraints given below :
$2x + 3y \le 6$, $3x - 2y \le 6$, $y \le 1$; $x, y \ge 0$

SECTION - D

33. Draw a rough sketch of the given curve $y = 1 + |x + 1|$, $x = -3$, $x = 3$, $y = 0$ and find the area of the region bounded by them, using integration.

OR

Sketch the graph of $y = |x + 3|$ and evaluate $\int_{-6}^{0} |x+3|\, dx$.

34. Write the vector equations of the following lines and hence determine the distance between them.

$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$ and $\frac{x-3}{4} = \frac{y-3}{6} = \frac{z+5}{12}$.

35. If $A = \begin{bmatrix} 8 & -4 & 1 \\ 10 & 0 & 6 \\ 8 & 1 & 6 \end{bmatrix}$, find A^{-1}. Hence solve the following system of equations :

$8x - 4y + z = 5$, $10x + 6z = 4$, $8x + y + 6z = \frac{5}{2}$

OR

Solve the system of the following equations

$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$, $\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$, $\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$

36. Find the maximum area of an isosceles triangle inscribed in the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ with its vertex at one end of the major axis.

Detailed Solutions

1. (c) : As we know, if A and B are two non-empty finite sets containing m and n elements respectively, then the number of one-one and onto mapping from A to B is n, if $m = n$ and 0, if $m \ne n$
Given, $m = 5$ and $n = 6 \quad \therefore \quad m \ne n$
So, number of mapping = 0

2. (a) : Given, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40}\begin{bmatrix} 5 & 10 & -5 \\ -5 & -2 & 13 \\ 10 & -4 & 6 \end{bmatrix}\begin{bmatrix} 5 \\ 0 \\ 5 \end{bmatrix}$,

$= \frac{1}{40}\begin{bmatrix} 25+0-25 \\ -25+0+65 \\ 50+0+30 \end{bmatrix} = \frac{1}{40}\begin{bmatrix} 0 \\ 40 \\ 80 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

$\Rightarrow \quad x = 0, y = 1, z = 2$
$\therefore \quad x + y + z = 0 + 1 + 2 = 3$

3. (b) : Since, area of a triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is

$\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad \therefore \quad \Delta = \frac{1}{2}\begin{vmatrix} -3 & 0 & 1 \\ 3 & 0 & 1 \\ 0 & k & 1 \end{vmatrix}$

Expanding along C_2, we get

$9 = \frac{1}{2}[-k(-3-3)] \Rightarrow 9 = \frac{1}{2} \times 6k \Rightarrow \quad k = \frac{9}{3} = 3$

4. (c) : We may write the given series as

$y = e^{x+y} \Rightarrow \log y = (x + y)$ ...(i)

Differentiating both sides of (i) w.r.t. x, we get

$\frac{1}{y} \cdot \frac{dy}{dx} = 1 + \frac{dy}{dx} \Rightarrow \left(\frac{1}{y} - 1\right)\frac{dy}{dx} = 1$

$\Rightarrow \quad \frac{(1-y)}{y}\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{y}{(1-y)}$.

5. (b) : We have $2y + x^2 = 3$ ...(i)

Differentiating (i) w.r.t. x, we get

$2\frac{dy}{dx} + 2x = 0 \Rightarrow \frac{dy}{dx} = -x.$

$\because \quad \left.\frac{dy}{dx}\right|_{(1,1)} = -1.$

$\therefore$ Slope of normal = 1
$\therefore$ The equation of normal is
$y - 1 = 1 \cdot (x - 1) \Rightarrow x - y = 0.$

www.exambites.in

6. **(c) :** Let $I = \int\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)dx$

$= \int\left(x^{1/2} + x^{-1/2}\right)dx = \frac{2}{3}x^{3/2} + 2x^{1/2} + C$

7. **(b) :** Let $f(x) = \int_0^x t\sin t\,dt$

$f(x) = [t(-\cos t)]_0^x - \int_0^x 1\cdot(-\cos t)dt$

$= [-t\cos t + \sin t]_0^x$

$f(x) = -x\cos x + \sin x$

$\therefore\ f'(x) = -[x(-\sin x) + \cos x] + \cos x$

$= x\sin x - \cos x + \cos x = x\sin x$

8. **(a) :** If $\left(\frac{1}{2}, \frac{1}{3}, n\right)$ are the dc's of the line, then using the relation $l^2 + m^2 + n^2 = 1$, we get

$\left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 + n^2 = 1 \Rightarrow n = \frac{\sqrt{23}}{6}$

9. **(c) :** Since, $-3 \le \lambda \le 2$

$\Rightarrow 0 \le |\lambda| \le 3 \Rightarrow 0 \le 4|\lambda| \le 12$

$\Rightarrow |\lambda\vec{a}| \in [0, 12]. \quad [\because |\lambda\vec{a}| = |\lambda||\vec{a}| = 4|\lambda|]$

10. **(d) :** We have, $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$

Now, $P(A'|B) = \frac{P(A' \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)}$

$= \frac{\frac{9}{13} - \frac{4}{13}}{\frac{9}{13}} = \frac{5}{9}$

11. Null matrix is both symmetric and skew-symmetric matrix.

OR

Since, A is a symmetric matrix,

$\therefore\ A' = A$

Now, $(A^3)' = (A')^3$

$= A^3 \quad [\because (A')^n = (A^n)']$

Hence, A^3 is a symmetric matrix.

12. Here, $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$

$\Rightarrow |A| = \begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix} = 3\times 2 - 1\times 4 = 2$

$\Rightarrow 3|A| = 3 \times 2 = 6.$

13. Since $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$\Rightarrow P(A) = \frac{P(A \cap B)}{P(B)} \quad$ [Given, $P(A|B) = P(A)$]

$\Rightarrow P(A)\cdot P(B) = P(A \cap B)$

Therefore, A is independent of B.

OR

$\because$ P (exactly one of A, B occurs)

$= P(A \cap B') + P(B \cap A')$

$= P(A)P(B') + P(B)P(A')$

$[\because A$ and B are independent events$]$

14. In a LPP, the objective function is always linear.

15. Since, $\vec{n} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\therefore\ \hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{\hat{i} + 2\hat{j} + 3\hat{k}}{\sqrt{1^2 + 2^2 + 3^2}} = \frac{1}{\sqrt{14}}\hat{i} + \frac{2}{\sqrt{14}}\hat{j} + \frac{3}{\sqrt{14}}\hat{k}$

16. Let $\sec^{-1}(-2) = y$.

$\Rightarrow \sec y = -2 = -\sec\left(\frac{\pi}{3}\right) = \sec\left(\pi - \frac{\pi}{3}\right) = \sec\left(\frac{2\pi}{3}\right)$

We know that the range of principal value branch of $\sec^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ and $\sec\left(\frac{2\pi}{3}\right) = -2$

Hence, principal value of $\sec^{-1}(-2) = \frac{2\pi}{3}$

OR

As we know, domain of $\sec^{-1}x$ is $R - (-1, 1)$

$\therefore\ x \in (-\infty, -1] \cup [1, \infty)$

Therefore, there is no set of values exist for $\sec^{-1}\frac{1}{2}$.

Hence, ϕ is the answer.

17. $\int 2^x\,dx = \frac{2^x}{\log 2} + C$

18. Let $I = \int_{-\pi/2}^{\pi/2} \sin^5 x\,dx$

Let $f(x) = \sin^5 x \Rightarrow f(-x) = \sin^5(-x) = -\sin^5 x = -f(x)$

$\therefore\ I = 0 \quad [\because f(x)$ is an odd function$]$

19. Degree of the given differential equation is 1.

20. The given cartesian equation (in symmetrical form) is

$\frac{x-5}{3} = \frac{y-(-4)}{7} = \frac{z-6}{2}$

www.exambits.in

i.e., The line passes through the point (5, – 4, 6) and is parallel to vector $3\hat{i} + 7\hat{j} + 2\hat{k}$. Hence, the vector equation of the line is

$$\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} + 2\hat{k}).$$

21. Clearly, the plane passes through the point (0, 3, 0). The direction cosines of the normal to the plane are < 0, 1, 0 >

$\therefore$ The equation of the plane is

$0 \cdot (x - 0) + 1 \cdot (y - 3) + 0 \cdot (z - 0) = 0$

$\Rightarrow \; y - 3 = 0 \Rightarrow y = 3.$

OR

The given equation of the plane is

$$2x + y - z = 5 \Rightarrow \frac{x}{5/2} + \frac{y}{5} + \frac{z}{-5} = 1.$$

Comparing it with $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, we get the intercepts on the axes are $\frac{5}{2}$, 5 and –5.

22. Let $f(x) = \cos^{-1}(2x - 1)$

Since $-1 \le 2x - 1 \le 1 \Rightarrow 0 \le 2x \le 2 \;\Rightarrow\; 0 \le x \le 1$

23. Given $P(A \mid B) = \frac{2}{5} \;\Rightarrow\; \frac{P(A \cap B)}{P(B)} = \frac{2}{5}$

$$\Rightarrow \quad P(A \cap B) = \frac{2}{5}P(B) = \frac{2}{5} \times \frac{5}{13} = \frac{2}{13}$$

Hence,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= \frac{1}{2} \times \frac{5}{13} + \frac{5}{13} - \frac{2}{13} = \frac{11}{26}.$$

24. We observe that f is continuous at all real numbers $x < 1$ and $x > 1$.

Now, continuity at $x = 1$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (x - 5) = \lim_{h \to 0} (1 + h - 5)$$

$$= \lim_{h \to 0} (h - 4) = -4$$

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (x + 5) = \lim_{h \to 0} (1 - h + 5)$$

$$= \lim_{h \to 0} (6 - h) = 6$$

Thus, $\lim_{x \to 1^+} f(x) \ne \lim_{x \to 1^-} f(x)$

$\therefore$ $f(x)$ is not continuous at $x = 1$.

OR

At $x = \pi$

$$\lim_{x \to \pi^+} f(x) = \lim_{h \to 0} [(\pi + h)^2 - \sin(\pi + h) + 5]$$

$$= \lim_{h \to 0} \left[(\pi^2 + h^2 + 2\pi h) + (\sin h + 5) \right] = \pi^2 + 5$$

$$\lim_{x \to \pi^-} f(x) = \lim_{h \to 0} [(\pi - h)^2 - \sin(\pi - h) + 5]$$

$$= \lim_{h \to 0} [(\pi^2 + h^2 - 2\pi h) - \sin h + 5] = \pi^2 + 5$$

$f(\pi) = \pi^2 + 5$

$\because$ R.H.L. = L.H.L. = $f(\pi)$

$\therefore$ Function is continuous at $x = \pi$.

25. Let $\vec{c} = |\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ and $\vec{d} = |\vec{a}|\vec{b} - |\vec{b}|\vec{a}$

$\therefore \quad \vec{c} \cdot \vec{d} = (|\vec{a}|\vec{b} + |\vec{b}|\vec{a}) \cdot (|\vec{a}|\vec{b} - |\vec{b}|\vec{a})$

$= |\vec{a}|^2 \vec{b} \cdot \vec{b} - |\vec{a}||\vec{b}|\vec{b} \cdot \vec{a} + |\vec{b}||\vec{a}|\vec{a} \cdot \vec{b} - |\vec{b}|^2 \vec{a} \cdot \vec{a}$

$= |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}||\vec{b}|\vec{a} \cdot \vec{b} + |\vec{a}||\vec{b}|\vec{a} \cdot \vec{b} - |\vec{b}|^2 |\vec{a}|^2$

$= 0 \qquad [\because \; \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}]$

Hence, $\vec{c}$ is perpendicular to $\vec{d}$.

26. We have $f(x) = \log x$ ...(i)

Domain of f is $(0, \infty)$

Now, $f'(x) = \frac{1}{x} > 0$ for all $x \in (0, \infty)$

$\Rightarrow$ $f'(x) > 0$ for all $x \in (0, \infty)$

$\therefore$ f is strictly increasing on $(0, \infty)$

(Note that $\log x$ is defined only for $x > 0$)

27. We have, $f : X \to Y$ is a function

$R = \{(a, b) : f(a) = f(b)\}$

(i) Reflexivity : For any $a \in X$, we have

$f(a) = f(a) \Rightarrow (a, a) \in R \Rightarrow R$ is reflexive.

(ii) Symmetric : For any $a, b \in X$,

Let $(a, b) \in R \Rightarrow f(a) = f(b)$

$\Rightarrow$ $f(b) = f(a) \Rightarrow (b, a) \in R$

So, R is symmetric.

(iii) Transitive : For any $a, b, c \in X$,

Let $(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow$ $f(a) = f(b)$ and $f(b) = f(c)$

$\Rightarrow$ $f(a) = f(c) \Rightarrow (a, c) \in R$

So, R is transitive.

Hence, R is an equivalence relation on X.

28. L.H.L. $= \lim_{x \to 2^-} f(x) = \lim_{x \to 2} (2x^2 - x) = 6$

R.H.L. $= \lim_{x \to 2^+} f(x) = \lim_{x \to 2} (5x - 4) = 6$

Also, $f(2) = 2(2)^2 - 2 = 6$

As, $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = f(2)$

$\therefore$ $f(x)$ is continuous at $x = 2$.

www.exambites.in

Test of differentiability :

We have, $Lf'(2) = \lim_{h\to 0} \frac{f(2-h)-f(2)}{-h}$

$= \lim_{h\to 0} \frac{2(2-h)^2-(2-h)-6}{-h}$

$= \lim_{h\to 0} \frac{8+2h^2-8h+h-8}{-h}$

$= \lim_{h\to 0} \frac{2h^2-7h}{-h} = \lim_{h\to 0} \frac{-h(-2h+7)}{-h} = 7$

$Rf'(2) = \lim_{h\to 0} \frac{f(2+h)-f(2)}{h}$

$= \lim_{h\to 0} \frac{5(2+h)-4-6}{h}$

$= \lim_{h\to 0} \frac{10+5h-10}{h} = \lim_{h\to 0} \frac{5h}{h} = 5$

$\because \quad Lf'(2) \neq Rf'(2)$

Hence, $f(x)$ is not differentiable at $x = 2$.

OR

We have, $y = \cos^{-1}\left[\frac{3x+4\sqrt{1-x^2}}{5}\right]$

Putting $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$, we get

$y = \cos^{-1}\left[\frac{3\sin\theta+4\cos\theta}{5}\right]$

$\Rightarrow \quad y = \cos^{-1}\left[\frac{3}{5}\sin\theta+\frac{4}{5}\cos\theta\right]$

Let $\frac{3}{5} = \sin\alpha$ and $\frac{4}{5} = \cos\alpha$

$\Rightarrow \quad y = \cos^{-1}[\sin\alpha\sin\theta + \cos\alpha\cos\theta]$

$\Rightarrow \quad y = \cos^{-1}[\cos(\alpha - \theta)] \Rightarrow y = \alpha - \theta$

$\Rightarrow \quad y = \alpha - \sin^{-1}x$

$\therefore \quad \frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$

29. Let $I = \int e^x\left(\frac{\sin 4x-4}{1-\cos 4x}\right)dx$

$\Rightarrow \quad I = \int e^x\left(\frac{\sin 4x}{1-\cos 4x} - \frac{4}{1-\cos 4x}\right)dx$

$\Rightarrow \quad I = \int e^x\left(\frac{2\sin 2x\cos 2x}{2\sin^2 2x} - \frac{4}{2\sin^2 2x}\right)dx$

$\Rightarrow \quad I = \int e^x\left(\cot 2x - 2\operatorname{cosec}^2 2x\right)dx$

Consider, $f(x) = \cot 2x \Rightarrow f'(x) = -2\operatorname{cosec}^2 2x$

$\therefore \quad I = \int e^x(\cot 2x - 2\operatorname{cosec}^2 2x)dx = e^x\cot 2x + C$

30. We have, $y\,dx + x\log\left(\frac{y}{x}\right)dy - 2x\,dy = 0$

$\Rightarrow \quad \frac{y}{x} + \log\left(\frac{y}{x}\right)\frac{dy}{dx} - 2\frac{dy}{dx} = 0$

$\Rightarrow \quad \frac{dy}{dx} = \frac{y/x}{2-\log(y/x)}$ ...(1)

This is a homogeneous differential equation.

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$, then (1) becomes

$v + x\frac{dv}{dx} = \frac{v}{2-\log v}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{v}{2-\log v} - v$

$\Rightarrow \quad \frac{(2-\log v)dv}{v\log v - v} = \frac{dx}{x}$

$\Rightarrow \quad \frac{1-(\log v-1)}{v(\log v-1)}dv = \frac{dx}{x}$

Integrating both sides, we get

$\int\left(\frac{1}{v(\log v-1)} - \frac{1}{v}\right)dv = \int\frac{dx}{x}$

$\Rightarrow \quad \log|\log v-1| - \log|v| = \log|x| + C_1$

$\Rightarrow \quad \log\left|\frac{\log v-1}{vx}\right| = C_1 \Rightarrow \left|\frac{\log v-1}{vx}\right| = e^{C_1}$

$\Rightarrow \quad \frac{\log v-1}{vx} = \pm e^{C_1} = C(\text{say})$

$\Rightarrow \quad \frac{\log\frac{y}{x}-1}{y} = C \Rightarrow \log\frac{y}{x} - 1 = Cy$

$\Rightarrow \log\frac{y}{x} = 1 + Cy$

which is the required general solution.

31. Let E_1 and E_2 be the events to select bag I and bag II respectively.

$\therefore \quad P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$

Let E be the event that drawn ball is white.

$P(E|E_1) = \frac{4}{7}$ and $P(E|E_2) = \frac{3}{10}$

$\therefore \quad P$ (white ball was drawn from bag I)

www.exambites.in

$$= P(E_1|E) = \frac{P(E_1)P(E|E_1)}{P(E_1)P(E|E_1) + P(E_2)P(E|E_2)}$$

$$= \frac{\frac{1}{2}\cdot\frac{4}{7}}{\frac{1}{2}\cdot\frac{4}{7}+\frac{1}{2}\cdot\frac{3}{10}} = \frac{\frac{4}{7}}{\frac{4}{7}+\frac{3}{10}} = \frac{\frac{4}{7}}{\frac{40+21}{70}} = \frac{40}{61}$$

OR

Let A_1, A_2 and A be the following events :

A_1 : The girl gets 5 or 6 and hence tosses a coin 3-times.

A_2 : The girl gets 1, 2, 3 or 4 and hence tosses a coin 2-times.

A : The girl gets exactly two heads.

Now, $P(A_1) = P(5 \text{ or } 6) = P(5) + P(6) = \frac{1}{6}+\frac{1}{6} = \frac{1}{3}$

$P(A_2) = P(1) + P(2) + P(3) + P(4)$

$$= \frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6} = \frac{2}{3}$$

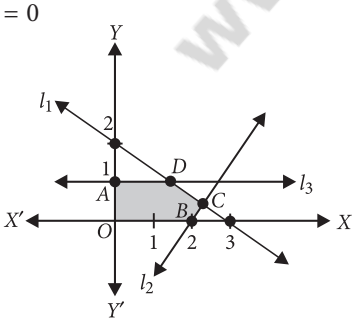
$$P(A|A_1) = \frac{3}{8},\ P(A|A_2) = \frac{1}{4}$$

$\therefore$ Required probability

$$= P(A_2|A) = \frac{P(A_2)\cdot P(A|A_2)}{P(A_1)\cdot P(A|A_1) + P(A_2)\cdot P(A|A_2)}$$

$$= \frac{\frac{2}{3}\times\frac{1}{4}}{\frac{1}{3}\times\frac{3}{8}+\frac{2}{3}\times\frac{1}{4}} = \frac{\frac{1}{6}}{\frac{1}{8}+\frac{1}{6}} = \frac{\frac{1}{6}}{\frac{3+4}{24}} = \frac{4}{7}$$

32. Let $l_1 : 2x + 3y = 6$, $l_2 : 3x - 2y = 6$, $l_3 : y = 1$; $x = 0$, $y = 0$



Solving l_1 and l_3, we get D (1.5, 1)

Solving l_1 and l_2, we get $C\left(\frac{30}{13},\frac{6}{13}\right)$

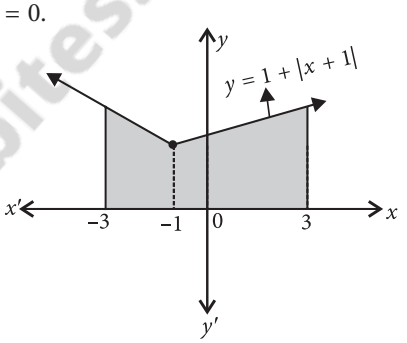
Shaded portion $OADCB$ is the feasible region, where coordinates of the corner points are $O(0, 0)$, $A(0, 1)$, $D(1.5, 1)$, $C\left(\frac{30}{13},\frac{6}{13}\right)$, $B(2, 0)$.

The value of the objective function at these points are :

Corner Points	Value of the objective function $Z = 8x + 9y$
O (0, 0)	$8 \times 0 + 9 \times 0 = 0$
A (0, 1)	$8 \times 0 + 9 \times 1 = 9$
D (1.5, 1)	$8 \times 1.5 + 9 \times 1 = 21$
$C\left(\frac{30}{13},\frac{6}{13}\right)$	$8\times\frac{30}{13}+9\times\frac{6}{13}=22.6$ (Maximum)
B (2, 0)	$8 \times 2 + 9 \times 0 = 16$

The maximum value of z is 22.6, which is at $C\left(\frac{30}{13},\frac{6}{13}\right)$

33. We have given, $y = 1 + |x + 1|$, $x = -3$, $x = 3$ and $y = 0$.



Now, $y = \begin{cases} -x, & \text{if } x < -1 \\ x + 2, & \text{if } x \geq -1 \end{cases}$

Thus the area of the shaded region

$$= \int_{-3}^{-1} -x\,dx + \int_{-1}^{3} (x+2)\,dx$$

$$= -\left[\frac{x^2}{2}\right]_{-3}^{-1} + \left[\frac{x^2}{2}+2x\right]_{-1}^{3}$$

$$= -\left[\frac{1}{2}-\frac{9}{2}\right] + \left[\frac{9}{2}+6-\frac{1}{2}+2\right]$$

$= -[-4] + [8 + 4]$

$= 4 + 12 = 16$ sq. units

OR

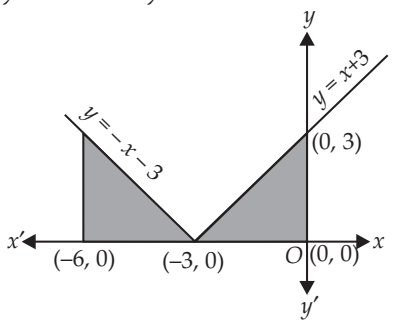
We have, $y = |x + 3|$

$$y = |x+3| = \begin{cases} x+3, & \text{if } x+3 \geq 0 \\ & \Rightarrow x \geq -3 \\ -(x+3), & \text{if } x+3 < 0 \\ & \Rightarrow x < -3 \end{cases}$$

www.exambites.in

$$= \begin{cases} x+3, & \text{if} \quad x \geq -3 \\ -x-3, & \text{if} \quad x < -3 \end{cases}$$

Hence there are two straight lines,
i.e., $y = x + 3$ and $y = -x - 3$



Now, $\int_{-6}^{0} |x+3|\,dx$

$$= \int_{-6}^{-3} (-x-3)dx + \int_{-3}^{0} (x+3)dx$$

$$= \left[-\frac{x^2}{2} - 3x\right]_{-6}^{-3} + \left[\frac{x^2}{2} + 3x\right]_{-3}^{0}$$

$$= \left[\left(-\frac{9}{2}+9\right) - \left(-\frac{36}{2}+18\right)\right] + \left[(0+0) - \left(\frac{9}{2}-9\right)\right]$$

$$= \left(\frac{9}{2}+0\right) + \left(0+\frac{9}{2}\right) = 9$$

34. Given lines are

$$l_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6} \text{ and } l_2 : \frac{x-3}{4} = \frac{y-3}{6} = \frac{z+5}{12}$$

Line l_1 passes through (1, 2, – 4) and has direction ratios 2, 3, 6.
So its vector equation is

$$\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k}) \quad \text{...(i)}$$

Similarly, vector equation of line l_2 is

$$\vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(4\hat{i} + 6\hat{j} + 12\hat{k}) \quad \text{...(ii)}$$

Now, we can see that the lines are parallel.

So, $\vec{a}_1 = \hat{i} + 2\hat{j} - 4\hat{k}$

$\vec{a}_2 = 3\hat{i} + 3\hat{j} - 5\hat{k}$

$\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$

$\therefore$ Shortest distance between l_1 and l_2

$$= \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$$

$$= \frac{1}{\sqrt{2^2 + 3^2 + 6^2}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ (3-1) & (3-2) & (-5+4) \\ 2 & 3 & 6 \end{vmatrix}$$

$$= \frac{1}{\sqrt{49}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 2 & 3 & 6 \end{vmatrix}$$

$$= \frac{1}{7}\{\hat{i}(6+3) - \hat{j}(12+2) + \hat{k}(6-2)\}$$

$$= \frac{1}{7}\{9\hat{i} - 14\hat{j} + 4\hat{k}\}$$

35. We have, $A = \begin{bmatrix} 8 & -4 & 1 \\ 10 & 0 & 6 \\ 8 & 1 & 6 \end{bmatrix}$

$$\Rightarrow |A| = \begin{vmatrix} 8 & -4 & 1 \\ 10 & 0 & 6 \\ 8 & 1 & 6 \end{vmatrix}$$

$= 8(0 - 6) + 4(60 - 48) + 1(10 - 0)$
$= -48 + 48 + 10 = 10 \neq 0$
$\therefore$ A^{-1} exists.
Now, $A_{11} = -6$, $A_{12} = -12$, $A_{13} = 10$, $A_{21} = 25$, $A_{22} = 40$, $A_{23} = -40$, $A_{31} = -24$, $A_{32} = -38$, $A_{33} = 40$

$$\therefore \quad \text{adj } A = \begin{bmatrix} -6 & 25 & -24 \\ -12 & 40 & -38 \\ 10 & -40 & 40 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{10}\begin{bmatrix} -6 & 25 & -24 \\ -12 & 40 & -38 \\ 10 & -40 & 40 \end{bmatrix}$$

Now the given equations are
$8x - 4y + z = 5$
$10x + 6z = 4$

$$8x + y + 6z = \frac{5}{2}$$

Given system of equations can be written as
$AX = B$

where, $A = \begin{bmatrix} 8 & -4 & 1 \\ 10 & 0 & 6 \\ 8 & 1 & 6 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 5 \\ 4 \\ 5/2 \end{bmatrix}$

Since A^{-1} exists. So, system of equations has a unique solution given by $X = A^{-1}B$

www.exambres.in

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10}\begin{bmatrix} -6 & 25 & -24 \\ -12 & 40 & -38 \\ 10 & -40 & 40 \end{bmatrix}\begin{bmatrix} 5 \\ 4 \\ 5/2 \end{bmatrix}$$

$$= \frac{1}{10}\begin{bmatrix} 10 \\ 5 \\ -10 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \\ -1 \end{bmatrix} \Rightarrow x = 1, y = \frac{1}{2}, z = -1.$$

OR

The equations can be written in the form of $AX = B$, where

$$A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}, X = \begin{bmatrix} 1/x \\ 1/y \\ 1/z \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{vmatrix}$$

$= 2(120 - 45) - 3(-80 - 30) + 10(36 + 36)$

$= 150 + 330 + 720 = 1200 \neq 0$

$\therefore$ A^{-1} exists.

$A_{11} = 75, A_{12} = 110, A_{13} = 72, A_{21} = 150, A_{22} = -100,$
$A_{23} = 0, A_{31} = 75, A_{32} = 30, A_{33} = -24,$

$$\therefore \text{ adj } A = \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{|A|}\text{adj}\, A = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

As $AX = B \Rightarrow X = A^{-1}B$

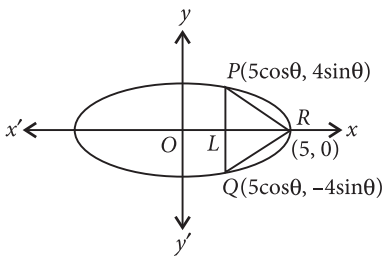
$$\Rightarrow \begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \\ \frac{1}{z} \end{bmatrix} = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 300 + 150 + 150 \\ 440 - 100 + 60 \\ 288 + 0 - 48 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \\ \frac{1}{z} \end{bmatrix} = \frac{1}{1200}\begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$

Hence, $x = 2, y = 3, z = 5$

36. Equation of the ellipse is $\frac{x^2}{5^2} + \frac{y^2}{4^2} = 1$, then any point P on the ellipse is $(5\cos\theta, 4\sin\theta)$.



From P, draw a line parallel to y-axis and produce it to meet the ellipse at Q, then PRQ is an isosceles triangle. Let A be its area, then

$$A = \frac{1}{2}PQ \cdot RL = \frac{1}{2}(2 \cdot 4\sin\theta)(5 - 5\cos\theta)$$

$\Rightarrow$ $A = (5 - 5\cos\theta) \times 4\sin\theta$

$\Rightarrow$ $A = 20(\sin\theta - \sin\theta\cos\theta)$

$$\Rightarrow A = 20\left(\sin\theta - \frac{1}{2}\sin 2\theta\right) \quad \text{...(i)}$$

Differentiating (i) w.r.t. θ, we get

$$\frac{dA}{d\theta} = 20(\cos\theta - \cos 2\theta) \quad \text{...(ii)}$$

For maxima or minima,

$$\frac{dA}{d\theta} = 0 \Rightarrow \cos\theta = \cos 2\theta$$

$\Rightarrow$ $\cos 2\theta = \cos(2\pi - \theta)$

$$\Rightarrow 2\theta = 2\pi - \theta \Rightarrow \theta = \frac{2\pi}{3} \quad (\because \theta \neq 0)$$

Differentiating (ii) w.r.t. θ, we get

$$\frac{d^2A}{d\theta^2} = 20(-\sin\theta + 2\sin 2\theta) \quad \text{...(iii)}$$

$$\text{Now, } \left(\frac{d^2A}{d\theta^2}\right)_{\theta=\frac{2\pi}{3}} = 20\left(-\sin\frac{2\pi}{3} + 2\sin\frac{4\pi}{3}\right)$$

$$= 20\left(-\frac{\sqrt{3}}{2} - \sqrt{3}\right) = \frac{-3\sqrt{3}}{2} \cdot 20 < 0$$

$\therefore$ A is maximum, when $\theta = \frac{2\pi}{3}$

Also, the maximum area is

$$A = 20\left(\sin\frac{2\pi}{3} - \frac{1}{2}\sin\frac{4\pi}{3}\right)$$

$$= 20\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\left(-\frac{\sqrt{3}}{2}\right)\right) = 15\sqrt{3} \text{ sq. units.}$$

www.exambites.in

PRACTICE PAPER 2
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) This question paper comprises four Sections A, B, C and D. This question paper carries 36 questions. All questions are compulsory.

(ii) Section A : Questions no. 1 to 20 comprises of 20 questions of 1 mark each.

(iii) Section B : Questions no. 21 to 26 comprises of 6 questions of 2 marks each.

(iv) Section C : Questions no. 27 to 32 comprises of 6 questions of 4 marks each.

(v) Section D : Questions no. 33 to 36 comprises of 4 questions of 6 marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 3 questions of one mark, 2 questions of two marks, 2 questions of four marks and 2 questions of six marks. Only one of the choices in such questions have to be attempted.

(vii) In addition to this, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

(Q.1-Q.10) are multiple choice questions. Select the correct option.

1. If $P(A) = \frac{4}{5}$ and $P(A \cap B) = \frac{7}{10}$, then $P(B|A)$ is equal to

(a) $\frac{1}{10}$ (b) $\frac{1}{8}$ (c) $\frac{7}{8}$ (d) $\frac{17}{20}$

2. The point of intersection of the lines

$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$; $\frac{x+3}{-36} = \frac{y-3}{2} = \frac{z-6}{4}$ is

(a) (2, 10, 4) (b) $\left(21, \frac{5}{3}, \frac{10}{3}\right)$

(c) (5, 7, –2) (d) (–3, 3, 6)

3. $\int \frac{x^3}{x+1}\,dx$ is equal to

(a) $x + \frac{x^2}{2} + \frac{x^3}{3} - \log|1-x| + C$

(b) $x + \frac{x^2}{2} - \frac{x^3}{3} - \log|1-x| + C$

(c) $x - \frac{x^2}{2} - \frac{x^3}{3} - \log|1+x| + C$

(d) $x - \frac{x^2}{2} + \frac{x^3}{3} - \log|1+x| + C$

4. $\int e^x(1 - \cot x + \cot^2 x)dx$ equals

(a) $e^x \cot x + C$ (b) $e^x \operatorname{cosec} x + C$

(c) $-e^x \cot x + C$ (d) $-e^x \operatorname{cosec} x + C$

5. Region represented by $x \geq 0, y \geq 0$ is

(a) first quadrant (b) second quadrant

(c) third quadrant (d) fourth quadrant

6. Let $\vec{a}$ and $\vec{b}$ be two vectors of equal magnitude inclined at an angle θ, then $\vec{a}\sin\frac{\theta}{2}$ is equal to

(a) $\frac{|\vec{a}-\vec{b}|}{2}$ (b) $\frac{|\vec{a}+\vec{b}|}{2}$

(c) $|\vec{a}-\vec{b}|$ (d) $|\vec{a}+\vec{b}|$

7. If $f(x) = x^3 - 6x^2 + 9x + 3$ be a decreasing function, then x lies in

(a) $(-\infty, -1) \cap (3, \infty)$ (b) (1, 3)

(c) (3, ∞) (d) None of these

8. If A and B are any two events such that $P(A) + P(B) - P(A \text{ and } B) = P(A)$, then

(a) $P(B \mid A) = 1$ (b) $P(A \mid B) = 1$

(c) $P(B \mid A) = 0$ (d) $P(A \mid B) = 0$

9. If $f(x) = -\sqrt{25-x^2}$, then $\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}$ is equal to

(a) $\frac{1}{24}$ (b) $\frac{1}{5}$ (c) $-\sqrt{24}$ (d) $\frac{1}{\sqrt{24}}$

10. Which of the following is false ?
(a) 30°, 45°, 60° can be the direction angles of a line in space.
(b) 90°, 135°, 45° can be the direction angles of a line in space.
(c) 120°, 60°, 45° can be the direction angles of a line in space.
(d) 60°, 45°, 60° can be the direction angles of a line in space.

(Q.11-Q.15) Fill in the blanks.

11. If $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$, $(\vec{a}\neq 0, \vec{b}\neq 0)$ then angle between $\vec{a}$ and $\vec{b}$ is _____ .

OR

If $\overrightarrow{AB}\times\overrightarrow{AC}=2\hat{i}-4\hat{j}+4\hat{k}$, then the area of ΔABC is _____ sq. units.

12. The total number of possible matrices of order 3×3 with each entry 2 or 0 is _____ .

13. The value of $\int_{-\pi}^{\pi} \sin^3 x\cos^2 x\,dx$ is _____ .

OR

The value of $\int_{-\infty}^{\infty} xe^{-x^2}dx$ is equal to _____ .

14. If A is non-singular matrix, then A^{-1} = ___ .

15. The degree of the differential equation $y\sqrt{1+\left(\frac{dy}{dx}\right)^2}=c$ is _____ .

(Q.16-Q.20) Answer the following questions.

16. Show that the signum function $f: R\to R$, given by

$$f(x)=\begin{cases}1, \text{ if } x>0\\ 0, \text{ if } x=0\\ -1, \text{ if } x<0\end{cases}$$ is neither one-one nor onto.

OR

Let $f: N\to N$ be defined by

$$f(n)=\begin{cases}\frac{n+1}{2}, \text{ if } n \text{ is odd}\\ \frac{n}{2}, \text{ if } n \text{ is even}\end{cases}$$

for all $n\in N$. State whether the function f is bijective. Justify your answer.

17. If $\begin{bmatrix}2x+y & 4x\\ 5x-7 & 4x\end{bmatrix}=\begin{bmatrix}7 & 7y-13\\ y & x+6\end{bmatrix}$, then find the value of $x+y$.

18. Evaluate $\int\frac{dx}{1+\cos x}$.

19. Write the principal value branch of $\text{cosec}^{-1}x$.

20. Find the maximum value of

$$\Delta=\begin{vmatrix}1 & 1 & 1\\ 1 & 1+\sin\theta & 1\\ 1+\cos\theta & 1 & 1\end{vmatrix}$$, where θ is real number.

SECTION - B

21. Find the domain of the function defined by $f(x)=\sin^{-1}\sqrt{x-1}$.

22. Let $f(x)=\frac{\ln(1+ax)-\ln(1-bx)}{x}$, $x\neq 0$.
If $f(x)$ is continuous at $x=0$, then find the value of $f(0)$.

23. Show that the tangents to the curve $y=7x^3+11$ at the points where $x=2$ and $x=-2$ are parallel.

OR

Find the points on the curve $x^2+y^2-2x-3=0$ at which the tangents are parallel to the x-axis.

24. Find the vector equation of a plane which is at a distance of 7 units from the origin and normal to the vector $3\hat{i}+5\hat{j}-6\hat{k}$

25. Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{i}-\hat{j}+2\hat{k}$ and $4\hat{i}-\hat{j}+3\hat{k}$.

OR

Find a vector of magnitude 5 units, and parallel to the resultant of the vectors $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$.

26. A fair coin and an unbiased die are tossed. Let A be the event 'head appears on the coin' and B be the event '3 on the die'. Check whether A and B are independent events or not ?

www.exambites.in

SECTION - C

27. Let R be relation defined on the set of natural number N as follows, $R = \{(x, y) : x \in N, y \in N, 2x + y = 41\}$. Find the domain and range of the relation R. Also verify whether R is reflexive, symmetric and transitive.

28. If $y = (\cot^{-1} x)^2$, then show that

$$(x^2 + 1)^2 \frac{d^2y}{dx^2} + 2x(x^2 + 1)\frac{dy}{dx} = 2.$$

OR

If $y = (\sin x - \cos x)^{\sin x - \cos x}$, $\frac{\pi}{4} < x < \frac{3\pi}{4}$, then find $\frac{dy}{dx}$.

29. Minimise and maximise $Z = 5x + 10y$ subject to $x + 2y \le 120$, $x + y \ge 60$, $x - 2y \ge 0$, $x, y \ge 0$.

30. A coin is biased so that the head is 3 times as likely to occur as tail. If the coin is tossed twice, find the probability distribution of number of tails.

31. Evaluate : $\int \frac{dx}{(x^2+1)(x^2+2)}$

OR

Evaluate : $\int \frac{x+2}{\sqrt{(x-2)(x-3)}}dx$

32. Solve the following differential equation :

$$(x^2 - 1)\frac{dy}{dx} + 2xy = \frac{1}{x^2-1}; |x| \neq 1$$

SECTION - D

33. A metal box with a square base and vertical sides is to contain 1024 cm^3. The material for the top and bottom costs ₹ 5 per cm^2 and the material for the sides costs ₹ 2.50 per cm^2. Then, find the least cost of the box.

OR

AB is a diameter of a circle and C is any point on the circle. Show that the area of ΔABC is maximum, when it is isosceles.

34. Using integration, find the area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

35. Using matrix method, solve the system of equations $3x + 2y - 2z = 3$, $x + 2y + 3z = 6$ and $2x - y + z = 2$.

OR

Given $A = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$, then find BA and use this to solve the system of equations $y + 2z = 7$, $x - y = 3$, $2x + 3y + 4z = 17$.

36. Find the shortest distance between the lines given by

$\vec{r} = (8 + 3\lambda)\hat{i} - (9 + 16\lambda)\hat{j} + (10 + 7\lambda)\hat{k}$

and $\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$.

www.exambites.in

Detailed Solutions

1. (c) : Given $P(A) = \frac{4}{5}$, $P(A \cap B) = \frac{7}{10}$

Now, $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{7/10}{4/5} = \frac{7}{8}$

2. (b) : The given equation of lines are

$$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1} = k \text{ (say)} \quad ...(1)$$

and $\frac{x+3}{-36} = \frac{y-3}{2} = \frac{z-6}{4}$ ... (2)

$\therefore$ Any point on line (1) is $(3k + 5, 7 - k, k - 2)$.

It should lie on line (2)

$\therefore \quad \frac{3k+5+3}{-36} = \frac{7-k-3}{2} = \frac{k-2-6}{4}$

On solving, we get $k = \frac{16}{3}$

$\therefore$ Coordinates of intersection point are $\left(21, \frac{5}{3}, \frac{10}{3}\right)$.

3. (d) : We have, $I = \int \frac{x^3}{x+1}dx = \int \frac{x^3 - 1 + 1}{x+1}dx$

$$= \int \left(\frac{(x+1)(x^2+1-x)}{x+1} - \frac{1}{x+1} \right) dx$$

$$= \int (x^2 + 1 - x)\, dx - \int \frac{dx}{x+1}$$

$= \frac{x^3}{3} + x - \frac{x^2}{2} - \log|x+1| + C$

4. (c) : $\int e^x(1-\cot x+\cot^2 x)dx$

$= \int e^x(-\cot x+\text{cosec}^2 x)dx$

$= -\int e^x \cot x\ dx + \int e^x \text{cosec}^2 x\ dx$

$= -e^x \cot x - \int e^x \text{cosec}^2 x\ dx + \int e^x \text{cosec}^2 x\ dx + C$

$= -e^x \cot x + C$

5. (a)

6. (a) : Here, $|\vec{a}| = |\vec{b}|$

$\therefore \quad |\vec{a}-\vec{b}| = \sqrt{\vec{a}^2+\vec{a}^2-2\vec{a}^2\cos\theta}$

$= \sqrt{2\vec{a}^2(1-\cos\theta)} = \sqrt{2\vec{a}^2 \cdot 2\sin^2\frac{\theta}{2}} = 2\vec{a}\sin\frac{\theta}{2}$

$\Rightarrow \quad |\vec{a}-\vec{b}| = 2\vec{a}\sin\frac{\theta}{2} \quad \Rightarrow \quad \vec{a}\sin\frac{\theta}{2} = \frac{|\vec{a}-\vec{b}|}{2}$

7. (b) : $\because f(x) = x^3 - 6x^2 + 9x + 3$

$\therefore \quad f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3)$

For decreasing, $f'(x) < 0$

$\Rightarrow \quad (x-3)(x-1) < 0 \quad \therefore \quad x \in (1, 3)$

8. (b) : $P(A) + P(B) - P(A \cap B) = P(A)$

$\Rightarrow \quad P(B) - P(A \cap B) = 0 \quad \Rightarrow \quad P(A \cap B) = P(B)$

$\Rightarrow \quad \frac{P(A \cap B)}{P(B)} = 1 \quad \Rightarrow \quad P(A \mid B) = 1.$

9. (d) : $\lim_{x \to 1} \frac{f(x)-f(1)}{x-1} = f'(1)$

Now, $f'(x) = -\frac{1}{2} \cdot \frac{-2x}{\sqrt{25-x^2}}$

$\therefore \quad f'(1) = -\frac{1}{2} \cdot \frac{-2 \times 1}{\sqrt{25-1^2}} = \frac{1}{\sqrt{24}}$

10. (a) : Let $\alpha = 30°$, $\beta = 45°$ and $\gamma = 60°$

Now, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2} \neq 1$

Thus 30°, 45°, 60° cannot be the direction angles of any line in space.

11. (d) : Since, $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$

$\Rightarrow \quad |\vec{a}+\vec{b}|^2 = |\vec{a}-\vec{b}|^2$

$\Rightarrow \quad |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b} = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}\cdot\vec{b}$

$\Rightarrow \quad 4\vec{a}\cdot\vec{b} = 0 \Rightarrow \quad \vec{a}\cdot\vec{b} = 0$

$\therefore$ Angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{2}$.

OR

Area of ΔABC

$= \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2}|2\hat{i} - 4\hat{j} + 4\hat{k}|$

$= \frac{1}{2}\left[\sqrt{(2)^2 + (-4)^2 + (4)^2}\right]$ = 3 sq. units

12. (d) : As there are 9 elements in 3 × 3 order matrix and each element is filled by either 2 or 0 in 2 ways.

$\therefore$ Total number of possible matrices of order 3 × 3 with each entry 2 or 0 is 2^9 *i.e.*, 512.

13. Let $f(x) = \int_{-\pi}^{\pi} \sin^3 x \cos^2 x\, dx$

Now, $f(-x) = \int_{-\pi}^{\pi} \sin^3(-x)\cos^2(-x)dx$

$= \int_{-\pi}^{\pi} -\sin^3 x \cos^2 x\, dx = -f(x)$

Since, $f(x)$ is an odd function.

$\therefore \quad \int_{-\pi}^{\pi} \sin^3 x \cos^2 x dx = 0$

OR

Let $I = \int_{-\infty}^{\infty} f(x)dx$, where $f(x) = xe^{-x^2}$

Now, $f(-x) = -xe^{-x^2}$.

Since, $f(x)$ is an odd function $\therefore \quad I = 0$.

14. $A^{-1} = \frac{adj(A)}{|A|}$

15. (d) : We have, $y\sqrt{1+\left(\frac{dy}{dx}\right)^2} = c$

$\Rightarrow \quad y^2\left\{1+\left(\frac{dy}{dx}\right)^2\right\} = c^2 \Rightarrow \quad y^2\left(\frac{dy}{dx}\right)^2 + y^2 = c^2$

Clearly, it is a differential equation of degree 2.

16. $f(x) = \begin{cases} 1, & \text{if} \quad x > 0 \\ 0, & \text{if} \quad x = 0 \\ -1 & \text{if} \quad x < 0 \end{cases}$

(a) We have $f(1) = f(2) = 1$ ($\because$ 1, 2 > 0)

But $1 \neq 2$

www.exambrees.in

Also $f(-2) = f(-3) = -1$. But $-2 \neq -3$
$\Rightarrow$ f is not one-one.
(b) Except the number $-1, 0, 1$, no other members of co-domain of f has any pre-image in its domain.
$\therefore$ f is not onto.
Hence f is neither one-one nor onto.

OR

Here $f(1) = \frac{1+1}{2} = 1$, $f(2) = \frac{2}{2} = 1$
$\Rightarrow$ $f(1) = f(2)$ but $1 \neq 2$
$\Rightarrow$ f is not one-one.
But f is onto because range of $f = N$
[$\because$ For any $x \in N$, $2x \in N$ such that $f(2x) = \frac{2x}{2} = x$]
$\Rightarrow$ f is onto.
Hence f is not bijective.

17. We have, $4x = x + 6 \Rightarrow x = 2$
and $4x = 7y - 13 \Rightarrow 8 = 7y - 13$
$\Rightarrow$ $7y = 21 \Rightarrow y = 3$
$\therefore$ $x + y = 2 + 3 = 5$

18. Let $I = \int \frac{dx}{1+\cos x} = \frac{1}{2}\int \frac{1}{\cos^2 \frac{x}{2}} dx$
$= \frac{1}{2}\int \sec^2 \frac{x}{2} dx = \frac{1}{2}\tan\frac{x}{2} \cdot 2 + C = \tan\frac{x}{2} + C$

19. The principal value branch of $\text{cosec}^{-1} x$ is $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] - \{0\}$.

20. We have, $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1+\cos\theta & 1 & 1 \end{vmatrix}$
$= 1\,(1 + \sin\theta - 1) - 1\,(1 - 1 - \cos\theta)$
$+ 1\,(1 - (1 + \sin\theta)(1 + \cos\theta))$
$= \sin\theta + \cos\theta + [1 - (1 + \sin\theta + \cos\theta + \sin\theta\cos\theta)]$
$= \sin\theta + \cos\theta - \sin\theta - \cos\theta - \sin\theta\cos\theta$
$= \frac{-1}{2}\sin 2\theta$
$\therefore$ max. $\Delta = \frac{-1}{2}(-1) = \frac{1}{2}$ ($\because -1 \leq \sin\theta \leq 1$)

21. Given, $f(x) = \sin^{-1}\sqrt{x-1}$
Since, $x - 1 \geq 0$ and $-1 \leq \sqrt{x-1} \leq 1$
$\therefore$ $0 \leq x - 1 \leq 1$
$\Rightarrow$ $1 \leq x \leq 2$

22. $f(0) = \lim_{x\to 0} f(x)$
$= \lim_{x\to 0} \frac{\ln(1+ax) - \ln(1-bx)}{x}$ ($\frac{0}{0}$ form)
$= \lim_{x\to 0}\left[\frac{a}{1+ax} + \frac{b}{1-bx}\right]$ (L'Hospital Rule)
$= a + b$.

23. We have $y = 7x^3 + 11$ ...(i)
Differentiating (i) w.r.t. x, we get
$\Rightarrow$ $\frac{dy}{dx} = 7 \cdot 3x^2 = 21x^2$
$\therefore$ Slope of tangent at $x = 2$ is
$\left(\frac{dy}{dx}\right)_{x=2} = 21(2)^2 = 84$
and slope of tangent at $x = -2$ is
$\left(\frac{dy}{dx}\right)_{x=-2} = 21(-2)^2 = 84$
Since, the slope of tangents at $x = 2$ and $x = -2$ are equal, therefore, these tangents are parallel.

OR

We have $x^2 + y^2 - 2x - 3 = 0$ ...(i)
Differentiating (i) w.r.t. x, we get
$2x + 2y\frac{dy}{dx} - 2 - 0 = 0$
$\Rightarrow$ $\frac{dy}{dx} = \frac{2(1-x)}{2y} = \frac{1-x}{y}$ ...(ii)
For tangents parallel to x-axis, we must have
$\frac{dy}{dx} = 0 \Rightarrow \frac{1-x}{y} = 0 \Rightarrow x = 1, y \neq 0$
Substituting $x = 1$ in (i), we get
$1^2 + y^2 - 2\cdot 1 - 3 = 0 \Rightarrow y^2 - 4 = 0 \Rightarrow y = \pm 2$
Hence, the required points are $(1, 2)$ and $(1, -2)$.

24. Let $\vec{n} = 3\hat{i} + 5\hat{j} - 6\hat{k}$
Then $|\vec{n}| = \sqrt{9 + 25 + 36} = \sqrt{70}$
$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{1}{\sqrt{70}}(3\hat{i} + 5\hat{j} - 6\hat{k})$
Hence, the required equation of the plane is
$\vec{r} \cdot \frac{1}{\sqrt{70}}(3\hat{i} + 5\hat{j} - 6\hat{k}) = 7$
$\Rightarrow$ $\vec{r} \cdot \left(\frac{3\hat{i} + 5\hat{j} - 6\hat{k}}{\sqrt{70}}\right) = 7$

www.examrites.in

25. Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = 4\hat{i} - \hat{j} + 3\hat{k}$

Let $\vec{c}$ be a vector perpendicular to both the vectors $\vec{a}$ and $\vec{b}$. Then,

$$\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 4 & -1 & 3 \end{vmatrix}$$

$$= \hat{i}(-3+2) - \hat{j}(6-8) + \hat{k}(-2+4) = -\hat{i} + 2\hat{j} + 2\hat{k}$$

So, a vector of magnitude 6 in the direction of $\vec{c}$

$$= \frac{\vec{c}}{|\vec{c}|} \cdot 6 = \frac{-\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{1^2 + 2^2 + 2^2}} \cdot 6$$

$$= -2\hat{i} + 4\hat{j} + 4\hat{k}$$

OR

The resultant of given vectors $= \vec{a} + \vec{b}$

$$= (2\hat{i} + 3\hat{j} - \hat{k}) + (\hat{i} - 2\hat{j} + \hat{k}) = 3\hat{i} + \hat{j}$$

$\therefore$ Required vector of magnitude 5 units

$$= 5\left(\frac{3\hat{i} + \hat{j}}{\sqrt{9+1}}\right) = \frac{3}{2}\sqrt{10}\,\hat{i} + \frac{\sqrt{10}}{2}\hat{j}.$$

26. Let S be the sample space of the given experiment.

$$S = \begin{Bmatrix} (H,1), (H,2), (H,3), (H,4), (H,5), (H,6), \\ (T,1), \ (T,2), (T,3), (T,4), (T,5), (T,6) \end{Bmatrix}$$

Given that A : 'head appears on the coin'

$\Rightarrow$ $A = \{(H,1), (H,2), (H,3), (H,4), (H,5), (H,6)\}$

and B : '3 appears on the die'

$\Rightarrow$ $B = \{(H,3), (T,3)\}$

$\Rightarrow$ $A \cap B = \{(H,3)\}$

Hence, $P(A) = \frac{6}{12} = \frac{1}{2}$, $P(B) = \frac{2}{12} = \frac{1}{6}$

and $P(A \cap B) = \frac{1}{12}$.

$$P(A) \times P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12} = P(A \cap B).$$

Hence, A and B are independent events.

27. We have given, $R = \{(x, y): x \in N, y \in N, 2x + y = 41\}$.

Domain = {1, 2, 3, ..., 20}

Range = {1, 3, 5, 7, ..., 37, 39}

$R = \{(1, 39), (2, 37), (3, 35), ..., (19, 3), (20, 1)\}$

R is not reflexive as $(3, 3) \notin R$

$\therefore$ $2 \times 3 + 3 \neq 41$

As $(1, 39) \in R$ but $(39, 1) \notin R$ ($\because$ for (39, 1), $(2 \times 39) + 1 \neq 41$)

So, R is not symmetric.

As $(11, 19) \in R$, $(19, 3) \in R$

But $(11, 3) \notin R$

So, R is not transitive.

Therefore, R is neither reflexive, nor symmetric and nor transitive.

28. We have, $y = (\cot^{-1} x)^2$

$$\Rightarrow \frac{dy}{dx} = 2(\cot^{-1} x) \times \frac{-1}{1+x^2}$$

$$\Rightarrow (x^2+1)\frac{dy}{dx} = -2\cot^{-1} x$$

$$\Rightarrow (x^2+1)^2\left(\frac{dy}{dx}\right)^2 = 4(\cot^{-1} x)^2$$

$$\Rightarrow (x^2+1)^2\left(\frac{dy}{dx}\right)^2 = 4y$$

Again differentiating w.r.t. x, we get

$$(x^2+1)^2\left(2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2}\right) + 2(x^2+1)(2x)\left(\frac{dy}{dx}\right)^2 = 4\frac{dy}{dx}$$

$$\Rightarrow (x^2+1)^2\frac{d^2y}{dx^2} + 2x(x^2+1)\frac{dy}{dx} = 2$$

OR

We have, $y = (\sin x - \cos x)^{(\sin x - \cos x)}$

Taking log on both sides, we get

$\log y = (\sin x - \cos x) \log (\sin x - \cos x)$

Differentiating w.r.t. x, we get

$$\frac{1}{y}\cdot\frac{dy}{dx} = (\sin x - \cos x)\frac{(\cos x + \sin x)}{(\sin x - \cos x)} + \log(\sin x - \cos x)\cdot(\cos x + \sin x)$$

$$\Rightarrow \frac{1}{y}\cdot\frac{dy}{dx} = (\cos x + \sin x)[1 + \log(\sin x - \cos x)]$$

$$\therefore \frac{dy}{dx} = (\sin x - \cos x)^{(\sin x - \cos x)} \times [(\cos x + \sin x)(1 + \log(\sin x - \cos x))]$$

29. The system of constraints is :

$x + 2y \leq 120$ ...(1)

$x + y \geq 60$ ...(2)

$x - 2y \geq 0$ ...(3)

and $x, y \geq 0$ ...(4)

Let $l_1 : x + 2y = 120$

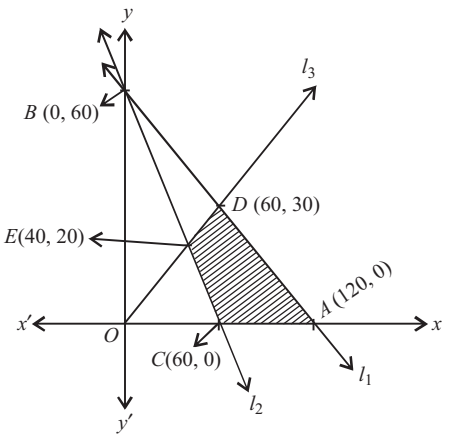
$l_2 : x + y = 60$

$l_3 : x - 2y = 0$

www.exambits.in

It is observed that the feasible region *CADE* is bounded.

The co-ordinates of *C, A, D, E* are (60, 0), (120, 0), (60, 30), (40, 20) respectively.



Thus we use corner point method to determine the maximum and minimum values of *Z*.

We have : $Z = 5x + 10y$

Corner Point	Corresponding values of Z
(60, 0)	300 (Minimum)
(120, 0)	600
(60, 30)	600
(40, 20)	400

Hence Z_{min} = 300 at (60, 0) and Z_{max} = 600 at all points on the line segment joining the points (120,0) and (60, 30).

30. Let *X* denote the random variable which represents the number of tails on the biased coin tossed twice.

$\therefore$ *X* can assume values 0, 1, 2

Let *p* be the probability of tail appears $= \frac{1}{4}$ and

q be the probability of head appears $= \frac{3}{4}$

$\therefore$ $P(X = 0)$ = Probability that both head appears

$$= q \times q = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$$

$P(X = 1)$ = Probability that one head & one tail appears $= 2(p \times q) = 2\left(\frac{1}{4} \times \frac{3}{4}\right) = \frac{3}{8}$

and $P(X = 2)$ = Probability that both tails appears

$$= p \times p = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}.$$

X	0	1	2
P(X)	9/16	3/8	1/16

31. Let $I = \int \frac{dx}{(x^2+1)(x^2+2)}$

Put $x^2 = y$

Let $\frac{1}{(x^2+1)(x^2+2)} = \frac{1}{(y+1)(y+2)} = \frac{A}{(y+1)} + \frac{B}{(y+2)}$

$\Rightarrow$ $1 = A(y + 2) + B(y + 1)$ ...(1)

Putting $y = -1$ in (1), we get $1 = A(-1 + 2) \Rightarrow A = 1$

Putting $y = -2$ in (1), we get $1 = B(-2 + 1) \Rightarrow B = -1$

$$\therefore \quad I = \int \frac{dx}{x^2+1} - \int \frac{dx}{x^2+2}$$

$$= \tan^{-1} x - \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C$$

OR

Let $I = \int \frac{x+2}{\sqrt{(x-2)(x-3)}} dx = \int \frac{x+2}{\sqrt{x^2-5x+6}} dx$

We write,

$$x + 2 = A\left[\frac{d}{dx}(x^2 - 5x + 6)\right] + B = A(2x - 5) + B$$

Equating coefficients of x and constant terms, we get

$2A = 1 \Rightarrow A = \frac{1}{2}$ and $-5A + B = 2 \Rightarrow B = \frac{9}{2}$

$$\therefore \quad I = \frac{1}{2}\int \frac{2x-5}{\sqrt{x^2-5x+6}} dx + \frac{9}{2}\int \frac{dx}{\sqrt{x^2-5x+\frac{25}{4}-\frac{25}{4}+6}}$$

$$= \frac{1}{2} \cdot 2 \cdot \sqrt{x^2-5x+6} + \frac{9}{2}\int \frac{dx}{\sqrt{\left(x-\frac{5}{2}\right)^2 - \left(\frac{1}{2}\right)^2}}$$

$$= \sqrt{x^2-5x+6} + \frac{9}{2}\log\left|\left(x-\frac{5}{2}\right) + \sqrt{x^2-5x+6}\right| + C$$

32. We have, $\frac{dy}{dx} + \frac{2xy}{x^2-1} = \frac{1}{(x^2-1)^2}$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \frac{2x}{x^2-1}, Q = \frac{1}{(x^2-1)^2}$

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{2x}{x^2-1} dx} = e^{\log|x^2-1|} = x^2 - 1$$

www.exambites.in

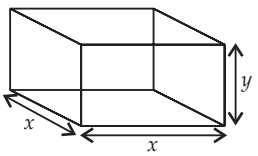
Required solution is

$$y(x^2-1)=\int \frac{1}{(x^2-1)^2}.(x^2-1)dx+C$$

$$y(x^2-1)=\int \frac{dx}{x^2-1}+C \Rightarrow y(x^2-1)=\frac{1}{2}\log\left|\frac{x-1}{x+1}\right|+C$$

$$\therefore \quad y=\frac{1}{2(x^2-1)}\log\left|\frac{x-1}{x+1}\right|+\frac{C}{x^2-1}$$

33. Let length and height of the square box be x cm and y cm respectively.



Then volume of the box (V) = x^2y = 1024 (given)

Since, $x^2y=1024 \Rightarrow y=\frac{1024}{x^2}$

If C denotes the cost of the box.

Then $C = (2x^2 \times 5) + (4xy \times 2.50)$

$= 10x^2 + 10xy = 10x(x + y)$

$$=10x\left(x+\frac{1024}{x^2}\right)=\frac{10x}{x^2}(x^3+1024)$$

$$\Rightarrow \quad C=10x^2+\frac{10240}{x} \qquad \text{... (i)}$$

On differentiating (i) both sides with respect to x, we get

$$\frac{dC}{dx}=20x-10240(x)^{-2}$$

$$=20x-\frac{10240}{x^2} \qquad \text{... (ii)}$$

Put $\frac{dC}{dx}=0$

$$\Rightarrow 20x=\frac{10240}{x^2} \Rightarrow 20x^3=10240$$

$\Rightarrow x^3 = 512 = 8^3 \quad \Rightarrow \quad x = 8$

Again, differentiating (ii) with respect to x, we get

$$\frac{d^2C}{dx^2}=20-10240(-2)\cdot\frac{1}{x^3}$$

$$=20+\frac{20480}{x^3}$$

$$\therefore \quad \left(\frac{d^2C}{dx^2}\right)_{x=8}=20+\frac{20480}{512}=60>0$$

Hence, cost is minimum for x = 8 and the corresponding least cost of the box is

$$C(8)=10\cdot 8^2+\frac{10240}{8} \qquad \text{(from (i))}$$

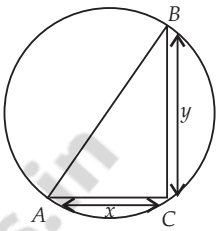
$$= 640 + 1280 = ₹\ 1920$$

OR

We have, $AB = 2r$

and $\angle ACB = 90°$

(since, angle in the semicircle is always 90°)



Let $AC = x$ and $BC = y$

$\therefore \quad (2r)^2 = x^2 + y^2$

$\Rightarrow \quad y^2 = 4r^2 - x^2$

$$\Rightarrow \quad y=\sqrt{4r^2-x^2} \qquad \text{... (i)}$$

Now, area of ΔABC, $A=\frac{1}{2}\times x\times y$

$$=\frac{1}{2}\times x\times(4r^2-x^2)^{1/2} \qquad \text{(from (i))}$$

Now, differentiating both sides with respect to x, we get

$$\frac{dA}{dx}=\frac{1}{2}\left[x\cdot\frac{1}{2}(4r^2-x^2)^{-1/2}\cdot(0-2x)+(4r^2-x^2)^{1/2}\cdot 1\right]$$

$$=\frac{1}{2}\left[\frac{-x^2}{\sqrt{4r^2-x^2}}+\sqrt{4r^2-x^2}\right]$$

$$=\frac{1}{2}\left[\frac{-x^2+4r^2-x^2}{\sqrt{4r^2-x^2}}\right]=\frac{1}{2}\left[\frac{-2x^2+4r^2}{\sqrt{4r^2-x^2}}\right]$$

$$\Rightarrow \quad \frac{dA}{dx}=\left[\frac{-x^2+2r^2}{\sqrt{4r^2-x^2}}\right] \qquad \text{... (ii)}$$

Put $\frac{dA}{dx}=0$

$\Rightarrow \quad -x^2 + 2r^2 = 0$

$$\Rightarrow \quad r^2=\frac{1}{2}x^2$$

$$\Rightarrow \quad r=\frac{1}{\sqrt{2}}x$$

$$\therefore \quad x=r\sqrt{2}$$

www.exambits.in

Again, differentiating (ii) both sides with respect to x, we get

$$\frac{d^2A}{dx^2}=\frac{\sqrt{4r^2-x^2}\cdot(-2x)-(2r^2-x^2)\times\frac{1}{2}(4r^2-x^2)^{-1/2}(-2x)}{(\sqrt{4r^2-x^2})^2}$$

$$=\frac{-2x\left[\sqrt{4r^2-x^2}-(2r^2-x^2)\cdot\frac{1}{2\sqrt{4r^2-x^2}}\right]}{(\sqrt{4r^2-x^2})^2}$$

$$=\frac{-2x\left[2(4r^2-x^2)-(2r^2-x^2)\right]}{2(4r^2-x^2)^{3/2}}$$

$$=\frac{-2x(6r^2-x^2)}{2(4r^2-x^2)^{3/2}}$$

At $x=r\sqrt{2}$

$$\frac{d^2A}{dx^2}=\frac{-(6r^2-2r^2)\sqrt{2}r}{(4r^2-2r^2)^{3/2}}$$

$$=\frac{-4r^2\times\sqrt{2}r}{2r^2\sqrt{2}r}=-2<0$$

Hence, at $x=r\sqrt{2}$, the area of triangle is maximum.

For $x=r\sqrt{2}$, $y=\sqrt{4r^2-(r\sqrt{2})^2}=\sqrt{2r^2}=r\sqrt{2}$

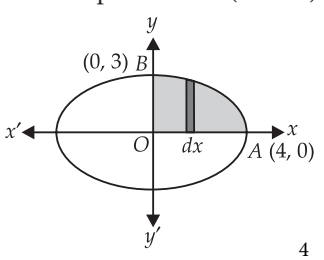
Hence, $x=r\sqrt{2}=y$

So, the triangle is isosceles.

34. The given ellipse is $\frac{x^2}{16}+\frac{y^2}{9}=1$

Since given curve is symmetrical about both axes

$\therefore$ Area of the ellipse = 4 area ($OABO$)



$\therefore$ Required area = area $(OABO)=4\int_0^4 y\,dx$

[By taking vertical strips]

$$=4\int_0^4\frac{3}{4}\sqrt{16-x^2}\,dx$$

$$\left[\because\frac{x^2}{16}+\frac{y^2}{9}=1\Rightarrow\frac{y^2}{9}=1-\frac{x^2}{16}\right.$$

$$\left.\Rightarrow y=\frac{3}{4}\sqrt{16-x^2}\ (\because y>0)\right]$$

$$=3\left[\frac{x\sqrt{16-x^2}}{2}+\frac{16}{2}\sin^{-1}\frac{x}{4}\right]_0^4$$

$$=3\left[0+\frac{16}{2}\sin^{-1}(1)-0-0\right]$$

$$=3\left[8\left(\frac{\pi}{2}\right)\right]=12\pi\ \text{sq.units}$$

35. Given system of equations

$3x+2y-2z=3$

$x+2y+3z=6$ and $2x-y+z=2$

can be written in the form of $AX=B$.

$$\therefore\quad\begin{bmatrix}3&2&-2\\1&2&3\\2&-1&1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}3\\6\\2\end{bmatrix}$$

For A^{-1}, $|A|=3(5)-2(1-6)+(-2)(-5)$

$=15+10+10=35\neq0$

$\therefore$ $A_{11}=5$, $A_{12}=5$, $A_{13}=-5$, $A_{21}=0$, $A_{22}=7$, $A_{23}=7$, $A_{31}=10$, $A_{32}=-11$ and $A_{33}=4$

$$\text{Hence, adj } A=\begin{bmatrix}5&5&-5\\0&7&7\\10&-11&4\end{bmatrix}^T=\begin{bmatrix}5&0&10\\5&7&-11\\-5&7&4\end{bmatrix}$$

$$\text{Now, } A^{-1}=\frac{\text{adj } A}{|A|}=\frac{1}{35}\begin{vmatrix}5&0&10\\5&7&-11\\-5&7&4\end{vmatrix}$$

For $X=A^{-1}B$,

$$\begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{1}{35}\begin{bmatrix}5&0&10\\5&7&-11\\-5&7&4\end{bmatrix}\begin{bmatrix}3\\6\\2\end{bmatrix}$$

$$=\frac{1}{35}\begin{bmatrix}15+20\\15+42-22\\-15+42+8\end{bmatrix}=\frac{1}{35}\begin{bmatrix}35\\35\\35\end{bmatrix}=\begin{bmatrix}1\\1\\1\end{bmatrix}$$

Hence, $x=1$, $y=1$ and $z=1$

OR

We have given,

$$A=\begin{bmatrix}2&2&-4\\-4&2&-4\\2&-1&5\end{bmatrix}\text{ and }B=\begin{bmatrix}1&-1&0\\2&3&4\\0&1&2\end{bmatrix}$$

www.exambites.in

$$\therefore \quad BA = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = 6I$$

$$\therefore \quad B^{-1} = \frac{A}{6} = \frac{1}{6}A = \frac{1}{6}\begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix} \qquad ...(i)$$

Also, for given system of equations, $x - y = 3$, $2x + 3y + 4z = 17$ and $y + 2z = 7$

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}^{-1}\begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$= \frac{1}{6}\begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}\begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix} \qquad \text{[using (i)]}$$

$$= \frac{1}{6}\begin{bmatrix} 6+34-28 \\ -12+34-28 \\ 6-17+35 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 12 \\ -6 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

Hence, $x = 2$, $y = -1$ and $z = 4$

36. We have given,

$$\vec{r} = (8 + 3\lambda)\hat{i} - (9 + 16\lambda)\hat{j} + (10 + 7\lambda)\hat{k}$$

$$= 8\hat{i} - 9\hat{j} + 10\hat{k} + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

Compare with $\vec{r}_1 = \vec{a}_1 + \lambda\vec{b}_1$, we get

$$\vec{a}_1 = 8\hat{i} - 9\hat{j} + 10\hat{k} \text{ and } \vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k} \qquad ...(i)$$

Also, $\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$

$$\Rightarrow \quad \vec{a}_2 = 15\hat{i} + 29\hat{j} + 5\hat{k} \text{ and } \vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k} \qquad ...(ii)$$

Now, shortest distance between two lines is given by

$$\left|\frac{(\vec{b}_1 \times \vec{b}_2)\cdot(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$$

Since, $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix}$

$$= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48)$$

$$= 24\hat{i} + 36\hat{j} + 72\hat{k}$$

$$\Rightarrow \quad |\vec{b}_1 \times \vec{b}_2| = \sqrt{(24)^2 + (36)^2 + (72)^2}$$

$$= 12\sqrt{2^2 + 3^2 + 6^2} = 84$$

and $(\vec{a}_2 - \vec{a}_1) = (15 - 8)\hat{i} + (29 + 9)\hat{j} + (5 - 10)\hat{k}$

$$= 7\hat{i} + 38\hat{j} - 5\hat{k}$$

Hence, shortest distance

$$= \left|\frac{(24\hat{i} + 36\hat{j} + 72\hat{k})\cdot(7\hat{i} + 38\hat{j} - 5\hat{k})}{84}\right|$$

$$= \left|\frac{168 + 1368 - 360}{84}\right| = \left|\frac{1176}{84}\right| = 14 \text{ units}$$

www.examkites.in

PRACTICE PAPER 3
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) This question paper comprises four Sections A, B, C and D. This question paper carries 36 questions. All questions are compulsory.

(ii) Section A : Questions no. 1 to 20 comprises of 20 questions of 1 mark each.

(iii) Section B : Questions no. 21 to 26 comprises of 6 questions of 2 marks each.

(iv) Section C : Questions no. 27 to 32 comprises of 6 questions of 4 marks each.

(v) Section D : Questions no. 33 to 36 comprises of 4 questions of 6 marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 3 questions of one mark, 2 questions of two marks, 2 questions of four marks and 2 questions of six marks. Only one of the choices in such questions have to be attempted.

(vii) In addition to this, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

SECTION - A

(Q.1-Q.10) are multiple choice type questions. Select the correct option.

1. Find the range of the function $f(x)=\frac{|x-1|}{(x-1)}$.

(a) {1, 3} (b) {1, 2} (c) {–1, 1} (d) {–1, 3}

2. Find the values of a, b, c and d from the following equation :

$$\begin{bmatrix} 2a+b & a-2b \\ 5c-d & 4c+3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$$

(a) $a = 2, b = 1, c = 4, d = 3$

(b) $a = 1, b = 2, c = 3, d = 4$

(c) $a = 3, b = 4, c = 1, d = 2$

(d) $a = 4, b = 3, c = 2, d = 1$

3. Evaluate : $\int (e^{x\log a}+e^{a\log x}+e^{a\log a})dx$

(a) $\frac{a^x}{\log a}+\frac{x^{a+1}}{a+1}+a^a x+C$

(b) $\frac{a^x}{\log a}+\frac{x^{a+1}}{a-1}+ax^a+C$

(c) $\frac{a^x}{\log a}+\frac{x^a}{a+1}+ax^a+C$

(d) $\frac{a^x}{\log x}+\frac{x^{a+1}}{a+1}+a^a x+C$

4. Find the order and degree of the given differential equation.

$$\sqrt{x+\left(\frac{dy}{dx}\right)^2}=a\left(\frac{d^2y}{dx^2}\right)^{\frac{1}{3}}.$$

(a) 1, 1 (b) 2, 2 (c) 2, 1 (d) 1, 2

5. $(\vec{a}\cdot\hat{i})^2+(\vec{a}\cdot\hat{j})^2+(\vec{a}\cdot\hat{k})^2$ is equal to

(a) 1 (b) $\vec{a}$ (c) $-\vec{a}$ (d) $|\vec{a}|^2$

6. Find the minor of the element of second row and third column in the following determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$.

(a) 13 (b) 4 (c) 5 (d) 0

7. An urn contains 10 black and 5 white balls. Two balls are drawn from the urn one after the other without replacement, then the probability that both drawn balls are black, is

(a) $\frac{2}{7}$ (b) $\frac{1}{7}$ (c) $\frac{5}{7}$ (d) $\frac{3}{7}$

8. Corner points of feasible region of inequalities gives

(a) optimal solution of LPP

(b) objective function

(c) constraints

(d) linear assumption

9. Find the principal value of $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$.

www.exambites.in

(a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{5\pi}{6}$ (d) $\frac{\pi}{4}$

10. Evaluate : $\int 2^{2^{2^x}} 2^{2^x} 2^x \, dx$

(a) $\frac{1}{(\log 2)^3} 2^{2^{2^x}} + C$ (b) $\frac{1}{(\log 2)^3} 2^{2^x} + C$

(c) $\frac{1}{(\log 2)^2} 2^{2^x} + C$ (d) $\frac{1}{(\log 2)^4} 2^{2^{2^x}} + C$

(Q.11-Q.15) Fill in the blanks.

11. If a matrix has 12 elements, then the number of all possible orders it can have is _____.

12. If a line makes angles 90°, 60° and 30° with the positive directions of x, y and z-axis respectively, then its direction cosines are _____.

13. $\int_{-1}^{0} \frac{dx}{2x+3}$ = _____.

14. If $\vec{a} = 3\hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = 5\hat{i} + 5\hat{j} - \hat{k}$, then $\vec{a}$ and $\vec{b}$ are _____ vectors.

OR

The projection of the vector $7\hat{i} + \hat{j} - 4\hat{k}$ on $2\hat{i} + 6\hat{j} + 3\hat{k}$ is _____ .

15. The function $f(x) = \frac{x}{\log x}$ increases in the interval _____ .

OR

The equation of the tangent to the curve $y = (4 - x^2)^{2/3}$ at $x = 2$ is _____ .

(Q.16-Q.20) Answer the following questions.

16. If the points (2, –3), (k, – 1) and (0, 4) are collinear, then find the value of k.

17. Two dice are thrown together. What is the probability that the sum of the numbers on the two faces is neither 9 nor 11 ?

18. If $y = \log_a x + \log_x a + \log_x x + \log_a a$, then find $\frac{dy}{dx}$.

OR

Find $\frac{dy}{dx}$, when $3x + 2y = \sin y$.

19. Find the value of k, if the lines $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ and $\frac{x-4}{1} = \frac{y+3}{k} = \frac{z+1}{7}$ are coplanar.

20. Find the value of $\int \frac{1+\tan x}{x+\log \sec x} dx$.

SECTION - B

21. Find the value of $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$.

22. Differentiate $x^{\sin^{-1} x}$ w.r.t. x.

OR

Differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$.

23. If $y = a \log x + bx^2 + x$ has its extreme values at $x = -1$ and $x = 2$, then find a and b.

24. Find a unit vector perpendicular to both $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = -2\hat{i} - \hat{j} - \hat{k}$.

OR

If $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$, $\vec{a} \neq 0$ and $\vec{b} \neq \vec{c}$, then show that $\vec{b} = \vec{c} + \lambda\vec{a}$ for some scalar λ.

25. Given $P(A) = 0.4$, $P(B) = 0.7$ and $P(B|A) = 0.6$. Find $P(A \cup B)$.

26. Find the vector equation of the line passing through the point (2, –1, –1) which is parallel to the line $6x - 2 = 3y + 1 = 2z - 2$.

SECTION - C

27. Find the value of $\int_{1/e}^{e} |\log x| \, dx$.

28. Evaluate : $\int \frac{xe^{2x}}{(2x+1)^2} dx$

OR

Evaluate : $\int \frac{dx}{\sqrt[3]{x} + \sqrt[4]{x}}$

29. Using integration, find the area of the triangular region whose sides have the equations $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

30. Show that $y = be^x + ce^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$.

31. A bag contains 5 white, 7 red and 8 black balls. If four balls are drawn one by one without replacement, then find the probability of getting all white balls.

OR

A certain team wins with the probability 0.7, loses with probability 0.2 and ties with

www.exambites.in

probability 0.1. The team plays three games. Find the probability that the team wins at least two of the games, but not lose any game.

32. Solve the following linear programming problem graphically :

Minimize $Z = 200x + 500y$ subject to the constraints :

$x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$

SECTION - D

33. Find the adjoint of the matrix $A = \begin{bmatrix} 4 & -6 & 1 \\ -1 & -1 & 1 \\ -4 & 11 & -1 \end{bmatrix}$ and verify that $A(\text{adj } A) = (\text{adj } A)\, A = |A| I_3$.

OR

Find A^{-1}, where $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$. Hence, solve the system of equations $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$.

34. Find the intervals in which the function f given by

$$f(x) = \frac{4\sin x - 2x - x\cos x}{2 + \cos x}, 0 \leq x \leq 2\pi \text{ is}$$

(i) increasing (ii) decreasing.

OR

Find the shortest distance of the point $(0, c)$ from the parabola $y = x^2$.

35. Find the shortest distance and the vector equation of the line of shortest distance between the lines given by

$\vec{r} = (3\hat{i} + 8\hat{j} + 3\hat{k}) + \lambda(3\hat{i} - \hat{j} + \hat{k})$ and

$\vec{r} = (-3\hat{i} - 7\hat{j} + 6\hat{k}) + \mu(-3\hat{i} + 2\hat{j} + 4\hat{k})$.

36. Let $f(x) = \begin{cases} \dfrac{1 - \sin^3 x}{3\cos^2 x}, & \text{if } x < \dfrac{\pi}{2} \\ a, & \text{if } x = \dfrac{\pi}{2} \\ \dfrac{b(1 - \sin x)}{(\pi - 2x)^2}, & \text{if } x > \dfrac{\pi}{2} \end{cases}$.

If $f(x)$ is a continuous function at $x = \dfrac{\pi}{2}$, find a and b.

Detailed Solutions

1. (c) : We have, $|x-1| = \begin{cases} x-1, & x \geq 1 \\ 1-x, & x < 1 \end{cases}$

$\therefore \quad f(x) = \dfrac{|x-1|}{(x-1)} = \begin{cases} 1, & x \geq 1 \\ -1, & x < 1 \end{cases}$

$\therefore$ Range $(f) = \{-1, 1\}$

2. (b) : By equality of two matrices, equating the corresponding elements, we get

$2a + b = 4$, $5c - d = 11$

$a - 2b = -3$, $4c + 3d = 24$

Solving these equations, we get

$a = 1$, $b = 2$, $c = 3$ and $d = 4$.

3. (a) : We have,

$$I = \int \left(e^{x\log a} + e^{a\log x} + e^{a\log a}\right) dx$$

Then, $I = \int \left(e^{\log a^x} + e^{\log x^a} + e^{\log a^a}\right) dx$

$\Rightarrow \quad I = \int (a^x + x^a + a^a) dx \qquad [\because e^{\log \lambda} = \lambda]$

$\Rightarrow \quad I = \int a^x dx + \int x^a dx + \int a^a dx$

$\Rightarrow \quad I = \dfrac{a^x}{\log a} + \dfrac{x^{a+1}}{a+1} + a^a x + C$

4. (b) : $\sqrt{x + \left(\dfrac{dy}{dx}\right)^2} = a\left(\dfrac{d^2y}{dx^2}\right)^{\frac{1}{3}}$

$\Rightarrow \quad \left[x + \left(\dfrac{dy}{dx}\right)^2\right]^3 = a^6 \left(\dfrac{d^2y}{dx^2}\right)^2$.

$\therefore$ Order of the differential equation is 2 and degree is 2.

5. (d) : Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k} \quad \Rightarrow (\vec{a} \cdot \hat{i})^2 = x^2$

Similarly, $(\vec{a} \cdot \hat{j})^2 = y^2$ and $(\vec{a} \cdot \hat{k})^2 = z^2$

$\therefore \quad (\vec{a} \cdot \hat{i})^2 + (\vec{a} \cdot \hat{j})^2 + (\vec{a} \cdot \hat{k})^2 = x^2 + y^2 + z^2 = |\vec{a}|^2$

6. (a) : $M_{23} = \begin{vmatrix} 2 & -3 \\ 1 & 5 \end{vmatrix} = 10 + 3 = 13$

www.xambrics.in

7. (d) : Let E and F denote respectively the events that first and second ball drawn are black. We have to find $P(E \cap F)$.

Now, $P(E) = \frac{10}{15}$, $P(F \mid E) = \frac{9}{14}$

By multiplication rule of probability, we have

$P(E \cap F) = P(E).P(F|E) = \frac{10}{15} \times \frac{9}{14} = \frac{3}{7}$

8. (a)

9. (c) : Let $x = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$, then

$\cos x = \frac{-\sqrt{3}}{2}$

We know that the range of principal value of $\cos^{-1}x$ is $[0, \pi]$

$\Rightarrow \cos x = \cos\left(\pi - \frac{\pi}{6}\right) \Rightarrow x = \frac{5\pi}{6} \in [0, \pi]$

$\therefore$ Principal value is $\frac{5\pi}{6}$.

10. (a) : Let $I = \int 2^{2^{2^x}} 2^{2^x} 2^x dx$

Let $2^{2^{2^x}} = t \Rightarrow 2^{2^{2^x}} 2^{2^x} 2^x (\log 2)^3 dx = dt$

$\Rightarrow I = \int \frac{1}{(\log 2)^3} dt$

$= \frac{1}{(\log 2)^3} t + C = \frac{1}{(\log 2)^3} 2^{2^{2^x}} + C$

11. All possible orders of matrix having 12 elements are

$1 \times 12, 12 \times 1, 2 \times 6, 6 \times 2, 3 \times 4, 4 \times 3$.

$\therefore$ Total number of possible orders = 6

12. Let the direction cosines of the line be l, m, n.

Then, $l = \cos 90° = 0$, $m = \cos 60° = \frac{1}{2}$ and $n = \cos 30° = \frac{\sqrt{3}}{2}$.

So, direction cosines are $\left(0, \frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

13. We have, $I = \int_{-1}^{0} \frac{dx}{2x+3}$

$= \left[\frac{\log(2x+3)}{2}\right]_{-1}^{0} = \left[\frac{\log 3}{2} - \frac{\log 1}{2}\right] = \frac{\log 3}{2}$

14. We have, $\vec{a} = 3\hat{i} - 2\hat{j} + 5\hat{k}, \vec{b} = 5\hat{i} + 5\hat{j} - \hat{k}$

$\therefore \quad \vec{a} \cdot \vec{b} = (3\hat{i} - 2\hat{j} + 5\hat{k}) \cdot (5\hat{i} + 5\hat{j} - \hat{k})$

$= 15 - 10 - 5 = 0$

So, $\vec{a}$ and $\vec{b}$ are perpendicular vectors.

OR

Let $\vec{a} = 7\hat{i} + \hat{j} - 4\hat{k}$ and $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$.

Now, $\vec{a} \cdot \vec{b} = (7\hat{i} + \hat{j} - 4\hat{k}) \cdot (2\hat{i} + 6\hat{j} + 3\hat{k})$

$= 14 + 6 - 12 = 8$

Also, $|\vec{b}| = \sqrt{2^2 + 6^2 + 3^2} = 7$

$\therefore$ Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{8}{7}$

15. $f(x) = \frac{x}{\log x}$ is defined for $x > 0$ and $x \neq 1$

Also, $f'(x) = \frac{\log x \cdot 1 - x \cdot \frac{1}{x}}{(\log x)^2} = \frac{\log x - 1}{(\log x)^2}$

$\therefore \quad f'(x) > 0 \Rightarrow \log x > 1 \Rightarrow x > e \quad \therefore \quad x \in (e, \infty)$.

OR

$\frac{dy}{dx} = \frac{-4x}{3}(4 - x^2)^{-1/3}$

$\Rightarrow \left(\frac{dy}{dx}\right)_{(2,0)} = \infty$. Hence there is a vertical tangent to the given curve at $x = 2$. Hence, the equation of the tangent is $x = 2$ (the vertical line through (2, 0)).

16. Since, the given points are collinear.

$\therefore \quad \frac{1}{2}\begin{vmatrix} 2 & -3 & 1 \\ k & -1 & 1 \\ 0 & 4 & 1 \end{vmatrix} = 0$

$\Rightarrow \quad 2(-1 - 4) + 3(k - 0) + 1(4k - 0) = 0$

$\Rightarrow \quad -10 + 3k + 4k = 0 \Rightarrow k = \frac{10}{7}$

17. If two dice are thrown, then total number of cases = 36

Number of cases of total of 9 or 11

= {(3, 6), (4, 5), (6, 3), (5, 4), (6, 5), (5, 6)}

$P(\text{total is 9 or 11}) = \frac{6}{36} = \frac{1}{6}$

P(total is neither 9 nor 11) = 1 – P(total is 9 or 11)

$= 1 - \frac{1}{6} = \frac{5}{6}$

18. $\because \quad y = \log_a x + \frac{\log a}{\log x} + 1 + 1$ $\{\because \log_x x = 1\}$

www.examitesin

$$\Rightarrow \frac{dy}{dx}=\frac{1}{x}\log_a e-\log a\left(\frac{1}{\log x}\right)^2\frac{1}{x}$$

$$=\frac{1}{x\log a}-\frac{\log a}{x(\log x)^2}$$

OR

We have, $3x + 2y = \sin y$

Differentiating w.r.t. x, we get

$$3+2\frac{dy}{dx}=\cos y\frac{dy}{dx}$$

$$\Rightarrow (\cos y-2)\frac{dy}{dx}=3 \Rightarrow \frac{dy}{dx}=\frac{3}{\cos y-2}$$

19. Since, the given lines are coplanar.

$$\therefore \begin{vmatrix} 4-1 & -3-(-1) & -1-(-10) \\ 2 & -3 & 8 \\ 1 & k & 7 \end{vmatrix}=0$$

$\Rightarrow 3(-21 - 8k) + 2(14 - 8) + 9(2k + 3) = 0$

$\Rightarrow 6k = -24 \therefore k = -4.$

20. Let $I=\int\frac{1+\tan x}{x+\log\sec x}dx$

Put $x + \log \sec x = t \Rightarrow (1 + \tan x)dx = dt$

$$\therefore I=\int\frac{dt}{t}=\log|t|+C=\log|x+\log\sec x|+C$$

21. Let $\cos^{-1}\left(\frac{1}{2}\right)=\theta\Rightarrow\cos\theta=\frac{1}{2}=\cos\frac{\pi}{3}$

$$\Rightarrow \theta=\frac{\pi}{3}\in[0,\pi]$$

$\therefore$ Principal value of $\cos^{-1}\frac{1}{2}$ is $\frac{\pi}{3}$

Let $\sin^{-1}\left(\frac{1}{2}\right)=\phi \Rightarrow \sin\phi=\frac{1}{2}=\sin\frac{\pi}{6}$

$$\Rightarrow \phi=\frac{\pi}{6}\in\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$$

$\therefore$ Principal value of $\sin^{-1}\frac{1}{2}$ is $\frac{\pi}{6}$.

So, the value of $\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$

$$=\frac{\pi}{3}+2\times\frac{\pi}{6}=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

22. Let $y = x^{\sin^{-1}x}$

Taking log on both sides, we get

$$\log y=\log x^{\sin^{-1}x} \Rightarrow \log y=\sin^{-1}x\cdot\log x$$

Differentiating w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx}=\frac{\sin^{-1}x}{x}+\frac{1}{\sqrt{1-x^2}}\log x$$

$$\Rightarrow \frac{dy}{dx}=x^{\sin^{-1}x}\left(\frac{\sin^{-1}x}{x}+\frac{\log x}{\sqrt{1-x^2}}\right)$$

OR

Let $u(x) = \sin^2 x$ and $v(x) = e^{\cos x}$

Now, $\frac{du}{dx}=2\sin x\cos x$

and $\frac{dv}{dx}=e^{\cos x}(-\sin x)=-(\sin x)e^{\cos x}$

Thus, $\frac{du}{dv}=\frac{2\sin x\cos x}{-\sin x\cdot e^{\cos x}}=-\frac{2\cos x}{e^{\cos x}}$

23. Given, $y = a\log x + bx^2 + x, x \neq 0$

$$\Rightarrow \frac{dy}{dx}=a\cdot\frac{1}{x}+b\cdot 2x+1$$

For extreme points, $\frac{dy}{dx}=0$

$$\Rightarrow \frac{a}{x}+2bx+1=0$$

At $x = -1$, we get $-a - 2b + 1 = 0$ ...(i)

At $x = 2$, we get $\frac{a}{2}+4b+1=0$ ... (ii)

On solving (i) and (ii), we get $a = 2$, $b=-\frac{1}{2}$.

24. Here, $\vec{a}=-\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=-2\hat{i}-\hat{j}-\hat{k}$

$$\Rightarrow \vec{a}\times\vec{b}=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 1 \\ -2 & -1 & -1 \end{vmatrix}$$

$$=(-1+1)\hat{i}-(1+2)\hat{j}+(1+2)\hat{k}=-3\hat{j}+3\hat{k}$$

$$\Rightarrow |\vec{a}\times\vec{b}|=\sqrt{(-3)^2+(3)^2}=3\sqrt{2}$$

$$\therefore \text{Required unit vector}=\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}=\frac{-3\hat{j}+3\hat{k}}{3\sqrt{2}}$$

$$=-\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}$$

OR

We have, $\vec{a}\times\vec{b}=\vec{a}\times\vec{c}$

$\Rightarrow \vec{a}\times\vec{b}-\vec{a}\times\vec{c}=0 \Rightarrow \vec{a}\times(\vec{b}-\vec{c})=0$

$\Rightarrow \vec{a}\,||\,\vec{b}-\vec{c}$ $[\because \vec{a}\neq 0$ and $\vec{b}\neq\vec{c}]$

www.exambites.in

$\Rightarrow \quad \vec{b} - \vec{c} = \lambda\vec{a}$, for some scalar λ

$\Rightarrow \quad \vec{b} = \vec{c} + \lambda\vec{a}$

25. We have, $P(A) = 0.4$,
$P(B) = 0.7$ and $P(B \mid A) = 0.6$

Now, $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$

$\Rightarrow \quad P(A \cap B) = P(B \mid A) \times P(A) = 0.6 \times 0.4 = 0.24$
Now, we have
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$\Rightarrow \quad P(A \cup B) = 0.4 + 0.7 - 0.24 \Rightarrow P(A \cup B) = 0.86$

26. We have, $6x - 2 = 3y + 1 = 2z - 2$

$$\Rightarrow \quad 6\left(x - \frac{1}{3}\right) = 3\left(y + \frac{1}{3}\right) = 2(z - 1)$$

$$\Rightarrow \quad \frac{x - \frac{1}{3}}{1} = \frac{y + \frac{1}{3}}{2} = \frac{z - 1}{3}$$

Direction ratios of the given line are 1, 2, 3.
Vector equation of line passing through the point (2, –1, –1) and parallel to the given line is

$\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \lambda\,(\hat{i} + 2\hat{j} + 3\hat{k})$.

27. Let, $I = \int_{1/e}^{e} |\log x|\, dx$

Let $f(x) = \log x$

$$|\log x| = \begin{cases} -\log x, & 0 < x < 1 \\ \log x, & x > 1 \end{cases}$$

$$\therefore \quad I = \int_{1/e}^{e} |\log x|\, dx = \int_{1/e}^{1} -(\log x)\, dx + \int_{1}^{e} \log x\, dx$$

$$= -\left[x \log x - x\right]_{1/e}^{1} + \left[x \log x - x\right]_{1}^{e}$$

$$= -\left[(0 - 1) - \left(-\frac{1}{e} - \frac{1}{e}\right)\right] + \left[(e - e) - (0 - 1)\right]$$

$$= -\frac{2}{e} + 2$$

28. Let $I = \int \dfrac{x\, e^{2x}}{(2x+1)^2}\, dx$

Put $2x = t \;\Rightarrow\; dx = \dfrac{dt}{2}$

$$\therefore \quad I = \frac{1}{4}\int \frac{t\, e^t}{(t+1)^2}\, dt = \frac{1}{4}\int e^t \left[\frac{t+1}{(t+1)^2} - \frac{1}{(t+1)^2}\right] dt$$

$$= \frac{1}{4}\int e^t \left(\frac{1}{t+1} - \frac{1}{(t+1)^2}\right) dt$$

If $f(t) = \dfrac{1}{t+1} \;\Rightarrow\; f'(t) = \dfrac{-1}{(t+1)^2}$

So, $I = \dfrac{1}{4}\dfrac{e^{2x}}{(2x+1)} + C$

OR

We have, $I = \int \dfrac{dx}{\sqrt[3]{x} + \sqrt[4]{x}} \Rightarrow I = \int \dfrac{dx}{(x)^{1/3} + (x)^{1/4}}$

Put $x = t^{12} \Rightarrow dx = 12t^{11}dt$

$$\Rightarrow \quad I = \int \frac{12t^{11}}{t^4 + t^3}\, dt = 12\int \frac{t^8}{t+1}\, dt$$

$$\Rightarrow \quad I = 12\int \left(\frac{t^8 - 1 + 1}{t+1}\right) dt$$

$$= 12\int \frac{t^8 - 1}{t+1}\, dt + 12\int \frac{1}{t+1}\, dt$$

$$= 12\int \left[\frac{(t+1)(t-1)(t^2+1)(t^4+1)}{t+1}\right] dt + 12\log|t+1| + C$$

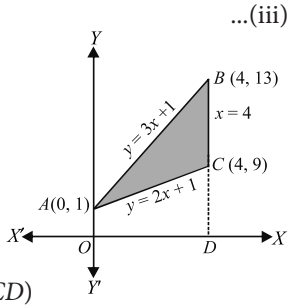
$$= 12\int (t^4+1)(t^2+1)(t-1)\, dt + 12\log|t+1| + C$$

$$\Rightarrow \quad I = 12\int \left(t^7 - t^6 + t^5 - t^4 + t^3 - t^2 + t - 1\right) dt + 12\log|t+1| + C$$

$$= 12\left[\frac{t^8}{8} - \frac{t^7}{7} + \frac{t^6}{6} - \frac{t^5}{5} + \frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2} - t + \log|t+1|\right] + C$$

$$= 12\left[\frac{1}{8}x^{\frac{2}{3}} - \frac{1}{7}x^{\frac{7}{12}} + \frac{1}{6}x^{\frac{1}{2}} - \frac{1}{5}x^{\frac{5}{12}} + \frac{1}{4}x^{\frac{1}{3}} - \frac{1}{3}x^{\frac{1}{4}} + \frac{1}{2}x^{\frac{1}{6}} - x^{\frac{1}{12}} + \log\left|x^{\frac{1}{12}} + 1\right|\right] + C$$

29. The given lines are $y = 2x + 1$, ...(i)
$y = 3x + 1$ ...(ii)
and $x = 4$ ...(iii)
Solving (i) and (ii), we get $x = 0$, $y = 1$
Solving (ii) and (iii), we get $x = 4$, $y = 13$
Solving (iii) and (i), we get $x = 4$, $y = 9$
$\therefore$ Required area
= area (ΔABC)
= $ar(OABD) - ar(OACD)$



www.exambites.in

$$= \int_0^4 (3x+1)\,dx - \int_0^4 (2x+1)\,dx = \left[\frac{3x^2}{2} + x\right]_0^4 - \left[x^2 + x\right]_0^4$$

$$= \left[\left(\frac{3}{2}(16) + 4\right) - (0+0)\right] - [(16+4) - (0+0)]$$

= 28 – 20 = 8 sq. units

30. We have, $y = be^x + ce^{2x}$ ...(i)

Differentiating (i) with respect to x, we get

$$\frac{dy}{dx} = be^x + 2ce^{2x} \quad \text{...(ii)}$$

Again differentiating (ii) with respect to x, we get

$$\frac{d^2y}{dx^2} = be^x + 4ce^{2x}$$

$$\therefore \quad \frac{d^2y}{dx^2} - \frac{3dy}{dx} + 2y = be^x + 4ce^{2x} - 3(be^x + 2ce^{2x}) + 2(be^x + ce^{2x})$$

$$= be^x + 4ce^{2x} - 3be^x - 6ce^{2x} + 2be^x + 2ce^{2x} = 0$$

So, $y = be^x + ce^{2x}$ satisfies the given differential equation. Hence, it is a solution of the given differential equation.

31. Let A, B, C, D denote events of getting a white ball in first, second, third and fourth draw respectively.

Required probability = $P(A \cap B \cap C \cap D)$

$$= P(A)P(B \mid A)P(C \mid A \cap B)P(D \mid A \cap B \cap C)$$

Now, $P(A) = \frac{5}{20} = \frac{1}{4}$. ...(i)

When a white ball is drawn in the first draw, there are 19 balls left in the bag out of which 4 are white.

$$\therefore \quad P(B \mid A) = \frac{4}{19}$$

Since the ball drawn is not replaced, therefore after drawing a white ball in second draw there are 18 balls left in the bag out which 3 are white.

$$\therefore \quad P(C \mid A \cap B) = \frac{3}{18} = \frac{1}{6}$$

After drawing a white ball in third draw, there are 17 balls left in the bag, out of which 2 are white.

$$\therefore \quad P(D \mid A \cap B \cap C) = \frac{2}{17}$$

$$\therefore \quad \text{Required probability} = \frac{1}{4} \times \frac{4}{19} \times \frac{1}{6} \times \frac{2}{17} = \frac{1}{969}$$

OR

Let W, L and D denote respectively the events that the team wins, loses and draws the game. Then, $P(W) = 0.7$, $P(L) = 0.2$, $P(D) = 0.1$

Now, required probability

= P(Team wins at least two games and does not lose any game)

= $P(WWW) + P(WWD) + P(WDW) + P(DWW)$

= $(0.7)^3 + (0.7)^2(0.1) + (0.7)^2(0.1) + (0.7)^2(0.1)$

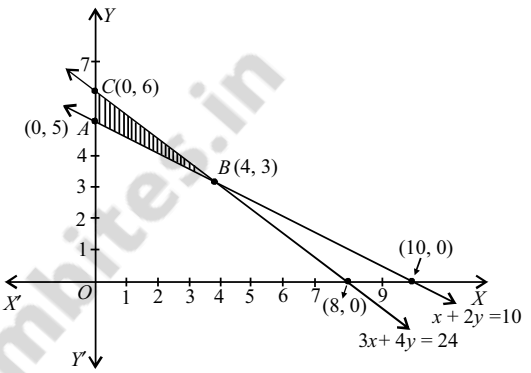
= $(0.7)^2(0.7 + 0.1 + 0.1 + 0.1) = 0.49$

32. Convert the given inequalities into equations.

Let $x + 2y = 10$ ...(i), $3x + 4y = 24$ ...(ii)

and $x = 0$, $y = 0$

The shaded region in the figure is the feasible region ABC determined by the given system of constraints which is bounded.



The coordinates of corner points A, B and C are (0, 5), (4, 3) and (0, 6) respectively. Now, we evaluate $Z = 200x + 500y$ at these points.

Corner point	Corresponding value of Z
$A(0, 5)$	2500
$B(4, 3)$	2300 ← Minimum
$C(0, 6)$	3000

Hence, minimum value of Z is 2300 attained at the point $B(4, 3)$.

33. Here, $A = \begin{bmatrix} 4 & -6 & 1 \\ -1 & -1 & 1 \\ -4 & 11 & -1 \end{bmatrix}$

$\therefore \quad |A| = 4(1 - 11) - (-6)(1 + 4) + 1(-11 - 4)$

$= -40 + 30 - 15 = -25 \neq 0$

Now, $A_{11} = -10$, $A_{12} = -5$, $A_{13} = -15$, $A_{21} = 5$, $A_{22} = 0$, $A_{23} = -20$, $A_{31} = -5$, $A_{32} = -5$, $A_{33} = -10$

$$\therefore \text{ adj}A = \begin{bmatrix} -10 & -5 & -15 \\ 5 & 0 & -20 \\ -5 & -5 & -10 \end{bmatrix}' = \begin{bmatrix} -10 & 5 & -5 \\ -5 & 0 & -5 \\ -15 & -20 & -10 \end{bmatrix}.$$

$$\text{Now, } A(\text{adj}A) = \begin{bmatrix} 4 & -6 & 1 \\ -1 & -1 & 1 \\ -4 & 11 & -1 \end{bmatrix} \begin{bmatrix} -10 & 5 & -5 \\ -5 & 0 & -5 \\ -15 & -20 & -10 \end{bmatrix}$$

www.exambites.in

$$= \begin{bmatrix} -40+30-15 & 20-0-20 & -20+30-10 \\ 10+5-15 & -5-0-20 & 5+5-10 \\ 40-55+15 & -20+0+20 & 20-55+10 \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 0 & 0 \\ 0 & -25 & 0 \\ 0 & 0 & -25 \end{bmatrix} = -25 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = (-25)\, I_3$$

$= |A|\, I_3$

Similarly, we can show that (adj A) $A = |A|\, I_3$

Hence, (adj A) $A = |A|\, I_3 = A$ (adj A).

OR

We have, $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$

$$\therefore \quad |A| = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{vmatrix} = -6+28+45 = 67 \neq 0$$

So, A is invertible.

Now, $A_{11} = -6$, $A_{12} = 14$, $A_{13} = -15$, $A_{21} = 17$, $A_{22} = 5$, $A_{23} = 9$, $A_{31} = 13$, $A_{32} = -8$, $A_{33} = -1$

$$\therefore \text{ adj } A = \begin{bmatrix} -6 & 14 & -15 \\ 17 & 5 & 9 \\ 13 & -8 & -1 \end{bmatrix}' = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

$$\text{So, } A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

Now, the given system of equations is

$x + 2y - 3z = -4$

$2x + 3y + 2z = 2$

$3x - 3y - 4z = 11$

or $AX = B$, where

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

The solution of the given system of equations is given by

$X = A^{-1} B$.

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix} \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

$$= \frac{1}{67} \begin{bmatrix} 24+34+143 \\ -56+10-88 \\ 60+18-11 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

$\Rightarrow$ $x = 3$, $y = -2$ and $z = 1$ is the required solution.

34. We have, $f(x) = \dfrac{4\sin x - 2x - x\cos x}{2 + \cos x}$

$$\Rightarrow \quad f'(x) = \left[\frac{\begin{matrix}(2+\cos x)(4\cos x - 2 + x\sin x - \cos x) \\ + (4\sin x - 2x - x\cos x)(\sin x)\end{matrix}}{(2+\cos x)^2} \right]$$

$$\Rightarrow \quad f'(x) = \frac{\cos x\,(4 - \cos x)}{(2+\cos x)^2}$$

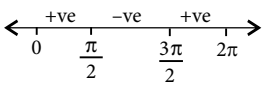
For critical points, put $f'(x) = 0$

$$\Rightarrow \quad \frac{\cos x\,(4 - \cos x)}{(2 + \cos x)^2} = 0$$

$\Rightarrow$ $\cos x = 0$ and $\cos x \neq 4$ $\quad [-1 \leq \cos x \leq 1]$

$\Rightarrow$ $x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$ $\quad [0 \leq x \leq 2\pi]$

The points $x = \dfrac{\pi}{2}$ and $x = \dfrac{3\pi}{2}$ divide the given interval into disjoint intervals $\left(0, \dfrac{\pi}{2}\right)$, $\left(\dfrac{\pi}{2}, \dfrac{3\pi}{2}\right)$ and $\left(\dfrac{3\pi}{2}, 2\pi\right)$



Hence, $f(x)$ is increasing in $\left[0, \dfrac{\pi}{2}\right) \cup \left(\dfrac{3\pi}{2}, 2\pi\right]$ and decreasing in $\left(\dfrac{\pi}{2}, \dfrac{3\pi}{2}\right)$.

OR

Let $P(x, y)$ be any point on the parabola and $Q(0, c)$ be the given point. Then,

$PQ^2 = x^2 + (y - c)^2$

$\Rightarrow$ $PQ^2 = x^2 + (x^2 - c)^2 \Rightarrow PQ^2 = x^4 - x^2(2c - 1) + c^2$

Let $Z = PQ^2$. Then, $Z = x^4 - x^2(2c - 1) + c^2$

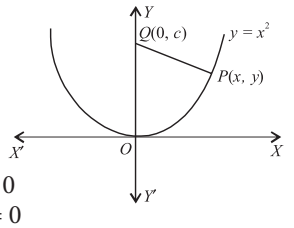
Clearly, Z will be minimum when PQ is minimum.

$$\Rightarrow \quad \frac{dZ}{dx} = 4x^3 - 2x(2c-1)$$

$$\text{and} \quad \frac{d^2Z}{dx^2} = 12x^2 - 2(2c-1)$$

www.exambites.in

For maximum or minimum value of Z, we must have



$$\frac{dZ}{dx} = 0$$

$\Rightarrow \quad 4x^3 - 2x(2c-1) = 0$

$\Rightarrow \quad 2x\{2x^2 - (2c-1)\} = 0$

$\Rightarrow \quad x = 0, \; x = \pm\sqrt{\frac{2c-1}{2}}$

$\Rightarrow \quad x = 0, x = \pm\alpha$ where $\alpha = \sqrt{\frac{2c-1}{2}}$

Now, $\left(\frac{d^2Z}{dx^2}\right)_{x=\pm\alpha} = 12\alpha^2 - 4\alpha^2 = 8\alpha^2 > 0$

So, Z is minimum at $x = \pm\alpha$.

The minimum value of PQ is given by

$PQ^2 = \alpha^2 + (\alpha^2 - c)^2$

$$\Rightarrow PQ^2 = \frac{2c-1}{2} + \left(\frac{2c-1}{2} - c\right)^2$$

$$= \frac{2c-1}{2} + \frac{1}{4} = \frac{4c-1}{4} \Rightarrow PQ = \frac{\sqrt{4c-1}}{2}$$

Hence, the minimum distance is $\frac{\sqrt{4c-1}}{2}$ units.

35. Given lines are

$\vec{r} = 3\hat{i} + 8\hat{j} + 3\hat{k} + \lambda(3\hat{i} - \hat{j} + \hat{k})$ and

$\vec{r} = -3\hat{i} - 7\hat{j} + 6\hat{k} + \mu(-3\hat{i} + 2\hat{j} + 4\hat{k})$

In cartesian form, we have

$$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = \lambda \quad ...(i)$$

and $$\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} = \mu \quad ...(ii)$$

Any point on (i) has coordinates

$P(3\lambda + 3, -\lambda + 8, \lambda + 3)$ and on (ii)

$Q(-3\mu - 3, 2\mu - 7, 4\mu + 6)$

Direction ratios of PQ are

$a = -3\mu - 3 - 3\lambda - 3 = -3\mu - 3\lambda - 6$

$b = 2\mu - 7 + \lambda - 8 = 2\mu + \lambda - 15$

$c = 4\mu + 6 - \lambda - 3 = 4\mu - \lambda + 3$

As PQ is perpendicular to (i) and (ii).

$\therefore \quad -9\mu - 9\lambda - 18 - 2\mu - \lambda + 15 + 4\mu - \lambda + 3 = 0$

$\Rightarrow \quad -7\mu - 11\lambda = 0$

$\Rightarrow \quad 7\mu + 11\lambda = 0 \quad ...(iii)$

and $9\mu + 9\lambda + 18 + 4\mu + 2\lambda - 30 + 16\mu - 4\lambda + 12 = 0$

$\Rightarrow \quad 29\mu + 7\lambda = 0 \quad ...(iv)$

Solving (iii) and (iv), we get $\mu = 0, \lambda = 0$.

$\therefore$ Coordinates of $P(3, 8, 3)$, $Q(-3, -7, 6)$ and direction ratios of PQ are $<-6, -15, 3>$ or $<2, 5, -1>$

So, shortest distance $PQ = \sqrt{6^2 + 15^2 + (-3)^2}$

$= \sqrt{270} = 3\sqrt{30}$ units.

Vector equation of PQ is

$\vec{r} = 3\hat{i} + 8\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 5\hat{j} - \hat{k})$

36. We have,

$$\text{L.H.L.} = \lim_{x \to \frac{\pi}{2}^-} f(x) = \lim_{x \to \frac{\pi}{2}^-} \frac{1 - \sin^3 x}{3\cos^2 x}$$

$$= \lim_{x \to \frac{\pi}{2}^-} \frac{(1 - \sin x)(1 + \sin x + \sin^2 x)}{3(1 - \sin^2 x)}$$

$$= \lim_{x \to \frac{\pi}{2}^-} \frac{(1 - \sin x)(1 + \sin x + \sin^2 x)}{3(1 - \sin x)(1 + \sin x)}$$

$$= \lim_{x \to \frac{\pi}{2}^-} \frac{1 + \sin x + \sin^2 x}{3(1 + \sin x)} = \frac{1 + 1 + 1}{3(2)} = \frac{1}{2}$$

$$\text{R.H.L.} = \lim_{x \to \frac{\pi}{2}^+} f(x) = \lim_{x \to \frac{\pi}{2}^+} \frac{b(1 - \sin x)}{(\pi - 2x)^2}$$

$$= \lim_{x \to \frac{\pi}{2}^+} \frac{b(1 - \sin x)}{(\pi - 2x)^2} = \lim_{h \to 0} \frac{b\left(1 - \sin\left(\frac{\pi}{2} + h\right)\right)}{\left(\pi - 2\left(\frac{\pi}{2} + h\right)\right)^2}$$

$$= \lim_{h \to 0} \frac{b(1 - \cos h)}{(0 - 2h)^2} = \lim_{h \to 0} \frac{b(1 - \cos h)}{4h^2}$$

$$= \lim_{h \to 0} \frac{b\left(2\sin^2 \frac{h}{2}\right)}{4h^2}$$

$$= \frac{b}{2} \lim_{h \to 0} \left(\frac{\sin \frac{h}{2}}{\frac{h}{2}}\right)^2 \times \frac{1}{4} = \frac{b}{8} \quad \text{and } f\left(\frac{\pi}{2}\right) = a$$

Since $f(x)$ is continuous at $x = \frac{\pi}{2}$

$$\therefore \lim_{x \to \frac{\pi}{2}^-} f(x) = \lim_{x \to \frac{\pi}{2}^+} f(x) = f\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \frac{1}{2} = \frac{b}{8} = a \Rightarrow a = \frac{1}{2} \text{ and } b = 4$$

www.exambites.in

PRACTICE PAPER 4
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) This question paper comprises four Sections A, B, C and D. This question paper carries 36 questions. All questions are compulsory.

(ii) Section A : Questions no. 1 to 20 comprises of 20 questions of 1 mark each.

(iii) Section B : Questions no. 21 to 26 comprises of 6 questions of 2 marks each.

(iv) Section C : Questions no. 27 to 32 comprises of 6 questions of 4 marks each.

(v) Section D : Questions no. 33 to 36 comprises of 4 questions of 6 marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 3 questions of one mark, 2 questions of two marks, 2 questions of four marks and 2 questions of six marks. Only one of the choices in such questions have to be attempted.

(vii) In addition to this, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

SECTION - A

(Q.1-Q.10) are multiple choice type questions. Select the correct option.

1. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$, then find the value of α for which $A^2 = B$.

(a) 1 (b) -1

(c) 4 (d) No real value

2. Let R be an equivalence relation on a finite set A having n elements. Then, the number of ordered pairs in R is

(a) Less than n

(b) Greater than or equal to n

(c) Less than or equal to n

(d) None of these

3. If $y = \log_{10}x + \log_e y$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{y}{y-1}$ (b) $\frac{y}{x}$

(c) $\frac{\log_{10} e}{x}\left(\frac{y}{y-1}\right)$ (d) None of these

4. Domain of $\cos^{-1}[x]$, where $[\cdot]$ denotes the greatest integer function, is

(a) $[-1, 2]$ (b) $[-1, 2)$

(c) $(-1, 2]$ (d) None of these

5. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3+\left(\frac{dy}{dx}\right)^2+\sin\left(\frac{dy}{dx}\right)+1=0$ is

(a) 3 (b) 2

(c) 1 (d) not defined

6. The distance of the plane $\vec{r}\cdot\left(\frac{2}{7}\hat{i}+\frac{3}{7}\hat{j}-\frac{6}{7}\hat{k}\right)=1$ from the origin is

(a) 1 (b) 7

(c) $\frac{1}{7}$ (d) None of these

7. Find x, if $\begin{bmatrix}1 & 2 & x\\ 1 & 1 & 1\\ 2 & 1 & -1\end{bmatrix}$ is singular.

(a) 1 (b) 2

(c) 3 (d) 4

8. Evaluate : $\int_0^1 e^{2-3x}dx$

(a) $(e^2 - e^{-1})$ (b) $\frac{1}{3}(e^2 - e^{-1})$

(c) $\frac{1}{2}(e^2 - e^{-1})$ (d) None of these

9. If E and F are events such that $0 < P(F) < 1$, then

(a) $P(E\,|\,F)+P(\bar{E}\,|\,F)=1$

(b) $P(E\,|\,F)+P(E\,|\,\bar{F})=1$

(c) $P(\bar{E}\,|\,F)+P(E\,|\,\bar{F})=1$

(d) $P(E\,|\,\bar{F})+P(\bar{E}\,|\,\bar{F})=0$

www.exambites.in

10. The function $f(x) = \tan x - x$
(a) always increases (b) always decreases
(c) never increases
(d) sometimes increases and sometimes decreases

(Q.11-Q.15) Fill in the blanks.

11. The feasible region for a L.P.P. is always a _____ polygon.

OR

In a L.P.P., the value of objective function (if exists) is always _____ .

12. The area of the parallelogram whose adjacent sides are determined by the vector $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$, is _____ .

13. If A and B are independent events, then P(neither A nor B) = _____ .

OR

If A' and B' are independent events, then $P(A' \cup B)$ = _____ .

14. The reflection of the point (α, β, γ) in the xy-plane is _____ .

15. $\int f'(ax+b)[f(ax+b)]^n dx$ = _____ .

(Q.16-Q.20) Answer the following questions.

16. Evaluate : $\int \frac{x^3 - 1}{x^2} dx$

OR

Find the value of $\int \frac{2 - 3\sin x}{\cos^2 x} dx$.

17. Given that $\vec{a} \cdot \vec{b} = 0$ and $\vec{a} \times \vec{b} = \vec{0}$. What can you conclude about the vectors $\vec{a}$ and $\vec{b}$?

18. Find values of k if area of triangle is 4 sq. units and vertices are $(k, 0)$, $(4, 0)$, $(0, 2)$

19. Prove that $\int_0^{\pi/2} \sin^3 x \, dx = \frac{2}{3}$.

20. If a matrix has 18 elements, then what are the possible orders it can have?

SECTION - B

21. Using principal value, find the value of

$$\tan^{-1}(1) + \cos^{-1}\left(\frac{-1}{2}\right)$$

22. If $x = \sin t$, $y = \cos 2t$, then find $\frac{dy}{dx}$.

23. Find the points on the line $\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2}$ at a distance of 5 units from the point $P(1, 3, 3)$.

24. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$, then find the angle between $\vec{a}$ and $\vec{b}$.

OR

If $\vec{a} = \vec{b} + \vec{c}$, then is it true that $|\vec{a}| = |\vec{b}| + |\vec{c}|$? Justify your answer.

25. Find the least value of a such that the function f given by $f(x) = x^2 + ax + 1$ is strictly increasing on $(1, 2)$.

26. An instructor has a question bank consisting of 300 easy True/False questions, 200 difficult True/False questions, 500 easy multiple choice questions and 400 difficult multiple choice questions. If a question is selected at random from the question bank, what is the probability that it will be an easy question given that it is a multiple choice question?

OR

Consider the experiment of throwing a die, if a multiple of 3 comes up, throw the die again and if any other number comes, toss a coin. Find the conditional probability of the event 'the coin shows a tail', given that 'at least one die shows a 3'.

SECTION - C

27. Let L be the set of all lines in XY-plane and R be the relation in L defined as $R = \{(L_1, L_2) : L_1 \text{ is parallel to } L_2\}$. Show that R is an equivalence relation.

28. At any point (x, y) of a curve, the slope of the tangent is twice the slope of the line segment joining the point of contact to the point $(-4, -3)$. Find the equation of the curve given that it passes through $(-2, 1)$.

OR

In a bank, principal increases continuously at the rate of 5% per year. An amount of ₹ 1000 is deposited with this bank, how much will it worth after 10 years? ($e^{0.5} = 1.648$).

www.examcues.in

29. Maximize $Z = x + y$ subject to $x - y \leq -1$, $-x + y \leq 0$, $x, y \geq 0$.

30. Find the values of a and b such that the function defined as follows is continuous :

$$f(x) = \begin{cases} x+2, & x \leq 2 \\ ax+b, & 2 < x < 5 \\ 3x-2, & x \geq 5 \end{cases}$$

31. Evaluate $\int_0^{\pi} \frac{x}{1+\sin x} dx$.

OR

Evaluate $\int_0^{\pi} x \sin x \cos^2 x \, dx$.

32. A laboratory blood test is 99% effective in detecting a certain disease when it is in fact, present. However, the test also yields a false positive result for 0.5% of the healthy person tested (*i.e.* if a healthy person is tested, then, with probability 0.005, the test will imply he has the disease). If 0.1 percent of the population actually has the disease, what is the probability that a person has the disease given that his test result is positive ?

SECTION - D

33. A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is 10 m. Find the dimensions of the window to admit maximum light through the whole opening.

OR

A telephone company in a town has 500 subscribers on its list and collects fixed charges of ₹ 300 per subscriber per year. The company proposes to increase the annual subscription and it is believed that for every increase of ₹ 1 per one subscriber will discontinue the service. Find what increase will bring maximum profit?

34. Find the area of the smaller part of the circle $x^2 + y^2 = a^2$ cut off by the line $x = \frac{a}{\sqrt{2}}$.

35. Find the coordinates of the point where the line through $(3, -4, -5)$ and $(2, -3, 1)$ crosses the plane $2x + y + z = 7$.

OR

Find the equation of the plane passing through the intersection of the planes $x + 3y + 6 = 0$ and $3x - y - 4z = 0$ and whose perpendicular distance from the origin is unity.

36. Solve the following system of linear equation, using matrix method.

$2x + y + z = 1$, $x - 2y - z = \frac{3}{2}$, $3y - 5z = 9$

Detailed Solutions

1. (d) : Given that, $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$

$$\Rightarrow A^2 = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$$

Also, $B = A^2 \Rightarrow \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$

Clearly, this is not satisfied by any real value of α.

2. (b) : As, R is an equivalence relation on set A. Hence, R has atleast n ordered pairs.

3. (c): We have, $y = \log_{10} x + \log_e y$

$$\therefore \quad \frac{dy}{dx} = \frac{1}{x}\log_{10} e + \frac{1}{y}\frac{dy}{dx}$$

$$\Rightarrow \left(\frac{y-1}{y}\right)\frac{dy}{dx} = \frac{1}{x}\log_{10} e \Rightarrow \frac{dy}{dx} = \frac{\log_{10} e}{x}\left(\frac{y}{y-1}\right)$$

4. (b) : $0 \leq \cos^{-1}[x] \leq \pi$

$\Rightarrow -1 \leq [x] \leq 1 \Rightarrow -1 \leq x < 2 \Rightarrow x \in [-1, 2)$

5. (d) : Since, the given differential equation is not a polynomial in $\frac{dy}{dx}$.

Therefore, its degree is not defined.

6. (a): Since, given equation

$\vec{r} \cdot \left(\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}\right) = 1$ is in the form of $\vec{r} \cdot \hat{n} = d$.

$\therefore$ The distance of the given plane from the origin is 1.

www.exambits.in

7. (d) : For singular matrix, $|A| = 0$

$$\therefore \quad \begin{vmatrix} 1 & 2 & x \\ 1 & 1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 0$$

$\Rightarrow$ $1(-1 - 1) - 2(-1 - 2) + x(1 - 2) = 0$

$\Rightarrow$ $-2 + 6 - x = 0 \Rightarrow x = 4$

8. (b) : We have, $\int_0^1 e^{2-3x}dx = \left[\frac{e^{2-3x}}{-3}\right]_0^1$

$$= \frac{e^{2-3}}{-3} - \frac{e^2}{-3} = \frac{1}{3}[e^2 - e^{-1}]$$

9. (a): $P(E|F) + P(\bar{E}|F)$

$$= \frac{P(E \cap F) + P(\bar{E} \cap F)}{P(F)} = \frac{P((E \cup \bar{E}) \cap F)}{P(F)} = \frac{P(F)}{P(F)} = 1$$

10. (a): We have, $f(x) = \tan x - x$

$\therefore$ $f'(x) = \sec^2 x - 1$

$\Rightarrow$ $f'(x) > 0, \forall\, x \in R$ $(\because \sec^2 x \geq 1)$

Hence, $f(x)$ always increases for all x.

11. The feasible region for a L.P.P. is always a convex polygon.

OR

In a L.P.P., the value of objective function (if exists) is always finite.

12. Here, $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 3 \\ 2 & -7 & 1 \end{vmatrix}$

$$= (-1 + 21)\hat{i} - (1 - 6)\hat{j} + (-7 + 2)\hat{k} = 20\hat{i} + 5\hat{j} - 5\hat{k}$$

$\therefore$ Area of parallelogram $= |\vec{a} \times \vec{b}|$

$$= |20\hat{i} + 5\hat{j} - 5\hat{k}| = \sqrt{(20)^2 + (5)^2 + (-5)^2}$$

$$= \sqrt{400 + 25 + 25} = \sqrt{450} = 15\sqrt{2} \text{ sq. units.}$$

13. We know, if A and B are independent events, then A' and B' are also independent.

$\therefore$ $P(\text{neither } A \text{ nor } B) = P(A' \cap B') = P(A') \cdot P(B)'$

OR

We know, if A and B are independent events, then A and B' are also independent.

$\therefore$ $P(A' \cup B) = P((A \cap B')')$

$= 1 - P(A \cap B') = 1 - P(A)P(B')$

14. The reflection of the point (α, β, γ) in xy-plane is $(\alpha, \beta, -\gamma)$.

15. Let $I = \int f'(ax + b)[f(ax + b)]^n dx$

Putting $f(ax + b) = t \Rightarrow af'(ax + b)dx = dt$

$$\therefore \quad I = \frac{1}{a}\int t^n dt = \frac{1}{a} \times \frac{t^{n+1}}{n+1} + C$$

$$= \frac{1}{a(n+1)}\left[f(ax + b)\right]^{n+1} + C$$

16. We have, $\int \frac{x^3 - 1}{x^2} dx$

$$= \int \left(x - \frac{1}{x^2}\right) dx = \frac{x^2}{2} + \frac{1}{x} + C$$

OR

$$\int \frac{2 - 3\sin x}{\cos^2 x} dx = \int \left(\frac{2}{\cos^2 x} - \frac{3\sin x}{\cos^2 x}\right) dx$$

$$= \int (2\sec^2 x - 3\sec x \tan x) dx = 2\tan x - 3\sec x + C$$

17. $\vec{a}.\vec{b} = \vec{0}$ and $\vec{a} \times \vec{b} = \vec{0}$

$\Rightarrow$ $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a} \perp \vec{b}$

and $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a} \,||\, \vec{b}$

$\Rightarrow$ Either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ [$\because$ $\vec{a} \perp \vec{b}$ and $\vec{a} \,||\, \vec{b}$ are not valid at the same time]

18. Area of $\Delta = \frac{1}{2}\begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$

$$= \frac{1}{2}[k(0 - 2) - 0 + 1(8 - 0)] = \frac{1}{2}[-2k + 8] = -k + 4$$

Now, $-k + 4 = \pm 4$ (Given)

$\Rightarrow$ $-k + 4 = 4$ or $-k + 4 = -4$

$\Rightarrow$ $k = 0$ or $k = 8 \Rightarrow k = 0, 8$

19. We have, $\int_0^{\pi/2} \sin^3 x\, dx = \frac{1}{4}\int_0^{\pi/2} (3\sin x - \sin 3x) dx$

[$\because$ $\sin 3x = 3\sin x - 4\sin^3 x$]

$$= \frac{1}{4}\left[-3\cos x + \frac{\cos 3x}{3}\right]_0^{\pi/2}$$

$$= \frac{1}{4}\left[-3\cos\frac{\pi}{2} + \frac{1}{3}\cos\frac{3\pi}{2}\right] - \frac{1}{4}\left[-3\cos 0 + \frac{\cos 0}{3}\right]$$

$$= \frac{1}{4}\left[0 + 0 + 3 - \frac{1}{3}\right] = \frac{1}{4}\left(\frac{8}{3}\right) = \frac{2}{3}$$

20. We know that a matrix of order $m \times n$ has mn elements. Thus, the possible orders are 1×18, 18×1, 2×9, 9×2, 3×6, 6×3.

www.examrules.in

21. We know that the range of principal value branch of $\tan^{-1}$ and $\cos^{-1}$ are $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and $[0, \pi]$ respectively.

Let $\tan^{-1}(1) = x \Rightarrow 1 = \tan x$

$\Rightarrow \quad 1 = \tan\left(\frac{\pi}{4}\right)$, where $\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Let $\cos^{-1}\left(-\frac{1}{2}\right) = y \Rightarrow -\frac{1}{2} = \cos y$

$\Rightarrow \quad -\frac{1}{2} = -\cos\left(\frac{\pi}{3}\right) = \cos\left(\pi - \frac{\pi}{3}\right) = \cos\frac{2\pi}{3}$,

where $\frac{2\pi}{3} \in [0, \pi]$

So, $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{4} + \frac{2\pi}{3} = \frac{11\pi}{12}$

22. Here, $x = \sin t$ ...(i)

and $y = \cos 2t$ ...(ii)

Differentiating (i) & (ii) w.r.t. t, we get

$\therefore \quad \frac{dx}{dt} = \cos t$ and $\frac{dy}{dt} = -\sin 2t.\, 2 = -2\sin 2t$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2\sin 2t}{\cos t} = \frac{-2.2\sin t \cos t}{\cos t}$$

$= -4\sin t$

23. Any point on the line

$\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2} = \lambda$ (say) is of the form

$Q(3\lambda - 2, 2\lambda - 1, 2\lambda + 3)$ ...(i)

Now, distance PQ, where P is (1, 3, 3), is 5.

So, $(3\lambda - 2 - 1)^2 + (2\lambda - 1 - 3)^2 + (2\lambda + 3 - 3)^2 = 5^2$

$\Rightarrow \quad 9\lambda^2 + 9 - 18\lambda + 4\lambda^2 + 16 - 16\lambda + 4\lambda^2 = 25$

$\Rightarrow \quad 17\lambda^2 - 34\lambda = 0 \Rightarrow 17\lambda(\lambda - 2) = 0$

$\Rightarrow \quad \lambda = 0$ or $\lambda = 2$

From (i), the required points are : (–2, –1, 3) and (4, 3, 7).

24. Given, $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$

$\Rightarrow |\vec{a}||\vec{b}||\cos\theta| = |\vec{a}||\vec{b}||\sin\theta|$

$\Rightarrow |\cos\theta| = |\sin\theta| \Rightarrow |\tan\theta| = 1$

$\Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$

OR

We have, $\vec{a} = \vec{b} + \vec{c}$

$\therefore \quad |\vec{a}| = |\vec{b} + \vec{c}|$

Squaring, $|\vec{a}|^2 = |\vec{b} + \vec{c}|^2$

$= (\vec{b} + \vec{c}) \cdot (\vec{b} + \vec{c}) = \vec{b} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{b} + \vec{c} \cdot \vec{c}$

$= |\vec{b}|^2 + 2\vec{b} \cdot \vec{c} + |\vec{c}|^2$ $[\because \ \vec{b}.\vec{c} = \vec{c}.\vec{b}]$

$= |\vec{b}|^2 + |\vec{c}|^2 + 2|\vec{b}||\vec{c}|\cos\theta$

where 'θ' is the angle between $\vec{b}$ and $\vec{c}$.

When θ = 0, then

$|\vec{a}|^2 = |\vec{b}|^2 + |\vec{c}|^2 + 2|\vec{b}||\vec{c}| = \left(|\vec{b}| + |\vec{c}|\right)^2$

$\Rightarrow \quad |\vec{a}| = |\vec{b}| + |\vec{c}|$

When $\theta \neq 0$, then $|\vec{a}| \neq |\vec{b}| + |\vec{c}|$

25. We have, $f(x) = x^2 + ax + 1$ ...(i)

$\Rightarrow \quad f'(x) = 2x + a$

If $1 < x < 2 \Rightarrow 2 < 2x < 4$

$\Rightarrow \quad 2 + a < 2x + a < 4 + a$

$\Rightarrow \quad 2 + a < f'(x) < 4 + a$

Now $f(x)$ is strictly increasing on (1, 2) only if $f'(x) > 0$ for $1 < x < 2 \Rightarrow 2 + a \geq 0$

$\Rightarrow \quad a \geq -2$

$\therefore$ Required least value of a is –2.

26. Let E : 'it is an easy question' and F : 'it is multiple choice question' then $E \cap F$: 'it is an easy multiple choice question.'

Total number of questions

$= 300 + 200 + 500 + 400 = 1400$

$\therefore \quad P(E \cap F) = \frac{500}{1400} = \frac{5}{14}$

and $P(F) = \frac{500 + 400}{1400} = \frac{9}{14}$

Hence, required probability = $P(E \mid F)$

$= \frac{P(E \cap F)}{P(F)} = \frac{5/14}{9/14} = \frac{5}{9}$

OR

The sample space S is given by

$$S = \begin{Bmatrix} (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6), \\ (1,H), (1,T), (2,H), (2,T), (4,H), (4,T), \\ (5,H), (5,T) \end{Bmatrix}$$

Let E : 'the coin shows a tail' and F : 'atleast one die shows a 3',

$E = \{(1, T), (2, T), (4, T), (5, T)\}$ and

$F = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (6, 3)\}$

$\Rightarrow \quad E \cap F = \phi$

$\therefore \quad P(E) = 4/20$, $P(F) = 7/20$, $P(E \cap F) = 0/20$

Hence, the required probability = $P(E \mid F)$

$= \frac{P(E \cap F)}{P(F)} = \frac{0}{P(F)} = 0$

www.examrace.in

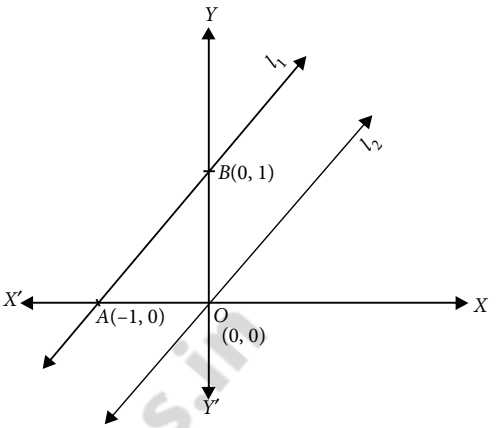
27. $R = \{(L_1, L_2) : L_1$ is parallel to $L_2\}$
(i) Reflexive
Let $L_1 \in L$
$L_1 \parallel L_1$ *i.e.*, $(L_1, L_1) \in R$. Thus, R is reflexive.
(ii) Symmetric
Let $L_1, L_2 \in L$
Let $(L_1, L_2) \in R \Rightarrow L_1 \parallel L_2$ then $L_2 \parallel L_1$
$\Rightarrow (L_2, L_1) \in R$
Thus, R is symmetric.
(iii) Transitive
Let $L_1, L_2, L_3 \in L$
Let $(L_1, L_2) \in R$ and $L_2, L_3 \in R$
$\Rightarrow L_1 \parallel L_2$ and $L_2 \parallel L_3 \Rightarrow L_1 \parallel L_3$
Clearly, R is transitive.
Hence, R is an equivalence relation.

28. Given, $\frac{dy}{dx} = 2\left(\frac{y+3}{x+4}\right)$ ...(i)
Integrating (i) both sides, we get
$$\int \frac{dy}{y+3} = 2\int \frac{dx}{x+4}$$
$\Rightarrow \log(y + 3) = 2 \log (x + 4) + C$
As, point (–2, 1) lies on it.
So, $\log 4 = 2 \log 2 + C$
$\Rightarrow 2 \log 2 = 2 \log 2 + C \Rightarrow C = 0$
$\therefore$ Required equation of the curve is
$\log (y + 3) = 2 \log (x + 4) \Rightarrow y + 3 = (x + 4)^2$

OR

Let P be the principal at any time t.
According to the given problem,
$$\frac{dP}{dt} = \left(\frac{5}{100}\right) \times P \Rightarrow \frac{dP}{dt} = \frac{P}{20} \quad ...(i)$$
Separating the variable in equation (i), we get
$$\Rightarrow \frac{dP}{P} = \frac{dt}{20} \quad ...(ii)$$
Integrating both sides of equation (ii), we get
$$\log P = \frac{t}{20} + C_1 \Rightarrow P = e^{\frac{t}{20}} e^{C_1}$$
$$\Rightarrow P = Ce^{t/20} \text{ (where } e^{C_1} = C) \quad ...(iii)$$
Now, $P = 1000$, when $t = 0$
Substituting the values of P and t in (iii), we get $C = 1000$
Therefore, equation (iii) gives $P = 1000\, e^{t/20}$
Let P_1 be the principal at $t = 10$ years.
Then, $P_1 = 1000\, e^{1/2} = 1000 \times 1.648 =$ ₹ 1648

29. We have, maximize $Z = x + y$
Subject to $x - y \le -1$; $-x + y \le 0$; $x, y \ge 0$
Let $l_1 : x - y = -1$; $l_2 : -x + y = 0$
Let us draw the graph of these equations.



Clearly, there is no feasible region.
[$\because$ There is no common region]
Hence, there is no maximum value of Z.

30. $\because f(x)$ is continuous at $x = 2$ and $x = 5$.
$\therefore \lim_{x \to 2^+} f(x) = f(2)$ and $\lim_{x \to 5^-} f(x) = f(5)$
$\therefore \lim_{x \to 2} (ax + b) = 4$ and $\lim_{x \to 5} (ax + b) = 13$
$\Rightarrow 2a + b = 4$ ...(i)
and $5a + b = 13$ ...(ii)
Solving these equations, we get $a = 3$ and $b = -2$

31. Let $I = \int_0^{\pi} \frac{x}{1 + \sin x} dx$ ... (i)
Using property $\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$, we get
$$I = \int_0^{\pi} \frac{\pi - x}{1 + \sin(\pi - x)} dx = \int_0^{\pi} \frac{\pi - x}{1 + \sin x} dx \quad ... (ii)$$
On adding (i) and (ii), we get
$$2I = \pi \int_0^{\pi} \frac{1}{1 + \sin x} dx = \pi \int_0^{\pi} \frac{(1 - \sin x)}{(1 + \sin x)(1 - \sin x)} dx$$
$$= \pi \int_0^{\pi} \frac{(1 - \sin x)\, dx}{\cos^2 x} \quad [\because \cos^2 x = 1 - \sin^2 x]$$
$$= \pi \int_0^{\pi} (\sec^2 x - \tan x \cdot \sec x) dx$$
$$= \pi \left[\tan x\right]_0^{\pi} - \pi \left[\sec x\right]_0^{\pi}$$
$= \pi[\tan \pi - \sec \pi - \tan 0 + \sec 0]$
$\Rightarrow 2I = \pi[0 + 1 - 0 + 1] = 2\pi \therefore I = \pi$

www.exambites.in

OR

Let $I = \int_0^{\pi} x \sin x \cos^2 x \, dx$ ... (i)

Using property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, we get

$$I = \int_0^{\pi} (\pi - x)\sin(\pi - x)\cos^2(\pi - x)dx$$

$$\Rightarrow \quad I = \int_0^{\pi} (\pi - x)\sin x \cos^2 x \, dx \qquad \text{... (ii)}$$

On adding (i) and (ii), we get

$$2I = \int_0^{\pi} \pi \sin x \cos^2 x \, dx$$

Substitute $\cos x = t \Rightarrow -\sin x\, dx = dt$

$x \to 0 \Rightarrow t \to 1$

and $x \to \pi \Rightarrow t \to -1$

$$\Rightarrow \quad 2I = -\pi \int_1^{-1} t^2 dt \quad \Rightarrow \quad 2I = -\pi \left[\frac{t^3}{3}\right]_1^{-1}$$

$$\Rightarrow \quad 2I = -\frac{\pi}{3}[-1-1] \quad \Rightarrow \quad 2I = \frac{2\pi}{3} \quad \therefore \quad I = \frac{\pi}{3}$$

32. Let E_1 : 'the person has the disease' and E_2 : 'the person is healthy',

$$\Rightarrow \quad P(E_1) = 0.1\% = \frac{0.1}{100} = \frac{1}{1000} = 0.001$$

and $P(E_2) = 1 - \frac{1}{1000} = \frac{999}{1000} = 0.999$

Let A : 'test is positive',

then $P(A \mid E_1) = \frac{99}{100} = 0.99$

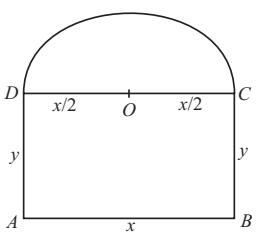
and $P(A \mid E_2) = 0.005$

Hence the required probability

$$= P(E_1 \mid A) = \frac{P(A \mid E_1)P(E_1)}{P(A \mid E_1)P(E_1) + P(A \mid E_2)P(E_2)}$$

$$= \frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.005 \times 0.999} = \frac{22}{133}$$

33. Let x and y be the length and breadth of the rectangle.



Radius of the semi-circle $= \frac{x}{2}$

Circumference of the semi-circle $= \frac{\pi x}{2}$

Perimeter of the window

$= AB + BC + AD + \overset{\frown}{DC}$

$$\Rightarrow \quad x + 2y + \frac{\pi x}{2} = 10 \qquad \text{[Given]}$$

$$\Rightarrow \quad 2x + 4y + \pi x = 20$$

$$\Rightarrow \quad y = \frac{20 - (2+\pi)x}{4} \qquad \text{... (i)}$$

Area of the window = Area of rectangle + Area of semi-circle

$$\Rightarrow \quad A = xy + \frac{1}{2}\pi\left(\frac{x}{2}\right)^2 = x\left(\frac{20-(2+\pi)x}{4}\right) + \frac{\pi x^2}{8} \qquad \text{[From (i)]}$$

$$= \frac{20x - (2+\pi)x^2}{4} + \frac{\pi x^2}{8}$$

$$\therefore \quad \frac{dA}{dx} = \frac{20 - (2+\pi)2x}{4} + \frac{2\pi x}{8}$$

For maxima/minima of A,

$$\frac{dA}{dx} = 0 \Rightarrow \frac{20-(2+\pi)2x}{4} + \frac{2\pi x}{8} = 0$$

$$\Rightarrow \quad 20 - (2+\pi)2x + \pi x = 0$$

$$\Rightarrow \quad 20 + x(\pi - 4 - 2\pi) = 0$$

$$\Rightarrow \quad 20 - x(4+\pi) = 0 \quad \Rightarrow \quad x = \frac{20}{4+\pi}$$

$$\frac{d^2A}{dx^2} = \frac{-(2+\pi)2}{4} + \frac{2\pi}{8} = \frac{-4-2\pi+\pi}{4} = \frac{-4-\pi}{4}$$

$$\Rightarrow \quad \frac{d^2A}{dx^2} < 0$$

Hence, the window admit the maximum light, when $x = \text{length} = \frac{20}{4+\pi}$

and $\text{breadth } y = \frac{20 - (2+\pi)\frac{20}{4+\pi}}{4}$

$$= \frac{80 + 20\pi - 40 - 20\pi}{4(4+\pi)} = \frac{40}{4(4+\pi)} = \frac{10}{4+\pi}$$

OR

Suppose company increases the annual subscription by ₹ x.

Then, x subscribes will discontinue the service.

$\therefore$ Total revenue of company after the increment $= R(x)$

$= (500 - x)(300 + x) = -x^2 + 200x + 150000$

www.exambites.in

On differentiating both sides w.r.t. 'x', we get

$R'(x) = -2x + 200$

Put $R'(x) = 0$

$\Rightarrow \quad 2x = 200 \Rightarrow x = 100$

Since $R''(x) = -2 < 0$

Hence, $R(x)$ is maximum when $x = 100$

Therefore, the company should increase the subscription fee by ₹ 100, so that it has maximum profit.

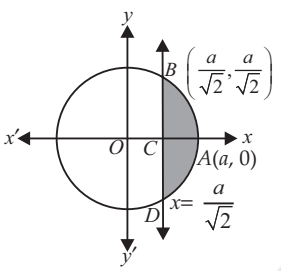
34. The given circle is $x^2 + y^2 = a^2$ ...(i)

and the line $x = \frac{a}{\sqrt{2}}$ ...(ii)

Solving (i) and (ii), we get

$$\frac{a^2}{2} + y^2 = a^2 \Rightarrow y^2 = \frac{a^2}{2} \Rightarrow y = \frac{a}{\sqrt{2}}$$

$\Rightarrow$ Line (ii) meets the circle (i) at $B\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ in the first quadrant.



$\therefore$ Required area = 2 Area $(CBAC) = 2\int_{\frac{a}{\sqrt{2}}}^{a} y\,dx$

$$= 2\int_{\frac{a}{\sqrt{2}}}^{a} \sqrt{a^2 - x^2}\,dx$$

$[\because \quad x^2 + y^2 = a^2 \Rightarrow y = \sqrt{a^2 - x^2} \ (\because \ y > 0)]$

$$= 2\left[x\frac{\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_{a/\sqrt{2}}^{a}$$

$$= 2\left[\left(0 + \frac{a^2}{2}\sin^{-1}(1)\right) - \left(\frac{a}{2\sqrt{2}}\sqrt{a^2 - \frac{a^2}{2}} + \frac{a^2}{2}\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\right)\right]$$

$$= 2\left[\frac{a^2}{2}\cdot\frac{\pi}{2} - \frac{a}{2\sqrt{2}}\frac{a}{\sqrt{2}} - \frac{a^2}{2}\frac{\pi}{4}\right]$$

$$= \frac{\pi a^2}{2} - \frac{a^2}{2} - \frac{\pi a^2}{4} = \frac{\pi a^2}{4} - \frac{a^2}{2}$$

$$= \frac{a^2}{2}\left(\frac{\pi}{2} - 1\right) \text{sq. units}$$

35. The equation of the line through (3, –4, –5) and (2, –3, 1) is

$$\frac{x-3}{2-3} = \frac{y+4}{-3+4} = \frac{z+5}{1+5} \ \textit{i.e.} \ \frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6}$$ ...(i)

Any point on (i) is $(3 - k, -4 + k, -5 + 6k)$ ...(ii)

This point lies on $2x + y + z = 7$

$\therefore \quad 2(3 - k) + (-4 + k) + (-5 + 6k) = 7$

$\Rightarrow \quad 6 - 2k - 4 + k - 5 + 6k = 7$

$\Rightarrow \quad 5k = 10 \Rightarrow k = 2$

Putting in (ii), the required point is

(3 –2, – 4 + 2, – 5 + 12) *i.e.*, (1, – 2, 7).

OR

Equation of plane passing through the intersection of given planes is

$x + 3y + 6 + \lambda(3x - y - 4z) = 0,$ ...(i)

which can be written as $(1 + 3\lambda)x + (3 - \lambda)y + (-4\lambda)z + 6 = 0$

$\because$ Distance of plane from origin is unity.

$$\therefore \quad \left|\frac{6}{\sqrt{(1+3\lambda)^2 + (3-\lambda)^2 + 16\lambda^2}}\right| = 1$$

$$\Rightarrow \quad \frac{6}{\sqrt{26\lambda^2 + 10}} = 1 \Rightarrow 36 = 26\lambda^2 + 10$$

(On squaring both sides)

$\Rightarrow \quad 26 = 26\lambda^2 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$

Putting the values of $\lambda = \pm 1$ in (i), we get equation of the required plane is

$x + 3y + 6 + (3x - y - 4z) = 0$ and

$x + 3y + 6 - 3x + y + 4z = 0$

or $\quad 4x + 2y - 4z + 6 = 0$ and

$\quad -2x + 4y + 4z + 6 = 0$

or $\quad 2x + y - 2z + 3 = 0$ and

$\quad x - 2y - 2z - 3 = 0$

36. The given system of equations can be written as $AX = B$, where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -2 & -1 \\ 0 & 3 & -5 \end{bmatrix}, \ X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 9 \end{bmatrix}$$

www.examrites.in

Now, $|A| = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -2 & -1 \\ 0 & 3 & -5 \end{vmatrix}$

$= 2\begin{vmatrix} -2 & -1 \\ 3 & -5 \end{vmatrix} - 1\begin{vmatrix} 1 & -1 \\ 0 & -5 \end{vmatrix} + 1\begin{vmatrix} 1 & -2 \\ 0 & 3 \end{vmatrix}$

$= 2(10 + 3) - 1(-5) + 1(3) = 26 + 5 + 3 = 34 \neq 0$

$\Rightarrow$ A^{-1} exists and hence the given equations have a unique solution.

Here,

$A_{11} = (-1)^{1+1}\begin{vmatrix} -2 & -1 \\ 3 & -5 \end{vmatrix} = (10 + 3) = 13,$

$A_{12} = (-1)^{1+2}\begin{vmatrix} 1 & -1 \\ 0 & -5 \end{vmatrix} = -(-5 - 0) = 5,$

$A_{13} = (-1)^{1+3}\begin{vmatrix} 1 & -2 \\ 0 & 3 \end{vmatrix} = (3 - 0) = 3,$

$A_{21} = (-1)^{2+1}\begin{vmatrix} 1 & 1 \\ 3 & -5 \end{vmatrix} = -(-5 - 3) = 8,$

$A_{22} = (-1)^{2+2}\begin{vmatrix} 2 & 1 \\ 0 & -5 \end{vmatrix} = (-10 - 0) = -10,$

$A_{23} = (-1)^{2+3}\begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} = -(6 - 0) = -6,$

$A_{31} = (-1)^{3+1}\begin{vmatrix} 1 & 1 \\ -2 & -1 \end{vmatrix} = (-1 + 2) = 1,$

$A_{32} = (-1)^{3+2}\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -(-2 - 1) = 3,$

$A_{33} = (-1)^{3+3}\begin{vmatrix} 2 & 1 \\ 1 & -2 \end{vmatrix} = -4 - 1 = -5$

$\therefore$ Adj $A = \begin{bmatrix} 13 & 5 & 3 \\ 8 & -10 & -6 \\ 1 & 3 & -5 \end{bmatrix}' = \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix}$

and $A^{-1} = \frac{1}{|A|}\text{Adj } A = \frac{1}{34}\begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix}$

Solution of given system is given by $X = A^{-1}B$.

$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34}\begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix}\begin{bmatrix} 1 \\ 3/2 \\ 9 \end{bmatrix}$

$= \frac{1}{34}\begin{bmatrix} 13 + 12 + 9 \\ 5 - 15 + 27 \\ 3 - 9 - 45 \end{bmatrix} = \frac{1}{34}\begin{bmatrix} 34 \\ 17 \\ -51 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \\ \frac{-3}{2} \end{bmatrix}$

Hence, $x = 1$, $y = \frac{1}{2}$ and $z = \frac{-3}{2}$.

www.exambites.in

PRACTICE PAPER 5
(Solved)

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) This question paper comprises four Sections A, B, C and D. This question paper carries 36 questions. All questions are compulsory.

(ii) Section A : Questions no. 1 to 20 comprises of 20 questions of 1 mark each.

(iii) Section B : Questions no. 21 to 26 comprises of 6 questions of 2 marks each.

(iv) Section C : Questions no. 27 to 32 comprises of 6 questions of 4 marks each.

(v) Section D : Questions no. 33 to 36 comprises of 4 questions of 6 marks each.

(vi) There is no overall choice in the question paper. However, an internal choice has been provided in 3 questions of one mark, 2 questions of two marks, 2 questions of four marks and 2 questions of six marks. Only one of the choices in such questions have to be attempted.

(vii) In addition to this, separate instructions are given with each section and question, wherever necessary.

(viii) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

SECTION - A

(Q.1-Q.10) are multiple choice type questions. Select the correct option.

1. Set A has three elements and set B has four elements. The number of injections that can be defined from A to B is

(a) 144 (b) 12 (c) 24 (d) 64

2. Which of the following is the principal value branch of $\sec^{-1} x$?

(a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (b) $(0, \pi)$

(c) $[0, \pi]$ (d) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$

3. If $A^2 - A + I = O$, then the inverse of A is

(a) $I - A$ (b) $A - I$

(c) A (d) $A + I$

4. If A_{ij} denotes the co-factor of the element a_{ij} of the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$, then value of $a_{12}A_{32}$ is

(a) -90 (b) -66 (c) 102 (d) -55

5. The function f defined by $f(x) = 4x^4 - 2x + 1$ is increasing for

(a) $x < 1$ (b) $x > 0$

(c) $x < \frac{1}{2}$ (d) $x > \frac{1}{2}$

6. $\int \sqrt{1+x^2}\, dx$ is equal to

(a) $\frac{x}{2}\sqrt{1+x^2} + \frac{1}{2}\log\left|(x + \sqrt{1+x^2})\right| + C$

(b) $\frac{2}{3}(1+x^2)^{3/2} + C$

(c) $\frac{2}{3}x(1+x^2)^{3/2} + C$

(d) $\frac{x^2}{2}\sqrt{1+x^2} + \frac{1}{2}x^2 \log\left|x + \sqrt{1+x^2}\right| + C$

7. The vector having initial and terminal points as (2, 5, 0) and (–3, 7, 4), respectively is

(a) $-\hat{i} + 12\hat{j} + 4\hat{k}$ (b) $5\hat{i} + 2\hat{j} - 4\hat{k}$

(c) $-5\hat{i} + 2\hat{j} + 4\hat{k}$ (d) $\hat{i} + \hat{j} + \hat{k}$

8. If A and B are 2 events such that $P(A) > 0$ and $P(B) \neq 1$, then $P(\overline{A} \mid \overline{B}) =$

(a) $1 - P(A|B)$ (b) $1 - P(A \mid \overline{B})$

(c) $\frac{1 - P(A \cup B)}{P(B)}$ (d) $\frac{P(\overline{A})}{P(B)}$

9. The conditions $x \geq 0$ and $y \geq 0$ are called

(a) non-positive restrictions

(b) negative restrictions

(c) non-negative restrictions

(d) none of these

10. $\int_0^{2/3} \frac{dx}{4 + 9x^2}$ equals to

www.exambites.in

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{12}$ (c) $\frac{\pi}{24}$ (d) $\frac{\pi}{4}$

(Q.11-Q.15) Fill in the blanks.

11. A plane passes through the points (2, 0, 0), (0, 3, 0) and (0, 0, 4), the equation of plane is _____ .

OR

The direction cosines of the normal to the plane $5y + 8 = 0$ are _____ .

12. The integrating factor of $\frac{dy}{dx} + y = \frac{1+y}{x}$ is _____ .

13. A real valued function whose domain is the sample space of a random experiment is called _____ .

OR

If $P(A) = \frac{1}{2}, P(B) = 0,$ then $P(A|B)$ = _____ .

14. $\int_0^{\pi/4} \tan x\,dx =$ _____.

15. If $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$ and $A = A^T$, then x and y are _____.

(Q.16-Q.20) Answer the following questions.

16. Find the value of λ such that the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$ are orthogonal.

OR

If $|\vec{a}| = 10, |\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 12,$ then find the value of $|\vec{a} \times \vec{b}|$.

17. Find the differential coefficient of $\sqrt{\sec\sqrt{x}}$.

18. Evaluate : $\int \sec^2(7 - 4x)dx$

19. Find the equation of the plane through the points (2, 1, 0), (3, –2, –2) and (3, 1, 7).

20. Find the adjoint of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

SECTION - B

21. Find two branches other than the principal value branch of $\tan^{-1} x$.

22. Find the probability that both children are males, if it is known that at least one of the children is male.

23. Find points at which the tangent to the curve $y = x^3 - 3x^2 - 9x + 7$ is parallel to the x-axis.

24. If $x = 3\sin t - \sin 3t$, $y = 3\cos t - \cos 3t$, then find $\frac{dy}{dx}$ at $t = \frac{\pi}{3}$.

OR

Differentiate $\frac{x}{\sin x}$ w.r.t. $\sin x$.

25. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$, then find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.

OR

Find λ and μ, if

$(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + \lambda\hat{j} + \mu\hat{k}) = \vec{0}$.

26. Show that the points (2, 3, 4), (–1, –2, 1), (5, 8, 7) are collinear.

SECTION - C

27. For what value of k is the function defined by

$$f(x) = \begin{cases} k(x^2 + 2), & \text{if} \quad x \le 0 \\ 3x + 1, & \text{if} \quad x > 0 \end{cases}$$

continuous at $x = 0$? Also, write whether the function is continuous at $x = 1$.

28. Show that the relation S in the set $A = \{x \in Z : 0 \le x \le 12\}$ given by $S = \{(a, b) : a, b \in Z, |a - b| \text{ is divisible by } 4\}$ is an equivalence relation.

OR

Show that the relation S in the set R of real numbers defined as $S = \{(a, b) : a, b \in R$ and $a \le b_3\}$ is neither reflexive, nor symmetric, nor transitive.

29. Minimise and maximise $Z = x + 2y$ subject to $x + 2y \ge 100$, $2x - y \le 0$, $2x + y \le 200$; $x, y \ge 0$.

30. In a class, 5% of the boys and 10% of the girls have an I.Q. of more than 150. In the class, 60% of the students are boys and rest are girls. If a student is selected at random found to have an I.Q. of more than 150, then find the probability that the student is a boy.

31. Evaluate : $\int_1^2 \frac{5x^2}{x^2 + 4x + 3}dx$

OR

Evaluate : $\int \frac{1 - x^2}{x(1 - 2x)}dx$

www.examfiles.in

32. Solve : $\frac{dy}{dx} = \tan(x + y)$

SECTION - D

33. Draw a rough sketch of the curve $y = \sqrt{x-1}$ in the interval [1, 5]. Find the area under the curve and between the lines $x = 1$ and $x = 5$.

OR

Sketch the region $\{(x,0) : y = \sqrt{4 - x^2}\}$ and x-axis. Find the area of the region using integration.

34. Find the distance of the point (– 1, – 5, – 10) from the point of intersection of the line $\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$.

35. A tank with rectangular base and rectangular sides, open at the top is to be constructed so that its depth is 2 m and volume is 8 m^3. If building of tank costs ₹ 70 per sq. metres for the base and ₹ 45 per square metre for sides. What is the cost of least expensive tank?

OR

Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\frac{4r}{3}$.

36. For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$.

Show that $A^3 - 6A^2 + 5A + 11I = O$. Hence, find A^{-1}.

Detailed Solutions

1. (c) : Since 3 < 4, injective functions from A to B are defined and the total number of such functions is

$${}^4P_3 = \frac{4!}{(4-3)!} = 4 \times 3 \times 2 \times 1 = 24$$

2. (d)

3. (a) : If A is any square matrix, then $AA^{-1} = I$ and $A^{-1}I = A^{-1}$.

Since, $A^2 - A + I = O$

$\Rightarrow A^{-1}A^2 - A^{-1}A + A^{-1}I = O$

$\Rightarrow (A^{-1}A)A - (A^{-1}A) + A^{-1} = O$

$\Rightarrow IA - I + A^{-1} = O \Rightarrow A - I + A^{-1} = O$

$\Rightarrow A^{-1} = I - A$

4. (b) : $M_{32} = \begin{vmatrix} 2 & 5 \\ 6 & 4 \end{vmatrix} = 8 - 30 = -22$

$\Rightarrow A_{32} = -M_{32} = 22$

$\therefore a_{12}A_{32} = (-3)(22) = -66$

5. (d) : We have, $f(x) = 4x^4 - 2x + 1$

$\Rightarrow f'(x) = 16x^3 - 2$

The function is increasing if $f'(x) > 0$

$\Rightarrow 16x^3 - 2 > 0 \Rightarrow x > \frac{1}{2}$

6. (a) : Let $I = \int \sqrt{1 + x^2}\, dx$

$$= \frac{x}{2}\sqrt{1+x^2} + \frac{1}{2}\log\left|x + \sqrt{1+x^2}\right| + C$$

$$\left[\because \int \sqrt{a^2 + x^2}\, dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\log\left|(x + \sqrt{a^2+x^2})\right| + C\right]$$

7. (c) : Required vector

$= (-3 - 2)\hat{i} + (7 - 5)\hat{j} + (4 - 0)\hat{k}$

$= -5\hat{i} + 2\hat{j} + 4\hat{k}$

8. (b) : $P(A \mid \bar{B}) + P(\bar{A} \mid \bar{B}) = 1$

$\Rightarrow P(\bar{A} \mid \bar{B}) = 1 - P(A \mid \bar{B})$

9. (c)

10. (c) : Let $I = \int_0^{2/3} \frac{dx}{4 + 9x^2} = \frac{1}{9}\int_0^{2/3} \frac{dx}{\left(\frac{2}{3}\right)^2 + x^2}$

$$= \frac{1}{9} \times \frac{1}{\frac{2}{3}}\left[\tan^{-1}\left(\frac{3x}{2}\right)\right]_0^{2/3}$$

$$= \frac{1}{6}\left[\tan^{-1}\left(\frac{3x}{2}\right)\right]_0^{2/3}$$

$$= \frac{1}{6}[\tan^{-1}(1) - \tan^{-1}(0)] = \frac{1}{6} \times \frac{\pi}{4} = \frac{\pi}{24}$$

11. As we know, equation of a plane that cut the coordinate axes at $(a, 0, 0)$ $(0, b, 0)$ and $(0, 0, c)$ is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

www.exambites.in

Therefore, the equation of plane passes through the points (2, 0, 0), (0, 3, 0) and (0, 0, 4) is $\frac{x}{2}+\frac{y}{3}+\frac{z}{4}=1$.

OR

The given plane is $5y + 8 = 0$.
Since, direction ratios of the normal to the plane are $< 0, 5, 0 >$
$\therefore$ Direction cosines of the normal to the plane are $< 0, 1, 0 >$.

12. We have, $\frac{dy}{dx}+y=\frac{1+y}{x}$

$\Rightarrow \quad \frac{dy}{dx}+y=\frac{1}{x}+\frac{y}{x} \Rightarrow \frac{dy}{dx}+y\left(1-\frac{1}{x}\right)=\frac{1}{x},$

which is linear differential equation of the type $\frac{dy}{dx}+Py=Q,$

where $P=1-\frac{1}{x}, Q=\frac{1}{x}$

$\therefore$ I.F. $= e^{\int P dx} = e^{\int\left(1-\frac{1}{x}\right)dx}$

$= e^{x-\log x} = e^{x}\cdot e^{-\log x} = \frac{e^x}{x}$

13. A real valued function whose domain is the sample space of a random experiment is called random variable.

OR

Since, $P(B) = 0$, therefore $P(A \mid B)=P\frac{(A\cap B)}{P(B)}$ is not defined.

14. $\int_0^{\pi/4}\tan x\,dx=\left[\log\sec x\right]_0^{\pi/4}$

$=\log\sec\frac{\pi}{4}-\log\sec 0=\log\sqrt{2}=\frac{1}{2}\log 2$

15. We have, $A=\begin{bmatrix}5 & x\\ y & 0\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}5 & y\\ x & 0\end{bmatrix}$

Now, $A = A^T$

$\Rightarrow \begin{bmatrix}5 & x\\ y & 0\end{bmatrix}=\begin{bmatrix}5 & y\\ x & 0\end{bmatrix} \Rightarrow x=y$

16. Since, two non-zero vectors $\vec{a}$ and $\vec{b}$ are orthogonal

$\therefore\ \vec{a}\cdot\vec{b}=0 \Rightarrow (2\hat{i}+\lambda\hat{j}+\hat{k})\cdot(\hat{i}+2\hat{j}+3\hat{k})=0$

$\Rightarrow\ 2+2\lambda+3=0 \Rightarrow \lambda=\frac{-5}{2}$

OR

We know that, $(\vec{a}\times\vec{b})^2=|\vec{a}|^2|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2$

$\Rightarrow \quad |\vec{a}\times\vec{b}|=\sqrt{(10)^2\cdot(2)^2-(12)^2}$

$=\sqrt{400-144}=\sqrt{256} \Rightarrow |\vec{a}\times\vec{b}|=16$

17. Let $y=\sqrt{\sec\sqrt{x}}$
Differentiating w.r.t. x, we get

$\frac{dy}{dx}=\frac{1}{2\sqrt{\sec\sqrt{x}}}\cdot\sec\sqrt{x}\cdot\tan\sqrt{x}\cdot\frac{1}{2\sqrt{x}}$

$=\frac{1}{4\sqrt{x}}(\sec\sqrt{x})^{1/2}\frac{\sin\sqrt{x}}{\cos\sqrt{x}}$

$=\frac{1}{4\sqrt{x}}(\sec\sqrt{x})^{3/2}\cdot\sin\sqrt{x}$

18. Let $I=\int\sec^2(7-4x)dx$

Put $7 - 4x = t \Rightarrow dx = \frac{-1}{4}dt$

$\therefore\ I=\int\frac{\sec^2 t}{-4}dt=\frac{\tan t}{-4}+C=\frac{\tan(7-4x)}{-4}+C$

19. The required equation of the plane passing through three non-collinear points is,

$\Rightarrow \begin{vmatrix}x-2 & y-1 & z-0\\ 3-2 & -2-1 & -2-0\\ 3-2 & 1-1 & 7-0\end{vmatrix}=0$

$\Rightarrow \begin{vmatrix}x-2 & y-1 & z\\ 1 & -3 & -2\\ 1 & 0 & 7\end{vmatrix}=0$

$\Rightarrow\ (x-2)(-21+0)-(y-1)(7+2)+z(3)=0$
$\Rightarrow\ 7x+3y-z=17$

20. We have, $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$

$\therefore\quad \text{adj}\,(A)=\begin{bmatrix}4 & -2\\ -3 & 1\end{bmatrix}$

21. We know, principal value branch of $\tan^{-1}x$ is $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

Now, the other two branches can be obtained by adding or subtracting π in each element of $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

$\therefore$ Other branches are $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$ and $\left(-\frac{3\pi}{2},-\frac{\pi}{2}\right)$.

www.exambites.in

22. Sample space is $S = \{ff, fm, mf, mm\}$ where f = female, m = male

(i) Let A = both are male, *i.e.* $\{mm\}$

$\Rightarrow P(A) = \frac{1}{4}$

B = atleast one is a male, *i.e.* $\{mm, fm, mf\}$

$\Rightarrow \quad P(B) = \frac{3}{4}$

$A \cap B = \{mm\} \quad \Rightarrow \quad P(A \cap B) = \frac{1}{4}$

$\therefore$ Required probability

$$= P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/4} = \frac{1}{3}$$

23. We have $y = x^3 - 3x^2 - 9x + 7$ ...(i)

Differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx} = 3x^2 - 6x - 9$$

Now, tangent to (i) is parallel to x-axis

$$\Rightarrow \frac{dy}{dx} = 0$$

$\Rightarrow \quad 3x^2 - 6x - 9 = 0 \quad \Rightarrow \quad x^2 - 2x - 3 = 0$

$\Rightarrow \quad (x - 3)(x + 1) = 0 \quad \Rightarrow \quad x = 3, -1.$

When $x = 3$, then from (i),

$$y = 3^3 - 3 \cdot (3^2) - 93 + 7 = 27 - 27 - 27 + 7 = -20$$

When $x = -1$, then from (i)

$$y = (-1)^3 - 3(-1)^2 - 9(-1) + 7$$

$$= -1 - 3 + 9 + 7 = 12$$

Hence, the required points are $(3, -20)$ and $(-1, 12)$.

24. We have, $x = 3\sin t - \sin 3t$

and $y = 3\cos t - \cos 3t$

$$\therefore \quad \frac{dx}{dt} = 3\cos t - 3\cos 3t \qquad ...(i)$$

and $$\frac{dy}{dt} = 3\sin 3t - 3\sin t \qquad ...(ii)$$

$$\therefore \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3(\sin 3t - \sin t)}{3(\cos t - \cos 3t)} = \frac{\sin 3t - \sin t}{\cos t - \cos 3t}$$

Now, $$\left(\frac{dy}{dx}\right)_{t=\pi/3}$$

$$= \frac{\sin\frac{3\pi}{3} - \sin\frac{\pi}{3}}{\left(\cos\frac{\pi}{3} - \cos 3\frac{\pi}{3}\right)} = \frac{0 - \left(\sqrt{3}/2\right)}{\frac{1}{2} - (-1)}$$

$$= \frac{-\sqrt{3}/2}{3/2} = \frac{-\sqrt{3}}{3} = \frac{-1}{\sqrt{3}}$$

OR

Let $u = \frac{x}{\sin x}$ and $v = \sin x$

$$\therefore \quad \frac{du}{dx} = \frac{\sin x \cdot \frac{d}{dx}x - x \cdot \frac{d}{dx}\sin x}{(\sin x)^2}$$ [By quotient rule]

$$= \frac{\sin x - x\cos x}{\sin^2 x} \quad \text{and} \quad \frac{dv}{dx} = \frac{d}{dx}\sin x = \cos x$$

$$\therefore \quad \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{(\sin x - x\cos x)/\sin^2 x}{\cos x}$$

$$= \frac{\frac{\sin x - x\cos x}{\sin^2 x}}{\cos x} = \frac{\sin x - x\cos x}{\sin^2 x \cos x} = \frac{\tan x - x}{\sin^2 x}$$

25. We have, $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ ...(i)

We know that $\vec{a} + \vec{b} + \vec{c} = 0$ ...(ii)

Squaring (ii), we get $(\vec{a} + \vec{b} + \vec{c})^2 = 0$

$\Rightarrow \quad |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$

$\Rightarrow \quad (1)^2 + (1)^2 + (1)^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$

Hence, $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a} = -\frac{3}{2}$

OR

Let $\vec{a} = 2\hat{i} + 6\hat{j} + 27\hat{k}$ and $\vec{b} = \hat{i} + \lambda\hat{j} + \mu\hat{k}$

$$\therefore \quad \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 27 \\ 1 & \lambda & \mu \end{vmatrix}$$

$$= (6\mu - 27\lambda)\hat{i} + (27 - 2\mu)\hat{j} + (2\lambda - 6)\hat{k}$$

Given, $\vec{a} \times \vec{b} = \vec{0}$

$\Rightarrow \quad (6\mu - 27\lambda)\hat{i} + (27 - 2\mu)\hat{j} + (2\lambda - 6)\hat{k} = \vec{0}$

$\Rightarrow \quad 6\mu - 27\lambda = 0, (2\lambda - 6) = 0, (27 - 2\mu) = 0$

$\Rightarrow \quad \lambda = 3$ and $\mu = \frac{27}{2}$

26. Let $A = (2, 3, 4)$, $B = (-1, -2, 1)$ and $C = (5, 8, 7)$

Direction ratios of AB are $< (-1 - 2), (-2 - 3), (1 - 4) >$ *i.e.*, $< -3, -5, -3 >$

Direction ratios of AC are $< (5 - 2), (8 - 3), (7 - 4) >$ *i.e.*, $< 3, 5, 3 >$

It is clear that the direction ratios of AB and AC are proportional. Hence, AB and AC are parallel, but these have a point A in common. Therefore A, B and C are collinear.

27. We have, $\lim_{x \to 0^-} f(x) = \lim_{h \to 0} k(h^2 + 2) = 2k$

www.exambuts.com

$\lim_{x\to 0^+} f(x) = \lim_{h\to 0} (3h+1) = 1$ and $f(0) = 2k$

As $f(x)$ is continuous at $x = 0$

$\therefore \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0)$

$\Rightarrow 2k = 1 \Rightarrow k = \frac{1}{2}$

$\therefore$ $f(x)$ is continuous at $x = 0$, if $k = \frac{1}{2}$.

Also, $f(x)$ is continuous at $x = 1$ as $f(x) = 3x + 1$ is a polynomial function.

28. We have, $A = \{x \in Z : 0 \le x \le 12\}$

$\therefore$ $A = \{0, 1, 2, 3, ..., 12\}$

and $S = \{(a, b) : |a - b|$ is a multiple of $4\}$

(i) Reflexive : For any $a \in A$,

$|a - a| = 0$, which is a multiple of 4.

Thus, $(a, a) \in S$

$\therefore$ S is reflexive.

(ii) Symmetric : For any $a, b \in A$,

Let $(a, b) \in S$

$\Rightarrow$ $|a - b|$ is a multiple of 4

$\Rightarrow$ $|b - a|$ is a multiple of 4 $\Rightarrow$ $(b, a) \in S$

$\therefore$ S is symmetric.

(iii) Transitive : For any $a, b, c \in A$,

Let $(a, b) \in S$ and $(b, c) \in S$

$\Rightarrow$ $|a - b|$ is a multiple of 4 and $|b - c|$ is a multiple of 4

$\Rightarrow$ $a - b = \pm 4k_1$ and $b - c = \pm 4k_2$; $k_1, k_2 \in N$

$\Rightarrow$ $(a - b) + (b - c) = \pm 4 (k_1 + k_2)$; $k_1, k_2 \in N$

$\Rightarrow$ $a - c = \pm 4 (k_1 + k_2)$; $k_1, k_2 \in N$

$\Rightarrow$ $|a - c|$ is a multiple of $4 \Rightarrow (a , c) \in S$

$\therefore$ S is transitive.

Hence, S is an equivalence relation.

OR

We have $S = \{(a, b) : a \le b^3\}$, where $a, b \in$ R.

(i) Reflexive : We observe that, $\frac{1}{2} \le \left(\frac{1}{2}\right)^3$ is not true.

$\therefore$ $\left(\frac{1}{2}, \frac{1}{2}\right) \notin S$. So, S is not reflexive.

(ii) Symmetric : We observe that, $1 \le 3^3$ but $3 \not\le 1^3$ *i.e.*, $(1, 3) \in S$ but $(3, 1) \notin S$. So, S is not symmetric.

(iii) Transitive : We observe that, $10 \le 3^3$ and $3 \le 2^3$ but $10 \not\le 2^3$

i.e., $(10, 3) \in S$ and $(3, 2) \in S$ but $(10, 2) \notin S$

So, S is not transitive.

$\therefore$ S is neither reflexive, nor symmetric, nor transitive.

29. We have, minimize and maximize $Z = x + 2y$.

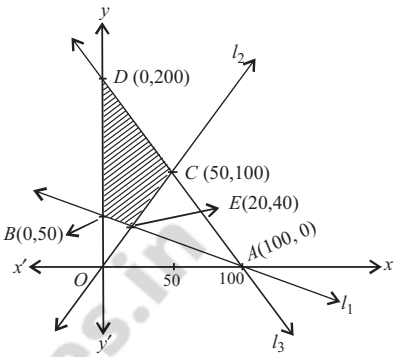
Subject to constraints :

$x + 2y \ge 100$; $2x - y \le 0$; $2x + y \le 200$;

and $x, y \ge 0$;

Let $l_1 : x + 2y = 100$, $l_2 : 2x - y = 0$, $l_3 : 2x + y = 200$, $l_4 : x = 0$ and $l_5 : y = 0$

Let us draw the graph of these equations.



It is observed that the feasible region $ECDB$ is bounded, where coordinates of E, C, D and B are (20, 40), (50, 100), (0, 200) and (0, 50) respectively.

Corner Point	Corresponding values of Z
(20, 40)	100
(50, 100)	250
(0, 200)	400
(0, 50)	100

Hence, $Z_{max} = 400$ at (0, 200) and $Z_{min} = 100$ at all points on the line segment joining the points (0, 50) and (20, 40).

30. Consider the following events:

E_1 : Selected student is a boy.

E_2 : Selected student is a girl.

A : The student has an I.Q. of more than 150.

Now, $P(E_1) = \frac{60}{100} = \frac{3}{5}$

$P(E_2) = 1 - P(E_1) = 1 - \frac{3}{5} = \frac{2}{5}$

$P(A|E_1) = \frac{5}{100} = \frac{1}{20}$

$P(A|E_2) = \frac{10}{100} = \frac{1}{10}$

$\therefore$ Required probability

$$= P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$$

$$= \frac{\frac{3}{5} \times \frac{1}{20}}{\frac{3}{5} \times \frac{1}{20} + \frac{2}{5} \times \frac{1}{10}} = \frac{3}{3+4} = \frac{3}{7}$$

www.exambees.in

31. Let $I = \int_1^2 \frac{5x^2}{x^2+4x+3}\,dx$

$= 5\int_1^2 \left(1 + \frac{-4x-3}{x^2+4x+3}\right)dx = 5\int_1^2 dx - 5\int_1^2 \frac{4x+3}{x^2+4x+3}\,dx$

$= 5[x]_1^2 - 5\int_1^2 \frac{4x+3}{(x+3)(x+1)}\,dx$

Let $\frac{4x+3}{(x+3)(x+1)} = \frac{A}{x+3} + \frac{B}{x+1}$

$\Rightarrow\ 4x + 3 = A(x+1) + B(x+3)$

Putting $x = -3$ and $x = -1$, we get $A = \frac{9}{2};\ B = -\frac{1}{2}$

$\therefore\quad I = 5(2-1) - 5\int_1^2 \left(\frac{9/2}{x+3} + \frac{-1/2}{x+1}\right)dx$

$= 5 - 5\left[\frac{9}{2}\log|x+3| - \frac{1}{2}\log|x+1|\right]_1^2$

$= 5 - 5\left[\left(\frac{9}{2}\log 5 - \frac{1}{2}\log 3\right) - \left(\frac{9}{2}\log 4 - \frac{1}{2}\log 2\right)\right]$

$= 5 - 5\left[\frac{9}{2}(\log 5 - \log 4) - \frac{1}{2}(\log 3 - \log 2)\right]$

$= 5 - \frac{45}{2}\log\frac{5}{4} + \frac{5}{2}\log\frac{3}{2}$

OR

Since, $\frac{1-x^2}{x(1-2x)} = \frac{1-x^2}{x-2x^2}$ is an improper fraction, therefore, we convert it into a proper fraction by long division method. We get

$\frac{x^2-1}{2x^2-x} = \frac{1}{2} + \frac{\frac{x}{2}-1}{2x^2-x}$

$\Rightarrow\ \int \frac{(-1+x^2)}{-x+2x^2}dx = \frac{1}{2}\int dx + \frac{1}{2}\int \frac{x-2}{2x^2-x}\,dx$

Let $\frac{x-2}{2x^2-x} = \frac{x-2}{x(2x-1)} = \frac{A}{x} + \frac{B}{2x-1}$

$\Rightarrow\ x - 2 = A(2x-1) + Bx$ ...(i)

Putting $x = 0$ in (i), we get

$-2 = A(-1) \Rightarrow A = 2$

Putting $x = \frac{1}{2}$ in (i), we get

$\frac{1}{2} - 2 = B\left(\frac{1}{2}\right) \Rightarrow 1 - 4 = B \Rightarrow B = -3$

$\therefore\quad \frac{x-2}{2x^2-x} = \frac{2}{x} - \frac{3}{2x-1} = \frac{2}{x} + \frac{3}{1-2x}$

$\Rightarrow\quad I = \int \frac{1-x^2}{x(1-2x)}dx$

$= \frac{1}{2}\int (1)\,dx + \frac{1}{2}\int\left(\frac{2}{x} + \frac{3}{1-2x}\right)dx$

$= \frac{1}{2}x + \log|x| - \frac{3}{4}\log|1-2x| + C$

32. We have, $\frac{dy}{dx} = \tan(x+y)$

Put $x + y = v \Rightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx}$

$\therefore\quad \frac{dv}{dx} - 1 = \tan v \ \Rightarrow\ \frac{dv}{dx} = 1 + \tan v$

$\Rightarrow\quad \int \frac{dv}{1+\tan v} = \int dx$

$\Rightarrow\quad \int \frac{\cos v}{\sin v + \cos v}dv = x + C_1$

$\Rightarrow\quad \frac{1}{2}\int \frac{\sin v + \cos v}{\sin v + \cos v}\,dv + \frac{1}{2}\int \frac{\cos v - \sin v}{\sin v + \cos v}\,dv = x + C_1$

$\Rightarrow\quad \frac{1}{2}v + \frac{1}{2}\log|\sin v + \cos v| = x + C_1$

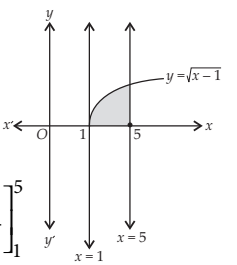
$\Rightarrow\quad \frac{1}{2}(x+y) + \frac{1}{2}\log|\sin(x+y) + \cos(x+y)| = x + C_1$

$\Rightarrow\quad y - x + \log|\sin(x+y) + \cos(x+y)| = C$ [where $C = 2C_1$]

33. We have given equation of the curve

$y = \sqrt{x-1} \Rightarrow y^2 = x - 1$

Thus the area of the shaded region



$= \int_1^5 (x-1)^{1/2}dx = \left[\frac{2\cdot(x-1)^{3/2}}{3}\right]_1^5$

$= \left[\frac{2}{3}(5-1)^{3/2} - 0\right] = \frac{16}{3}$ sq. units

OR

The given region

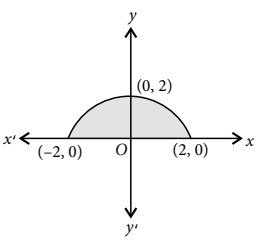
$= \{(x,0) : y = \sqrt{4-x^2}\}$

and x-axis.

We have $y = \sqrt{4-x^2}$

$\Rightarrow\quad y^2 = 4 - x^2$

$\Rightarrow\quad x^2 + y^2 = 4$



www.examrace.in

Thus the area of the shaded region

$$= \int_{-2}^{2} \sqrt{4 - x^2}\,dx = \int_{-2}^{2} \sqrt{2^2 - x^2}\,dx$$

$$= \left[\frac{x}{2}\sqrt{2^2 - x^2} + \frac{2^2}{2} \cdot \sin^{-1}\frac{x}{2}\right]_{-2}^{2}$$

$$= \frac{2}{2} \cdot 0 + 2 \cdot \frac{\pi}{2} - 0 - 2\sin^{-1}(-1)$$

$$= 2 \cdot \frac{\pi}{2} + 2 \cdot \frac{\pi}{2} = 2\pi \text{ sq. units}$$

34. The given line is

$\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ ...(1)

and the given plane is $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$ ...(2)

Solving (1) and (2), we get

$[2\hat{i} - \hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})] \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$

$\Rightarrow [(2+3\lambda)\hat{i} + (4\lambda - 1)\hat{j} + (2\lambda + 2)\hat{k}] \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$

$\Rightarrow$ $(2 + 3\lambda)(1) + (4\lambda - 1)(-1) + (2\lambda + 2)(1) = 5$

$\Rightarrow$ $2 + 3\lambda - 4\lambda + 1 + 2\lambda + 2 = 5 \Rightarrow \lambda = 0$

$\therefore$ The point of intersection of (1) and (2) is

$(2\hat{i} - \hat{j} + 2\hat{k})$ *i.e* $(2, -1, 2)$.

The other point is $(-1, -5, -10)$.

$\therefore$ Required distance =

$$\sqrt{(-1-2)^2 + (-5+1)^2 + (-10-2)^2}$$

$$= \sqrt{9 + 16 + 144} = \sqrt{169} = 13$$

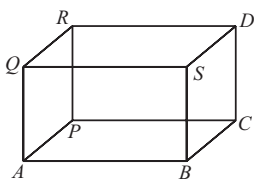
35. Let the dimensions of the base be x metres and y metres. The total expenses be ₹ S and the volume of the tank be V m^3, then

$V = 2xy$ but $V = 8$ (given)

$$\Rightarrow \quad 2xy = 8 \quad \Rightarrow \quad y = \frac{4}{x} \qquad \text{...(i)}$$

Also area of the base $ABCP$

$$= xy = x \times \frac{4}{x} = 4 \text{ m}^2 \qquad \text{(from (i))}$$



and area of the four sides = $(2x + 2x + 2y + 2y)$ m^2

$$= \left(4x + 4\left(\frac{4}{x}\right)\right) \text{m}^2 \qquad \text{(from (i))}$$

$$= 4\left\{x + \frac{4}{x}\right\} \text{ m}^2$$

$$\therefore \quad S = 4 \times 70 + 45 \times 4\left\{x + \frac{4}{x}\right\}$$

$$= 280 + 180\left\{x + \frac{4}{x}\right\} \qquad \text{...(ii)}$$

Differentiating (ii) w.r.t. x, we get

$$\frac{dS}{dx} = 180\left(1 - \frac{4}{x^2}\right) \qquad \text{...(iii)}$$

For maximum/minimum expenses, put $\frac{dS}{dx} = 0$

$$\Rightarrow 180\left(1 - \frac{4}{x^2}\right) = 0 \Rightarrow x^2 = 4 \Rightarrow x = 2$$

Differentiating (iii) w.r.t. x, we get

$$\frac{d^2S}{dx^2} = 180\left(0 + \frac{8}{x^3}\right) = \frac{1440}{x^3} \qquad \text{...(iv)}$$

Substituting the value of $x = 2$ in (iv), we get

$$\left(\frac{d^2S}{dx^2}\right)_{x=2} = \frac{1440}{2^3} > 0$$

$\therefore$ S is least when $x = 2$

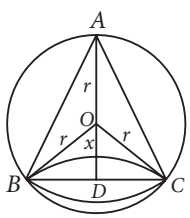
$\therefore$ Cost of least expensive tank = ₹ S

$$= ₹\left\{280 + 180\left(2 + \frac{4}{2}\right)\right\} \qquad \text{(from (ii))}$$

= ₹ (280 + 720) = ₹ 1000.

OR

Let the distance of the base BC of the cone from the centre O of the sphere be x, radius of the sphere is r.



Height of the cone, $H = AD = r + x$ and radius of the base of cone,

$$R = BD = \sqrt{r^2 - x^2}.$$

Let, V be the volume of the cone, then

$$V = \frac{1}{3}\pi R^2 H$$

$$= \frac{1}{3}\pi(r^2 - x^2)(r + x)$$

www.exambites.in

$$\therefore \ \frac{dV}{dx} = \frac{\pi}{3}[(r^2 - x^2)(0+1) + (r+x)(0-2x)]$$

$$= \frac{\pi}{3}[(r-x)(r+x) - 2x(r+x)]$$

$$= \frac{\pi}{3}[(r+x)(r-x-2x)] = \frac{\pi}{3}(r+x)(r-3x).$$

For maximum/minimum,

$$\frac{dV}{dx} = 0 \ \Rightarrow \ \frac{\pi}{3}(r+x)(r-3x) = 0$$

$$\Rightarrow \ (r+x)(r-3x) = 0 \ \Rightarrow \ x = -r, \ \frac{r}{3}.$$

But $x \neq -r$, $\therefore \ x = \frac{r}{3}$.

Now $\frac{d^2V}{dx^2} = \frac{\pi}{3}[(r+x)(-3) + (r-3x)(1)]$

$$= \frac{\pi}{3}(-3r - 3x + r - 3x) = \frac{\pi}{3}(-2r - 6x)$$

$$\therefore \ \left[\frac{d^2V}{dx^2}\right]_{x=\frac{r}{3}} = \frac{\pi}{3}\left[-2r - 6\left(\frac{r}{3}\right)\right]$$

$$= \frac{\pi}{3}(-2r - 2r) = -\frac{4r\pi}{3} < 0$$

Hence V is maximum when $x = \frac{r}{3}$ and the altitude $= r + x = r + \frac{r}{3} = \frac{4r}{3}$.

36. We have, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$

$$\Rightarrow \ A^2 = A \cdot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1+2 & 1+2-1 & 1-3+3 \\ 1+2-6 & 1+4+3 & 1-6-9 \\ 2-1+6 & 2-2-3 & 2+3+9 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4+2+2 & 4+4-1 & 4-6+3 \\ -3+8-28 & -3+16+14 & -3-24-42 \\ 7-3+28 & 7-6-14 & 7+9+42 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix}$$

Now, L.H.S. $= A^3 - 6A^2 + 5A + 11I$

$$= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - 6\begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$$

$$+ 5\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} + 11\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8-24+5+11 & 7-12+5+0 & 1-6+5+0 \\ -23+18+5+0 & 27-48+10+11 & -69+84-15+0 \\ 32-42+10+0 & -13+18-5+0 & 58-84+15+11 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O = \text{R.H.S.}$$

Now, $A^3 - 6A^2 + 5A + 11I = O$

$\Rightarrow \ 11I = -A^3 + 6A^2 - 5A$ ... (i)

Multiplying (i) by A^{-1}, we get

$11A^{-1}I = -A^{-1}A^3 + 6A^{-1}A^2 - 5A^{-1}A$

$\Rightarrow \ 11A^{-1} = -A^2 + 6A - 5I$

$$\Rightarrow A^{-1} = -\frac{1}{11}A^2 + \frac{6}{11}A - \frac{5}{11}I$$

$$= -\frac{1}{11}\begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$$

$$+ \frac{6}{11}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} - \frac{5}{11}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{11}\begin{bmatrix} -4+6-5 & -2+6+0 & -1+6+0 \\ 3+6+0 & -8+12-5 & 14-18+0 \\ -7+12+0 & 3-6+0 & -14+18-5 \end{bmatrix}$$

$$= \frac{1}{11}\begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

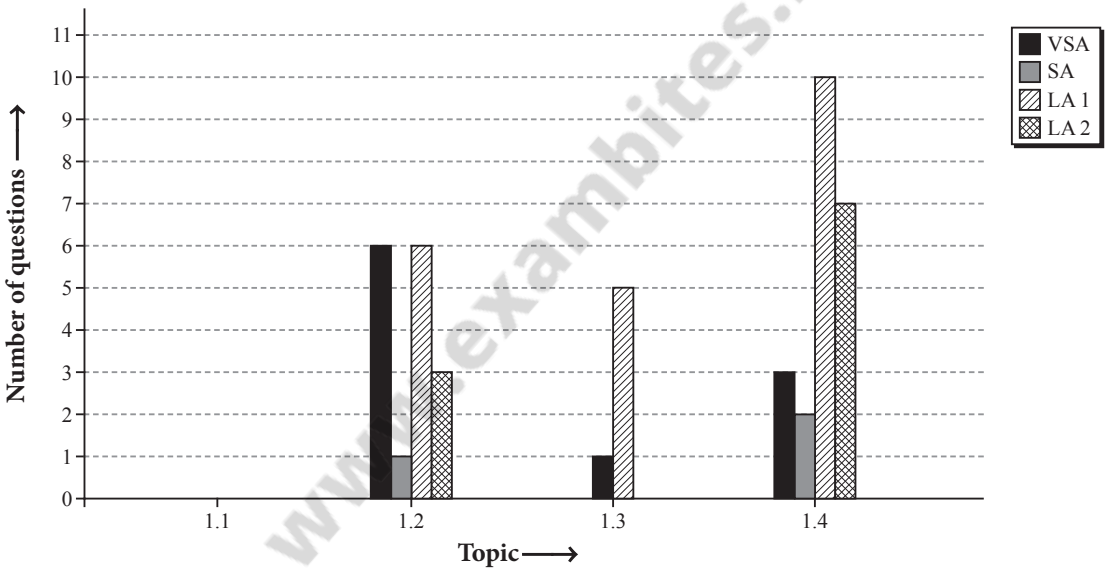
www.exambites.in

CHAPTER
1

Relations and Functions

1.1 Introduction
1.2 Types of Relations
1.3 Types of Functions
1.4 Composition of Functions and Invertible Function

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Composition of Functions and Invertible Function*
- Maximum LA 1 & LA 2 type questions were asked from *Composition of Functions and Invertible Function*

QUICK RECAP

RELATION

A relation R from a set A to a set B is a subset of $A \times B$. So, we say $R \subseteq A \times B$. A relation from a set A to itself is called a relation in A.

Empty Relation

- If no element of A is related to any element of A, then relation R on A is called an empty relation *i.e.*, $R = \phi \subset A \times A$.

Universal Relation

- If each element of A is related to every element of A, then relation R in A is called universal relation *i.e.*, $R = A \times A$.

- A relation R in a set A is called
  (i) reflexive, if $(a, a) \in R$, for all $a \in A$
  (ii) symmetric, if $(a, b) \in R \Rightarrow (b, a) \in R$, for all $a, b \in A$
  (iii) transitive, if $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$, for all $a, b, c \in A$
- A relation R in a set A is called an **equivalence relation**, if it is reflexive, symmetric and transitive.
- In a relation R in a set A, the set of all elements related to any element $a \in A$ is denoted by $[a]$ *i.e.*, $[a] = \{x \in A : (x, a) \in R\}$
  Here, $[a]$ is called an equivalence class of $a \in A$.

FUNCTION

- A relation f from a set A to a set B is called a function if
  (i) for each $a \in A$, there exists some $b \in B$ such that $(a, b) \in f$ *i.e.*, $f(a) = b$
  (ii) $(a, b) \in f$ and $(a, c) \in f \Rightarrow b = c$
- A function $f: A \to B$ is called
  (i) **one-one or injective function**, if distinct elements of A have distinct images in B *i.e.*, for $a, b \in A$, $f(a) = f(b) \Rightarrow a = b$
  (ii) **onto or surjective function**, if for every element $b \in B$, there exists some $a \in A$ such that $f(a) = b$.
- A function $f : A \to B$ is called **bijective function**, if it is both one-one and onto function.

Composition of Functions

- Let $f: A \to B$ and $g: B \to C$ be any two functions, then the function $gof : A \to C$ defined as $gof(x) = g(f(x))$, for all $x \in A$, is called the composition of f and g.

Invertible Functions

- A function $f : A \to B$ is said to be invertible, if there exists a function $g : B \to A$ such that $gof = I_A$ and $fog = I_B$. Here, g is called the inverse of f.
- Also, f is an invertible function iff it is a bijective function.

www.examtie.in

Previous Years' CBSE Board Questions

1.2 Types of Relations

VSA (1 mark)

1. A relation R in a set A is called _____, if $(a_1, a_2) \in R$ implies $(a_2, a_1) \in R$, for all $a_1, a_2 \in A$. *(2020)*

2. A relation in a set A is called _____ relation, if each element of A is related to itself. *(2020)*

3. If $R = \{(x, y) : x + 2y = 8\}$ is a relation on N, write the range of R. *(AI 2014)*

4. Let $R = \{(a, a^3) : a$ is a prime number less than 5$\}$ be a relation. Find the range of R. *(Foreign 2014)*

5. Let R be the equivalence relation in the set $A = \{0, 1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : 2$ divides $(a - b)\}$. Write the equivalence class [0]. *(Delhi 2014 C)*

6. State the reason for the relation R in the set $\{1, 2, 3\}$ given by $R = \{(1, 2), (2, 1)\}$ not to be transitive. *(Delhi 2011)*

SA (2 marks)

7. Check if the relation R in the set $\mathbb{R}$ of real numbers defined as $R = \{(a, b) : a < b\}$ is (i) symmetric, (ii) transitive. *(2020)*

LA 1 (4 marks)

8. Let N be the set of natural numbers and R be the relation on $N \times N$ defined by $(a, b)\ R\ (c, d)$ iff $ad = bc$ for all $a, b, c, d \in N$. Show that R is an equivalence relation. *(2020)*

9. Show that the relation R in the set $A = \{1, 2, 3, 4, 5, 6\}$ given by $R = \{(a, b) : |a - b|$ is divisible by 2$\}$ is an equivalence relation. *(2020)*

10. Show that the relation R on $\mathbb{R}$ defined as $R = \{(a, b) : a \leq b\}$, is reflexive and transitive but not symmetric. *(Delhi 2019)*

11. Show that the relation S in the set $A = \{x \in Z : 0 \leq x \leq 12\}$ given by $S = \{(a, b) : a, b \in Z, |a - b|$ is divisible by 3$\}$ is an equivalence relation. *(AI 2019)*

12. Let $A = \{1, 2, 3, ..., 9\}$ and R be the relation in $A \times A$ defined by $(a, b)\ R\ (c, d)$ if $a + d = b + c$ for $(a, b), (c, d)$ in $A \times A$. Prove that R is an equivalence relation. Also obtain the equivalence class $[(2, 5)]$. *(Delhi 2014)*

13. Let R be a relation defined on the set of natural numbers N as follow :
$R = \{(x, y) \mid x \in N, y \in N$ and $2x + y = 24\}$
Find the domain and range of the relation R. Also, find if R is an equivalence relation or not. *(Delhi 2014 C)*

LA 2 (6 marks)

14. Let $A = \{x \in Z : 0 \leq x \leq 12\}$. Show that $R = \{(a, b) : a, b \in A, |a - b|$ is divisible by 4$\}$, is an equivalence relation. Find the set of all elements related to 1. Also write the equivalence class [2]. *(2018)*

15. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a, b)\ R(c, d)$ if $ad(b + c) = bc(a + d)$. Show that R is an equivalence relation. *(Delhi 2015)*

16. Show that the relation R in the set $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : |a - b|$ is divisible by 2$\}$ is an equivalence relation. Write all the equivalence classes of R. *(AI 2015 C)*

1.3 Types of Functions

VSA (1 mark)

17. Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and let $f = \{(1, 4), (2, 5), (3, 6)\}$ be a function from A to B, state whether f is one-one or not. *(AI 2011)*

www.examkit.in

LA 1 (4 marks)

18. Show that the function $f: (-\infty, 0) \to (-1, 0)$ defined by $f(x) = \frac{x}{1+|x|}$, $x \in (-\infty, 0)$ is one-one and onto. *(2020)*

19. Show that $f: N \to N$, given by
$$f(x) = \begin{cases} x+1, & \text{if } x \text{ is odd} \\ x-1, & \text{if } x \text{ is even} \end{cases}$$
is both one-one and onto. *(AI 2012)*

20. Let $f: N \to N$ be defined by
$$f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases} \quad \text{for all } n \in N$$
Find whether the function f is bijective. *(Delhi 2012 C)*

21. If $f: R \to R$ be the function defined by $f(x) = 4x^3 + 7$, show that f is a bijection. *(Delhi 2011 C)*

22. Show that the function $f: W \to W$ defined by
$$f(n) = \begin{cases} n+1, & \text{if } n \text{ is even} \\ n-1, & \text{if } n \text{ is odd} \end{cases}$$
is a bijective function. *(AI 2011 C)*

1.4 Composition of Functions and Invertible Function

VSA (1 mark)

23. If f and g are two functions from R to R defined as $f(x) = |x| + x$ and $g(x) = |x| - x$, then $fog(x)$ for $x < 0$ is
(a) $4x$ (b) $2x$
(c) 0 (d) $-4x$ *(2020)*

24. Let $f: \{1, 3, 4\} \to \{1, 2, 5\}$ and $g: \{1, 2, 5\} \to \{1, 3\}$ given by $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(1, 3), (2, 3), (5, 1)\}$. Write down gof. *(AI 2014 C)*

25. If $f: R \to R$ defined as $f(x) = \frac{2x-7}{4}$ is an invertible function, write $f^{-1}(x)$. *(Delhi 2012 C)*

SA (2 marks)

26. Consider a bijective function $f: R_+ \to (7, \infty)$ given by $f(x) = 16x^2 + 24x + 7$, where R_+ is the set of all positive real numbers. Find the inverse function of f. *(2020)*

27. If $f(x) = \frac{4x+3}{6x-4}, x \neq \frac{2}{3}$, then show that $(fof)(x) = x$, for all $x \neq \frac{2}{3}$. Also, write inverse of f. *(2020)*

LA 1 (4 marks)

28. Prove that the function $f: N \to N$, defined by $f(x) = x^2 + x + 1$ is one-one but not onto. Find inverse of $f: N \to S$, where S is range of f. *(Delhi 2019)*

29. Let $A = R - \{2\}$ and $B = R - \{1\}$. If $f: A \to B$ is a function defined by $f(x) = \frac{x-1}{x-2}$, show that f is one-one and onto. Hence, find f^{-1}. *(AI 2019, Delhi 2013C)*

30. If the function $f: R \to R$ be given by $f(x) = x^2 + 2$ and $g: R \to R$ be given by $g(x) = \frac{x}{x-1}, x \neq 1$, find fog and gof and hence find fog (2) and gof (–3). *(AI 2014)*

31. Let $f: W \to W$, be defined as $f(x) = x - 1$, if x is odd and $f(x) = x + 1$, if x is even. Show that f is invertible. Find the inverse of f, where W is the set of all whole numbers. *(Foreign 2014)*

32. Let $A = R - \{3\}$, $B = R - \{1\}$. Let $f: A \to B$ be defined by $f(x) = \left(\frac{x-2}{x-3}\right)$, for all $x \in A$. Then show that f is bijective. Hence find $f^{-1}(x)$. *(Delhi 2014 C, 2012)*

33. Let $f, g: R \to R$ be two functions defined as $f(x) = |x| + x$ and $g(x) = |x| - x$, for all $x \in R$. Then find fog and gof. *(AI 2014 C)*

34. Show that the function f in $A = R - \left\{\frac{2}{3}\right\}$ defined as $f(x) = \frac{4x+3}{6x-4}$ is one-one and onto. Hence find f^{-1}. *(Delhi 2013)*

www.exambites.in

35. Consider $f: R_+ \to [4, \infty)$ given by $f(x) = x^2 + 4$. Show that f is invertible with the inverse f^{-1} of f given by $f^{-1}(y) = \sqrt{y-4}$, where R_+ is the set of all non-negative real numbers. *(AI 2013)*

36. Let $A = R - \{3\}$ and $B = R - \{1\}$. Consider the function $f: A \to B$ defined by $f(x) = \left(\frac{x-2}{x-3}\right)$. Is f one-one and onto ? Justify your answer. *(AI 2012C)*

37. Let $f: R \to R$ be defined as $f(x) = 10x + 7$. Find the function $g: R \to R$ such that $gof = fog = I_R$. *(AI 2011)*

LA 2 (6 marks)

38. Show that the function $f: R \to R$ defined by $f(x) = \frac{x}{x^2+1}$, $\forall\ x \in R$ is neither one-one nor onto. Also, if $g: R \to R$ is defined as $g(x) = 2x - 1$, find $fog(x)$. *(2018)*

39. Consider $f: R_+ \to [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$. Show that f is invertible with $f^{-1}(y) = \left(\frac{\sqrt{y+6}-1}{3}\right)$. Hence find

(i) $f^{-1}(10)$ (ii) y if $f^{-1}(y) = \frac{4}{3}$

where R_+ is the set of all non-negative real numbers. *(Delhi 2017)*

40. Consider $f: R - \left\{-\frac{4}{3}\right\} \to R - \left\{\frac{4}{3}\right\}$ given by $f(x) = \frac{4x+3}{3x+4}$. Show that f is bijective. Find the inverse of f and hence find $f^{-1}(0)$ and x such that $f^{-1}(x) = 2$. *(AI 2017)*

41. Let $f: N \to N$ be a function defined as $f(x) = 9x^2 + 6x - 5$. Show that $f: N \to S$, where S is the range of f, is invertible. Find the inverse of f and hence find $f^{-1}(43)$ and $f^{-1}(163)$. *(Delhi 2016)*

42. If $f, g: R \to R$ be two functions defined as $f(x) = |x| + x$ and $g(x) = |x| - x$, $\forall x \in R$. Then find fog and gof. Hence find $fog(-3)$, $fog(5)$, and $gof(-2)$. *(Foreign 2016)*

43. Consider $f: R_+ \to [-9, \infty[$ given by $f(x) = 5x^2 + 6x - 9$. Prove that f is invertible with $f^{-1}(y) = \left(\frac{\sqrt{54+5y}-3}{5}\right)$. *(AI 2015)*

44. Let $f: N \to R$ be a function defined as $f(x) = 4x^2 + 12x + 15$. Show that $f: N \to S$, where S is the range of f, is invertible. Also find the inverse of f. *(Foreign 2015, AI 2013 C)*

Detailed Solutions

1. A relation R in a set A is called symmetric, if $(a_1, a_2) \in R$ implies $(a_2, a_1) \in R$, for all $a_1, a_2 \in A$.

2. A relation in a set A is called reflexive relation, if each element of A is related to itself.

3. Here, $R = \{(x, y) : x + 2y = 8\}$, where $x, y \in N$.
For $x = 1, 3, 5, ...$
$x + 2y = 8$ has no solution in N.
For $x = 2$, we have $2 + 2y = 8 \Rightarrow y = 3$
For $x = 4$, we have $4 + 2y = 8 \Rightarrow y = 2$
For $x = 6$, we have $6 + 2y = 8 \Rightarrow y = 1$
For $x = 8, 10, ...$
$x + 2y = 8$ has no solution in N.
$\therefore$ Range of $R = \{y : (x, y) \in R\} = \{1, 2, 3\}$

4. Given relation is
$R = \{(a, a^3) : a$ is a prime number less than $5\}$.
$\therefore$ $R = \{(2, 8), (3, 27)\}$
So, the range of R is $\{8, 27\}$.

5. Here, $R = \{(a, b) \in A \times A : 2$ divides $(a - b)\}$
This is the given equivalence relation, where
$A = \{0, 1, 2, 3, 4, 5\}$
$\therefore$ Equivalence class of $[0]$ is $\{0, 2, 4\}$.

6. For transitivity of a relation,
if $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$
We have, $R = \{(1, 2), (2, 1)\}$
$(1, 2) \in R$ and $(2, 1) \in R$ but $(1, 1) \notin R$
$\therefore$ R is not transitive.

www.examhits.in

7. We have, $R = \{(a, b) : a < b\}$, where $a, b \in \mathbb{R}$
(i) Symmetry : Let $(x, y) \in R$, *i.e.*, $x\,R\,y \Rightarrow x < y$
But $y \not< x$, so $(x, y) \in R \Rightarrow (y, x) \notin R$
Thus, R is not symmetric.
(ii) Transitive : Let $(x, y) \in R$ and $(y, z) \in R$
$\Rightarrow$ $x < y$ and $y < z \Rightarrow x < z$
$\Rightarrow$ $(x, z) \in R$. Thus, R is transitive.

8. (i) Reflexivity : Let (a, b) be an arbitrary element of $N \times N$. Then, we have $(a, b) \in N \times N \forall$ $a, b \in N$
$\Rightarrow$ $ab = ba$ (By commutativity of multiplication)
$\Rightarrow$ $(a, b)\,R\,(a, b)$
Thus, $(a, b)\,R\,(a, b)$ for all $(a, b) \in N \times N$
So, R is reflexive.
(ii) Symmetry : Let $(a, b), (c, d) \in N \times N$ such that $(a, b)\,R\,(c, d) \Rightarrow ad = bc \Rightarrow bc = ad \Rightarrow cb = da$
(By commutativity of multiplication)
$\Rightarrow$ $(c, d)\,R\,(a, b)$
Thus, $(a, b)\,R\,(c, d)$
$\Rightarrow$ $(c, d)\,R\,(a, b)$ for all $(a, b), (c, d) \in N \times N$
So, R is symmetric.
(iii) Transitivity : Let $(a, b), (c, d), (e, f) \in N \times N$ such that $(a, b)\,R\,(c, d)$ and $(c, d)\,R\,(e, f)$
Now, $(a, b)\,R\,(c, d) \Rightarrow ad = bc$ ...(i)
and $(c, d)\,R\,(e, f) \Rightarrow cf = de$ ...(iii)
Multiplying (i) and (ii), we get $ad \cdot cf = bc \cdot de$
$\Rightarrow$ $af = be \Rightarrow (a, b)\,R\,(e, f)$
Thus, $(a, b)\,R\,(c, d)$ and $(c, d)\,R\,(e, f) \Rightarrow (a, b)\,R\,(e, f)$
So R is transitive.
$\therefore$ R is an equivalence relation.

9. We have, $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(a, b) : |a - b|$ is divisible by $2\}$
(i) Reflexive : For any $a \in A$
$|a - a| = 0$, which is divisible by 2.
Thus, $(a, a) \in R$. So, R is reflexive.
(ii) Symmetric : For any $a, b \in A$
Let $(a, b) \in R$
$\Rightarrow$ $|a - b|$ is divisible by 2
$\Rightarrow$ $|b - a|$ is divisible by 2
$\Rightarrow$ $(b, a) \in R$
$\therefore$ $(a, b) \in R \Rightarrow (b, a) \in R$
$\therefore$ R is symmetric.
(iii) Transitive : For any $a, b, c \in A$
Let $(a, b) \in R$ and $(b, c) \in R$
$\Rightarrow$ $|a - b|$ is divisible by 2 and $|b - c|$ is divisible by 2.
$\Rightarrow$ $a - b = \pm 2\,k_1$ and $b - c = \pm 2k_2$, $k_1, k_2 \in N$
$\Rightarrow$ $a - b + b - c = \pm 2(k_1 + k_2)$
$\Rightarrow$ $a - c = \pm 2\,k_3$, $k_3 \in N$
$\Rightarrow$ $|a - c|$ is divisible by 2 $\Rightarrow$ $(a, c) \in R$
$\therefore$ R is transitive.
Hence, R is an equivalence relation.

10. We have, $R = \{(a,b) : a \le b,\ a, b \in \mathbb{R}\}$
(i) Reflexive : Since $a \le a$
$\therefore$ $a\,R\,a\ \forall\, a \in R$
Hence, R is reflexive.
(ii) Symmetric : Let $a, b \in R$ such that
$aRb \Rightarrow a \le b \not\Rightarrow b \le a$
Hence R is not symmetric.
(iii) Transitive : Let $a, b, c \in R$
such that aRb and bRc
Now, $aRb \Rightarrow a \le b$ ... (i)
and $bRc \Rightarrow b \le c$ ... (ii)
From (i) and (ii), we have
$a \le b \le c \Rightarrow a \le c$
$\therefore$ aRc
Hence, relation R is transitive.

11. We have, $A = \{x \in Z : 0 \le x \le 12\}$
$\therefore$ $A = \{0, 1, 2, 3, ..., 12\}$
Also, $S = \{(a, b) : a, b \in Z, |a - b|$ is divisible by $3\}$
(i) Reflexive : For any $a \in A$,
$|a - a| = 0$, which is divisible by 3
Thus, $(a, a) \in S$ $\therefore$ S is reflexive.
(ii) Symmetric : For any $a, b \in A$, let $(a, b) \in S$
$\Rightarrow$ $|a - b|$ is divisible by 3
$\Rightarrow$ $|b - a|$ is divisible by 3 $\Rightarrow$ $(b, a) \in S$
i.e. $(a, b) \in S \Rightarrow (b, a) \in S$
$\therefore$ S is symmetric.
(iii) Transitive : For any $a, b, c \in A$
Let $(a, b) \in S$ and $(b, c) \in S$
$\Rightarrow$ $|a - b|$ is divisible by 3 and $|b - c|$ is divisible by 3
$\Rightarrow$ $(a - b) = \pm 3k_1$ and $(b - c) = \pm 3k_2$; $k_1, k_2 \in N$
$\Rightarrow$ $(a - b) + (b - c) = \pm 3(k_1 + k_2)$
$\Rightarrow$ $(a - c) = \pm 3(k_1 + k_2)$; $k_1, k_2 \in N$
$\Rightarrow$ $|a - c|$ is a multiple of 3
$\Rightarrow$ $|a - c|$ is divisible by 3 $\Rightarrow$ $(a, c) \in S$
$\therefore$ S is transitive.
Hence, S is an equivalence relation.

12. Given $A = \{1, 2, 3, 4,...,9\} \subset N$, the set of natural numbers.
To show : R is an equivalence relation.
(i) Reflexivity : Let (a, b) be an arbitrary element of $A \times A$. Then, we have $(a, b) \in A \times A \Rightarrow a, b \in A$

www.examrites.in

$\Rightarrow \quad a + b = b + a$
(by commutativity of addition on $A \subset N$)
$\Rightarrow \quad (a, b)\ R\ (a, b)$
Thus, $(a, b)\ R\ (a, b)$ for all $(a, b) \in A \times A$
So, R is reflexive.
(ii) Symmetry: Let $(a, b), (c, d) \in A \times A$ such that
$(a, b)\ R\ (c, d) \Rightarrow a + d = b + c \Rightarrow b + c = a + d$
$\Rightarrow \quad c + b = d + a$
(by commutativity of addition on $A \subset N$)
$\Rightarrow \quad (c, d)\ R\ (a, b)$.
Thus, $(a, b)\ R\ (c, d)$
$\Rightarrow (c, d)\ R\ (a, b)$ for all $(a, b), (c, d) \in A \times A$.
So, R is symmetric.
(iii) Transitivity: Let $(a, b), (c, d), (e, f) \in A \times A$ such that $(a, b)\ R\ (c, d)$ and $(c, d)\ R\ (e, f)$
Now, $(a, b)\ R\ (c, d) \Rightarrow a + d = b + c$ ...(i)
and $(c, d)\ R\ (e, f) \Rightarrow c + f = d + e$ ...(ii)
Adding (i) and (ii), we get
$(a + d) + (c + f) = (b + c) + (d + e)$
$\Rightarrow \quad a + f = b + e \Rightarrow (a, b)\ R\ (e, f)$
Thus, $(a, b)\ R\ (c, d)$ and $(c, d)\ R\ (e, f)$
$\Rightarrow \quad (a, b)\ R\ (e, f)$.
So, R is transitive.
$\therefore \quad R$ is an equivalence relation.
Equivalence class for $[(2, 5)]$ is $\{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$.

13. Here, $R = \{(x, y) \mid x \in N, y \in N \text{ and } 2x + y = 24\}$
Domain of $R = \{1, 2, 3, 4, ..., 11\}$
Range of $R = \{2, 4, 6, 8, 10, 12, ..., 22\}$
R is not reflexive as if $(2, 2) \in R$
$\Rightarrow \quad 2 \times 2 + 2 = 6 \neq 24$
In fact R is neither symmetric nor transitive.
$\Rightarrow \quad R$ is not an equivalence relation.

14. We have, $A = \{x \in Z : 0 \leq x \leq 12\}$
$\therefore \quad A = \{0, 1, 2, 3, ..., 12\}$
and $S = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$
(i) Reflexive : For any $a \in A$,
$|a - a| = 0$, which is a multiple of 4.
Thus, $(a, a) \in S$
$\therefore \quad S$ is reflexive.
(ii) Symmetric : For any $a, b \in A$,
Let $(a, b) \in S$
$\Rightarrow \quad |a - b|$ is a multiple of 4
$\Rightarrow \quad |b - a|$ is a multiple of $4 \Rightarrow (b, a) \in S$
i.e., $(a, b) \in S \Rightarrow (b, a) \in S$
$\therefore \quad S$ is symmetric.
(iii) Transitive : For any $a, b, c \in A$,
Let $(a, b) \in S$ and $(b, c) \in S$
$\Rightarrow \quad |a - b|$ is a multiple of 4 and $|b - c|$ is a multiple of 4
$\Rightarrow \quad a - b = \pm 4k_1$ and $b - c = \pm 4k_2$; $k_1, k_2 \in N$
$\Rightarrow \quad (a - b) + (b - c) = \pm 4 (k_1 + k_2)$; $k_1, k_2 \in N$
$\Rightarrow \quad a - c = \pm 4 (k_1 + k_2)$; $k_1, k_2 \in N$
$\Rightarrow \quad |a - c|$ is a multiple of $4 \Rightarrow (a, c) \in S$
$\therefore \quad S$ is transitive.
Hence, S is an equivalence relation.
The set of elements related to 1 is $\{5, 9\}$.
Equivalence class for $[2]$ is $\{2, 6, 10\}$.

15. (i) Reflexivity : Let (a, b) be an arbitrary element of $N \times N$. Then, $(a, b) \in N \times N$
$\Rightarrow \quad ab(b + a) = ba(a + b)$
[by commutativity of addition and multiplication on N]
$\Rightarrow \quad (a, b)\ R\ (a, b)$
So, R is reflexive on $N \times N$.
(ii) Symmetry : Let $(a, b), (c, d) \in N \times N$ be such that $(a, b)\ R\ (c, d)$.
$\Rightarrow \quad ad(b + c) = bc(a + d)$
$\Rightarrow \quad cb(d + a) = da(c + b)$
[by commutativity of addition and multiplication on N]
Thus, $(a, b)\ R\ (c, d) \Rightarrow (c, d)\ R\ (a, b)$ for all $(a, b), (c, d) \in N \times N$.
So, R is symmetric on $N \times N$.
(iii) Transitivity : Let $(a, b), (c, d), (e, f) \in N \times N$ be such that
$(a, b)\ R\ (c, d)$ and $(c, d)\ R\ (e, f)$. Then,
$(a, b)\ R\ (c, d) \Rightarrow ad(b + c) = bc(a + d)$

$$\Rightarrow \quad \frac{b+c}{bc} = \frac{a+d}{ad} \Rightarrow \frac{1}{b} + \frac{1}{c} = \frac{1}{a} + \frac{1}{d} \qquad \text{...(i)}$$

and $(c, d)\ R\ (e, f) \Rightarrow cf(d + e) = de(c + f)$

$$\Rightarrow \quad \frac{d+e}{de} = \frac{c+f}{cf} \Rightarrow \frac{1}{d} + \frac{1}{e} = \frac{1}{c} + \frac{1}{f} \qquad \text{...(ii)}$$

Adding (i) and (ii), we get

$$\left(\frac{1}{b} + \frac{1}{c}\right) + \left(\frac{1}{d} + \frac{1}{e}\right) = \left(\frac{1}{a} + \frac{1}{d}\right) + \left(\frac{1}{c} + \frac{1}{f}\right)$$

$$\Rightarrow \quad \frac{1}{b} + \frac{1}{e} = \frac{1}{a} + \frac{1}{f} \Rightarrow \frac{b+e}{be} = \frac{a+f}{af}$$

$\Rightarrow \quad af(b + e) = be(a + f)$
$\Rightarrow \quad (a, b)\ R\ (e, f)$
So, R is transitive on $N \times N$.
Hence, R is an equivalence relation.

16. *Refer to answer 14.*
Further R has only two equivalence classes, namely $[1] = [3] = [5] = \{1, 3, 5\}$ and $[2] = [4] = \{2, 4\}$.

www.examrace.in

17. $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and $f = \{(1, 4), (2, 5), (3, 6)\}$
We have, $f(1) = 4$, $f(2) = 5$ and $f(3) = 6$. Distinct elements of A have distinct images in B. Hence, f is a one-one function.

18. Given, $f(x) = \dfrac{x}{1+|x|}, x \in (-\infty, 0)$

$= \dfrac{x}{1-x}$ $(\because x \in (-\infty, 0), |x| = -x)$

For one-one : Let $f(x_1) = f(x_2)$, $x_1 x_2 \in (-\infty, 0)$

$\Rightarrow \quad \dfrac{x_1}{1-x_1} = \dfrac{x_2}{1-x_2}$

$\Rightarrow \quad x_1(1 - x_2) = x_2(1 - x_1)$

$\Rightarrow \quad x_1 - x_1x_2 = x_2 - x_1x_2 \Rightarrow x_1 = x_2$

Hence, if $f(x_1) = f(x_2)$, then $x_1 = x_2$

$\therefore \quad f$ is one-one

For onto : Let $f(x) = y$

$\Rightarrow \quad y = \dfrac{x}{1-x} \Rightarrow y(1 - x) = x \Rightarrow y - xy = x$

$\Rightarrow \quad x + xy = y \Rightarrow x(1 + y) = y \Rightarrow x = \dfrac{y}{1+y}$

Here, $y \in (-1, 0)$

So, x is defined for all values of y.

$\therefore \quad f$ is onto

19. Here, $f : N \to N$ s.t.

$$f(x) = \begin{cases} x+1, & \text{if } x \text{ is odd} \\ x-1, & \text{if } x \text{ is even} \end{cases}$$

Let $x, y \in N$ s.t. $f(x) = f(y)$

We shall show that $x = y$

(i) If x and y both are even

$f(x) = f(y) \Rightarrow x - 1 = y - 1 \Rightarrow x = y$

(ii) If x and y both are odd

$f(x) = f(y) \Rightarrow x + 1 = y + 1 \Rightarrow x = y$

(iii) If x is odd and y is even

$f(x) = f(y) \Rightarrow x + 1 = y - 1$

$\Rightarrow \quad y - x = 2$ ...(1)

R.H.S. is even but L.H.S. is odd.

$\Rightarrow$ Equation (1) in N is not possible.

$\Rightarrow$ (iii) does not arise.

(iv) If x is even and y is odd, does not arise.

In any case, $f(x) = f(y) \Rightarrow x = y$

$\Rightarrow \quad f$ is one-one

For any $y \in N$ (co-domain), y can be even or odd

When y is odd, $y + 1$ is even, so

$f(y + 1) = (y + 1) - 1 = y$

When y is even, $y - 1$ is odd, so

$f(y - 1) = (y - 1) + 1 = y$

$\Rightarrow \quad f : N \to N$ is onto.

Hence, f is both one-one and onto.

20. (i) Injectivity :

Here, $f(1) = \dfrac{1+1}{2} = 1, f(2) = \dfrac{2}{2} = 1,$

$f(3) = \dfrac{3+1}{2} = 2, f(4) = \dfrac{4}{2} = 2$

Thus, $f(2k-1) = \dfrac{(2k-1)+1}{2} = k$ and

$f(2k) = \dfrac{2k}{2} = k$

$\Rightarrow \quad f(2k - 1) = f(2k)$, where $k \in N$

But, $2k - 1 \neq 2k, \Rightarrow f$ is not one-one.

Hence, f is not bijective.

21. Here, $f(x) = 4x^3 + 7$

Let $x_1, x_2 \in R$ s.t.

$f(x_1) = f(x_2)$

$\Rightarrow \quad 4x_1^3 + 7 = 4x_2^3 + 7 \Rightarrow 4x_1^3 = 4x_2^3 \Rightarrow x_1^3 = x_2^3$

$\Rightarrow \quad x_1^3 - x_2^3 = 0$

$\Rightarrow \quad (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 0$

$\Rightarrow \quad x_1 - x_2 = 0$

$[\because \ x_1^2 + x_1x_2 + x_2^2 = 0$ has no real roots$]$

$\Rightarrow \quad x_1 = x_2$

$\therefore \quad f : R \to R$ is one-one.

Again $\forall \ y \in R$ (co-domain), we find $x \in R$ (domain) s.t. $f(x) = y$

$\Leftrightarrow \quad 4x^3 + 7 = y \Leftrightarrow 4x^3 = y - 7$

$\Leftrightarrow \quad x^3 = \dfrac{y-7}{4} \Leftrightarrow x = \sqrt[3]{\dfrac{y-7}{4}} \in R$

$(\because \ x^3 = \alpha, \alpha \in R$ has always one real root)

Hence, f is onto

So, $f : R \to R$ is a bijection.

22. *Refer to answer 19.*

23. **(d) :** Here, $f, g : R \to R$ s.t.

$$f(x) = |x| + x = \begin{cases} 2x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

and $g(x) = |x| - x = \begin{cases} 0 & \text{if } x \geq 0 \\ -2x & \text{if } x < 0 \end{cases}$

$\therefore \quad (fog)(x) = f(g(x)) = \begin{cases} f(0) & \text{if } x \geq 0 \\ f(-2x) & \text{if } x < 0 \end{cases}$

$= \begin{cases} 0 & \text{if } x \geq 0 \\ 2(-2x) & \text{if } x < 0 \end{cases} = \begin{cases} 0 & \text{if } x \geq 0 \\ -4x & \text{if } x < 0 \end{cases}$

24. Here, $f = \{(1,2), (3, 5), (4, 1)\}$

$\Leftrightarrow \quad f(1) = 2; f(3) = 5; f(4) = 1$ ...(1)

$g = \{(1, 3), (2, 3), (5, 1)\}$

$\Leftrightarrow \quad g(1) = 3; g(2) = 3, g(5) = 1$ ...(2)

www.examchits.in

Now, $gof: \{1, 3, 4\} \to \{1, 3\}$
Using (1) and (2), we get
$(gof)(1) = g(f(1)) = g(2) = 3$
$(gof)(3) = g(f(3)) = g(5) = 1$
$(gof)(4) = g(f(4)) = g(1) = 3$
$\therefore \quad gof = \{(1, 3), (3, 1), (4, 3)\}$.

25. Let $y = f(x) = \dfrac{2x-7}{4}$

$\Rightarrow \quad 4y = 2x - 7 \quad \Rightarrow \quad x = \dfrac{4y+7}{2}$

As $y = f(x)$ is an invertible function, so

$$x = f^{-1}(y) \; i.e., \; f^{-1}(y) = \frac{4y+7}{2}$$

$$\Rightarrow \quad f^{-1} : R \to R \text{ s.t. } f^{-1}(x) = \frac{4x+7}{2}, \forall x \in R.$$

26. Let $y = f(x) = 16x^2 + 24x + 7$
$\Rightarrow \quad y = (4x + 3)^2 - 2 \Rightarrow (4x + 3)^2 = y + 2$

$$\Rightarrow \quad 4x + 3 = \sqrt{y+2} \Rightarrow x = \frac{\sqrt{y+2}-3}{4}$$

$$\therefore \quad f^{-1}(y) = \frac{\sqrt{y+2}-3}{4} \Rightarrow f^{-1}(x) = \frac{\sqrt{x+2}-3}{4}$$

27. Given, $f(x) = \dfrac{4x+3}{6x-4}, x \neq \dfrac{2}{3}$

$$\therefore \quad fof(x) = f(f(x)) = f\left(\frac{4x+3}{6x-4}\right) = \frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}$$

$$= \frac{16x+12+18x-12}{24x+18-24x+16} = \frac{34x}{34} = x$$

Now, let $y = f(x) = \dfrac{4x+3}{6x-4}$

$\Rightarrow \quad 6xy - 4y = 4x + 3 \Rightarrow 6xy - 4x = 4y + 3$

$$\Rightarrow \quad x = \frac{4y+3}{6y-4}$$

$\therefore$ Inverse of f is given by

$$f^{-1}(x) = \frac{4x+3}{6x-4}, x \neq \frac{2}{3}$$

28. We have $f: N \to N$ as $y = f(x) = x^2 + x + 1$
Let $x, y \in N$ such that $f(x) = f(y)$
$\Rightarrow \quad x^2 + x + 1 = y^2 + y + 1$
$\Rightarrow \quad (x^2 - y^2) + (x - y) = 0$
$\Rightarrow \quad (x + y)(x - y) + (x - y) = 0$
$\Rightarrow \quad (x - y)(x + y + 1) = 0$
$\Rightarrow \quad x = y \qquad (\because \; x + y + 1 \neq 0)$
Hence, $f(x)$ is one-one.

Now, for every y in N there does not exist x in N such that $y = x^2 + x + 1$.
As $y = 1 \in N$ gives $1 = x^2 + x + 1$
$\Rightarrow \quad x(x + 1) = 0 \quad \Rightarrow \quad x = 0$ or $x = -1$,
which is impossible.
Hence, $f: N \to N$ defined by $f(x) = x^2 + x + 1$ is not onto.
If S is the range of f, then $f: N \to S$ is onto and inverse is given by

$$y = f(x) = x^2 + x + 1 \Rightarrow y = \left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$$

$$\Rightarrow y - \frac{3}{4} = \left(x + \frac{1}{2}\right)^2 \Rightarrow \sqrt{\left(y - \frac{3}{4}\right)} - \frac{1}{2} = x = f^{-1}(y)$$

Hence inverse of $f: N \to S$ is given by

$$f^{-1}(x) = \sqrt{x - \frac{3}{4}} - \frac{1}{2}$$

29. Here, $f: A \to B$ is given by $f(x) = \dfrac{x-1}{x-2}$,
where $A = R - \{2\}$ and $B = R - \{1\}$
Let $f(x_1) = f(x_2)$, where $x_1, x_2 \in A$ (*i.e.*, $x_1 \neq 2, x_2 \neq 2$)

$$\Rightarrow \quad \frac{x_1 - 1}{x_1 - 2} = \frac{x_2 - 1}{x_2 - 2}$$

$\Rightarrow \quad (x_1 - 1)(x_2 - 2) = (x_1 - 2)(x_2 - 1)$
$\Rightarrow \quad x_1x_2 - 2x_1 - x_2 + 2 = x_1x_2 - x_1 - 2x_2 + 2$
$\Rightarrow \quad -2x_1 - x_2 = -x_1 - 2x_2$
$\Rightarrow \quad x_1 = x_2 \Rightarrow f$ is one-one.
Let $y \in B = R - \{1\}$ *i.e.*, $y \in R$ and $y \neq 1$
such that $f(x) = y$

$$\Leftrightarrow \quad \frac{x-1}{x-2} = y \Leftrightarrow (x-2)y = x - 1$$

$$\Leftrightarrow \quad xy - 2y = x - 1 \Leftrightarrow x(y - 1) = 2y - 1$$

$$\Leftrightarrow \quad x = \frac{2y-1}{y-1} \qquad \text{...(i)}$$

$$\therefore \quad f(x) = y \text{ when } x = \frac{2y-1}{y-1} \in A \quad (\text{as } y \neq 1)$$

Hence, f is onto.
Thus, f is one–one and onto.
From (i), $f^{-1}: B \to A$ is given by $x = f^{-1}(y)$

$$i.e., \quad f^{-1}(y) = \frac{2y-1}{y-1}.$$

30. Here $f: R \to R$ s.t. $f(x) = x^2 + 2$

and $g: R \to R$ s.t. $g(x) = \dfrac{x}{x-1}, x \neq 1$

www.exambites.in

Now $fog : R \to R$ s.t.

$$(fog)(x) = f(g(x)) = f\left(\frac{x}{x-1}\right) = \left(\frac{x}{x-1}\right)^2 + 2, x \neq 1$$

$$\Rightarrow \quad (fog)(2) = \left(\frac{2}{2-1}\right)^2 + 2 = 6$$

Also, $gof : R \to R$ s.t.

$$(gof)(x) = g(f(x)) = g(x^2+2) = \frac{x^2+2}{x^2+1}$$

$$\therefore \quad (gof)(-3) = \frac{(-3)^2+2}{(-3)^2+1} = \frac{11}{10}.$$

31. *Refer to answer 19.*

since f is bijective function. So, it is invertible.

Now, let $x, y \in W$ such that $f(x) = y$

$\Rightarrow$ $x + 1 = y$, if x is even and $x - 1 = y$, if x is odd.

$$\Rightarrow \quad x = \begin{cases} y-1, \text{ if } y \text{ is odd} \\ y+1, \text{ if } y \text{ is even} \end{cases}$$

$$\Rightarrow \quad f^{-1}(y) = \begin{cases} y-1, \text{ if } y \text{ is odd} \\ y+1, \text{ if } y \text{ is even} \end{cases}$$

Hence, $f^{-1}(x) = \begin{cases} x-1, \text{ if } x \text{ is odd} \\ x+1, \text{ if } x \text{ is even} \end{cases}$

32. $A = R - \{3\}$; $B = R - \{1\}$

and $f : A \to B$ defined as

$$f(x) = \frac{x-2}{x-3} \; \forall \; x \in A.$$

Here, f is defined $\forall \; x \in A$, as $3 \notin A$.

Also, $f(x) \neq 1$ $\quad [\because \; f(x) = 1 \Leftrightarrow \frac{x-2}{x-3} = 1$

$\Leftrightarrow x - 2 = x - 3 \Leftrightarrow -2 = -3$, which is absurd]

Let $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$

$$\Rightarrow \quad \frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$$

$\Rightarrow$ $(x_1 - 2)(x_2 - 3) = (x_1 - 3)(x_2 - 2)$

$\Rightarrow$ $x_1x_2 - 3x_1 - 2x_2 + 6 = x_1x_2 - 2x_1 - 3x_2 + 6$

$\Rightarrow$ $-3x_1 - 2x_2 = -2x_1 - 3x_2 \Rightarrow x_1 = x_2$

$\therefore$ f is a one-one function.

Let $y \in B = R - \{1\} \Rightarrow y \neq 1$.

We want to solve $y = f(x)$ for some $x \in A$

$$\Leftrightarrow \quad y = \frac{x-2}{x-3} \quad \Leftrightarrow \quad xy - 3y = x - 2$$

$$\Leftrightarrow \quad x(y-1) = -2 + 3y$$

$$\Leftrightarrow \quad x = \frac{3y-2}{y-1} \in A \text{ (as } y \neq 1)$$

$\therefore$ f is onto also.

$\Rightarrow$ $f : A \to B$ is a bijective function.

Now, $f^{-1}(y) = x = \frac{3y-2}{y-1} \quad \Leftrightarrow \quad f^{-1}(x) = \frac{3x-2}{x-1}$

33. *Refer to answer 23.*

34. Here, $f(x) = \frac{4x+3}{6x-4}$ where $x \in A = R - \left\{\frac{2}{3}\right\}$.

(i) Let $f(x_1) = f(x_2)$ $(\forall \; x_1, x_2 \in A)$

$$\Rightarrow \quad \frac{4x_1+3}{6x_1-4} = \frac{4x_2+3}{6x_2-4}$$

$\Rightarrow$ $(4x_1 + 3)(6x_2 - 4) = (6x_1 - 4)(4x_2 + 3)$

$\Rightarrow$ $24x_1x_2 - 16x_1 + 18x_2 - 12 = 24x_1x_2 + 18x_1 - 16x_2 - 12$

$\Rightarrow$ $-34x_1 = -34x_2 \Rightarrow x_1 = x_2$

$\therefore$ f is one-one.

(ii) For $y \in A = R - \left\{\frac{2}{3}\right\}$.

Let $f(x) = y$

$$\Leftrightarrow \quad \frac{4x+3}{6x-4} = y \Leftrightarrow (6x-4)y = 4x + 3$$

$$\Leftrightarrow \quad x = \frac{4y+3}{6y-4} \in A \left(\text{as } y \neq \frac{2}{3}\right)$$

$\Leftrightarrow$ f is onto.

Refer to answer 27.

35. $f : R_+ \to [4, \infty)$ given by $f(x) = x^2 + 4$

(i) Let $x_1, x_2 \in R_+$ s.t. $f(x_1) = f(x_2)$

$\Rightarrow$ $x_1^2 + 4 = x_2^2 + 4 \Rightarrow x_1^2 = x_2^2$

$\Rightarrow$ $x_1 = x_2$ $\qquad (\because \; x_1, x_2 \in R_+)$

$\Rightarrow$ f is one-one

(ii) $y = f(x) \; \forall \; y \in [4, \infty), y \geq 4$

$\Rightarrow$ $x^2 + 4 = y \Rightarrow x = \sqrt{y-4}$

Now, x is defined if

$$y - 4 \geq 0 \Rightarrow \sqrt{y-4} \in R_+ \text{ and}$$

$$f\left(\sqrt{y-4}\right) = \left(\sqrt{y-4}\right)^2 + 4 = (y-4) + 4 = y$$

$\Rightarrow$ f is onto.

$\therefore$ f is one–one and onto.

$\Rightarrow$ f is invertible and f^{-1} exists.

$f^{-1}(y) = \sqrt{y-4}$.

36. *Refer to answer 32.*

37. Let $y \in R$ (co-domain) be arbitrary.

By definition, $y = 10x + 7$ for some $x \in R$

$$\Rightarrow \quad x = \frac{y-7}{10}$$

So, we define, $g : R \to R$ by $g(y) = \frac{y-7}{10}$

www.examples.in

Now, $(gof)(x) = g(f(x)) = g(10x+7)$

$$= \frac{(10x+7)-7}{10} = x$$

and $(fog)(y) = f(g(y)) = f\left(\frac{y-7}{10}\right)$

$$= 10\left(\frac{y-7}{10}\right)+7 = y$$

Thus, $gof = fog = I_R$.

38. Here, $f: R \to R$ is given by

$$f(x) = \frac{x}{x^2+1}, \forall x \in R$$

Let $f(x_1) = f(x_2) \Rightarrow \frac{x_1}{x_1^2+1} = \frac{x_2}{x_2^2+1}$

$\Rightarrow$ $x_1(x_2^2 + 1) = x_2(x_1^2 + 1)$
$\Rightarrow$ $x_1x_2^2 + x_1 = x_2x_1^2 + x_2$
$\Rightarrow$ $x_1x_2(x_2 - x_1) = x_2 - x_1$
$\Rightarrow$ $x_1 = x_2$ or $x_1x_2 = 1$

Hence f is not one one as there are points for which $x_1 \neq x_2$ but $x_1x_2 = 1$.

Also, f is not onto as when $1 \in R$ $\exists$ $x \in R$ such that $f(x) = 1$ which gives $\frac{x}{x^2+1} = 1$. But there is no such x in the domain R, since the equation $x^2 - x + 1 = 0$ does not give any real value of x.

Now, $g: R \to R$ s.t. $g(x) = 2x - 1$

$fog(x) = f(g(x)) = f(2x - 1)$

$$= \frac{(2x-1)}{(2x-1)^2+1} = \frac{2x-1}{4x^2-4x+2} = \frac{2x-1}{2(2x^2-2x+1)}$$

39. We have $f: R_+ \to [-5, \infty)$ is given by

$f(x) = 9x^2 + 6x - 5$

Let $x_1, x_2 \in R_+$ such that $f(x_1) = f(x_2)$

$$\Rightarrow \quad 9x_1^2+6x_1-5 = 9x_2^2+6x_2-5$$

$$\Rightarrow \quad 9(x_1^2-x_2^2)+6(x_1-x_2) = 0$$

$\Rightarrow$ $(x_1 - x_2)(9(x_1 + x_2) + 6) = 0$

$$\Rightarrow \quad x_1 = x_2 \text{ or } x_1+x_2 = -\frac{6}{9}$$

(which is not possible as $x_1, x_2 \in R_+$)

$\Rightarrow$ f is one – one.

Let $y = f(x)$ $\forall$ $y \in [-5, \infty)$

$\Rightarrow$ $9x^2 + 6x - 5 = y \Rightarrow (3x + 1)^2 - 1 - 5 = y$

$$\Rightarrow \quad 3x+1 = \sqrt{y+6} \Rightarrow x = \frac{\sqrt{y+6}-1}{3}$$

Now, x is defined and $x \in R_+$ if $y + 6 \geq 1$

$\Rightarrow$ $y \geq -5$
$\Rightarrow$ f is onto
$\therefore$ f is one-one and onto.
$\Rightarrow$ f is invertible and f^{-1} exists.

$$f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$$

(i) $$f^{-1}(10) = \frac{\sqrt{10+6}-1}{3} = \frac{4-1}{3} = 1$$

(ii) $$f^{-1}(y) = \frac{4}{3} = x$$

$$\therefore \quad y = f(x) = f\left(\frac{4}{3}\right) = 9\left(\frac{4}{3}\right)^2 + 6\left(\frac{4}{3}\right) - 5$$

$= 16 + 8 - 5 = 19$

40. Injectivity : Let x, y be any two elements of $R - \left\{-\frac{4}{3}\right\}$ and consider, $f(x) = f(y)$

$$\Rightarrow \frac{4x+3}{3x+4} = \frac{4y+3}{3y+4}$$

$\Rightarrow$ $(4x + 3)(3y + 4) = (4y + 3)(3x + 4)$
$\Rightarrow$ $12xy + 16x + 9y + 12 = 12xy + 16y + 9x + 12$
$\Rightarrow$ $16x - 9x = 16y - 9y \Rightarrow x = y$

So, f is an injective.

Surjectivity : Let y be an arbitrary element of $R - \left\{\frac{4}{3}\right\}$. Then, $f(x) = y$

$$\Rightarrow \frac{4x+3}{3x+4} = y \Rightarrow 4x+3 = y(3x+4)$$

$\Rightarrow$ $4x + 3 = 3xy + 4y \Rightarrow x(4 - 3y) = 4y - 3$

$$\Rightarrow \quad x = \frac{4y-3}{4-3y}$$

Clearly, $x = \frac{4y-3}{4-3y}$ is a real number for all $y \neq \frac{4}{3}$

Also $\frac{4y-3}{4-3y} \neq \frac{-4}{3}$ for any y, if we take,

$\frac{4y-3}{4-3y} = \frac{-4}{3}$, then we get $3(4y - 3) = -4(4 - 3y)$

$\Rightarrow$ $12y - 9 = -16 + 12y$
$\Rightarrow$ $-9 \neq -16$, which is wrong.

Thus, every element y in $R - \left\{\frac{4}{3}\right\}$ has its pre-image x in $R - \left\{-\frac{4}{3}\right\}$ given by $x = \frac{4y-3}{4-3y}$.

www.examites.in

So, f is a surjective.
Hence, f is a bijective.

$\Rightarrow \quad f^{-1}(x) = \dfrac{4x-3}{4-3x}$

Now, $f^{-1}(0) = \dfrac{0-3}{4-0} = -\dfrac{3}{4}$

$f^{-1}(x) = 2 \Rightarrow \dfrac{4x-3}{4-3x} = 2 \Rightarrow 4x - 3 = 8 - 6x$

$\Rightarrow \quad 10x = 11 \Rightarrow x = \dfrac{11}{10}$

41. Let $f : N \to S$, $f(x) = 9x^2 + 6x - 5$
Consider, $f(x_1) = f(x_2)$
$\Rightarrow \quad 9x_1^2 + 6x_1 - 5 = 9x_2^2 + 6x_2 - 5$
$\Rightarrow \quad 9(x_1^2 - x_2^2) + 6(x_1 - x_2) = 0$
$\Rightarrow \quad (x_1 - x_2)[9x_1 + 9x_2 + 6] = 0$
$\Rightarrow \quad x_1 = x_2 \qquad [\because \ x_1, x_2 \in N]$
$\Rightarrow \quad f$ is one-one.
Since, S is the range of f.
$\therefore \quad f$ is onto.
Since, f is one-one and onto.
So, f is invertible.
Let $y \in S$ be arbitrary number.
Consider, $y = f(x) \Rightarrow x = f^{-1}(y)$
$\Rightarrow \quad y = 9x^2 + 6x - 5 \Rightarrow y = (3x + 1)^2 - 6$

$\Rightarrow \quad \sqrt{y+6} = 3x + 1 \Rightarrow x = \dfrac{\sqrt{y+6} - 1}{3}$

Also, $f^{-1}(y) = \dfrac{\sqrt{y+6} - 1}{3}$ or $f^{-1}(x) = \dfrac{\sqrt{x+6} - 1}{3}$

Now, $f^{-1}(43) = \dfrac{\sqrt{49} - 1}{3} = \dfrac{7-1}{3} = 2$

and $f^{-1}(163) = \dfrac{\sqrt{169} - 1}{3} = \dfrac{13-1}{3} = 4$

42. *Refer to answer 23.*

We get $fog(x) = \begin{cases} 0, & x \geq 0 \\ -4x, & x < 0 \end{cases}$

and $gof(x) = 0 \ \forall \ x \in R$
Now, $fog(-3) = -4(-3) = 12$
and $fog(5) = 0$, $gof(-2) = 0$

43. Here $f : R_+ \to [-9, \infty[$ as
$f(x) = 5x^2 + 6x - 9$
First we shall show that f is one-one.
Let $f(x) = f(y)$, for $x, y \in R_+$
$\Rightarrow \quad 5x^2 + 6x - 9 = 5y^2 + 6y - 9$
$\Rightarrow \quad 5(x^2 - y^2) + 6(x - y) = 0$
$\Rightarrow \quad (x - y)[5(x + y) + 6] = 0$
$\Rightarrow \quad x = y \qquad [\because$ for $x, y \in R_+, 5(x + y) + 6 \neq 0]$
$\Rightarrow \quad f$ is one-one.
Let $y \in [-9, \infty[$ be such that $f(x) = y$
$\Leftrightarrow \quad 5x^2 + 6x - 9 = y$
$\Rightarrow \quad 5x^2 + 6x - (9 + y) = 0$

$\Rightarrow \quad x = \dfrac{-6 \pm \sqrt{6^2 + 4 \cdot 5(9+y)}}{2 \cdot 5} = \dfrac{-6 \pm \sqrt{216 + 20y}}{10}$

$= \dfrac{-3 \pm \sqrt{54 + 5y}}{5}$

Taking only +ve sign (as for –ve sign, $x \notin R_+$)

We get $x = \dfrac{-3 + \sqrt{54 + 5y}}{5} \in R_+$ for which
$f(x) = y$
$\Rightarrow \quad f$ is onto.
$\Rightarrow \quad f$ is both one-one and onto.
$\Rightarrow \quad f$ is invertible and f^{-1} is given by

$f^{-1}(y) = x = \dfrac{-3 + \sqrt{54 + 5y}}{5}$

44. Let $f : N \to S$, $f(x) = 4x^2 + 12x + 15$
Consider, $f(x_1) = f(x_2)$
$\Rightarrow \quad 4x_1^2 + 12x_1 + 15 = 4x_2^2 + 12x_2 + 15$
$\Rightarrow \quad 4(x_1^2 - x_2^2) + 12(x_1 - x_2) = 0$
$\Rightarrow \quad (x_1 - x_2)[4x_1 + 4x_2 + 12] = 0$
$\Rightarrow \quad x_1 = x_2 \qquad [\because \ x_1, x_2 \in N]$
$\therefore \quad f$ is one-one
Since, S is the range of f.
$\therefore \quad f$ is onto.
Since, f is one-one and onto.
Therefore, f is invertible.
Let $y \in S$ be arbitrary number such that $f(x) = y$
$\Rightarrow \quad y = 4x^2 + 12x + 15$
$\Rightarrow \quad y = (2x + 3)^2 + 6$

$\Rightarrow \quad \sqrt{y-6} = 2x + 3 \Rightarrow x = \dfrac{\sqrt{y-6} - 3}{2}$

Also, $f(x) = y \Rightarrow x = f^{-1}(y)$

$\Rightarrow \quad f^{-1}(y) = \dfrac{\sqrt{y-6} - 3}{2}$ or $f^{-1}(x) = \dfrac{\sqrt{x-6} - 3}{2}$

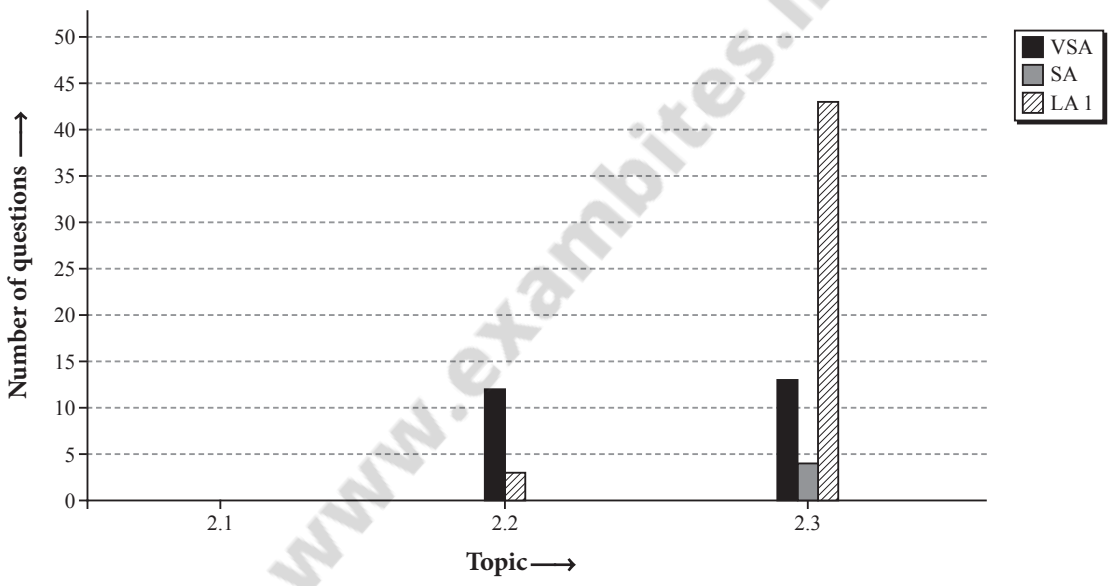
www.examstest.in

CHAPTER
2

Inverse Trigonometric Functions

2.1 Introduction
2.2 Basic Concepts
2.3 Properties of Inverse Trigonometric Functions

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Properties of Inverse Trigonometric Functions*
- Maximum LA 1 type questions were asked from *Properties of Inverse Trigonometric Functions*
- No LA 2 type questions were asked till now

QUICK RECAP

INVERSE TRIGONOMETRIC FUNCTIONS

- Trigonometric functions are not one-one and onto over their natural domains and ranges *i.e.*, R(real numbers). But some restrictions on domains and ranges of trigonometric function ensures the existence of their inverses.
  Let $y = f(x) = \cos x$, then its inverse is $x = \cos^{-1} y$
- The domains and ranges (principal value branches) of inverse trigonometric functions are as follows :

Functions	Domain	Range
$y = \sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \tan^{-1} x$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$y = \cot^{-1} x$	R	$(0, \pi)$
$y = \operatorname{cosec}^{-1} x$	$R-(-1, 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]-\{0\}$
$y = \sec^{-1} x$	$R-(-1, 1)$	$[0,\pi]-\left\{\frac{\pi}{2}\right\}$

▶ The value of the inverse trigonometric functions which lies in its principal value branch is called the principal value of inverse trigonometric function.

PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTIONS

$\sin^{-1}(\sin x) = x, \ \forall -\frac{\pi}{2} \le x \le \frac{\pi}{2}$

$\cos^{-1}(\cos x) = x, \ \forall \ 0 \le x \le \pi$

$\tan^{-1}(\tan x) = x, \ \forall -\frac{\pi}{2} < x < \frac{\pi}{2}$

$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x \ , \forall \ x \ge 1 \text{ or } x \le -1$

$\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x \ , \forall \ x \ge 1 \text{ or } x \le -1$

$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x, \forall \ x > 0$

$= -\pi + \cot^{-1} x, \forall \ x < 0$

$\sin(\sin^{-1} x) = x, \ \forall -1 \le x \le 1$
$\cos(\cos^{-1} x) = x, \ \forall -1 \le x \le 1$
$\tan(\tan^{-1} x) = x, \ \forall \ x \in R$

$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \ \forall -1 \le x \le 1$

$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \ \forall \ x \in R$

$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2} \ , \forall \ x \le -1 \text{ or } x \ge 1$

$\sin^{-1} x = \cos^{-1}\sqrt{1-x^2} \ , \forall \ 0 \le x \le 1$

$\sin^{-1} x = -\cos^{-1}\sqrt{1-x^2} \ , \forall -1 \le x < 0$

$\cos^{-1} x = \sin^{-1}\sqrt{1-x^2} \ , \forall \ 0 \le x \le 1$

$\cos^{-1} x = \pi - \sin^{-1}\sqrt{1-x^2} \ , \forall -1 \le x < 0$

$\sin^{-1}(-x) = -\sin^{-1} x, \ \forall -1 \le x \le 1$
$\cos^{-1}(-x) = \pi - \cos^{-1} x \ , \forall -1 \le x \le 1$
$\tan^{-1}(-x) = -\tan^{-1} x, \ \forall \ x \in R$
$\cot^{-1}(-x) = \pi - \cot^{-1} x, \ \forall \ x \in R$
$\sec^{-1}(-x) = \pi - \sec^{-1} x, \ \forall \ |x| \ge 1$
$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x, \ \forall \ |x| \ge 1$

$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \ , \ \forall \ xy < 1$

$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right),$
$-1 \le x, y \le 1, x^2 + y^2 \le 1$

$\cos^{-1} x + \cos^{-1} y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right),$
$-1 \le x, y \le 1, x + y \ge 0$

$\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), xy > -1$

$\sin^{-1} x - \sin^{-1} y = \sin^{-1}\left\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\right\},$
$-1 \le x, y \le 1$ and $x^2 + y^2 \le 1$

$\cos^{-1} x - \cos^{-1} y = \cos^{-1}\left\{xy + \sqrt{1-x^2}\sqrt{1-y^2}\right\},$
$-1 \le x, y \le 1$ and $x \le y$.

$2\tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right), \ \forall -1 \le x \le 1$

$2\tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), \ \forall \ x \ge 0$

$2\tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right), \ \forall |x| < 1$

$2\sin^{-1} x = \sin^{-1}\left(2x\sqrt{1-x^2}\right), \ \forall -\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}}$

$2\cos^{-1} x = \sin^{-1}(2x\sqrt{1-x^2}), \ \forall \frac{1}{\sqrt{2}} \le x \le 1$

$2\cos^{-1} x = \cos^{-1}(2x^2 - 1), \ \forall \ 0 \le x \le 1$

$3\sin^{-1} x = \sin^{-1}(3x - 4x^3), \ \forall -\frac{1}{2} \le x \le \frac{1}{2}$

$3\cos^{-1} x = \cos^{-1}(4x^3 - 3x), \ \forall \frac{1}{2} \le x \le 1$

$3\tan^{-1} x = \tan^{-1}\left(\frac{3x - x^3}{1-3x^2}\right), \ \forall -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$

www.exambites.in

Previous Years' CBSE Board Questions

2.2 Basic Concepts

VSA (1 mark)

1. Write the value of $\cos^{-1}\left(-\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$.
(Foreign 2014)

2. Write the principal value of $\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$.
(AI 2014C)

3. Find the value of the following :
$\cot\left(\frac{\pi}{2}-2\cot^{-1}\sqrt{3}\right)$ *(AI 2014C)*

4. Write the principal value of
$\tan^{-1}(1)+\cos^{-1}\left(-\frac{1}{2}\right)$. *(Delhi 2013)*

5. Write the value of $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$.
(AI 2013)

6. Write the principal value of
$\left[\cos^{-1}\frac{\sqrt{3}}{2}+\cos^{-1}\left(-\frac{1}{2}\right)\right]$. *(Delhi 2013C)*

7. Write the principal value of
$[\tan^{-1}(-\sqrt{3})+\tan^{-1}(1)]$. *(AI 2013C)*

8. Write the principal value of
$\cos^{-1}\left(\frac{1}{2}\right)-2\sin^{-1}\left(-\frac{1}{2}\right)$. *(Delhi 2012)*

9. Using principal values, write the value of
$\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$. *(AI 2012C)*

10. Evaluate : $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(\frac{-1}{2}\right)\right]$. *(Delhi 2011)*

11. Write the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$.
(Delhi 2011C)

12. Using principal values, write the value of
$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$. *(AI 2011C)*

LA 1 (4 marks)

13. Prove that :
$\cos^{-1}\left(\frac{12}{13}\right)+\sin^{-1}\left(\frac{3}{5}\right)=\sin^{-1}\left(\frac{56}{65}\right)$
(AI 2019, AI 2012)

14. Prove that
$\cos^{-1}\frac{12}{13}+\cos^{-1}\frac{4}{5}=\tan^{-1}\frac{56}{33}$. *(AI 2013C)*

15. Prove that :
$\cos^{-1}\left(\frac{4}{5}\right)+\cos^{-1}\left(\frac{12}{13}\right)=\cos^{-1}\left(\frac{33}{65}\right)$
(AI 2012)

2.3 Properties of Inverse Trigonometric Functions

VSA (1 mark)

16. The principal value of $\tan^{-1}\left(\tan\frac{3\pi}{5}\right)$ is

(a) $\frac{2\pi}{5}$ (b) $\frac{-2\pi}{5}$

(c) $\frac{3\pi}{6}$ (d) $\frac{-3\pi}{6}$ *(2020)*

17. $\left(\tan^{-1}\frac{7}{9}+\tan^{-1}\frac{1}{8}\right)$ is equal to

(a) $\tan^{-1}\left(\frac{65}{72}\right)$ (b) $\tan^{-1}\left(\frac{63}{65}\right)$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$ *(2020)*

18. Find the value of $\sin^{-1}\left[\sin\left(-\frac{17\pi}{8}\right)\right]$.
(2020)

19. Find the value of $\tan^{-1}\sqrt{3}-\cot^{-1}(-\sqrt{3})$.
(2018)

www.exambites.in

20. If $\sin\left(\sin^{-1}\frac{1}{5}+\cos^{-1}x\right)=1$, then find the value of x. *(Delhi 2014)*

21. If $\tan^{-1}x+\tan^{-1}y=\frac{\pi}{4}$, $xy<1$, then write the value of $x+y+xy$. *(AI 2014, Delhi 2012C)*

22. Write the value of $\tan\left(2\tan^{-1}\frac{1}{5}\right)$. *(Delhi 2013)*

23. Write the principal value of $\tan^{-1}\left(\sqrt{3}\right)-\cot^{-1}\left(-\sqrt{3}\right)$. *(AI 2013)*

24. Evaluate $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$. *(AI 2013C)*

25. Find the principal value of $\tan^{-1}\sqrt{3}-\sec^{-1}(-2)$. *(AI 2012)*

26. Find the value of $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$. *(Delhi 2011)*

27. Write the principal value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$. *(Delhi 2011)*

28. Using principal values, evaluate the following.

$\cos^{-1}\left(\cos\frac{2\pi}{3}\right)+\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$. *(AI 2011)*

SA (2 marks)

29. Prove that

$\sin^{-1}(2x\sqrt{1-x^2})=2\cos^{-1}x, \frac{1}{\sqrt{2}}\le x\le 1$. *(2020)*

30. Solve for x : $\sin^{-1}4x+\sin^{-1}3x=-\frac{\pi}{2}$. *(2020)*

31. Express $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right), -\frac{3\pi}{2}<x<\frac{\pi}{2}$ in the simplest form. *(2020)*

32. Prove that :

$3\sin^{-1}x=\sin^{-1}(3x-4x^3)$, $x\in\left[-\frac{1}{2},\frac{1}{2}\right]$ *(2018)*

LA 1 (4 marks)

33. Solve the equation for x :

$\sin^{-1}\left(\frac{5}{x}\right)+\sin^{-1}\left(\frac{12}{x}\right)=\frac{\pi}{2}(x\ne 0)$. *(2020)*

34. Solve : $\tan^{-1}4x+\tan^{-1}6x=\frac{\pi}{4}$. *(Delhi 2019)*

35. Prove that $\tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}=\frac{2b}{a}$. *(Delhi 2017)*

36. If $\tan^{-1}\frac{x-3}{x-4}+\tan^{-1}\frac{x+3}{x+4}=\frac{\pi}{4}$, then find the value of x. *(AI 2017)*

37. Solve for x : $2\tan^{-1}(\cos x)=\tan^{-1}(2\operatorname{cosec} x)$ *(Delhi 2016, 2014C, Foreign 2015)*

38. Prove that

$\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{8}\right)=\frac{\pi}{4}$. *(Delhi 2016)*

39. Solve the equation for x :

$\sin^{-1}x+\sin^{-1}(1-x)=\cos^{-1}x$ *(AI 2016)*

40. If $\cos^{-1}\frac{x}{a}+\cos^{-1}\frac{y}{b}=\alpha$, prove that

$\frac{x^2}{a^2}-2\frac{xy}{ab}\cos\alpha+\frac{y^2}{b^2}=\sin^2\alpha$. *(AI 2016)*

41. Solve for x :

$\tan^{-1}\left(\frac{x-2}{x-1}\right)+\tan^{-1}\left(\frac{x+2}{x+1}\right)=\frac{\pi}{4}$ *(Foreign 2016)*

42. Prove that :

$\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right)=\frac{x}{2}, x\in\left(0,\frac{\pi}{4}\right)$ *(Foreign 2016, Delhi 2014, 2014C, 2011)*

www.exambites.in

43. If $\sin[\cot^{-1}(x+1)] = \cos(\tan^{-1}x)$, then find x.
(Delhi 2015)

44. If $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$, then find x.
(Delhi 2015)

45. Prove the following:
$$\cot^{-1}\left(\frac{xy+1}{x-y}\right)+\cot^{-1}\left(\frac{yz+1}{y-z}\right)+\cot^{-1}\left(\frac{zx+1}{z-x}\right)=0,$$
$(0 < xy, yz, zx < 1)$ *(AI 2015)*

46. Solve for x :
$$\tan^{-1}(x+1)+\tan^{-1}(x-1)=\tan^{-1}\frac{8}{31}.$$
(AI 2015)

47. If $\tan^{-1}\left(\frac{1}{1+1\cdot 2}\right)+\tan^{-1}\left(\frac{1}{1+2\cdot 3}\right)$
$$+\ldots+\tan^{-1}\left(\frac{1}{1+n\cdot(n+1)}\right)=\tan^{-1}\theta,$$
then find the value of θ. *(Foreign 2015)*

48. Solve for x : $\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}$.
(Delhi 2015C, 2013C, AI 2012C)

49. Prove that: $\tan^{-1}\frac{63}{16}=\sin^{-1}\frac{5}{13}+\cos^{-1}\frac{3}{5}$
(Delhi 2015C)

50. Prove that
$$2\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{7}\right)=\sin^{-1}\left(\frac{31}{25\sqrt{2}}\right).$$
(AI 2015C)

51. Solve for x :
$$\tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\tan^{-1}x,\ x>0.$$
(AI 2015C, 2014C)

52. Prove that
$$2\tan^{-1}\left(\frac{1}{5}\right)+\sec^{-1}\left(\frac{5\sqrt{2}}{7}\right)+2\tan^{-1}\left(\frac{1}{8}\right)=\frac{\pi}{4}.$$
(Delhi 2014)

53. Prove that :
$$\tan^{-1}\left[\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right]=\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x,\ \frac{-1}{\sqrt{2}}\le x\le 1$$
(AI 2014, 2011)

54. If $\tan^{-1}\left(\frac{x-2}{x-4}\right)+\tan^{-1}\left(\frac{x+2}{x+4}\right)=\frac{\pi}{4}$; find the value of x. *(AI 2014)*

55. Solve for x : $\cos(\tan^{-1}x)=\sin\left(\cot^{-1}\frac{3}{4}\right)$.
(Foreign 2014, AI 2013)

56. Prove that: $\cot^{-1}7+\cot^{-1}8+\cot^{-1}18=\cot^{-1}3$.
(Foreign 2014)

57. Prove that :
$$\cos^{-1}(x)+\cos^{-1}\left\{\frac{x}{2}+\frac{\sqrt{3-3x^2}}{2}\right\}=\frac{\pi}{3}.$$
(AI 2014C)

58. Solve for x : $\tan^{-1}x+2\cot^{-1}x=\frac{2\pi}{3}$.
(AI 2014C)

59. Prove that : $\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{8}{17}=\cos^{-1}\frac{36}{85}$
(AI 2014C, Delhi 2012)

60. Find the value of the following :
$$\tan\frac{1}{2}\left[\sin^{-1}\frac{2x}{1+x^2}+\cos^{-1}\frac{1-y^2}{1+y^2}\right],$$
$|x| < 1$, $y > 0$ and $xy < 1$ *(Delhi 2013)*

61. Prove that,
$$\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{1}{8}\right)=\frac{\pi}{4}.$$
(Delhi 2013, 2012C, AI 2011)

62. Show that:
$$\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right)=\frac{4-\sqrt{7}}{3}.$$ *(AI 2013)*

63. Write the value of the following :
$$\tan^{-1}\left(\frac{a}{b}\right)-\tan^{-1}\left(\frac{a-b}{a+b}\right)$$ *(Delhi 2013C)*

64. Prove that: $\sin^{-1}\frac{8}{17}+\sin^{-1}\frac{3}{5}=\tan^{-1}\frac{77}{36}$
(Delhi 2013C)

65. Solve for x :
$$\sin^{-1}(1-x)-2\sin^{-1}x=\frac{\pi}{2}$$ *(AI 2013C)*

66. Prove that
$$\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)=\frac{\pi}{4}-\frac{x}{2},\ x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right).$$
(Delhi 2012)

www.exambites.in

67. Prove the following:

$$\cos\left(\sin^{-1}\frac{3}{5}+\cot^{-1}\frac{3}{2}\right)=\frac{6}{5\sqrt{13}} \quad \textit{(AI 2012)}$$

68. Solve for x :

$$\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}.$$

(Delhi 2012C)

69. Prove that :

$$\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{3}{5}\right)-\tan^{-1}\left(\frac{8}{19}\right)=\frac{\pi}{4}$$

(Delhi 2012C)

70. Find the value of

$$\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left(\frac{x-y}{x+y}\right). \quad \textit{(Delhi 2011)}$$

71. Prove that :

$$2\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{7}\right)=\tan^{-1}\left(\frac{31}{17}\right).$$

(AI 2011)

72. Prove that:

$$2\tan^{-1}\frac{3}{4}-\tan^{-1}\frac{17}{31}=\frac{\pi}{4} \quad \textit{(Delhi 2011C)}$$

73. Solve for x:

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\cot^{-1}\left(\frac{1-x^2}{2x}\right)=\frac{\pi}{3},\ -1<x<1$$

(Delhi 2011C)

74. Prove that :

$$\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{2}{9}=\frac{1}{2}\tan^{-1}\frac{4}{3} \quad \textit{(AI 2011C)}$$

75. Solve for x : $\cos(2\sin^{-1}x)=\frac{1}{9},\ x>0$

(AI 2011C)

Detailed Solutions

1. Given $\cos^{-1}\left(-\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$

$$=\cos^{-1}\left(\cos\frac{2\pi}{3}\right)+2\sin^{-1}\left(\sin\frac{\pi}{6}\right)=\frac{2\pi}{3}+2\times\frac{\pi}{6}=\pi$$

$[\because \text{Range of } \cos^{-1}x \text{ is } [0,\pi] \text{ \& of } \sin^{-1}x \text{ is } [-\pi/2,\pi/2]]$

2. Here, $\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]=\tan^{-1}(-1)=-\frac{\pi}{4}.$

This is the required principal value as it is lie in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

3. $\cot\left(\frac{\pi}{2}-2\cot^{-1}\sqrt{3}\right)$

$$=\cot\left(\frac{\pi}{2}-2\cot^{-1}\left(\cot\frac{\pi}{6}\right)\right)=\cot\left(\frac{\pi}{2}-2.\frac{\pi}{6}\right)$$

$$=\cot\left(\frac{\pi}{2}-\frac{\pi}{3}\right)=\cot\frac{\pi}{6}=\sqrt{3}$$

4. $\tan^{-1}(1)+\cos^{-1}\left(\frac{-1}{2}\right)$

$$=\tan^{-1}\left(\tan\frac{\pi}{4}\right)+\cos^{-1}\left(\cos\left(\frac{2\pi}{3}\right)\right)=\frac{\pi}{4}+\frac{2\pi}{3}=\frac{11\pi}{12}$$

5. $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$

$$=\tan^{-1}\left[2\sin\left(2.\frac{\pi}{6}\right)\right]=\tan^{-1}\left[2\sin\frac{\pi}{3}\right]$$

$$=\tan^{-1}.\left[2.\frac{\sqrt{3}}{2}\right]=\tan^{-1}\left(\sqrt{3}\right)=\frac{\pi}{3}.$$

6. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)+\cos^{-1}\left(-\frac{1}{2}\right)$

$$=\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)+\cos^{-1}\left(\cos\frac{2\pi}{3}\right)=\frac{\pi}{6}+\frac{2\pi}{3}=\frac{5\pi}{6}$$

7. $\tan^{-1}\left(-\sqrt{3}\right)+\tan^{-1}(1)$

$$=\tan^{-1}\left(-\tan\frac{\pi}{3}\right)+\tan^{-1}\left(\tan\frac{\pi}{4}\right)$$

$$=\tan^{-1}\left(\tan\left(-\frac{\pi}{3}\right)\right)+\frac{\pi}{4}=-\frac{\pi}{3}+\frac{\pi}{4}=-\frac{\pi}{12}.$$

8. $\cos^{-1}\left(\frac{1}{2}\right)-2\sin^{-1}\left(-\frac{1}{2}\right)$

$$=\cos^{-1}\left(\cos\frac{\pi}{3}\right)-2\sin^{-1}\left(\sin\left(-\frac{\pi}{6}\right)\right)=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

www.exambites.in

9. Principal value of

$$\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}+2.\frac{\pi}{6}=\frac{2\pi}{3}.$$

10. $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]=\sin\left[\frac{\pi}{3}-\sin^{-1}\left(\sin\left(\frac{-\pi}{6}\right)\right)\right]$

$$=\sin\left(\frac{\pi}{3}+\frac{\pi}{6}\right)=\sin\frac{\pi}{2}=1$$

11. Let $\sin^{-1}\left(\frac{-1}{2}\right)=\theta$

Then, $\sin\theta=\frac{-1}{2}=\sin\left(-\frac{\pi}{6}\right)$, where $\frac{-\pi}{6}\in\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

12. The principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

$$=\sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right)=-\frac{\pi}{3},\text{where }\frac{-\pi}{3}\in\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$$

13. Let $x=\cos^{-1}\left(\frac{12}{13}\right)$ and $y=\sin^{-1}\left(\frac{3}{5}\right)$

or $\cos x=\frac{12}{13}$ and $\sin y=\frac{3}{5}$

Now, $\sin x=\sqrt{1-\cos^2 x}$ and $\cos y=\sqrt{1-\sin^2 y}$

$\Rightarrow$ $\sin x=\sqrt{1-\frac{144}{169}}$ and $\cos y=\sqrt{1-\frac{9}{25}}$

$\Rightarrow$ $\sin x=\frac{5}{13}$ and $\cos y=\frac{4}{5}$

We know that,

$\sin(x+y)=\sin x\cos y+\cos x\sin y$

$$=\frac{5}{13}\times\frac{4}{5}+\frac{12}{13}\times\frac{3}{5}=\frac{20}{65}+\frac{36}{65}=\frac{56}{65}$$

$\Rightarrow$ $x+y=\sin^{-1}\left(\frac{56}{65}\right)$

or, $\cos^{-1}\left(\frac{12}{13}\right)+\sin^{-1}\left(\frac{3}{5}\right)=\sin^{-1}\left(\frac{56}{65}\right)$

14. Let $x=\cos^{-1}\left(\frac{4}{5}\right)$ and $y=\cos^{-1}\left(\frac{12}{13}\right)$

$\Rightarrow$ $\cos x=\frac{4}{5}$ and $\cos y=\frac{12}{13}$

Now, $\sin x=\sqrt{1-\cos^2 x}$ and $\sin y=\sqrt{1-\cos^2 y}$

$\Rightarrow$ $\sin x=\sqrt{1-\frac{16}{25}}$ and $\sin y=\sqrt{1-\frac{144}{169}}$

$\Rightarrow$ $\sin x=\frac{3}{5}$ and $\sin y=\frac{5}{13}$

We know that, $\cos(x+y)=\cos x\cos y-\sin x\sin y$

$$=\frac{4}{5}\times\frac{12}{13}-\frac{3}{5}\times\frac{5}{13}$$

$\Rightarrow$ $\cos(x+y)=\frac{48}{65}-\frac{15}{65}=\frac{33}{65}$

$\Rightarrow$ $x+y=\cos^{-1}\left(\frac{33}{65}\right)$

$\therefore$ $\cos^{-1}\left(\frac{4}{5}\right)+\cos^{-1}\left(\frac{12}{13}\right)=\cos^{-1}\left(\frac{33}{65}\right)$

Now, $\cos^{-1}x=\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

$$\therefore\ \cos^{-1}\left(\frac{33}{65}\right)=\tan^{-1}\left(\frac{\sqrt{1-\left(\frac{33}{65}\right)^2}}{\frac{33}{65}}\right)=\tan^{-1}\left(\frac{56}{35}\right)$$

$\therefore$ $\cos^{-1}\left(\frac{4}{5}\right)+\cos^{-1}\left(\frac{12}{13}\right)=\tan^{-1}\left(\frac{56}{35}\right)$

15. *Refer to answer 14.*

16. **(b) :** $\tan^{-1}\left(\tan\frac{3\pi}{5}\right)\neq\frac{3\pi}{5}$ as the principal value branch of $\tan^{-1}\theta$ is $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$.

$$\therefore\ \tan^{-1}\left(\tan\frac{3\pi}{5}\right)=\tan^{-1}\left(\tan\left(\pi-\frac{2\pi}{5}\right)\right)$$

$$=\tan^{-1}\left[-\tan\left(\frac{2\pi}{5}\right)\right]\qquad[\because\ \tan(\pi-\theta)]=-\tan\theta]$$

$$=\tan^{-1}\left(\tan\left(-\frac{2\pi}{5}\right)\right)\qquad[\because\ -\tan\theta=\tan(-\theta)]$$

$$=-\frac{2\pi}{5}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$$

17. **(c) :** We have, $\tan^{-1}\frac{7}{9}+\tan^{-1}\frac{1}{8}$

$$=\tan^{-1}\left(\frac{\frac{7}{9}+\frac{1}{8}}{1-\frac{7}{9}\times\frac{1}{8}}\right)=\tan^{-1}\left(\frac{\frac{56+9}{72}}{\frac{72-7}{72}}\right)$$

$$=\tan^{-1}\left(\frac{65}{65}\right)=\tan^{-1}(1)=\frac{\pi}{4}$$

www.examhits.in

18. We have, $\sin^{-1}\left[\sin\left(-\frac{17\pi}{8}\right)\right]$

$= \sin^{-1}\left[\sin\left(-2\pi-\frac{\pi}{8}\right)\right] = \sin^{-1}\left[-\sin\frac{\pi}{8}\right]$

$= -\sin^{-1}\sin\frac{\pi}{8} = -\frac{\pi}{8}$

19. $\tan^{-1}\sqrt{3} - \cot^{-1}(-\sqrt{3})$

$= \tan^{-1}\sqrt{3} - (\pi - \cot^{-1}\sqrt{3})$

$= \tan^{-1}\sqrt{3} + \cot^{-1}\sqrt{3} - \pi = \frac{\pi}{2} - \pi = -\frac{\pi}{2}$

20. $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$

or $\sin^{-1}\frac{1}{5} + \cos^{-1}x = \sin^{-1}1$

$\Rightarrow \sin^{-1}\frac{1}{5} + \frac{\pi}{2} - \sin^{-1}x = \frac{\pi}{2}$

$\left[\because \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}\right]$

$\Rightarrow \sin^{-1}\frac{1}{5} = \sin^{-1}x \Rightarrow x = \sin\left(\sin^{-1}\frac{1}{5}\right) = \frac{1}{5}$

21. Given: $\tan^{-1}x + \tan^{-1}y = \frac{\pi}{4}$ $(xy < 1)$

$\Rightarrow \tan^{-1}\left(\frac{x+y}{1-xy}\right) = \frac{\pi}{4} \Rightarrow \frac{x+y}{1-xy} = \tan\frac{\pi}{4} = 1$

$\Rightarrow x + y = 1 - xy \Rightarrow x + y + xy = 1$

22. Since $2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, for $|x| < 1$

So, $2\tan^{-1}\left(\frac{1}{5}\right) = \tan^{-1}\left(\frac{2\times\frac{1}{5}}{1-\left(\frac{1}{5}\right)^2}\right)$

$= \tan^{-1}\left(\frac{\frac{2}{5}}{\frac{24}{25}}\right) = \tan^{-1}\left(\frac{5}{12}\right)$

$\therefore \tan\left(2\tan^{-1}\frac{1}{5}\right) = \tan\left(\tan^{-1}\frac{5}{12}\right) = \frac{5}{12}$

23. *Refer to answer 19.*

24. We know that, $\sin^{-1}(\sin x) = x$

Therefore, $\sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \frac{3\pi}{5}$

But $\frac{3\pi}{5} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, which is the principal value of $\sin^{-1}x$.

So, $\sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \sin^{-1}\left[\sin\left(\pi - \frac{2\pi}{5}\right)\right]$

$= \sin^{-1}\left[\sin\left(\frac{2\pi}{5}\right)\right] = \frac{2\pi}{5}$ and $\frac{2\pi}{5} \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\therefore \sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \frac{2\pi}{5}$

25. $\tan^{-1}\sqrt{3} - \sec^{-1}(-2) = \tan^{-1}\sqrt{3} - \sec^{-1}\left(\sec\frac{2\pi}{3}\right)$

$= \frac{\pi}{3} - \frac{2\pi}{3} = -\frac{\pi}{3}$

$\therefore$ Principal value of $\tan^{-1}\sqrt{3} - \sec^{-1}(-2) = -\frac{\pi}{3}$

26. $\tan^{-1}\left(\tan\frac{3\pi}{4}\right) \neq \frac{3\pi}{4}$ as the principal value branch of $\tan^{-1}\theta$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

So, $\tan^{-1}\left(\tan\frac{3\pi}{4}\right) = \tan^{-1}\left(\tan\left(\pi - \frac{\pi}{4}\right)\right)$

$= \tan^{-1}\left[-\tan\left(\frac{\pi}{4}\right)\right]$

$= \tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right)$ $[\because -\tan\theta = \tan(-\theta)]$

$= -\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Hence, $\tan^{-1}\left(\tan\frac{3\pi}{4}\right) = -\frac{\pi}{4}$

27. $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) \neq \frac{7\pi}{6}$ as principal value branch of $\cos^{-1}\theta$ is $[0, \pi]$

So, $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \cos^{-1}\left[\cos\left(\pi + \frac{\pi}{6}\right)\right]$

$= \cos^{-1}\left(-\cos\frac{\pi}{6}\right) = \cos^{-1}\left(\cos\left(\pi - \frac{\pi}{6}\right)\right) = \frac{5\pi}{6}$

where $\frac{5\pi}{6} \in [0, \pi]$

Hence, $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \frac{5\pi}{6}$

www.exambites.in

28. We know that the range of principal value branch of $\cos^{-1}\theta$ is $[0, \pi]$ and $\sin^{-1}\theta$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Then, $\cos^{-1}\left(\cos\frac{2\pi}{3}\right)+\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$

$=\frac{2\pi}{3}+\sin^{-1}\left(\sin\left(\pi-\frac{\pi}{3}\right)\right)$

$=\frac{2\pi}{3}+\sin^{-1}\left(\sin\frac{\pi}{3}\right)=\frac{2\pi}{3}+\frac{\pi}{3}=\pi$

29. Put $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$ ...(i)

$\therefore$ L.H.S. $= \sin^{-1}(2x\sqrt{1-x^2})$

$= \sin^{-1}(2\cos\theta\sqrt{1-\cos^2\theta})$

$= \sin^{-1}(2\sin\theta\cos\theta) = \sin^{-1}(\sin 2\theta)$

$= 2\theta = 2\cos^{-1}x$ (Using (i))

= R.H.S.

30. We have, $\sin^{-1}4x + \sin^{-1}3x = -\frac{\pi}{2}$ ...(i)

$\Rightarrow \sin^{-1}4x = -\frac{\pi}{2} - \sin^{-1}3x$

$\Rightarrow 4x = \sin\left(-\frac{\pi}{2}-\sin^{-1}3x\right)$

$\Rightarrow 4x = -\sin\left(\frac{\pi}{2}+\sin^{-1}3x\right)$

$\Rightarrow 4x = -\cos(\sin^{-1}3x)$

$\Rightarrow 4x = -\cos(\cos^{-1}\sqrt{1-9x^2})$

$\Rightarrow 4x = -\sqrt{1-9x^2}$

Squaring both sides, we get

$16x^2 = 1 - 9x^2$

$\Rightarrow 25x^2 = 1 \Rightarrow x^2 = \frac{1}{25} \Rightarrow x = \pm\frac{1}{5}$

But $x = \frac{1}{5}$ does not satisfy (i)

$\therefore$ $x = -\frac{1}{5}$ is the only solution of given equation.

31. We have, $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$

$= \tan^{-1}\left(\frac{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2}\right)$

$= \tan^{-1}\left(\frac{\cos\frac{x}{2}+\sin\frac{x}{2}}{\cos\frac{x}{2}-\sin\frac{x}{2}}\right)$ $\left[\because \frac{-3\pi}{2} < x < \frac{\pi}{2} \Rightarrow \frac{-3\pi}{4} < \frac{x}{2} < \frac{\pi}{4}\right]$

$= \tan^{-1}\left(\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right)$

$\left[\because \frac{-3\pi}{2} < x < \frac{\pi}{2}, \therefore \left(\frac{\pi}{4}+\frac{x}{2}\right) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right]$

$= \frac{\pi}{4}+\frac{x}{2}$

32. Put $\sin^{-1}x = \theta$. Then $x = \sin\theta$

Now, $\sin 3\theta = (3\sin\theta - 4\sin^3\theta) = (3x - 4x^3)$

$\Rightarrow 3\theta = \sin^{-1}(3x - 4x^3)$

$\Rightarrow 3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$ [$\because \theta = \sin^{-1}x$]

Hence, $3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$

33. We have, $\sin^{-1}\left(\frac{5}{x}\right)+\sin^{-1}\left(\frac{12}{x}\right)=\frac{\pi}{2}$ ...(i)

$\Rightarrow \sin^{-1}\left(\frac{5}{x}\right)=\frac{\pi}{2}-\sin^{-1}\left(\frac{12}{x}\right)$

$\Rightarrow \frac{5}{x}=\sin\left(\frac{\pi}{2}-\sin^{-1}\left(\frac{12}{x}\right)\right)$

$\Rightarrow \frac{5}{x}=\cos\left(\sin^{-1}\left(\frac{12}{x}\right)\right)$

$\Rightarrow \frac{5}{x}=\cos\left(\cos^{-1}\sqrt{1-\frac{144}{x^2}}\right)$

$(\because \sin^{-1}x = \cos^{-1}\sqrt{1-x^2})$

$\Rightarrow \frac{25}{x^2}=\frac{x^2-144}{x^2} \Rightarrow x = \pm 13$

For $x = -13$, L.H.S. of (i)

$= \sin^{-1}\left(\frac{5}{-13}\right)+\sin^{-1}\left(\frac{12}{-13}\right)$

$= -\left[\sin^{-1}\left(\frac{5}{13}\right)+\sin^{-1}\left(\frac{12}{13}\right)\right]$

$= -\sin^{-1}\left(\frac{5}{13}\sqrt{1-\left(\frac{12}{13}\right)^2}+\frac{12}{13}\sqrt{1-\left(\frac{5}{13}\right)^2}\right)$

www.examfiles.in

$$= -\sin^{-1}\left(\frac{5}{13}\cdot\frac{5}{13}+\frac{12}{13}\cdot\frac{12}{13}\right)$$

$$= -\sin^{-1}\left(\frac{25+144}{13\cdot 13}\right)$$

$$= -\sin^{-1}\left(\frac{169}{169}\right) = -\sin^{-1}(1) = -\frac{\pi}{2} \neq \frac{\pi}{2}$$

$\therefore$ $x = -13$ is not a solution of (i).

Hence, $x = 13$ is the only solution of the given equation.

34. We have, $\tan^{-1}(4x)+\tan^{-1}(6x)=\frac{\pi}{4}$

$$\Rightarrow \tan^{-1}\left[\frac{4x+6x}{1-(4x)(6x)}\right]=\tan^{-1}(1)$$

$$\Rightarrow \frac{10x}{1-24x^2}=1$$

$\Rightarrow$ $10x = 1 - 24x^2 \Rightarrow 24x^2 + 10x - 1 = 0$

$\Rightarrow$ $24x^2 + 12x - 2x - 1 = 0$

$\Rightarrow$ $12x(2x + 1) - 1(2x + 1) = 0$

$\Rightarrow$ $(2x + 1)(12x - 1) = 0$

$$\Rightarrow x=-\frac{1}{2} \text{ or } x=\frac{1}{12}$$

$\therefore$ $x=\frac{1}{12}$ is the only solution.

[$\because$ $x = -1/2$ does not satisfy the given equation]

35. We have,

$$\text{L.H.S.} = \tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}$$

Let $\cos^{-1}\frac{a}{b}=\theta \Rightarrow \frac{a}{b}=\cos\theta$

$$\therefore \text{L.H.S.} = \tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)+\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)$$

$$=\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}+\frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}}$$

$$=\frac{1+\tan^2\frac{\theta}{2}+1+\tan^2\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}=2\left(\frac{1+\tan^2\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}\right)$$

$$=\frac{2}{\cos\left(2\cdot\frac{\theta}{2}\right)}=\frac{2}{\cos\theta}=\frac{2}{a/b}=\frac{2b}{a}=\text{R.H.S.}$$

36. We have, $\tan^{-1}\left(\frac{x-3}{x-4}\right)+\tan^{-1}\left(\frac{x+3}{x+4}\right)=\frac{\pi}{4}$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{x-3}{x-4}+\frac{x+3}{x+4}}{1-\frac{(x-3)(x+3)}{(x-4)(x+4)}}\right)=\frac{\pi}{4}$$

$$\left[\because \tan^{-1}A+\tan^{-1}B=\tan^{-1}\left(\frac{A+B}{1-A\times B}\right)\right]$$

$$\Rightarrow \tan^{-1}\left[\frac{(x-3)(x+4)+(x+3)(x-4)}{(x^2-16)-(x^2-9)}\right]=\frac{\pi}{4}$$

$$\Rightarrow \frac{x^2+x-12+x^2-x-12}{x^2-16-x^2+9}=\tan\frac{\pi}{4}$$

$$\Rightarrow \frac{2x^2-24}{-7}=1 \Rightarrow 2x^2-24=-7 \Rightarrow 2x^2=17$$

$$\Rightarrow x^2=\frac{17}{2} \Rightarrow x=\pm\sqrt{\frac{17}{2}}$$

37. $2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$

$\Rightarrow$ $2\tan^{-1}(\cos x) - \tan^{-1}(2\operatorname{cosec} x) = 0$

$$\Rightarrow \tan^{-1}\left(\frac{2\cos x}{1-\cos^2 x}\right)-\tan^{-1}(2\operatorname{cosec} x)=0$$

$$\Rightarrow \tan^{-1}\left(\frac{2\cos x}{\sin^2 x}\right)-\tan^{-1}(2\operatorname{cosec} x)=0$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{2\cos x}{\sin^2 x}-\frac{2}{\sin x}}{1+\left(\frac{2\cos x}{\sin^2 x}\right)\left(\frac{2}{\sin x}\right)}\right)=0$$

$$\Rightarrow \tan^{-1}\left(\frac{2\cos x\sin x-2\sin^2 x}{\sin^3 x+4\cos x}\right)=0$$

$$\Rightarrow \frac{2\cos x\sin x-2\sin^2 x}{\sin^3 x+4\cos x}=0$$

$\Rightarrow$ $2\cos x\sin x = 2\sin^2 x \Rightarrow \tan x = 1$

$$\Rightarrow x=\tan^{-1}(1)=\tan^{-1}\left(\tan\frac{\pi}{4}\right)=\frac{\pi}{4}$$

38. $$\text{L.H.S.} = \left(\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{5}\right)+\left(\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}\right)$$

$$=\tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{5}}{1-\frac{1}{3}\times\frac{1}{5}}\right)+\tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right)$$

www.exambites.in

$$= \tan^{-1}\left(\frac{\frac{8}{15}}{\frac{14}{15}}\right) + \tan^{-1}\left(\frac{\frac{15}{56}}{\frac{55}{56}}\right) = \tan^{-1}\left(\frac{4}{7}\right) + \tan^{-1}\left(\frac{3}{11}\right)$$

$$= \tan^{-1}\left[\frac{\frac{4}{7}+\frac{3}{11}}{1-\frac{4}{7}\times\frac{3}{11}}\right] = \tan^{-1}\left(\frac{\frac{65}{77}}{\frac{65}{77}}\right)$$

$$= \tan^{-1}1 = \frac{\pi}{4} = \text{R.H.S.}$$

39. $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$

$$\Rightarrow \sin^{-1}x + \frac{\pi}{2} - \cos^{-1}(1-x) = \cos^{-1}x \qquad (\forall\ -1 \le x \le 1)$$

$$\Rightarrow \sin^{-1}x + \frac{\pi}{2} - \cos^{-1}x = \cos^{-1}(1-x)$$

$$\Rightarrow \sin^{-1}x + \sin^{-1}x = \cos^{-1}(1-x)$$

$\Rightarrow$ $2\sin^{-1}x = \cos^{-1}(1-x)$

$\Rightarrow$ $\cos(2\sin^{-1}x) = (1-x)$

$\Rightarrow$ $1 - 2\sin^2(\sin^{-1}x) = (1-x)$

$\Rightarrow$ $2\sin^2(\sin^{-1}x) = x$

$\Rightarrow$ $2x^2 = x \Rightarrow 2x^2 - x = 0 \Rightarrow x(2x-1) = 0$

$\Rightarrow$ $x = 0$ or $2x - 1 = 0 \Rightarrow x = 0$ or $x = 1/2$

40. We have, $\cos^{-1}\frac{x}{a} + \cos^{-1}\frac{y}{b} = \alpha$

$$\Rightarrow \cos^{-1}\left[\frac{xy}{ab} - \sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}\right] = \alpha$$

$$\Rightarrow \frac{xy}{ab} - \sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}} = \cos\alpha \qquad \ldots (1)$$

Squaring on both sides, we get

$$\frac{x^2y^2}{a^2b^2} + \left(1-\frac{x^2}{a^2}\right)\left(1-\frac{y^2}{b^2}\right) - \frac{2xy}{ab}\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}} = \cos^2\alpha$$

$$\Rightarrow \frac{x^2y^2}{a^2b^2} + 1 - \frac{y^2}{b^2} - \frac{x^2}{a^2} + \frac{x^2y^2}{a^2b^2} - \frac{2xy}{ab}\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}} = 1 - \sin^2\alpha$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\left[\frac{xy}{ab} - \sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}\right] = \sin^2\alpha$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{2xy}{ab}\cos\alpha + \frac{y^2}{b^2} = \sin^2\alpha \qquad \text{[From (1)]}$$

41. $\tan^{-1}\left(\frac{x-2}{x-1}\right) + \tan^{-1}\left(\frac{x+2}{x+1}\right) = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1}\left(\frac{\left(\frac{x-2}{x-1}\right)+\left(\frac{x+2}{x+1}\right)}{1-\left(\frac{x-2}{x-1}\right)\left(\frac{x+2}{x+1}\right)}\right) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{(x-2)(x+1)+(x+2)(x-1)}{(x^2-1-(x^2-2^2)}\right) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{x^2+x-2x-2+x^2-x+2x-2}{x^2-1-x^2+4}\right) = \frac{\pi}{4}$$

$$\Rightarrow \left(\frac{2x^2-4}{3}\right) = \tan\left(\frac{\pi}{4}\right) \Rightarrow \frac{2x^2-4}{3} = 1$$

$\Rightarrow$ $2x^2 - 4 = 3 \Rightarrow 2x^2 = 3 + 4$

$\Rightarrow$ $x^2 = \frac{7}{2} \Rightarrow x = \pm\sqrt{\frac{7}{2}}$

42. $\text{L.H.S.} = \cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}} \times \frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}+\sqrt{1-\sin x}}\right)$

$$= \cot^{-1}\left(\frac{(1+\sin x)+(1-\sin x)+2\sqrt{1-\sin^2 x}}{(1+\sin x)-(1-\sin x)}\right)$$

$$= \cot^{-1}\left(\frac{2(1+\cos x)}{2\sin x}\right) = \cot^{-1}\left(\frac{1+\cos x}{\sin x}\right)$$

$$= \cot^{-1}\left(\frac{2\cos^2(x/2)}{2\sin(x/2)\cos(x/2)}\right)$$

$$= \cot^{-1}\left(\cot\frac{x}{2}\right) = \frac{x}{2} = \text{R.H.S.}$$

43. We have, $\sin[\cot^{-1}(x+1)] = \cos(\tan^{-1}x)$ ... (1)

Let $\cot^{-1}(x+1) = A$ and $\tan^{-1}x = B$

$\Rightarrow$ $x + 1 = \cot A \Rightarrow \sin A = \frac{1}{\sqrt{(x+1)^2+1}}$

Also, $x = \tan B \quad \therefore \quad \cos B = \frac{1}{\sqrt{x^2+1}}$

Now, $\sin A = \cos B$ [From (1)]

$$\Rightarrow \frac{1}{\sqrt{(x+1)^2+1}} = \frac{1}{\sqrt{x^2+1}} \Rightarrow (x+1)^2 + 1 = x^2 + 1$$

$\Rightarrow$ $1 + 2x = 0 \Rightarrow x = -\frac{1}{2}$

www.exambites.in

44. $(\tan^{-1}x)^2+(\cot^{-1}x)^2=\frac{5\pi^2}{8}$ [Given]

$\Rightarrow (\tan^{-1}x)^2+\left(\frac{\pi}{2}-\tan^{-1}x\right)^2=\frac{5\pi^2}{8}$

Putting $\tan^{-1}x=\theta$, we get

$\theta^2+\left(\frac{\pi}{2}-\theta\right)^2=\frac{5\pi^2}{8} \Rightarrow \theta^2+\frac{\pi^2}{4}+\theta^2-\pi\theta=\frac{5\pi^2}{8}$

$\Rightarrow 2\theta^2-\pi\theta+\left(\frac{\pi^2}{4}-\frac{5\pi^2}{8}\right)=0$

$\Rightarrow 2\theta^2-\pi\theta-\frac{3}{8}\pi^2=0$

$\Rightarrow 16\theta^2-8\pi\theta-3\pi^2=0$

$\Rightarrow 4\theta(4\theta-3\pi)+\pi(4\theta-3\pi)=0$

$\Rightarrow (4\theta+\pi)(4\theta-3\pi)=0$

$\Rightarrow$ Either $4\theta=3\pi$ or $4\theta=-\pi$

$\Rightarrow \theta=\frac{3\pi}{4}$ or $\theta=-\frac{\pi}{4}$

Hence, $\tan^{-1}x=\frac{3\pi}{4}$ or $-\frac{\pi}{4} \Rightarrow x=-1$

45. L.H.S. $=\cot^{-1}\left(\frac{xy+1}{x-y}\right)+\cot^{-1}\left(\frac{yz+1}{y-z}\right)+\cot^{-1}\left(\frac{zx+1}{z-x}\right)$

$=\tan^{-1}\left(\frac{x-y}{1+xy}\right)+\tan^{-1}\left(\frac{y-z}{1+yz}\right)+\tan^{-1}\left(\frac{z-x}{1+zx}\right)$

$\left[\because \cot^{-1}x=\tan^{-1}\frac{1}{x}\right]$

$=(\tan^{-1}x-\tan^{-1}y)+(\tan^{-1}y-\tan^{-1}z)+(\tan^{-1}z-\tan^{-1}x)$

$=0=$ R.H.S.

46. We have, $\tan^{-1}(x+1)+\tan^{-1}(x-1)=\tan^{-1}\frac{8}{31}$

$\Rightarrow \tan^{-1}\left(\frac{(x+1+x-1)}{1-(x+1)(x-1)}\right)=\tan^{-1}\frac{8}{31}$

for $(x+1)(x-1)<1$

$\Rightarrow \frac{2x}{1-(x^2-1)}=\frac{8}{31} \Rightarrow \frac{2x}{2-x^2}=\frac{8}{31}$

$\Rightarrow 31x=8-4x^2 \Rightarrow 4x^2+31x-8=0$

$\Rightarrow (4x-1)(x+8)=0$

$\Rightarrow 4x-1=0$ or $x+8=0$

$\Rightarrow x=\frac{1}{4}$ or $x=-8$

But $x=-8$ does not satisfy the equation.

Hence, $x=\frac{1}{4}$ is the only solution.

47. $\tan^{-1}\left(\frac{1}{1+1\cdot 2}\right)+\tan^{-1}\left(\frac{1}{1+2\cdot 3}\right)+\ldots+\tan^{-1}\left(\frac{1}{1+n\cdot(n+1)}\right)=\tan^{-1}\theta$

$\Rightarrow \tan^{-1}\left(\frac{2-1}{1+1\cdot 2}\right)+\tan^{-1}\left(\frac{3-2}{1+2\cdot 3}\right)+\ldots+\tan^{-1}\left(\frac{(n+1)-n}{1+n(n+1)}\right)=\tan^{-1}\theta$

$\Rightarrow \tan^{-1}2-\tan^{-1}1+\tan^{-1}3-\tan^{-1}2+\ldots+\tan^{-1}(n+1)-\tan^{-1}(n)=\tan^{-1}\theta$

$\Rightarrow \tan^{-1}(n+1)-\tan^{-1}1=\tan^{-1}\theta$

$\Rightarrow \tan^{-1}\left(\frac{(n+1)-1}{1+(n+1)(1)}\right)=\tan^{-1}\theta$

$\Rightarrow \tan^{-1}\left(\frac{n}{n+2}\right)=\tan^{-1}\theta \Rightarrow \frac{n}{n+2}=\theta$

48. We have, $\tan^{-1}2x+\tan^{-1}3x=\frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left(\frac{2x+3x}{1-2x\times 3x}\right)=\frac{\pi}{4}$ (for $2x\cdot 3x<1$)

$\Rightarrow \tan^{-1}\left(\frac{5x}{1-6x^2}\right)=\frac{\pi}{4}$

Therefore, $\frac{5x}{1-6x^2}=\tan\frac{\pi}{4}=1$

$\Rightarrow 6x^2+5x-1=0 \Rightarrow (6x-1)(x+1)=0$

which gives $x=\frac{1}{6}$ or $x=-1$.

Since $x=-1$ does not satisfy the equation as the L.H.S. of the equation becomes negative.

$\therefore$ $x=\frac{1}{6}$ is the only solution of the given equation.

49. Consider R.H.S. $=\sin^{-1}\frac{5}{13}+\cos^{-1}\frac{3}{5}$

$=\tan^{-1}\frac{5}{12}+\tan^{-1}\frac{4}{3}=\tan^{-1}\left(\frac{\frac{5}{12}+\frac{4}{3}}{1-\frac{5}{12}\cdot\frac{4}{3}}\right)$

$=\tan^{-1}\left(\frac{15+48}{36-20}\right)=\tan^{-1}\left(\frac{63}{16}\right)=$ L.H.S.

www.exambits.in

50. L.H.S. $= 2\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{7}\right)$

$=\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{7}\right)$

$=\tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{2}}{1-\frac{1}{2}.\frac{1}{2}}\right)+\tan^{-1}\left(\frac{1}{7}\right)$

$=\tan^{-1}\left(\frac{4}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)$

$=\tan^{-1}\left(\frac{\frac{4}{3}+\frac{1}{7}}{1-\frac{4}{3}.\frac{1}{7}}\right)=\tan^{-1}\left(\frac{28+3}{21-4}\right)=\tan^{-1}\left(\frac{31}{17}\right)$

Now, $\tan^{-1}\left(\frac{31}{17}\right)=\theta$ (say) ...(1)

$\Rightarrow \quad \tan\theta=\frac{31}{17}$

$\therefore \quad \sin\theta=\frac{1}{\operatorname{cosec}\theta}=\frac{1}{\sqrt{1+\cot^2\theta}}$

$=\frac{1}{\sqrt{1+\left(\frac{17}{31}\right)^2}} \quad =\frac{31}{\sqrt{31^2+17^2}}=\frac{31}{\sqrt{1250}}=\frac{31}{25\sqrt{2}}$

$\Rightarrow \quad \theta=\sin^{-1}\left(\frac{31}{25\sqrt{2}}\right)$ ...(2)

From (1) & (2), L.H.S $=\sin^{-1}\left(\frac{31}{25\sqrt{2}}\right)=$ R.H.S.

51. We have, $\tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\tan^{-1}x, \quad (x>0)$

$\Rightarrow \quad \tan^{-1}1-\tan^{-1}x=\frac{1}{2}\tan^{-1}x$

$\Rightarrow \quad \frac{3}{2}\tan^{-1}x=\tan^{-1}1=\frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}x=\frac{\pi}{4}\times\frac{2}{3}=\frac{\pi}{6}$

$\Rightarrow \quad x=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}$

$\Rightarrow \quad x=\frac{1}{\sqrt{3}}$

52. L.H.S. $=2\tan^{-1}\left(\frac{1}{5}\right)+\sec^{-1}\left(\frac{5\sqrt{2}}{7}\right)+2\tan^{-1}\left(\frac{1}{8}\right)$

$=2\left[\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{1}{8}\right)\right]+\sec^{-1}\left(\frac{5\sqrt{2}}{7}\right)$

$=2\tan^{-1}\left(\frac{\frac{1}{5}+\frac{1}{8}}{1-\frac{1}{5}\times\frac{1}{8}}\right)+\sec^{-1}\left(\frac{5\sqrt{2}}{7}\right)$

$=2\tan^{-1}\left(\frac{\frac{13}{40}}{\frac{39}{40}}\right)+\sec^{-1}\left(\frac{5\sqrt{2}}{7}\right)$

$=2\tan^{-1}\left(\frac{1}{3}\right)+\sec^{-1}\left(\frac{5\sqrt{2}}{7}\right)$

$=2\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\sqrt{\left(\frac{5}{7}\sqrt{2}\right)^2-1}$

$=2\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)$

$=\tan^{-1}\left(\frac{\frac{2}{3}}{1-\frac{1}{9}}\right)+\tan^{-1}\left(\frac{1}{7}\right)=\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{1}{7}\right)$

$=\tan^{-1}\left(\frac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}\times\frac{1}{7}}\right)=\tan^{-1}(1)=\frac{\pi}{4}=$ R.H.S.

53. Putting $x = \cos\theta$, we get

$\text{L.H.S.}=\tan^{-1}\left\{\frac{\sqrt{1+\cos\theta}-\sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta}+\sqrt{1-\cos\theta}}\right\}$

$=\tan^{-1}\left\{\frac{\sqrt{2\cos^2(\theta/2)}-\sqrt{2\sin^2(\theta/2)}}{\sqrt{2\cos^2(\theta/2)}+\sqrt{2\sin^2(\theta/2)}}\right\}$

$=\tan^{-1}\left\{\frac{\cos(\theta/2)-\sin(\theta/2)}{\cos(\theta/2)+\sin(\theta/2)}\right\}$

$=\tan^{-1}\left\{\frac{1-\tan(\theta/2)}{1+\tan(\theta/2)}\right\}$

[Dividing numerator and denominator by $\cos(\theta/2)$]

$=\tan^{-1}\left\{\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right\}=\frac{\pi}{4}-\frac{\theta}{2}$

$=\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)=$ R.H.S.

www.exambites.in

54. *Refer to answer 41.*

55. We have $\cos(\tan^{-1}x) = \sin\left(\cot^{-1}\frac{3}{4}\right)$

Let $\tan^{-1}x = \theta \Rightarrow \tan\theta = x$

$$\Rightarrow \cos\theta = \frac{1}{\sqrt{1+x^2}} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$$

Also, let $\cot^{-1}\frac{3}{4} = \beta \Rightarrow \cot\beta = \frac{3}{4}$

$$\Rightarrow \sin\beta = \frac{4}{5} \Rightarrow \beta = \sin^{-1}\frac{4}{5}$$

So, $\cos(\tan^{-1}x) = \sin\left(\cot^{-1}\frac{3}{4}\right)$

$$\Rightarrow \cos\left(\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right) = \sin\left(\sin^{-1}\frac{4}{5}\right)$$

$$\Rightarrow \frac{1}{\sqrt{1+x^2}} = \frac{4}{5}$$

$\Rightarrow 16 + 16x^2 = 25 \Rightarrow x = \pm 3/4.$

Hence, value of x is 3/4.

[$\because$ $-3/4$ does not satisfy the given equation]

56. L.H.S. $= \cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18$

$$= \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} + \tan^{-1}\frac{1}{18}$$

$$= \tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right) + \tan^{-1}\frac{1}{18} = \tan^{-1}\frac{3}{11} + \tan^{-1}\frac{1}{18}$$

$$= \tan^{-1}\left(\frac{\frac{3}{11}+\frac{1}{18}}{1-\frac{3}{11}\times\frac{1}{18}}\right) = \tan^{-1}\left(\frac{65}{195}\right) = \tan^{-1}\frac{1}{3}$$

$= \cot^{-1}3$ = R.H.S.

57. L.H.S $= \cos^{-1}x + \cos^{-1}\left\{\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right\}$

$$= \cos^{-1}x + \cos^{-1}\left\{\frac{1}{2}\cdot x + \frac{\sqrt{3}}{2}\cdot\sqrt{1-x^2}\right\}$$

$$= \cos^{-1}x + \cos^{-1}\left\{\frac{1}{2}\cdot x + \sqrt{1-\left(\frac{1}{2}\right)^2}\cdot\sqrt{1-x^2}\right\}$$

$$= \cos^{-1}x + \cos^{-1}\frac{1}{2} - \cos^{-1}x.$$

$$= \cos^{-1}\frac{1}{2} = \frac{\pi}{3} = \text{R.H.S.}$$

58. $\tan^{-1}x + 2\cot^{-1}x = \frac{2\pi}{3}$

$$\Rightarrow \frac{\pi}{2} - \cot^{-1}x + 2\cot^{-1}x = \frac{2\pi}{3}$$

$$\Rightarrow \cot^{-1}x = \frac{2\pi}{3} - \frac{\pi}{2} \Rightarrow \cot^{-1}x = \frac{4\pi - 3\pi}{6}$$

$$\Rightarrow \cot^{-1}x = \frac{\pi}{6} \Rightarrow x = \cot\frac{\pi}{6} \Rightarrow x = \sqrt{3}$$

59. L.H.S. $= \sin^{-1}\frac{3}{5} + \sin^{-1}\frac{8}{17}$

$$= \sin^{-1}\left(\frac{3}{5}\sqrt{1-\left(\frac{8}{17}\right)^2} + \frac{8}{17}\sqrt{1-\left(\frac{3}{5}\right)^2}\right)$$

$$= \sin^{-1}\left(\frac{3}{5}\times\frac{15}{17} + \frac{8}{17}\times\frac{4}{5}\right)$$

$$= \sin^{-1}\left(\frac{45}{85} + \frac{32}{85}\right) = \sin^{-1}\left(\frac{77}{85}\right)$$

$$= \cos^{-1}\sqrt{1-\left(\frac{77}{85}\right)^2} = \cos^{-1}\sqrt{\frac{7225-5929}{7225}}$$

$$= \cos^{-1}\sqrt{\frac{1296}{7225}} = \cos^{-1}\frac{36}{85} = \text{R.H.S.}$$

60. $\tan\frac{1}{2}\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right]$

$$= \tan\frac{1}{2}[2\tan^{-1}x + 2\tan^{-1}y] = \tan(\tan^{-1}x + \tan^{-1}y)$$

$$= \tan\left\{\tan^{-1}\left(\frac{x+y}{1-xy}\right)\right\} = \frac{x+y}{1-xy}, y > 0 \,\&\, xy < 1$$

61. L.H.S. $= \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{8}\right)$

$$= \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{5}}{1-\frac{1}{2}\times\frac{1}{5}}\right) + \tan^{-1}\frac{1}{8}$$

$$= \tan^{-1}\left(\frac{5+2}{10-1}\right) + \tan^{-1}\frac{1}{8}$$

www.examiten.in

$$= \tan^{-1}\frac{7}{9} + \tan^{-1}\frac{1}{8} \quad = \tan^{-1}\left[\frac{\left(\frac{7}{9}+\frac{1}{8}\right)}{\left(1-\frac{7}{9}\times\frac{1}{8}\right)}\right]$$

$$= \tan^{-1}\left(\frac{56+9}{72-7}\right) = \tan^{-1}\left(\frac{65}{65}\right)$$

$$= \tan^{-1}1 = \frac{\pi}{4} = \text{R.H.S.}$$

62. Put $\sin^{-1}\frac{3}{4} = \theta \Rightarrow \sin\theta = \frac{3}{4}$

$$\Rightarrow \quad \cos\theta \quad = \sqrt{1-\left(\frac{3}{4}\right)^2} \quad = \frac{\sqrt{7}}{4}$$

Now, $\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \tan\frac{\theta}{2}$

$$= \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \sqrt{\frac{1-\frac{\sqrt{7}}{4}}{1+\frac{\sqrt{7}}{4}}} = \sqrt{\frac{4-\sqrt{7}}{4+\sqrt{7}}\times\frac{4-\sqrt{7}}{4-\sqrt{7}}}$$

$$= \frac{4-\sqrt{7}}{3}.$$

63. $\tan^{-1}\left(\frac{a}{b}\right) - \tan^{-1}\left(\frac{a-b}{a+b}\right)$

$$= \tan^{-1}\left[\frac{\frac{a}{b}-\frac{a-b}{a+b}}{1+\frac{a}{b}.\frac{a-b}{a+b}}\right]$$

$$= \tan^{-1}\left[\frac{a(a+b)-b(a-b)}{b(a+b)+a(a-b)}\right]$$

$$= \tan^{-1}\left[\frac{a^2+b^2}{a^2+b^2}\right] \quad = \tan^{-1}(1) = \frac{\pi}{4}$$

64. *Refer to answer 59.*

65. We have, $\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$ ...(1)

$$\Rightarrow \quad \sin^{-1}(1-x) = \frac{\pi}{2} + 2\sin^{-1}x$$

$$\Rightarrow \quad 1-x = \sin\left(\frac{\pi}{2} + 2\sin^{-1}x\right)$$

$\Rightarrow$ $1 - x = \cos(2\sin^{-1}x) = \cos 2\theta$,

where $\theta = \sin^{-1}x \Rightarrow x = \sin\theta$

$\therefore$ $1 - x = 1 - 2\sin^2\theta \Rightarrow 1 - x = 1 - 2x^2$

$\Rightarrow$ $x = 2x^2 \Rightarrow x(2x - 1) = 0 \Rightarrow x = 0, \frac{1}{2}$

For $x = \frac{1}{2}$,

L.H.S. of (1) $= \sin^{-1}\left(1-\frac{1}{2}\right) - 2\sin^{-1}\frac{1}{2}$

$$= -\sin^{-1}\frac{1}{2} \quad = -\frac{\pi}{6} \neq \frac{\pi}{2}$$

$\therefore$ $x = \frac{1}{2}$ is not a solution of (1).

Hence, $x = 0$ is the only solution of (1).

66. L.H.S. $= \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$

$$= \tan^{-1}\left(\frac{\sin\left(\frac{\pi}{2}-x\right)}{1+\cos\left(\frac{\pi}{2}-x\right)}\right)$$

$$= \tan^{-1}\left(\frac{2\sin\left(\frac{\pi}{4}-\frac{x}{2}\right)\cos\left(\frac{\pi}{4}-\frac{x}{2}\right)}{2\cos^2\left(\frac{\pi}{4}-\frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right) = \frac{\pi}{4}-\frac{x}{2} = \text{R.H.S.}$$

67. Let $\sin^{-1}\frac{3}{5} = \theta$ and $\cot^{-1}\frac{3}{2} = \phi$

$$\Rightarrow \quad \sin\theta = \frac{3}{5} \text{ and } \cot\phi = \frac{3}{2}$$

$$\Rightarrow \quad \tan\theta = \frac{3}{4} \text{ and } \tan\phi = \frac{2}{3}$$

$$\therefore \quad \theta = \tan^{-1}\frac{3}{4} \text{ and } \phi = \tan^{-1}\frac{2}{3}$$

Thus, $\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{2}{3}\right) = \tan^{-1}\left[\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}\times\frac{2}{3}}\right]$$

$$= \tan^{-1}\left(\frac{17}{6}\right) = \alpha \text{ (say)} \qquad \ldots(1)$$

$$\Rightarrow \quad \tan\alpha = \frac{17}{6} \Rightarrow \cos\alpha = \frac{6}{\sqrt{6^2+17^2}} = \frac{6}{5\sqrt{13}}$$

Now, L.H.S. $= \cos\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$

$$= \cos\alpha \quad = \frac{6}{5\sqrt{13}} = \text{R.H.S.} \qquad \text{[Using (1)]}$$

www.exambites.in

68. *Refer to answer 41.*

69. L.H.S. $= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{3}{5} - \tan^{-1}\frac{8}{19}$

$= \tan^{-1}\left[\frac{\frac{3}{4}+\frac{3}{5}}{1-\frac{3}{4}\times\frac{3}{5}}\right] - \tan^{-1}\frac{8}{19}$

$= \tan^{-1}\left(\frac{15+12}{20-9}\right) - \tan^{-1}\frac{8}{19}$

$= \tan^{-1}\left(\frac{27}{11}\right) - \tan^{-1}\frac{8}{19} = \tan^{-1}\left[\frac{\frac{27}{11}-\frac{8}{19}}{1+\frac{27}{11}\times\frac{8}{19}}\right]$

$= \tan^{-1}\left(\frac{513-88}{209+216}\right)$

$= \tan^{-1}\left(\frac{425}{425}\right) = \tan^{-1}1 = \frac{\pi}{4} = \text{R.H.S.}$

70. *Refer to answer 63.*

71. *Refer to answer 50.*

72. L.H.S. $= 2\tan^{-1}\left(\frac{3}{4}\right) - \tan^{-1}\left(\frac{17}{31}\right)$

$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{4}\right) - \tan^{-1}\left(\frac{17}{31}\right)$

$= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{3}{4}}{1-\frac{3}{4}\cdot\frac{3}{4}}\right) - \tan^{-1}\left(\frac{17}{31}\right)$

$= \tan^{-1}\left(\frac{24}{7}\right) - \tan^{-1}\left(\frac{17}{31}\right) = \tan^{-1}\left(\frac{\frac{24}{7}-\frac{17}{31}}{1+\frac{24}{7}\cdot\frac{17}{31}}\right)$

$= \tan^{-1}\left(\frac{24\times31-17\times7}{7\times31+24\times17}\right)$

$= \tan^{-1}\left(\frac{744-119}{217+408}\right) = \tan^{-1}\left(\frac{625}{625}\right)$

$= \tan^{-1}(1) = \frac{\pi}{4} = \text{R.H.S.}$

73. We have, $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{\pi}{3}$

$\left[\because \cot^{-1}x = \tan^{-1}\frac{1}{x}\right]$

$\Rightarrow \quad \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{\pi}{6} \Rightarrow \frac{2x}{1-x^2} = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad 2\sqrt{3}x = 1-x^2 \Rightarrow x^2 + 2\sqrt{3}x - 1 = 0$

$\Rightarrow \quad x = \frac{-2\sqrt{3} \pm \sqrt{12+4}}{2} = -\sqrt{3} \pm 2$

$\Rightarrow \quad x = 2-\sqrt{3}$ (Reject $-\sqrt{3}-2$ as $-1 < x < 1$).

74. To prove : $\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9} = \frac{1}{2}\tan^{-1}\frac{4}{3}$

$\Rightarrow \quad 2\left(\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9}\right) = \tan^{-1}\frac{4}{3}$

Now L.H.S. $= 2\left(\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9}\right)$

$= 2\tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right)$

$= 2\tan^{-1}\frac{1}{2} = \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{2}$

$= \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{2}}{1-\frac{1}{2}\times\frac{1}{2}}\right) = \tan^{-1}\frac{4}{3} = \text{R.H.S.}$

75. The given equation is

$\cos(2\sin^{-1}x) = \frac{1}{9} \qquad (x > 0) \qquad \ldots(1)$

Put $\sin^{-1}x = \theta \Rightarrow x = \sin\theta$

$\therefore \quad$ Eq. (1) $\Rightarrow \cos 2\theta = \frac{1}{9} \Rightarrow 1 - 2\sin^2\theta = \frac{1}{9}$

$\Rightarrow \quad 2\sin^2\theta = 1 - \frac{1}{9} = \frac{8}{9}$

$\Rightarrow \quad x^2 = \frac{4}{9} \Rightarrow x = \frac{2}{3} \quad (\because x > 0)$

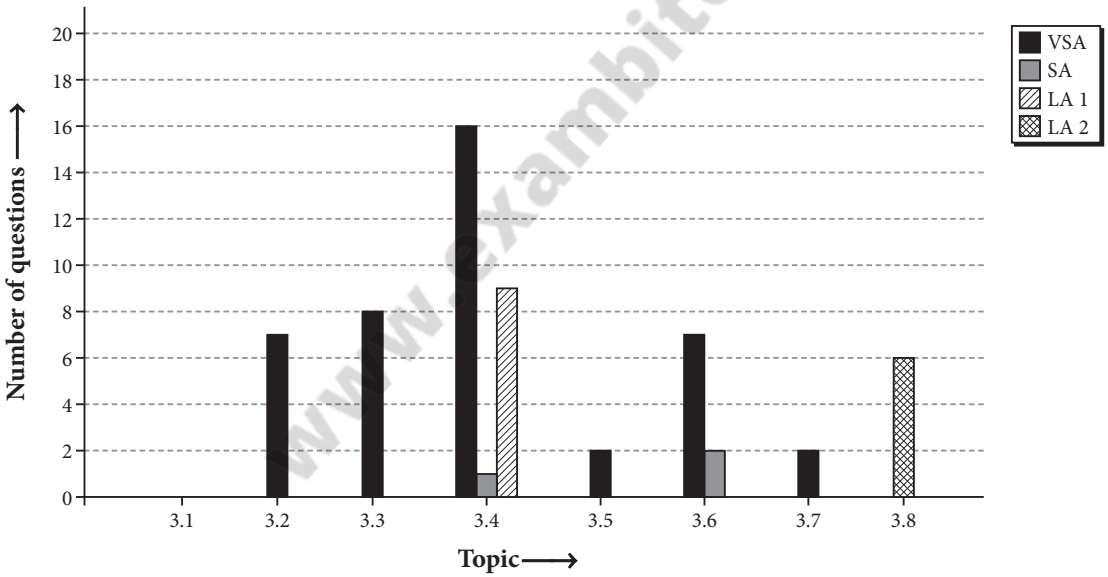
www.exambites.in

CHAPTER
3

Matrices

3.1 Introduction
3.2 Matrix
3.3 Types of Matrices
3.4 Operations on Matrices
3.5 Transpose of a Matrix
3.6 Symmetric and Skew Symmetric Matrices
3.7 Elementary Operation (Transformation) of a Matrix
3.8 Invertible Matrices

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum weightage is of *Operations on Matrices*
- Maximum VSA and LA 1 type questions were asked from *Operations on Matrices*
- LA 2 type questions were asked from only *Invertible Matrices*

QUICK RECAP

MATRIX

A matrix is any rectangular array of numbers or functions in m rows and n columns within brackets.

A matrix of m rows and n columns is usually written as

$$A=\begin{bmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22} & ..... & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & ..... & a_{mn} \end{bmatrix}_{m\times n}$$

The above matrix is also represented by $A = [a_{ij}]_{m\times n}$ or, $A = [a_{ij}]$

Order of a Matrix

- A matrix having m rows and n columns has order $m\times n$.

Types of Matrices

- **Row Matrix :** A matrix having only one row.
- **Column Matrix :** A matrix having only one column.
- **Square Matrix :** A matrix in which number of rows is equal to the number of columns.
- **Diagonal Matrix :** A square matrix whose all the non-diagonal elements are zero.

Ex. $A=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is a diagonal matrix and it can also be written as A = diag (1 2 3)

- **Scalar Matrix :** A diagonal matrix in which all the diagonal elements are equal.
- **Identity or Unit Matrix :** A square matrix is said to be identity matrix if all its diagonal entries are equal to 1 and rest are zero.
- **Zero or Null Matrix :** A matrix whose all the elements are zero.

EQUALITY OF MATRICES

Two matrices are said to be equal, if their order is same and their corresponding elements are also equal.

OPERATIONS ON MATRICES

Operations	Definition	Properties
Addition of two Matrices	Let A and B be two matrices each of order $m \times n$. Then, $A + B = [a_{ij} + b_{ij}]$ for $i = 1, 2, ..., m$ and $j = 1, 2, ..., n$	(i) **Commutative Law :** For any two matrices A & B, $A + B = B + A$ (ii) **Associative Law :** For any three matrices A, B and C, $A + (B + C) = (A + B) + C$ (iii) **Existence of Additive Identity :** For any matrix A, if there exists a zero matrix O such that $A + O = A = O + A$. Then O is called additive identity. (iv) **Existence of Additive Inverse :** For any matrix A, if there exists a matrix $(-A)$ such that $A + (-A) = O = (-A) + A$. Then $(-A)$ is called additive inverse of A.
Multiplication of a Matrix by a Scalar	Let A be a matrix of order $m\times n$. Then, for any scalar k, $kA = [k \cdot a_{ij}]_{m \times n}$	Let A and B be two matrices each of order $m\times n$. Then, for any scalars k and l, we have (i) $k(A + B) = kA + kB$ (ii) $(k + l)\, A = kA + lA$
Multiplication of two Matrices	Let A and B be any two matrices of orders $m \times n$ and $n \times p$ respectively. Then $AB = C = [c_{ik}]_{m\times p}$ where $c_{ik} = \sum_{r=1}^{n} a_{ir} b_{rk}$	(i) Multiplication of two matrices is not commutative *i.e.*, $AB \neq BA$. (ii) **Associative Law :** For any three matrices A, B, and C • $(AB)C = A(BC)$ (iii) **Distributive Law :** For any three matrices A, B and C, • $A(B + C) = AB + AC$ • $(A + B)C = AC + BC$ (iv) **Existence of Multiplicative Identity :** For any square matrix, there exists a matrix I such that $AI = A = IA$, where I is called the identity matrix.

www.exambites.in

TRANSPOSE OF A MATRIX

The matrix obtained by interchanging the rows and columns of matrix A. It is represented by A' or A^T. In other words, if $A=[a_{ij}]_{m\times n}$, then $A' = [a_{ji}]_{n\times m}$

Properties of Transpose

- For any matrices A and B of suitable orders,
  (i) $(A + B)' = A' + B'$
  (ii) $(A')' = A$
  (iii) $(kA)' = kA'$, where k is any constant.
  (iv) $(AB)' = B'A'$

Symmetric Matrix : A square matrix $A = [a_{ij}]$ is called a symmetric matrix, if $a_{ij} = a_{ji}$ for all i, j or we say, if $A = A'$.

Skew-Symmetric Matrix : A square matrix $A = [a_{ij}]$ is called a skew symmetric matrix, if $a_{ji} = -a_{ij}$ for all i,j or we say, if $A' = -A$.

Note :

(i) For any square matrix A with real entries, $A + A'$ is a symmetric matrix and $A - A'$ is a skew symmetric matrix.

(ii) A matrix which is both symmetric and skew symmetric, is zero matrix.

ELEMENTARY OPERATION (TRANSFORMATION) OF A MATRIX

(i) Interchange of any two rows (or columns) : If i^{th} row (or column) is interchanged with j^{th} row (or column), we write $R_i \leftrightarrow R_j$ (or $C_i \leftrightarrow C_j$).

(ii) Multiplying the elements of a row (or column) by a non-zero scalar : If the elements of ith row (or column) are multiplied by a non-zero scalar k, we write $R_i \to kR_i$ (or $C_i \to kC_i$).

(iii) Adding the elements of a row (or column), to the constant times the corresponding elements of another row (or column) : If k times the elements of jth row (or column) are added to the corresponding elements of the ith row (or column), we write $R_i \to R_i + kR_j$ (or $C_i \to C_i + kC_j$).

INVERTIBLE MATRICES

If A is a square matrix of order m and if there exists another square matrix B of the same order m, such that $AB = BA = I$, then B is called the inverse matrix of A and is denoted by A^{-1}. Inverse of a square matrix, if it exists, is unique.

Inverse of a matrix by elementary operations

- Let A is a square matrix of order n. To find A^{-1} using elementary row (or column) operations, write $A = IA$ (or $A = AI$)and apply a sequence of row (or column) operations on $A = IA$ (or $A = AI$) till we get, $I = BA$(or AB). The matrix B will be the inverse of A.

www.exambites.in

Previous Years' CBSE Board Questions

3.2 Matrix

VSA (1 mark)

1. Construct a 2 × 2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = |(i)^2 - j|$. *(2020)*

2. Write the number of all possible matrices of order 2 × 2 with each entry 1, 2 or 3. *(AI 2016)*

3. Write the element a_{23} of a 3 × 3 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by $a_{ij} = \frac{|i-j|}{2}$. *(Delhi 2015)*

4. The elements a_{ij} of a 3 × 3 matrix are given by $a_{ij} = \frac{1}{2}|-3i + j|$. Write the value of element a_{32}. *(AI 2014C)*

5. For a 2 × 2 matrix $A = [a_{ij}]$, whose elements are given by $a_{ij} = \frac{(i+2j)^2}{4}$, write the value of a_{21}. *(Delhi 2012C)*

6. For a 2 × 2 matrix, $A = (a_{ij})$ whose elements are given by $a_{ij} = \frac{i}{j}$, write the value of a_{12}. *(Delhi 2011)*

7. If a matrix has 5 elements, then write all possible orders it can have. *(AI 2011)*

3.3 Types of Matrices

VSA (1 mark)

8. If $\begin{bmatrix} x-y & z \\ 2x-y & w \end{bmatrix} = \begin{bmatrix} -1 & 4 \\ 0 & 5 \end{bmatrix}$, find the value of $x + y$. *(AI 2014)*

9. If $\begin{pmatrix} a+4 & 3b \\ 8 & -6 \end{pmatrix} = \begin{pmatrix} 2a+2 & b+2 \\ 8 & a-8b \end{pmatrix}$, write the value of $a - 2b$. *(Foreign 2014)*

10. If $\begin{bmatrix} x \cdot y & 4 \\ z+6 & x+y \end{bmatrix} = \begin{bmatrix} 8 & w \\ 0 & 6 \end{bmatrix}$, write the value of $(x + y + z)$. *(Delhi 2014C)*

11. Find the value of a if $\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$ *(Delhi 2013)*

12. Find the value of b if $\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$ *(Delhi 2013)*

13. If $\begin{bmatrix} x-y & 2y \\ 2y+z & x+y \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 9 & 5 \end{bmatrix}$, then write the value of $(x + y + z)$. *(AI 2013C)*

14. If $\begin{bmatrix} 2x+1 & 2y \\ 0 & y^2+1 \end{bmatrix} = \begin{bmatrix} x+3 & 10 \\ 0 & 26 \end{bmatrix}$, write the value of $(x + y)$. *(AI 2012C)*

15. If $\begin{bmatrix} x & x-y \\ 2x+y & 7 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 8 & 7 \end{bmatrix}$, then find the value of y. *(Delhi 2011C)*

3.4 Operations on Matrices

VSA (1 mark)

16. If A is a square matrix such that $A^2 = A$, then $(I - A)^3 + A$ is equal to

(a) I (b) 0

(c) $I - A$ (d) $I + A$ *(2020)*

17. If A is a square matrix such that $A^2 = I$, then find the simplified value of $(A - I)^3 + (A + I)^3 - 7A$. *(Delhi 2016)*

18. If $[2\ 1\ 3]\begin{bmatrix} -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = A$, then write the order of matrix A. *(Foreign 2016)*

19. Solve the following matrix equation for x: $[x\ 1]\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = O$ *(Delhi 2014)*

20. If $2\begin{bmatrix} 3 & 4 \\ 5 & x \end{bmatrix} + \begin{bmatrix} 1 & y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 10 & 5 \end{bmatrix}$, find $(x - y)$. *(Delhi 2014)*

21. If A is a square matrix such that $A^2 = A$, then write the value of $7A - (I + A)^3$, where I is an identity matrix. *(AI 2014)*

www.examfree.in

22. If $(2x \quad 4)\begin{pmatrix} x \\ -8 \end{pmatrix} = O$, find the positive value of x. *(AI 2014C)*

23. If $\begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} = A + \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$, then find the matrix A. *(Delhi 2013)*

24. If matrix $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and $A^2 = kA$, then write the value of k. *(AI 2013)*

25. If matrix $A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ and $A^2 = pA$, then write the value of p. *(AI 2013)*

26. If matrix $A = \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix}$ and $A^2 = \lambda A$, then write the value of λ. *(AI 2013)*

27. If $2\begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$, then write the value of $(x + y)$. *(Delhi 2013C, AI 2012)*

28. Simplify :

$$\cos\theta\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \sin\theta\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$$

(Delhi 2012)

29. If $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}\begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$, write the value of x. *(Delhi 2012)*

30. Find the value of $x + y$ from the following equation : $2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ *(AI 2012)*

31. If $3A - B = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$, then find the matrix A. *(Delhi 2012C)*

SA (2 marks)

32. Find a matrix A such that $2A - 3B + 5C = O$, where $B = \begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$. *(Delhi 2019)*

LA 1 (4 marks)

33. Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 2 \\ 7 & 4 \end{pmatrix}$, $C = \begin{pmatrix} 2 & 5 \\ 3 & 8 \end{pmatrix}$, find a matrix D such that $CD - AB = O$. *(Delhi 2017)*

34. Find matrix A such that

$$\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix} A = \begin{bmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{bmatrix}$$ *(AI 2017)*

35. If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$, find $A^2 - 5A + 4I$ and hence find a matrix X such that $A^2 - 5A + 4I + X = O$ *(Delhi 2015)*

36. Three schools A, B and C organized a mela for collecting funds for helping the rehabilitation of flood victims. They sold hand made fans, mats and plates from recycled material at a cost of ₹ 25, ₹ 100 and ₹ 50 each. The number of articles sold are given below.

Article/School	***A***	***B***	***C***
Hand-fans	40	25	35
Mats	50	40	50
Plates	20	30	40

Find the funds collected by each school separately by selling the above articles. Also, find the total funds collected for the purpose. Write one value generated by the above situation. *(Delhi 2015)*

37. To promote the making of toilets for women, an organisation tried to generate awareness through (i) house calls (ii) letters and (iii) announcements. The cost for each mode per attempt is given below :

(i) ₹ 50 (ii) ₹ 20 (iii) ₹ 40

The number of attempts made in three villages X, Y and Z are given below:

	(i)	(ii)	(iii)
X	400	300	100
Y	300	250	75
Z	500	400	150

Find the total cost incurred by the organisation for the three villages separately, using matrices. Write one value generated by the organisation in the society. *(AI 2015)*

www.exambits.in

38. If $A=\begin{bmatrix}1 & -1\\ 2 & -1\end{bmatrix}$ and $B=\begin{bmatrix}a & 1\\ b & -1\end{bmatrix}$ and $(A+B)^2 = A^2 + B^2$, then find the values of a and b. *(Foreign 2015)*

39. In a parliament election, a political party hired a public relations firm to promote its candidates in three ways-telephone, house calls and letters. The cost per contact (in paise) is given in matrix A as

$$A=\begin{bmatrix}140\\ 200\\ 150\end{bmatrix}\begin{matrix}\text{Telephone}\\ \text{House call}\\ \text{Letters}\end{matrix}$$

The number of contacts of each type made in two cities X and Y is given in matrix B as

$$B=\begin{matrix}\\ \text{City } X\\ \text{City } Y\end{matrix}\begin{matrix}\begin{matrix}\text{Telephone} & \text{House call} & \text{Letters}\end{matrix}\\ \begin{bmatrix}1000 & 500 & 5000\\ 3000 & 1000 & 10000\end{bmatrix}\end{matrix}$$

Find the total amount spent by the party in the two cities. What should one consider before casting his/her vote-party's promotional activity or their social activities? *(Foreign 2015)*

40. If $\begin{bmatrix}2x & 3\end{bmatrix}\begin{bmatrix}1 & 2\\ -3 & 0\end{bmatrix}\begin{bmatrix}x\\ 3\end{bmatrix}=O$, find x. *(Delhi 2015C)*

41. A trust fund, ₹ 35,000 is to be invested in two different types of bonds. The first bond pays 8% interest per annum which will be given to orphanage and second bond pays 10% interest per annum which will be given to an N.G.O. (Cancer Aid Society). Using matrix multiplication, determine how to divide ₹ 35,000 among two types of bonds if the trust fund obtains an annual total interest of ₹ 3,200. What are the values reflected in this question? *(AI 2015C)*

3.5 Transpose of a Matrix

VSA (1 mark)

42. If A is a matrix of order 3×2, then the order of the matrix A' is _____. *(2020)*

43. If $A^T=\begin{bmatrix}3 & 4\\ -1 & 2\\ 0 & 1\end{bmatrix}$ and $B=\begin{bmatrix}-1 & 2 & 1\\ 1 & 2 & 3\end{bmatrix}$, then find $A^T - B^T$. *(AI 2012)*

3.6 Symmetric and Skew Symmetric Matrices

VSA (1 mark)

44. A square matrix A is said to be skew-symmetric, if _____. *(2020)*

45. If the matrix $A=\begin{bmatrix}0 & a & -3\\ 2 & 0 & -1\\ b & 1 & 0\end{bmatrix}$ is skew symmetric, find the values of 'a' and 'b'. *(2018)*

46. Matrix $A=\begin{bmatrix}0 & 2b & -2\\ 3 & 1 & 3\\ 3a & 3 & -1\end{bmatrix}$ is given to be symmetric, find values of a and b. *(Delhi 2016)*

47. If $A=\begin{bmatrix}3 & 5\\ 7 & 9\end{bmatrix}$ is written as $A = P + Q$, where P is a symmetric matrix and Q is a skew symmetric matrix, then write the matrix P. *(Foreign 2016)*

48. Express the matrix $A=\begin{bmatrix}2 & 4 & -6\\ 7 & 3 & 5\\ 1 & -2 & 4\end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrix. *(AI 2015C)*

49. Write a 2×2 matrix which is both symmetric and skew symmetric. *(Delhi 2014C)*

50. For what value of x, is the matrix $A=\begin{bmatrix}0 & 1 & -2\\ -1 & 0 & 3\\ x & -3 & 0\end{bmatrix}$, a skew-symmetric matrix? *(AI 2013)*

SA (2 marks)

51. If A and B are symmetric matrices, such that

www.exambites.in

AB and BA are both defined, then prove that $AB - BA$ is a skew symmetric matrix. *(AI 2019)*

52. Show that all the diagonal elements of a skew symmetric matrix are zero. *(Delhi 2017)*

3.7 Elementary Operation (Transformation) of a Matrix

VSA (1 mark)

53. Use elementary column operation $C_2 \to C_2 + 2C_1$ in the following matrix equation:

$$\begin{pmatrix} 2 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$ *(AI 2016)*

54. Use elementary column operation $C_2 \to C_2 - 2C_1$ in the matrix equation

$$\begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}.$$ *(Foreign 2014)*

3.8 Invertible Matrices

LA 2 (6 marks)

55. Obtain the inverse of the following matrix using elementary operations :

$$A = \begin{bmatrix} 2 & 1 & -3 \\ -1 & -1 & 4 \\ 3 & 0 & 2 \end{bmatrix}$$ *(2020)*

56. Find the inverse of the following matrix using elementary operations $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$. *(Delhi 2019)*

57. Using elementary row transformations, find the inverse of the matrix $\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$. *(AI 2019)*

58. Using elementary row transformations, find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}.$$ *(2018)*

59. Using elementary operations, find the inverse of the following matrix :

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$ *(Delhi 2012)*

60. Using elementary transformations, find the inverse of the matrix.

$$\begin{pmatrix} 1 & 3 & -2 \\ -3 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}$$ *(Delhi 2011)*

Detailed Solutions

1. Here, $a_{11} = |(1)^2 - 1| = 0$, $a_{12} = |(1)^2 - 2| = 1$, $a_{21} = |(2)^2 - 1| = 3$ and $a_{22} = |(2)^2 - 2| = 2$

$\therefore$ Required matrix $= \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$.

2. As matrix is of order 2×2, so there are 4 entries possible.

Each entry has 3 choices *i.e.* 1, 2 or 3. So, the number of ways to make such matrices is $3 \times 3 \times 3 \times 3 = 81$.

3. Here, $a_{ij} = \frac{|i-j|}{2}$

$\therefore \quad a_{23} = \frac{|2-3|}{2} = \frac{1}{2}$ [For $i = 2, j = 3$]

4. Here, $a_{ij} = \frac{1}{2}|-3i + j|$

$\therefore \quad a_{32} = \frac{1}{2}|-3 \cdot 3 + 2|$ [For $i = 3, j = 2$]

$= \frac{1}{2}|-9 + 2| = \frac{1}{2}|-7| = \frac{7}{2}$

5. Here, $a_{ij} = \frac{(i+2j)^2}{4}$

$\therefore \quad a_{21} = \frac{(2+2\cdot 1)^2}{4} = 4$ [For $i = 2, j = 1$]

6. Here, $a_{ij} = \frac{i}{j} \Rightarrow a_{12} = \frac{1}{2}$ [For $i = 1, j = 2$]

www.examfree.in

7. The possible orders are 1×5 or 5×1.

8. Here, $\begin{bmatrix} x-y & z \\ 2x-y & w \end{bmatrix} = \begin{bmatrix} -1 & 4 \\ 0 & 5 \end{bmatrix}$

$\Rightarrow$ $x - y = -1$, $z = 4$, $2x - y = 0$, $w = 5$

Solving these equations for x and y, we get

$x = 1$, $y = 2$ $\therefore$ $x + y = 1 + 2 = 3$.

9. Given, $\begin{pmatrix} a+4 & 3b \\ 8 & -6 \end{pmatrix} = \begin{pmatrix} 2a+2 & b+2 \\ 8 & a-8b \end{pmatrix}$

By equality of matrices, we get

$a + 4 = 2a + 2$, $3b = b + 2$, $-6 = a - 8b$

On solving these equations, we get $a = 2$, $b = 1$.

So $a - 2b = 0$.

10. Here, $\begin{bmatrix} x \cdot y & 4 \\ z+6 & x+y \end{bmatrix} = \begin{bmatrix} 8 & w \\ 0 & 6 \end{bmatrix}$

$\Rightarrow$ $x \cdot y = 8$, $w = 4$

$z + 6 = 0$, $x + y = 6$

$\Rightarrow$ $z = -6$, $x + y = 6$

$\Rightarrow$ $x + y + z = 6 - 6 = 0$.

11. Given, $\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$

On comparing the corresponding elements, we get

$a - b = -1$, $2a - b = 0$

On subtraction, we get $a = 1$.

12. *Refer to answer 11.*

Now, we get $b = 2$

13. $\begin{bmatrix} x-y & 2y \\ 2y+z & x+y \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 9 & 5 \end{bmatrix}$

On comparing the corresponding elements, we get

$x - y = 1$, $2y = 4$, $2y + z = 9$, $x + y = 5$

$\Rightarrow$ $y = 2$, $x = 3$, $z = 5$

$\therefore$ $x + y + z = 3 + 2 + 5 = 10$.

14. Given, $\begin{bmatrix} 2x+1 & 2y \\ 0 & y^2+1 \end{bmatrix} = \begin{bmatrix} x+3 & 10 \\ 0 & 26 \end{bmatrix}$

$\Rightarrow$ $2x + 1 = x + 3$, $2y = 10$, $y^2 + 1 = 26$

From first two equations, we get $x = 2$, $y = 5$

which also satisfies $y^2 + 1 = 26$

$\therefore$ $x + y = 2 + 5 = 7$

15. Given, $\begin{bmatrix} x & x-y \\ 2x+y & 7 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 8 & 7 \end{bmatrix}$

$\Rightarrow$ $x = 3$, $x - y = 1$, $2x + y = 8 \Rightarrow x = 3$, $y = 2$.

Also, $2x + y = 8$ is satisfied by $x = 3$ and $y = 2$

$\therefore$ $x = 3$ and $y = 2$.

16. (a) : We have, $A^2 = A$

Now, $(I - A)^3 + A = (I - A)(I - A)(I - A) + A$

$= (I \cdot I - I \cdot A - A \cdot I + A \cdot A)(I - A) + A$

$= (I - A - A + A)(I - A) + A$

$[\because\ I \cdot A = A \cdot I = A$ and $A^2 = A]$

$= (I - A)(I - A) + A$

$= (I \cdot I - I \cdot A - A \cdot I + A \cdot A) + A$

$= (I - A - A + A) + A = (I - A) + A = I$

17. Given, $A^2 = I$

Consider, $(A - I)^3 + (A + I)^3 - 7A$

$= A^3 - I^3 - 3A^2I + 3AI^2 + A^3 + I^3 + 3A^2I + 3AI^2 - 7A$

$= 2A^3 + 6AI^2 - 7A = 2AA^2 + 6AI - 7A$

$= 2AI + 6A - 7A = 2A + 6A - 7A = A$

18. Given, $A = [2 \quad 1 \quad 3]\begin{bmatrix} -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$= [-2-1 \quad 1+3 \quad -2+3]\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = [-3 \quad 4 \quad 1]\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$= [-3 + 0 - 1] = [-4]$

$\therefore$ The order of matrix $A = 1 \times 1$

19. Given, $[x \ \ 1]\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} = O \Rightarrow [x - 2 \quad 0] = [0 \quad 0]$

$\Rightarrow x - 2 = 0 \Rightarrow x = 2$

20. We have, $2\begin{bmatrix} 3 & 4 \\ 5 & x \end{bmatrix} + \begin{bmatrix} 1 & y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 10 & 5 \end{bmatrix}$

$\Rightarrow$ $\begin{bmatrix} 6 & 8 \\ 10 & 2x \end{bmatrix} + \begin{bmatrix} 1 & y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 10 & 5 \end{bmatrix}$

$\Rightarrow$ $\begin{bmatrix} 7 & 8+y \\ 10 & 2x+1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 10 & 5 \end{bmatrix}$

On comparing corresponding elements of the matrices, we get

$8 + y = 0$ and $2x + 1 = 5$ $\Rightarrow$ $y = -8$ and $x = 2$

$\therefore$ $x - y = 2 + 8 = 10$

21. Here $A^2 = A$

Now, $7A - (I + A)^3$

$= 7A - (I + A)(I + A)(I + A)$

$= 7A - (I + A)(I \cdot I + I \cdot A + A \cdot I + (A \cdot A))$

$= 7A - (I + A)(I + A + A + A)$

$(\because\ I \cdot A = A \cdot I = A$ and $A^2 = A)$

$= 7A - (I + A)(I + 3A)$

$= 7A - (I \cdot I + I \cdot (3A) + A \cdot I + A \cdot (3A))$

$= 7A - (I + 3A + A + 3A) = 7A - I - 7A = -I$.

www.exambites.in

22. Here, $(2x \quad 4)\begin{pmatrix} x \\ -8 \end{pmatrix} = O$

$\Rightarrow \quad 2x \cdot x + 4 \cdot (-8) = 0$
$\Rightarrow \quad 2x^2 - 32 = 0$
$\Rightarrow \quad x^2 = 16 = 4^2 \Rightarrow x = 4$
which is the required positive value of x.

23. $A = \begin{bmatrix} 9 & -1 & 4 \\ -2 & 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 9 \end{bmatrix}$

$\Rightarrow A = \begin{bmatrix} 9-1 & -1-2 & 4-(-1) \\ -2-0 & 1-4 & 3-9 \end{bmatrix} = \begin{bmatrix} 8 & -3 & 5 \\ -2 & -3 & -6 \end{bmatrix}$

24. Here, $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and $A^2 = kA$

$\Rightarrow \quad \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = k\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = k\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$\Rightarrow \quad 2\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = k\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow k = 2$

25. *Refer to answer 24.*

26. *Refer to answer 24.*

27. Given, $2\begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} 2 & 6 \\ 0 & 2x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} 2+y & 6 \\ 1 & 2x+2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$

$\Rightarrow$ $2 + y = 5, 2x + 2 = 8 \Rightarrow y = 3, x = 3$
$\therefore \quad x + y = 6$

28. $\cos\theta\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \sin\theta\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

$= \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ -\sin\theta\cos\theta & \cos^2\theta \end{bmatrix} + \begin{bmatrix} \sin^2\theta & -\cos\theta\sin\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}$

$= \begin{bmatrix} \cos^2\theta + \sin^2\theta & \cos\theta\sin\theta - \cos\theta\sin\theta \\ -\sin\theta\cos\theta + \sin\theta\cos\theta & \cos^2\theta + \sin^2\theta \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

29. Given, $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}\begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} 2-6 & -6+12 \\ 5-14 & -15+28 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} -4 & 6 \\ -9 & 13 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix} \Rightarrow x = 13$

30. Given, $2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} 2x & 10 \\ 14 & 2y-6 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$

$\Rightarrow \quad \begin{bmatrix} 2x+3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$

$\Rightarrow$ $2x + 3 = 7, 2y - 4 = 14$
$\Rightarrow$ $2x = 4, 2y = 18 \Rightarrow x = 2, y = 9$
$\therefore \quad x + y = 2 + 9 = 11$

31. Here, $3A - B = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$

$\therefore \quad (3A - B) + B = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$

$\Rightarrow \quad 3A = \begin{bmatrix} 5+4 & 0+3 \\ 1+2 & 1+5 \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3 & 6 \end{bmatrix}$

$\Rightarrow \quad 3A = 3\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

32. Let $A = \begin{bmatrix} x & y & z \\ p & q & r \end{bmatrix}$ [$\because$ B and C are 2×3 matrices]

Given, $2A - 3B + 5C = O$

$\Rightarrow \quad 2A = 3B - 5C \Rightarrow A = \frac{1}{2}[3B - 5C]$ ...(i)

Now, $3B - 5C = 3\begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix} - 5\begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$

$= \begin{bmatrix} -6 & 6 & 0 \\ 9 & 3 & 12 \end{bmatrix} - \begin{bmatrix} 10 & 0 & -10 \\ 35 & 5 & 30 \end{bmatrix}$

$= \begin{bmatrix} -16 & 6 & 10 \\ -26 & -2 & -18 \end{bmatrix}$

From (i), we get $A = \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix}$.

www.examhits.in

33. We have,

$$A=\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix},\ B=\begin{bmatrix}5 & 2\\ 7 & 4\end{bmatrix},\ C=\begin{bmatrix}2 & 5\\ 3 & 8\end{bmatrix}$$

Let, $D=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

Now, $CD - AB = O$

$$\therefore \quad \begin{bmatrix}2 & 5\\ 3 & 8\end{bmatrix}\begin{bmatrix}a & b\\ c & d\end{bmatrix}-\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}\begin{bmatrix}5 & 2\\ 7 & 4\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}2a+5c & 2b+5d\\ 3a+8c & 3b+8d\end{bmatrix}-\begin{bmatrix}10-7 & 4-4\\ 15+28 & 6+16\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}2a+5c-3 & 2b+5d\\ 3a+8c-43 & 3b+8d-22\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

On comparing the corresponding elements of the matrices, we get

$2a + 5c - 3 = 0$ ...(i) and $3a + 8c - 43 = 0$ ...(ii)

Also, $2b + 5d = 0$ ...(iii) and $3b + 8d - 22 = 0$ ...(iv)

Solving (i) and (ii), we get $a = -191$, $c = 77$

Solving (iii) and (iv), we get $b = -110$, $d = 44$

$$\therefore \quad D=\begin{bmatrix}-191 & -110\\ 77 & 44\end{bmatrix}$$

34. Given that, $\begin{bmatrix}2 & -1\\ 1 & 0\\ -3 & 4\end{bmatrix}A=\begin{bmatrix}-1 & -8\\ 1 & -2\\ 9 & 22\end{bmatrix}$

Let $X=\begin{bmatrix}2 & -1\\ 1 & 0\\ -3 & 4\end{bmatrix}_{3\times 2}$ and $Y=\begin{bmatrix}-1 & -8\\ 1 & -2\\ 9 & 22\end{bmatrix}_{3\times 2}$

As order of X is 3×2, then A should be of order 2×2, so that we get Y matrix of order 3×2.

Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

Now, $\begin{bmatrix}2 & -1\\ 1 & 0\\ -3 & 4\end{bmatrix}\begin{bmatrix}a & b\\ c & d\end{bmatrix}=\begin{bmatrix}-1 & -8\\ 1 & -2\\ 9 & 22\end{bmatrix}$

$$\Rightarrow \begin{bmatrix}2a-c & 2b-d\\ a+0 & b+0\\ -3a+4c & -3b+4d\end{bmatrix}=\begin{bmatrix}-1 & -8\\ 1 & -2\\ 9 & 22\end{bmatrix}$$

On comparing the corresponding elements, we get

$2a - c = -1$ ...(i), $2b - d = -8$ ...(ii),

$a = 1$ ...(iii) and $b = -2$ ...(iv)

Substituting $a = 1$ in (i), we get $c = 3$

and substituting $b = -2$ in (ii), we get $d = 4$

So, $A=\begin{bmatrix}1 & -2\\ 3 & 4\end{bmatrix}$

35. $A^2 - 5A + 4I$

$$=\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}-5\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}+4\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}5 & -1 & 2\\ 9 & -2 & 5\\ 0 & -1 & -2\end{bmatrix}-\begin{bmatrix}10 & 0 & 5\\ 10 & 5 & 15\\ 5 & -5 & 0\end{bmatrix}+\begin{bmatrix}4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\end{bmatrix}$$

$$=\begin{bmatrix}9 & -1 & 2\\ 9 & 2 & 5\\ 0 & -1 & 2\end{bmatrix}-\begin{bmatrix}10 & 0 & 5\\ 10 & 5 & 15\\ 5 & -5 & 0\end{bmatrix}=\begin{bmatrix}-1 & -1 & -3\\ -1 & -3 & -10\\ -5 & 4 & 2\end{bmatrix}$$

Since, $A^2 - 5A + 4I + X = O \Rightarrow X = -(A^2 - 5A + 4I)$

$$\therefore \quad X=\begin{bmatrix}1 & 1 & 3\\ 1 & 3 & 10\\ 5 & -4 & -2\end{bmatrix}$$

36. The number of articles sold by each school can be written in the matrix form as

$$X=\begin{bmatrix}40 & 25 & 35\\ 50 & 40 & 50\\ 20 & 30 & 40\end{bmatrix}$$

The cost of each article can be written in the matrix form as $Y = [25 \quad 100 \quad 50]$

The fund collected by each school is given by

$$YX=[25\ \ 100\ \ 50]\begin{bmatrix}40 & 25 & 35\\ 50 & 40 & 50\\ 20 & 30 & 40\end{bmatrix}$$

$= [7000 \quad 6125 \quad 7875]$

Therefore, the funds collected by schools A, B and C are ₹ 7000, ₹ 6125 and ₹ 7875 respectively

Thus, the total fund collected

= ₹ (7000 + 6125 + 7875) = ₹ 21000

The situation highlights the helping nature of the students.

37. Let ₹ A, ₹ B and ₹ C be the cost incurred by the organisation for villages X, Y and Z respectively.

www.examrites.in

Then we get the matrix eq. as

$$\begin{bmatrix}400 & 300 & 100\\ 300 & 250 & 75\\ 500 & 400 & 150\end{bmatrix}\begin{bmatrix}50\\20\\40\end{bmatrix}=\begin{bmatrix}A\\B\\C\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}A\\B\\C\end{bmatrix}=\begin{bmatrix}400\times50+300\times20+100\times40\\300\times50+250\times20+75\times40\\500\times50+400\times20+150\times40\end{bmatrix}$$

$$=\begin{bmatrix}20{,}000+6{,}000+4{,}000\\15{,}000+5{,}000+3{,}000\\25{,}000+8{,}000+6{,}000\end{bmatrix}=\begin{bmatrix}30{,}000\\23{,}000\\39{,}000\end{bmatrix}$$

$\therefore$ A = ₹ 30,000; B = ₹ 23,000 and C = ₹ 39,000

These are respectively the costs incurred by the organisation on villages X, Y and Z respectively.

The value generated by the organisation in the society is cleanliness.

38. We have, $A=\begin{bmatrix}1 & -1\\2 & -1\end{bmatrix}$ and $B=\begin{bmatrix}a & 1\\b & -1\end{bmatrix}$

Consider, $(A+B)=\begin{bmatrix}1 & -1\\2 & -1\end{bmatrix}+\begin{bmatrix}a & 1\\b & -1\end{bmatrix}=\begin{bmatrix}1+a & 0\\2+b & -2\end{bmatrix}$

Now, $(A+B)^2=\begin{bmatrix}1+a & 0\\2+b & -2\end{bmatrix}\begin{bmatrix}1+a & 0\\2+b & -2\end{bmatrix}$

$$=\begin{bmatrix}(1+a)^2 & 0\\(2+b)(1+a-2) & 4\end{bmatrix}=\begin{bmatrix}(1+a)^2 & 0\\(2+b)(a-1) & 4\end{bmatrix}$$

Now, consider $A^2=\begin{bmatrix}1 & -1\\2 & -1\end{bmatrix}\begin{bmatrix}1 & -1\\2 & -1\end{bmatrix}$

$$=\begin{bmatrix}1-2 & -1+1\\2-2 & -2+1\end{bmatrix}=\begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}$$

and $B^2=\begin{bmatrix}a & 1\\b & -1\end{bmatrix}\begin{bmatrix}a & 1\\b & -1\end{bmatrix}=\begin{bmatrix}a^2+b & a-1\\ab-b & b+1\end{bmatrix}$

$$\therefore\quad A^2+B^2=\begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}+\begin{bmatrix}a^2+b & a-1\\ab-b & b+1\end{bmatrix}$$

$$=\begin{bmatrix}a^2+b-1 & a-1\\ab-b & b\end{bmatrix}$$

It is given that $(A+B)^2=A^2+B^2$

$$\therefore\quad \begin{bmatrix}(1+a)^2 & 0\\(2+b)(a-1) & 4\end{bmatrix}=\begin{bmatrix}a^2+b-1 & a-1\\ab-b & b\end{bmatrix}$$

By equality of matrices, comparing the corresponding elements, we get

$a-1=0\Rightarrow a=1$ and $b=4$

Also, $(1+a)^2=a^2+b-1$ and $(2+b)(a-1)=ab-b$ satisfied by $a=1$ and $b=4$

Therefore, $a=1$ and $b=4$.

39. The total amount spent by the party in two cities X and Y is represented in the matrix equation by matrix C as,

$$C=BA$$

$$\Rightarrow \begin{bmatrix}X\\Y\end{bmatrix}=\begin{bmatrix}1000 & 500 & 5000\\3000 & 1000 & 10000\end{bmatrix}\begin{bmatrix}140\\200\\150\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}X\\Y\end{bmatrix}=\begin{bmatrix}1000\times140+500\times200+5000\times150\\3000\times140+1000\times200+10000\times150\end{bmatrix}$$

$$=\begin{bmatrix}990000\\2120000\end{bmatrix}$$

$\Rightarrow$ X = 990000 paise, Y = 2120000 paise

$\therefore$ X = ₹ 9900 and Y = ₹ 21200

i.e., Amount spent by the party in city X and Y is ₹ 9900 and ₹ 21200 respectively. One should consider about the social activities of a political party before casting his/her vote.

40. Here, $\begin{bmatrix}2x & 3\end{bmatrix}\begin{bmatrix}1 & 2\\-3 & 0\end{bmatrix}\begin{bmatrix}x\\3\end{bmatrix}=O$

$$\Rightarrow \begin{bmatrix}2x & 3\end{bmatrix}\begin{bmatrix}x+6\\-3x\end{bmatrix}=O$$

$\Rightarrow$ $2x(x+6)+3(-3x)=0\ \Rightarrow\ 2x^2+12x-9x=0$

$\Rightarrow$ $2x^2+3x=0\Rightarrow x(2x+3)=0\ \Rightarrow\ x=0,\frac{-3}{2}$.

41. Trust fund = ₹ 35,000.

Let ₹ x be invested in the first bond and then ₹ $(35{,}000-x)$ will be invested in the second bond.

Interest paid on the first bond = 8% = 0.08

Interest paid on the second bond = 10% = 0.10

Total annual interest = ₹ 3,200.

$$\therefore\quad \text{In matrices, } [x\ \ 35{,}000-x]\begin{bmatrix}0.08\\0.10\end{bmatrix}=[3{,}200]$$

$\Rightarrow$ $x\times0.08+(35{,}000-x)\times0.10=3{,}200$

$$\Rightarrow\quad x\times\frac{8}{100}+(35{,}000-x)\times\frac{10}{100}=3{,}200$$

$\Rightarrow$ $8x+3{,}50{,}000-10x=3{,}20{,}000$

$\Rightarrow$ $2x=30{,}000\ \Rightarrow\ x=15{,}000$

$\therefore$ ₹ 15,000 should be invested in the first bond and ₹ 35,000 – ₹ 15,000 = ₹ 20,000 should be invested in the second bond.

www.examples.in

The values reflected in this question are :
(i) Spirit of investment.
(ii) Giving charity to cancer patients.
(iii) Helping the orphans living in the society.

42. If A is a matrix of order 3×2, then the order of the matrix A' is 2×3.

43. Given, $A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$

$\Rightarrow \quad B^T = \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$

$\therefore \quad A^T - B^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix}$

44. A square matrix A is said to be skew-symmetric, if $A' = -A$.

45. A square matrix A is said to be skew symmetric matrix if $A = -A'$ ...(i)

Now, $A = \begin{bmatrix} 0 & a & -3 \\ 2 & 0 & -1 \\ b & 1 & 0 \end{bmatrix} \quad \therefore \quad A' = \begin{bmatrix} 0 & 2 & b \\ a & 0 & 1 \\ -3 & -1 & 0 \end{bmatrix}$

From (i), $A + A' = O$

$\Rightarrow \begin{bmatrix} 0 & a & -3 \\ 2 & 0 & -1 \\ b & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 & b \\ a & 0 & 1 \\ -3 & -1 & 0 \end{bmatrix} = O$

$\Rightarrow \begin{bmatrix} 0 & 2+a & b-3 \\ a+2 & 0 & 0 \\ b-3 & 0 & 0 \end{bmatrix} = O$

$\Rightarrow \quad a + 2 = 0 \;\&\; b - 3 = 0$

$\therefore \quad a = -2 \;\&\; b = 3$

46. Given, $A = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$

$\because$ A is symmetric.

$\therefore \quad A' = A$

$\Rightarrow \begin{bmatrix} 0 & 3 & 3a \\ 2b & 1 & 3 \\ -2 & 3 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$

On comparing the corresponding elements of the matrices, we get $a = \frac{-2}{3}$ and $b = \frac{3}{2}$.

47. Given, $A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} \Rightarrow A' = \begin{bmatrix} 3 & 7 \\ 5 & 9 \end{bmatrix}$

$\because$ P is symmetric matrix. So, $P = \frac{1}{2}(A + A')$

$\therefore \quad P = \frac{1}{2}\left(\begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} 3 & 7 \\ 5 & 9 \end{bmatrix}\right)$

$= \frac{1}{2}\begin{bmatrix} 3+3 & 5+7 \\ 7+5 & 9+9 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 6 & 12 \\ 12 & 18 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$

Hence, the matrix $P = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$

48. We know that a square matrix A can be written as

$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

Out of which $\frac{1}{2}(A + A^T)$ is symmetric and $\frac{1}{2}(A - A^T)$ is skew symmetric matrix.

$\therefore$ For the given matrix

$A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}, A^T = \begin{bmatrix} 2 & 7 & 1 \\ 4 & 3 & -2 \\ -6 & 5 & 4 \end{bmatrix}$

$\therefore \quad A + A^T = \begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix}$ and

$A - A^T = \begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$

Hence, $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

$= \begin{bmatrix} 2 & 11/2 & -5/2 \\ 11/2 & 3 & 3/2 \\ -5/2 & 3/2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & -3/2 & -7/2 \\ 3/2 & 0 & 7/2 \\ 7/2 & -7/2 & 0 \end{bmatrix}$

First of which is symmetric and the second is skew symmetric matrix.

49. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is a 2×2 symmetric as well as skew symmetric matrix.

www.exambites.in

50. The matrix $A=\begin{bmatrix}0 & 1 & -2\\ -1 & 0 & 3\\ x & -3 & 0\end{bmatrix}$ is skew-symmetric.

$\therefore \quad A'=-A \Rightarrow \begin{bmatrix}0 & -1 & x\\ 1 & 0 & -3\\ -2 & 3 & 0\end{bmatrix}=\begin{bmatrix}0 & -1 & 2\\ 1 & 0 & -3\\ -x & 3 & 0\end{bmatrix}$

$\Rightarrow \quad x=2$

51. Given, A and B are symmetric matrices
$\therefore \quad A'=A$ and $B'=B$

$$\text{Now, } (AB-BA)'=(AB)'-(BA)'$$
$$=(B'A')-(A'B')$$
$$=(BA-AB) \quad [\because A'=A \text{ and } B'=B]$$
$$=-(AB-BA)$$
$$\text{Thus, } (AB-BA)'=-(AB-BA)$$

Hence,$(AB - BA)$ is a skew symmetric matrix.

52. Let $A = [a_{ij}]$ be a skew symmetric matrix
Then, $a_{ji} = - a_{ij} \ \forall \ i, j$

$\Rightarrow \quad a_{ii} = - a_{ii} \ \forall \ i \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0 \ \forall \ i$
$\Rightarrow \quad a_{11} = a_{22} = a_{33} = ... = a_{nn} = 0$

53. We have, $\begin{pmatrix}2 & 1\\ 2 & 0\end{pmatrix}=\begin{pmatrix}3 & 1\\ 2 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ -1 & 1\end{pmatrix}$

On applying $C_2 \to C_2 + 2C_1$, we get

$$\begin{pmatrix}2 & 5\\ 2 & 4\end{pmatrix}=\begin{pmatrix}3 & 1\\ 2 & 0\end{pmatrix}\begin{pmatrix}1 & 2\\ -1 & -1\end{pmatrix}$$

54. We have, $\begin{pmatrix}4 & 2\\ 3 & 3\end{pmatrix}=\begin{pmatrix}1 & 2\\ 0 & 3\end{pmatrix}\begin{pmatrix}2 & 0\\ 1 & 1\end{pmatrix}$

Applying $C_2 \to C_2 - 2C_1$, we get

$$\begin{pmatrix}4 & -6\\ 3 & -3\end{pmatrix}=\begin{pmatrix}1 & 2\\ 0 & 3\end{pmatrix}\begin{pmatrix}2 & -4\\ 1 & -1\end{pmatrix}$$

55. We have, $A=\begin{bmatrix}2 & 1 & -3\\ -1 & -1 & 4\\ 3 & 0 & 2\end{bmatrix}$

We know that $A = IA$

$$\therefore \quad \begin{bmatrix}2 & 1 & -3\\ -1 & -1 & 4\\ 3 & 0 & 2\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}A$$

Applying $R_1 \leftrightarrow R_2$, we get

$$\begin{bmatrix}-1 & -1 & 4\\ 2 & 1 & -3\\ 3 & 0 & 2\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}A$$

Applying $R_2 \to R_2 + 2R_1$, $R_3 \to R_3 + 3R_1$, we get

$$\begin{bmatrix}-1 & -1 & 4\\ 0 & -1 & 5\\ 0 & -3 & 14\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 1 & 2 & 0\\ 0 & 3 & 1\end{bmatrix}A$$

Applying $R_1 \to (-1)R_1$ and $R_2 \to (-1)R_2$, we get

$$\begin{bmatrix}1 & 1 & -4\\ 0 & 1 & -5\\ 0 & -3 & 14\end{bmatrix}=\begin{bmatrix}0 & -1 & 0\\ -1 & -2 & 0\\ 0 & 3 & 1\end{bmatrix}A$$

Applying $R_3 \to R_3 + 3R_2$, we get

$$\begin{bmatrix}1 & 1 & -4\\ 0 & 1 & -5\\ 0 & 0 & -1\end{bmatrix}=\begin{bmatrix}0 & -1 & 0\\ -1 & -2 & 0\\ -3 & -3 & 1\end{bmatrix}A$$

Applying $R_1 \to R_1 - R_2$, we get

$$\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & -5\\ 0 & 0 & -1\end{bmatrix}=\begin{bmatrix}1 & 1 & 0\\ -1 & -2 & 0\\ -3 & -3 & 1\end{bmatrix}A$$

Applying $R_1 \to R_1 + R_3$, we get

$$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & -5\\ 0 & 0 & -1\end{bmatrix}=\begin{bmatrix}-2 & -2 & 1\\ -1 & -2 & 0\\ -3 & -3 & 1\end{bmatrix}A$$

Applying $R_2 \to R_2 - 5R_3$, we get

$$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & -1\end{bmatrix}=\begin{bmatrix}-2 & -2 & 1\\ 14 & 13 & -5\\ -3 & -3 & 1\end{bmatrix}A$$

Applying $R_3 \to (-1)R_3$, we get

$$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}-2 & -2 & 1\\ 14 & 13 & -5\\ 3 & 3 & -1\end{bmatrix}A$$

Hence, $A^{-1}=\begin{bmatrix}-2 & -2 & 1\\ 14 & 13 & -5\\ 3 & 3 & -1\end{bmatrix}$

56. $A=\begin{bmatrix}1 & 2 & -2\\ -1 & 3 & 0\\ 0 & -2 & 1\end{bmatrix}$

www.exambites.in

Since, $A = IA$

$$\therefore \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_2 \to R_2 + R_1$, we get

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & 5 & -2 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_2 \to \frac{1}{5}R_2$, we get

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -\frac{2}{5} \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_1 \to R_1 - 2R_2$, $R_3 \to R_3 + 2R_2$, we get

$$\begin{bmatrix} 1 & 0 & -\frac{6}{5} \\ 0 & 1 & -\frac{2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & 0 \\ \frac{2}{5} & \frac{2}{5} & 1 \end{bmatrix} A$$

Applying $R_3 \to 5R_3$, we get

$$\begin{bmatrix} 1 & 0 & -\frac{6}{5} \\ 0 & 1 & -\frac{2}{5} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & 0 \\ 2 & 2 & 5 \end{bmatrix} A$$

Applying $R_1 \to R_1 + \frac{6}{5}R_3$, $R_2 \to R_2 + \frac{2}{5}R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} A$$

Hence, $A^{-1} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$

57. Let $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$.

We know, $A = IA$

$$\therefore \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_1 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 1 & 1 & -2 \\ 3 & 2 & -4 \\ 2 & -3 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

Applying $R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - 2R_1$, we get

$$\begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 2 \\ 0 & -5 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & 0 & -2 \end{bmatrix} A$$

Applying $R_3 \to R_3 - 5R_2$, we get

$$\begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -5 & 13 \end{bmatrix} A$$

Applying $R_1 \to R_1 + R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 1 & -3 \\ 1 & -5 & 13 \end{bmatrix} A$$

Applying $R_2 \to R_2 + 2R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} A$$

Applying $R_2 \to (-1)\, R_2$ and $R_3 \to (-1)\, R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} A$$

Hence, $A^{-1} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$

58. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}$

Since, $A = IA$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 + 2R_1$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} A$$

www.examrites.in

Applying $R_1 \to R_1 - 2R_2$, we get

$$\begin{bmatrix}1&0&1\\0&1&1\\0&0&1\end{bmatrix}=\begin{bmatrix}5&-2&0\\-2&1&0\\2&0&1\end{bmatrix}A$$

Applying $R_1 \to R_1 - R_3$, $R_2 \to R_2 - R_3$, we get

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}3&-2&-1\\-4&1&-1\\2&0&1\end{bmatrix}A$$

$$\therefore \quad A^{-1}=\begin{bmatrix}3&-2&-1\\-4&1&-1\\2&0&1\end{bmatrix}$$

59. We have $A = IA$

$$\Rightarrow \quad \begin{bmatrix}-1&1&2\\1&2&3\\3&1&1\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}A$$

Applying $R_2 \to R_2 + R_1$, $R_3 \to R_3 + 3R_1$, we get

$$\begin{bmatrix}-1&1&2\\0&3&5\\0&4&7\end{bmatrix}=\begin{bmatrix}1&0&0\\1&1&0\\3&0&1\end{bmatrix}A$$

Applying $R_2 \to R_2 - R_3$, we get

$$\begin{bmatrix}-1&1&2\\0&-1&-2\\0&4&7\end{bmatrix}=\begin{bmatrix}1&0&0\\-2&1&-1\\3&0&1\end{bmatrix}A$$

Applying $R_1 \to R_1 + R_2$, we get

$$\begin{bmatrix}-1&0&0\\0&-1&-2\\0&4&7\end{bmatrix}=\begin{bmatrix}-1&1&-1\\-2&1&-1\\3&0&1\end{bmatrix}A$$

Applying $R_3 \to R_3 + 4R_2$, we get

$$\begin{bmatrix}-1&0&0\\0&-1&-2\\0&0&-1\end{bmatrix}=\begin{bmatrix}-1&1&-1\\-2&1&-1\\-5&4&-3\end{bmatrix}A$$

Applying $R_2 \to R_2 - 2R_3$, we get

$$\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}=\begin{bmatrix}-1&1&-1\\8&-7&5\\-5&4&-3\end{bmatrix}A$$

Applying $R_1 \to (-1)R_1$, $R_2 \to (-1)R_2$, $R_3 \to (-1)R_3$

$$\Rightarrow \quad \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}1&-1&1\\-8&7&-5\\5&-4&3\end{bmatrix}A$$

$$\text{Hence, } A^{-1}=\begin{bmatrix}1&-1&1\\-8&7&-5\\5&-4&3\end{bmatrix}$$

60. Consider $A = \begin{pmatrix}1&3&-2\\-3&0&-1\\2&1&0\end{pmatrix}$

We have $A = IA$

$$\begin{pmatrix}1&3&-2\\-3&0&-1\\2&1&0\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}A$$

Applying $R_2 \to R_2 + 3R_1$ and $R_3 \to R_3 - 2R_1$, we get

$$\begin{pmatrix}1&3&-2\\0&9&-7\\0&-5&4\end{pmatrix}=\begin{pmatrix}1&0&0\\3&1&0\\-2&0&1\end{pmatrix}A$$

Applying $R_2 \to \frac{R_2}{9}$, we get

$$\begin{pmatrix}1&3&-2\\0&1&-7/9\\0&-5&4\end{pmatrix}=\begin{pmatrix}1&0&0\\1/3&1/9&0\\-2&0&1\end{pmatrix}A$$

Applying $R_3 \to R_3 + 5R_2$, we get

$$\begin{pmatrix}1&3&-2\\0&1&-7/9\\0&0&1/9\end{pmatrix}=\begin{pmatrix}1&0&0\\1/3&1/9&0\\-1/3&5/9&1\end{pmatrix}A$$

Applying $R_3 \to 9R_3$, we get

$$\begin{pmatrix}1&3&-2\\0&1&-7/9\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\1/3&1/9&0\\-3&5&9\end{pmatrix}A$$

Applying $R_1 \to R_1 - 3R_2$, we get

$$\begin{pmatrix}1&0&1/3\\0&1&-7/9\\0&0&1\end{pmatrix}=\begin{pmatrix}0&-1/3&0\\1/3&1/9&0\\-3&5&9\end{pmatrix}A$$

Applying $R_1 \to R_1 - \frac{1}{3}R_3, R_2 \to R_2 + \frac{7}{9}R_3$, we get

$$\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&-2&-3\\-2&4&7\\-3&5&9\end{pmatrix}A$$

$$\therefore \quad A^{-1}=\begin{pmatrix}1&-2&-3\\-2&4&7\\-3&5&9\end{pmatrix}$$

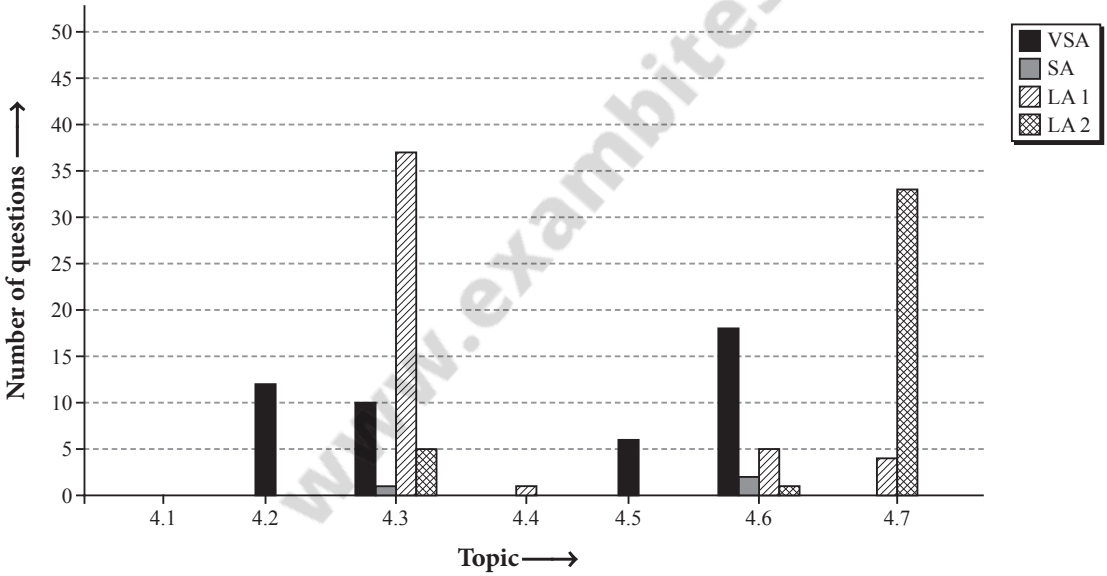
www.examples.in

CHAPTER
4

Determinants

4.1 Introduction
4.2 Determinant
4.3 Properties of Determinants
4.4 Area of a Triangle
4.5 Minors and Cofactors
4.6 Adjoint and Inverse of a Matrix
4.7 Applications of Determinants and Matrices

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Properties of Determinants*
- Maximum LA 1 type questions were asked from *Properties of Determinants*
- Maximum LA 2 type questions were asked from *Applications of Determinants and Matrices*

QUICK RECAP

DETERMINANT

- If $A = [a_{ij}]$ is a square matrix of order n, then a number (real or complex) associated to matrix A is called determinant of A.

  It is denoted by det A or $|A|$ or Δ.

- If $A = [a]$ be a matrix of order 1, then det $(A) = a$.
- If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ be a matrix of order 2, then $|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$

- If $A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a matrix of order 3, then

$$|A|=\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}=(-1)^{1+1}a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$+(-1)^{1+2}a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}+(-1)^{1+3}a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{11}(a_{22}\,a_{33}-a_{32}\,a_{23})-a_{12}\,(a_{21}\,a_{33}-a_{31}\,a_{23}) + a_{13}\,(a_{21}\,a_{32}-a_{31}\,a_{22})$$

Note : $|KA| = K^n|A|$, where A is of order n.

Properties of Determinants

- The value of the determinant remains unchanged if its rows and columns are interchanged.
- If any two rows (or columns) of a determinant are interchanged, then the value of the determinant is multiplied by –1.
- If any two rows (or columns) of a determinant are identical then the value of determinant is zero.
- If the elements of a row (or column) of a determinant are multiplied by a scalar, then the value of the new determinant is equal to same scalar times the value of the original determinant.
- If each element of any row (or column) of a determinant is the sum of two (or more) terms then the determinant is expressible as the sum of two (or more) determinants of the same order.
- The value of a determinant does not change when any row (or column) is multiplied by a number or an expression and then added to or subtracted from any other row (or column).

AREA OF A TRIANGLE

The area of a triangle whose vertices are (x_1, y_1), (x_2, y_2) and (x_3, y_3) is given by

$$\Delta=\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$=\frac{1}{2}\{x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\}$$

Note :

(i) Since area is always a positive quantity, therefore we always take the absolute value of the determinant for the area.

(ii) Area of a triangle formed by three collinear points is always zero.

MINOR OF AN ELEMENT

Minor of an element a_{ij} of the determinant of matrix A is the determinant obtained by deleting i^{th} row and j^{th} column. Minor of a_{ij} is denoted by M_{ij}.

- Minor of an element of a determinant of order $n(n \geq 2)$ is a determinant of order $n - 1$.

COFACTOR OF AN ELEMENT

Cofactor of an element a_{ij} of determinant of matrix A is, $A_{ij} = (-1)^{i+j} M_{ij}$.

- The determinant of a matrix A can also be obtained by using cofactors *i.e.*, sum of product of elements of a row (or column) with their corresponding cofactors.
  $\therefore\ |A| = a_{11}\,A_{11} + a_{12}\,A_{12} + a_{13}\,A_{13}$
- If the elements of one row (or column) are multiplied with the cofactors of elements of any other row (or column), then their sum is zero *i.e.*, $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 0$

ADJOINT OF A MATRIX

The adjoint of a square matrix is the transpose of the matrix of cofactors.

- Adjoint of A is denoted by adj$A$.

Remark : For a matrix A of order n, $A(\text{adj}A) = (\text{adj}A)A = |A|I_n$

SINGULAR AND NON-SINGULAR MATRIX

Let A be a square matrix, then A is called

- Singular matrix, iff $|A| = 0$
- Non-singular matrix, iff $|A| \neq 0$

Note : If A and B are non-singular matrices of same order, then AB and BA are also non-singular matrices of same order.

www.exambres.in

INVERSE OF A MATRIX

A square matrix A is invertible iff A is non-singular matrix and

$$A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

SOLUTION OF A SYSTEM OF LINEAR EQUATIONS

For a square matrix A, a system of equations $AX = B$ is said to be

(i) Consistent, if it has one or more solutions.

(ii) Inconsistent, if its solution doesn't exist.

Conditions for Consistency

For a system of equations, $AX = B$.

(i) If $|A| \neq 0$, then the given system of equations is consistent and has a unique solution.

(ii) If $|A| = 0$ and $(\text{adj}A)B \neq O$, then the solution does not exist and the given system is inconsistent.

(iii) If $|A| = 0$ and $(\text{adj}A)B = O$, then the given system may be either consistent or inconsistent, according as the system have either infinitely many solutions or no solution.

www.exambites.in

Previous Years' CBSE Board Questions

4.2 Determinant

VSA (1 mark)

1. Find the maximum value of
$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$. *(Delhi 2016)*

2. If $x \in N$ and $\begin{vmatrix} x+3 & -2 \\ -3x & 2x \end{vmatrix} = 8$, then find the value of x. *(AI 2016)*

3. If $\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix} = 8$, write the value of x. *(Foreign 2016)*

4. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$, write the value of $|AB|$. *(Delhi 2015C)*

5. If $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, write the value of x. *(Delhi 2014)*

6. If $\begin{vmatrix} 3x & 7 \\ -2 & 4 \end{vmatrix} = \begin{vmatrix} 8 & 7 \\ 6 & 4 \end{vmatrix}$, find the value of x. *(AI 2014)*

7. Write the value of the determinant $\begin{vmatrix} p & p+1 \\ p-1 & p \end{vmatrix}$. *(Delhi 2014C)*

8. Write the value of $\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$. *(AI 2014C)*

9. If $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$, then write the value of x. *(Delhi 2013)*

10. If $\begin{vmatrix} 2x & x+3 \\ 2(x+1) & x+1 \end{vmatrix} = \begin{vmatrix} 1 & 5 \\ 3 & 3 \end{vmatrix}$, then write the value of x. *(Delhi 2013C)*

11. Evaluate : $\begin{vmatrix} \cos 15^\circ & \sin 15^\circ \\ \sin 75^\circ & \cos 75^\circ \end{vmatrix}$. *(AI 2011)*

12. If $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$, find the value of $3|A|$. *(AI 2011C)*

4.3 Properties of Determinants

VSA (1 mark)

13. If A is a square matrix of order 3 and $|A| = 5$, then the value of $|2A'|$ is
(a) -10 (b) 10
(c) -40 (d) 40 *(2020)*

14. If A is a skew-symmetric matrix of order 3, then the value of $|A|$ is
(a) 3 (b) 0
(c) 9 (d) 27 *(2020)*

15. If A is a 3×3 matrix such that $|A| = 8$, then $|3A|$ equals
(a) 8 (b) 24
(c) 72 (d) 216 *(2020)*

16. If A and B are square matrices each of order 3 and $|A| = 5$, $|B| = 3$, then the value of $|3AB|$ is _____. *(2020)*

17. If A and B are square matrices of the same order 3, such that $|A| = 2$ and $AB = 2I$, write the value of $|B|$. *(Delhi 2019)*

18. Write the value of $\Delta = \begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ -3 & -3 & -3 \end{vmatrix}$. *(AI 2015)*

19. If A is a 3×3 matrix, $|A| \neq 0$ and $|3A| = k|A|$, then write the value of k. *(Foreign 2014)*

20. Let A be a square matrix of order 3×3. Write the value of $|2A|$, where $|A| = 4$. *(AI 2012, Delhi 2011C)*

21. The value of the determinant of a matrix A of order 3×3 is 4. Find the value of $|5A|$. *(Delhi 2012C)*

www.exambites.in

22. If the determinant of matrix A of order 3×3 is of value 4, write the value of $|3A|$.
(AI 2012C)

SA (2 marks)

23. If A is a skew-symmetric matrix of order 3, then prove that $\det A = 0$. *(AI 2017)*

LA 1 (4 marks)

24. Using properties of determinants, prove that
$$\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3.$$
(Delhi 2019, AI 2017)

25. Using properties of determinants, find the value of x for which $\begin{vmatrix} 4-x & 4+x & 4+x \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0.$
(AI 2019)

26. Using properties of determinants, prove that
$$\begin{vmatrix} 1 & 1 & 1+3x \\ 1+3y & 1 & 1 \\ 1 & 1+3z & 1 \end{vmatrix} = 9(3xyz + xy + yz + zx)$$
(2018)

27. Using properties of determinants, prove that
$$\begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix} = 9y^2(x+y).$$
(Delhi 2017, AI 2013)

28. If $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, using properties of determinants find the value of $f(2x) - f(x)$.
(Delhi 2015)

29. Using properties of determinants, prove the following :
$$\begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix} = 4a^2b^2c^2.$$
(AI 2015, Foreign 2014)

30. Using properties of determinants, prove the following:
$$\begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix} = (1-a^3)^2$$
(Foreign 2015)

31. Using properties of determinants, prove that
$$\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix} = -2.$$
(Delhi 2015C)

32. Using properties of determinants, solve for x :
$$\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0.$$
(AI 2015C, 2011)

33. Using properties of determinants, prove that
$$\begin{vmatrix} 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix} = (x+y+z)^3$$
(Delhi 2014)

34. Prove the following using properties of determinants :
$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3.$$
(Delhi 2014, 2012C)

35. Using properties of determinants, prove the following:
$$\begin{vmatrix} x^2+1 & xy & xz \\ xy & y^2+1 & yz \\ xz & yz & z^2+1 \end{vmatrix} = 1+x^2+y^2+z^2$$
(Delhi 2014)

36. Using properties of determinants, prove the following :
$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = ab+bc+ca+abc$$
(AI 2014, Delhi 2012)

www.exambites.in

37. Using properties of determinants, prove that

$$\begin{vmatrix} b+c & c+a & a+b \\ q+r & r+p & p+q \\ y+z & z+x & x+y \end{vmatrix} = 2\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}.$$

(AI 2014)

38. Using properties of determinants, show that

$$\begin{vmatrix} x+y & x & x \\ 5x+4y & 4x & 2x \\ 10x+8y & 8x & 3x \end{vmatrix} = x^3.$$ *(AI 2014)*

39. Using properties of determinants, prove that :

$$\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z).$$

(Foreign 2014)

40. Using properties of determinants, prove that :

$$\begin{vmatrix} x+\lambda & 2x & 2x \\ 2x & x+\lambda & 2x \\ 2x & 2x & x+\lambda \end{vmatrix} = (5x+\lambda)(\lambda-x)^2.$$

(Foreign 2014)

41. Using properties of determinants, prove the following :

$$\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} = (1-x^3)^2.$$ *(Delhi 2014C, 2013)*

42. Using properties of determinants, prove the following :

$$\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(bc+ca+ab)$$

(Delhi 2014C)

43. Using properties of determinants, prove the following :

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc.$$ *(AI 2014C, 2012)*

44. Show that $\Delta = \Delta_1$, where

$$\Delta = \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}, \Delta_1 = \begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$$

(AI 2014C)

45. Using properties of determinants, prove the following :

$$\begin{vmatrix} 3x & -x+y & -x+z \\ x-y & 3y & z-y \\ x-z & y-z & 3z \end{vmatrix} = 3(x+y+z)(xy+yz+zx)$$

(AI 2013)

46. Using properties of determinants, prove that

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

(Delhi 2013C)

47. Using properties of determinants, prove that

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

(AI 2013C, Delhi 2011C)

48. Using properties of determinants, prove that

$$\begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} = 2\begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}.$$

(Delhi 2012)

49. Using properties of determinants, prove the following :

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

(Delhi 2012, 2011C)

50. Using properties of determinants, prove the following :

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{vmatrix}$$

$= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\alpha + \beta + \gamma)$

(Delhi 2012C)

51. Using properties of determinants, prove the following :

$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix} = a^3+b^3+c^3-3abc.$$

(Delhi 2012C)

www.exambites.in

52. Using properties of determinants, prove the following :

$$\begin{vmatrix} b+c & a-b & a \\ c+a & b-c & b \\ a+b & c-a & c \end{vmatrix} = 3abc - a^3 - b^3 - c^3$$

(AI 2012C)

53. Using properties of determinants, prove the following :

$$\begin{vmatrix} a^2 & a^2-(b-c)^2 & bc \\ b^2 & b^2-(c-a)^2 & ca \\ c^2 & c^2-(a-b)^2 & ab \end{vmatrix}$$

$= (a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$

(AI 2012C)

54. Using properties of determinants, prove the following :

$$\begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix} = (a+b+c)(a^2+b^2+c^2).$$

(AI 2012C)

55. Using properties of determinants, prove that

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2b^2c^2.$$

(Delhi 2011, AI 2011C)

56. Using properties of determinants, prove that

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x-y)(y-z)(z-x).$$

(Delhi 2011)

57. Using properties of determinants, prove that

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2.$$

(Delhi 2011)

58. Using properties of determinants, solve the following for x.

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0.$$ *(AI 2011)*

59. Using properties of determinants, solve the following for x.

$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0, a \neq 0$$ *(AI 2011)*

60. Using properties of determinants, prove the following :

$$\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix} = 1+a^2+b^2+c^2$$

(AI 2011C)

LA 2 (6 marks)

61. Using properties of determinants, prove that

$$\begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$$

(2020)

62. If a, b, c are p^{th}, q^{th} and r^{th} terms respectively of a G.P., then prove that

$$\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} = 0.$$ *(2020)*

63. Prove that $\begin{vmatrix} yz-x^2 & zx-y^2 & xy-z^2 \\ zx-y^2 & xy-z^2 & yz-x^2 \\ xy-z^2 & yz-x^2 & zx-y^2 \end{vmatrix}$ is divisible by $(x+y+z)$, and hence find the quotient. *(Delhi 2016)*

64. Using properties of determinants, show that ΔABC is isosceles, if

$$\begin{vmatrix} 1 & 1 & 1 \\ 1+\cos A & 1+\cos B & 1+\cos C \\ \cos^2 A+\cos A & \cos^2 B+\cos B & \cos^2 C+\cos C \end{vmatrix} = 0$$

(AI 2016)

65. If a, b and c are all non-zero and

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$$, then prove that

$$\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+1=0$$ *(Foreign 2016)*

www.exambites.in

4.4 Area of a Triangle

LA 1 (4 marks)

66. Find the equation of the line joining $A(1, 3)$ and $B(0, 0)$ using determinants and find the value of k if $D(k, 0)$ is a point such that area of ΔABD is 3 square units. *(AI 2013C)*

4.5 Minors and Cofactors

VSA (1 mark)

67. Find the cofactors of all the elements of $\begin{bmatrix} 1 & -2 \\ 4 & 3 \end{bmatrix}$. *(2020)*

68. If $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, then write the cofactor of the element a_{21} of its 2nd row. *(Foreign 2015)*

69. If A_{ij} is the cofactor of the element a_{ij} of the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$, then write the value of $a_{32} \cdot A_{32}$. *(AI 2013)*

70. If $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$, write the minor of the element a_{23}. *(Delhi 2012)*

71. If $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$, write the cofactor of the element a_{32}. *(Delhi 2012)*

72. If $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 5 & 3 & 8 \end{vmatrix}$, write the minor of element a_{22}. *(Delhi 2012)*

4.6 Adjoint and Inverse of a Matrix

VSA (1 mark)

73. If $A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 3 \\ 3 & 3 & 5 \end{bmatrix}$, then find A (adj A). *(2020)*

74. If $A = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$, then the value of $|\text{adj } A|$ is

(a) 64 (b) 16

(c) 0 (d) −8 *(2020)*

75. If A is a square matrix of order 3 with $|A| = 9$, then write the value of $|2 \cdot \text{adj } A|$. *(AI 2019)*

76. If A is a 3×3 invertible matrix, then what will be the value of k if $\det(A^{-1}) = (\det A)^k$? *(Delhi 2017)*

77. If for any 2×2 square matrix A, $A(\text{adj } A) = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$, then write the value of $|A|$. *(AI 2017)*

78. In the interval $\pi/2 < x < \pi$, find the value of x for which the matrix $\begin{bmatrix} 2\sin x & 3 \\ 1 & 2\sin x \end{bmatrix}$ is singular. *(AI 2015C)*

79. Find (adj A), if $A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$. *(Delhi 2014C)*

80. If A is a square matrix of order 3 such that $|\text{adj } A| = 64$, find $|A|$. *(Delhi 2013C)*

81. If A is an invertible square matrix of order 3 and $|A| = 5$, then find the value of $|\text{adj } A|$. *(AI 2013C, 2011C)*

82. For what value of x, is the given matrix $A = \begin{bmatrix} 3-2x & x+1 \\ 2 & 4 \end{bmatrix}$ singular? *(AI 2013C)*

83. For what value of x, the matrix $\begin{bmatrix} 1+x & 7 \\ 3-x & 8 \end{bmatrix}$ is a singular matrix? *(AI 2012C)*

84. Write A^{-1} for $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$. *(Delhi 2011)*

85. For what value of x, the matrix $\begin{bmatrix} 5-x & x+1 \\ 2 & 4 \end{bmatrix}$ is singular ? *(Delhi 2011)*

86. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, then write A^{-1} in terms of A. *(AI 2011)*

www.exambites.in

87. For what value of x is the matrix $\begin{bmatrix} 6-x & 4 \\ 3-x & 1 \end{bmatrix}$ singular? *(Delhi 2011C)*

88. For what value of x is the matrix $\begin{bmatrix} 2x & 4 \\ x+2 & 3 \end{bmatrix}$ singular? *(Delhi 2011C)*

89. For what value of x is the matrix $\begin{bmatrix} 2x+4 & 4 \\ x+5 & 3 \end{bmatrix}$ a singular matrix? *(AI 2011C)*

90. For what value of x is $A = \begin{bmatrix} 2(x+1) & 2x \\ x & x-2 \end{bmatrix}$ a singular matrix? *(AI 2011C)*

SA (2 marks)

91. Find $(AB)^{-1}$ if $A = \begin{bmatrix} 1 & 0 \\ -4 & 2 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$. *(2020)*

92. Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute A^{-1} and show that $2A^{-1} = 9I - A$. *(2018)*

LA 1 (4 marks)

93. If $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix}$, find $(A')^{-1}$. *(Delhi 2015)*

94. Find the adjoint of the matrix $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ and hence show that $A \cdot (\text{adj } A) = |A| I_3$. *(AI 2015)*

95. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the identity matrix of order 2, then show that $A^2 = 4A - 3I$. Hence find A^{-1}. *(Foreign 2015)*

96. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. *(Delhi 2015C)*

97. If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}, B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1} A^{-1}$. *(AI 2015C)*

LA 2 (6 marks)

98. If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, find adj A and verify that $A(\text{adj } A) = (\text{adj } A)A = |A| I_3$. *(Foreign 2016)*

4.7 Applications of Determinants and Matrices

LA 1 (4 marks)

99. The monthly incomes of Aryan and Babban are in the ratio 3 : 4 and their monthly expenditures are in the ratio 5 : 7. If each saves ₹ 15,000 per month, find their monthly incomes using matrix method. This problem reflects which value? *(Delhi 2016)*

100. A trust invested some money in two type of bonds. The first bond pays 10% interest and second bond pays 12% interest. The trust received ₹ 2,800 as interest. However, if trust had interchanged money in bonds they would have got ₹ 100 less as interest. Using matrix method, find the amount invested by the trust. Interest received on this amount will be given to Helpage India as donation. Which value is reflected in this question? *(AI 2016)*

101. A coaching institute of English (Subject) conducts classes in two batches I and II and fees for rich and poor children are different. In batch I, it has 20 poor and 5 rich children and total monthly collection is ₹ 9,000, whereas in batch II, it has 5 poor and 25 rich children and total monthly collection is ₹ 26,000. Using matrix method, find monthly fees paid by each child of two types. What values the coaching institute is inculcating in the society? *(Foreign 2016)*

www.exambites.in

102. Two schools A and B decided to award prizes to their students for three values, team spirit, truthfulness and tolerance at the rate of ₹ x, ₹ y and ₹ z per student respectively. School A, decided to award a total of ₹ 1,100 for the three values to 3, 1 and 2 students respectively while school B decided to award ₹ 1,400 for the three values to 1, 2 and 3 students respectively. If one prize for all the three values together amount to ₹ 600 then
(i) Represent the above situation by a matrix equation after forming linear equations.
(ii) Is it possible to solve the system of equations so obtained using matrices?
(iii) Which value you prefer to be rewarded most and why? *(Delhi 2015C)*

LA 2 (6 marks)

103. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 2 & -2 \\ 2 & -1 & 1 \end{bmatrix}$, then find A^{-1} and use it to solve the following system of the equations :
$x + 2y - 3z = 6$,
$3x + 2y - 2z = 3$
$2x - y + z = 2$ *(2020)*

104. Solve the following system of equations by matrix method :
$x - y + 2z = 7$
$2x - y + 3z = 12$
$3x + 2y - z = 5$ *(2020)*

105. If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, then find A^{-1}.
Using A^{-1}, solve the following system of equations :
$2x - 3y + 5z = 11$
$3x + 2y - 4z = -5$
$x + y - 2z = -3$ *(2020, 2018, AI 2012C)*

106. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix}$, find A^{-1}. Hence, solve the system of equations $x + y + z = 6$, $x + 2z = 7$, $3x + y + z = 12$. *(Delhi 2019)*

107. Using matrices, solve the following system of linear equations:
$x + 2y - 3z = -4$
$2x + 3y + 2z = 2$
$3x - 3y - 4z = 11$ *(AI 2019, 2011)*

108. Use product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}\begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$ to solve the system of equations $x + 3z = 9$, $-x + 2y - 2z = 4$, $2x - 3y + 4z = -3$ *(Delhi 2017)*

109. Determine the product $\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$ and use it to solve the system of equations $x - y + z = 4$, $x - 2y - 2z = 9$, $2x + y + 3z = 1$. *(AI 2017, Delhi 2012C)*

110. Using elementary transformations, find the inverse of the matrix $A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ and use it to solve the following system of linear equations:
$8x + 4y + 3z = 19$; $2x + y + z = 5$;
$x + 2y + 2z = 7$ *(Delhi 2016)*

111. A shopkeeper has 3 varieties of pens 'A', 'B' and 'C'. Meenu purchased 1 pen of each variety for a total of ₹ 21. Jeevan purchased 4 pens of 'A' variety, 3 pens of 'B' variety and 2 pens of 'C' variety for ₹ 60. While Shikha purchased 6 pens of 'A' variety, 2 pens of 'B' variety and 3 pens of 'C' variety for ₹ 70. Using matrix method, find cost of each variety of pen. *(AI 2016)*

112. Two schools P and Q want to award their selected students on the values of discipline, politeness and punctuality. The school P wants to award ₹ x each, ₹ y each and ₹ z each for the three respective values to its 3, 2 and 1 students with a total award money of ₹ 1,000. School Q wants to spend ₹ 1,500 to award its 4, 1 and 3 students on the respective values (by giving the same award

www.exambites.in

money for the three values as before). If the total amount of awards for one prize on each value is ₹ 600, using matrices, find the award money for each value.

Apart from the above three values, suggest one more value for awards. *(Delhi 2014)*

113. Two schools A and B want to award their selected students on the values of sincerity, truthfulness and helpfulness. The school A wants to award ₹ x each, ₹ y each and ₹ z each for the three respective values to 3, 2 and 1 students respectively with a total award money of ₹ 1,600. School B wants to spend ₹ 2,300 to award its, 4, 1 and 3 students on the respective values (by giving the same award money to the three values as before). If the total amount of award for one prize on each value is ₹ 900, using matrices, find the award money for each value. Apart from these three values, suggest one more value which should be considered for award. *(AI 2014)*

114. Two schools P and Q want to award their selected students on the values of tolerance, kindness and leadership. The school P wants to award ₹ x each, ₹ y each and ₹ z each for the three respective values to its 3, 2 and 1 students respectively with a total award money of ₹ 2200. School Q wants to spend ₹ 3100 to award its 4, 1 and 3 students on the respective values (by giving the same award money for the three values as school P). If the total amount of award for one prize on each value is ₹ 1200, using matrices, find the award money for each value.

Apart from the above these three values, suggest one more value which should be considered for award. *(Foreign 2014)*

115. A total amount of ₹ 7,000 is deposited in three different savings bank accounts with annual interest rates of 5%, 8% and $8\frac{1}{2}\%$ respectively. The total annual interest from these three accounts is ₹ 550. Equal amounts have been deposited in the 5% and 8% savings accounts. Find the amount deposited in each of the three accounts, with the help of matrices. *(Delhi 2014C)*

116. Two schools, P and Q, want to award their selected students for the values of sincerity, truthfulness and hard work at the rate of ₹ x, ₹ y and ₹ z for each respective value per student. School P awards its 2, 3 and 4 students on the above respective values with a total prize money of ₹ 4,600. School Q wants to award its 3, 2 and 3 students on the respective values with a total award money of ₹ 4,100. If the total amount of award money for one prize on each value is ₹ 1,500, using matrices find the award money for each value. Suggest one other value which the school can consider for awarding the students. *(AI 2014C)*

117. A school wants to award its students for the value of honesty, regularity and hard work with a total cash award of ₹ 6,000. Three times the award money for hard work added to that given for honesty amounts to ₹ 11,000. The award money given for honesty and hard work together is double the one given for regularity. Represent the above situation algebraically and find the award money for each value, using matrix method. Apart from these values, namely, honesty, regularity and hard work, suggest one more value which the school must include for awards. *(Delhi 2013)*

118. The management committee of a residential colony decided to award some of its members (say x) for honesty, some (say y) for helping others and some others (say z) for supervising the workers to keep the colony neat and clean. The sum of all the awardees is 12. Three times the sum of awardees for cooperation and supervision added to two times the numbers of awardees for honesty is 33. If the sum of the number of awardees for honesty and supervision is twice the number of awardees for helping others, using matrix method, find the number of awardees of each category. Apart from these values, namely, honesty, cooperation and supervision, suggest one more value which the management of the colony must include for awards. *(AI 2013)*

www.examtes.in

119. Two institutions decided to award their employees for the three values of resourcefulness, competence and determination in the form of prizes at the rate of ₹ x, ₹ y and ₹ z respectively per person. The first institution decided to award respectively 4, 3 and 2 employees with a total prize money of ₹ 37,000 and the second institution decided to award respectively 5, 3 and 4 employees with a total prize money of ₹ 47,000. If all the three prizes per person together amount to ₹ 12,000 then using matrix method find the value of x, y and z. What values are described in the question?
(Delhi 2013C)

120. Two factories decided to award their employees for three values of (a) adaptable to new techniques, (b) careful and alert in difficult situations and (c) keeping calm in tense situations, at the rate of ₹ x, ₹ y and ₹ z per person respectively. The first factory decided to honour respectively 2, 4 and 3 employees with a total prize money of ₹ 29,000. The second factory decided to honour respectively 5, 2 and 3 employees with the prize money of ₹ 30,500. It the three prizes per person together cost ₹ 9,500; then
(i) Represent the above situation by a matrix equation and form linear equations using matrix multiplication.
(ii) Solve these equations using matrices.
(iii) Which values are reflected in this question? *(AI 2013C)*

121. Using matrices, solve the following system of linear equations :
$x - y + 2z = 7$, $3x + 4y - 5z = -5$,
$2x - y + 3z = 12$ *(Delhi 2012)*

122. Using matrices, solve the following system of equations :
$2x + 3y + 3z = 5$, $x - 2y + z = -4$,
$3x - y - 2z = 3$ *(AI 2012)*

123. Using matrices, solve the following system of equations :
$3x + 4y + 7z = 4$; $2x - y + 3z = -3$;
$x + 2y - 3z = 8$ *(A I 2012)*

124. Using matrices, solve the following system of equations :
$x + y - z = 3$; $2x + 3y + z = 10$;
$3x - y - 7z = 1$ *(AI 2012)*

125. If $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix}$, find A^{-1} and hence solve the system of equations :
$x + 2y + z = 4$, $-x + y + z = 0$,
$x - 3y + z = 4$ *(Delhi 2012C)*

126. Find A^{-1}, where $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$.
Hence solve the system of equations
$x + 2y - 3z = -4$; $2x + 3y + 2z = 2$;
$3x - 3y - 4z = 11$. *(Delhi 2012C)*

127. If $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$ are two square matrices, find AB and hence solve the system of equations
$x - y = 3$; $2x + 3y + 4z = 17$; $y + 2z = 7$.
(AI 2012C)

128. If $A = \begin{bmatrix} 1 & 2 & 5 \\ 1 & -1 & -1 \\ 2 & 3 & -1 \end{bmatrix}$, find A^{-1}. Hence solve the following system of equations :
$x + 2y + 5z = 10$, $x - y - z = -2$,
$2x + 3y - z = -11$ *(AI 2012C)*

129. Using matrix method, solve the following system of equations :
$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4, \frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1, \frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$;
$x, y, z \neq 0$ *(Delhi 2011)*

130. Using matrix method, solve the following system of equations:
$x + 2y + z = 7$, $x + 3z = 11$, $2x - 3y = 1$.
(AI 2011)

131. Using matrices, solve the following system of equations :
$4x + 3y + 2z = 60$, $x + 2y + 3z = 45$,
$6x + 2y + 3z = 70$. *(AI 2011)*

www.exambites.in

132. If $A=\begin{bmatrix}3 & -4 & 2\\ 2 & 3 & 5\\ 1 & 0 & 1\end{bmatrix}$, find A^{-1} and hence solve the following system of equations :
$3x - 4y + 2z = -1$, $2x + 3y + 5z = 7$ and $x + z = 2$ *(Delhi 2011C)*

133. If $A=\begin{bmatrix}1 & -2 & 0\\ 2 & 1 & 3\\ 0 & -2 & 1\end{bmatrix}$ and $B=\begin{bmatrix}7 & 2 & -6\\ -2 & 1 & -3\\ -4 & 2 & 5\end{bmatrix}$, find AB. Hence solve the system of equations :
$x - 2y = 10$, $2x + y + 3z = 8$ and $-2y + z = 7$. *(Delhi 2011C)*

134. If $A=\begin{bmatrix}1 & -2 & 1\\ 0 & -1 & 1\\ 2 & 0 & -3\end{bmatrix}$, find A^{-1} and hence solve the system of equations
$x - 2y + z = 0$, $-y + z = -2$, $2x - 3z = 10$. *(AI 2011C)*

135. If $A=\begin{bmatrix}2 & -1 & 1\\ 3 & 0 & -1\\ 2 & 6 & 0\end{bmatrix}$, find A^{-1}. Using A^{-1}, solve the following system of equations:
$2x - y + z = -3$, $3x - z = 0$, $2x + 6y - 2 = 0$ *(AI 2011C)*

Detailed Solutions

1. Let $\Delta=\begin{vmatrix}1 & 1 & 1\\ 1 & 1+\sin\theta & 1\\ 1 & 1 & 1+\cos\theta\end{vmatrix}$

$\Rightarrow \Delta = 1[(1+\sin\theta)(1+\cos\theta) - 1] - 1(1+\cos\theta - 1) + 1(1 - 1 - \sin\theta)$
$= 1 + \cos\theta + \sin\theta + \sin\theta\cos\theta - 1 - \cos\theta - \sin\theta$
$= \sin\theta\cos\theta$

$\therefore$ Maximum value of Δ is $\frac{1}{2}$.

2. Given, $\begin{vmatrix}x+3 & -2\\ -3x & 2x\end{vmatrix} = 8$

$\Rightarrow (x+3)(2x) - (-2)(-3x) = 8$
$\Rightarrow 2x^2 + 6x - 6x = 8 \Rightarrow 2x^2 = 8$
$\Rightarrow x^2 = 4 \Rightarrow x = 2$ $[x \neq -2 \ \because x \in N]$

3. Given, $\begin{vmatrix}x & \sin\theta & \cos\theta\\ -\sin\theta & -x & 1\\ \cos\theta & 1 & x\end{vmatrix} = 8$

$\Rightarrow x(-x^2 - 1) - \sin\theta(-x\sin\theta - \cos\theta) + \cos\theta(-\sin\theta + x\cos\theta) = 8$
$\Rightarrow -x^3 - x + x\sin^2\theta + \sin\theta\cos\theta - \sin\theta\cos\theta + x\cos^2\theta = 8$
$\Rightarrow -x^3 - x + x(\sin^2\theta + \cos^2\theta) = 8$
$\Rightarrow -x^3 - x + x = 8 \Rightarrow x^3 + 8 = 0$
$\Rightarrow (x+2)(x^2 - 2x + 4) = 0 \Rightarrow x + 2 = 0$ $[\because x^2 - 2x + 4 > 0 \ \forall x]$
$\Rightarrow x = -2$

4. Given that $A=\begin{bmatrix}1 & 2\\ 3 & -1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 3\\ -1 & 1\end{bmatrix}$

$\therefore AB=\begin{bmatrix}1 & 2\\ 3 & -1\end{bmatrix}\cdot\begin{bmatrix}1 & 3\\ -1 & 1\end{bmatrix}=\begin{bmatrix}-1 & 5\\ 4 & 8\end{bmatrix}$

$\therefore |AB| = \begin{vmatrix}-1 & 5\\ 4 & 8\end{vmatrix} = (-1)\cdot 8 - 4\cdot 5 = -28.$

5. Given, $\begin{vmatrix}2x & 5\\ 8 & x\end{vmatrix} = \begin{vmatrix}6 & -2\\ 7 & 3\end{vmatrix}$

$\Rightarrow 2x^2 - 40 = 18 + 14$
$\Rightarrow 2x^2 = 72 \Rightarrow x^2 = 36 \Rightarrow x = \pm 6$

6. Given, $\begin{vmatrix}3x & 7\\ -2 & 4\end{vmatrix} = \begin{vmatrix}8 & 7\\ 6 & 4\end{vmatrix}$

$\Rightarrow 12x + 14 = 32 - 42$
$\Rightarrow 12x = -10 - 14 = -24 \Rightarrow x = -2.$

7. $\begin{vmatrix}p & p+1\\ p-1 & p\end{vmatrix} = p^2 - (p-1)(p+1)$
$= p^2 - (p^2 - 1) = 1.$

8. Let $\Delta=\begin{vmatrix}2 & 7 & 65\\ 3 & 8 & 75\\ 5 & 9 & 86\end{vmatrix}$

$\Rightarrow \Delta = 2[8(86) - 9(75)] - 7[3(86) - 5(75)] + 65[3(9) - 5(8)]$
$= 2(688 - 675) - 7(258 - 375) + 65(27 - 40)$
$= 2(13) - 7(-117) + 65(-13) = 26 + 819 - 845 = 0$

www.exambites.in

9. Given, $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$

$\Rightarrow (x+1)(x+2) - (x-3)(x-1) = 4 \times 3 - (1) \times (-1)$

$\Rightarrow x^2 + x + 2x + 2 - (x^2 - 3x - x + 3) = 12 + 1$

$\Rightarrow x^2 + 3x + 2 - x^2 + 4x - 3 = 13$

$\Rightarrow 7x = 13 + 1 \Rightarrow x = 2$

10. Given, $\begin{vmatrix} 2x & x+3 \\ 2(x+1) & x+1 \end{vmatrix} = \begin{vmatrix} 1 & 5 \\ 3 & 3 \end{vmatrix}$

$\Rightarrow 2x(x+1) - 2(x+1)(x+3) = 3 - 15$

$\Rightarrow 2x^2 + 2x - 2(x^2 + 4x + 3) = -12$

$\Rightarrow 2x^2 + 2x - 2x^2 - 8x - 6 = -12$

$\Rightarrow -6x = -12 + 6 \Rightarrow x = \dfrac{-6}{-6} = 1$

11. $\begin{vmatrix} \cos 15^\circ & \sin 15^\circ \\ \sin 75^\circ & \cos 75^\circ \end{vmatrix}$

$= \cos 15^\circ \cos 75^\circ - \sin 75^\circ \sin 15^\circ$

$= \cos(15^\circ + 75^\circ) = \cos 90^\circ = 0$

12. Here, $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$

$\Rightarrow |A| = \begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix} = 3 \times 2 - 1 \times 4 = 2$

$\Rightarrow 3|A| = 3 \times 2 = 6.$

13. (d) : Given, A is a 3×3 matrix and $|A| = 5$

Now, $|2A'| = 2^3|A'|$

$= 2^3|A|$ [$\because |A'| = |A|$]

$= 8 \times 5 = 40$

14. (b) : We have, $A^T = -A$

[$\because$ A is skew-symmetric matrix]

$\therefore |A^T| = |-A| \Rightarrow |A| = (-1)^3|A|$ [$\because$ A is of order 3]

$\Rightarrow |A| = -|A| \Rightarrow 2|A| = 0 \Rightarrow |A| = 0$

15. (d) : We have, $|3A| = 3^3|A| = 3^3 \cdot 8$

[Given $|A| = 8$]

$= 27 \cdot 8 = 216$

16. We have, $|A| = 5$, $|B| = 3$

Now, $|3AB| = 3^3|AB|$ (As order of AB is 3)

$= 3^3|A||B| = 3^3 \times 5 \times 3 = 405$

17. We have, $AB = 2I \Rightarrow |AB| = |2I| \Rightarrow |A||B| = 2^3|I|$

$\Rightarrow 2|B| = 8$ [$\because |A| = 2$ (given)]

$\Rightarrow |B| = 4$

18. Here, $\Delta = \begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ -3 & -3 & -3 \end{vmatrix}$

Applying $R_1 \to R_1 + R_2$, we get

$\Delta = \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ z & x & y \\ -3 & -3 & -3 \end{vmatrix}$

Taking -3 common from R_3 and $x + y + z$ common from R_1, we get

$\Delta = -3(x+y+z)\begin{vmatrix} 1 & 1 & 1 \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix}$

$= -3(x + y + z) \cdot 0$ ($\because R_1$ and R_3 are identical)

$= 0.$

19. We have, $|3A| = k|A|$

$\Rightarrow 3^3|A| = k|A|$

[Using $|mA| = m^n|A|$, where n is order of A]

$\Rightarrow k = 27.$

20. *Refer to answer 15.*

21. *Refer to answer 15.*

22. *Refer to answer 15.*

23. *Refer to answer 14.*

24. Let $\Delta = \begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$

Applying $R_1 \to R_1 - R_2$, we get

$\Delta = \begin{vmatrix} a^2-1 & a-1 & 0 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$

Taking out common $(a - 1)$ from R_1, we get

$\Delta = (a-1)\begin{vmatrix} a+1 & 1 & 0 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$

Applying $R_2 \to R_2 - R_3$, we get

$\Delta = (a-1)\begin{vmatrix} a+1 & 1 & 0 \\ 2a-2 & a-1 & 0 \\ 3 & 3 & 1 \end{vmatrix}$

Taking out common $(a - 1)$ from R_2, we get

$\Delta = (a-1)^2\begin{vmatrix} a+1 & 1 & 0 \\ 2 & 1 & 0 \\ 3 & 3 & 1 \end{vmatrix}$

www.examfiles.in

Now, expanding along C_3, we get
$\Delta = (a-1)^2 (a+1-2) = (a-1)^2 (a-1) = (a-1)^3$

25. Given, $\begin{vmatrix} 4-x & 4+x & 4+x \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} 12+x & 4+x & 4+x \\ 12+x & 4-x & 4+x \\ 12+x & 4+x & 4-x \end{vmatrix} = 0$$

Applying $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} 12+x & 4+x & 4+x \\ 0 & -2x & 0 \\ 0 & 0 & -2x \end{vmatrix} = 0$$

$\Rightarrow \quad (12+x)(-2x)^2 = 0 \Rightarrow 4x^2(12+x) = 0$

$\Rightarrow \quad x = 0, -12$

26. L.H.S. $= \begin{vmatrix} 1 & 1 & 1+3x \\ 1+3y & 1 & 1 \\ 1 & 1+3z & 1 \end{vmatrix}$

Applying $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$, we get

$$\begin{vmatrix} 1 & 0 & 3x \\ 1+3y & -3y & -3y \\ 1 & 3z & 0 \end{vmatrix}$$

Taking out common 3 from both C_2 and C_3, we get

$$9\begin{vmatrix} 1 & 0 & x \\ 1+3y & -y & -y \\ 1 & z & 0 \end{vmatrix}$$

$= 9[1(0 + zy) - 0 + x(z + 3yz + y)]$
$= 9(zy + xz + 3xyz + xy) = 9(3xyz + xy + yz + zx)$
$=$ R.H.S.

27. L.H.S. $= \begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} 3x+3y & x+y & x+2y \\ 3x+3y & x & x+y \\ 3x+3y & x+2y & x \end{vmatrix}$$

Taking $3(x + y)$ common from C_1, we get

$$3(x+y)\begin{vmatrix} 1 & x+y & x+2y \\ 1 & x & x+y \\ 1 & x+2y & x \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_2$, we get

$$3(x+y)\begin{vmatrix} 1 & x+y & x+2y \\ 0 & -y & -y \\ 0 & 2y & -y \end{vmatrix}$$

Taking y common from R_2 and R_3 both, we get

$$3(x+y)\cdot y \cdot y\begin{vmatrix} 1 & x+y & x+2y \\ 0 & -1 & -1 \\ 0 & 2 & -1 \end{vmatrix}$$

$= 3y^2 (x + y) \cdot 1(1 + 2) = 9y^2 (x + y) =$ R.H.S.

28. $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$

$$\Rightarrow \quad f(x) = a\begin{vmatrix} 1 & -1 & 0 \\ x & a & -1 \\ x^2 & ax & a \end{vmatrix}$$

Applying $C_2 \to C_2 + C_1$, we get

$$f(x) = a\begin{vmatrix} 1 & 0 & 0 \\ x & x+a & -1 \\ x^2 & x^2+ax & a \end{vmatrix}$$

$\Rightarrow \quad f(x) = a[a(x + a) + (x^2 + ax)]$
$\Rightarrow \quad f(x) = a(a^2 + ax + ax + x^2) = a(a^2 + 2ax + x^2)$
Now, $f(2x) = a\{a^2 + 2a(2x) + (2x)^2\} = a(a^2 + 4ax + 4x^2)$
$\therefore \quad f(2x) - f(x) = a(a^2 + 4ax + 4x^2 - a^2 - 2ax - x^2)$
$= ax(3x + 2a)$

29. L.H.S. $= \begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix}$

Taking a, b, c common from C_1, C_2, C_3 respectively, we get

$$abc\begin{vmatrix} a & c & a+c \\ a+b & b & a \\ b & b+c & c \end{vmatrix}$$

www.exambites.in

Applying $C_3 \to C_3 - C_1 - C_2$, we get

$$abc\begin{vmatrix} a & c & 0 \\ a+b & b & -2b \\ b & b+c & -2b \end{vmatrix}$$

Applying $R_2 \to R_2 - R_3$, we get

$$abc\begin{vmatrix} a & c & 0 \\ a & -c & 0 \\ b & b+c & -2b \end{vmatrix}$$

$= abc(-2b)(-ac - ac) = -2ab^2c(-2ac)$

$= 4a^2b^2c^2$ = R.H.S.

30. L.H.S. $= \begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} 1+a+a^2 & a & a^2 \\ 1+a+a^2 & 1 & a \\ 1+a+a^2 & a^2 & 1 \end{vmatrix}$$

Taking $(1 + a + a^2)$ common from C_1, we get

$$(1+a+a^2)\begin{vmatrix} 1 & a & a^2 \\ 1 & 1 & a \\ 1 & a^2 & 1 \end{vmatrix}$$

Applying $C_1 \to C_1 - C_2$, we get

$$(1+a+a^2)\begin{vmatrix} 1-a & a & a^2 \\ 0 & 1 & a \\ 1-a^2 & a^2 & 1 \end{vmatrix}$$

Taking $(1 - a)$ common from C_1, we get

$$(1+a+a^2)(1-a)\begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & a \\ 1+a & a^2 & 1 \end{vmatrix}$$

$= (1 - a^3)[(1 - a^3) + (1 + a)(a^2 - a^2)]$

$= (1 - a^3)^2$ = R.H.S.

31. L.H.S. $= \begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$

$$= \begin{vmatrix} a^2+3a+2 & a+2 & 1 \\ a^2+5a+6 & a+3 & 1 \\ a^2+7a+12 & a+4 & 1 \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_2$, we get

$$\begin{vmatrix} a^2+3a+2 & a+2 & 1 \\ 2a+4 & 1 & 0 \\ 2a+6 & 1 & 0 \end{vmatrix}$$

Applying $R_3 \to R_3 - R_2$, we get

$$\begin{vmatrix} a^2+3a+2 & a+2 & 1 \\ 2a+4 & 1 & 0 \\ 2 & 0 & 0 \end{vmatrix} = 0 - 2 = -2 = \text{R.H.S.}$$

32. Given, $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} 3a-x & a-x & a-x \\ 3a-x & a+x & a-x \\ 3a-x & a-x & a+x \end{vmatrix} = 0$$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} 3a-x & a-x & a-x \\ 0 & 2x & 0 \\ 0 & 0 & 2x \end{vmatrix} = 0$$

$\Rightarrow \quad (3a - x)\cdot(2x)^2 = 0 \quad \Rightarrow \quad x = 0, 3a.$

33. L.H.S. $= \begin{vmatrix} 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix}$

Applying $R_1 \to R_1 + R_2 + R_3$, we get

$$\begin{vmatrix} x+y+z & x+y+z & x+y+z \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix}$$

Taking $(x + y + z)$ common from R_1, we get

$$(x+y+z)\begin{vmatrix} 1 & 1 & 1 \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix}$$

www.exambtes.in

Applying $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_2$, we get

$$(x+y+z)\begin{vmatrix} 1 & 0 & 0 \\ 2z & 0 & -(x+y+z) \\ x-y-z & x+y+z & 0 \end{vmatrix}$$

$= (x + y + z) \cdot 1 \cdot (x + y + z)^2 = (x + y + z)^3$ = R.H.S.

34. L.H.S. $= \begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$ and taking $2(a + b + c)$ common from C_1, we get

$$2(a+b+c)\begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$, we get

$$2(a+b+c)\begin{vmatrix} 1 & a & b \\ 0 & b+c+a & 0 \\ 0 & 0 & c+a+b \end{vmatrix}$$

$= 2(a + b + c) \cdot 1 \cdot (b + c + a) \cdot (c + a + b)$
$= 2(a + b + c)^3$ = R.H.S.

35. L.H.S. $= \begin{vmatrix} x^2+1 & xy & xz \\ xy & y^2+1 & yz \\ xz & yz & z^2+1 \end{vmatrix}$

Applying $R_1 \to \frac{1}{x}R_1$, $R_2 \to \frac{1}{y}R_2$, $R_3 \to \frac{1}{z}R_3$, we get

$$xyz\begin{vmatrix} x+\frac{1}{x} & y & z \\ x & y+\frac{1}{y} & z \\ x & y & z+\frac{1}{z} \end{vmatrix}$$

Applying $C_1 \to xC_1$, $C_2 \to yC_2$, $C_3 \to zC_3$, we get

$$\frac{xyz}{xyz}\begin{vmatrix} x^2+1 & y^2 & z^2 \\ x^2 & y^2+1 & z^2 \\ x^2 & y^2 & z^2+1 \end{vmatrix}$$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} x^2+y^2+z^2+1 & y^2 & z^2 \\ x^2+y^2+z^2+1 & y^2+1 & z^2 \\ x^2+y^2+z^2+1 & y^2 & z^2+1 \end{vmatrix}$$

Taking $(x^2 + y^2 + z^2 + 1)$ common from C_1, we get

$$(x^2+y^2+z^2+1)\begin{vmatrix} 1 & y^2 & z^2 \\ 1 & y^2+1 & z^2 \\ 1 & y^2 & z^2+1 \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$, we get

$$(1+x^2+y^2+z^2)\begin{vmatrix} 1 & y^2 & z^2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$= (1 + x^2 + y^2 + z^2)\ (1)\ (1 - 0)$
$= (1 + x^2 + y^2 + z^2)$ = R.H.S.

36. L.H.S. $= \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} 1+a & 1 & 1 \\ -a & b & 0 \\ -a & 0 & c \end{vmatrix}$$

$= -a[0 - (b)] + c[b(1 + a) + a]$
$= ab + bc + abc + ac$
$= ab + bc + ca + abc$ = R.H.S.

37. L.H.S. $= \begin{vmatrix} b+c & c+a & a+b \\ q+r & r+p & p+q \\ y+z & z+x & x+y \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$ and taking 2 common from C_1, we get

$$2\begin{vmatrix} a+b+c & c+a & a+b \\ p+q+r & r+p & p+q \\ x+y+z & z+x & x+y \end{vmatrix}$$

Applying $C_2 \to C_2 - C_1$; $C_3 \to C_3 - C_1$, we get

$$2\begin{vmatrix} a+b+c & -b & -c \\ p+q+r & -q & -r \\ x+y+z & -y & -z \end{vmatrix} = 2(-1)(-1)\begin{vmatrix} a+b+c & b & c \\ p+q+r & q & r \\ x+y+z & y & z \end{vmatrix}$$

Applying $C_1 \to C_1 - C_2 - C_3$, we get

$$2\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = \text{R.H.S.}$$

www.exambites.in

38. L.H.S. $= \begin{vmatrix} x+y & x & x \\ 5x+4y & 4x & 2x \\ 10x+8y & 8x & 3x \end{vmatrix}$.

Since each element in the first column of determinant is the sum of two elements, therefore, determinant can be expressed as the sum of two determinants given by

$$\begin{vmatrix} x & x & x \\ 5x & 4x & 2x \\ 10x & 8x & 3x \end{vmatrix} + \begin{vmatrix} y & x & x \\ 4y & 4x & 2x \\ 8y & 8x & 3x \end{vmatrix}$$

Taking x common from R_1, R_2, R_3 in first determinant and x common from C_2, C_3, y common from C_1 in second determinant, we get

$$x^3 \begin{vmatrix} 1 & 1 & 1 \\ 5 & 4 & 2 \\ 10 & 8 & 3 \end{vmatrix} + yx^2 \begin{vmatrix} 1 & 1 & 1 \\ 4 & 4 & 2 \\ 8 & 8 & 3 \end{vmatrix}$$

$$= x^3 \begin{vmatrix} 1 & 1 & 1 \\ 5 & 4 & 2 \\ 10 & 8 & 3 \end{vmatrix} + yx^2 \cdot 0$$

($\because$ C_1 and C_2 are identical in the second determinant)

Applying $C_1 \to C_1 - C_3$ and $C_2 \to C_2 - C_3$, we get

$$x^3 \begin{vmatrix} 0 & 0 & 1 \\ 3 & 2 & 2 \\ 7 & 5 & 3 \end{vmatrix} = x^3 \cdot 1 \cdot (15 - 14) = x^3 = \text{R.H.S.}$$

39. L.H.S. $= \begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} a+x+y+z & y & z \\ a+x+y+z & a+y & z \\ a+x+y+z & y & a+z \end{vmatrix}$$

Taking $(a + x + y + z)$ common from C_1, we get

$$(a+x+y+z) \begin{vmatrix} 1 & y & z \\ 1 & a+y & z \\ 1 & y & a+z \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2$, $R_2 \to R_2 - R_3$, we get

$$(a+x+y+z) \begin{vmatrix} 0 & -a & 0 \\ 0 & a & -a \\ 1 & y & a+z \end{vmatrix}$$

$= (a + x + y + z) 1 (a^2 - 0)$

$= a^2 (a + x + y + z) = \text{R.H.S.}$

40. L.H.S. $= \begin{vmatrix} x+\lambda & 2x & 2x \\ 2x & x+\lambda & 2x \\ 2x & 2x & x+\lambda \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} 5x+\lambda & 2x & 2x \\ 5x+\lambda & x+\lambda & 2x \\ 5x+\lambda & 2x & x+\lambda \end{vmatrix}$$

Taking $(5x + \lambda)$ common from C_1, we get

$$(5x+\lambda) \begin{vmatrix} 1 & 2x & 2x \\ 1 & x+\lambda & 2x \\ 1 & 2x & x+\lambda \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$, we get

$$(5x+\lambda) \begin{vmatrix} 1 & 2x & 2x \\ 0 & -x+\lambda & 0 \\ 0 & 0 & -x+\lambda \end{vmatrix}$$

Taking $(\lambda - x)$ common from R_2 and R_3 both, we get

$$(5x+\lambda)(\lambda-x)^2 \begin{vmatrix} 1 & 2x & 2x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$= (5x + \lambda)(\lambda - x)^2 \, 1 \cdot (1 - 0)$

$= (5x + \lambda)(\lambda - x)^2 = \text{R.H.S.}$

41. *Refer to answer 30.*

42. L.H.S. $= \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix}$

Applying $R_1 \to aR_1$, $R_2 \to bR_2$, $R_3 \to cR_3$, we get

$$\frac{1}{abc} \begin{vmatrix} a^2 & a^3 & abc \\ b^2 & b^3 & abc \\ c^2 & c^3 & abc \end{vmatrix} = \frac{1}{abc} \cdot abc \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2$; $R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix} a^2-b^2 & a^3-b^3 & 0 \\ b^2-c^2 & b^3-c^3 & 0 \\ c^2 & c^3 & 1 \end{vmatrix}$$

www.examkites.in

$$= \begin{vmatrix} (a-b)(a+b) & (a-b)(a^2+ab+b^2) & 0 \\ (b-c)(b+c) & (b-c)(b^2+bc+c^2) & 0 \\ c^2 & c^3 & 1 \end{vmatrix}$$

$$= (a-b)(b-c)\begin{vmatrix} a+b & a^2+ab+b^2 & 0 \\ b+c & b^2+bc+c^2 & 0 \\ c^2 & c^3 & 1 \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$, we get

$$(a-b)(b-c)\begin{vmatrix} a+b & a^2+ab+b^2 & 0 \\ c-a & (bc-ab)+c^2-a^2 & 0 \\ c^2 & c^3 & 1 \end{vmatrix}$$

$$= (a-b)(b-c)(c-a)\begin{vmatrix} a+b & a^2+ab+b^2 & 0 \\ 1 & b+c+a & 0 \\ c^2 & c^3 & 1 \end{vmatrix}$$

$= (a - b)\,(b - c)\,(c - a)$
$\quad [(a + b)\,(a + b + c) - (a^2 + ab + b^2)]$
$= (a - b)\,(b - c)\,(c - a)$
$\quad [a^2 + ab + ac + ba + b^2 + bc - a^2 - ab - b^2]$
$= (a - b)\,(b - c)\,(c - a)\,(bc + ca + ab)$ = R.H.S.

43. L.H.S. $= \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$

Applying $R_1 \to R_1 - R_2 - R_3$, we get

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Taking –2 common from R_1, we get

$$(-2)\begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Applying $C_2 \to \frac{1}{c}C_2, C_3 \to \frac{1}{b}C_3$, we get

$$(-2)\,bc\begin{vmatrix} 0 & 1 & 1 \\ b & 1+a/c & 1 \\ c & 1 & a/b+1 \end{vmatrix}$$

Applying $C_2 \to C_2 - C_3$, we get

$$-2bc\begin{vmatrix} 0 & 0 & 1 \\ b & a/c & 1 \\ c & -a/b & a/b+1 \end{vmatrix}$$

$= -2bc\,(-a - a) = 4abc$ = R.H.S.

44. Here, $\Delta_1 = \begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$

Multiplying C_1, C_2 and C_3 by x, y and z respectively, we get

$$\Delta_1 = \frac{1}{xyz}\begin{vmatrix} Ax & By & Cz \\ x^2 & y^2 & z^2 \\ xzy & xyz & xyz \end{vmatrix}$$

Taking xyz common from R_3, we get

$$\Delta_1 = \frac{1}{xyz}\cdot xyz\begin{vmatrix} Ax & By & Cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$$

Interchanging corresponding rows and columns, we get

$$\Delta_1 = \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix} = \Delta.$$

45. L.H.S. $= \begin{vmatrix} 3x & -x+y & -x+z \\ x-y & 3y & z-y \\ x-z & y-z & 3z \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix} x+y+z & -x+y & -x+z \\ x+y+z & 3y & z-y \\ x+y+z & y-z & 3z \end{vmatrix}$$

Taking $(x + y + z)$ common from C_1, we get

$$(x+y+z)\begin{vmatrix} 1 & y-x & z-x \\ 1 & 3y & z-y \\ 1 & y-z & 3z \end{vmatrix}$$

www.exambites.in

Applying $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$, we get

$$(x+y+z)\begin{vmatrix}1 & y-x & z-x\\ 0 & 2y+x & x-y\\ 0 & x-z & 2z+x\end{vmatrix}$$

$= (x + y + z)\ [(2y + x)\cdot(2z + x) - (x - z)\cdot(x - y)]$
$= (x + y + z) \cdot [(4yz + 2xy + 2zx + x^2)$
$\qquad - (x^2 - xy - zx + yz)]$
$= 3(x + y + z)\ (xy + yz + zx) =$ R.H.S.

46. L.H.S. $= \begin{vmatrix}1 & a & a^3\\ 1 & b & b^3\\ 1 & c & c^3\end{vmatrix}$

Applying $R_1 \to R_1 - R_2$; $R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix}0 & a-b & a^3-b^3\\ 0 & b-c & b^3-c^3\\ 1 & c & c^3\end{vmatrix}$$

Taking $(a - b)$ and $(b - c)$ common from R_1 and R_2 respectively, we get

$$(a-b)(b-c)\begin{vmatrix}0 & 1 & a^2+ab+b^2\\ 0 & 1 & b^2+bc+c^2\\ 1 & c & c^3\end{vmatrix}$$

$= (a - b)\ (b - c)\ [(b^2 + bc + c^2) - (a^2 + ab + b^2)]$
$= (a - b)\ (b - c)\ [(c^2 - a^2) + b(c - a)]$
$= (a - b)\ (b - c)\ (c - a)\ (c + a + b) =$ R.H.S.

47. *Refer to answer 42.*

48. *Refer to answer 37.*

49. *Refer to answer 46.*

50. L.H.S. $= \begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \beta+\gamma & \gamma+\alpha & \alpha+\beta\end{vmatrix}$

Applying $R_3 \to R_3 + R_1$, we get

$$\begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \alpha+\beta+\gamma & \alpha+\beta+\gamma & \alpha+\beta+\gamma\end{vmatrix}$$

Taking $(\alpha + \beta + \gamma)$ common from R_3, we get

$$(\alpha+\beta+\gamma)\begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ 1 & 1 & 1\end{vmatrix}$$

Applying $C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$, we get

$$(\alpha+\beta+\gamma)\begin{vmatrix}\alpha-\beta & \beta-\gamma & \gamma\\ \alpha^2-\beta^2 & \beta^2-\gamma^2 & \gamma^2\\ 0 & 0 & 1\end{vmatrix}$$

Taking $(\alpha - \beta)$ and $(\beta - \gamma)$ common from C_1 and C_2 respectively, we get

$$(\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)\begin{vmatrix}1 & 1 & \gamma\\ \alpha+\beta & \beta+\gamma & \gamma^2\\ 0 & 0 & 1\end{vmatrix}$$

$= (\alpha + \beta + \gamma)\ (\alpha - \beta)\ (\beta - \gamma)\ [(\beta + \gamma) - (\alpha + \beta)]$
$= (\alpha - \beta)\ (\beta - \gamma)\ (\gamma - \alpha)\ (\alpha + \beta + \gamma) =$ R.H.S.

51. L.H.S. $= \begin{vmatrix}a & b & c\\ a-b & b-c & c-a\\ b+c & c+a & a+b\end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix}a+b+c & b & c\\ 0 & b-c & c-a\\ 2(a+b+c) & c+a & a+b\end{vmatrix}$$

Taking $(a + b + c)$ common from C_1, we get

$$(a+b+c)\begin{vmatrix}1 & b & c\\ 0 & b-c & c-a\\ 2 & c+a & a+b\end{vmatrix}$$

Applying $R_3 \to R_3 - 2R_1$, we get

$$(a+b+c)\begin{vmatrix}1 & b & c\\ 0 & b-c & c-a\\ 0 & c+a-2b & a+b-2c\end{vmatrix}$$

$= (a + b + c)\ [(b - c)(a + b - 2c) - (c - a)(c + a - 2b)]$
$= (a + b + c)\ (a^2 + b^2 + c^2 - ab - bc - ca)$
$= a^3 + b^3 + c^3 - 3\ abc =$ R.H.S.

52. *Refer to answer 51.*

53. L.H.S. $= \begin{vmatrix}a^2 & a^2-(b-c)^2 & bc\\ b^2 & b^2-(c-a)^2 & ca\\ c^2 & c^2-(a-b)^2 & ab\end{vmatrix}$

Applying $C_2 \to C_2 - C_1$, we get

$$\begin{vmatrix}a^2 & -(b-c)^2 & bc\\ b^2 & -(c-a)^2 & ca\\ c^2 & -(a-b)^2 & ab\end{vmatrix}$$

www.exambites.in

Taking (–1) common from C_2, we get

$$-\begin{vmatrix} a^2 & b^2+c^2-2bc & bc \\ b^2 & c^2+a^2-2ca & ca \\ c^2 & a^2+b^2-2ab & ab \end{vmatrix}$$

Applying $C_2 \to C_2 + C_1 + 2C_3$, we get

$$-\begin{vmatrix} a^2 & a^2+b^2+c^2 & bc \\ b^2 & a^2+b^2+c^2 & ca \\ c^2 & a^2+b^2+c^2 & ab \end{vmatrix}$$

Taking $(a^2 + b^2 + c^2)$ common from C_2, we get

$$-(a^2+b^2+c^2)\begin{vmatrix} a^2 & 1 & bc \\ b^2 & 1 & ca \\ c^2 & 1 & ab \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2$; $R_2 \to R_2 - R_3$, we get

$$-(a^2+b^2+c^2)\begin{vmatrix} a^2-b^2 & 0 & c(b-a) \\ b^2-c^2 & 0 & a(c-b) \\ c^2 & 1 & ab \end{vmatrix}$$

Taking $(a - b)$ and $(b - c)$ common from R_1 and R_2 respectively, we get

$$-(a-b)(b-c)(a^2+b^2+c^2)\begin{vmatrix} a+b & 0 & -c \\ b+c & 0 & -a \\ c^2 & 1 & ab \end{vmatrix}$$

$$= -(a-b)(b-c)(a^2+b^2+c^2)[(a+b)a - c(b+c)]$$

$$= -(a-b)(b-c)(a^2+b^2+c^2)[a^2+ab-bc-c^2+ac-ac]$$

$$= -(a-b)(b-c)(a^2+b^2+c^2)[a(a+b+c) - c(a+b+c)]$$

$$= -(a-b)(b-c)(a^2+b^2+c^2)(a+b+c)(a-c)$$

$$= (a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$$

= R.H.S.

54. L.H.S. $= \begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix}$

Applying $C_1 \to aC_1$, we get

$$\frac{1}{a}\begin{vmatrix} a^2 & b-c & c+b \\ a^2+ac & b & c-a \\ a^2-ab & b+a & c \end{vmatrix}$$

Applying $C_1 \to C_1 + bC_2 + cC_3$, we get

$$\frac{1}{a}\begin{vmatrix} a^2+b^2+c^2 & b-c & c+b \\ a^2+b^2+c^2 & b & c-a \\ a^2+b^2+c^2 & b+a & c \end{vmatrix}$$

Taking $(a^2 + b^2 + c^2)$ common from C_1, we get

$$\frac{a^2+b^2+c^2}{a}\begin{vmatrix} 1 & b-c & c+b \\ 1 & b & c-a \\ 1 & b+a & c \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_2$, we get

$$\frac{a^2+b^2+c^2}{a}\begin{vmatrix} 1 & b-c & c+b \\ 0 & c & -a-b \\ 0 & a & a \end{vmatrix}$$

$$= \frac{a^2+b^2+c^2}{a} \cdot 1(ac + a^2 + ab)$$

$$= \frac{a^2+b^2+c^2}{a} \cdot a(a+b+c)$$

$= (a + b + c)(a^2 + b^2 + c^2)$ = R.H.S.

55. L.H.S. $= \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix}$

Taking a, b, c common from R_1, R_2 and R_3, respectively, we get

$$abc\begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$$

Taking a, b, c common from C_1, C_2 and C_3, respectively, we get

$$a^2b^2c^2\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

Applying $C_1 \to C_1 + C_2$, we get

$$a^2b^2c^2\begin{vmatrix} 0 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}$$

$= a^2b^2c^2 \cdot 2(1 + 1) = 4a^2b^2c^2$ = R.H.S.

56. L.H.S. $= \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix}$

www.examhites.in

Taking x, y and z common from C_1, C_2 and C_3 respectively, we get

$$xyz\begin{vmatrix}1 & 1 & 1\\ x & y & z\\ x^2 & y^2 & z^2\end{vmatrix}$$

Applying $C_2 \to C_2 - C_1$; $C_3 \to C_3 - C_1$, we get

$$xyz\begin{vmatrix}1 & 0 & 0\\ x & y-x & z-x\\ x^2 & y^2-x^2 & z^2-x^2\end{vmatrix}$$

Taking $(y - x)$ and $(z - x)$ common from C_2 and C_3 respectively, we get

$$xyz(y-x)(z-x)\begin{vmatrix}1 & 0 & 0\\ x & 1 & 1\\ x^2 & y+x & z+x\end{vmatrix}$$

$= xyz\,(y - x)\,(z - x) \cdot 1[(z + x) - (y + x)]$
$= xyz\,(y - x)\,(z - x)\,[z + x - y - x]$
$= xyz\,(y - x)\,(z - x)\,(z - y)$
$= xyz\,(x - y)\,(y - z)\,(z - x)$ = R.H.S.

57. *Refer to answer 40.*

58. Given, $\begin{vmatrix}x-2 & 2x-3 & 3x-4\\ x-4 & 2x-9 & 3x-16\\ x-8 & 2x-27 & 3x-64\end{vmatrix} = 0$

Applying $R_1 \to R_1 - R_2$; $R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix}2 & 6 & 12\\ 4 & 18 & 48\\ x-8 & 2x-27 & 3x-64\end{vmatrix} = 0$$

Taking 2 common from R_1 and R_2 both, we get

$$2\times 2\begin{vmatrix}1 & 3 & 6\\ 2 & 9 & 24\\ x-8 & 2x-27 & 3x-64\end{vmatrix} = 0$$

Applying $C_2 \to C_2 - 2C_1$ and $C_3 \to C_3 - 3C_1$, we get

$$4\begin{vmatrix}1 & 1 & 3\\ 2 & 5 & 18\\ x-8 & -11 & -40\end{vmatrix} = 0$$

Applying $R_2 \to R_2 - 2R_1$, we get

$$4\begin{vmatrix}1 & 1 & 3\\ 0 & 3 & 12\\ x-8 & -11 & -40\end{vmatrix} = 0$$

$\Rightarrow$ $4[(-120 + 132) + (x - 8)\,(12 - 9)] = 0$
$\Rightarrow$ $4(12 + 3x - 24) = 0$
$\Rightarrow$ $3x - 12 = 0$
$\Rightarrow$ $x = 4$.

59. Given, $\begin{vmatrix}x+a & x & x\\ x & x+a & x\\ x & x & x+a\end{vmatrix} = 0$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\begin{vmatrix}3x+a & x & x\\ 3x+a & x+a & x\\ 3x+a & x & x+a\end{vmatrix} = 0$$

Taking $(3x + a)$ common from C_1, we get

$$(3x+a)\begin{vmatrix}1 & x & x\\ 1 & x+a & x\\ 1 & x & x+a\end{vmatrix} = 0$$

Applying $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$, we get

$$(3x+a)\begin{vmatrix}1 & x & x\\ 0 & a & 0\\ 0 & 0 & a\end{vmatrix} = 0$$

$\Rightarrow$ $(3x + a)\,(1 \cdot a \cdot a) = 0$
$\Rightarrow$ $a^2\,(3x + a) = 0$
$\Rightarrow$ $x = -\dfrac{a}{3}$. $[\because a \neq 0]$

60. *Refer to answer 35.*

61. We have, $\begin{vmatrix}(b+c)^2 & a^2 & bc\\ (c+a)^2 & b^2 & ca\\ (a+b)^2 & c^2 & ab\end{vmatrix}$

Applying $R_1 \to R_1 - R_3$, $R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix}(b+c)^2-(a+b)^2 & a^2-c^2 & bc-ab\\ (c+a)^2-(a+b)^2 & b^2-c^2 & ca-ab\\ (a+b)^2 & c^2 & ab\end{vmatrix}$$

$$= \begin{vmatrix}(c-a)(a+2b+c) & -(c+a)(c-a) & b(c-a)\\ -(b-c)(2a+b+c) & (b+c)(b-c) & -a(b-c)\\ (a+b)^2 & c^2 & ab\end{vmatrix}$$

$$= (b-c)(c-a)\begin{vmatrix}a+2b+c & -c-a & b\\ -2a-b-c & b+c & -a\\ (a+b)^2 & c^2 & ab\end{vmatrix}$$

www.exambres.in

Applying $R_1 \to R_1 + R_2$, we get

$$(b-c)(c-a)\begin{vmatrix} -(a-b) & -(a-b) & -(a-b) \\ -2a-b-c & b+c & -a \\ (a+b)^2 & c^2 & ab \end{vmatrix}$$

$$= (b-c)(c-a)(a-b)\begin{vmatrix} -1 & -1 & -1 \\ -2a-b-c & b+c & -a \\ (a+b)^2 & c^2 & ab \end{vmatrix}$$

Applying $C_1 \to C_1 - C_3$, $C_2 \to C_2 - C_3$, we get

$$(a-b)(b-c)(c-a)\begin{vmatrix} 0 & 0 & -1 \\ -(a+b+c) & (a+b+c) & -a \\ (a+b)^2 - ab & c^2 - ab & ab \end{vmatrix}$$

$= (a - b)(b - c)(c - a)(-1)[-(a + b + c)(c^2 - ab)$
$- (a + b + c)((a+ b)^2 - ab)]$
$= (a - b)(b - c)(c - a)(a + b + c)$
$[c^2 - ab + a^2 + b^2 + 2ab - ab]$
$= (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2)$

62. We have, a, b, c are p^{th}, q^{th} and r^{th} terms of G.P.
Then $A_p = AR^{p-1} = a$, $A_q = AR^{q-1} = b$, $A_r = AR^{r-1} = c$, where A and R be the first term and common ratio of the G.P. respectively.

$$\text{Now, L.H.S.} = \begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} = \begin{vmatrix} \log(AR^{p-1}) & p & 1 \\ \log(AR^{q-1}) & q & 1 \\ \log(AR^{r-1}) & r & 1 \end{vmatrix}$$

$$\text{L.H.S.} = \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log A + (q-1)\log R & q & 1 \\ \log A + (r-1)\log R & r & 1 \end{vmatrix}$$

Applying $C_1 \to C_1 - (\log A)C_3$, we get

$$\text{L.H.S.} = \begin{vmatrix} (p-1)\log R & p & 1 \\ (q-1)\log R & q & 1 \\ (r-1)\log R & r & 1 \end{vmatrix} = \log R \begin{vmatrix} (p-1) & p & 1 \\ (q-1) & q & 1 \\ (r-1) & r & 1 \end{vmatrix}$$

Applying $C_1 \to C_1 + C_3$, we get

$$\text{L.H.S.} = \begin{vmatrix} p & p & 1 \\ q & q & 1 \\ r & r & 1 \end{vmatrix} = 0$$

(Since, C_1 and C_2 are identical)
= R.H.S.

63. Let $\Delta = \begin{vmatrix} yz - x^2 & zx - y^2 & xy - z^2 \\ zx - y^2 & xy - z^2 & yz - x^2 \\ xy - z^2 & yz - x^2 & zx - y^2 \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$$\Delta = \begin{vmatrix} -(x^2 + y^2 + z^2 - xy - yz - zx) & zx - y^2 & xy - z^2 \\ -(x^2 + y^2 + z^2 - xy - yz - zx) & xy - z^2 & yz - x^2 \\ -(x^2 + y^2 + z^2 - xy - yz - zx) & yz - x^2 & zx - y^2 \end{vmatrix}$$

Taking $-(x^2 + y^2 + z^2 - xy - yz - zx)$ common from C_1, we get

$$\Delta = -(x^2 + y^2 + z^2 - xy - yz - zx)\begin{vmatrix} 1 & zx - y^2 & xy - z^2 \\ 1 & xy - z^2 & yz - x^2 \\ 1 & yz - x^2 & zx - y^2 \end{vmatrix}$$

Applying $R_1 \to R_1 - R_3$, $R_2 \to R_2 - R_3$, we get

$$\Delta = -(x^2 + y^2 + z^2 - xy - yz - zx)\begin{vmatrix} 0 & (x-y)(x+y+z) & (y-z)(x+y+z) \\ 0 & (x-z)(x+y+z) & (y-x)(x+y+z) \\ 1 & yz - x^2 & zx - y^2 \end{vmatrix}$$

Taking $(x + y + z)$ common from R_1 and R_2 both, we get

$$\Delta = -(x^2 + y^2 + z^2 - xy - yz - zx)(x + y + z)^2\begin{vmatrix} 0 & x-y & y-z \\ 0 & x-z & y-x \\ 1 & yz - x^2 & zx - y^2 \end{vmatrix}$$

$$\Rightarrow \Delta = -(x + y + z)(x^3 + y^3 + z^3 - 3xyz)\begin{vmatrix} 0 & x-y & y-z \\ 0 & x-z & y-x \\ 1 & yz - x^2 & zx - y^2 \end{vmatrix}$$

$\Rightarrow \Delta = -(x + y + z)(x^3 + y^3 + z^3 - 3xyz)$
$[(x - y)(y - x) - (x - z)(y - z)]$
$\Rightarrow \Delta = -(x + y + z)(x^3 + y^3 + z^3 - 3xyz)$
$(xy + yz + zx - x^2 - y^2 - z^2)$
$\Rightarrow \Delta = (x + y + z)(x^3 + y^3 + z^3 - 3xyz)$
$(x^2 + y^2 + z^2 - xy - yz - zx)$

Hence, Δ is divisible by $(x + y + z)$ and quotient is $(x^3 + y^3 + z^3 - 3xyz)(x^2 + y^2 + z^2 - xy - yz - zx)$

www.exambites.in

64. We have,

$$\begin{vmatrix} 1 & 1 & 1 \\ 1+\cos A & 1+\cos B & 1+\cos C \\ \cos^2 A+\cos A & \cos^2 B+\cos B & \cos^2 C+\cos C \end{vmatrix} = 0$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$, we get

$$\begin{vmatrix} 1 & 0 & 0 \\ 1+\cos A & \cos B-\cos A & \cos C-\cos A \\ \cos^2 A & (\cos B-\cos A)\times & (\cos C-\cos A)\times \\ +\cos A & (\cos A+\cos B+1) & (\cos C+\cos A+1) \end{vmatrix} = 0$$

Taking common $(\cos B - \cos A)$ from C_2 and $(\cos C - \cos A)$ from C_3, we get

$(\cos B - \cos A)(\cos C - \cos A) \times$

$$\begin{vmatrix} 1 & 0 & 0 \\ 1+\cos A & 1 & 1 \\ \cos^2 A & (\cos A+\cos B+1) & (\cos C+\cos A+1) \\ +\cos A & & \end{vmatrix} = 0$$

$\Rightarrow$ $(\cos B - \cos A)(\cos C - \cos A)(\cos C - \cos B) = 0$

$\Rightarrow$ $\cos B = \cos A \Rightarrow B = A$

or $\cos C = \cos A \Rightarrow C = A$ or $\cos C = \cos B \Rightarrow C = B$

$\therefore$ ΔABC is an isosceles triangle.

65. It is given that, $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$

Applying $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix} a & -b & 0 \\ 0 & b & -c \\ 1 & 1 & 1+c \end{vmatrix} = 0$$

$\Rightarrow$ $a(b + bc + c) + b(0 + c) = 0$

$\Rightarrow$ $ab + bc + ac + abc = 0$

Dividing both sides by abc, we get

$$\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+1 = 0$$

66. Using determinants, the line joining $A(1, 3)$ and $B(0,0)$ is given by $\begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0$

$\Rightarrow$ $1(3x - y) = 0 \Rightarrow y = 3x$

Now, $D(k, 0)$ is a point s.t. area of $\Delta ABD = 3$ sq. units

$$\therefore \quad \frac{1}{2}\begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = 3$$

$\Rightarrow$ $(0 + 3k) = \pm 6 \Rightarrow k = \pm 2$

67. Let $A = \begin{bmatrix} 1 & -2 \\ 4 & 3 \end{bmatrix}$

Cofactor of 1 = 3, Cofactor of –2 = –4.

Cofactor of 4 = 2, Cofactor of 3 = 1.

68. We have, $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$

$$\therefore \quad \text{Cofactor of } a_{21} = (-1)^{2+1}\begin{vmatrix} 6 & -3 \\ -7 & 3 \end{vmatrix}$$

$$= -1(18 - 21) = 3$$

69. Let $\Delta = \begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$

Now, $a_{32} = 5$

$$A_{32} = \text{cofactor of } a_{32} \text{ in } \Delta = (-1)^{3+2}\begin{vmatrix} 2 & 5 \\ 6 & 4 \end{vmatrix}$$

$$= -(8 - 30) = 22$$

$\therefore$ $a_{32} \cdot A_{32} = 5 \cdot 22 = 110.$

70. Here, $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

$$\therefore \quad \text{Minor of } a_{23} = \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} = 10 - 3 = 7$$

71. Here, $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

$$\therefore \quad \text{Cofactor of } a_{32} = (-1)^{3+2}\begin{vmatrix} 5 & 8 \\ 2 & 1 \end{vmatrix} = -(5-16) = 11$$

72. We have, $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 5 & 3 & 8 \end{vmatrix}$

$$\therefore \quad \text{Minor of } a_{22} = \begin{vmatrix} 1 & 3 \\ 5 & 8 \end{vmatrix} = 8 - 15 = -7$$

73. Given, $A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 3 \\ 3 & 3 & 5 \end{bmatrix}$

Now, $A(\text{adj } A) = |A| = \begin{vmatrix} 2 & 0 & 0 \\ -1 & 2 & 3 \\ 3 & 3 & 5 \end{vmatrix}$

$= 2(10 - 9) - 0 + 0 = 2(1) = 2$

www.exambites.in

74. **(a) :** We have, $|A| = \begin{vmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{vmatrix}$

$= -2(4 - 0) - 0 + 0 = -8$

Now, we know that $|\text{adj } A| = |A|^{n-1}$, where n is the order of A.

$\therefore \quad |\text{adj } A| = (-8)^2 = 64$

75. Given, $|A| = 9$

We know that, $|k \text{ adj } A| = k^n |A|^{n-1}$

where n is the order of the matrix A.

$\therefore \quad |2 \cdot \text{adj } A| = 2^3 (9)^2 = 8 \times 81 = 648.$

76. Given that, $\det(A^{-1}) = (\det A)^k$

i.e., $|A^{-1}| = |A|^k$

We know that $|A^{-1}| = \dfrac{1}{|A|} = |A|^{-1}$

$\therefore \quad k = -1$

77. We have, $A(\text{adj } A) = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$

$|A| \cdot I = 8 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad [\because A(\text{adj } A) = |A| \cdot I]$

$\Rightarrow \quad |A| = 8$

78. For the matrix $\begin{bmatrix} 2\sin x & 3 \\ 1 & 2\sin x \end{bmatrix}$ to be singular, its determinant = 0

$\therefore \quad \begin{vmatrix} 2\sin x & 3 \\ 1 & 2\sin x \end{vmatrix} = 0$

$\Rightarrow \quad 4\sin^2 x - 3 = 0 \Rightarrow \sin^2 x = \dfrac{3}{4} \Rightarrow \sin x = \pm \dfrac{\sqrt{3}}{2}$

$\therefore \quad x = \dfrac{2\pi}{3} \left(\because \dfrac{\pi}{2} < x < \pi \right)$

79. Here, $A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$

Cofactor of matrix $A = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}$

$\therefore \quad \text{adj } A = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}^t = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$

80. A is a 3×3 matrix s.t. $|\text{adj } A| = 64$.

We know that

$|\text{adj } A| = |A|^{n-1}$

$\Rightarrow \quad 64 = |A|^{3-1} \Rightarrow |A|^2 = 64 \Rightarrow |A| = \pm 8.$

81. Given, $|A| = 5$ and A is invertible matrix of order 3.

We know that $|\text{adj } A| = |A|^{n-1}$

$\Rightarrow \quad |\text{adj } A| = (5)^{3-1} \Rightarrow |\text{adj } A| = 25.$

82. Given, $A = \begin{bmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{bmatrix}$.

Matrix A is singular, iff $|A| = 0$

$\Rightarrow \quad \begin{vmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{vmatrix} = 0$

$\Rightarrow \quad 4(3 - 2x) - 2(x + 1) = 0 \Rightarrow 12 - 8x - 2x - 2 = 0$

$\Rightarrow \quad -10x + 10 = 0 \Rightarrow x = 1$

83. Let $A = \begin{bmatrix} 1 + x & 7 \\ 3 - x & 8 \end{bmatrix}$

For A to be singular, $|A| = 0$

$\Rightarrow \quad \begin{vmatrix} 1 + x & 7 \\ 3 - x & 8 \end{vmatrix} = 0$

$\Rightarrow \quad 8 + 8x - 21 + 7x = 0 \Rightarrow 15x = 13 \Rightarrow x = \dfrac{13}{15}.$

84. We have, $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$

$\therefore \quad |A| = \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} = 6 - 5 = 1 \neq 0$, so A^{-1} exists.

$\Rightarrow \quad \text{adj } A = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$

Now, $A^{-1} = \dfrac{\text{adj } A}{|A|} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$

85. Let $A = \begin{bmatrix} 5 - x & x + 1 \\ 2 & 4 \end{bmatrix}$

Matrix A is singular, iff $|A| = 0$

$\Rightarrow \quad \begin{vmatrix} 5 - x & x + 1 \\ 2 & 4 \end{vmatrix} = 0$

$\Rightarrow \quad 20 - 4x - 2x - 2 = 0$

$\Rightarrow \quad -6x + 18 = 0 \Rightarrow x = 3$

86. We have, $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$

$\therefore \quad |A| = \begin{vmatrix} 2 & 3 \\ 5 & -2 \end{vmatrix} = -4 - 15 = -19 \neq 0$, so A^{-1} exists.

$\text{adj } A = \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix}$

www.exambites.in

Now, $A^{-1} = \frac{\text{adj}A}{|A|} = -\frac{1}{19}\begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix} = \frac{1}{19}\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$

$\Rightarrow A^{-1} = \frac{1}{19}A.$

87. *Refer to answer 83.*

88. Let $A = \begin{bmatrix} 2x & 4 \\ x+2 & 3 \end{bmatrix}$

For the matrix A to be singular, $|A| = 0$

$\Rightarrow \begin{vmatrix} 2x & 4 \\ x+2 & 3 \end{vmatrix} = 0$

$\Rightarrow 6x - 4x - 8 = 0 \Rightarrow 2x = 8 \Rightarrow x = 4.$

89. *Refer to answer 88.*

90. For the given matrix, $A = \begin{bmatrix} 2(x+1) & 2x \\ x & x-2 \end{bmatrix}$

to be singular, $|A| = 0$

$\Rightarrow \begin{vmatrix} 2(x+1) & 2x \\ x & x-2 \end{vmatrix} = 0$

$\Rightarrow 2(x + 1)(x - 2) - 2x^2 = 0$

$\Rightarrow 2(x^2 - x - 2) - 2x^2 = 0 \Rightarrow 2(-x - 2) = 0$

$\Rightarrow x = -2$

91. Given, $A = \begin{bmatrix} 1 & 0 \\ -4 & 2 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$

Now, $|A| = \begin{vmatrix} 1 & 0 \\ -4 & 2 \end{vmatrix} = 2$ and adj $A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$

$\therefore A^{-1} = \frac{1}{|A|}(\text{adj } A) = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$

Now, $(AB)^{-1} = B^{-1}A^{-1}$

$= \frac{1}{2}\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 10 & 1 \\ 18 & 2 \end{bmatrix}$

92. We have, $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$

$\therefore |A| = \begin{vmatrix} 2 & -3 \\ -4 & 7 \end{vmatrix} = 14 - 12 = 2 \neq 0$

So, A is a non-singular matrix and therefore it is invertible.

$\therefore$ adj $A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$

Hence, $A^{-1} = \frac{1}{|A|}\text{adj } A = \frac{1}{2}\begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$

$\Rightarrow 2A^{-1} = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$ ...(i)

Now, $9I - A = 9\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$

$= \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} = 2A^{-1}$ [From (i)]

93. $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix} \Rightarrow A' = \begin{bmatrix} 1 & 0 & -2 \\ -2 & -1 & 2 \\ 3 & 4 & 1 \end{bmatrix}$

$|A'| = \begin{vmatrix} 1 & 0 & -2 \\ -2 & -1 & 2 \\ 3 & 4 & 1 \end{vmatrix} = 1(-1 - 8) - 2(-8 + 3)$

$= -9 + 10 = 1 \neq 0$. So, $(A')^{-1}$ exists.

Let the cofactors of a_{ij}'s are A_{ij} in A'

Now, $A_{11} = -9$, $A_{12} = 8$, $A_{13} = -5$,

$A_{21} = -8$, $A_{22} = 7$, $A_{23} = -4$,

$A_{31} = -2$, $A_{32} = 2$, $A_{33} = -1$

$\therefore \text{adj}(A') = \begin{bmatrix} -9 & -8 & -2 \\ 8 & 7 & 2 \\ -5 & -4 & -1 \end{bmatrix}$

$\therefore (A')^{-1} = \frac{\text{adj}(A')}{|A'|} = \begin{bmatrix} -9 & -8 & -2 \\ 8 & 7 & 2 \\ -5 & -4 & -1 \end{bmatrix}$

94. Here, $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

$\Rightarrow |A| = \begin{vmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{vmatrix}$

$= -1(1 - 4) - (-2)(2 + 4) - 2(-4 - 2)$

$= 3 + 12 + 12 = 27$

Now, $A_{11} = -3$, $A_{12} = -6$, $A_{13} = -6$,

$A_{21} = 6$, $A_{22} = 3$, $A_{23} = -6$,

$A_{31} = 6$, $A_{32} = -6$, $A_{33} = 3$

$\therefore \text{adj } A = \begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix}$

www.exambites.in

$$\therefore \quad A\cdot(\text{adj } A)=\begin{bmatrix}-1 & -2 & -2\\ 2 & 1 & -2\\ 2 & -2 & 1\end{bmatrix}\cdot\begin{bmatrix}-3 & 6 & 6\\ -6 & 3 & -6\\ -6 & -6 & 3\end{bmatrix}$$

$$=\begin{bmatrix}27 & 0 & 0\\ 0 & 27 & 0\\ 0 & 0 & 27\end{bmatrix}=27\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=|A|\,I_3.$$

95. $A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix} \Rightarrow A^2=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$

$$\Rightarrow \quad A^2=\begin{bmatrix}5 & -4\\ -4 & 5\end{bmatrix} \qquad ...(i)$$

Now, $4A-3I=4\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}-3\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$$\Rightarrow \quad 4A-3I=\begin{bmatrix}5 & -4\\ -4 & 5\end{bmatrix} \qquad ...(ii)$$

From (i) and (ii), we get

$A^2 = 4A - 3I$

Pre-multiplying by A^{-1} on both sides, we get

$A^{-1}(A^2) = 4A^{-1}A - 3A^{-1}I$

$\Rightarrow \quad A = 4I - 3A^{-1} \qquad [\because AA^{-1} = I]$

$\Rightarrow \quad 3A^{-1} = 4I - A$

$$\Rightarrow \quad A^{-1}=\frac{4}{3}I-\frac{1}{3}A$$

$$=\frac{4}{3}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}-\frac{1}{3}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}=\begin{bmatrix}2/3 & 1/3\\ 1/3 & 2/3\end{bmatrix}$$

96. Given $A=\begin{bmatrix}1 & 3 & 3\\ 1 & 4 & 3\\ 1 & 3 & 4\end{bmatrix}$

Now, $A_{11} = 7$, $A_{12} = -1$, $A_{13} = -1$,
$A_{21} = -3$, $A_{22} = 1$, $A_{23} = 0$,
$A_{31} = -3$, $A_{32} = 0$, $A_{33} = 1$

$$\therefore \quad \text{adj } A=\begin{bmatrix}7 & -3 & -3\\ -1 & 1 & 0\\ -1 & 0 & 1\end{bmatrix} \text{ and } |A|=\begin{vmatrix}1 & 3 & 3\\ 1 & 4 & 3\\ 1 & 3 & 4\end{vmatrix}$$

$= 1(16 - 9) - 3(4 - 3) + 3(3 - 4)$
$= 7 - 3 - 3 = 1 \neq 0$

So, A^{-1} exists and it is given by

$$A^{-1}=\frac{1}{|A|}\cdot\text{adj } A=\begin{bmatrix}7 & -3 & -3\\ -1 & 1 & 0\\ -1 & 0 & 1\end{bmatrix}.$$

97. Here, $A=\begin{bmatrix}2 & 3\\ 1 & -4\end{bmatrix}$ and $B=\begin{bmatrix}1 & -2\\ -1 & 3\end{bmatrix}$

$\Rightarrow \quad |A| = -11, |B| = 1$

$$\therefore \quad A^{-1}=\frac{1}{|A|}\text{adj } A=-\frac{1}{11}\begin{bmatrix}-4 & -3\\ -1 & 2\end{bmatrix} \text{ and } B^{-1}=\begin{bmatrix}3 & 2\\ 1 & 1\end{bmatrix}$$

$\therefore$ R.H.S. $= B^{-1}A^{-1}$

$$=\begin{bmatrix}3 & 2\\ 1 & 1\end{bmatrix}\cdot\left(-\frac{1}{11}\right)\begin{bmatrix}-4 & -3\\ -1 & 2\end{bmatrix}=\left(-\frac{1}{11}\right)\begin{bmatrix}-14 & -5\\ -5 & -1\end{bmatrix} \qquad ...(i)$$

Now, $A\cdot B=\begin{bmatrix}2 & 3\\ 1 & -4\end{bmatrix}\cdot\begin{bmatrix}1 & -2\\ -1 & 3\end{bmatrix}=\begin{bmatrix}-1 & 5\\ 5 & -14\end{bmatrix}$

$\Rightarrow \quad |AB| = 14 - 25 = -11$

$$\therefore \quad \text{L.H.S.}=(AB)^{-1}=\left(-\frac{1}{11}\right)\begin{bmatrix}-14 & -5\\ -5 & -1\end{bmatrix} \qquad ...(ii)$$

From (i) and (ii), $(AB)^{-1} = B^{-1}A^{-1}$.

98. Given, $A=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$

Now, $A_{11} = \cos\alpha$, $A_{12} = -\sin\alpha$, $A_{13} = 0$,
$A_{21} = \sin\alpha$, $A_{22} = \cos\alpha$, $A_{23} = 0$,
$A_{31} = 0$, $A_{32} = 0$, $A_{33} = 1$

$$\therefore \quad \text{adj}(A)=\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$A.\text{adj}(A)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \qquad ...(i)$$

$$\text{adj}(A)\cdot A=\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \qquad ...(ii)$$

$$|A|=\begin{vmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{vmatrix}$$

www.exambitos.in

$= \cos\alpha(\cos\alpha - 0) + \sin\alpha(\sin\alpha - 0) + 0 = 1$ ...(iii)

From (i), (ii) and (iii), we get

$A(\text{adj } A) = (\text{adj } A)\, A = |A|\, I_3$.

99. Let the monthly income of Aryan be ₹ $3x$ and that of Babban be ₹ $4x$

Also, let monthly expenditure of Aryan be ₹ $5y$ and that of Babban be ₹ $7y$

According to question,

$3x - 5y = 15000$

$4x - 7y = 15000$

These equations can be written as

$AX = B$

where, $A = \begin{bmatrix} 3 & -5 \\ 4 & -7 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 15000 \\ 15000 \end{bmatrix}$

$$|A| = \begin{vmatrix} 3 & -5 \\ 4 & -7 \end{vmatrix} = (-21 + 20) = -1 \neq 0$$

Thus, A^{-1} exists. So, system of equations has a unique solution and given by $X = A^{-1} B$

$$\text{Now, adj}(A) = \begin{bmatrix} -7 & 5 \\ -4 & 3 \end{bmatrix}$$

$$\therefore \quad A^{-1} = \frac{\text{adj}(A)}{|A|} = \begin{bmatrix} 7 & -5 \\ 4 & -3 \end{bmatrix}$$

$$\text{Now, } X = A^{-1}B = \begin{bmatrix} 7 & -5 \\ 4 & -3 \end{bmatrix}\begin{bmatrix} 15000 \\ 15000 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 30000 \\ 15000 \end{bmatrix} \Rightarrow x = 30000 \text{ and } y = 15000$$

So, monthly income of Aryan = 3×30000

= ₹90000

Monthly income of Babban = 4×30000 = ₹120000

From this question we are encouraged to save a part of money every month.

100. Let ₹ x be invested in the first bond and ₹ y be invested in the second bond.

According to question,

$$\frac{10x}{100} + \frac{12y}{100} = 2800 \Rightarrow 10x + 12y = 280000 \quad \text{...(i)}$$

If the rate of interest had been interchanged, then the total interest earned is ₹ 100 less than the previous interest *i.e.*, ₹ 2700.

$$\therefore \quad \frac{12x}{100} + \frac{10y}{100} = 2700 \Rightarrow 12x + 10y = 270000 \text{...(ii)}$$

The system of equations (i) and (ii) can be represented as

$AX = B$ where $A = \begin{bmatrix} 10 & 12 \\ 12 & 10 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 280000 \\ 270000 \end{bmatrix}$

$$|A| = \begin{vmatrix} 10 & 12 \\ 12 & 10 \end{vmatrix} = 100 - 144 = -44 \neq 0$$

Thus A^{-1} exists. So, system of equations has a unique solution and given by $X = A^{-1} B$

$$\text{adj } A = \begin{bmatrix} 10 & -12 \\ -12 & 10 \end{bmatrix}$$

$$\text{Now, } X = A^{-1}B \Rightarrow X = \frac{\text{adj } A}{|A|} B$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{(-44)}\begin{bmatrix} 10 & -12 \\ -12 & 10 \end{bmatrix}\begin{bmatrix} 280000 \\ 270000 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{(-44)}\begin{bmatrix} -440000 \\ -660000 \end{bmatrix} = \begin{bmatrix} 10000 \\ 15000 \end{bmatrix}$$

$\Rightarrow$ $x = 10000$ and $y = 15000$

Therefore, ₹ 10,000 be invested in the first bond and ₹ 15,000 be invested in the second bond. Thus, the total amount invested by the trust

= 10,000 + 15,000 = ₹ 25,000.

The interest received will be given to Helpage India as donation reflects the helping and caring nature of the trust.

101. Let the monthly fees paid by poor and rich children be ₹ x and ₹ y respectively.

For batch I : $20x + 5y = 9000$ ...(i)

For batch II : $5x + 25y = 26000$ ...(ii)

The system of equations (i) and (ii) can be written as

$AX = B$

where, $A = \begin{bmatrix} 20 & 5 \\ 5 & 25 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 9000 \\ 26000 \end{bmatrix}$

$$|A| = \begin{vmatrix} 20 & 5 \\ 5 & 25 \end{vmatrix} = 500 - 25 = 475 \neq 0$$

Thus, A^{-1} exists. So, the given system has a unique solution and it is given by $X = A^{-1}B$.

$$\text{adj } A = \begin{bmatrix} 25 & -5 \\ -5 & 20 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{475}\begin{bmatrix} 25 & -5 \\ -5 & 20 \end{bmatrix}$$

$$\text{Now, } X = A^{-1}B$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{475}\begin{bmatrix} 25 & -5 \\ -5 & 20 \end{bmatrix}\begin{bmatrix} 9000 \\ 26000 \end{bmatrix}$$

www.examfear.in

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{475}\begin{bmatrix} 95000 \\ 475000 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 200 \\ 1000 \end{bmatrix}$$

$\Rightarrow$ $x = 200, y = 1000$

Hence, the monthly fees paid by each poor child is ₹ 200 and the monthly fees paid by each rich child is ₹ 1000.

By offering discount to the poor children, the coaching institute offers an unbiased chance for the development and enhancement of the weaker section of our society.

102. (i) Given, value of prize for team spirit = ₹ x

Value of prize for truthfulness = ₹ y

Value of prize for tolerance = ₹ z

Linear equation for School A is $3x + y + 2z = 1100$

Linear equation for School B is $x + 2y + 3z = 1400$

Linear equation for Prize is $x + y + z = 600$

The corresponding matrix equation is $PX = Q$

$$\text{where, } P = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } Q = \begin{bmatrix} 1100 \\ 1400 \\ 600 \end{bmatrix}$$

$$\text{(ii) Now } |P| = \begin{vmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix}$$

$= 3\cdot(2 - 3) - 1\cdot(1 - 3) + 2\cdot(1 - 2)$

$= -3 + 2 - 2 = -3 \neq 0$

Thus, P^{-1} exists. So, system of equations has unique solution and it is given by $X = P^{-1}Q$

Now, cofactors of elements of P are

$A_{11} = -1, A_{12} = 2, A_{13} = -1,$

$A_{21} = 1, A_{22} = 1, A_{23} = -2,$

$A_{31} = -1, A_{32} = -7, A_{33} = 5$

$$\therefore \quad \text{adj } P = \begin{bmatrix} -1 & 1 & -1 \\ 2 & 1 & -7 \\ -1 & -2 & 5 \end{bmatrix}$$

$$P^{-1} = \frac{1}{|P|}\cdot \text{adj } P = -\frac{1}{3}\begin{bmatrix} -1 & 1 & -1 \\ 2 & 1 & -7 \\ -1 & -2 & 5 \end{bmatrix}$$

Now, $X = P^{-1}Q$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{3}\begin{bmatrix} -1 & 1 & -1 \\ 2 & 1 & -7 \\ -1 & -2 & 5 \end{bmatrix}\begin{bmatrix} 1100 \\ 1400 \\ 600 \end{bmatrix}$$

$$= -\frac{1}{3}\begin{bmatrix} -300 \\ -600 \\ -900 \end{bmatrix} = \begin{bmatrix} 100 \\ 200 \\ 300 \end{bmatrix} \Rightarrow x = 100, y = 200, z = 300.$$

Thus, the above system of equations is solvable.

(iii) The value truthfulness should be rewarded the most because a student who is truthful will be also tolerant and will work with a team spirit in the school.

103. Given, $A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 2 & -2 \\ 2 & -1 & 1 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 2 & -3 \\ 3 & 2 & -2 \\ 2 & -1 & 1 \end{vmatrix} = 1(2 - 2) - 2(3 + 4) - 3(-3 - 4)$$

$= -14 + 21 = 7 \neq 0$

$\therefore$ A^{-1} exists

Now, $A_{11} = 0, A_{12} = -7, A_{13} = -7, A_{21} = 1, A_{22} = 7,$ $A_{23} = 5, A_{31} = 2, A_{32} = -7, A_{33} = -4$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 0 & 1 & 2 \\ -7 & 7 & -7 \\ -7 & 5 & -4 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{|A|}\text{adj } A = \frac{1}{7}\begin{bmatrix} 0 & 1 & 2 \\ -7 & 7 & -7 \\ -7 & 5 & -4 \end{bmatrix}$$

The given system of equations is

$x + 2y - 3z = 6$

$3x + 2y - 2z = 3$

$2x - y + z = 2$

The system of equations can be written as $AX = B$

$$\text{where } A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 2 & -2 \\ 2 & -1 & 1 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$$

$\because$ A^{-1} exists, so system of equations has a unique solution given by $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 0 & 1 & 2 \\ -7 & 7 & -7 \\ -7 & 5 & -4 \end{bmatrix}\begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 7 \\ -35 \\ -35 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ -5 \end{bmatrix}$$

$\Rightarrow$ $x = 1, y = -5, z = -5$

104. We have, $x - y + 2z = 7$

$2x - y + 3z = 12$

$3x + 2y - z = 5$

www.examples.in

The given system of equations can be written as $AX = B$, where

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 \\ 12 \\ 5 \end{bmatrix}$$

Here, $|A| = \begin{vmatrix} 1 & -1 & 2 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{vmatrix}$

$= 1(1 - 6) + 1(-2 - 9) + 2(4 + 3)$

$= -5 - 11 + 14 = -2 \neq 0$

$\therefore$ A^{-1} exists. So system of equations has a unique solution given by $X = A^{-1}B$

$$\therefore \quad \text{adj } A = \begin{bmatrix} -5 & 11 & 7 \\ 3 & -7 & -5 \\ -1 & 1 & 1 \end{bmatrix}' = \begin{bmatrix} -5 & 3 & -1 \\ 11 & -7 & 1 \\ 7 & -5 & 1 \end{bmatrix}$$

$$\text{Now, } A^{-1} = \frac{1}{|A|}(\text{adj } A) = \frac{-1}{2}\begin{bmatrix} -5 & 3 & -1 \\ 11 & -7 & 1 \\ 7 & -5 & 1 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} -5 & 3 & -1 \\ 11 & -7 & 1 \\ 7 & -5 & 1 \end{bmatrix}\begin{bmatrix} 7 \\ 12 \\ 5 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} -4 \\ -2 \\ -6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$\Rightarrow$ $x = 2, y = 1, z = 3$

105. We have, $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$

$$\Rightarrow \quad |A| = \begin{vmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{vmatrix}$$

$= 2(-4 + 4) + 3(-6 + 4) + 5(3 - 2)$

$= -6 + 5 = -1 \neq 0$

$\therefore$ A^{-1} exists.

Now, $A_{11} = 0, A_{12} = 2, A_{13} = 1, A_{21} = -1, A_{22} = -9, A_{23} = -5, A_{31} = 2, A_{32} = 23, A_{33} = 13$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

$$\text{Now, } A^{-1} = \frac{1}{|A|}\text{adj } A = (-1)\begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

The given system of equations is

$2x - 3y + 5z = 11, 3x + 2y - 4z = -5, x + y - 2z = -3$

The system of equations can be written as $AX = B$,

where $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$

Since A^{-1} exists, therefore, system of equations has a unique solution given by

$$X = A^{-1}B = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}\begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\Rightarrow$ $x = 1, y = 2$ and $z = 3$.

106. We have $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix}$

Now, $|A| = 1(0 - 2) - 1(1 - 6) + 1(1)$

$= -2 + 5 + 1 = 4 \neq 0$

$\therefore$ A^{-1} exists.

Now, $A_{11} = -2, A_{12} = 5, A_{13} = 1, A_{21} = 0, A_{22} = -2, A_{23} = 2, A_{31} = 2, A_{32} = -1, A_{33} = -1$

$$\therefore \quad \text{adj } A = \begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\therefore \quad A^{-1} = \frac{1}{|A|}\text{adj } A = \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

Now the given equations are

$x + y + z = 6$

$x + 0y + 2z = 7$

$3x + y + z = 12$

The given system of equations can be written as $AX = B$ where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix}$$

$\because A^{-1}$ exists, so system has a unique solution given by $X = A^{-1}B$.

www.exambites.in

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}\begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} -12+0+24 \\ 30-14-12 \\ 6+14-12 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 12 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

$\Rightarrow x = 3, y = 1, z = 2$

107. Here, $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$

$$\Rightarrow |A| = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{vmatrix}$$

$= 1(-12 + 6) - 2(-8 - 6) - 3(-6 - 9)$
$= -6 + 28 + 45 = 67 \neq 0$
$\therefore$ A^{-1} exists.
Now, $A_{11} = -6, A_{12} = 14, A_{13} = -15, A_{21} = 17, A_{22} = 5,$ $A_{23} = 9, A_{31} = 13, A_{32} = -8, A_{33} = -1$

$$\therefore \quad \text{adj } A = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

The given system of equations is
$x + 2y - 3z = -4$
$2x + 3y + 2z = 2$
$3x - 3y - 4z = 11$
The system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$

$\therefore$ A^{-1} exists, so system of equations has a unique solution given by $X = A^{-1} B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}\begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

$$= \frac{1}{67}\begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \Rightarrow x = 3, y = -2, z = 1.$$

108. We have, $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}\begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} 1 & 0 & 3 \\ -1 & 2 & -2 \\ 2 & -3 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 9 & 6 \\ 0 & 2 & 1 \\ 1 & -3 & -2 \end{bmatrix}$$

Now the given system of equations is
$x + 3z = 9$
$-x + 2y - 2z = 4$
$2x - 3y + 4z = -3$
The system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 0 & 3 \\ -1 & 2 & -2 \\ 2 & -3 & 4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 9 \\ 4 \\ -3 \end{bmatrix}$

Since, A^{-1} exists, so system of equations has a unique solution, given by
$X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 & 9 & 6 \\ 0 & 2 & 1 \\ 1 & -3 & -2 \end{bmatrix}\begin{bmatrix} 9 \\ 4 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} -18+36-18 \\ 8-3 \\ 9-12+6 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \end{bmatrix}$$

$\Rightarrow x = 0, y = 5, z = 3$

109. We have, $\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$

$$= \begin{bmatrix} -4+4+8 & 4-8+4 & -4-8+12 \\ -7+1+6 & 7-2+3 & -7-2+9 \\ 5-3-2 & -5+6-1 & 5+6-3 \end{bmatrix}$$

www.examrules.in

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} = 8I$$

$$\Rightarrow \frac{1}{8}\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix} = I$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}^{-1} = \frac{1}{8}\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}$$

The given system of equations is
$x - y + z = 4$, $x - 2y - 2z = 9$, $2x + y + 3z = 1$ and it can be written as
$AX = B$, where

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 9 \\ 1 \end{bmatrix}$$

Here, $|A| = 1(-6 + 2) + 1(3 + 4) + 1(1 + 4)$
$= -4 + 7 + 5 = 8 \neq 0$

So, the given system of equations has a unique solution given by $X = A^{-1}B$.

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}\begin{bmatrix} 4 \\ 9 \\ 1 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -16 + 36 + 4 \\ -28 + 9 + 3 \\ 20 - 27 - 1 \end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix} 24 \\ -16 \\ -8 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} \Rightarrow x = 3, y = -2, z = -1$$

110. $A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

Since, $AA^{-1} = I$

$$\Rightarrow \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 2 \end{bmatrix}A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Applying $R_1 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 8 & 4 & 3 \end{bmatrix}A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 8R_1$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -3 & -3 \\ 0 & -12 & -13 \end{bmatrix}A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ 1 & 0 & -8 \end{bmatrix}$$

Applying $R_3 \to (-1)\, R_3$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -3 & -3 \\ 0 & 12 & 13 \end{bmatrix}A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ -1 & 0 & 8 \end{bmatrix}$$

Applying $R_3 \to R_3 + 4R_2$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 1 \end{bmatrix}A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ -1 & 4 & 0 \end{bmatrix}$$

Applying $R_2 \to \left(-\frac{1}{3}\right) R_2$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1/3 & 2/3 \\ -1 & 4 & 0 \end{bmatrix}$$

Applying $R_2 \to R_2 - R_3$ and $R_1 \to R_1 - 2R_3$, we get

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}A^{-1} = \begin{bmatrix} 2 & -8 & 1 \\ 1 & -13/3 & 2/3 \\ -1 & 4 & 0 \end{bmatrix}$$

Applying $R_1 \to R_1 - 2R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}A^{-1} = \begin{bmatrix} 0 & 2/3 & -1/3 \\ 1 & -13/3 & 2/3 \\ -1 & 4 & 0 \end{bmatrix}$$

So, $A^{-1} = \begin{bmatrix} 0 & 2/3 & -1/3 \\ 1 & -13/3 & 2/3 \\ -1 & 4 & 0 \end{bmatrix}$ ...(i)

The given system of linear equations can be written as $AX = B$, where

$$A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 19 \\ 5 \\ 7 \end{bmatrix}$$

$\therefore$ The solution of above equation is $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 2/3 & -1/3 \\ 1 & -13/3 & 2/3 \\ -1 & 4 & 0 \end{bmatrix}\begin{bmatrix} 19 \\ 5 \\ 7 \end{bmatrix} \quad \text{[From (i)]}$$

www.exambites.in

$$= \begin{bmatrix} 0+10/3-7/3 \\ 19-65/3+14/3 \\ -19+20+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \Rightarrow x = 1, y = 2, z = 1$$

111. Let one pen of variety '*A*' costs ₹ *x*, one pen of variety '*B*' costs ₹ *y* and one pen of variety '*C*' costs ₹ *z*.

According to question,

$x + y + z = 21$ (For Meenu)

$4x + 3y + 2z = 60$ (For Jeevan)

$6x + 2y + 3z = 70$ (For Shikha)

The given system of equations can be written as $AX = B$

$$\text{where, } A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & 2 \\ 6 & 2 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 21 \\ 60 \\ 70 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 2 \\ 6 & 2 & 3 \end{vmatrix} = 1(9-4) - 1(12-12) + 1(8-18) = -5 \neq 0$$

$\therefore$ A^{-1} exists and system of equations has a unique solution given by $X = A^{-1}B$.

Now, $A_{11} = 5$, $A_{12} = 0$, $A_{13} = -10$,

$A_{21} = -1$, $A_{22} = -3$, $A_{23} = 4$,

$A_{31} = -1$, $A_{32} = 2$, $A_{33} = -1$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 5 & -1 & -1 \\ 0 & -3 & 2 \\ -10 & 4 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj } A) = \frac{1}{(-5)} \begin{bmatrix} 5 & -1 & -1 \\ 0 & -3 & 2 \\ -10 & 4 & -1 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{(-5)} \begin{bmatrix} 5 & -1 & -1 \\ 0 & -3 & 2 \\ -10 & 4 & -1 \end{bmatrix} \begin{bmatrix} 21 \\ 60 \\ 70 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{(-5)} \begin{bmatrix} -25 \\ -40 \\ -40 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix} \Rightarrow x = 5, y = 8, z = 8$$

$\therefore$ Cost of 1 pen of variety '*A*' = ₹ 5

Cost of 1 pen of variety '*B*' = ₹ 8

Cost of 1 pen of variety '*C*' = ₹ 8

112. According to question, we have,

$3x + 2y + z = 1000$ ...(i)

$4x + y + 3z = 1500$ ...(ii)

$x + y + z = 600$ ...(iii)

The given system of equations can be written as $AX = B$

$$\text{where, } A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 1000 \\ 1500 \\ 600 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 1 & 1 & 1 \end{vmatrix} = -5 \neq 0$$

$\therefore$ A is invertible and system of equations has a unique solution given by $X = A^{-1}B$

Now, $A_{11} = -2$, $A_{12} = -1$, $A_{13} = 3$,

$A_{21} = -1$, $A_{22} = 2$, $A_{23} = -1$,

$A_{31} = 5$, $A_{32} = -5$, $A_{33} = -5$

$$\therefore \quad \text{adj } A = \begin{bmatrix} -2 & -1 & 5 \\ -1 & 2 & -5 \\ 3 & -1 & -5 \end{bmatrix}$$

$$\therefore \quad A^{-1} = \frac{\text{adj}(A)}{|A|} = \frac{-1}{5} \begin{bmatrix} -2 & -1 & 5 \\ -1 & 2 & -5 \\ 3 & -1 & -5 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{-1}{5} \begin{bmatrix} -2 & -1 & 5 \\ -1 & 2 & -5 \\ 3 & -1 & -5 \end{bmatrix} \begin{bmatrix} 1000 \\ 1500 \\ 600 \end{bmatrix} = \frac{-1}{5} \begin{bmatrix} -500 \\ -1000 \\ -1500 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 100 \\ 200 \\ 300 \end{bmatrix} \Rightarrow x = 100, y = 200, z = 300$$

Hence the money awarded for discipline, politeness and punctuality are ₹ 100, ₹ 200 and ₹ 300 respectively.

Apart from the above three values schools can award children for sincerity.

113. *Refer to answer 112.*

114. *Refer to answer 112.*

115. Let ₹ *x*, ₹ *y* and ₹ *z* be deposited at the rates of interest 5%, 8% and $8\frac{1}{2}$% respectively.

According to question,

$x + y + z = 7000$

$x - y = 0$

www.exambites.in

$$x\cdot\frac{5}{100}+y\cdot\frac{8}{100}+z\cdot\frac{17}{2}\times\frac{1}{100}=550$$

$\Rightarrow \quad 10x + 16y + 17z = 110000$

The system of equations can be written as $AX = B$

where, $A=\begin{bmatrix}1&1&1\\1&-1&0\\10&16&17\end{bmatrix}, X=\begin{bmatrix}x\\y\\z\end{bmatrix}$ and $B=\begin{bmatrix}7000\\0\\110000\end{bmatrix}$

$$|A| = \begin{vmatrix}1&1&1\\1&-1&0\\10&16&17\end{vmatrix} = 1(-17) - (17) + 1(16 + 10)$$

$= -8 \neq 0$

$\therefore \quad A^{-1}$ exists. So, system of equations has a unique solution and it is given by $X = A^{-1}B$

Now, $A_{11} = -17, A_{12} = -17, A_{13} = 26,$

$A_{21} = -1, A_{22} = 7, A_{23} = -6,$

$A_{31} = 1, A_{32} = 1, A_{33} = -2$

$$\therefore \quad \text{adj } A=\begin{bmatrix}-17&-1&1\\-17&7&1\\26&-6&-2\end{bmatrix}$$

$$\text{and } A^{-1}=\frac{1}{|A|}\text{adj } A=\frac{1}{-8}\begin{bmatrix}-17&-1&1\\-17&7&1\\26&-6&-2\end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix}x\\y\\z\end{bmatrix}=-\frac{1}{8}\begin{bmatrix}-17&-1&1\\-17&7&1\\26&-6&-2\end{bmatrix}\cdot\begin{bmatrix}7000\\0\\110000\end{bmatrix}$$

$$=-\frac{1}{8}\begin{bmatrix}-9000\\-9000\\-38000\end{bmatrix}=\begin{bmatrix}1125\\1125\\4750\end{bmatrix}$$

$\Rightarrow \quad x = 1125 = y, z = 4750$

116. According to question, we have

$x + y + z = 1500$

$2x + 3y + 4z = 4600$

$3x + 2y + 3z = 4100$

The system of equations can be written as $AX = B$

where, $A=\begin{bmatrix}1&1&1\\2&3&4\\3&2&3\end{bmatrix}; X=\begin{bmatrix}x\\y\\z\end{bmatrix}$ and $B=\begin{bmatrix}1500\\4600\\4100\end{bmatrix}$

$$|A|=\begin{vmatrix}1&1&1\\2&3&4\\3&2&3\end{vmatrix}$$

$= 1(9 - 8) - 1(6 - 12) + 1(4 - 9)$

$= 1 + 6 - 5 = 2 \neq 0$

$\therefore \quad A^{-1}$ exists and so, system of equations has a unique solution given by $X = A^{-1}B$

Now, $A_{11} = 1, A_{12} = 6, A_{13} = -5,$

$A_{21} = -1, A_{22} = 0, A_{23} = 1,$

$A_{31} = 1, A_{32} = -2, A_{33} = 1$

$$\therefore \quad \text{adj } A=\begin{bmatrix}1&-1&1\\6&0&-2\\-5&1&1\end{bmatrix}$$

$$A^{-1}=\frac{1}{|A|}\cdot\text{adj } A=\frac{1}{2}\begin{bmatrix}1&-1&1\\6&0&-2\\-5&1&1\end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{1}{2}\begin{bmatrix}1&-1&1\\6&0&-2\\-5&1&1\end{bmatrix}\begin{bmatrix}1500\\4600\\4100\end{bmatrix}$$

$$=\frac{1}{2}\begin{bmatrix}1000\\800\\1200\end{bmatrix}=\begin{bmatrix}500\\400\\600\end{bmatrix}\Rightarrow x = 500; y = 400; z = 600.$$

Apart from sincerity, truthfulness and hard work, the schools can include an award for regularity.

117. Let the award money for honesty = ₹ x

Let the award money for regularity = ₹ y

Let the award money for hard work = ₹ z

According to question, we have

$x + y + z = 6000; 3z + x = 11000; x + z - 2y = 0$

The system of equations can be written as $AX = B$

where, $A=\begin{bmatrix}1&1&1\\1&0&3\\1&-2&1\end{bmatrix}, X=\begin{bmatrix}x\\y\\z\end{bmatrix}, B=\begin{bmatrix}6000\\11000\\0\end{bmatrix}$

$$\text{Now, } |A|=\begin{vmatrix}1&1&1\\1&0&3\\1&-2&1\end{vmatrix}$$

$= 1(0 + 6) - 1(1 - 3) + 1(-2 - 0)$

$= 6 - (-2) - 2 = 6 \neq 0$

$\therefore \quad A$ is invertible. So, the given system has a unique solution given by $X = A^{-1}B$

Now, $A_{11} = 6, A_{12} = 2, A_{13} = -2$

$A_{21} = -3, A_{22} = 0, A_{23} = 3$

$A_{31} = 3, A_{32} = -2, A_{33} = -1$

www.examples.in

$$\therefore \quad \text{adj}\,(A) = \begin{bmatrix} 6 & -3 & 3 \\ 2 & 0 & -2 \\ -2 & 3 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}\,\text{adj}\,(A) = \frac{1}{6}\begin{bmatrix} 6 & -3 & 3 \\ 2 & 0 & -2 \\ -2 & 3 & -1 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 6 & -3 & 3 \\ 2 & 0 & -2 \\ -2 & 3 & -1 \end{bmatrix}\begin{bmatrix} 6000 \\ 11000 \\ 0 \end{bmatrix}$$

$$= \frac{1}{6}\begin{bmatrix} 3000 \\ 12000 \\ 21000 \end{bmatrix} = \begin{bmatrix} 500 \\ 2000 \\ 3500 \end{bmatrix} \Rightarrow x = 500, y = 2000, z = 3500$$

One more value which the school can include for awards is discipline.

118. According to question, we have $x + y + z = 12$

$2x + 3(y + z) = 33 \Rightarrow 2x + 3y + 3z = 33$

$x + z = 2y \Rightarrow x - 2y + z = 0$

The system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 1 & -2 & 1 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} 12 \\ 33 \\ 0 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 1 & -2 & 1 \end{vmatrix} = 1(3 + 6) - 1(2 - 3) + 1(-4 - 3)$$

$= 9 + 1 - 7 = 3 \neq 0$

$\therefore$ A^{-1} exists. So, system of equations has a unique solution and it is given by $X = A^{-1}B$

Now, $A_{11} = 9, A_{12} = 1, A_{13} = -7,$

$A_{21} = -3, A_{22} = 0, A_{23} = 3, A_{31} = 0, A_{32} = -1, A_{33} = 1$

$$\therefore \quad \text{adj}\,A = \begin{bmatrix} 9 & -3 & 0 \\ 1 & 0 & -1 \\ -7 & 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{3}\begin{bmatrix} 9 & -3 & 0 \\ 1 & 0 & -1 \\ -7 & 3 & 1 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 9 & -3 & 0 \\ 1 & 0 & -1 \\ -7 & 3 & 1 \end{bmatrix}\begin{bmatrix} 12 \\ 33 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 9 \\ 12 \\ 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

$\Rightarrow$ $x = 3, y = 4, z = 5.$

The management of the colony can include the awards for those members of the colony who help for keeping the environment of the colony free from pollution.

119. According to question, we have

$x + y + z = 12000$

$4x + 3y + 2z = 37000$

$5x + 3y + 4z = 47000$

The system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & 2 \\ 5 & 3 & 4 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} 12000 \\ 37000 \\ 47000 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 2 \\ 5 & 3 & 4 \end{vmatrix}$$

$= 1(12 - 6) - 1(16 - 10) + 1(12 - 15)$

$= 6 - 6 - 3 = -3 \neq 0$

$\therefore$ A^{-1} exists. So, system of equations has a unique solution and it is given by $X = A^{-1}B$

Now, $A_{11} = 6, A_{12} = -6, A_{13} = -3,$

$A_{21} = -1, A_{22} = -1, A_{23} = 2,$

$A_{31} = -1, A_{32} = 2, A_{33} = -1$

$$\therefore \quad \text{adj}\,A = \begin{bmatrix} 6 & -1 & -1 \\ -6 & -1 & 2 \\ -3 & 2 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}\,A = -\frac{1}{3}\begin{bmatrix} 6 & -1 & -1 \\ -6 & -1 & 2 \\ -3 & 2 & -1 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{3}\begin{bmatrix} 6 & -1 & -1 \\ -6 & -1 & 2 \\ -3 & 2 & -1 \end{bmatrix}\begin{bmatrix} 12000 \\ 37000 \\ 47000 \end{bmatrix}$$

$$= -\frac{1}{3}\begin{bmatrix} -12000 \\ -15000 \\ -9000 \end{bmatrix} = \begin{bmatrix} 4000 \\ 5000 \\ 3000 \end{bmatrix}$$

$\Rightarrow$ $x = 4000, y = 5000; z = 3000$

The values described in this question are resourcefulness, competence and determination.

120. (i) According to question, we have

$x + y + z = 9500$

$2x + 4y + 3z = 29000$

$5x + 2y + 3z = 30500$

www.examrace.in

The system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 3 \\ 5 & 2 & 3 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B = \begin{bmatrix} 9500 \\ 29000 \\ 30500 \end{bmatrix}$

(ii) $|A| = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 3 \\ 5 & 2 & 3 \end{vmatrix}$

$= 1(12 - 6) - 1(6 - 15) + 1(4 - 20)$
$= 6 + 9 - 16 = -1 \neq 0$
$\therefore$ A^{-1} exists. So, system of equations has a unique solution and it is given by $X = A^{-1}B$
Now, $A_{11} = 6$, $A_{12} = 9$, $A_{13} = -16$,
$A_{21} = -1$, $A_{22} = -2$, $A_{23} = 3$,
$A_{31} = -1$, $A_{32} = -1$, $A_{33} = 2$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 6 & -1 & -1 \\ 9 & -2 & -1 \\ -16 & 3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = (-1)\begin{bmatrix} 6 & -1 & -1 \\ 9 & -2 & -1 \\ -16 & 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & 1 & 1 \\ -9 & 2 & 1 \\ 16 & -3 & -2 \end{bmatrix}$$

$$\text{Now, } X = A^{-1}B = \begin{bmatrix} -6 & 1 & 1 \\ -9 & 2 & 1 \\ 16 & -3 & -2 \end{bmatrix}\begin{bmatrix} 9500 \\ 29000 \\ 30500 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2500 \\ 3000 \\ 4000 \end{bmatrix}$$

$\Rightarrow$ $x = 2500$, $y = 3000$, $z = 4000$.
(iii) The factories honours the most, those employees who are keeping calm in tense situations.

121. We have, $x - y + 2z = 7$
$3x + 4y - 5z = -5$
$2x - y + 3z = 12$
The given system of equations can be written as
$AX = B$

$$\text{where, } A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$$

$$\text{Here } |A| = \begin{vmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{vmatrix}$$

$= 1(12 - 5) - 3(-3 + 2) + 2(5 - 8)$
$= 7 + 3 - 6 = 4 \neq 0$.
$\therefore$ A^{-1} exists. So, system of equations has a unique solution given by $X = A^{-1}B$
Now, $A_{11} = 7$, $A_{12} = -19$, $A_{13} = -11$, $A_{21} = 1$, $A_{22} = -1$, $A_{23} = -1$, $A_{31} = -3$, $A_{32} = 11$, $A_{33} = 7$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj } A) = \frac{1}{4}\begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}\begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 8 \\ 4 \\ 12 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$\Rightarrow$ $x = 2$, $y = 1$, $z = 3$

122. We have,
$2x + 3y + 3z = 5$
$x - 2y + z = -4$
$3x - y - 2z = 3$
The given system of equations can be written as
$AX = B$

$$\text{where, } A = \begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{vmatrix}$$

$= 2(4 + 1) - 3(-2 - 3) + 3(-1 + 6)$
$= 10 + 15 + 15 = 40 \neq 0$
$\therefore$ A^{-1} exists. So, system of equations has a unique solution and it is given by $X = A^{-1}B$
Now, $A_{11} = 5$, $A_{12} = 5$, $A_{13} = 5$, $A_{21} = 3$, $A_{22} = -13$, $A_{23} = 11$, $A_{31} = 9$, $A_{32} = 1$, $A_{33} = -7$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

www.exambytes.in

$$A^{-1}=\frac{1}{|A|}\cdot \text{adj } A=\frac{1}{40}\begin{bmatrix}5 & 3 & 9\\ 5 & -13 & 1\\ 5 & 11 & -7\end{bmatrix}$$

$$\text{Now, } X=A^{-1}B=\frac{1}{40}\begin{bmatrix}5 & 3 & 9\\ 5 & -13 & 1\\ 5 & 11 & -7\end{bmatrix}\begin{bmatrix}5\\ -4\\ 3\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}x\\ y\\ z\end{bmatrix}=\frac{1}{40}\begin{bmatrix}40\\ 80\\ -40\end{bmatrix}=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$$

$\Rightarrow \quad x=1, y=2, z=-1.$

123. We have,

$3x+4y+7z=4$

$2x-y+3z=-3$

$x+2y-3z=8$

The system of equations can be written as $AX=B$

$$\text{where, } A=\begin{bmatrix}3 & 4 & 7\\ 2 & -1 & 3\\ 1 & 2 & -3\end{bmatrix}; X=\begin{bmatrix}x\\ y\\ z\end{bmatrix} \text{ and } B=\begin{bmatrix}4\\ -3\\ 8\end{bmatrix}$$

$$\text{Now, } |A|=\begin{vmatrix}3 & 4 & 7\\ 2 & -1 & 3\\ 1 & 2 & -3\end{vmatrix}$$

$= 3(3-6)-4(-6-3)+7(4+1)$

$= -9+36+35=62\neq 0$

$\therefore \quad A^{-1}$ exists. So, system of equations has a unique solution given by $X=A^{-1}B$

Now, $A_{11}=-3, A_{12}=9, A_{13}=5, A_{21}=26, A_{22}=-16, A_{23}=-2, A_{31}=19, A_{32}=5, A_{33}=-11$

$$\therefore \quad \text{adj } A=\begin{bmatrix}-3 & 26 & 19\\ 9 & -16 & 5\\ 5 & -2 & -11\end{bmatrix}$$

$$A^{-1}=\frac{1}{|A|}\cdot \text{adj } A=\frac{1}{62}\begin{bmatrix}-3 & 26 & 19\\ 9 & -16 & 5\\ 5 & -2 & -11\end{bmatrix}$$

$$\text{Now, } X=A^{-1}B=\frac{1}{62}\begin{bmatrix}-3 & 26 & 19\\ 9 & -16 & 5\\ 5 & -2 & -11\end{bmatrix}\begin{bmatrix}4\\ -3\\ 8\end{bmatrix}=\frac{1}{62}\begin{bmatrix}62\\ 124\\ -62\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix} \Rightarrow x=1, y=2, z=-1.$$

124. We have,

$x+y-z=3$

$2x+3y+z=10$

$3x-y-7z=1$

The system of equations can be written as $AX=B$

$$\text{where, } A=\begin{bmatrix}1 & 1 & -1\\ 2 & 3 & 1\\ 3 & -1 & -7\end{bmatrix}; X=\begin{bmatrix}x\\ y\\ z\end{bmatrix} \text{ and } B=\begin{bmatrix}3\\ 10\\ 1\end{bmatrix}$$

$$\text{Now, } |A|=\begin{vmatrix}1 & 1 & -1\\ 2 & 3 & 1\\ 3 & -1 & -7\end{vmatrix}$$

$= 1(-21+1)-1(-14-3)-1(-2-9)$

$= -20+17+11=8\neq 0$

$\therefore \quad A^{-1}$ exists. So, system of equation has a unique solution given by $X=A^{-1}B$

Now, $A_{11}=-20, A_{12}=17, A_{13}=-11, A_{21}=8, A_{22}=-4, A_{23}=4, A_{31}=4, A_{32}=-3, A_{33}=1$

$$\therefore \quad \text{adj}A=\begin{bmatrix}-20 & 8 & 4\\ 17 & -4 & -3\\ -11 & 4 & 1\end{bmatrix}$$

$$A^{-1}=\frac{1}{|A|}\cdot \text{adj } A=\frac{1}{8}\begin{bmatrix}-20 & 8 & 4\\ 17 & -4 & -3\\ -11 & 4 & 1\end{bmatrix}$$

$$\text{Now, } X=A^{-1}B=\frac{1}{8}\begin{bmatrix}-20 & 8 & 4\\ 17 & -4 & -3\\ -11 & 4 & 1\end{bmatrix}\begin{bmatrix}3\\ 10\\ 1\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}x\\ y\\ z\end{bmatrix}=\frac{1}{8}\begin{bmatrix}24\\ 8\\ 8\end{bmatrix}=\begin{bmatrix}3\\ 1\\ 1\end{bmatrix} \Rightarrow x=3, y=1, z=1$$

125. Here, $A=\begin{bmatrix}1 & 2 & 1\\ -1 & 1 & 1\\ 1 & -3 & 1\end{bmatrix}$

$$\Rightarrow \quad |A|=\begin{vmatrix}1 & 2 & 1\\ -1 & 1 & 1\\ 1 & -3 & 1\end{vmatrix}$$

$= 1(1+3)-2(-1-1)+1(3-1)=10\neq 0$

$\therefore \quad A^{-1}$ exists.

Now, $A_{11}=4, A_{12}=2, A_{13}=2, A_{21}=-5, A_{22}=0, A_{23}=5, A_{31}=1, A_{32}=-2, A_{33}=3$

www.exambits.in

$$\therefore \quad \text{adj } A = \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{10}\begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

The given system of equations is

$x + 2y + z = 4$; $-x + y + z = 0$; $x - 3y + z = 4$

The system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix}$

Since A^{-1} exists, therefore, system of equations has a unique solution given by

$$X = A^{-1}B = \frac{1}{10}\begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}\begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10}\begin{bmatrix} 20 \\ 0 \\ 20 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \Rightarrow x = 2, y = 0, z = 2.$$

126. *Refer to answer 107.*

127. Here, $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$; $B = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$

$$\therefore \quad AB = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = 6I$$

$$\Rightarrow \quad A \cdot \left(\frac{1}{6}B\right) = I \Rightarrow A \text{ is invertible and } A^{-1} = \frac{1}{6}B$$

Now the given system of equations is

$x - y = 3$

$2x + 3y + 4z = 17$

$y + 2z = 7$

The system of equations can be written as

$AX = P$

where, $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $P = \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$

Since A^{-1} exists, so system of equations has a unique solution given by $X = A^{-1} P$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6}BP = \frac{1}{6}\begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}\begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$= \frac{1}{6}\begin{bmatrix} 12 \\ -6 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$\Rightarrow \quad x = 2, y = -1, z = 4.$

128. Here, $A = \begin{bmatrix} 1 & 2 & 5 \\ 1 & -1 & -1 \\ 2 & 3 & -1 \end{bmatrix}$

$$\Rightarrow \quad |A| = \begin{vmatrix} 1 & 2 & 5 \\ 1 & -1 & -1 \\ 2 & 3 & -1 \end{vmatrix}$$

$= 1(1 + 3) - 2(-1 + 2) + 5(3 + 2) = 4 - 2 + 25 = 27 \neq 0$

$\therefore \quad A^{-1}$ exists.

Now, $A_{11} = 4, A_{12} = -1, A_{13} = 5, A_{21} = 17, A_{22} = -11, A_{23} = 1, A_{31} = 3, A_{32} = 6, A_{33} = -3$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 4 & 17 & 3 \\ -1 & -11 & 6 \\ 5 & 1 & -3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}\text{adj } A = \frac{1}{27}\begin{bmatrix} 4 & 17 & 3 \\ -1 & -11 & 6 \\ 5 & 1 & -3 \end{bmatrix}$$

The given system of equations is

$x + 2y + 5z = 10$

$x - y - z = -2$

$2x + 3y - z = -11$

The given system of equations can be written as

$AX = B$

where, $A = \begin{bmatrix} 1 & 2 & 5 \\ 1 & -1 & -1 \\ 2 & 3 & -1 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 10 \\ -2 \\ -11 \end{bmatrix}$

$\therefore \quad A^{-1}$ exists. So, system of equations has a unique solution given by $X = A^{-1} B$

www.exambites.in

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{27}\begin{bmatrix} 4 & 17 & 3 \\ -1 & -11 & 6 \\ 5 & 1 & -3 \end{bmatrix} \cdot \begin{bmatrix} 10 \\ -2 \\ -11 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{27}\begin{bmatrix} -27 \\ -54 \\ 81 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}$$

$\Rightarrow \quad x = -1, y = -2, z = 3$

129. The given equations are

$$\frac{2}{x}+\frac{3}{y}+\frac{10}{z}=4, \frac{4}{x}-\frac{6}{y}+\frac{5}{z}=1, \frac{6}{x}+\frac{9}{y}-\frac{20}{z}=2$$

Given equations can be written as $AX = B$

where, $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}, X = \begin{bmatrix} 1/x \\ 1/y \\ 1/z \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{vmatrix}$$

$= [2(120 - 45) - 3(-80 - 30) + 10(36 + 36)]$

$= [150 + 330 + 720] = 1200 \neq 0$

Since A^{-1} exists, therefore system of equations has a unique solution given by $X = A^{-1} B$

$A_{11} = 75, A_{12} = 110, A_{13} = 72, A_{21} = 150, A_{22} = -100, A_{23} = 0, A_{31} = 75, A_{32} = 30, A_{33} = -24$

$$\therefore \text{ adj}A = \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

Now, $X = A^{-1} B$.

$$\Rightarrow \begin{bmatrix} 1/x \\ 1/y \\ 1/z \end{bmatrix} = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/3 \\ 1/5 \end{bmatrix}$$

$$\Rightarrow \quad \frac{1}{x}=\frac{1}{2} \Rightarrow x=2, \frac{1}{y}=\frac{1}{3} \Rightarrow y=3, \frac{1}{z}=\frac{1}{5} \Rightarrow z=5$$

130. The given equations are

$x + 2y + z = 7$

$x + 3z = 11$

$2x - 3y = 1$

The given system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix}$

Now, $|A| = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix}$

$= 2(6 - 0) + 3(3 - 1) = 12 + 6 = 18 \neq 0$

$\therefore \quad A^{-1}$ exists. So, system of equations has a unique solution given by $X = A^{-1}B$

Now, $A_{11} = 9, A_{12} = 6, A_{13} = -3, A_{21} = -3, A_{22} = -2, A_{23} = 7, A_{31} = 6, A_{32} = -2, A_{33} = -2$

$$\therefore \text{ adj}A = \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}\text{adj } A = \frac{1}{18}\begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18}\begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}\begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix} = \frac{1}{18}\begin{bmatrix} 36 \\ 18 \\ 54 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$\Rightarrow \quad x = 2, y = 1, z = 3$

131. The given system of equations are

$4x + 3y + 2z = 60,$

$x + 2y + 3z = 45,$

$6x + 2y + 3z = 70$

The given system of equations can be written as $AX = B$

where, $A = \begin{bmatrix} 4 & 3 & 2 \\ 1 & 2 & 3 \\ 6 & 2 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 60 \\ 45 \\ 70 \end{bmatrix}$

Now, $|A| = \begin{vmatrix} 4 & 3 & 2 \\ 1 & 2 & 3 \\ 6 & 2 & 3 \end{vmatrix}$

www.exambits.in

$= 4(6-6) - 3(3-18) + 2(2-12)$
$= 0 + 45 - 20 = 25 \neq 0$
$\therefore$ A^{-1} exists. So system of equations has a unique solution $X = A^{-1}B$
Now,
$A_{11} = 0, A_{12} = 15, A_{13} = -10, A_{21} = -5, A_{22} = 0,$
$A_{23} = 10, A_{31} = 5, A_{32} = -10, A_{33} = 5,$

$$\therefore \quad \text{adj}A = \begin{bmatrix} 0 & -5 & 5 \\ 15 & 0 & -10 \\ -10 & 10 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}\text{adj } A = \frac{1}{25}\begin{bmatrix} 0 & -5 & 5 \\ 15 & 0 & -10 \\ -10 & 10 & 5 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\therefore \quad X = \frac{1}{25}\begin{bmatrix} 0 & -5 & 5 \\ 15 & 0 & -10 \\ -10 & 10 & 5 \end{bmatrix}\begin{bmatrix} 60 \\ 45 \\ 70 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 0-225+350 \\ 900+0-700 \\ -600+450+350 \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 125 \\ 200 \\ 200 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix}$$

$\therefore$ $x = 5, y = 8, z = 8.$

132. Here, $A = \begin{bmatrix} 3 & -4 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$

$$\Rightarrow \quad |A| = \begin{vmatrix} 3 & -4 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{vmatrix}$$

$= 3(3-0) + 4(2-5) + 2(0-3)$
$= 9 - 12 - 6 = -9 \neq 0$
$\therefore$ A^{-1} exists.
Now, $A_{11} = 3, A_{12} = 3, A_{13} = -3, A_{21} = 4, A_{22} = 1,$
$A_{23} = -4, A_{31} = -26, A_{32} = -11, A_{33} = 17$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 3 & 4 & -26 \\ 3 & 1 & -11 \\ -3 & -4 & 17 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}\cdot\text{adj } A = -\frac{1}{9}\begin{bmatrix} 3 & 4 & -26 \\ 3 & 1 & -11 \\ -3 & -4 & 17 \end{bmatrix}$$

The given system of equations is
$3x - 4y + 2z = -1,$
$2x + 3y + 5z = 7$
$x + z = 2$
The given system of equations can be written as
$AX = B$

$$\text{where, } A = \begin{bmatrix} 3 & -4 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} -1 \\ 7 \\ 2 \end{bmatrix}$$

Since A^{-1} exists, so, system of equations has a unique solution given by $X = A^{-1}B$

$$\Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{9}\begin{bmatrix} 3 & 4 & -26 \\ 3 & 1 & -11 \\ -3 & -4 & 17 \end{bmatrix}\begin{bmatrix} -1 \\ 7 \\ 2 \end{bmatrix}$$

$$= -\frac{1}{9}\begin{bmatrix} -27 \\ -18 \\ 9 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

$\Rightarrow$ $x = 3, y = 2, z = -1.$

133. Here, $A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 3 \\ 0 & -2 & 1 \end{bmatrix}; B = \begin{bmatrix} 7 & 2 & -6 \\ -2 & 1 & -3 \\ -4 & 2 & 5 \end{bmatrix}$

$$\therefore \quad AB = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 3 \\ 0 & -2 & 1 \end{bmatrix}\cdot\begin{bmatrix} 7 & 2 & -6 \\ -2 & 1 & -3 \\ -4 & 2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = 11\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 11I$$

$$\Rightarrow \quad A \times \left(\frac{1}{11}B\right) = I \Rightarrow A^{-1} = \frac{1}{11}B$$

Now the given system of equations is
$x - 2y = 10$
$2x + y + 3z = 8$
$-2y + z = 7$
The system of equations can be written as $AX = P$

$$\text{where, } A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 3 \\ 0 & -2 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, P = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

$\therefore$ A^{-1} exists, so given system of equations has a unique solution given by $X = A^{-1}P$

www.exambites.in

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{11}BP = \frac{1}{11}\begin{bmatrix} 7 & 2 & -6 \\ -2 & 1 & -3 \\ -4 & 2 & 5 \end{bmatrix}\begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

$$= \frac{1}{11}\begin{bmatrix} 44 \\ -33 \\ 11 \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$$

$\Rightarrow$ $x = 4, y = -3, z = 1.$

134. Here, $A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 1 \\ 2 & 0 & -3 \end{bmatrix}$

$$\Rightarrow |A| = \begin{vmatrix} 1 & -2 & 1 \\ 0 & -1 & 1 \\ 2 & 0 & -3 \end{vmatrix} = 1(3-0) + 2(-2+1) = 1 \neq 0$$

$\therefore$ A^{-1} exists.

Now, $A_{11} = 3, A_{12} = 2, A_{13} = 2, A_{21} = -6, A_{22} = -5, A_{23} = -4, A_{31} = -1, A_{32} = -1, A_{33} = -1$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ 2 & -4 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}\text{adj } A = \begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ 2 & -4 & -1 \end{bmatrix}$$

Now the given linear equations are

$x - 2y + z = 0$

$-y + z = -2$

$2x - 3z = 10$

The system of equations can be written as $AX = B$

$$\text{where, } A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 1 \\ 2 & 0 & -3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 0 \\ -2 \\ 10 \end{bmatrix}$$

Since A^{-1} exists, so given system of equations has a unique solution given by

$X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ 2 & -4 & -1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -2 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}$$

$\Rightarrow$ $x = 2, y = 0, z = -2.$

135. Here, $A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 0 & -1 \\ 2 & 6 & 0 \end{bmatrix}$ $\Rightarrow$ $|A| = \begin{vmatrix} 2 & -1 & 1 \\ 3 & 0 & -1 \\ 2 & 6 & 0 \end{vmatrix}$

$= 2(0+6) + 1(0+2) + 1(18-0) = 12 + 2 + 18 = 32 \neq 0$

$\therefore$ A^{-1} exists.

Now, $A_{11} = 6, A_{12} = -2, A_{13} = 18, A_{21} = 6, A_{22} = -2, A_{23} = -14, A_{31} = 1, A_{32} = 5, A_{33} = 3$

$$\therefore \quad \text{adj } A = \begin{bmatrix} 6 & 6 & 1 \\ -2 & -2 & 5 \\ 18 & -14 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{32}\begin{bmatrix} 6 & 6 & 1 \\ -2 & -2 & 5 \\ 18 & -14 & 3 \end{bmatrix}$$

Now the given equations are

$2x - y + z = -3,$

$3x - z = 0,$

$2x + 6y - 2 = 0$

The given system of equations can be written as $AX = B$

$$\text{where, } A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 0 & -1 \\ 2 & 6 & 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$$

$\therefore$ A^{-1} exists, so, the system has a unique solution given by $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{32}\begin{bmatrix} 6 & 6 & 1 \\ -2 & -2 & 5 \\ 18 & -14 & 3 \end{bmatrix}\begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix} = \frac{1}{32}\begin{bmatrix} -16 \\ 16 \\ -48 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ -3/2 \end{bmatrix}$$

$$\Rightarrow \quad x = -\frac{1}{2}, y = \frac{1}{2}, z = -\frac{3}{2}$$

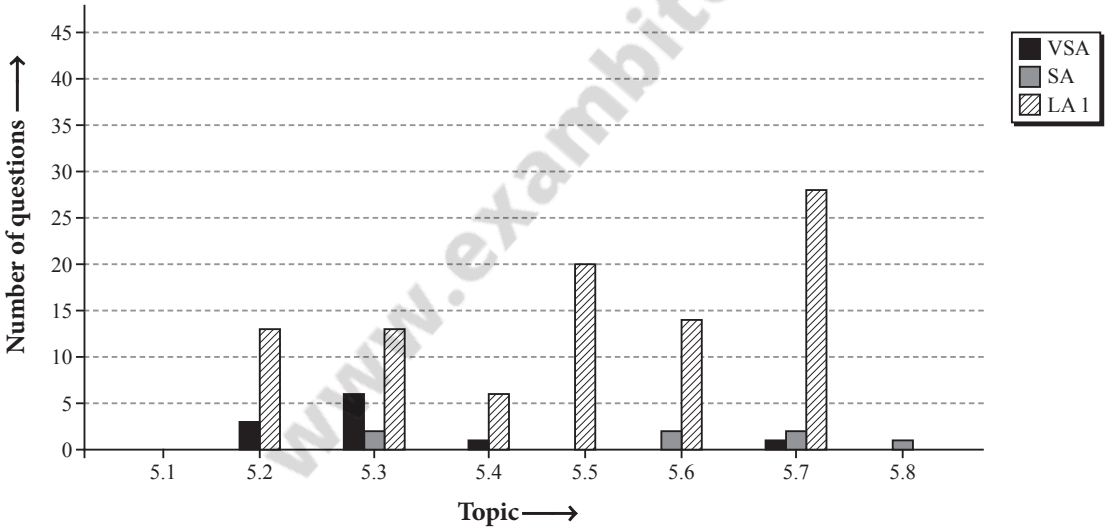
www.exambites.in

CHAPTER
5

Continuity and Differentiability

5.1 Introduction
5.2 Continuity
5.3 Differentiability
5.4 Exponential and Logarithmic Functions
5.5 Logarithmic Differentiation
5.6 Derivatives of Functions in Parametric Forms
5.7 Second Order Derivative
5.8 Mean Value Theorem

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Second Order Derivative*
- Only VSA, SA and LA 1 type questions were asked till now

QUICK RECAP

CONTINUITY

A real valued function f is said to be continuous at a point $x = c$, if the function is defined at $x = c$ and $\lim_{x \to c} f(x) = f(c)$ or we say f is continuous at $x = c$ iff $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$

- **Discontinuity of a Function :** A real function f is said to be discontinuous at $x = c$, if it is not continuous at $x = c$.

  i.e., f is discontinuous if any of the following reasons arise:

  (i) $\lim_{x \to c^-} f(x)$ or $\lim_{x \to c^+} f(x)$ or both does not exist.

(ii) $\lim_{x\to c^-} f(x) \neq \lim_{x\to c^+} f(x)$

(iii) $\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) \neq f(c)$

- A function f is said to be continuous in an interval (a, b) iff f is continuous at every point in the interval (a, b) ; and f is said to be continuous in the interval $[a, b]$ iff f is continuous in the interval (a, b) and it is continuous at a from the right and at b from the left.
- A function f is said to be discontinuous in the interval (a, b) if it is not continuous at atleast one point in the given interval.
- **Algebra of Continuous Functions :** If f and g be two real valued functions, continuous at $x = c$, then
  (i) $f + g$ is continuous at $x = c$.
  (ii) $f - g$ is continuous at $x = c$.
  (iii) $f \cdot g$ is continuous at $x = c$.
  (iv) $\left(\frac{f}{g}\right)$ is continuous at $x = c$, (provided $g(c) \neq 0$).
- Composition of two continuous functions is continuous *i.e.,* if f and g are two real valued functions and g is continuous at c and f is continuous at $g(c)$, then fog is continuous at c.
- The following functions are continuous everywhere.
  (i) Constant function
  (ii) Identity function
  (iii) Polynomial function
  (iv) Modulus function
  (v) Sine and cosine functions
  (vi) Exponential function

DIFFERENTIABILITY

Let $f(x)$ be a real function and a be any real number. Then, we define

(i) **Right-hand derivative :**
$\lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}$, if it exists, is called the right-hand derivative of $f(x)$ at $x = a$ and is denoted by $Rf'(a)$.

(ii) **Left-hand derivative :**
$\lim_{h\to 0^-} \frac{f(a-h)-f(a)}{-h}$, if it exists, is called the left-hand derivative of $f(x)$ at $x = a$ and is denoted by $Lf'(a)$.

A function $f(x)$ is said to be differentiable at $x = a$, if $Rf'(a) = Lf'(a)$.

The common value of $Rf'(a)$ and $Lf'(a)$ is denoted by $f'(a)$ and is known as the derivative of $f(x)$ at $x = a$. If, however, $Rf'(a) \neq Lf'(a)$ we say that $f(x)$ is not differentiable at $x = a$.

- A function is said to be differentiable in (a, b), if it is differentiable at every point of (a, b).
- Every differentiable function is continuous but the converse is not necessarily true.

SOME GENERAL DERIVATIVES

Function	Derivative	Function	Derivative	Function	Derivative
x^n	nx^{n-1}	$\sin x$	$\cos x$	$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$	$\cot x$	$-\operatorname{cosec}^2 x$	$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$	e^{ax}	ae^{ax}	e^x	e^x
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}};\ x \in (-1,1)$	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}};x \in (-1,1)$	$\tan^{-1} x$	$\frac{1}{1+x^2};\ x \in R$
$\cot^{-1} x$	$-\frac{1}{1+x^2};\ x \in R$	$\sec^{-1} x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}};x \in R-[-1,1]$	$\operatorname{cosec}^{-1} x$	$-\frac{1}{\lvert x\rvert\sqrt{x^2-1}};x \in R-[-1,1]$
$\log_e x$	$\frac{1}{x};\ x > 0$	a^x	$a^x \log_e a;\ a > 0$	$\log_a x$	$\frac{1}{x \log_e a};\ x > 0$ and $a > 0$

www.examrace.in

EXPONENTIAL FUNCTION

If a is any positive real number, then the function f defined by $f(x) = a^x$ is called the exponential function.

LOGARITHMIC FUNCTION

Let $a > 1$ be a real number. The logarithmic function of x to the base a is the function $y = f(x) = \log_a x$ *i.e.,* $\log_a x = b$, if $x = a^b$

- The logarithm function, with base $a = 10$, is called common logarithm and with base $a = e$, is called natural logarithm.
- The function $\log_a x$ $(a > 0, \neq 1)$ has the following properties :

(i) $\log_a(mn) = \log_a m + \log_a n$; $m, n > 0$

(ii) $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$; $m, n > 0$

(iii) $\log_a m^n = n\log_a m$; $m > 0$

(iv) $\log_a x = \frac{\log x}{\log a}$; $x > 0$

(v) $\log_a a = 1, \log_a 1 = 0$

SOME PROPERTIES OF DERIVATIVES

1.	Sum or Difference	$(u \pm v)' = u' \pm v'$
2.	Product Rule	$(uv)' = u'v + uv'$
3.	Quotient Rule	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$, $v \neq 0$
4.	Composite Function (Chain Rule)	(a) Let $y = f(t)$ and $t = g(x)$, then $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ (b) Let $y = f(t)$, $t = g(u)$ and $u = m(x)$, then $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{du} \times \frac{du}{dx}$
5.	Implicit Function	Here, we differentiate the function of type $f(x, y) = 0$.
6.	Logarithmic Function	If $y = u^v$, where u and v are the functions of x, then $\log y = v \log u$. Differentiating w.r.t. x, we get $\frac{d}{dx}(u^v) = u^v\left[\frac{v}{u}\frac{du}{dx} + \log u \frac{dv}{dx}\right]$
7.	Parametric Function	If $x = f(t)$ and $y = g(t)$, then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}$, $f'(t) \neq 0$
8.	Second Order Derivative	Let $y = f(x)$, then $\frac{dy}{dx} = f'(x)$ If $f'(x)$ is differentiable, then $\frac{d}{dx}\left(\frac{dy}{dx}\right) = f''(x)$ or $\frac{d^2y}{dx^2} = f''(x)$

ROLLE'S THEOREM

Let $f : [a, b] \to R$ be a continuous function on $[a, b]$ and differentiable on (a, b) such that $f(a) = f(b)$, then there exists some $c \in (a, b)$ such that $f'(c) = 0$

MEAN VALUE THEOREM

Let $f : [a, b] \to R$ be a continuous function on $[a, b]$ and differentiable on (a, b), then there exists some $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

www.eiambites.in

Previous Years' CBSE Board Questions

5.2 Continuity

VSA (1 mark)

1. The value of λ so that the function f defined by $f(x) = \begin{cases} \lambda x, \text{ if } x \le \pi \\ \cos x, \text{ if } x > \pi \end{cases}$ is continuous at $x = \pi$ is _____. *(2020)*

2. Determine the value of the constant 'k' so that the function $f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \ge 0 \end{cases}$ is continuous at $x = 0$. *(Delhi 2017)*

3. Determine the value of 'k' for which the following function is continuous at $x = 3$.

$$f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \ne 3 \\ k, & x = 3 \end{cases}$$ *(AI 2017)*

LA 1 (4 marks)

4. Find the values of p and q, for which

$$f(x) = \begin{cases} \frac{1-\sin^3 x}{3\cos^2 x}, & \text{if } x < \pi/2 \\ p, & \text{if } x = \pi/2 \\ \frac{q(1-\sin x)}{(\pi - 2x)^2}, & \text{if } x > \pi/2 \end{cases}$$

is continuous at $x = \pi/2$. *(Delhi 2016)*

5. Find the value of the constant k so that the function f, defined below, is continuous at $x = 0$, where

$$f(x) = \begin{cases} \left(\frac{1-\cos 4x}{8x^2}\right), & \text{if } x \ne 0 \\ k, & \text{if } x = 0 \end{cases}$$ *(AI 2014C)*

6. Find the value of k, for which

$$f(x) = \begin{cases} \frac{\sqrt{1+kx} - \sqrt{1-kx}}{x}, & \text{if } -1 \le x < 0 \\ \frac{2x+1}{x-1}, & \text{if } 0 \le x < 1 \end{cases}$$

is continuous at $x = 0$. *(AI 2013)*

7. If $f(x) = \begin{cases} \frac{1-\cos 4x}{x^2}, & \text{when } x < 0 \\ a, & \text{when } x = 0 \\ \frac{\sqrt{x}}{\left(\sqrt{16+\sqrt{x}}\right) - 4}, & \text{when } x > 0 \end{cases}$

and f is continuous at $x = 0$, find the value of a. *(Delhi 2013C, AI 2012C)*

8. If $f(x) = \begin{cases} 1, & \text{if } x \le 3 \\ ax + b, & \text{if } 3 < x < 5, \\ 7, & \text{if } x \ge 5 \end{cases}$

find the values of a and b so that $f(x)$ is a continuous function. *(AI 2013C, Delhi 2012C)*

9. Find the value of k so that the following function is continuous at $x = 2$.

$$f(x) = \begin{cases} \frac{x^3 + x^2 - 16x + 20}{(x-2)^2}; & x \ne 2 \\ k; & x = 2 \end{cases}$$ *(Delhi 2012C)*

10. Find the value of k so that the following function is continuous at $x = \frac{\pi}{2}$:

$$f(x) = \begin{cases} \frac{k\cos x}{\pi - 2x}, & \text{if } x \ne \frac{\pi}{2} \\ 5, & \text{if } x = \frac{\pi}{2} \end{cases}$$ *(Delhi 2012C)*

11. If the function $f(x)$ given by

$$f(x) = \begin{cases} 3ax + b, & \text{if } x > 1 \\ 11, & \text{if } x = 1 \\ 5ax - 2b, & \text{if } x < 1 \end{cases}$$

is continuous at $x = 1$, find the values of a and b. *(Delhi 2012C, 2011)*

12. Find the values of a and b such that the following function $f(x)$ is a continuous function :

www.exambites.in

$$f(x)=\begin{cases}5, & x\le 1\\ ax+b, & 2<x<10\\ 21, & x\ge 10\end{cases}$$

(Delhi 2011)

13. For what value of a is the function f defined by

$$f(x)=\begin{cases}a\sin\frac{\pi}{2}(x+1), & x\le 0\\ \frac{\tan x-\sin x}{x^3}, & x>0\end{cases}$$

continuous at $x = 0$? *(Delhi 2011)*

14. Find the relationship between a and b so that the function 'f' defined by

$$f(x)=\begin{cases}ax+1, & \text{if } x\le 3\\ bx+3, & \text{if } x>3\end{cases}$$

is continuous at $x = 3$. *(AI 2011)*

15. Discuss the continuity of the function $f(x)$ at $x=\frac{1}{2}$, when $f(x)$ is defined as follows:

$$f(x)=\begin{cases}\frac{1}{2}+x, & 0\le x<\frac{1}{2}\\ 1, & x=\frac{1}{2}\\ \frac{3}{2}+x, & \frac{1}{2}<x\le 1\end{cases}$$

(Delhi 2011C)

16. Find the value of 'a' if the function $f(x)$ defined by

$$f(x)=\begin{cases}2x-1, & x<2\\ a, & x=2\\ x+1, & x>2\end{cases}$$ is continuous at $x = 2$.

(AI 2011C)

5.3 Differentiability

VSA (1 mark)

17. The greatest integer function defined by $f(x) = [x]$, $0 < x < 2$ is not differentiable at $x =$ _____. *(2020)*

18. If $y = \tan^{-1} x + \cot^{-1} x$, $x \in R$, then $\frac{dy}{dx}$ is equal to _____. *(2020)*

19. If $\cos(xy) = k$, where k is a constant and $xy \ne n\pi$, $n \in Z$, then $\frac{dy}{dx}$ is equal to _____. *(2020)*

20. Differentiate $\sin^2(\sqrt{x})$ with respect to x. *(2020)*

21. Let $f(x) = x|x|$, for all $x \in R$ check its differentiability at $x = 0$. *(2020)*

22. If $f(x) = x + 1$, find $\frac{d}{dx}(fof)(x)$. *(Delhi 2019)*

SA (2 marks)

23. Differentiate $\tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$ with respect to x. *(2018)*

24. Find $\frac{dy}{dx}$ at $x = 1$, $y = \frac{\pi}{4}$ if $\sin^2 y + \cos xy = K$. *(Delhi 2017)*

LA 1 (4 marks)

25. Find the values of a and b, if the function f defined by $f(x)=\begin{cases}x^2+3x+a, & x\le 1\\ bx+2, & x>1\end{cases}$ is differentiable at $x = 1$. *(Foreign 2016)*

26. If $y=\tan^{-1}\left(\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right)$, $x^2 \le 1$ then find $\frac{dy}{dx}$. *(Delhi 2015)*

27. If $f(x)=\sqrt{x^2+1}; g(x)=\frac{x+1}{x^2+1}$ and $h(x) = 2x–3$, then find $f'[h'(g'(x))]$. *(AI 2015)*

28. Show that the function $f(x) = |x – 1| + |x + 1|$, for all $x \in R$, is not differentiable at the points $x = –1$ and $x = 1$. *(AI 2015)*

29. Find whether the following function is differentiable at $x = 1$ and $x = 2$ or not.

$$f(x)=\begin{cases}x, & x<1\\ 2-x, & 1\le x\le 2\\ -2+3x-x^2, & x>2\end{cases}$$

(Foreign 2015)

www.exambites.in

30. For what value λ of the function defined by $f(x)=\begin{cases}\lambda(x^2+2), & \text{if } x\le 0\\ 4x+6, & \text{if } x>0\end{cases}$ is continuous at $x = 0$? Hence check the differentiability of $f(x)$ at $x = 0$. *(AI 2015C)*

31. If $\cos y = x\cos(a + y)$, where $\cos a \neq \pm 1$, prove that $\frac{dy}{dx}=\frac{\cos^2(a+y)}{\sin a}$. *(Foreign 2014)*

32. If $y=\sin^{-1}\left\{x\sqrt{1-x}-\sqrt{x}\sqrt{1-x^2}\right\}$ and $0 < x < 1$, then find $\frac{dy}{dx}$. *(AI 2014C)*

33. Show that the function $f(x) = |x - 3|$, $x \in R$, is continuous but not differentiable at $x = 3$. *(Delhi 2013, AI 2012C)*

34. If $\sin y = x \sin(a + y)$, then prove that $\frac{dy}{dx}=\frac{\sin^2(a+y)}{\sin a}$. *(Delhi 2012, 2011C)*

35. Differentiate $\tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right]$ with respect to x. *(AI 2012)*

36. If $x\sqrt{1+y}+y\sqrt{1+x}=0$ for $x \neq y$. Prove the following :
$\frac{dy}{dx}=\frac{-1}{(1+x)^2}$. *(Delhi 2011C)*

37. If $y = a \sin x + b \cos x$, prove that
$y^2+\left(\frac{dy}{dx}\right)^2=a^2+b^2$. *(AI 2011C)*

5.4 Exponential and Logarithmic Functions

VSA (1 mark)

38. If $y = \log(\cos e^x)$, then find $\frac{dy}{dx}$. *(AI 2019)*

LA 1 (4 marks)

39. If $y = e^{x^2\cos x} + (\cos x)^x$, then find $\frac{dy}{dx}$. *(2020)*

40. If $\log(x^2 + y^2) = 2\tan^{-1}\left(\frac{y}{x}\right)$, show that $\frac{dy}{dx}=\frac{x+y}{x-y}$. *(Delhi 2019)*

41. If $y=\frac{x\cos^{-1}x}{\sqrt{1-x^2}}-\log\sqrt{1-x^2}$, then prove that $\frac{dy}{dx}=\frac{\cos^{-1}x}{(1-x^2)^{3/2}}$. *(Delhi 2015C)*

42. If $e^x + e^y = e^{x+y}$, prove that $\frac{dy}{dx}+e^{y-x}=0$ *(Foreign 2014)*

43. If $y=\tan^{-1}\left(\frac{a}{x}\right)+\log\sqrt{\frac{x-a}{x+a}}$, prove that $\frac{dy}{dx}=\frac{2a^3}{x^4-a^4}$. *(AI 2014C)*

44. If $\log(\sqrt{1+x^2}-x)=y\sqrt{1+x^2}$, show that $(1+x^2)\frac{dy}{dx}+xy+1=0$ *(AI 2011C)*

5.5 Logarithmic Differentiation

LA 1 (4 marks)

45. If $y = x^3 (\cos x)^x + \sin^{-1}\sqrt{x}$, find $\frac{dy}{dx}$. *(2020)*

46. If $y = (\log x)^x + x^{\log x}$, then find $\frac{dy}{dx}$. *(2020, Delhi 2013)*

47. If $x^y - y^x = a^b$, find $\frac{dy}{dx}$. *(Delhi 2019)*

48. If $y = (x)^{\cos x} + (\cos x)^{\sin x}$, then find $\frac{dy}{dx}$. *(AI 2019)*

49. Differentiate the function $(\sin x)^x + \sin^{-1}\sqrt{x}$ with respect to x. *(Delhi 2017)*

50. If $x^y + y^x = a^b$, then find $\frac{dy}{dx}$. *(AI 2017)*

51. Differentiate $x^{\sin x} + (\sin x)^{\cos x}$ with respect to x. *(AI 2016)*

52. If $y=(\sin x)^x+\sin^{-1}\sqrt{x}$, then find $\frac{dy}{dx}$. *(Delhi 2015C, 2013C)*

53. If $x^m y^n = (x + y)^{m+n}$, prove that $\frac{dy}{dx}=\frac{y}{x}$. *(Foreign 2014)*

www.examrites.in

54. If $(x-y)\cdot e^{\frac{x}{x-y}} = a$, prove that $y\frac{dy}{dx} + x = 2y$.
(Delhi 2014C)

55. If $(\tan^{-1}x)^y + y^{\cot x} = 1$, then find $\frac{dy}{dx}$.
(AI 2014C)

56. If $y^x = e^{y-x}$, prove that $\frac{dy}{dx} = \frac{(1+\log y)^2}{\log y}$.
(AI 2013)

57. Differentiate the following with respect to x:
$\sin^{-1}\left(\frac{2^{x+1}\cdot 3^x}{1+(36)^x}\right)$ *(AI 2013)*

58. If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.
(AI 2013)

59. Find $\frac{dy}{dx}$, if $y = \sin^{-1}\left(\frac{2^{x+1}}{1+4^x}\right)$. *(AI 2013C)*

60. If $(\cos x)^y = (\cos y)^x$, find $\frac{dy}{dx}$. *(Delhi 2012)*

61. If $y = x^{\sin x - \cos x} + \frac{x^2-1}{x^2+1}$, find $\frac{dy}{dx}$.
(Delhi 2012C)

62. Find $\frac{dy}{dx}$ when $y = x^{\cot x} + \frac{2x^2-3}{x^2+x+2}$.
(AI 2012C)

63. Differentiate $x^{x\cos x} + \frac{x^2+1}{x^2-1}$ w.r.t. x.
(Delhi 2011)

64. If $x^y = e^{x-y}$, show that $\frac{dy}{dx} = \frac{\log x}{\{\log(xe)\}^2}$.
(AI 2011)

5.6 Derivatives of Functions in Parametric Forms

SA (2 marks)

65. If $x = a\sec\theta$, $y = b\tan\theta$, then find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$. *(2020)*

66. Find the differential of $\sin^2 x$ w.r.t. $e^{\cos x}$.
(2020)

LA 1 (4 marks)

67. Differentiate $\tan^{-1}\left[\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}\right]$ with respect to $\cos^{-1}x^2$. *(AI 2019)*

68. If $x = a(2\theta - \sin 2\theta)$ and $y = a(1-\cos 2\theta)$, find $\frac{dy}{dx}$ when $\theta = \frac{\pi}{3}$. *(2018)*

69. If $x = a\sin 2t(1+\cos 2t)$ and $y = b\cos 2t(1-\cos 2t)$, find the values of $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ and $t = \frac{\pi}{3}$.
(Delhi 2016, AI 2016)

70. Differentiate
$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ w.r.t. $\sin^{-1}\frac{2x}{1+x^2}$, if $x \in (-1, 1)$
(Foreign 2016, Delhi 2014)

71. If $x = ae^t(\sin t + \cos t)$ and $y = ae^t(\sin t - \cos t)$, prove that $\frac{dy}{dx} = \frac{x+y}{x-y}$. *(AI 2015C)*

72. Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ with respect to $\cos^{-1}\left(2x\sqrt{1-x^2}\right)$, when $x \neq 0$. *(Delhi 2014)*

73. Differentiate $\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ with respect to $\sin^{-1}(2x\sqrt{1-x^2})$. *(Delhi 2014)*

74. Find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$, if $x = ae^{\theta}(\sin\theta - \cos\theta)$ and $y = ae^{\theta}(\sin\theta + \cos\theta)$.
(AI 2014)

75. If $x = a\sin 2t(1+\cos 2t)$ and $y = b\cos 2t(1-\cos 2t)$, show that at $t = \frac{\pi}{4}, \left(\frac{dy}{dx}\right) = \frac{b}{a}$. *(AI 2014)*

76. If $x = \cos t(3 - 2\cos^2 t)$ and $y = \sin t(3 - 2\sin^2 t)$, find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$. *(AI 2014)*

www.exambites.in

77. If $x = 2\cos\theta - \cos 2\theta$ and $y = 2\sin\theta - \sin 2\theta$, then prove that $\frac{dy}{dx} = \tan\left(\frac{3\theta}{2}\right)$. *(Delhi 2013C)*

78. If $x = \sqrt{a^{\sin^{-1}t}}$, $y = \sqrt{a^{\cos^{-1}t}}$, show that $\frac{dy}{dx} = -\frac{y}{x}$. *(AI 2012)*

79. If $x = a(\theta - \sin\theta)$ and $y = a(1 + \cos\theta)$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$. *(Delhi 2011C)*

80. If $x = a(\cos t + \log\tan\frac{t}{2})$ and $y = a\sin t$, find $\frac{dy}{dx}$. *(Delhi 2011C)*

5.7 Second Order Derivative

VSA (1 mark)

81. If $y = \log_e\left(\frac{x^2}{e^2}\right)$, then $\frac{d^2y}{dx^2}$ equals

(a) $-\frac{1}{x}$ (b) $-\frac{1}{x^2}$

(c) $\frac{2}{x^2}$ (d) $-\frac{2}{x^2}$ *(2020)*

SA (2 marks)

82. If $x = at^2$, $y = 2at$, then find $\frac{d^2y}{dx^2}$. *(2020)*

83. If $x = a\cos\theta$; $y = b\sin\theta$, then find $\frac{d^2y}{dx^2}$. *(2020)*

LA 1 (4 marks)

84. If $y = (\sin^{-1}x)^2$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 2 = 0$. *(Delhi 2019)*

85. If $x = \sin t$, $y = \sin pt$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + p^2y = 0$. *(AI 2019, Foreign 2016)*

86. If $y = \sin(\sin x)$, prove that $\frac{d^2y}{dx^2} + \tan x\frac{dy}{dx} + y\cos^2 x = 0$. *(2018)*

87. If $x^m y^n = (x + y)^{m+n}$, prove that $\frac{d^2y}{dx^2} = 0$. *(Delhi 2017)*

88. If $e^y(x + 1) = 1$, then show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$. *(AI 2017)*

89. If $y = x^x$, prove that $\frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0$. *(Delhi 2016, 2014)*

90. If $y = 2\cos(\log x) + 3\sin(\log x)$, prove that $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$ *(AI 2016)*

91. If $x = a\cos\theta + b\sin\theta$, $y = a\sin\theta - b\cos\theta$, show that $y^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ *(Delhi 2015, Foreign 2014, AI 2013C)*

92. If $y = e^{m\sin^{-1}x}$, $-1 \le x \le 1$, then show that $(1-x^2)\frac{d^2y}{dx^2} - \frac{xdy}{dx} - m^2y = 0$. *(AI 2015)*

93. If $y = (x + \sqrt{1+x^2})^n$, then show that $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = n^2y$. *(Foreign 2015, Delhi 2013C)*

94. If $x = a\sec^3\theta$, $y = a\tan^3\theta$, find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$. *(Delhi 2015C)*

95. If $y = Ae^{mx} + Be^{nx}$, show that $\frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny = 0$. *(AI 2015C, 2014)*

96. If $x = a(\cos t + t\sin t)$ and $y = a(\sin t - t\cos t)$, then find the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$. *(Delhi 2014C)*

97. If $x = a\left(\cos t + \log\tan\frac{t}{2}\right)$, $y = a\sin t$, evaluate $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{3}$. *(Delhi 2014C)*

www.exambites.in

98. If $x = a \sin t$ and $y = a(\cos t + \log \tan(t/2))$, find $\frac{d^2y}{dx^2}$. *(Delhi 2013)*

99. If $y = \log\left[x + \sqrt{x^2 + a^2}\right]$, show that

$$(x^2 + a^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0.$$

(Delhi 2013, 2013C)

100. If $x = a\cos^3\theta$ and $y = a\sin^3\theta$, then find the value of $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{6}$. *(AI 2013)*

101. If $y = x\log\left(\frac{x}{a+bx}\right)$, then prove that

$$x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2.$$ *(Delhi 2013C)*

102. If $x = \tan\left(\frac{1}{a}\log y\right)$, then show that

$$(1+x^2)\frac{d^2y}{dx^2} + (2x - a)\frac{dy}{dx} = 0.$$

(AI 2013C, 2011)

103. If $x = \cos\theta$ and $y = \sin^3\theta$, then prove that

$$y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 3\sin^2\theta(5\cos^2\theta - 1).$$

(AI 2013C)

104. If $y = \sin^{-1}x$, show that

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$$ *(Delhi 2012)*

105. If $y = (\tan^{-1}x)^2$, show that

$$(x^2+1)^2\frac{d^2y}{dx^2} + 2x(x^2+1)\frac{dy}{dx} = 2.$$

(Delhi 2012, AI 2012)

106. If $y = 3\cos(\log x) + 4\sin(\log x)$, show that

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0.$$ *(Delhi 2012)*

107. If $x = a(\cos t + t\sin t)$ and $y = a(\sin t - t\cos t)$, $0 < t < \frac{\pi}{2}$, find $\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$.

(AI 2012, 2011C, Delhi 2012C)

108. If $x = a\left(\cos t + \log\tan\frac{t}{2}\right), y = a\sin t$, find $\frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$. *(AI 2012)*

109. If $x = \cos t + \log\tan\frac{t}{2}, y = \sin t$, then find the value of $\frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

(AI 2012C)

110. If $x = a(\theta - \sin\theta)$, $y = a(1 + \cos\theta)$, find $\frac{d^2y}{dx^2}$.

(Delhi 2011)

111. If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, find $\frac{d^2y}{dx^2}$. *(AI 2011C)*

5.8 Mean Value Theorem

SA (2 marks)

112. Find the value of c in Rolle's theorem for the function $f(x) = x^3 - 3x$ in $[-\sqrt{3}, 0]$.

(AI 2017)

Detailed Solutions

1. $\because$ $f(x)$ is continuous at $x = \pi$

$\therefore$ $f(\pi) = \lim\limits_{x\to\pi^-} f(x) = \lim\limits_{x\to\pi^+} f(x)$ ...(i)

Here, $f(\pi) = \lambda\pi$ ...(ii)

And $\lim\limits_{x\to\pi^+} f(x) = \lim\limits_{h\to 0} f(\pi + h)$

$= \lim\limits_{h\to 0}\cos(\pi + h) = -1$ ...(iii)

From (i), (ii) and (iii), we get

$\lambda\pi = -1 \Rightarrow \lambda = -\frac{1}{\pi}$

2. We have, $f(x) = \begin{cases} \frac{kx}{|x|}, & x < 0 \\ 3, & x \geq 0 \end{cases}$

L.H.L. $= \lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0}\frac{kx}{-x} = -k$

www.exambites.in

R.H.L. = $\lim_{x\to 0^+} f(x) = \lim_{x\to 0} 3 = 3$

Since, $f(x)$ is continuous at $x = 0$.

$\therefore$ L.H.L. = R.H.L. = $f(0)$

$\Rightarrow$ $-k = 3 \Rightarrow k = -3$

3. Given, $f(x)$ is continuous at $x = 3$.

So, $\lim_{x\to 3} f(x) = f(3) \Rightarrow \lim_{x\to 3} \frac{(x+3)^2 - 36}{x-3} = k$

$\Rightarrow \lim_{x\to 3} \frac{(x+3)^2 - 6^2}{x-3} = k$

$\Rightarrow \lim_{x\to 3} \frac{(x+3+6)(x+3-6)}{x-3} = k$

$\Rightarrow$ $3 + 3 + 6 = k$

$\Rightarrow$ $k = 12$

4. $\because$ $f(x)$ is continuous at $\pi/2$.

$\therefore \lim_{x\to \pi/2^-} f(x) = \lim_{x\to \pi/2^+} f(x) = f(\pi/2)$ ...(1)

Now, $\lim_{x\to \pi/2^-} f(x) = \lim_{h\to 0} f\left(\frac{\pi}{2} - h\right)$

$= \lim_{h\to 0} \frac{1 - \sin^3\left(\frac{\pi}{2} - h\right)}{3\cos^2\left(\frac{\pi}{2} - h\right)} = \lim_{h\to 0} \frac{1 - \cos^3 h}{3\sin^2 h}$

$= \lim_{h\to 0} \frac{(1-\cos h)(1+\cos^2 h + \cos h)}{3(1-\cos h)(1+\cos h)}$

$= \lim_{h\to 0} \frac{(1+\cos^2 h + \cos h)}{3(1+\cos h)} = \frac{1+1+1}{3(1+1)} = \frac{1}{2}$

and $\lim_{x\to \pi/2^+} f(x) = \lim_{h\to 0} f\left(\frac{\pi}{2} + h\right)$

$= \lim_{h\to 0} \frac{q\left[1 - \sin\left(\frac{\pi}{2} + h\right)\right]}{\left[\pi - 2\left(\frac{\pi}{2} + h\right)\right]^2} = \lim_{h\to 0} \frac{q(1-\cos h)}{4h^2}$

$= \frac{q}{4} \times \lim_{h\to 0} \frac{2\sin^2 \frac{h}{2}}{\frac{h^2}{4} \times 4} = \frac{q}{4} \times \frac{2}{4} = \frac{q}{8}$

and $f(\pi/2) = p$ $\therefore$ $\frac{1}{2} = \frac{q}{8} = p$

$\Rightarrow$ $p = \frac{1}{2}$ and $q = 4$ [From (1)]

5. $\because$ $f(x)$ is continuous at $x = 0$.

$\therefore$ $f(0) = k$

and $\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{1 - \cos 4x}{8x^2}$

$= \lim_{x\to 0} \frac{2 \cdot \sin^2 2x}{8x^2} = \lim_{x\to 0} \left(\frac{\sin 2x}{2x}\right)^2 = 1$

$\because$ f is continuous at $x = 0$

$\therefore$ $f(0) = \lim_{x\to 0} f(x) \Rightarrow k = 1$

6. $\because$ $f(x)$ is continuous at $x = 0$

$\therefore \lim_{x\to 0^+} f(x) = f(0) = \lim_{x\to 0^-} f(x)$ ...(1)

Now $f(0) = \frac{2\times 0 + 1}{0 - 1} = -1$

$\lim_{x\to 0^+} f(x) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} \frac{2h+1}{h-1} = -1$

$\lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} \frac{\sqrt{1-kh} - \sqrt{1+kh}}{-h}$

$= \lim_{h\to 0} \frac{\sqrt{1-kh} - \sqrt{1+kh}}{-h} \times \frac{\sqrt{1-kh} + \sqrt{1+kh}}{\sqrt{1-kh} + \sqrt{1+kh}}$

$= \lim_{h\to 0} \frac{(1-kh) - (1+kh)}{-h[\sqrt{1-kh} + \sqrt{1+kh}]}$

$= \lim_{h\to 0} \frac{2k}{\sqrt{1-kh} + \sqrt{1-kh}} = \frac{2k}{2} = k$

$\therefore$ From (1), we get $k = -1$

7. $\because$ $f(x)$ is continuous at $x = 0$.

$\therefore \lim_{x\to 0^+} f(x) = f(0) = \lim_{x\to 0^-} f(x)$ ...(1)

Now $f(0) = a$

$\lim_{x\to 0^+} f(x) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} \frac{\sqrt{h}}{\sqrt{16+\sqrt{h}} - 4}$

$= \lim_{h\to 0} \frac{\sqrt{h}}{\sqrt{16+\sqrt{h}} - 4} \times \frac{\sqrt{16+\sqrt{h}} + 4}{\sqrt{16+\sqrt{h}} + 4}$

$= \lim_{h\to 0} \frac{\sqrt{h}\left(\sqrt{16+\sqrt{h}} + 4\right)}{16 + \sqrt{h} - 4^2} = \lim_{h\to 0} \sqrt{16+\sqrt{h}} + 4 = 8$

$\lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} \frac{1 - \cos 4(-h)}{(-h)^2}$

$= \lim_{h\to 0} \frac{1 - \cos 4h}{h^2} = \lim_{h\to 0} \frac{2\sin^2 2h}{h^2}$

$= 8 . \lim_{h\to 0} \left(\frac{\sin 2h}{2h}\right)^2 = 8$

$\therefore$ From (1), we get $a = 8$

www.exambites.in

8. Continuity at $x = 3$

$\because$ $f(x)$ is continuous at $x = 3$

$\therefore \quad f(3) = \lim\limits_{x\to 3^-} f(x) = \lim\limits_{x\to 3^+} f(x)$ ...(1)

Now, $f(3) = 1$

$\lim\limits_{x\to 3^-} f(x) = \lim\limits_{h\to 0} f(3-h) = \lim\limits_{h\to 0} 1 = 1$

$\lim\limits_{x\to 3^+} f(x) = \lim\limits_{h\to 0} f(3+h)$

$= \lim\limits_{h\to 0} [a(3+h)+b] = 3a+b$

$\therefore$ From (1), $3a + b = 1$ ...(2)

Continuity at $x = 5$

$\because$ $f(x)$ is continuous at $x = 5$

$\therefore \quad f(5) = \lim\limits_{x\to 5^-} f(x) = \lim\limits_{x\to 5^+} f(x)$ ...(3)

Now $f(5) = 7$

$\lim\limits_{x\to 5^+} f(x) = \lim\limits_{h\to 0} f(5+h) = \lim\limits_{h\to 0} 7 = 7$

$\lim\limits_{x\to 5^-} f(x) = \lim\limits_{h\to 0} f(5-h)$

$= \lim\limits_{h\to 0} [a(5-h)+b] = 5a+b$

$\therefore$ From (3), $5a + b = 7$ ...(4)

Solving (2) and (4) we get

$a = 3, b = -8$.

9. $\because$ $f(x)$ is continuous at $x = 2$

$\therefore \quad \lim\limits_{x\to 2} f(x) = f(2)$ ...(1)

Here $f(2) = k$

$$\lim_{x\to 2} f(x) = \lim_{x\to 2} \frac{x^3+x^2-16x+20}{(x-2)^2}$$

$$= \lim_{x\to 2} \frac{(x-2)^2(x+5)}{(x-2)^2}$$

$$= \lim_{x\to 2} (x+5) = 7$$

$\therefore$ From (1), we get $k = 7$

10. $\because$ $f(x)$ is continuous at $x = \frac{\pi}{2}$,

$\therefore \quad f\left(\frac{\pi}{2}\right) = \lim\limits_{x\to \frac{\pi}{2}} f(x)$ ...(1)

Here $f\left(\frac{\pi}{2}\right) = 5$

$$\lim_{x\to \frac{\pi}{2}} f(x) = \lim_{x\to \frac{\pi}{2}} \frac{k\cos x}{\pi - 2x} = \lim_{h\to 0} \frac{k\cos\left(\frac{\pi}{2}+h\right)}{\pi - 2\left(\frac{\pi}{2}+h\right)}$$

$$= \lim_{h\to 0} \frac{-k\sin h}{-2h} = \frac{k}{2}\lim_{h\to 0}\frac{\sin h}{h} = \frac{k}{2}.1 = \frac{k}{2}$$

$\therefore$ From (1), we get $k = 10$

11. $\because$ $f(x)$ is continuous at $x = 1$

$\therefore \quad \lim\limits_{x\to 1^-} f(x) = f(1) = \lim\limits_{x\to 1^+} f(x)$ ...(1)

Here $f(1) = 11$

$\lim\limits_{x\to 1^-} f(x) = \lim\limits_{h\to 0} f(1-h) = \lim\limits_{h\to 0} [5a(1-h)-2b]$

$= 5a - 2b$

$\lim\limits_{x\to 1^+} f(x) = \lim\limits_{h\to 0} f(1+h) = \lim\limits_{h\to 0} [3a(1+h)+b] = 3a+b$

$\therefore$ From (1), we get

$5a - 2b = 11$ and $3a + b = 11$

Solving these, we get

$a = 3, b = 2$.

12. *Refer to answer 8.*

13. $\because$ $f(x)$ is continuous at $x = 0$,

$\therefore \quad \lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0^+} f(x) = f(0)$ ...(1)

Here, $f(0) = a\sin\frac{\pi}{2} = a$

$\therefore \quad \lim\limits_{x\to 0^-} f(x) = \lim\limits_{h\to 0} a\ \sin\left(\frac{\pi}{2}(-h+1)\right)$

$= \lim\limits_{h\to 0} a\sin\left(\frac{\pi}{2} - h\frac{\pi}{2}\right) = a$

$$\lim_{x\to 0^+} f(x) = \lim_{h\to 0} \frac{\tan h - \sin h}{h^3}$$

$$= \lim_{h\to 0} \frac{\frac{\sin h}{\cos h} - \sin h}{h^3} = \lim_{h\to 0} \frac{\sin h\left(\frac{1}{\cos h} - 1\right)}{h^3}$$

$$= \lim_{h\to 0} \frac{\sin h}{h} \times \lim_{h\to 0} \frac{1-\cos h}{\cos h \cdot h^2}$$

$$= 1\times \lim_{h\to 0}\frac{1}{\cos h} \times \lim_{x\to 0} \frac{2\sin^2\left(\frac{h}{2}\right)}{4\times\left(\frac{h}{2}\right)^2} = 1\times\frac{1}{2} = \frac{1}{2}$$

$\Rightarrow \quad a = \frac{1}{2}$

Hence, $f(x)$ is continuous at $x = 0$, if $a = \frac{1}{2}$.

14. $\because$ $f(x)$ is continuous at $x = 3$

$\therefore \quad \lim\limits_{x\to 3^-} f(x) = \lim\limits_{x\to 3^+} f(x) = f(3)$ ...(1)

www.examrites.in

$$\lim_{x\to 3^-} f(x) = \lim_{h\to 0}(a(3-h)+1) = 3a+1$$

$$\lim_{x\to 3^+} f(x) = \lim_{h\to 0}(b(3+h)+3) = 3b+3$$

Also, $f(3) = 3a + 1$

From (1), $3a + 1 = 3b + 3 \Rightarrow a - b = \frac{2}{3}$,

which is the required relation between a and b.

15. Here, $f\left(\frac{1}{2}\right) = 1$.

$$\lim_{x\to\left(\frac{1}{2}\right)^-} f(x) = \lim_{h\to 0} f\left(\frac{1}{2}-h\right)$$

$$= \lim_{h\to 0}\left[\frac{1}{2}+\left(\frac{1}{2}-h\right)\right] = 1$$

$$\lim_{x\to\left(\frac{1}{2}\right)^+} f(x) = \lim_{h\to 0} f\left(\frac{1}{2}+h\right) = \lim_{h\to 0}\left[\frac{3}{2}+\left(\frac{1}{2}+h\right)\right] = 2$$

Since $\lim_{x\to\left(\frac{1}{2}\right)^+} f(x) \neq \lim_{x\to\left(\frac{1}{2}\right)^-} f(x)$

$\Rightarrow$ f is not continuous at $x = \frac{1}{2}$.

16. For f to be continuous at $x = 2$, we must have

$$\lim_{x\to 2^-} f(x) = f(2) = \lim_{x\to 2^+} f(x) \quad \ldots (1)$$

Now $f(2) = a$

$$\lim_{x\to 2^-} f(x) = \lim_{h\to 0} f(2-h) = \lim_{h\to 0}[2(2-h)-1] = 3$$

$$\lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(2+h) = \lim_{h\to 0}[(2+h)+1] = 3$$

$\therefore$ From (1), we get $a = 3$

17. The greatest integer function defined by $f(x) = [x]$, $0 < x < 2$ is not differentiable at $x = 1$.

18. We have, $y = \tan^{-1} x + \cot^{-1} x$

$\Rightarrow$ $y = \frac{\pi}{2}$ $\quad (\because \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2})$

$\Rightarrow$ $\frac{dy}{dx} = 0$

19. We have, $\cos(xy) = k$

$\Rightarrow$ $-\sin(xy)\left(y + x\frac{dy}{dx}\right) = 0$

$\Rightarrow$ $y + x\frac{dy}{dx} = 0$ $\quad [\because xy \neq n\pi]$

$\Rightarrow$ $\frac{dy}{dx} = -\frac{y}{x}$

20. Let $y = \sin^2(\sqrt{x})$

$\therefore$ $\frac{dy}{dx} = 2\sin(\sqrt{x})\cdot\cos(\sqrt{x})\cdot\frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}}\sin\left(2\sqrt{x}\right)$

21. To check the differentiability of $f(x) = x|x|$ at $x = 0$.

Consider, $\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$

$$= \lim_{h\to 0}\frac{f(h)-f(0)}{h} = \lim_{h\to 0}\frac{h|h|-0}{h} = \lim_{h\to 0}|h| = 0$$

Hence, $f'(0)$ exists, so $f(x) = x|x|$ is differentiable at $x = 0$.

22. Given, $f(x) = x + 1$

Now, $(fof)(x) = f(f(x)) = f(x+1)$

$= (x + 1) + 1 = x + 2$

$\therefore$ $\frac{d}{dx}(fof)(x) = \frac{d}{dx}(x+2) = 1$

23. Let, $y = \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left[\tan^{-1}\left(\frac{2\cos^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}\right)\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\tan^{-1}\left(\frac{\cos\frac{x}{2}}{\sin\frac{x}{2}}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\tan^{-1}\left(\cot\frac{x}{2}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\tan^{-1}\left[\tan\left(\frac{\pi}{2}-\frac{x}{2}\right)\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{2}-\frac{x}{2}\right) = -\frac{1}{2}$$

24. We have, $\sin^2 y + \cos xy = K$

Differentiating w.r.t. x on both sides, we get

$$2\sin y\cos y\,\frac{dy}{dx} + (-\sin xy)\left(x\frac{dy}{dx} + y\right) = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{y\sin xy}{\sin 2y - x\sin xy}$$

www.exambres.in

$$\Rightarrow \quad \left[\frac{dy}{dx}\right]_{\left(1,\frac{\pi}{4}\right)} = \frac{\frac{\pi}{4}\sin\frac{\pi}{4}}{\sin\frac{\pi}{2}-\sin\frac{\pi}{4}} = \frac{\pi}{4(\sqrt{2}-1)}$$

25. Given that $f(x)$ is differentiable at $x = 1$. Therefore, $f(x)$ is continuous at $x = 1$

$$\Rightarrow \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1)$$

$$\Rightarrow \lim_{x\to 1}(x^2+3x+a) = \lim_{x\to 1}(bx+2)$$

$$\Rightarrow \quad 1+3+a = b+2$$

$$\Rightarrow \quad a-b+2=0 \qquad ...(1)$$

Again, $f(x)$ is differentiable at $x = 1$. So,

(L.H.D. at $x = 1$) = (R.H.D. at $x = 1$)

$$\Rightarrow \lim_{x\to 1^-}\frac{f(x)-f(1)}{x-1} = \lim_{x\to 1^+}\frac{f(x)-f(1)}{x-1}$$

$$\Rightarrow \lim_{x\to 1}\frac{(x^2+3x+a)-(4+a)}{x-1} = \lim_{x\to 1}\frac{(bx+2)-(4+a)}{x-1}$$

$$\Rightarrow \lim_{x\to 1}\frac{x^2+3x-4}{x-1} = \lim_{x\to 1}\frac{bx-2-a}{x-1}$$

$$\Rightarrow \lim_{x\to 1}\frac{(x+4)(x-1)}{x-1} = \lim_{x\to 1}\frac{bx-b}{x-1} \qquad \text{[From (1)]}$$

$$\Rightarrow \lim_{x\to 1}(x+4) = \lim_{x\to 1}\frac{b(x-1)}{x-1} \Rightarrow 5 = b$$

Putting $b = 5$ in (1), we get $a = 3$

Hence, $a = 3$ and $b = 5$

26. We have,

$$y = \tan^{-1}\left(\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right), x^2 \le 1$$

Putting $x^2 = \cos\theta \Rightarrow \theta = \cos^{-1}(x^2)$ we get

$$y = \tan^{-1}\left(\frac{\sqrt{1+\cos\theta}+\sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta}-\sqrt{1-\cos\theta}}\right)$$

$$= \tan^{-1}\left(\frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}\right) = \tan^{-1}\left(\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right) = \frac{\pi}{4}+\frac{\theta}{2}$$

$$\Rightarrow \quad y = \frac{\pi}{4}+\frac{1}{2}\cos^{-1}(x^2)$$

Differentiating w.r.t x on both sides, we get

$$\frac{dy}{dx} = -\frac{1\times 2x}{2\sqrt{1-x^4}} = \frac{-x}{\sqrt{1-x^4}}$$

27. Here $f(x) = \sqrt{x^2+1} = (x^2+1)^{\frac{1}{2}}$

$$\Rightarrow \quad f'(x) = \frac{1}{2}\cdot(x^2+1)^{-\frac{1}{2}}\cdot 2x = \frac{x}{\sqrt{x^2+1}}, \qquad ...(1)$$

$$g(x) = \frac{x+1}{x^2+1}$$

$$\Rightarrow \quad g'(x) = \frac{(x^2+1)\cdot 1-(x+1)\cdot 2x}{(x^2+1)^2} = \frac{-x^2-2x+1}{(x^2+1)^2} \qquad ...(2)$$

and $h(x) = 2x - 3$

$$\Rightarrow \quad h'(x) = 2 \qquad ...(3)$$

$$\therefore \quad f'[h'(g'(x))] = f'\left[h'\left(\frac{-x^2-2x+1}{(x^2+1)^2}\right)\right] \text{ [Using (2)]}$$

$$= f'(2) \qquad \text{[Using (3)]}$$

$$= \frac{2}{\sqrt{2^2+1}} = \frac{2}{\sqrt{5}} \qquad \text{[Using (1)]}$$

28. The given function is $f(x) = |x-1| + |x+1|$

$$= \begin{cases} -(x-1)-(x+1), & x<-1 \\ -(x-1)+x+1, & -1\le x\le 1 \\ x-1+x+1, & x>1 \end{cases} = \begin{cases} -2x, & x<-1 \\ 2, & -1\le x\le 1 \\ 2x, & x>1 \end{cases}$$

At $x = 1$,

$$f'(1^-) = \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0}\frac{2-2}{-h} = 0$$

$$f'(1^+) = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h\to 0}\frac{2(1+h)-2}{h} = \lim_{h\to 0}\frac{2h}{h} = 2$$

$\therefore \quad f'(1^-) \ne f'(1^+)$

$\Rightarrow$ f is not differentiable at $x = 1$.

At $x = -1$,

$$f'(-1^-) = \lim_{h\to 0}\frac{f(-1-h)-f(-1)}{-h}$$

$$= \lim_{h\to 0}\frac{-2(-1-h)-(2)}{-h} = \lim_{h\to 0}\frac{2h}{-h} = -2$$

$$f'(-1^+) = \lim_{h\to 0}\frac{f(-1+h)-f(-1)}{h} = \lim_{h\to 0}\frac{2-2}{h} = 0$$

$\therefore \quad f'(-1^-) \ne f'(-1^+)$

$\Rightarrow$ f is not differentiable at $x = -1$.

29. At $x = 1$:

$$f'(1^-) = \lim_{x\to 1^-}\frac{f(x)-f(1)}{x-1} = \lim_{x\to 1}\frac{x-1}{x-1} = 1$$

www.exambites.in

$$f'(1^+) = \lim_{x\to 1^+} \frac{f(x)-f(1)}{x-1}$$

$$= \lim_{x\to 1} \frac{2-x-1}{x-1} = \lim_{x\to 1} \frac{1-x}{x-1} = -1$$

Since, $f'(1^-) \neq f'(1^+)$

$\therefore$ $f(x)$ is not differentiable at $x = 1$.

At $x = 2$:

$$f'(2^-) = \lim_{x\to 2^-} \frac{f(x)-f(2)}{x-2} = \lim_{x\to 2} \frac{2-x-0}{x-2} = -1$$

$$f'(2^+) = \lim_{x\to 2^+} \frac{f(x)-f(2)}{x-2}$$

$$= \lim_{x\to 2} \frac{-2+3x-x^2-0}{x-2} = \lim_{x\to 2} \frac{(1-x)(x-2)}{x-2} = -1$$

Since, $f'(2^-) = f'(2^+)$

$\therefore$ $f(x)$ is differentiable at $x = 2$.

30. Here $f(x) = \begin{cases} \lambda(x^2+2), & \text{if } x \le 0 \\ 4x+6, & \text{if } x > 0 \end{cases}$

At $x = 0$, $f(0) = \lambda(0^2 + 2) = 2\lambda$

$$\text{L.H. L.} = \lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h)$$

$$= \lim_{h\to 0} \lambda[(0-h)^2+2] = 2\lambda$$

$$\text{R.H. L.} = \lim_{x\to 0^+} f(x) = \lim_{h\to 0} f(0+h)$$

$$= \lim_{h\to 0} [4(0+h)+6] = 6$$

$\therefore$ For f to be continuous at $x = 0$

$2\lambda = 6 \Rightarrow \lambda = 3$.

Hence the function becomes

$$f(x) = \begin{cases} 3(x^2+2), & \text{if } x \le 0 \\ 4x+6 & \text{if } x > 0 \end{cases}$$

$$f'(0^-) = \lim_{h\to 0} \frac{f(0-h)-f(0)}{0-h}$$

$$= \lim_{h\to 0} \frac{3(h^2+2)-6}{-h} = \lim_{h\to 0} (-3h) = 0$$

$$\text{and } f'(0^+) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{0+h} = \lim_{h\to 0} \frac{4h+6-6}{h} = 4$$

$\Rightarrow$ $f'(0^-) \neq f'(0^+)$

$\therefore$ f is not differentiable at $x = 0$.

31. We have $\cos y = x\cos(a+y)$

$$\Rightarrow \quad x = \frac{\cos y}{\cos(a+y)}$$

Differentiating w.r.t. y on both sides, we get

$$\frac{dx}{dy} = \frac{\cos(a+y)\left(\frac{d}{dy}\cos y\right) - \cos y\left(\frac{d}{dy}\cos(a+y)\right)}{\cos^2(a+y)}$$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\cos(a+y)(-\sin y) + \cos y \sin(a+y)}{\cos^2(a+y)}$$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\cos y \sin(a+y) - \cos(a+y)\sin y}{\cos^2(a+y)}$$

$$= \frac{\sin[(a+y)-y]}{\cos^2(a+y)} = \frac{\sin a}{\cos^2(a+y)}$$

$$\therefore \quad \frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$$

32. We have, $y = \sin^{-1}\left(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right)$

$$\Rightarrow \quad y = \sin^{-1}\left(x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2}\right)$$

$$\Rightarrow \quad y = \sin^{-1} x - \sin^{-1}\sqrt{x}$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{d}{dx}\left(\sqrt{x}\right)$$

$$= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{x-x^2}}$$

33. $f(x) = |x-3| = \begin{cases} x-3, & \text{if } x \ge 3 \\ -(x-3), & \text{if } x < 3 \end{cases}$

We have $f(3) = |3-3| = 0$

$$\lim_{x\to 3^+} f(x) = \lim_{h\to 0} f(3+h) = \lim_{h\to 0} (3+h)-3 = \lim_{h\to 0} h = 0$$

$$\lim_{x\to 3^-} f(x) = \lim_{h\to 0} f(3-h) = \lim_{h\to 0} [-(3-h-3)] = \lim_{h\to 0} h = 0$$

$\because$ $\lim_{x\to 3^+} f(x) = \lim_{x\to 3^-} f(x) = f(3) = 0$

So, $f(x)$ is continuous at $x = 3$.

$$\text{Now, } Rf'(3) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{h}$$

$$= \lim_{h\to 0} \frac{(3+h-3)-0}{h} = \lim_{h\to 0} \frac{h}{h} = 1$$

$$\text{And } Lf'(3) = \lim_{h\to 0} \frac{f(3-h)-f(3)}{-h}$$

$$= \lim_{h\to 0} \frac{[-(3-h-3)]-0}{-h} = \lim_{h\to 0} \frac{h}{-h} = -1$$

Thus, $Rf'(3) \neq Lf'(3)$

$\therefore$ $f(x)$ is not differentiable at $x = 3$.

www.exambites.in

34. *Refer to answer 31.*

35. Let $y = \tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right]$

Put $x = \tan\theta \Rightarrow \theta = \tan^{-1} x$

$\therefore \quad y = \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right) = \tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right)$

$\Rightarrow \quad y = \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) = \tan^{-1}\left(\tan\frac{\theta}{2}\right)$

$\Rightarrow \quad y = \frac{1}{2}\theta = \frac{1}{2}\tan^{-1} x \quad \therefore \quad \frac{dy}{dx} = \frac{1}{2}\cdot\left(\frac{1}{1+x^2}\right)$

36. We have, $x\sqrt{1+y} + y\sqrt{1+x} = 0$

$\Rightarrow \quad x\sqrt{1+y} = -y\sqrt{1+x} \Rightarrow x^2(1+y) = y^2(1+x)$

$\Rightarrow \quad (x^2 - y^2) + xy(x-y) = 0 \Rightarrow y = \frac{-x}{x+1}$

$\therefore \quad \frac{dy}{dx} = \frac{(x+1)(-1)-(-x)(1)}{(x+1)^2} = \frac{-x-1+x}{(x+1)^2} = \frac{-1}{(x+1)^2}$

37. Here, $y = a\sin x + b\cos x$

$\Rightarrow \quad \frac{dy}{dx} = a\cos x - b\sin x$

Now, L.H.S. $= y^2 + \left(\frac{dy}{dx}\right)^2$

$= (a\sin x + b\cos x)^2 + (a\cos x - b\sin x)^2$

$= a^2\sin^2 x + b^2\cos^2 x + 2ab\sin x\cos x + a^2\cos^2 x + b^2\sin^2 x - 2ab\sin x\cos x$

$= a^2(\sin^2 x + \cos^2 x) + b^2(\cos^2 x + \sin^2 x)$

$= a^2 + b^2 =$ R.H.S.

38. Given $y = \log(\cos e^x)$

On differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\cos e^x}(-\sin e^x \cdot e^x) = -e^x \tan e^x$$

39. We have, $y = e^{x^2\cos x} + (\cos x)^x$

$= e^{x^2\cos x} + e^{x(\ln\cos x)}$

$$\therefore \quad \frac{dy}{dx} = e^{x^2\cos x}\frac{d}{dx}(x^2\cos x) + e^{x\ln\cos x}\frac{d}{dx}(x\ln\cos x)$$

$$= e^{x^2\cos x}(2x\cos x - x^2\sin x) + e^{x\ln\cos x}\left(\ln\cos x - \frac{x}{\cos x}\sin x\right)$$

$$= e^{x^2\cos x}(2x\cos x - x^2\sin x) + (\cos x)^x(\ln\cos x - x\tan x)$$

40. Given, $\log(x^2+y^2) = 2\tan^{-1}\left(\frac{y}{x}\right)$

On differentiating w.r.t. x on both sides, we get

$$\frac{1}{x^2+y^2}\left(2x + 2y\frac{dy}{dx}\right) = 2\times\frac{1}{1+\frac{y^2}{x^2}}\frac{d}{dx}\left(\frac{y}{x}\right)$$

$$\Rightarrow \quad \frac{2x}{x^2+y^2} + \frac{2y}{x^2+y^2}\frac{dy}{dx} = \frac{2x^2}{x^2+y^2}\left(\frac{1}{x}\frac{dy}{dx} + y\left(\frac{-1}{x^2}\right)\right)$$

$$\Rightarrow \quad \frac{dy}{dx}\left[\frac{2y}{x^2+y^2} - \frac{2x}{x^2+y^2}\right] = \frac{2x^2}{x^2+y^2}\left[\frac{-y}{x^2} - \frac{1}{x}\right]$$

$$\Rightarrow \quad \frac{2(y-x)}{x^2+y^2}\frac{dy}{dx} = \frac{-2x^2}{x^2+y^2}\left(\frac{y+x}{x^2}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{x+y}{x-y}$$

41. Here $y = \frac{x\cos^{-1}x}{\sqrt{1-x^2}} - \log\sqrt{1-x^2}$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{\sqrt{1-x^2}\cdot\frac{d}{dx}(x\cos^{-1}x) - x\cos^{-1}x\cdot\frac{d}{dx}\sqrt{1-x^2}}{1-x^2} - \frac{1}{\sqrt{1-x^2}}\cdot\frac{d}{dx}\left(\sqrt{1-x^2}\right)$$

$$= \frac{\sqrt{1-x^2}\cdot\left(1\cdot\cos^{-1}x - \frac{x}{\sqrt{1-x^2}}\right) - x\cos^{-1}x\left(\frac{-x}{\sqrt{1-x^2}}\right)}{1-x^2} - \frac{1}{\sqrt{1-x^2}}\left(\frac{-x}{\sqrt{1-x^2}}\right)$$

$$= \frac{\sqrt{1-x^2}\cos^{-1}x - x + \frac{x^2\cos^{-1}x}{\sqrt{1-x^2}}}{1-x^2} + \frac{x}{1-x^2}$$

$$= \frac{(1-x^2)\cos^{-1}x + x^2\cos^{-1}x}{(1-x^2)\sqrt{1-x^2}} = \frac{\cos^{-1}x}{(1-x^2)^{3/2}}$$

42. Given $e^x + e^y = e^{x+y} \Rightarrow 1 + e^{y-x} = e^y$ ...(1)

Differentiating (1) w.r.t. x, we get

$$e^{y-x}\cdot\frac{d}{dx}(y-x) = e^y\frac{dy}{dx}$$

$$\Rightarrow \quad e^{y-x}\left(\frac{dy}{dx} - 1\right) = e^y\frac{dy}{dx}$$

www.exambites.in

$$\Rightarrow \frac{dy}{dx}(e^{y-x}-e^{y})=e^{y-x}$$

$$\Rightarrow \frac{dy}{dx}(-1)=e^{y-x} \quad \text{[Using (1)]}$$

$$\Rightarrow \frac{dy}{dx}+e^{y-x}=0$$

43. Here, $y=\tan^{-1}\left(\frac{a}{x}\right)+\log\sqrt{\frac{x-a}{x+a}}$

$$=\tan^{-1}\left(\frac{a}{x}\right)+\frac{1}{2}\log\left(\frac{x-a}{x+a}\right)$$

$$=\tan^{-1}\left(\frac{a}{x}\right)+\frac{1}{2}[\log(x-a)-\log(x+a)]$$

Differentiating w.r.t x, we get

$$\frac{dy}{dx}=\frac{1}{1+\frac{a^2}{x^2}}\cdot\frac{d}{dx}\left(\frac{a}{x}\right)+\frac{1}{2}\left[\frac{1}{x-a}-\frac{1}{x+a}\right]$$

$$=\frac{x^2}{x^2+a^2}\cdot a\cdot\left(-\frac{1}{x^2}\right)+\frac{1}{2}\left[\frac{(x+a)-(x-a)}{x^2-a^2}\right]$$

$$=\frac{-a}{x^2+a^2}+\frac{a}{x^2-a^2}=\frac{-a(x^2-a^2)+a(x^2+a^2)}{x^4-a^4}$$

$$=\frac{2a^3}{x^4-a^4}$$

44. We have, $\log(\sqrt{1+x^2}-x)=y\sqrt{1+x^2}$

Differentiating w.r.t. x, we get

$$\frac{1}{\sqrt{1+x^2}-x}\cdot\left[\frac{1}{\sqrt{1+x^2}}\cdot x-1\right]=\frac{dy}{dx}\sqrt{1+x^2}+y\cdot\frac{x}{\sqrt{1+x^2}}$$

$$\Rightarrow \frac{\sqrt{1+x^2}}{\sqrt{1+x^2}-x}\cdot\frac{x-\sqrt{1+x^2}}{\sqrt{1+x^2}}=(1+x^2)\frac{dy}{dx}+xy$$

$$\Rightarrow (1+x^2)\frac{dy}{dx}+xy+1=0$$

45. We have $y=x^3(\cos x)^x+\sin^{-1}\sqrt{x}$ ...(i)

Let $z=(\cos x)^x=e^{x\log\cos x}$

$$\therefore \frac{dz}{dx}=e^{x\log\cos x}\left[\frac{x(-\sin x)}{\cos x}+\log\cos x\right]$$

$$=(\cos x)^x\times[-x\tan x+\log\cos x] \quad ...(ii)$$

Now, differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx}=3x^2(\cos x)^x+x^3(\cos x)^x[-x\tan x+\log\cos x]+\frac{1}{\sqrt{1-x}}\left(\frac{1}{2\sqrt{x}}\right) \quad \text{[Using (ii)]}$$

$$=x^2(\cos x)^x[3-x^2\tan x+x\log\cos x]+\frac{1}{2\sqrt{x}}\left(\frac{1}{\sqrt{1-x}}\right)$$

46. Let $y=(\log x)^x+x^{\log x}$

$$\therefore \quad y=e^{x\log(\log x)}+e^{(\log x)^2}$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx}=(\log x)^x\frac{d}{dx}\{x\log(\log x)\}+x^{\log x}\frac{d}{dx}\{(\log x)^2\}$$

$$=(\log x)^x\left\{x\left(\frac{1}{\log x}\right)\frac{1}{x}+\log(\log x)\right\}+x^{\log x}\left(2(\log x)\frac{1}{x}\right)$$

$$=(\log x)^x\left\{\frac{1}{\log x}+\log(\log x)\right\}+2\left(\frac{\log x}{x}\right)x^{\log x}$$

47. We have, $x^y-y^x=a^b$

Taking log on both sides, we get

$$y\log x-x\log y=b\log a \quad ...(i)$$

Now, differentiating (i) w.r.t. x on both sides, we get

$$\frac{y}{x}+(\log x)\frac{dy}{dx}-\log y-\frac{x\cdot1}{y}\frac{dy}{dx}=0$$

$$\Rightarrow \left(\frac{y}{x}-\log y\right)=\frac{dy}{dx}\left(\frac{x}{y}-\log x\right)$$

$$\Rightarrow \frac{1}{x}(y-x\log y)=\frac{dy}{dx}\left(\frac{x-y\log x}{y}\right)$$

$$\Rightarrow \frac{dy}{dx}=\frac{y}{x}\frac{(y-x\log y)}{(x-y\log x)}$$

48. Given $y=(x)^{\cos x}+(\cos x)^{\sin x}$

Let $u=(x)^{\cos x}$, $v=(\cos x)^{\sin x}$

$$\therefore \quad y=u+v$$

$$\Rightarrow \frac{dy}{dx}=\frac{du}{dx}+\frac{dv}{dx} \quad ...(i)$$

Now, $u=x^{\cos x} \Rightarrow \log u=\cos x\log x$

Differentiating with respect to x, we get

$$\frac{1}{u}\frac{du}{dx}=\cos x\cdot\frac{1}{x}+\log x(-\sin x)$$

$$\Rightarrow \frac{du}{dx}=x^{\cos x}\left[\frac{\cos x}{x}-\sin x\log x\right] \quad ...(ii)$$

www.exambites.in

Now, $v = (\cos x)^{\sin x} \Rightarrow \log v = \sin x \log\cos x$
Differentiating with respect to x, we get

$$\frac{1}{v}\frac{dv}{dx} = \sin x \cdot \frac{1}{\cos x}(-\sin x) + \log\cos x \cdot \cos x$$

$$\Rightarrow \frac{dv}{dx} = (\cos x)^{\sin x}\left[\frac{-\sin^2 x + \cos^2 x \log\cos x}{\cos x}\right] \quad \text{... (iii)}$$

Putting value of (ii) and (iii) into (i), we get

$$\frac{dy}{dx} = x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right] + (\cos x)^{\sin x}\left[\frac{-\sin^2 x + \cos^2 x \log\cos x}{\cos x}\right]$$

49. Let $y = (\sin x)^x + \sin^{-1}\sqrt{x}$

$\Rightarrow y = u + v$ [where $u = (\sin x)^x$ and $v = \sin^{-1}\sqrt{x}$]

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \quad \text{...(i)}$$

Now, $u = (\sin x)^x$
Taking logarithm on both sides, we get

$\log u = x \log \sin x$

$$\Rightarrow \frac{1}{u}\frac{du}{dx} = x\frac{1}{\sin x}(\cos x) + \log \sin x$$

$$\Rightarrow \frac{du}{dx} = (\sin x)^x \ (x \cot x + \log \sin x) \quad \text{...(ii)}$$

$v = \sin^{-1}\sqrt{x}$

$$\Rightarrow \frac{dv}{dx} = \left(\frac{1}{\sqrt{1-x}}\right)\cdot\frac{1}{2\sqrt{x}} \quad \text{...(iii)}$$

From (i), (ii) and (iii), we get

$$\frac{dy}{dx} = (\sin x)^x (x\cot x + \log\sin x) + \frac{1}{2\sqrt{x}}\left(\frac{1}{\sqrt{1-x}}\right)$$

50. *Refer to answer 47.*

51. *Refer to answer 48.*

52. *Refer to answer 49.*

53. Given $x^m y^n = (x+y)^{m+n}$
Taking log on both the sides, we get
$\log x^m + \log y^n = (m+n)\log(x+y)$
$\Rightarrow m \log x + n \log y = (m+n)\log(x+y)$
Differentiating w.r.t. x, we get

$$m\cdot\frac{1}{x} + n\cdot\frac{1}{y}\cdot\frac{dy}{dx} = (m+n)\cdot\frac{1}{x+y}\left(1+\frac{dy}{dx}\right)$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{n}{y} - \frac{m+n}{x+y}\right) = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{nx+ny-my-ny}{y(x+y)}\right) = \frac{mx+nx-mx-my}{x(x+y)}$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{nx-my}{y(x+y)}\right) = \frac{nx-my}{x(x+y)} \quad \therefore \quad \frac{dy}{dx} = \frac{y}{x}.$$

54. Here, $(x-y)\cdot e^{\frac{x}{x-y}} = a$
Taking log on both sides, we get

$$\Rightarrow \log\left\{(x-y)\cdot e^{\frac{x}{x-y}}\right\} = \log a$$

$$\Rightarrow \log(x-y) + \frac{x}{x-y} = \log a$$

Differentiating w.r.t. x, we get

$$\frac{1}{x-y}\cdot\left(1-\frac{dy}{dx}\right) + \frac{(x-y)\cdot 1 - x\left(1-\frac{dy}{dx}\right)}{(x-y)^2} = 0$$

$$\Rightarrow (x-y)\left(1-\frac{dy}{dx}\right) + x - y - x\left(1-\frac{dy}{dx}\right) = 0$$

$$\Rightarrow -y\left(1-\frac{dy}{dx}\right) + x - y = 0 \Rightarrow y\frac{dy}{dx} + x = 2y$$

55. Here, $(\tan^{-1}x)^y + y^{\cot x} = 1$
$\Rightarrow u + v = 1$ where $u = (\tan^{-1}x)^y$ and $v = y^{\cot x}$
Differentiating w.r.t. x, we get

$$\frac{du}{dx} + \frac{dv}{dx} = 0 \quad \text{...(1)}$$

Now, $u = (\tan^{-1}x)^y$
$\Rightarrow \log u = y \log(\tan^{-1}x)$
Differentiating w.r.t. x, we get

$$\frac{1}{u}\cdot\frac{du}{dx} = \frac{dy}{dx}\log(\tan^{-1}x) + y\cdot\frac{1}{\tan^{-1}x}\cdot\frac{1}{1+x^2}$$

$$\Rightarrow \frac{du}{dx} = (\tan^{-1}x)^y \times \left[\frac{dy}{dx}\log(\tan^{-1}x) + \frac{y}{(1+x^2)\tan^{-1}x}\right] \quad \text{...(2)}$$

And $v = y^{\cot x}$
$\Rightarrow \log v = \cot x \cdot \log y$
Differentiating w.r.t. x, we get

$$\frac{1}{v}\frac{dv}{dx} = \cot x\cdot\frac{1}{y}\cdot\frac{dy}{dx} - \text{cosec}^2 x\cdot\log y$$

$$\Rightarrow \frac{dv}{dx} = y^{\cot x}\left[\frac{\cot x}{y}\cdot\frac{dy}{dx} - \text{cosec}^2 x\cdot\log y\right] \quad \text{...(3)}$$

www.exambites.in

From (1), (2) and (3), we get

$$(\tan^{-1}x)^y\left[\frac{dy}{dx}\log(\tan^{-1}x)+\frac{y}{(1+x^2)\tan^{-1}x}\right]$$
$$+y^{\cot x}\left[\frac{\cot x}{y}\cdot\frac{dy}{dx}-\text{cosec}^2x\cdot\log y\right]=0$$

$$\Rightarrow \frac{dy}{dx}[(\tan^{-1}x)^y\cdot\log(\tan^{-1}x)+y^{\cot x-1}\cdot\cot x]$$
$$=y^{\cot x}\cdot\text{cosec}^2x\cdot\log y-(\tan^{-1}x)^{y-1}\cdot\frac{y}{(1+x^2)}$$

$$\therefore \frac{dy}{dx}=\frac{y^{\cot x}\cdot\text{cosec}^2x\cdot\log y-(\tan^{-1}x)^{y-1}\cdot\frac{y}{(1+x^2)}}{(\tan^{-1}x)^y.\log(\tan^{-1}x)+y^{\cot x-1}.\cot x}$$

56. Here $y^x=e^{y-x}$

Taking log on both sides, we get

$x\log y=(y-x)\log e=y-x$

$\Rightarrow x(1+\log y)=y$

$$\Rightarrow x=\frac{y}{1+\log y}$$

Differentiating w.r.t. y, we get

$$\frac{dx}{dy}=\frac{(1+\log y)\cdot1-y\cdot\frac{1}{y}}{(1+\log y)^2}=\frac{\log y}{(1+\log y)^2}$$

$$\Rightarrow \frac{dy}{dx}=\frac{(1+\log y)^2}{\log y}$$

57. Let $y=\sin^{-1}\left[\frac{2^{x+1}\cdot3^x}{1+(36)^x}\right]$

$$\Rightarrow y=\sin^{-1}\left[\frac{2\cdot2^x\cdot3^x}{1+(36)^x}\right]=\sin^{-1}\left[\frac{2\cdot6^x}{1+(6^x)^2}\right]$$

Put $6^x=\tan\theta\Rightarrow\theta=\tan^{-1}6^x$

$$\therefore \quad y=\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)=\sin^{-1}(\sin2\theta)$$

$\Rightarrow y=2\theta=2\tan^{-1}(6^x)$

Now differentiating w.r.t. x, we get

$$\frac{dy}{dx}=2\cdot\frac{1}{1+(6^x)^2}\cdot\frac{d}{dx}(6^x)$$

$$=\frac{2}{1+(36)^x}\cdot6^x\log6=\frac{2\log6\cdot6^x}{1+(36)^x}$$

58. *Refer to answer 56.*

59. *Refer to answer 57.*

60. We have, $(\cos x)^y=(\cos y)^x$

Taking log on both sides, we get

$y\log(\cos x)=x\log(\cos y)$ ... (i)

Differentiating w.r.t. x, we get

$$y\cdot\frac{1}{\cos x}\cdot(-\sin x)+\log(\cos x)\cdot\frac{dy}{dx}$$
$$=x\cdot\frac{1}{\cos y}\cdot(-\sin y)\cdot\frac{dy}{dx}+\log(\cos y)\cdot1$$

$$\Rightarrow \frac{dy}{dx}(x\tan y+\log(\cos x))=y\tan x+\log(\cos y)$$

$$\Rightarrow \frac{dy}{dx}=\frac{y\tan x+\log(\cos y)}{x\tan y+\log(\cos x)}$$

61. Here $y=x^{\sin x-\cos x}+\frac{x^2-1}{x^2+1}$

Let $u=x^{\sin x-\cos x}$ and $v=\frac{x^2-1}{x^2+1}$

$\therefore \quad y=u+v$

$$\Rightarrow \frac{dy}{dx}=\frac{du}{dx}+\frac{dv}{dx} \quad ...(1)$$

Now, $u=x^{\sin x-\cos x}$

$\Rightarrow \log u=(\sin x-\cos x)\log x$

Differentiating w.r.t. x, we get

$$\frac{1}{u}\cdot\frac{du}{dx}=(\sin x-\cos x)\cdot\frac{1}{x}+(\cos x+\sin x)\cdot\log x$$

$$\Rightarrow \frac{du}{dx}=x^{\sin x-\cos x}$$
$$\times\left[\frac{\sin x-\cos x}{x}+(\cos x+\sin x)\cdot\log x\right] \quad ...(2)$$

Now, $v=\frac{x^2-1}{x^2+1}=1-\frac{2}{x^2+1}$

$$\Rightarrow \frac{dv}{dx}=0-2\cdot(-1)(x^2+1)^{-2}\cdot2x=\frac{4x}{(x^2+1)^2} \quad ...(3)$$

From (1), (2) and (3), we get

$$\frac{dy}{dx}=x^{\sin x-\cos x}\cdot\left[\frac{\sin x-\cos x}{x}+(\sin x+\cos x)\cdot\log x\right]$$
$$+\frac{4x}{(x^2+1)^2}$$

62. Here, $y=x^{\cot x}+\frac{2x^2-3}{x^2+x+2}$

Let $u=x^{\cot x}, v=\frac{2x^2-3}{x^2+x+2}$

$$\Rightarrow y=u+v\Rightarrow\frac{dy}{dx}=\frac{du}{dx}+\frac{dv}{dx} \quad ...(1)$$

www.exambites.in

Now, $u = x^{\cot x} \Rightarrow \log u = \cot x \cdot \log x$

Differentiating w.r.t. x, we get

$$\frac{1}{u}\cdot\frac{du}{dx} = \cot x \cdot \frac{1}{x} - \text{cosec}^2 x \cdot \log x$$

$$\therefore \quad \frac{du}{dx} = x^{\cot x}\left(\frac{\cot x}{x} - \text{cosec}^2 x \cdot \log x\right) \qquad ...(2)$$

Also, $v = \dfrac{2x^2 - 3}{x^2 + x + 2}$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{(x^2 + x + 2)\cdot 4x - (2x^2 - 3)\cdot(2x + 1)}{(x^2 + x + 2)^2}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{(4x^3 + 4x^2 + 8x) - (4x^3 + 2x^2 - 6x - 3)}{(x^2 + x + 2)^2}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{2x^2 + 14x + 3}{(x^2 + x + 2)^2} \qquad ...(3)$$

From (1), (2) and (3), we get

$$\frac{dy}{dx} = x^{\cot x}\left(\frac{\cot x}{x} - \text{cosec}^2 x \cdot \log x\right) + \frac{2x^2 + 14x + 3}{(x^2 + x + 2)^2}$$

63. Let $y = x^{x\cos x} + \dfrac{x^2+1}{x^2-1}$

Let $u = x^{x\cos x}$ and $v = \dfrac{x^2+1}{x^2-1}$

$\therefore \quad y = u + v$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \qquad ...(1)$$

Now, $u = x^{x\cos x}$

Taking log on both sides, we get

$\log u = x\cos x \cdot \log x$

Differentiating w.r.t. x, we get

$$\frac{1}{u}\cdot\frac{du}{dx} = x\cos x\frac{d}{dx}(\log x) + x\log x\frac{d}{dx}(\cos x) + \cos x \log x \frac{d}{dx}(x)$$

$$\Rightarrow \quad \frac{1}{u}\cdot\frac{du}{dx} = x\cos x\cdot\frac{1}{x} + x\log x(-\sin x) + \cos x\log x$$

$$= \cos x - x\sin x\log x + \cos x\log x$$

$$\therefore \quad \frac{du}{dx} = x^{x\cos x}[\cos x - x\sin x\log x + \cos x\log x] \qquad ...(2)$$

Also, $v = \dfrac{x^2+1}{x^2-1}$

Differentiating w.r.t. x, we get

$$\frac{dv}{dx} = \frac{(x^2-1)\frac{d}{dx}(x^2+1) - (x^2+1)\frac{d}{dx}(x^2-1)}{(x^2-1)^2}$$

$$= \frac{(x^2-1)(2x) - (x^2+1)(2x)}{(x^2-1)^2}$$

$$= \frac{2x[x^2 - 1 - x^2 - 1]}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2} \qquad ...(3)$$

From (1), (2) and (3), we get

$$\frac{dy}{dx} = x^{x\cos x}[\cos x - x\sin x\log x + \cos x\log x] - \frac{4x}{(x^2-1)^2}$$

$$= x^{x\cos x}[\cos x(1 + \log x) - x\sin x\log x] - \frac{4x}{(x^2-1)^2}$$

64. *Refer to answer 58.*

We have,

$$\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2} = \frac{\log x}{(\log e + x)^2} = \frac{\log x}{(\log(xe))^2}$$

65. We have, $x = a\sec\theta$, $y = b\tan\theta$

$$\therefore \quad \frac{dx}{d\theta} = a\sec\theta\tan\theta \text{ and } \frac{dy}{d\theta} = b\sec^2\theta$$

$$\text{Now, } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{b\sec^2\theta}{a\sec\theta\tan\theta} = \frac{b}{a}\cdot\frac{1}{\cos\theta}\cdot\frac{\cos\theta}{\sin\theta}$$

$$= \frac{b}{a}\cdot\frac{1}{\sin\theta}$$

$$\therefore \quad \left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{3}} = \frac{b}{a}\cdot\frac{1}{\sin\frac{\pi}{3}} = \frac{b}{a}\cdot\frac{1}{\frac{\sqrt{3}}{2}} = \frac{2b}{a\sqrt{3}}$$

66. Let $u = \sin^2 x$, $v = e^{\cos x}$

$$\therefore \quad \frac{du}{dx} = 2\sin x\cos x \text{ and } \frac{dv}{dx} = e^{\cos x}(-\sin x)$$

$$\text{Now, } \frac{du}{dv} = \frac{du}{dx}\cdot\frac{dx}{dv} = \frac{2\sin x\cos x}{e^{\cos x}(-\sin x)} = \frac{-2\cos x}{e^{\cos x}}$$

67. Let $u = \tan^{-1}\left[\dfrac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}}\right]$ ...(i)

Putting $x^2 = \cos 2\theta$ in (i), we get

$$u = \tan^{-1}\left[\frac{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}\right]$$

$$= \tan^{-1}\left[\frac{\sqrt{2\cos^2\theta} - \sqrt{2\sin^2\theta}}{\sqrt{2\cos^2\theta} + \sqrt{2\sin^2\theta}}\right]$$

www.examfree.in

$$= \tan^{-1}\left[\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}\right] = \tan^{-1}\left[\frac{1-\tan\theta}{1+\tan\theta}\right]$$

$$= \tan^{-1}\left[\tan\left(\frac{\pi}{4} - \theta\right)\right] = \frac{\pi}{4} - \theta$$

$$= \frac{\pi}{4} - \frac{\cos^{-1}(x^2)}{2}$$

$$\Rightarrow \quad \frac{du}{dx} = \frac{1}{2}\frac{1}{\sqrt{1-x^4}}2x = \frac{x}{\sqrt{1-x^4}} \qquad \text{... (ii)}$$

Let $v = \cos^{-1}(x^2) \Rightarrow \frac{dv}{dx} = \frac{-2x}{\sqrt{1-x^4}}$ ... (iii)

Dividing (ii) by (iii), we get $\frac{du}{dv} = -\frac{1}{2}$.

68. We have, $x = a(2\theta - \sin 2\theta)$ ...(i)

and $y = a(1 - \cos 2\theta)$ ...(ii)

Differentiating (i) w.r.t. θ, we get

$$\frac{dx}{d\theta} = a(2 - 2\cos 2\theta) \qquad \text{...(iii)}$$

Differentiating (ii) w.r.t. θ, we get

$$\frac{dy}{d\theta} = 2a\sin 2\theta \qquad \text{...(iv)}$$

Now, $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{2a\sin 2\theta}{a(2-2\cos 2\theta)} = \frac{\sin 2\theta}{1-\cos 2\theta}$

$$\therefore \quad \left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{3}} = \frac{\sin 2\left(\frac{\pi}{3}\right)}{1-\cos\frac{2\pi}{3}} = \frac{\sin\left(\pi - \frac{\pi}{3}\right)}{1-\cos\left(\pi - \frac{\pi}{3}\right)}$$

$$= \frac{\sin\left(\frac{\pi}{3}\right)}{1+\cos\left(\frac{\pi}{3}\right)} = \frac{\frac{\sqrt{3}}{2}}{1+\frac{1}{2}} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

69. $x = a\sin 2t\,(1 + \cos 2t)$, $y = b\cos 2t\,(1 - \cos 2t)$

Now, $\frac{dx}{dt} = 2a\cos 2t(1+\cos 2t) + a\sin 2t(-2\sin 2t)$

$= 2a\cos 2t + 2a[\cos^2 2t - \sin^2 2t]$

$= 2a\cos 2t + 2a\cos 4t$

Also, $\frac{dy}{dt} = -2b\sin 2t(1-\cos 2t) + b\cos 2t(2\sin 2t)$

$= -2b\sin 2t + 4b\,(\sin 2t\cos 2t)$

$= -2b\sin 2t + 2b\sin 4t$

So, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2b(\sin 4t - \sin 2t)}{2a(\cos 4t + \cos 2t)}$

$$\therefore \quad \left(\frac{dy}{dx}\right)_{\text{at } t=\pi/4} = \frac{b}{a}\left[\frac{\sin\pi - \sin(\pi/2)}{\cos\pi + \cos(\pi/2)}\right]$$

$$= \frac{b}{a}\left[\frac{0-1}{-1+0}\right] = \frac{b}{a}$$

$$\left(\frac{dy}{dx}\right)_{\text{at } t=\pi/3} = \frac{b}{a}\left[\frac{\sin(4\pi/3) - \sin(2\pi/3)}{\cos(4\pi/3) + \cos(2\pi/3)}\right]$$

$$= \frac{b}{a}\left[\frac{\frac{-\sqrt{3}}{2} - \frac{\sqrt{3}}{2}}{\frac{-1}{2} - \frac{1}{2}}\right] = \frac{\sqrt{3}b}{a}$$

70. Let $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Put $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$

$$\therefore \quad u = \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$

$$\Rightarrow \quad u = \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right) \Rightarrow u = \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$\Rightarrow u = \tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right) \Rightarrow u = \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$\therefore \quad u = \frac{\theta}{2} \Rightarrow u = \frac{1}{2}\tan^{-1}x$$

Differentiating w.r.t. x, we get

$$\frac{du}{dx} = \frac{1}{2(1+x^2)}$$

Also, let $v = \sin^{-1}\left(\frac{2x}{1+x^2}\right) \Rightarrow v = 2\tan^{-1}x$

Differentiating w.r.t. x, we get

$$\frac{dv}{dx} = \frac{2}{1+x^2}$$

$$\therefore \quad \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{\frac{1}{2(1+x^2)}}{\frac{2}{1+x^2}} \Rightarrow \frac{du}{dv} = \frac{1}{4}$$

71. We have $x = ae^t(\sin t + \cos t)$

$$\Rightarrow \quad \frac{dx}{dt} = ae^t(\sin t + \cos t) + ae^t(\cos t - \sin t) = 2ae^t\cos t$$

and $y = ae^t(\sin t - \cos t)$

$$\Rightarrow \quad \frac{dy}{dt} = ae^t(\sin t - \cos t) + ae^t(\cos t + \sin t) = 2ae^t\sin t$$

$$\therefore \quad \text{L.H.S.} = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2ae^t\sin t}{2ae^t\cos t} = \tan t$$

www.exambites.in

Also, R.H.S$=\dfrac{x+y}{x-y}$

$$=\frac{ae^t(\sin t+\cos t)+ae^t(\sin t-\cos t)}{ae^t(\sin t+\cos t)-ae^t(\sin t-\cos t)}$$

$$=\frac{2ae^t\sin t}{2ae^t\cos t}=\tan t=\text{L.H.S.}$$

72. Let $u=\tan^{-1}\left(\dfrac{\sqrt{1-x^2}}{x}\right)$

Put $x=\cos\theta$

$$\therefore\ u=\tan^{-1}\left[\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}\right]=\tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)$$

$$=\tan^{-1}(\tan\theta)=\theta\ \Rightarrow\ \frac{du}{d\theta}=1$$

Also let,

$v=\cos^{-1}(2x\sqrt{1-x^2})\Rightarrow v=\cos^{-1}(2\cos\theta\sqrt{1-\cos^2\theta})$

$=\cos^{-1}(2\cos\theta\sin\theta)=\cos^{-1}(\sin 2\theta)$

$$=\cos^{-1}\left(\cos\left(\frac{\pi}{2}-2\theta\right)\right)=\frac{\pi}{2}-2\theta\ \Rightarrow\ \frac{dv}{d\theta}=-2$$

Now $\dfrac{du}{dv}=\dfrac{du/d\theta}{dv/d\theta}=\dfrac{-1}{2}$

73. *Refer to answer 72.*

74. *Refer to answer 71.*

$$\therefore\ \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{2ae^\theta\cos\theta}{2ae^\theta\sin\theta}=\cot\theta$$

$$\Rightarrow\ \left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{4}}=\cot\frac{\pi}{4}=1$$

75. *Refer to answer 69.*

76. Here, $x=\cos t(3-2\cos^2 t)$, $y=\sin t(3-2\sin^2 t)$

$$\Rightarrow\ \frac{dx}{dt}=-\sin t(3-2\cos^2 t)+\cos t\,[2\cdot 2\cos t\sin t]$$

$$=-3\sin t+6\cos^2 t\sin t$$

and $\dfrac{dy}{dt}=\cos t(3-2\sin^2 t)+\sin t(-2\cdot 2\sin t\cos t)$

$$=3\cos t-6\sin^2 t\cos t$$

$$\therefore\ \frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{3\cos t-6\sin^2 t\cos t}{-3\sin t+6\cos^2 t\sin t}$$

$$=\frac{3\cos t\cdot\cos 2t}{3\sin t\cdot\cos 2t}=\cot t$$

$$\Rightarrow\ \left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}}=\cot\frac{\pi}{4}=1$$

77. Here, $x=2\cos\theta-\cos 2\theta$, $y=2\sin\theta-\sin 2\theta$

$$\Rightarrow\ \frac{dx}{d\theta}=-2\sin\theta+2\sin 2\theta$$

and $\dfrac{dy}{d\theta}=2\cos\theta-2\cos 2\theta$

$$\therefore\ \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{2(\cos\theta-\cos 2\theta)}{2(\sin 2\theta-\sin\theta)}$$

$$=\frac{2\sin\left(\frac{3\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)}{2\cos\left(\frac{3\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)}=\tan\left(\frac{3\theta}{2}\right).$$

78. Here $x=\sqrt{a^{\sin^{-1}t}}$

$$\Rightarrow\ \log x=\frac{1}{2}\sin^{-1}t\cdot\log a$$

Differentiating w.r.t. t, we get

$$\frac{1}{x}\cdot\frac{dx}{dt}=\frac{1}{2}\log a\cdot\frac{1}{\sqrt{1-t^2}}\qquad\ldots(1)$$

Also, $y=\sqrt{a^{\cos^{-1}t}}$

$$\Rightarrow\ \log y=\frac{1}{2}\cos^{-1}t\cdot\log a$$

Differentiating w.r.t. t, we get

$$\frac{1}{y}\cdot\frac{dy}{dt}=\frac{1}{2}\log a\cdot\frac{-1}{\sqrt{1-t^2}}\qquad\ldots(2)$$

Now dividing (2) by (1), we get

$$\frac{\frac{1}{y}\cdot\frac{dy}{dt}}{\frac{1}{x}\cdot\frac{dx}{dt}}=-1\Rightarrow\frac{dy}{dx}=-\frac{y}{x}$$

79. Here , $x=a(\theta-\sin\theta)\ \Rightarrow\ \dfrac{dx}{d\theta}=a(1-\cos\theta)$

and $y=a(1+\cos\theta)\ \Rightarrow\ \dfrac{dy}{d\theta}=a(-\sin\theta)$

$$\therefore\ \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{-a\sin\theta}{a(1-\cos\theta)}=\frac{-\sin\theta}{(1-\cos\theta)}$$

$$\therefore\ \left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{3}}=\frac{-\sin\frac{\pi}{3}}{1-\cos\frac{\pi}{3}}=\frac{-\sqrt{3}/2}{1-(1/2)}=-\sqrt{3}$$

80. Here, $x=a(\cos t+\log\tan\dfrac{t}{2})$

$$\Rightarrow\ \frac{dx}{dt}=a\left(-\sin t+\frac{1}{\tan\frac{t}{2}}\cdot\frac{1}{2}\sec^2\frac{t}{2}\right)$$

www.examples.in

$$= a\left(-\sin t + \frac{\cos\frac{t}{2}}{\sin\frac{t}{2}} \cdot \frac{1}{2\cos^2\frac{t}{2}}\right) = a\left(-\sin t + \frac{1}{\sin t}\right)$$

$$= a\frac{(-\sin^2 t + 1)}{\sin t} = \frac{a\cos^2 t}{\sin t}$$

Also, $y = a \sin t \Rightarrow \frac{dy}{dt} = a\cos t$

$$\therefore \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = a\cos t \cdot \frac{\sin t}{a\cos^2 t} = \tan t.$$

81. (d) : We have, $y = \log_e\left(\frac{x^2}{e^2}\right)$

$$\therefore \quad \frac{dy}{dx} = \frac{e^2}{x^2} \cdot \frac{1}{e^2} \cdot 2x = \frac{2}{x} \Rightarrow \frac{d^2y}{dx^2} = -\frac{2}{x^2}$$

82. Given, $x = at^2$, $y = 2at$

$$\therefore \quad \frac{dx}{dt} = 2at, \frac{dy}{dt} = 2a$$

$$\therefore \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t}$$

So, $\frac{d^2y}{dx^2} = -\frac{1}{t^2} \cdot \frac{dt}{dx} = -\frac{1}{t^2} \cdot \frac{1}{2at} = -\frac{1}{2at^3}$

83. We have, $x = a \cos\theta$, $y = b \sin\theta$

$$\therefore \quad \frac{dx}{d\theta} = -a\sin\theta, \frac{dy}{d\theta} = b\cos\theta$$

So, $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} = \frac{b\cos\theta}{-a\sin\theta} = -\frac{b}{a}\cot\theta$

Now, $\frac{d^2y}{dx^2} = -\frac{b}{a}(-\text{cosec}^2\theta)\frac{d\theta}{dx}$

$$= \frac{b}{a}\text{cosec}^2\theta\left(-\frac{1}{a}\text{cosec}\,\theta\right) = -\frac{b}{a^2}\text{cosec}^3\theta$$

84. We have, $y = (\sin^{-1}x)^2$ ...(i)

$$\Rightarrow \quad \frac{dy}{dx} = 2(\sin^{-1}x) \times \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)^2 = \frac{4(\sin^{-1}x)^2}{(1-x^2)}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)^2 = \frac{4y}{1-x^2} \qquad \text{[From (i)]}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 (1-x^2) = 4y \qquad \text{...(ii)}$$

Again, differentiating (ii) w.r.t. x on both sides, we get

$$2\frac{dy}{dx}\frac{d^2y}{dx^2}(1-x^2) + \left(\frac{dy}{dx}\right)^2(-2x) = 4\left(\frac{dy}{dx}\right)$$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 2 = 0$$

85. We have, $x = \sin t$ and $y = \sin pt$

$$\frac{dx}{dt} = \cos t \text{ and } \frac{dy}{dt} = p\cos pt$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{p\cos pt}{\cos t}$$

Differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = \frac{-p^2\sin pt\cos t + p\cos pt\sin t}{\cos^2 t} \times \frac{dt}{dx}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \frac{-p^2\sin pt\cos t + p\cos pt\sin t}{\cos^3 t}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = -\frac{p^2\sin pt\cos t}{\cos^3 t} + \frac{p\cos pt\sin t}{\cos^3 t}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \frac{-p^2y}{\cos^2 t} + \frac{x\frac{dy}{dx}}{\cos^2 t}$$

$$\Rightarrow \quad \cos^2 t\frac{d^2y}{dx^2} = -p^2y + x\frac{dy}{dx}$$

$$\Rightarrow \quad (1-\sin^2 t)\frac{d^2y}{dx^2} = -p^2y + x\frac{dy}{dx}$$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + p^2y = 0$$

86. Here, $y = \sin(\sin x)$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \cos(\sin x) \cdot \cos x$$

Again differentiating w.r.t. x both sides, we get

$$\frac{d^2y}{dx^2} = -\sin(\sin x) \cdot \cos x \cdot \cos x + (-\sin x)\cos(\sin x)$$

$$= -\cos^2 x \cdot \sin(\sin x) - \sin x \cdot \cos(\sin x)$$

Now, L.H.S. $= \frac{d^2y}{dx^2} + \tan x\frac{dy}{dx} + y\cos^2 x$

$$= -\cos^2 x \cdot \sin(\sin x) - \sin x \cdot \cos(\sin x) + \tan x(\cos x \cdot \cos(\sin x)) + \cos^2 x \cdot \sin(\sin x)$$

$$= -\sin x \cdot \cos(\sin x) + \sin x \cdot \cos(\sin x)$$

$$= 0 = \text{R.H.S.}$$

www.exambites.in

87. *Refer to answer 53.*
Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=\frac{x\dfrac{dy}{dx}-y}{x^2}=\frac{x\left(\dfrac{y}{x}\right)-y}{x^2}=0 \;\therefore\; \frac{d^2y}{dx^2}=0$$

88. Given that $e^y\cdot(x+1)=1$ ...(i)
Differentiating (i) w.r.t. x, we get

$$e^y\frac{d}{dx}(x+1)+(x+1)\frac{d}{dx}e^y=\frac{d}{dx}(1)$$

$$\Rightarrow\quad e^y+(x+1)e^y\frac{dy}{dx}=0$$

$$\Rightarrow\quad e^y\left[1+(x+1)\frac{dy}{dx}\right]=0\;\Rightarrow\;(x+1)\frac{dy}{dx}=-1$$

and $\dfrac{dy}{dx}=\dfrac{-1}{x+1}$ ...(ii)

or $\left(\dfrac{dy}{dx}\right)^2=\dfrac{1}{(x+1)^2}$ ...(iii)

Again differentiating (ii) w.r.t. x, we get

$$\frac{d^2y}{dx^2}=\frac{1}{(x+1)^2}\;\Rightarrow\frac{d^2y}{dx^2}=\left(\frac{dy}{dx}\right)^2 \qquad \text{[From (iii)]}$$

89. We have, $y=x^x\Rightarrow y=e^{x\log x}$
Differentiating w.r.t. x, we get

$$\frac{dy}{dx}=e^{x\log x}\left(x\times\frac{1}{x}+\log x\right)$$

$$\Rightarrow\quad \frac{dy}{dx}=x^x(1+\log x)\;\Rightarrow\;\frac{dy}{dx}=y(1+\log x) \quad ...(1)$$

Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=(1+\log x)\cdot\frac{dy}{dx}+y\times\frac{1}{x}$$

$$\Rightarrow\quad \frac{d^2y}{dx^2}=\frac{1}{y}\left(\frac{dy}{dx}\right)^2+\frac{y}{x} \qquad \text{[From (1)]}$$

$$\Rightarrow\quad \frac{d^2y}{dx^2}-\frac{1}{y}\left(\frac{dy}{dx}\right)^2-\frac{y}{x}=0$$

90. We have, $y=2\cos(\log x)+3\sin(\log x)$
Differentiating w.r.t. x, we get

$$\frac{dy}{dx}=-2\sin(\log x)\times\frac{1}{x}+3\cos(\log x)\times\frac{1}{x}$$

$$\Rightarrow\quad x\frac{dy}{dx}=-2\sin(\log x)+3\cos(\log x) \quad ...(1)$$

Again differentiating w.r.t. x, we get

$$x\frac{d^2y}{dx^2}+\frac{dy}{dx}=-2\cos(\log x)\times\frac{1}{x}-3\sin(\log x)\times\frac{1}{x}$$

$$\Rightarrow\quad x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}=-[2\cos(\log x)+3\sin(\log x)]$$

$$\Rightarrow\; x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}=-y\;\Rightarrow x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+y=0$$

91. Given, $x=a\cos\theta+b\sin\theta$, $y=a\sin\theta-b\cos\theta$
$\Rightarrow\quad x^2=a^2\cos^2\theta+b^2\sin^2\theta+2ab\cos\theta\sin\theta$
and $y^2=a^2\sin^2\theta+b^2\cos^2\theta-2ab\sin\theta\cos\theta$
Adding (1) and (2), we get $x^2+y^2=a^2+b^2$
Differentiating w.r.t x, we get

$$2x+2y\frac{dy}{dx}=0\;\Rightarrow\;x+y\frac{dy}{dx}=0 \quad ...(1)$$

Again differentiating w.r.t. x, we get

$$1+y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$$

Multiplying by y on both sides, we get

$$y^2\frac{d^2y}{dx^2}+\left(y\cdot\frac{dy}{dx}\right)\frac{dy}{dx}+y=0$$

$$\Rightarrow\quad y^2\frac{d^2y}{dx^2}-x\frac{dy}{dx}+y=0 \qquad \text{[From (1)]}$$

92. We have, $y=e^{m\sin^{-1}x}$
Differentiating w.r.t. x, we get

$$\frac{dy}{dx}=e^{m\sin^{-1}x}\left(\frac{m}{\sqrt{1-x^2}}\right)=\frac{my}{\sqrt{1-x^2}} \quad ...(1)$$

Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=m\left[\frac{\sqrt{1-x^2}\,\dfrac{dy}{dx}-\dfrac{y}{2\sqrt{1-x^2}}\cdot(-2x)}{(1-x^2)}\right]$$

$$\Rightarrow\quad (1-x^2)\frac{d^2y}{dx^2}=m\left[my+\frac{xy}{\sqrt{1-x^2}}\right] \qquad \text{[From (1)]}$$

$$\Rightarrow\quad (1-x^2)\frac{d^2y}{dx^2}=m\left[my+x\cdot\left(\frac{1}{m}\cdot\frac{dy}{dx}\right)\right]$$

$$\Rightarrow\quad (1-x^2)\frac{d^2y}{dx^2}=m^2y+x\frac{dy}{dx}$$

$$\Rightarrow\quad (1-x^2)\frac{d^2y}{dx^2}-x\frac{dy}{dx}-m^2y=0$$

93. We have, $y=(x+\sqrt{1+x^2})^n$
Differentiating w.r.t. x, we get

$$\frac{dy}{dx}=n(x+\sqrt{1+x^2})^{n-1}\left(1+\frac{2x}{2\sqrt{1+x^2}}\right)$$

$$\Rightarrow\quad \frac{dy}{dx}=n(x+\sqrt{1+x^2})^{n-1}\left(\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}}\right)$$

www.examrites.in

$$\Rightarrow \quad \frac{dy}{dx}=\frac{n(x+\sqrt{1+x^2})^n}{\sqrt{1+x^2}}=\frac{ny}{\sqrt{1+x^2}} \qquad ...(1)$$

Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=n\left[\frac{\sqrt{1+x^2}\cdot\frac{dy}{dx}-\frac{2(xy)}{2\sqrt{1+x^2}}}{1+x^2}\right]$$

$$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2}=n\left[\sqrt{1+x^2}\times\frac{ny}{\sqrt{1+x^2}}-\frac{xy}{\sqrt{1+x^2}}\right]$$

$$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2}=n^2y-\frac{nxy}{\sqrt{1+x^2}}$$

$$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2}=n^2y-x\frac{dy}{dx} \qquad \text{[From (1)]}$$

$$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=n^2y$$

94. Here $x = a\sec^3\theta$

$$\Rightarrow \quad \frac{dx}{d\theta}=a\cdot 3\sec^2\theta\cdot\sec\theta\tan\theta=3a\sec^3\theta\tan\theta$$

and $y = a\tan^3\theta$

$$\Rightarrow \quad \frac{dy}{d\theta}=a\cdot 3\tan^2\theta\cdot\sec^2\theta$$

$$\therefore \quad \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{3a\tan^2\theta\sec^2\theta}{3a\sec^3\theta\tan\theta}=\frac{\tan\theta}{\sec\theta}=\sin\theta$$

On differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=\cos\theta\cdot\frac{d\theta}{dx}=\frac{\cos\theta}{3a\sec^3\theta\tan\theta}=\frac{1}{3a}\cos^4\theta\cdot\cot\theta$$

$$\therefore \quad \left.\frac{d^2y}{dx^2}\right|_{\theta=\frac{\pi}{4}}=\frac{1}{3a}\cos^4\frac{\pi}{4}\cdot\cot\frac{\pi}{4}=\frac{1}{3a}\cdot\left(\frac{1}{\sqrt{2}}\right)^4\cdot 1$$

$$=\frac{1}{3a}\cdot\frac{1}{4}=\frac{1}{12a}$$

95. Given $y = Ae^{mx} + Be^{nx}$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx}=Ae^{mx}\cdot m+Be^{nx}\cdot n$$

$$\Rightarrow \quad \frac{d^2y}{dx^2}=m^2Ae^{mx}+n^2Be^{nx}$$

Now, L.H.S. $=\frac{d^2y}{dx^2}-(m+n)\frac{dy}{dx}+mny$

$= m^2Ae^{mx} + n^2Be^{nx} - (m + n)\,(mAe^{mx} + nBe^{nx})$

$\quad + mn(Ae^{mx} + Be^{nx})$

$=Ae^{mx}[m^2 - (m + n)m + mn]$

$\quad + Be^{nx}\,[n^2 - (m + n)n + mn]$

$= Ae^{mx} \times 0 + Be^{nx} \times 0 = 0 = \text{R.H.S.}$

96. Here, $x = a(\cos t + t \sin t)$

$$\Rightarrow \quad \frac{dx}{dt}=a[-\sin t+1\cdot\sin t+t\cos t]=at\cos t \qquad ...(1)$$

and $y = a(\sin t - t \cos t)$

$$\Rightarrow \quad \frac{dy}{dt}=a[\cos t-(1\cdot\cos t-t\sin t)]=a\,t\sin t$$

$$\therefore \quad \frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{a\,t\sin t}{a\,t\cos t}=\tan t$$

$$\Rightarrow \quad \frac{d^2y}{dx^2}=\sec^2 t\cdot\frac{dt}{dx}=\frac{\sec^2 t}{a\,t\cos t} \qquad \text{[Using (1)]}$$

$$=\frac{1}{a}\cdot\frac{1}{t\cos^3 t}$$

$$\left.\frac{d^2y}{dx^2}\right|_{t=\frac{\pi}{4}}=\frac{1}{a}\cdot\frac{1}{\frac{\pi}{4}\cos^3\frac{\pi}{4}}=\frac{4}{\pi a}\cdot\left(\sqrt{2}\right)^3=\frac{8\sqrt{2}}{\pi a}$$

97. *Refer to answer 80.*

We get $\frac{dy}{dx}=\tan t$

Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=\sec^2 t\cdot\frac{dt}{dx}=\frac{\sec^2 t}{a\cos^2 t/\sin t}=\frac{\sin t}{a\cos^4 t}$$

$$\therefore \quad \left.\frac{d^2y}{dx^2}\right|_{t=\frac{\pi}{3}}=\frac{\sin\frac{\pi}{3}}{a\cos^4\frac{\pi}{3}}=\frac{\sqrt{3}/2}{a(1/2)^4}=\frac{8\sqrt{3}}{a}$$

98. *Refer to answer 97.*

99. Given that $y=\log(x+\sqrt{x^2+a^2})$ ...(1)

Differentiating (1) w.r.t. 'x' on both sides, we get

$$\frac{dy}{dx}=\frac{1}{x+\sqrt{x^2+a^2}}\cdot\frac{d}{dx}(x+\sqrt{x^2+a^2})$$

$$=\frac{1}{x+\sqrt{x^2+a^2}}\left(1+\frac{1}{2\sqrt{x^2+a^2}}\cdot 2x\right)$$

$$=\frac{1}{x+\sqrt{x^2+a^2}}\cdot\frac{(\sqrt{x^2+a^2}+x)}{\sqrt{x^2+a^2}}$$

$$\Rightarrow \frac{dy}{dx}=\frac{1}{\sqrt{x^2+a^2}}\Rightarrow\sqrt{x^2+a^2}\,\frac{dy}{dx}=1 \qquad ...(2)$$

www.exambites.in

Again differentiating (2) on both sides w.r.t. x, we get

$$\sqrt{x^2+a^2}\,\frac{d^2y}{dx^2}+\frac{2x}{2\sqrt{x^2+a^2}}\frac{dy}{dx}=0$$

$$\Rightarrow (x^2+a^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=0$$

100. Here $x = a\cos^3\theta$, $y = a\sin^3\theta$

$$\Rightarrow \frac{dx}{d\theta}=3a\cos^2\theta\cdot(-\sin\theta) \text{ and } \frac{dy}{d\theta}=3a\sin^2\theta\cdot\cos\theta$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta}=-\tan\theta$$

Differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=-\sec^2\theta\frac{d\theta}{dx}=-\frac{\sec^2\theta}{-3a\cos^2\theta\sin\theta}$$

$$=\frac{1}{3a}\cdot\frac{1}{\cos^4\theta\cdot\sin\theta}$$

$$\therefore \quad \left.\frac{d^2y}{dx^2}\right|_{\theta=\frac{\pi}{6}}=\frac{1}{3a}\cdot\frac{1}{\cos^4\frac{\pi}{6}.\sin\frac{\pi}{6}}$$

$$=\frac{1}{3a}\cdot\frac{1}{\left(\frac{\sqrt{3}}{2}\right)^4\cdot\frac{1}{2}}=\frac{32}{27a}$$

101. Here, $y=x\log\left(\frac{x}{a+bx}\right)$ ... (1)

$$\Rightarrow y=x[\log x-\log(a+bx)]=x\log x-x\log(a+bx)$$

$$\Rightarrow \quad \frac{dy}{dx}=x\cdot\frac{1}{x}+1\cdot\log x-\left[1\cdot\log(a+bx)+x\cdot\frac{1}{a+bx}\cdot b\right]$$

$$=1-\frac{bx}{a+bx}+\log x-\log(a+bx)$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{a}{a+bx}+\log\left(\frac{x}{a+bx}\right) \qquad ...(2)$$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{a}{a+bx}+\frac{y}{x} \qquad \text{[Using (1)]}$$

Again differentiating (2) w.r.t. x, we get

$$\frac{d^2y}{dx^2}=a\cdot(-1)(a+bx)^{-2}\cdot b+\frac{a(a+bx)}{x(a+bx)^2}$$

$$=\frac{-ab}{(a+bx)^2}+\frac{a}{x(a+bx)}=\frac{-abx+a(a+bx)}{x(a+bx)^2}$$

$$=\frac{a^2}{x(a+bx)^2}$$

Now, R.H.S. $=\left(x\frac{dy}{dx}-y\right)^2$

$$=\left\{x\cdot\left[\frac{a}{a+bx}+\frac{y}{x}\right]-y\right\}^2=\left(\frac{ax}{a+bx}\right)^2$$

and L.H.S. $=x^3\frac{d^2y}{dx^2}$

$$=\frac{a^2x^2}{(a+bx)^2}=\left[\frac{ax}{a+bx}\right]^2=\text{R.H.S.}$$

102. We have, $x=\tan\left(\frac{1}{a}\log y\right)$

$$\Rightarrow \quad \frac{1}{a}\log y=\tan^{-1}x$$

Differentiating w.r.t. x, we get

$$\frac{1}{a}\cdot\frac{1}{y}\cdot\frac{dy}{dx}=\frac{1}{1+x^2} \Rightarrow (1+x^2)\frac{dy}{dx}=ay$$

Again differentiating w.r.t. x, we get

$$(1+x^2)\frac{d^2y}{dx^2}+2x\cdot\frac{dy}{dx}=a\frac{dy}{dx}$$

$$\Rightarrow \quad (1+x^2)\frac{d^2y}{dx^2}+(2x-a)\frac{dy}{dx}=0$$

103. Given $x = \cos\theta$ and $y = \sin^3\theta$

$$\Rightarrow \quad \frac{dx}{d\theta}=-\sin\theta \text{ and } \frac{dy}{d\theta}=3\sin^2\theta\cos\theta$$

$$\therefore \quad \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{3\sin^2\theta\cos\theta}{-\sin\theta}=-3\sin\theta\cos\theta$$

Differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2}=[-3\cos^2\theta-3\sin\theta(-\sin\theta)]\frac{d\theta}{dx}$$

$$=(-3\cos^2\theta+3\sin^2\theta)\cdot\frac{-1}{\sin\theta}$$

Now, L.H.S. $=y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2$

$$=\sin^3\theta\left(\frac{3\cos^2\theta-3\sin^2\theta}{\sin\theta}\right)+(-3\sin\theta\cos\theta)^2$$

$$= \sin^2\theta(3\cos^2\theta - 3\sin^2\theta) + 9\sin^2\theta\cos^2\theta$$

www.examrites.in

$= 3\sin^2\theta(\cos^2\theta - \sin^2\theta + 3\cos^2\theta)$
$= 3\sin^2\theta(4\cos^2\theta - \sin^2\theta)$
$= 3\sin^2\theta(4\cos^2\theta - 1 + \cos^2\theta)$
$= 3\sin^2\theta(5\cos^2\theta - 1) = \text{R.H.S.}$

104. Given $y = \sin^{-1} x$
Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

Again differentiating w.r.t. x, we get

$$\frac{d^2y}{dx^2} = \frac{\sqrt{1-x^2}\cdot 0 - 1\cdot\dfrac{1(-2x)}{2\sqrt{1-x^2}}}{1-x^2}$$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} = \frac{x}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\cdot\frac{1}{\sqrt{1-x^2}} = 0$$

$$\Rightarrow \quad (1-x^2)\frac{d^2y}{dx^2} - x\cdot\frac{dy}{dx} = 0$$

105. $y = (\tan^{-1}x)^2$
Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = 2\cdot\tan^{-1}x\cdot\frac{d}{dx}(\tan^{-1}x) = 2\tan^{-1}x\cdot\frac{1}{1+x^2}$$

$$\Rightarrow \quad (x^2+1)\frac{dy}{dx} = 2\tan^{-1}x$$

Again differentiating. w.r.t. x, we get

$$(x^2+1)\frac{d^2y}{dx^2} + 2x\cdot\frac{dy}{dx} = 2\cdot\frac{1}{1+x^2}$$

$$\Rightarrow \quad (x^2+1)^2\frac{d^2y}{dx^2} + 2x(x^2+1)\frac{dy}{dx} = 2$$

106. *Refer to answer 90.*

107. *Refer to answer 96.*

Since, we have $\dfrac{dx}{dt} = at\cos t$

$$\therefore \quad \frac{d^2x}{dt^2} = a[t(-\sin t) + \cos t] = -at\sin t + a\cos t$$

and $\dfrac{dy}{dt} = at\sin t \Rightarrow \dfrac{d^2y}{dt^2} = a[t\cos t + \sin t]$

108. *Refer to answer 97.*

$\because \quad y = a\sin t \quad \therefore \quad \dfrac{dy}{dt} = a\cos t$

Again differentiating w.r.t. t, we get

$$\frac{d^2y}{dt^2} = -a\sin t$$

109. *Refer to answer 97.*

110. *Refer to answer 79.*

We have, $\dfrac{dy}{dx} = \dfrac{-\sin\theta}{(1-\cos\theta)} = \dfrac{-2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}}{2\sin^2\dfrac{\theta}{2}}$

$$\Rightarrow \frac{dy}{dx} = -\cot\frac{\theta}{2} \Rightarrow \frac{d^2y}{dx^2} = \frac{1}{2}\text{cosec}^2\frac{\theta}{2}\times\frac{d\theta}{dx}$$

$$= \frac{1}{2}\times\text{cosec}^2\frac{\theta}{2}\times\frac{1}{2a\sin^2\dfrac{\theta}{2}} = \frac{\text{cosec}^4\dfrac{\theta}{2}}{4a}$$

111. *Refer to answer 110.*

112. We have, $f(x) = x^3 - 3x$

$f(-\sqrt{3}) = (-\sqrt{3})^3 - 3(-\sqrt{3}) = -3\sqrt{3} + 3\sqrt{3} = 0 = f(0).$

Since, $f(x)$ is continuous in $[-\sqrt{3}, 0]$ and differentiable on $(-\sqrt{3}, 0)$ then $\exists$ some c in $(-\sqrt{3}, 0)$ such that $f'(c) = 0$

Now, $f'(x) = 3x^2 - 3$
$\Rightarrow \quad 3c^2 - 3 = 0 \Rightarrow c^2 = 1$
$\Rightarrow \quad c = \pm 1 \Rightarrow c = -1$
as $c \in (-\sqrt{3}, 0)$.

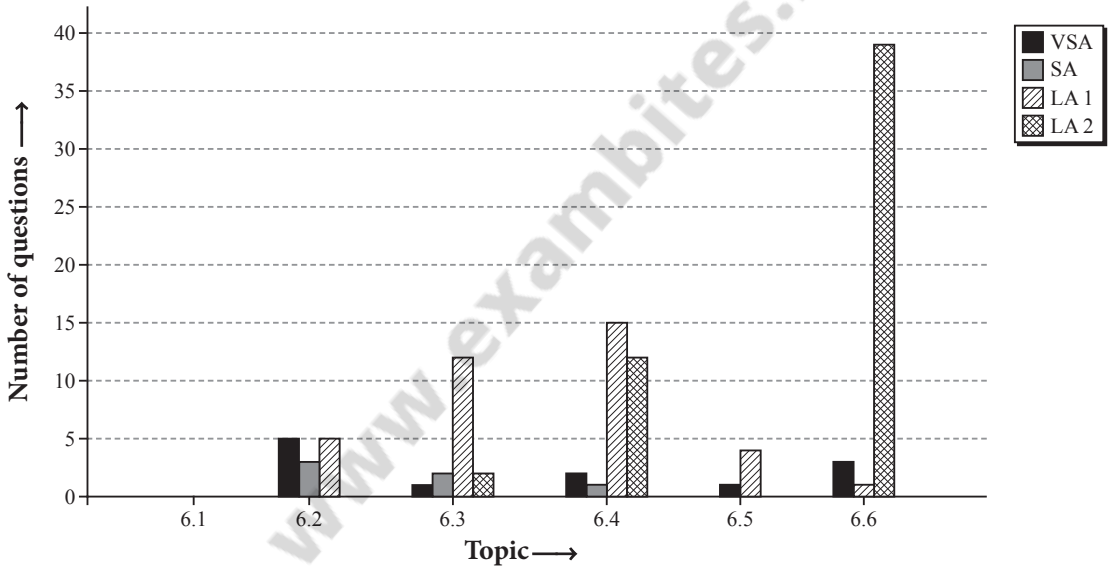
www.examrites.in

CHAPTER
6

Application of Derivatives

6.1 Introduction
6.2 Rate of Change of Quantities
6.3 Increasing and Decreasing Functions
6.4 Tangents and Normals
6.5 Approximations
6.6 Maxima and Minima

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Maxima and Minima*
- Maximum LA 1 type questions were asked from *Tangents and Normals*
- Maximum LA 2 type questions were asked from *Maxima and Minima*

QUICK RECAP

RATE OF CHANGE OF QUANTITIES

- Let $y = f(x)$ be a function. If the change in one quantity y varies with another quantity x, then $\frac{dy}{dx}$ or $f'(x)$ denotes the rate of change of y (or $f(x)$) with respect to x.

INCREASING AND DECREASING FUNCTIONS

A function f is said to be

- an increasing function on an interval (a, b) if, for all $x_1, x_2 \in (a, b)$, $x_2 > x_1 \Rightarrow f(x_2) \geq f(x_1)$ and strictly increasing if $x_2 > x_1 \Rightarrow f(x_2) > f(x_1)$

- a decreasing function on an interval (a, b) if, for all $x_1, x_2 \in (a, b), x_2 > x_1 \Rightarrow f(x_2) \le f(x_1)$ and strictly decreasing if $x_2 > x_1 \Rightarrow f(x_2) < f(x_1)$

  Note : If f is a continuous function on $[a, b]$ and differentiable on (a, b), then for each $x \in (a, b)$

  (i) $f'(x) > 0 \Rightarrow f$ is strictly increasing on (a, b)

  (ii) $f'(x) < 0 \Rightarrow f$ is strictly decreasing on (a, b)

  (iii) $f'(x) = 0 \Rightarrow f$ is a constant function on (a, b).

TANGENTS AND NORMALS

- Let $y = f(x)$ be a curve and let $P(x_1, y_1)$ be a point on it. Then, $\left(\frac{dy}{dx}\right)_P = \tan\psi$ = Slope of the tangent at P, where ψ is the angle which the tangent at $P(x_1, y_1)$ makes with the positive direction of x-axis.
- The equation of the tangent at $P(x_1, y_1)$ to the curve $y = f(x)$ is $y - y_1 = \left(\frac{dy}{dx}\right)_P (x - x_1)$
- The normal to a curve at $P(x_1, y_1)$ is a line perpendicular to the tangent at P and passing through P.

  $\therefore$ Slope of the normal at P

  $$= -\frac{1}{\text{Slope of the tangent at } P} = -\frac{1}{\left(\frac{dy}{dx}\right)_P}$$

- The equation of the normal at $P(x_1, y_1)$ to the curve $y = f(x)$ is $y - y_1 = -\frac{1}{\left(\frac{dy}{dx}\right)_P}(x - x_1)$

  Note :

  (i) If $\left(\frac{dy}{dx}\right)_P = \infty$, then the tangent at $P(x_1, y_1)$ is parallel to y-axis and its equation is $x = x_1$.

  (ii) If $\left(\frac{dy}{dx}\right)_P = 0$, then the tangent at $P(x_1, y_1)$ is parallel to x-axis and its equation is $y = y_1$.

APPROXIMATIONS

- Let $y = f(x)$ be any function of x. Let Δx be the change in x and Δy be the corresponding change in y. Then, $\Delta y = \frac{dy}{dx}.\Delta x$

  Note :

  (i) $f(x + \Delta x) \approx f(x) + \Delta x\, f'(x)$ is the approximate value.

  (ii) The change Δx is an error in x.

  (iii) $\frac{\Delta x}{x}$ is the relative error in x.

  (iv) $\left(\frac{\Delta x}{x}\right) \times 100$ is the percentage of error in x.

MAXIMA AND MINIMA

- Let f be a real valued function defined on I (subset of R), then
- f is said to have a maximum value in I, if there exists a point c in I such that $f(c) > f(x)$ for all $x \in I$. The number $f(c)$ is called the (absolute) maximum value of f in I and the point c is called point of maxima of f in I.
- f is said to have a minimum value in I, if there exists a point d in I such that $f(d) < f(x)$ for all $x \in I$. The number $f(d)$ is called the (absolute) minimum value of f in I and the point d is called point of minima of f in I.

  Note : If $f : I \to R$ be a differentiable function and c be any interior point of I, then $f'(c) = 0$ if f attains its absolute maximum (minimum) value at c.

- A point c is local maximum of a function $f(x)$ if there is an open interval I containing c such that $f(x) < f(c)$ for all $x \in I$.
- A point c is local minimum of a function $f(x)$ if there is an open interval I containing c such that $f(c) < f(x)$ for all $x \in I$.

  Note : If a function f is either increasing or decreasing in an interval I, then f is said to be a monotonic function.

- **Critical Point** : If $f : I \to R$, then a point $c \in I$ is called the critical point of f, if either $f'(c) = 0$ or f is not differentiable at c.

www.exambits.in

Methods of Finding Local Maxima and Local Minima

- **First Derivative Test :** Let $f(x)$ be a function defined on an open interval I and $f(x)$ be continuous at a critical point c in I. Then,
  - (i) If $f'(x)$ changes sign from positive to negative as x increases through c, then $x = c$ is a point of local maxima.
  - (ii) If $f'(x)$ changes sign from negative to positive as x increases through c, then $x = c$ is a point of local minima.
  - (iii) If $f'(x)$ does not change sign as x increases through c, we say that c is neither a point of local maxima nor a point of local minima. In this case, $x = c$ is called a point of inflection.
- **Second Derivative Test :** Let $f(x)$ be a function defined on an interval I and $c \in I$. Let f be twice differentiable at c. Then,
  - (i) If $f'(c) = 0$ and $f''(c) < 0$, then $x = c$ is a point of local maxima.
  - (ii) If $f'(c) = 0$ and $f''(c) > 0$, then $x = c$ is a point of local minima.
  - (iii) If $f'(c) = 0$ and $f''(c) = 0$, then use the first-derivative test.

Working Rule for Finding Absolute Maxima and Minima

- (i) Find all critical points of f in the interval $I = [a, b]$ *i.e.,* find all points x where either $f'(x) = 0$ or f is not differentiable.
- (ii) Find the end points of the interval *i.e.,* a and b.
- (iii) Find the values of f at all its critical points in interval I and at the end points of the interval I.
- (iv) Absolute maximum value of f in I = greatest of values of f at end points and at all critical points.
- (v) Absolute minimum value of f in I = least of values of f at end points and at all critical points.

www.exambites.in

Previous Years' CBSE Board Questions

6.2 Rate of Change of Quantities

VSA (1 mark)

1. The radius of a circle is increasing at the uniform rate of 3 cm/sec. At the instant when the radius of the circle is 2 cm, its area increases at the rate of _____ cm^2/s.
(*2020*)

2. The amount of pollution content added in air in a city due to x-diesel vehicles is given by $P(x) = 0.005x^3 + 0.02x^2 + 30x$. Find the marginal increase in pollution content when 3 diesel vehicles are added and write which value is indicated in the above question.
(*Delhi 2013*)

3. The money to be spent for the welfare of the employees of a firm is proportional to the rate of change of its total revenue (marginal revenue). If the total revenue (in rupees) received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$, find the marginal revenue, when $x = 5$, and write which value does the question indicate.
(AI 2013)

4. The total cost $C(x)$ associated with provision of free mid-day meals to x students of a school in primary classes is given by
$C(x) = 0.005x^3 - 0.02x^2 + 30x + 50$
If the marginal cost is given by rate of change $\frac{dC}{dx}$ of total cost, write the marginal cost of food for 300 students. What value is shown here? *(Delhi 2013C)*

5. The total expenditure (in ₹) required for providing the cheap edition of a book for poor and deserving students is given by $R(x) = 3x^2 + 36x$, where x is the number of sets of books. If the marginal expenditure is defined as $\frac{dR}{dx}$, write the marginal expenditure required for 1200 such sets. What value is reflected in this question?
(AI 2013C)

SA (2 marks)

6. The total cost $C(x)$ associated with the production of x units of an item is given by $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$. Find the marginal cost when 3 units are produced, where by marginal cost we mean the instantaneous rate of change of total cost at any level of output. *(2018)*

7. The volume of a sphere is increasing at the rate of 3 cubic centimeter per second. Find the rate of increase of its surface area, when the radius is 2 cm. *(Delhi 2017)*

8. The volume of a cube is increasing at the rate of 9 cm^3/s. How fast is its surface area increasing when the length of an edge is 10 cm?
(AI 2017)

LA 1 (4 marks)

9. A ladder 13 m long is leaning against a vertical wall. The bottom of the ladder is dragged away from the wall along the ground at the rate of 2 cm/sec. How fast is the height on the wall decreasing when the foot of the ladder is 5 m away from the wall? *(AI 2019)*

10. The side of an equilateral triangle is increasing at the rate of 2 cm/s. At what rate is its area increasing when the side of the triangle is 20 cm? (*Delhi 2015*)

11. The sides of an equilateral triangle are increasing at the rate of 2 cm/sec. Find the rate at which the area increases, when the side is 10 cm. *(AI 2014C)*

12. A ladder 5 m long is leaning against a wall. The bottom of the ladder is pulled along the ground, away from the wall, at the rate of 2 cm/s. How fast is its height on the wall decreasing when the foot of the ladder is 4 m away from the wall? *(AI 2012)*

www.exambits.in

13. Sand is pouring from a pipe at the rate of 12 cm^3/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm ?
(Delhi 2011)

6.3 Increasing and Decreasing Functions

VSA (1 mark)

14. Find the interval in which the function f given by $f(x) = 7 - 4x - x^2$ is strictly increasing. *(2020)*

SA (2 marks)

15. Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is always increasing on R. *(Delhi 2017)*

16. Show that the function $f(x) = x^3 - 3x^2 + 6x - 100$ is increasing on R. *(AI 2017)*

LA 1 (4 marks)

17. Find the intervals in which the function

$$f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12 \text{ is}$$

(a) strictly increasing
(b) strictly decreasing *(2018)*

18. Find the intervals in which the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is
(a) strictly increasing
(b) strictly decreasing *(Delhi 2014)*

19. Find the value(s) of x for which $y = [x(x-2)]^2$ is an increasing function. *(AI 2014)*

20. Find the intervals in which the function

$$f(x) = \frac{3}{2}x^4 - 4x^3 - 45x^2 + 51 \text{ is}$$

(i) strictly increasing
(ii) strictly decreasing *(Foreign 2014)*

21. Find the intervals in which the function

$$f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36}{5}x + 11 \text{ is}$$

(a) strictly increasing
(b) strictly decreasing. *(AI 2014C)*

22. Find the intervals in which the function given by $f(x) = \sin x + \cos x$, $0 \le x \le 2\pi$ is (a) increasing, (b) decreasing. *(Delhi 2012C)*

23. Find the intervals in which the following function is (a) increasing (b) decreasing :
$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$ *(AI 2012C)*

24. Prove that $y = \frac{4\sin\theta}{2+\cos\theta} - \theta$ is an increasing function in $\left[0, \frac{\pi}{2}\right]$. *(AI 2011)*

25. Find the intervals in which the following function is (a) increasing (b) decreasing :
$f(x) = 2x^3 - 9x^2 + 12x - 15$ *(Delhi 2011C)*

26. Find the intervals in which the following function is (a) increasing (b) decreasing :
$f(x) = 2x^3 + 9x^2 + 12x + 20$ *(Delhi 2011C)*

27. Find the intervals in which the function $f(x) = (x-1)^3(x-2)^2$ is (a) increasing (b) decreasing. *(AI 2011C)*

28. Show that the function f given by $f(x) = x^3 - 3x^2 + 4x$, $x \in R$ is strictly increasing on R. *(AI 2011C)*

LA 2 (6 marks)

29. Find the intervals in which $f(x) = \sin 3x - \cos 3x$, $0 < x < \pi$, is strictly increasing or strictly decreasing.
(Delhi 2016)

30. Prove that the function f defined by $f(x) = x^2 - x + 1$ is neither increasing nor decreasing in $(-1, 1)$. Hence, find the intervals in which $f(x)$ is (i) strictly increasing (ii) strictly decreasing. *(Delhi 2014C)*

6.4 Tangents and Normals

VSA (1 mark)

31. The equation of the normal to the curve $y^2 = 8x$ at the origin is _____. *(2020)*

32. Find the slope of the tangent to the curve $y = 2\sin^2(3x)$ at $x = \frac{\pi}{6}$. *(2020)*

SA (2 marks)

33. Find the points on the curve $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel to the line $4x + y - 3 = 0$. *(2020)*

www.examples.com

LA 1 (4 marks)

34. Find the equation of tangent to the curve $y = \sqrt{3x-2}$ which is parallel to the line $4x - 2y + 5 = 0$. Also, write the equation of normal to the curve at the point of contact. *(Delhi 2019)*

35. Find the equations of the tangent and the normal, to the curve $16x^2 + 9y^2 = 145$ at the point (x_1, y_1), where $x_1 = 2$ and $y_1 > 0$. *(2018)*

36. Show that the equation of normal at any point t on the curve $x = 3\cos t - \cos^3 t$ and $y = 3\sin t - \sin^3 t$ is $4(y\cos^3 t - x\sin^3 t) = 3\sin 4t$. *(Delhi 2016)*

37. The equation of tangent at (2, 3) on the curve $y^2 = ax^3 + b$ is $y = 4x - 5$. Find the values of a and b. *(AI 2016)*

38. Find the angle of intersection of the curves $y^2 = 4ax$ and $x^2 = 4by$ *(Foreign 2016)*

39. Find the point on the curve $9y^2 = x^3$, where the normal to the curve makes equal intercepts on the axes. *(Foreign 2015)*

40. Find the equation of the tangent and normal to the curve $x = a\sin^3\theta$ and $y = a\cos^3\theta$ at $\theta = \frac{\pi}{4}$. *(Delhi 2014)*

41. Find the equations of the tangent and normal to the curve $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $(\sqrt{2}a, b)$. *(AI 2014)*

42. Find the point on the curve $y = x^3 - 11x + 5$ at which the equation of tangent is $y = x - 11$. *(Delhi 2012)*

43. Show that the equation of tangent to the parabola $y^2 = 4ax$ at (x_1, y_1) is $yy_1 = 2a(x + x_1)$ *(AI 2012C)*

44. Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel to x-axis. *(Delhi 2011)*

45. Find the equation of the tangent to the curve $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at the point $x = 1$. *(Delhi 2011C)*

46. Find the equation of the tangent to the curve $4x^2 + 9y^2 = 36$ at the point $(3\cos\theta, 2\sin\theta)$. *(Delhi 2011C)*

47. Find the equations of tangents to the curve $y = (x^2 - 1)(x - 2)$ at the points where the curve cuts the x-axis. *(AI 2011C)*

48. Find the equation of tangent to the curve $x = \sin 3t$, $y = \cos 2t$ at $t = \frac{\pi}{4}$. *(AI 2011C)*

LA 2 (6 marks)

49. Find the points on the curve $9y^2 = x^3$, where the normal to the curve makes equal intercepts with both the axes. Also find the equation of the normals. *(2020)*

50. Find the equation of tangents to the curve $y = \cos(x + y)$, $-2\pi \le x \le 2\pi$ that are parallel to the line $x + 2y = 0$. *(Foreign 2016)*

51. Find the value of p for which the curves $x^2 = 9p(9 - y)$ and $x^2 = p(y + 1)$ cut each other at right angles. *(AI 2015)*

52. Find the equation of the tangent line to the curve $y = x^2 - 2x + 7$ which is (i) parallel to the line $2x - y + 9 = 0$, (ii) perpendicular to the line $5y - 15x = 13$. *(Delhi 2014C)*

53. Find the equation of the normal at a point on the curve $x^2 = 4y$ which passes through the point (1, 2). Also find the equation of the corresponding tangent. *(Delhi 2013)*

54. Find the equations of tangents to the curve $3x^2 - y^2 = 8$, which pass through the point $\left(\frac{4}{3}, 0\right)$. *(AI 2013)*

55. For the curve $y = 4x^3 - 2x^5$, find all the points on the curve at which the tangent passes through the origin. *(Delhi 2013C, 2012C)*

56. Find the equation of the tangent to the curve $y = \sqrt{3x-2}$ which is parallel to the line $4x - 2y + 5 = 0$ *(Delhi 2013C)*

57. Prove that the curve $x = y^2$ and $xy = k$ cut at the right angles if $8k^2 = 1$. *(Delhi 2013C)*

58. Prove that all normals to the curve $x = a\cos t + at\sin t$, $y = a\sin t - at\cos t$ are at a constant distance 'a' from the origin. *(AI 2013C)*

www.exambites.in

59. Find the equations of the tangent and normal to the parabola $y^2 = 4ax$ at the point $(at^2, 2at)$. *(AI 2013C)*

60. Find the equations of the normals to the curve $y = x^3 + 2x + 6$ which are parallel to the line $x + 14y + 4 = 0$. *(AI 2013C)*

6.5 Approximations

VSA (1 mark)

61. If $f(x) = x^4 - 10$, then find the approximate value of $f(2.1)$. *(2020)*

LA 1 (4 marks)

62. Find the approximate value of $f(3.02)$, upto 2 places of decimals, where $f(x) = 3x^2 + 5x + 3$. *(Foreign 2014)*

63. Using differentials, find the approximate value of $(3.968)^{3/2}$. *(Delhi 2014C)*

64. Using differentials, find the approximate value of $\sqrt{49.5}$. *(Delhi 2012)*

65. If the radius of a sphere is measured as 9 cm with an error of 0.03 cm, then find the approximate error in calculating its surface area. *(AI 2011)*

6.6 Maxima and Minima

VSA (1 mark)

66. The maximum value of slope of the curve $y = -x^3 + 3x^2 + 12x - 5$ is

(a) 15 (b) 12

(c) 9 (d) 0 *(2020)*

67. The absolute minimum value of $f(x) = 2 \sin x$ in $\left[0, \frac{3\pi}{2}\right]$ is _____. *(2020)*

68. The least value of the function $f(x) = ax + \frac{b}{x}$ $(a > 0, b > 0, x > 0)$ is _____. *(2020)*

LA 1 (4 marks)

69. An open tank with a square base and vertical sides is to be constructed from a metal sheet so as to hold a given quantity of water. Show that the cost of material will be least when depth of the tank is half of its width. If the cost is to be borne by nearby settled lower income families, for whom water will be provided, what kind of value is hidden in this question? *(2018)*

LA 2 (6 marks)

70. Show that the height of the right circular cylinder of greatest volume which can be inscribed in a right circular cone of height h and radius r is one-third of the height of the cone, and the greatest volume of the cylinder is $\frac{4}{9}$ times the volume of the cone. *(2020)*

71. Find the minimum value of $(ax + by)$, where $xy = c^2$. *(2020, Foreign 2015)*

72. A tank with rectangular base and rectangular sides, open at the top is to be constructed so that its depth is 2 m and volume is 8 m^3. If building of tank costs ₹ 70 per square metre for the base and ₹ 45 per square metre for the sides, what is the cost of least expensive tank? *(Delhi 2019)*

73. Find the area of the greatest rectangle that can be inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. *(AI 2019, AI 2013)*

74. If the sum of lengths of the hypotenuse and a side of a right angled triangle is given, show that the area of the triangle is maximum, when the angle between them is $\frac{\pi}{3}$. *(Delhi 2017, AI 2016, 2014)*

75. Show that the surface area of a closed cuboid with square base and given volume is minimum, when it is a cube. *(AI 2017)*

76. Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\frac{4r}{3}$. Also find maximum volume in terms of volume of the sphere. *(Delhi 2016, AI 2014)*

77. Prove that the least perimeter of an isosceles triangle in which a circle of radius r can be inscribed is $6\sqrt{3}r$. *(AI 2016)*

www.examhites.in

78. The sum of the surface areas of a cuboid with sides x, $2x$ and $\frac{x}{3}$ and a sphere is given to be constant. Prove that the sum of their volumes is minimum, if x is equal to three times the radius of sphere. Also find the minimum value of the sum of their volumes.
(Foreign 2016)

79. Find the local maxima and local minima of the function $f(x) = \sin x - \cos x, 0 < x < 2\pi$. Also find the local maximum and local minimum values. *(Delhi 2015)*

80. Find the coordinates of a point of the parabola $y = x^2 + 7x + 2$ which is closest to the straight line $y = 3x - 3$. *(Foreign 2015)*

81. A tank with rectangular base and rectangular sides open at the top is to be constructed so that its depth is 3 m and volume is 75 m^3. If building of tank costs ₹ 100 per square metre for the base and ₹ 50 per square metre for the sides, find the cost of least expensive tank. *(Delhi 2015C)*

82. A point on the hypotenuse of a right triangle is at distance 'a' and 'b' from the sides of the triangle. Show that the minimum length of the hypotenuse is $(a^{2/3} + b^{2/3})^{3/2}$.
(Delhi 2015C)

83. The sum of surface areas of a sphere and a cuboid with sides $\frac{x}{3}, x$ and $2x$, is constant. Show that the sum of their volumes is minimum if x is equal to three times the radius of sphere. *(AI 2015C)*

84. Of all the closed right circular cylindrical cans of volume 128 π cm^3, find the dimensions of the can which has minimum surface area.
(Delhi 2014)

85. Show that the semi vertical angle of the cone of the maximum volume and of given slant height is $\cos^{-1}\frac{1}{\sqrt{3}}$. *(Delhi 2014)*

86. Prove that the semi vertical angle of the right circular cone of given volume and least curved surface area is $\cot^{-1}\sqrt{2}$.
(Delhi 2014)

87. Prove that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume.
(AI 2014, 2012C, Delhi 2013, 2012C)

88. The sum of the perimeters of a circle and a square is k, where k is some constant. Prove that the sum of their areas is least when the side of the square is equal to the diameter of the circle. *(Foreign 2014, Delhi 2014C)*

89. Show that a cylinder of a given volume which is open at the top has minimum total surface area, when its height is equal to the radius of its base. *(Foreign 2014, Delhi 2011C)*

90. A window is of the form of a semi-circle with a rectangle on its diameter. The total perimeter of the window is 10 m. Find the dimension of the window to admit maximum light through the whole opening.
(Foreign 2014)

91. AB is a diameter of a circle and C is any point on the circle. Show that the area of ΔABC is maximum , when it is isosceles.
(AI 2014C)

92. Find the point P on the curve $y^2 = 4ax$ which is nearest to the point $(11a, 0)$. *(AI 2014C)*

93. If the length of three sides of a trapezium other than base is 10 cm each, then find the area of the trapezium when it is maximum.
(AI 2014C, Delhi 2013C)

94. Prove that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of a radius r is $\frac{4r}{3}$.
(Delhi 2013C)

95. Prove that of all the rectangles inscribed in a given circle, the square has the maximum area. *(Delhi 2013C, 2011)*

96. Prove that the radius of the right circular cylinder of greatest curved surface area which can be inscribed in a given cone, is half that of the cone. *(AI 2013C, 2012)*

www.exambites.in

97. Show that the right circular cone of least curved surface and given volume has an altitude equal to $\sqrt{2}$ times the radius of the base. *(AI 2013C, 2011)*

98. Prove that the volume of the largest cone that can be inscribed in a sphere of radius R, is $\frac{8}{27}$ of the volume of the sphere. *(AI 2013C)*

99. Show that the height of a closed right circular cylinder of given surface and maximum volume, is equal to the diameter of its base. *(Delhi 2012)*

100. An open box with a square base is to be made out of a given quantity of cardboard of area c^2 square units. Show that the maximum volume of the box is $\frac{c^3}{6\sqrt{3}}$ cubic units. *(AI 2012, 2012C)*

101. Prove that the area of a right angled triangle of given hypotenuse is maximum when the triangle is isosceles. *(Delhi 2012C)*

102. Given the sum of the perimeters of a square and a circle. Show that sum of their areas is least when the side of the square is equal to the diameter of the circle. *(Delhi 2012C)*

103. Find the volume of the largest right circular cylinder that can be inscribed in a sphere of radius r cm. *(AI 2012C, 2011C)*

104. Show that of all the rectangles with a given perimeter, the square has the largest area. *(Delhi 2011)*

105. Show that of all the rectangles of given area, the square has the smallest perimeter. *(Delhi 2011)*

106. A window has the shape of a rectangle surmounted by an equilateral triangle. If the perimeter of the window is 12 m, find the dimensions of the rectangle that will produce the largest area of the window. *(AI 2011)*

107. Find the point on the curve $y^2 = 2x$ which is at a minimum distance from the point (1, 4). *(Delhi 2011C)*

108. Show that the semi-vertical angle of a right circular cone of maximum volume and given slant height is $\tan^{-1}\sqrt{2}$. *(AI 2011C)*

Detailed Solutions

1. Let r be the radius and A be the area of circle.

Given that $\frac{dr}{dt} = 3$ cm/sec ...(i)

We know that, area of circle $A = \pi r^2$

$\therefore \quad \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi r \cdot 3$ [Using (i)]

$= 6\pi r$

$\therefore \quad \left(\frac{dA}{dt}\right)_{r=2\text{ cm}} = 12\pi \text{ cm}^2/\text{s}$

2. We have, $P(x) = 0.005x^3 + 0.02x^2 + 30x$

$\Rightarrow \quad \frac{dP}{dx} = 0.015x^2 + 0.04x + 30$

Now, $\left(\frac{dP}{dx}\right)_{x=3} = 0.015 \times 3^2 + 0.04 \times 3 + 30$

$= 30.255$

Value indicated in the question is that the increase in pollution is due to the increase in use of diesel vehicles.

3. Total revenue is given by

$R(x) = 3x^2 + 36x + 5$

$\therefore \quad$ Marginal revenue $= \frac{dR}{dx} = 6x + 36$

Now, $\left(\frac{dR}{dx}\right)_{x=5} = 6 \times 5 + 36 = 66$

Value indicated in the question is that more amount of money is spent for the welfare of the employees with the increase in marginal revenue.

4. Given, $C(x) = 0.005x^3 - 0.02x^2 + 30x + 50$

Marginal cost $= \frac{dC}{dx} = 0.015x^2 - 0.04x + 30$

www.examdres.in

Now, $\left(\frac{dC}{dx}\right)_{x=300} = 0.015 \times 300^2 - 0.04 \times 300 + 30$

$= 1350 - 12 + 30 = 1368.$

The value indicated here is that a kind of care and concern is shown towards the health of students of primary classes by providing free mid-day meal to them.

5. Here, $R(x) = 3x^2 + 36x$,

$\therefore$ Marginal expenditure $= \frac{dR}{dx} = 6x + 36$

$\left(\frac{dR}{dx}\right)_{x=1200} = 6 \times 1200 + 36 = 7236$

The value indicated here is that a kind of help is provided to poor and deserving students who want to study but they don't have sources to purchase books.

6. *Refer to answer 2.*

7. Let r, S and V respectively be the radius, surface area and volume of sphere at any time t.

Then, $\frac{dV}{dt} = 3\,\text{cm}^3/\text{sec}$

Now, $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

$\Rightarrow \quad \frac{dr}{dt} = \frac{3}{4\pi r^2}$

Also, $S = 4\pi r^2 \Rightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$

$\Rightarrow \quad \frac{dS}{dt} = \frac{6}{r} = \frac{6}{2} = 3\ \text{cm}^2/\text{sec}$

8. We have, $\frac{dv}{dt} = 9\,\text{cm}^3/\text{s}$ and $l = 10$ cm

Volume of cube $(v) = l^3$

$\therefore \quad \frac{dv}{dt} = \frac{d}{dt}(l)^3 = 9 \Rightarrow 3l^2 \frac{dl}{dt} = 9$

$\Rightarrow \quad \frac{dl}{dt} = \frac{3}{l^2}$ ...(i)

And, surface area of cube $(A) = 6l^2$

$\therefore \quad \frac{dA}{dt} = \frac{d}{dt}(6l^2) = 12l\frac{dl}{dt} = 12l \times \frac{3}{l^2}$ (From (i))

$= \frac{36}{l} = \frac{36}{10} = 3.6\,\text{cm}^2/\text{s}$

9. Let foot of the ladder is at a distance x from the wall and height on the wall is y.

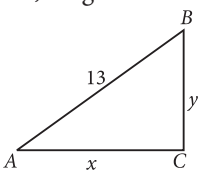
Here, $x^2 + y^2 = (13)^2$

[Using Pythagoras theorem]

Differentiating with respect to t, we get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$\Rightarrow \quad \frac{dy}{dt} = \frac{-x}{y}\frac{dx}{dt}$$



When $x = 5$ m, $y^2 = (13)^2 - (5)^2 = 169 - 25 = 144$

$\therefore \quad y = 12$ m

Also, $\frac{dx}{dt} = 2$ cm/sec [Given]

$\therefore \quad \frac{dy}{dt} = \frac{-5}{12} \times 2 = \frac{-5}{6}$ cm/sec

10. Let 'a' be the side of an equilateral triangle.

Then $\frac{da}{dt} = 2\,\text{cm/ sec}$

Let 'A' be the area of an equilateral triangle, then

$A = \frac{\sqrt{3}}{4}a^2 \Rightarrow \quad \frac{dA}{dt} = 2 \times \frac{\sqrt{3}}{4}a\frac{da}{dt} = \frac{\sqrt{3}}{2}a\frac{da}{dt}$

$\therefore \quad \left(\frac{dA}{dt}\right)_{a=20} = \frac{\sqrt{3}}{2} \times 20 \times 2 = 20\sqrt{3}\,\text{cm}^2/\text{sec}$

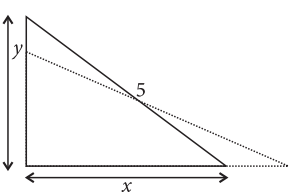
11. *Refer to answer 10.*

12. Let foot of the ladder is at a distance of 'x' from the wall and height of the wall is 'y'. Here, $x^2 + y^2 = 5^2$

Differentiating w.r.t. t, we get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$\Rightarrow \quad \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$



When $x = 4$ m ; $y^2 = 5^2 - x^2 = 5^2 - 4^2 = 3^2 \Rightarrow y = 3$ m

and $\frac{dx}{dt} = 2$ cm/sec (Given)

$\therefore \quad \frac{dy}{dt} = -\frac{400}{300} \times 2 = -\frac{8}{3}$ cm/sec.

$\therefore$ The height of the ladder on the wall is decreasing at the rate of $\frac{8}{3}$ cm/sec.

www.exambites.in

13. Let at any instant of time t, the radius of the base of the cone be r, its height be h and the volume of cone be V, then

$$h = \frac{1}{6}r \Rightarrow r = 6h$$

and $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi (6h)^2 h = 12\pi h^3$ ...(i)

Differentiating (i) w.r.t. t, we get

$$\frac{dV}{dt} = (12\pi)\left(3h^2 \frac{dh}{dt}\right) \Rightarrow 12 = 36\pi (4)^2 \frac{dh}{dt}$$

$$\left(\because \quad h = 4 \text{ cm} \text{ and } \frac{dV}{dt} = 12 \text{ cm}^3 / \text{sec}\right)$$

$$\Rightarrow \quad \frac{dh}{dt} = \frac{12}{36\pi \times 16} = \frac{1}{48\pi} \text{ cm / sec}$$

$\therefore$ Height of the sand cone is increasing at the rate of $\frac{1}{48\pi}$ cm / sec.

14. Let $y = f(x) = 7 - 4x - x^2$

$\therefore \quad \frac{dy}{dx} = -4 - 2x$

For strictly increasing, $\frac{dy}{dx} > 0$

$\Rightarrow \quad -4 - 2x > 0 \Rightarrow x < -2$

$\therefore$ Required interval is $(-\infty, -2)$

15. We have, $f(x) = 4x^3 - 18x^2 + 27x - 7$

$\Rightarrow \quad f'(x) = 12x^2 - 36x + 27$

$$= 12\left(x^2 - 3x + \frac{9}{4} - \frac{9}{4}\right) + 27$$

$$= 12\left(x - \frac{3}{2}\right)^2 - 27 + 27 = 12\left(x - \frac{3}{2}\right)^2 \geq 0 \ \forall x \in R$$

Hence, $f(x)$ is always increasing on R.

16. We have, $f(x) = x^3 - 3x^2 + 6x - 100$ ...(i)

Differentiating (i) w.r.t. x, we get

$f'(x) = 3x^2 - 6x + 6$

$= 3(x^2 - 2x + 1) + 3$

$= 3(x - 1)^2 + 3 > 0$

($\because$ For all values of x, $(x - 1)^2$ is always positive)

$\therefore \quad f'(x) > 0$

So, $f(x)$ is an increasing function on R.

17. We have, $f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12$ ...(i)

$f(x)$ being polynomial function is continuous and derivable on R.

Differentiating (i) w.r.t. x

$$f'(x) = \frac{4x^3}{4} - 3x^2 - 10x + 24$$

$= x^3 - 3x^2 - 10x + 24$

$= (x - 2)(x^2 - x - 12) = (x - 2)(x - 4)(x + 3)$

(a) For increasing, $f'(x) > 0$

$\Rightarrow \quad (x - 2)(x - 4)(x + 3) > 0$

$\Rightarrow \quad x \in (4, \infty) \cup (-3, 2)$

(b) For decreasing, $f'(x) < 0$

$\Rightarrow \quad (x - 2)(x - 4)(x + 3) < 0$

$\Rightarrow \quad x \in (2, 4) \cup (-\infty, -3)$

18. We have, $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2)$

$\Rightarrow \quad f'(x) = 12x(x + 1)(x - 2)$

Now, $f'(x) = 0$

$\Rightarrow \quad 12x(x + 1)(x - 2) = 0$

$\Rightarrow \quad x = -1, x = 0$ or $x = 2$

Hence these points divide the whole real line into four disjoint open intervals namely $(-\infty, -1)$, $(-1, 0)$, $(0, 2)$ and $(2, \infty)$

Interval	Sign of $f'(x)$	Nature of function
$(-\infty, -1)$	$(-)(-)(-) < 0$	Strictly decreasing
$(-1, 0)$	$(-)(+)(-) > 0$	Strictly increasing
$(0, 2)$	$(+)(+)(-) < 0$	Strictly decreasing
$(2, \infty)$	$(+)(+)(+) > 0$	Strictly increasing

(a) $f(x)$ is strictly increasing in $(-1, 0) \cup (2, \infty)$

(b) $f(x)$ is strictly decreasing in $(-\infty, -1) \cup (0, 2)$

19. Here, $y = [x(x - 2)]^2 = x^2(x - 2)^2$

$$\Rightarrow \quad \frac{dy}{dx} = 2x(x-2)^2 + 2x^2(x-2)$$

$$= 2x(x - 2)(x - 2 + x) = 4x(x - 1)(x - 2)$$

For y to be an increasing function, $\frac{dy}{dx} > 0$

$\Rightarrow \quad x(x - 1)(x - 2) > 0$

Case 1: When $-\infty < x < 0$

$\frac{dy}{dx} < 0 \Rightarrow y$ is a decreasing function.

Case 2: When $0 < x < 1$

$\frac{dy}{dx} > 0 \Rightarrow y$ is an increasing function.

Case 3: When $1 < x < 2$

$\frac{dy}{dx} < 0 \Rightarrow y$ is a decreasing function.

www.examrace.in

Case 4: When $2 < x < \infty$

$\frac{dy}{dx} > 0 \Rightarrow y$ is an increasing function.

$\therefore$ y is an increasing function in $[0, 1] \cup [2, \infty)$

20. *Refer to answer 17.*

21. Here, $f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36}{5}x + 11$

Differentiating w.r.t. x, we get

$$f'(x) = \frac{3}{10}\cdot 4x^3 - \frac{4}{5}\cdot 3x^2 - 3\cdot 2x + \frac{36}{5}\cdot 1$$

$$= \frac{6}{5}(x^3 - 2x^2 - 5x + 6)$$

$$= \frac{6}{5}(x-1)(x^2 - x - 6)$$

$$= \frac{6}{5}(x-1)(x+2)(x-3)$$

$\therefore$ $f'(x) = 0 \Rightarrow x = -2, 1, 3.$

Hence, the points divide the real line into disjoint intervals $(-\infty, -2)$, $(-2, 1)$, $(1, 3)$ and $(3, \infty)$

Interval	Sign of $f'(x)$	Nature of function
$(-\infty, -2)$	$(-)(-)(-) < 0$	Strictly Decreasing
$(-2, 1)$	$(-)(+)(-) > 0$	Strictly Increasing
$(1, 3)$	$(+)(+)(-) < 0$	Strictly Decreasing
$(3, \infty)$	$(+)(+)(+) > 0$	Strictly Increasing

(a) $f(x)$ is strictly increasing in $(-2, 1) \cup (3, \infty)$

(b) $f(x)$ is strictly decreasing in $(-\infty, -2) \cup (1, 3)$

22. The given function is

$f(x) = \sin x + \cos x; 0 \le x \le 2\pi$

$\Rightarrow$ $f'(x) = \cos x - \sin x$

Now $f'(x) = 0 \Rightarrow \cos x - \sin x = 0$

$\Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$

Thus $f'(x) > 0$ in $\left[0, \frac{\pi}{4}\right)$,

$f'(x) < 0$ in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ and $f'(x) > 0$ in $\left(\frac{5\pi}{4}, 2\pi\right]$

$\Rightarrow$ The function f is decreasing in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ and it is increasing in $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$.

23. The given function is

$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$

$\Rightarrow$ $f'(x) = 4x^3 - 24x^2 + 44x - 24$

$= 4(x^3 - 6x^2 + 11x - 6)$

$= 4(x-1)(x^2 - 5x + 6)$

$= 4(x-1)(x-2)(x-3)$

Thus $f'(x) = 0 \Rightarrow x = 1, 2, 3.$

Hence, possible disjoint intervals are $(-\infty, 1)$, $(1, 2)$, $(2, 3)$ and $(3, \infty)$.

In the interval $(-\infty, 1)$, $f'(x) < 0$

In the interval $(1, 2)$, $f'(x) > 0$

In the interval $(2, 3)$, $f'(x) < 0$

In the interval $(3, \infty)$, $f'(x) > 0$

$\therefore$ f is increasing in $(1, 2) \cup (3, \infty)$ and f is decreasing in $(-\infty, 1) \cup (2, 3)$.

24. We have, $y = \frac{4\sin\theta}{2 + \cos\theta} - \theta, \theta \in \left[0, \frac{\pi}{2}\right]$

$$\Rightarrow \frac{dy}{d\theta} = 4\left\{\frac{(2+\cos\theta)\cos\theta - \sin\theta(-\sin\theta)}{(2+\cos\theta)^2}\right\} - 1$$

$$= \frac{4(2\cos\theta + 1)}{(2+\cos\theta)^2} - 1$$

$$= \frac{8\cos\theta + 4 - (2+\cos\theta)^2}{(2+\cos\theta)^2}$$

$$= \frac{4\cos\theta - \cos^2\theta}{(2+\cos\theta)^2} = \frac{\cos\theta(4 - \cos\theta)}{(2+\cos\theta)^2}$$

Now, $\cos\theta \ge 0$ in $\left[0, \frac{\pi}{2}\right]$; $4 - \cos\theta > 0$ in $\left[0, \frac{\pi}{2}\right]$ $(\because -1 \le \cos\theta \le 1)$

and $(2+\cos\theta)^2 > 0$ in $\left[0, \frac{\pi}{2}\right]$ (Being a perfect square)

$\therefore$ $\frac{dy}{d\theta} > 0$ for all $\theta \in \left[0, \frac{\pi}{2}\right]$

Hence y is increasing function in $\left[0, \frac{\pi}{2}\right]$.

25. Here, $f(x) = 2x^3 - 9x^2 + 12x - 15$

$\Rightarrow$ $f'(x) = 6x^2 - 18x + 12$

$= 6(x^2 - 3x + 2) = 6(x-2)(x-1)$

(a) For f to be increasing,

$f'(x) > 0 \Rightarrow 6(x-2)(x-1) > 0$

$\Rightarrow$ $(x-2)(x-1) > 0 \Rightarrow x - 2 > 0, x - 1 > 0$

or $x - 2 < 0, x - 1 < 0$

www.exambites.in

$\Rightarrow \quad x > 2$ or $x < 1 \Rightarrow x \in (2, \infty)$ or $x \in (-\infty, 1)$

$\therefore$ f is increasing in $(-\infty, 1) \cup (2, \infty)$

(b) For f to be decreasing,

$f'(x) < 0 \Rightarrow (x - 2)(x - 1) < 0$

$\Rightarrow \quad x < 2$ and $x > 1 \Rightarrow 1 < x < 2 \Rightarrow x \in (1, 2)$

$\therefore$ f is decreasing in $(1, 2)$

26. Given : $f(x) = 2x^3 + 9x^2 + 12x + 20$

$\Rightarrow \quad f'(x) = 6x^2 + 18x + 12$

$= 6(x^2 + 3x + 2) = 6(x + 1)(x + 2)$

(a) For f to be increasing,

$f'(x) > 0 \Rightarrow 6(x + 1)(x + 2) > 0$

$\Rightarrow \quad (x + 1)(x + 2) > 0 \Rightarrow x + 1 > 0, x + 2 > 0$

or $\quad x + 1 < 0, x + 2 < 0$

$\Rightarrow \quad x > -1$ or $x < -2$

$\Rightarrow \quad x \in (-1, \infty)$ or $(-\infty, -2)$

$\therefore$ f is increasing in $(-\infty, -2) \cup (-1, \infty)$

(b) For f to be decreasing,

$f'(x) < 0 \Rightarrow (x + 1)(x + 2) < 0$

$\Rightarrow \quad x + 1 < 0$ and $x + 2 > 0$

$\Rightarrow \quad x < -1$ and $x > -2$

$\Rightarrow \quad -2 < x < -1 \Rightarrow x \in (-2, -1)$

$\therefore$ f is decreasing in $(-2, -1)$

27. Here, $f(x) = (x - 1)^3 \cdot (x - 2)^2$

$\Rightarrow \quad f'(x) = 3(x - 1)^2(x - 2)^2 + (x - 1)^3 \cdot 2(x - 2)$

$= (x - 1)^2(x - 2)[3(x - 2) + 2(x - 1)]$

$= (x - 1)^2 (x - 2)(5x - 8)$

(a) For f to be an increasing function, $f'(x) > 0$

$\Rightarrow \quad (x - 1)^2 (x - 2)(5x - 8) > 0$

$\Rightarrow \quad (x - 2)(5x - 8) > 0 \quad [\because (x - 1)^2 \geq 0 \ \forall \ x \in R]$

$\Rightarrow \quad x - 2 > 0, 5x - 8 > 0$ or $x - 2 < 0, 5x - 8 < 0$

$\Rightarrow \quad x > 2$ or $x < \frac{8}{5} \Rightarrow \quad x \in (2, \infty)$ or $x \in \left(-\infty, \frac{8}{5}\right)$

(b) For f to be decreasing function, $f'(x) < 0$

$\Rightarrow \quad (x - 1)^2 (x - 2)(5x - 8) < 0$

$\Rightarrow \quad (x - 2)(5x - 8) < 0$

$\Rightarrow \quad x - 2 < 0, 5x - 8 > 0$ ($\because$ The other case is absurd)

$\Rightarrow \quad \frac{8}{5} < x < 2 \Rightarrow x \in \left(\frac{8}{5}, 2\right)$

28. Here $f(x) = x^3 - 3x^2 + 4x$

$\Rightarrow \quad f'(x) = 3x^2 - 6x + 4 = 3(x^2 - 2x) + 4$

$= 3(x^2 - 2x + 1) - 3 + 4$

$= 3(x - 1)^2 + 1 > 0 \ \forall x \in R$

$\Rightarrow \quad f$ is strictly increasing on R.

29. $f(x) = \sin 3x - \cos 3x \Rightarrow f'(x) = 3\cos 3x + 3\sin 3x$

$f'(x) = 0 \Rightarrow 3 \cos 3x = -3 \sin 3x$

$\Rightarrow \quad \cos 3x = -\sin 3x \ \Rightarrow \ \tan 3x = -1$

which gives $3x = \frac{3\pi}{4}$ or $\frac{7\pi}{4}$ or $\frac{11\pi}{4}$

$\Rightarrow \quad x = \frac{\pi}{4}$ or $\frac{7\pi}{12}$ or $\frac{11\pi}{12}$ $\quad [\because 0 < x < \pi]$

The points $x = \frac{\pi}{4}$, $x = \frac{7\pi}{12}$ and $x = \frac{11\pi}{12}$ divides the interval $(0, \pi)$ into four disjoint intervals,

$\left(0, \frac{\pi}{4}\right), \left(\frac{\pi}{4}, \frac{7\pi}{12}\right), \left(\frac{7\pi}{12}, \frac{11\pi}{12}\right), \left(\frac{11\pi}{12}, \pi\right)$

Now, $f'(x) > 0$ in $\left(0, \frac{\pi}{4}\right)$

$\Rightarrow \quad f$ is strictly increasing in $\left(0, \frac{\pi}{4}\right)$

$f'(x) < 0$ in $\left(\frac{\pi}{4}, \frac{7\pi}{12}\right)$

$\Rightarrow \quad f$ is strictly decreasing in $\left(\frac{\pi}{4}, \frac{7\pi}{12}\right)$

$f'(x) > 0$ in $\left(\frac{7\pi}{12}, \frac{11\pi}{12}\right)$

$\Rightarrow \quad f$ is strictly increasing in $\left(\frac{7\pi}{12}, \frac{11\pi}{12}\right)$

$f'(x) < 0$ in $\left(\frac{11\pi}{12}, \pi\right)$

$\Rightarrow \quad f$ is strictly decreasing in $\left(\frac{11\pi}{12}, \pi\right)$

Hence, f is strictly increasing in the intervals

$\left(0, \frac{\pi}{4}\right) \cup \left(\frac{7\pi}{12}, \frac{11\pi}{12}\right)$

and f is strictly decreasing in the intervals

$\left(\frac{\pi}{4}, \frac{7\pi}{12}\right) \cup \left(\frac{11\pi}{12}, \pi\right)$

30. Here, $f(x) = x^2 - x + 1; x \in (-1, 1)$

$\Rightarrow \quad f'(x) = 2x - 1$

$f'(x) = 0 \Rightarrow x = \frac{1}{2}$

Now $f'(x) = 2\left(x - \frac{1}{2}\right) > 0$ for $\frac{1}{2} < x < 1$

$\Rightarrow \quad f$ is strictly increasing in $\left(\frac{1}{2}, 1\right)$

Also $f'(x) = 2\left(x - \frac{1}{2}\right) < 0$ for $-1 < x < \frac{1}{2}$

$\Rightarrow \quad f$ is strictly decreasing in $\left(-1, \frac{1}{2}\right)$.

Thus f is neither increasing nor decreasing in $(-1, 1)$.

www.exambites.in

31. The given parabola is $y^2 = 8x$ ...(i)

$\Rightarrow \quad 2y\frac{dy}{dx} = 8 \Rightarrow \frac{dy}{dx} = \frac{4}{y}$

Now, slope of normal $= \left(\frac{-dx}{dy}\right)_{(0,0)} = -\left(\frac{0}{4}\right) = 0$

Now, equation of normal at (0, 0) is

$y - 0 = 0\,(x - 0)$

$\Rightarrow \quad y = 0$

32. We have, $y = 2\sin^2(3x)$ ...(i)

Slope of tangent to (i) is given by

$$\frac{dy}{dx} = 4\sin(3x)\cdot\cos(3x)\cdot 3 = 6\sin 6x$$

Now, $\left[\frac{dy}{dx}\right]_{x=\pi/6} = 6\sin 6\cdot\frac{\pi}{6} = 6\sin\pi = 6\times 0 = 0$

33. The given curve is $y = x^3 - 3x^2 - 4x$ ...(i)

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = 3x^2 - 6x - 4$$

Since, the tangent is parallel to $4x + y - 3 = 0$

$\therefore \quad \frac{dy}{dx}$ = Slope of line $= -4$

$\Rightarrow \quad 3x^2 - 6x - 4 = -4$

$\Rightarrow \quad 3x^2 - 6x = 0 \Rightarrow 3x(x - 2) = 0 \Rightarrow x = 0, 2$

Putting the value of $x = 0, 2$, in (i), we get

$y = 0$ and $y = -12$

$\therefore$ The points are (0, 0) and (2, –12).

34. The given curve is $y = \sqrt{3x-2}$ ...(i)

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2}\cdot(3x-2)^{-1/2}.3 = \frac{3}{2}\cdot\frac{1}{\sqrt{3x-2}}$$

We want the tangent to (i) that is parallel to the line $4x - 2y + 5 = 0$ ...(ii)

$\therefore$ Slope of tangent = slope of line $4x - 2y + 5 = 0$

$$\Rightarrow \frac{3}{2}\cdot\frac{1}{\sqrt{3x-2}} = -\frac{4}{-2} = 2$$

$$\Rightarrow \quad 3 = 4\sqrt{3x-2} \Rightarrow 9 = 16(3x-2)$$

$$\Rightarrow 3x - 2 = \frac{9}{16} \Rightarrow 3x = 2 + \frac{9}{16} = \frac{41}{16} \Rightarrow x = \frac{41}{48}$$

From (i), $y = \sqrt{3\cdot\frac{41}{48} - 2} = \frac{3}{4}$

$\therefore$ The equation of the tangent to (i) at $\left(\frac{41}{48}, \frac{3}{4}\right)$ that have slope = 2 is

$$y - \frac{3}{4} = 2\left(x - \frac{41}{48}\right) \Rightarrow \quad 4y - 3 = 8x - \frac{41}{6}$$

$$\Rightarrow \quad 8x - 4y - \frac{41}{6} + 3 = 0 \Rightarrow \quad 8x - 4y - \frac{23}{6} = 0$$

The equation of normal to given curve at $\left(\frac{41}{48}, \frac{3}{4}\right)$ having slope –1/2 is given by

$$\left(y - \frac{3}{4}\right) = \frac{-1}{2}\left(x - \frac{41}{48}\right) \Rightarrow \quad \frac{x}{2} + y = \frac{113}{96}$$

or $48x + 96y - 113 = 0$

35. The given curve is, $16x^2 + 9y^2 = 145$ ...(i)

Differentiating (i) both sides w.r.t. x, we get

$32x + 18yy' = 0$

$$\Rightarrow \quad y' = \frac{-32x}{18y} = \frac{-16}{9}\frac{x}{y}$$ ...(ii)

Now, (i) passes through (x_1, y_1)

$\therefore \quad 16x_1^2 + 9y_1^2 = 145$

$\Rightarrow \quad 16(2)^2 + 9y_1^2 = 145 \qquad (\because \quad x_1 = 2)$

$\Rightarrow \quad 9y_1^2 = 145 - 64$

$\Rightarrow \quad y_1^2 = 9 \Rightarrow y_1 = \pm 3$

But $y_1 > 0 \quad \therefore y_1 = 3$

$\therefore$ From (ii), we get

$$y' = \frac{-16}{9}\frac{x_1}{y_1} \Rightarrow \quad y' = \frac{-16}{9}\times\frac{2}{3} = \frac{-32}{27}$$

So, equation of tangent passing through (2, 3) and having slope $\left(\frac{-32}{27}\right)$ is

$$y - 3 = \frac{-32}{27}(x - 2)$$

$\Rightarrow \quad 27y - 81 = -32x + 64 \Rightarrow 32x + 27y - 145 = 0$

Also, equation of normal passing through (2, 3) is

$$(y - 3) = \frac{-1}{\left(\frac{-32}{27}\right)}\,(x - 2)$$

$\Rightarrow \quad 32y - 96 = 27x - 54 \Rightarrow 27x - 32y + 42 = 0$

36. $x = 3\cos t - \cos^3 t$ and $y = 3\sin t - \sin^3 t$

Now,

$$\frac{dx}{dt} = -3\sin t + 3\cos^2 t\sin t = -3\sin t(1 - \cos^2 t)$$

$= -3\sin t\sin^2 t = -3\sin^3 t$

Also, $\frac{dy}{dt} = 3\cos t - 3\sin^2 t\cos t = 3\cos t(1 - \sin^2 t)$

$= 3\cos t\cos^2 t = 3\cos^3 t$

So, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3\cos^3 t}{-3\sin^3 t} = -\frac{\cos^3 t}{\sin^3 t}$

www.exambee.com

Slope of normal $= \dfrac{-1}{\dfrac{dy}{dx}} = \dfrac{\sin^3 t}{\cos^3 t}$

Required equation of normal is

$$y-(3\sin t-\sin^3 t)=\frac{\sin^3 t}{\cos^3 t} \quad [x-(3\cos t-\cos^3 t)]$$

$\Rightarrow$ $y\cos^3 t - 3\sin t\cos^3 t + \sin^3 t\cos^3 t$
$= x\sin^3 t - 3\cos t\sin^3 t + \sin^3 t\cos^3 t$

$\Rightarrow$ $y\cos^3 t - x\sin^3 t = 3\sin t\cos t(\cos^2 t - \sin^2 t)$

$\Rightarrow$ $y\cos^3 t - x\sin^3 t = \dfrac{3\sin 2t\cdot\cos 2t}{2}$

$\Rightarrow$ $y\cos^3 t - x\sin^3 t = \dfrac{3}{2}\cdot\dfrac{\sin 4t}{2}$

$\Rightarrow$ $4(y\cos^3 t - x\sin^3 t) = 3\sin 4t.$

37. We have, $y^2 = ax^3 + b$

Differentiating w.r.t. x, we get

$$2y\frac{dy}{dx} = 3ax^2 \Rightarrow \frac{dy}{dx} = \frac{3ax^2}{2y}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(2,3)} = \left(\frac{3ax^2}{2y}\right)_{(2,3)} = 2a$$

So, equation of tangent at the point (2, 3) is

$y - 3 = 2a(x - 2)$

$\Rightarrow$ $y = 2ax - 4a + 3$ ...(i)

But we are given that equation of tangent at (2, 3) is

$y = 4x - 5$ ...(ii)

$\therefore$ On comparing (i) and (ii), we get

$2a = 4 \Rightarrow a = 2$

$\because$ Point (2, 3) lies on the curve $y^2 = ax^3 + b$,

$\therefore$ $(3)^2 = (2)^3 a + b \Rightarrow 9 = 8a + b$

$\Rightarrow$ $9 = 8\times 2 + b \Rightarrow b = -7$

38. The given curves are

$y^2 = 4ax$ ... (i)

$x^2 = 4by$ ...(ii)

Solving (i) and (ii), we get

$$\left(\frac{x^2}{4b}\right)^2 = 4ax$$

$\Rightarrow$ $x^4 - 64ab^2x = 0 \Rightarrow x(x^3 - 64ab^2) = 0$

$\Rightarrow$ $x = 0, 4a^{1/3}b^{2/3}$

When $x = 0$, $y = 0$

When $x = 4a^{1/3}b^{2/3}$, $y = \dfrac{(4a^{1/3}b^{2/3})^2}{4b} = 4a^{2/3}b^{1/3}$

Thus, the given curves intersect at (0, 0) and $(4a^{1/3}b^{2/3}, 4a^{2/3}b^{1/3})$

At (0, 0) the angle between the curves is 90°

Differentiating (i) with respect to x, we get

$$2y\frac{dy}{dx} = 4a \Rightarrow \frac{dy}{dx} = \frac{2a}{y}$$

$$\Rightarrow m_1 = \left(\frac{dy}{dx}\right)_{(4a^{1/3}b^{2/3},\,4a^{2/3}b^{1/3})} = \frac{1}{2}\left(\frac{a}{b}\right)^{1/3}$$

Differentiating (ii) with respect to x, we get

$$2x = 4b\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2b}$$

$$\Rightarrow m_2 = \left(\frac{dy}{dx}\right)_{(4a^{1/3}b^{2/3},\,4a^{2/3}b^{1/3})}$$

$$= \frac{4a^{1/3}b^{2/3}}{2b} = 2\left(\frac{a}{b}\right)^{1/3}$$

Let θ be the angle between the two curves. Then,

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$\Rightarrow \tan\theta = \left|\frac{\frac{1}{2}\left(\frac{a}{b}\right)^{1/3} - 2\left(\frac{a}{b}\right)^{1/3}}{1 + \frac{1}{2}\left(\frac{a}{b}\right)^{1/3}\times 2\left(\frac{a}{b}\right)^{1/3}}\right|$$

$$\Rightarrow \tan\theta = \left|\frac{\frac{-3}{2}\left(\frac{a}{b}\right)^{1/3}}{1 + \left(\frac{a}{b}\right)^{2/3}}\right| = \left|\frac{-3a^{1/3}b^{1/3}}{2(a^{2/3} + b^{2/3})}\right|$$

$$\Rightarrow \theta = \tan^{-1}\left|\frac{-3(ab)^{1/3}}{2(a^{2/3} + b^{2/3})}\right|$$

39. We have, $9y^2 = x^3$ ...(i)

Differentiating (i) w.r.t. x, we get

$$18y\frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = \frac{x^2}{6y}$$

Let $P(x_1, y_1)$ be the point on (i) where the normal makes equal intercepts on the axes

$\therefore$ $9y_1^2 = x_1^3$ ...(ii)

Now, slope of tangent at (x_1, y_1) is $\dfrac{x_1^2}{6y_1}$

$\therefore$ Slope of normal at (x_1, y_1) is $-\dfrac{6y_1}{x_1^2}$

Since the normal makes equal intercepts on the axes,

www.exambites.in

$\therefore$ Its slope $= \pm 1 \Rightarrow \dfrac{6y_1}{x_1^2} = \pm 1$ ...(iii)

Now from (ii) and (iii), we get $9\left(\dfrac{x_1^2}{6}\right)^2 = x_1^3$

$\Rightarrow \dfrac{x_1^4}{4} = x_1^3 \Rightarrow x_1 = 4$

Putting $x_1 = 4$ in (iii), we get $y_1 = \pm\dfrac{16}{6} = \pm\dfrac{8}{3}$

Hence, the points are $\left(4, \pm\dfrac{8}{3}\right)$.

40. We have, $x = a\sin^3\theta$; $y = a\cos^3\theta$

$\dfrac{dx}{d\theta} = 3a\sin^2\theta\cos\theta$ and $\dfrac{dy}{d\theta} = 3a\cos^2\theta(-\sin\theta)$

$\therefore \quad \dfrac{dy}{dx} = \dfrac{-3a\sin\theta\cos^2\theta}{3a\sin^2\theta\cos\theta} = -\dfrac{\cos\theta}{\sin\theta} = -\cot\theta$

$\therefore \quad \left[\dfrac{dy}{dx}\right]_{\theta=\frac{\pi}{4}} = -\cot\dfrac{\pi}{4} = -1$

Hence, the equation of the tangent at $\theta = \dfrac{\pi}{4}$ is

$y - a\cos^3\theta = (-1)(x - a\sin^3\theta)$

$\Rightarrow \quad x + y = a\left\{\left(\dfrac{1}{\sqrt{2}}\right)^3 + \left(\dfrac{1}{\sqrt{2}}\right)^3\right\} = \dfrac{a}{\sqrt{2}}$

$\Rightarrow \quad \sqrt{2}x + \sqrt{2}y = a$

and the equation of the normal is

$y - a\cos^3\theta = \dfrac{1}{-(-1)}(x - a\sin^3\theta)$

$\therefore \quad y - a\cos^3\theta = (x - a\sin^3\theta)$

$\Rightarrow \quad x - y = a\left(\left(\dfrac{1}{\sqrt{2}}\right)^3 - \left(\dfrac{1}{\sqrt{2}}\right)^3\right)$

$\Rightarrow \quad x - y = 0$

41. The given curve is $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$

Differentiating w.r.t. x, we get

$\dfrac{2x}{a^2} - \dfrac{2y}{b^2}\cdot\dfrac{dy}{dx} = 0 \Rightarrow \dfrac{dy}{dx} = \dfrac{x/a^2}{y/b^2} = \dfrac{b^2}{a^2}\cdot\dfrac{x}{y}$

At $P(\sqrt{2}a, b)$

Slope of the tangent $= \dfrac{b^2}{a^2}\cdot\dfrac{\sqrt{2}a}{b} = \sqrt{2}\cdot\dfrac{b}{a}$

and slope of the normal $= -\dfrac{a}{b\sqrt{2}}$

$\therefore$ Equation of the tangent at P is

$y - b = \sqrt{2}\dfrac{b}{a}(x - \sqrt{2}a) \Rightarrow \quad y = \sqrt{2}\dfrac{b}{a}x - b$

Also equation of the normal at P is

$y - b = -\dfrac{a}{\sqrt{2}b}(x - \sqrt{2}a) \Rightarrow \quad y = -\dfrac{ax}{\sqrt{2}b} + b + \dfrac{a^2}{b}$

42. $y = x^3 - 11x + 5$ ...(i)

Differentiating (i) w.r.t. x, we get

$\dfrac{dy}{dx} = 3x^2 - 11$ = slope of tangent

Also, equation of tangent is $y = x - 11$

$\therefore$ its slope = 1.

So $3x^2 - 11 = 1 \Rightarrow x^2 = 4 \quad \therefore \quad x = \pm 2$

Putting the values of x in (i), we get

$y = 2^3 - 11(2) + 5 = 8 - 22 + 5 = -9$

$y = (-2)^3 - 11(-2) + 5 = -8 + 22 + 5 = 19$

So points are (2, –9) and (–2, 19).

But only (2, –9) satisfies the equation of tangent.

So required point is (2, –9).

43. The given parabola is

$y^2 = 4ax$ ... (i)

$\Rightarrow \quad 2y\dfrac{dy}{dx} = 4a \Rightarrow \dfrac{dy}{dx} = \dfrac{2a}{y}$

$\therefore$ Slope of the tangent to (i) at (x_1, y_1) is $\dfrac{2a}{y_1}$

Equation of the tangent to (i) at (x_1, y_1) is

$y - y_1 = \dfrac{2a}{y_1}(x - x_1)$

$\Rightarrow \quad yy_1 - y_1^2 = 2ax - 2ax_1$

$\Rightarrow \quad yy_1 = 2ax + (y_1^2 - 2ax_1)$ ...(ii)

Now (x_1, y_1) lies on (i),

$\therefore \quad y_1^2 = 4ax_1 \Rightarrow y_1^2 - 2ax_1 = 2ax_1$ ...(iii)

From (ii) and (iii), we get

$yy_1 = 2ax + 2ax_1 = 2a(x + x_1)$

This is the equation of the tangent to (i) at (x_1, y_1).

44. The given curve is

$x^2 + y^2 - 2x - 3 = 0.$ ...(i)

Differentiating with respect to x, we get

$2x + 2y\dfrac{dy}{dx} - 2 = 0 \Rightarrow \dfrac{dy}{dx} = \dfrac{1-x}{y}$

Since the tangent is parallel to x-axis,

$\therefore \quad \dfrac{dy}{dx} = 0 \Rightarrow \dfrac{1-x}{y} = 0 \Rightarrow x = 1$

www.examblues.in

Putting the value of $x = 1$ in (i), we get
$(1)^2 + y^2 - 2(1) - 3 = 0 \Rightarrow y^2 = 4$
$\Rightarrow \quad y = \pm 2$
$\therefore$ The points are (1, 2) and (1, –2).

45. Here, $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ ...(i)
When $x = 1$, $y = 1 - 6 + 13 - 10 + 5 = 3$.
We want to find tangent to (i) at $P(1, 3)$.
Differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx} = 4x^3 - 18x^2 + 26x - 10$$

$$\therefore \quad \left(\frac{dy}{dx}\right)_P = 4 - 18 + 26 - 10 = 2$$

Hence the equation of the tangent to (i) at $P(1, 3)$ is
$y - 3 = 2(x - 1) \Rightarrow y = 2x + 1$

46. The given curve is
$4x^2 + 9y^2 = 36$ ...(i)

$$\Rightarrow \quad 4 \cdot 2x + 9 \cdot 2y \frac{dy}{dx} = 0 \Rightarrow \quad \frac{dy}{dx} = -\frac{4x}{9y}$$

Let $P \equiv (3\cos\theta, 2\sin\theta)$

$$\therefore \quad \left[\frac{dy}{dx}\right]_P = -\frac{4}{9} \cdot \frac{3\cos\theta}{2\sin\theta} = -\frac{2\cos\theta}{3\sin\theta}$$

$\therefore$ Equation of the tangent to (i) at P is

$$y - 2\sin\theta = -\frac{2\cos\theta}{3\sin\theta}(x - 3\cos\theta)$$

$\Rightarrow \quad 3y \sin\theta - 6\sin^2\theta = -2x\cos\theta + 6\cos^2\theta$
$\Rightarrow \quad 2x\cos\theta + 3y\sin\theta = 6$

47. The given curve is
$y = (x^2 - 1)(x - 2)$ ...(i)
It meet x – axis, where $y = 0$
$\therefore \quad (x^2 - 1)(x - 2) = 0 \Rightarrow x = 1, -1, 2$
$\therefore$ We want to find tangents at (1, 0), (–1, 0) and (2, 0)
Now from equation (i), we get
$y = x^3 - 2x^2 - x + 2$

$$\Rightarrow \quad \frac{dy}{dx} = 3x^2 - 4x - 1$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{x=1} = -2; \left(\frac{dy}{dx}\right)_{x=-1} = 6; \left(\frac{dy}{dx}\right)_{x=2} = 3$$

Now, tangent at $(1, 0) : y - 0 = -2(x - 1) \Rightarrow 2x + y = 2$
tangent at $(-1, 0) : y - 0 = 6(x + 1) \Rightarrow y = 6x + 6$
tangent at $(2, 0) : y - 0 = 3(x - 2) \Rightarrow y = 3x - 6$

48. The given curve is $x = \sin 3t$; $y = \cos 2t$

$$\Rightarrow \quad \frac{dx}{dt} = 3\cos 3t; \frac{dy}{dt} = -2\sin 2t$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{2\sin 2t}{3\cos 3t}$$

At $t = \frac{\pi}{4}$, $x = \sin\frac{3\pi}{4} = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$$y = \cos 2t = \cos\frac{\pi}{2} = 0$$

and $$\frac{dy}{dx} = -\frac{2\sin\frac{\pi}{2}}{3\cos\frac{3\pi}{4}} = -\frac{2 \cdot 1}{3} = \frac{2\sqrt{2}}{3}$$

$\therefore$ Equation of the tangent to the given curve at $t = \frac{\pi}{4}$ is $y - 0 = \frac{2\sqrt{2}}{3}\left(x - \frac{1}{\sqrt{2}}\right) \Rightarrow 3y = 2\sqrt{2}x - 2$

49. *Refer to answer 39.*
The equations of normal is $y - \frac{8}{3} = -1(x - 4)$

$$\Rightarrow \quad y + x = 4 + \frac{8}{3} \Rightarrow \quad y + x = \frac{20}{3}$$

And $y + \frac{8}{3} = 1(x - 4)$

$$\Rightarrow \quad y - x = -4 - \frac{8}{3} \Rightarrow \quad y - x = \frac{-20}{3}$$

50. Let the point of contact of one of the tangents be (x_1, y_1).
Then (x_1, y_1) lies on $y = \cos(x + y)$
$\therefore \quad y_1 = \cos(x_1 + y_1)$ ...(i)
Since the tangents are parallel to the line $x + 2y = 0$. Therefore,
Slope of tangent at (x_1, y_1) = slope of line $x + 2y = 0$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -\frac{1}{2}$$

Since, the equation of curve is $y = \cos(x + y)$
Differentiating with respect to x, we get

$$\frac{dy}{dx} = -\sin(x + y)\left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -\sin(x_1 + y_1) \left\{1 + \left(\frac{dy}{dx}\right)_{(x_1, y_1)}\right\}$$

$$\Rightarrow \quad -\frac{1}{2} = -\sin(x_1 + y_1)\left(1 - \frac{1}{2}\right)$$

$\Rightarrow \quad \sin(x_1 + y_1) = 1$ ...(ii)
Squaring (i) and (ii), then adding, we get
$\cos^2(x_1 + y_1) + \sin^2(x_1 + y_1) = y_1^2 + 1$
$\Rightarrow \quad y_1^2 + 1 = 1 \Rightarrow y_1 = 0$

www.exambites.in

Put $y_1 = 0$ in (i) and (ii),
$\cos x_1 = 0$ and $\sin x_1 = 1$

$\Rightarrow \quad x_1 = \frac{\pi}{2}, -\frac{3\pi}{2}$

Hence, the points of contact are

$$\left(\frac{\pi}{2}, 0\right) \text{and} \left(-\frac{3\pi}{2}, 0\right)$$

The slope of the tangent is $-\frac{1}{2}$

Therefore, equation of tangents at

$$\left(\frac{\pi}{2}, 0\right) \text{and} \left(-\frac{3\pi}{2}, 0\right) \text{ are}$$

$$y - 0 = -\frac{1}{2}\left(x - \frac{\pi}{2}\right) \text{and } y - 0 = -\frac{1}{2}\left(x + \frac{3\pi}{2}\right)$$

or $\quad 2x + 4y - \pi = 0$ and $2x + 4y + 3\pi = 0$

51. The given curves are

$x^2 = 9p(9 - y)$ ... (i)

$x^2 = p(y + 1)$ ... (ii)

Clearly $p \neq 0$ (as $p = 0 \Rightarrow$ both curves become same)

Solving (i) and (ii), we get

$9p(9 - y) = p(y + 1)$

$\Rightarrow \quad 9(9 - y) = y + 1 \qquad (\because p \neq 0)$

$\Rightarrow \quad y = 8$ and then $x^2 = 9p$ ...(iii)

Diff. (i) w.r.t. y, $2x\frac{dx}{dy} = 9p(-1)$

$$\Rightarrow \quad m_1 = \frac{dy}{dx} = -\frac{2x}{9p}$$

Differentiating (ii) w.r.t. y, $2x \cdot \frac{dx}{dy} = p \Rightarrow \quad m_2 = \frac{2x}{p}$

For curves (i) and (ii) to cut each other at right angles,

$m_1 \cdot m_2 = -1$

$$\Rightarrow \quad -\frac{2x}{9p} \cdot \frac{2x}{p} = -1$$

$\Rightarrow \quad 9p^2 = 4x^2 = 4 \cdot 9p$ [Using (iii)]

$\Rightarrow \quad p(p - 4) = 0 \Rightarrow p = 4$ as $p \neq 0$

52. The given curve is $y = x^2 - 2x + 7$ ...(i)

$$\Rightarrow \quad \frac{dy}{dx} = 2x - 2$$

= Slope of the tangent to (i) at (x, y) ...(ii)

(i) The tangent is parallel to the line $2x - y + 9 = 0$

Its slope = 2

$\therefore$ From (ii), $2x - 2 = 2 \Rightarrow x = 2$

From (i), $y = 2^2 - 2 \times 2 + 7 = 7$

$\therefore$ Equation of the tangent to (i) at (2, 7) whose slope = 2, is

$y - 7 = 2(x - 2) \Rightarrow 2x - y + 3 = 0$

(ii) The tangent is perpendicular to the line $5y - 15x = 13$,

its slope $= \frac{15}{5} = 3$

$\therefore$ From (ii), $(2x - 2) \times 3 = -1$

$$\Rightarrow \quad 2x - 2 = -\frac{1}{3} \Rightarrow x = \frac{5}{6}$$

From (i), $y = \left(\frac{5}{6}\right)^2 - 2 \times \frac{5}{6} + 7 = \frac{217}{36}$

$\therefore$ Equation of the tangent to (i) at $\left(\frac{5}{6}, \frac{217}{36}\right)$ whose slope $= -\frac{1}{3}$, is

$$y - \frac{217}{36} = -\frac{1}{3}\left(x - \frac{5}{6}\right)$$

$\Rightarrow \quad 36y - 217 = -12x + 10 \Rightarrow 12x + 36y = 227$

53. We have, $x^2 = 4y$

Differentiating w.r.t. x, we get

$$2x = 4\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2}$$

Let $P(x_1, y_1)$ be the point on the given curve.

$$\left[\frac{dy}{dx}\right]_P = \frac{x_1}{2}$$

Equation of normal at $P(x_1, y_1)$

$$(y - y_1) = -\frac{2}{x_1}(x - x_1) \qquad \text{... (i)}$$

Equation (i) passes through the point (1, 2)

$$(2 - y_1) = -\frac{2}{x_1}(1 - x_1) \Rightarrow 2 - y_1 = \frac{-2}{x_1} + 2$$

$\Rightarrow \quad x_1 y_1 = 2$ ...(ii)

Also $P(x_1, y_1)$ lies on $x^2 = 4y$

$\therefore \quad x_1^2 = 4y_1$

$$\Rightarrow \quad x_1^2 = \frac{4 \times 2}{x_1} \qquad \text{(Using (ii))}$$

$\Rightarrow \quad x_1^3 = 8 \Rightarrow x_1 = 2,$

From (ii), $2 \times y_1 = 2 \Rightarrow y_1 = 1$

Now, putting the value of x_1 and y_1 in equation (i), we get

$$(y - 1) = -\frac{2}{2}(x - 2)$$

$\Rightarrow \quad y - 1 = -x + 2 \Rightarrow x + y = 3$

Equation of tangent

$$(y - y_1) = \left[\frac{dy}{dx}\right]_P (x - x_1)$$

www.exambrains.in

$\Rightarrow \quad y-1=\frac{2}{2}(x-2) \Rightarrow x-y=1$

54. We have, $3x^2 - y^2 = 8$ ...(i)

Differentiating w.r.t. x, we get

$$6x-2y\frac{dy}{dx}=0 \Rightarrow \frac{dy}{dx}=\frac{3x}{y}.$$

Let $P(x_1, y_1)$ be a point of the curve

$\therefore \quad 3x_1^2 - y_1^2 = 8$ ...(ii)

and $\left(\frac{dy}{dx}\right)_P = \frac{3x_1}{y_1}$

$\therefore$ Equation of tangent to (i) at (x_1, y_1) is

$$y-y_1=\left[\frac{dy}{dx}\right]_P (x-x_1)$$

$$\Rightarrow \quad y-y_1=\frac{3x_1}{y_1}(x-x_1)$$

The tangent is passes through $\left(\frac{4}{3},0\right)$

$$\therefore \quad 0-y_1=\frac{3x_1}{y_1}\left(\frac{4}{3}-x_1\right)$$

$\Rightarrow \quad -y_1^2 = 4x_1 - 3x_1^2$

$\Rightarrow \quad 4x_1 = 3x_1^2 - y_1^2 = 8$ [Using (ii)]

$\Rightarrow \quad x_1 = 2$

Again from (ii), $3x_1^2 - y_1^2 = 8$

$\Rightarrow \quad y_1 = \pm 2$

$\therefore$ The two points on (i), at which tangents pass through $\left(\frac{4}{3},0\right)$ are (2, 2) and (2, –2).

$\therefore$ Equations of tangents are

$$y-2=\frac{3\times 2}{2}(x-2) \Rightarrow y=3x-4$$

and $y+2=\frac{3\times 2}{-2}(x-2) \Rightarrow y=-3x+4$

55. The given curve is

$y = 4x^3 - 2x^5$ ...(i)

$$\Rightarrow \quad \frac{dy}{dx}=12x^2-10x^4$$

Let $P(x_1, y_1)$ be a point on the curve

$\therefore \quad y_1 = 4x_1^3 - 2x_1^5$ ...(ii)

and $\left[\frac{dy}{dx}\right]_P = 12x_1^2 - 10x_1^4$

Now tangent to (i) at P is

$y - y_1 = (12x_1^2 - 10x_1^4).(x - x_1)$

This will pass through (0, 0) when

$0 - y_1 = (12x_1^2 - 10x_1^4)(0 - x_1)$

$\Rightarrow \quad y_1 = 12x_1^3 - 10x_1^5$

$\Rightarrow \quad 4x_1^3 - 2x_1^5 = 12x_1^3 - 10x_1^5$ [Using (ii)]

$\Rightarrow \quad 8x_1^3 - 8x_1^5 = 0 \Rightarrow x_1^3(1 - x_1^2) = 0$

$\Rightarrow \quad x_1 = 0, 1, -1.$

When, $x_1 = 0$, then from (i), $y_1 = 0$

When, $x_1 = 1$, then from (i), $y_1 = 2$

When, $x_1 = -1$, then from (i), $y_1 = -2$

$\therefore$ Tangents to given curve will pass through the origin at the point (0, 0), (1, 2) and (– 1, –2).

56. *Refer to answer 34.*

57. The given curves are

$y^2 = x$ ... (i)

$xy = k$ ...(ii)

Differentiating (i) w.r.t. x, we get

$$2y\frac{dy}{dx}=1 \Rightarrow \frac{dy}{dx}=\frac{1}{2y}=m_1 \text{ (say)}$$

Differentiating (ii) w.r.t. x, we get

$$x\cdot\frac{dy}{dx}+1\cdot y=0 \Rightarrow \frac{dy}{dx}=-\frac{y}{x}=m_2 \text{ (say)}$$

Now (i) and (ii) will intersect orthogonally if

$m_1 \cdot m_2 = -1$

$$\Rightarrow \quad \frac{1}{2y}\cdot\left(-\frac{y}{x}\right)=-1 \;\Rightarrow\; 2x=1 \Rightarrow x=\frac{1}{2}$$

From (ii), $y=\frac{k}{x}=2k \qquad \left(\because x=\frac{1}{2}\right)$

Substituting the values of x and y in (i), we get

$$(2k)^2=\frac{1}{2} \Rightarrow 8k^2=1$$

which is the required condition for the orthogonality of the curves (i) and (ii).

58. Here, $x = a \cos t + at \sin t$

$y = a \sin t - at \cos t$

Now, $\frac{dx}{dt}=-a\sin t+a.1\sin t+at\cos t=at\cos t$

and $\frac{dy}{dt}=a\cos t-a[1.\cos t-t\sin t]=at\sin t$

$\therefore$ Slope of the tangent $=\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\tan t$

$\Rightarrow$ Slope of the normal $=-\frac{1}{\tan t}=-\frac{\cos t}{\sin t}$

$\therefore$ Equation of the normal to the given curve

$y - (a \sin t - at \cos t)$

www.exambites.in

$= -\frac{\cos t}{\sin t}(x - a\cos t - at\sin t)$

$\Rightarrow$ $y \sin t - a \sin^2 t + at \sin t \cos t$
$= -x \cos t + a \cos^2 t + at \sin t \cos t$

$\Rightarrow$ $x \cos t + y \sin t - a = 0$

$\therefore$ Perpendicular distance from (0, 0) on it
$= \frac{|0 \cdot \cos t + 0 \cdot \sin t - a|}{\sqrt{\cos^2 t + \sin^2 t}} = a$

59. The given parabola is
$y^2 = 4ax$ ...(i)
and the point is $P(at^2, 2at)$.
Differentiating (i) w.r.t. x, we get

$2y\frac{dy}{dx} = 4a \Rightarrow \frac{dy}{dx} = \frac{2a}{y}$ ... (ii)

$\therefore$ Slope of the tangent at $P = \frac{2a}{2at} = \frac{1}{t}$

Equation of the tangent to (i) at P is

$y - 2at = \frac{1}{t}(x - at^2)$

$\Rightarrow$ $ty - 2at^2 = x - at^2 \Rightarrow ty = x + at^2$

From (ii), $-\frac{dx}{dy} = -\frac{y}{2a}$

$\therefore$ Slope of the normal at $P = -\frac{2at}{2a} = -t$

$\therefore$ Equation of the normal to (i) at P is
$y - 2at = -t(x - at^2) \Rightarrow y + tx - 2at - at^3 = 0$

60. The given curve is
$y = x^3 + 2x + 6$ ... (i)

$\Rightarrow \frac{dy}{dx} = 3x^2 + 2$

$\therefore$ Slope of the normal $= -\frac{dx}{dy} = \frac{-1}{3x^2 + 2}$

The normal to (i) is parallel to the line $x + 14y + 4 = 0$
which have slope $= -\frac{1}{14}$

$\therefore$ $\frac{-1}{3x^2 + 2} = -\frac{1}{14} \Rightarrow 3x^2 + 2 = 14$

$\Rightarrow$ $3x^2 = 14 - 2 = 12 \Rightarrow x^2 = 4 \Rightarrow x \pm 2$

From (i),
$x = 2 \Rightarrow y = 2^3 + 2 \cdot 2 + 6 = 18$
$x = -2 \Rightarrow y = (-2)^3 + 2 \cdot (-2) + 6 = -6$
Equation of the normal at (2, 18) is

$y - 18 = -\frac{1}{14}(x - 2) \Rightarrow x + 14y = 254$

Equation of the normal at (–2, –6) is

$y + 6 = -\frac{1}{14}(x + 2) \Rightarrow x + 14y + 86 = 0$

61. Given, $f(x) = x^4 - 10 \Rightarrow f'(x) = 4x^3$
$\because$ $f(x + \Delta x) = f(x) + \Delta x f'(x)$, where $\Delta x \to 0$
Replace $x = 2$ and $\Delta x = 0.1$ in above relation
$\therefore$ $f(2 + 0.1) = f(2) + (0.1)f'(2)$
$\Rightarrow$ $f(2.1) = (2^4 - 10) + (0.1)(4 \times 2^3)$
$= (16 - 10) + (0.1) \times 32 = 6 + 3.2 = 9.2$

62. Given $f(x) = 3x^2 + 5x + 3 \Rightarrow f'(x) = 6x + 5$
$\because$ $f(x + \Delta x) \approx f(x) + \Delta x f'(x)$, where $\Delta x \to 0$
Replace $x = 3$ and $\Delta x = 0.02$ in above relation
$\therefore$ $f(3 + 0.2) \approx f(3) + (0.02)f'(3)$
$\Rightarrow$ $f(3.02) \approx [3(3)^2 + 5(3) + 3] + (0.02)[6(3) + 5]$
$\Rightarrow$ $f(3.02) \approx 45.46$

63. Let $y = f(x) = x^{3/2} \Rightarrow f'(x) = \frac{3}{2}x^{1/2}$
Let $x = 4$ and $\Delta x = 3.968 - 4 = -0.032$
$f(x + \Delta x) \approx f(x) + \Delta x f'(x)$

$f(3.968) = (4)^{3/2} - (0.032) \times \frac{3}{2}(4)^{1/2} = 8 - 0.096$
$= 7.904$

64. Let $f(x) = \sqrt{x}$ so $f'(x) = \frac{1}{2\sqrt{x}}$

Hence, $f(x + \Delta x) \approx f(x) + \Delta x f'(x)$

$\Rightarrow$ $\sqrt{x + \Delta x} \approx \sqrt{x} + \frac{\Delta x}{2\sqrt{x}}$

Now, taking $x = 49$ and $\Delta x = 0.5$, we get

$\sqrt{49.5} \approx 7 + \frac{0.5}{2 \times 7} = 7 + 0.0357 = 7.0357$

65. Let r be the radius of sphere and Δr be the error in measuring the radius. Then, $r = 9$ cm, $\Delta r = 0.03$ cm
Now surface area S of the sphere is $S = 4\pi r^2$

$\Rightarrow$ $\frac{dS}{dr} = 8\pi r$

$\therefore$ $\Delta S = \frac{dS}{dr}\Delta r = 8\pi r \Delta r = 8\pi \times 9 \times 0.03 = 2.16\,\pi$ cm^2

This is the approximate error in calculating surface area.

66. **(a) :** Given, $y = -x^3 + 3x^2 + 12x - 5$

$\Rightarrow$ $\frac{dy}{dx} = -3x^2 + 6x + 12$
$= -3[x^2 - 2x - 4] = -3[(x - 1)^2 - 5]$
$= 15 - 3(x - 1)^2$

So, maximum value of slope of the curve is 15.

www.examfree.in

67. Here, $f(x) = 2\sin x$

$\Rightarrow \quad f'(x) = 2\cos x$

Putting $f'(x) = 0 \Rightarrow 2\cos x = 0 \Rightarrow \cos x = 0$

$\Rightarrow \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$

$\therefore \quad \frac{\pi}{2}, \frac{3\pi}{2}$ are the critical points

At $x = \frac{\pi}{2}$, $f(x) = 2 \times 1 = 2$

At $x = \frac{3\pi}{2}$, $f(x) = 2\sin\left(\pi + \frac{\pi}{2}\right) = -2\sin\frac{\pi}{2} = -2$

Hence, absolute minimum value of $f(x)$ is –2.

68. As we know that, A.M. ≥ G.M.

$\therefore \quad \frac{1}{2}\left(ax + \frac{b}{x}\right) \geq \sqrt{ax.\frac{b}{x}} \Rightarrow \left(ax + \frac{b}{x}\right) \geq 2\sqrt{ab}$

So, least value of $f(x)$ is $2\sqrt{ab}$.

69. Let x be the side of square base and y be the height of the open tank.

$\therefore \quad l = x$, $b = x$ and $h = y$

where l, b and h be the length, breadth and height of tank respectively.

Volume of tank $V = x^2y \Rightarrow y = \frac{V}{x^2}$

The cost of the material will be least if the total surface area is least.

Total surface area of tank $(S) = x^2 + 4xy$

$$\Rightarrow \quad S = x^2 + 4x\frac{V}{x^2} \qquad \left(\because \quad y = \frac{V}{x^2}\right)$$

$$\Rightarrow \quad S = x^2 + \frac{4V}{x} \Rightarrow \frac{dS}{dx} = 2x - \frac{4V}{x^2}$$

For maxima or minima, $\frac{dS}{dx} = 0$

$$\Rightarrow \quad 2x - \frac{4V}{x^2} = 0 \Rightarrow x^3 = 2V \Rightarrow x = 2y$$

Also, $\frac{d^2S}{dx^2} = 2 + \frac{8V}{x^3} > 0 \qquad (\because V = x^2y)$

$\therefore \quad$ Cost of material is least, when $y = \frac{x}{2}$

i.e., the depth of the tank is half of its width.

As the cost is borne by nearby settled lower income families it shows that they are spending money on social welfare so that no body will face the water problem in future. It shows social responsibility.

70. We have, $\tan\theta = \frac{r}{h} = \frac{R}{h-H}$

$$\Rightarrow \quad R = r\left(1 - \frac{H}{h}\right)$$

Now, volume of cylinder $(V) = \pi R^2 H$

$$= \pi r^2\left(1 - \frac{H}{h}\right)^2 H$$

For maximum volume, $\frac{dV}{dH} = 0$

$$\Rightarrow \quad \pi r^2\left[H \times 2\left(1 - \frac{H}{h}\right) \times -\frac{1}{h} + \left(1 - \frac{H}{h}\right)^2\right] = 0$$

$$\Rightarrow \quad \pi r^2\left[-\frac{4H}{h} + \frac{3H^2}{h^2} + 1\right] = 0$$

$$\Rightarrow \quad \pi r^2\left(\frac{H}{h} - 1\right)\left(\frac{3H}{h} - 1\right) = 0$$

$$\Rightarrow \quad H = \frac{h}{3} \qquad [\because \ H \neq h]$$

So, height of cylinder = $\frac{1}{3}$ of height of cone

Also, maximum volume of cylinder

$$= \pi r^2\left(1 - \frac{1}{3}\right)^2 \times \frac{h}{3} = \left(\frac{1}{3}\pi r^2 h\right) \times \frac{4}{9}$$

$= \frac{4}{9}$ of volume of cone.

71. Let $u = ax + by$, where $xy = c^2$

$$\Rightarrow \quad u = ax + b\left(\frac{c^2}{x}\right) \qquad \text{... (i)}$$

Differentiating w.r.t. x, we get

$$\frac{du}{dx} = a - \frac{bc^2}{x^2} \quad \text{and} \quad \frac{d^2u}{dx^2} = \frac{2bc^2}{x^3}$$

For critical points, $\frac{du}{dx} = 0$

$$\Rightarrow \quad \frac{ax^2 - bc^2}{x^2} = 0 \Rightarrow x^2 = \frac{bc^2}{a}$$

$$\therefore \quad x = \pm\sqrt{\frac{b}{a}}c$$

At $x = \sqrt{\frac{b}{a}}c$, $\frac{d^2u}{dx^2} = 2bc^2\left(\sqrt{\frac{a}{b}} \times \frac{1}{c}\right)^3$

$$= \frac{2bc^2}{c^3}\left(\frac{a\sqrt{a}}{b\sqrt{b}}\right) = 2\frac{a}{c}\sqrt{\frac{a}{b}} > 0$$

$\Rightarrow \quad u$ is minimum at $x = c\sqrt{\frac{b}{a}}$

www.examrace.in

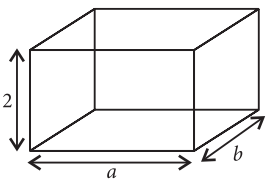
At $x=-\sqrt{\frac{b}{a}}c, \frac{d^2u}{dx^2}=-2bc^2\left(\sqrt{\frac{a}{b}}\times\frac{1}{c}\right)<0$

$\Rightarrow$ u is maximum at $x=-\sqrt{\frac{b}{a}}c$

The minimum value of u at $x=\sqrt{\frac{b}{a}}c$ is

$$u=a\left(\sqrt{\frac{b}{a}}c\right)+bc^2\left(\sqrt{\frac{a}{b}}\times\frac{1}{c}\right)$$
$$=c\sqrt{ab}+\sqrt{ba}c=2c\sqrt{ab}$$

72. Let a m and b m be the length and breadth of rectangular tank respectively.



$\therefore$ Volume of tank = $2ab = 8$ [Given]

$\Rightarrow$ $ab=4 \Rightarrow b=\frac{4}{a}$ ...(i)

If C is the total cost in rupees, then

$C = 70(ab) + 45(2a + 2b) \times 2$

$\Rightarrow$ $C = 70ab + 90(2a + 2b)$

$\Rightarrow C=70(a)\left(\frac{4}{a}\right)+180\left(a+\frac{4}{a}\right)$ [Using (i)]

$\Rightarrow C=280+180a+\frac{720}{a}$ ...(ii)

Differentiating (ii) w.r.t. 'a', we get

$$\frac{dC}{da}=180-\frac{720}{a^2} \text{ and } \frac{d^2C}{da^2}=\frac{720\times2}{a^3}$$

For maximum or minimum cost,

$\frac{dC}{da}=0 \Rightarrow 180-\frac{720}{a^2}=0 \Rightarrow a^2=4 \Rightarrow a=2$

For $a=2, \frac{d^2C}{da^2}>0 \Rightarrow C$ is least

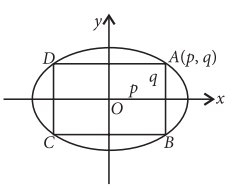
Using (i), $a=2$ and $b=2$

Hence,cost of least expensive tank is

$C = 280 + 360 + 360 = ₹\ 1000$

73. Let $ABCD$ be a rectangle inscribed in the ellipse,

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



Let $AB = 2q$, $DA = 2p$.

Then coordinates of A are (p, q).

As A lies on the ellipse so

$$\frac{p^2}{a^2}+\frac{q^2}{b^2}=1\Rightarrow q^2=b^2\left(1-\frac{p^2}{a^2}\right)$$

Now area A of the rectangle $ABCD = 2p \cdot 2q = 4pq$

$\Rightarrow$ $A^2 = 16p^2q^2$

$$=16p^2\cdot b^2\left(1-\frac{p^2}{a^2}\right)=16b^2\left(p^2-\frac{p^4}{a^2}\right)$$

$\therefore$ $\frac{dA^2}{dp}=16b^2\left(2p-\frac{4p^3}{a^2}\right)$ and

$$\frac{d^2A^2}{dp^2}=16b^2\left(2-\frac{12p^2}{a^2}\right)$$

For A to be max. or min. so is A^2,

$\frac{dA^2}{dp}=0\Rightarrow 2p-\frac{4p^3}{a^2}=0 \Rightarrow p^2=\frac{a^2}{2}$ $(\because p\neq 0)$

For this value of p,

$\frac{d^2A^2}{dp^2}=16b^2(2-12\times\frac{1}{2})=16b^2(-4)<0$

Hence A^2 is max. $\Rightarrow$ A is max.

$\Rightarrow$ The area of the greatest rectangle inscribed in the ellipse

$=4pq=4\cdot\sqrt{\frac{a^2}{2}}\cdot\sqrt{b^2\left(1-\frac{1}{2}\right)}=2ab$ sq.units

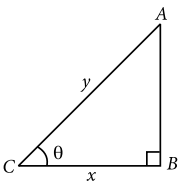
74. Let ABC be a right angled triangle with $BC = x$, $AC = y$ such that $x + y = k$, where k is any constant.

Let θ be the angle between the base and the hypotenuse.

Let P be the area of the triangle.

$P=\frac{1}{2}\times BC\times AB=\frac{1}{2}\times x\sqrt{y^2-x^2}$

$\Rightarrow P^2=\frac{x^2}{4}(y^2-x^2)$



$\Rightarrow P^2=\frac{x^2}{4}[(k-x)^2-x^2]$

$\Rightarrow P^2=\frac{k^2x^2-2kx^3}{4}$

Let $Q = P^2$ *i.e.* $Q=\frac{k^2x^2-2kx^3}{4}$

$\therefore$ P is maximum when Q is maximum.

www.examcbres.in

Differentiating Q w.r.t. x, we get

$$\frac{dQ}{dx} = \frac{2k^2x - 6kx^2}{4} \quad ...(i)$$

For maximum or minimum area,

$$\frac{dQ}{dx} = 0 \Rightarrow k^2x - 3kx^2 = 0 \Rightarrow x = \frac{k}{3}$$

Differentiating (i) w.r.t. x, we get

$$\frac{d^2Q}{dx^2} = \frac{2k^2 - 12kx}{4}$$

$$\therefore \quad \left[\frac{d^2Q}{dx^2}\right]_{x=\frac{k}{3}} = \frac{-k^2}{2} < 0$$

Thus, Q is maximum when $x = \frac{k}{3}$

$\Rightarrow$ P is maximum at $x = \frac{k}{3}$

Now, $x = \frac{k}{3} \Rightarrow y = k - \frac{k}{3} = \frac{2k}{3}$ $[\because \ x + y = k]$

$$\therefore \quad \cos\theta = \frac{x}{y} = \frac{k/3}{2k/3} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

So, the area of ΔABC is maximum when angle between the hypotenuse and base is $\frac{\pi}{3}$.

75. Let V and S be the volume and the surface area of a closed cuboid of length = x units, breadth = x units and height = y units respectively.

Then, $V = x^2y$ ...(i)

and $S = 2(x^2 + xy + xy) = 2x^2 + 4xy$ ...(ii)

$$\Rightarrow \quad S = 2x^2 + 4x\left(\frac{V}{x^2}\right) \quad \text{[From (i)]}$$

$$\Rightarrow \quad S = 2x^2 + \frac{4V}{x} \quad \Rightarrow \quad \frac{dS}{dx} = 4x - \frac{4V}{x^2} \quad ...(iii)$$

For maximum or minimum of S, $\frac{dS}{dx} = 0$

$$\therefore \quad \frac{dS}{dx} = 0 \quad \Rightarrow \quad 4x - \frac{4V}{x^2} = 0 \Rightarrow V = x^3$$

$\Rightarrow$ $x^2y = x^3$ $[\because V = x^2y]$

$\Rightarrow$ $x = y$

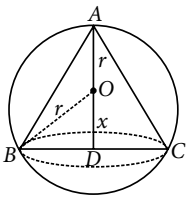
Differentiating (iii) with respect to x, we get

$$\frac{d^2S}{dx^2} = 4 + \frac{8V}{x^3} = 4 + \frac{8x^2y}{x^3} = 4 + \frac{8y}{x}$$

$$\Rightarrow \quad \left(\frac{d^2S}{dx^2}\right)_{y=x} = 12 > 0.$$

Thus, S is minimum when $x = y$.

76. Let ABC be a cone of maximum volume inscribed in the sphere.



Let $OD = x$

$$\therefore \quad BD = \sqrt{r^2 - x^2}$$

and $AD = AO + OD$

$= r + x$ = altitude of cone.

Let V be the volume of cone.

$$V = \frac{1}{3}\pi(BD)^2(AD) = \frac{1}{3}\pi(r^2 - x^2)(r + x)$$

$$\Rightarrow \quad \frac{dV}{dx} = \frac{1}{3}\pi\left[(r^2 - x^2) + (r + x)(-2x)\right]$$

$$= \frac{\pi}{3}\left[r^2 - 3x^2 - 2rx\right]$$

and $\frac{d^2V}{dx^2} = \frac{\pi}{3}\left[-6x - 2r\right]$

For maximum or minimum value $\frac{dV}{dx} = 0$

$\Rightarrow$ $r^2 - 3x^2 - 2rx = 0 \Rightarrow r^2 - 3rx + rx - 3x^2 = 0$

$\Rightarrow (r - 3x)(r + x) = 0$

$\Rightarrow$ $r = 3x$ $[\because \ r + x \neq 0]$

$\Rightarrow$ $x = \frac{r}{3}$

Also, $\left(\frac{d^2V}{dx^2}\right)_{x=\frac{r}{3}} = \frac{\pi}{3}\left[-6\left(\frac{r}{3}\right) - 2r\right]$

$$= \frac{\pi}{3}[-2r - 2r] = \frac{-4}{3}r\pi < 0$$

$\Rightarrow$ V is maximum when $x = \frac{r}{3}$

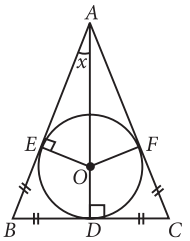
and altitude of cone = $AD = r + x = r + \frac{r}{3} = \frac{4r}{3}$

Also, maximum volume of cone when $x = \frac{r}{3}$

$$= \frac{1}{3}\pi\left(r^2 - \frac{r^2}{9}\right)\left(r + \frac{r}{3}\right) = \frac{\pi}{3}\left(\frac{8}{9}r^2\right)\left(\frac{4}{3}r\right)$$

$$= \frac{8}{27}\left(\frac{4}{3}\pi r^3\right) = \frac{8}{27} \text{ (Volume of sphere)}$$

77. Let ΔABC be the given triangle and AD is the altitude of the isosceles triangle ABC. Since, 'r' be the radius of the inscribed circle.



So, $OD = OE = OF = r$, where O is the centre of the inscribed circle.

www.examrites.in

AB and AC are the equal sides.

$BD = DC$ ...(i)

$BD = BE$ and $CD = CF$ ...(ii)

From (i) and (ii), $BD = BE = DC = CF$ ...(iii)

Similarly, $AE = AF$ ...(iv)

Perimeter of the triangle $ABC = AB + BC + AC$

$= AE + BE + BD + DC + CF + AF$

$= 2AE + 4BD$ (Using (iii) and (iv))

In right triangle OEA, $AE = \frac{OE}{\tan x} = \frac{r}{\tan x}$ and $AO = \frac{r}{\sin x}$

In right triangle ABD, $BD = AD \tan x$

$$= (AO + OD)\tan x = \left(\frac{r}{\sin x} + r\right)\tan x$$

Let P be the perimeter of a triangle ABC.

So perimeter, $(P) = 2AE + 4BD$

$$= \frac{2r}{\tan x} + 4\left(\frac{r}{\sin x} + r\right)\tan x$$

$\Rightarrow \quad P(x) = r(2\cot x + 4\sec x + 4\tan x)$

For maximum or minimum perimeter, $\frac{dP(x)}{dx} = 0$

$$\Rightarrow \quad \frac{dP(x)}{dx} = r(-2\operatorname{cosec}^2 x + 4\sec x\tan x + 4\sec^2 x) = 0$$

$$\Rightarrow \quad r\left(-\frac{2}{\sin^2 x} + \frac{4\sin x}{\cos^2 x} + \frac{4}{\cos^2 x}\right) = 0$$

$$\Rightarrow \quad r\left(\frac{-2\cos^2 x + 4\sin^3 x + 4\sin^2 x}{\sin^2 x \cos^2 x}\right) = 0$$

$\Rightarrow \quad -2(1 - \sin^2 x) + 4\sin^3 x + 4\sin^2 x = 0$

$\Rightarrow \quad 2\sin^3 x + 3\sin^2 x - 1 = 0$

$\Rightarrow \quad (\sin x + 1)(2\sin^2 x + \sin x - 1) = 0$

$\sin x$ cannot be -1 because 'x' cannot be more than 90°.

So, $2\sin^2 x + \sin x - 1 = 0$

$\Rightarrow \quad (2\sin x - 1)(\sin x + 1) = 0$

Again $\sin x$ cannot be -1.

So $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2} \Rightarrow x = 30°$

$$\frac{d^2P(x)}{dx^2} = r[4\operatorname{cosec}^2 x\cot x + 4\sec x\tan^2 x + 4\sec^3 x + 8\sec^2 x\tan x]$$

$$\therefore \quad \left[\frac{d^2P}{dx^2}\right]_{x=30°} > 0$$

So it is a point of minima for $P(x)$

Hence, least perimeter $= [P(x)]_{x = 30°}$

$= r(2\cot 30° + 4\sec 30° + 4\tan 30°)$

$$= r\left(2\sqrt{3} + 4 \times \frac{2}{\sqrt{3}} + 4 \times \frac{1}{\sqrt{3}}\right) = r\left(\frac{18}{\sqrt{3}}\right) = 6\sqrt{3}r$$

78. Surface area of cuboid $= 2(lb + bh + hl)$

$$= 2\left(2x^2 + \frac{2x^2}{3} + \frac{x^2}{3}\right) = 6x^2$$

Let radius of the sphere be r

Surface area of sphere $= 4\pi r^2$

Therefore, $6x^2 + 4\pi r^2 = k$ (constant) ...(i)

Now, sum of volumes of cuboid and sphere is

$$V = \frac{2}{3}x^3 + \frac{4}{3}\pi r^3 \quad ...(ii)$$

Putting the value of r from (i) into (ii), we get

$$V = \frac{2}{3}x^3 + \frac{4}{3}\pi\left(\frac{k - 6x^2}{4\pi}\right)^{3/2} \quad ...(iii)$$

Differentiating (iii) w.r.t. 'x', we get

$$\frac{dV}{dx} = 2x^2 + \frac{4}{3}\pi\left(\frac{1}{4\pi}\right)^{3/2}\frac{3}{2}(k - 6x^2)^{1/2}(-12x) \quad ...(iv)$$

For minimum or maximum value, $\frac{dV}{dx} = 0$

$$\Rightarrow \quad \frac{dV}{dx} = 2x^2 + \frac{4}{3}\pi\left(\frac{1}{4\pi}\right)^{3/2} \cdot \frac{3}{2}(k - 6x^2)^{1/2}(-12x) = 0$$

$$\Rightarrow \quad 2x^2 = \left(\frac{1}{4\pi}\right)^{1/2}(k - 6x^2)^{1/2}(6x)$$

$$\Rightarrow \quad 2x^2 = \left(\frac{1}{4\pi}\right)^{1/2}(4\pi r^2)^{1/2}(6x) \quad \text{[From (i)]}$$

$\Rightarrow \quad x = 3r$

Differentiating (iv) w.r.t 'x', we get

$$\frac{d^2V}{dx^2} = 4x - \left(\frac{1}{4\pi}\right)^{\frac{1}{2}}\left[(6)[k - 6x^2]^{1/2} + (6x)\frac{(-12x)}{2(k - 6x^2)^{1/2}}\right]$$

Now, $\left[\frac{d^2V}{dx^2}\right]_{x=3r} = \frac{24\pi r^2 + 324r^2}{4\pi r} > 0$

Thus, V is minimum at $x = 3r$.

Further, minimum value of sum of their volume

$$= \frac{2}{3}x^3 + \frac{4}{3}\pi r^3 = \frac{2}{3}x^3 + \frac{4}{3}\pi\left(\frac{x}{3}\right)^3 \quad \left[\because r = \frac{x}{3}\right]$$

$$= \frac{2}{3}x^3 + \frac{4}{3}\pi\frac{x^3}{27} = \frac{2}{3}x^3\left(1 + \frac{2\pi}{27}\right) = \frac{2}{3}x^3\left(1 + \frac{44}{189}\right)$$

$$= \frac{2}{3}x^3 \cdot \frac{233}{189} = \frac{466}{567}x^3$$

www.exambites.in

79. We have, $f(x) = \sin x - \cos x$

$\Rightarrow \quad f'(x) = \cos x + \sin x$

For maxima or minima, $f'(x) = 0$

$\Rightarrow \quad \cos x + \sin x = 0 \Rightarrow \tan x = -1$

$\Rightarrow \quad x = \frac{3\pi}{4}, \frac{7\pi}{4}$

$f''(x) = -\sin x + \cos x$

At $x = \frac{3\pi}{4}$, $f''(x) = -\sin\frac{3\pi}{4} + \cos\frac{3\pi}{4}$

$= -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{-2}{\sqrt{2}} = -\sqrt{2}$

At $x = \frac{7\pi}{4}$, $f''(x) = -\sin\frac{7\pi}{4} + \cos\frac{7\pi}{4}$

$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

Since $(f''(x)) < 0$ when $x = \frac{3\pi}{4}$

$\therefore \quad f(x)$ has local maxima at $x = \frac{3\pi}{4}$

Since $(f''(x)) > 0$ when $x = \frac{7\pi}{4}$

$\therefore \quad f(x)$ has local minima at $x = \frac{7\pi}{4}$

$\therefore \quad$ Local maximum value at $x = \frac{3\pi}{4}$ is

$$f(x) = \sin\frac{3\pi}{4} - \cos\frac{3\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Local minimum value at $x = \frac{7\pi}{4}$ is

$$f(x) = \sin\frac{7\pi}{4} - \cos\frac{7\pi}{4} = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

$$= \frac{-2}{\sqrt{2}} = -\sqrt{2}$$

80. Let $P(h, k)$ be the coordinates of the point on given parabola.

$\therefore \quad k = h^2 + 7h + 2$ ...(i)

The distance S of P from the straight line $-3x + y + 3 = 0$ is

$$S = \left|\frac{-3h + k + 3}{\sqrt{10}}\right| = \left|\frac{-3h + h^2 + 7h + 2 + 3}{\sqrt{10}}\right| \quad \text{[From (i)]}$$

$$= \left|\frac{h^2 + 4h + 5}{\sqrt{10}}\right| \qquad \therefore \quad S = \frac{f(h)}{\sqrt{10}}$$

$\Rightarrow \quad S$ will be maximum or minimum according as $f(h)$ is maximum or minimum.

Since, $f(h) = h^2 + 4h + 5$

$f'(h) = 2h + 4$

For maxima or minima, $f'(h) = 0$

$\Rightarrow \quad 2h + 4 = 0 \Rightarrow h = -2$

Also, $f''(h) = 2 > 0$ when $h = -2$

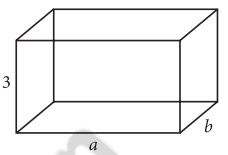
S is minimum at $h = -2$

Putting this value in (i), we get

$k = (-2)^2 + 7(-2) + 2 = 4 - 14 + 2 = -8$

$\therefore \quad$ The required coordinates are $(-2, -8)$

81. Let a m and b m be the sides of the base of the tank.



$\therefore \quad$ Volume of the tank $= a \cdot b \cdot 3 = 75$ m^3(given)

$\Rightarrow \quad ab = 25 \Rightarrow b = \frac{25}{a}$ ... (i)

If C is the total cost in rupees, then

$C = a \times b \times 100 + 2 \times 3 \times a \times 50 + 2 \times 3 \times b \times 50$

$= 100\,ab + 300(a + b) = 100 \times 25 + 300\left(a + \frac{25}{a}\right)$

$\Rightarrow \quad C = 2500 + 300\left(a + \frac{25}{a}\right)$

Differentiating w.r.t. a, we get

$$\frac{dC}{da} = 300\left(1 - \frac{25}{a^2}\right) \text{and}$$

$$\frac{d^2C}{da^2} = 300\left(0 + \frac{25 \times 2}{a^3}\right) = \frac{300 \times 50}{a^3}$$

For maximum or minimum cost,

$$\frac{dC}{da} = 0 \ \Rightarrow \ 1 - \frac{25}{a^2} = 0 \Rightarrow a = 5 \text{ m}$$

and from (i) $b = 5$ m

At $a = 5$; $\frac{d^2C}{da^2} > 0 \Rightarrow C$ is minimum.

Hence the least cost of the tank is

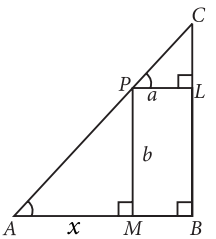
$$C = \left[2500 + 300\left(5 + \frac{25}{5}\right)\right] = [2500 + 3000] = 5500.$$

82. Let P be any point on the hypotenuse of the given right triangle.

Let $PL = a$, $PM = b$

and $AM = x$.

Clearly, ΔCPL and



www.exambites.in

ΔPAM are similar

$$\therefore \quad \frac{PL}{CL}=\frac{AM}{PM} \Rightarrow CL=\frac{PL\cdot PM}{AM}=\frac{a\cdot b}{x}$$

Now $AB = x + a$ and $BC = b + CL = b+\frac{ab}{x}$.

From right ΔABC, $AC^2 = AB^2 + BC^2$

Taking $l = AC^2$

$$\therefore \quad l=(x+a)^2+\left(b+\frac{ab}{x}\right)^2$$

$$=(x+a)^2+b^2\left(1+\frac{a}{x}\right)^2$$

Differentiating w.r.t. x, we get

$$\frac{dl}{dx}=2(x+a)+b^2 2\left(1+\frac{a}{x}\right)\cdot\frac{-a}{x^2}$$

$$=2(x+a)-\frac{2ab^2(x+a)}{x^3}=2(x+a)\left[1-\frac{ab^2}{x^3}\right]$$

and $\frac{d^2l}{dx^2}=2\cdot 1\left[1-\frac{ab^2}{x^3}\right]+2(x+a)\cdot\frac{3ab^2}{x^4}$

$\therefore$ For maximum and minimum value of l,

$$\frac{dl}{dx}=0\Rightarrow x+a=0 \text{ or } 1-\frac{ab^2}{x^3}=0$$

As $x = AM \neq 0$ $\therefore$ Reject $x + a = 0$

$\therefore \quad x^3 = ab^2 \Rightarrow x = a^{1/3}b^{2/3}$

For this value of x, clearly $\frac{d^2l}{dx^2}>0$

$\therefore$ l and consequently the hypotenuse AC is minimum (least).

Hence the least value of AC is given by

$$AC=\sqrt{(x+a)^2+b^2\left(1+\frac{a}{x}\right)^2} \text{ where } x=a^{1/3}b^{2/3}$$

$$=\sqrt{(x+a)^2+\frac{b^2}{x^2}(x+a)^2}=(x+a)\sqrt{1+\frac{b^2}{x^2}}$$

$$=\left(\frac{x+a}{x}\right)\sqrt{b^2+x^2}$$

$$=\left(\frac{a^{1/3}b^{2/3}+a}{a^{1/3}b^{2/3}}\right)\sqrt{b^2+a^{2/3}b^{4/3}}$$

$$=\frac{a^{1/3}(b^{2/3}+a^{2/3})}{a^{1/3}b^{2/3}}\cdot b^{2/3}\sqrt{b^{2/3}+a^{2/3}}$$

$$=(a^{2/3}+b^{2/3})^{3/2}$$

83. *Refer to answer 78.*

84. Let r and h be the radius and height of the cylindrical can respectively.

Therefore, the total surface area of the closed cylinder is given by

$S = 2\pi rh + 2\pi r^2 = 2\pi r(r + h)$ ...(i)

Given volume of the can = 128π cm^3

Also volume $(V) = \pi r^2 h$ ...(ii)

$$\Rightarrow \quad \pi r^2 h = 128\pi \Rightarrow h=\frac{128}{r^2} \quad \text{...(iii)}$$

Putting the value of h in equation (i), we get

$$S=2\pi r\left(r+\frac{128}{r^2}\right)=2\pi r^2+\frac{256}{r}\pi \quad \text{...(iv)}$$

Differentiating (iv) w.r.t. r, we get

$$\frac{dS}{dr}=4\pi r-\frac{256\pi}{r^2} \quad \text{...(v)}$$

Substituting $\frac{dS}{dr}=0$ for critical points, we get

$$4\pi r-\frac{256\pi}{r^2}=0 \Rightarrow r^3=64 \Rightarrow r=4 \text{ cm}$$

Differentiating (v) w.r.t. r, we get

$$\frac{d^2S}{dr^2}=4\pi-256\pi(-2r^{-3})=4\pi+\frac{512}{r^3}\pi$$

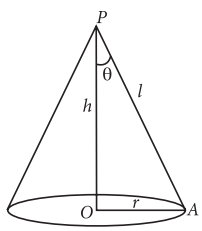
$$\therefore \quad \left[\frac{d^2S}{dr^2}\right]_{r=4}>0$$

Thus the total surface area of the cylinder is minimum when $r = 4$.

From (iii), we have $h=\frac{128}{r^2}=\frac{128}{16}=8$

Thus radius = 4 cm and height = 8 cm.

85. Let θ be the semi-vertical angle of the cone, V its volume, h its height, r base radius and slant height l.



Then from ΔOAP,

$r = l \sin\theta$, $h = l \cos\theta$

Now

$$V=\frac{1}{3}\pi r^2 h=\frac{1}{3}\pi l^2\sin^2\theta\cdot l\cos\theta$$

$$=\frac{1}{3}\pi l^3\sin^2\theta\cdot\cos\theta$$

$$\Rightarrow \quad \frac{dV}{d\theta}=\frac{1}{3}\pi l^3(2\sin\theta\cdot\cos\theta\cdot\cos\theta-\sin^2\theta\cdot\sin\theta)$$

www.examtics.in

$$= \frac{1}{3}\pi l^3 \sin\theta(2\cos^2\theta - \sin^2\theta)$$

and $\dfrac{d^2V}{d\theta^2} = \dfrac{1}{3}\pi l^3$ [cos θ(2cos²θ – sin²θ)

+ sin θ (–4 cos θ sin θ –2 sin θ cos θ)]

$$= \frac{1}{3}\pi l^3[\cos\theta(2\cos^2\theta - \sin^2\theta) - 6\sin^2\theta\cos\theta]$$

For maximum or minimum value of θ,

$\dfrac{dV}{d\theta} = 0 \Rightarrow \sin\theta(2\cos^2\theta - \sin^2\theta) = 0$

$\Rightarrow$ sin θ = 0 or 2 cos² θ – sin²θ = 0

$\Rightarrow$ 2cos² θ – (1 – cos²θ) = 0

[Note : sin θ ≠ 0 as θ ≠ 0]

$\Rightarrow \cos^2\theta = \dfrac{1}{3} \Rightarrow \cos\theta = \dfrac{1}{\sqrt{3}}$

$\Rightarrow \theta = \cos^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$

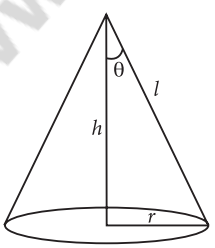
For $\cos\theta = \dfrac{1}{\sqrt{3}} \Rightarrow \sin\theta = \dfrac{\sqrt{2}}{\sqrt{3}}$

$$\therefore \quad \frac{d^2V}{d\theta^2} = \frac{1}{3}\pi l^3\left[\frac{1}{\sqrt{3}}\left(2\cdot\frac{1}{3} - \frac{2}{3}\right) - 6\cdot\frac{2}{3}\cdot\frac{1}{\sqrt{3}}\right]$$

$$= \frac{1}{3}\pi l^3\left(-\frac{4}{\sqrt{3}}\right) < 0$$

$\therefore$ *V* is maximum for $\theta = \cos^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$.

86. Let *r*, *h*, *l*, *V* and *S* be respectively the base radius, height, slant height, volume and curved surface area of the cone. Then,



$l^2 = r^2 + h^2$,

$$V = \frac{1}{3}\pi r^2 h \quad \text{... (i)}$$

and $S = \pi r l = \pi r\sqrt{r^2 + h^2}$

$\Rightarrow$ $S^2 = \pi^2 r^2(r^2 + h^2)$

$$= \pi^2\frac{3V}{\pi h}\left(\frac{3V}{\pi h} + h^2\right) \qquad \text{[Using (i)]}$$

$$= 3\pi V\left(\frac{3V}{\pi h^2} + h\right)$$

For *S* to be least, S^2 is also least.

$$\therefore \quad \frac{dS^2}{dh} = 3\pi V.\left(\frac{-6V}{\pi h^3} + 1\right) \text{ and}$$

$$\frac{d^2S^2}{dh^2} = 3\pi V\left(\frac{-6V}{\pi}\right)\cdot\frac{-3}{h^4} = \frac{54V^2}{h^4}$$

For maximum or minimum *S* (and so S^2),

$\dfrac{dS^2}{dh} = 0 \Rightarrow 6V = \pi h^3$

$$\Rightarrow \quad h = \left(\frac{6V}{\pi}\right)^{1/3} \qquad \text{... (ii)}$$

For this value of *h*, $\dfrac{d^2S^2}{dh^2} = \dfrac{54V^2}{h^4} > 0$

$\Rightarrow$ S^2 and therefore *S* is least.

$$\cot\theta = \frac{h}{r} = \frac{h}{\sqrt{3V/\pi h}} = \sqrt{\frac{\pi}{3V}}\cdot h^{3/2} \qquad \text{[From (i)]}$$

$$\Rightarrow \quad \cot\theta = \sqrt{\frac{\pi}{3V}}\sqrt{\frac{6V}{\pi}} = \sqrt{2} \qquad \text{[From (ii)]}$$

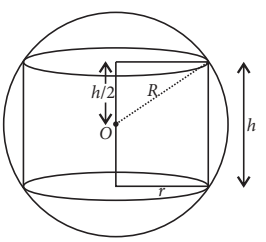
$\Rightarrow$ The semi vertical angle, $\theta = \cot^{-1}\sqrt{2}$

87. Let *r* and *h* be the base radius and height of cylinder respectively.

$$\therefore \quad \left(\frac{h}{2}\right)^2 + r^2 = R^2 \qquad \text{... (i)}$$

Now, *V* = Volume of the cylinder inscribed in a sphere

$= \pi r^2 h$



$$\Rightarrow \quad V = \pi h\left(R^2 - \frac{h^2}{4}\right) \qquad \text{[Using (i)]}$$

$$\Rightarrow \quad V = \pi\left(R^2 h - \frac{h^3}{4}\right)$$

Now differentiating w.r.t. *h*, we get

$$\frac{dV}{dh} = \pi\left(R^2 - \frac{3h^2}{4}\right) \text{ and } \frac{d^2V}{dh^2} = \pi\left(0 - \frac{3}{4}\cdot 2h\right)$$

For maximum or minimum,

www.exambites.in

$$\frac{dV}{dh}=0\Rightarrow R^2-\frac{3}{4}h^2=0$$

$$\Rightarrow\quad h^2=\frac{4}{3}R^2\ \Rightarrow\ \ h=\frac{2R}{\sqrt{3}}$$

For this value of h,

$$\frac{d^2V}{dh^2}=-\frac{3}{2}\pi\cdot\frac{2R}{\sqrt{3}}=-\sqrt{3}\pi R<0$$

$\Rightarrow$ V is maximum

Also maximum value of V

$$=\pi\cdot\frac{2R}{\sqrt{3}}\left(R^2-\frac{1}{4}\cdot\frac{4}{3}R^2\right)=\pi\cdot\frac{2R}{\sqrt{3}}\cdot\frac{2}{3}R^2$$

$$=\frac{4\pi}{3\sqrt{3}}R^3\ \text{cu.units}$$

88. Let a be the side of the given square and r be the radius of the circle.

By hypothesis

$4a + 2\pi r = k$

$$\Rightarrow\quad a=\frac{k-2\pi r}{4}\qquad\text{...(i)}$$

Let A = Sum of areas of the circle and the square

$$\Rightarrow A=\pi r^2+a^2=\pi r^2+\frac{1}{16}(k-2\pi r)^2\quad\text{[Using (i)]}$$

$$\Rightarrow\quad\frac{dA}{dr}=2\pi r+\frac{1}{16}\cdot 2(k-2\pi r)\cdot(-2\pi)$$

$$=2\pi r-\frac{\pi}{4}(k-2\pi r)$$

and $$\frac{d^2A}{dr^2}=2\pi-\frac{\pi}{4}(0-2\pi)$$

For maxima or minima,

$$\frac{dA}{dr}=0\Rightarrow 2\pi r-\frac{\pi}{4}(k-2\pi r)=0$$

$$\Rightarrow\quad 8r-k+2\pi r=0\Rightarrow(8+2\pi)r=k$$

$$\Rightarrow\quad r=\frac{k}{2\pi+8}$$

For this value of r, $\dfrac{d^2A}{dr^2}=2\pi+\dfrac{\pi^2}{2}>0$

$\therefore$ A is minimum (least), when $r=\dfrac{k}{2\pi+8}$

From (i), $$a=\frac{k-2\pi\cdot\dfrac{k}{2\pi+8}}{4}$$

$$=\frac{k}{4}\cdot\left(\frac{2\pi+8-2\pi}{2\pi+8}\right)=\frac{2k}{2\pi+8}=2r$$

$\therefore$ Area is least, when $a = 2r$.

89. Let r and h be the base radius and height of the cylinder respectively and volume of cylinder, $V = \pi r^2 h$

$$\Rightarrow\quad h=\frac{V}{\pi r^2}\qquad\text{... (i)}$$

Total surface area of the cylinder, $S = 2\pi rh + \pi r^2$

$$\Rightarrow\quad S=2\pi r\left(\frac{V}{\pi r^2}\right)+\pi r^2\qquad\text{[By using (i)]}$$

$$\Rightarrow\quad S=\frac{2V}{r}+\pi r^2$$

On diff. w.r.t. r both sides, $\dfrac{dS}{dr}=-\dfrac{2V}{r^2}+2\pi r$

Again diff. w.r.t. r both sides, $\dfrac{d^2S}{dr^2}=\dfrac{4V}{r^3}+2\pi$

For maxima or minima, $\dfrac{dS}{dr}=0$

$$\Rightarrow\quad -\frac{2V}{r^2}+2\pi r=0\quad\Rightarrow\quad 2\pi r=\frac{2V}{r^2}$$

$$\Rightarrow\quad \pi r^3=V\quad\Rightarrow\quad r=\left(\frac{V}{\pi}\right)^{1/3}$$

$$\therefore\quad\left[\frac{d^2V}{dh^2}\right]_{r=\left(\frac{V}{\pi}\right)^{1/3}}=4V\left(\frac{\pi}{V}\right)+2\pi>0$$

So, S is minimum at $r=\left(\dfrac{V}{\pi}\right)^{1/3}$

Now, $\pi r^3 = V \Rightarrow \pi r^3 = \pi r^2 h \Rightarrow r = h$

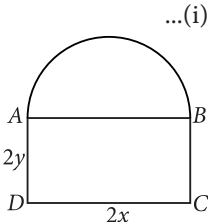
Hence, the cylinder of a given volume which is open at the top has minimum total surface area, when its height is equal to the radius of its base.

90. Let $ABCD$ be a rectangle and let the semi-circle is described on the side AB as its diameter. Let $AB = 2x$ and $AD = 2y$. Let $P = 10$ m be the given perimeter of window.

Therefore, $10 = 2x + 4y + \pi x$

$$\Rightarrow\quad 4y=10-2x-\pi x\qquad\text{...(i)}$$

Area of the window,



$$A=(2x)(2y)+\frac{1}{2}\pi x^2$$

$$\Rightarrow\quad A=4xy+\frac{1}{2}\pi x^2$$

$$\Rightarrow\quad A=10x-2x^2-\pi x^2+\frac{1}{2}\pi x^2\qquad\text{[using (i)]}$$

$$\Rightarrow\quad A=10x-2x^2-\frac{1}{2}\pi x^2$$

www.examfree.in

On diff. w.r.t. x both sides, $\frac{dA}{dx} = 10 - 4x - \pi x$

Again diff. w.r.t. x both sides, $\frac{d^2A}{dx^2} = -(4+\pi)$

For maxima or minima,

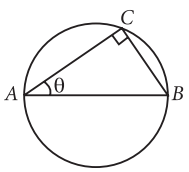
$$\frac{dA}{dx} = 0 \Rightarrow 10 - 4x - \pi x = 0 \Rightarrow x = \frac{10}{4+\pi}$$

$$\left[\frac{d^2A}{dx^2}\right]_{x=\frac{10}{4+\pi}} = -(4+\pi) < 0$$

So, A is maximum at $x = \left(\frac{10}{4+\pi}\right)$m.

Now, length of the window is $2x = \left(\frac{20}{4+\pi}\right)$m and width is $2y = \left(\frac{10}{4+\pi}\right)$m.

91. Here BA is a diameter of the given circle, of radius = r.



Let $\angle CAB = \theta$

Also $\angle ACB = \frac{\pi}{2}$

Now $AC = AB\cos\theta = 2r\cos\theta$

$BC = AB\sin\theta = 2r\sin\theta$

Let Area of $\Delta ABC = \frac{1}{2}\cdot AC \cdot BC$

$$= \frac{1}{2}\cdot 2r\cos\theta \cdot 2r\sin\theta = r^2 \cdot \sin 2\theta$$

$$\Rightarrow \quad \frac{d\Delta}{d\theta} = r^2 \cdot 2\cos 2\theta \text{ and } \frac{d^2\Delta}{d\theta^2} = -r^2 \cdot 4\sin 2\theta$$

For maxima or minima,

$$\frac{d\Delta}{d\theta} = 0 \Rightarrow \cos 2\theta = 0 \Rightarrow 2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$$

and $\left[\frac{d^2\Delta}{d\theta^2}\right]_{\theta=\frac{\pi}{4}} = -4r^2\sin\frac{\pi}{2} = -4r^2 < 0$

Hence area of ΔABC is maximum when

$$\angle CAB = \theta = \frac{\pi}{4} = \angle ABC \qquad \left[\because \quad \angle ACB = \frac{\pi}{2}\right]$$

$\Rightarrow$ ΔABC is isosceles.

92. The given parabola is

$y^2 = 4ax$ ...(i)

Let $Q(11a, 0)$.

Any point on (i) is $P(at^2, 2at)$

$\therefore$ $PQ^2 = (at^2 - 11a)^2 + (2at - 0)^2$

Let $l = PQ^2 = a^2t^4 - 18\,a^2t^2 + 121a^2$

$$\Rightarrow \quad \frac{dl}{dt} = 4a^2t^3 - 36a^2t \text{ and } \frac{d^2l}{dt^2} = 12a^2t^2 - 36a^2$$

For maximum or minimum value of l,

$$\frac{dl}{dt} = 0 \Rightarrow 4a^2t(t^2-9) = 0 \Rightarrow t = 0, 3, -3$$

For $t = 0, \frac{d^2l}{dt^2} = -36a^2 < 0$

This corresponds to a maximum value of l.

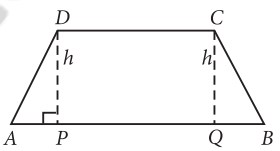
Both for $t = 3$ and -3,

$$\frac{d^2l}{dt^2} = 12a^2\cdot 9 - 36a^2 = 72a^2 > 0$$

$\therefore$ This corresponds to a minimum value of l *i.e.*, of PQ^2 and therefore of PQ.

Thus, there are two such points P with coordinates, $(9a, 6a)$and $(9a, -6a)$ nearest to the given point Q.

93. Let $ABCD$ be the given trapezium.



Then $AD = DC = CB = 10$ cm

In ΔAPD and ΔBQC

$DP = CQ = h$

$AD = BC = 10$ cm

$\angle DPA = \angle CQB = 90°$

$\therefore$ $\Delta APD \cong \Delta BQC$ (by R.H.S. congruency)

$\Rightarrow$ $AP = QB = x$ cm (Say)

$\therefore$ $AB = AP + PQ + QB$

$= x + 10 + x = (2x + 10)$cm

Also from ΔAPD, $AP^2 + PD^2 = AD^2$

$\Rightarrow$ $x^2 + h^2 = 10^2$

$$\Rightarrow \quad h = \sqrt{100 - x^2} \qquad \text{...(i)}$$

Now, area A of this trapezium is given by

$$A = \frac{1}{2}(AB + DC)\cdot h = \frac{1}{2}(2x + 10 + 10)h$$

$$= (x+10)\cdot\sqrt{100 - x^2} \qquad \text{...(ii)} \quad \text{[Using (i)]}$$

Differentiating w.r.t. x, we get

$$\frac{dA}{dx} = 1\cdot\sqrt{100-x^2} + (x+10)\cdot\frac{1}{2\sqrt{100-x^2}}\cdot(-2x)$$

$$= \frac{100 - x^2 - x^2 - 10x}{\sqrt{100-x^2}} = \frac{-2(x^2+5x-50)}{\sqrt{100-x^2}}$$

www.exambites.in

$$= \frac{-2(x+10)(x-5)}{\sqrt{100-x^2}}$$

For max. or min. value of A, $\frac{dA}{dx} = 0$

$\Rightarrow$ $(x + 10)(x - 5) = 0 \Rightarrow x = 5$

(Reject $x = -10$ as $x \neq 0$)

For this value of x, $\frac{dA}{dx}$ changes sign from positive to negative.

$\therefore$ A is maximum at $x = 5$.

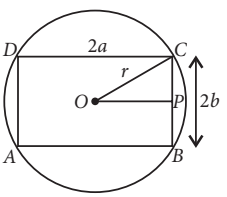
From (ii), the max. value of $A = (5+10)\cdot\sqrt{100-5^2}$

$= 15\sqrt{75} = 75\sqrt{3}$ sq.cm.

94. *Refer to answer 76.*

95. Let the radius of given circle = r

Let $2a$ and $2b$ are the lengths of the sides of any inscribed rectangle in the given circle. Then from ΔOPC,



$OC^2 = OP^2 + PC^2$

$\Rightarrow$ $r^2 = a^2 + b^2$

$\Rightarrow$ $b = \sqrt{r^2 - a^2}$ ...(i)

Now, A = Area of rectangle $ABCD$

$$= 2a \cdot 2b = 4ab = 4a\sqrt{r^2 - a^2}$$

$$\therefore \quad \frac{dA}{da} = 4\left[1\cdot\sqrt{r^2-a^2} + a\cdot\frac{1}{2}(r^2-a^2)^{-1/2}(-2a)\right]$$

$$= 4\left[\sqrt{r^2-a^2} - \frac{a^2}{\sqrt{r^2-a^2}}\right] \quad \text{...(ii)}$$

For area A to be max. or min.

$$\frac{dA}{da} = 0 \Rightarrow \sqrt{r^2-a^2} - \frac{a^2}{\sqrt{r^2-a^2}} = 0$$

$\Rightarrow$ $r^2 - a^2 - a^2 = 0 \Rightarrow r = \sqrt{2}a$ $(\because r > 0, a > 0)$

$\therefore$ $a = \frac{r}{\sqrt{2}}$

From (ii), $\frac{dA}{da} = 4\cdot\frac{r^2-2a^2}{\sqrt{r^2-a^2}}$

$$\Rightarrow \frac{d^2A}{da^2} = 4\cdot\frac{\sqrt{r^2-a^2}\cdot(-4a) - (r^2-2a^2)\cdot\frac{-a}{\sqrt{r^2-a^2}}}{(r^2-a^2)}$$

$$= -4\cdot\frac{(r^2-a^2)4a - (r^2-2a^2)a}{(r^2-a^2)^{3/2}} = -4a\cdot\frac{3r^2-2a^2}{(r^2-a^2)^{3/2}}$$

Now, $\left[\frac{d^2A}{da^2}\right]_{a=\frac{r}{\sqrt{2}}} = -4\cdot\frac{r}{\sqrt{2}}\cdot\frac{3r^2-r^2}{\left(r^2-\frac{r^2}{2}\right)^{3/2}} < 0$

$\therefore$ The area A is max. when $a = \frac{r}{\sqrt{2}}$

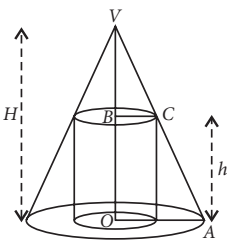
Also, from (i), $b = \sqrt{r^2-a^2} = \sqrt{r^2-\frac{r^2}{2}} = \frac{r}{\sqrt{2}} = a$

$\Rightarrow$ $2a = 2b = \sqrt{2}r$

$\therefore$ The rectangle of max. area inscribed in the circle is a square.

96. Let R and H be the base radius and height of the given cone and r, h be the same for the inscribed cylinder.

Clearly, ΔVBC is similar to ΔVOA.



$$\therefore \quad \frac{BV}{BC} = \frac{OV}{OA} \Rightarrow \frac{H-h}{r} = \frac{H}{R}$$

$$\Rightarrow H - h = H\frac{r}{R} \Rightarrow h = H - H\frac{r}{R}.$$

Now the curved surface area, S of the inscribed cylinder is

$$S = 2\pi rh = 2\pi r\left(H - H\frac{r}{R}\right) = \frac{2\pi H}{R}(rR - r^2)$$

$$\Rightarrow \frac{dS}{dr} = \frac{2\pi H}{R}(1\cdot R - 2r)$$

For greatest (max.) or least surface area,

$\frac{dS}{dr} = 0 \Rightarrow R = 2r$

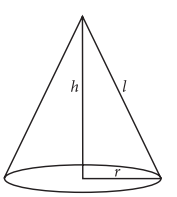
Also $\frac{d^2S}{dr^2} = \frac{2\pi H}{R}\cdot(-2) < 0$

Hence, S is greatest, when $r = \frac{R}{2}$.

97. Let r be the base radius of the cone, l be the slant height and h be its height. Let V be its volume and S be its curved surface.

Now $V = \frac{1}{3}\pi r^2 h$

$\Rightarrow$ $h = \frac{3V}{\pi r^2}$



www.examcrack.in

We have, $l^2 = h^2 + r^2$

Now S = curved surface area of the cone $= \pi rl = \pi r\sqrt{h^2 + r^2}$

S will be least $\Leftrightarrow S^2$ will be least.

$$\therefore \quad S^2 = \pi^2 r^2 (h^2 + r^2) = \pi^2 r^2 \left(\frac{9V^2}{\pi^2 r^4} + r^2 \right)$$

$$\Rightarrow S^2 = \frac{9V^2}{r^2} + \pi^2 r^4$$

Differentiating w.r.t. r, we get

$$\frac{dS^2}{dr} = -\frac{18V^2}{r^3} + 4\pi^2 r^3 \text{ and } \frac{d^2S^2}{dr^2} = \frac{54V^2}{r^4} + 12\pi^2 r^2$$

For max. or min., $\frac{dS^2}{dr} = 0$

$$\Rightarrow \quad 4\pi^2 r^3 = \frac{18V^2}{r^3} = \frac{18}{r^3}\left(\frac{1}{3}\pi r^2 h\right)^2 = 2\pi^2 r h^2$$

$$\Rightarrow \quad h^2 = 2r^2 \Rightarrow \quad h = \sqrt{2}\, r \qquad (\because r \neq 0)$$

For this value of r, $\frac{d^2S^2}{dr^2} > 0$

$\therefore$ Curved surface area of the cone is least when $h = \sqrt{2}\, r$.

98. *Refer to answer 76.*

99. Let r be the radius of the circular base, h be the height and S be the total surface area of a right circular cylinder, then

$S = 2\pi r^2 + 2\pi rh$ is given to be a constant.

Let V be the volume of the cylinder, then

$$V = \pi r^2 h = \pi r^2 \left(\frac{S - 2\pi r^2}{2\pi r} \right) = \frac{r}{2}(S - 2\pi r^2)$$

$$\Rightarrow \quad V = \frac{Sr}{2} - \pi r^3 \qquad \text{...(i)}$$

Differentiating (i),w.r.t r, we get

$$\frac{dV}{dr} = \frac{S}{2} - 3\pi r^2 \quad \text{and} \quad \frac{d^2V}{dr^2} = -6\pi r$$

Now for maxima or minima,

$$\frac{dV}{dr} = 0 \Rightarrow \frac{S}{2} - 3\pi r^2 = 0 \Rightarrow r^2 = \frac{S}{6\pi}$$

$$\Rightarrow \quad r = \sqrt{\frac{S}{6\pi}}$$

Also, $\left(\frac{d^2V}{dr^2}\right)_{r=\sqrt{S/(6\pi)}} = -6\pi\sqrt{\frac{S}{6\pi}} < 0$

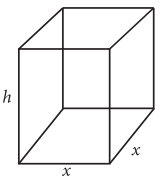
$\Rightarrow$ V has a maximum value at $r = \sqrt{\frac{S}{6\pi}}$

Now, $h = \frac{S - 2\pi r^2}{2\pi r} = \frac{S - 2\pi\left(\frac{S}{6\pi}\right)}{2\pi\sqrt{\frac{S}{6\pi}}} = \frac{2S}{6\pi}\sqrt{\frac{6\pi}{S}}$

i.e., $h = 2\sqrt{\frac{S}{6\pi}} = 2r$

So, volume is maximum when the height is equal to the diameter.

100. Let h be height and x be the side of the square base of the open box. Then its area $= x \times x + 4h \times x = c^2$ (given)



$$\Rightarrow \quad h = \frac{c^2 - x^2}{4x}$$

Now V = volume of the box

$$= x^2 h = x^2 \cdot \frac{c^2 - x^2}{4x} = \frac{1}{4}(c^2 x - x^3)$$

$$\Rightarrow \quad \frac{dV}{dx} = \frac{1}{4}(c^2 - 3x^2) \text{ and } \frac{d^2V}{dx^2} = \frac{1}{4}(-6x) = \frac{-3}{2}x$$

For maxima or minima $\frac{dV}{dx} = 0 \Rightarrow x^2 = \frac{c^2}{3}$

$$\Rightarrow \quad x = \frac{c}{\sqrt{3}} \quad (\because x \neq 0)$$

For this value of x, $\frac{d^2V}{dx^2} < 0$

$\Rightarrow$ V is maximum at $x = \frac{c}{\sqrt{3}}$ and its maximum value is,

$$V = \frac{1}{4}x(c^2 - x^2) = \frac{1}{4} \cdot \frac{c}{\sqrt{3}}\left(c^2 - \frac{c^2}{3}\right)$$

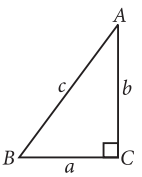
$$= \frac{c^3}{6\sqrt{3}} \text{ cubic units.}$$

101. Let ΔABC be right angled triangle with sides a, b and hypotenuse c.

We have, $a^2 + b^2 = c^2$

Area of $\Delta ABC = A = \frac{1}{2}a \cdot b$

$$= \frac{1}{2}a\sqrt{c^2 - a^2}$$



www.exambits.in

$$\Rightarrow \quad \frac{dA}{da} = \frac{1}{2}\cdot 1\cdot\sqrt{c^2-a^2} + \frac{1}{2}a\cdot\frac{1}{2}\frac{(-2a)}{\sqrt{c^2-a^2}}$$

$$= \frac{1}{2}\left[\sqrt{c^2-a^2} - \frac{a^2}{\sqrt{c^2-a^2}}\right]$$

and $\dfrac{d^2A}{da^2} = \dfrac{1}{2}\left[\dfrac{-a(3c^2-2a^2)}{(c^2-a^2)^{3/2}}\right]$

For maximum or minimum value, $\dfrac{dA}{da} = 0$

$$\Rightarrow \quad \frac{c^2-a^2-a^2}{\sqrt{c^2-a^2}} = 0$$

$$\Rightarrow \quad c^2 = 2a^2 \Rightarrow a = \frac{c}{\sqrt{2}} \ (\because a \neq 0)$$

For this value of a, we have

$$\frac{d^2A}{da^2} = -2 < 0$$

$\therefore$ Area of ΔABC is max. at $a = \dfrac{c}{\sqrt{2}}$ and

$$b = \sqrt{c^2-a^2} = \sqrt{2a^2-a^2} = a$$

$\Rightarrow$ ΔABC is isosceles right angled triangle.

102. *Refer to answer 88.*

103. *Refer to answer 87.*

104. Let x and y be the length and breadth of the rectangle whose perimeter (P) is given.

$$\therefore \quad P = 2(x+y) \Rightarrow y = \frac{P}{2} - x$$

Area of rectangle (A) = xy

$$\Rightarrow \quad A = x\left(\frac{P}{2} - x\right)$$

$$\Rightarrow \quad A = \frac{Px}{2} - x^2 \Rightarrow \frac{dA}{dx} = \frac{P}{2} - 2x$$

For maxima or minima, $\dfrac{dA}{dx} = 0$

$$\therefore \quad \frac{P}{2} - 2x = 0 \Rightarrow x = \frac{P}{4}$$

Also, $\dfrac{d^2A}{dx^2} = -2 < 0$

$\therefore$ Area is maximum, when $x = \dfrac{P}{4}$.

Now, $y = \dfrac{P}{2} - \dfrac{P}{4} = \dfrac{P}{4}$

$\therefore$ Area is maximum, when $x = y$ *i.e.* rectangle is a square.

105. Let x and y be the length and breadth of rectangle of given area A, then we have,

$$A = xy \Rightarrow y = \frac{A}{x}$$

Now, perimeter $P = 2(x+y) = 2\left(x + \dfrac{A}{x}\right)$

$$\Rightarrow \quad \frac{dP}{dx} = 2\left(1 - \frac{A}{x^2}\right) \text{and } \frac{d^2P}{dx^2} = \frac{4A}{x^3}$$

For maxima or minima, $\dfrac{dP}{dx} = 0$

$$\Rightarrow \quad 2\left(1 - \frac{A}{x^2}\right) = 0 \Rightarrow A = x^2.$$

Also, $\dfrac{d^2P}{dx^2} = \dfrac{4A}{A^{3/2}} > 0$

$\therefore$ Perimeter of rectangle is minimum, when $x = \sqrt{A}$ and $y = \dfrac{A}{x} = \dfrac{x^2}{x} = x$

So, perimeter of rectangle is minimum, when $y = x$ *i.e.*, rectangle is a square.

106. Let a be the breadth and b be the length of rectangle and b be the side of equilateral triangle.

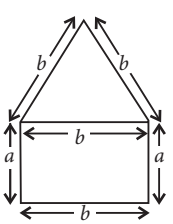
Total perimeter of the window

$= b + 2a + 2b = 12$ (Given)

$$\Rightarrow \quad 2a + 3b = 12$$

$$\Rightarrow \quad a = \frac{12-3b}{2} \qquad \text{...(i)}$$

Now, area A of the window is,



$$A = ab + \frac{\sqrt{3}}{4}b^2$$

$$= \frac{12-3b}{2}b + \frac{\sqrt{3}}{4}b^2$$

$$= 6b - \frac{3}{2}b^2 + \frac{\sqrt{3}}{4}b^2$$

$$\Rightarrow \quad \frac{dA}{db} = 6 - 3b + \frac{\sqrt{3}}{2}b$$

For maximum or minimum area,

$$\frac{dA}{db} = 0 \Rightarrow \left(3 - \frac{\sqrt{3}}{2}\right)b = 6 \Rightarrow b = \frac{12}{6-\sqrt{3}}$$

Also $\dfrac{d^2A}{db^2} = -3 + \dfrac{\sqrt{3}}{2} < 0$

www.exambites.in

$\therefore$ Area is maximum, when $b = \dfrac{12}{6-\sqrt{3}}$

From (i), $a = \dfrac{12-3b}{2} = 6 - \dfrac{3}{2}.\left(\dfrac{12}{6-\sqrt{3}}\right) = \dfrac{18-6\sqrt{3}}{6-\sqrt{3}}$

107. Let the point on $y^2 = 2x$, which is at a min. distance from $Q(1, 4)$, be $P(x, y)$.

$\therefore$ We have to minimise

$s = PQ^2 = (x-1)^2 + (y-4)^2$

$= \left(\dfrac{y^2}{2} - 1\right)^2 + (y-4)^2 \quad [\because \ y^2 = 2x \Rightarrow x = \dfrac{1}{2}y^2]$

$\Rightarrow \quad \dfrac{ds}{dy} = 2\left(\dfrac{y^2}{2} - 1\right).y + 2(y-4) = y^3 - 8$

and $\dfrac{d^2s}{dy^2} = 3y^2$

For maxima or minima $\dfrac{ds}{dy} = 0$

$\Rightarrow \quad y^3 - 8 = 0$

$\Rightarrow \quad y^3 - 2^3 = 0$

$\Rightarrow \quad (y-2)(y^2 + 2y + 4) = 0$

$\Rightarrow \quad y = 2 \qquad (\because y^2 + 2y + 4 = 0$ has no real roots)

For this value of y, $\dfrac{d^2s}{dy^2} = 3 \times 2^2 = 12 > 0$

$\therefore$ s is min. and from $y^2 = 2x$,

$x = \dfrac{y^2}{2} = \dfrac{2^2}{2} = 2$

$\therefore$ The required point is (2, 2)

108. *Refer to answer 85.*

We get, $\theta = \cos^{-1}\dfrac{1}{\sqrt{3}} \Rightarrow \theta = \tan^{-1}\sqrt{2}$

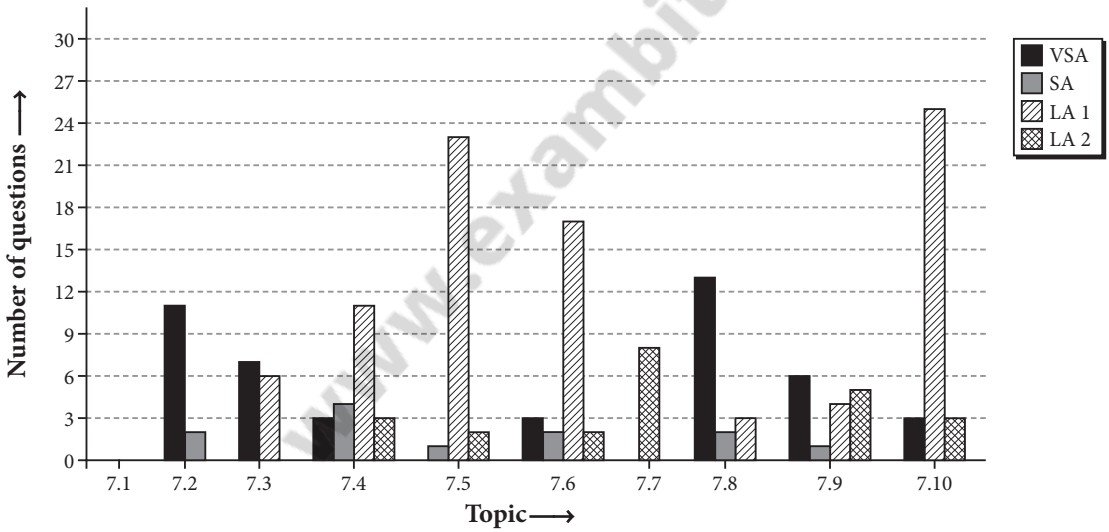
www.exambites.in

CHAPTER
7

Integrals

7.1 Introduction
7.2 Integration as an Inverse Process of Differentiation
7.3 Methods of Integration
7.4 Integrals of Some Particular Functions
7.5 Integration by Partial Fractions
7.6 Integration by Parts
7.7 Definite Integral
7.8 Fundamental Theorem of Calculus
7.9 Evaluation of Definite Integrals by Substitution
7.10 Some Properties of Definite Integrals

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambits.in

- Maximum weightage is of *Some Properties of Definite Integrals.*
- Maximum VSA type questions were asked from *Fundamental Theorem of Calculus.*
- Maximum LA 1 type questions were asked from *Some Properties of Definite Integrals.*

QUICK RECAP

INDEFINITE INTEGRAL

- Integration is the inverse process of differentiation.

$$i.e., \frac{d}{dx}F(x) = f(x) \Rightarrow \int f(x)\,dx = F(x) + C,$$

where C is the constant of integration.
Integrals are also known as antiderivatives.

Some Standard Integrals

- $\int dx = x + C$, where 'C' is the constant of integration
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, where $n \neq -1$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\log_e a} + C$, where $a > 0$
- $\int \frac{1}{x} dx = \log_e |x| + C$, where $x \neq 0$
- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$
- $\int \sec^2 x \, dx = \tan x + C$
- $\int \operatorname{cosec}^2 x \, dx = -\cot x + C$
- $\int \sec x \tan x \, dx = \sec x + C$
- $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C$
- $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C = -\cos^{-1} x + C$, where $|x| < 1$
- $\int \frac{dx}{1+x^2} = \tan^{-1} x + C = -\cot^{-1} x + C$
- $\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + C = -\operatorname{cosec}^{-1} x + C$, where $|x| > 1$
- $\int \tan x \, dx = \log|\sec x| + C = -\log|\cos x| + C$
- $\int \cot x \, dx = \log|\sin x| + C$
- $\int \sec x \, dx = \log|\sec x + \tan x| + C = \log\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right| + C$
- $\int \operatorname{cosec} x \, dx = \log|\operatorname{cosec} x - \cot x| + C = \log\left|\tan\frac{x}{2}\right| + C$

Properties of Indefinite Integral

(i) $\int f'(x)dx = f(x) + C$

(ii) $\int f(x)dx = \int g(x)dx + C$, f and g are indefinite integrals with the same derivative.

(iii) $\int [f(x) + g(x)]dx = \int f(x)\,dx + \int g(x)\,dx$

(iv) $\int k \cdot f(x)\,dx = k\int f(x)\,dx$, k being any real number.

METHODS OF INTEGRATION

Integration by Substitution

The given integral $\int f(x)\,dx$ can be transformed into another form by changing the independent variable x to t by substituting $x = g(t)$.

Integrals	Substitution
$\int f(ax+b)dx$	$ax + b = t$
$\int f(g(x))g'(x)dx$	$g(x) = t$
$\int \frac{f'(x)}{f(x)}dx$	$f(x) = t$
$\int (f(x))^n f'(x)dx$	$f(x) = t$
$\int (px+q)\sqrt{cx+d}\,dx$ or $\int \frac{px+q}{\sqrt{cx+d}}dx$	$px + q = A(cx + d) + B$. Find A and B by equating coefficients of like powers of x on both sides.
$\int \frac{1}{(px+q)\sqrt{cx+d}}dx$ or $\int \frac{1}{(px^2+qx+r)\sqrt{cx+d}}dx$	$cx + d = t^2$
$\int \frac{1}{(px+q)\sqrt{cx^2+dx+e}}dx$	$px + q = \frac{1}{t}$
$\int \frac{1}{(px^2+q)\sqrt{cx^2+d}}dx$	$x = \frac{1}{t}$ and then $c + dt^2 = u^2$
$\int \frac{px+q}{ax^2+bx+c}dx$ or $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$ or $\int (px+q)\sqrt{ax^2+bx+c}\,dx$	$(px + q) = A\frac{d}{dx}(ax^2+bx+c) + B$

www.examples.in

▶ **Integration using Trigonometric Identities**
When the integrand consists of trigonometric functions, we use known identities to convert it into a form which can be easily integrated. Some of the identities useful for this purpose are given below :

(i) $2\sin^2\left(\frac{x}{2}\right)=(1-\cos x)$

(ii) $2\cos^2\left(\frac{x}{2}\right)=(1+\cos x)$

(iii) $2\sin x\cos y=\sin(x+y)+\sin(x-y)$

(iv) $2\cos x\sin y=\sin(x+y)-\sin(x-y)$

(v) $2\cos x\cos y=\cos(x+y)+\cos(x-y)$

(vi) $2\sin x\sin y=\cos(x-y)-\cos(x+y)$

▶ **Some Special Substitutions**

Expression	Substitution
$\sqrt{a^2-x^2}$	$x=a\sin\theta$ or $a\cos\theta$
$\sqrt{a^2+x^2}$ or (a^2+x^2)	$x=a\tan\theta$ or $a\cot\theta$
$\sqrt{x^2-a^2}$	$x=a\sec\theta$ or $a\operatorname{cosec}\theta$
$\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	$x=a\cos2\theta$
$\sqrt{\frac{x}{a-x}}$ or $\sqrt{\frac{a-x}{x}}$	$x=a\sin^2\theta$ or $a\cos^2\theta$
$\sqrt{\frac{x}{a+x}}$ or $\sqrt{\frac{a+x}{x}}$	$x=a\tan^2\theta$ or $a\cot^2\theta$
$\sqrt{\frac{a-x}{x-b}}$ or $\sqrt{\frac{x-b}{a-x}}$ or $\sqrt{(a-x)(x-b)}$	$x=a\cos^2\theta+b\sin^2\theta$

▶ **Integrals of Some Particular Functions**

(i) $\int\frac{1}{a^2-x^2}dx=\frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+C$

(ii) $\int\frac{1}{\sqrt{a^2-x^2}}dx=\sin^{-1}\left(\frac{x}{a}\right)+C$

(iii) $\int\frac{1}{\sqrt{x^2-a^2}}dx=\log\left|x+\sqrt{x^2-a^2}\right|+C$

(iv) $\int\frac{1}{\sqrt{x^2+a^2}}dx=\log\left|x+\sqrt{x^2+a^2}\right|+C$

(v) $\int\frac{1}{x^2-a^2}dx=\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+C$

(vi) $\int\frac{1}{x^2+a^2}dx=\frac{1}{a}\tan^{-1}\frac{x}{a}+C$

Integration by Partial Fractions

▶ If $f(x)$ and $g(x)$ are two polynomials such that deg $f(x)\geq$ deg $g(x)$, then we divide $f(x)$ by $g(x)$.

$\therefore\ \frac{f(x)}{g(x)}=\text{Quotient}+\frac{\text{Remainder}}{g(x)}$

▶ If $f(x)$ and $g(x)$ are two polynomials such that the degree of $f(x)$ is less than the degree of $g(x)$, then we can evaluate $\int\frac{f(x)}{g(x)}dx$ by decomposing $\frac{f(x)}{g(x)}$ into partial fraction.

Form of the Rational Function	Form of the Partial Fraction
$\frac{px+q}{(x-a)(x-b)}, a\neq b$	$\frac{A}{x-a}+\frac{B}{x-b}$
$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}$
$\frac{px+q}{(x-a)^3}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{(x-a)^3}$
$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$
$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{(x-b)}$
$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$ where x^2+bx+c can not be factorised further	$\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$

Integration by Parts

If u and v are two differentiable functions of x, then

$$\int(uv)\,dx=\left[u\cdot\int v\,dx\right]-\int\left\{\frac{du}{dx}\cdot\int v\,dx\right\}dx.$$

In order to choose 1st function, we take the letter which comes first in the word ILATE.

www.examtips.in

I – Inverse Trigonometric Function
L – Logarithmic Function
A – Algebraic Function
T – Trigonometric Function
E – Exponential Function

- **Integral of the type**

$$\int e^x (f(x)+f'(x))dx = e^x f(x)+C$$

INTEGRALS OF SOME MORE TYPES

(i) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+C$

(ii) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+C$

(iii) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+C$

DEFINITE INTEGRAL

Let $F(x)$ be integral of $f(x)$, then for any two values of the independent variable x, say a and b, the difference $F(b) - F(a)$ is called the definite integral of $f(x)$ from a to b and is denoted by $\int_a^b f(x)dx$.

Here, $x = a$ is the lower limit and $x = b$ is the upper limit of the integral.

Definite Integral as a Limit of Sum

Let $f(x)$ be a continuous real valued function defined on the closed interval $[a, b]$. Then

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+ \ldots + f(a+(n-1)h)],$$

where $h = \frac{b-a}{n} \to 0$ as $n \to \infty$

FUNDAMENTAL THEOREM OF CALCULUS

First Fundamental Theorem : Let $f(x)$ be a continuous function in the closed interval $[a, b]$ and let $A(x)$ be the area function. Then $A'(x) = f(x)$, for all $x \in [a, b]$.

Second Fundamental Theorem : Let $f(x)$ be a continuous function in the closed interval $[a, b]$ and $F(x)$ be an integral of $f(x)$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

EVALUATION OF DEFINITE INTEGRAL BY SUBSTITUTION

When definite integral is to be found by substitution, change the lower and upper limits of integration. If substitution is $t = f(x)$ and lower limit of integration is a and upper limit is b, then new lower and upper limits will be $f(a)$ and $f(b)$ respectively.

SOME PROPERTIES OF DEFINITE INTEGRALS

(i) $\int_a^b f(x)dx = \int_a^b f(t)dt$

(ii) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

In Particular $\int_a^a f(x)dx = 0$

(iii) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$, where $a < c < b$

(iv) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

(v) $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

(vi) $\int_{-a}^a f(x)dx = \begin{cases} 0, & \text{if } f(-x) = -f(x) \\ 2\int_0^a f(x)dx, & \text{if } f(-x) = f(x) \end{cases}$

(vii) $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$

(viii) $\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$

www.examkiser.in

Previous Years' CBSE Board Questions

7.2 Integration as an Inverse Process of Differentiation

VSA (1 mark)

1. Find : $\int \frac{\sin^2 x - \cos^2 x}{\sin x \cos x} dx$ *(AI 2017)*

2. Write the antiderivative of $\left(3\sqrt{x} + \frac{1}{\sqrt{x}}\right)$. *(Delhi 2014)*

3. Evaluate : $\int \cos^{-1}(\sin x) dx$ *(Delhi 2014)*

4. Evaluate : $\int \frac{dx}{\sin^2 x \cos^2 x}$ *(Foreign 2014, Delhi 2014 C)*

5. Find : $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} dx$ *(Delhi 2014 C)*

6. Evaluate : $\int (1-x)\sqrt{x}\, dx$ *(Delhi 2012)*

7. Write the value of $\int \frac{\sec^2 x}{\text{cosec}^2 x} dx$. *(Delhi 2012C, 2011)*

8. Write the value of $\int \frac{2 - 3\sin x}{\cos^2 x} dx$. *(Delhi 2011)*

9. Write the value of $\int \sec x(\sec x + \tan x) dx$. *(Delhi 2011)*

10. Write the value of $\int (ax+b)^3 dx$. *(AI 2011)*

11. Evaluate : $\int \frac{x^3 - x^2 + x - 1}{x-1} dx$ *(Delhi 2011 C)*

SA (2 marks)

12. Find : $\int \sqrt{1 - \sin 2x}\, dx, \frac{\pi}{4} < x < \frac{\pi}{2}$ *(Delhi 2019)*

13. Evaluate : $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} dx$ *(2018)*

7.3 Methods of Integration

VSA (1 mark)

14. Find : $\int \frac{2x}{\sqrt[3]{x^2+1}} dx$ *(2020)*

15. $\int x^2 e^{x^3} dx$ equals

(a) $\frac{1}{3}e^{x^3} + C$ (b) $\frac{1}{3}e^{x^4} + C$

(c) $\frac{1}{2}e^{x^3} + C$ (d) $\frac{1}{2}e^{x^2} + C$ *(2020)*

16. Find : $\int \frac{\sin^6 x}{\cos^8 x} dx$ *(AI 2014 C)*

17. Write the value of : $\int \frac{x + \cos 6x}{3x^2 + \sin 6x} dx$ *(AI 2012 C)*

18. Evaluate : $\int \frac{(\log x)^2}{x} dx$ *(AI 2011)*

19. Evaluate : $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$ *(AI 2011)*

20. Evaluate : $\int \frac{2\cos x}{3\sin^2 x} dx$ *(AI 2011 C)*

LA1 (4 marks)

21. Integrate the function $\frac{\cos(x+a)}{\sin(x+b)}$ with respect to x. *(AI 2019)*

22. Find $\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta} d\theta$. *(Delhi 2016, 2013 C)*

23. Evaluate : $\int \frac{\sin(x-a)}{\sin(x+a)} dx$ *(Foreign 2015, Delhi 2013)*

24. Evaluate : $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$ *(Delhi 2014)*

25. Evaluate : $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos\alpha} dx$ *(AI 2013)*

26. Evaluate : $\int \sin x \sin 2x \sin 3x\, dx$ *(Delhi 2012)*

www.exambfree.in

7.4 Integrals of Some Particular Functions

VSA (1 mark)

27. Find : $\int \frac{dx}{\sqrt{9-4x^2}}$ *(2020)*

28. Find : $\int \frac{dx}{9+4x^2}$ *(2020)*

29. Write the value of $\int \frac{dx}{x^2+16}$. *(Delhi 2011)*

SA (2 marks)

30. Find : $\int \frac{\sec^2 x}{\sqrt{\tan^2 x+4}}dx$ *(Delhi 2019)*

31. Find : $\int \frac{dx}{\sqrt{5-4x-2x^2}}$ *(AI 2019)*

32. Find : $\int \frac{dx}{x^2+4x+8}$ *(Delhi 2017)*

33. Find : $\int \frac{dx}{5-8x-x^2}$ *(AI 2017)*

LA1 (4 marks)

34. Find $\int \frac{\sqrt{x}}{\sqrt{a^3-x^3}}dx$. *(Delhi 2016)*

35. Find $\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$.

(AI 2015, 2014, 2013C)

36. Evaluate : $\int \frac{x+2}{2x^2+6x+5}dx$ *(Delhi 2015C)*

37. Find $\int \frac{x+3}{\sqrt{5-4x-2x^2}}dx$. *(AI 2015C)*

38. Evaluate : $\int \frac{x+2}{\sqrt{x^2+5x+6}}dx$ *(AI 2014)*

39. Evaluate : $\int \frac{5x-2}{1+2x+3x^2}dx$

(Delhi 2014C, 2013)

40. Evaluate : $\int \frac{x+2}{\sqrt{x^2+2x+3}}dx$ *(AI 2013)*

41. Evaluate : $\int \frac{5x+3}{\sqrt{x^2+4x+10}}dx$

(AI 2012C, Delhi 2011)

42. Evaluate : $\int \frac{\sin x-\cos x}{\sqrt{\sin 2x}}dx$ *(Delhi 2011C)*

43. Evaluate : $\int \frac{x^2+1}{x^4+1}dx$ *(Delhi 2011C)*

44. Evaluate : $\int \frac{x^2+4}{x^4+16}dx$ *(AI 2011C)*

LA 2 (6 marks)

45. Evaluate : $\int \frac{1}{\cos^4 x+\sin^4 x}dx$ *(AI 2014)*

46. Evaluate : $\int \frac{1}{\sin^4 x+\sin^2 x\cos^2 x+\cos^4 x}dx$

(AI 2014)

47. Evaluate : $\int \frac{6x+7}{\sqrt{(x-5)(x-4)}}dx$ *(AI 2011)*

7.5 Integration by Partial Fractions

SA (2 marks)

48. Find $\int \frac{x+1}{(x+2)(x+3)}dx$. *(2020)*

LA 1 (4 marks)

49. Find : $\int \frac{3x+5}{x^2+3x-18}dx$ *(Delhi 2019)*

50. Find : $\int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}dx$ *(2018)*

51. Find : $\int \frac{2x}{(x^2+1)(x^2+2)^2}dx$. *(Delhi 2017)*

52. Find : $\int \frac{\cos\theta}{(4+\sin^2\theta)(5-4\cos^2\theta)}d\theta$

(AI 2017)

53. Find : $\int \frac{x^2}{x^4+x^2-2}dx$ *(AI 2016)*

54. Find : $\int \frac{(x^2+1)(x^2+4)}{(x^2+3)(x^2-5)}dx$ *(Foreign 2016)*

55. Find : $\int \frac{dx}{\sin x+\sin 2x}$ *(Delhi 2015)*

56. Evaluate : $\int \frac{x^2+x+1}{(x^2+1)(x+2)}dx$

(AI 2015, 2013C)

www.exambites.in

57. Evaluate : $\int \frac{x^2}{(x^2+4)(x^2+9)}dx$
(Foreign 2015, Delhi 2013)

58. Find : $\int \frac{x}{(x-1)^2(x+2)}dx$ *(Delhi 2015C)*

59. Find : $\int \frac{x}{(x^2+1)(x-1)}dx$
(AI 2015C, Delhi 2013C)

60. Evaluate : $\int \frac{x^2}{(x^2+1)(x^2+4)}dx$
(Delhi 2014C, AI 2013C)

61. Find : $\int \frac{x^3}{x^4+3x^2+2}dx$ *(AI 2014C)*

62. Evaluate : $\int \frac{2x^2+1}{x^2(x^2+4)}dx$ *(Delhi 2013)*

63. Evaluate : $\int \frac{x^2+1}{(x^2+4)(x^2+25)}dx$
(Delhi 2013)

64. Evaluate : $\int \frac{dx}{x(x^5+3)}$ *(AI 2013)*

65. Evaluate : $\int \frac{dx}{x(x^3+8)}$ *(AI 2013)*

66. Evaluate : $\int \frac{dx}{x(x^3+1)}$ *(AI 2013)*

67. Evaluate : $\int \frac{3x+1}{(x+1)^2(x+3)}dx$ *(Delhi 2013C)*

68. Evaluate : $\int \frac{3x+5}{x^3-x^2-x+1}dx$ *(Delhi 2013C)*

69. Evaluate : $\int \frac{8}{(x+2)(x^2+4)}dx$ *(AI 2013C)*

70. Evaluate : $\int \frac{2}{(1-x)(1+x^2)}dx$ *(Delhi 2012)*

71. Evaluate : $\int \frac{2x\,dx}{(x^2+1)(x^2+3)}$ *(Delhi 2011)*

LA 2 (6 marks)

72. Find $\int \frac{x^2+x+1}{(x+1)^2(x+2)}dx.$ *(Delhi 2014C)*

73. Evaluate : $\int \frac{x^2+1}{(x-1)^2(x+3)}dx$ *(AI 2012)*

7.6 Integration by Parts

VSA (1 mark)

74. Find : $\int x^4 \log x\,dx$ *(2020)*

75. $\int e^x\left(\frac{x\log x+1}{x}\right)dx$ is equal to
(a) $\log(e^x \log x)+c$ (b) $\frac{e^x}{x}+c$
(c) $x\log x+e^x+c$ (d) $e^x\log x+c$
(2020)

76. Given $\int e^x(\tan x+1)\sec x\,dx = e^x f(x)+C$
Write $f(x)$ satisfying above. *(AI 2012)*

SA (2 marks)

77. Find : $\int \sin^{-1}(2x)dx$ *(Delhi 2019)*

78. Find : $\int x\cdot\tan^{-1}x\,dx$ *(AI 2019)*

LA 1 (4 marks)

79. Find : $\int \sec^3 x\,dx$ *(2020)*

80. Find : $\int (3x+1)\sqrt{4-3x-2x^2}\,dx$ *(AI 2016)*

81. Find : $\int \frac{x\sin^{-1}x}{\sqrt{1-x^2}}dx$ *(Foreign 2016, AI 2012)*

82. Find : $\int (2x+5)\sqrt{10-4x-3x^2}\,dx$
(Foreign 2016)

83. Find : $\int (x+3)\sqrt{3-4x-x^2}\,dx.$
(Delhi 2015, 2014C)

84. Integrate the following w.r.t. x : $\frac{x^2-3x+1}{\sqrt{1-x^2}}$
(Delhi 2015)

85. Evaluate : $\int (3-2x)\cdot\sqrt{2+x-x^2}\,dx$ *(AI 2015)*

86. Find : $\int \frac{\log x}{(x+1)^2}dx$ *(AI 2015)*

87. Evaluate : $\int e^{2x}\cdot\sin(3x+1)dx$ *(Foreign 2015)*

www.exambites.in

88. Find : $\int \frac{(x^2+1)e^x}{(x+1)^2}dx$ *(Delhi 2015C)*

89. Evaluate : $\int (x-3)\sqrt{x^2+3x-18}\,dx$ *(Delhi 2014)*

90. Evaluate : $\int \frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$ *(Foreign 2014)*

91. Evaluate : $\int (3x-2)\sqrt{x^2+x+1}\,dx$ *(Foreign 2014, AI 2014C)*

92. Evaluate : $\int e^{2x}\left(\frac{1-\sin 2x}{1-\cos 2x}\right)dx$ *(Delhi 2013C)*

93. Evaluate : $\int \frac{\sqrt{1-\sin x}}{1+\cos x}\cdot e^{\frac{-x}{2}}dx$ *(AI 2013C)*

94. Evaluate : $\int \left(\frac{1+\sin x}{1+\cos x}\right)e^x dx$ *(Delhi 2012C)*

95. Evaluate : $\int \frac{x^2}{(x\sin x+\cos x)^2}dx$ *(Delhi 2012C)*

LA 2 (6 marks)

96. Find : $\int \frac{\sqrt{x^2+1}[\log(x^2+1)-2\log x]}{x^4}dx$ *(AI 2014C, 2012C)*

97. Find : $\int \frac{\sin^{-1}\sqrt{x}-\cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}}dx,\ x\in[0,1]$ *(AI 2014C)*

7.7 Definite Integral

LA 2 (6 marks)

98. Evaluate the following integral as the limit of sums $\int_1^4 (x^2-x)dx$. *(2020, AI 2012C)*

99. Evaluate $\int_1^3 (x^2+3x+e^x)dx$, as the limit of the sum. *(2018)*

100. Evaluate $\int_1^3 (e^{2-3x}+x^2+1)dx$ as a limit of a sum. *(Delhi 2015)*

101. Evaluate $\int_1^3 (3x^2+1)dx$ by the method of limit of sums. *(Delhi 2014C)*

102. Evaluate $\int_1^3 (2x^2+5x)\,dx$ as limit of sums. *(Delhi 2012)*

103. Evaluate $\int_2^5 (x^2+3)dx$ as limit of sums. *(Delhi 2012C)*

104. Evaluate $\int_0^2 (x^2-x)dx$ as limit of sums. *(Delhi 2011C)*

105. Evaluate $\int_0^2 (3x^2-2)dx$ as limit of sums. *(AI 2011C)*

7.8 Fundamental Theorem of Calculus

VSA (1 mark)

106. $\int_{-\pi/4}^{\pi/4} \sec^2 x\,dx$ is equal to

(a) -1 (b) 0

(c) 1 (d) 2 *(2020)*

107. Evaluate : $\int_2^3 3^x dx$ *(Delhi 2017)*

108. Evaluate : $\int_0^3 \frac{dx}{9+x^2}$ *(Delhi 2014)*

109. Evaluate : $\int_0^{\pi/2} e^x(\sin x-\cos x)dx$ *(Delhi 2014)*

110. If $f(x)=\int_0^x t\sin t\,dt$, then write the value of $f'(x)$. *(AI 2014)*

111. If $\int_0^a \frac{1}{4+x^2}dx=\frac{\pi}{8}$, find the value of a. *(AI 2014)*

112. Evaluate : $\int_0^{\pi/4} \tan x\,dx$ *(Foreign 2014)*

www.exambites.in

113. Evaluate : $\int_0^{\pi/4} \sin 2x\,dx$ *(Foreign 2014)*

114. Evaluate : $\int_0^1 \frac{1}{\sqrt{1-x^2}}dx$ *(AI 2014C, 2011C)*

115. Evaluate : $\int_1^2 \frac{x^3-1}{x^2}dx$ *(AI 2014C)*

116. Evaluate : $\int_2^3 \frac{1}{x}dx$ *(Delhi 2012)*

117. Evaluate : $\int_0^2 \sqrt{4-x^2}\,dx$ *(AI 2012)*

118. Evaluate : $\int_0^1 \frac{dx}{1+x^2}$ *(Delhi 2011C)*

SA (2 marks)

119. Evaluate $\int_1^2 \left[\frac{1}{x}-\frac{1}{2x^2}\right]e^{2x}dx$. *(2020)*

120. Find the value of $\int_0^1 \tan^{-1}\left(\frac{1-2x}{1+x-x^2}\right)dx$. *(2020)*

LA 1 (4 marks)

121. Evaluate : $\int_0^{\pi/2} x^2 \sin x\,dx$ *(Delhi 2014C)*

122. Evaluate : $\int_0^{\frac{\pi}{2}} \frac{x+\sin x}{1+\cos x}dx$ *(AI 2011)*

123. Evaluate : $\int_0^1 \frac{x^4+1}{x^2+1}dx$ *(AI 2011C)*

7.9 Evaluation of Definite Integrals by Substitution

VSA (1 mark)

124. Evaluate : $\int_2^4 \frac{x}{x^2+1}dx$ *(AI 2014)*

125. Evaluate : $\int_e^{e^2} \frac{dx}{x\log x}$. *(AI 2014)*

126. Evaluate : $\int_0^1 xe^{x^2}dx$ *(Foreign 2014)*

127. Evaluate : $\int_0^1 \frac{\tan^{-1}x}{1+x^2}dx$ *(AI 2014C)*

128. Write the value of $\int_0^1 \frac{e^x}{1+e^{2x}}dx$. *(Delhi 2012 C)*

129. Write the value of : $\int_0^1 \frac{2x}{1+x^2}dx$ *(AI 2012C, 2011C)*

SA (2 marks)

130. Find : $\int_{-\frac{\pi}{4}}^0 \frac{1+\tan x}{1-\tan x}dx$ *(AI 2019)*

LA 1 (4 marks)

131. Evaluate $\int_0^{\pi} e^{2x}\cdot\sin\left(\frac{\pi}{4}+x\right)dx$. *(Delhi 2016)*

132. Find : $\int_0^{\pi/4} \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$ *(AI 2015)*

133. Evaluate : $\int_{\pi/6}^{\pi/3} \frac{\sin x+\cos x}{\sqrt{\sin 2x}}dx$ *(AI 2014C*

134. Evaluate : $\int_0^{\pi/4} \frac{\sin 2\theta}{\sin^4\theta+\cos^4\theta}d\theta$ *(AI 2013C)*

LA 2 (6 marks)

135. Evaluate : $\int_0^{\pi/4} \frac{\sin x+\cos x}{16+9\sin 2x}dx$ *(2018)*

136. Evaluate : $\int_0^{\pi/4} \frac{\sin x+\cos x}{9+16\sin 2x}dx$

(Foreign 2014, Delhi 2014C)

137. Prove that $\int_0^{\pi/4} (\sqrt{\tan x}+\sqrt{\cot x})\,dx = \sqrt{2}\cdot\frac{\pi}{2}$

(Delhi 2012)

www.exambites.in

138. Evaluate : $\int_{\pi/4}^{\pi/2} \cos 2x \log \sin x \, dx$ *(Delhi 2012C)*

139. Evaluate : $\int_{0}^{\pi/2} 2\sin x \cos x \tan^{-1}(\sin x) dx$ *(Delhi 2011)*

7.10 Some Properties of Definite Integrals

VSA (1 mark)

140. Evaluate : $\int_{1}^{3} |2x-1| dx$ *(2020)*

141. Evaluate : $\int_{-2}^{2} |x| \, dx$ *(2020)*

142. Find the value of $\int_{1}^{4} |x-5| dx$. *(2020)*

LA 1 (4 marks)

143. Evaluate : $\int_{-1}^{2} |x^3 - x| dx$

(2020, Delhi 2016, AI 2012, 2011)

144. Prove that $\int_{0}^{a} f(x)dx = \int_{0}^{a} f(a-x)dx$, hence evaluate $\int_{0}^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$. *(Delhi 2019)*

145. Prove that $\int_{0}^{a} f(x)\, dx = \int_{0}^{a} f(a-x)dx$ and hence evaluate $\int_{0}^{1} x^2 (1-x)^n \, dx$. *(AI 2019)*

146. Evaluate : $\int_{0}^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$

(Delhi 2017, AI 2013, 2012, Delhi 2011C)

147. Evaluate : $\int_{0}^{3/2} | x \sin \pi x | \, dx$ *(Delhi 2017)*

148. Evaluate : $\int_{0}^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$

(AI 2017, Foreign 2014, Delhi 14C)

149. Evaluate : $\int_{1}^{4} \{| x-1 | + | x-2 | + | x-4 |\} dx$

(AI 2017, Delhi 2011 C)

150. Evaluate : $\int_{0}^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} dx$ *(AI 2016)*

151. Evaluate : $\int_{0}^{3/2} | x \cos \pi x | dx$ *(AI 2016)*

152. Evaluate : $\int_{0}^{\pi} \frac{x}{1+\sin \alpha \sin x} dx$ *(Foreign 2016)*

153. Evaluate : $\int_{-\pi}^{\pi} (\cos ax - \sin bx)^2 dx$ *(Delhi 2015)*

154. Evaluate : $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x} dx$ *(Foreign 2015)*

155. Evaluate : $\int_{0}^{\pi/2} \frac{dx}{1+\sqrt{\tan x}}$ *(Delhi 2015C)*

156. Evaluate : $\int_{0}^{\pi/4} \log(1+\tan x) dx$ *(AI 2015C, 2011)*

157. Evaluate : $\int_{0}^{\pi} \frac{x \tan x}{\sec x \operatorname{cosec} x} dx$

(Delhi 2014, 2011C)

158. Evaluate : $\int_{0}^{\pi} \frac{4x \sin x}{1+\cos^2 x} dx$ *(AI 2014)*

159. Show that : $\int_{0}^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} dx = \frac{1}{\sqrt{2}} \log(\sqrt{2}+1)$

(AI 2014C)

160. Evaluate : $\int_{0}^{4} (| x | + | x-2 | + | x-4 |) dx$

(Delhi 2013)

161. Evaluate : $\int_{2}^{5} [| x-2 | + | x-3 | + | x-5 |] dx$

(Delhi 2013)

www.exambites.in

162. Evaluate : $\int_1^3 [|x-1|+|x-2|+|x-3|]\,dx$ *(Delhi 2013)*

163. Evaluate : $\int_0^{2\pi} \frac{1}{1+e^{\sin x}}dx$ *(AI 2013)*

164. Evaluate : $\int_0^{\pi} \frac{x}{1+\sin x}dx$ *(AI 2012C, 2011C)*

165. Evaluate : $\int_0^1 \log\left(\frac{1}{x}-1\right)dx$ *(AI 2011)*

166. Using properties of definite integrals, evaluate the following : $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{1+\sqrt{\tan x}}dx$ *(AI 2011)*

167. Evaluate : $\int_0^1 \frac{\log(1+x)}{1+x^2}dx$ *(AI 2011C)*

LA 2 (6 marks)

168. Evaluate : $\int_0^{\pi/2} \frac{x\sin x\cos x}{\sin^4 x+\cos^4 x}dx$ *(Delhi 2014, 2011)*

169. Evaluate : $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}}$ *(Delhi 2014)*

170. Evaluate : $\int_0^{\pi} \frac{xdx}{a^2\cos^2 x+b^2\sin^2 x}$ *(Foreign 2014)*

Detailed Solutions

1. We have, $\int \frac{\sin^2 x}{\sin x\cos x}dx - \int \frac{\cos^2 x}{\sin x\cos x}dx$

$= \int \tan x dx - \int \cot x dx$

$= \ln|\sec x| - \ln|\sin x| + C = \ln\left|\frac{1}{\sin x\cos x}\right| + C$

$= \ln\left|\frac{2}{2\sin x\cos x}\right| + C = \ln|2\,\text{cosec}\,2x| + C$

2. The antiderivative of $3\sqrt{x}+\frac{1}{\sqrt{x}}$

$= \int\left(3\sqrt{x}+\frac{1}{\sqrt{x}}\right)dx = 3\int x^{1/2}dx + \int x^{-1/2}dx$

$= 3\cdot\frac{x^{3/2}}{3/2} + \frac{x^{1/2}}{1/2} + C = 2x\sqrt{x} + 2\sqrt{x} + C$

$= 2\sqrt{x}(x+1) + C$

3. $\int \cos^{-1}(\sin x)dx = \int \cos^{-1}\left[\cos\left(\frac{\pi}{2}-x\right)\right]dx$

$= \int\left(\frac{\pi}{2}-x\right)dx = \frac{\pi}{2}x - \frac{x^2}{2} + C$

4. We have $\int \frac{dx}{\sin^2 x\cos^2 x} = \int \frac{(\sin^2 x+\cos^2 x)}{\sin^2 x\cos^2 x}dx$

$= \int(\sec^2 x + \text{cosec}^2 x)dx = \tan x - \cot x + C$

5. *Refer to answer 4.*

6. $\int(1-x)\sqrt{x}\,dx = \int(\sqrt{x}-x^{3/2})dx$

$= \frac{2}{3}x^{3/2} - \frac{2}{5}x^{5/2} + C$

7. $\int \frac{\sec^2 x}{\text{cosec}^2 x}dx = \int \frac{\sin^2 x}{\cos^2 x}dx = \int \tan^2 x\,dx$

$= \int(\sec^2 x - 1)dx = \tan x - x + C$

8. $\int \frac{2-3\sin x}{\cos^2 x}dx = \int\left(\frac{2}{\cos^2 x} - \frac{3\sin x}{\cos^2 x}\right)dx$

$= \int(2\sec^2 x - 3\sec x\tan x)dx = 2\tan x - 3\sec x + C$

9. $\int \sec x(\sec x+\tan x)dx$

$= \int(\sec^2 x + \sec x\tan x)dx = \tan x + \sec x + C$

10. $\int(ax+b)^3\,dx = \frac{(ax+b)^4}{4a} + C$

11. Let $I = \int \frac{x^3-x^2+x-1}{x-1}dx$

www.examtes.in

$$= \int \frac{x^2(x-1)+1(x-1)}{x-1}dx = \int \frac{(x^2+1)(x-1)}{x-1}dx$$

$$= \int (x^2+1)dx = \frac{1}{3}x^3 + x + C$$

12. Let $I = \int (\sqrt{1-\sin 2x})\, dx$

$$= \int \sqrt{\cos^2 x + \sin^2 x - 2\sin x \cos x}\, dx$$

$$= \pm \int (\cos x - \sin x)\, dx$$

Since, $\frac{\pi}{4} < x < \frac{\pi}{2}$, so we get $I = \int (\sin x - \cos x)dx$

$= -(\cos x + \sin x) + C$

13. Let $I = \int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x}\, dx$

$$= \int \frac{\cos^2 x - \sin^2 x + 2\sin^2 x}{\cos^2 x}\, dx$$

$$= \int \frac{\cos^2 x + \sin^2 x}{\cos^2 x}\, dx = \int \frac{1}{\cos^2 x}\, dx = \int \sec^2 x\, dx$$

$= \tan x + C$

14. Let $I = \int \frac{2x}{\sqrt[3]{x^2+1}}dx$

Put $x^2 + 1 = z \Rightarrow 2x\, dx = dz$

$$\therefore \quad I = \int \frac{dz}{(z)^{1/3}} = \int z^{-1/3}dz = \frac{z^{(-1/3)+1}}{(-1/3)+1} + C$$

$$= \frac{3}{2}(x^2+1)^{2/3} + C$$

15. **(a) :** Let $I = \int x^2 e^{x^3} dx$

Put $x^3 = t \;\Rightarrow\; 3x^2 dx = dt$

$$\therefore \quad I = \int e^t \frac{dt}{3} = \frac{1}{3}e^t + C = \frac{1}{3}e^{x^3} + C$$

16. Let $I = \int \frac{\sin^6 x}{\cos^8 x}dx = \int \frac{\sin^6 x}{\cos^6 x \cdot \cos^2 x}dx$

$= \int \tan^6 x \sec^2 x dx$

Put $\tan x = t \Rightarrow \sec^2 x\, dx = dt$

$$\therefore \quad I = \int t^6 dt = \frac{t^7}{7} + C = \frac{1}{7}\tan^7 x + C$$

17. Let $I = \int \frac{x + \cos 6x}{3x^2 + \sin 6x}dx$

Put $3x^2 + \sin 6\,x = t \Rightarrow (6x + 6\cos 6x)dx = dt$

$$\Rightarrow \quad (x + \cos 6x)dx = \frac{1}{6}dt \quad \therefore \; I = \int \frac{\frac{1}{6}dt}{t} = \frac{1}{6}\log t + C$$

$$= \frac{1}{6}\log(3x^2 + \sin 6x) + C$$

18. Let $I = \int \frac{(\log x)^2}{x}dx$

Put $\log x = t \;\Rightarrow\; \frac{1}{x}dx = dt$

$$\therefore \quad I = \int t^2 dt = \frac{t^3}{3} + C = \frac{(\log x)^3}{3} + C$$

19. Let $I = \int \frac{e^{\tan^{-1}x}}{1+x^2}dx$

Put $\tan^{-1} x = t \;\Rightarrow\; \frac{1}{1+x^2}\cdot dx = dt$

$$\therefore \quad I = \int e^t dt = e^t + C = e^{\tan^{-1}x} + C$$

20. Let $I = \int \frac{2\cos x}{3\sin^2 x}dx$

Put $\sin x = t \Rightarrow \cos x dx = dt$

$$\therefore \quad I = \frac{2}{3}\int \frac{dt}{t^2} = \frac{2}{3}\int t^{-2}dt = \frac{2}{3}\cdot\frac{t^{-1}}{-1} + C = \frac{-2}{3\sin x} + C$$

21. $\int \frac{\cos(x+a)}{\sin(x+b)}dx = \int \frac{\cos(x+b+a-b)}{\sin(x+b)}dx$

$$= \int \frac{\cos(x+b)\cos(a-b) - \sin(x+b)\sin(a-b)}{\sin(x+b)}dx$$

$$= \cos(a-b)\int \frac{\cos(x+b)}{\sin(x+b)}dx - \sin(a-b)\int dx$$

$= \cos(a - b) \log \sin(x + b) - x \sin(a - b) + C$

22. Let $I = \int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta}d\theta$

$$= 3\int \frac{\sin\theta\cos\theta}{4 + \sin^2\theta - 4\sin\theta}d\theta - 2\int \frac{\cos\theta}{4 + \sin^2\theta - 4\sin\theta}d\theta$$

$= 3I_1 - 2I_2$ (say) …(i)

Now, $I_1 = \int \frac{\sin\theta\cos\theta}{4 + \sin^2\theta - 4\sin\theta}d\theta$

Put $\sin^2\theta = t \Rightarrow 2\sin\theta\cos\theta\, d\theta = dt$

$$\therefore \quad I_1 = \frac{1}{2}\int \frac{dt}{4 + t - 4\sqrt{t}} = \frac{1}{2}\int \frac{dt}{(\sqrt{t} - 2)^2}$$

Put $\sqrt{t} - 2 = u \Rightarrow \sqrt{t} = u + 2$

www.exambites.in

$\Rightarrow \quad \frac{1}{2\sqrt{t}}dt = du \Rightarrow dt = 2(u+2)du$

$\therefore \quad I_1 = \int \frac{(u+2)}{u^2}du = \int \frac{du}{u} + 2\int \frac{du}{u^2}$

$= \log u - \frac{2}{u} + C_1 = \log(\sqrt{t}-2) - \frac{2}{\sqrt{t}-2} + C_1$

$= \log(\sin\theta - 2) - \frac{2}{\sin\theta - 2} + C_1 \quad \ldots(ii)$

Also, $I_2 = \int \frac{\cos\theta}{4 + \sin^2\theta - 4\sin\theta}\, d\theta$

Put $\sin\theta = m \Rightarrow \cos\theta\, d\theta = dm$

$\therefore \quad I_2 = \int \frac{dm}{4 + m^2 - 4m} = \int \frac{dm}{(m-2)^2}$

$= \frac{-1}{m-2} + C_2 = \frac{-1}{\sin\theta - 2} + C_2 \quad \ldots(iii)$

From (i), (ii) and (iii), we get

$I = 3\log(\sin\theta - 2) - \frac{6}{\sin\theta - 2} + \frac{2}{\sin\theta - 2} + C,$

where $C = 3C_1 - 2C_2$

$\Rightarrow \quad I = 3\log(\sin\theta - 2) - \frac{4}{\sin\theta - 2} + C$

23. Let $I = \int \frac{\sin(x-a)}{\sin(x+a)}dx = \int \frac{\sin(x+a-2a)}{\sin(x+a)}dx$

$= \int \left(\frac{\sin(x+a)\cos 2a - \cos(x+a)\sin 2a}{\sin(x+a)} \right) dx$

$\Rightarrow \quad I = \cos 2a \int dx - \sin 2a \int \frac{\cos(x+a)}{\sin(x+a)}dx$

Put $\sin(x+a) = t \Rightarrow \cos(x+a)dx = dt$

$\Rightarrow \quad I = \cos 2a \int dx - \sin 2a \int \frac{dt}{t}$

$= x\cos 2a - \sin 2a \log|\sin(x+a)| + C$

24. $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x}dx$

$= \int \frac{\sin^6 x}{\sin^2 x \cdot \cos^2 x}dx + \int \frac{\cos^6 x}{\sin^2 x \cdot \cos^2 x}dx$

$= \int \frac{\sin^4 x}{\cos^2 x}dx + \int \frac{\cos^4 x}{\sin^2 x}dx$

$= \int \frac{\sin^2 x(1-\cos^2 x)}{\cos^2 x}dx + \int \frac{\cos^2 x(1-\sin^2 x)}{\sin^2 x}dx$

$= \int \tan^2 x\, dx - \int \sin^2 x\, dx + \int \cot^2 x\, dx - \int \cos^2 x\, dx$

$= \int (\sec^2 x - 1)dx + \int (\operatorname{cosec}^2 x - 1)\, dx - \int \sin^2 x\, dx - \int (1 - \sin^2 x)\, dx$

$= \tan x - x + (-\cot x) - x - x + C = \tan x - \cot x - 3x + C$

25. Here $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos\alpha}dx$

$= \int \frac{(2\cos^2 x - 1) - (2\cos^2\alpha - 1)}{\cos x - \cos\alpha}dx$

$= 2\int \frac{\cos^2 x - \cos^2\alpha}{\cos x - \cos\alpha}dx$

$= 2\int \frac{(\cos x + \cos\alpha)\cdot(\cos x - \cos\alpha)}{\cos x - \cos\alpha}dx$

$= 2\int (\cos x + \cos\alpha)dx = 2[\sin x + x\cos\alpha] + C$

26. $\int \sin x \sin 2x \sin 3x\, dx$

$= \int \sin 3x \sin x \sin 2x\, dx$

$= \frac{1}{2}\int (\cos 2x - \cos 4x)\sin 2x\, dx$

$= \frac{1}{2}\int \sin 2x \cos 2x\, dx - \frac{1}{2}\int \cos 4x \sin 2x\, dx$

$= \frac{1}{4}\int \sin 4x\, dx - \frac{1}{4}\int (\sin 6x - \sin 2x)\, dx$

$= \frac{1}{4}\int \sin 4x\, dx - \frac{1}{4}\int \sin 6x\, dx + \frac{1}{4}\int \sin 2x\, dx$

$= \frac{1}{4}\left[\frac{-\cos 4x}{4} - \frac{(-\cos 6x)}{6} + \frac{(-\cos 2x)}{2} \right] + C$

$= \frac{1}{4}\left[\frac{\cos 6x}{6} - \frac{\cos 4x}{4} - \frac{\cos 2x}{2} \right] + C$

27. Let $I = \int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2}\int \frac{dx}{\sqrt{\frac{9}{4} - x^2}}$

$= \frac{1}{2}\int \frac{dx}{\sqrt{\left(\frac{3}{2}\right)^2 - x^2}} = \frac{1}{2}\sin^{-1}\left(\frac{x}{\frac{3}{2}}\right) + C$

$= \frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right) + C$

28. Let $I = \int \frac{dx}{9+4x^2} = \frac{1}{4}\int \frac{dx}{x^2 + \frac{9}{4}} = \frac{1}{4}\int \frac{dx}{x^2 + \left(\frac{3}{2}\right)^2}$

$= \frac{1}{4}\cdot\frac{2}{3}\tan^{-1}\left(\frac{2x}{3}\right) + C = \frac{1}{6}\tan^{-1}\left(\frac{2x}{3}\right) + C$

www.examfotes.in

29. $\int \frac{dx}{x^2+(4)^2} = \frac{1}{4}\tan^{-1}\frac{x}{4} + C$

30. Let $I = \int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}}\,dx$

Put $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$\therefore \quad I = \int \frac{dt}{\sqrt{t^2+4}} = \log|t + \sqrt{t^2+4}| + C$

$= \log|\tan x + \sqrt{\tan^2 x + 4}| + C$

31. Let $I = \int \frac{dx}{\sqrt{5-4x-2x^2}} = \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\frac{5}{2}-2x-x^2}}$

$= \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\frac{7}{2}-1-2x-x^2}} = \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\left(\sqrt{\frac{7}{2}}\right)^2-(x+1)^2}}$

$= \frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x+1}{\sqrt{\frac{7}{2}}}\right) + C = \frac{1}{\sqrt{2}}\sin^{-1}\left[\sqrt{\frac{2}{7}}(x+1)\right] + C$

32. We have, $\int \frac{dx}{x^2+4x+8} = \int \frac{dx}{x^2+4x+4+4}$

$= \int \frac{dx}{(x+2)^2+(2)^2} = \frac{1}{2}\tan^{-1}\left(\frac{x+2}{2}\right) + C$

33. Let $I = \int \frac{dx}{5-8x-x^2}$

$= \int \frac{dx}{5+16-16-8x-x^2} = \int \frac{dx}{21-(x+4)^2}$

$= \int \frac{dx}{(\sqrt{21})^2-(x+4)^2}$

$= \frac{1}{2\sqrt{21}}\log\left|\frac{\sqrt{21}+x+4}{\sqrt{21}-x-4}\right| + C$

$\left[\because \quad \int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + C\right]$

34. Let $I = \int \frac{\sqrt{x}}{\sqrt{a^3-x^3}}\,dx$

Put $x^{3/2} = t \Rightarrow \frac{3}{2}x^{1/2}dx = dt$

$\therefore \quad I = \frac{2}{3}\int \frac{dt}{\sqrt{a^3-t^2}} = \frac{2}{3}\int \frac{dt}{\sqrt{(a^{3/2})^2-t^2}}$

$= \frac{2}{3}\left[\sin^{-1}\left(\frac{t}{a^{3/2}}\right)\right] + C = \frac{2}{3}\left[\sin^{-1}\left(\frac{x^{3/2}}{a^{3/2}}\right)\right] + C$

$= \frac{2}{3}\sin^{-1}\left(\frac{x}{a}\right)^{3/2} + C$

35. Let $I = \int(\sqrt{\tan x} + \sqrt{\cot x})\,dx$

$= \int\left(\sqrt{\frac{\sin x}{\cos x}} + \sqrt{\frac{\cos x}{\sin x}}\right)dx = \int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\,dx$

$= \sqrt{2}\int \frac{\sin x + \cos x}{\sqrt{2\sin x\cos x}}\,dx = \sqrt{2}\int \frac{\sin x + \cos x}{\sqrt{\sin 2x + 1 - 1}}\,dx$

$= \sqrt{2}\int \frac{\sin x + \cos x}{\sqrt{1-(1-\sin 2x)}}\,dx = \sqrt{2}\int \frac{\sin x + \cos x}{\sqrt{1-(\sin x - \cos x)^2}}\,dx$

Put $\sin x - \cos x = t \Rightarrow (\cos x + \sin x)\,dx = dt$

$\therefore \quad I = \sqrt{2}\int \frac{dt}{\sqrt{1-t^2}} = \sqrt{2}\sin^{-1} t + C$

$= \sqrt{2}\sin^{-1}(\sin x - \cos x) + C$

36. Let $I = \int \frac{x+2}{2x^2+6x+5}\,dx$

We write, $x + 2 = A\left[\frac{d}{dx}(2x^2+6x+5)\right] + B$

$\Rightarrow \quad x + 2 = A(4x + 6) + B$

Equating coefficients of x and constant terms, we get

$4A = 1 \Rightarrow A = 1/4$ and $6A + B = 2 \Rightarrow B = 1/2$

$\therefore \quad I = \frac{1}{4}\int \frac{4x+6}{2x^2+6x+5}\,dx + \frac{1}{2}\int \frac{dx}{2x^2+6x+5}$

$= \frac{1}{4}\log\left|2x^2+6x+5\right| + \frac{1}{4}\int \frac{dx}{x^2+3x+\frac{5}{2}}$

$= \frac{1}{4}\log\left|2x^2+6x+5\right| + \frac{1}{4}\int \frac{dx}{x^2+3x+\frac{9}{4}-\frac{9}{4}+\frac{5}{2}}$

$= \frac{1}{4}\log\left|2x^2+6x+5\right| + \frac{1}{4}\int \frac{dx}{\left(x+\frac{3}{2}\right)^2+\left(\frac{1}{2}\right)^2}$

$= \frac{1}{4}\log\left|2x^2+6x+5\right| + \frac{1}{2}\tan^{-1}\left(\frac{x+\frac{3}{2}}{\frac{1}{2}}\right) + C$

$= \frac{1}{4}\log\left|2x^2+6x+5\right| + \frac{1}{2}\tan^{-1}(2x+3) + C$

www.exambits.in

37. Let $I=\int\frac{x+3}{\sqrt{5-4x-2x^2}}dx$

Let $x+3=A\frac{d}{dx}(5-4x-2x^2)+B=A(-4-4x)+B$

On comparing the like coefficients, we get

$x+3=-\frac{1}{4}(-4-4x)+2$

$\Rightarrow \quad I=\int\frac{-\frac{1}{4}(-4-4x)+2}{\sqrt{5-4x-2x^2}}dx$

$=\frac{-1}{4}\int\frac{-4-4x}{\sqrt{5-4x-2x^2}}dx+2\cdot\int\frac{1}{\sqrt{5-4x-2x^2}}dx$

$\Rightarrow \quad -\frac{1}{4}I_1+2I_2$ ...(1)

where $I_1=\int\frac{-4-4x}{\sqrt{5-4x-2x^2}}dx$

Put $5-4x-2x^2=t\Rightarrow(-4-4x)dx=dt$

$\therefore \quad I_1=\int\frac{dt}{\sqrt{t}}=\int t^{-1/2}dt=2\sqrt{t}+C_1$

$=2\sqrt{5-4x-2x^2}+C_1$ ...(2)

and $I_2=\int\frac{dx}{\sqrt{5-4x-2x^2}}=\frac{1}{\sqrt{2}}\int\frac{dx}{\sqrt{\frac{5}{2}-2x-x^2}}$

$=\frac{1}{\sqrt{2}}\int\frac{dx}{\sqrt{\left(\sqrt{\frac{7}{2}}\right)^2-(x+1)^2}}=\frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x+1}{\sqrt{7/2}}\right)+C_2$ ...(3)

From (1), (2) and (3), we get

$I=-\frac{1}{4}\cdot2\sqrt{5-4x-2x^2}+2\cdot\frac{1}{\sqrt{2}}\sin^{-1}\left[\sqrt{\frac{2}{7}}(x+1)\right]+C$

where $C=C_1+C_2$

$\Rightarrow \quad I=-\frac{1}{2}\sqrt{5-4x-2x^2}+\sqrt{2}\sin^{-1}\left[\sqrt{\frac{2}{7}}(x+1)\right]+C$

38. Let $I=\int\frac{x+2}{\sqrt{x^2+5x+6}}dx=\int\frac{\frac{1}{2}(2x+5)-\frac{1}{2}}{\sqrt{x^2+5x+6}}dx$

$=\frac{1}{2}\int(x^2+5x+6)^{-1/2}(2x+5)dx-\frac{1}{2}\int\frac{dx}{\sqrt{x^2+5x+6}}$

Put $x^2+5x+6=t \Rightarrow (2x+5)\,dx=dt$

$\Rightarrow \quad I=\frac{1}{2}\int t^{-1/2}dt-\frac{1}{2}\int\frac{dx}{\sqrt{\left(x+\frac{5}{2}\right)^2-\left(\frac{1}{2}\right)^2}}$

$=\frac{1}{2}\frac{t^{1/2}}{\frac{1}{2}}-\frac{1}{2}\log\left|\left(x+\frac{5}{2}\right)+\sqrt{\left(x+\frac{5}{2}\right)^2-\left(\frac{1}{2}\right)^2}\right|+C$

$=\sqrt{x^2+5x+6}-\frac{1}{2}\log\left|x+\frac{5}{2}+\sqrt{x^2+5x+6}\right|+C$

39. $I=\int\frac{5x-2}{1+2x+3x^2}dx$

Let $5x-2=A\frac{d}{dx}(1+2x+3x^2)+B=A(2+6x)+B$

On comparing the like coefficients, we get

$5x-2=\frac{5}{6}(2+6x)-\frac{5}{3}-2=\frac{5}{6}\frac{d}{dx}(1+2x+3x^2)-\frac{11}{3}$

$\therefore \quad I=\int\frac{\frac{5}{6}\frac{d}{dx}(1+2x+3x^2)-\frac{11}{3}}{1+2x+3x^2}dx$

$=\frac{5}{6}\int\frac{\frac{d}{dx}(1+2x+3x^2)}{1+2x+3x^2}dx-\frac{11}{3}\int\frac{dx}{1+2x+3x^2}$

$=\frac{5}{6}\log|1+2x+3x^2|-\frac{11}{3\cdot3}\int\frac{dx}{x^2+\frac{2}{3}x+\frac{1}{3}}$

$=\frac{5}{6}\log|1+2x+3x^2|-\frac{11}{9}\int\frac{dx}{\left(x+\frac{1}{3}\right)^2+\left(\frac{\sqrt{2}}{3}\right)^2}$

$=\frac{5}{6}\log|1+2x+3x^2|-\frac{11}{9}\cdot\frac{1}{\sqrt{2}/3}\tan^{-1}\left(\frac{x+\frac{1}{3}}{\sqrt{2}/3}\right)+C$

$=\frac{5}{6}\log|1+2x+3x^2|-\frac{11}{3\sqrt{2}}\tan^{-1}\left(\frac{3x+1}{\sqrt{2}}\right)+C$

40. Let $I=\int\frac{x+2}{\sqrt{x^2+2x+3}}dx=\int\frac{(x+1)+1}{\sqrt{x^2+2x+3}}dx$

Put $x^2+2x+3=t$

$\Rightarrow \quad (2x+2)dx=dt \Rightarrow (x+1)dx=\frac{1}{2}dt$

$\therefore \quad I=\frac{1}{2}\int t^{-1/2}dt+\int\frac{1}{\sqrt{(x+1)^2+(\sqrt{2})^2}}dx$

www.exambites.in

$$= \frac{1}{2}\frac{t^{1/2}}{(1/2)} + \log\left|(x+1)+\sqrt{x^2+2x+3}\right| + C$$

$$\left[\because \int \frac{1}{\sqrt{x^2+a^2}}dx = \log\left|x+\sqrt{x^2+a^2}\right| + C\right]$$

$$= \sqrt{x^2+2x+3} + \log\left|(x+1)+\sqrt{x^2+2x+3}\right| + C$$

41. Let $I = \int \frac{5x+3}{\sqrt{x^2+4x+10}}dx = \int \frac{\frac{5}{2}(2x+4)-7}{\sqrt{x^2+4x+10}}dx$

$$= \frac{5}{2}\int \frac{2x+4}{\sqrt{x^2+4x+10}}dx - 7\int \frac{dx}{\sqrt{x^2+4x+10}}$$

$= I_1 + I_2$ (say) ...(1)

where $I_1 = \frac{5}{2}\int \frac{2x+4}{\sqrt{x^2+4x+10}}dx$

Put $x^2 + 4x + 10 = t \Rightarrow (2x+4)dx = dt$

$$\therefore \quad I_1 = \frac{5}{2}\int t^{-1/2}dt = \frac{5}{2}\cdot\frac{t^{1/2}}{(1/2)} = 5\sqrt{t}$$

$$= 5\sqrt{x^2+4x+10} + C_1 \quad ...(2)$$

and $I_2 = -7\int \frac{dx}{\sqrt{x^2+4x+10}}$

$$= -7\int \frac{dx}{\sqrt{(x+2)^2+(\sqrt{6})^2}}$$

$$= -7\log\left|x+2+\sqrt{x^2+4x+10}\right| + C_2 \quad ...(3)$$

From (1), (2) and (3), we get

$I = 5\sqrt{x^2+4x+10} - 7\log\left|x+2+\sqrt{x^2+4x+10}\right| + C$,

where $C = C_1 + C_2$

42. Let $I = \int \frac{\sin x - \cos x}{\sqrt{\sin 2x}}dx = \int \frac{\sin x - \cos x}{\sqrt{1+\sin 2x - 1}}dx$

$$= \int \frac{\sin x - \cos x}{\sqrt{\sin^2 x + \cos^2 x + 2\sin x\cos x - 1}}dx$$

$$= \int \frac{\sin x - \cos x}{\sqrt{(\sin x + \cos x)^2 - 1}}\cdot dx$$

Put $\sin x + \cos x = t \Rightarrow (\cos x - \sin x)\,dx = dt$

$$\therefore \quad I = \int \frac{-dt}{\sqrt{t^2-1}} = -\log\left|t+\sqrt{t^2-1}\right| + C$$

(where $t = \sin x + \cos x$)

$$= -\log\left|\sin x + \cos x + \sqrt{\sin 2x}\right| + C$$

43. Let $I = \int \frac{x^2+1}{x^4+1}dx$

$$= \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx = \int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}dx = \int \frac{dt}{t^2+\left(\sqrt{2}\right)^2}$$

$$\left[\because x - \frac{1}{x} = t \Rightarrow \left(1+\frac{1}{x^2}\right)dx = dt\right]$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\frac{t}{\sqrt{2}} + C = \frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(x-\frac{1}{x}\right)}{\sqrt{2}} + C$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) + C$$

44. *Refer to answer 43.*

45. Let $I = \int \frac{1}{\cos^4 x + \sin^4 x}dx = \int \frac{\sec^4 x}{1+\tan^4 x}dx$

$$= \int \frac{(\tan^2 x + 1)\sec^2 x}{1+\tan^4 x}dx$$

Put $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$$\therefore \quad I = \int \frac{t^2+1}{t^4+1}dt = \int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt = \int \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2}dt$$

Put $t - \frac{1}{t} = y \Rightarrow \left(1+\frac{1}{t^2}\right)dt = dy$

$$\therefore \quad I = \int \frac{dy}{y^2+(\sqrt{2})^2} = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{y}{\sqrt{2}}\right) + C$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan x - \cot x}{\sqrt{2}}\right) + C$$

46. Let $I = \int \frac{1}{\sin^4 x + \sin^2 x\cos^2 x + \cos^4 x}dx$

$$= \int \frac{\sec^4 x\,dx}{\tan^4 x + \tan^2 x + 1} = \int \frac{(1+\tan^2 x)\sec^2 x dx}{\tan^4 x + \tan^2 x + 1}$$

Put $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$$I = \int \frac{1+t^2}{t^4+t^2+1}dt = \int \frac{1+\frac{1}{t^2}}{t^2+1+\frac{1}{t^2}}dt$$

www.exambites.in

$$= \int \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+3}dt$$

Put $t-\frac{1}{t}=y \Rightarrow \left(1+\frac{1}{t^2}\right)dt = dy$

Thus, $I = \int \frac{dy}{y^2+(\sqrt{3})^2} = \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{y}{\sqrt{3}}\right)+C$

$$= \frac{1}{\sqrt{3}}\tan^{-1}\frac{\left(t-\frac{1}{t}\right)}{\sqrt{3}}+C = \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{\tan x-\cot x}{\sqrt{3}}\right)+C$$

47. Let $I = \int \frac{6x+7}{\sqrt{(x-5)(x-4)}}dx = \int \frac{6x+7}{\sqrt{x^2-9x+20}}dx$

Let $6x+7 = A\left[\frac{d}{dx}(x^2-9x+20)\right]+B$

$\therefore \quad 6x + 7 = A[2x - 9] + B$

Equating the coefficients of like terms from both sides, we get

$2A = 6$ and $-9A + B = 7$

$\Rightarrow \quad A = 3$ and $-9(3) + B = 7$

$\Rightarrow \quad B = 7 + 27 = 34$

$$\therefore \quad I = \int \frac{3(2x-9)}{\sqrt{x^2-9x+20}}dx + \int \frac{34}{\sqrt{x^2-9x+20}}dx$$

Put $x^2 - 9x + 20 = t$ in first integral

$$\therefore \quad I = \int \frac{3}{\sqrt{t}}dt + 34\int \frac{dx}{\sqrt{\left(x-\frac{9}{2}\right)^2+20-\frac{81}{4}}}$$

$$= 3\int t^{-1/2}dt + 34\int \frac{dx}{\sqrt{\left(x-\frac{9}{2}\right)^2-\frac{1}{4}}}$$

$$= 3\frac{t^{1/2}}{1/2} + 34\int \frac{dx}{\sqrt{\left(x-\frac{9}{2}\right)^2-\left(\frac{1}{2}\right)^2}}$$

$$= 6\sqrt{t} + 34\log\left|\left(x-\frac{9}{2}\right)+\sqrt{\left(x-\frac{9}{2}\right)^2-\left(\frac{1}{2}\right)^2}\right|+C$$

$$= 6\sqrt{x^2-9x+20} + 34\log\left|\left(x-\frac{9}{2}\right)+\sqrt{x^2-9x+20}\right|+C$$

48. Let $I = \int \frac{(x+1)}{(x+2)(x+3)}dx$

Also let, $\frac{(x+1)}{(x+2)(x+3)} = \frac{A}{(x+2)} + \frac{B}{(x+3)}$

$\Rightarrow \quad x + 1 = A(x + 3) + B(x + 2)$ ...(i)

Putting $x = -3$ in (i), we get

$-B = -3 + 1 = -2 \Rightarrow B = 2$

Putting $x = -2$ in (i), we get

$A = -2 + 1 = -1$

$$\therefore \quad I = \int \frac{-1}{(x+2)}dx + 2\int \frac{1}{(x+3)}dx$$

$$= -\log(x + 2) + 2\log(x + 3) + C$$

49. Let $I = \int \frac{3x+5}{x^2+3x-18}dx = \int \frac{3x+5}{(x+6)(x-3)}dx$

Let $\frac{3x+5}{(x+6)(x-3)} = \frac{A}{(x+6)} + \frac{B}{(x-3)}$

$\Rightarrow \quad 3x + 5 = A(x - 3) + B(x + 6)$ ...(i)

Putting $x = 3$ in (i), we get $9B = 14 \quad \Rightarrow \quad B = \frac{14}{9}$

Putting $x = -6$ in (i), we get $-9A = -13 \Rightarrow A = \frac{13}{9}$

$$\therefore \quad I = \frac{13}{9}\int \frac{1}{(x+6)}dx + \frac{14}{9}\int \frac{1}{(x-3)}dx$$

$$= \frac{13}{9}\log(x+6) + \frac{14}{9}\log(x-3) + C$$

50. Let $I = \int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}dx$

Put $\sin x = t \Rightarrow \cos x\, dx = dt$

$$\therefore \quad I = 2\int \frac{1}{(1-t)(1+t^2)}dt \qquad ...(i)$$

Now, $\frac{1}{(1-t)(1+t^2)} = \frac{A}{(1-t)} + \frac{Bt+C}{(1+t^2)}$

$\Rightarrow \quad 1 = A(1 + t^2) + (Bt + C)(1 - t)$

$\Rightarrow \quad 1 = A + At^2 + Bt - Bt^2 + C - Ct$

$\Rightarrow \quad 1 = (A + C) + t^2(A - B) + t(B - C)$

On equating coefficients of like terms on both sides, we get

$A + C = 1, A - B = 0 \Rightarrow A = B$

$B - C = 0 \Rightarrow B = C$

$\therefore \quad A = B = C$

$\therefore \quad A = \frac{1}{2}, B = \frac{1}{2}, C = \frac{1}{2}$

$$\therefore \quad \frac{1}{(1-t)(1+t^2)} = \frac{1}{2(1-t)} + \frac{1}{2}\frac{t}{(1+t^2)} + \frac{1}{2(1+t^2)}$$

www.exambites.in

Put above equation in (i), we get

$$I = 2\left[\int \frac{1}{2(1-t)}dt + \int \frac{1}{2}\frac{t}{(1+t^2)}dt + \int \frac{1}{2(1+t^2)}dt\right]$$

$$= \int \frac{dt}{(1-t)} + \int \frac{t}{(1+t^2)}dt + \int \frac{dt}{(1+t^2)}$$

$$= -\log(1-t) + \frac{1}{2}\log(1+t^2) + \tan^{-1}(t) + C$$

$$= -\log(1-\sin x) + \frac{1}{2}\log(1+\sin^2 x) + \tan^{-1}(\sin x) + C$$

$$= -\log(1-\sin x) + \log(1+\sin^2 x)^{1/2} + \tan^{-1}(\sin x) + C$$

$$= \log\frac{\sqrt{1+\sin^2 x}}{1-\sin x} + \tan^{-1}(\sin x) + C$$

51. Let $I = \int \frac{2x}{(x^2+1)(x^2+2)^2}dx$

Put $x^2 = y \Rightarrow 2x\,dx = dy$

$$\therefore \quad I = \int \frac{dy}{(y+1)(y+2)^2}$$

Let $\frac{1}{(y+1)(y+2)^2} = \frac{A}{y+1} + \frac{B}{y+2} + \frac{C}{(y+2)^2}$

$\Rightarrow \quad 1 = A(y+2)^2 + B(y+1)(y+2) + C(y+1)$

For $y = -1$, $1 = A$

For $y = -2$, $1 = -C \Rightarrow C = -1$

For $y = 0$, $1 = 4A + 2B + C \Rightarrow B = \frac{1-4+1}{2} = -1$

$$\therefore \quad I = \int\left[\frac{1}{y+1} - \frac{1}{y+2} - \frac{1}{(y+2)^2}\right]dy$$

$$= \log(y+1) - \log(y+2) + \frac{1}{y+2} + c$$

$$= \log\left(\frac{y+1}{y+2}\right) + \frac{1}{y+2} + c$$

$$= \log\left(\frac{x^2+1}{x^2+2}\right) + \frac{1}{x^2+2} + c \qquad [\because y = x^2]$$

52. $I = \int \frac{\cos\theta}{(4+\sin^2\theta)(5-4\cos^2\theta)}d\theta$

$$= \int \frac{\cos\theta}{(4+\sin^2\theta)(1+4\sin^2\theta)}d\theta$$

Let $\sin\theta = t \Rightarrow \cos\theta\, d\theta = dt$

$$\therefore \quad I = \int \frac{1}{(4+t^2)(1+4t^2)}dt$$

Consider, $\frac{1}{(4+t^2)(1+4t^2)} = \frac{At+B}{4+t^2} + \frac{Ct+D}{1+4t^2}$

(Using partial fraction)

$1 = (At + B)(1 + 4t^2) + (Ct + D)(4 + t^2)$

$= At + B + 4At^3 + 4Bt^2 + 4Ct + Ct^3 + 4D + Dt^2$

$= (4A + C)t^3 + (4B + D)t^2 + (A + 4C)t + (B + 4D)$

Comparing coeff. of like powers, we get

$4A + C = 0$ ...(i)

$4B + D = 0$ ...(ii)

$A + 4C = 0$ ...(iii)

$B + 4D = 1$ ...(iv)

Solving (i) & (iii), we get

$A = 0$ and $C = 0$

Solving (ii) & (iv), we get

$B = \frac{-1}{15}$ and $D = \frac{4}{15}$

$$\therefore \quad \frac{1}{(4+t^2)(1+4t^2)} = \frac{-1/15}{4+t^2} + \frac{4/15}{1+4t^2}$$

$$\therefore \quad I = -\frac{1}{15}\int \frac{1}{4+t^2}dt + \frac{4}{15}\times\frac{1}{4}\int \frac{1}{\frac{1}{4}+t^2}dt$$

$$= -\frac{1}{15}\times\frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right) + \frac{1}{15}\times\frac{1}{1/2}\tan^{-1}\left(\frac{t}{1/2}\right) + C$$

$$= -\frac{1}{30}\tan^{-1}\left(\frac{t}{2}\right) + \frac{2}{15}\tan^{-1}(2t) + C$$

$$= \frac{2}{15}\tan^{-1}(2\sin\theta) - \frac{1}{30}\tan^{-1}\left(\frac{\sin\theta}{2}\right) + C$$

53. Let $I = \int \frac{x^2}{x^4+x^2-2}dx = \int \frac{x^2}{(x^2-1)(x^2+2)}dx$

Let $x^2 = z$

$$\therefore \quad \frac{x^2}{(x^2-1)(x^2+2)} = \frac{z}{(z-1)(z+2)}$$

Using partial fractions, we have

$$\frac{z}{(z-1)(z+2)} = \frac{A}{z-1} + \frac{B}{z+2}$$

$\Rightarrow \quad z = A(z+2) + B(z-1)$

when $z = 1$, we get $A = \frac{1}{3}$

and when $z = -2$, we get $B = \frac{2}{3}$

$$\therefore \quad I = \int \frac{x^2}{(x^2-1)(x^2+2)}dx$$

www.exambites.in

$$= \int \frac{1/3}{(x^2-1)}\,dx + \int \frac{2/3}{(x^2+2)}\,dx$$

$$= \frac{1}{3}\int \frac{1}{x^2-1}\,dx + \frac{2}{3}\int \frac{1}{x^2+(\sqrt{2})^2}\,dx$$

$$= \frac{1}{3}\times\frac{1}{2}\log\left|\frac{x-1}{x+1}\right| + \frac{2}{3}\times\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right)+C$$

$$= \frac{1}{6}\log\left|\frac{x-1}{x+1}\right| + \frac{\sqrt{2}}{3}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right)+C$$

54. Let $I = \int \frac{(x^2+1)(x^2+4)}{(x^2+3)(x^2-5)}dx$

Let $x^2 = t$

$$\therefore \quad \frac{(x^2+1)(x^2+4)}{(x^2+3)(x^2-5)} = \frac{(t+1)(t+4)}{(t+3)(t-5)}$$

$$= \frac{t^2+5t+4}{(t+3)(t-5)} = 1 + \frac{7t+19}{(t+3)(t-5)}$$

Let $\frac{7t+19}{(t+3)(t-5)} = \frac{A}{t+3} + \frac{B}{t-5}$

$\Rightarrow \quad 7t+19 = A(t-5) + B(t+3)$

Putting $t = 5$, we get $B = \frac{27}{4}$

Putting $t = -3$, we get $A = \frac{1}{4}$

$$\therefore \quad \frac{t^2+5t+4}{(t+3)(t-5)} = 1 + \frac{1}{4(t+3)} + \frac{27}{4(t-5)}$$

$$\Rightarrow \quad I = \int \frac{(x^2+1)(x^2+4)}{(x^2+3)(x^2-5)}dx = \int dx + \frac{1}{4}\int \frac{1}{(x^2+3)}dx + \frac{27}{4}\int \frac{1}{(x^2-5)}dx$$

$$= x + \frac{1}{4\sqrt{3}}\tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + \frac{27}{4}\times\frac{1}{2\sqrt{5}}\log\left|\frac{x-\sqrt{5}}{x+\sqrt{5}}\right| + C$$

$$= x + \frac{1}{4\sqrt{3}}\tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + \frac{27}{8\sqrt{5}}\log\left|\frac{x-\sqrt{5}}{x+\sqrt{5}}\right| + C$$

55. Let $I = \int \frac{1}{\sin x + \sin 2x}dx$

$$= \int \frac{1}{\sin x + 2\sin x\cos x}dx = \int \frac{1}{\sin x(1+2\cos x)}dx$$

$$= \int \frac{\sin x}{\sin^2 x(1+2\cos x)}dx$$

Let $u = \cos x \Rightarrow du = -\sin x\,dx$

Also, $\sin^2 x = 1 - \cos^2 x = 1 - u^2$

$$\therefore \quad I = \int \frac{-1}{(1-u^2)(1+2u)}du$$

$$= \int \frac{-1}{(1+u)(1-u)(1+2u)}du$$

Using partial fractions, we have

$$\frac{-1}{(1+u)(1-u)(1+2u)} = \frac{A}{(1+u)} + \frac{B}{(1-u)} + \frac{C}{(1+2u)}$$

$\Rightarrow \quad -1 = A(1-u)(1+2u) + B(1+u)(1+2u) + C(1+u)(1-u)$

Put $u = 1$, we get $B = -1/6$

Put $u = -1$, we get $A = 1/2$

Put $u = -\frac{1}{2}$, we get $C = \frac{-4}{3}$

So, $\frac{-1}{(1+u)(1-u)(1+2u)}$

$$= \frac{1}{2(1+u)} - \frac{1}{6(1-u)} - \frac{4}{3(1+2u)}$$

$$\Rightarrow \quad I = \int\left[\frac{1}{2(1+u)} - \frac{1}{6(1-u)} - \frac{4}{3(1+2u)}\right]du$$

$$= \frac{1}{2}\log(1+u) + \frac{1}{6}\log(1-u) - \frac{4}{3\times 2}\log(1+2u) + C_1$$

$$= \frac{1}{2}\log(1+\cos x) + \frac{1}{6}\log(1-\cos x) - \frac{2}{3}\log(1+2\cos x) + C_1$$

56. Let $I = \int \frac{x^2+x+1}{(x^2+1)(x+2)}dx$ ...(1)

Let $\frac{x^2+x+1}{(x^2+1)(x+2)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+2}$ ...(2)

$\Rightarrow \quad x^2 + x + 1 = (Ax+B)(x+2) + C(x^2+1)$

Put $x = 0, 1, -2$ in it to get

$1 = 2B + C$; $3 = 3(A+B) + 2C$ and $3 = 5C$

$\Rightarrow \quad C = \frac{3}{5}, B = \frac{1}{5}$ and $A = \frac{2}{5}$

Hence, from (2)

$$\frac{x^2+x+1}{(x^2+1)(x+2)} = \frac{\left(\frac{2}{5}x + \frac{1}{5}\right)}{x^2+1} + \frac{\frac{3}{5}}{x+2}$$

$$= \frac{1}{5}\cdot\frac{2x+1}{x^2+1} + \frac{3}{5}\cdot\frac{1}{x+2}$$

$$\therefore \quad I = \frac{1}{5}\int \frac{2x+1}{x^2+1}dx + \frac{3}{5}\int \frac{1}{x+2}dx$$

www.exambites.in

$$=\frac{1}{5}\left[\int\frac{2x}{x^2+1}dx+\int\frac{dx}{x^2+1}\right]+\frac{3}{5}\int\frac{dx}{x+2}$$

$$=\frac{1}{5}\left[\log|x^2+1|+\tan^{-1}x\right]+\frac{3}{5}\log|x+2|+C_1$$

57. Let $I=\int\frac{x^2}{(x^2+4)(x^2+9)}dx$

Put $x^2=y$. Then $\frac{x^2}{(x^2+4)(x^2+9)}=\frac{y}{(y+4)(y+9)}$

Let $\frac{y}{(y+4)(y+9)}=\frac{A}{y+4}+\frac{B}{y+9}$ ...(1)

$\Rightarrow$ $y=A(y+9)+B(y+4)$ ...(2)

Putting $y=-4$ and $y=-9$ successively in (2), we get

$A=\frac{-4}{5}$ and $B=\frac{9}{5}$

Substituting the values of A and B in (1), we get

$$\frac{y}{(y+4)(y+9)}=\frac{-4/5}{(y+4)}+\frac{9/5}{(y+9)}$$

$$\Rightarrow\quad\frac{x^2}{(x^2+4)(x^2+9)}=\frac{-4}{5(x^2+4)}+\frac{9}{5(x^2+9)}$$

$$\therefore\quad I=\int\frac{x^2}{(x^2+4)(x^2+9)}dx$$

$$=\frac{-4}{5}\int\frac{1}{(x^2+4)}dx+\frac{9}{5}\int\frac{1}{(x^2+9)}dx$$

$$=\frac{-4}{5}\times\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)+\frac{9}{5}\times\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)+C$$

$$=\frac{-2}{5}\tan^{-1}\left(\frac{x}{2}\right)+\frac{3}{5}\tan^{-1}\left(\frac{x}{3}\right)+C$$

58. Let $I=\int\frac{x}{(x-1)^2(x+2)}dx$. ...(1)

Let $\frac{x}{(x-1)^2(x+2)}=\frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{x+2}$ ...(2)

$\Rightarrow$ $x=A(x-1)(x+2)+B(x+2)+C(x-1)^2$ ...(3)

Comparing coeffs. of x^2, x and constants in (3), we get $0=A+C$; $1=A+B-2C$; $0=-2A+2B+C$

Solving these, we get

$A=\frac{2}{9}$, $B=\frac{1}{3}$, $C=-\frac{2}{9}$

$$\therefore\quad\frac{x}{(x-1)^2(x+2)}=\frac{2}{9}\cdot\frac{1}{x-1}+\frac{1}{3}\cdot\frac{1}{(x-1)^2}-\frac{2}{9}\cdot\frac{1}{x+2}$$

$$\therefore\quad I=\frac{2}{9}\int\frac{1}{x-1}dx+\frac{1}{3}\int\frac{1}{(x-1)^2}dx-\frac{2}{9}\int\frac{1}{x+2}dx$$

$$=\frac{2}{9}\log\left|\frac{x-1}{x+2}\right|-\frac{1}{3}\cdot\frac{1}{x-1}+C_1$$

59. Let $I=\int\frac{x}{(x^2+1)(x-1)}dx$

Let $\frac{x}{(x^2+1)(x-1)}=\frac{Ax+B}{x^2+1}+\frac{C}{x-1}$ ...(1)

$\Rightarrow$ $x=(Ax+B)(x-1)+C(x^2+1)$ ...(2)

Comparing coefficients of x^2, x and constant terms, we get

$0=A+C$; $1=B-A$; $0=-B+C$

Solving these, we get

$A=-\frac{1}{2}, C=\frac{1}{2}, B=\frac{1}{2}$

$\therefore$ From (1), we get

$$\frac{x}{(x^2+1)(x-1)}=\frac{-\frac{1}{2}(x-1)}{x^2+1}+\frac{1}{2}\cdot\frac{1}{x-1}$$

$$=-\frac{1}{2}\cdot\frac{x}{x^2+1}+\frac{1}{2}\cdot\frac{1}{x^2+1}+\frac{1}{2}\cdot\frac{1}{x-1}$$

$$\therefore\quad I=-\frac{1}{4}\int\frac{2x}{x^2+1}dx+\frac{1}{2}\int\frac{dx}{x^2+1}+\frac{1}{2}\int\frac{dx}{x-1}$$

$$\Rightarrow\quad I=-\frac{1}{4}\log|x^2+1|+\frac{1}{2}\tan^{-1}x+\frac{1}{2}\log|x-1|+C_1$$

60. *Refer to answer 57.*

61. $I=\int\frac{x^3}{x^4+3x^2+2}dx$

Put $x^2=t\Rightarrow xdx=\frac{1}{2}dt$

$$\therefore\quad I=\frac{1}{2}\int\frac{t}{t^2+3t+2}dt=\frac{1}{2}\int\frac{t}{(t+2)(t+1)}dt$$

Let $\frac{t}{(t+2)(t+1)}=\frac{A}{(t+2)}+\frac{B}{(t+1)}$

$\Rightarrow$ $t=A(t+1)+B(t+2)$

Put $t=-1, -2$ in it, we get $A=2$, $B=-1$

$$\therefore\quad\frac{t}{(t+2)(t+1)}=\frac{2}{t+2}-\frac{1}{t+1}$$

www.exambits.in

$$\Rightarrow \quad I=\frac{1}{2}\int\left[\frac{2}{(t+2)}-\frac{1}{(t+1)}\right]dt$$

$$=\frac{1}{2}\left[2\log|t+2|-\log|t+1|\right]+C$$

$$=\frac{1}{2}[2\log|x^2+2|-\log|x^2+1|]+C$$

62. Let $I=\int\frac{2x^2+1}{x^2(x^2+4)}dx$

Let $x^2=y$, then $\frac{2x^2+1}{x^2(x^2+4)}=\frac{2y+1}{y(y+4)}$

$$\Rightarrow \quad \frac{2y+1}{y(y+4)}=\frac{A}{y}+\frac{B}{y+4}$$

$\Rightarrow \quad 2y+1=A(y+4)+By$

Put $y=0, -4$ in it to get

$$A=\frac{1}{4} \text{ and } B=\frac{7}{4}$$

$$\therefore \quad \frac{2y+1}{y(y+4)}=\frac{1}{4}\cdot\frac{1}{y}+\frac{7}{4}\cdot\frac{1}{y+4}$$

$$\Rightarrow \quad \frac{2x^2+1}{x^2(x^2+4)}=\frac{1}{4}\cdot\frac{1}{x^2}+\frac{7}{4}\cdot\frac{1}{x^2+4}$$

Integrating both sides w.r.t. x, we get

$$\int\frac{2x^2+1}{x^2(x^2+4)}dx=\frac{1}{4}\int x^{-2}dx+\frac{7}{4}\int\frac{dx}{x^2+2^2}$$

$$=\frac{1}{4}\cdot\frac{x^{-1}}{-1}+\frac{7}{4}\cdot\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)+C$$

$$=-\frac{1}{4x}+\frac{7}{8}\tan^{-1}\left(\frac{x}{2}\right)+C$$

63. Let $I=\int\frac{x^2+1}{(x^2+4)(x^2+25)}dx$

Put $x^2=y$. Then,

$$\frac{x^2+1}{(x^2+4)(x^2+25)}=\frac{y+1}{(y+4)(y+25)}$$

Let $\frac{y+1}{(y+4)(y+25)}=\frac{A}{y+4}+\frac{B}{y+25}$ ...(1)

$\Rightarrow \quad y+1=A(y+25)+B(y+4)$ ...(2)

Putting $y=-4$ and $y=-25$ successively in (2), we get $A=-\frac{1}{7}$ and $B=\frac{8}{7}$

Substituting the values of A and B in (1), we get

$$\frac{y+1}{(y+4)(y+25)}=\frac{-1/7}{(y+4)}+\frac{8/7}{(y+25)}$$

$$\Rightarrow \quad \frac{x^2+1}{(x^2+4)(x^2+25)}=\frac{-1}{7(x^2+4)}+\frac{8}{7(x^2+25)}$$

$$\therefore \quad I=\int\frac{x^2+1}{(x^2+4)(x^2+25)}dx$$

$$=-\frac{1}{7}\int\frac{1}{x^2+2^2}dx+\frac{8}{7}\int\frac{1}{x^2+5^2}dx$$

$$=-\frac{1}{7}\times\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)+\frac{8}{7}\times\frac{1}{5}\tan^{-1}\left(\frac{x}{5}\right)+C$$

$$=-\frac{1}{14}\tan^{-1}\left(\frac{x}{2}\right)+\frac{8}{35}\tan^{-1}\left(\frac{x}{5}\right)+C$$

64. Let $I=\int\frac{dx}{x(x^5+3)}$

Put $x^5+3=t \Rightarrow 5x^4dx=dt$

$$\Rightarrow \quad dx=\frac{dt}{5x^4}$$

$$\therefore \quad I=\frac{1}{5}\int\frac{dt}{t(t-3)}=\frac{1}{5}\int\left[\frac{1}{3(t-3)}-\frac{1}{3t}\right]dt$$

$$=\frac{1}{15}[\log|t-3|-\log|t|]+C=\frac{1}{15}\log\left|\frac{t-3}{t}\right|+C$$

$$=\frac{1}{15}\log\left|\frac{x^5}{x^5+3}\right|+C$$

65. Let $I=\int\frac{dx}{x(x^3+8)}$

Put $x^3+8=t \Rightarrow 3x^2dx=dt$

$$\Rightarrow \quad dx=\frac{dt}{3x^2} \quad \therefore \quad I=\frac{1}{3}\int\frac{dt}{t(t-8)}$$

$$=\frac{1}{3}\int\frac{1}{8}\left[\frac{1}{t-8}-\frac{1}{t}\right]dt=\frac{1}{24}[\log|t-8|-\log|t|]+C$$

$$=\frac{1}{24}\log\left|\frac{t-8}{t}\right|+C=\frac{1}{24}\log\left|\frac{x^3}{x^3+8}\right|+C$$

66. *Refer to answer 65.*

67. Let $I=\int\frac{3x+1}{(x+1)^2(x+3)}dx$

Let $\frac{3x+1}{(x+1)^2(x+3)}=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{x+3}$ ...(1)

www.exambites.in

$\Rightarrow \quad 3x+1 = A(x+1)(x+3) + B(x+3) + C(x+1)^2$ ...(2)

Put $x = -1, -3, 0$ in (2), we get

$B = -1$; $C = -2$; $A = 2$

$\therefore$ From (1),

$$\frac{3x+1}{(x+1)^2(x+3)} = \frac{2}{x+1} - \frac{1}{(x+1)^2} - \frac{2}{x+3}$$

Integrating both sides w.r.t. x, we get

$$I = \int \frac{2}{x+1}dx - \int \frac{1}{(x+1)^2}dx - \int \frac{2}{(x+3)}dx$$

$$= 2\log|x+1| + \frac{1}{x+1} - 2\log|x+3| + C_1$$

$$= 2\log\left|\frac{x+1}{x+3}\right| + \frac{1}{x+1} + C_1$$

68. Let $I = \int \frac{3x+5}{x^3 - x^2 - x + 1}dx$

Here, $x^3 - x^2 - x + 1 = x^2(x-1) - 1(x-1)$

$= (x^2-1)(x-1) = (x-1)(x+1)(x-1) = (x-1)^2(x+1)$

Let $\frac{3x+5}{x^3 - x^2 - x + 1} = \frac{3x+5}{(x-1)^2(x+1)}$

$$= \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} \quad ...(1)$$

$\Rightarrow \quad 3x+5 = A(x-1)(x+1) + B(x+1) + C(x-1)^2$ ...(2)

Put $x = 1, -1, 0$ in (2) to get

$B = 4$; $C = \frac{1}{2}$; $A = -\frac{1}{2}$

From (1),

$$\frac{3x+5}{x^3 - x^2 - x + 1} = -\frac{1}{2}\cdot\frac{1}{x-1} + \frac{4}{(x-1)^2} + \frac{1}{2}\cdot\frac{1}{x+1}$$

Integrating, we get

$$I = -\frac{1}{2}\int \frac{dx}{x-1} + 4\int \frac{dx}{(x-1)^2} + \frac{1}{2}\int \frac{dx}{x+1}$$

$$= -\frac{1}{2}\log|x-1| - \frac{4}{x-1} + \frac{1}{2}\log|x+1| + C_1$$

$$= \frac{1}{2}\log\left|\frac{x+1}{x-1}\right| - \frac{4}{x-1} + C_1$$

69. Let $I = \int \frac{8}{(x+2)(x^2+4)}dx$

Let $\frac{8}{(x+2)(x^2+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+4}$ ...(1)

$\Rightarrow \quad 8 = A(x^2+4) + (Bx+C)(x+2)$ ...(2)

Put $x = -2, 0, 1$ in (2), we get

$A = 1$; $C = 2$; $B = -1$

$\therefore$ From (1),

$$\frac{8}{(x+2)(x^2+4)} = \frac{1}{x+2} + \frac{-x+2}{x^2+4}$$

Integrating, we get

$$I = \int \frac{8}{(x+2)(x^2+4)}dx = \int \frac{dx}{x+2} + \int \frac{-x+2}{x^2+4}dx$$

$$= \log(x+2) - \frac{1}{2}\int \frac{2x}{x^2+4}dx + 2\int \frac{dx}{x^2+2^2} + C_1$$

$$= \log(x+2) - \frac{1}{2}\log(x^2+4) + 2\cdot\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C_1$$

$$= \log\left(\frac{x+2}{\sqrt{x^2+4}}\right) + \tan^{-1}\left(\frac{x}{2}\right) + C_1$$

70. Let $I = \int \frac{2}{(1-x)(1+x^2)}dx$

Let $\frac{2}{(1-x)(1+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x^2}$

$\Rightarrow \quad 2 = A(1+x^2) + (Bx+C)(1-x)$

$\Rightarrow \quad 2 = (A-B)x^2 + (B-C)x + (A+C)$

Comparing coefficients of x^2, x and constant term, we get $A = B = C = 1$

Hence $\frac{2}{(1-x)(1+x^2)} = \frac{1}{1-x} + \frac{x+1}{1+x^2}$

$$\therefore \quad I = \int \frac{-1}{x-1}dx + \frac{1}{2}\int \frac{2x}{1+x^2}dx + \int \frac{1}{1+x^2}dx$$

$$= -\log(x-1) + \frac{1}{2}\log(1+x^2) + \tan^{-1}x + C_1$$

71. Let $I = \int \frac{2x}{(x^2+1)(x^2+3)}dx$

Put $x^2 = y \Rightarrow 2xdx = dy \quad \therefore \quad I = \int \frac{dy}{(y+1)(y+3)}$

We write, $\frac{1}{(y+1)(y+3)} = \frac{A}{y+1} + \frac{B}{y+3}$

$\Rightarrow \quad 1 = A(y+3) + B(y+1)$ ...(1)

Putting $y = -1$ in (1), we get $1 = 2A \Rightarrow A = \frac{1}{2}$

Putting $y = -3$ in (1), we get $1 = -2B \Rightarrow B = -\frac{1}{2}$

$$\Rightarrow \quad \frac{1}{(y+1)(y+3)} = \frac{1}{2(y+1)} - \frac{1}{2(y+3)}$$

www.exambites.in

$$\therefore \quad I=\int\left[\frac{1}{2(y+1)}-\frac{1}{2(y+3)}\right]dy$$

$$=\frac{1}{2}\int\frac{dy}{y+1}-\frac{1}{2}\int\frac{dy}{y+3}$$

$$=\frac{1}{2}\log|y+1|-\frac{1}{2}\log|y+3|+C$$

$$=\frac{1}{2}\log\left|\frac{y+1}{y+3}\right|+C=\frac{1}{2}\log\left|\frac{x^2+1}{x^2+3}\right|+C$$

72. Let $I=\int\frac{x^2+x+1}{(x+1)^2(x+2)}\,dx$

Let $\frac{x^2+x+1}{(x+1)^2(x+2)}=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{x+2}$ ...(1)

$\Rightarrow \quad x^2+x+1=A(x+1)(x+2)+B(x+2)+C(x+1)^2$

Put $x = -1, -2, 0$ in it, to get

$B = 1; C = 3; A = -2$

From (1), we get

$$\frac{x^2+x+1}{(x+1)^2(x+2)}=\frac{-2}{x+1}+\frac{1}{(x+1)^2}+\frac{3}{x+2}$$

Integrating both sides, we get

$$I=\int\frac{x^2+x+1}{(x+1)^2(x+2)}\,dx$$

$$=-2\int\frac{1}{x+1}dx+\int\frac{1}{(x+1)^2}dx+3\int\frac{1}{x+2}dx$$

$$=-2\log|x+1|-\frac{1}{x+1}+3\log|x+2|+C_1$$

73. Let $I=\int\frac{x^2+1}{(x-1)^2(x+3)}dx$

Let $\frac{x^2+1}{(x-1)^2(x+3)}=\frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{x+3}$

$\Rightarrow \quad x^2+1=A(x-1)(x+3)+B(x+3)+C(x-1)^2$ ...(1)

Put $x = 1$ in (1) we get $B=\frac{1}{2}$

Put $x = -3$ in (1) we get $C=\frac{5}{8}$

Put $x = 0$ in (1) we get $A=\frac{3}{8}$

$$\therefore \quad \frac{x^2+1}{(x-1)^2(x+3)}=\frac{3}{8}\cdot\frac{1}{x-1}+\frac{1}{2}\cdot\frac{1}{(x-1)^2}+\frac{5}{8}\cdot\frac{1}{x+3}$$

Integrating both sides, we get

$$I=\int\frac{x^2+1}{(x-1)^2(x+3)}\,dx=\frac{3}{8}\int\frac{dx}{(x-1)}+\frac{1}{2}\int\frac{dx}{(x-1)^2}+\frac{5}{8}\int\frac{dx}{x+3}$$

$$=\frac{3}{8}\log|x-1|-\frac{1}{2}\cdot\frac{1}{(x-1)}+\frac{5}{8}\log|x+3|+C_1$$

74. Let $I=\int x^4\log x\,dx=\log x\cdot\frac{x^5}{5}-\int\frac{1}{x}\cdot\frac{x^5}{5}dx$

$$=\frac{x^5}{5}\log x-\frac{1}{5}\int x^4dx=\frac{1}{5}x^5\log x-\frac{x^5}{25}+C$$

75. **(d) :** Let $I=\int e^x\left(\log x+\frac{1}{x}\right)dx$

$\Rightarrow \quad I = e^x \log x + c$

$\left(\because \int e^x[f(x)+f'(x)]dx=e^xf(x)+c,\right)$

76. Given : $\int e^x(\tan x+1)\sec x\,dx = e^x f(x) + C$ ...(1)

$$\text{L.H.S.}=\int e^x(\tan x+1)\sec x\,dx$$

$$=\int e^x(\sec x+\sec x\tan x)dx$$

$$=\int e^x\sec x\,dx+\int e^x\sec x\tan x\,dx$$

(Integrating first integral by parts)

$$=e^x\sec x-\int e^x\sec x\tan x\,dx+\int e^x\sec x\tan x\,dx+C$$

$= e^x \sec x + C = e^x f(x) + C$ (by (1))

On comparing, we get $f(x) = \sec x$

77. Let $I=\int\sin^{-1}(2x)dx=\int 1\cdot\sin^{-1}(2x)\,dx$

$$=\sin^{-1}(2x)x-\int\left(\frac{1}{\sqrt{1-4x^2}}\frac{d}{dx}(2x)\cdot x\right)dx$$

$$=x\sin^{-1}(2x)-\int\frac{2x}{\sqrt{1-4x^2}}\,dx$$

$$=x\sin^{-1}(2x)+\int\frac{dt}{4\sqrt{t}}$$

(Putting $1-4x^2 = t \Rightarrow -8x\,dx = dt$)

$$=x\sin^{-1}(2x)+\frac{2}{4}(t)^{1/2}+C$$

$$=x\sin^{-1}(2x)+\frac{1}{2}\sqrt{1-4x^2}+C$$

78. Let $I=\int x\cdot\tan^{-1}x\,dx$

www.exambits.in

Integrating by parts, we get

$$I=\tan^{-1}x\int x\,dx-\int\left\{\frac{d}{dx}(\tan^{-1}x)\int x\,dx\right\}dx$$

$$=(\tan^{-1}x)\frac{x^2}{2}-\int\frac{1}{(1+x^2)}\frac{x^2}{2}dx$$

$$=\frac{x^2\tan^{-1}x}{2}-\frac{1}{2}\int\left(1-\frac{1}{1+x^2}\right)dx$$

$$=\frac{x^2\tan^{-1}x}{2}-\frac{1}{2}\int dx+\frac{1}{2}\int\frac{1}{1+x^2}dx$$

$$=\frac{x^2\tan^{-1}x}{2}-\frac{x}{2}+\frac{1}{2}\tan^{-1}x+C$$

$$=\frac{1}{2}(1+x^2)\tan^{-1}x-\frac{x}{2}+C$$

79. Let $I=\int\sec^3x\,dx=\int\sec x\cdot\sec^2x\,dx$

$=\sec x\tan x-\int\tan x\cdot\sec x\tan x\,dx$

$=\sec x\tan x-\int\sec x(\sec^2x-1)\,dx$

$=\sec x\tan x-\int\sec^3x\,dx+\int\sec x\,dx$

$\Rightarrow\quad 2I=\sec x\tan x-I+\ln|\sec x+\tan x|+C_1,$

$=\sec x\tan x+\ln|\sec x+\tan x|+C_1$

$\therefore\quad I=\frac{1}{2}(\sec x\tan x)+\frac{1}{2}\ln|\sec x+\tan x|+C$

80. Let $I=\int(3x+1)\sqrt{4-3x-2x^2}dx$

Let $3x+1=\lambda\frac{d}{dx}(4-3x-2x^2)+\mu$

$\Rightarrow\quad 3x+1=\lambda(-3-4x)+\mu$

$\Rightarrow\quad 3x+1=-3\lambda+\mu-4\lambda x$

$\Rightarrow\quad 3=-4\lambda,\ -3\lambda+\mu=1$

$\Rightarrow\quad \lambda=\frac{-3}{4},\mu=\frac{-5}{4}$

$\therefore\quad I=\int(3x+1)\sqrt{4-3x-2x^2}\,dx$

$=\int\left[-\frac{3}{4}(-3-4x)-\frac{5}{4}\right]\sqrt{4-3x-2x^2}\,dx$

$=-\frac{3}{4}\int(-3-4x)\sqrt{4-3x-2x^2}dx$

$$-\frac{5}{4}\int\sqrt{4-3x-2x^2}\,dx$$

Let $4-3x-2x^2=t$ in the first integral

$\Rightarrow\quad (-3-4x)dx=dt$

$$\therefore\quad I=\frac{-3}{4}\int\sqrt{t}\,dt-\frac{5}{4}\int\sqrt{-2\left(x^2+\frac{3}{2}x-2\right)}dx$$

$$=\frac{-3}{4}\times\frac{2}{3}t^{3/2}+C_1$$

$$-\frac{5}{4}\int\sqrt{-2\left(x^2+\frac{3}{2}x-2+\frac{9}{16}-\frac{9}{16}\right)}dx$$

$$=\frac{-1}{2}(4-3x-2x^2)^{\frac{3}{2}}+C_1$$

$$-\frac{5}{4}\int\sqrt{-2\left[\left(x+\frac{3}{4}\right)^2-\left(\frac{\sqrt{41}}{4}\right)^2\right]}dx$$

$$=-\frac{1}{2}(4-3x-2x^2)^{\frac{3}{2}}+C_1$$

$$-\frac{5\times\sqrt{2}}{4}\int\sqrt{\left(\frac{\sqrt{41}}{4}\right)^2-\left(x+\frac{3}{4}\right)^2}\,dx$$

$$=-\frac{1}{2}(4-3x-2x^2)^{\frac{3}{2}}+C_1$$

$$-\frac{5\sqrt{2}}{4}\left[\frac{1}{2}\left(x+\frac{3}{4}\right)\sqrt{\left(\frac{41}{16}\right)-\left(x+\frac{3}{4}\right)^2}\right.$$

$$\left.+\frac{1}{2}\left(\frac{41}{16}\right)\sin^{-1}\left(\frac{x+\frac{3}{4}}{\frac{\sqrt{41}}{4}}\right)+C_2\right]$$

$$=\frac{-1}{2}(4-3x-2x^2)^{\frac{3}{2}}-\frac{5}{4\sqrt{2}}\left(x+\frac{3}{4}\right)\sqrt{\left(\frac{41}{16}\right)-\left(x+\frac{3}{4}\right)^2}$$

$$-\frac{205}{64\sqrt{2}}\sin^{-1}\left(\frac{4x+3}{\sqrt{41}}\right)+C,$$

where $C=C_1-\frac{5\sqrt{2}}{4}C_2$

81. Let $I=\int\frac{x\sin^{-1}x}{\sqrt{1-x^2}}dx$

$$=\sin^{-1}x\int\frac{x}{\sqrt{1-x^2}}dx-\int\left[\frac{d}{dx}(\sin^{-1}x)\int\frac{x}{\sqrt{1-x^2}}dx\right]dx$$

(Applying integration by parts)

Firstly, let us evaluate the integral $\int\frac{x}{\sqrt{1-x^2}}dx$

Put $t=1-x^2$ and $dt=-2x\,dx$. So,

$$\int\frac{x}{\sqrt{1-x^2}}dx=-\frac{1}{2}\int\frac{dt}{\sqrt{t}}=-\sqrt{t}=-\sqrt{1-x^2}$$

www.exambites.in

$$\therefore \quad I=\sin^{-1}x\left(-\sqrt{1-x^2}\right)-\int\frac{1}{\sqrt{1-x^2}}\left(-\sqrt{1-x^2}\right)dx$$

$$=-\sqrt{1-x^2}\sin^{-1}x+\int dx=-\sqrt{1-x^2}\sin^{-1}x+x+C$$

82. Let $I=\int(2x+5)\sqrt{10-4x-3x^2}\,dx$

Let $2x+5=\lambda\frac{d}{dx}(10-4x-3x^2)+\mu$

$\Rightarrow \quad 2x+5=\lambda(-4-6x)+\mu$

$\Rightarrow \quad 2x+5=-6\lambda x-4\lambda+\mu$

$\Rightarrow \quad -6\lambda=2,\ -4\lambda+\mu=5$

$$\Rightarrow \quad \lambda=-\frac{1}{3},\mu=\frac{11}{3}$$

$$\therefore \quad I=\int(2x+5)\sqrt{10-4x-3x^2}\,dx$$

$$=\int\left[-\frac{1}{3}(-4-6x)+\frac{11}{3}\right]\sqrt{10-4x-3x^2}\,dx$$

$$=\int-\frac{1}{3}\int(-4-6x)\sqrt{10-4x-3x^2}\,dx+\frac{11}{3}\int\sqrt{10-4x-3x^2}\,dx$$

Put $10-4x-3x^2=t$ in the first integral.

$\therefore \quad (-4-6x)\,dx=dt$

$$\Rightarrow \quad I=-\frac{1}{3}\int\sqrt{t}dt+\frac{11}{3}\int\sqrt{-3\left(x^2+\frac{4}{3}x-\frac{10}{3}\right)}dx$$

$$\Rightarrow \quad I=-\frac{1}{3}\times\frac{2}{3}t^{\frac{3}{2}}+C_1+\frac{11}{3}\int\sqrt{-3\left(x^2+\frac{4}{3}x-\frac{10}{3}+\frac{4}{9}-\frac{4}{9}\right)}dx$$

$$\Rightarrow \quad I=-\frac{2}{9}\left(10-4x-3x^2\right)^{3/2}+C_1+\frac{11}{3}\int\sqrt{-3\left[\left(x+\frac{2}{3}\right)^2-\left(\frac{\sqrt{34}}{3}\right)^2\right]}dx$$

$$\Rightarrow \quad I=-\frac{2}{9}\left(10-4x-3x^2\right)^{3/2}+C_1+\frac{11\times\sqrt{3}}{3}\int\sqrt{\left(\frac{\sqrt{34}}{3}\right)^2-\left(x+\frac{2}{3}\right)^2}dx$$

$$\Rightarrow \quad I=-\frac{2}{9}\left(10-4x-3x^2\right)^{3/2}+C_1+\frac{11\sqrt{3}}{3}\left[\frac{1}{2}\left(x+\frac{2}{3}\right)\sqrt{\frac{34}{9}-\left(x+\frac{2}{3}\right)^2}+\frac{\frac{34}{9}}{2}\sin^{-1}\left(\frac{x+\frac{2}{3}}{\frac{\sqrt{34}}{3}}\right)+C_2\right]$$

$$\Rightarrow \quad I=-\frac{2}{9}\left(10-4x-3x^2\right)^{3/2}+\frac{11}{2\sqrt{3}}\left(x+\frac{2}{3}\right)\sqrt{\left(\frac{34}{9}\right)-\left(x+\frac{2}{3}\right)^2}+\frac{187}{9\sqrt{3}}\sin^{-1}\left(\frac{3x+2}{\sqrt{34}}\right)+C$$

where $C=C_1+\frac{11}{\sqrt{3}}C_2$

83. Let $I=\int(x+3)\sqrt{3-4x-x^2}\,dx$

$$=\int(x+2+1)\sqrt{3-4x-x^2}\,dx$$

$$=\frac{-1}{2}\int-2(x+2)\sqrt{3-4x-x^2}\,dx+\int\sqrt{3-4x-x^2}\,dx$$

$= I_1 + I_2$ (say)

$$\therefore \quad I_1=-\frac{1}{2}\int-2(x+2)\sqrt{3-4x-x^2}\,dx$$

Put $3-4x-x^2=t \Rightarrow (-4-2x)dx=dt$

$$\Rightarrow \quad I_1=-\frac{1}{2}\int\sqrt{t}\,dt=-\frac{1}{2}\times\frac{2}{3}(t)^{3/2}+C_1$$

$$=-\frac{1}{3}\times(3-4x-x^2)^{3/2}+C_1 \qquad ...(1)$$

and $I_2=\int\sqrt{3-4x-x^2}\,dx$

$$=\int\sqrt{-(x^2+4x-3+4-4)}\,dx$$

$$=\int\sqrt{-((x+2)^2-7)}\,dx=\int\sqrt{7-(x+2)^2}\,dx$$

$$=\frac{(x+2)\sqrt{3-4x-x^2}}{2}+\frac{7}{2}\sin^{-1}\frac{(x+2)}{\sqrt{7}}+C_2 \qquad ...(2)$$

From (1) & (2), we get

$$I=-\frac{1}{3}(3-4x-x^2)^{3/2}+\frac{(x+2)\sqrt{3-4x-x^2}}{2}+\frac{7}{2}\sin^{-1}\left(\frac{x+2}{\sqrt{7}}\right)+C$$

where $C = C_1 + C_2$

84. Let $I=\int\frac{x^2-3x+1}{\sqrt{1-x^2}}dx$

$$=-\int\frac{(-x^2+3x-1)}{\sqrt{1-x^2}}dx=-\int\frac{(1-x^2)+3x-2}{\sqrt{1-x^2}}dx$$

$$i.e., I=-\int\sqrt{1-x^2}\,dx+\int\frac{-3x+2}{\sqrt{1-x^2}}dx$$

www.exambites.in

$$= -\int \sqrt{1-x^2}\,dx + \frac{3}{2}\int \frac{-2x}{\sqrt{1-x^2}}dx + 2\int \frac{1}{\sqrt{1-x^2}}dx$$

$$= -\int \sqrt{1-x^2}\,dx + \frac{3\times 2}{2}\sqrt{1-x^2} + 2\sin^{-1}x + C$$

$$= -\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right] + 3\sqrt{1-x^2} + 2\sin^{-1}x + C$$

$$= -\frac{x}{2}\sqrt{1-x^2} + \frac{3}{2}\sin^{-1}x + 3\sqrt{1-x^2} + C$$

85. Let $I = \int (3-2x)\sqrt{2+x-x^2}\,dx$

Here $\frac{d}{dx}(2+x-x^2) = 1-2x$

We can write

$$I = \int \sqrt{2+x-x^2}\,(1-2x+2)dx$$

$$= \int \sqrt{2+x-x^2}\cdot(1-2x)dx + 2\int \sqrt{2+x-x^2}\,dx$$

$= I_1 + 2I_2$ (say) ...(1)

where, $I_1 = \int \sqrt{2+x-x^2}\,(1-2x)dx$

Take $2 + x - x^2 = t \Rightarrow (1 - 2x)\,dx = dt$

$$\therefore\ I_1 = \int t^{1/2}dt = \frac{2}{3}t^{3/2} + C_1 = \frac{2}{3}(2+x-x^2)^{3/2} + C_1$$

...(2)

and $I_2 = \int \sqrt{2+x-x^2}\,dx = \int \sqrt{\left(\frac{3}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2}\,dx$

$$= \frac{\left(x-\frac{1}{2}\right)\sqrt{\left(\frac{3}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2}}{2} + \frac{(3/2)^2}{2}\sin^{-1}\left(\frac{x-\frac{1}{2}}{3/2}\right) + C'$$

$$= \frac{1}{2}\left(x-\frac{1}{2}\right)\sqrt{2+x-x^2} + \frac{9}{8}\sin^{-1}\left(\frac{2x-1}{3}\right) + C' \quad ...(3)$$

From (1), (2) and (3), we get

$$I = \frac{2}{3}(2+x-x^2)^{3/2} + \left(x-\frac{1}{2}\right)\sqrt{2+x-x^2} + \frac{9}{4}\sin^{-1}\left(\frac{2x-1}{3}\right) + C$$

where $C = C_1 + 2C'$

86. Let $I = \int \frac{\log x}{(x+1)^2}dx = \int (x+1)^{-2}\cdot \log x\,dx$

Integrating by parts, taking $\log x$ as first function.

$$I = \frac{(x+1)^{-1}}{-1}\cdot \log x - \int \frac{(x+1)^{-1}}{-1}\cdot \frac{1}{x}\,dx$$

$$= \frac{-\log x}{x+1} + \int \frac{dx}{x(x+1)} = \frac{-\log x}{x+1} + \int\left[\frac{1}{x} - \frac{1}{x+1}\right]dx$$

$$= -\frac{\log x}{x+1} + \log x - \log(x+1) + C$$

$$= -\frac{\log x}{x+1} + \log\left(\frac{x}{x+1}\right) + C$$

87. Let $I = \int e^{2x}\sin(3x+1)\,dx$

$$= e^{2x}\int \sin(3x+1)\,dx - \int\left(\frac{d(e^{2x})}{dx}\cdot \int \sin(3x+1)dx\right)dx$$

$$= e^{2x}\,\frac{[-\cos(3x+1)]}{3} - \int 2e^{2x}\cdot\frac{[-\cos(3x+1)]}{3}dx$$

$$= \frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{3}\int e^{2x}\cos(3x+1)dx$$

$$= \frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{3}\left[e^{2x}\int \cos(3x+1)dx - \int\left(\frac{d}{dx}(e^{2x})\cdot\int\cos(3x+1)dx\right)dx\right]$$

$$= \frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{9}e^{2x}\sin(3x+1) - \frac{4}{9}\int e^{2x}\sin(3x+1)dx$$

$$= \frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{9}e^{2x}\sin(3x+1) - \frac{4}{9}I + C_1$$

$$\therefore\quad I + \frac{4}{9}I = \frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{9}e^{2x}\sin(3x+1) + C_1$$

$$\Rightarrow\quad \frac{13I}{9} = \frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{9}e^{2x}\sin(3x+1) + C_1$$

$$\Rightarrow\quad I = \frac{9}{13}\left[\frac{-e^{2x}\cos(3x+1)}{3} + \frac{2}{9}e^{2x}\sin(3x+1) + C_1\right]$$

$$= \frac{9}{13}e^{2x}\left[\frac{2\sin(3x+1) - 3e^{2x}\cos(3x+1)}{9}\right] + \frac{9}{13}C_1$$

$$= \frac{1}{13}e^{2x}[2\sin(3x+1) - 3\cos(3x+1)] + C$$

88. Let $I = \int \frac{x^2+1}{(x+1)^2}e^x dx$

www.exambites.in

$$= \int e^x \cdot \left[\frac{(x^2-1)+2}{(x+1)^2}\right] dx$$

$$= \int e^x \cdot \left[\frac{x^2-1}{(x+1)^2} + \frac{2}{(x+1)^2}\right] dx$$

$$= \int e^x \cdot \left[\frac{x-1}{x+1} + \frac{2}{(x+1)^2}\right] dx \quad \ldots(1)$$

Put $f(x) = \frac{x-1}{x+1}$

$\Rightarrow \quad f'(x) = \frac{(x+1)\cdot 1 - (x-1)\cdot 1}{(x+1)^2} = \frac{2}{(x+1)^2}$

From (1), we get

$$I = \int e^x [f(x) + f'(x)] dx$$

$$= e^x f(x) + C = e^x . \left[\frac{x-1}{x+1}\right] + C$$

89. $I = \int (x-3)\sqrt{x^2+3x-18}\ dx$

$$= \int x\sqrt{x^2+3x-18}\ dx - 3\int \sqrt{x^2+3x-18}\ dx$$

$$= \int \frac{1}{2}(2x+3-3)\sqrt{x^2+3x-18}\ dx - 3\int \sqrt{x^2+3x-18}\, dx$$

$$= \frac{1}{2}\int (2x+3)\sqrt{x^2+3x-18}\, dx - \frac{3}{2}\int \sqrt{x^2+3x-18}\ dx - 3\int \sqrt{x^2+3x-18}\ dx$$

$$= \frac{1}{2}\int (2x+3)\times\sqrt{x^2+3x-18}\, dx - \frac{9}{2}\int \sqrt{x^2+3x-18}\, dx$$

$$= \frac{1}{2} I_1 - \frac{9}{2} I_2 \text{ (say)}$$

For I_1, put $x^2 + 3x - 18 = z \Rightarrow (2x + 3)dx = dz$

$\therefore \quad I_1 = \int \sqrt{z}\, dz = \frac{2}{3} z^{3/2} + C_1$

$\therefore \quad I_1 = \frac{2}{3}(x^2+3x-18)^{\frac{3}{2}} + C_1$

and $I_2 = \int \sqrt{x^2+3x-18}\, dx$

$$= \int \sqrt{x^2+3x+\left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 - 18}\, dx$$

$$= \int \sqrt{\left(x+\frac{3}{2}\right)^2 - \frac{81}{4}}\, dx = \int \sqrt{\left(x+\frac{3}{2}\right)^2 - \left(\frac{9}{2}\right)^2}\, dx$$

$$= \frac{x+\frac{3}{2}}{2}\sqrt{\left(x+\frac{3}{2}\right)^2 - \left(\frac{9}{2}\right)^2} - \frac{81/4}{2}\log\left|\left(x+\frac{3}{2}\right) + \sqrt{\left(x+\frac{3}{2}\right)^2 - \frac{81}{4}}\right| + C_2$$

$$= \frac{2x+3}{4}\sqrt{x^2+3x-18} - \frac{81}{8}\log\left|\left(x+\frac{3}{2}\right) + \sqrt{x^2+3x-18}\right| + C_2$$

The given integral is written as

$$I = \frac{1}{3}(x^2+3x-18)^{3/2} - \frac{9}{8}(2x+3)\sqrt{x^2+3x-18} + \frac{729}{16}\log\left|\frac{(2x+3)}{2} + \sqrt{x^2+3x-18}\right| + C,$$

where $C = \frac{1}{2} C_1 - \frac{9}{2} C_2$ is a constant

90. Let $I = \int \frac{x\cos^{-1} x}{\sqrt{1-x^2}} dx$

Put $\cos^{-1} x = \theta \Rightarrow x = \cos\theta \Rightarrow dx = -\sin\theta\, d\theta$

$\Rightarrow \quad I = \int \frac{\cos\theta(\theta)}{\sqrt{1-\cos^2\theta}}(-\sin\theta) d\theta$

$\Rightarrow \quad I = -\int \theta\cos\theta d\theta$

$\Rightarrow \quad -I = \theta\int \cos\theta\, d\theta - \int\left(\frac{d}{d\theta}\theta \int \cos\theta d\theta\right) d\theta$

$\Rightarrow \quad -I = \theta\sin\theta - \int \sin\theta d\theta$

$\Rightarrow \quad -I = \theta\sin\theta + \cos\theta + C$

$\Rightarrow \quad I = -[\cos^{-1} x\sqrt{1-\cos^2\theta} + x] + C$

$\therefore \quad I = -\left[\sqrt{1-x^2}\cos^{-1} x + x\right] + C$

91. Let $I = \int (3x-2)\sqrt{x^2+x+1}\, dx$

Let $3x - 2 = A\frac{d}{dx}(x^2+x+1) + B$

$\Rightarrow \quad 3x - 2 = A(2x+1) + B$

On equating the coefficients of like terms, we get $A = 3/2$, $B = -7/2$.

$\therefore \quad I = \int \left[\frac{3}{2}(2x+1) - \frac{7}{2}\right]\sqrt{x^2+x+1}\, dx$

$\Rightarrow \quad I = \frac{3}{2}\int (2x+1)\sqrt{x^2+x+1}\, dx - \frac{7}{2}\int \sqrt{x^2+x+1}\, dx$

www.exambites.in

Put $x^2 + x + 1 = t$ in 1st integral $\Rightarrow (2x+1)dx = dt$

i.e., $I = \frac{3}{2}\int \sqrt{t}\,dt - \frac{7}{2}\int \sqrt{(x+(1/2))^2 + (\sqrt{3}/2)^2}\,dx$

$$I = \frac{3}{2}\left(\frac{t^{3/2}}{3/2}\right) - \frac{7}{2}\left[\frac{x+\frac{1}{2}}{2}\sqrt{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right.$$
$$\left. + \frac{\left(\frac{\sqrt{3}}{2}\right)^2}{2}\log\left|\left(x+\frac{1}{2}\right) + \sqrt{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right|\right] + C$$

$$\Rightarrow \quad I = (x^2+x+1)^{3/2} - \frac{7}{2}\left[\frac{2x+1}{4}\sqrt{x^2+x+1}\right.$$
$$\left. + \frac{3}{8}\log|x+\frac{1}{2}+\sqrt{x^2+x+1}|\right] + C$$

92. Let $I = \int e^{2x}\left(\frac{1-\sin 2x}{1-\cos 2x}\right)dx$

Put $2x = t \Rightarrow \quad dx = \frac{1}{2}dt$

$$\therefore \quad I = \frac{1}{2}\int e^t \frac{1-\sin t}{1-\cos t}dt$$

$$= \frac{1}{2}\int e^t \left(\frac{1-2\sin\frac{t}{2}\cos\frac{t}{2}}{2\sin^2\frac{t}{2}}\right)dt$$

$$= \frac{1}{2}\int e^t \left(\frac{1}{2}\operatorname{cosec}^2\frac{t}{2} - \cot\frac{t}{2}\right)dt$$

$$= \frac{1}{2}\int e^t \left(-\cot\frac{t}{2} + \frac{1}{2}\operatorname{cosec}^2\frac{t}{2}\right)dt$$

[Compare with $\int e^x[f(x)+f'(x)]dx = e^x f(x)$

Take $f(t) = -\cot\frac{t}{2} \Rightarrow f'(t) = \frac{1}{2}\operatorname{cosec}^2\frac{t}{2}$]

$$\therefore \quad I = \frac{1}{2}e^t\left(-\cot\frac{t}{2}\right) + C = \frac{-1}{2}e^{2x}\cot x + C$$

93. Let $I = \int \frac{\sqrt{1-\sin x}}{1+\cos x}\cdot e^{\frac{-x}{2}}dx$

Put $-\frac{x}{2} = t \Rightarrow x = -2t \Rightarrow dx = -2dt$

$$\therefore \quad I = \int \frac{\sqrt{1-\sin(-2t)}}{1+\cos(-2t)}\cdot e^t(-2)dt$$

$$= -2\int \frac{\sqrt{1+\sin 2t}}{1+\cos 2t}\cdot e^t dt$$

$$= -2\int \frac{\sqrt{\cos^2 t + \sin^2 t + 2\sin t\cos t}}{2\cos^2 t}\cdot e^t dt$$

$$= -\int \frac{\cos t + \sin t}{\cos^2 t}\cdot e^t dt \; = -\int (\sec t + \sec t\tan t)e^t dt$$

$$= -e^t\sec t + C = -e^{-x/2}\cdot\sec\frac{x}{2} + C$$

94. Let $I = \int\left(\frac{1+\sin x}{1+\cos x}\right)e^x dx$

$$= \int \frac{1+2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\cdot e^x dx$$

$$= \int\left(\frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}\right)e^x dx$$

$$= \int e^x\cdot\left(\tan\frac{x}{2} + \frac{1}{2}\sec^2\frac{x}{2}\right)dx$$

Compare it with

$\int e^x[f(x)+f'(x)]dx = e^x\cdot f(x)$

Here $f(x) = \tan\frac{x}{2} \Rightarrow f'(x) = \frac{1}{2}\sec^2\frac{x}{2}$

$$\therefore \quad I = e^x\cdot\tan\frac{x}{2} + C$$

95. Let $I = \int \frac{x^2}{(x\sin x + \cos x)^2}dx$

$$= \int \frac{x\cos x}{(x\sin x+\cos x)^2}\cdot x\sec x\,dx \qquad ...(1)$$

Let $I_1 = \int \frac{x\cos x}{(x\sin x+\cos x)^2}dx$

Put $x\sin x + \cos x = t$

$\Rightarrow$ $(x\cos x + 1\cdot\sin x - \sin x)dx = dt$

$\Rightarrow$ $x\cos x\,dx = dt$

$$\therefore \quad I_1 = \int\frac{dt}{t^2} = -\frac{1}{t} = \frac{-1}{x\sin x+\cos x}$$

Now integrating (1) by parts taking $x\sec x$ as the first function and using I_1, we get

$$I = x\sec x\cdot\frac{-1}{x\sin x+\cos x}$$

www.examrites.in

$$-\int(1\cdot\sec x + x\sec x\tan x)\cdot\frac{-dx}{x\sin x+\cos x}$$

$$=\frac{-x\sec x}{x\sin x+\cos x}+\int\sec x\left(1+\frac{x\sin x}{\cos x}\right)\frac{dx}{x\sin x+\cos x}$$

$$=\frac{-x\sec x}{x\sin x+\cos x}+\int\sec^2 x\,dx$$

$$=\frac{-x\sec x}{x\sin x+\cos x}+\tan x+C$$

96. Let $I=\int\frac{\sqrt{x^2+1}[\log(x^2+1)-2\log x]}{x^4}\,dx$

$$=\int\frac{\sqrt{x^2+1}\,\log\left(\frac{x^2+1}{x^2}\right)}{x^4}\,dx$$

Put $\frac{x^2+1}{x^2}=t\Rightarrow x^2=\frac{1}{t-1}$

$$\Rightarrow\ 2x\,dx=\frac{-1}{(t-1)^2}dt\ \Rightarrow\ dx=-\frac{1}{2x}\cdot\frac{1}{(t-1)^2}\,dt$$

$$=-\frac{1}{2}\cdot\sqrt{t-1}\cdot\frac{1}{(t-1)^2}\,dt\ =-\frac{dt}{2(t-1)^{3/2}}$$

Also $\sqrt{x^2+1}=\sqrt{\frac{1}{t-1}+1}=\sqrt{\frac{t}{t-1}}$

$$\therefore\quad I=\int\sqrt{\frac{t}{t-1}}\cdot\log t\cdot\frac{1}{1/(t-1)^2}\times\frac{-dt}{2(t-1)^{3/2}}$$

$$=-\frac{1}{2}\int\sqrt{t}.\log t\,dt$$

$$=-\frac{1}{2}\left[\frac{t^{3/2}}{3/2}\cdot\log t-\int\frac{t^{3/2}}{3/2}\cdot\frac{1}{t}dt\right]+C$$

$$=-\frac{1}{2}\left[t^{3/2}\log t-\int t^{1/2}dt\right]+C$$

$$=\frac{-1}{3}\left[t^{3/2}\log t-\frac{2}{3}t^{3/2}\right]+C$$

$$=\frac{-1}{3}\left[\left(\frac{x^2+1}{x^2}\right)^{3/2}\log\left(\frac{x^2+1}{x^2}\right)-\frac{2}{3}\left(\frac{x^2+1}{x^2}\right)^{3/2}\right]+C$$

97. Let $I=\int\frac{\sin^{-1}\sqrt{x}-\cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}}\,dx, x\in[0,1]$

We know that $\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}=\frac{\pi}{2}$

$$\Rightarrow\ \sin^{-1}\sqrt{x}=\frac{\pi}{2}-\cos^{-1}\sqrt{x}$$

$$\therefore\quad I=\int\frac{\frac{\pi}{2}-2\cos^{-1}\sqrt{x}}{\pi/2}\,dx$$

$$=\int1\cdot dx-\frac{4}{\pi}\int1\cdot\cos^{-1}\sqrt{x}\,dx$$

$$=x-\frac{4}{\pi}\left[x\cdot\cos^{-1}\sqrt{x}-\int x\cdot\frac{-1}{\sqrt{1-x}}\cdot\frac{1}{2\sqrt{x}}dx\right]+C$$

Put $x=\sin^2\theta\Rightarrow dx=2\sin\theta\cos\theta\,d\theta$

$$\therefore\ I=x-\frac{4}{\pi}x\cos^{-1}\sqrt{x}-\frac{2}{\pi}\int\sqrt{\frac{\sin^2\theta}{1-\sin^2\theta}}\cdot2\sin\theta\cos\theta d\theta+C$$

$$=x-\frac{4}{\pi}x\cos^{-1}\sqrt{x}-\frac{2}{\pi}\int\frac{\sin\theta}{\cos\theta}\cdot2\sin\theta\cos\theta d\theta+C$$

$$=x-\frac{4}{\pi}x\cos^{-1}\sqrt{x}-\frac{2}{\pi}\int(1-\cos2\theta)d\theta+C$$

$$=x-\frac{4}{\pi}x\cos^{-1}\sqrt{x}-\frac{2}{\pi}\left[\theta-\frac{\sin2\theta}{2}\right]+C$$

$$=x-\frac{4}{\pi}x\cos^{-1}\sqrt{x}-\frac{2}{\pi}[\theta-\sin\theta\cos\theta]+C$$

$$=x-\frac{4}{\pi}x\cos^{-1}\sqrt{x}-\frac{2}{\pi}\left[\sin^{-1}\sqrt{x}-\sqrt{x}\sqrt{1-x}\right]+C$$

98. We have, by definition

$$\int_a^b f(x)dx=\lim_{h\to0}h[f(a)+f(a+h)+f(a+2h)+\ldots.+f(a+(n-1)h)],$$

where $nh=b-a$

Here, $a=1$, $b=4$, $f(x)=x^2-x$ and $nh=3$

$$\therefore\ \int_1^4(x^2-x)dx=\lim_{h\to0}h[\{1^2-1\}+\{(1+h)^2-(1+h)\}+\ldots+\{(1+(n-1)h)^2-(1+(n-1)h)\}]$$

$$=\lim_{h\to0}h[h^2(1^2+2^2+\ldots+(n-1)^2)+2h(1+2+\ldots+(n-1))-h(1+2+\ldots+(n-1))]$$

$$=\lim_{h\to0}h[h^2\{1+2^2+3^2+\ldots+(n-1)^2\}+h\{1+2+3+\ldots+(n-1)\}]$$

$$=\lim_{h\to0}h\left[h^2\frac{(n-1)n(2n-1)}{6}+\frac{h(n-1)n}{2}\right]$$

www.exambites.in

$$= \lim_{h\to 0}\left[\frac{(nh-h)nh(2nh-h)}{6}+\frac{(nh-h)nh}{2}\right]$$

$$= \lim_{h\to 0}\left[\frac{(3-h)3(6-h)}{6}+\frac{(3-h)3}{2}\right]$$

$$= \frac{3(3)(6)}{6}+\frac{3(3)}{2} = 9+\frac{9}{2} = \frac{27}{2}$$

99. Let $I = \int_1^3 (x^2+3x+e^x)\,dx$

We know that,

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+...+f(a+(n-1)h)]$$

Here, $f(x) = x^2 + 3x + e^x$, $a = 1$, $b = 3$, $nh = 2$

Then, $I = \int_1^3 (x^2+3x+e^x)\,dx$

$$= \lim_{h\to 0} h[(1+3+e)+\{(1+h)^2+3(1+h)+e^{(1+h)}\}$$

$$+ .... + \{(1+(n-1)h)^2+3(1+(n-1)h) + e^{(1+(n-1)h)}\}$$

$$= \lim_{h\to 0} h\{n+h^2(1+2^2+3^2+...+(n-1)^2)$$

$$+2h(1+2+3+....+(n-1))+[3n+3h(1+2+...+(n-1))]+e(e^0+e^h+e^{2h}+...+e^{(n-1)h})\}$$

$$= \lim_{h\to 0} h\left[n+\frac{h^2(n-1)n(2n-1)}{6}+2h\frac{n(n-1)}{2} +3n+\frac{3h(n-1)n}{2}+e\left\{\frac{1(1-e^{hn})}{1-e^h}\right\}\right]$$

$$= \lim_{h\to 0}\left[nh+\frac{(nh-h)nh(2nh-h)}{6}+nh(nh-h) +3nh+\frac{3}{2}nh(nh-h)+\frac{e(1-e^2)}{\left(\frac{1-e^h}{h}\right)}\right]$$

$$= 2+\frac{2\times 2\times 4}{6}+(2\times 2)+6+\left(\frac{3}{2}\times 2\times 2\right) - \frac{e(1-e^2)}{\lim_{h\to 0}\left(\frac{e^h-1}{h}\right)}$$

$$= 2+\frac{8}{3}+4+6+6-e(1-e^2) = \frac{62}{3}-e+e^3$$

100. We have, by definition

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+....+f(a+(n-1)h)]$$

where, $nh = b-a$

Here, $a = 1$, $b = 3$, $f(x) = e^{2-3x} + x^2 + 1$ and $nh = 2$

$$\therefore \int_1^3 (e^{2-3x}+x^2+1)\,dx = \lim_{h\to 0} h[\{e^{2-3}+1^2+1\} + \{e^{2-3(1+h)}+(1+h)^2+1\} + ... + \{e^{2-3(1+(n-1)h)}+(1+(n-1)h)^2+1\}]$$

$$= \lim_{h\to 0} h[e^2e^{-3}+e^2e^{-3(1+h)}+e^2e^{-3(1+2h)}+.... + e^2e^{-3(1+(n-1)h)}+1^2+(1+h)^2+(1+2h)^2 + ... + (1+(n-1)h)^2+n]$$

$$= \lim_{h\to 0} h\,[e^2e^{-3}(1+e^{-3h}+e^{-6h}+....+e^{-3(n-1)h}) + n+h^2\times\{1^2+2^2+....+(n-1)^2\} + 2h(1+2+3+....+(n-1))+n]$$

$$= \lim_{h\to 0} h\left[\frac{e^{-1}}{1-e^{-3h}}\frac{(1-(e^{-3hn}))}{1-e^{-3h}}+h^2\times\frac{(n-1)n(2n-1)}{6} + 2h\times\frac{n(n-1)}{2}+2n\right]$$

$$= \lim_{h\to 0}\left[he^{-1}\frac{(1-e^{-6})}{1-e^{-3h}}+\frac{(nh-h)nh(2nh-h)}{6} + nh\,(nh-h)+2nh\right]$$

$$= \lim_{h\to 0}\left[\frac{h(e^{-1}-e^{-7})}{(1-e^{-3h})}+\frac{(2-h)2(4-h)}{6}+2(2-h)+4\right]$$

$$= \frac{(e^{-1}-e^{-7})}{\lim_{h\to 0}\left(\frac{1-\frac{1}{e^{3h}}}{h}\right)}+\frac{8}{3}+4+4$$

$$= \frac{(e^{-1}-e^{-7})}{\lim_{h\to 0}\left(\frac{e^{3h}-1}{3h}\right)\times\frac{3h}{e^{3h}}\times\frac{1}{h}}+\frac{32}{3} = \frac{e^{-1}-e^{-7}}{3}+\frac{32}{3}$$

101. We have, by definition

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+...+f(a+(n-1)h)]$$

where, $nh = b - a$

Here, $f(x) = 3x^2 + 1$, $a = 1$, $b = 3$ and $nh = 2$

www.exambites.in

$$\therefore \int_1^3 (3x^2+1)dx = \lim_{h\to 0} h[(3\cdot 1^2+1)+\{3(1+h)^2+1\}+...$$
$$....+3\{(1+(n-1)\cdot h)^2+1\}]$$
$$= \lim_{h\to 0} h[n+3\cdot n+6h\cdot\{1+2+3+....+(n-1)\}$$
$$+3h^2\{1^2+2^2+....+(n-1)^2\}]$$
$$= \lim_{h\to 0} h[4n+6h\cdot\frac{(n-1)n}{2}+3h^2\cdot\frac{1}{6}(n-1)n(2n-1)]$$
$$= \lim_{h\to 0}[4nh+3nh\,(nh-h)+\frac{1}{2}nh(nh-h)(2nh-h)]$$
$$= 4\times 2+3\times 2(2-0)+\frac{1}{2}\times 2\times(2-0)(2\times 2-0)$$
$$= 8+12+\frac{1}{2}\times 16 = 28$$

102. We have, by definition

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+f(a+2h)+$$
$$...+f(a+(n-1)h)], \text{ where } nh = b-a$$

Here, $a = 1$, $b = 3$, $f(x) = 2x^2 + 5x$ and $nh = 2$

$$\therefore \int_1^3 (2x^2+5x)dx = \lim_{h\to 0} h[\{2(1)^2+5\times 1\}+\{2(1+h)^2$$
$$+5(1+h)\}+\{2(1+2h)^2+5(1+2h)\}$$
$$+.....+\{2(1+(n-1)h)^2+5(1+(n-1)h)\}]$$
$$= \lim_{h\to 0} h[2\{1^2+(1+h)^2+(1+2h)^2$$
$$+...+(1+(n-1)h)^2\}+5\{1+(1+h)+(1+2h)$$
$$+...+(1+(n-1)h)\}]$$
$$= \lim_{h\to 0} h[2\{n+2h(1+2+3+...+(n-1))$$
$$+h^2(1^2+2^2+...+(n-1)^2)\}$$
$$+5\{n+h(1+2+....+(n-1))\}]$$
$$= \lim_{h\to 0} h[7n+9h(1+2+3+...+(n-1))$$
$$+2h^2(1^2+2^2+...+(n-1)^2)]$$
$$= \lim_{h\to 0} h\left\{7n+9h\times\frac{n(n-1)}{2}+2h^2\times\frac{n(n-1)(2n-1)}{6}\right\}$$
$$= \lim_{h\to 0}\left\{7nh+\frac{9}{2}nh\,(nh-h)+\frac{1}{3}nh\,(nh-h)(2nh-h)\right\}$$
$$= 7\times 2+\frac{9}{2}\times 2(2-0)+\frac{1}{3}\times 2(2-0)(2\times 2-0)$$
$$= 14+18+\frac{16}{3} = \frac{112}{3}$$

103. We have by definition,

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+....+f(a+(n-1)h)]$$

where $nh = b - a$

Here, $f(x) = x^2 + 3$, $a = 2$, $b = 5$ and $nh = 3$

$$\therefore \int_2^5 (x^2+3)dx = \lim_{h\to 0} h[(2^2+3)+\{(2+h)^2+3\}$$
$$+\{(2+2h)^2+3\}+.....+\{(2+(n-1)h)^2+3\}]$$
$$= \lim_{h\to 0} h[3n+2^2\cdot n+h^2(1^2+2^2+.....+(n-1)^2)$$
$$+4h(1+2+.....+(n-1))]$$
$$= \lim_{h\to 0}\left[7nh+h^3\cdot\frac{(n-1)\cdot n\cdot(2n-1)}{6}+4h^2\frac{(n-1)n}{2}\right]$$
$$= \lim_{h\to 0}[7nh+\frac{1}{6}(nh-h)nh(2nh-h)+2(nh-h)nh]$$
$$= 7\times 3+\frac{1}{6}(3-0)\times 3\times(2\times 3-0)+2(3-0)\times 3$$
$$= 21 + 9 + 18 = 48$$

104. *Refer to answer 98.*

105. We have, by definition

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a)+f(a+h)+...+f(a+(n-1)h)]$$

where $nh = b - a$

Here $f(x) = 3x^2 - 2$, $a = 0$, $b = 2$ and $nh = b - a = 2$

$$\Rightarrow \int_0^2 (3x^2-2)dx = \lim_{h\to 0} h[(0-2)+(3h^2-2)$$
$$+\{3(2h)^2-2\}+....+\{3((n-1)h)^2-2\}]$$
$$= \lim_{h\to 0} h[-2n+3h^2\{1^2+2^2+....+(n-1)^2\}]$$
$$= \lim_{h\to 0} h\left[-2n+3h^2\frac{(n-1)n(2n-1)}{6}\right]$$
$$= \lim_{h\to 0}\left[-2nh+\frac{1}{2}(nh-h)nh(2nh-h)\right]$$
$$= [-2\times 2+\frac{1}{2}(2-0)2(2\times 2-0)] = -4+8 = 4$$

106. (d) : Let $I = \int_{-\pi/4}^{\pi/4} \sec^2 x\,dx = [\tan x]_{-\pi/4}^{\pi/4}$

www.examrites.in

$= \tan\frac{\pi}{4} - \tan\left(-\frac{\pi}{4}\right) = 1 + 1 = 2$

107. We have, $\int_2^3 3^x dx = \left[\frac{3^x}{\log 3}\right]_2^3 = \frac{3^3 - 3^2}{\log 3} = \frac{18}{\log 3}$

108. $\int \frac{dx}{9+x^2} = \int \frac{dx}{x^2+3^2} = \left[\frac{1}{3}\tan^{-1}\frac{x}{3}\right] = F(x)$

$\therefore$ By second fundamental theorem of integral calculus, we have

$\int_0^3 \frac{dx}{9+x^2} = F(3) - F(0) = \frac{1}{3}[\tan^{-1}1 - \tan^{-1}0]$

$= \frac{1}{3}\left[\frac{\pi}{4} - 0\right] = \frac{\pi}{12}$

109. Let $I = \int_0^{\pi/2} e^x(\sin x - \cos x)dx$

$= \int_0^{\pi/2} e^x \sin x\, dx - \int_0^{\pi/2} e^x \cos x\, dx$

Integrating II integral by parts, we get

$I = \int_0^{\pi/2} e^x \sin x\, dx - \left[\{e^x \cos x\}_0^{\pi/2} - \int_0^{\pi/2} e^x(-\sin x)dx\right]$

$= \int_0^{\pi/2} e^x \sin x\, dx - [e^{\pi/2}\cdot 0 - e^0 \cdot 1] - \int_0^{\pi/2} e^x \sin x\, dx = 1$

110. By the first fundamental theorem of integral calculus

$f'(x) = \frac{d}{dx}\int_0^x t\sin t\ dt = x\sin x.$

111. Here, $\int_0^a \frac{1}{4+x^2}dx = \frac{\pi}{8}$

$\Rightarrow \int_0^a \frac{1}{x^2+2^2}dx = \frac{\pi}{8} \Rightarrow \left[\frac{1}{2}\tan^{-1}\frac{x}{2}\right]_0^a = \frac{\pi}{8}$

$\Rightarrow \frac{1}{2}\tan^{-1}\frac{a}{2} = \frac{\pi}{8} \Rightarrow \tan^{-1}\frac{a}{2} = \frac{\pi}{4}$

$\Rightarrow \frac{a}{2} = \tan\frac{\pi}{4} = 1 \Rightarrow a = 2$

112. We have $\int_0^{\pi/4} \tan x\, dx = [\log|\sec x|]_0^{\pi/4}$

$= \log\left|\sec\frac{\pi}{4}\right| - \log|\sec 0| = \log|\sqrt{2}| - \log|1|$

$\therefore \int_0^{\pi/4} \tan x\, dx = \frac{1}{2}\log 2$

113. We have, $\int \sin 2x\, dx = -\frac{1}{2}[\cos 2x] = F(x)$

$\therefore$ By second fundamental theorem of integral calculus, we have $\int_0^{\pi/4} \sin 2x\ dx = F\left(\frac{\pi}{4}\right) - F(0)$

$\Rightarrow F(x) = -\frac{1}{2}[\cos(\pi/2) - \cos(0)] = \frac{1}{2}$

114. We have, $\int_0^1 \frac{1}{\sqrt{1-x^2}}dx = [\sin^{-1}x]_0^1$

$= \sin^{-1}1 - \sin^{-1}0 = \frac{\pi}{2} - 0 = \frac{\pi}{2}$

115. Here, $\int_1^2 \frac{x^3-1}{x^2}dx = \int_1^2 (x - x^{-2})dx$

$= \left[\frac{x^2}{2} - \frac{x^{-1}}{-1}\right]_1^2 = \left[\frac{x^2}{2} + \frac{1}{x}\right]_1^2$

$= \left[\frac{4}{2} + \frac{1}{2}\right] - \left[\frac{1}{2} + 1\right] = \frac{5}{2} - \frac{3}{2} = 1$

116. $\int_2^3 \frac{1}{x}dx = [\log x]_2^3 = \log 3 - \log 2 = \log\left(\frac{3}{2}\right)$

117. $\int_0^2 \sqrt{4-x^2}\ dx = \int_0^2 \sqrt{2^2-x^2}\ dx$

$= \left[\frac{x}{2}\sqrt{2^2-x^2} + \frac{2^2}{2}\sin^{-1}\frac{x}{2}\right]_0^2 = [2\sin^{-1}1 - 0] = \pi$

118. $\int_0^1 \frac{dx}{1+x^2} = [\tan^{-1}x]_0^1$

$= \tan^{-1}1 - \tan^{-1}0 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$

119. Let $I = \int_1^2 \left[\frac{1}{x} - \frac{1}{2x^2}\right]e^{2x}dx$

Putting $2x = y \Rightarrow 2dx = dy$

As $x \to 1 \Rightarrow y \to 2$ and $x \to 2 \Rightarrow y \to 4$

$\therefore I = \frac{1}{2}\int_2^4 \left[\frac{2}{y} - \frac{2}{y^2}\right]e^y dy = \int_2^4 \left[\frac{1}{y} - \frac{1}{y^2}\right]e^y dy$

www.exambites.in

$$= \left[e^y \cdot \frac{1}{y} \right]_2^4 = \frac{1}{4}e^4 - \frac{1}{2}e^2 = \frac{e^2}{2}\left(\frac{e^2}{2} - 1 \right)$$

120. Let $I = \int_0^1 \tan^{-1}\left(\frac{1-2x}{1+x-x^2} \right) dx$

$$= \int_0^1 \tan^{-1}\left[\frac{(1-x)-x}{1+x(1-x)} \right] dx$$

$$I = \int_0^1 [\tan^{-1}(1-x) - \tan^{-1} x]\, dx \quad \text{...(i)}$$

$$I = \int_0^1 [\tan^{-1} x - \tan^{-1}(1-x)]\, dx \quad \text{...(ii)}$$

$$\left[\text{Using properly } \int_0^a f(x)dx = \int_0^a f(a-x)dx \right]$$

Adding (i) and (ii), we get

$$2I = \int_0^1 [\tan^{-1}(1-x) - \tan^{-1} x + \tan^{-1} x - \tan^{-1}(1-x)]\, dx$$

$$= 0$$

$$\Rightarrow \quad I = 0$$

121. Let $I = \int_0^{\pi/2} x^2 \sin x\, dx$

Integrating by parts

$$I = \left[x^2(-\cos x) \right]_0^{\pi/2} - \int_0^{\pi/2} 2x(-\cos x)dx$$

$$= -\frac{\pi^2}{4}.0 + 0 + 2\int_0^{\pi/2} x\cos x\, dx = 2\int_0^{\pi/2} x\cos x\, dx$$

Again integrating by parts

$$I = 2\left[[x\sin x]_0^{\pi/2} - \int_0^{\pi/2} 1 \cdot \sin x\, dx \right]$$

$$= 2\left\{ \frac{\pi}{2} \cdot 1 - 0 - [-\cos x]_0^{\pi/2} \right\} = 2\left[\frac{\pi}{2} + (0-1) \right] = \pi - 2$$

122. Let $I = \int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx$

$$\Rightarrow \quad I = \int_0^{\frac{\pi}{2}} \frac{x}{1+\cos x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{x}{2\cos^2\frac{x}{2}} dx + \int_0^{\frac{\pi}{2}} \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}} dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} x\sec^2\frac{x}{2} dx + \int_0^{\frac{\pi}{2}} \tan\frac{x}{2} dx$$

$$= \frac{1}{2}\left[\left[2x\tan\frac{x}{2} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \left[\frac{d(x)}{dx}\int \sec^2\frac{x}{2} dx \right] dx \right] + \int_0^{\frac{\pi}{2}} \tan\frac{x}{2} dx$$

$$= \frac{2}{2}\left[\frac{\pi}{2} \right] - \frac{1}{2}\int_0^{\frac{\pi}{2}} 2\tan\frac{x}{2} dx + \int_0^{\frac{\pi}{2}} \tan\frac{x}{2} dx = \frac{\pi}{2}$$

123. Let $I = \int_0^1 \frac{x^4+1}{x^2+1} dx = \int_0^1 \frac{(x^4-1)+2}{x^2+1} dx$

Consider

$$\int \frac{(x^4-1)+2}{x^2+1} dx = \int \left[x^2 - 1 + \frac{2}{x^2+1} \right] dx$$

$$= \left[\frac{x^3}{3} - x + 2\tan^{-1} x \right] = F(x)$$

$\therefore$ By second fundamental theorem of calculus, we have, $I = F(1) - F(0)$

$$= \frac{1}{3} - 1 + 2\tan^{-1} 1 - 0 = -\frac{2}{3} + \frac{\pi}{2} = \frac{3\pi - 4}{6}$$

124. Let $I = \int_2^4 \frac{x}{x^2+1} dx$

Put $x^2 + 1 = t \Rightarrow x\,dx = \frac{1}{2}dt$

Also $x = 2 \Rightarrow t = 5$ and $x = 4 \Rightarrow t = 17$

$$\therefore \quad I = \frac{1}{2}\int_5^{17} \frac{dt}{t} = \frac{1}{2}[\log t]_5^{17} = \frac{1}{2}[\log 17 - \log 5]$$

$$= \frac{1}{2}\log\left(\frac{17}{5} \right).$$

125. Let $I = \int_e^{e^2} \frac{dx}{x\log x}$

Put $\log x = t \Rightarrow \frac{1}{x}dx = dt$

Also $x = e \Rightarrow t = \log e = 1$

www.exambits.in

and $x = e^2 \Rightarrow\ t = \log e^2 = 2 \log e = 2$

$$\therefore \quad I = \int_1^2 \frac{dt}{t} = [\log t]_1^2 = \log 2 - \log 1 = \log 2$$

126. Let $I = \int_0^1 xe^{x^2}\, dx$

Put $x^2 = t \Rightarrow xdx = \frac{1}{2}dt.$

Also $x = 0 \Rightarrow t = 0$ and $x = 1 \Rightarrow t = 1.$

$$\therefore \quad I = \frac{1}{2}\int_0^1 e^t\, dt \Rightarrow I = \frac{1}{2}\left[e^t\right]_0^1 = \frac{1}{2}(e-1)$$

127. Let $I = \int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$

Put $\tan^{-1} x = t \Rightarrow \frac{1}{1+x^2}dx = dt$

Also $x = 0 \Rightarrow t = 0$ and $x = 1 \ \Rightarrow\ t = \frac{\pi}{4}$

$$\therefore \quad I = \int_0^{\pi/4} t\, dt = \left[\frac{1}{2}t^2\right]_0^{\pi/4} = \frac{1}{2}\cdot\left[\left(\frac{\pi}{4}\right)^2 - 0\right] = \frac{\pi^2}{32}$$

128. Let $I = \int_0^1 \frac{e^x}{1+e^{2x}} dx$

Put $e^x = t \Rightarrow e^x\, dx = dt$
Also $x = 0 \Rightarrow t = e^0 = 1$
and $x = 1 \Rightarrow t = e^1 = e$

$$\therefore \quad I = \int_1^e \frac{dt}{(1+t^2)} = [\tan^{-1} t]_1^e = \tan^{-1} e - \tan^{-1} 1$$

$$= \tan^{-1}\left(\frac{e-1}{1+e}\right)$$

129. *Refer to answer 124.*

130. Let $I = \int_{\frac{-\pi}{4}}^{0} \frac{(1+\tan x)}{(1-\tan x)} dx = \int_{\frac{-\pi}{4}}^{0} \frac{\left(1+\frac{\sin x}{\cos x}\right)}{\left(1-\frac{\sin x}{\cos x}\right)} dx$

$$= \int_{\frac{-\pi}{4}}^{0} \frac{\cos x + \sin x}{\cos x - \sin x} dx$$

Put $\cos x - \sin x = t \Rightarrow -(\sin x + \cos x)\, dx = dt$

When $x = 0, t = 1$, when $x = \frac{-\pi}{4}, t = \sqrt{2}$

$$\therefore \quad I = \int_{\sqrt{2}}^{1} -\frac{dt}{t} = \int_1^{\sqrt{2}} \frac{dt}{t} = [\log t]_1^{\sqrt{2}}$$

$$= \log\sqrt{2} - \log 1 = \frac{1}{2}\log 2$$

131. Let $I = \int_0^{\pi} e^{2x} \cdot \sin\left(\frac{\pi}{4} + x\right) dx$

Put $\frac{\pi}{4} + x = t \Rightarrow x = t - \frac{\pi}{4} \Rightarrow dx = dt$

When $x = 0,\ t = \frac{\pi}{4}$ and when $x = \pi,\ t = \frac{5\pi}{4}$

$$\therefore \quad I = \int_{\pi/4}^{5\pi/4} e^{2\left(t-\frac{\pi}{4}\right)} \sin t\, dt = e^{-\pi/2} \int_{\pi/4}^{5\pi/4} e^{2t} \sin t\, dt$$

$$= e^{-\pi/2}\left[\left(\sin t \frac{e^{2t}}{2}\right)_{\pi/4}^{5\pi/4} - \int_{\pi/4}^{5\pi/4} \cos t \frac{e^{2t}}{2} dt\right]$$

$$= e^{-\pi/2}\left[\frac{1}{2}\left(e^{5\pi/2}\sin\frac{5\pi}{4} - e^{\pi/2}\sin\frac{\pi}{4}\right)\right.$$

$$\left. -\left(\frac{e^{2t}}{4}\cos t\right)_{\pi/4}^{5\pi/4} - \int_{\pi/4}^{5\pi/4} \frac{e^{2t}}{4}\sin t\, dt\right]$$

$$= e^{-\pi/2}\left[\frac{1}{2}\left(\frac{-1}{\sqrt{2}}e^{5\pi/2} - \frac{1}{\sqrt{2}}e^{\pi/2}\right)\right.$$

$$\left. -\frac{1}{4}\left(-\frac{1}{\sqrt{2}}e^{5\pi/2} - \frac{1}{\sqrt{2}}e^{\pi/2}\right)\right] - \frac{I}{4}$$

$$\Rightarrow \quad I + \frac{1}{4}I = -\frac{1}{2\sqrt{2}}\left[e^{2\pi}+1\right] + \frac{1}{4\sqrt{2}}[e^{2\pi}+1]$$

$$\Rightarrow \quad \frac{5}{4}I = \frac{(e^{2\pi}+1)}{2\sqrt{2}}\left[\frac{1}{2}-1\right] = -\frac{1}{4\sqrt{2}}[e^{2\pi}+1]$$

$$\Rightarrow \quad I = \frac{-1}{5\sqrt{2}}(1+e^{2\pi})$$

132. Let $I = \int_0^{\pi/4} \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$

$$= \int_0^{\pi/4} \frac{dx}{\cos^3 x\sqrt{2\cdot 2\sin x\cos x}}$$

$$= \frac{1}{2}\int \frac{dx}{\cos^{\left(3+\frac{1}{2}\right)} x \cdot \sin^{\frac{1}{2}} x}$$

www.exambites.in

$$= \frac{1}{2}\int_0^{\pi/4} \frac{dx}{\cos^{\frac{7}{2}} x \cdot \tan^{\frac{1}{2}} x \cdot \cos^{\frac{1}{2}} x}$$

$$= \frac{1}{2}\int_0^{\pi/4} \frac{dx}{\cos^4 x\sqrt{\tan x}} = \frac{1}{2}\int_0^{\pi/4} \frac{\sec^4 x}{\sqrt{\tan x}}\,dx$$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

Also $x = 0 \Rightarrow t = 0$ and $x = \frac{\pi}{4} \Rightarrow t = 1$

$$\therefore \quad I = \frac{1}{2}\int_0^1 \frac{(1+t^2)dt}{\sqrt{t}} = \frac{1}{2}\int_0^1 (t^{-\frac{1}{2}} + t^{\frac{3}{2}})\,dt$$

$$= \frac{1}{2}\left[\frac{t^{1/2}}{1/2} + \frac{t^{5/2}}{5/2}\right]_0^1 = 1 + \frac{1}{5} = \frac{6}{5}$$

133. Let $I = \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}}dx$

$$= \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{1-(1-\sin 2x)}}dx$$

$$= \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{1-(\sin x - \cos x)^2}}dx$$

Put $\sin x - \cos x = t \Rightarrow (\cos x + \sin x)dx = dt$

Also $x = \frac{\pi}{3} \Rightarrow t = \frac{\sqrt{3}-1}{2} = \alpha$

and $x = \frac{\pi}{6} \Rightarrow t = \frac{1-\sqrt{3}}{2} = -\alpha$

$$\therefore \quad I = \int_{-\alpha}^{\alpha} \frac{dt}{\sqrt{1-t^2}}$$

$$= \left[\sin^{-1} t\right]_{-\alpha}^{\alpha} = 2\sin^{-1}\alpha = 2\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$$

134. Let $I = \int_0^{\pi/4} \frac{\sin 2\theta}{\sin^4\theta + \cos^4\theta}\,d\theta$

$$= \int_0^{\pi/4} \frac{2\sin\theta\cos\theta}{\sin^4\theta + (1-\sin^2\theta)^2}\,d\theta$$

Put $\sin^2\theta = t \Rightarrow 2\sin\theta\cos\theta\,d\theta = dt$

Also $\theta = 0 \Rightarrow t = 0$ and $\theta = \frac{\pi}{4} \Rightarrow t = \frac{1}{2}$

$$\therefore \quad I = \int_0^{1/2} \frac{dt}{t^2 + (1-t)^2} = \int_0^{1/2} \frac{dt}{2t^2 - 2t + 1}$$

$$= \frac{1}{2}\int_0^{1/2} \frac{dt}{\left(t^2 - t + \frac{1}{2}\right)}$$

$$= \frac{1}{2}\int_0^{1/2} \frac{dt}{\left(t-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{2}\cdot\frac{1}{\frac{1}{2}}\left[\tan^{-1}\left(\frac{t-\frac{1}{2}}{\frac{1}{2}}\right)\right]_0^{1/2}$$

$$= \tan^{-1}\left[\frac{2t-1}{1}\right]_0^{1/2}$$

$$= \tan^{-1} 0 - \tan^{-1}(-1) = \frac{\pi}{4}$$

135. Let $I = \int_0^{\pi/4} \frac{\sin x + \cos x}{16 + 9\sin 2x}dx$

Put $\sin x - \cos x = t$

$\Rightarrow \quad (\cos x + \sin x)\,dx = dt$

and $1 - 2\sin x \cos x = t^2$

$\Rightarrow \quad 1 - \sin 2x = t^2$

When $x = \frac{\pi}{4}, t = \sin\frac{\pi}{4} - \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$

When $x = 0$, $t = \sin 0 - \cos 0 = -1$

$$\therefore \quad \int_0^{\pi/4} \frac{\sin x + \cos x}{16 + 9\sin 2x}\,dx = \int_{-1}^{0} \frac{dt}{16 + 9(1-t^2)}$$

$$= \int_{-1}^{0} \frac{dt}{25 - 9t^2} = \frac{1}{9}\int_{-1}^{0} \frac{dt}{\left(\frac{5}{3}\right)^2 - t^2}$$

$$= \frac{1}{9}\cdot\frac{1}{2\times\frac{5}{3}}\left[\log\left|\frac{\frac{5}{3}+t}{\frac{5}{3}-t}\right|\right]_{-1}^{0} + C$$

$$= \frac{1}{30}[\log 1 - (\log 1 - \log 4)] + C = \frac{1}{30}\log 4 + C$$

136. *Refer to answer 135.*

137. L.H.S. $= \int_0^{\pi/4} (\sqrt{\tan x} + \sqrt{\cot x})\,dx$

$$= \int_0^{\pi/4} \left(\sqrt{\frac{\sin x}{\cos x}} + \sqrt{\frac{\cos x}{\sin x}}\right)dx = \int_0^{\pi/4} \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\,dx$$

www.examtics.in

$$= \sqrt{2}\int_0^{\pi/4} \frac{(\sin x + \cos x)}{\sqrt{2\sin x \cos x}}\,dx$$

$$= \sqrt{2}\int_0^{\pi/4} \frac{\sin x + \cos x}{\sqrt{1-(\sin x - \cos x)^2}}\,dx$$

Let $\sin x - \cos x = t$ then $(\cos x + \sin x)\,dx = dt$

Also, $x = 0 \Rightarrow t = -1$ and $x = \pi/4 \Rightarrow t = 0$.

$$\therefore \int_0^{\pi/4} (\sqrt{\tan x} + \sqrt{\cot x})\,dx = \sqrt{2}\int_{-1}^{0} \frac{dt}{\sqrt{1-t^2}}$$

$$= \sqrt{2}\left[\sin^{-1} t\right]_{-1}^{0} = \sqrt{2}\,[\sin^{-1} 0 - \sin^{-1}(-1)]$$

$$= \sqrt{2}\cdot \sin^{-1} 1 = \sqrt{2}\cdot\frac{\pi}{2} = \text{R.H.S.}$$

138. Let $I = \int_{\pi/4}^{\pi/2} \cos 2x \log \sin x\,dx$

Integrating by parts taking $\cos 2x$ as 2nd function

$$\therefore I = \left[\frac{\sin 2x}{2}\cdot \log \sin x\right]_{\pi/4}^{\pi/2} - \int_{\pi/4}^{\pi/2} \frac{\sin 2x}{2}\cdot\frac{1}{\sin x}\cdot \cos x\,dx$$

$$= \frac{1}{2}\left[0 - 1\cdot \log \sin\frac{\pi}{4}\right] - \frac{1}{2}\int_{\pi/4}^{\pi/2} 2\sin x\cdot\cos x\cdot\frac{\cos x}{\sin x}\,dx$$

$$= -\frac{1}{2}\log\frac{1}{\sqrt{2}} - \int_{\pi/4}^{\pi/2} \cos^2 x\,dx$$

$$= -\frac{1}{2}(\log 1 - \log\sqrt{2}) - \int_{\pi/4}^{\pi/2} \frac{1+\cos 2x}{2}\,dx$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left[\frac{\pi}{2} + 0 - \left(\frac{\pi}{4} + \frac{1}{2}\right)\right] = \frac{1}{4}\log 2 - \frac{\pi}{8} + \frac{1}{4}$$

139. Let $I = \int_0^{\pi/2} 2\sin x\cdot\cos x\cdot \tan^{-1}(\sin x)dx$

Put $\sin x = t \Rightarrow \cos x\cdot dx = dt$

Also, $x = 0 \Rightarrow t = 0$ and $x = \frac{\pi}{2} \Rightarrow t = 1$

$$\therefore \quad I = 2\int_0^1 t \times \tan^{-1} t\cdot dt$$

$$= 2\left[\frac{t^2}{2}\times\tan^{-1} t\right]_0^1 - 2\int_0^1 \frac{1}{1+t^2}\times\frac{t^2}{2}\cdot dt$$

$$= 2\times\frac{1}{2}\times\tan^{-1}\left(\tan\frac{\pi}{4}\right) - \int_0^1 \frac{1+t^2-1}{1+t^2}\cdot dt$$

$$= 1\times\frac{\pi}{4} - \int_0^1\left(1 - \frac{1}{1+t^2}\right)\cdot dt = \frac{\pi}{4} - \left[t - \tan^{-1} t\right]_0^1$$

$$= \frac{\pi}{4} - 1 + \tan^{-1}\left(\tan\frac{\pi}{4}\right) = \frac{2\pi}{4} - 1 = \left(\frac{\pi}{2} - 1\right)$$

140. Let $I = \int_1^3 |2x-1|dx$

$$= \int_1^3 (2x-1)dx = \left[\frac{2x^2}{2} - x\right]_1^3$$

$$= [(3^2 - 3) - (1^2 - 1)] = [(9-3) - (1-1)] = 6$$

141. Let $I = \int_{-2}^{2} |x|dx$

$$\therefore \quad I = \int_{-2}^{0} (-x)dx + \int_0^2 x\,dx = \left[-\frac{x^2}{2}\right]_{-2}^{0} + \left[\frac{x^2}{2}\right]_0^2$$

$$= 2 + 2 = 4$$

142. Let $I = \int_1^4 |x-5|\,dx$

$$= -\int_1^4 (x-5)\,dx = \left[-\frac{x^2}{2} + 5x\right]_1^4$$

$$= -\frac{16}{2} + 5\cdot 4 + \frac{1}{2} - 5 = -8 + 20 - 5 + \frac{1}{2} = 7 + \frac{1}{2} = \frac{15}{2}$$

143. Let $I = \int_{-1}^{2} |\,x^3 - x\,|\,dx$

$$\text{Since, } |\,x^3 - x\,| = \begin{cases} x^3 - x, & x \in (-1,0)\cup(1,2) \\ -(x^3 - x), & x \in (0,1) \end{cases}$$

$$\therefore \quad I = \int_{-1}^{0} (x^3 - x)dx + \int_0^1 -(x^3 - x)dx + \int_1^2 (x^3 - x)dx$$

$$= \left(\frac{x^4}{4} - \frac{x^2}{2}\right)_{-1}^{0} + \left(\frac{-x^4}{4} + \frac{x^2}{2}\right)_0^1 + \left(\frac{x^4}{4} - \frac{x^2}{2}\right)_1^2$$

$$= \left[0 - \left(\frac{1}{4} - \frac{1}{2}\right)\right] + \left[-\frac{1}{4} + \frac{1}{2} - 0\right] + \left[(4-2) - \left(\frac{1}{4} - \frac{1}{2}\right)\right]$$

$$= \frac{1}{4} + \frac{1}{4} + 2 + \frac{1}{4} = \frac{11}{4}$$

144. In the R.H.S. integral, put $(a - x) = t$ so that $dx = -dt$.

Now, when $x = 0 \Rightarrow t = a$

and, when $x = a \Rightarrow t = 0$

www.exambites.in

$$\therefore \int_0^a f(a-x)dx = -\int_a^0 f(t)\,dt = \int_0^a f(t)\,dt = \int_0^a f(x)dx$$

Hence, $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ ...(i)

Let $I = \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$ ... (ii)

$$= \int_0^{\pi} \frac{(\pi-x)\sin(\pi-x)}{1+[\cos(\pi-x)]^2}dx \quad \text{[Using (i)]}$$

$$\Rightarrow \quad I = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+\cos^2 x}dx \quad ...(iii)$$

Adding (ii) and (iii), we get

$$2I = \int_0^{\pi} \frac{\pi\sin x}{1+\cos^2 x}dx$$

Put $\cos x = t \Rightarrow -\sin x\,dx = dt$

Also, $x = 0 \Rightarrow t = 1$ and $x = \pi \Rightarrow t = -1$

$$\therefore \quad 2I = \int_1^{-1} \frac{-\pi dt}{1+t^2} \Rightarrow I = \frac{\pi}{2}\int_{-1}^{1} \frac{dt}{1+t^2}$$

$$\therefore \quad I = \frac{\pi}{2}[\tan^{-1}t]_{-1}^{1} = \frac{\pi}{2}[\tan^{-1}(1) - \tan^{-1}(-1)]$$

$$= \frac{\pi}{2}\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \frac{\pi^2}{4}.$$

145. *Refer to answer 144.*

We have, $\int_0^a f(x)\,dx = \int_0^a f(a-x)dx$ ...(i)

Now, $\int_0^1 x^2(1-x)^n dx = \int_0^1 (1-x)^2(1-(1-x))^n dx$ [using (i)]

$$= \int_0^1 (1-x)^2 x^n dx = \int_0^1 (1+x^2-2x)x^n dx$$

$$= \int_0^1 (x^n + x^{n+2} - 2x^{n+1})\,dx$$

$$= \int_0^1 x^n dx + \int_0^1 x^{n+2}dx - 2\int_0^1 x^{n+1}dx$$

$$= \left[\frac{x^{n+1}}{n+1}\right]_0^1 + \left[\frac{x^{n+3}}{n+3}\right]_0^1 - 2\left[\frac{x^{n+2}}{n+2}\right]_0^1$$

$$= \frac{1}{n+1} + \frac{1}{n+3} - \frac{2}{n+2}.$$

146. *Refer to answer 144.*

147. Let $I = \int_0^{3/2} |x\,\sin \pi x|dx$

$$= \int_0^1 x\,\sin \pi x\,dx - \int_1^{3/2} x\sin \pi x\,dx$$

$$= \left[\frac{-x\cos \pi x}{\pi}\right]_0^1 + \int_0^1 \frac{\cos \pi x}{\pi}dx + \left[\frac{x\cos \pi x}{\pi}\right]_1^{3/2} - \int_1^{3/2} \frac{\cos \pi x}{\pi}dx$$

$$= \frac{1}{\pi} - 0 + \left[\frac{\sin \pi x}{\pi^2}\right]_0^1 + 0 - \frac{(-1)}{\pi} - \left[\frac{\sin \pi x}{\pi^2}\right]_1^{3/2}$$

$$= \frac{2}{\pi} + 0 - \frac{(-1)}{\pi^2} + 0 = \frac{2}{\pi} + \frac{1}{\pi^2}$$

148. Let $I = \int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}dx$

$$= \int_0^{\pi} \frac{x\frac{\sin x}{\cos x}}{\frac{1}{\cos x} + \frac{\sin x}{\cos x}}dx = \int_0^{\pi} \frac{x\sin x}{1+\sin x}dx \quad ...(i)$$

$$= \int_0^{\pi} \frac{(\pi-x)\sin(\pi-x)}{1+\sin(\pi-x)}dx$$

$$\left[\text{Using the property } \int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$\Rightarrow \quad I = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+\sin x}dx \quad ...(ii)$$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi} \frac{\pi\sin x}{1+\sin x}dx$$

$$I = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{1+\sin x}dx = \frac{\pi}{2}\int_0^{\pi} \frac{1+\sin x-1}{1+\sin x}dx$$

$$= \frac{\pi}{2}\int_0^{\pi} \left(1 - \frac{1}{1+\sin x}\right)dx$$

www.exambites.in

$$= \frac{\pi}{2}\left[\int_0^{\pi} dx - \int_0^{\pi} \frac{1}{1+\sin x}\,dx\right]$$

$$= \frac{\pi}{2}\left[|\,x\,|_0^{\pi} - \int_0^{\pi} \frac{1-\sin x}{\cos^2 x}\,dx\right]$$

$$= \frac{\pi}{2}\left[|\,x\,|_0^{\pi} - \int_0^{\pi} \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x}\right)dx\right]$$

$$= \frac{\pi}{2}\left[(\pi - 0) - \int_0^{\pi} (\sec^2 x - \tan x \cdot \sec x)\,dx\right]$$

$$= \frac{\pi}{2}\left[\pi - (\tan x - \sec x)_0^{\pi}\right]$$

$$= \frac{\pi}{2}\left[\pi - [(0-(-1)) - (0-1)]\right] = \frac{\pi}{2}[\pi - 2]$$

149. Let $I = \int_1^4 (|\,x-1\,| + |\,x-2\,| + |\,x-4\,|)\,dx$

Also, let $f(x) = |x-1| + |x-2| + |x-4|$

We have three critical points $x = 1, 2, 4$.

$$f(x) = \begin{cases} (x-1)-(x-2)-(x-4) \text{ if } 1 \le x < 2 \\ (x-1)+(x-2)-(x-4) \text{ if } 2 \le x < 4 \end{cases}$$

$$\therefore f(x) = \begin{cases} -x+5 \text{ if } 1 \le x < 2 \\ x+1 \text{ if } 2 \le x < 4 \end{cases}$$

$$\therefore I = \int_1^4 f(x)\,dx = \int_1^2 f(x)\,dx + \int_2^4 f(x)\,dx$$

$$= \int_1^2 (-x+5)\,dx + \int_2^4 (x+1)\,dx$$

$$= \left[-\frac{x^2}{2} + 5x\right]_1^2 + \left[\frac{x^2}{2} + x\right]_2^4$$

$$= \left(-\frac{4}{2} + 10\right) - \left(-\frac{1}{2} + 5\right) + \left(\frac{16}{2} + 4\right) - \left(\frac{4}{2} + 2\right)$$

$$= 8 - \frac{9}{2} + 12 - 4 = 16 - \frac{9}{2} = \frac{23}{2}$$

150. Let $I = \int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx$ ...(1)

$$\Rightarrow I = \int_0^{\pi/2} \frac{\sin^2\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}\,dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\cos^2 x}{\cos x + \sin x}\,dx \quad \text{...(2)}$$

Adding (1) and (2), we get

$$2I = \int_0^{\pi/2} \left(\frac{\sin^2 x}{\sin x + \cos x} + \frac{\cos^2 x}{\sin x + \cos x}\right)dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{1}{\sin x + \cos x}\,dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{1}{\frac{2\tan(x/2)}{1+\tan^2(x/2)} + \frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}}\,dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{1+\tan^2(x/2)}{2\tan\frac{x}{2} + 1 - \tan^2\frac{x}{2}}\,dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sec^2\frac{x}{2}}{2\tan\frac{x}{2} + 1 - \tan^2\frac{x}{2}}\,dx$$

Put $\tan\frac{x}{2} = t \Rightarrow \sec^2\frac{x}{2} \cdot \frac{1}{2}dx = dt$

When $x = 0 \Rightarrow t = 0$ and when $x = \frac{\pi}{2} \Rightarrow t = 1$

$$\therefore 2I = \int_0^1 \frac{2\,dt}{2t + 1 - t^2} = 2\int_0^1 \frac{1}{(\sqrt{2})^2 - (t-1)^2}\,dt$$

$$\Rightarrow 2I = 2 \times \frac{1}{2\sqrt{2}}\left[\log\left|\frac{\sqrt{2}+t-1}{\sqrt{2}-t+1}\right|\right]_0^1$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}}\left\{\log\left(\frac{\sqrt{2}}{\sqrt{2}}\right) - \log\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)\right\}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}}\left\{0 - \log\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)\right\} = \frac{1}{\sqrt{2}}\log\frac{\sqrt{2}+1}{\sqrt{2}-1}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}}\log\left\{\frac{(\sqrt{2}+1)^2}{(\sqrt{2}-1)(\sqrt{2}+1)}\right\}$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}}\log(\sqrt{2}+1)^2 = \frac{2}{\sqrt{2}}\log(\sqrt{2}+1)$$

$$\Rightarrow I = \frac{1}{\sqrt{2}}\log(\sqrt{2}+1)$$

www.exambites.in

151. When $0 < x < \frac{1}{2} \Rightarrow 0 < \pi x < \frac{\pi}{2} \Rightarrow \cos \pi x > 0$

When $\frac{1}{2} < x < \frac{3}{2} \Rightarrow \frac{\pi}{2} < \pi x < \frac{3\pi}{2} \Rightarrow \cos \pi x < 0$

$$\therefore \ |x \cos \pi x| = \begin{cases} x \cos \pi x, \text{ if } 0 < x < \frac{1}{2} \\ -x \cos \pi x, \text{ if } \frac{1}{2} < x < \frac{3}{2} \end{cases}$$

Let $I = \int_0^{3/2} |x \cos \pi x| \, dx$

$$= \int_0^{1/2} x \cos \pi x \, dx + \int_{1/2}^{3/2} -(x \cos \pi x) \, dx$$

$$\therefore \ I = \left[\frac{x}{\pi} \sin \pi x + \frac{\cos \pi x}{\pi^2}\right]_0^{1/2} - \left[\frac{x}{\pi} \sin \pi x + \frac{\cos \pi x}{\pi^2}\right]_{1/2}^{3/2}$$

(Using by parts)

$$= \left[\frac{1}{\pi}\left(\frac{1}{2} - 0\right) + \frac{1}{\pi^2}(0 - 1)\right] - \left[\frac{1}{\pi}\left(\frac{3}{2}(-1) - \frac{1}{2}(1)\right) + \frac{1}{\pi^2}(0 - 0)\right]$$

$$= \left(\frac{1}{2\pi} - \frac{1}{\pi^2}\right) - \left(\frac{-2}{\pi}\right) = \left(\frac{5\pi - 2}{2\pi^2}\right)$$

152. Let $I = \int_0^{\pi} \frac{x}{1 + \sin \alpha \sin x} dx$

$$\Rightarrow \quad I = \int_0^{\pi} \frac{\pi - x}{1 + \sin \alpha \sin(\pi - x)} dx$$

$$\left[\because \int_0^a f(x) dx = \int_0^a f(a - x) dx\right]$$

$$\Rightarrow \quad I = \int_0^{\pi} \frac{\pi}{1 + \sin \alpha \sin x} dx - \int_0^{\pi} \frac{x}{1 + \sin \alpha \sin x} dx$$

$$\Rightarrow \quad I = \int_0^{\pi} \frac{\pi}{1 + \sin \alpha \sin x} dx - I$$

$$\Rightarrow \quad 2I = \int_0^{\pi} \frac{\pi}{1 + \sin \alpha \sin x} dx$$

$$\Rightarrow \quad 2I = \pi \int_0^{\pi} \frac{1}{1 + \sin \alpha \sin x} dx$$

$$\Rightarrow \quad 2I = \pi \int_0^{\pi} \frac{1 + \tan^2 \frac{x}{2}}{\left(1 + \tan^2 \frac{x}{2} + \sin \alpha \times 2 \tan \frac{x}{2}\right)} dx$$

$$\therefore \quad I = \frac{\pi}{2} \int_0^{\pi} \frac{\sec^2 \frac{x}{2}}{\left(1 + \tan^2 \frac{x}{2} + \sin \alpha \times 2 \tan \frac{x}{2}\right)} dx$$

Let $\tan \frac{x}{2} = t \Rightarrow \sec^2 \frac{x}{2} dx = 2dt$

Also, when $x \to 0$, $t \to \tan 0 = 0$;

when $x \to \pi$, $t \to \tan \frac{\pi}{2} = \infty$

$$\therefore \quad I = \frac{\pi}{2} \int_0^{\infty} \frac{2dt}{t^2 + 2t \sin \alpha + 1}$$

$$\Rightarrow \quad I = \pi \int_0^{\infty} \frac{1}{(t + \sin \alpha)^2 + \cos^2 \alpha} dt$$

$$\Rightarrow \quad I = \frac{\pi}{\cos \alpha} \left[\tan^{-1}\left(\frac{t + \sin \alpha}{\cos \alpha}\right)\right]_0^{\infty}$$

$$\Rightarrow \quad I = \frac{\pi}{\cos \alpha} \left[\tan^{-1} \infty - \tan^{-1}(\tan \alpha)\right]$$

$$\Rightarrow \quad I = \frac{\pi}{\cos \alpha} \left(\frac{\pi}{2} - \alpha\right)$$

153. Let $I = \int_{-\pi}^{\pi} (\cos ax - \sin bx)^2 \, dx$

$$= \int_{-\pi}^{\pi} (\cos^2 ax + \sin^2 bx - 2 \cos ax \sin bx) \, dx$$

$$= \int_{-\pi}^{\pi} \cos^2 ax \, dx + \int_{-\pi}^{\pi} \sin^2 bx \, dx - 2 \int_{-\pi}^{\pi} \cos ax \sin bx \, dx$$

$$= 2\left[\int_0^{\pi} \cos^2 ax \, dx + \int_0^{\pi} \sin^2 bx \, dx\right]$$

$$\left[\because \int_{-a}^{a} f(x) dx = \begin{cases} 0, \text{ if } f(x) \text{ is odd} \\ 2 \int_0^a f(x) dx, \text{if } f(x) \text{ is even} \end{cases}\right]$$

$$= 2\left[\int_0^{\pi} \left(\frac{1 + \cos 2ax}{2}\right) dx + \int_0^{\pi} \left(\frac{1 - \cos 2bx}{2}\right) dx\right]$$

$$= \int_0^{\pi} (1 + \cos 2ax) dx + \int_0^{\pi} (1 - \cos 2bx) dx$$

$$= |2x|_0^{\pi} + \left|\frac{1}{2a}\right| \sin 2ax \, |_0^{\pi} - \frac{1}{2b} |\sin 2bx|_0^{\pi}$$

www.exambites.in

$$= 2\pi + \frac{\sin 2a\pi}{2a} - \frac{\sin 2b\pi}{2b}$$

154. Let $I = \int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x} dx = \int_{-\pi/2}^{\pi/2} \frac{\cos x}{e^{-x}+1} e^{-x} dx$

$$= \int_{-\pi/2}^{\pi/2} \frac{\cos x(e^{-x}+1-1)}{e^{-x}+1} dx$$

$$= \int_{-\pi/2}^{\pi/2} \cos x \, dx - \int_{-\pi/2}^{\pi/2} \frac{\cos x}{e^{-x}+1} dx$$

Now, put $x = -z$ in 2nd integral,

$\therefore \quad dx = -dz$

Also, $x = \frac{-\pi}{2} \Rightarrow z = \frac{\pi}{2}$ and $x = \frac{\pi}{2} \Rightarrow z = \frac{-\pi}{2}$

$$\therefore \quad I = \int_{-\pi/2}^{\pi/2} \cos x \, dx + \int_{\pi/2}^{-\pi/2} \frac{\cos z}{e^z+1} dz$$

$$I = [\sin x]_{-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2} \frac{\cos x}{e^x+1} dx$$

$$I = \left[\sin\frac{\pi}{2} + \sin\frac{\pi}{2}\right] - I \Rightarrow 2I = 2 \Rightarrow I = 1$$

155. Let $I = \int_0^{\pi/2} \frac{dx}{1+\sqrt{\tan x}} = \int_0^{\pi/2} \frac{dx}{1+\sqrt{\frac{\sin x}{\cos x}}}$

$$\Rightarrow \quad I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}} dx \qquad ...(1)$$

By the property, $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, we get

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}+\sqrt{\sin\left(\frac{\pi}{2}-x\right)}} dx$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx \qquad ...(2)$$

Adding (1) and (2), we get

$$2I = \int_0^{\pi/2} \left[\frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}} + \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\right] dx$$

$$= \int_0^{\pi/2} 1 \cdot dx = [x]_0^{\pi/2} = \frac{\pi}{2} \quad \Rightarrow \quad I = \frac{\pi}{4}$$

156. Let $I = \int_0^{\pi/4} \log(1+\tan x) dx$

By using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, we get

$$I = \int_0^{\pi/4} \log\left[1+\tan\left(\frac{\pi}{4}-x\right)\right] dx$$

$$= \int_0^{\pi/4} \log\left[1+\frac{1-\tan x}{1+\tan x}\right] dx = \int_0^{\pi/4} \log\left(\frac{2}{1+\tan x}\right) dx$$

$$= \int_0^{\pi/4} [\log 2 - \log(1+\tan x)] dx$$

$$= \log 2 \int_0^{\pi/4} 1 \cdot dx - \int_0^{\pi/4} \log(1+\tan x) dx$$

$$= \log 2 [x]_0^{\pi/4} - I$$

$$\Rightarrow \quad 2I = \log 2\left[\frac{\pi}{4} - 0\right] \Rightarrow I = \frac{\pi}{8}\log 2$$

157. Let $I = \int_0^{\pi} \frac{x \tan x}{\sec x \operatorname{cosec} x} dx$ ...(1)

Using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, we get

$$I = \int_0^{\pi} \frac{(\pi-x)\tan(\pi-x)}{\sec(\pi-x)\operatorname{cosec}(\pi-x)} dx = \int_0^{\pi} \frac{(\pi-x)\tan x}{\sec x \operatorname{cosec} x}$$

...(2)

Adding (1) and (2), we get

$$2I = \int_0^{\pi} \left[\frac{x\tan x}{\sec x \operatorname{cosec} x} + \frac{(\pi-x)\tan x}{\sec x \operatorname{cosec} x}\right] dx$$

$$= \pi \int_0^{\pi} \frac{\tan x}{\sec x \operatorname{cosec} x} dx = \pi \int_0^{\pi} \frac{\sin x / \cos x}{\frac{1}{\cos x} \cdot \frac{1}{\sin x}} dx$$

$$= \pi \int_0^{\pi} \sin^2 x \, dx = \pi \int_0^{\pi} \frac{1-\cos 2x}{2} dx$$

$$= \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\pi} = \frac{\pi^2}{2} \quad \Rightarrow \quad I = \frac{\pi^2}{4}$$

www.exambites.in

158. Let $I=\int_0^{\pi}\frac{4x\sin x}{1+\cos^2 x}dx$ ... (1)

$$I=\int_0^{\pi}\frac{4(\pi-x)\sin(\pi-x)}{1+\cos^2(\pi-x)}dx$$

$$\left[\text{By }\int_0^a f(x)dx=\int_0^a f(a-x)dx\right]$$

$$\Rightarrow\quad I=\int_0^{\pi}\frac{4(\pi-x)\sin x}{1+\cos^2 x}dx \qquad \text{... (2)}$$

Adding (1) and (2), we get

Hence, $2I=4\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx$

Put $\cos x = t \Rightarrow \sin x\,dx = -dt$

Also $x = 0 \Rightarrow t = 1$ and $x = \pi \Rightarrow t = -1$

$$\therefore\quad I=2\pi\int_1^{-1}\frac{-dt}{1+t^2}=2\pi\int_{-1}^{1}\frac{dt}{t^2+1^2}=2\pi\left[\tan^{-1}t\right]_{-1}^{1}$$

$$=2\pi\left[\tan^{-1}1-\tan^{-1}(-1)\right]$$

$$=2\pi\left[\frac{\pi}{4}-\left(-\frac{\pi}{4}\right)\right]=\pi^2$$

159. *Refer to answer 150.*

160. Let $I=\int_0^4(|x|+|x-2|+|x-4|)dx$

$$=\int_0^2[x-(x-2)-(x-4)]dx+\int_2^4[x+(x-2)-(x-4)]dx$$

$$=\int_0^2(-x+6)dx+\int_2^4(x+2)dx$$

$$=\left[-\frac{x^2}{2}+6x\right]_0^2+\left[\frac{x^2}{2}+2x\right]_2^4$$

$= (-2 + 12 - 0) + [8 + 8 - (2 + 4)]$

$= 10 + 10 = 20$

161. Let $I=\int_2^5[|x-2|+|x-3|+|x-5|]dx$

$$=\int_2^3[(x-2)-(x-3)-(x-5)]dx$$

$$+\int_3^5[(x-2)+(x-3)-(x-5)]dx$$

$$=\int_2^3(-x+6)dx+\int_3^5(x)dx=\left[-\frac{x^2}{2}+6x\right]_2^3+\left[\frac{x^2}{2}\right]_3^5$$

$$=\left[\left(-\frac{9}{2}+18\right)-(-2+12)\right]+\frac{1}{2}(25-9)$$

$$=-\frac{9}{2}+8+8=\frac{23}{2}$$

162. *Refer to answer 149.*

163. Let $I=\int_0^{2\pi}\frac{1}{1+e^{\sin x}}dx$

$$=\int_0^{\pi}\frac{dx}{1+e^{\sin x}}+\int_{\pi}^{2\pi}\frac{dx}{1+e^{\sin x}}=\int_0^{\pi}\frac{dx}{1+e^{\sin x}}+I_1 \qquad ...(1)$$

where $I_1=\int_{\pi}^{2\pi}\frac{dx}{1+e^{\sin x}}$

$$=\int_0^{\pi}\frac{dy}{1+e^{-\sin y}} \qquad \text{(Put : } x=\pi+y\text{)}$$

$$=\int_0^{\pi}\frac{e^{\sin y}\,dy}{e^{\sin y}+1}=\int_0^{\pi}\frac{e^{\sin x}}{1+e^{\sin x}}dx$$

$\therefore$ From (1),

$$I=\int_0^{\pi}\frac{dx}{1+e^{\sin x}}+\int_0^{\pi}\frac{e^{\sin x}}{1+e^{\sin x}}dx$$

$$\Rightarrow\quad I=\int_0^{\pi}\frac{1+e^{\sin x}}{1+e^{\sin x}}dx=\int_0^{\pi}1\cdot dx=[x]_0^{\pi}=\pi$$

164. Let $I=\int_0^{\pi}\frac{x}{1+\sin x}dx$ ...(1)

Using $\int_0^a f(x)dx=\int_0^a f(a-x)dx$, we get

$$I=\int_0^{\pi}\frac{\pi-x}{1+\sin(\pi-x)}dx=\int_0^{\pi}\frac{\pi-x}{1+\sin x}dx \qquad ...(2)$$

Adding (1) and (2), we get

$$2I=\int_0^{\pi}\frac{\pi}{1+\sin x}dx=\pi\int_0^{\pi}\frac{dx}{1+\sin x}$$

www.exambites.in

$$= \pi\int_0^{\pi} \frac{1}{1+\sin x} \times \frac{1-\sin x}{1-\sin x} dx = \pi\int_0^{\pi} \frac{1-\sin x}{\cos^2 x} dx$$

$$= \pi\int_0^{\pi} (\sec^2 x - \tan x \sec x) dx = \pi[\tan x - \sec x]_0^{\pi}$$

$= \pi[0 - \sec\pi - (0 - \sec 0)] = \pi[1 + 1] = 2\pi \Rightarrow I = \pi$

165. Let $I = \int_0^1 \log\left(\frac{1}{x} - 1\right) dx$ ... (1)

$$\Rightarrow \quad I = \int_0^1 \log\left(\frac{1}{1-x} - 1\right) dx$$

$$\left[\because \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$$

$$\Rightarrow \quad I = \int_0^1 \log\left(\frac{1-1+x}{1-x}\right) dx$$

$$\Rightarrow \quad I = \int_0^1 \log\left(\frac{x}{1-x}\right) dx \qquad \text{... (2)}$$

Adding (1) and (2), we get

$$2I = \int_0^1 \left[\log\left(\frac{1-x}{x}\right) + \log\left(\frac{x}{1-x}\right)\right] dx$$

$$= \int_0^1 \log 1 \cdot dx = 0 \quad \Rightarrow I = 0$$

166. Let $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{\left(1+\sqrt{\tan x}\right)}$

$$\Rightarrow \quad I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{\left(1+\sqrt{\frac{\sin x}{\cos x}}\right)} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}} dx \quad ...(1)$$

$$\Rightarrow \quad I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)}}{\sqrt{\cos\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)}+\sqrt{\sin\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)}} dx$$

$$\left[\because \int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$\Rightarrow \quad I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}+\sqrt{\sin\left(\frac{\pi}{2}-x\right)}} dx$$

$$\Rightarrow \quad I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx \qquad ...(2)$$

Adding (1) and (2), we get

$$2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x}+\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} dx$$

$$\Rightarrow \quad 2I = [x]_{\pi/6}^{\pi/3} = \left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \frac{\pi}{6} \Rightarrow 2I = \frac{\pi}{6}$$

$$\Rightarrow \quad I = \frac{\pi}{12}$$

167. $I = \int_0^1 \frac{\log(1+x)}{1+x^2} dx$

Put $x = \tan\theta \Rightarrow dx = \sec^2\theta d\theta$

Also, $x = 0 \Rightarrow \theta = 0$ and $x = 1 \Rightarrow \theta = \frac{\pi}{4}$

$$\therefore \quad I = \int_0^{\pi/4} \frac{\log(1+\tan\theta)}{1+\tan^2\theta} \cdot \sec^2\theta d\theta$$

$$= \int_0^{\pi/4} \log(1+\tan\theta) d\theta$$

Refer to answer 156.

168. Let $I = \int_0^{\pi/2} \frac{x\sin x\cos x}{\sin^4 x + \cos^4 x} dx$ ...(1)

$$\therefore \quad I = \int_0^{\pi/2} \frac{\left(\frac{\pi}{2}-x\right)\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)}{\sin^4\left(\frac{\pi}{2}-x\right)+\cos^4\left(\frac{\pi}{2}-x\right)} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2}-x\right)\sin x\cos x dx}{\cos^4 x + \sin^4 x} \qquad ...(2)$$

Adding (1) and (2), we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{\sin x\cos x dx}{\sin^4 x + \cos^4 x}$$

www.exambites.in

Dividing numerator and denominator by $\cos^4 x$, we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{\tan x \sec^2 x dx}{1+\tan^4 x}.$$

Put $\tan^2 x = t \Rightarrow 2\tan x \sec^2 x dx = dt$

When $x = 0$, $t = 0$ and when $x = \frac{\pi}{2}$, $t = \infty$

$$\therefore \quad I = \frac{\pi}{8}\int_0^{\infty} \frac{dt}{1+t^2} = \frac{\pi}{8}\left[\tan^{-1} t\right]_0^{\infty} = \frac{\pi}{8}\times\frac{\pi}{2} = \frac{\pi^2}{16}$$

169. Let $I = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$

Refer to answer 166.

170. Let $I = \int_0^{\pi} \frac{x\,dx}{a^2\cos^2 x + b^2 \sin^2 x}$ ...(1)

$$\Rightarrow \quad I = \int_0^{\pi} \frac{(\pi - x)dx}{a^2\cos^2(\pi - x) + b^2\sin^2(\pi - x)}$$

$$\left[\text{Using} \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$$

$$\Rightarrow \quad I = \int_0^{\pi} \frac{(\pi - x)dx}{a^2\cos^2 x + b^2\sin^2 x} \quad ...(2)$$

Adding (1) and (2), we get

$$I = \frac{\pi}{2}\int_0^{\pi} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x}$$

Let $f(x) = \frac{1}{a^2\cos^2 x + b^2\sin^2 x}$

$$\Rightarrow \quad f(\pi - x) = \frac{1}{a^2\cos^2(\pi - x) + b^2\sin^2(\pi - x)}$$

$$\Rightarrow \quad f(\pi - x) = \frac{1}{a^2\cos^2 x + b^2\sin^2 x} = f(x)$$

$$\left[\therefore \text{ using} \int_0^{2a} f(x)dx = 2\int_0^a f(x)dx, \text{ if } f(2a-x) = f(x)\right]$$

$$\therefore \quad I = \frac{\pi}{2}\left(2\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x}\right)$$

$$\Rightarrow \quad I = \pi\int_0^{\pi/2} \frac{\sec^2 x dx}{a^2 + b^2\tan^2 x}$$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$.

Also when $x = 0 \Rightarrow t = \tan 0 = 0$.

And when $x = \frac{\pi}{2} \Rightarrow t = \tan\frac{\pi}{2} = \infty$

$$\therefore \quad I = \int_0^{\infty} \frac{dt}{a^2 + b^2t^2} \Rightarrow I = \frac{\pi}{b^2}\int_0^{\infty} \frac{dt}{\left(\frac{a}{b}\right)^2 + t^2}$$

$$\Rightarrow \quad I = \frac{\pi}{b^2}\left[\frac{b}{a}\tan^{-1}\left(\frac{bt}{a}\right)\right]_0^{\infty}$$

$$= \quad I = \frac{\pi}{ab}\left[\tan^{-1}\infty - \tan^{-1} 0\right] = \frac{\pi^2}{2ab}$$

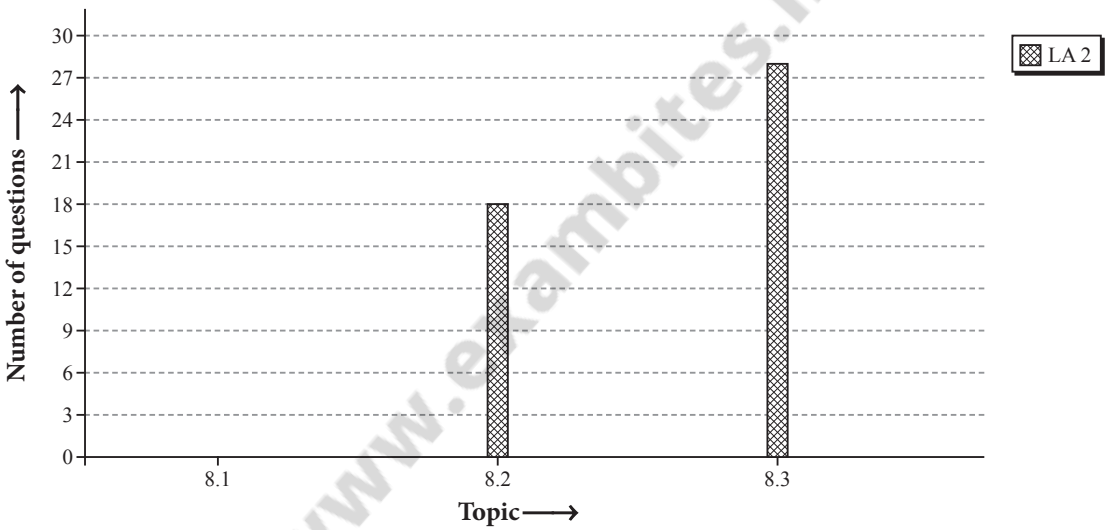
www.exambites.in

CHAPTER
8

Application of Integrals

8.1 Introduction
8.2 Area under Simple Curves
8.3 Area between Two Curves

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

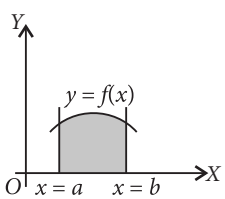
- Maximum weightage is of *Area between Two Curves*
- Maximum LA 2 type questions were asked from *Area between Two Curves*
- No VSA, SA and LA 1 type questions were asked till now

QUICK RECAP

AREA UNDER SIMPLE CURVES

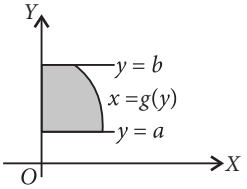
- Area of the region bounded by the curve $y = f(x)$, x-axis and the lines $x = a$ and $x = b$ $(b > a)$ is,

$$\text{Area} = \int_a^b y\,dx = \int_a^b f(x)\,dx$$



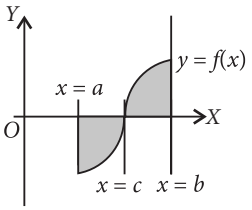
- Area of the region bounded by the curve $x = g(y)$, y-axis and the lines $y = a$ and $y = b$ $(b > a)$

is, Area $= \int_a^b x\,dy = \int_a^b g(y)\,dy$



- Area of the region bounded by the curve $y = f(x)$, some portion of which is above the x-axis and some below the x-axis is,

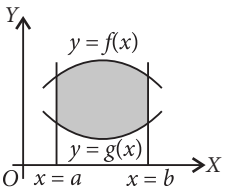
$$\text{Area} = \left|\int_a^c f(x)dx\right| + \int_c^b f(x)dx$$



AREA BETWEEN TWO CURVES

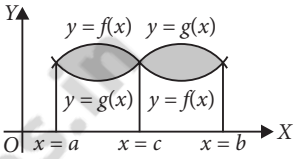
- Area of the region between two curves $y = f(x)$, $y = g(x)$ and the lines $x = a$, $x = b$ is,

$$\text{Area} = \int_a^b [f(x) - g(x)]\,dx,\ f(x) \geq g(x) \text{ in } [a,b]$$



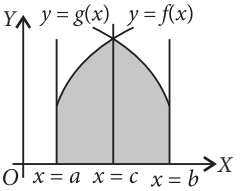
If $f(x) \geq g(x)$ in $[a, c]$ and $f(x) \leq g(x)$ in $[c, b]$, where $a < c < b$, then

- $$\text{Area} = \int_a^c [f(x) - g(x)]\,dx + \int_c^b [g(x) - f(x)]\,dx$$



- Area of shaded portion as shown in figure,

$$\text{Area} = \int_a^c f(x)dx + \int_c^b g(x)dx$$



www.exambites.in

Previous Years' CBSE Board Questions

8.2 Area under Simple Curves

LA 2 (6 marks)

1. Using integration, find the smaller area enclosed by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$. *(2020)*

2. Using integration, find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$. *(2018, Delhi 2014)*

3. Find the area bounded by the circle $x^2 + y^2 = 16$ and the line $\sqrt{3}y = x$ in the first quadrant, using integration. *(Delhi 2017)*

4. Find the area enclosed between the parabola $4y = 3x^2$ and the straight line $3x - 2y + 12 = 0$. *(AI 2017, 2015C)*

5. Using integration, find the area of the region bounded by the line $x - y + 2 = 0$, the curve $x = \sqrt{y}$ and y – axis. *(Foreign 2015)*

6. Find the area of the region in the first quadrant enclosed by the y-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$, using integration. *(Delhi 2015C)*

7. Find the area of the smaller region bounded by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the line $\frac{x}{3} + \frac{y}{2} = 1$. *(Foreign 2014)*

8. Using integration, find the area of the region bounded by the curves :
$y = |x + 1| + 1, x = -3, x = 3, y = 0$
(Delhi 2014C)

9. Using integration, find the area bounded by the curve $x^2 = 4y$ and the line $x = 4y - 2$. *(Delhi 2014C, 2013, 2013C)*

10. Using integration, find the area of the region in the first quadrant enclosed by the x – axis, the line $y = x$ and the circle $x^2 + y^2 = 18$. *(AI 2014C)*

11. Find the area of the region bounded by the parabola $y = x^2$ and $y = |x|$. *(AI 2013)*

12. Using integration, find the area of the region bounded by the curves $y = x^2$ and $y = x$. *(Delhi 2013C)*

13. Using integration, find the area of the region enclosed by the curves $y^2 = 4x$ and $y = x$. *(Delhi 2013C)*

14. Find the area of the region $\{(x, y) : x^2 + y^2 \le 4, x + y \ge 2\}$. *(AI 2012)*

15. Draw the graph of $y = |x + 1|$and using integration, find the area below $y = |x + 1|$, above x - axis and between $x = -4$ to $x = 2$. *(Delhi 2012C)*

16. Using integration, find the area of the region given by $\{(x, y) : x^2 \le y \le |x|\}$ *(AI 2012C, Delhi 2011C)*

17. Sketch the graph of $y = | x + 3 |$ and evaluate the area under the curve $y = | x + 3|$,above x-axis and between $x = -6$ to $x = 0$. *(AI 2011)*

18. Find the area of the region $\{(x, y) : x^2 + y^2 \le 1 \le x + y\}$. *(AI 2011C)*

8.3 Area between Two Curves

LA 2 (6 marks)

19. Using integration, find the area of the region bounded by the triangle whose vertices are (2, –2), (4, 5) and (6, 2). *(2020)*

20. Using integration, find the area of the region bounded by the lines
$x - y = 0$, $3x - y = 0$ and $x + y = 12$. *(2020)*

21. Using integration find the area of the region bounded between the two circles $x^2 + y^2 = 9$ and $(x - 3)^2 + y^2 = 9$. *(2020)*

22. Using integration, find the area of triangle ABC, whose vertices are $A(2, 5)$, $B(4, 7)$ and $C(6, 2)$. *(Delhi 2019)*

www.exambites.in

23. Find the area of the region lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and inside of the parabola $y^2 = 4x$.
(Delhi 2019)

24. Using integration, find the area of the region bounded by the parabola $y^2 = 4x$ and the circle $4x^2 + 4y^2 = 9$. *(AI 2019)*

25. Using the method of integration, find the area of the region bounded by the lines $3x - 2y + 1 = 0$, $2x + 3y - 21 = 0$ and $x - 5y + 9 = 0$. *(AI 2019, Delhi 2012)*

26. Using integration, find the area of region bounded by the triangle whose vertices are $(-2, 1)$, $(0, 4)$ and $(2, 3)$. *(Delhi 2017)*

27. Using the method of integration, find the area of the triangle ABC, coordinates of whose vertices are $A(4, 1)$, $B(6, 6)$ and $C(8, 4)$.
(AI 2017)

28. Using integration find the area of the region $\{(x, y) : x^2 + y^2 \le 2ax, y^2 \ge ax; x, y \ge 0\}$.
(Delhi 2016)

29. Prove that the curves $y^2 = 4x$ and $x^2 = 4y$ divide the area of the square bounded by $x = 0$, $x = 4$, $y = 4$ and $y = 0$ into three equal parts.
(AI 2016, 2015)

30. Using integration, find the area of the region bounded by the curves
$y = \sqrt{4 - x^2}, x^2 + y^2 - 4x = 0$ and the x – axis.
(Foreign 2016)

31. Using integration, find the area of the triangle formed by positive x-axis and tangent and normal to the circle $x^2 + y^2 = 4$ at $(1, \sqrt{3})$.
(Delhi 2015)

32. Find the area of the region $\{(x, y) : y^2 \le 4x, 4x^2 + 4y^2 \le 9\}$, using integration.
(AI 2015C, 2013)

33. Using integration, find the area of the region bounded by the triangle whose vertices are $(-1, 2)$, $(1, 5)$ and $(3, 4)$. *(AI 2014)*

34. Using the method of integration, find the area of the region bounded by the lines $2x + y = 4$, $3x - 2y = 6$ and $x - 3y + 5 = 0$.
(AI 2014C)

35. Using integration, find the area of the triangle PQR, coordinates of whose vertices are $P(2, 0)$, $Q(4, 5)$ and $R(6, 3)$. *(AI 2014C)*

36. Using integration, find the area of the region enclosed between the two circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$.
(Delhi 2013, AI 2013C, 2012C)

37. Find the area of the region $\{(x, y) : y^2 \le 6ax$ and $x^2 + y^2 \le 16a^2\}$ using method of integration. *(AI 2013)*

38. Find the area of the region enclosed between the two circles $x^2 + y^2 = 9$ and $(x - 3)^2 + y^2 = 9$.
(AI 2013C)

39. Using integration, find the area of the region enclosed between two circles $x^2 + y^2 = 1$ and $(x - 1)^2 + y^2 = 1$. *(AI 2013C)*

40. Using the method of integration, find the area of the region bounded by the following lines : $3x - y - 3 = 0$, $2x + y - 12 = 0$, $x - 2y - 1 = 0$ *(Delhi 2012)*

41. Using the method of integration, find the area of the region bounded by the following lines : $5x - 2y - 10 = 0$, $x + y - 9 = 0$, $2x - 5y - 4 = 0$ *(Delhi 2012)*

42. Using integration, find the area of the triangle ABC where A is $(2, 3)$, B is $(4, 7)$ and C is $(6, 2)$.
(Delhi 2012C)

43. Using integration, find the area of the region bounded by the two parabolas $y^2 = 4x$ and $x^2 = 4y$. *(Delhi 2012C)*

44. Using integration, find the area of the circle $x^2 + y^2 = 16$, which is exterior to the parabola $y^2 = 6x$. *(AI 2012C)*

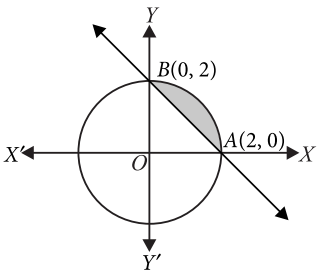
45. Using integration, find the area of the triangular region whose sides have equations $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.
(Delhi 2011, AI 2011C)

46. Using integration, find the area of the region bounded by the triangle whose vertices are $(2, 5)$, $(4, 7)$ and $(6, 2)$. *(Delhi 2011 C)*

www.exambites.in

Detailed Solutions

1. The given curves are
$x^2 + y^2 = 4$ ...(i) and $x + y = 2$ ...(ii)



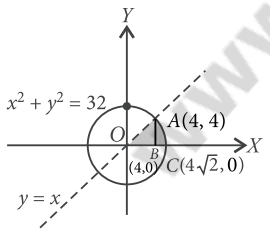
$\therefore$ Required area $= \int_0^2 \left[\sqrt{4-x^2} - (2-x)\right] dx$

$= \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2}\sin^{-1}\frac{x}{2} - 2x + \frac{x^2}{2}\right]_0^2$

$= 0 + 2\sin^{-1}(1) - 4 + 2 - 0$

$= 2\cdot\frac{\pi}{2} - 2 = (\pi - 2)$ sq.units.

2. The given equation of the circle is $x^2 + y^2 = 32$ and the line is $y = x$
These intersect at $A(4, 4)$ in the first quadrant. The required area is shown shaded in the figure. Points $B(4, 0)$ and $C\left(4\sqrt{2}, 0\right)$



$\therefore$ Required area = Area $BACB$ + Area $OABO$

$= \int_4^{4\sqrt{2}} y_1\,dx + \int_0^4 y_2\,dx = \int_4^{4\sqrt{2}} \sqrt{32-x^2}\,dx + \int_0^4 x\,dx$

$= \int_4^{4\sqrt{2}} \sqrt{\left(4\sqrt{2}\right)^2 - x^2}\,dx + \int_0^4 x\,dx$

$= \left[\frac{x\sqrt{32-x^2}}{2} + \frac{32}{2}\sin^{-1}\left(\frac{x}{4\sqrt{2}}\right)\right]_4^{4\sqrt{2}} + \left[\frac{x^2}{2}\right]_0^4$

$= \frac{4\sqrt{2}\times 0}{2} + 16\sin^{-1}1 - \left(\frac{4\times 4}{2} + 16\sin^{-1}\frac{1}{\sqrt{2}}\right) + \frac{1}{2}(4^2 - 0)$

$= 16\cdot\frac{\pi}{2} - \left(8 + 16\cdot\frac{\pi}{4}\right) + 8 = 16\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = 4\pi$ sq.units

3. We have curves, $y = \frac{1}{\sqrt{3}}x$ ...(i)

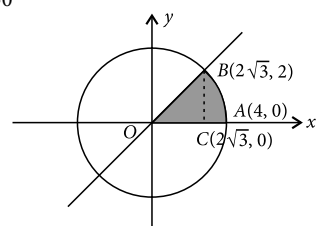
and $x^2 + y^2 = 16$ ...(ii)

Curves (i) and (ii) intersect at $(2\sqrt{3}, 2)$ and $(-2\sqrt{3}, -2)$.

$\therefore$ Required area = Area of region $OBAO$
= area ΔOBC + area of region $BCAB$

$= \int_0^{2\sqrt{3}} \frac{x}{\sqrt{3}}\,dx + \int_{2\sqrt{3}}^4 \sqrt{16-x^2}\,dx$

$= \left[\frac{x^2}{2\sqrt{3}}\right]_0^{2\sqrt{3}} + \left[\frac{x}{2}\sqrt{16-x^2} + \frac{16}{2}\sin^{-1}\left(\frac{x}{4}\right)\right]_{2\sqrt{3}}^4$



$= 2\sqrt{3} + 8\left(\frac{\pi}{2}\right) - 2\sqrt{3} - \frac{8\pi}{3}$

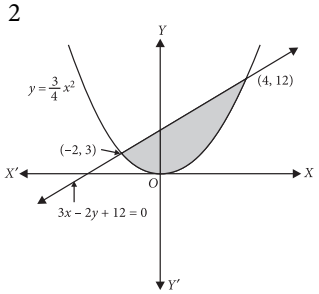
$= \frac{12\pi - 8\pi}{3} = \frac{4\pi}{3}$ sq.units

4. Given equations are $y = \frac{3x^2}{4}$ ...(i)

and $3x - 2y + 12 = 0 \Rightarrow y = \frac{3x+12}{2}$ ...(ii)

Solving equations (i) and (ii), we get

$\frac{3x^2}{4} = \frac{3x+12}{2}$



www.examples.in

$\Rightarrow \quad x^2 - 2x - 8 = 0 \Rightarrow (x + 2)(x - 4) = 0$
$\Rightarrow \quad x = -2, 4$
When $x = -2 \Rightarrow y = 3$
When $x = 4 \Rightarrow y = 12$

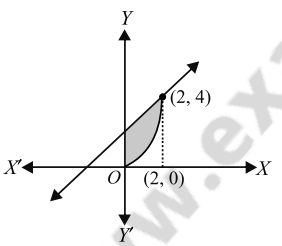
$\therefore$ Required area $= \int_{-2}^{4} \left(\frac{3x+12}{2} - \frac{3}{4}x^2\right)dx$

$= \left[\frac{3}{4}x^2 + 6x - \frac{x^3}{4}\right]_{-2}^{4}$

$= \left[\frac{3\times 16}{4} + 6\times 4 - \frac{64}{4}\right] - \left[\frac{3}{4}\times 4 - 6\times 2 + \frac{8}{4}\right]$

= 27 sq. units.

5. We have curves $x - y + 2 = 0$ and $x = \sqrt{y}$.
$x = \sqrt{y} \Rightarrow y = x^2$, which is a parabola with vertex at origin.
From the given equations, we get
$x - x^2 + 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$
$\Rightarrow x = 2$ or $x = -1$
$\Rightarrow x = 2$ [$\because x \neq -1$, x is positive]
When $x = 2$, $y = 4$
So, the point of intersection is (2, 4)
$\therefore$ Required area



$= \int_0^2 (x+2)dx - \int_0^2 x^2 dx$

$= \int_0^2 (x + 2 - x^2)dx$

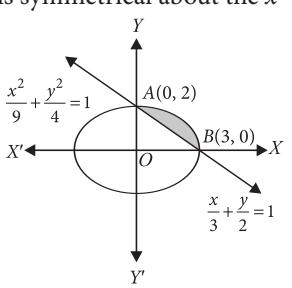
$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_0^2 = 2 + 4 - \frac{8}{3} = \frac{10}{3}$ sq. units

6. *Refer to answer 2.*

7. We have $\frac{x^2}{9} + \frac{y^2}{4} = 1$ ...(i) and $\frac{x}{3} + \frac{y}{2} = 1$ ...(ii)

Curve (i) is an ellipse of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

That means its major axis is along x – axis. Also this ellipse is symmetrical about the x – axis.



Required area $= \frac{2}{3}\int_0^3 \sqrt{(3)^2 - x^2}\,dx - \frac{2}{3}\int_0^3 (3 - x)dx$

$= \frac{2}{3}\left[\frac{x}{2}\sqrt{9 - x^2} + \frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right)\right]_0^3 - \frac{2}{3}\left[\frac{(3-x)^2}{-2}\right]_0^3$

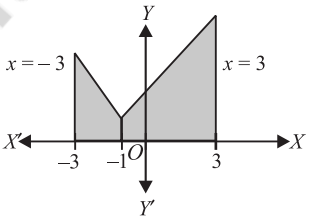
$= \frac{2}{3}\left[\left(0 + \frac{9}{2}\sin^{-1}(1)\right) - \left(0 + \frac{9}{2}\sin^{-1}(0)\right)\right] + \frac{1}{3}\left[0^2 - 9\right]$

$= \frac{3\pi}{2} - 3$ sq. units.

8. Here, $y = |x + 1| + 1$

$$y = \begin{cases} x + 2 & \text{if} \quad x \geq -1 \\ -x & \text{if} \quad x < -1 \end{cases}$$

We now draw the lines : $y = 0$, $x = 3$, $x = -3$ and
$y = x + 2$ if $x \geq -1$ ...(i)
$y = -x$ if $x < -1$ ...(ii)
Lines (i) and (ii) intersect at (– 1, 1)



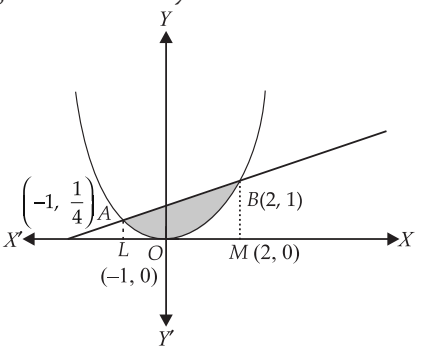
$\therefore$ Required area $= \int_{-3}^{-1} (-x)dx + \int_{-1}^{3} (x + 2)dx$

$= -\left[\frac{x^2}{2}\right]_{-3}^{-1} + \left[\frac{x^2}{2} + 2x\right]_{-1}^{3}$

$= -\frac{1}{2}(1 - 9) + \frac{1}{2}(9 - 1) + 2(3 + 1)$

= 4 + 4 + 8 = 16 sq. units.

9. The given curve is $x^2 = 4y$ ...(i)
The given line is $x = 4y - 2$ ...(ii)



www.exambites.in

Putting $4y = (x + 2)$ from (ii) in (i), we get $(x + 2) = x^2$

$\Rightarrow \quad x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = 2, -1$

Putting $x = 2$ in (i), we get $y = 1$

Putting $x = -1$ in (i), we get $y = \frac{1}{4}$

Thus the points of intersection of the given curve and line are $A\left(-1, \frac{1}{4}\right)$ and $B\ (2, 1)$

$\therefore$ Required area

$$= \int_{-1}^{2}\left(\frac{x+2}{4}\right)dx - \int_{-1}^{2}\frac{x^2}{4}dx = \int_{-1}^{2}\left(\frac{x}{4}+\frac{1}{2}-\frac{x^2}{4}\right)dx$$

$$= \frac{1}{4}\left[\frac{x^2}{2}\right]_{-1}^{2} + \frac{1}{2}[x]_{-1}^{2} - \frac{1}{4}\left[\frac{x^3}{3}\right]_{-1}^{2}$$

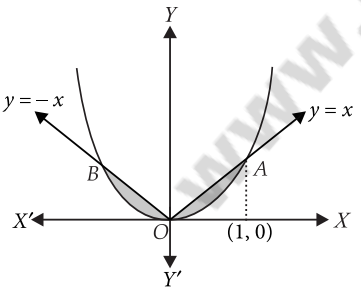
$$= \frac{1}{8}[4-1] + \frac{1}{2}[2+1] - \frac{1}{12}[8+1]$$

$$= \frac{3}{8} + \frac{3}{2} - \frac{3}{4} = \frac{3}{2}\left[\frac{1}{4}+1-\frac{1}{2}\right] = \frac{3}{2}\left[\frac{3}{4}\right] = \frac{9}{8}\text{sq. units}$$

10. *Refer to answer 2.*

11. The given curves are $y = x^2$ ...(i)

$$y = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases} \quad \text{...(ii)}$$



Their points of intersection are $A\ (1, 1)$, $O(0, 0)$ and $B\ (-1, 1)$.

In view of symmetry, the required area

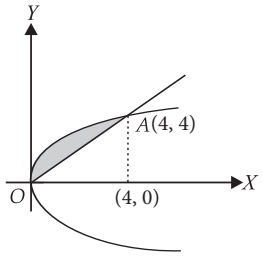
$$= 2\int_0^1 (x - x^2)dx = 2\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$$

$$= 2\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{1}{3}\text{sq.unit.}$$

12. *Refer to answer 11.*

13. The given curves are $y^2 = 4x$, $y = x$

They intersect at $O(0, 0)$ and $A\ (4, 4)$.

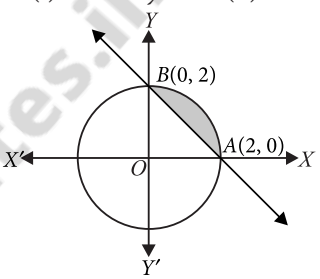


$\therefore$ Required area $= \int_0^4 \left(\sqrt{4x} - x\right)dx = \left[2 \cdot \frac{x^{3/2}}{3/2} - \frac{x^2}{2}\right]_0^4$

$$= \frac{4}{3} \cdot 4^{3/2} - \frac{1}{2} \cdot 4^2 = \frac{32}{3} - 8 = \frac{8}{3}\text{sq.units.}$$

14. The given curves are

$x^2 + y^2 = 4$ ...(i) and $x + y = 2$ ...(ii)



$\therefore$ Required area $= \int_0^2 \left[\sqrt{4 - x^2} - (2 - x)\right]dx$

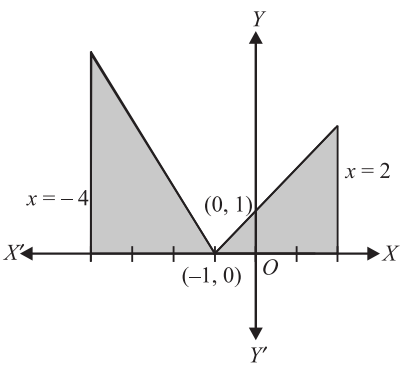
$$= \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2}\sin^{-1}\frac{x}{2} - 2x + \frac{x^2}{2}\right]_0^2$$

$$= 0 + 2\sin^{-1}(1) - 4 + 2 - 0 = 2 \cdot \frac{\pi}{2} - 2 = (\pi - 2)\text{ sq.units.}$$

15. Here $y = |x + 1| = \begin{cases} x + 1, & \text{if} \quad x \geq -1 \\ -x - 1, & \text{if} \quad x < -1 \end{cases}$

Thus we get two lines $-x + y = 1$ ...(i), $x + y = -1$...(ii)

Their graphs are as shown and the area to be calculated is shaded.



www.exambites.in

Hence, the required area

$$= \int_{-1}^{2} (x+1)dx + \int_{-4}^{-1} (-x-1)dx$$

$$= \left[\frac{x^2}{2}+x\right]_{-1}^{2} - \left[\frac{x^2}{2}+x\right]_{-4}^{-1}$$

$$= (2+2) - \left(\frac{1}{2}-1\right) - \left[\left(\frac{1}{2}-1\right) - (8-4)\right] = \frac{9}{2}+\frac{9}{2}$$

= 9 sq. units.

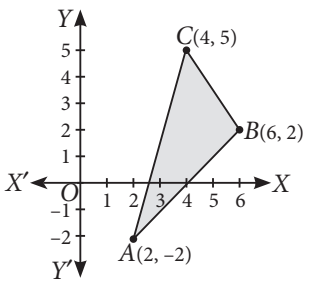
16. *Refer to answer 11.*

17. *Refer to answer 15.*

18. *Refer to answer 14.*

19. Equation of AC is

$$\frac{y-5}{x-4} = \frac{7}{2} \Rightarrow y = \frac{7x}{2} - 9$$



Equation of BC is

$$\frac{y-5}{x-4} = -\frac{3}{2} \Rightarrow y = -\frac{3x}{2} + 11$$

Equation of AB is

$$\frac{y-2}{x-6} = \frac{4}{4} = 1 \Rightarrow y = x - 4$$

∴ Required area

$$= \int_2^4 \left(\frac{7x}{2}-9\right)dx + \int_4^6 \left(-\frac{3x}{2}+11\right)dx - \int_2^6 (x-4)dx$$

$$= \left[\frac{7x^2}{4}-9x\right]_2^4 + \left[\frac{-3x^2}{4}+11x\right]_4^6 - \left[\frac{x^2}{2}-4x\right]_2^6$$

$$= [(28-36)-(7-18)] + [(-27+66)-(-12+44)] - [(18-24)-(2-8)]$$

= 10 sq. units

20. The given lines are $x - y = 0$ ...(i)

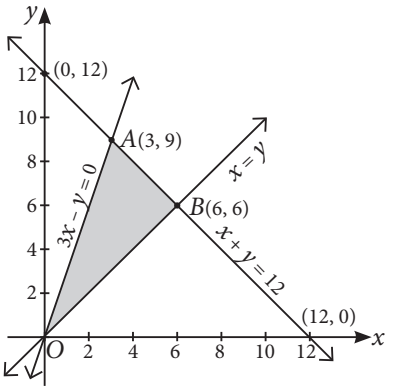
$3x - y = 0$ ...(ii)

$x + y = 12$ ...(iii)

Solving (i) and (ii), we get $x = 0, y = 0$

Solving (ii) and (iii), we get $x = 3, y = 9$

Solving (iii) and (i), we get $x = 6, y = 6$



∴ Required area = area (ΔOAB)

$$= \int_0^3 3x\,dx + \int_3^6 (12-x)\,dx - \int_0^6 x\,dx$$

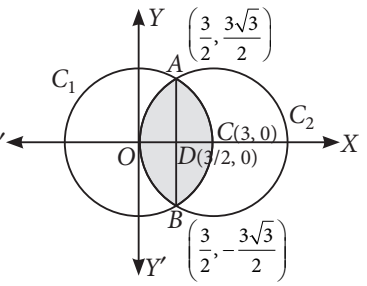
$$= \left[\frac{3x^2}{2}\right]_0^3 + \left[12x - \frac{x^2}{2}\right]_3^6 - \left[\frac{x^2}{2}\right]_0^6$$

$$= \frac{3}{2}\cdot 9 + 12\cdot 6 - \frac{6^2}{2} - 12\cdot 3 + \frac{3^2}{2} - \frac{6^2}{2}$$

$$= \frac{27}{2} + 72 - 36 - 36 + \frac{9}{2}$$

$$= \frac{27}{2} + \frac{9}{2} = \frac{27+9}{2} = 18 \text{ sq. units}$$

21.



The given circles are

$C_1 : x^2 + y^2 = 9$ ...(i)

and $C_2 : (x-3)^2 + y^2 = 9$ ...(ii)

Eliminating y from (i) and (ii), we get

$9 - x^2 = 9 - (x-3)^2$

$6x = 9 \Rightarrow x = \frac{9}{6} = \frac{3}{2}$

Putting $x = \frac{3}{2}$ in (i), we get $y^2 = \frac{27}{4} = \pm\frac{3\sqrt{3}}{2}$

⇒ Points of intersection of the two circles are

$$A\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right) \text{ and } B\left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$$

Required area = 2(area $AOCA$)

= 2(area $AODA$ + area $ADCA$)

www.exambits.in

$$= 2\int_0^{3/2}\sqrt{9-(x-3)^2}\,dx + 2\int_{3/2}^{3}\sqrt{9-x^2}\,dx$$

$$= 2\left[\frac{x-3}{2}\sqrt{9-(x-3)^2}+\frac{9}{2}\sin^{-1}\left(\frac{x-3}{3}\right)\right]_0^{3/2} + 2\left[\frac{x}{2}\sqrt{9-x^2}+\frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right)\right]_{3/2}^{3}$$

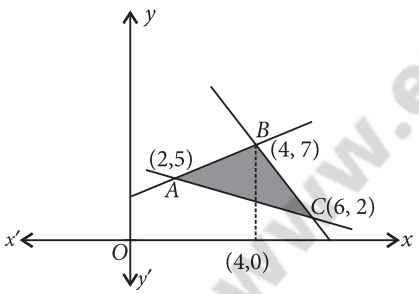
$$= 2\left[-\frac{3}{4}\sqrt{9-\frac{9}{4}}+\frac{9}{2}\left(-\frac{\pi}{6}\right)-\frac{9}{2}\left(-\frac{\pi}{2}\right)\right] + 2\left[\frac{9}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\sqrt{9-\frac{9}{4}}-\frac{9}{2}\left(\frac{\pi}{6}\right)\right]$$

$$= 2\left[-\frac{3}{4}\cdot\frac{3\sqrt{3}}{2}-\frac{9\pi}{12}+\frac{9\pi}{4}\right]+2\left[\frac{9\pi}{4}-\frac{3}{4}\cdot\frac{3\sqrt{3}}{2}-\frac{9\pi}{12}\right]$$

$$= 2\left[-\frac{9\sqrt{3}}{8}-\frac{9\pi}{12}+\frac{9\pi}{4}+\frac{9\pi}{4}-\frac{9\sqrt{3}}{8}-\frac{9\pi}{12}\right]$$

$$= 2\left[-\frac{2\times 9\sqrt{3}}{8}+3\pi\right]=\left(6\pi-\frac{9\sqrt{3}}{2}\right)\text{ sq. units}$$

22.



Equation of AB is

$$\frac{y-5}{x-2}=\frac{7-5}{4-2}=1 \quad\Rightarrow\quad y=x+3 \qquad ...(i)$$

Equation of BC is

$$\frac{y-7}{x-4}=\frac{2-7}{6-4}=\frac{-5}{2} \quad\Rightarrow\quad y=\frac{-5}{2}x+17 \qquad ...(ii)$$

Equation of AC is

$$\frac{y-5}{x-2}=\frac{2-5}{6-2}=\frac{-3}{4} \quad\Rightarrow\quad y=\frac{-3x}{4}+\frac{13}{2} \qquad ...(iii)$$

$$\therefore \text{ Required area} = \int_2^4 (x+3)dx + \int_4^6\left(\frac{-5}{2}x+17\right)dx - \int_2^6\left(\frac{-3x}{4}+\frac{13}{2}\right)dx$$

$$= \left[\frac{x^2}{2}+3x\right]_2^4+\left[\frac{-5x^2}{4}+17x\right]_4^6-\left[\frac{-3x^2}{8}+\frac{13x}{2}\right]_2^6$$

$$= \left[\frac{1}{2}(16-4)+3(4-2)\right]+\left[\frac{-5}{4}(36-16)+17(6-4)\right] - \left[\frac{-3}{8}(36-4)+\frac{13}{2}(6-2)\right]$$

= 12 + 9 – 14 = 7 sq. units

23. Given, $x^2 + y^2 = 8x$ ....(i)

and $y^2 = 4x$ ....(ii)

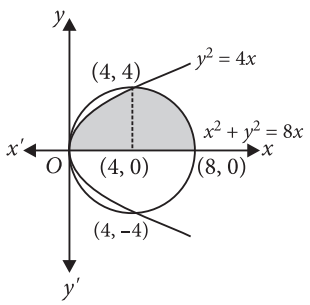
Solving (i) and (ii), we get

$$x^2 + 4x = 8x \quad\Rightarrow\quad x^2 - 4x = 0$$

$$\Rightarrow\quad x(x-4)=0 \quad\Rightarrow\quad x=0 \text{ or } x=4$$

when $x = 0 \Rightarrow y = 0$

when $x = 4 \Rightarrow y = \pm 4$



$$\therefore \text{ Required area} = \int_0^4 2\sqrt{x}dx + \int_4^8\sqrt{8x-x^2}\,dx$$

$$= 2\int_0^4\sqrt{x}dx + \int_4^8(\sqrt{(4)^2-(x-4)^2})dx$$

$$= 2\times\frac{2}{3}[(x)^{3/2}]_0^4+\left[\frac{(x-4)}{2}\sqrt{(4)^2-(x-4)^2} + \frac{16}{2}\sin^{-1}\left(\frac{x-4}{4}\right)\right]_4^8$$

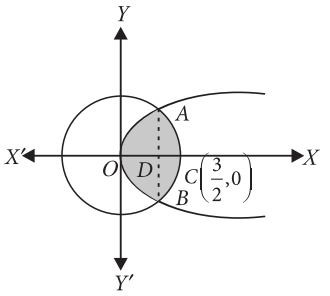
$$= \frac{4}{3}\times 8+\frac{16}{2}\sin^{-1}\left(\frac{4}{4}\right)=\frac{32}{3}+4\pi$$

24. Let $R = \{(x, y) : y^2 \le 4x,\ 4x^2 + 4y^2 \le 9\}$

$= \{(x, y): y^2 \le 4x\} \cap \{(x, y) : 4x^2 + 4y^2 \le 9\}$

$= R_1 \cap R_2$.

www.examfree.in



Let $4x^2 + 4y^2 = 9$ ...(i)

$\Rightarrow x^2 + y^2 = \left(\frac{3}{2}\right)^2$

and $y^2 = 4x$ ...(ii)

R_1 is the region lying inside $y^2 = 4x$

R_2 is the region lying inside $x^2 + y^2 = \frac{9}{4}$

Curves (i) and (ii) intersect at $A\left(\frac{1}{2}, \sqrt{2}\right)$ and $B\left(\frac{1}{2}, -\sqrt{2}\right)$

As both the curves are symmteric about x – axis.

$\therefore$ Reqd. area = 2 (Area of the shaded region above x – axis) = 2 (Area $OADO$ + $ADCA$)

$$= 2\left[\int_0^{1/2} 2\sqrt{x}\,dx + \int_{1/2}^{3/2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2}\,dx\right] = 2\left[2\left(\frac{x^{3/2}}{3/2}\right)_0^{1/2} + \left(\frac{x\sqrt{(3/2)^2 - x^2}}{2} + \frac{(3/2)^2}{2}\sin^{-1}\left(\frac{x}{3/2}\right)\right)_{1/2}^{3/2}\right]$$

$$= 2\left[\frac{4}{3}\left(\frac{1}{2}\right)^{3/2} + \frac{9}{8}\sin^{-1}(1) - \frac{1}{4}\sqrt{2} - \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right)\right]$$

$$= 2\left[\frac{4}{3}\cdot\frac{1}{2\sqrt{2}} - \frac{\sqrt{2}}{4} + \frac{9\pi}{16} - \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right)\right]$$

$$= \left[\frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right)\right] \text{sq.units.}$$

25. The given lines are

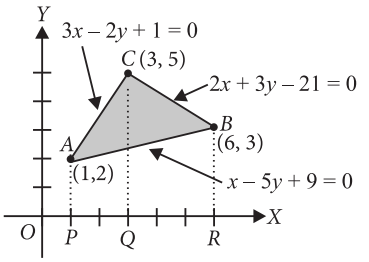
$3x - 2y + 1 = 0$ ...(i), $2x + 3y - 21 = 0$ ...(ii)

and $x - 5y + 9 = 0$ ...(iii)

Solving (i) and (ii), we get $x = 3$, $y = 5$

Solving (ii) and (iii), we get $x = 6$, $y = 3$

Solving (i) and (iii), we get $x = 1$, $y = 2$



$\therefore$ Area of ΔABC

= area $PQCA$ + area $QRBC$ – area $PRBA$

$$= \int_1^3 \frac{3x+1}{2}\,dx + \int_3^6 \frac{21-2x}{3}\,dx - \int_1^6 \frac{x+9}{5}\,dx$$

$$= \left[\frac{3}{4}x^2 + \frac{x}{2}\right]_1^3 + \left[7x - \frac{x^2}{3}\right]_3^6 - \left[\frac{x^2}{10} + \frac{9x}{5}\right]_1^6$$

$$= \left\{\frac{27}{4} + \frac{3}{2} - \left(\frac{3}{4} + \frac{1}{2}\right)\right\} + \{42 - 12 - (21 - 3)\} - \left\{\frac{18}{5} + \frac{54}{5} - \left(\frac{1}{10} + \frac{9}{5}\right)\right\}$$

$$= \left\{\frac{33}{4} - \frac{5}{4}\right\} + 12 - \left\{\frac{72}{5} - \frac{19}{10}\right\}$$

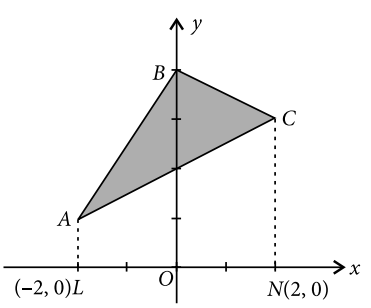
$$= 7 + 12 - \frac{25}{2} = \frac{13}{2} \text{ sq.units.}$$

26. Let $A(-2, 1)$, $B(0, 4)$ and $C(2, 3)$

Eq. of AB is $y - 4 = \left(\frac{4-1}{2}\right)(x - 0) \Rightarrow y = \frac{3}{2}x + 4$

Eq. of BC is $y - 3 = \left(\frac{3-4}{2}\right)(x - 2) \Rightarrow y = \frac{-1}{2}x + 4$

Eq. of AC is $y - 1 = \left(\frac{3-1}{2+2}\right)(x + 2) \Rightarrow y = \frac{x}{2} + 2$



Area of required region = Area of trap. $ALOB$ + Area of trap. $BONC$ – Area of trap. $ALNC$

www.exambits.in

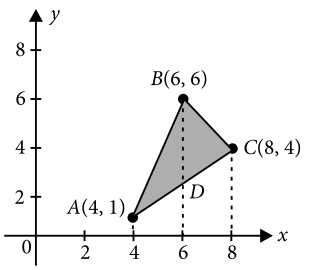
$$= \int_{-2}^{0}\left(\frac{3}{2}x+4\right)dx + \int_{0}^{2}\left(\frac{-1}{2}x+4\right)dx - \int_{-2}^{2}\left(\frac{x}{2}+2\right)dx$$

$$= \left[\frac{3}{4}x^2+4x\right]_{-2}^{0} + \left[\frac{-x^2}{4}+4x\right]_{0}^{2} - \left[\frac{x^2}{4}+2x\right]_{-2}^{2}$$

$= (-3 + 8) + (-1 + 8) - (5 + 3) = 4$ sq. units

27. Let us first find the equations of the sides of triangle ABC by using two points formula.

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1).$$



Equation of AB : $y-1=\frac{6-1}{6-4}(x-4)$

$\Rightarrow \; y-1=\frac{5}{2}(x-4) \;\Rightarrow\; y=\frac{5x}{2}-9$

Equation of BC : $y-6=\frac{4-6}{8-6}(x-6)$

$\Rightarrow \; y-6=\frac{-2}{2}(x-6) \Rightarrow y=-x+12$

Equation of AC : $y-1=\frac{4-1}{8-4}(x-4)$

$\Rightarrow \; y-1=\frac{3}{4}(x-4) \;\Rightarrow\; y=\frac{3x}{4}-2$

$$\therefore \quad \text{Area of } \Delta ABC = \int_{4}^{6} AB\,dx + \int_{6}^{8} BC\,dx - \int_{4}^{8} AC\,dx$$

$$= \int_{4}^{6}\left(\frac{5x}{2}-9\right)dx + \int_{6}^{8}(-x+12)dx - \int_{4}^{8}\left(\frac{3x}{4}-2\right)dx$$

$$= \left[\frac{5x^2}{4}-9x\right]_{4}^{6} + \left[-\frac{x^2}{2}+12x\right]_{6}^{8} - \left[\frac{3x^2}{8}-2x\right]_{4}^{8}$$

$$= \left[\left(\frac{5(36)}{4}-54\right)-\left(\frac{5(16)}{4}-36\right)\right] + \left[\left(\frac{-64}{2}+96\right) - \left(-\frac{36}{2}+72\right)\right] - \left[(24-16)-\left(\frac{48}{8}-8\right)\right]$$

= 7 sq. units.

28. Let $R = \{(x, y) : x^2 + y^2 \le 2ax, y^2 \ge ax ; x, y \ge 0\}$

$\Rightarrow R = R_1 \cap R_2 \cap R_3$

where $R_1 = \{(x, y) : x^2 + y^2 \le 2ax\}$,

$R_2 = \{(x, y) : y^2 \ge ax\}$ and $R_3 = \{(x, y) : x \ge 0, y \ge 0\}$

Region R_1 : $(x - a)^2 + y^2 = a^2$ represents a circle with centre at $(a, 0)$ and radius a.

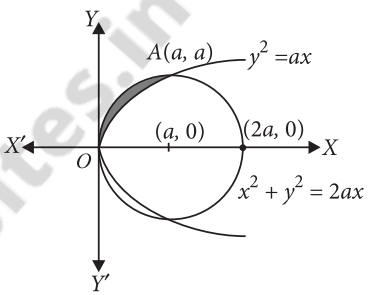
Region R_2 : $y^2 = ax$ represents a parabola with vertex at $(0, 0)$ and its axis along x-axis.

Region $R_3 : x \ge 0, y \ge 0$ represents the first quadrant.

$\Rightarrow \quad R = R_1 \cap R_2 \cap R_3$ is the shaded portion in the figure.

Since, given curves are $x^2 + y^2 = 2ax$ and $y^2 = ax$

So, point of intersection of the curves are $(0, 0)$ and (a, a).



$$\therefore \quad \text{Required area} = \int_{0}^{a}\left(\sqrt{a^2-(x-a)^2} - \sqrt{ax}\right)dx$$

$$= \left[\frac{1}{2}(x-a)\sqrt{a^2-(x-a)^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x-a}{a}\right) - \frac{2}{3}\sqrt{a}x^{3/2}\right]_{0}^{a}$$

$$= \left\{\left(-\frac{2\sqrt{a}}{3}a^{3/2}\right) - \left(\frac{1}{2}a^2\sin^{-1}(-1)\right)\right\}$$

$$= \left\{-\frac{2a^2}{3} - \frac{a^2}{2}\left(-\frac{\pi}{2}\right)\right\} = \left(-\frac{2a^2}{3}+\frac{a^2\pi}{4}\right)$$

$$= \left(\frac{\pi}{4}-\frac{2}{3}\right)a^2 \text{ sq. units}$$

29. We have $y^2 = 4x \;\Rightarrow\; x = \frac{y^2}{4}$

Also, $x^2 = 4y \;\Rightarrow\; \left(\frac{y^2}{4}\right)^2 = 4y$

$\Rightarrow \quad y^4 = 64y \Rightarrow y = 0$ or $y = 4$

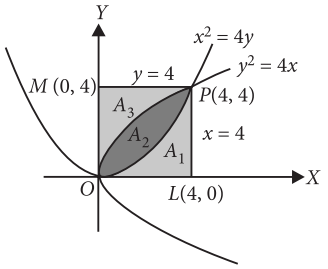
When $y = 0$, $x = 0$ and when $y = 4$, $x = 4$

So, the points of intersection are $O(0, 0)$ and $P(4, 4)$

Let A_1, A_2, A_3 be the area denoted in the figure.

We need to prove $A_1 = A_2 = A_3$.

www.exambites.in



$$A_1 = \int_0^4 \frac{x^2}{4}dx = \frac{1}{4}\left[\frac{x^3}{3}\right]_0^4 = \frac{16}{3}\text{ sq. units}$$

$$A_2 = \int_0^4 \left(\sqrt{4x} - \frac{x^2}{4}\right)dx$$

$$= \left[\frac{4}{3}x^{3/2} - \frac{x^3}{12}\right]_0^4 = \frac{16}{3}\text{ sq. units}$$

$$A_3 = \int_0^4 \frac{y^2}{4}dy = \frac{1}{4}\left[\frac{y^3}{3}\right]_0^4 = \frac{16}{3}\text{ sq. units}$$

Therefore, $A_1 = A_2 = A_3 = \frac{16}{3}$ sq. units

Thus $y^2 = 4x$ and $x^2 = 4y$ divide the area of square bounded by $x = 0$, $x = 4$, $y = 4$ and $y = 0$ into three equal parts.

30. The given curves are $y = \sqrt{4 - x^2}$ and $x^2 + y^2 - 4x = 0$

$y = \sqrt{4 - x^2} \Rightarrow x^2 + y^2 = 4$ ...(i)

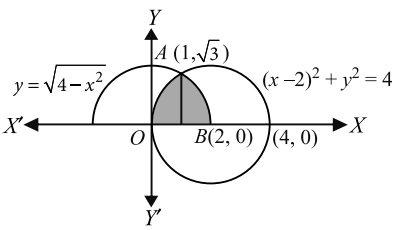
This represents a circle with centre $O(0, 0)$ and radius = 2 units.

$x^2 + y^2 - 4x = 0 \Rightarrow (x - 2)^2 + y^2 = 4$ ...(ii)

This represents a circle with centre $B(2, 0)$ and radius = 2 units.

Solving (i) and (ii), we get $(x - 2)^2 = x^2 \Rightarrow x = 1$

$\therefore\ y^2 = 3 \Rightarrow y = \pm\sqrt{3}$



Thus, the given curves intersect at $A\left(1, \sqrt{3}\right)$

$\therefore$ Required area = Area of the region $OABO$

$$= \int_0^1 \sqrt{4 - (x-2)^2}\,dx + \int_1^2 \sqrt{4 - x^2}\,dx$$

$$= \left[\frac{1}{2}(x-2)\sqrt{4-(x-2)^2} + \frac{4}{2}\sin^{-1}\left(\frac{x-2}{2}\right)\right]_0^1$$

$$+ \left[\frac{1}{2}x\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right)\right]_1^2$$

$$= \left[-\frac{\sqrt{3}}{2} + 2\sin^{-1}\left(-\frac{1}{2}\right)\right] - [0 + 2\sin^{-1}(-1)]$$

$$+ \left[0 + 2\sin^{-1}(1)\right] - \left[\frac{\sqrt{3}}{2} + 2\sin^{-1}\left(\frac{1}{2}\right)\right]$$

$$= -\frac{\sqrt{3}}{2} - 2\times\frac{\pi}{6} + 2\times\frac{\pi}{2} + 2\times\frac{\pi}{2} - \frac{\sqrt{3}}{2} - 2\times\frac{\pi}{6}$$

$$= \left(\frac{4\pi}{3} - \sqrt{3}\right)\text{ sq. units}$$

31. Given equation of circle is $x^2 + y^2 = 4$

Differentiate w.r.t. 'x' on both sides, we get

$$2x + 2y\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y} \Rightarrow \left(\frac{dy}{dx}\right)_{(1,\sqrt{3})} = \frac{-1}{\sqrt{3}}$$

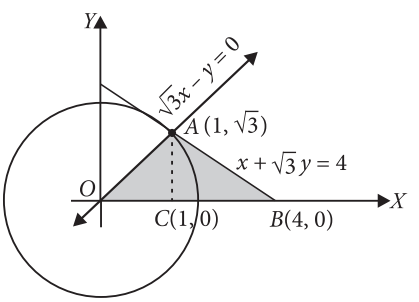
$\Rightarrow$ Equation of tangent at $(1, \sqrt{3})$ is

$$y - \sqrt{3} = \frac{-1}{\sqrt{3}}(x-1) \Rightarrow x + \sqrt{3}y = 4$$

Equation of normal at $(1, \sqrt{3})$ is

$$y - \sqrt{3} = \sqrt{3}(x-1) \Rightarrow \sqrt{3}x - y = 0$$

So, point of intersection of $x + \sqrt{3}y = 4$ and $\sqrt{3}x - y = 0$ is $A(1, \sqrt{3})$



Required area = area of ΔOAB

$$= \int_0^1 \sqrt{3}x\,dx + \int_1^4 \left(\frac{4-x}{\sqrt{3}}\right)dx$$

www.exambites.in

$$= \left[\frac{\sqrt{3}x^2}{2}\right]_0^1 + \frac{1}{\sqrt{3}}\left[4x - \frac{x^2}{2}\right]_1^4$$

$$= \frac{\sqrt{3}}{2}\times 1 + \frac{4}{\sqrt{3}}(4-1) - \frac{1}{\sqrt{3}\times 2}(4^2 - 1^2)$$

$$= \frac{\sqrt{3}}{2} + \frac{12}{\sqrt{3}} - \frac{1}{2\sqrt{3}}\times 15 = \frac{6\sqrt{3}}{3} = 2\sqrt{3} \text{ sq. units}$$

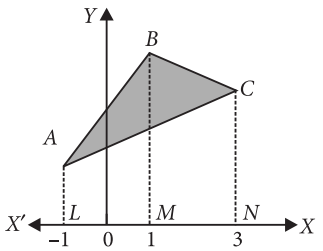
32. *Refer to answer 24.*

33. Let $A(-1, 2)$, $B(1, 5)$ and $C(3, 4)$

Eq. of AB is $y - 5 = \frac{5-2}{1+1}(x-1) \Rightarrow y = \frac{3}{2}x + \frac{7}{2}$

Eq. of BC is $y - 4 = \frac{4-5}{3-1}(x-3) \Rightarrow y = -\frac{1}{2}x + \frac{11}{2}$

Eq. of AC is $y - 2 = \frac{4-2}{3+1}(x+1) \Rightarrow y = \frac{1}{2}x + \frac{5}{2}$



Area of the reqd. triangular region, ABC
= Area of trap. $ALMB$ + Area of trap. $BMNC$ − Area of trap. $ALNC$

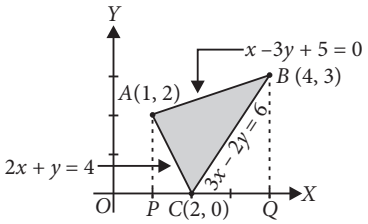
$$= \int_{-1}^{1}\left(\frac{3x}{2} + \frac{7}{2}\right)dx + \int_{1}^{3}\left(\frac{-x}{2} + \frac{11}{2}\right)dx - \int_{-1}^{3}\left(\frac{x}{2} + \frac{5}{2}\right)dx$$

$$= \left[\frac{3x^2}{4} + \frac{7x}{2}\right]_{-1}^{1} + \left[\frac{-x^2}{4} + \frac{11x}{2}\right]_1^3 - \left[\frac{x^2}{4} + \frac{5x}{2}\right]_{-1}^{3}$$

$$= \left(\frac{3}{4} + \frac{7}{2}\right) - \left(\frac{3}{4} - \frac{7}{2}\right) + \left(\frac{-9}{4} + \frac{33}{2}\right) - \left(\frac{-1}{4} + \frac{11}{2}\right) - \left(\frac{9}{4} + \frac{15}{2}\right) + \left(\frac{1}{4} - \frac{5}{2}\right)$$

= 7 − 2 + 11 − 2 − 10 = 4 sq. units.

34. The given lines are $2x + y = 4$ ...(i)
$3x - 2y = 6$ ...(ii) and $x - 3y + 5 = 0$ ...(iii)
Solving (i) and (ii), we get $x = 2, y = 0$
Solving (ii) and (iii), we get $x = 4, y = 3$
Solving (i) and (iii), we get $x = 1, y = 2$



∴ Required area = area (ΔABC)
= area ($ABQP$) − area (ΔAPC) − area (ΔBCQ)

$$= \int_1^4 \frac{x+5}{3}dx - \int_1^2 (4-2x)dx - \int_2^4 \frac{3x-6}{2}dx$$

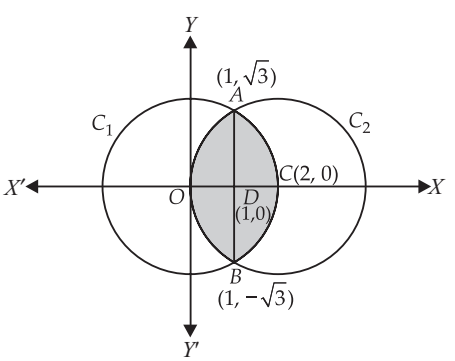
$$= \frac{1}{3}\left[\frac{x^2}{2} + 5x\right]_1^4 - [4x - x^2]_1^2 - \frac{1}{2}\left[\frac{3x^2}{2} - 6x\right]_2^4$$

$$= \frac{1}{3}\left[(8+20) - \left(\frac{1}{2} + 5\right)\right] - [(8-4) - (4-1)] - \frac{1}{2}[(24-24) - (6-12)]$$

$$= \frac{1}{3}\left[28 - \frac{1}{2} - 5\right] - [4-3] - \frac{1}{2}[0+6] = \frac{7}{2} \text{ sq. units}$$

35. *Refer to answer 33.*

36. The given circles are
$C_1 : x^2 + y^2 = 4$ ...(i) and $C_2: (x-2)^2 + y^2 = 4$ ...(ii)
Eliminating y from (i) and (ii), we get
$4 - x^2 = 4 - (x-2)^2 \Rightarrow 4x = 4 \Rightarrow x = 1$
Putting $x = 1$ in (i), we get $y^2 = 3 \Rightarrow y = \pm\sqrt{3}$
$\Rightarrow$ Points of intersection of the two circles are $A(1, \sqrt{3})$ and $B(1, -\sqrt{3})$



Required area = 2(area $AOCA$)
= 2(area $AODA$ + area $ADCA$)

$$= 2\int_0^1 \sqrt{4-(x-2)^2}dx + 2\int_1^2 \sqrt{4-x^2}dx$$

www.examrites.in

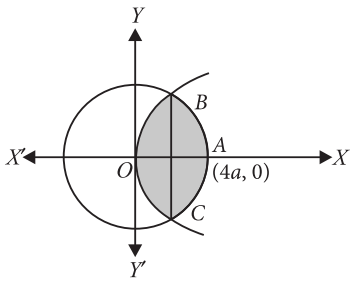
$$= 2\left[\frac{x-2}{2}\sqrt{4-(x-2)^2} + \frac{4}{2}\sin^{-1}\left(\frac{x-2}{2}\right)\right]_0^1$$

$$+2\left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right)\right]_1^2$$

$$= 2\left[-\frac{\sqrt{3}}{2} + 2\left(\frac{-\pi}{6}\right) - 2\left(-\frac{\pi}{2}\right) + 2\cdot\frac{\pi}{2} - \frac{\sqrt{3}}{2} - \frac{2\times\pi}{6}\right]$$

$$= 2\left[-\sqrt{3} + 2\pi - \frac{2\pi}{3}\right] = 2\left[\frac{-3\sqrt{3}+4\pi}{3}\right] \text{ sq. units}$$

37. Let $R = \{(x, y) : y^2 \le 6ax$ and $x^2 + y^2 \le 16a^2\}$
Let us draw the curves
$y^2 = 6ax$ ...(i) and $x^2 + y^2 = 16a^2$ ...(ii)



From (i) and (ii),
$x^2 + 6ax = 16a^2 \Rightarrow (x + 8a)(x - 2a) = 0 \Rightarrow x = 2a$
(Note : $x \neq -8a$ as curve (i) lies in 1st and 4th quadrants only).
Curves intersect at point
$B(2a, 2\sqrt{3}a)$ and $C(2a, -2\sqrt{3}a)$
Now in view of symmetry, the required area

$$= 2\left[\int_0^{2a}\sqrt{6ax}\,dx + \int_{2a}^{4a}\sqrt{16a^2-x^2}\,dx\right]$$

$$= 2\left[\sqrt{6a}\cdot\left\{\frac{x^{3/2}}{3/2}\right\}_0^{2a}\right.$$

$$\left.+\left\{\frac{x\sqrt{16a^2-x^2}}{2} + \frac{16a^2}{2}\sin^{-1}\left(\frac{x}{4a}\right)\right\}_{2a}^{4a}\right]$$

$$= 2\left[\frac{2}{3}\sqrt{6a}(2\sqrt{2}a^{3/2} - 0)\right.$$

$$\left.+\left\{8a^2\sin^{-1}(1) - \frac{2a\sqrt{12}a}{2} - 8a^2\sin^{-1}\left(\frac{1}{2}\right)\right\}\right]$$

$$= \frac{16\sqrt{3}}{3}a^2 + 16a^2\cdot\frac{\pi}{2} - 4\sqrt{3}a^2 - 16a^2\cdot\frac{\pi}{6}$$

$$= \left(\frac{4\sqrt{3}}{3} + 16\cdot\frac{\pi}{3}\right)a^2 \text{ sq.units.}$$

38. *Refer to answer 36.*

39. *Refer to answer 36.*

40. *Refer to answer 34.*

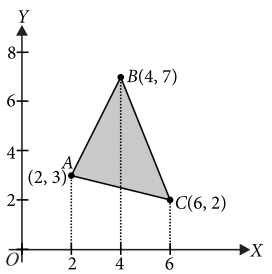
41. *Refer to answer 34.*

42. Here $A(2, 3)$, $B(4, 7)$ and $C(6, 2)$

Eq. of AB is $y - 3 = \frac{7-3}{4-2}(x-2) \Rightarrow y = 2x - 1$

Eq. of BC is $y - 7 = \frac{7-2}{4-6}(x-4) \Rightarrow y = -\frac{5}{2}x + 17$

Eq. of AC is $y - 3 = \frac{2-3}{6-2}(x-2) \Rightarrow y = -\frac{x}{4} + \frac{7}{2}$



$\therefore$ The required area of ΔABC

$$= \int_2^4 (2x-1)dx + \int_4^6\left(-\frac{5}{2}x + 17\right)dx - \int_2^6\left(-\frac{x}{4} + \frac{7}{2}\right)dx$$

$$= \left[x^2 - x\right]_2^4 + \left[-\frac{5}{4}x^2 + 17x\right]_4^6 - \left[-\frac{x^2}{8} + \frac{7}{2}x\right]_2^6$$

$$= [16 - 4 - (4 - 2)] + [-45 + 102 - (-20 + 68)]$$

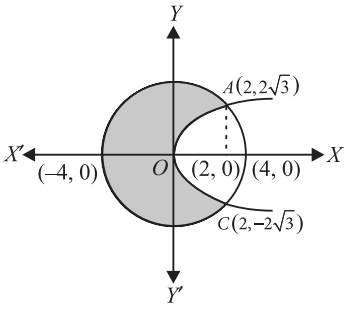
$$-\left[-\frac{9}{2} + 21 - \left(-\frac{1}{2} + 7\right)\right]$$

$= 10 + 9 - 10 = 9$ sq. units.

43. *Refer to answer 29.*

44. We have $y^2 = 6x$ which is a parabola and $x^2 + y^2 = 16$ which is a circle with centre at (0, 0) and radius 4.
Solving both, we get $x^2 + 6x - 16 = 0$
$\Rightarrow$ $(x + 8)(x - 2) = 0$
$\Rightarrow$ $x = 2$ ($\because$ $x = -8$ is not possible)

www.exambites.in



The required area

$$= 2\left[\int_{-4}^{2} \sqrt{16-x^2}\,dx - \int_{0}^{2} \sqrt{6x}\,dx\right]$$

$$= 2\left[\left(\frac{x}{2}\sqrt{16-x^2} + \frac{16}{2}\sin^{-1}\frac{x}{4}\right)_{-4}^{2} - \left(\sqrt{6}\times\frac{2}{3}(x)^{3/2}\right)_{0}^{2}\right]$$

$$= 2\left[\left(2\sqrt{3} + 8\cdot\frac{\pi}{6}\right) - 8\sin^{-1}(-1)\right] - 2\left[\frac{2}{3}\sqrt{6}\cdot 2\sqrt{2}\right]$$

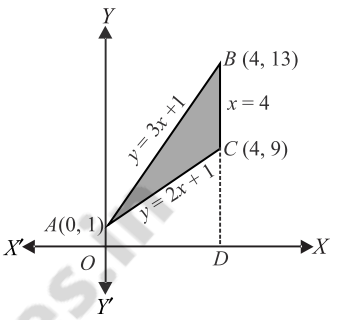
$$= 4\sqrt{3} + \frac{8\pi}{3} + 8\pi - \frac{16}{3}\sqrt{3} = \frac{32\pi}{3} - \frac{4\sqrt{3}}{3}$$

$$= \frac{4}{3}(8\pi - \sqrt{3}) \text{ sq. units}$$

45. The given lines are $y = 2x + 1$ ...(i)
$y = 3x + 1$ ...(ii) and $x = 4$ ...(iii)

Solving (i) and (ii), we get $x = 0$, $y = 1$
Solving (ii) and (iii), we get $x = 4$, $y = 13$
Solving (iii) and (i), we get $x = 4$, $y = 9$
$\therefore$ Required area = area (ΔABC)
= area ($OABD$) – area ($OACD$)

$$= \int_{0}^{4} (3x+1)dx - \int_{0}^{4} (2x+1)dx$$



$$= \left[\frac{3x^2}{2} + x\right]_0^4 - \left[x^2 + x\right]_0^4$$

$$= \left[\left(\frac{3}{2}(16) + 4\right) - (0+0)\right] - [(16+4) - (0+0)]$$

= 28 – 20 = 8 sq. units

46. *Refer to answer 42.*

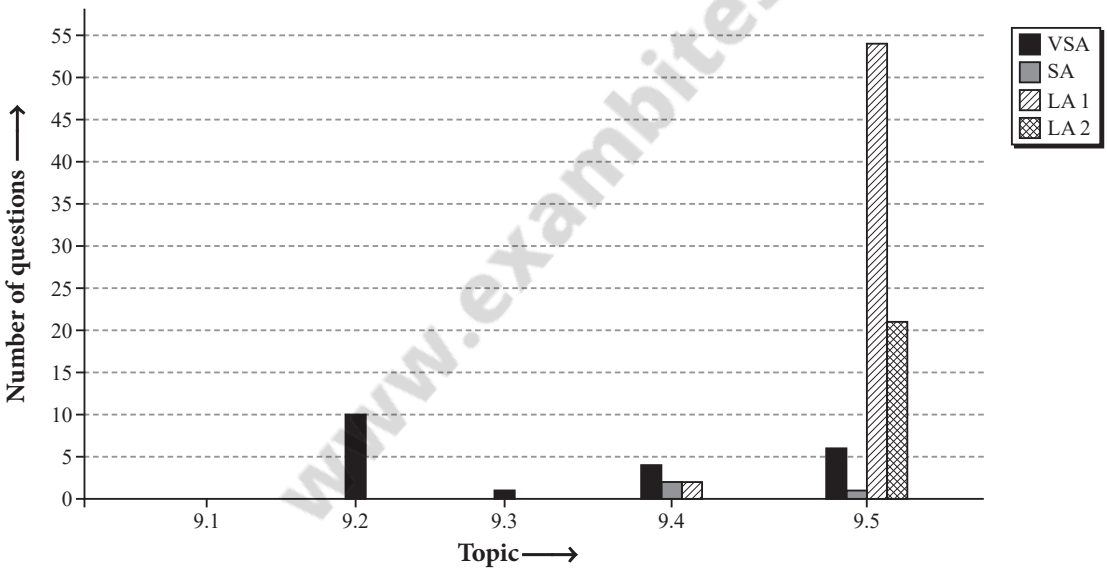
www.exambites.in

CHAPTER
9

Differential Equations

9.1 Introduction
9.2 Basic Concepts
9.3 General and Particular Solutions of a Differential Equation
9.4 Formation of a Differential Equation whose General Solution is given
9.5 Methods of Solving First Order, First Degree Differential Equations

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum weightage is of *Methods of Solving First Order, First Degree Differential Equations*
- Maximum LA 1 and LA 2 type questions were asked from *Methods of Solving First Order, First Degree Differential Equations*

QUICK RECAP

DIFFERENTIAL EQUATION

- An equation involving an independent variable, a dependent variable and the derivatives of the dependent variable is called differential equation.
- A differential equation involving derivatives of the dependent variable with respect to only one independent variable is called an ordinary differential equation.
- A differential equation involving derivatives with respect to more than one independent

www.exambites.in

variables is called a partial differential equation.

Order and Degree of a Differential Equation

- The order of highest derivative appearing in a differential equation is called order of the differential equation.
- The power of the highest order derivative appearing in a differential equation, after it is made free from radicals and fractions, is called degree of the differential equation.

Note : Order and degree (if defined) of a differential equation are always positive integers.

HOMOGENEOUS DIFFERENTIAL EQUATIONS

A differential equation of the form

$$\frac{dy}{dx}=\frac{f(x,y)}{g(x,y)}$$

where, $f(x, y)$ and $g(x, y)$ are homogeneous functions of x and y of the same degree.

LINEAR DIFFERENTIAL EQUATIONS

An equation of the form $\frac{dy}{dx}+Py=Q$ where P and Q are functions of x only (or constants) is called a linear differential equation of the first order.

SOLUTION OF A DIFFERENTIAL EQUATION

Solution of a differential equation is a function of the form $y = f(x) + C$ which satisfies the given differential equation.

General Solution : The solution of a differential equation which contains a number of arbitrary constants equal to the order of the differential equation.

Particular Solution : A solution obtained by giving particular values to arbitrary constants in the general solution.

FORMATION OF A DIFFERENTIAL EQUATION

Differential equation of the equation of a family of curves is obtained by eliminating arbitrary constants occurring in it with the help of equation of the curves and equations formed by its differentiation.

METHODS OF SOLVING DIFFERENTIAL EQUATIONS

Equation in Variable Separable Form : If the differential equation is of the form $f(x)\,dx = g(y)\,dy$, then the variables are separable and such equations can be solved by integrating on both sides. The solution is given by

$\int f(x)\,dx = \int g(y)\,dy + C$, where C is an arbitrary constant.

Equation Reducible to Homogeneous Form : If the equation is of the form $\frac{dy}{dx}=\frac{f(x,y)}{g(x,y)}$, where $f(x, y)$ and $g(x, y)$ are homogeneous functions of the same degree in x and y, then put $y = vx$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$ so that the dependent variable y is changed to another variable v, then apply variable separable method.

Solution of Linear Differential Equation : A differential equation of the form $\frac{dy}{dx}+Py=Q$, where P and Q are functions of x (or constants) can be solved as :

1. Find Integrating Factor $(\text{I.F.}) = e^{\int P\,dx}$
2. The solution of the differential equation is $y(\text{I.F.}) = \int Q\,(\text{I.F.})\,dx + C$, where C is constant of integration.

www.examfears.in

Previous Years' CBSE Board Questions

9.2 Basic Concepts

VSA (1 mark)

1. The degree of the differential equation $1+\left(\frac{dy}{dx}\right)^2 = x$ is _____. *(2020)*

2. Find the order and the degree of the differential equation $x^2\frac{d^2y}{dx^2} = \left\{1+\left(\frac{dy}{dx}\right)^2\right\}^4$. *(Delhi 2019)*

3. Write the sum of the order and degree of the following differential equation

$$\frac{d}{dx}\left\{\left(\frac{dy}{dx}\right)^3\right\} = 0. \quad \textit{(AI 2015)}$$

4. Write the sum of the order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 + x^4 = 0$. *(Foreign 2015)*

5. Write the sum of the order and degree of the differential equation

$$1+\left(\frac{dy}{dx}\right)^4 = 7\left(\frac{d^2y}{dx^2}\right)^3. \quad \textit{(Delhi 2015C)}$$

6. Write the degree of the differential equation

$$\left(\frac{dy}{dx}\right)^4 + 3x\frac{d^2y}{dx^2} = 0. \quad \textit{(Delhi 2013)}$$

7. Write the degree of differential equation

$$x^3\left(\frac{d^2y}{dx^2}\right)^2 + x\left(\frac{dy}{dx}\right)^4 = 0. \quad \textit{(Delhi 2013)}$$

8. Write the degree of the differential equation

$$x\left(\frac{d^2y}{dx^2}\right)^3 + y\left(\frac{dy}{dx}\right)^4 + x^3 = 0. \quad \textit{(Delhi 2013)}$$

9. Write the degree of the differential equation :

$$\left(\frac{dy}{dx}\right)^4 + 3y\frac{d^2y}{dx^2} = 0 \quad \textit{(Delhi 2013C)}$$

10. Write the degree of the differential equation

$$\left(\frac{d^2s}{dt^2}\right)^2 + \left(\frac{ds}{dt}\right)^3 + 4 = 0. \quad \textit{(AI 2013C)}$$

9.3 General and Particular Solutions of a Differential Equation

VSA (1 mark)

11. The number of arbitrary constants in the particular solution of a differential equation of second order is (are)

(a) 0 (b) 1
(c) 2 (d) 3 *(2020)*

9.4 Formation of a Differential Equation whose General Solution is given

VSA (1 mark)

12. Form the differential equation representing the family of curves $y = \frac{A}{x} + 5$, by eliminating the arbitrary constant A. *(AI 2019)*

13. Find the differential equation representing the family of curves $v = \frac{A}{r} + B$, where A and B are arbitrary constants. *(Delhi 2015)*

14. Write the differential equation obtained by eliminating the arbitrary constant C in the equation representing the family of curves $xy = C\cos x$. *(Delhi 2015C)*

15. Write the differential equation representing the family of curves $y = mx$, where m is an arbitrary constant. *(AI 2013)*

SA (2 marks)

16. Form the differential equation representing the family of curves $y = e^{2x}(a + bx)$, where 'a' and 'b' are arbitrary constants. *(Delhi 2019)*

www.exambites.in

17. Find the differential equation representing the family of curves $y = ae^{bx+5}$, where a and b are arbitrary constants. *(2018)*

LA 1 (4 marks)

18. Form the differential equation of the family of circles in the second quadrant and touching the coordinate axes. *(AI 2016, 2012)*

19. Form the differential equation of the family of parabolas having vertex at origin and axis along positive y-axis. *(Delhi 2011)*

9.5 Methods of Solving First Order, First Degree Differential Equations

VSA (1 mark)

20. The integrating factor of the differential equation $x\frac{dy}{dx}+2y=x^2$ is _____. *(2020)*

21. The integrating factor of the differential equation $(x+3y^2)\frac{dy}{dx}=y$ is

(a) y (b) $-y$

(c) $\frac{1}{y}$ (d) $-\frac{1}{y}$ *(2020)*

22. Find the integrating factor of the differential equation $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}}-\frac{y}{\sqrt{x}}\right)\frac{dx}{dy}=1.$ *(Delhi 2015, AI 2015C)*

23. Write the integrating factor of the following differential equation :

$$\left(1+y^2\right)+\left(2xy-\cot y\right)\frac{dy}{dx}=0$$ *(AI 2015)*

24. Write the solution of the differential equation $\frac{dy}{dx}=2^{-y}$ *(Foreign 2015)*

25. Find the solution of the differential equation $\frac{dy}{dx}=x^3\,e^{-2y}$. (AI 2015C)

SA (2 marks)

26. Solve the following differential equation:

$$\frac{dy}{dx}+y=\cos x-\sin x$$ *(AI 2019)*

LA 1 (4 marks)

27. Find the general solution of the differential equation $ye^y\,dx = (y^3 + 2x\,e^y)dy$. *(2020)*

28. Find the particular solution of the differential equation $x\frac{dy}{dx}=y-x\tan\left(\frac{y}{x}\right)$, given that $y=\frac{\pi}{4}$ at $x = 1$. *(2020)*

29. Find the particular solution of the differential equation $\cos y\,dx + (1 + e^{-x})\sin y\,dy = 0$ given that $y=\frac{\pi}{4}$ when $x = 0$. *(2020)*

30. Find the general solution of the differential equation $ye^{x/y}\,dx = (xe^{x/y} + y^2)dy$, $y \neq 0$ *(2020)*

31. Solve the differential equation :

$xdy-ydx=\sqrt{x^2+y^2}\,dx$, given that $y = 0$ when $x = 1$. *(Delhi 2019)*

32. Solve the differential equation :

$(1+x^2)\frac{dy}{dx}+2xy-4x^2=0,$ subject to the initial condition $y(0) = 0$. *(Delhi 2019)*

33. Solve the differential equation :

$\frac{dy}{dx}=1+x^2+y^2+x^2y^2$, given that $y = 1$ when $x = 0$. *(AI 2019)*

34. Find the particular solution of the differential equation $\frac{dy}{dx}=\frac{xy}{x^2+y^2}$, given that $y = 1$ when $x = 0$. *(AI 2019, Delhi 2015)*

35. Find the particular solution of the differential equation $e^x \tan y\,dx + (2 - e^x)\sec^2 y\,dy = 0$, given that $y=\frac{\pi}{4}$ when $x = 0$. *(2018)*

36. Find the particular solution of the differential equation $\frac{dy}{dx}+2y\tan x=\sin x,$ given that $y = 0$ when $x=\frac{\pi}{3}$. *(2018, Foreign 2014)*

37. Prove that $x^2 - y^2 = C(x^2 + y^2)^2$ is the general solution of the differential equation $(x^3 - 3xy^2)dx = (y^3 - 3x^2y)\,dy$, where C is a parameter. *(Delhi 2017)*

www.examrbtes.in

38. Solve the differential equation $(\tan^{-1}x - y)dx = (1 + x^2)dy$. *(AI 2017)*

39. Find the general solution of the following differential equation :

$$(1+y^2)+(x-e^{\tan^{-1}y})\frac{dy}{dx}=0 \quad \textit{(Delhi 2016)}$$

40. Find the particular solution of the differential equation $(1 - y^2)(1 + \log x)dx + 2xy\,dy = 0$, given that $y = 0$ when $x = 1$. *(Delhi 2016)*

41. Solve the differential equation :

$$y+x\frac{dy}{dx}=x-y\frac{dy}{dx} \quad \textit{(AI 2016)}$$

42. Solve the following differential equation $y^2dx + (x^2 - xy + y^2)dy = 0$ *(Foreign 2016)*

43. Solve the following differential equation $(\cot^{-1} y + x)dy = (1 + y^2)dx$ *(Foreign 2016)*

44. Find the particular solution of the differential equation $\frac{dy}{dx}=\frac{x(2\log x+1)}{\sin y + y\cos y}$, given that $y=\frac{\pi}{2}$, when $x = 1$. *(Delhi 2014)*

45. Solve the following differential equation :

$$(x^2-1)\frac{dy}{dx}+2xy=\frac{2}{x^2-1},\ |x|\neq 1 \quad \textit{(Delhi 2014)}$$

46. Find the particular solution of the differential equation $e^x\sqrt{1-y^2}\,dx+\frac{y}{x}dy=0$ given that $y = 1$ when $x = 0$. *(Delhi 2014)*

47. Solve the following differential equation :

$$\operatorname{cosec} x\log y\frac{dy}{dx}+x^2y^2=0. \quad \textit{(Delhi 2014)}$$

48. Find the particular solution of the differential equation $\frac{dy}{dx}=1+x+y+xy$, given that $y = 0$ when $x = 1$ *(AI 2014)*

49. Solve the differential equation

$$(1+x^2)\frac{dy}{dx}+y=e^{\tan^{-1}x} \quad \textit{(AI 2014)}$$

50. Find the particular solution of the differential equation $x(1 + y^2)\,dx - y(1 + x^2)dy = 0$, given that $y = 1$ when $x = 0$. *(AI 2014)*

51. Find the particular solution of the differential equation $\log\left(\frac{dy}{dx}\right)=3x+4y$, given that $y = 0$ when $x = 0$. *(AI 2014)*

52. Solve the differential equation $(x^2 - yx^2)dy + (y^2 + x^2y^2)dx = 0$, given that $y = 1$ when $x = 1$. *(Foreign 2014)*

53. Solve the differential equation

$$\frac{dy}{dx}+y\cot x=2\cos x, \text{ given that } y=0 \text{ when } x=\frac{\pi}{2}. \quad \textit{(Foreign 2014)}$$

54. Solve the differential equation

$$x\log x\frac{dy}{dx}+y=\frac{2}{x}\log x. \quad \textit{(Foreign 2014)}$$

55. If $y(x)$ is a solution of the differential equation $\left(\frac{2+\sin x}{1+y}\right)\frac{dy}{dx}=-\cos x$ and $y(0)=1$, then find the value of $y\left(\frac{\pi}{2}\right)$. *(Delhi 2014C)*

56. Find the general solution of the differential equation $(x-y)\frac{dy}{dx}=x+2y$. *(Delhi 2014C)*

57. Find the particular solution of the differential equation $x\frac{dy}{dx}-y+x\operatorname{cosec}\left(\frac{y}{x}\right)=0$; given that $y = 0$ when $x = 1$. *(AI 2014C, 2011C)*

58. Solve the differential equation

$$x\frac{dy}{dx}+y=x\cos x+\sin x, \text{given } y\left(\frac{\pi}{2}\right)=1. \quad \textit{(AI 2014C)}$$

59. Solve the following differential equation :

$$x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x;\ x\neq 0. \quad \textit{(AI 2014C, Delhi 2012C)}$$

60. Find the particular solution of the following differential equation :

$$xy\frac{dy}{dx}=(x+2)(y+2);\ y=-1 \text{ when } x=1. \quad \textit{(Delhi 2012)}$$

61. Solve the following differential equation :

$$2x^2\frac{dy}{dx}-2xy+y^2=0 \quad \textit{(Delhi 2012)}$$

www.examhits.in

62. Find the particular solution of the following differential equation; $\frac{dy}{dx} = 1+x^2+y^2+x^2y^2$, given that $y = 1$ when $x = 0$. *(Delhi 2012)*

63. Find the particular solution of the following differential equation : $(x+1)\frac{dy}{dx} = 2e^{-y} - 1$; $y = 0$ when $x = 0$. *(Delhi 2012)*

64. Find the particular solution of the following differential equation : $x\left(x^2-1\right)\frac{dy}{dx} = 1$; $y = 0$ when $x = 2$ *(AI 2012)*

65. Solve the following differential equation : $(1+x^2)\,dy + 2xy\,dx = \cot x\,dx$; $x \neq 0$ *(AI 2012, 2012C, Delhi 2011C)*

66. Find the particular solution of the differential equation : $\frac{dy}{dx} + y\cot x = 4x \operatorname{cosec} x$, $(x \neq 0)$, given that $y = 0$ when $x = \frac{\pi}{2}$. *(AI 2012)*

67. Find the particular solution of the following differential equation : $x\frac{dy}{dx} - y + x\sin\left(\frac{y}{x}\right) = 0$, given that when $x = 2$, $y = \pi$. *(AI 2012)*

68. Solve the following differential equation : $\frac{dy}{dx} - y = \cos x$, given that if $x = 0$, $y = 1$. *(Delhi 2012C)*

69. Find the particular solution of the following differential equation, given that $x = 2$, $y = 1$: $x\frac{dy}{dx} + 2y = x^2, (x \neq 0)$ *(Delhi 2012C)*

70. Find the particular solution of the differential equation : $\frac{dy}{dx} + y\cot x = 2x + x^2\cot x$, $x \neq 0$, given that $y = 0$, when $x = \frac{\pi}{2}$. *(Delhi 2012C)*

71. Solve the following differential equation : $\frac{dy}{dx} + \sec x \cdot y = \tan x, \left(0 \leq x < \frac{\pi}{2}\right)$. *(AI 2012C)*

72. Solve the following differential equation : $x\frac{dy}{dx} + y - x + xy\cot x = 0$, $x \neq 0$. *(AI 2012C, Delhi 2011C)*

73. Solve the following differential equation : $\cos^2 x\frac{dy}{dx} + y = \tan x$. *(Delhi 2011)*

74. Solve the following differential equation : $(1 + y^2)\,(1 + \log x)\,dx + xdy = 0$. *(Delhi 2011)*

75. Solve the following differential equation : $e^x \tan y dx + (1 - e^x)\sec^2 y dy = 0$. *(Delhi 2011)*

76. Solve the differential equation : $xdy + (y - x^3)dx = 0$. *(AI 2011)*

77. Solve the differential equation : $xdy - (y + 2x^2)dx = 0$. *(AI 2011)*

78. Solve the differential equation : $(y + 3x^2)\frac{dx}{dy} = x$. *(AI 2011)*

79. Solve the following differential equation : $xdy - ydx = \sqrt{x^2 + y^2}\,dx$ *(AI 2011)*

80. Solve the following differential equation : $\left[x\sin^2\left(\frac{y}{x}\right) - y\right]dx + x\,dy = 0$ *(Delhi 2011C)*

LA 2 (6 marks)

81. Solve the differential equation $x\frac{dy}{dx} + y = x\cos x + \sin x$, given that $y = 1$ when $x = \frac{\pi}{2}$. *(Delhi 2017)*

82. Find the particular solution of the differential equation $(x - y)\frac{dy}{dx} = (x + 2y)$, given that $y = 0$ when $x = 1$. *(AI 2017)*

83. Solve the differential equation : $(\tan^{-1}y - x)dy = (1 + y^2)dx$. *(Delhi 2015)*

84. Show that the differential equation $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$ is homogeneous and also solve it. *(AI 2015)*

85. Find the particular solution of the differential equation $(\tan^{-1}y - x)\,dy = (1+y^2)dx$, given that $x = 1$ when $y = 0$. *(AI 2015)*

www.examrites.in

86. Solve the following differential equation :

$$\left[y - x\cos\left(\frac{y}{x}\right)\right]dy + \left[y\cos\left(\frac{y}{x}\right) - 2x\sin\left(\frac{y}{x}\right)\right]dx = 0$$

(Foreign 2015)

87. Solve the following differential equation :

$$\left(\sqrt{1+x^2+y^2+x^2y^2}\right)dx + xy\,dy = 0$$

(Foreign 2015)

88. Find the particular solution of the differential equation $x\frac{dy}{dx} + y - x + xy\cot x = 0;\ x \neq 0$, given that when $x = \frac{\pi}{2}, y = 0$. *(Delhi 2015C)*

89. Solve the differential equation $x^2\,dy + (xy+y^2)dx = 0$ given $y = 1$, when $x = 1$ *(Delhi 2015C, 2013C)*

90. Solve the differential equation

$$\left(x\sin^2\left(\frac{y}{x}\right) - y\right)dx + x\,dy = 0 \text{ given } y = \frac{\pi}{4}$$

when $x = 1$ *(AI 2015C, 2014C, 2013C, 2013)*

91. Solve the differential equation

$$\frac{dy}{dx} - 3y\cot x = \sin 2x \text{ given } y = 2 \text{ when } x = \frac{\pi}{2}.$$

(AI 2015C)

92. Show that the differential equation

$$x\frac{dy}{dx}\sin\left(\frac{y}{x}\right) + x - y\sin\left(\frac{y}{x}\right) = 0$$

is homogeneous. Find the particular solution of this differential equation, given that $x = 1$ when $y = \frac{\pi}{2}$. *(Delhi 2013)*

93. Show that the differential equation $2ye^{x/y}\,dx + (y - 2x\,e^{x/y})\,dy = 0$ is homogeneous. Find the particular solution of this differential equation, given that $x = 0$ when $y = 1$. *(Delhi 2013, AI 2012C)*

94. Show that the differential equation $(xe^{x/y} + y)\,dx = xdy$ is homogeneous. Find the particular solution of this differential equation, given that $x = 1$ when $y = 1$. *(Delhi 2013)*

95. Find the particular solution of the differential equation $(\tan^{-1} y - x)\,dy = (1 + y^2)\,dx$, given that when $x = 0, y = 0$. *(AI 2013)*

96. Find the particular solution of the differential equation $\frac{dx}{dy} + x\cot y = 2y + y^2\cot y, (y \neq 0)$, given that $x = 0$ when $y = \frac{\pi}{2}$. *(AI 2013)*

97. Find the particular solution of the differential equation $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$; for $x = 1, y = 1$. *(Delhi 2013C)*

98. Find the particular solution of the following differential equation given that $y = 0$ when $x = 1$; $(x^2 + xy)\,dy = (x^2 + y^2)\,dx$. *(Delhi 2013C)*

99. Find the particular solution of the differential equation $(x - y)\frac{dy}{dx} = x + 2y$, given that when $x = 1, y = 0$. *(AI 2013C)*

100. Find the particular solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$, given that when $x = 1$, $y = \frac{\pi}{4}$. *(AI 2013C)*

101. Solve the following differential equation :

$$x\cos\left(\frac{y}{x}\right)(y\,dx + x\,dy) = y\sin\left(\frac{y}{x}\right)(xdy - ydx).$$

(AI 2013C)

Detailed Solutions

1. The degree of the differential equation $1 + \left(\frac{dy}{dx}\right)^2 = x$ is 2.

2. The given differential equation is

$$x^2\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^4$$

∴ Its order is 2 and degree is 1.

3. The given differential equation is

$$\frac{d}{dx}\left\{\left(\frac{dy}{dx}\right)^3\right\} = 0 \Rightarrow 3.\left(\frac{dy}{dx}\right)^2 \cdot \frac{d^2y}{dx^2} = 0$$

Order = 2 and Degree = 1

∴ Order + Degree = 2 + 1 = 3

www.examkites.in

4. Order = 2, Degree = 2.
∴ Order + Degree = 2 + 2 = 4

5. Order = 2, Degree = 3
∴ Order + Degree = 2 + 3 = 5

6. Degree of the given differential equation is 1.

7. Degree of the given differential equation is 2.

8. Degree of the given differential equation is 3.

9. The degree of the differential equation is 1.

10. Degree of given differential equation is 2.

11. **(a) :** In the particular solution of a differential equation of any order, there is no arbitrary constant because in the particular solution of any differential equation, we remove all the arbitrary constant by substituting some particular values.

12. Given, $y = \frac{A}{x} + 5$ ...(i)

Differentiating (i) w.r.t. x, we get, $\frac{dy}{dx} = \frac{-A}{x^2}$ ...(ii)

Putting value of A from (ii) into (i), we get

$$y = -x^2 \frac{dy}{dx} \cdot \frac{1}{x} + 5$$

$$\Rightarrow \quad y + x\frac{dy}{dx} = 5 \Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{5}{x}$$

which is required differential equation.

13. $v = \frac{A}{r} + B \Rightarrow \frac{dv}{dr} = -\frac{A}{r^2} \Rightarrow \frac{d^2v}{dr^2} = \frac{2A}{r^3}$

Now, $\frac{d^2v}{dr^2} \div \frac{dv}{dr} = \frac{2A}{r^3} \div \left(\frac{-A}{r^2}\right)$

$$\Rightarrow \frac{d^2v}{dr^2} \div \frac{dv}{dr} = \frac{-2}{r} \Rightarrow \frac{d^2v}{dr^2} = \frac{-2}{r} \cdot \frac{dv}{dr}$$

$$\Rightarrow \frac{d^2v}{dr^2} + \frac{2}{r}\frac{dv}{dr} = 0 \text{ is the required D.E.}$$

14. Here, $xy = C \cos x$ ...(i)

Differentiating (i) w.r.t. x, we get

$$1 \cdot y + x \cdot \frac{dy}{dx} = -C \sin x \qquad ...(ii)$$

∴ Eliminating C from (i) and (ii), we get

$$\frac{y + x\frac{dy}{dx}}{xy} = -\frac{\sin x}{\cos x} \Rightarrow x.\frac{dy}{dx} + y = -xy \tan x, \text{ is}$$

the required differential equation.

15. Here, $y = mx$ ...(i)

Differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx} = m \qquad ...(ii)$$

Eliminating m from (i) and (ii), we get

$y = x \cdot \frac{dy}{dx} \Rightarrow x\frac{dy}{dx} - y = 0$, is the required differential equation.

16. We have $y = e^{2x}(a + bx)$ ...(i)

On differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx} = 2e^{2x}(a + bx) + be^{2x} = 2y + be^{2x} \qquad ...(ii)$$

Again differentiating (ii) w.r.t. x, we get

$$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2be^{2x} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right)$$

[Using (ii)]

$$= 4\frac{dy}{dx} - 4y$$

Hence, the required differential equation is

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0.$$

17. We have, $y = a\, e^{bx+5}$ ...(i)

Differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx} = ae^{bx+5} \cdot b \Rightarrow e^{bx+5} = \frac{1}{ab}\frac{dy}{dx} \qquad ...(ii)$$

Differentiating (ii) w.r.t. x, we get

$$b \cdot e^{bx+5} = \frac{1}{ab}\frac{d^2y}{dx^2} \Rightarrow e^{bx+5} = \frac{1}{ab^2}\frac{d^2y}{dx^2} \qquad ...(iii)$$

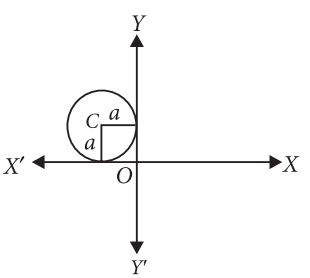
From (ii) & (iii), we have

$$\frac{1}{ab}\frac{dy}{dx} = \frac{1}{ab^2}\frac{d^2y}{dx^2} \Rightarrow \frac{d^2y}{dx^2} = b\frac{dy}{dx} \qquad ...(iv)$$

From (i) and (ii), we have, $\frac{1}{y}\frac{dy}{dx} = b$ ...(v)

From (iv) and (v), we get $\frac{d^2y}{dx^2} = \frac{1}{y}\left(\frac{dy}{dx}\right)^2$

18. The equation of the circles in second quadrant touching co-ordinate axes is



www.exambites.in

$(x + a)^2 + (y - a)^2 = a^2$ ...(i)

[Here C is $(-a, a)$ and radius = a]

which has only one arbitrary constant a.

Differentiating (i) w.r.t. x, we get

$$2(x+a)+2(y-a)\frac{dy}{dx}=0 \Rightarrow a=\frac{x+yy'}{y'-1}$$

Substituting for a in (i), we get

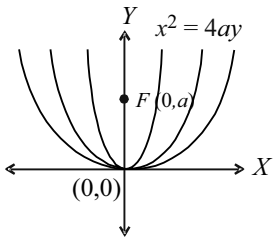
$$\left(x+\frac{x+y'y}{y'-1}\right)^2+\left(y-\frac{x+y'y}{y'-1}\right)^2=\left(\frac{x+y'y}{y'-1}\right)^2$$

$$\Rightarrow [x(y'-1)+x+y'y]^2+[y(y'-1)-x-y'y]^2=(x+y'y)^2$$

$$\Rightarrow (x+y)^2(y')^2+(x+y)^2=(x+y'y)^2$$

$$\Rightarrow (x+y)^2\left[\left(\frac{dy}{dx}\right)^2+1\right]=\left(x+y\frac{dy}{dx}\right)^2,$$ is the required differential equation.

19.



Equation of parabola having vertex at origin and axis along positive y-axis is

$x^2 = 4ay$, where a is the parameter. ...(i)

Differentiating (i) w.r.t. x, we get

$$2x = 4ay_1 \Rightarrow \frac{2x}{y_1}=4a \quad ...(ii)$$

Substituting the value of $4a$ from equation (ii) in equation (i), we get

$$x^2=\frac{2x}{y_1}y \Rightarrow x^2y_1-2xy=0 \Rightarrow xy_1-2y=0,$$

is the required differential equation.

20. We have, $x\frac{dy}{dx}+2y=x^2 \Rightarrow \frac{dy}{dx}+2\frac{y}{x}=x$

$\therefore$ I.F. $= e^{\int\frac{2}{x}dx} = e^{2\log x} = e^{\log x^2} = x^2$

21. **(c) :** We have, $(x+3y^2)\frac{dy}{dx}=y$

$$\Rightarrow \frac{x+3y^2}{y}=\frac{dx}{dy} \Rightarrow \frac{dx}{dy}-\frac{x}{y}=3y$$

This is a linear differential equation.

$\therefore$ I.F. $= e^{-\int\frac{dy}{y}} = e^{-\log y} = e^{\log y^{-1}} = \frac{1}{y}$

22. We have, $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}}-\frac{y}{\sqrt{x}}\right)\frac{dx}{dy}=1$

or $\frac{dy}{dx}+\frac{1}{\sqrt{x}}y=\frac{e^{-2\sqrt{x}}}{\sqrt{x}}$

This is a linear differential equation of the form $\frac{dy}{dx}+Py=Q$, where $P=\frac{1}{\sqrt{x}}, Q=\frac{e^{-2\sqrt{x}}}{\sqrt{x}}$

$\therefore$ I.F. $= e^{\int P dx} \Rightarrow$ I.F. $= e^{\int\frac{1}{\sqrt{x}}dx} = e^{2\sqrt{x}}$

23. The given differential equation is

$$(1+y^2)+(2xy-\cot y)\frac{dy}{dx}=0$$

$$\Rightarrow (1+y^2)\frac{dx}{dy}+2xy-\cot y=0$$

$$\Rightarrow \frac{dx}{dy}+\frac{2y}{1+y^2}\cdot x=\frac{\cot y}{1+y^2}$$

This is a linear differential equation of the form $\frac{dx}{dy}+Px=Q$, where, $P=\frac{2y}{1+y^2}$ and $Q=\frac{\cot y}{1+y^2}$

$\therefore$ I.F. $= e^{\int P\,dy} = e^{\int\frac{2y}{1+y^2}dy} = e^{\log(1+y^2)} = 1 + y^2.$

24. We have, $\frac{dy}{dx}=2^{-y}$

$$\Rightarrow \frac{dy}{dx}=\frac{1}{2^y} \Rightarrow 2^y\,dy=dx \quad ...(i)$$

Integrating both sides of (i), we get

$$\frac{2^y}{\log 2}=x+C \Rightarrow 2^y=(C+x)\log 2$$

Taking log on both sides to the base 2, we get

$\log_2 2^y = \log_2 [(C + x) \log 2]$

$\Rightarrow y = \log_2 [(C + x) \log 2]$,

which is the required solution.

25. We have, $\frac{dy}{dx}=x^3e^{-2y} \Rightarrow e^{2y}\,dy=x^3dx$

On integrating, we get $\frac{e^{2y}}{2}=\frac{x^4}{4}+C'$

$\Rightarrow 2e^{2y}=x^4+C$, where $C=4C'$

26. We have, $\frac{dy}{dx}+y=\cos x-\sin x$, which is a linear differential equation of the form

www.exambites.in

$\frac{dy}{dx} + Py = Q$, where $P = 1$, $Q = \cos x - \sin x$

$\therefore$ I.F. $= e^{\int dx} = e^x$

The solution of given differential equation is

$ye^x = \int e^x(\cos x - \sin x)dx + C$

$\Rightarrow \quad ye^x = e^x \cos x + C \Rightarrow y = \cos x + Ce^{-x}$

27. We have, $ye^y dx = (y^3 + 2xe^y)dy$

$\Rightarrow \quad \frac{dx}{dy} = \frac{y^3 + 2xe^y}{ye^y} \Rightarrow \frac{dx}{dy} - \frac{2}{y}x = y^2 e^{-y}$ ...(i)

This is a linear D.E. of the form $\frac{dx}{dy} + Px = Q$

$\therefore$ I.F. $= e^{-\int \frac{2}{y}dy} = e^{-2\log y} = e^{\log y^{-2}} = \frac{1}{y^2}$

So, the solution of (i) is

$$x \cdot \frac{1}{y^2} = \int \frac{1}{y^2} \cdot y^2 e^{-y} dy$$

$\Rightarrow \quad \frac{x}{y^2} = -e^{-y} + C \Rightarrow x = -y^2 e^{-y} + Cy^2$

28. We have, $x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$

$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right)$,

which is homogeneous differential equation.

Now, put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x\frac{dv}{dx} = v - \tan v$

$\Rightarrow \quad x\frac{dv}{dx} = -\tan v \Rightarrow \frac{dv}{\tan v} = -\frac{dx}{x}$

$\Rightarrow \quad \cot v\, dv + \frac{dx}{x} = 0$

Integrating both sides,we get

$\log|\sin v| + \log x = \log C$

$\Rightarrow \quad x \sin v = C \Rightarrow x\sin\left(\frac{y}{x}\right) = C$

When $x = 1$, $y = \frac{\pi}{4}$, we get

$$1 \cdot \sin\left(\frac{\pi}{4}\right) = C \Rightarrow C = \frac{1}{\sqrt{2}}$$

So, $x\sin\left(\frac{y}{x}\right) = \frac{1}{\sqrt{2}}$ is the required particular solution.

29. We have, $\cos y\, dx + (1 + e^{-x})\sin y\, dy = 0$

$\Rightarrow \quad dx + (1 + e^{-x})\tan y\, dy = 0$

$\Rightarrow \quad \frac{dx}{1 + e^{-x}} + \tan y\, dy = 0$

Integrating both sides, we get

$$\int \frac{e^x}{1 + e^x} dx + \int \tan y\, dy = 0$$

$\Rightarrow \quad \log(1 + e^x) + \log|\sec y| = \log C$

$\Rightarrow \quad \sec y\,(1 + e^x) = C$

When, $x = 0$, $y = \frac{\pi}{4}$, we get $C = 2\sqrt{2}$

$\therefore$ Particular solution of the differential equation is, $\sec y(1 + e^x) = 2\sqrt{2}$

30. We have, $ye^{x/y}\,dx = (xe^{x/y} + y^2)dy$, $y \neq 0$

$\Rightarrow \quad \frac{dx}{dy} = \frac{x}{y} + \frac{y}{e^{x/y}}$ ...(i)

Putting $\frac{x}{y} = t \Rightarrow x = yt \Rightarrow \frac{dx}{dy} = t + y\frac{dt}{dy}$

$\therefore$ Equation (i) becomes, $t + y\frac{dt}{dy} = t + \frac{y}{e^t}$

$\Rightarrow \quad y\frac{dt}{dy} = ye^{-t} \Rightarrow \frac{dt}{dy} = e^{-t} \Rightarrow dy = e^t dt$

Integrating both sides, we get

$y = e^t + C \Rightarrow y = e^{x/y} + C$

31. We have, $x\frac{dy}{dx} - y = \sqrt{x^2 + y^2}$

$\Rightarrow x\frac{dy}{dx} = y + \sqrt{x^2 + y^2}$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}$ ...(i)

This is a linear homogeneous differential equation.

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ Eq. (i) becomes

$v + x\frac{dv}{dx} = v + \sqrt{1 + v^2} \Rightarrow x\frac{dv}{dx} = \sqrt{1 + v^2}$

$\Rightarrow \quad \frac{dx}{x} = \frac{dv}{\sqrt{1 + v^2}} \Rightarrow \int \frac{dx}{x} = \int \frac{dv}{\sqrt{1 + v^2}}$

$\Rightarrow \quad \log x + \log C_1 = \log|v + \sqrt{1 + v^2}|$

$\Rightarrow \log x + \log C_1 = \log\left|\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}}\right|$

$\Rightarrow \log C_1 x = \log|y + \sqrt{x^2 + y^2}| - \log x$

$\Rightarrow \pm C_1 x^2 = y + \sqrt{x^2 + y^2}$

$\Rightarrow \quad Cx^2 = y + \sqrt{x^2 + y^2}$ [where $C = \pm C_1$]

www.exambites.in

General solution of the given equation is

$Cx^2 = y + \sqrt{x^2 + y^2}$ ...(ii)

Now, putting $y = 0$ and $x = 1$ in (ii), we get $C = 1$

$\therefore$ Required solution is $x^2 = y + \sqrt{x^2 + y^2}$.

32. We have $(1+x^2)\frac{dy}{dx} + 2xy = 4x^2$

$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{4x^2}{1+x^2}$

This is a linear differential equation of the form

$\frac{dy}{dx} + Py = Q$ where $P = \frac{2x}{1+x^2}$ and $Q = \frac{4x^2}{1+x^2}$

$\therefore$ I.F. $= e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx}$

$= e^{\log(1+x^2)} = 1 + x^2$

Hence, the required solution is

$$y(1+x^2) = \int \frac{4x^2}{1+x^2}(1+x^2)\,dx + C$$

$\Rightarrow y(1+x^2) = 4\int x^2 dx + C$

$\Rightarrow y(1+x^2) = \frac{4x^3}{3} + C$

Given that $y(0) = 0$

$\therefore 0(1+0) = 0 + C \Rightarrow C = 0$

Thus, $y = \frac{4x^3}{3(1+x^2)}$ is the required solution.

33. We have, $\frac{dy}{dx} = 1 + x^2 + y^2 + x^2y^2$

$\therefore \frac{dy}{dx} = 1 + x^2 + y^2(1+x^2) = (1+x^2)\,.\,(1+y^2)$

$\Rightarrow \frac{dy}{1+y^2} = (1+x^2)dx$

Integrating both sides, we get $\tan^{-1} y = x + \frac{x^3}{3} + C$

when $x = 0$, $y = 1$

$\tan^{-1} 1 = 0 + 0 + C \Rightarrow C = \frac{\pi}{4}$

$\therefore \tan^{-1} y = x + \frac{1}{3}x^3 + \frac{\pi}{4}$ is the required solution.

34. We have, $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$

This is a homogeneous linear differential equation

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x\frac{dv}{dx} = \frac{x \cdot vx}{x^2 + v^2x^2}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{v}{1+v^2} \Rightarrow x\frac{dv}{dx} = \frac{v}{1+v^2} - v$

$\Rightarrow x\frac{dv}{dx} = \frac{-v^3}{1+v^2} \Rightarrow \frac{dx}{x} = -\left(\frac{1+v^2}{v^3}\right)dv$

Integrating both sides, we get

$\int \frac{dx}{x} = -\int v^{-3}dv - \int \frac{1}{v}dv$

$\Rightarrow \log x = \frac{1}{2v^2} - \log v + C$

$\Rightarrow \log x = \frac{x^2}{2y^2} - \log y + \log x + C$

$\Rightarrow \log y = \frac{x^2}{2y^2} + C$

When $y = 1$, $x = 0 \Rightarrow \log 1 = 0 + C \Rightarrow C = 0$

$\therefore$ Particular solution is $y = e^{\frac{x^2}{2y^2}}$

35. The given differential equation is,

$e^x \tan y\,dx + (2 - e^x)\sec^2 y\,dy = 0$

$\Rightarrow (2 - e^x)\sec^2 y\,dy = -e^x \tan y\,dx$

$\Rightarrow \frac{\sec^2 y}{\tan y}dy = \frac{-e^x}{2-e^x}dx$

Integrating both sides, we get

$\int \frac{\sec^2 y}{\tan y}dy = \int \frac{-e^x}{2-e^x}dx$

$\Rightarrow \log \tan y = \log(2 - e^x) + C$

When $y = \frac{\pi}{4}, x = 0$

$\therefore \log \tan\frac{\pi}{4} = \log(2 - e^0) + C$

$\Rightarrow 0 = \log 1 + C \Rightarrow C = 0$

$\therefore$ Particular solution is $\log \tan y = \log(2 - e^x)$

$\Rightarrow e^x + \tan y - 2 = 0$

36. We have, $\frac{dy}{dx} + 2y\tan x = \sin x$

It is linear differential equation of the form

$\frac{dy}{dx} + Py = Q$

where $P = 2\tan x$, and $Q = \sin x$

Now, I.F. $= e^{\int 2\tan x\,dx} = e^{2\log|\sec x|} = \sec^2 x$

www.exambrics.in

$\therefore \quad y(\sec^2 x) = \int (\sec^2 x)(\sin x)\,dx$

$\Rightarrow y(\sec^2 x) = \int \sec x \tan x\,dx$

$\Rightarrow y(\sec^2 x) = \sec x + C$

When $x = \dfrac{\pi}{3}$, $y = 0$

$(0)[\sec^2(\pi/3)] = \sec(\pi/3) + C \Rightarrow C = -2$

$\therefore \quad y(\sec^2 x) = \sec x - 2$ *i.e.,* $y = \cos x - 2\cos^2 x$ is the required solution.

37. We have, $(x^3 - 3xy^2)dx = (y^3 - 3x^2y)dy$

$$\Rightarrow \frac{dy}{dx} = \frac{x^3 - 3xy^2}{y^3 - 3x^2y} \qquad ...(i)$$

Put, $y = vx \Rightarrow \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$\therefore$ (i) becomes

$$v + x\frac{dv}{dx} = \frac{1 - 3v^2}{v^3 - 3v}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{1 - 3v^2 - v^4 + 3v^2}{v^3 - 3v} = \frac{1 - v^4}{v(v^2 - 3)}$$

$$\Rightarrow \quad \frac{v(v^2 - 3)dv}{1 - v^4} = \frac{dx}{x}$$

$$\Rightarrow \quad \int \frac{(v^3 - 3v)dv}{(1 - v^2)(1 + v^2)} = \int \frac{dx}{x} \qquad ...(ii)$$

$$\text{Now, let } \frac{v^3 - 3v}{(1 - v^2)(1 + v^2)} = \frac{Av + B}{1 - v^2} + \frac{Cv + D}{1 + v^2} \qquad ...(iii)$$

$\Rightarrow \quad v^3 - 3v = (Av + B)(1 + v^2) + (Cv + D)(1 - v^2)$

Comparing coeff. of like powers, we get

$A - C = 1$, $A + C = -3$, $B - D = 0$ and $B + D = 0$

Solving these equations, we get $A = -1$, $B = 0$, $C = -2$, $D = 0$

From (ii) and (iii), we have

$$\int \frac{-v}{1 - v^2}dv - \int \frac{2v}{1 + v^2}dv = \int \frac{dx}{x}$$

$$\Rightarrow \quad \frac{1}{2}\log(1 - v^2) - \log(1 + v^2) = \log x + \log C_1$$

$$\Rightarrow \quad \frac{\sqrt{1 - v^2}}{1 + v^2} = C_1 x \Rightarrow x\frac{\left(\sqrt{x^2 - y^2}\right)}{x^2 + y^2} = C_1 x$$

$\Rightarrow \quad x^2 - y^2 = C_1^2(x^2 + y^2)^2$

i.e., $x^2 - y^2 = C(x^2 + y^2)^2$ (where $C_1^2 = C$)

which is the required solution.

38. We have, $\dfrac{dy}{dx} = \dfrac{(\tan^{-1} x - y)}{1 + x^2}$

$$\Rightarrow \quad \frac{dy}{dx} + \frac{y}{1 + x^2} = \frac{\tan^{-1} x}{1 + x^2}$$

which is a linear differential equation

where $P = \dfrac{1}{1 + x^2}$, $Q = \dfrac{\tan^{-1} x}{1 + x^2}$

$$\therefore \quad \text{I.F} = e^{\int P dx} = e^{\int \frac{1}{1 + x^2}dx} = e^{\tan^{-1} x}$$

$\therefore$ Solution is $y.(\text{I.F}) = \int Q(\text{I.F})\,dx$

$$\Rightarrow \quad ye^{\tan^{-1} x} = \int \frac{\tan^{-1} x}{1 + x^2} \cdot e^{\tan^{-1} x}dx \qquad ...(i)$$

Let $I = \int \dfrac{\tan^{-1} x}{1 + x^2} e^{\tan^{-1} x}dx$

Put $\tan^{-1} x = t \Rightarrow \dfrac{dx}{1 + x^2} = dt$

$$\therefore \quad I = \int t \cdot e^t dt = t\int e^t dt - \int \left(\frac{d}{dt}(t)\int e^t dt\right)dt$$

$$= te^t - \int e^t dt = te^t - e^t + C$$

$$= e^t(t - 1) + C$$

$$\Rightarrow \quad I = e^{\tan^{-1} x}(\tan^{-1} x - 1) + C \qquad ...(ii)$$

Putting (ii) in (i), we get

$$ye^{\tan^{-1} x} = e^{\tan^{-1} x}(\tan^{-1} x - 1) + C$$

$$\Rightarrow \quad y = \tan^{-1} x - 1 + Ce^{-\tan^{-1} x}$$

39. We have, $(1 + y^2) + (x - e^{\tan^{-1} y})\dfrac{dy}{dx} = 0$

$$\Rightarrow \quad (x - e^{\tan^{-1} y})\frac{dy}{dx} = -(1 + y^2)$$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{x - e^{\tan^{-1} y}}{-(1 + y^2)} \Rightarrow \frac{dx}{dy} + \frac{1}{1 + y^2} \cdot x = \frac{e^{\tan^{-1} y}}{1 + y^2}$$

This is a linear differential equation of the form $\dfrac{dx}{dy} + Px = Q$, where $P = \dfrac{1}{1 + y^2}$ and $Q = \dfrac{e^{\tan^{-1} y}}{1 + y^2}$

$$\therefore \quad \text{I.F.} = e^{\int \frac{dy}{1 + y^2}} = e^{\tan^{-1} y}$$

www.exambites.in

$\therefore$ Solution is

$$x \cdot e^{\tan^{-1} y} = \int \frac{(e^{\tan^{-1} y})^2}{1+y^2}\, dy + C$$

$$= \int \frac{e^{2\tan^{-1} y}}{1+y^2}\, dy + C$$

$$\Rightarrow \quad x \cdot e^{\tan^{-1} y} = \frac{e^{2\tan^{-1} y}}{2} + C_1$$

$$\Rightarrow \quad x = \frac{e^{\tan^{-1} y}}{2} + C_1 e^{-\tan^{-1} y}$$

40. We have, $(1 - y^2)(1 + \log x)\, dx + 2xy\, dy = 0$

$\Rightarrow$ $(1 - y^2)(1 + \log x)\, dx = -2xy\, dy$

$$\Rightarrow \quad \frac{(1+\log x)}{x} dx = -\frac{2y}{1-y^2} dy$$

On integrating both sides, we get

$$\frac{(1+\log x)^2}{2} = \log|1-y^2| + C$$

When $x = 1$, $y = 0$

$$\therefore \quad \frac{(1+\log 1)^2}{2} = \log(1) + C \Rightarrow C = \frac{1}{2}$$

$$\Rightarrow \quad \frac{(1+\log x)^2}{2} = \log|1-y^2| + \frac{1}{2}$$

$\Rightarrow$ $(1 + \log x)^2 = 2 \log |1 - y^2| + 1$ is the required solution.

41. We have, $y + x\frac{dy}{dx} = x - y\frac{dy}{dx}$

$$\Rightarrow \quad x\frac{dy}{dx} + y\frac{dy}{dx} = x - y \Rightarrow \quad \frac{dy}{dx} = \frac{x-y}{x+y} \qquad \text{...(i)}$$

This is a linear homogeneous D.E.

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ Equation (i) becomes

$$v + x\frac{dv}{dx} = \frac{x - vx}{x + vx} = \frac{1-v}{1+v}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{1-v}{1+v} - v = \frac{1-v-v^2-v}{1+v} = \frac{1-2v-v^2}{1+v}$$

$$\Rightarrow \quad \frac{(1+v)}{v^2+2v-1} dv = -\frac{dx}{x}$$

Integrating both sides, we get

$$\frac{1}{2}\log|v^2+2v-1| = -\log|x| + \log C$$

$$\Rightarrow \quad \frac{1}{2}\log|v^2+2v-1| + \log|x| = \log C$$

$$\Rightarrow \quad \frac{1}{2}\log\left|\frac{y^2}{x^2} + \frac{2y}{x} - 1\right| + \log|x| = \log C$$

$$\Rightarrow \quad \log\left|\frac{y^2}{x^2} + \frac{2y}{x} - 1\right| + 2\log|x| = 2\log C$$

$$\Rightarrow \quad \log\left|\frac{y^2+2xy-x^2}{x^2} \times x^2\right| = \log C^2$$

$\Rightarrow$ $y^2 + 2xy - x^2 = \pm C^2$

$\Rightarrow$ $y^2 + 2xy - x^2 = C_1$ (where $C_1 = \pm C^2$)

42. We have, $y^2\, dx + (x^2 - xy + y^2)\, dy = 0$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-y^2}{x^2 - xy + y^2}$$

This is homogeneous differential equation.

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$, we get

$$v + x\frac{dv}{dx} = \frac{-v^2x^2}{x^2 - vx^2 + v^2x^2}$$

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{-v^2}{1-v+v^2}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{-v^2}{1-v+v^2} - v$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{-v-v^3}{1-v+v^2} \Rightarrow \frac{1-v+v^2}{v(1+v^2)} dv = -\frac{1}{x} dx$$

Integrating both sides, we get

$$\int \frac{1+v^2}{v(1+v^2)} dv - \int \frac{v}{v(1+v^2)} dv = -\int \frac{1}{x} dx$$

$$\Rightarrow \quad \int \frac{1}{v} dv - \int \frac{1}{1+v^2} dv = -\int \frac{1}{x} dx$$

$\Rightarrow$ $\log|v| - \tan^{-1} v = -\log|x| + \log C$

$$\Rightarrow \quad \log\left|\frac{vx}{C}\right| = \tan^{-1} v \qquad \Rightarrow \quad \left|\frac{vx}{C}\right| = e^{\tan^{-1} v}$$

$\Rightarrow$ $|y| = Ce^{\tan^{-1} y/x}$ is the required solution.

www.exambites.in

43. We have, $(\cot^{-1}y + x)\,dy = (1 + y^2)dx$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\cot^{-1}y + x}{1+y^2}$$

$$\Rightarrow \quad \frac{dx}{dy} + \left(-\frac{1}{1+y^2}\right)x = \frac{\cot^{-1}y}{1+y^2}$$

This is a linear differential equation of the form

$$\frac{dx}{dy} + Px = Q, \text{ where, } P = -\frac{1}{1+y^2} \text{ and } Q = \frac{\cot^{-1}y}{1+y^2}$$

$$\therefore \quad \text{I.F.} = e^{-\int \frac{1}{1+y^2}dy} = e^{\cot^{-1}y}$$

$\therefore$ Solution is,

$$xe^{\cot^{-1}y} = \int \frac{\cot^{-1}y}{(1+y^2)} e^{\cot^{-1}y}\,dy$$

[Put $t = \cot^{-1}y \Rightarrow dt = -\dfrac{1}{1+y^2}\,dy$]

$$xe^{\cot^{-1}y} = -\int te^t\,dt$$

$$\Rightarrow \quad xe^{\cot^{-1}y} = -e^t(t-1) + C$$

$$\Rightarrow \quad xe^{\cot^{-1}y} = e^{\cot^{-1}y}(1 - \cot^{-1}y) + C$$

44. We have, $\dfrac{dy}{dx} = \dfrac{x(2\log x + 1)}{\sin y + y\cos y}$

$\Rightarrow (\sin y + y\cos y)dy = x(2\log x + 1)dx$

On integrating both sides, we get

$$-\cos y + y\sin y - (-\cos y) = 2\left[\log x \times \frac{x^2}{2} - \int \frac{1}{x} \times \frac{x^2}{2}\,dx\right] + \frac{x^2}{2} + C$$

$$\Rightarrow \quad y\sin y = x^2\log x - \frac{x^2}{2} + \frac{x^2}{2} + C$$

$$\Rightarrow \quad y\sin y = x^2\log x + C$$

when $x = 1$, $y = \dfrac{\pi}{2}$

$$\therefore \quad \frac{\pi}{2}\sin\frac{\pi}{2} = 1\cdot\log(1) + C \quad \Rightarrow \quad \frac{\pi}{2} = C$$

$\therefore$ $y\sin y = x^2\log x + \pi/2$ is the required solution.

45. We have, $(x^2-1)\dfrac{dy}{dx} + 2xy = \dfrac{2}{x^2-1}, |x| \neq 1$

$$\Rightarrow \quad \frac{dy}{dx} + \frac{2xy}{x^2-1} = \frac{2}{(x^2-1)^2}$$

This is a linear differential equation of the form,

$$\frac{dy}{dx} + Py = Q, \text{ where } P = \frac{2x}{x^2-1} \text{ and } Q = \frac{2}{(x^2-1)^2}$$

$$\therefore \quad \text{I.F.} = e^{\int Pdx} = e^{\int \frac{2x}{x^2-1}dx} = e^{\log(x^2-1)} = x^2 - 1$$

Hence, solution of differential equation is given by

$$y(x^2-1) = \int \frac{2(x^2-1)}{(x^2-1)^2}dx$$

$$\Rightarrow \quad y(x^2-1) = 2\int \frac{dx}{x^2-1}$$

$$\Rightarrow \quad y(x^2-1) = 2\times\frac{1}{2}\log\left|\frac{x-1}{x+1}\right| + C$$

$$\Rightarrow \quad y(x^2-1) = \log\left|\frac{x-1}{x+1}\right| + C$$

46. We have, $e^x\sqrt{1-y^2}\,dx + \dfrac{y}{x}dy = 0$

$$x\,e^x dx + \frac{y}{\sqrt{1-y^2}}dy = 0$$

Integrating both sides, we get

$$x\cdot e^x - \int 1.e^x dx - \frac{1}{2}\int (1-y^2)^{-\frac{1}{2}}(-2y)\,dy = C$$

$$\Rightarrow x\,e^x - e^x - \frac{1}{2}\frac{(1-y^2)^{\frac{1}{2}}}{1/2} = C$$

$$\Rightarrow e^x(x-1) - \sqrt{1-y^2} = C$$

When $x = 0, y = 1, e^0(0-1) - \sqrt{1-1} = C$

$\Rightarrow \quad C = -1$

$\therefore$ $e^x(x-1) - \sqrt{1-y^2} = -1$ is the required solution.

47. We have, $\text{cosec}\,x\log y\dfrac{dy}{dx} + x^2y^2 = 0$

$$\Rightarrow \quad \frac{\log y}{y^2}dy + \frac{x^2}{\text{cosec}\,x}dx = 0$$

Integrating both sides, we get

$$\int \frac{\log y}{y^2}dy + \int x^2\sin x\,dx = 0$$

[Put $\log y = t \Rightarrow \dfrac{1}{y}dy = dt$ and $y = e^t$]

www.exambites.in

$\Rightarrow \int t.e^{-t}dt + \int x^2 \sin x\,dx = 0$

$\Rightarrow t.\frac{e^{-t}}{-1} - \int 1.\frac{e^{-t}}{-1}dt + x^2(-\cos x) - \int 2x(-\cos x)\,dx = C$

$\Rightarrow \quad -te^{-t} - e^{-t} - x^2\cos x + 2x\sin x - 2\int 1.\sin x\,dx = C$

$\Rightarrow -\frac{1+\log y}{y} - x^2\cos x + 2x\sin x + 2\cos x = C$

is the required solution.

48. We have, $\frac{dy}{dx} = 1 + x + y + xy$

$\Rightarrow \frac{dy}{dx} = (1+x) + (1+x)y = (1+x)(1+y)$

$\Rightarrow \frac{dy}{1+y} = (1+x)dx$

Integrating both sides, we get

$\int \frac{dy}{1+y} = \int (1+x)dx + C$

$\Rightarrow \log(1+y) = x + \frac{x^2}{2} + C$ ...(i)

When $x = 1, y = 0$

$\therefore \quad \log 1 = 1 + \frac{1}{2} + C \Rightarrow C = -\frac{3}{2}$

$\therefore$ The particular solution of (i) is

$\log(1+y) = x + \frac{x^2}{2} - \frac{3}{2}.$

49. *Refer to answer 39.*

50. We have, $x(1+y^2)\,dx - y(1+x^2)\,dy = 0$

$\Rightarrow \frac{x}{1+x^2}dx - \frac{y}{1+y^2}dy = 0$

$\Rightarrow \frac{2x}{1+x^2}dx = \frac{2y}{1+y^2}dy$

Integrating both sides, we get

$\log(1+y^2) = \log(1+x^2) + \log C$

$\Rightarrow \quad 1 + y^2 = C(1+x^2)$

When $x = 0, y = 1$

$\therefore \quad 1 + 1 = C(1+0) \Rightarrow C = 2$

$\therefore \quad 1 + y^2 = 2(1+x^2)$ is the required particular solution.

51. We have, $\log\left(\frac{dy}{dx}\right) = 3x + 4y$

$\frac{dy}{dx} = e^{3x+4y} = e^{3x}\cdot e^{4y} \Rightarrow e^{-4y}\,dy = e^{3x}\,dx$

Integrating both sides, we get

$\int e^{3x}dx - \int e^{-4y}dy = 0 \Rightarrow \frac{e^{3x}}{3} - \frac{e^{-4y}}{-4} = C$

When $x = 0, y = 0$

$\therefore \quad \frac{1}{3} + \frac{1}{4} = C \Rightarrow C = \frac{7}{12}$

$\Rightarrow \frac{e^{3x}}{3} + \frac{e^{4y}}{4} = \frac{7}{12}$

$\Rightarrow 4e^{3x} + 3e^{4y} = 7$ is the required particular solution.

52. We have, $(x^2 - yx^2)dy + (y^2 + x^2y^2)dx = 0$

$\Rightarrow x^2(1-y)dy + y^2(1+x^2)dx = 0$

$\Rightarrow \int \frac{(1-y)}{y^2}dy + \int \frac{(1+x^2)}{x^2}dx = 0$

$\Rightarrow \int\left(\frac{1}{y^2} - \frac{1}{y}\right)dy + \int\left(\frac{1}{x^2} + 1\right)dx = 0$

$\Rightarrow -\frac{1}{y} - \log|y| - \frac{1}{x} + x = C$

$\Rightarrow -x - xy\log|y| - y + x^2y = C(xy)$ ...(i)

when $x = 1, y = 1$

$\therefore \quad -(1) - (1)(1)\log|1| - (1) + (1)^2(1) = C(1)$

$\Rightarrow \quad C = -1$

Equation (i) becomes

$x^2y = x + xy\log|y| + y - xy$

53. We have, $\frac{dy}{dx} + y\cot x = 2\cos x$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where $P = \cot x$, $Q = 2\cos x$

$\therefore \quad$ I.F. $= e^{\int \cot x\,dx} = e^{\log|\sin x|} = |\sin x|$

$\therefore \quad y|\sin x| = \int |\sin x|(2\cos x)\,dx$

$\Rightarrow y|\sin x| = \int \sin 2x\,dx$

$\Rightarrow y(\sin x) = -\frac{1}{2}\cos 2x + C$

when $x = \frac{\pi}{2}, y = 0$

$0(\sin\frac{\pi}{2}) = -\frac{1}{2}\cos 2\left(\frac{\pi}{2}\right) + C \Rightarrow C = -\frac{1}{2}$

$\therefore \quad y(\sin x) = -\frac{1}{2}\cos 2x - \frac{1}{2}$

i.e., $2y\sin x + \cos 2x + 1 = 0$ is the required solution.

www.examkites.in

54. We have, $x\log x\frac{dy}{dx}+y=\frac{2}{x}\log x$

$\Rightarrow \frac{dy}{dx}+\frac{1}{x\log x}y=\frac{2}{x^2}$

It is a linear differential equation of the form

$\frac{dy}{dx}+Py=Q,$ where $P=\frac{1}{x\log x}, Q=\frac{2}{x^2}$

$\therefore$ I.F. $=e^{\int\frac{1}{x\log x}dx}=e^{\log\log x}=\log x$

$\therefore \quad y(\log x)=\int(\log x)\frac{2}{x^2}dx$

$\Rightarrow y(\log x)=\log x\int\frac{2}{x^2}dx-\int\left(\frac{d}{dx}(\log x)\int\frac{2}{x^2}dx\right)dx$

$\Rightarrow y(\log x)=\log x\left(-\frac{2}{x}\right)+\int\frac{2}{x^2}dx$

$\Rightarrow y(\log x)=\log x\left(-\frac{2}{x}\right)-\frac{2}{x}+C$

55. We have, $\left(\frac{2+\sin x}{1+y}\right)\frac{dy}{dx}=-\cos x$

$\Rightarrow \frac{dy}{1+y}=-\frac{\cos x}{2+\sin x}dx$

Integrating both sides, we get

$\log(y+1)=-\log(2+\sin x)+\log C$

$\Rightarrow \log(y+1)=\log\frac{C}{2+\sin x}$

$\Rightarrow y+1=C/(2+\sin x) \Rightarrow (y+1)(2+\sin x)=C$

Given : $y(0)=1 \Leftrightarrow x=0, y=1$

$\therefore \quad (1+1).(2+\sin 0)=C \Rightarrow C=4$

$\therefore \quad (y+1)(2+\sin x)=4$

$\Rightarrow y=\frac{4}{2+\sin x}-1$ ...(i)

Put $x=\frac{\pi}{2}$ in (i), $y\left(\frac{\pi}{2}\right)=\frac{4}{2+1}-1=\frac{1}{3}.$

56. We have, $(x-y)\frac{dy}{dx}=x+2y$

$\Rightarrow \frac{dy}{dx}=\frac{x+2y}{x-y}$ ...(i)

This is a linear homogeneous differential equation.

$\therefore$ Put $y=vx \Rightarrow \frac{dy}{dx}=v.1+x\frac{dv}{dx}$

$\therefore$ Equation (i) becomes

$v+x\frac{dv}{dx}=\frac{x+2vx}{x-vx}=\frac{1+2v}{1-v}$

$\Rightarrow x\frac{dv}{dx}=\frac{1+2v}{1-v}-v=\frac{1+v+v^2}{1-v}$

$\Rightarrow \frac{1-v}{1+v+v^2}dv=\frac{dx}{x}$

Integrating both sides, we get

$\int\frac{-\frac{1}{2}(2v+1)+\frac{3}{2}}{v^2+v+1}dv=\log x+C$

$\Rightarrow -\frac{1}{2}\int\frac{2v+1}{v^2+v+1}dv+\frac{3}{2}\int\frac{dv}{\left(v+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$

$=\log x+C$

$\Rightarrow -\frac{1}{2}\log(v^2+v+1)+\frac{3}{2}\cdot\frac{1}{\sqrt{3}/2}\tan^{-1}\left[\frac{v+\frac{1}{2}}{\sqrt{3}/2}\right]=\log x+C$

$\Rightarrow -\frac{1}{2}\log\left(\frac{y^2}{x^2}+\frac{y}{x}+1\right)+\sqrt{3}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}\cdot x}\right)=\log x+C$

$\Rightarrow -\frac{1}{2}\log(y^2+xy+x^2)+\sqrt{3}\tan^{-1}\left(\frac{x+2y}{\sqrt{3}\cdot x}\right)=C.$

57. We have, $x\frac{dy}{dx}-y+x\operatorname{cosec}\left(\frac{y}{x}\right)=0$

$\Rightarrow \frac{dy}{dx}-\frac{y}{x}=-\operatorname{cosec}\left(\frac{y}{x}\right)$ ...(i)

Put $y=vx \Rightarrow \frac{dy}{dx}=v.1+x.\frac{dv}{dx}$

$\therefore$ Equation (i) becomes

$v+x\frac{dv}{dx}-v=-\operatorname{cosec}v \Rightarrow x\frac{dv}{dx}=-\operatorname{cosec}v$

$\Rightarrow -\sin v\,dv=\frac{dx}{x}$

Integrating both sides, we get $\cos v=\log x+C$

$\Rightarrow \cos\left(\frac{y}{x}\right)=\log x+C$

When $x=1, y=0$

$\Rightarrow \cos\left(\frac{0}{1}\right)=\log 1+C \Rightarrow C=1$

$\therefore \cos\left(\frac{y}{x}\right)=\log x+1$

is the required particular solution.

www.exambites.in

58. We have, $x\frac{dy}{dx}+y=x\cos x+\sin x$

$\Rightarrow \quad \frac{dy}{dx}+\frac{1}{x}.y=\frac{x\cos x+\sin x}{x}$

This is a linear differential equation of the form $\frac{dy}{dx}+Py=Q$, where $P=\frac{1}{x}$, $Q=\frac{x\cos x+\sin x}{x}$

$\therefore \quad$ I.F. $=e^{\int P\,dx}=e^{\int \frac{1}{x}dx}=e^{\log x}=x.$

$\therefore \quad y\cdot x=\int \frac{x\cos x+\sin x}{x}\cdot x\,dx+C$

$\Rightarrow \quad xy=\int x\cos x\,dx+\int \sin x\;dx+C$

$=x\cdot\sin x-\int 1\cdot\sin x\,dx+\int \sin x dx+C$

$=x\sin x+C$

Given $y\left(\frac{\pi}{2}\right)=1$

$\therefore \quad \frac{\pi}{2}.1=\frac{\pi}{2}\sin\frac{\pi}{2}+C\Rightarrow C=0$

$\therefore \quad xy=x\sin x$

$\Rightarrow \quad y=\sin x$ is the required solution.

59. We have, $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x;x\neq 0$

$\Rightarrow \cos\left(\frac{y}{x}\right).\frac{dy}{dx}=\left(\frac{y}{x}\right)\cos\left(\frac{y}{x}\right)+1$ ...(i)

This is a linear homogeneous differential equation

Put $y=vx \quad\Rightarrow\quad \frac{dy}{dx}=v\cdot 1+x\cdot\frac{dv}{dx}$

Now (i) becomes

$\cos v.\left[v+x\frac{dv}{dx}\right]=v\cos v+1$

$\Rightarrow \quad x\cos v\frac{dv}{dx}=1 \quad\Rightarrow\quad \cos v\,dv=\frac{dx}{x}$

Integrating both sides, we get

$\sin v=\log x+C\Rightarrow \sin\left(\frac{y}{x}\right)=\log x+C$

is the required solution.

60. We have, $xy\frac{dy}{dx}=(x+2)(y+2)$

$\Rightarrow \quad \frac{y\,dy}{(y+2)}=\left(\frac{x+2}{x}\right)dx$

$\Rightarrow \quad dy-\frac{2}{(y+2)}dy=dx+\frac{2}{x}dx$

Integrating both sides, we get

$y-2\log(y+2)=x+2\log x+C$

when $x=1, y=-1$

So, $-1-2\log(-1+2)=1+2\log 1+C$

$\Rightarrow \quad C=-1-1=-2$

So, we have $y-2\log(y+2)=x+2\log x-2$

$\Rightarrow \quad y-x+2=2\log x(y+2).$

61. We have, $2x^2\frac{dy}{dx}-2xy+y^2=0$

$\Rightarrow \quad \frac{dy}{dx}-\frac{y}{x}+\frac{y^2}{2x^2}=0$ ...(i)

This is a homogeneous linear differential equation

Put $y=vx\Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$

$\therefore$ (i) becomes

$v+x\frac{dv}{dx}-v+\frac{v^2}{2}=0 \;\Rightarrow\; x\frac{dv}{dx}=\frac{-v^2}{2}$

$\Rightarrow \frac{dv}{-v^2}=\frac{1}{2}\frac{dx}{x}$

Integrating both sides, we get

$\frac{1}{v}=\frac{1}{2}\log x+C_1\Rightarrow \frac{x}{y}=\frac{1}{2}\log x+C_1$

$\Rightarrow \frac{2x}{y}=\log x+C.$ [where $C=2C_1$]

62. *Refer to answer 33.*

63. We have, $(x+1)\frac{dy}{dx}=2e^{-y}-1$ ...(i)

$\Rightarrow \quad \frac{dy}{2e^{-y}-1}=\frac{dx}{x+1}\Rightarrow \frac{e^y}{2-e^y}dy=\frac{dx}{x+1}$

Integrating both sides, we get

$-\log(2-e^y)=\log(x+1)+C$ ...(ii)

(Taking $2-e^y=t$ on L.H.S.)

when $x=0, y=0$

$\therefore \quad -\log(2-1)=\log(0+1)+C\Rightarrow C=0$

$\therefore \quad$ Eq. (ii) becomes

$-\log(2-e^y)=\log(x+1)$

$\Rightarrow \quad \log(x+1)+\log(2-e^y)=0$

$\Rightarrow \quad \log[(x+1)(2-e^y)]=0$

$\Rightarrow \quad (x+1)(2-e^y)=1$

is the required particular solution.

www.examrites.in

64. We have, $x(x^2-1)\dfrac{dy}{dx}=1$ ...(i)

$$dy=\frac{1}{x(x^2-1)}dx$$

Integrating both sides, we get

$$\int dy=\int\frac{1}{x(x-1)(x+1)}dx+C$$

$$\Rightarrow y=\int\left[-\frac{1}{x}+\frac{1}{2}.\frac{1}{x-1}+\frac{1}{2}.\frac{1}{x+1}\right]dx+C$$

$$=-\log x+\frac{1}{2}\log(x-1)+\frac{1}{2}\log(x+1)+C \quad ...(ii)$$

when $x=2, y=0$

$$\therefore 0=-\log 2+\frac{1}{2}\log 1+\frac{1}{2}\log 3+C$$

$$\Rightarrow C=\log 2-\frac{1}{2}\log 3=\frac{1}{2}\log\frac{4}{3}$$

$\therefore$ From (ii),

$$y=-\log x+\frac{1}{2}\log(x-1)+\frac{1}{2}\log(x+1)+\frac{1}{2}\log\frac{4}{3}$$

$$=\frac{1}{2}\log\frac{4(x^2-1)}{3x^2}$$ is the required particular solution

65. We have, $(1+x^2)\,dy+2xy\,dx=\cot x\,dx; x\neq 0$

$$\Rightarrow \frac{dy}{dx}+\frac{2x}{1+x^2}.y=\frac{\cot x}{1+x^2} \quad ...(i)$$

This is a linear differential equation of the form $\dfrac{dy}{dx}+Py=Q$, where $P=\dfrac{2x}{1+x^2}, Q=\dfrac{\cot x}{1+x^2}$

$$\therefore \text{ I.F.}=e^{\int P dx}=e^{\int\frac{2x}{1+x^2}dx}=e^{\log(1+x^2)}=1+x^2$$

$\therefore$ The solution of (i) is

$$y\cdot(1+x^2)=\int\frac{\cot x}{1+x^2}\cdot(1+x^2)dx+C$$

$$=\int\cot x\,dx+C=\log|\sin x|+C$$

$$\Rightarrow y=\frac{\log|\sin x|+C}{1+x^2}$$ is the solution of the given differential equation.

66. We have $\dfrac{dy}{dx}+y\cot x=4x\operatorname{cosec}x(x\neq 0)$ ...(i)

This is a linear differential equation of the form $\dfrac{dy}{dx}+Py=Q$,

where, $P=\cot x$, $Q=4x\operatorname{cosec}x$

$$\therefore \text{ I.F.}=e^{\int P dx}=e^{\int\cot x\,dx}=|\sin x|$$

$\therefore$ The solution of (i) is

$$y\cdot|\sin x|=\int 4x\operatorname{cosec}x\cdot|\sin x|dx=\int 4x\,dx$$

$$=2x^2+C \quad ...(ii)$$

when $x=\dfrac{\pi}{2}, y=0$

$\therefore$ From (ii), $0.\sin\dfrac{\pi}{2}=2\cdot\left(\dfrac{\pi}{2}\right)^2+C\Rightarrow C=-\dfrac{\pi^2}{2}$

$\Rightarrow$ $y\cdot\sin x=2x^2-\dfrac{\pi^2}{2}$ is the required particular solution.

67. We have $x\dfrac{dy}{dx}-y+x\sin\left(\dfrac{y}{x}\right)=0$

$$\Rightarrow \frac{dy}{dx}-\frac{y}{x}+\sin\left(\frac{y}{x}\right)=0$$

This is a linear homogeneous differential equation.

Put $y=vx \Rightarrow \dfrac{dy}{dx}=v\cdot 1+x\dfrac{dv}{dx}$

$$\therefore v+x\frac{dv}{dx}-v+\sin v=0$$

$$\Rightarrow x\frac{dv}{dx}+\sin v=0 \Rightarrow \operatorname{cosec}v\,dv+\frac{dx}{x}=0$$

Integrating both sides, we get

$\log|\operatorname{cosec}v-\cot v|+\log x=\log C$

$\Rightarrow$ $x(\operatorname{cosec}v-\cot v)=C$

$$\Rightarrow x\left[\operatorname{cosec}\left(\frac{y}{x}\right)-\cot\left(\frac{y}{x}\right)\right]=C$$

when $x=2, y=\pi$

$$\therefore 2\left[\operatorname{cosec}\frac{\pi}{2}-\cot\frac{\pi}{2}\right]=C\Rightarrow C=2$$

$$\Rightarrow x\left[\operatorname{cosec}\left(\frac{y}{x}\right)-\cot\left(\frac{y}{x}\right)\right]=2$$

is the required particular solution.

68. We have, $\dfrac{dy}{dx}-y=\cos x$ ... (i)

This is a linear differential equation of the form $\dfrac{dy}{dx}+Py=Q$, where $P=-1$, $Q=\cos x$

www.exambites.in

$\therefore$ I.F. $= e^{\int P dx} = e^{-x}$

Hence the solution of (i) is

$y.e^{-x} = \int e^{-x}.\cos x\,dx$

$\int e^{-x}\cos x\,dx = -e^{-x}\cos x + e^{-x}\sin x - \int e^{-x}\cos x\,dx$

$= \frac{1}{2}e^{-x}(\sin x - \cos x)$

Thus the required solution of (i) is

$y.e^{-x} = \frac{1}{2}e^{-x}(\sin x - \cos x) + C$ ...(ii)

Putting, $x = 0, y = 1$ in (ii), we get

$1 \cdot e^0 = \frac{1}{2}e^0(\sin 0 - \cos 0) + C$

$\Rightarrow 1 = \frac{1}{2}(-1) + C \Rightarrow C = \frac{3}{2}$

From (ii), $y = \frac{1}{2}(\sin x - \cos x) + \frac{3}{2}e^x$

is the required particular solution.

69. We have, $x\frac{dy}{dx} + 2y = x^2, (x \neq 0)$

$\Rightarrow \quad \frac{dy}{dx} + \frac{2}{x}.y = x$ ...(i)

This is linear differential equation of the form

$\frac{dy}{dx} + Py = Q$, where $P = \frac{2}{x}, Q = x$

$\therefore$ I.F. $= e^{\int P dx} = e^{\int (2/x)dx}$

$= e^{2\log x} = e^{\log x^2} = x^2$

$\therefore$ The solution of (i) is

$y \cdot x^2 = \int x \cdot x^2 dx = \int x^3\,dx = \frac{x^4}{4} + C$

$\Rightarrow y = \frac{x^2}{4} + C\,x^{-2}$ ... (ii)

When $x = 2, y = 1$ in (ii), we get

$\therefore \quad 1 = \frac{2^2}{4} + \frac{C}{2^2} = 1 + \frac{C}{4} \Rightarrow \frac{C}{4} = 0 \Rightarrow C = 0$

$y = \frac{x^2}{4}$ is the required particular solution.

70. We have, $\frac{dy}{dx} + y\cot x = 2x + x^2\cot x, (x \neq 0)$

This is a linear D.E of the form $\frac{dy}{dx} + Py = Q$

where $P = \cot x$, $Q = 2x + x^2\cot x$

$\therefore \quad$ I.F. $= e^{\int P dx} = e^{\int \cot x\,dx} = e^{\log|\sin x|} = \sin x$

$\Rightarrow \quad y \cdot \sin x = \int (2x + x^2 \cdot \cot x)\sin x\,dx$

$= 2\int x\sin x\,dx + \int x^2\cos x\,dx$

$= 2\int x\sin x\,dx + x^2\sin x - \int 2x\sin x\,dx + C$

$= x^2\sin x + C$

When $x = \frac{\pi}{2}, y = 0$

$\therefore \quad 0 \cdot \sin\frac{\pi}{2} = \left(\frac{\pi}{2}\right)^2 \cdot \sin\frac{\pi}{2} + C \Rightarrow 0 = \frac{\pi^2}{4} + C$

$\Rightarrow C = -\frac{\pi^2}{4} \Rightarrow y \cdot \sin x = x^2\sin x - \frac{\pi^2}{4}$

$\Rightarrow y = x^2 - \frac{\pi^2}{4}\operatorname{cosec} x$

71. We have, $\frac{dy}{dx} + \sec x.y = \tan x, \left(0 \leq x < \frac{\pi}{2}\right)$ ... (i)

This is a linear D.E. of the form $\frac{dy}{dx} + Py = Q$,

where $P = \sec x$, $Q = \tan x$

$\therefore$ I.F. $= e^{\int P dx} = e^{\int \sec x\,dx}$

$= e^{\log|\sec x + \tan x|} = |\sec x + \tan x|$

$\therefore \quad$ The solution of (i) is

$y \cdot |\sec x + \tan x|$

$= \int |\sec x + \tan x| \cdot \tan x\,dx$

$= \int |\sec x\,\tan x + \tan^2 x|\,dx$

$= \int |\sec x\,\tan x|\,dx + \int |\sec^2 x - 1|\,dx$

$= \sec x + \tan x - x + C$

72. We have, $x\frac{dy}{dx} + y - x + xy\cot x = 0, (x \neq 0)$

$\Rightarrow x\frac{dy}{dx} + (1 + x\cot x).y = x$

$\Rightarrow \frac{dy}{dx} + \frac{1 + x\cot x}{x}.y = 1$ ...(i)

This is linear D.E. of the form $\frac{dy}{dx} + Py = Q$

where $P = \frac{1 + x\cot x}{x} = \frac{1}{x} + \cot x, Q = 1$

$\therefore \quad$ Now I.F. $= e^{\int P dx} = e^{\log(x\sin x)} = x\sin x$

www.exambites.in

$\therefore$ The solution. of (i) is

$$y \cdot x \sin x = \int 1 \cdot x \sin x \, dx + C$$

$$= x(-\cos x) + \int 1 \cdot \cos x \, dx + C$$

$$\Rightarrow \quad xy \sin x = -x \cos x + \sin x + C$$

73. We have, $\cos^2 x \cdot \dfrac{dy}{dx} + y = \tan x$

Dividing by $\cos^2 x$ on both sides, we get

$$\frac{dy}{dx} + \frac{y}{\cos^2 x} = \frac{\tan x}{\cos^2 x}$$

This is a linear D.E. of the form $\dfrac{dy}{dx} + Py = Q$,

where $P = \dfrac{1}{\cos^2 x} = \sec^2 x, Q = \dfrac{\tan x}{\cos^2 x}$

$$\text{I.F.} = e^{\int P dx} = e^{\int \sec^2 x . dx} = e^{\tan x}$$

$$y . e^{\tan x} = \int e^{\tan x} \times \tan x \times \sec^2 x . dx$$

Put $\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$

$$y \times e^{\tan x} = \int e^t \times t \cdot dt$$

$$\Rightarrow \quad y e^{\tan x} = t \int e^t \, dt - \int \left[\frac{d}{dt}(t) \int e^t \, dt \right] \cdot dt$$

$$\Rightarrow \quad y \, e^{\tan x} = t \times e^t - e^t + C$$

$$\Rightarrow \quad y \, e^{\tan x} = \tan x \, e^{\tan x} - e^{\tan x} + C$$

$$\Rightarrow \quad y \, e^{\tan x} = e^{\tan x} [\tan x - 1] + C$$

$$\Rightarrow \quad y = (\tan x - 1) + Ce^{-\tan x}$$

74. We have, $(1 + y^2)(1 + \log x)dx + xdy = 0$

$$\Rightarrow \quad (1 + y^2)(1 + \log x)dx = -xdy$$

$$\Rightarrow \quad \left(\frac{1 + \log x}{x} \right) dx = \frac{-dy}{1 + y^2}$$

Integrating both sides, we get

$$\frac{(1 + \log x)^2}{2} = -\tan^{-1} y + C$$

$$\Rightarrow \quad \frac{(1 + \log x)^2}{2} + \tan^{-1} y = C$$

which is the required general solution.

75. *Refer to answer 35.*

76. We have, $xdy + (y - x^3)dx = 0$

$$\Rightarrow \quad \frac{dy}{dx} + \frac{y}{x} = x^2$$

This is a linear differential equation of the form

$\dfrac{dy}{dx} + Py = Q$, where $P = \dfrac{1}{x}$, $Q = x^2$

$\therefore$ I.F. $= e^{\int \frac{1}{x} dx} = e^{\log x} = x$.

$\therefore$ The solution of differential equation is

$$y \cdot x = \int x^2 \cdot x \, dx$$

$$\Rightarrow \quad yx = \frac{x^4}{4} + C \Rightarrow y = \frac{x^3}{4} + \frac{C}{x}.$$

77. We have, $xdy - (y + 2x^2)dx = 0$

$$\Rightarrow \quad \frac{dy}{dx} - \frac{y}{x} = 2x.$$

This is a linear differential equation of the form

$\dfrac{dy}{dx} + Py = Q$, where $P = \dfrac{-1}{x}$, $Q = 2x$

$\therefore$ I.F. $= e^{-\int \frac{1}{x} dx} = e^{-\log x} = \dfrac{1}{x}$

$\therefore$ The solution of differential equation is

$$y\left(\frac{1}{x}\right) = \int 2x \frac{1}{x} \, dx + C$$

$$\Rightarrow \quad \frac{y}{x} = 2x + C \Rightarrow y = 2x^2 + Cx$$

78. *Refer to answer 77.*

79. *Refer to answer 31.*

80. We have, $\left[x \sin^2 \left(\dfrac{y}{x} \right) - y\right] dx + x \, dy = 0$

$$\Rightarrow \sin^2 \left(\frac{y}{x} \right) - \frac{y}{x} + \frac{dy}{dx} = 0 \qquad \text{... (i)}$$

This is a linear homogeneous differential equation

$\therefore$ Put $y = vx \Rightarrow \dfrac{dy}{dx} = v + x \dfrac{dv}{dx}$

$\therefore$ (i) becomes

$$\sin^2 v - v + v + x \frac{dv}{dx} = 0$$

$$\Rightarrow x \frac{dv}{dx} + \sin^2 v = 0 \Rightarrow \operatorname{cosec}^2 v dv + \frac{dx}{x} = 0$$

Integrating both sides, we get

$$-\cot v + \log x = C \Rightarrow -\cot \left(\frac{y}{x} \right) + \log x = C$$

is the required solution.

81. We have, $x \dfrac{dy}{dx} + y = x \cos x + \sin x$

$$\Rightarrow \quad \frac{dy}{dx} + \frac{y}{x} = \cos x + \frac{\sin x}{x}$$

It is a linear differential equation.

www.examrites.in

$$\text{I.F.} = e^{\int \frac{1}{x}dx} = e^{\log x} = x$$

$$\therefore \quad y \cdot x = \int x\left(\cos x + \frac{\sin x}{x}\right)dx + c$$

$$= \int [x\cos x + \sin x]dx + c$$

$$= x\sin x - \int \sin x dx + \int \sin x dx + c \; = x\sin x + c$$

$$\Rightarrow \quad y = \sin x + \frac{c}{x}$$

Given that, $y = 1$ when $x = \frac{\pi}{2}$

$$\therefore \quad 1 = 1 + \frac{c}{\pi/2} \quad \Rightarrow \quad c = 0$$

$\therefore$ $y = \sin x$ is the required solution.

82. We have, $(x - y)\frac{dy}{dx} = x + 2y$

$$\frac{dy}{dx} = \frac{x + 2y}{x - y} \qquad \ldots(i)$$

Put $y = Vx \quad \Rightarrow \quad \frac{dy}{dx} = V + x\frac{dV}{dx}$

Putting $\frac{dy}{dx} = V + x\frac{dv}{dx}$ in (i), we get

$$V + x\frac{dV}{dx} = \frac{x + 2Vx}{x - Vx} = \frac{1 + 2V}{1 - V}$$

$$\Rightarrow \quad x\frac{dV}{dx} = \frac{1 + 2V}{1 - V} - V = \frac{1 + 2V - V + V^2}{1 - V}$$

$$\Rightarrow \quad \int \frac{1 - V}{V^2 + V + 1}dV = \int \frac{1}{x}dx$$

$$\Rightarrow \quad \int \frac{2 - 2V}{V^2 + V + 1}dV = 2\log|x| + c$$

$$\Rightarrow \quad \int \frac{3 - (2V + 1)}{V^2 + V + 1}dV = 2\log|x| + c$$

$$\Rightarrow \quad \int \frac{3}{V^2 + V + 1}dV - \int \frac{2V + 1}{V^2 + V + 1}dV = \log|x|^2 + c$$

$$\Rightarrow \quad 3\int \frac{1}{\left(V + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}dV - \log|V^2 + V + 1| = \log|x^2| + c$$

$$\Rightarrow \quad \frac{3}{\left(\frac{\sqrt{3}}{2}\right)}\tan^{-1}\left(\frac{V + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) = \log|x^2(V^2 + V + 1)| + c$$

$$\Rightarrow \quad 2\sqrt{3}\tan^{-1}\left(\frac{2V + 1}{\sqrt{3}}\right) = \log|x^2(V^2 + V + 1)| + c \qquad \ldots(ii)$$

Substituting $V = \frac{y}{x}$ in (ii), we get

$$2\sqrt{3}\tan^{-1}\left(\frac{2y + x}{\sqrt{3}x}\right) = \log\left|x^2\frac{(y^2 + yx + x^2)}{x^2}\right| + c \qquad \ldots(iii)$$

Now, at $y = 0$ and $x = 1$, we have

$$2\sqrt{3}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \log|1| + c$$

$$\Rightarrow \quad c = 2\sqrt{3}\cdot\frac{\pi}{6} = \frac{\pi}{\sqrt{3}}$$

Substituting $c = \frac{\pi}{\sqrt{3}}$ in (iii), we get

$$2\sqrt{3}\tan^{-1}\left(\frac{2y + x}{\sqrt{3}x}\right) = \log|y^2 + xy + x^2| + \frac{\pi}{\sqrt{3}}$$

$$\Rightarrow \quad 6\tan^{-1}\left(\frac{2y + x}{\sqrt{3}x}\right) = \sqrt{3}\log(x^2 + xy + y^2) + \pi$$

83. We have, $(\tan^{-1}y - x)dy = (1 + y^2)dx$

$$\Rightarrow \quad \frac{dx}{dy} = \frac{\tan^{-1}y - x}{1 + y^2}$$

$$\Rightarrow \quad \frac{dx}{dy} + \frac{1}{1 + y^2}\cdot x = \frac{\tan^{-1}y}{1 + y^2}$$

This is a linear differential equation of the form $\frac{dx}{dy} + Px = Q$, where $P = \frac{1}{1 + y^2}$ and $Q = \frac{\tan^{-1}y}{1 + y^2}$

Refer to answer 38.

84. We have, $\frac{dy}{dx} = \frac{y^2}{xy - x^2} = \frac{y^2/x^2}{(xy - x^2)/x^2}$ ... (i)

$\Rightarrow$ It is a homogeneous differential equation

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v\cdot 1 + x\frac{dv}{dx}$

$\therefore$ Equation (i) becomes

$$v + x\frac{dv}{dx} = \frac{v^2}{v - 1} \quad \Rightarrow x\frac{dv}{dx} = \frac{v^2}{v - 1} - v = \frac{v}{v - 1}$$

$$\Rightarrow \frac{v - 1}{v}dv = \frac{dx}{x} \quad \Rightarrow \left(1 - \frac{1}{v}\right)dv = \frac{dx}{x}$$

www.exambites.in

Integrating, we get

$v - \log v = \log x + C \quad \Rightarrow \quad v = \log vx + C$

$\Rightarrow \quad \frac{y}{x} = \log y + C$

$\Rightarrow \quad y = x(\log y + C)$ is the required solution.

85. *Refer to answer 83.*

We get the solution as

$x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$ ...(i)

Now, putting $x = 1, y = 0$ in (i), we get

$1 = \tan^{-1} 0 - 1 + Ce^{-\tan^{-1} 0} \Rightarrow C = 2$

So, required particular solution is

$x = \tan^{-1} y - 1 + 2e^{-\tan^{-1} y}$.

86. We have,

$$\left[y - x\cos\left(\frac{y}{x}\right)\right]dy + \left[y\cos\left(\frac{y}{x}\right) - 2x\sin\left(\frac{y}{x}\right)\right]dx = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x\sin\left(\frac{y}{x}\right) - y\cos\left(\frac{y}{x}\right)}{y - x\cos\left(\frac{y}{x}\right)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2\sin\left(\frac{y}{x}\right) - \frac{y}{x}\cos\left(\frac{y}{x}\right)}{\frac{y}{x} - \cos\left(\frac{y}{x}\right)} \quad ...(i)$$

This is a homogeneous differential equation.

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ Equation (i) becomes,

$$v + x\frac{dv}{dx} = \frac{2\sin v - v\cos v}{v - \cos v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{2\sin v - v\cos v}{v - \cos v} - v$$

$$\Rightarrow x\frac{dv}{dx} = \frac{2\sin v - v^2}{v - \cos v} \Rightarrow \frac{v - \cos v}{2\sin v - v^2}dv = \frac{dx}{x}$$

$$\Rightarrow \frac{-1}{2}\frac{(2\cos v - 2v)}{2\sin v - v^2}dv = \frac{dx}{x}$$

Integrating both sides, we get

$$\frac{-1}{2}\log(2\sin v - v^2) = \log x + C_1$$

$$\Rightarrow \log x^2 + 2C_1 + \log\left(2\sin\frac{y}{x} - \frac{y^2}{x^2}\right) = 0$$

$$\Rightarrow \log\left[x^2\left(2\sin\frac{y}{x} - \frac{y^2}{x^2}\right)\right] = -2C_1$$

$$\Rightarrow 2x^2\sin\frac{y}{x} - y^2 = e^{-2C_1} = C \text{ (say)}$$

which is the required solution.

87. We have, $\sqrt{1 + x^2 + y^2 + x^2y^2} + xy\frac{dy}{dx} = 0$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-\sqrt{(1+x^2)(1+y^2)}}{xy}$$

$$\Rightarrow \quad \int \frac{y}{\sqrt{1+y^2}}dy = -\int \frac{\sqrt{1+x^2}}{x^2}xdx$$

$$\Rightarrow \quad \frac{1}{2}\int \frac{2y}{\sqrt{1+y^2}}dy = -\int \frac{v^2}{v^2-1}dv$$

[putting $1 + x^2 = v^2 \Rightarrow 2xdx = 2vdv$]

$$\Rightarrow \quad \sqrt{1+y^2} = -\int\left(1 + \frac{1}{v^2-1}\right)dv$$

$$\Rightarrow \quad \sqrt{1+y^2} = -v - \frac{1}{2}\log\left|\frac{v-1}{v+1}\right| + C$$

$$\Rightarrow \quad \sqrt{1+y^2} + \sqrt{1+x^2} + \frac{1}{2}\log\left|\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right| = C$$

88. *Refer to answer 72.*

The required solution

$y \cdot x \sin x = x(-\cos x) + \sin x + C$ ...(i)

Putting $x = \frac{\pi}{2}, y = 0$ in (i), we get

$\therefore \quad 0 = -\frac{\pi}{2}\cos\frac{\pi}{2} + \sin\frac{\pi}{2} + C \Rightarrow C = -1$

$xy \sin x = \sin x - x\cos x - 1$ is the required particular solution.

89. We have, $x^2\,dy + (xy + y^2)\,dx = 0$

$$\Rightarrow \frac{dy}{dx} = -\frac{xy + y^2}{x^2} \quad ...(i)$$

This is a homogeneous linear differential equation

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v \cdot 1 + x \cdot \frac{dv}{dx}$

$\therefore$ (i) becomes $v + x\frac{dv}{dx} = -\frac{x \cdot vx + v^2x^2}{x^2}$

www.exambites.in

$$\Rightarrow \quad x\frac{dv}{dx} = -(2v + v^2)$$

Separating the variables, we get

$$\frac{dv}{2v+v^2} + \frac{dx}{x} = 0 \quad \Rightarrow \quad \frac{dv}{v(v+2)} + \frac{dx}{x} = 0$$

$$\Rightarrow \quad \frac{1}{2}\left[\frac{1}{v} - \frac{1}{v+2}\right]dv + \frac{dx}{x} = 0$$

Integrating, we get

$$\frac{1}{2}\left[\log v - \log(v+2)\right] + \log x = \log C$$

$$\Rightarrow \quad \log\left(\frac{v}{v+2}\right) + 2\log x = \log C$$

$$\Rightarrow \quad \log\left(\frac{v}{v+2}\right) + \log x^2 = \log C$$

$$\Rightarrow \quad \log\left(\frac{vx^2}{v+2}\right) = \log C \quad \Rightarrow \quad \frac{vx^2}{v+2} = C$$

$$\Rightarrow \quad \frac{\frac{y}{x}.x^2}{\frac{y}{x}+2} = C \quad \Rightarrow x^2y = C\,(2x + y) \qquad \text{...(ii)}$$

Putting $x = 1$, $y = 1$ in (ii), we get

$$1^2.1 = C(2 \cdot 1 + 1) \Rightarrow C = \frac{1}{3}$$

$\therefore$ The required particular solution is

$$3x^2y = 2x + y \Leftrightarrow y = \frac{2x}{3x^2 - 1}$$

90. We have, $\left(x\sin^2\left(\frac{y}{x}\right) - y\right)dx + x\,dy = 0$

$$\Rightarrow \quad \frac{dy}{dx} + \sin^2\left(\frac{y}{x}\right) - \frac{y}{x} = 0 \qquad \text{...(i)}$$

Put $y = vx \Rightarrow \frac{dy}{dx} = v \cdot 1 + x\frac{dv}{dx}$ $\therefore$ (i) becomes

$$v + x\frac{dv}{dx} + \sin^2 v - v = 0$$

$$\Rightarrow \quad x\frac{dv}{dx} + \sin^2 v = 0 \quad \Rightarrow \quad \text{cosec}^2 v\,dv + \frac{dx}{x} = 0$$

Integrating both sides, we get

$$\int \text{cosec}^2 v\,dv + \int \frac{dx}{x} = C \quad \Rightarrow -\cot v + \log x = C$$

$$\Rightarrow \quad -\cot\left(\frac{y}{x}\right) + \log x = C \qquad \text{...(ii)}$$

Put $x = 1$, $y = \pi/4$ in (ii), we get

$$-\cot\frac{\pi}{4} + \log 1 = C \quad \Rightarrow \quad C = -1$$

$\therefore$ $-\cot\left(\frac{y}{x}\right) + \log x + 1 = 0$ is the required particular solution.

91. We have, $\frac{dy}{dx} - 3y\cot x = \sin 2x$ ...(i)

This is a linear differential equation of the form

$$\frac{dy}{dx} + Py = Q$$

where $P = -3\cot x$, $Q = \sin 2x$

$$\therefore \quad \text{I.F.} = e^{\int P dx} = e^{-3\int \cot x dx}$$

$$= e^{-3\log|\sin x|} = |\sin^{-3} x|$$

$$\therefore \quad y \cdot \sin^{-3} x = \int \sin 2x \cdot \sin^{-3} x\,dx + C$$

$$\Rightarrow \quad \frac{y}{\sin^3 x} = \int \frac{2\sin x \cos x}{\sin^3 x}dx + C$$

$$= \int \frac{2\cos x}{\sin^2 x}dx + C \quad (\text{Put } \sin x = t \Rightarrow \cos x\,dx = dt)$$

$$= 2\int \frac{dt}{t^2} + C = -\frac{2}{t} + C = -\frac{2}{\sin x} + C$$

$$\Rightarrow \quad y = -2\sin^2 x + C\sin^3 x \qquad \text{...(ii)}$$

Put $x = \frac{\pi}{2}, y = 2$ in (ii), we get

$$2 = -2 \cdot 1 + C \cdot 1 \Rightarrow C = 4$$

$y = 4\sin^3 x - 2\sin^2 x$ is the required particular solution.

92. We have, $x\frac{dy}{dx}\sin\left(\frac{y}{x}\right) + x - y\sin\left(\frac{y}{x}\right) = 0$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y\sin\left(\frac{y}{x}\right) - x}{x\sin\left(\frac{y}{x}\right)} = \frac{\frac{y}{x}\sin\left(\frac{y}{x}\right) - 1}{\sin\left(\frac{y}{x}\right)} \qquad \text{... (i)}$$

This is a linear homogeneous differential equation

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ $\therefore$ (i) becomes

$$v + x\frac{dv}{dx} = \frac{v\sin v - 1}{\sin v} \quad \Rightarrow \quad x\frac{dv}{dx} = \frac{v\sin v - 1}{\sin v} - v$$

$$\Rightarrow \quad x\frac{dv}{dx} = -\frac{1}{\sin v} \Rightarrow \sin v\,dv = -\frac{dx}{x}$$

www.exambites.in

Integrating both sides, we get

$-\cos v = -\log x + C$

$\Rightarrow \quad -\cos\left(\frac{y}{x}\right) + \log x = C$ ...(ii)

Put $y = \frac{\pi}{2}$, $x = 1$ in (ii), we get

$-\cos\left(\frac{\pi}{2}\right) + \log 1 = C \Rightarrow C = 0$

$\Rightarrow \log x = \cos\left(\frac{y}{x}\right)$ is the required solution.

93. We have, $2ye^{x/y}\,dx + (y - 2x\,e^{x/y})dy = 0$

$\Rightarrow \frac{dx}{dy} = \frac{2xe^{x/y} - y}{2ye^{x/y}} = \frac{x}{y} - \frac{1}{2e^{x/y}}.$ ...(i)

This is homogenous linear differential equation.

$\therefore$ Put $\frac{x}{y} = t \Rightarrow x = yt \Rightarrow \frac{dx}{dy} = 1 \cdot t + y \cdot \frac{dt}{dy}$

$\therefore$ (i) becomes

$t + y\,\frac{dt}{dy} = t - \frac{1}{2e^t} \quad \Rightarrow 2\,e^t\,dt + \frac{dy}{y} = 0$

Integrating both sides, we get

$2\,e^t + \log y = C \quad \Rightarrow \; 2\,e^{x/y} + \log y = C$ ...(ii)

Putting $x = 0$, $y = 1$ in (ii) we get $C = 2$

$2\,e^{x/y} + \log y = 2$, is the required particular solution.

94. We have, $(x\,e^{x/y} + y)\,dx = x\,dy$

$\Rightarrow \frac{dx}{dy} = \frac{x}{xe^{x/y} + y} = \frac{x/y}{\frac{x}{y}e^{x/y} + 1}$ ...(i)

This is a homogeneous linear differential equation

$\therefore$ Put $x = vy \Rightarrow \frac{dx}{dy} = v + y \cdot \frac{dv}{dy}$

$\therefore$ (i) becomes

$v + y\,\frac{dv}{dy} = \frac{v}{ve^v + 1} \Rightarrow y\frac{dv}{dy} = \frac{v}{ve^v + 1} - v = \frac{-v^2e^v}{ve^v + 1}$

$\Rightarrow \frac{ve^v + 1}{v^2e^v}dv = -\frac{dy}{y} \Rightarrow \left(\frac{1}{v} + \frac{1}{v^2e^v}\right)dv = -\frac{dy}{y}$

Integrating both sides, we get

$\log v + \int \frac{1}{v^2e^v}dv = -\log y + C$

$\Rightarrow \log vy + \int \frac{dv}{v^2e^v} = C \Rightarrow \log x + \int \frac{dv}{v^2e^v} = C$

which can't be integrated further.

95. *Refer to answer 83.*

We have, $x = \tan^{-1} y - 1 + C\,e^{-\tan^{-1}y}$ ...(i)

Putting $x = 0$, $y = 0$ in (i), we get

$0 = 0 - 1 + Ce^0 \Rightarrow C = 1$

$x = \tan^{-1} y - 1 + e^{-\tan^{-1} y}$, is the required particular solution.

96. We have, $\frac{dx}{dy} + x\cot y = 2y + y^2\cot y, (y \neq 0)$ ...(i)

This is a linear differential equation of the form

$\frac{dx}{dy} + Py = Q$, where $P = \cot y$, $Q = 2y + y^2 \cot y$

I.F. $= e^{\int P dy} = e^{\int \cot y\, dy} = e^{\log \sin y} = \sin y.$

$\therefore$ The required solution is

$x \cdot \sin y = \int (2y + y^2 \cot y)\sin y\, dy$

$= \int 2y \sin y\, dy + \int y^2 \cos y\, dy$

$= \int 2y \sin y dy + y^2 \sin y - \int 2y \sin y dy + C$

$= y^2 \sin y + C$ ...(ii)

Putting $y = \frac{\pi}{2}, x = 0$ in (ii), we get

$0 = \frac{\pi^2}{4}\sin\frac{\pi}{2} + C \Rightarrow C = -\frac{\pi^2}{4}$

Hence the required particular solution of (i) is

$x \sin y = y^2 \sin y - \frac{\pi^2}{4}.$

97. We have, $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$

$\Rightarrow \frac{dy}{dx} = -\frac{3xy + y^2}{x^2 + xy}$ ...(i)

This is a homogeneous linear differential equation

Put $y = vx \Rightarrow \frac{dy}{dx} = v \cdot 1 + x \cdot \frac{dv}{dx}$

$\therefore$ (i) becomes

$v + x\frac{dv}{dx} = -\frac{3x \cdot vx + v^2x^2}{x^2 + x \cdot vx} = -\frac{3v + v^2}{1 + v}$

$\Rightarrow x\frac{dv}{dx} = -v - \frac{3v + v^2}{1 + v} = \frac{-2v^2 - 4v}{v + 1}$

$= \frac{v + 1}{2v^2 + 4v}dv + \frac{dx}{x} = 0$

Integrating both sides, we get

$\frac{1}{4}\log|2v^2 + 4v| + \log|x| = \log C'$

www.exambites.in

$$\Rightarrow \left(2v^2+4v\right)^{\frac{1}{4}} x = C' \Rightarrow \left(\frac{2y^2}{x^2}+\frac{4y}{x}\right)^{\frac{1}{4}} x = C'$$

$$\Rightarrow \left(2x^2y^2+4x^3y\right)^{\frac{1}{4}} = C'$$

$\Rightarrow 2x^2y^2 + 4x^3y = C$ [where $C = (C')^4$] ...(ii)

Put $x = 1, y = 1$ in (ii), we get $C = 6$

Hence $2x^2y^2 + 4x^3y = 6 \Rightarrow x^2y^2 + 2x^3y = 3$

is the required particular solution.

98. We have, $(x^2 + xy)\, dy = (x^2 + y^2)\, dx$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2+y^2}{x^2+xy} \quad \text{... (i)}$$

This is a homogeneous linear differential equation

$\therefore$ Put $y = vx \Rightarrow \frac{dy}{dx} = v \cdot 1 + x\frac{dv}{dx}$

$\therefore$ (i) becomes

$$v + x\frac{dv}{dx} = \frac{x^2+v^2x^2}{x^2+x\cdot vx} = \frac{1+v^2}{1+v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{1+v^2}{1+v} - v = \frac{1-v}{1+v} \Rightarrow \frac{1+v}{1-v}dv = \frac{dx}{x}$$

Integrating both sides, we get

$$\Rightarrow \int\left[-1+\frac{2}{1-v}\right]dv = \log x + C$$

$\Rightarrow -v - 2\log(1-v) = \log x + C$

$$\Rightarrow \frac{-y}{x} - 2\log\left(1-\frac{y}{x}\right) = \log x + C \quad \text{...(ii)}$$

Putting $x = 1, y = 0$ in (ii), we get $C = 0$

$$\Rightarrow \frac{-y}{x} - 2\left[\log(x-y) - \log x\right] = \log x$$

$$\Rightarrow \frac{-y}{x} - 2\log(x-y) + \log x = 0$$

is the required particular solution.

99. *Refer to answer 56.*

The general solution is

$$\log x + C = \frac{-1}{2}\log\left(\frac{y^2}{x^2}+\frac{y}{x}+1\right) + \sqrt{3}\tan^{-1}\left(\frac{2y}{x}+1\right)/\sqrt{3} \quad \text{...(i)}$$

Putting $x = 1, y = 0$ in (i), we get

$$0 + C = -\frac{1}{2}\log(0+0+1) + \sqrt{3}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \Rightarrow C = \frac{\pi}{2\sqrt{3}}$$

$$\therefore \log x + \frac{\pi}{2\sqrt{3}} = -\frac{1}{2}[\log(y^2+xy+x^2) - \log x^2] + \sqrt{3}\tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right)$$

$$\Rightarrow \frac{\pi}{2\sqrt{3}} = -\frac{1}{2}\log(x^2+xy+y^2) + \sqrt{3}\tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right)$$

100. *Refer to answer 59.*

The general solution is $\sin\frac{y}{x} = \log x + C$ ...(i)

Putting $x = 1, y = \frac{\pi}{4}$ in (i), we get

$$\sin\frac{\pi}{4} = \log 1 + C \Rightarrow C = \frac{1}{\sqrt{2}}$$

$\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{\sqrt{2}}$, is the required particular solution.

101. We have,

$$x\cos\left(\frac{y}{x}\right)(ydx+xdy) = y\sin\left(\frac{y}{x}\right)(xdy-ydx)$$

$$\Rightarrow \cos\left(\frac{y}{x}\right)\left(y+x\frac{dy}{dx}\right) = \frac{y}{x}\sin\left(\frac{y}{x}\right)\left(x\frac{dy}{dx}-y\right) \quad \text{...(i)}$$

Putting $y = vx \Rightarrow \frac{dy}{dx} = v\cdot 1 + x\frac{dv}{dx}$

$\therefore$ (i) becomes

$$\cos v\left[vx + x\left(v + x\frac{dv}{dx}\right)\right] = v\sin v\left[x\left(v+x\frac{dv}{dx}\right) - vx\right]$$

$$\Rightarrow \cos v\left(2v + x\frac{dv}{dx}\right) = v\sin v\cdot x\frac{dv}{dx}$$

$$\Rightarrow 2v\cos v = x(v\sin v - \cos v)\frac{dv}{dx}$$

$$\Rightarrow \frac{v\sin v - \cos v}{v\cos v}dv = 2\frac{dx}{x}$$

$$\Rightarrow \left(\tan v - \frac{1}{v}\right)dv = 2\frac{dx}{x}$$

Integrating both sides, we get

$\log|\sec v| - \log v = 2\log x + \log C$

$$\Rightarrow \log\frac{|\sec v|}{v} = \log Cx^2$$

$$\Rightarrow |\sec v| = Cvx^2$$

$$\Rightarrow \sec\left(\frac{y}{x}\right) = Cxy$$

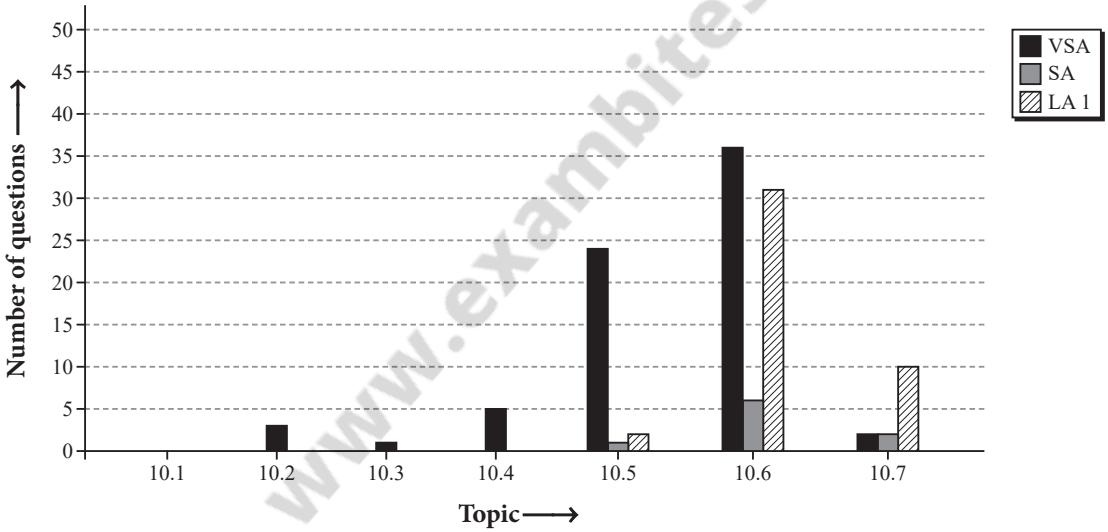
www.examfiles.in

CHAPTER
10

Vector Algebra

10.1 Introduction
10.2 Some Basic Concepts
10.3 Types of Vectors
10.4 Addition of Vectors
10.5 Multiplication of a Vector by a Scalar
10.6 Product of Two Vectors
10.7 Scalar Triple Product

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

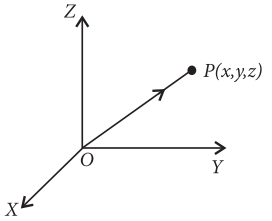
- Maximum weightage is of *Product of Two Vectors*
- Maximum VSA and LA 1 type questions were asked from *Product of Two Vectors*
- No LA 2 type questions were asked till now

QUICK RECAP

VECTOR

- A physical quantity having magnitude as well as direction is called a vector. A vector is represented by a line segment, denoted as $\overrightarrow{AB}$ or $\vec{a}$. Here, point A is the initial point and B is the terminal point of the vector $\overrightarrow{AB}$.
- **Magnitude :** The distance between the points A and B is called the magnitude of the directed line segment $\overrightarrow{AB}$. It is denoted by $|\overrightarrow{AB}|$.
- **Position Vector :** Let P be any point in space, having coordinates (x, y, z) with respect to some fixed point O (0, 0, 0) as origin, then

the vector $\overrightarrow{OP}$ having O as its initial point and P as its terminal point is called the position vector of the point P with respect to O. The vector $\overrightarrow{OP}$ is usually denoted by $\vec{r}$.



Magnitude of $\overrightarrow{OP}$ is, $|\overrightarrow{OP}| = \sqrt{x^2 + y^2 + z^2}$

i.e., $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.

In general, the position vectors of points A, B, C, etc. with respect to the origin O are denoted by $\vec{a}, \vec{b}, \vec{c}$, etc. respectively.

Direction Cosines and Direction Ratios : The angles α, β, γ made by the vector $\vec{r}$ with the positive directions of x, y and z-axes respectively are called its direction angles. The cosine values of these angles, *i.e.*, $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are called direction cosines of the vector $\vec{r}$, and usually denoted by l, m and n respectively.

Direction cosines of $\vec{r}$ are given as

$$l = \frac{x}{\sqrt{x^2 + y^2 + z^2}}, m = \frac{y}{\sqrt{x^2 + y^2 + z^2}},$$

$$n = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

The numbers lr, mr and nr, proportional to the direction cosines of vector $\vec{r}$ are called direction ratios of the vector $\vec{r}$ and denoted as a, b and c respectively.

i.e., $a = lr$, $b = mr$ and $c = nr$

Note : $l^2 + m^2 + n^2 = 1$ and $a^2 + b^2 + c^2 \neq 1$, (in general).

TYPES OF VECTORS

Zero vector : A vector whose initial and terminal points coincide is called a zero (or null) vector. It cannot be assigned a definite direction as it has zero magnitude and it is denoted by the $\vec{0}$.

Unit Vector : A vector whose magnitude is unity *i.e.*, $|\vec{a}| = 1$. It is denoted by $\hat{a}$.

Equal Vectors : Two vectors $\vec{a}$ and $\vec{b}$ are said to be equal, written as $\vec{a} = \vec{b}$, iff they have equal magnitudes and direction regardless of the positions of their initial points.

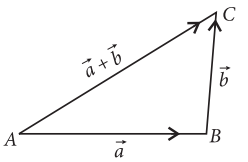
Coinitial Vectors : Vectors having same initial point are called co-initial vectors.

Collinear Vectors : Two or more vectors are called collinear if they have same or parallel supports, irrespective of their magnitudes and directions.

Negative of a Vector : A vector having the same magnitude as that of a given vector but directed in the opposite sense is called negative of the given vector *i.e.*, $\overrightarrow{BA} = -\overrightarrow{AB}$.

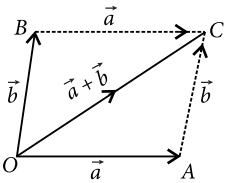
ADDITION OF VECTORS

Triangle law : Let the vectors be $\vec{a}$ and $\vec{b}$ so positioned such that initial point of one coincides with terminal point of the other. If $\vec{a} = \overrightarrow{AB}$, $\vec{b} = \overrightarrow{BC}$. Then the vector $\vec{a} + \vec{b}$ is represented by the third side of ΔABC



i.e., $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$

Parallelogram law : If the two vectors $\vec{a}$ and $\vec{b}$ are represented by the two adjacent sides OA and OB of a parallelogram $OACB$, then their sum $\vec{a} + \vec{b}$ is represented in magnitude and direction by the diagonal OC of parallelogram $OACB$ through their common point O *i.e.*, $\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$



Properties of Vector Addition

- Vector addition is commutative *i.e.*, $\vec{a} + \vec{b} = \vec{b} + \vec{a}$.
- Vector addition is associative *i.e.*, $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$.
- Existence of additive identity : The zero vector acts as additive identity *i.e.*, $\vec{a} + \vec{0} = \vec{a} = \vec{0} + \vec{a}$ for any vector $\vec{a}$.
- Existence of additive inverse : The negative of $\vec{a}$ *i.e.*, $-\vec{a}$ acts as additive inverse *i.e.*, $\vec{a} + (-\vec{a}) = \vec{0} = (-\vec{a}) + \vec{a}$ for any vector $\vec{a}$.

www.examples.in

MULTIPLICATION OF A VECTOR BY A SCALAR

Let $\vec{a}$ be a given vector and λ be a given scalar (a real number), then $\lambda\vec{a}$ is defined as the multiplication of vector $\vec{a}$ by the scalar λ. Its magnitude is $|\lambda|$ times the modulus of $\vec{a}$ *i.e.*, $|\lambda\vec{a}| = |\lambda|\,|\vec{a}|$.

Direction of $\lambda\vec{a}$ is same as that of $\vec{a}$ if $\lambda > 0$ and opposite to that of $\vec{a}$ if $\lambda < 0$.

Note : If $\lambda = \frac{1}{|\vec{a}|}$, provided that $\vec{a} \neq 0$, then $\lambda\vec{a}$ represents the unit vector in the direction of $\vec{a}$ *i.e.* $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$

COMPONENTS OF A VECTOR

Let O be the origin and $P(x, y, z)$ be any point in space. Let $\hat{i}, \hat{j}, \hat{k}$ be unit vectors along the X-axis, Y-axis and Z-axis respectively. Then $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$, which is called the component form of $\overrightarrow{OP}$. Here x, y and z are scalar components of $\overrightarrow{OP}$ and $x\hat{i}, y\hat{j}, z\hat{k}$ are vector components of $\overrightarrow{OP}$.

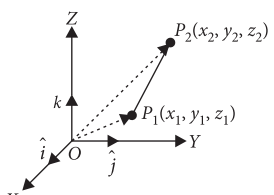
If $\vec{a}$ and $\vec{b}$ are two given vectors as $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and λ be any scalar, then

- $\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$
- $\vec{a} - \vec{b} = (a_1 - b_1)\hat{i} + (a_2 - b_2)\hat{j} + (a_3 - b_3)\hat{k}$
- $\lambda\vec{a} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$
- $\vec{a} = \vec{b} \Leftrightarrow a_1 = b_1, a_2 = b_2$ and $a_3 = b_3$
- $\vec{a}$ and $\vec{b}$ are collinear iff

$$\frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3} = \lambda.$$

VECTOR JOINING TWO POINTS

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are any two points in the space then the vector joining P_1 and P_2 is the vector $\overrightarrow{P_1P_2}$.



Applying triangle law in ΔOP_1P_2, we get

$\overrightarrow{OP_1} + \overrightarrow{P_1P_2} = \overrightarrow{OP_2}$

$\Rightarrow \quad \overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$

$= (x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$

$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

$\therefore |\overrightarrow{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

SECTION FORMULA

Let A, B be two points such that $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$.

- The position vector $\vec{r}$ of the point P which divides the line segment AB internally in the ratio $m : n$ is given by $\vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n}$.
- The position vector $\vec{r}$ of the point P which divides the line segment AB externally in the ratio $m : n$ is given by $\vec{r} = \frac{m\vec{b} - n\vec{a}}{m - n}$.
- The position vector $\vec{r}$ of the mid-point of the line segment AB is given by $\vec{r} = \frac{\vec{a} + \vec{b}}{2}$.

PRODUCT OF TWO VECTORS

Scalar (or dot) product : The scalar (or dot) product of two (non-zero) vectors $\vec{a}$ and $\vec{b}$, denoted by $\vec{a} \cdot \vec{b}$ (read as $\vec{a}$ dot $\vec{b}$), is defined as $\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta = ab\cos\theta$,

where, $a = |\vec{a}|, b = |\vec{b}|$ and $\theta (0 \le \theta \le \pi)$ is the angle between $\vec{a}$ and $\vec{b}$.

- **Properties of Scalar Product :**

(i) Scalar product is commutative : $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

(ii) $\vec{a} \cdot \vec{0} = 0$

(iii) Scalar product is distributive over addition :

- $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
- $(\vec{a} + \vec{b}) \cdot \vec{c} = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$

(iv) $\lambda(\vec{a} \cdot \vec{b}) = (\lambda\vec{a}) \cdot \vec{b} = \vec{a} \cdot (\lambda\vec{b})$, λ be any scalar.

(v) If $\hat{i}, \hat{j}, \hat{k}$ are three unit vectors along three mutually perpendicular lines, then $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$

(vi) Angle between two non-zero vectors $\vec{a}$ and $\vec{b}$ is given by $\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$

i.e., $\theta = \cos^{-1}\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}\,||\vec{b}\,|}\right)$

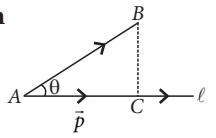
www.examBites.in

(vii) Two non-zero vectors $\vec{a}$ and $\vec{b}$ are mutually perpendicular if and only if $\vec{a}\cdot\vec{b}=0$

(viii) If $\theta = 0$, then $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|$

If $\theta = \pi$, then $\vec{a}\cdot\vec{b}=-|\vec{a}||\vec{b}|$

Projection of a vector on a line : Let the vector $\overrightarrow{AB}$ makes an angle θ with directed line ℓ.



Projection of $\overrightarrow{AB}$ on $\ell = |\overrightarrow{AB}|\cos\theta = \overrightarrow{AC} = \vec{p}$.

The vector $\vec{p}$ is called the projection vector. Its magnitude is $|\vec{p}|$, which is known as projection of vector $\overrightarrow{AB}$.

Projection of a vector $\vec{a}$ on $\vec{b}$, is given as $\vec{a}\cdot\hat{b}$ *i.e.*, $\frac{1}{|\vec{b}|}(\vec{a}\cdot\vec{b})$.

Vector (or Cross) Product : The vector (or cross) product of two (non-zero) vectors $\vec{a}$ and $\vec{b}$ (in an assigned order), denoted by $\vec{a}\times\vec{b}$ (read as $\vec{a}$ cross $\vec{b}$), is defined as $\vec{a}\times\vec{b}=|\vec{a}|\,|\vec{b}|\sin\theta\,\hat{n}$ where $\theta(0\le\theta\le\pi)$ is the angle between $\vec{a}$ and $\vec{b}$ and $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

▶ **Properties of Vector Product :**

(i) Non-commutative : $\vec{a}\times\vec{b}=-\vec{b}\times\vec{a}$

(ii) Vector product is distributive over addition :

$\vec{a}\times(\vec{b}+\vec{c})=\vec{a}\times\vec{b}+\vec{a}\times\vec{c}$

(iii) $\lambda(\vec{a}\times\vec{b})=(\lambda\vec{a})\times\vec{b}=\vec{a}\times(\lambda\vec{b})$, λ be any scalar.

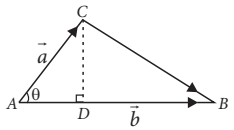
(iv) $(\lambda_1\vec{a})\times(\lambda_2\vec{b})=\lambda_1\lambda_2(\vec{a}\times\vec{b})$

(v) $\hat{i}\times\hat{j}=\hat{k},\ \hat{j}\times\hat{k}=\hat{i},\ \hat{k}\times\hat{i}=\hat{j}$

(vi) Two non-zero vectors $\vec{a}$, $\vec{b}$ are collinear if and only if $\vec{a}\times\vec{b}=\vec{0}$

Similarly, $\vec{a}\times\vec{a}=\vec{0}$ and $\vec{a}\times(-\vec{a})=\vec{0}$, since in the first situation $\theta = 0$ and in the second one, $\theta = \pi$, making the value of $\sin\theta$ to be 0.

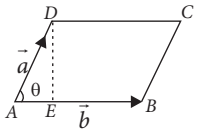
(vii) If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a triangle as given in the figure. Then,



Area of triangle $ABC = \frac{1}{2}AB\cdot CD$

$=\frac{1}{2}|\vec{b}||\vec{a}|\sin\theta = \frac{1}{2}|\vec{a}\times\vec{b}|$

(viii) If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a parallelogram as given in the figure.



Then, area of parallelogram $ABCD = AB\cdot DE$

$= |\vec{b}||\vec{a}|\sin\theta = |\vec{a}\times\vec{b}|$

(ix) If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}, \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$,

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}$$

$=(a_2b_3-a_3b_2)\hat{i}+(a_3b_1-a_1b_3)\hat{j}+(a_1b_2-a_2b_1)\hat{k}$

(x) Angle between two vectors $\vec{a}$ and $\vec{b}$ is given by $\sin\theta = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$

i.e., $\theta = \sin^{-1}\left(\frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}\right)$

Scalar Triple Product : The scalar triple product of any three vectors $\vec{a},\vec{b}$ and $\vec{c}$ is written as $\vec{a}\cdot(\vec{b}\times\vec{c})$ or $[\vec{a}\ \vec{b}\ \vec{c}]$.

▶ **Coplanarity of Three Vectors :** Three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar iff $\vec{a}\cdot(\vec{b}\times\vec{c})=0$.

▶ Volume of parallelopiped formed by adjacent sides given by the three vectors $\vec{a}=(a_1\hat{i}+a_2\hat{j}+a_3\hat{k})$, $\vec{b}=(b_1\hat{i}+b_2\hat{j}+b_3\hat{k})$, and $\vec{c}=(c_1\hat{i}+c_2\hat{j}+c_3\hat{k})$, is $|\vec{a}\cdot(\vec{b}\times\vec{c})|$.

i.e., $|\vec{a}\cdot(\vec{b}\times\vec{c})|=\begin{vmatrix}a_1&a_2&a_3\\b_1&b_2&b_3\\c_1&c_2&c_3\end{vmatrix}$

▶ For any three vectors $\vec{a},\vec{b}$ and $\vec{c}$,

(i) $[\vec{a}\ \vec{b}\ \vec{c}]=[\vec{b}\ \vec{c}\ \vec{a}]=[\vec{c}\ \vec{a}\ \vec{b}]$

(ii) $[\vec{a}\ \vec{b}\ \vec{c}]=-[\vec{a}\ \vec{c}\ \vec{b}]$ (iii) $[\vec{a}\ \vec{a}\ \vec{b}]=0$

www.exambites.in

Previous Years' CBSE Board Questions

10.2 Some Basic Concepts

VSA (1 mark)

1. Find a vector $\vec{a}$ of magnitude $5\sqrt{2}$, making an angle of $\frac{\pi}{4}$ with x-axis, $\frac{\pi}{2}$ with y-axis and an acute angle θ with z-axis. *(AI 2014)*

2. If a unit vector $\vec{a}$ makes angles $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and an acute angle θ with $\hat{k}$, then find the value of θ. *(Delhi 2013)*

3. Find the magnitude of the vector $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$. *(AI 2011C)*

10.3 Types of Vectors

VSA (1 mark)

4. The value of p for which $p(\hat{i} + \hat{j} + \hat{k})$ is a unit vector is

(a) 0 (b) $\frac{1}{\sqrt{3}}$

(c) 1 (d) $\sqrt{3}$ *(2020)*

10.4 Addition of Vectors

VSA (1 mark)

5. $ABCD$ is a rhombus, whose diagonals intersect at E. Then $\overrightarrow{EA} + \overrightarrow{EB} + \overrightarrow{EC} + \overrightarrow{ED}$ equals

(a) $\vec{0}$ (b) $\overrightarrow{AD}$

(c) $2\overrightarrow{BC}$ (d) $2\overrightarrow{AD}$ *(2020)*

6. Find the sum of the vectors $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$. *(Delhi 2012)*

7. Find the sum of the following vectors :
$\vec{a} = \hat{i} - 3\hat{k}$, $\vec{b} = 2\hat{j} - \hat{k}$, $\vec{c} = 2\hat{i} - 3\hat{j} + 2\hat{k}$ *(Delhi 2012)*

8. Find the sum of the following vectors :
$\vec{a} = \hat{i} - 2\hat{j}, \vec{b} = 2\hat{i} - 3\hat{j}, \vec{c} = 2\hat{i} + 3\hat{k}$ *(Delhi 2012)*

9. If A, B and C are the vertices of a triangle ABC, then what is the value of $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}$?
(Delhi 2011C)

10.5 Multiplication of a Vector by a Scalar

VSA (1 mark)

10. The position vector of two points A and B are $\overrightarrow{OA} = 2\hat{i} - \hat{j} - \hat{k}$ and $\overrightarrow{OB} = 2\hat{i} - \hat{j} + 2\hat{k}$, respectively. The position vector of a point P which divides the line segment joining A and B in the ratio 2 : 1 is _____. *(2020)*

11. Find the position vector of a point which divides the join of points with position vectors $\vec{a} - 2\vec{b}$ and $2\vec{a} + \vec{b}$ externally in the ratio 2 : 1. *(Delhi 2016)*

12. Write the position vector of the point which divides the join of points with position vectors $3\vec{a} - 2\vec{b}$ and $2\vec{a} + 3\vec{b}$ in the ratio 2 : 1. *(AI 2016)*

13. Find the unit vector in the direction of the sum of the vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $4\hat{i} - 3\hat{j} + 2\hat{k}$. *(Foreign 2015)*

14. Find a vector in the direction of $\vec{a} = \hat{i} - 2\hat{j}$ that has magnitude 7 units. *(Delhi 2015C)*

15. Write the direction ratios of the vector $3\vec{a} + 2\vec{b}$ where $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$. *(AI 2015C)*

16. Write a unit vector in the direction of the sum of the vectors $\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - 7\hat{k}$. *(Delhi 2014)*

17. Find the value of 'p' for which the vectors $3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\hat{i} - 2p\hat{j} + 3\hat{k}$ are parallel. *(AI 2014)*

18. Find a vector in the direction of vector $2\hat{i} - 3\hat{j} + 6\hat{k}$ which has magnitude 21 units. *(Foreign 2014)*

www.exambee.in

19. Write a unit vector in the direction of vector $\overrightarrow{PQ}$, where P and Q are the points(1, 3, 0) and (4, 5, 6) respectively. *(Foreign 2014)*

20. Write a vector in the direction of the vector $\hat{i}-2\hat{j}+2\hat{k}$ that has magnitude 9 units. *(Delhi 2014C)*

21. If $\vec{a}=x\hat{i}+2\hat{j}-z\hat{k}$ and $\vec{b}=3\hat{i}-y\hat{j}+\hat{k}$ are two equal vectors, then write the value of $x+y+z$. *(Delhi 2013)*

22. Write a unit vector in the direction of the sum of vectors $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+\hat{j}+3\hat{k}$. *(Delhi 2013)*

23. P and Q are two points with position vectors $3\vec{a}-2\vec{b}$ and $\vec{a}+\vec{b}$ respectively. Write the position vector of a point R which divides the line segment PQ in the ratio 2 : 1 externally. *(AI 2013)*

24. A and B are two points with position vectors $2\vec{a}-3\vec{b}$ and $6\vec{b}-\vec{a}$ respectively. Write the position vector of a point P which divides the line segment AB internally in the ratio 1 : 2. *(AI 2013)*

25. L and M are two points with position vectors $2\vec{a}-\vec{b}$ and $\vec{a}+2\vec{b}$ respectively. Write the position vector of a point N which divides the line segment LM in the ratio 2 : 1 externally. *(AI 2013)*

26. Find the scalar components of the vector $\overrightarrow{AB}$ with initial point $A(2, 1)$ and terminal point $B(-5, 7)$. *(AI 2012)*

27. Find a unit vector parallel to the sum of the vectors $\hat{i}+\hat{j}+\hat{k}$ and $2\hat{i}-3\hat{j}+5\hat{k}$. *(Delhi 2012C)*

28. Find a unit vector in the direction of the vector $\vec{a}=3\hat{i}-2\hat{j}+6\hat{k}$. *(AI 2012C)*

29. Write the direction cosines of the vector $-2\hat{i}+\hat{j}-5\hat{k}$. *(Delhi 2011)*

30. For what value of 'a', the vectors $2\hat{i}-3\hat{j}+4\hat{k}$ and $a\hat{i}+6\hat{j}-8\hat{k}$ are collinear ? *(Delhi 2011)*

31. Write a unit vector in the direction of the vector $\vec{a}=2\hat{i}+\hat{j}+2\hat{k}$. *(AI 2011)*

32. Find a unit vector in the direction of $\vec{a}=2\hat{i}-3\hat{j}+6\hat{k}$. *(Delhi 2011C)*

33. Find a unit vector in the direction of the vector $\vec{a}=2\hat{i}+3\hat{j}+6\hat{k}$. *(AI 2011C)*

SA (2 marks)

34. X and Y are two points with position vectors $3\vec{a}+\vec{b}$ and $\vec{a}-3\vec{b}$ respectively. Write the position vector of a point Z which divides the line segment XY in the ratio 2 : 1 externally. *(AI 2019)*

LA 1 (4 marks)

35. The two vectors $\hat{j}+\hat{k}$ and $3\hat{i}-\hat{j}+4\hat{k}$ represent the two sides AB and AC, respectively of a ΔABC. Find the length of the median through A. *(Delhi 2016, Foreign 2015)*

36. Find a vector of magnitude 5 units and parallel to the resultant of the vectors $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$. *(Delhi 2011)*

10.6 Product of Two Vectors

VSA (1 mark)

37. If the projection of $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$ on $\vec{b}=2\hat{i}+\lambda\hat{k}$ is zero, then the value of λ is

(a) 0 (b) 1

(c) $\frac{-2}{3}$ (d) $\frac{-3}{2}$ *(2020)*

38. The area of the parallelogram whose diagonals are $2\hat{i}$ and $-3\hat{k}$ is _____ square units.

39. The value of λ for which the vectors $2\hat{i}-\lambda\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}-\hat{k}$ are orthogonal is _____. *(2020)*

40. If $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along three mutually perpendicular directions, then

(a) $\hat{i}\cdot\hat{j}=1$ (b) $\hat{i}\times\hat{j}=1$

(c) $\hat{i}\cdot\hat{k}=0$ (d) $\hat{i}\times\hat{k}=0$ *(2020)*

www.exambites.in

41. Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude such that the angle between them is 60° and their scalar product is $\frac{9}{2}$. *(2018)*

42. Write the number of vectors of unit length perpendicular to both the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. *(AI 2016)*

43. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then write the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$. *(Foreign 2016)*

44. If $\left|\vec{a} \times \vec{b}\right|^2 + \left|\vec{a} \cdot \vec{b}\right|^2 = 400$ and $|\vec{a}| = 5$ then write the value of $|\vec{b}|$. *(Foreign 2016)*

45. If $\vec{a} = 7\hat{i} + \hat{j} - 4\hat{k}$ and $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$, then find the projection of $\vec{a}$ on $\vec{b}$. *(Delhi 2015, 2013C)*

46. If $\hat{a}$, $\hat{b}$ and $\hat{c}$ are mutually perpendicular unit vectors, then find the value of $|2\hat{a} + \hat{b} + \hat{c}|$. *(AI 2015)*

47. Write a unit vector perpendicular to both the vectors $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. *(AI 2015)*

48. Find the area of a parallelogram whose adjacent sides are represented by the vectors $2\hat{i} - 3\hat{k}$ and $4\hat{j} + 2\hat{k}$. *(Foreign 2015)*

49. If $\vec{a}$ and $\vec{b}$ are unit vectors, then what is the angle between $\vec{a}$ and $\vec{b}$ so that $\sqrt{2}\,\vec{a} - \vec{b}$ is a unit vector ? *(Delhi 2015C)*

50. Find the projection of the vector $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on the vector $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$. *(AI 2015C)*

51. Find the projection of vector $\hat{i} + 3\hat{j} + 7\hat{k}$ on the vector $2\hat{i} - 3\hat{j} + 6\hat{k}$. *(Delhi 2014)*

52. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + \vec{b}$ is also a unit vector, then find the angle between $\vec{a}$ and $\vec{b}$. *(Delhi 2014)*

53. If vectors $\vec{a}$ and $\vec{b}$ are such that, $|\vec{a}| = 3, |\vec{b}| = \frac{2}{3}$ and $\vec{a} \times \vec{b}$ is a unit vector, then write the angle between $\vec{a}$ and $\vec{b}$. *(Delhi 2014)*

54. If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, $|\vec{a} + \vec{b}| = 13$ and $|\vec{a}| = 5$, find the value of $|\vec{b}|$. *(AI 2014)*

55. Write the projection of the vector $\hat{i} + \hat{j} + \hat{k}$ along the vector $\hat{j}$. *(Foreign 2014)*

56. Write the value of $\hat{i} \times (\hat{j} + \hat{k}) + \hat{j} \times (\hat{k} + \hat{i}) + \hat{k} \times (\hat{i} + \hat{j})$. *(Foreign 2014)*

57. Write the projection of the vector $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ on the vector $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$. *(Delhi 2014C)*

58. If $\vec{a}$ and $\vec{b}$ are unit vectors, then find the angle between $\vec{a}$ and $\vec{b}$, given that $(\sqrt{3}\,\vec{a} - \vec{b})$ is a unit vector. *(Delhi 2014C)*

59. Write the value of cosine of the angle which the vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ makes with *y*-axis. *(Delhi 2014C)*

60. If $|\vec{a}| = 8, |\vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 12$, find the angle between $\vec{a}$ and $\vec{b}$. *(AI 2014C)*

61. Find the angle between *x*-axis and the vector $\hat{i} + \hat{j} + \hat{k}$. *(AI 2014C)*

62. Find $|\vec{x}|$, if for a unit vector $\vec{a}$, $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 15$ *(AI 2013)*

63. Write the value of λ so that the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular to each other. *(Delhi 2013C, AI 2012C)*

64. For what value of λ are the vectors $\hat{i} + 2\lambda\hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} - 3\hat{k}$ are perpendicular? *(AI 2013C, 2011C, Delhi 2012C)*

65. Write the projection of $\vec{b} + \vec{c}$ on $\vec{a}$, where $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. *(AI 2013C)*

66. Find 'λ' when the projection of $\vec{a} = \lambda\hat{i} + \hat{j} + 4\hat{k}$ on $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$ is 4 units. *(Delhi 2012)*

67. Write the value of $(\hat{i} \times \hat{j}) \cdot \hat{k} + \hat{i} \cdot \hat{j}$. *(AI 2012)*

www.exambites.in

68. Write the value of $(\hat{k}\times\hat{j})\cdot\hat{i}+\hat{j}\cdot\hat{k}$. *(AI 2012)*

69. Write the value of $(\hat{k}\times\hat{i})\cdot\hat{j}+\hat{i}\cdot\hat{k}$. *(AI 2012)*

70. Write the angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes $\sqrt{3}$ and 2 respectively having $\vec{a}\cdot\vec{b}=\sqrt{6}$. *(AI 2011)*

71. Write the projection of the vector $\hat{i}-\hat{j}$ on the vector $\hat{i}+\hat{j}$. *(AI 2011)*

72. If $|\vec{a}|=\sqrt{3}, |\vec{b}|=2$ and angle between $\vec{a}$ and $\vec{b}$ is 60°, find $\vec{a}\cdot\vec{b}$. *(Delhi 2011C)*

SA (2 marks)

73. Find a unit vector perpendicular to each of the vectors $\vec{a}$ and $\vec{b}$ and where $\vec{a}=5\hat{i}+6\hat{j}-2\hat{k}$ and $\vec{b}=7\hat{i}+6\hat{j}+2\hat{k}$. *(2020)*

74. Show that for any two non-zero vectors $\vec{a}$ and $\vec{b}$, $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$ iff $\vec{a}$ and $\vec{b}$ are perpendicular vectors. *(2020)*

75. Show that the vectors $2\hat{i}-\hat{j}+\hat{k}$, $3\hat{i}+7\hat{j}+\hat{k}$ and $5\hat{i}+6\hat{j}+2\hat{k}$ form the sides of a right-angled triangle. *(2020)*

76. If the sum of two unit vectors is a unit vector, prove that the magnitude of their difference is $\sqrt{3}$. *(Delhi 2019)*

77. Let $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b}=3\hat{i}-\hat{j}+2\hat{k}$ be two vectors. Show that the vectors $(\vec{a}+\vec{b})$ and $(\vec{a}-\vec{b})$ are perpendicular to each other. *(AI 2019)*

78. If θ is the angle between two vectors $\hat{i}-2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$, find sin θ. *(2018)*

LA 1 (4 marks)

79. If $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ represent two adjacent sides of a parallelogram, find unit vectors parallel to the diagonals of the parallelogram. *(2020)*

80. Using vectors, find the area of the triangle *ABC* with vertices *A*(1, 2, 3), *B*(2, –1, 4) and *C*(4, 5, –1) *(2020, Delhi 2013, AI 2013)*

81. If $\hat{i}+\hat{j}+\hat{k}, 2\hat{i}+5\hat{j}, 3\hat{i}+2\hat{j}-3\hat{k}$ and $\hat{i}-6\hat{j}-\hat{k}$ respectively are the position vectors of points *A*, *B*, *C* and *D*, then find the angle between the straight lines *AB* and *CD*. Find whether $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear or not. *(Delhi 2019)*

82. Let $\vec{a}=4\hat{i}+5\hat{j}-\hat{k}, \vec{b}=\hat{i}-4\hat{j}+5\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}-\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{c}$ and $\vec{b}$ and $\vec{d}\cdot\vec{a}=21$. *(2018)*

83. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are mutually perpendicular vectors of equal magnitudes, show that the vector $\vec{a}+\vec{b}+\vec{c}$ is equally inclined to $\vec{a}$, $\vec{b}$ and $\vec{c}$. Also, find the angle which $\vec{a}+\vec{b}+\vec{c}$ makes with $\vec{a}$ or $\vec{b}$ or $\vec{c}$. *(Delhi 2017)*

84. Show that the points *A*, *B*, *C* with position vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$ respectively, are the vertices of a right-angled triangle. Hence find the area of the triangle. *(AI 2017)*

85. The two adjacent sides of a parallelogram are $2\hat{i}-4\hat{j}-5\hat{k}$ and $2\hat{i}+2\hat{j}+3\hat{k}$. Find the two unit vectors parallel to its diagonals. Using the diagonal vectors, find the area of the parallelogram. *(AI 2016)*

86. If $\vec{a}\times\vec{b}=\vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c}=\vec{b}\times\vec{d}$, show that $\vec{a}-\vec{d}$ is parallel to $\vec{b}-\vec{c}$, where $\vec{a}\neq\vec{d}$ and $\vec{b}\neq\vec{c}$. *(Foreign 2016)*

87. If $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$, find $(\vec{r}\times\hat{i})\cdot(\vec{r}\times\hat{j})+xy$. *(Delhi 2015)*

88. If $\vec{a}=\hat{i}+2\hat{j}+\hat{k}, \vec{b}=2\hat{i}+\hat{j}$ and $\vec{c}=3\hat{i}-4\hat{j}-5\hat{k}$, then find a unit vector perpendicular to both of the vectors $(\vec{a}-\vec{b})$ and $(\vec{c}-\vec{b})$. *(AI 2015)*

89. Vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ and $|\vec{a}|=3$, $|\vec{b}|=5$ and $|\vec{c}|=7$. Find the angle between $\vec{a}$ and $\vec{b}$. *(Delhi 2014)*

90. The scalar product of the vector $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ with a unit vector along the sum of vectors $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\vec{c}=\lambda\hat{i}+2\hat{j}+3\hat{k}$ is

www.exambites.in

equal to one. Find the value of λ and hence find the unit vector along $\vec{b} + \vec{c}$. *(AI 2014)*

91. Find a unit vector perpendicular to both of the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ where $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$. *(Foreign 2014)*

92. If $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}, \vec{b} = -\hat{i} + \hat{k}$, $\vec{c} = 2\hat{j} - \hat{k}$ are three vectors, find the area of the parallelogram having diagonals $(\vec{a} + \vec{b})$ and $(\vec{b} + \vec{c})$.
(Delhi 2014C)

93. Find the vector $\vec{p}$ which is perpendicular to both $\vec{\alpha} = 4\hat{i} + 5\hat{j} - \hat{k}$ and $\vec{\beta} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{p} \cdot \vec{q} = 21$, where $\vec{q} = 3\hat{i} + \hat{j} - \hat{k}$. *(AI 2014C)*

94. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a} + \vec{b}| = |\vec{a}|$, then prove that vector $2\vec{a} + \vec{b}$ is perpendicular to vector $\vec{b}$. *(Delhi 2013)*

95. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$, find a vector $\vec{c}$, such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$.
(Delhi 2013)

96. If $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$, then find the value of λ, so that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors. *(AI 2013)*

97. If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors of the same magnitude, prove that $\vec{a} + \vec{b} + \vec{c}$ is equally inclined with the vectors $\vec{a}, \vec{b}$ and $\vec{c}$. *(Delhi 2013C)*

98. Dot product of a vector with vectors $\hat{i} - \hat{j} + \hat{k}$, $2\hat{i} + \hat{j} - 3\hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$ are respectively 4, 0 and 2. Find the vector.
(Delhi 2013C)

99. Find the values of λ for which the angle between the vectors $\vec{a} = 2\lambda^2\,\hat{i} + 4\lambda\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + \lambda\hat{k}$ is obtuse. *(AI 2013C)*

100. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors such that each one is perpendicular to the vector obtained by sum of the other two and $|\vec{a}| = 3, |\vec{b}| = 4$ and $|\vec{c}| = 5$, then prove that $|\vec{a} + \vec{b} + \vec{c}| = 5\sqrt{2}$. *(AI 2013C)*

101. If $\vec{a} = 3\hat{i} - \hat{j}$ and $\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$ then express $\vec{b}$ in the form $\vec{b} = \vec{b}_1 + \vec{b}_2$ where $\vec{b}_1 \parallel \vec{a}$ and $\vec{b}_2 \perp \vec{a}$. *(AI 2013C)*

102. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = 5, |\vec{b}| = 12$ and $|\vec{c}| = 13$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.
(Delhi 2012)

103. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{p}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{p} \cdot \vec{c} = 18$. *(AI 2012)*

104. If the sum of two unit vectors $\hat{a}$ and $\hat{b}$ is a unit vector, show that the magnitude of their difference is $\sqrt{3}$. *(Delhi 2012C)*

105. Find a unit vector perpendicular to each of the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$.
(Delhi 2011)

106. If two vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = 2, |\vec{b}| = 1$ and $\vec{a} \cdot \vec{b} = 1$, then find the value of $(3\vec{a} - 5\vec{b}) \cdot (2\vec{a} + 7\vec{b})$. *(Delhi 2011)*

107. Using vectors, find the area of the triangle with vertices $A(1, 1, 2)$, $B(2, 3, 5)$ and $C(1, 5, 5)$.
(AI 2011)

108. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors of equal magnitude, show that $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}, \vec{b}$ and $\vec{c}$. Also find the angle. *(Delhi 2011C)*

109. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors such that $|\vec{a}| = 3, |\vec{b}| = 4$ and $|\vec{c}| = 5$ and each one of them is perpendicular to the sum of the other two, then find $|\vec{a} + \vec{b} + \vec{c}|$. *(AI 2011C)*

10.7 Scalar Triple Product

VSA (1 mark)

110. Find λ, if the vectors $\vec{a} = \hat{i} + 3\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} - \hat{k}$ and $\vec{c} = \lambda\hat{j} + 3\hat{k}$ are coplanar.
(Delhi 2015)

111. Find $\vec{a} \cdot (\vec{b} \times \vec{c})$, if $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}, \vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} + 2\hat{k}$. *(AI 2014)*

www.kambres.in

SA (2 marks)

112. Find the volume of the parallelepiped whose adjacent edges are represented by $2\vec{a}, -\vec{b}$ and $3\vec{c}$, where $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 3\hat{k}$. *(2020)*

113. If $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}, \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = -3\hat{i} + \hat{j} + 2\hat{k}$, find $[\vec{a}\ \vec{b}\ \vec{c}]$. *(Delhi 2019)*

LA 1 (4 marks)

114. Find the value of x, for which the four points $A(x, -1, -1)$, $B(4, 5, 1)$, $C(3, 9, 4)$ and $D(-4, 4, 4)$ are coplanar. *(AI 2019)*

115. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, then

(a) Let $c_1 = 1$ and $c_2 = 2$, find c_3 which makes $\vec{a}, \vec{b}$ and $\vec{c}$ coplanar.

(b) If $c_2 = -1$ and $c_3 = 1$, show that no value of c_1 can make $\vec{a}, \vec{b}$ and $\vec{c}$ coplanar. *(Delhi 2017)*

116. Find the value of λ, if four points with position vectors $3\hat{i} + 6\hat{j} + 9\hat{k}$, $\hat{i} + 2\hat{j} + 3\hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $4\hat{i} + 6\hat{j} + \lambda\hat{k}$ are coplanar. *(AI 2017)*

117. Show that the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar if $\vec{a} + \vec{b}, \vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$ are coplanar. *(Delhi 2016)*

118. Find the value of λ so that the four points A, B, C and D with position vectors $4\hat{i} + 5\hat{j} + \hat{k}$, $-\hat{j} - \hat{k}$, $3\hat{i} + \lambda\hat{j} + 4k$ and $-4\hat{i} + 4\hat{j} + 4\hat{k}$ respectively are coplanar. *(Delhi 2015C)*

119. Prove that : $[\vec{a}, \vec{b} + \vec{c}, \vec{d}] = [\vec{a}, \vec{b}, \vec{d}] + [\vec{a}, \vec{c}, \vec{d}]$. *(AI 2015C)*

120. Prove that, for any three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ $[\vec{a} + \vec{b}\ \ \vec{b} + \vec{c}\ \ \vec{c} + \vec{a}] = 2[\vec{a}\ \vec{b}\ \vec{c}]$ *(Delhi 2014)*

121. Show that the four points A, B, C and D with position vectors $4\hat{i} + 5\hat{j} + \hat{k}$, $-\hat{j} - \hat{k}$, $3\hat{i} + 9\hat{j} + 4\hat{k}$ and $4(-\hat{i} + \hat{j} + \hat{k})$ respectively are coplanar. *(AI 2014)*

122. Show that the vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar if and only if $\vec{a} + \vec{b}, \vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$ are coplanar. *(Foreign 2014)*

123. If the three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, prove that the vectors $\vec{a} + \vec{b}, \vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$ are also coplanar. *(Delhi 2014C, 2013C)*

Detailed Solutions

1. Here, $l = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$, $m = \cos\frac{\pi}{2} = 0$, $n = \cos\theta$

Since, $l^2 + m^2 + n^2 = 1$

$\Rightarrow \frac{1}{2} + 0 + \cos^2\theta = 1 \Rightarrow \cos^2\theta = 1 - \frac{1}{2} = \frac{1}{2}$

$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4} \quad \therefore n = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\therefore$ The vector of magnitude $5\sqrt{2}$ is

$\vec{a} = 5\sqrt{2}(l\hat{i} + m\hat{j} + n\hat{k})$

$= 5\sqrt{2}\left(\frac{1}{\sqrt{2}}\hat{i} + 0\hat{j} + \frac{1}{\sqrt{2}}\hat{k}\right) = 5(\hat{i} + \hat{k})$

2. $l = \cos\frac{\pi}{3} = \frac{1}{2}, m = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ and $n = \cos\theta$

Now, $l^2 + m^2 + n^2 = 1$

$\Rightarrow \left(\frac{1}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + n^2 = 1$

$\Rightarrow \frac{1}{4} + \frac{1}{2} + n^2 = 1 \Rightarrow n^2 = \frac{1}{4} \Rightarrow n = \pm\frac{1}{2}$

$\Rightarrow \cos\theta = \pm\frac{1}{2}$

But θ is an acute angle (given).

$\therefore \quad \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

3. Here, $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$

$\therefore$ Its magnitude = $|\vec{a}|$

$= \sqrt{3^2 + (-2)^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7.$

www.examfree.in

4. **(b) :** Let $\vec{a} = (\hat{i} + \hat{j} + \hat{k})$

So, unit vector of $\vec{a} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{1+1+1}} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

$\therefore$ The value of p is $\frac{1}{\sqrt{3}}$.

5. **(a) :** $\overrightarrow{EA} + \overrightarrow{EB} + \overrightarrow{EC} + \overrightarrow{ED}$

$= \overrightarrow{EA} + \overrightarrow{EB} - \overrightarrow{EA} - \overrightarrow{EB}$

[As diagonals of a rhombus bisect each other]

$= \vec{0}$

6. The given vectors are

$\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$, $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$

$\therefore$ Their sum $= \vec{a} + \vec{b} + \vec{c}$

$= (\hat{i} - 2\hat{j} + \hat{k}) + (-2\hat{i} + 4\hat{j} + 5\hat{k}) + (\hat{i} - 6\hat{j} - 7\hat{k})$

$= -4\hat{j} - \hat{k}$.

7. Required sum $= \vec{a} + \vec{b} + \vec{c}$

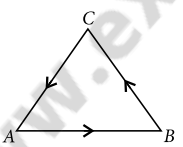
$= (\hat{i} - 3\hat{k}) + (2\hat{j} - \hat{k}) + (2\hat{i} - 3\hat{j} + 2\hat{k})$

$= 3\hat{i} - \hat{j} - 2\hat{k}$.

8. Required sum $= \vec{a} + \vec{b} + \vec{c}$

$= (\hat{i} - 2\hat{j}) + (2\hat{i} - 3\hat{j}) + (2\hat{i} + 3\hat{k}) = 5\hat{i} - 5\hat{j} + 3\hat{k}$.

9. Let ABC be the given triangle.



Now $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$

(By Triangle law)

$\Rightarrow \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{AC} + \overrightarrow{CA} = \vec{0}$

10. Required position vector of point P

$= \frac{1(2\hat{i} - \hat{j} - \hat{k}) + 2(2\hat{i} - \hat{j} + 2\hat{k})}{2+1}$

$= \frac{2\hat{i} - \hat{j} - \hat{k} + 4\hat{i} - 2\hat{j} + 4\hat{k}}{3}$

$= \frac{1}{3}(6\hat{i} - 3\hat{j} + 3\hat{k}) = 2\hat{i} - \hat{j} + \hat{k}$

11. Required position vector

$= \frac{2\cdot(2\vec{a} + \vec{b}) - 1(\vec{a} - 2\vec{b})}{2-1} = \frac{4\vec{a} + 2\vec{b} - \vec{a} + 2\vec{b}}{1}$

$= 3\vec{a} + 4\vec{b}$

12. Required position vector

$= \frac{2(2\vec{a} + 3\vec{b}) + 1(3\vec{a} - 2\vec{b})}{2+1} = \frac{7\vec{a} + 4\vec{b}}{3} = \frac{7}{3}\vec{a} + \frac{4}{3}\vec{b}$

13. Let $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = 4\hat{i} - 3\hat{j} + 2\hat{k}$.

Then, the sum of the given vectors is

$\vec{c} = \vec{a} + \vec{b} = (2+4)\hat{i} + (3-3)\hat{j} + (-1+2)\hat{k} = 6\hat{i} + \hat{k}$

and $|\vec{c}| = |\vec{a} + \vec{b}| = \sqrt{6^2 + 1^2} = \sqrt{36+1} = \sqrt{37}$

$\therefore$ Unit vector, $\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{6\hat{i} + \hat{k}}{\sqrt{37}} = \frac{6}{\sqrt{37}}\hat{i} + \frac{1}{\sqrt{37}}\hat{k}$

14. A unit vector in the direction of $\vec{a} = \hat{i} - 2\hat{j}$ is $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 2\hat{j}}{\sqrt{1^2 + (-2)^2}} = \frac{1}{\sqrt{5}}(\hat{i} - 2\hat{j})$

$\therefore$ The required vector of magnitude 7 in the direction of $\vec{a} = 7 \cdot \hat{a} = \frac{7}{\sqrt{5}}(\hat{i} - 2\hat{j})$.

15. $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$; $\vec{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$

$\therefore$ $3\vec{a} + 2\vec{b} = 3(\hat{i} + \hat{j} - 2\hat{k}) + 2(2\hat{i} - 4\hat{j} + 5\hat{k})$

$= (3\hat{i} + 3\hat{j} - 6\hat{k}) + (4\hat{i} - 8\hat{j} + 10\hat{k}) = 7\hat{i} - 5\hat{j} + 4\hat{k}$

$\therefore$ The direction ratios of the vector $3\vec{a} + 2\vec{b}$ are $7, -5, 4$.

16. *Refer to answer 13.*

17. Let $\vec{a} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\vec{b} = \hat{i} - 2p\hat{j} + 3\hat{k}$

For $\vec{a}$ and $\vec{b}$ to be parallel, $\vec{b} = \lambda\vec{a}$.

$\Rightarrow \hat{i} - 2p\hat{j} + 3\hat{k} = \lambda(3\hat{i} + 2\hat{j} + 9\hat{k}) = 3\lambda\hat{i} + 2\lambda\hat{j} + 9\lambda\hat{k}$

$\Rightarrow 1 = 3\lambda; -2p = 2\lambda, 3 = 9\lambda$

$\Rightarrow \lambda = \frac{1}{3}$ and $p = -\lambda = -\frac{1}{3}$

18. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$.

The vector in the direction of $\vec{a}$ with a magnitude of $21 = 21 \times \hat{a}$

$\therefore$ Required vector $= 21 \times \frac{2\hat{i} - 3\hat{j} + 6\hat{k}}{\sqrt{2^2 + (-3)^2 + 6^2}}$

$= 21 \times \frac{2\hat{i} - 3\hat{j} + 6\hat{k}}{7} = 6\hat{i} - 9\hat{j} + 18\hat{k}$

19. We have $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$

$= (4\hat{i} + 5\hat{j} + 6\hat{k}) - (\hat{i} + 3\hat{j}) = 3\hat{i} + 2\hat{j} + 6\hat{k}$

Required unit vector $= \frac{3\hat{i} + 2\hat{j} + 6\hat{k}}{\sqrt{3^2 + 2^2 + 6^2}} = \frac{3\hat{i} + 2\hat{j} + 6\hat{k}}{7}$

www.exambites.in

20. *Refer to answer 18.*

21. Given, $\vec{a} = \vec{b}$

$\Rightarrow \quad x\hat{i} + 2\hat{j} - z\hat{k} = 3\hat{i} - y\hat{j} + \hat{k}$

$\therefore \quad x = 3, y = -2, z = -1$

Hence, the value of $x + y + z = 0$

22. *Refer to answer 13.*

23. *Refer to answer 11.*

24. *Refer to answer 12.*

25. *Refer to answer 11.*

26. Vector $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$= (-5\hat{i} + 7\hat{j}) - (2\hat{i} + \hat{j}) = -7\hat{i} + 6\hat{j}$

So, its scalar components are (–7, 6).

27. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$

$\therefore \vec{a} + \vec{b} = (\hat{i} + \hat{j} + \hat{k}) + (2\hat{i} - 3\hat{j} + 5\hat{k})$

$= 3\hat{i} - 2\hat{j} + 6\hat{k}$

Any vector parallel to $\vec{a} + \vec{b}$

$= \lambda(\vec{a} + \vec{b}) = \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$

$\therefore$ The unit vector in this direction

$$= \frac{\lambda(3\hat{i} - 2\hat{j} + 6\hat{k})}{\sqrt{(3\lambda)^2 + (-2\lambda)^2 + (6\lambda)^2}}$$

$$= \frac{\lambda(3\hat{i} - 2\hat{j} + 6\hat{k})}{|\lambda| \cdot 7} = \pm\frac{1}{7}(3\hat{i} - 2\hat{j} + 6\hat{k})$$

28. The given vector is $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$

$\Rightarrow |\vec{a}| = \sqrt{3^2 + (-2)^2 + 6^2} = 7$

$\therefore$ A unit vector in the direction of vector $\vec{a}$ is

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{7}(3\hat{i} - 2\hat{j} + 6\hat{k})$$

29. We have, $\vec{a} = -2\hat{i} + \hat{j} - 5\hat{k}$

Direction cosines of the given vector are

$$\left(\frac{-2}{\sqrt{(-2)^2 + (1)^2 + (-5)^2}}, \frac{1}{\sqrt{(-2)^2 + (1)^2 + (-5)^2}}, \frac{-5}{\sqrt{(-2)^2 + (1)^2 + (-5)^2}}\right)$$

$$= \left(\frac{-2}{\sqrt{4+1+25}}, \frac{1}{\sqrt{4+1+25}}, \frac{-5}{\sqrt{4+1+25}}\right)$$

$\therefore$ Direction cosines are $\left(\frac{-2}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{-5}{\sqrt{30}}\right)$

30. We have, $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $a\hat{i} + 6\hat{j} - 8\hat{k}$

Two vectors are collinear if and only if,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \lambda \Rightarrow \frac{2}{a} = \frac{-3}{6} = \frac{4}{-8} = \frac{-1}{2} = \lambda$$

$$\Rightarrow \quad \frac{2}{a} = \frac{-1}{2} \Rightarrow a = -4$$

31. $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$

$|\vec{a}| = \sqrt{(2)^2 + (1)^2 + (2)^2} = \sqrt{4+1+4} = \sqrt{9} = 3.$

Required unit vector is $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{2\hat{i} + \hat{j} + 2\hat{k}}{3}$

$$= \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$$

32. *Refer to answer 31.*

33. *Refer to answer 31.*

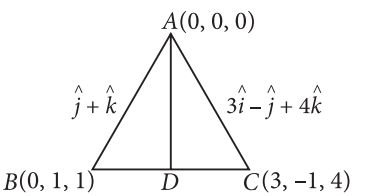
34. Position vector which divides the line segment joining points with position vectors $3\vec{a} + \vec{b}$ and $\vec{a} - 3\vec{b}$ in the ratio 2 : 1 externally is given by

$$\frac{2(\vec{a} - 3\vec{b}) - 1(3\vec{a} + \vec{b})}{2-1} = \frac{2\vec{a} - 6\vec{b} - 3\vec{a} - \vec{b}}{1}$$

$$= -\vec{a} - 7\vec{b}$$

35. Take A to be as origin (0, 0, 0).

$\therefore$ Coordinates of B are (0, 1, 1) and coordinates of C are (3, –1, 4).



Let D be the mid point of BC and AD is a median of ΔABC.

$\therefore$ Coordinates of D are $\left(\frac{3}{2}, 0, \frac{5}{2}\right)$

So, length of $AD = \sqrt{\left(\frac{3}{2} - 0\right)^2 + (0)^2 + \left(\frac{5}{2} - 0\right)^2}$

$$= \sqrt{\frac{9}{4} + \frac{25}{4}} = \frac{\sqrt{34}}{2} \text{ units}$$

36. $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$

$\therefore \quad \vec{a} + \vec{b} = (2\hat{i} + 3\hat{j} - \hat{k}) + (\hat{i} - 2\hat{j} + \hat{k}) = 3\hat{i} + \hat{j}$

www.examides.in

$|\vec{a}+\vec{b}| = \sqrt{3^2+1^2} = \sqrt{10}$

$\therefore$ A vector of magnitude 5 in the direction of $\vec{a}+\vec{b}$ is $\dfrac{5(\vec{a}+\vec{b})}{|\vec{a}+\vec{b}|} = \dfrac{5(3\hat{i}+\hat{j})}{\sqrt{10}}$

37. (c) : Here, $\vec{a} = \hat{i}-2\hat{j}+3\hat{k}, \vec{b} = 2\hat{i}+\lambda\hat{k}$

Since, projection of $\vec{a}$ on $\vec{b} = 0$

$\Rightarrow \quad \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = 0 \Rightarrow \dfrac{(\hat{i}-2\hat{j}+3\hat{k})\cdot(2\hat{i}+\lambda\hat{k})}{\sqrt{2^2+\lambda^2}} = 0$

$\Rightarrow \quad \dfrac{2+3\lambda}{\sqrt{4+\lambda^2}} = 0 \Rightarrow 2+3\lambda = 0 \Rightarrow \lambda = -\dfrac{2}{3}$

38. Given, two diagonals $\vec{d}_1$ and $\vec{d}_2$ are $2\hat{i}$ and $-3\hat{k}$ respectively.

$\therefore \quad \vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 0 & 0 & -3 \end{vmatrix} = \hat{i}(0)-\hat{j}(-6-0)+\hat{k}(0) = 6\hat{j}$

So, area of the parallelogram $= \dfrac{1}{2}|\vec{d}_1 \times \vec{d}_2|$

$= \dfrac{1}{2}\times 6 = 3$ sq. units

39. Let $\vec{a} = 2\hat{i}-\lambda\hat{j}+\hat{k}$ and $\vec{b} = \hat{i}+2\hat{j}-\hat{k}$

We know, $\vec{a}$ and $\vec{b}$ are orthogonal iff $\vec{a}\cdot\vec{b} = 0$

$\Rightarrow \quad (2\hat{i}-\lambda\hat{j}+\hat{k})\cdot(\hat{i}+2\hat{j}-\hat{k}) = 0$

$\Rightarrow \quad 2 - 2\lambda - 1 = 0 \Rightarrow 1 - 2\lambda = 0 \Rightarrow \lambda = \dfrac{1}{2}$

40. (c) : Since, $\hat{i}, \hat{j}, \hat{k}$ are mutually perpendicular.

$\therefore \quad \hat{i}\cdot\hat{k} = 0$

41. Given, $|\vec{a}| = |\vec{b}|, \theta = 60°$ and $\vec{a}\cdot\vec{b} = \dfrac{9}{2}$

Now, $\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

$\Rightarrow \quad \cos 60° = \dfrac{9/2}{|\vec{a}|^2} \Rightarrow \dfrac{1}{2} = \dfrac{9/2}{|\vec{a}|^2}$

$\Rightarrow \quad |\vec{a}|^2 = 9 \Rightarrow |\vec{a}| = 3 \quad \therefore \quad |\vec{a}| = |\vec{b}| = 3$

42. Given, $\vec{a} = 2\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b} = \hat{j}+\hat{k}$

Unit vectors perpendicular to $\vec{a}$ and $\vec{b}$ are $\pm\left(\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}\right)$.

Now, $\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{i}-2\hat{j}+2\hat{k}$

$\therefore$ Unit vectors perpendicular to $\vec{a}$ and $\vec{b}$ are

$\pm\dfrac{(-\hat{i}-2\hat{j}+2\hat{k})}{\sqrt{(-1)^2+(-2)^2+(2)^2}} = \pm\left(-\dfrac{1}{3}\hat{i}-\dfrac{2}{3}\hat{j}+\dfrac{2}{3}\hat{k}\right)$

So, there are two unit vectors perpendicular to the given vectors.

43. We have $\vec{a}, \vec{b}, \vec{c}$ are unit vectors.

Therefore, $|\vec{a}| = 1, |\vec{b}| = 1$ and $|\vec{c}| = 1$

Also, $\vec{a}+\vec{b}+\vec{c} = \vec{0}$ (given)

$\Rightarrow \quad |\vec{a}+\vec{b}+\vec{c}|^2 = 0$

$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}) = 0$

$\Rightarrow 1+1+1+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}) = 0$

$\Rightarrow 3+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}) = 0$

$\Rightarrow (\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}) = -\dfrac{3}{2}$

44. $|\vec{a}\times\vec{b}|^2 + |\vec{a}\cdot\vec{b}|^2 = 400$

$\Rightarrow \{|\vec{a}||\vec{b}|\sin\theta\}^2 + \{|\vec{a}||\vec{b}|\cos\theta\}^2 = 400$

$\Rightarrow |\vec{a}|^2|\vec{b}|^2\sin^2\theta + |\vec{a}|^2|\vec{b}|^2\cos^2\theta = 400$

$\Rightarrow |\vec{a}|^2|\vec{b}|^2 = 400 \Rightarrow 25\times|\vec{b}|^2 = 400 \quad [\because |\vec{a}| = 5]$

$\Rightarrow |\vec{b}|^2 = 16 \Rightarrow |\vec{b}| = 4$

45. Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

$= \dfrac{(7\hat{i}+\hat{j}-4\hat{k})\cdot(2\hat{i}+6\hat{j}+3\hat{k})}{\sqrt{(2)^2+(6)^2+(3)^2}} = \dfrac{14+6-12}{7} = \dfrac{8}{7}$

46. Here $\hat{a}, \hat{b}$ and $\hat{c}$ are mutually perpendicular unit vectors.

$\Rightarrow \quad |\hat{a}| = |\hat{b}| = |\hat{c}| = 1$ and $\hat{a}\cdot\hat{b} = \hat{b}\cdot\hat{c} = \hat{c}\cdot\hat{a} = 0 \quad ...(1)$

$\therefore \quad |2\hat{a}+\hat{b}+\hat{c}|^2 = (2\hat{a}+\hat{b}+\hat{c}).(2\hat{a}+\hat{b}+\hat{c})$

$= 4\hat{a}\cdot\hat{a} + 2\hat{a}\cdot\hat{b} + 2\hat{a}\cdot\hat{c} + 2\hat{b}\cdot\hat{a} + \hat{b}\cdot\hat{b} + \hat{b}\cdot\hat{c} + 2\hat{c}\cdot\hat{a} + \hat{c}\cdot\hat{b} + \hat{c}\cdot\hat{c}$

$= 4|\hat{a}|^2 + |\hat{b}|^2 + |\hat{c}|^2 + 4\hat{a}\cdot\hat{b} + 2\hat{b}\cdot\hat{c} + 4\hat{a}\cdot\hat{c}$

$(\because \ b\cdot a = a\cdot b, c\cdot a = a\cdot c, c\cdot b = b\cdot c)$

www.exambite.in

$= 4 \cdot 1^2 + 1^2 + 1^2$ [Using (1)]

$= 6$

$\therefore \ |2\hat{a} + \hat{b} + \hat{c}| = \sqrt{6}.$

47. Here, $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$

Vector perpendicular to both $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -\hat{i} + \hat{j} + 0\hat{k} = -\hat{i} + \hat{j}$$

$\therefore$ Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{-\hat{i} + \hat{j}}{\sqrt{(-1)^2 + 1^2}} = \frac{1}{\sqrt{2}}\left(-\hat{i} + \hat{j}\right).$$

48. Let $\vec{a} = 2\hat{i} - 3\hat{k}$ and $\vec{b} = 4\hat{j} + 2\hat{k}$

The area of a parallelogram with $\vec{a}$ and $\vec{b}$ as its adjacent sides is given by $|\vec{a} \times \vec{b}|$.

Now, $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & -3 \\ 0 & 4 & 2 \end{vmatrix} = 12\hat{i} - 4\hat{j} + 8\hat{k}$

$\therefore \quad |\vec{a} \times \vec{b}| = \sqrt{(12)^2 + (-4)^2 + (8)^2} = \sqrt{144 + 16 + 64}$

$= \sqrt{224} = 4\sqrt{14}$ sq. units.

49. Let θ be the angle between the unit vectors $\vec{a}$ and $\vec{b}$.

$\therefore \ \cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \vec{a} \cdot \vec{b} \qquad (\because \ |\vec{a}| = 1 = |\vec{b}|) \ \ldots(1)$

Now $1 = \left|\sqrt{2}\vec{a} - \vec{b}\right|$

$\Rightarrow \ 1 = \left|\sqrt{2}\vec{a} - \vec{b}\right|^2 = \left(\sqrt{2}\vec{a} - \vec{b}\right) \cdot \left(\sqrt{2}\vec{a} - \vec{b}\right)$

$= 2|\vec{a}|^2 - \sqrt{2}\vec{a} \cdot \vec{b} - \vec{b} \cdot \sqrt{2}\vec{a} + |\vec{b}|^2 = 2 - 2\sqrt{2}\vec{a} \cdot \vec{b} + 1$

$(\because \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a})$

$= 3 - 2\sqrt{2}\ \vec{a} \cdot \vec{b}$

$\Rightarrow \ \vec{a} \cdot \vec{b} = \dfrac{1}{\sqrt{2}} \ \Rightarrow \ \cos\theta = \dfrac{1}{\sqrt{2}}$ [By using (1)]

$\therefore \quad \theta = \pi/4$

50. *Refer to answer 45.*

51. *Refer to answer 45.*

52. Given $|\vec{a}| = 1 = |\vec{b}|$, $|\vec{a} + \vec{b}| = 1$

$\Rightarrow |\vec{a} + \vec{b}|^2 = 1 \Rightarrow (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = 1$

$\Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = 1$

$\Rightarrow |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 = 1 \Rightarrow 1 + 2\vec{a} \cdot \vec{b} + 1 = 1$

$\Rightarrow 2\vec{a} \cdot \vec{b} = -1 \Rightarrow 2|\vec{a}| \cdot |\vec{b}| \cos\theta = -1$

$\Rightarrow 2 \cdot 1 \cdot 1 \cos\theta = -1$

$\Rightarrow \ \cos\theta = -\dfrac{1}{2} \Rightarrow \theta = 120°$

53. Given, $|\vec{a}| = 3, |\vec{b}| = \dfrac{2}{3}, |\vec{a} \times \vec{b}| = 1$

$\Rightarrow |\vec{a}||\vec{b}| \sin\theta = 1 \ \Rightarrow 3.\dfrac{2}{3}\sin\theta = 1$

$\Rightarrow \sin\theta = \dfrac{1}{2} \ \Rightarrow \theta = \dfrac{\pi}{6}$

54. Given: $\vec{a} \perp \vec{b} \ \Rightarrow \vec{a} \cdot \vec{b} = 0$

Also, $|\vec{a}| = 5$ and $|\vec{a} + \vec{b}| = 13$

$\Rightarrow \ |\vec{a} + \vec{b}|^2 = 13^2 \Rightarrow \ (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = 169$

$\Rightarrow \ \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = 169$

$\Rightarrow \ |\vec{a}|^2 + 0 + 0 + |\vec{b}|^2 = 169$

$\Rightarrow \ |\vec{b}|^2 = 169 - |\vec{a}|^2 = 169 - 5^2 = 144$

$\Rightarrow \ |\vec{b}| = 12$

55. The projection of the vector $\left(\hat{i} + \hat{j} + \hat{k}\right)$ along the vector $\hat{j}$ is $(\hat{i} + \hat{j} + \hat{k}) \cdot \left(\dfrac{\hat{j}}{\sqrt{0^2 + 1^2 + 0^2}}\right) = 1$

56. We have,

$\hat{i} \times \left(\hat{j} + \hat{k}\right) + \hat{j} \times \left(\hat{k} + \hat{i}\right) + \hat{k} \times \left(\hat{i} + \hat{j}\right)$

$= \hat{i} \times \hat{j} + \hat{i} \times \hat{k} + \hat{j} \times \hat{k} + \hat{j} \times \hat{i} + \hat{k} \times \hat{i} + \hat{k} \times \hat{j}$

$= \hat{k} - \hat{j} + \hat{i} - \hat{k} + \hat{j} - \hat{i} = \vec{0}.$

57. *Refer to answer 45.*

58. *Refer to answer 49.*

59. Let θ be the angle between the vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and y-axis *i.e.*, $\vec{b} = \hat{j}$

$$\therefore \cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{\left(\hat{i} + \hat{j} + \hat{k}\right).\hat{j}}{\left|\hat{i} + \hat{j} + \hat{k}\right||\hat{j}|}$$

$$= \frac{1}{\sqrt{1^2 + 1^2 + 1^2}\sqrt{1^2}} = \frac{1}{\sqrt{3}}$$

60. Let angle between the vectors $\vec{a}$ and $\vec{b}$ be θ.

Given: $|\vec{a}| = 8, |\vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 12$

$\Rightarrow |\vec{a}| \cdot |\vec{b}| \cdot |\sin\theta| = 12 \ \Rightarrow 8 \times 3\sin\theta = 12$

$\Rightarrow \sin\theta = \dfrac{12}{24} = \dfrac{1}{2} \Rightarrow \theta = \dfrac{\pi}{6}.$

www.examcores.in

61. Here, $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ and vector along x-axis is $\hat{i}$.

$\therefore$ Angle between $\vec{a}$ and $\hat{i}$ is given by

$$\cos\theta=\frac{\vec{a}.\hat{i}}{\sqrt{1^2+1^2+1^2}.\sqrt{1^2}}=\frac{(\hat{i}+\hat{j}+\hat{k})\cdot\hat{i}}{\sqrt{3}\cdot 1}=\frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta=\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

62. Here $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=15$, where $\vec{a}$ is unit vector.

$\Rightarrow \vec{x}\cdot\vec{x}+\vec{x}\cdot\vec{a}-\vec{a}\cdot\vec{x}-\vec{a}\cdot\vec{a}=15$

$\Rightarrow |\vec{x}|^2-|\vec{a}|^2=15 \qquad (\because \vec{x}\cdot\vec{a}=\vec{a}\cdot\vec{x})$

$\Rightarrow |\vec{x}|^2-1=15 \qquad (\because |\vec{a}|=1)$

$\Rightarrow |\vec{x}|^2=16=4^2 \Rightarrow |\vec{x}|=4$

63. Here, $\vec{a}=2\hat{i}+\lambda\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$

For $\vec{a}$ is perpendicular to $\vec{b}$, $\vec{a}\cdot\vec{b}=0$

$\Rightarrow (2\hat{i}+\lambda\hat{j}+\hat{k})\cdot(\hat{i}-2\hat{j}+3\hat{k})=0$

$\Rightarrow 2\times 1+\lambda(-2)+1\times 3=0$

$\Rightarrow 2-2\lambda+3=0$

$\Rightarrow \lambda=\frac{5}{2}$

64. *Refer to answer 63.*

65. Here, $\vec{a}=2\hat{i}-2\hat{j}+\hat{k}$

$\vec{b}=\hat{i}+2\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\hat{j}+4\hat{k}$

$\Rightarrow \quad \vec{b}+\vec{c}=3\hat{i}+\hat{j}+2\hat{k}$

$\therefore$ Projection of $\vec{b}+\vec{c}$ on $\vec{a}$

$$=\frac{(\vec{b}+\vec{c}).\vec{a}}{|\vec{a}|}=\frac{(3\hat{i}+\hat{j}+2\hat{k}).(2\hat{i}-2\hat{j}+\hat{k})}{|2\hat{i}-2\hat{j}+\hat{k}|}$$

$$=\frac{3\times 2+1\times(-2)+2\times 1}{\sqrt{2^2+(-2)^2+1^2}}=\frac{6}{3}=2$$

66. Here, $\vec{a}=\lambda\hat{i}+\hat{j}+4\hat{k}$, $\vec{b}=2\hat{i}+6\hat{j}+3\hat{k}$

Given : Projection of $\vec{a}$ on $\vec{b}$ $=4$

$$\Rightarrow \quad \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=4\Rightarrow\frac{(\lambda\hat{i}+\hat{j}+4\hat{k})\cdot(2\hat{i}+6\hat{j}+3\hat{k})}{|2\hat{i}+6\hat{j}+3\hat{k}|}=4$$

$$\Rightarrow \quad \frac{2\lambda+6+12}{\sqrt{2^2+6^2+3^2}}=4$$

$\Rightarrow \quad 2\lambda+18=4\times 7$

$\Rightarrow \quad 2\lambda=28-18=10\Rightarrow\lambda=5.$

67. $(\hat{i}\times\hat{j})\cdot\hat{k}+\hat{i}\cdot\hat{j}=\hat{k}\cdot\hat{k}+\hat{i}\cdot\hat{j}=1+0=1$

68. $(\hat{k}\times\hat{j})\cdot\hat{i}+\hat{j}\cdot\hat{k}=-\hat{i}\cdot\hat{i}+\hat{j}\cdot\hat{k}=-1+0=-1$

69. $(\hat{k}\times\hat{i})\cdot\hat{j}+\hat{i}\cdot\hat{k}=\hat{j}\cdot\hat{j}+0 \quad =1+0=1$

70. Let θ be the angle between $\vec{a}$ and $\vec{b}$, then

$$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}=\frac{\sqrt{6}}{\sqrt{3}\cdot 2}\times\frac{\sqrt{3}}{\sqrt{3}}$$

$$=\frac{\sqrt{18}}{3\times 2}=\frac{3\sqrt{2}}{3\times 2}=\frac{\sqrt{2}}{2}=\frac{1}{\sqrt{2}}$$

$\therefore \quad \theta=\cos^{-1}\frac{1}{\sqrt{2}}\Rightarrow\theta=\frac{\pi}{4}.$

71. Let $\vec{a}=\hat{i}-\hat{j}$ and $\vec{b}=\hat{i}+\hat{j}$

$|\vec{b}|=\sqrt{1^2+1^2}=\sqrt{1+1}=\sqrt{2}$

Also, $\vec{a}\cdot\vec{b}=(\hat{i}-\hat{j})\cdot(\hat{i}+\hat{j})=\hat{i}\cdot\hat{i}+\hat{i}\cdot\hat{j}-\hat{j}\cdot\hat{i}-\hat{j}\cdot\hat{j}=1-1=0$

$\therefore$ Projection of $\vec{a}$ on $\vec{b}=\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=\frac{0}{\sqrt{2}}=0$

72. Here, $|\vec{a}|=\sqrt{3}, |\vec{b}|=2$ and angle between $\vec{a}$ and $\vec{b}$ is 60°.

Now $\vec{a}\cdot\vec{b}=|\vec{a}|\cdot|\vec{b}|\cdot\cos 60^\circ=\sqrt{3}\times 2\times\frac{1}{2}=\sqrt{3}.$

73. Here, $\vec{a}=5\hat{i}+6\hat{j}-2\hat{k}$ and $\vec{b}=7\hat{i}+6\hat{j}+2\hat{k}$

Vector perpendicular to both $\vec{a}$ and $\vec{b}$ is

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\5&6&-2\\7&6&2\end{vmatrix}$$

$$=\hat{i}(12+12)-\hat{j}(10+14)+\hat{k}(30-42)$$

$$=24\hat{i}-24\hat{j}-12\hat{k}=12(2\hat{i}-2\hat{j}-\hat{k})$$

$\therefore$ Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$=\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}=\frac{24\hat{i}-24\hat{j}-12\hat{k}}{\sqrt{576+576+144}}=\frac{12(2\hat{i}-2\hat{j}-\hat{k})}{\sqrt{1296}}$$

$$=\frac{12(2\hat{i}-2\hat{j}-\hat{k})}{36}=\frac{1}{3}(2\hat{i}-2\hat{j}-\hat{k})$$

74. For any two non-zero vectors $\vec{a}$ and $\vec{b}$, we have

$$|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}| \Rightarrow |\vec{a}+\vec{b}|^2=|\vec{a}-\vec{b}|^2$$

$\Rightarrow \quad \vec{a}^2+\vec{b}^2+2\vec{a}\cdot\vec{b}=\vec{a}^2+\vec{b}^2-2\vec{a}\cdot\vec{b}$

$\Rightarrow \quad 4\vec{a}\cdot\vec{b}=0 \Rightarrow \vec{a}\cdot\vec{b}=0$

So, $\vec{a}$ and $\vec{b}$ are perpendicular vectors.

75. Let $A(2\hat{i}-\hat{j}+\hat{k}), B(3\hat{i}+7\hat{j}+\hat{k})$ and $C(5\hat{i}+6\hat{j}+2\hat{k})$

Then, $\overrightarrow{AB}=(3-2)\hat{i}+(7+1)\hat{j}+(1-1)\hat{k}=\hat{i}+8\hat{j}$

www.exambres.in

$\overrightarrow{AC} = (5-2)\hat{i} + (6+1)\hat{j} + (2-1)\hat{k} = 3\hat{i} + 7\hat{j} + \hat{k}$

$\overrightarrow{BC} = (5-3)\hat{i} + (6-7)\hat{j} + (2-1)\hat{k} = 2\hat{i} - \hat{j} + \hat{k}$

Now, angle between $\overrightarrow{AC}$ and $\overrightarrow{BC}$ is given by

$\Rightarrow \quad \cos\theta = \dfrac{\overrightarrow{AC}\cdot\overrightarrow{BC}}{|\overrightarrow{AC}||\overrightarrow{BC}|} = \dfrac{6-7+1}{\sqrt{9+49+1}\sqrt{4+1+1}}$

$\Rightarrow \quad \cos\theta = 0 \Rightarrow AC \perp BC$

So, A, B, C are the vertices of right angled triangle.

76. Given, $\hat{a} + \hat{b} = \hat{c}$

$\Rightarrow (\hat{a} + \hat{b})\cdot(\hat{a} + \hat{b}) = \hat{c}\cdot\hat{c}$

$\Rightarrow \hat{a}\cdot\hat{a} + \hat{a}\cdot\hat{b} + \hat{b}\cdot\hat{b} + \hat{b}\cdot\hat{a} = \hat{c}\cdot\hat{c}$

$\Rightarrow 1 + \hat{a}\cdot\hat{b} + 1 + \hat{a}\cdot\hat{b} = 1$

$\Rightarrow 2\,\hat{a}\cdot\hat{b} = -1$ ...(i)

Now $(\hat{a} - \hat{b})^2 = (\hat{a} - \hat{b})\cdot(\hat{a} - \hat{b})$

$= \hat{a}\cdot\hat{a} - \hat{a}\cdot\hat{b} - \hat{b}\cdot\hat{a} + \hat{b}\cdot\hat{b} = 1 - \hat{a}\cdot\hat{b} - \hat{a}\cdot\hat{b} + 1$

$= 2 - 2\hat{a}\cdot\hat{b} = 2 - (-1)$ [Using(i)]

$= 3$

$\therefore \quad |\hat{a} - \hat{b}| = \sqrt{3}$

77. Given, $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$

Now, $\vec{a} + \vec{b} = 4\hat{i} + \hat{j} - \hat{k}$

Also, $\vec{a} - \vec{b} = -2\hat{i} + 3\hat{j} - 5\hat{k}$

Now, $(\vec{a} + \vec{b})\cdot(\vec{a} - \vec{b}) = (4\hat{i} + \hat{j} - \hat{k})\cdot(-2\hat{i} + 3\hat{j} - 5\hat{k})$

$= (4)(-2) + (1)(3) + (-1)(-5) = -8 + 3 + 5 = 0$

Hence, $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular to each other.

78. Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$

Now, $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$

$\Rightarrow \quad (\hat{i} - 2\hat{j} + 3\hat{k})\cdot(3\hat{i} - 2\hat{j} + \hat{k}) = \sqrt{(1)^2 + (-2)^2 + (3)^2} \times \sqrt{(3)^2 + (-2)^2 + (1)^2}\cos\theta$

$\Rightarrow \quad 3 + 4 + 3 = \sqrt{14} \times \sqrt{14}\cos\theta$

$\Rightarrow \quad \cos\theta = \dfrac{10}{14}$

$\therefore \quad \sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \dfrac{100}{196}} = \sqrt{\dfrac{96}{196}}$

$\Rightarrow \quad \sin\theta = \dfrac{4\sqrt{6}}{14} = \dfrac{2\sqrt{6}}{7}$

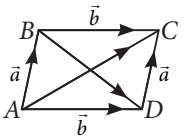
79. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$

Then diagonal $\overrightarrow{AC}$ of the parallelogram is

$\vec{p} = \vec{a} + \vec{b}$

$= \hat{i} + 2\hat{j} + 3\hat{k} + 2\hat{i} + 4\hat{j} - 5\hat{k}$

$= 3\hat{i} + 6\hat{j} - 2\hat{k}$



Therefore unit vector parallel to it is

$\dfrac{\vec{p}}{|\vec{p}|} = \dfrac{3\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{9+36+4}} = \dfrac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k})$

Now, diagonal $\overrightarrow{BD}$ of the parallelogram is

$\vec{p}' = \vec{b} - \vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k} - \hat{i} - 2\hat{j} - 3\hat{k} = \hat{i} + 2\hat{j} - 8\hat{k}$

Therefore unit vector parallel to it is

$\dfrac{\vec{p}'}{|\vec{p}'|} = \dfrac{\hat{i} + 2\hat{j} - 8\hat{k}}{\sqrt{1+4+64}} = \dfrac{1}{\sqrt{69}}(\hat{i} + 2\hat{j} - 8\hat{k})$

80. Given, ΔABC with vertices

$A(1,2,3) \equiv \hat{i} + 2\hat{j} + 3\hat{k}$, $B(2,-1,4) \equiv 2\hat{i} - \hat{j} + 4\hat{k}$,

$C(4,5,-1) \equiv 4\hat{i} + 5\hat{j} - \hat{k}$

Now $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (2\hat{i} - \hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$

$= \hat{i} - 3\hat{j} + \hat{k}$.

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = (4\hat{i} + 5\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$

$= 3\hat{i} + 3\hat{j} - 4\hat{k}$.

$\therefore \quad (\overrightarrow{AB} \times \overrightarrow{AC}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 3 & 3 & -4 \end{vmatrix} = 9\hat{i} + 7\hat{j} + 12\hat{k}$

Hence, area of ΔABC

$= \dfrac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \dfrac{1}{2}|9\hat{i} + 7\hat{j} + 12\hat{k}|$

$= \dfrac{1}{2}\sqrt{9^2 + 7^2 + 12^2} = \dfrac{1}{2}\sqrt{81 + 49 + 144}$

$= \dfrac{1}{2}\sqrt{274}$ sq. units

81. Given, position vector of $A = \hat{i} + \hat{j} + \hat{k}$

Position vector of $B = 2\hat{i} + 5\hat{j}$

Position vector of $C = 3\hat{i} + 2\hat{j} - 3\hat{k}$

Position vector of $D = \hat{i} - 6\hat{j} - \hat{k}$

$\therefore \quad \overrightarrow{AB} = (2\hat{i} + 5\hat{j}) - (\hat{i} + \hat{j} + \hat{k}) = \hat{i} + 4\hat{j} - \hat{k}$ and

$\overrightarrow{CD} = (\hat{i} - 6\hat{j} - \hat{k}) - (3\hat{i} + 2\hat{j} - 3\hat{k}) = -2\hat{i} - 8\hat{j} + 2\hat{k}$

Now $|\overrightarrow{AB}| = \sqrt{(1)^2 + (4)^2 + (1)^2} = \sqrt{18}$

www.examrace.in

$$|\overrightarrow{CD}| = \sqrt{(-2)^2 + (-8)^2 + (2)^2} = \sqrt{4+64+4}$$
$$= \sqrt{72} = 2\sqrt{18}$$

Let θ be the angle between $\overrightarrow{AB}$ and $\overrightarrow{CD}$.

$$\therefore \cos\theta = \frac{\overrightarrow{AB}\cdot\overrightarrow{CD}}{|\overrightarrow{AB}||\overrightarrow{CD}|} = \frac{(\hat{i}+4\hat{j}-\hat{k})\cdot(-2\hat{i}-8\hat{j}+2\hat{k})}{(\sqrt{18})(2\sqrt{18})}$$
$$= \frac{-2-32-2}{36} = \frac{-36}{36} = -1$$

$\Rightarrow \quad \cos\theta = -1 \quad \Rightarrow \quad \theta = \pi$

Since, angle between $\overrightarrow{AB}$ and $\overrightarrow{CD}$ is 180°.

$\therefore$ $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear.

82. Let $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$

Now, it is given that, $\vec{d}$ is perpendicular to $\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} - \hat{k}$

$\therefore \ \vec{d}\cdot\vec{b} = 0$ and $\vec{d}\cdot\vec{c} = 0$

$\Rightarrow \quad x - 4y + 5z = 0$ ...(i)

and $3x + y - z = 0$ ...(ii)

Also, $\vec{d}\cdot\vec{a} = 21$, where $\vec{a} = 4\hat{i} + 5\hat{j} - \hat{k}$

$\Rightarrow \quad 4x + 5y - z = 21$ ...(iii)

Eliminating z from (i) and (ii), we get

$16x + y = 0$ ...(iv)

Eliminating z from (ii) and (iii), we get

$x + 4y = 21$ ...(v)

Solving (iv) and (v), we get

$$x = \frac{-1}{3},\ y = \frac{16}{3}$$

Putting the values of x and y in (i), we get $z = \frac{13}{3}$

$\therefore \ \vec{d} = \frac{-1}{3}\hat{i} + \frac{16}{3}\hat{j} + \frac{13}{3}\hat{k}$ is the required vector.

83. $|\vec{a}| = |\vec{b}| = |\vec{c}|$ (Given) ...(i)

and $\vec{a}\cdot\vec{b} = 0,\ \vec{b}\cdot\vec{c} = 0,\ \vec{c}\cdot\vec{a} = 0$ ...(ii)

Let $(\vec{a}+\vec{b}+\vec{c})$ be inclined to vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ by angles α, β and γ respectively. Then

$$\cos\alpha = \frac{(\vec{a}+\vec{b}+\vec{c})\cdot\vec{a}}{|\vec{a}+\vec{b}+\vec{c}||\vec{a}|} = \frac{\vec{a}\cdot\vec{a}+\vec{b}\cdot\vec{a}+\vec{c}\cdot\vec{a}}{|\vec{a}+\vec{b}+\vec{c}||\vec{a}|}$$
$$= \frac{|\vec{a}|^2+0+0}{|\vec{a}+\vec{b}+\vec{c}||\vec{a}|} \qquad [\text{Using (ii)}]$$
$$= \frac{|\vec{a}|}{|\vec{a}+\vec{b}+\vec{c}|} \qquad \text{...(iii)}$$

Similarly, $\cos\beta = \frac{|\vec{b}|}{|\vec{a}+\vec{b}+\vec{c}|}$ ...(iv)

and $\cos\gamma = \frac{|\vec{c}|}{|\vec{a}+\vec{b}+\vec{c}|}$ ...(v)

From (i), (iii), (iv) and (v), we get

$\cos\alpha = \cos\beta = \cos\gamma \Rightarrow \alpha = \beta = \gamma$

Hence, the vector $\vec{a}+\vec{b}+\vec{c}$ is equally inclined to the vector $\vec{a}$, $\vec{b}$ and $\vec{c}$.

Also the angle between them is given as

$$\alpha = \cos^{-1}\left(\frac{|\vec{a}|}{|\vec{a}+\vec{b}+\vec{c}|}\right),\ \beta = \cos^{-1}\left(\frac{|\vec{b}|}{|\vec{a}+\vec{b}+\vec{c}|}\right),$$
$$\gamma = \cos^{-1}\left(\frac{|\vec{c}|}{|\vec{a}+\vec{b}+\vec{c}|}\right)$$

84. We have, $A(2\hat{i}-\hat{j}+\hat{k})$, $B(\hat{i}-3\hat{j}-5\hat{k})$ and $C(3\hat{i}-4\hat{j}-4\hat{k})$

Then, $\overrightarrow{AB} = (1-2)\hat{i} + (-3+1)\hat{j} + (-5-1)\hat{k}$
$$= -\hat{i} - 2\hat{j} - 6\hat{k}$$

$\overrightarrow{AC} = (3-2)\hat{i} + (-4+1)\hat{j} + (-4-1)\hat{k} = \hat{i} - 3\hat{j} - 5\hat{k}$

and $\overrightarrow{BC} = (3-1)\hat{i} + (-4+3)\hat{j} + (-4+5)\hat{k} = 2\hat{i} - \hat{j} + \hat{k}$

Now angle between $\overrightarrow{AC}$ and $\overrightarrow{BC}$ is given by

$$\cos\theta = \frac{(\overrightarrow{AC})(\overrightarrow{BC})}{|\overrightarrow{AC}||\overrightarrow{BC}|} = \frac{2+3-5}{\sqrt{1+9+25}\,.\,\sqrt{4+1+1}}$$

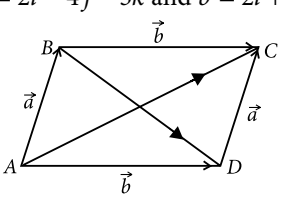
$\Rightarrow \quad \cos\theta = 0 \Rightarrow BC \perp AC$

So, A, B, C are vertices of right angled triangle.

Now area of $\Delta ABC = \frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{BC}|$

$$= \frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -3 & -5\\ 2 & -1 & 1\end{vmatrix} = \frac{1}{2}|(-3-5)\hat{i} - (1+10)\hat{j} + (-1+6)\hat{k}|$$
$$= \frac{1}{2}|-8\hat{i} - 11\hat{j} + 5\hat{k}|$$
$$= \frac{1}{2}\sqrt{64+121+25} = \frac{\sqrt{210}}{2} \text{ sq. units.}$$

85. Let $\vec{a} = 2\hat{i} - 4\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + 2\hat{j} + 3\hat{k}$



www.exambites.in

Then diagonal $\overrightarrow{AC}$ of the parallelogram is
$\vec{p} = \vec{a} + \vec{b}$
$= 2\hat{i} - 4\hat{j} - 5\hat{k} + 2\hat{i} + 2\hat{j} + 3\hat{k} = 4\hat{i} - 2\hat{j} - 2\hat{k}$
Therefore, unit vector parallel to it is
$$\frac{\vec{p}}{|\vec{p}|} = \frac{4\hat{i} - 2\hat{j} - 2\hat{k}}{\sqrt{16+4+4}} = \frac{2\hat{i} - \hat{j} - \hat{k}}{\sqrt{6}}$$
Now, diagonal $\overrightarrow{BD}$ of the parallelogram is
$\vec{p}' = \vec{b} - \vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k} - 2\hat{i} + 4\hat{j} + 5\hat{k} = 6\hat{j} + 8\hat{k}$
Therefore, unit vector parallel to it is
$$\frac{\vec{p}'}{|\vec{p}'|} = \frac{6\hat{j} + 8\hat{k}}{\sqrt{36+64}} = \frac{6\hat{j} + 8\hat{k}}{10} = \frac{3\hat{j} + 4\hat{k}}{5}$$
Now, $\vec{p} \times \vec{p}' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -2 & -2 \\ 0 & 6 & 8 \end{vmatrix}$
$= \hat{i}(-16+12) - \hat{j}(32-0) + \hat{k}(24-0)$
$= -4\hat{i} - 32\hat{j} + 24\hat{k}$
$\therefore$ Area of parallelogram $= \frac{|\vec{p} \times \vec{p}'|}{2}$
$= \frac{\sqrt{16+1024+576}}{2} = 2\sqrt{101}$ sq. units.

86. Two non zero vectors are parallel if and only if their cross product is zero vector.
So, we have to prove that cross product of $\vec{a} - \vec{d}$ and $\vec{b} - \vec{c}$ is zero vector.
$(\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c}) - (\vec{d} \times \vec{b}) + (\vec{d} \times \vec{c})$
Since, it is given that $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$
And, $\vec{d} \times \vec{b} = -\vec{b} \times \vec{d}$, $\vec{d} \times \vec{c} = -\vec{c} \times \vec{d}$
Therefore,
$(\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = (\vec{c} \times \vec{d}) - (\vec{b} \times \vec{d}) + (\vec{b} \times \vec{d}) - (\vec{c} \times \vec{d}) = \vec{0}$
Hence, $\vec{a} - \vec{d}$ is parallel to $\vec{b} - \vec{c}$, where
$\vec{a} \neq \vec{d}$ and $\vec{b} \neq \vec{c}$.

87. $(\vec{r} \times \hat{i}) \cdot (\vec{r} \times \hat{j}) + xy$
$= [(x\hat{i} + y\hat{j} + z\hat{k}) \times \hat{i}] \cdot [(x\hat{i} + y\hat{j} + z\hat{k}) \times \hat{j})] + xy$
$= (-y\hat{k} + z\hat{j}) \cdot (x\hat{k} - z\hat{i}) + xy = -xy + xy = 0$

88. Here, $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j}$, $\vec{c} = 3\hat{i} - 4\hat{j} - 5\hat{k}$
$\therefore$ $\vec{a} - \vec{b} = (\hat{i} + 2\hat{j} + \hat{k}) - (2\hat{i} + \hat{j}) = -\hat{i} + \hat{j} + \hat{k}$
$\vec{c} - \vec{b} = (3\hat{i} - 4\hat{j} - 5\hat{k}) - (2\hat{i} + \hat{j}) = \hat{i} - 5\hat{j} - 5\hat{k}$
Vector perpendicular to both $\vec{a} - \vec{b}$ and $\vec{c} - \vec{b}$ is
$$(\vec{a} - \vec{b}) \times (\vec{c} - \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 1 \\ 1 & -5 & -5 \end{vmatrix}$$
$= (-5+5)\hat{i} - (5-1)\hat{j} + (5-1)\hat{k} = -4\hat{j} + 4\hat{k}$
$\therefore$ Unit vector perpendicular to both $\vec{a} - \vec{b}$ and $\vec{c} - \vec{b}$
$= \frac{-4\hat{j} + 4\hat{k}}{|-4\hat{j} + 4\hat{k}|} = \frac{-4\hat{j} + 4\hat{k}}{\sqrt{(-4)^2 + 4^2}} = \frac{-4\hat{j} + 4\hat{k}}{4\sqrt{2}} = \frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$.

89. Given $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}| = 3$, $|\vec{b}| = 5$, $|\vec{c}| = 7$
We have $\vec{a} + \vec{b} + \vec{c} = 0$
$\Rightarrow \vec{a} + \vec{b} = -\vec{c} \Rightarrow |\vec{a} + \vec{b}|^2 = |-\vec{c}|^2$
$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b}) = |\vec{c}|^2$
$\Rightarrow 9 + 25 + 2|\vec{a}||\vec{b}|\cos\theta = 49$
$\Rightarrow 2 \times 3 \times 5 \times \cos\theta = 49 - 34 = 15$
$\Rightarrow \cos\theta = \frac{15}{30} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3} = 60°$

90. Here, $\vec{a} = \hat{i} + \hat{j} + \hat{k}$; $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = \lambda\hat{i} + 2\hat{j} + 3\hat{k}$
$\Rightarrow \vec{b} + \vec{c} = (2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}$
The unit vector along $\vec{b} + \vec{c}$ is $\vec{p} = \frac{\vec{b} + \vec{c}}{|\vec{b} + \vec{c}|}$
$= \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{(2+\lambda)^2 + 6^2 + (-2)^2}} = \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{\lambda^2 + 4\lambda + 44}}$
Also, $\vec{a} \cdot \vec{p} = 1$ (Given)
$\Rightarrow \frac{(2+\lambda) + 6 - 2}{\sqrt{\lambda^2 + 4\lambda + 44}} = 1$
$\Rightarrow \sqrt{\lambda^2 + 4\lambda + 44} = \lambda + 6$
$\Rightarrow \lambda^2 + 4\lambda + 44 = \lambda^2 + 12\lambda + 36$
$\Rightarrow 8\lambda = 8 \Rightarrow \lambda = 1$
$\therefore$ The required unit vector
$\vec{p} = \frac{(2+1)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{1+4+44}} = \frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k})$.

91. We have $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$
Let $\vec{r} = \vec{a} + \vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$
and $\vec{p} = \vec{a} - \vec{b} = -\hat{j} - 2\hat{k}$
A unit vector perpendicular to both $\vec{r}$ and $\vec{p}$ is given as $\pm\frac{\vec{r} \times \vec{p}}{|\vec{r} \times \vec{p}|}$.

www.exambites.in

Now, $\vec{r}\times\vec{p}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&4\\0&-1&-2\end{vmatrix}=-2\hat{i}+4\hat{j}-2\hat{k}$

So, the required unit vector is

$=\pm\dfrac{\left(-2\hat{i}+4\hat{j}-2\hat{k}\right)}{\sqrt{(-2)^2+4^2+(-2)^2}}=\mp\dfrac{\left(\hat{i}-2\hat{j}+\hat{k}\right)}{\sqrt{6}}.$

92. Here, $\vec{a}=2\hat{i}-3\hat{j}+\hat{k};\ \hat{b}=-\hat{i}+\hat{k};\ \vec{c}=2\hat{j}-\hat{k}$

$\therefore\quad \vec{a}+\vec{b}=\left(2\hat{i}-3\hat{j}+\hat{k}\right)+\left(-\hat{i}+\hat{k}\right)=\hat{i}-3\hat{j}+2\hat{k},$

$\vec{b}+\vec{c}=\left(-\hat{i}+\hat{k}\right)+\left(2\hat{j}-\hat{k}\right)=-\hat{i}+2\hat{j}$

$\therefore(\vec{a}+\vec{b})\times(\vec{b}+\vec{c})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-3&2\\-1&2&0\end{vmatrix}=-4\hat{i}-2\hat{j}-\hat{k}$

$\therefore$ Area of a parallelogram whose diagonals are $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$

$=\dfrac{1}{2}\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right|=\dfrac{1}{2}\left|-4\hat{i}-2\hat{j}-\hat{k}\right|$

$=\dfrac{1}{2}\sqrt{(-4)^2+(-2)^2+(-1)^2}=\dfrac{\sqrt{21}}{2}$ sq.units.

93. *Refer to answer 82.*

94. Here $|\vec{a}+\vec{b}|=|\vec{a}|$

$\Rightarrow|\vec{a}+\vec{b}|^2=|\vec{a}|^2\ \Rightarrow(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})=\vec{a}\cdot\vec{a}$

$\Rightarrow\vec{a}\cdot\vec{a}+\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b}=\vec{a}\cdot\vec{a}$

$\Rightarrow 2\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{b}=0\qquad[\because\ \vec{b}\cdot\vec{a}=\vec{a}\cdot\vec{b}]$

$\Rightarrow\left(2\vec{a}+\vec{b}\right)\cdot\vec{b}=0\ \Rightarrow\left(2\vec{a}+\vec{b}\right)\perp\vec{b}$

95. Given $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{j}-\hat{k}$

Let $\vec{c}=x\hat{i}+y\hat{j}+z\hat{k}$

Now we have, $\vec{a}\times\vec{c}=\vec{b}$

$\Rightarrow\ (\hat{i}+\hat{j}+\hat{k})\times(x\hat{i}+y\hat{j}+z\hat{k})=\hat{j}-\hat{k}$

$\Rightarrow\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&1\\x&y&z\end{vmatrix}=\hat{j}-\hat{k}$

$\Rightarrow\ \hat{i}(z-y)-\hat{j}(z-x)+\hat{k}(y-x)=\hat{j}-\hat{k}$

$\Rightarrow\ z-y=0,\ x-z=1$ and $y-x=-1$

$\Rightarrow\ y=z,\ x-z=1,\ x-y=1$ .....(i)

Also, we have $\vec{a}\cdot\vec{c}=3$

$\Rightarrow\ (\hat{i}+\hat{j}+\hat{k})\cdot(x\hat{i}+y\hat{j}+z\hat{k})=3$

$\Rightarrow\ x+y+z=3$

$\Rightarrow\ x+x-1+x-1=3$ [Using (i)]

$\Rightarrow\quad 3x-2=3\Rightarrow x=\dfrac{5}{3},\ y=\dfrac{2}{3},\ z=\dfrac{2}{3}$

Hence, $\vec{c}=\dfrac{5}{3}\hat{i}+\dfrac{2}{3}\hat{j}+\dfrac{2}{3}\hat{k}$

96. Here $\vec{a}=\hat{i}-\hat{j}+7\hat{k}\ ;\ \vec{b}=5\hat{i}-\hat{j}+\lambda\hat{k}$

$\therefore\ \vec{a}+\vec{b}=6\hat{i}-2\hat{j}+(7+\lambda)\hat{k}$

$\vec{a}-\vec{b}=-4\hat{i}+(7-\lambda)\hat{k}$

For $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ to be perpendicular,

$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=0$

$\Rightarrow\left[6\hat{i}-2\hat{j}+(7+\lambda)\hat{k}\right]\cdot\left[-4\hat{i}+(7-\lambda)\hat{k}\right]=0$

$\Rightarrow\quad 6\times(-4)+(7+\lambda)\times(7-\lambda)=0$

$\Rightarrow\quad -24+49-\lambda^2=0\ \Rightarrow\ \lambda^2=25\Rightarrow\ \lambda=\pm 5$

97. *Refer to answer 83.*

98. Let the required vector be $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}.$

Also let,

$\vec{a}=\hat{i}-\hat{j}+\hat{k},\ \vec{b}=2\hat{i}+\hat{j}-3\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}+\hat{k}$

$\vec{r}\cdot\vec{a}=4,\ \vec{r}\cdot\vec{b}=0,\ \vec{r}\cdot\vec{c}=2$ (Given)

$\Rightarrow\quad x-y+z=4$ ...(i)

$2x+y-3z=0$ ...(ii)

$x+y+z=2$ ...(iii)

Now (iii) – (i) $\Rightarrow 2y=-2\Rightarrow y=-1$

From (ii) and (iii)

$2x-3z-1=0,\ x+z-3=0\ \Rightarrow\ x=2,\ z=1$

$\therefore$ The required vector is $\vec{r}=2\hat{i}-\hat{j}+\hat{k}.$

99. Here, $\vec{a}=2\lambda^2\hat{i}+4\lambda\hat{j}+\hat{k}$ and $\vec{b}=7\hat{i}-2\hat{j}+\lambda\hat{k}$

If θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then $\cos\theta=\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

For θ to be obtuse, $\cos\theta<0\Rightarrow\vec{a}\cdot\vec{b}<0$

$\Rightarrow\ \left(2\lambda^2\hat{i}+4\lambda\hat{j}+\hat{k}\right)\cdot\left(7\hat{i}-2\hat{j}+\lambda\hat{k}\right)<0$

$\Rightarrow 2\lambda^2\cdot 7+4\lambda\cdot(-2)+1\cdot\lambda<0$

$\Rightarrow 14\lambda^2-7\lambda<0\Rightarrow\lambda(2\lambda-1)<0$

$\Rightarrow$ Either $\lambda<0,\ 2\lambda-1>0$ or $\lambda>0,\ 2\lambda-1<0$

$\Rightarrow$ Either $\lambda<0,\lambda>\dfrac{1}{2}$ or $\lambda>0,\lambda<\dfrac{1}{2}$

First alternative is impossible.

$\therefore\ \lambda>0,\lambda<\dfrac{1}{2}$ *i.e.*, $0<\lambda<\dfrac{1}{2}$ *i.e.*, $\lambda\in\left]0,\dfrac{1}{2}\right[$

100. Given, $|\vec{a}|=3,\ |\vec{b}|=4,|\vec{c}|=5$ ...(i)

and $\vec{a}\cdot(\vec{b}+\vec{c})=0,\ \vec{b}\cdot(\vec{c}+\vec{a})=0,\ \vec{c}\cdot(\vec{a}+\vec{b})=0$

www.exambites.in

$\therefore \quad \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{c} + \vec{b}\cdot\vec{a} + \vec{c}\cdot\vec{a} + \vec{c}\cdot\vec{b}$
$= 0 + 0 + 0 = 0$
$\Rightarrow 2(\vec{a}\cdot\vec{b}) + 2(\vec{b}\cdot\vec{c}) + 2(\vec{c}\cdot\vec{a}) = 0$ ...(ii)
Now $|\vec{a}+\vec{b}+\vec{c}|^2 = (\vec{a}+\vec{b}+\vec{c})\cdot(\vec{a}+\vec{b}+\vec{c})$
$= (\vec{a}+\vec{b}+\vec{c})\cdot\vec{a} + (\vec{a}+\vec{b}+\vec{c})\cdot\vec{b} + (\vec{a}+\vec{b}+\vec{c})\cdot\vec{c}$
$= \vec{a}\cdot\vec{a} + \vec{b}\cdot\vec{a} + \vec{c}\cdot\vec{a} + \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{b} + \vec{c}\cdot\vec{b} + \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{c}$
$= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b}) + 2(\vec{b}\cdot\vec{c}) + 2(\vec{c}\cdot\vec{a})$
$= 3^2 + 4^2 + 5^2 + 0$ [Using (i) and (ii)]
$= 50$
$\therefore |\vec{a}+\vec{b}+\vec{c}| = 5\sqrt{2}$.

101. Here $\vec{a} = 3\hat{i} - \hat{j},\ \vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$
We have to express : $\vec{b} = \vec{b}_1 + \vec{b}_2$, where
$\vec{b}_1 \parallel \vec{a}$ and $\vec{b}_2 \perp \vec{a}$
Let $\vec{b}_1 = \lambda\vec{a} = \lambda(3\hat{i} - \hat{j})$ and $\vec{b}_2 = x\hat{i} + y\hat{j} + z\hat{k}$
Now $\vec{b}_2 \perp \vec{a} \Rightarrow \vec{b}_2 \cdot \vec{a} = 0$
$\Rightarrow (x\hat{i} + y\hat{j} + z\hat{k})\cdot(3\hat{i} - \hat{j}) = 0$
$\Rightarrow \quad 3x - y = 0$ ...(i)
Now, $\vec{b} = \vec{b}_1 + \vec{b}_2$
$\Rightarrow 2\hat{i} + \hat{j} - 3\hat{k} = \lambda(3\hat{i} - \hat{j}) + (x\hat{i} + y\hat{j} + z\hat{k})$
On comparing, we get

$$\left.\begin{aligned} 2 &= 3\lambda + x \\ 1 &= -\lambda + y \end{aligned}\right] \Rightarrow x + 3y = 5 \quad \text{...(ii)}$$

and $-3 = z \Rightarrow z = -3$
Solving (i) and (ii), we get $x = \frac{1}{2}, y = \frac{3}{2}$
$\therefore 1 = -\lambda + y = -\lambda + \frac{3}{2} \Rightarrow \lambda = \frac{1}{2}$
Hence, $\vec{b}_1 = \lambda(3\hat{i} - \hat{j}) = \frac{3}{2}\hat{i} - \frac{1}{2}\hat{j}$
and $\vec{b}_2 = \frac{1}{2}\hat{i} + \frac{3}{2}\hat{j} - 3\hat{k}$

102. We have,
$|\vec{a}| = 5, |\vec{b}| = 12, |\vec{c}| = 13$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$
$\Rightarrow (\vec{a}+\vec{b}+\vec{c})^2 = |\vec{0}|^2$ (Squaring on both sides)
$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2[\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}] = 0$
$\Rightarrow 25 + 144 + 169 + 2[\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}] = 0$
$\Rightarrow 2[\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}] = -338$
$\Rightarrow \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = \frac{-338}{2} = -169$

103. *Refer to answer 82.*

104. *Refer to answer 76.*

105. *Refer to answer 91.*

106. We have $|\vec{a}| = 2$, $|\vec{b}| = 1$ and $\vec{a}\cdot\vec{b} = 1$
Now, $(3\vec{a} - 5\vec{b})\cdot(2\vec{a} + 7\vec{b})$
$= 6|\vec{a}|^2 + 21\ \vec{a}\cdot\vec{b} - 10\vec{a}\cdot\vec{b} - 35|\vec{b}|^2$
$= 6|\vec{a}|^2 + 11\vec{a}\cdot\vec{b} - 35|\vec{b}|^2$
$= 6(2)^2 + 11(1) - 35(1)^2 = 24 + 11 - 35 = 0$

107. *Refer to answer 80.*

108. *Refer to answer 97.*
Also the angle between them is given as

$$\alpha = \cos^{-1}\left(\frac{|\vec{a}|}{|\vec{a}+\vec{b}+\vec{c}|}\right),$$

$$\beta = \cos^{-1}\left(\frac{|\vec{b}|}{|\vec{a}+\vec{b}+\vec{c}|}\right),$$

$$\gamma = \cos^{-1}\left(\frac{|\vec{c}|}{|\vec{a}+\vec{b}+\vec{c}|}\right)$$

109. *Refer to answer 100.*

110. Since the vectors are coplanar.

$$\therefore \quad \begin{vmatrix} 1 & 3 & 1 \\ 2 & -1 & -1 \\ 0 & \lambda & 3 \end{vmatrix} = 0$$

$\Rightarrow \quad 1(-3 + \lambda) - 3(6 - 0) + 1(2\lambda - 0) = 0$
$\Rightarrow \quad -3 + \lambda - 18 + 2\lambda = 0$
$\Rightarrow \quad 3\lambda - 21 = 0 \ \Rightarrow \ \lambda = 7$

111. Here $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k},\ \vec{b} = -\hat{i} + 2\hat{j} + \hat{k},$
$\vec{c} = 3\hat{i} + \hat{j} + 2\hat{k}$

$$\text{Now, } \vec{b}\times\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 3\hat{i} + 5\hat{j} - 7\hat{k}$$

$\therefore \ \vec{a}\cdot(\vec{b}\times\vec{c}) = (2\hat{i} + \hat{j} + 3\hat{k})\cdot(3\hat{i} + 5\hat{j} - 7\hat{k})$
$= 2 \times 3 + 1 \times 5 + 3 \times (-7)$
$= 6 + 5 - 21 = -10$

112. Given, $\vec{a} = \hat{i} - \hat{j} + 2\hat{k},\ \vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and
$\vec{c} = 2\hat{i} - \hat{j} + 3\hat{k}$
$\therefore \quad 2\vec{a} = 2\hat{i} - 2\hat{j} + 4\hat{k}$
$-\vec{b} = -3\hat{i} - 4\hat{j} + 5\hat{k}$
$3\vec{c} = 6\hat{i} - 3\hat{j} + 9\hat{k}$

www.exambites.in

Now, $2\vec{a}\cdot(-\vec{b}\times 3\vec{c}) = \begin{vmatrix} 2 & -2 & 4 \\ -3 & -4 & 5 \\ 6 & -3 & 9 \end{vmatrix}$

$= 2(-36 + 15) + 2(-27 - 30) + 4(9 + 24)$

$= 2(-21) - 2(57) + 4(33)$

$= -42 - 114 + 132 = -24$

$\therefore$ Volume of parallelepiped

$|2\vec{a}\cdot(-\vec{b}\times 3\vec{c})| = |-24| = 24$ cubic units

113. Given, $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = -3\hat{i} + \hat{j} + 2\hat{k}$

Now, $\vec{b}\times\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ -3 & 1 & 2 \end{vmatrix}$

$= \hat{i}(-4-1) - \hat{j}(2+3) + \hat{k}(1-6)$

$= -5\hat{i} - 5\hat{j} - 5\hat{k}$

$\therefore\ \vec{a}\cdot(\vec{b}\times\vec{c}) = (2\hat{i} + 3\hat{j} + \hat{k})\cdot(-5\hat{i} - 5\hat{j} - 5\hat{k})$

$= -10 - 15 - 5 = -30$

114. Given points are $A(x, -1, -1)$, $B(4, 5, 1)$, $C(3, 9, 4)$ and $D(-4, 4, 4)$.

$\overrightarrow{AB} = (4-x)\hat{i} + (5+1)\hat{j} + (1+1)\hat{k}$

$= (4-x)\hat{i} + 6\hat{j} + 2\hat{k}$

$\overrightarrow{AC} = (3-x)\hat{i} + (9+1)\hat{j} + (4+1)\hat{k}$

$= (3-x)\hat{i} + 10\hat{j} + 5\hat{k}$

$\overrightarrow{AD} = (-4-x)\hat{i} + (4+1)\hat{j} + (4+1)\hat{k}$

$= -(4+x)\hat{i} + 5\hat{j} + 5\hat{k}$

The given points will be coplanar iff

$[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0$

Now, $[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0 \Rightarrow \begin{vmatrix} 4-x & 6 & 2 \\ 3-x & 10 & 5 \\ -(4+x) & 5 & 5 \end{vmatrix} = 0$

$\Rightarrow$ $(4 - x)(50 - 25) - 6(15 - 5x + 20 + 5x)$

$+ 2(15 - 5x + 40 + 10x) = 0$

$\Rightarrow$ $(4 - x)(25) - 6(35) + 2(55 + 5x) = 0$

$\Rightarrow$ $100 - 25x - 210 + 110 + 10x = 0$

$\Rightarrow$ $-15x = 0$

$\Rightarrow$ $x = 0$

115. We have, $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

(a) $c_1 = 1, c_2 = 2$ $\therefore$ $\vec{c} = \hat{i} + 2\hat{j} + c_3\hat{k}$

Given that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar

$\therefore$ $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 2 & c_3 \end{vmatrix} = 0$

$\Rightarrow$ $-1(c_3) + 1(2) = 0 \Rightarrow c_3 = 2$

(b) $c_2 = -1, c_3 = 1$ $\therefore$ $\vec{c} = c_1\hat{i} - \hat{j} + \hat{k}$

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar.

$\therefore$ $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ c_1 & -1 & 1 \end{vmatrix} = 0$

$\Rightarrow$ $-1(1) + 1(-1) = 0 \Rightarrow -2 = 0$, which is false.

So, no value of c_1 can make $\vec{a}$, $\vec{b}$ and $\vec{c}$ coplanar.

116. Let A, B, C, D be the given points. The given points will be coplanar iff any one of the following triads of vectors are coplanar.

$\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$; $\overrightarrow{BC}, \overrightarrow{BA}, \overrightarrow{BD}$ *etc.*

If $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ are coplanar, then their scalar triple product $[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0$ where, $A(3\hat{i} + 6\hat{j} + 9\hat{k})$, $B(\hat{i} + 2\hat{j} + 3\hat{k})$, $C(2\hat{i} + 3\hat{j} + \hat{k})$ and $D(4\hat{i} + 6\hat{j} + \lambda\hat{k})$.

Now, $\overrightarrow{AB} = (\hat{i} + 2\hat{j} + 3\hat{k}) - (3\hat{i} + 6\hat{j} + 9\hat{k})$

$= -2\hat{i} - 4\hat{j} - 6\hat{k}$

$\overrightarrow{AC} = (2\hat{i} + 3\hat{j} + \hat{k}) - (3\hat{i} + 6\hat{j} + 9\hat{k}) = -\hat{i} - 3\hat{j} - 8\hat{k}$

$\overrightarrow{AD} = (4\hat{i} + 6\hat{j} + \lambda\hat{k}) - (3\hat{i} + 6\hat{j} + 9\hat{k}) = \hat{i} + (\lambda - 9)\hat{k}$

$\therefore$ $[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = \begin{vmatrix} -2 & -4 & -6 \\ -1 & -3 & -8 \\ 1 & 0 & \lambda - 9 \end{vmatrix} = 0$

$\Rightarrow$ $-2(-3\lambda + 27) + 4(-\lambda + 9 + 8) - 6(0 + 3) = 0$

$\Rightarrow$ $2(3\lambda - 27) - 4(\lambda - 17) - 6(3) = 0$

$\Rightarrow$ $6\lambda - 54 - 4\lambda + 68 - 18 = 0$

$\Rightarrow$ $2\lambda - 4 = 0 \Rightarrow \lambda = 2$

117. Since, $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$ are coplanar.

$\therefore$ $(\vec{a} + \vec{b})\cdot\left[(\vec{b} + \vec{c})\times(\vec{c} + \vec{a})\right] = 0$

$\Rightarrow$ $(\vec{a} + \vec{b})\cdot\left[\vec{b}\times\vec{c} + \vec{b}\times\vec{a} + \vec{c}\times\vec{c} + \vec{c}\times\vec{a}\right] = 0$

$\Rightarrow$ $(\vec{a} + \vec{b})\cdot(\vec{b}\times\vec{c} + \vec{b}\times\vec{a} + \vec{c}\times\vec{a}) = 0$ $\left[\because\ \vec{c}\times\vec{c} = 0\right]$

www.exambites.in

$\Rightarrow \quad \vec{a}\cdot(\vec{b}\times\vec{c})+\vec{a}\cdot(\vec{b}\times\vec{a})+\vec{a}\cdot(\vec{c}\times\vec{a})+\vec{b}\cdot(\vec{b}\times\vec{c})$
$+\vec{b}\cdot(\vec{b}\times\vec{a})+\vec{b}\cdot(\vec{c}\times\vec{a})=0$

$\Rightarrow \quad 2\left[\vec{a}\cdot(\vec{b}\times\vec{c})\right]=0 \Rightarrow \vec{a}\cdot(\vec{b}\times\vec{c})=0$

$\Rightarrow \quad \vec{a}, \vec{b}$ and $\vec{c}$ are coplanar.

118. Here position vectors of *A*, *B*, *C* and *D* are $4\hat{i}+5\hat{j}+\hat{k}, -\hat{j}-\hat{k}, 3\hat{i}+\lambda\hat{j}+4\hat{k}$ and $-4\hat{i}+4\hat{j}+4\hat{k}$ respectively.

$\Rightarrow \quad \overrightarrow{AB}=-\hat{j}-\hat{k}-\left(4\hat{i}+5\hat{j}+\hat{k}\right)=-4\hat{i}-6\hat{j}-2\hat{k}$

$\overrightarrow{AC}=\left(3\hat{i}+\lambda\hat{j}+4\hat{k}\right)-\left(4\hat{i}+5\hat{j}+\hat{k}\right)=-\hat{i}+(\lambda-5)\hat{j}+3\hat{k}$

$\overrightarrow{AD}=\left(-4\hat{i}+4\hat{j}+4\hat{k}\right)-\left(4\hat{i}+5\hat{j}+\hat{k}\right)=-8\hat{i}-\hat{j}+3\hat{k}$

$\therefore$ For points *A, B, C, D* to be coplanar

$\Leftrightarrow$ Vectors $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ will be coplanar

$\Leftrightarrow \quad [\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}]=0$

$$\Rightarrow \begin{vmatrix} -4 & -6 & -2 \\ -1 & \lambda-5 & 3 \\ -8 & -1 & 3 \end{vmatrix}=0$$

$\Rightarrow \quad -4(3\lambda-15+3)+6(-3+24)-2(1+8\lambda-40)=0$
$\Rightarrow \quad -12\lambda+48+126+78-16\lambda=0$
$\Rightarrow \quad 28\lambda=252 \Rightarrow \lambda=9.$

119. We know that $[\vec{a},\vec{b},\vec{c}]=\vec{a}\cdot(\vec{b}\times\vec{c})$

$\therefore \quad$ L.H.S. $=[\vec{a},\vec{b}+\vec{c},\vec{d}]=\vec{a}\cdot[(\vec{b}+\vec{c})\times\vec{d}]$

$=\vec{a}\cdot[\vec{b}\times\vec{d}+\vec{c}\times\vec{d}]=\vec{a}\cdot(\vec{b}\times\vec{d})+\vec{a}\cdot(\vec{c}\times\vec{d})$

$=[\vec{a},\vec{b},\vec{d}]+[\vec{a},\vec{c},\vec{d}]=$ R.H.S.

120. We have, $[\vec{a}+\vec{b}\ \vec{b}+\vec{c}\ \vec{c}+\vec{a}]$

$=(\vec{a}+\vec{b})\cdot\{(\vec{b}+\vec{c})\times(\vec{c}+\vec{a})\}$

$=(\vec{a}+\vec{b})\cdot\{(\vec{b}\times\vec{c})+(\vec{b}\times\vec{a})+(\vec{c}\times\vec{c})+(\vec{c}\times\vec{a})\}$

$=(\vec{a}+\vec{b})\cdot\{(\vec{b}\times\vec{c})+(\vec{b}\times\vec{a})+(\vec{c}\times\vec{a})\}$ $[\because\ \vec{c}\times\vec{c}=0]$

$=\vec{a}\cdot(\vec{b}\times\vec{c})+\vec{a}\cdot(\vec{b}\times\vec{a})+\vec{a}\cdot(\vec{c}\times\vec{a})+$
$\vec{b}\cdot(\vec{b}\times\vec{c})+\vec{b}\cdot(\vec{b}\times\vec{a})+\vec{b}\cdot(\vec{c}\times\vec{a})$

$=[\vec{a}\ \vec{b}\ \vec{c}]+[\vec{b}\ \vec{c}\ \vec{a}]$

[$\because$ Scalar triple product with two equal vectors is 0]

$=[\vec{a}\ \vec{b}\ \vec{c}]+[\vec{a}\ \vec{b}\ \vec{c}]$ $(\because\ [\vec{b}\ \vec{c}\ \vec{a}]=[\vec{a}\ \vec{b}\ \vec{c}])$

$=2[\vec{a}\ \vec{b}\ \vec{c}]$

121. *Refer to answer 118.*

122. If the vectors $\vec{a}+\vec{b}, \vec{b}+\vec{c}$ and $\vec{c}+\vec{a}$ are coplanar, then $\left[\vec{a}+\vec{b}\ \vec{b}+\vec{c}\ \vec{c}+\vec{a}\right]=0$

$\Leftrightarrow (\vec{a}+\vec{b})\cdot\left[(\vec{b}+\vec{c})\times(\vec{c}+\vec{a})\right]=0$

$\Leftrightarrow (\vec{a}+\vec{b})\cdot\left[\vec{b}\times\vec{c}+\vec{b}\times\vec{a}+\vec{c}\times\vec{c}+\vec{c}\times\vec{a}\right]=0$

$\Leftrightarrow (\vec{a}+\vec{b})\cdot\left[\vec{b}\times\vec{c}+\vec{b}\times\vec{a}+\vec{0}+\vec{c}\times\vec{a}\right]=0$

$\Leftrightarrow \vec{a}\cdot(\vec{b}\times\vec{c})+\vec{a}\cdot(\vec{b}\times\vec{a})+\vec{a}\cdot(\vec{c}\times\vec{a})$
$+\vec{b}\cdot(\vec{b}\times\vec{c})+\vec{b}\cdot(\vec{b}\times\vec{a})+\vec{b}\cdot(\vec{c}\times\vec{a})=0$

$\Leftrightarrow \left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{a}\ \vec{b}\ \vec{a}\right]+\left[\vec{a}\ \vec{c}\ \vec{a}\right]+$
$\left[\vec{b}\ \vec{b}\ \vec{c}\right]+\left[\vec{b}\ \vec{b}\ \vec{a}\right]+\left[\vec{b}\ \vec{c}\ \vec{a}\right]=0$

$\Leftrightarrow \left[\vec{a}\ \vec{b}\ \vec{c}\right]+0+0+0+0+\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$

$\Leftrightarrow 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0 \Leftrightarrow \left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$

$\therefore$ The vectors $\vec{a},\vec{b},\vec{c}$ are coplanar.

Hence the vectors $\vec{a},\vec{b},\vec{c}$ are coplanar if and only if $\vec{a}+\vec{b}, \vec{b}+\vec{c}$ and $\vec{c}+\vec{a}$ are coplanar.

123. *Refer to answer 122.*

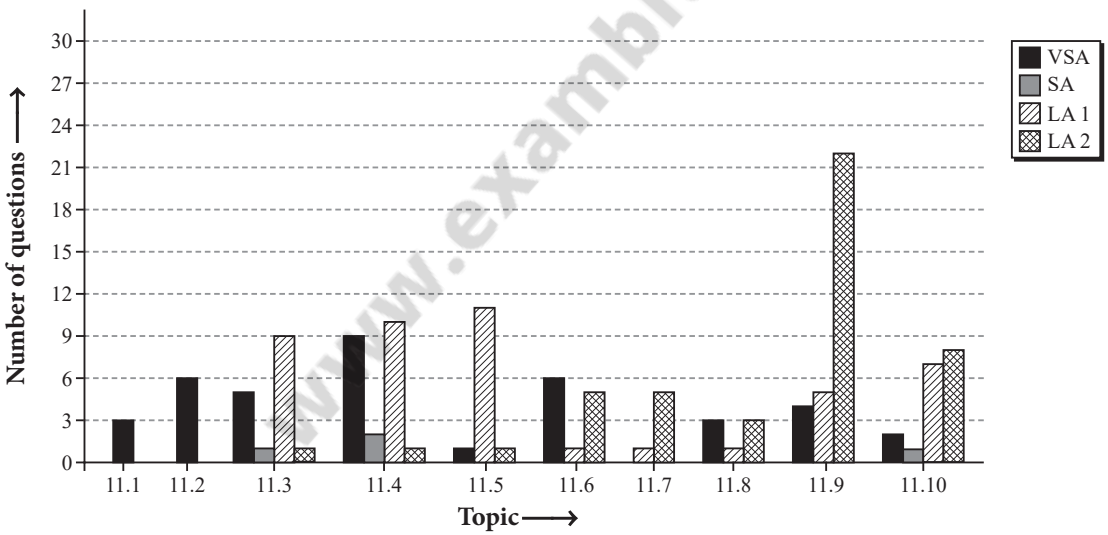
www.examin.in

CHAPTER
11

Three Dimensional Geometry

11.1 Introduction
11.2 Direction Cosines and Direction Ratios of a Line
11.3 Equation of a Line in Space
11.4 Angle between Two Lines
11.5 Shortest Distance between Two Lines
11.6 Plane
11.7 Coplanarity of Two Lines
11.8 Angle between Two Planes
11.9 Distance of a Point from a Plane
11.10 Angle between a Line and a Plane

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



- Maximum weightage is of *Distance of a Point from a Plane*.
- Maximum LA 1 type questions were asked from *Shortest Distance between Two lines*.
- Maximum LA 2 type questions were asked from *Distance of a Point from a Plane*.

QUICK RECAP

DIRECTION COSINES AND DIRECTION RATIOS OF A LINE

If α, β, γ are the angles which a vector $\overrightarrow{OP}$ makes with the positive directions of the co-ordinate axes OX, OY and OZ respectively, then $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are known as the direction

cosines of $\overrightarrow{OP}$ and are generally denoted by l, m and n respectively *i.e.* $l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$.

Let l, m, n be the direction cosines of a vector $\vec{r}$ and a, b, c be three numbers such that $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$.

Then a, b, c are known as direction ratios or direction numbers of vector $\vec{r}$.

Note :

(i) $l^2 + m^2 + n^2 = 1$ *i.e.*, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

(ii) Direction ratios of a line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are $x_2 - x_1$, $y_2 - y_1$, $z_2 - z_1$ and direction cosines are $\pm\frac{x_2 - x_1}{|\overrightarrow{PQ}|}, \pm\frac{y_2 - y_1}{|\overrightarrow{PQ}|}, \pm\frac{z_2 - z_1}{|\overrightarrow{PQ}|}$.

EQUATION OF A LINE IN SPACE

- Vector equation of a line passing through a given point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$, $\lambda \in R$.
- Cartesian equation of a line passing through a point $A(x_1, y_1, z_1)$ having position vector $\vec{a}$ and in the direction of a vector having a_1, b_1, c_1 as direction ratios, is $\frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$

Equation of Line Passing Through Two Given Points

- Let $\vec{a}$ and $\vec{b}$ be position vectors of two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ respectively. Let $\vec{r}$ be the position vector of $P(x, y, z)$. Then, equation of line is
  - **Vector form:** $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}), \lambda \in R$
  - **Cartesian form :** $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$

ANGLE BETWEEN TWO LINES

- **Vector form :** If θ is the angle between the lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$, then

$$\cos\theta = \left|\frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}\right|$$

- **Cartesian form :** If θ is the angle between the lines $l_1 : \frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$ and

$l_2 : \frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2}$

then $\cos\theta = \left|\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\,.\sqrt{a_2^2 + b_2^2 + c_2^2}}\right|$

Note : The lines are perpendicular to each other if $a_1a_2 + b_1b_2 + c_1c_2 = 0$ and parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

SHORTEST DISTANCE BETWEEN TWO LINES

- **Distance between skew lines**
  - **Vector form :** Let l_1 and l_2 be two lines whose equations are $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ respectively. Let $\overrightarrow{PQ}$ be the shortest distance vector between l_1 and l_2. Then,

$$PQ = \left|\frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$$

  - **Cartesian form :** Let two skew lines be

$l_1 : \frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$ and

$l_2 : \frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2}$.

The shortest distance between l_1 and l_2 is given by

$$\left|\frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - a_1c_2)^2 + (a_1b_2 - a_2b_1)^2}}\right|.$$

Note : If the shortest distance between two lines is zero, then the lines are intersecting.

- **Distance between parallel lines**
  - The shortest distance between two parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}$ is

$$d = \left|\frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right|.$$

www.exambites.in

PLANE

A plane is a surface such that if any two points are taken on it, the line segment joining them lies completely on the surface.

Equation of a plane in normal form

- **Vector form :** When a unit vector $(\hat{n})$ is normal to the plane and d is the perpendicular distance from the origin, then the equation of plane is $\vec{r}.\hat{n} = d$.
- **Cartesian form :** If unit vector $\hat{n} = l\hat{i} + m\hat{j} + n\hat{k}$, where l, m, n are direction cosines and d, the perpendicular distance from origin to plane, then equation of the plane is $lx + my + nz = d$.

Equation of a plane perpendicular to a given vector and passing through a given point

Equation of a plane passing through a point with position vector $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ and perpendicular to the vector $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ is

- **Vector form :** $(\vec{r} - \vec{a}).\vec{n} = 0$ or $\vec{r}.\vec{n} = \vec{a}.\vec{n}$
- **Cartesian form :**
  $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$.

Equation of a plane passing through three non-collinear points : If a plane passes through points $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ having position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively, and let $P(x, y, z)$ be any point on the plane having position vector $\vec{r}$, then equation of plane is

- **Vector form :** $(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$
- **Cartesian form :**

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

Intercept form of the equation of a plane : If a plane makes intercepts a, b and c on x, y and z-axes respectively, then equation of plane in intercept form is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Equation of a plane passing through the intersection of two planes

- **Vector form :** The equation of a plane passing through the intersection of the planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is $(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$ or we write $\vec{r} \cdot (\vec{n}_1 + \lambda\vec{n}_2) = d_1 + \lambda d_2$.
- **Cartesian form :** The equation of a plane passing through the intersection of planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is $(a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$, where λ is a constant.

COPLANARITY OF TWO LINES

Vector form : Let $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ be two lines, then these lines are coplanar iff $[\vec{a}_2 - \vec{a}_1 \ \ \vec{b}_1 \ \ \vec{b}_2] = 0$

Cartesian form :

Let $\frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$ and

$\frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2}$ are two lines, then these lines are coplanar iff

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

ANGLE BETWEEN TWO PLANES

Vector form : Let $\vec{r} \cdot \vec{n}_1 = \vec{d}_1$ and $\vec{r} \cdot \vec{n}_2 = \vec{d}_2$ be two planes making an angle θ with each other. Then θ is the angle between their normals $\vec{n}_1$ and $\vec{n}_2$ respectively.

Therefore, $\cos\theta = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|} \right|$

Cartesian form : If $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ be two planes and θ is the angle between them, where a_1, b_1, c_1 and a_2, b_2, c_2 are direction ratios of normals to the planes, then

$$\cos\theta = \left| \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

Note : The planes are perpendicular to each other if $\vec{n}_1 \cdot \vec{n}_2 = 0$ and parallel if $\vec{n}_1 = \lambda\vec{n}_2$, where λ is a scalar.

www.exambits.in

DISTANCE OF A POINT FROM A PLANE

- **Vector form :** The length of the perpendicular from a point having position vector $\vec{a}$ to the plane $\vec{r}.\vec{n} = d$ is given by $\frac{|\vec{a}\cdot\vec{n} - d|}{|\vec{n}|}$.
- **Cartesian form :** The length of the perpendicular from a point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by $\left|\frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}\right|$.

ANGLE BETWEEN A LINE AND A PLANE

- **Vector form :** If θ is the angle between a line $\vec{r} = \vec{a} + \lambda\vec{b}$ and normal to the plane $\vec{r}\cdot\vec{n} = d$, then $\cos\theta = \left|\frac{\vec{b}\cdot\vec{n}}{|\vec{b}||\vec{n}|}\right|$.

So, the angle ϕ between the line and the plane is 90° – θ *i.e.*, sin(90° – θ) = cos θ

$$\Rightarrow \quad \sin\phi = \left|\frac{\vec{b}\cdot\vec{n}}{|\vec{b}||\vec{n}|}\right|$$

- **Cartesian form :** Let a_1, a_2, a_3 are the direction ratios of line and $lx + my + nz + d = 0$ be the equation of plane. If θ is the angle between the line and plane, then

$$\sin\theta = \left|\frac{a_1 l + a_2 m + a_3 n}{\sqrt{a_1^2 + a_2^2 + a_3^2}\,\sqrt{l^2 + m^2 + n^2}}\right|$$

www.exambites.in

Previous Years' CBSE Board Questions

11.1 Introduction

VSA (1 mark)

1. The length of the perpendicular drawn from the point (4, –7, 3) on the y-axis is
(a) 3 units (b) 4 units
(c) 5 units (d) 7 units *(2020)*

2. The vector equation of XY-plane is
(a) $\vec{r}\cdot\hat{k}=0$ (b) $\vec{r}\cdot\hat{j}=0$
(c) $\vec{r}\cdot\hat{i}=0$ (d) $\vec{r}\cdot\vec{n}=1$ *(2020)*

3. Write the distance of a point $P(a, b, c)$ from x-axis. *(Delhi 2014C)*

11.2 Direction Cosines and Direction Ratios of a Line

VSA (1 mark)

4. If a line makes angles 90°, 135°, 45° with the x, y and z axes respectively, find its direction cosines. *(Delhi 2019)*

5. If a line makes angles 90° and 60° respectively with the positive directions of x and y axes, find the angle which it makes with the positive direction of z-axis. *(Delhi 2017)*

6. If a line makes angles 90°, 60° and θ with x, y and z-axis respectively, where θ is acute, then find θ. (*Delhi 2015*)

7. If a line makes angles α, β, γ with the positive direction of coordinate axes, then write the value of $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$. (*Delhi 2015C*)

8. If a line has direction ratios 2, –1, –2, then what are its direction cosines? (*Delhi 2012*)

9. Write the direction cosines of the line joining the points (1, 0, 0) and (0, 1, 1). *(AI 2011)*

11.3 Equation of a Line in Space

VSA (1 mark)

10. The vector equation of a line which passes through the points (3, 4, –7) and (1, –1, 6) is _____. *(2020)*

11. The equation of a line are $5x - 3 = 15y + 7 = 3 - 10z$. Write the direction cosines of the line. (*AI 2015*)

12. If the cartesian equation of a line is $\frac{3-x}{5}=\frac{y+4}{7}=\frac{2z-6}{4}$, write the vector equation for the line. (*AI 2014*)

13. Find the direction cosines of the line $\frac{4-x}{2}=\frac{y}{6}=\frac{1-z}{3}$. (*Delhi 2013C*)

14. Write the vector equation of a line given by $\frac{x-5}{3}=\frac{y+4}{7}=\frac{z-6}{2}$. *(AI 2011)*

SA (2 marks)

15. The x-coordinate of a point on the line joining the points $P(2, 2, 1)$ and $Q(5, 1, -2)$ is 4. Find its z-coordinate. *(AI 2017)*

LA 1 (4 marks)

16. Prove that the line through $A(0, -1, -1)$ and $B(4, 5, 1)$ intersects the line through $C(3, 9, 4)$ and $D(-4, 4, 4)$. *(Foreign 2016)*

17. Show that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ intersect. Also find their point of intersection. *(Delhi 2014)*

18. Show that lines $\vec{r}=(\hat{i}+\hat{j}-\hat{k})+\lambda(3\hat{i}-\hat{j})$ and $\vec{r}=(4\hat{i}-\hat{k})+\mu(2\hat{i}+3\hat{k})$ intersect. Also, find their point of intersection. *(Delhi 2014)*

19. Show that the lines
$\vec{r}=3\hat{i}+2\hat{j}-4\hat{k}+\lambda(\hat{i}+2\hat{j}+2\hat{k})$;
$\vec{r}=5\hat{i}-2\hat{j}+\mu(3\hat{i}+2\hat{j}+6\hat{k})$ are intersecting.
Hence find their point of intersection. *(AI 2013)*

20. Find the image of the point (2, –1, 5) in the line $\frac{x-11}{10}=\frac{y+2}{-4}=\frac{z+8}{-11}$. Also find the

www.examfree.in

equation of the line joining the given point and its image. Find the length of that line segment also. *(AI 2013C)*

21. Show that the lines $\vec{r} = \hat{i} + \hat{j} - \hat{k} + \lambda(3\hat{i} - \hat{j})$ and $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$ intersects. *(AI 2013C)*

22. Using vectors, show that the points $A(-2, 3, 5)$ $B(7, 0, -1)$ $C(-3, -2, -5)$ and $D(3, 4, 7)$ are such that AB and CD intersect at the point $P(1, 2, 3)$. *(AI 2012C)*

23. Find the perpendicular distance of the point $(1, 0, 0)$ from the line $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$. Also find the coordinates of the foot of the perpendicular and the equation of the perpendicular. *(Delhi 2011C)*

24. Find the equation of the perpendicular from the point $(3, -1, 11)$ to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Also find the coordinates of the foot of the perpendicular and the length of the perpendicular. *(AI 2011C)*

LA 2 (6 marks)

25. Find the coordinates of the foot of perpendicular and the length of the perpendicular drawn from the point $P(5, 4, 2)$ to the line, $\vec{r} = -\hat{i} + 3\hat{j} + \hat{k} + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$. Also find the image of P in this line. *(AI 2012)*

11.4 Angle between Two Lines

VSA (1 mark)

26. Find the vector equation of the line which passes through the point (3, 4, 5) and is parallel to the vector $2\hat{i} + 2\hat{j} - 3\hat{k}$. *(Delhi 2019)*

27. Find the angle between the lines

$\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ and

$\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$. *(Foreign 2014)*

28. Write the equation of the straight line through the point (α, β, γ) and parallel to z-axis. *(AI 2014C)*

29. Find the cartesian equation of the line which passes through the point $(-2, 4, -5)$ and is parallel to the line $\frac{x+3}{3} = \frac{4-y}{5} = \frac{z+8}{6}$. *(Delhi 2013)*

30. Write the vector equation of a line passing through the point $(1, -1, 2)$ and parallel to the line whose equation is $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+1}{-2}$. *(AI 2013C)*

31. If the equation of a line AB is $\frac{3-x}{-3} = \frac{y+2}{-2} = \frac{z+2}{6}$, find the direction cosines of a line parallel to AB. *(Delhi 2012C)*

32. The equation of a line is given by $\frac{4-x}{3} = \frac{y+3}{3} = \frac{z+2}{6}$. Write the direction cosines of a line parallel to the above line. *(AI 2012C)*

33. If the equation of a line AB is $\frac{3-x}{1} = \frac{y+2}{-2} = \frac{z-5}{4}$, write the direction ratios of a line parallel to AB. *(Delhi 2011C)*

34. The equation of a line is $\frac{4-x}{2} = \frac{y+3}{2} = \frac{z+2}{1}$. Find the direction cosines of a line parallel to the above line. *(AI 2011C)*

SA (2 marks)

35. Find the value of k so that the lines $x = -y = kz$ and $x - 2 = 2y + 1 = -z + 1$ are perpendicular to each other. *(2020)*

36. Find the vector equation of the line passing through the point $A(1, 2, -1)$ and parallel to the line $5x - 25 = 14 - 7y = 35z$. *(Delhi 2017)*

LA 1 (4 marks)

37. Find the value of λ, so that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. Also, find whether the lines are intersecting or not. *(Delhi 2019)*

www.exambites.in

38. Find the vector and cartesian equations of the line through the point (1, 2, –4) and perpendicular to the two lines
$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$ and
$\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$.
(Delhi 2016, AI 2015)

39. Find the vector and cartesian equations of the line passing through the point (2, 1, 3) and perpendicular to the lines
$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x}{-3} = \frac{y}{2} = \frac{z}{5}$.
(AI 2014)

40. Find the value of p, so that the lines
$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2}$ and
$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$ are perpendicular to each other. Also find the equation of a line passing through a point (3, 2, –4) and parallel to line l_1. *(AI 2014)*

41. A line passes through (2, –1, 3) and is perpendicular to the lines
$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(2\hat{i} - 2\hat{j} + \hat{k})$ and
$\vec{r} = (2\hat{i} - \hat{j} - 3\hat{k}) + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$.
Obtain its equation in vector and cartesian form. *(AI 2014, Delhi 2012)*

42. Find the direction cosines of the line $\frac{x+2}{2} = \frac{2y-7}{6} = \frac{5-z}{6}$. Also, find the vector equation of the line through the point $A(-1, 2, 3)$ and parallel to the given line. *(Delhi 2014C)*

43. The cartesian equations of a line are $6x - 2 = 3y + 1 = 2z - 2$. Find the direction cosines of the line. Write down the cartesian and vector equations of a line passing through (2, –1, –1) which is parallel to the given line. *(Delhi 2013C)*

44. Find the vector and cartesian equations of the line passing through the point (1, 2, –4) and perpendicular to the two lines
$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$
and $\frac{x-15}{3} = \frac{y+29}{8} = \frac{z-5}{-5}$ *(Delhi 2012)*

45. Find the equation of the line passing through the point (–1, 3, –2) and perpendicular to the lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$.
(Delhi 2012)

46. Find the angle between the following pair of lines $\frac{-x+2}{-2} = \frac{y-1}{7} = \frac{z+3}{-3}$ and $\frac{x+2}{-1} = \frac{2y-8}{4} = \frac{z-5}{4}$ and check whether the lines are parallel or perpendicular.
(Delhi 2011)

LA 2 (6 marks)

47. Find the vector and cartesian equations of a line passing through (1, 2, –4) and perpendicular to the two lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$. *(Delhi 2017)*

11.5 Shortest Distance between Two Lines

VSA (1 mark)

48. The line of shortest distance between two skew lines is _____ to both the lines.
(2020)

LA 1 (4 marks)

49. Find the shortest distance between the lines
$\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and
$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$. *(2018)*

50. Find the shortest distance between the lines
$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and
$\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$
(Delhi 2015C)

51. Find the shortest distance between the following lines :
$\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ and
$\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$ *(AI 2015C)*

www.exambites.in

52. Find the shortest distance between the lines whose vector equations are

$\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} - \hat{j} + \hat{k})$

and $\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$.

(Foreign 2014)

53. Find the shortest distance between the following lines:

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}, \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$$

(Foreign 2014)

54. Find the distance between the lines l_1 and l_2 given by

$l_1 : \vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

$l_2 : \vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(4\hat{i} + 6\hat{j} + 12\hat{k})$.

(Foreign 2014)

55. Find the shortest distance between the two lines whose vector equations are

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ and

$\vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$.

(Delhi 2014C)

56. Find the shortest distance between the two lines whose vector equations are

$\vec{r} = (6\hat{i} + 2\hat{j} + 2\hat{k}) + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$ and

$\vec{r} = (-4\hat{i} - \hat{k}) + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$.

(Delhi 2013C)

57. By computing the shortest distance between the following pair of lines, determine whether they intersect or not?

$\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} - \hat{k})$; $\vec{r} = 2\hat{i} - \hat{j} + \mu(\hat{i} - \hat{j} - \hat{k})$

(Delhi 2012C)

58. Find the shortest distance between the following pair of lines:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}; \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-5}{5}$$

(AI 2012C)

59. Find the shortest distance between the following lines whose vector equations are

$\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$ and

$\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}$. *(AI 2011)*

LA 2 (6 marks)

60. Find the vector equation of a line passing through the point (2, 3, 2) and parallel to the line $\vec{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$. Also, find the distance between these two lines.

(AI 2019)

11.6 Plane

VSA (1 mark)

61. Find the length of the intercept, cut off by the plane $2x + y - z = 5$ on the x-axis. *(AI 2019)*

62. Find the vector equation of a plane which is at a distance of 5 units from the origin and its normal vector is $2\hat{i} - 3\hat{j} + 6\hat{k}$. *(Delhi 2016)*

63. Find the vector equation of the plane with intercepts 3, –4 and 2 on x, y and z-axis respectively. *(AI 2016)*

64. Write the equation of a plane which is at a distance of $5\sqrt{3}$ units from origin and the normal to which is equally inclined to coordinate axes. *(Foreign 2016)*

65. Find the sum of the intercepts cut off by the plane $2x + y - z = 5$, on the coordinate axes.

(Foreign 2015)

66. Write the intercept cut off by the plane $2x + y - z = 5$ on x-axis. *(Delhi 2011)*

LA 1 (4 marks)

67. Find the unit vector perpendicular to the plane ABC where the position vectors of A, B and C are $2\hat{i} - \hat{j} + \hat{k}, \hat{i} + \hat{j} + 2\hat{k}$ and $2\hat{i} + 3\hat{k}$ respectively. *(AI 2014C)*

LA 2 (6 marks)

68. Find the coordinates of the point where the line through the points (3, –4, –5) and (2, –3, 1), crosses the plane determined by the points (1, 2, 3), (4, 2, –3) and (0, 4, 3).

(AI 2017)

69. Find the coordinate of the point P where the line through $A(3, -4, -5)$ and $B(2, -3, 1)$ crosses the plane passing through three points $L(2, 2, 1)$, $M(3, 0, 1)$ and $N(4, -1, 0)$.

www.examples.in

Also, find the ratio in which P divides the line segment AB. *(Delhi 2016)*

70. Find the equation of the plane which contains the line of intersection of the planes $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ and whose x-intercept is twice its z-intercept. *(Foreign 2016)*

71. Find the vector and cartesian equations of the plane passing through the line of intersection of the planes $\vec{r}\cdot(2\hat{i}+2\hat{j}-3\hat{k})=7$, $\vec{r}\cdot(2\hat{i}+5\hat{j}-3\hat{k})=9$ such that the intercepts made by the plane on x-axis and z-axis are equal. *(AI 2015C)*

72. Find the vector equation of the plane passing through three points with position vectors $\hat{i}+\hat{j}-2\hat{k}$, $2\hat{i}-\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$. Also find the coordinates of the point of intersection of this plane and the line $\vec{r}=3\hat{i}-\hat{j}-\hat{k}+\lambda(2\hat{i}-2\hat{j}+\hat{k})$. *(Delhi 2013)*

11.7 Coplanarity of Two Lines

LA 1 (4 marks)

73. Show that the lines $\frac{5-x}{-4}=\frac{y-7}{4}=\frac{z+3}{-5}$ and $\frac{x-8}{7}=\frac{2y-8}{2}=\frac{z-5}{3}$ are coplanar. *(Delhi 2014)*

LA 2 (6 marks)

74. Show that the lines $\frac{x-2}{1}=\frac{y-2}{3}=\frac{z-3}{1}$ and $\frac{x-2}{1}=\frac{y-3}{4}=\frac{z-4}{2}$ intersect.
Also, find the coordinates of the point of intersection. Find the equation of the plane containing the two lines. *(2020)*

75. If lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ intersect, then find the value of k and hence find the equation of the plane containing these lines. *(Delhi 2015)*

76. Show that the lines $\vec{r}=(\hat{i}+\hat{j}-\hat{k})+\lambda(3\hat{i}-\hat{j})$ and $\vec{r}=(4\hat{i}-\hat{k})+\mu(2\hat{i}+3\hat{k})$ are coplanar. Also find the equation of the plane containing them. *(Delhi 2013C)*

77. Show that the lines $\frac{x+3}{-3}=\frac{y-1}{1}=\frac{z-5}{5}$ and $\frac{x+1}{-1}=\frac{y-2}{2}=\frac{z-5}{5}$ are coplanar. Also find the equation of the plane containing these lines. *(AI 2013C)*

78. If the lines $\frac{x-1}{-3}=\frac{y-2}{-2k}=\frac{z-3}{2}$ and $\frac{x-1}{k}=\frac{y-2}{1}=\frac{z-3}{5}$ are perpendicular, find the value of k and hence find the equation of plane containing these lines. *(AI 2012)*

11.8 Angle between Two Planes

VSA (1 mark)

79. Find the acute angle between the planes $\vec{r}\cdot(\hat{i}-2\hat{j}-2\hat{k})=1$ and $\vec{r}\cdot(3\hat{i}-6\hat{j}+2\hat{k})=0$. *(AI 2019)*

80. Find the distance between the planes $2x - y + 2z = 5$ and $5x - 2.5y + 5z = 20$. *(AI 2017)*

81. Write the vector equation of the plane, passing through the point (a, b, c) and parallel to the plane $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})=2$. *(Delhi 2014)*

LA 1 (4 marks)

82. Find the vector equation of the plane through the points $(2, 1, -1)$ and $(-1, 3, 4)$ and perpendicular to the plane $x - 2y + 4z = 10$. *(AI 2013)*

LA 2 (6 marks)

83. Find the vector and cartesian equations of the plane passing through the points $(2, 2, -1)$, $(3, 4, 2)$ and $(7, 0, 6)$. Also find the vector equation of a plane passing through $(4, 3, 1)$ and parallel to the plane obtained above. *(Delhi 2019)*

www.exambites.in

84. Find the equation of the plane through the line of intersection of $\vec{r}\cdot(2\hat{i}-3\hat{j}+4\hat{k})=1$ and $\vec{r}\cdot(\hat{i}-\hat{j})+4=0$ and perpendicular to the plane $\vec{r}\cdot(2\hat{i}-\hat{j}+\hat{k})+8=0$. Hence find whether the plane thus obtained contains the line $x-1=2y-4=3z-12$. *(Delhi 2017)*

85. Find the vector equation of the plane which contains the line of intersection of the planes $\vec{r}\cdot(\hat{i}+2\hat{j}+3\hat{k})-4=0$ and $\vec{r}\cdot(2\hat{i}+\hat{j}-\hat{k})+5=0$ and which is perpendicular to the plane $\vec{r}\cdot(5\hat{i}+3\hat{j}-6\hat{k})+8=0$. *(Delhi 2013, 2011)*

11.9 Distance of a Point from a Plane

VSA (1 mark)

86. The distance of the origin (0, 0, 0) from the plane $-2x+6y-3z=-7$ is

(a) 1 unit (b) $\sqrt{2}$ units

(c) $2\sqrt{2}$ units (d) 3 units (2020)

87. The distance of the point (2, 3, 4) from the plane $\vec{r}\cdot(3\hat{i}-6\hat{j}+2\hat{k})=-11$ is

(a) 0 unit (b) 1 unit

(c) 2 units (d) $\frac{15}{7}$ units *(2020)*

88. Find the length of the perpendicular drawn from the origin to the plane $2x-3y+6z+21=0$. *(AI 2013)*

89. Find the distance of the plane $3x-4y+12z=3$ from the origin. *(AI 2012)*

LA 1 (4 marks)

90. Find the vector equation of the plane determined by the points $A(3,-1,2)$, $B(5,2,4)$ and $C(-1,-1,6)$. Hence, find the distance of the plane, thus obtained, from the origin. *(AI 2019)*

91. Find the distance between the point (–1, –5, –10) and the point of intersection of the line $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x-y+z=5$. *(Delhi 2015)*

92. Find the distance of the point (–1, –5, –10) from the point of intersection of the line $\vec{r}=2\hat{i}-\hat{j}+2\hat{k}+\lambda(3\hat{i}+4\hat{j}+2\hat{k})$ and the plane $\vec{r}\cdot(\hat{i}-\hat{j}+\hat{k})=5$. *(Delhi 2014, AI 2014C, 2011)*

93. A plane makes intercepts –6, 3, 4 respectively on the coordinate axes. Find the length of the perpendicular from the origin on it. *(Delhi 2014C)*

94. Find the image of the point having position vector $\hat{i}+3\hat{j}+4\hat{k}$ in the plane $\vec{r}\cdot(2\hat{i}-\hat{j}+\hat{k})+3=0$. *(AI 2014C)*

LA 2 (6 marks)

95. Find the distance of the point $P(3, 4, 4)$ from the point, where the line joining the points $A(3, -4, -5)$ and $B(2, -3, 1)$ intersects the plane $2x+y+z=7$. *(2020)*

96. Find the vector equation of the plane that contains the line $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and the point (–1, 3, –4). Also find the length of the perpendicular drawn from the point (2, 1, 4) to the plane, thus obtained. *(Delhi 2019)*

97. Find the coordinates of the foot of the perpendicular Q drawn from $P(3, 2, 1)$ to the plane $2x-y+z+1=0$. Also, find the distance PQ and the image of the point P treating this plane as a mirror. *(AI 2019)*

98. A variable plane which remains at a constant distance $3p$ from the origin cuts the coordinates axes at A, B, C. Show that the locus of the centroid of triangle ABC is $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$. *(AI 2017)*

99. Find the position vector of the foot of perpendicular and the perpendicular distance

www.exambits.in

from the point P with position vector $2\hat{i}+3\hat{j}+4\hat{k}$ to the plane $\vec{r}\cdot(2\hat{i}+\hat{j}+3\hat{k})-26=0$. Also find image of P in the plane. *(AI 2016)*

100. Find the distance of the point $(1,-2,3)$ from the plane $x-y+z=5$ measured parallel to the line whose direction cosines are proportional to 2, 3, –6. *(Foreign 2015)*

101. Find the equation of the plane through the line of intersection of the planes $x+y+z=1$ and $2x+3y+4z=5$ which is perpendicular to the plane $x-y+z=0$. Then find the distance of plane thus obtained from the point $A(1, 3, 6)$. *(Delhi 2015C)*

102. Find the distance between the point $(7,2,4)$ and the plane determined by the points $A(2,5,-3)$, $B(-2,-3,5)$ and $C(5,3,-3)$. *(Delhi 2014)*

103. Find the distance of the point (2, 12, 5) from the point of intersection of the line $\vec{r}=2\hat{i}-4\hat{j}+2\hat{k}+\lambda(3\hat{i}+4\hat{j}+2\hat{k})$ and the plane $\vec{r}\cdot(\hat{i}-2\hat{j}+\hat{k})=0$. *(AI 2014)*

104. Find the equation of the plane through the line of intersection of the planes $x+y+z=1$ and $2x+3y+4z=5$ which is perpendicular to the plane $x-y+z=0$. Also find the distance of the plane obtained above from the origin. *(AI 2014)*

105. Find the distance of the point (–1, –5, –10) from the point of intersection of the line joining the points $A(2,-1,2)$ and $B(5,3,4)$ with the plane $x-y+z=5$. *(Foreign 2014)*

106. Find the equation of the plane that contains the point (1, –1, 2) and is perpendicular to both the planes $2x+3y-2z=5$ and $x+2y-3z=8$. Hence find the distance of point $P(-2,5,5)$ from the plane obtained above. *(Foreign 2014)*

107. Find the vector equation of the plane determined by the points $A(3,-1,2)$, $B(5,2,4)$ and $C(-1,-1,6)$. Also find the distance of point $P(6,5,9)$ from this plane. *(Delhi 2013, 2012)*

108. Find the equation of the plane passing through the line of intersection of the planes $\vec{r}\cdot(\hat{i}+3\hat{j})-6=0$ and $\vec{r}\cdot(3\hat{i}-\hat{j}-4\hat{k})=0$, whose perpendicular distance from origin is unity. *(AI 2013)*

109. Find the distance of the point (1, –2, 3) from the plane $x-y+z=5$ measured parallel to the line $\frac{x-1}{2}=\frac{y-3}{3}=\frac{z+2}{-6}$. *(Delhi 2013C)*

110. Find the equation of the plane through the line of intersection of the planes $\vec{r}\cdot(\hat{i}+3\hat{j})+6=0$ and $\vec{r}\cdot(3\hat{i}-\hat{j}-4\hat{k})=0$ which is at unit distance from the origin. *(AI 2013C)*

111. Find the length and the foot of the perpendicular from the point $P(7, 14, 5)$ to the plane $2x+4y-z=2$. Also find the image of point P in the plane. *(AI 2012)*

112. Find the equation of the plane passing through the point $(3,-3,1)$ and perpendicular to the line joining the points (3, 4, –1) and (2, –1, 5). Also find the coordinates of foot of perpendicular, the equation of perpendicular line and the length of perpendicular drawn from origin to the plane. *(Delhi 2012C)*

113. Find the distance of the point (3, 4, 5) from the plane $x+y+z=2$ measured parallel to the line $2x=y=z$. *(Delhi 2012C)*

114. Find the coordinates of the foot of the perpendicular, the equation of the perpendicular and the perpendicular distance of the point $P(3, 2, 1)$ from the plane $2x-y+z+1=0$. Find also the image of the point P in the plane. *(AI 2012C)*

115. Find the vector equation of the plane that contains the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and $\vec{r}=(\hat{i}+\hat{j})+\mu(-\hat{i}+\hat{j}-2\hat{k})$. Also find the length of perpendicular drawn from the point (2, 1, 4) to the plane thus obtained. *(AI 2012C)*

116. Find the equation of the plane passing through the point (1, 2, 1) and perpendicular to the line joining the points (1, 4, 2) and (2, 3, 5).

www.exambites.in

Also find the coordinates of the foot of the perpendicular and the perpendicular distance of the point (4, 0, 3) from the above found plane. *(Delhi 2011C)*

11.10 Angle between a Line and a Plane

VSA (1 mark)

117. The vector equation of the line passing through the point (–1, 5, 4) and perpendicular to the plane $z = 0$ is

(a) $\vec{r} = -\hat{i} + 5\hat{j} + 4\hat{k} + \lambda(\hat{i} + \hat{j})$

(b) $\vec{r} = -\hat{i} + 5\hat{j} + (4 + \lambda)\hat{k}$

(c) $\vec{r} = \hat{i} - 5\hat{j} - 4\hat{i} + \lambda\hat{k}$

(d) $\vec{r} = \lambda\hat{k}$ *(2020)*

118. Write the vector equation of the line passing through (1, 2, 3) and perpendicular to the plane $\vec{r} \cdot (\hat{i} + 2\hat{j} - 5\hat{k}) + 9 = 0$. *(Delhi 2015C, AI 2015C)*

SA (2 marks)

119. Find the coordinates of the point where the line through (–1, 1, –8) and (5, –2, 10) crosses the *ZX*-plane. *(2020)*

LA 1 (4 marks)

120. Find the coordinates of the point where the line through the points $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the *XZ* plane. Also find the angle which this line makes with the *XZ* plane. *(AI 2016)*

121. Find the equation of the plane passing through the points (–1, 2, 0), (2, 2, –1) and parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$. *(AI 2015)*

122. Find the equation of a plane which passes through the point (3, 2, 0) and contains the line $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$. *(Foreign 2015)*

123. Find the coordinates of the point, where the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{2}$ intersects the plane $x - y + z - 5 = 0$. Also find the angle between the line and the plane. *(Delhi 2013)*

124. Find the coordinates of the point where the line through the points $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the *XY*-plane. *(AI 2012)*

125. Find the coordinates of the point where the line through the points $(3, -4, -5)$ and $(2, -3, 1)$ crosses the plane $2x + y + z = 7$. *(AI 2012)*

126. Find the coordinates of the point where the line through the points $(3, -4, -5)$ and $(2, -3, 1)$ crosses the plane $3x + 2y + z + 14 = 0$. *(AI 2012)*

LA 2 (6 marks)

127. Find the equation of the plane that contains the point $A(2, 1, -1)$ and is perpendicular to the line of intersection of the planes $2x + y - z = 3$ and $x + 2y + z = 2$. Also find the angle between the plane thus obtained and the *y*-axis. *(2020)*

128. Find the distance of the point $P(-2, -4, 7)$ from the point of intersection Q of the line $\vec{r} = (3\hat{i} - 2\hat{j} + 6\hat{k}) + \lambda(2\hat{i} - \hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 6$. Also write the vector equation of the line *PQ*. *(2020)*

129. Find the vector and cartesian forms of the equation of the plane passing through the point (1, 2, –4) and parallel to the lines $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = \hat{i} - 3\hat{j} + 5\hat{k} + \mu(\hat{i} + \hat{j} - \hat{k})$.
Also, find the distance of the point (9, –8, –10) from the plane thus obtained. *(Delhi 2014C)*

130. Find the equation of the plane passing through the line of intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0$ and parallel to *x*-axis. *(AI 2014C, 2011)*

131. Find the coordinates of the point where the line through (3, –4, –5) and (2, –3, 1) crosses the plane, passing through the points (2, 2, 1), (3, 0, 1) and (4, –1, 0). *(Delhi 2013)*

132. Find the vector equation of the line passing through the point (1, 2, 3) and

www.exambites.in

parallel to the planes $\vec{r}\cdot(\hat{i}-\hat{j}+2\hat{k})=5$ and $\vec{r}\cdot(3\hat{i}+\hat{j}+\hat{k})=6$. *(AI 2013)*

133. Find the equation of the plane through the line of intersection of the planes $2x + y - z = 3$ and $5x - 3y + 4z + 9 = 0$ and parallel to the line $\frac{x-1}{2}=\frac{y-3}{4}=\frac{z-5}{5}$. *(AI 2011)*

134. Find the vector and cartesian equations of a plane containing the two lines

$\vec{r}=2\hat{i}+\hat{j}-3\hat{k}+\lambda(\hat{i}+2\hat{j}+5\hat{k})$ and

$\vec{r}=3\hat{i}+3\hat{j}+2\hat{k}+\mu(3\hat{i}-2\hat{j}+5\hat{k})$

Also show that the line

$\vec{r}=(2\hat{i}+5\hat{j}+2\hat{k})+p(3\hat{i}-2\hat{j}+5\hat{k})$ lies in the plane. *(AI 2011C)*

Detailed Solutions

1. **(c) :** Let $P(4, -7, 3)$ be the given point and A be a point on y-axis s.t. $PA \perp$ to y-axis.

$\therefore$ $A \equiv (0, -7, 0)$

Now, $PA = \sqrt{(4-0)^2+(3-0)^2}$

$= \sqrt{4^2+3^2}=\sqrt{16+9}=\sqrt{25}=5$ units

2. **(a) :** Vector equation of XY-plane is $\vec{r}\cdot\hat{k}=0$.

3. We have eq. of x-axis as $y = 0, z = 0$

$\therefore$ Distance of $P(a, b, c)$ from x-axis

$= \sqrt{(a-a)^2+b^2+c^2}=\sqrt{b^2+c^2}$ units.

4. Since the line makes angles 90°, 135° and 45° with x, y and z axes respectively.

$\therefore$ $l = \cos 90° = 0$, $m=\cos 135°=-\frac{1}{\sqrt{2}}$

and $n=\cos 45°=\frac{1}{\sqrt{2}}$

Hence, direction cosines of the line are

$0,-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}$.

5. Let the line makes an angle α, β, γ with the positive direction of x, y, z axes respectively.

$\therefore$ $\alpha = 90°, \beta = 60°$

Now, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow$ $\cos^2 90° + \cos^2 60° + \cos^2\gamma = 1$

$\Rightarrow$ $\cos^2\gamma = 1 - 0 - \frac{1}{4} = \frac{3}{4}$

$\Rightarrow$ $\cos\gamma=\frac{\sqrt{3}}{2} \Rightarrow \gamma = 30°$

6. Since, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow$ $\cos^2 90° + \cos^2 60° + \cos^2\theta = 1$

$\Rightarrow \frac{1}{4}+\cos^2\theta=1 \Rightarrow \cos^2\theta=\frac{3}{4}$

$\Rightarrow \cos\theta=\frac{\sqrt{3}}{2} \Rightarrow \theta=\frac{\pi}{6}$ $(\because$ θ is acute)

7. Here, the direction cosines of the given line are $\cos\alpha$, $\cos\beta$, $\cos\gamma$ and $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow$ $(1 - \sin^2\alpha) + (1 - \sin^2\beta) + (1 - \sin^2\gamma) = 1$

$\Rightarrow$ $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

8. The direction ratios of the given line are 2, –1, –2. Let $a = 2$, $b = -1$, $c = -2$

$\Rightarrow \sqrt{a^2+b^2+c^2}=\sqrt{2^2+(-1)^2+(-2)^2}=3$

$\therefore$ The direction cosines of given line are

$\frac{2}{3},\frac{-1}{3},\frac{-2}{3}$.

9. The direction cosines of the line joining the points (1, 0, 0) and (0, 1, 1) are $\frac{-1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}$.

10. Vector equation of a line passes through the points (3, 4, –7) and (1, –1, 6) is given by

$\vec{r}=(3\hat{i}+4\hat{j}-7\hat{k})+\lambda[(\hat{i}-\hat{j}+6\hat{k})-(3\hat{i}+4\hat{j}-7\hat{k})]$

$\therefore$ $\vec{r}=3\hat{i}+4\hat{j}-7\hat{k}+\lambda(-2\hat{i}-5\hat{j}+13\hat{k})$

11. The given line is $5x - 3 = 15y + 7 = 3 - 10z$

$\Rightarrow \frac{x-\frac{3}{5}}{\frac{1}{5}}=\frac{y+\frac{7}{15}}{\frac{1}{15}}=\frac{z-\frac{3}{10}}{-\frac{1}{10}}$

Its direction ratios are $\frac{1}{5},\frac{1}{15},-\frac{1}{10}$

i.e., Its direction ratios are proportional to 6, 2, –3.

Now, $\sqrt{6^2+2^2+(-3)^2}=7$

$\therefore$ Its direction cosines are $\frac{6}{7},\frac{2}{7},-\frac{3}{7}$.

www.examfears.in

12. The cartesian equation of a line is

$$\frac{3-x}{5}=\frac{y+4}{7}=\frac{2z-6}{4} \quad ...(i)$$

$$\Rightarrow \quad \frac{x-3}{-5}=\frac{y-(-4)}{7}=\frac{z-3}{2}=\lambda \text{ (say)}$$

$\Rightarrow$ $x = 3 - 5\lambda$, $y = -4 + 7\lambda$, $z = 3 + 2\lambda$

Take $\vec{a}=3\hat{i}-4\hat{j}+3\hat{k}$ and $\vec{b}=-5\hat{i}+7\hat{j}+2\hat{k}$.

$\therefore$ The vector equation of the line (i) is $\vec{r}=\vec{a}+\lambda\vec{b}$

$$\Rightarrow \quad \vec{r}=(3\hat{i}-4\hat{j}+3\hat{k})+\lambda(-5\hat{i}+7\hat{j}+2\hat{k})$$

13. The eq. of given line is $\frac{4-x}{2}=\frac{y}{6}=\frac{1-z}{3}$ ...(i)

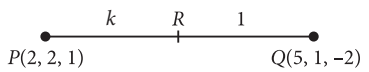
$$\Rightarrow \quad \frac{x-4}{-2}=\frac{y}{6}=\frac{z-1}{-3} \Rightarrow \quad \frac{x-4}{2}=\frac{y}{-6}=\frac{z-1}{3}$$

As $\sqrt{2^2+(-6)^2+3^2}=7$

$\therefore$ D.c's. of (i) are $\frac{2}{7}, \frac{-6}{7}, \frac{3}{7}$.

14. *Refer to answer 12.*

15. Given that $P(2, 2, 1)$ and $Q(5, 1, -2)$



Let the point R on the line PQ, divides the line in the ratio $k : 1$. And x-coordinate of point R on the line is 4.

So, by section formula

$$4=\frac{5k+2}{k+1} \Rightarrow k=2$$

Now, z-coordinate of point R,

$$z=\frac{-2k+1}{k+1}=\frac{-2\times 2+1}{2+1}=-1$$

$\Rightarrow$ z-coordinate of point $R = -1$

16. The equation of line AB is given by

$$\frac{x-0}{4-0}=\frac{y+1}{5+1}=\frac{z+1}{1+1}=\lambda \text{ (say)}$$

$\Rightarrow$ $x = 4\lambda$, $y = 6\lambda - 1$, $z = 2\lambda - 1$

The coordinates of a general point on AB are $(4\lambda, 6\lambda - 1, 2\lambda - 1)$

The equation of line CD is given by

$$\frac{x-3}{3+4}=\frac{y-9}{9-4}=\frac{z-4}{4-4}=\mu \text{ (say)}$$

$\Rightarrow$ $x = 7\mu + 3$, $y = 5\mu + 9$, $z = 4$

The coordinates of a general point on CD are $(7\mu + 3, 5\mu + 9, 4)$

If the line AB and CD intersect then they have a common point. So, for some values of λ and μ, we must have

$4\lambda = 7\mu + 3$, $6\lambda - 1 = 5\mu + 9$, $2\lambda - 1 = 4$

$\Rightarrow$ $4\lambda - 7\mu = 3$ ...(i), $6\lambda - 5\mu = 10$ ...(ii) and

$\lambda = \frac{5}{2}$ ...(iii)

Substituting $\lambda=\frac{5}{2}$ in (ii), we get $\mu = 1$

Since $\lambda=\frac{5}{2}$ and $\mu = 1$ satisfy (i), so the given lines AB and CD intersect.

17. Any point on the line

$$\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}=r \text{ (say)} \quad ...(i)$$

is $(3r - 1, 5r - 3, 7r - 5)$.

Any point on the line

$$\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}=k \text{ (say)} \quad ...(ii)$$

is $(k + 2, 3k + 4, 5k + 6)$

For lines (i) and (ii) to intersect, we must have

$3r - 1 = k + 2$, $5r - 3 = 3k + 4$, $7r - 5 = 5k + 6$

On solving these, we get $r=\frac{1}{2}$, $k=-\frac{3}{2}$

$\therefore$ Lines (i) and (ii) intersect and their point of intersection is $\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$

18. The given lines are

$$\vec{r}=(\hat{i}+\hat{j}-\hat{k})+\lambda(3\hat{i}-\hat{j})=(3\lambda+1)\hat{i}+(1-\lambda)\hat{j}-\hat{k} \quad ...(i)$$

$$\text{and } \vec{r}=(4\hat{i}-\hat{k})+\mu(2\hat{i}+3\hat{k})=(2\mu+4)\hat{i}+0\cdot\hat{j}+(3\mu-1)\hat{k} \quad ...(ii)$$

If the lines (i) & (ii) intersect, then they have a common point. So, we must have

$$(3\lambda+1)\hat{i}+(1-\lambda)\hat{j}-\hat{k}=(2\mu+4)\hat{i}+0\cdot\hat{j}+(3\mu-1)\hat{k}$$

$\Rightarrow$ $3\lambda + 1 = 2\mu + 4$, $1 - \lambda = 0$ and $-1 = 3\mu - 1$

On solving last two equations, we get $\lambda = 1$ and $\mu = 0$.

These values of λ and μ satisfy the first equation.

So, the given lines intersect.

Putting $\lambda = 1$ in (i), we get the position vector of the point of intersection.

Thus, the coordinates of the point of intersection are $(4, 0, -1)$.

www.examfree.in

19. The given lines are

$\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and

$\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$

$\Rightarrow \quad \vec{r} = (3+\lambda)\hat{i} + (2+2\lambda)\hat{j} + (2\lambda - 4)\hat{k}$ ...(i)

and $\vec{r} = (5+3\mu)\hat{i} + (2\mu - 2)\hat{j} + 6\mu\hat{k}$ ...(ii)

If these lines intersect, they must have a common point. So, we must have

$(3+\lambda)\hat{i} + (2+2\lambda)\hat{j} + (2\lambda - 4)\hat{k}$

$= (5+3\mu)\hat{i} + (2\mu - 2)\hat{j} + 6\mu\hat{k}$

$\Rightarrow \quad 3 + \lambda = 5 + 3\mu \quad \Rightarrow \quad \lambda - 3\mu = 2,$

$2 + 2\lambda = 2\mu - 2 \quad \Rightarrow \quad \lambda - \mu = -2,$

and $2\lambda - 4 = 6\mu \quad \Rightarrow \quad \lambda - 3\mu = 2$

$\Rightarrow \quad \lambda = -4, \mu = -2.$

$\therefore$ The given lines intersect and their point of intersection is $(-1, -6, -12)$.

20. The given line is

$$\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11} = \lambda \text{ (say)} \quad \text{...(i)}$$

Any point on it is

$M(10\lambda + 11, -4\lambda - 2, -11\lambda - 8)$.

Let the given point be $P(2, -1, 5)$

D.r's of PM are $10\lambda + 11 - 2, -4\lambda - 2 + 1, -11\lambda - 8 - 5$.

i.e., $10\lambda + 9, -4\lambda - 1, -11\lambda - 13$.

As line (i) is $\perp$ to PM, so

$(10\lambda + 9)\cdot 10 + (-4\lambda - 1)\cdot(-4)$

$+ (-11\lambda - 13)\cdot(-11) = 0$

$\Rightarrow \quad (100 + 16 + 121)\lambda + (90 + 4 + 143) = 0$

$\Rightarrow \quad 237\lambda + 237 = 0 \Rightarrow \lambda = -1.$

$\therefore$ D.r's of PM are $-1, 3, -2$ *i.e.*, $1, -3, 2$.

$\therefore$ Eq. of the line PM is

$$\frac{x-2}{1} = \frac{y+1}{-3} = \frac{z-5}{2} \quad \text{...(ii)}$$

If $Q(\alpha, \beta, \gamma)$ is the image of P in line (i), then M is the mid point of PQ.

Here, $M \equiv (1, 2, 3)$

$$\therefore \quad \frac{\alpha+2}{2} = 1, \frac{\beta-1}{2} = 2, \frac{\gamma+5}{2} = 3$$

$\Rightarrow \quad \alpha = 0, \beta = 5, \gamma = 1$

$\therefore$ $Q(0, 5, 1)$ is the image of P in line (i)

and $PQ = \sqrt{(2-0)^2 + (-1-5)^2 + (5-1)^2}$

$= \sqrt{56} = 2\sqrt{14}$ units

21. *Refer to answer 17.*

22. Here $\vec{a} = -2\hat{i} + 3\hat{j} + 5\hat{k}, \vec{b} = 7\hat{i} - \hat{k}$

$\therefore$ Equation of line joining A and B is,

$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

$\Rightarrow \quad \vec{r} = -2\hat{i} + 3\hat{j} + 5\hat{k} + \lambda(9\hat{i} - 3\hat{j} - 6\hat{k})$ ... (i)

Again, $\vec{c} = -3\hat{i} - 2\hat{j} - 5\hat{k}, \vec{d} = 3\hat{i} + 4\hat{j} + 7\hat{k}$

$\therefore$ Equation of line joining C and D is,

$\vec{r} = \vec{c} + \mu(\vec{d} - \vec{c})$

$\Rightarrow \quad \vec{r} = -3\hat{i} - 2\hat{j} - 5\hat{k} + \mu(6\hat{i} + 6\hat{j} + 12\hat{k})$ ... (ii)

Equations (i) and (ii) will intersect, when

$-2\hat{i} + 3\hat{j} + 5\hat{k} + \lambda(9\hat{i} - 3\hat{j} - 6\hat{k}) = -3\hat{i} - 2\hat{j} - 5\hat{k}$

$+\mu(6\hat{i} + 6\hat{j} + 12\hat{k})$

$\Rightarrow \quad -2 + 9\lambda = -3 + 6\mu\ ;\ 3 - 3\lambda = -2 + 6\mu\ ;$

$5 - 6\lambda = -5 + 12\mu$

$\Rightarrow \quad 9\lambda - 6\mu = -1\ ;\ 3\lambda + 6\mu = 5\ ;\ 6\lambda + 12\mu = 10$

Solving first two equations, we get $\lambda = \frac{1}{3}, \mu = \frac{2}{3}$ which also satisfy third equation.

Put $\lambda = \frac{1}{3}$ in (i), we get the point of intersection of lines is $\hat{i} + 2\hat{j} + 3\hat{k} \equiv (1,2,3)$.

23. Any point on the given line,

$$\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} = k \text{ (say)} \quad \text{...(i)}$$

is $R(2k + 1, -3k - 1, 8k - 10)$

If this is the foot of the $\perp$ from $P(1, 0, 0)$ on (i), then $(2k + 1 - 1)\cdot 2 + (-3k - 1 - 0)\cdot(-3)$

$+ (8k - 10 - 0)\cdot 8 = 0$

$\Rightarrow \quad 4k + 9k + 3 + 64k - 80 = 0$

$\Rightarrow \quad 77k = 77 \Rightarrow k = 1.$

$\therefore$ R is $(3, -4, -2)$.

This is the required foot of perpendicular.

Also, perpendicular distance = PR

$= \sqrt{(3-1)^2 + (-4-0)^2 + (-2-0)^2}$

$= \sqrt{24} = 2\sqrt{6}$ units.

Also eq. of PR is $\frac{x-1}{2} = \frac{y}{-4} = \frac{z}{-2}$

24. Any point on the given line

$$\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4} = k \text{ (say)} \quad \text{...(i)}$$

is $R(2k, 3k + 2, 4k + 3)$.

If R is the foot of the $\perp$ from $P(3, -1, 11)$ on (i), then line PR has d.r's $2k - 3, 3k + 2 + 1, 4k + 3 - 11$

$\therefore$ $2(2k - 3) + 3(3k + 3) + 4(4k - 8) = 0$

$\Rightarrow \quad 29k - 29 = 0 \Rightarrow k = 1.$

www.examfree.in

$\therefore$ D.r's of perpendicular line are –1, 6, –4.
As it is drawn through $P(3, -1, 11)$, so its eq. is

$$\frac{x-3}{-1} = \frac{y+1}{6} = \frac{z-11}{-4}$$

Now foot of $\perp$ is $R(2, 5, 7)$
and length of the perpendicular = PR

$$= \sqrt{(2-3)^2 + (5+1)^2 + (7-11)^2} = \sqrt{53} \text{ units.}$$

25. The given line is

$$\vec{r} = -\hat{i} + 3\hat{j} + \hat{k} + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$$

Its cartesian eq. is

$$\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1} = \lambda \text{ (say)} \quad \text{...(i)}$$

Any point Q on (i) is $(2\lambda - 1, 3\lambda + 3, -\lambda + 1)$
Also, the given point is $P(5, 4, 2)$.
Now d.r's of the line PQ are
$(2\lambda - 1 - 5, 3\lambda + 3 - 4, -\lambda + 1 - 2)$
$= (2\lambda - 6, 3\lambda - 1, -\lambda - 1)$.
For PQ to be $\perp$ to (i), we must have
$(2\lambda - 6)\cdot 2 + (3\lambda - 1)\cdot 3 + (-\lambda - 1)\cdot(-1) = 0$
$\Rightarrow \quad 14\lambda - 14 = 0 \Rightarrow \lambda = 1$
$\therefore$ Q is (1, 6, 0)
which is the foot of $\perp$ from P on line (i).

$$\text{Now, } PQ = \sqrt{(5-1)^2 + (4-6)^2 + (2-0)^2}$$

$$= \sqrt{24} = 2\sqrt{6} \text{ units.}$$

Further if $R(\alpha, \beta, \gamma)$ is the image of P in line (i), then

$$\frac{\alpha+5}{2} = 1, \frac{\beta+4}{2} = 6, \frac{\gamma+2}{2} = 0$$

$\Rightarrow \quad \alpha = -3, \beta = 8, \gamma = -2$
$\therefore$ Image of P in line (i) is $R(-3, 8, -2)$.

26. We know that vector equation of a line passing through point $\vec{a}$ and parallel to vector $\vec{b}$ is given by $\vec{r} = \vec{a} + \lambda\vec{b}$
Here $\vec{a} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{b} = 2\hat{i} + 2\hat{j} - 3\hat{k}$
$\therefore$ Required equation is

$$\vec{r} = (3\hat{i} + 4\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 2\hat{j} - 3\hat{k})$$

27. For $\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$,
$\vec{b}_1 = 3\hat{i} + 2\hat{j} + 6\hat{k}$
Also for

$$\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k}), \vec{b}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$$

Let θ be the angle between the lines.

$$\text{So, } \theta = \cos^{-1}\left|\frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}\right|$$

$$\Rightarrow \quad \theta = \cos^{-1}\left|\frac{(3\hat{i} + 2\hat{j} + 6\hat{k})\cdot(\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{3^2 + 2^2 + 6^2}\ \sqrt{1^2 + 2^2 + 2^2}}\right|$$

$$\Rightarrow \quad \theta = \cos^{-1}\left|\frac{3 + 4 + 12}{7 \times 3}\right| \Rightarrow \theta = \cos^{-1}\left(\frac{19}{21}\right)$$

28. Any line parallel to z-axis has direction ratios proportional to 0, 0, 1.
$\therefore$ The eqn. of a line through (α, β, γ) and parallel to z-axis is $\frac{x-\alpha}{0} = \frac{y-\beta}{0} = \frac{z-\gamma}{1}$

29. Equation of the line can be written as

$$\frac{x+3}{3} = \frac{y-4}{-5} = \frac{z+8}{6}$$

Direction ratios of this line are 3, –5, 6.
The required line passes through (–2, 4, –5) and its direction ratios are proportional to 3, –5, 6. So, its equation is $\frac{x+2}{3} = \frac{y-4}{-5} = \frac{z+5}{6}$

30. Vector eq. of the line passing through (1, –1, 2) and parallel to the line, $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+1}{-2}$ is

$$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{j} - 2\hat{k})$$

31. The line AB is given by

$$\frac{3-x}{-3} = \frac{y+2}{-2} = \frac{z+2}{6}$$

$$\Rightarrow \quad \frac{x-3}{3} = \frac{y+2}{-2} = \frac{z+2}{6}$$

Its direction ratios are 3, –2, 6.
Hence its d.c's are

$$\frac{3}{\sqrt{3^2 + (-2)^2 + 6^2}}, \frac{-2}{\sqrt{3^2 + (-2)^2 + 6^2}}, \frac{6}{\sqrt{3^2 + (-2)^2 + 6^2}}$$

i.e., $\frac{3}{7}, \frac{-2}{7}, \frac{6}{7}$

$\therefore$ D.c's of a line parallel to AB are proportional to $\frac{3}{7}, \frac{-2}{7}, \frac{6}{7}$

32. The given line is

$$\frac{4-x}{3} = \frac{y+3}{3} = \frac{z+2}{6} \quad \text{...(i)}$$

$$\Rightarrow \quad \frac{x-4}{-1} = \frac{y+3}{1} = \frac{z+2}{2}$$

www.examblaster.in

Here, $\sqrt{(-1)^2+1^2+2^2}=\sqrt{6}$

$\therefore$ The direction cosines of any line parallel to (i) are $\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$

33. *Refer to answer 31.*

34. *Refer to answer 32.*

35. The given lines are

$$l_1: \frac{x-0}{1}=\frac{y-0}{-1}=\frac{z-0}{\frac{1}{k}}$$

$$l_2: \frac{x-2}{1}=\frac{y+\frac{1}{2}}{\frac{1}{2}}=\frac{z-1}{-1}$$

$\because$ l_1 is perpendicular to l_2

$$\therefore \quad 1(1)+(-1)\left(\frac{1}{2}\right)+\left(\frac{1}{k}\right)(-1)=0$$

$$\Rightarrow \quad 1-\frac{1}{2}-\frac{1}{k}=0 \;\Rightarrow\; \frac{1}{2}=\frac{1}{k} \Rightarrow k=2$$

36. Vector equation of the line passing through (1, 2, –1) and parallel to the line

$5x - 25 = 14 - 7y = 35z$

i.e., $\frac{x-5}{1/5}=\frac{y-2}{-1/7}=\frac{z}{1/35}$ or $\frac{x-5}{7}=\frac{y-2}{-5}=\frac{z}{1}$

is $\vec{r}=(\hat{i}+2\hat{j}-\hat{k})+\lambda(7\hat{i}-5\hat{j}+\hat{k})$

37. The given lines are

$$l_1: \frac{x-1}{-3}=\frac{y-2}{\lambda/7}=\frac{z-3}{2}$$

and $l_2: \frac{x-1}{-3\lambda/7}=\frac{y-5}{1}=\frac{z-6}{-5}$

Now, $l_1 \perp l_2$ [Given]

$$\therefore \quad (-3)\left(-\frac{3\lambda}{7}\right)+\frac{\lambda}{7}-10=0$$

$$\Rightarrow \quad \frac{9\lambda}{7}+\frac{\lambda}{7}-10=0 \;\Rightarrow\; \frac{10\lambda}{7}=10 \;\Rightarrow\; \lambda=7$$

Since for $\lambda = 7$, given lines are at right angles.

$\therefore$ Lines are intersecting.

38. The given lines are

$$\vec{r}=(8\hat{i}-19\hat{j}+10\hat{k})+\lambda(3\hat{i}-16\hat{j}+7\hat{k})$$

and $\vec{r}=(15\hat{i}+29\hat{j}+5\hat{k})+\mu(3\hat{i}+8\hat{j}-5\hat{k})$

Equation of any line through (1, 2, –4) with d.r's l, m, n is

$$\vec{r}=(\hat{i}+2\hat{j}-4\hat{k})+p(l\hat{i}+m\hat{j}+n\hat{k}) \quad ...(i)$$

Since, the required line is perpendicular to both the given lines.

$\therefore$ $3l - 16m + 7n = 0$ and $3l + 8m - 5n = 0$

$$\Rightarrow \quad \frac{l}{80-56}=\frac{m}{21+15}=\frac{n}{24+48} \Rightarrow \frac{l}{2}=\frac{m}{3}=\frac{n}{6}$$

$\therefore$ From (i), the required line is

$$\vec{r}=(\hat{i}+2\hat{j}-4\hat{k})+t(2\hat{i}+3\hat{j}+6\hat{k})$$

Here, the position vector of passing point is $\vec{a}=\hat{i}+2\hat{j}-4\hat{k}$ and parallel vector is

$$\vec{b}=2\hat{i}+3\hat{j}+6\hat{k}$$

$\therefore$ Cartesian equation is given by

$$\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{6}$$

39. Let the eq. of line passing through (2, 1, 3) and perpendicular to the lines

$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x}{-3}=\frac{y}{2}=\frac{z}{5}$ be

$$\frac{x-2}{l}=\frac{y-1}{m}=\frac{z-3}{n} \quad ...(i)$$

$\therefore$ $l\cdot 1 + m\cdot 2 + n\cdot 3 = 0$ and

$l\cdot(-3) + m\cdot 2 + n\cdot 5 = 0$

$$\Rightarrow \quad \frac{l}{10-6}=\frac{m}{-9-5}=\frac{n}{2+6} \;\Rightarrow\; \frac{l}{2}=\frac{m}{-7}=\frac{n}{4}.$$

$\therefore$ The eq. of the required line is

$$\frac{x-2}{2}=\frac{y-1}{-7}=\frac{z-3}{4}.$$

Also its vector eq. is

$$\vec{r}=(2\hat{i}+\hat{j}+3\hat{k})+\lambda(2\hat{i}-7\hat{j}+4\hat{k}).$$

40. *Refer to answer 37.*

Now, eq. of the line passing through (3, 2, –4) and parallel to l_1 is

$$\frac{x-3}{-3}=\frac{y-2}{1}=\frac{z+4}{2}.$$

41. The given lines are

$$\vec{r}=(\hat{i}+\hat{j}-\hat{k})+\lambda(2\hat{i}-2\hat{j}+\hat{k}) \text{ and}$$

$$\vec{r}=(2\hat{i}-\hat{j}-3\hat{k})+\mu(\hat{i}+2\hat{j}+2\hat{k})$$

Eq. of any line through (2, –1, 3) with d.r's l, m, n is

$$\vec{r}=(2\hat{i}-\hat{j}+3\hat{k})+p(l\hat{i}+m\hat{j}+n\hat{k}) \quad ...(i)$$

Since, the required line is perpendicular to both the given lines.

www.exambites.in

$\therefore$ $2l - 2m + n = 0$ and $l + 2m + 2n = 0$

$\Rightarrow \quad \frac{l}{-4-2} = \frac{m}{1-4} = \frac{n}{4+2} \Rightarrow \frac{l}{2} = \frac{m}{1} = \frac{n}{-2}$

$\therefore$ From (i), the required line is

$\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} + \hat{j} - 2\hat{k})$.

42. The given line is $\frac{x+2}{2} = \frac{2y-7}{6} = \frac{5-z}{6}$

$\Rightarrow \quad \frac{x+2}{2} = \frac{y-\frac{7}{2}}{3} = \frac{z-5}{-6}$ ...(i)

Its d.r's are 2, 3, –6

$\because \quad \sqrt{2^2 + 3^2 + (-6)^2} = 7$

$\therefore$ Its d.c's are $\frac{2}{7}, \frac{3}{7}, -\frac{6}{7}$

Eq. of a line through (–1, 2, 3) and parallel to (i) is

$\frac{x+1}{2} = \frac{y-2}{3} = \frac{z-3}{-6} = \lambda$ (say)

$\therefore$ Vector equation of a line passing through (–1, 2, 3) and parallel to (i) is given by

$\vec{r} = (-\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} - 6\hat{k})$

43. The cartesian eqs. of the given line are

$6x - 2 = 3y + 1 = 2z - 2$

$\Rightarrow \quad 6\left(x - \frac{1}{3}\right) = 3\left(y + \frac{1}{3}\right) = 2(z-1)$

$\Rightarrow \quad \frac{x - \frac{1}{3}}{\frac{1}{6}} = \frac{y + \frac{1}{3}}{\frac{1}{3}} = \frac{z-1}{\frac{1}{2}}$

$\therefore$ D.r's of the line are $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$ *i.e.*, proportional to 1, 2, 3

$\Rightarrow$ D.c's of the line are

$\frac{1}{\sqrt{1^2+2^2+3^2}}, \frac{2}{\sqrt{1^2+2^2+3^2}}, \frac{3}{\sqrt{1^2+2^2+3^2}}$

i.e., $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$.

Cartesian eq. of the line through (2, –1, –1) and parallel to the given line is

$\frac{x-2}{1} = \frac{y+1}{2} = \frac{z+1}{3}$

Here, the position vector of the point (2, –1, –1) is $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}$ and the line is in the direction of $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\therefore$ The corresponding vector eq. of the required line is $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$.

44. *Refer to answer 39.*

45. Let l, m, n be the direction ratios of the line which is perpendicular to the lines

$\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$.

Then, $l + 2m + 3n = 0$ and $-3l + 2m + 5n = 0$

$\Rightarrow \quad \frac{l}{10-6} = \frac{m}{-9-5} = \frac{n}{2+6} \Rightarrow \frac{l}{2} = \frac{m}{-7} = \frac{n}{4}$

$\therefore$ Eq. of the required line through (–1, 3, –2) having d.r's proportional to 2, –7, 4 is

$\frac{x+1}{2} = \frac{y-3}{-7} = \frac{z+2}{4}$.

46. Writing given lines in standard form as

$\frac{x-2}{2} = \frac{y-1}{7} = \frac{z-(-3)}{-3}$ and $\frac{x-(-2)}{-1} = \frac{y-4}{2} = \frac{z-5}{4}$

Let $\vec{b}_1$ and $\vec{b}_2$ be vectors parallel to both the lines respectively, then $\vec{b}_1 = 2\hat{i} + 7\hat{j} - 3\hat{k}$ and

$\vec{b}_2 = -\hat{i} + 2\hat{j} + 4\hat{k}$

If θ is the angle between given lines, then

$\cos\theta = \left| \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|} \right|$

$\Rightarrow \quad \cos\theta = \left| \frac{2(-1) + 7(2) - 3(4)}{\sqrt{2^2+7^2+(-3)^2}\sqrt{(-1)^2+2^2+4^2}} \right|$

$\Rightarrow \quad \cos\theta = \left| \frac{-2+14-12}{\sqrt{62}\sqrt{21}} \right| = 0$

$\Rightarrow \quad \theta = \pi/2$

$\therefore$ Given lines are perpendicular.

47. Let the equation of line passing through (1, 2, –4) and perpendicular to the lines

$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$

and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$

be $\frac{x-1}{l} = \frac{y-2}{m} = \frac{z+4}{n}$ ...(i)

$\therefore \quad l(3) + m(-16) + n(7) = 0$ and

$l(3) + m(8) + n(-5) = 0$

$\Rightarrow \quad \frac{l}{80-56} = \frac{m}{21+15} = \frac{n}{24+48}$

www.examfires.in

$\Rightarrow \quad \frac{l}{24}=\frac{m}{36}=\frac{n}{72} \Rightarrow \frac{l}{2}=\frac{m}{3}=\frac{n}{6}$

$\therefore$ The equation of the required line is

$$\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{6}$$

and its vector equation is

$\vec{r}=(\hat{i}+2\hat{j}-4\hat{k})+\lambda(2\hat{i}+3\hat{j}+6\hat{k})$

48. The line of shortest distance between two skew lines is perpendicular to both the lines.

49. We have, $\vec{r}=(4\hat{i}-\hat{j})+\lambda(\hat{i}+2\hat{j}-3\hat{k})$ ...(i)

$\vec{r}=(\hat{i}-\hat{j}+2\hat{k})+\mu(2\hat{i}+4\hat{j}-5\hat{k})$ ...(ii)

Comparing with lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\lambda\vec{b}_2$, we get

$\vec{a}_1=(4\hat{i}-\hat{j}) \quad , \quad \vec{b}_1=\hat{i}+2\hat{j}-3\hat{k}$

$\vec{a}_2=\hat{i}-\hat{j}+2\hat{k} \quad , \quad \vec{b}_2=2\hat{i}+4\hat{j}-5\hat{k}$

Now, $\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}$

$=\hat{i}(-10+12)-\hat{j}(-5+6)+\hat{k}(4-4)$

$=2\hat{i}-\hat{j}$

and $|\vec{b}_1\times\vec{b}_2|=\sqrt{(2)^2+(1)^2}=\sqrt{5}$

$\vec{a}_2-\vec{a}_1=-3\hat{i}+2\hat{k}$

$\therefore$ Shortest distance, $d=\left|\frac{(\vec{b}_1\times\vec{b}_2)\cdot(\vec{a}_2-\vec{a}_1)}{|\vec{b}_1\times\vec{b}_2|}\right|$

$=\left|\frac{(2\hat{i}-\hat{j})\cdot(-3\hat{i}+2\hat{k})}{\sqrt{5}}\right| =\left|\frac{-6+0+0}{\sqrt{5}}\right|=\frac{6}{\sqrt{5}}$

50. The given lines are

$\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and

$\vec{r}=(2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k})$

We know that

S.D. between the lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$, and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ is given by $d=\left|\frac{(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|$

On comparing, we get

$\vec{a}_1=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}_1=\hat{i}-\hat{j}+\hat{k}$

$\vec{a}_2=2\hat{i}-\hat{j}-\hat{k}$, $\vec{b}_2=2\hat{i}+\hat{j}+2\hat{k}$

$\therefore \quad \vec{a}_2-\vec{a}_1=\hat{i}-3\hat{j}-2\hat{k}$

$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 1\\ 2 & 1 & 2\end{vmatrix}=-3\hat{i}+0\cdot\hat{j}+3\hat{k}$

$\Rightarrow \quad |\vec{b}_1\times\vec{b}_2|=\sqrt{(-3)^2+3^2}=3\sqrt{2}$

$\Rightarrow \quad (\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)=1(-3)-3(0)-2(3)=-9.$

$\therefore \quad d=\left|\frac{-9}{3\sqrt{2}}\right|=\frac{3}{\sqrt{2}}=\frac{3}{2}\sqrt{2}$ units

51. The given lines are

$\vec{r}=2\hat{i}-5\hat{j}+\hat{k}+\lambda(3\hat{i}+2\hat{j}+6\hat{k})$ and

$\vec{r}=7\hat{i}-6\hat{k}+\mu(\hat{i}+2\hat{j}+2\hat{k})$

S.D. between the lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ is given by

$d=\left|\frac{(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|$

On comparing, we get

$\vec{a}_1=2\hat{i}-5\hat{j}+\hat{k}$, $\vec{b}_1=3\hat{i}+2\hat{j}+6\hat{k}$

$\vec{a}_2=7\hat{i}-6\hat{k}$, $\vec{b}_2=\hat{i}+2\hat{j}+2\hat{k}$

$\therefore \quad \vec{a}_2-\vec{a}_1=5\hat{i}+5\hat{j}-7\hat{k}$

$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 2 & 6\\ 1 & 2 & 2\end{vmatrix}=-8\hat{i}+4\hat{k}$

$\therefore \quad |\vec{b}_1\times\vec{b}_2|=\sqrt{(-8)^2+4^2}=4\sqrt{5}$

Hence, $d=\left|\frac{(5\hat{i}+5\hat{j}-7\hat{k})\cdot(-8\hat{i}+4\hat{k})}{4\sqrt{5}}\right|$

$=\frac{|5(-8)-7(4)|}{4\sqrt{5}}=\frac{68}{4\sqrt{5}}=\frac{17\sqrt{5}}{5}$ units

52. $\vec{r}=\hat{i}+\hat{j}+\lambda(2\hat{i}-\hat{j}+\hat{k})$

$\therefore \quad \vec{a}_1=\hat{i}+\hat{j}, \vec{b}_1=2\hat{i}-\hat{j}+\hat{k}$

Also, $\vec{r}=2\hat{i}+\hat{j}-\hat{k}+\mu(3\hat{i}-5\hat{j}+2\hat{k})$

$\therefore \quad \vec{a}_2=2\hat{i}+\hat{j}-\hat{k}, \vec{b}_2=3\hat{i}-5\hat{j}+2\hat{k}$

www.examstres.in

So, $\vec{a}_2 - \vec{a}_1 = \hat{i} - \hat{k}$

And, $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} = 3\hat{i} - \hat{j} - 7\hat{k}$

Shortest distance between two skew lines is,

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$\Rightarrow \quad d = \left| \frac{(3\hat{i} - \hat{j} - 7\hat{k}) \cdot (\hat{i} - \hat{k})}{\sqrt{3^2 + (-1)^2 + (-7)^2}} \right|$$

$$\Rightarrow \quad d = \left| \frac{3+7}{\sqrt{59}} \right| = \frac{10}{\sqrt{59}} \text{ units.}$$

53. Let $l_1 : \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$

$$\Rightarrow \quad \frac{x-(-1)}{7} = \frac{y-(-1)}{-6} = \frac{z-(-1)}{1}$$

and $l_2 : \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$

$\therefore$ Vector equation of lines are

$\vec{r} = -\hat{i} - \hat{j} - \hat{k} + \lambda(7\hat{i} - 6\hat{j} + \hat{k})$

and $\vec{r} = 3\hat{i} + 5\hat{j} + 7\hat{k} + \mu(\hat{i} - 2\hat{j} + \hat{k})$

We get $\vec{a}_1 = -\hat{i} - \hat{j} - \hat{k}$, $\vec{b}_1 = 7\hat{i} - 6\hat{j} + \hat{k}$

and $\vec{a}_2 = 3\hat{i} + 5\hat{j} + 7\hat{k}$, $\vec{b}_2 = \hat{i} - 2\hat{j} + \hat{k}$

So, $\vec{a}_2 - \vec{a}_1 = (3\hat{i} + 5\hat{j} + 7\hat{k}) - (-\hat{i} - \hat{j} - \hat{k})$

$$= 4\hat{i} + 6\hat{j} + 8\hat{k}$$

And, $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix} = -4\hat{i} - 6\hat{j} - 8\hat{k}$

Shortest distance between two skew lines is,

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$\Rightarrow \quad d = \left| \frac{(-4\hat{i} - 6\hat{j} - 8\hat{k}).(4\hat{i} + 6\hat{j} + 8\hat{k})}{\sqrt{(-4)^2 + (-6)^2 + (-8)^2}} \right|$$

$$\Rightarrow \quad d = \left| \frac{-16 - 36 - 64}{\sqrt{116}} \right| \Rightarrow d = 2\sqrt{29} \text{ units}$$

54. Given lines are

$l_1 : \vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$;

$l_2 : \vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(4\hat{i} + 6\hat{j} + 12\hat{k})$

$\therefore$ We have $\vec{a}_1 = \hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{b}_1 = 2\hat{i} + 3\hat{j} + 6\hat{k}$

and $\vec{a}_2 = 3\hat{i} + 3\hat{j} - 5\hat{k}$, $\vec{b}_2 = 4\hat{i} + 6\hat{j} + 12\hat{k}$

So, $\vec{a}_2 - \vec{a}_1 = 2\hat{i} + \hat{j} - \hat{k}$

Also, $\vec{b}_2 = 4\hat{i} + 6\hat{j} + 12\hat{k} = 2\vec{b}_1 \Rightarrow \vec{b}_1 \parallel \vec{b}_2$

Hence l_1 and l_2 are parallel lines.

Shortest distance between two parallel lines is,

$$d = \left| \frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|} \right|$$

$$\Rightarrow \quad d = \left| \frac{(2\hat{i} + 3\hat{j} + 6\hat{k}) \times (2\hat{i} + \hat{j} - \hat{k})}{\sqrt{2^2 + 3^2 + 6^2}} \right|$$

$$\Rightarrow \quad d = \left| \frac{-9\hat{i} + 14\hat{j} - 4\hat{k}}{7} \right|$$

$$\Rightarrow \quad d = \frac{\sqrt{(-9)^2 + 14^2 + (-4)^2}}{7} = \frac{\sqrt{293}}{7} \text{ units.}$$

55. Here, the lines are

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ and

$\vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$

Here, $\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b}_1 = \hat{i} - 3\hat{j} + 2\hat{k}$ and

$\vec{a}_2 = 4\hat{i} + 5\hat{j} + 6\hat{k}$, $\vec{b}_2 = 2\hat{i} + 3\hat{j} + \hat{k}$

The shortest distance between the lines is given by

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$\vec{a}_2 - \vec{a}_1 = 3\hat{i} + 3\hat{j} + 3\hat{k}$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ 2 & 3 & 1 \end{vmatrix} = -9\hat{i} + 3\hat{j} + 9\hat{k}$$

$$\Rightarrow \quad |\vec{b}_1 \times \vec{b}_2| = \sqrt{(-9)^2 + 3^2 + 9^2} = 3\sqrt{19}$$

Also, $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$

$$= (3\hat{i} + 3\hat{j} + 3\hat{k}) \cdot (-9\hat{i} + 3\hat{j} + 9\hat{k})$$

$$= 3 \times (-9) + 3 \times 3 + 3 \times 9 = 9$$

$\therefore \quad d = \left| \frac{9}{3\sqrt{19}} \right| = \frac{3}{\sqrt{19}}$ unit.

56. The given lines are

$\vec{r} = (6\hat{i} + 2\hat{j} + 2\hat{k}) + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$ and

www.exambites.in

$\vec{r} = (-4\hat{i} - \hat{k}) + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$

Here, $\vec{a}_1 = 6\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b}_1 = \hat{i} - 2\hat{j} + 2\hat{k}$

$\vec{a}_2 = -4\hat{i} - \hat{k}$, $\vec{b}_2 = 3\hat{i} - 2\hat{j} - 2\hat{k}$

$\therefore$ Shortest distance between the lines is

$$d = \left|\frac{(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}\right|$$

Now, $\vec{a}_2 - \vec{a}_1 = -10\hat{i} - 2\hat{j} - 3\hat{k}$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} = 8\hat{i} + 8\hat{j} + 4\hat{k}$$

$\therefore$ $|\vec{b}_1 \times \vec{b}_2| = \sqrt{8^2 + 8^2 + 4^2} = \sqrt{64 + 64 + 16} = 12$

and $(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2) = (-10)(8) + (-2)(8) + (-3)(4) = -108$

$\therefore$ $d = \left|\frac{-108}{12}\right| = 9$ units.

57. The eqns. of the lines are

$\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} - \hat{k})$, $\vec{r} = 2\hat{i} - \hat{j} + \mu(\hat{i} - \hat{j} - \hat{k})$

Here, $\vec{a}_1 = \hat{i} - \hat{j}$, $\vec{b}_1 = 2\hat{i} - \hat{k}$ and

$\vec{a}_2 = 2\hat{i} - \hat{j}$, $\vec{b}_2 = \hat{i} - \hat{j} - \hat{k}$.

The S.D. between the given lines $= \left|\frac{(\vec{b}_1 \times \vec{b}_2)\cdot(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$

Now, $\vec{a}_2 - \vec{a}_1 = \hat{i}$

$$\Rightarrow \quad \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & -1 \\ 1 & -1 & -1 \end{vmatrix} = -\hat{i} + \hat{j} - 2\hat{k}$$

$$\Rightarrow \quad |\vec{b}_1 \times \vec{b}_2| = \sqrt{(-1)^2 + 1^2 + (-2)^2} = \sqrt{6}$$

$$\therefore \quad \text{S.D.} = \left|\frac{(-\hat{i} + \hat{j} - 2\hat{k})\cdot\hat{i}}{\sqrt{6}}\right| = \frac{1}{\sqrt{6}} \ (\neq 0)$$

$\Rightarrow$ The given lines do not intersect.

58. The given lines are

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ and } \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-5}{5}$$

Their vector eqns. are

$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ and

$\vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k} + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$

Here, $\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{a}_2 = 2\hat{i} + 3\hat{j} + 5\hat{k}$, $\vec{b}_2 = 3\hat{i} + 4\hat{j} + 5\hat{k}$

Now, $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = -\hat{i} + 2\hat{j} - \hat{k}$.

$$\Rightarrow \quad |\vec{b}_1 \times \vec{b}_2| = \sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{6}$$

and $\vec{a}_2 - \vec{a}_1 = \hat{i} + \hat{j} + 2\hat{k}$.

$$\therefore \quad \text{S.D.} = \left|\frac{(\vec{b}_1 \times \vec{b}_2)\cdot(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$$

$$= \left|\frac{(-\hat{i} + 2\hat{j} - \hat{k})\cdot(\hat{i} + \hat{j} + 2\hat{k})}{\sqrt{6}}\right| = \left|\frac{-1 + 2 - 2}{\sqrt{6}}\right| = \frac{1}{\sqrt{6}} \text{ unit}.$$

59. From the given equations, we get

$\vec{r} = \hat{i} - 2\hat{j} + 3\hat{k} + t(-\hat{i} + \hat{j} - 2\hat{k})$

and $\vec{r} = \hat{i} - \hat{j} - \hat{k} + s(\hat{i} + 2\hat{j} - 2\hat{k})$

Here, $\vec{a}_1 = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{a}_2 = \hat{i} - \hat{j} - \hat{k}$

$\vec{b}_1 = -\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b}_2 = \hat{i} + 2\hat{j} - 2\hat{k}$

Here, $\vec{a}_2 - \vec{a}_1 = (\hat{i} - \hat{j} - \hat{k}) - (\hat{i} - 2\hat{j} + 3\hat{k}) = \hat{j} - 4\hat{k}$

Also, $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & -2 \end{vmatrix} = 2\hat{i} - 4\hat{j} - 3\hat{k}$

$|\vec{b}_1 \times \vec{b}_2| = \sqrt{2^2 + (-4)^2 + (-3)^2} = \sqrt{4 + 16 + 9} = \sqrt{29}$

$$\therefore \quad \text{Shortest distance} = \left|\frac{(\vec{b}_1 \times \vec{b}_2)\cdot(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$$

$$= \left|\frac{(2\hat{i} - 4\hat{j} - 3\hat{k})\cdot(\hat{j} - 4\hat{k})}{\sqrt{29}}\right| = \left|\frac{-4 + 12}{\sqrt{29}}\right| = \frac{8}{\sqrt{29}} \text{ units}$$

60. Vector equation of a line passing through (2, 3, 2) and parallel to the line

$\vec{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$ is given by

$\vec{r} = (2\hat{i} + 3\hat{j} + 2\hat{k}) + \mu(2\hat{i} - 3\hat{j} + 6\hat{k})$

Now, $\vec{a}_1 = -2\hat{i} + 3\hat{j}$, $\vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

$\vec{a}_2 = 2\hat{i} + 3\hat{j} + 2\hat{k}$

Distance between given parallel lines

$$= \left|\frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right| = \left|\frac{(2\hat{i} - 3\hat{j} + 6\hat{k})\cdot(4\hat{i} + 0\hat{j} + 2\hat{k})}{|\sqrt{4 + 9 + 36}|}\right|$$

$$= \left|\frac{8 + 0 + 12}{\sqrt{49}}\right| = \frac{20}{7} \text{ units}$$

www.exambire.in

61. We have, $2x + y - z = 5$

$\Rightarrow \quad \frac{x}{5/2} + \frac{y}{5} + \frac{z}{-5} = 1$

which is the equation of plane in intercept form.

$\therefore$ Intercepts on x, y and z-axis respectively are $\frac{5}{2}, 5, -5$.

$\therefore$ Required length of intercept $= \frac{5}{2}$

62. Let $\vec{n} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

$\therefore \quad \hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{2\hat{i} - 3\hat{j} + 6\hat{k}}{\sqrt{4+9+36}} = \frac{2\hat{i} - 3\hat{j} + 6\hat{k}}{7}$

So, the required equation of the plane is

$\vec{r} \cdot \left(\frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}\right) = 5$

$\Rightarrow \quad \vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) = 35$

63. Intercepts are $a = 3$, $b = -4$, $c = 2$

$\therefore$ The intercept form of the plane is

$\frac{x}{3} + \left(\frac{y}{-4}\right) + \frac{z}{2} = 1 \Rightarrow 4x - 3y + 6z = 12$

So, the vector equation of the plane is

$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (4\hat{i} - 3\hat{j} + 6\hat{k}) = 12$

$\Rightarrow \quad \vec{r} \cdot (4\hat{i} - 3\hat{j} + 6\hat{k}) = 12$

64. Let α, β and γ be the angles made by $\vec{n}$ with x, y and z-axis, respectively.

Given $\alpha = \beta = \gamma \Rightarrow \cos\alpha = \cos\beta = \cos\gamma$

$\Rightarrow l = m = n$, where l, m, n are direction cosines of $\vec{n}$.

But $l^2 + m^2 + n^2 = 1 \Rightarrow l^2 + l^2 + l^2 = 1 \Rightarrow l = \pm\frac{1}{\sqrt{3}}$

So, $l = m = n = \pm\frac{1}{\sqrt{3}}$

$\therefore$ The normal form of the plane is $lx + my + nz = d$

$\Rightarrow \quad \pm\frac{1}{\sqrt{3}}x \pm \frac{1}{\sqrt{3}}y \pm \frac{1}{\sqrt{3}}z = 5\sqrt{3}$

$\Rightarrow \quad \pm x \pm y \pm z = 15$

65. *Refer to answer 61.*

66. *Refer to answer 61.*

67. Here the position vectors of A, B and C are respectively $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + \hat{j} + 2\hat{k}$ and $2\hat{i} + 3\hat{k}$.

$\therefore \quad \overrightarrow{AB} = (\hat{i} + \hat{j} + 2\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) = -\hat{i} + 2\hat{j} + \hat{k}$

$\overrightarrow{AC} = (2\hat{i} + 3\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) = 0 \cdot \hat{i} + \hat{j} + 2\hat{k}$

A vector normal to the plane containing points A, B and C is

$$\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix} = 3\hat{i} + 2\hat{j} - \hat{k}$$

$\therefore$ The required unit vector

$= \frac{\vec{n}}{|\vec{n}|} = \frac{3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{3^2 + 2^2 + (-1)^2}} = \frac{1}{\sqrt{14}}(3\hat{i} + 2\hat{j} - \hat{k})$

68. Equation of the line passing through $A(3, -4, -5)$ and $B(2, -3, 1)$ is

$\frac{x-3}{2-3} = \frac{y+4}{-3+4} = \frac{z+5}{1+5}$

$\Rightarrow \quad \frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6} = \lambda(\text{say}) \quad \ldots(i)$

The coordinates of any point on this line are

$x = -\lambda + 3$, $y = \lambda - 4$, $z = 6\lambda - 5$

Now, equation of the plane passing through given three points (1, 2, 3), (4, 2, –3) and (0, 4, 3) is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 3 & 0 & -6 \\ -1 & 2 & 0 \end{vmatrix} = 0$$

$\Rightarrow \quad (x - 1)\,12 + (y - 2)\,6 + (z - 3)6 = 0$

$\Rightarrow \quad 12x - 12 + 6y - 12 + 6z - 18 = 0$

$\Rightarrow \quad 12x + 6y + 6z - 42 = 0$

$\Rightarrow \quad 2x + y + z - 7 = 0 \quad \ldots(ii)$

Since, line through the points (3, –4, –5) and (2, –3, 1) crosses the plane (ii).

$\therefore \quad 2(-\lambda + 3) + 1(\lambda - 4) + (6\lambda - 5) - 7 = 0$

$\Rightarrow \quad -2\lambda + 6 + \lambda - 4 + 6\lambda - 5 - 7 = 0$

$\Rightarrow \quad 5\lambda = 10 \Rightarrow \lambda = 2$

So, the coordinates of any point on the line (i) are,

$x = -2 + 3$, $y = 2 - 4$, $z = 12 - 5$

$\Rightarrow \quad x = 1$, $y = -2$, $z = 7$.

69. General equation of a plane passing through $L(2, 2, 1)$ is $a(x - 2) + b(y - 2) + c(z - 1) = 0 \quad \ldots(i)$

It will pass through $M(3, 0, 1)$ and $N(4, -1, 0)$ if

$a(3 - 2) + b(0 - 2) + c(1 - 1) = 0$

$\Rightarrow a - 2b + 0 \cdot c = 0 \quad \ldots(ii)$

$a(4 - 2) + b(-1 - 2) + c(0 - 1) = 0$

$\Rightarrow 2a - 3b - c = 0 \quad \ldots(iii)$

Solving (ii) and (iii), we get

$\frac{a}{2+0} = \frac{b}{0+1} = \frac{c}{-3+4} \Rightarrow \frac{a}{2} = \frac{b}{1} = \frac{c}{1} = \lambda(\text{say})$

$\Rightarrow \quad a = 2\lambda$, $b = \lambda$, $c = \lambda$

www.exambites.in

$\therefore$ From (i), we get
$2\lambda(x-2)+\lambda(y-2)+\lambda(z-1)=0$
$\Rightarrow \quad 2x+y+z-7=0$ ...(iv)
Equation of line passing through $A(3,-4,-5)$ and $B(2,-3,1)$ is $\frac{x-3}{2-3}=\frac{y+4}{-3+4}=\frac{z+5}{1+5}$

$\Rightarrow \quad \frac{x-3}{-1}=\frac{y+4}{1}=\frac{z+5}{6}=r(\text{say})$
$\Rightarrow \quad x=-r+3, y=r-4, z=6r-5$
Any point on the line AB is $P(-r+3, r-4, 6r-5)$
$\because$ It lies on the plane (iv)
$\therefore \quad 2(-r+3)+(r-4)+(6r-5)-7=0$
$\Rightarrow \quad 5r=10 \Rightarrow r=2$
So, coordinates of point P are $(1,-2,7)$.
Let P divides the line segment AB in $k:1$.

$$\therefore \left(\frac{3\times1+2\times k}{k+1}, \frac{1\times(-4)+k\times(-3)}{k+1}, \frac{-5\times1+k\times1}{k+1}\right) = (1,-2,7)$$

$$\Rightarrow \quad \frac{3+2k}{k+1}=1 \Rightarrow 3+2k=k+1 \Rightarrow k=-2$$

Hence, P divides the line segment AB in the ratio $2:1$ externally.

70. Equation of the plane passing through the line of intersection of the given plane is
$(x+2y+3z-4)+\lambda(2x+y-z+5)=0$
$\Rightarrow \quad (2\lambda+1)x+(\lambda+2)y+(3-\lambda)z=4-5\lambda$
This equation of the plane can be written in the intercept form as

$$\frac{x}{\left(\frac{4-5\lambda}{2\lambda+1}\right)}+\frac{y}{\left(\frac{4-5\lambda}{\lambda+2}\right)}+\frac{z}{\left(\frac{4-5\lambda}{3-\lambda}\right)}=1$$

It is given that, x-intercept $= 2\times z$-intercept

$$\therefore \quad \frac{4-5\lambda}{2\lambda+1}=2\left(\frac{4-5\lambda}{3-\lambda}\right)$$

$$\Rightarrow \quad 4\lambda+2=3-\lambda \Rightarrow \lambda=\frac{1}{5}$$

Therefore, the equation of the plane is

$$\frac{7}{5}x+\frac{11}{5}y+\frac{14}{5}z=3$$

$\Rightarrow \quad 7x+11y+14z=15$

71. The given planes are
$\vec{r}\cdot(2\hat{i}+2\hat{j}-3\hat{k})=7$ and $\vec{r}\cdot(2\hat{i}+5\hat{j}-3\hat{k})=9$
$\Rightarrow \quad 2x+2y-3z=7$ and $2x+5y-3z=9$
$\therefore$ Equation of the plane passing through their line of intersection is
$(2x+2y-3z-7)+\lambda(2x+5y-3z-9)=0$
$\Rightarrow (2+2\lambda)x+(2+5\lambda)y+(-3\lambda-3)z-(7+9\lambda)=0$ ...(i)
Given, x-intercept = z-intercept

$$\therefore \quad \frac{7+9\lambda}{2+2\lambda}=\frac{7+9\lambda}{-3\lambda-3} \Rightarrow -3\lambda-3=2+2\lambda.$$

$\Rightarrow \quad \lambda=-1$
Put $\lambda=-1$ in (i), we get, $3y=2$
This is the cartesian eq. of the plane.
Corresponding vector eq. is $\vec{r}\cdot(3\hat{j})=2$

72. The given points are
$P(\hat{i}+\hat{j}-2\hat{k})\equiv(1,1,-2)$, $Q(2\hat{i}-\hat{j}+\hat{k})\equiv(2,-1,1)$
and $R(\hat{i}+2\hat{j}+\hat{k})\equiv(1,2,1)$

Eq. of any plane through P is
$a(x-1)+b(y-1)+c(z+2)=0$ ...(i)
As it is also passing through Q and R, so
$a-2b+3c=0$ ... (ii)
$0\cdot a+b+3c=0$ ... (iii)
Solving (ii) and (iii), we get

$$\frac{a}{-6-3}=\frac{b}{0-3}=\frac{c}{1-0} \Rightarrow \frac{a}{9}=\frac{b}{3}=\frac{c}{-1}$$

$\therefore$ From (i), equation of plane is
$9(x-1)+3(y-1)-(z+2)=0$
$\Rightarrow \quad 9x+3y-z-14=0 \Rightarrow \vec{r}\cdot(9\hat{i}+3\hat{j}-\hat{k})=14$
Now any point on the line
$\vec{r}=3\hat{i}-\hat{j}-\hat{k}+\lambda(2\hat{i}-2\hat{j}+\hat{k})$

i.e., $\frac{x-3}{2}=\frac{y+1}{-2}=\frac{z+1}{1}=\lambda$ (say) is
$(2\lambda+3, -2\lambda-1, \lambda-1)$
If it lies on plane, then
$9(2\lambda+3)+3(-2\lambda-1)-(\lambda-1)-14=0$
$\Rightarrow \quad 11\lambda+11=0 \Rightarrow \lambda=-1$
$\therefore$ Their point of intersection is
$(-2+3, 2-1, -1-1)=(1,1,-2)$

73. The given lines are

$$\frac{5-x}{-4}=\frac{y-7}{4}=\frac{z+3}{-5} \text{ and } \frac{x-8}{7}=\frac{2y-8}{2}=\frac{z-5}{3}$$

$$\Rightarrow \quad \frac{x-5}{4}=\frac{y-7}{4}=\frac{z+3}{-5} \text{ and } \frac{x-8}{7}=\frac{y-4}{1}=\frac{z-5}{3}$$

The condition of coplanarity of these lines is

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$$

$$\text{Here, L.H.S.}=\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

www.examrites.net

$$= \begin{vmatrix} 8-5 & 4-7 & 5-(-3) \\ 4 & 4 & -5 \\ 7 & 1 & 3 \end{vmatrix} = \begin{vmatrix} 3 & -3 & 8 \\ 4 & 4 & -5 \\ 7 & 1 & 3 \end{vmatrix}$$

$= 3(12 + 5) + 3(12 + 35) + 8(4 - 28)$
$= 51 + 141 - 192 = 0 = \text{R.H.S.}$
Hence the given lines are coplanar.

74. Any point on the line

$$\frac{x-2}{1} = \frac{y-2}{3} = \frac{z-3}{1} = r(\text{say}) \quad \text{...(i)}$$

is $(r + 2, 3r + 2, r + 3)$
Any point on the line

$$\frac{x-2}{1} = \frac{y-3}{4} = \frac{z-4}{2} = k(\text{say}) \quad \text{...(ii)}$$

is $(k + 2, 4k + 3, 2k + 4)$
For lines (i) and (ii) to intersect, we must have $r + 2 = k + 2, 3r + 2 = 4k + 3, r + 3 = 2k + 4$
On solving these, we get, $r = k = -1$
$\therefore$ Lines (i) and (ii) intersect and their point of intersection is $(1, -1, 2)$
Now, required equation of plane is

$$\begin{vmatrix} x-2 & y-2 & z-3 \\ 1 & 3 & 1 \\ 1 & 4 & 2 \end{vmatrix} = 0$$

$\Rightarrow \quad (x-2)(6-4) - (y-2)(2-1) + (z-3)(4-3) = 0$
$\Rightarrow \quad 2x - 4 - y + 2 + z - 3 = 0$
$\Rightarrow \quad 2x - y + z - 5 = 0$

75. We have, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} = \lambda$ (say)

$\Rightarrow \quad x = 2\lambda + 1, y = 3\lambda - 1, z = 4\lambda + 1$
Since, given both lines intersect.
$\therefore$ Point $(2\lambda + 1, 3\lambda - 1, 4\lambda + 1)$ satisfies

$$\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$$

$$\Rightarrow \quad \frac{2\lambda+1-3}{1} = \frac{3\lambda-1-k}{2} = \frac{4\lambda+1}{1} \quad \text{...(i)}$$

$$\Rightarrow \quad 2\lambda - 2 = 4\lambda + 1 \Rightarrow 2\lambda = -3 \Rightarrow \lambda = \frac{-3}{2}$$

Putting value of λ in (i), we get

$$2\left(\frac{-3}{2}\right) - 2 = \frac{3\left(\frac{-3}{2}\right) - 1 - k}{2}$$

$$\Rightarrow \quad -5 \times 2 = -\frac{11}{2} - k \Rightarrow \quad k = \frac{-11}{2} + 10 = \frac{9}{2}$$

Now, required equation of plane is

$$\begin{vmatrix} x-1 & y+1 & z-1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$\Rightarrow \quad -5x + 5 + 2y + 2 + z - 1 = 0$
$\Rightarrow \quad 5x - 2y - z - 6 = 0$

76. The given lines are

$$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(3\hat{i} - \hat{j}) \text{ and } \vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + 3\hat{k})$$

Here, $\vec{a}_1 = \hat{i} + \hat{j} - \hat{k}$, $\vec{b}_1 = 3\hat{i} - \hat{j}$

$$\vec{a}_2 = 4\hat{i} - \hat{k}, \vec{b}_2 = 2\hat{i} + 3\hat{k}$$

We know that the lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ are coplanar if $[\vec{a}_2 - \vec{a}_1\ \vec{b}_1\ \vec{b}_2] = 0$
or $[\vec{a}_1\ \vec{b}_1\ \vec{b}_2] = [\vec{a}_2\ \vec{b}_1\ \vec{b}_2]$.

$$\text{Now, } [\vec{a}_1\ \vec{b}_1\ \vec{b}_2] = \begin{vmatrix} 1 & 1 & -1 \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix}$$

$= 1(-3) - 1(9) - 1(2) = -14$

$$\text{and } [\vec{a}_2\ \vec{b}_1\ \vec{b}_2] = \begin{vmatrix} 4 & 0 & -1 \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix} = 4(-3) - 1(2) = -14$$

Hence $[\vec{a}_1\ \vec{b}_1\ \vec{b}_2] = [\vec{a}_2\ \vec{b}_1\ \vec{b}_2]$
$\Rightarrow$ The given lines are coplanar.
Now the eq. of the plane containing the given lines is

$$(\vec{r} - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0, \text{ or } [\vec{r}\ \vec{b}_1\ \vec{b}_2] = [\vec{a}_1\ \vec{b}_1\ \vec{b}_2]$$

$$\Rightarrow \quad \vec{r} \cdot (\vec{b}_1 \times \vec{b}_2) = -14. \quad \text{...(i)}$$

$$\text{Since, } \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix} = -3\hat{i} - 9\hat{j} + 2\hat{k}$$

$\therefore$ Eqn (i) becomes, $\vec{r} \cdot (-3\hat{i} - 9\hat{j} + 2\hat{k}) = -14$

$$\Rightarrow \quad \vec{r} \cdot (3\hat{i} + 9\hat{j} - 2\hat{k}) = 14.$$

77. The given lines are

$$\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5} \text{ and } \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$$

These lines will be coplanar, if

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0 \quad \text{...(i)}$$

www.exambites.in

Here, $x_1 = -3, y_1 = 1, z_1 = 5$;
$a_1 = -3, b_1 = 1, c_1 = 5$;
$x_2 = -1, y_2 = 2, z_2 = 5$;
$a_2 = -1, b_2 = 2, c_2 = 5$

$$\therefore \quad \text{L.H.S. of (i)} = \begin{vmatrix} -1+3 & 2-1 & 5-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = 2(-5) - 1(-10) = -10 + 10 = 0$$

$\therefore$ Given lines are coplanar.

Also eqn. of the plane containing the lines is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x+3 & y-1 & z-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = 0$$

$\Rightarrow$ $(x+3)(5-10)-(y-1)(-15+5)+(z-5)(-6+1)=0$
$\Rightarrow$ $-5x - 15 + 10y - 10 - 5z + 25 = 0$
$\Rightarrow$ $5x - 10y + 5z = 0 \Rightarrow x - 2y + z = 0$

78. The given lines are

$$\frac{x-1}{-3} = \frac{y-2}{-2k} = \frac{z-3}{2} \quad \text{...(i)}$$

$$\text{and } \frac{x-1}{k} = \frac{y-2}{1} = \frac{z-3}{5} \quad \text{...(ii)}$$

Since, both the lines are perpendicular.

$\therefore$ $-3 \cdot k - 2k \cdot 1 + 2 \cdot 5 = 0$
$\Rightarrow$ $-5k + 10 = 0 \Rightarrow k = 2.$

Now eqn. of the plane containing lines (i) & (ii) is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ -3 & -4 & 2 \\ 2 & 1 & 5 \end{vmatrix} = 0$$

$\Rightarrow$ $(x-1)(-20-2)-(y-2)(-15-4)+(z-3)(-3+8)=0$
$\Rightarrow$ $-22\,(x-1) + 19(y-2) + 5(z-3) = 0$
$\Rightarrow$ $-22x + 19y + 5z = 31.$

79. The given planes are

$$\vec{r} \cdot (\hat{i} - 2\hat{j} - 2\hat{k}) = 1 \text{ and } \vec{r} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 0$$

If θ is the angle between given planes then

$$\cos\theta = \left| \frac{(\hat{i} - 2\hat{j} - 2\hat{k}) \cdot (3\hat{i} - 6\hat{j} + 2\hat{k})}{|\hat{i} - 2\hat{j} - 2\hat{k}||3\hat{i} - 6\hat{j} + 2\hat{k}|} \right|$$

$$= \left| \frac{(1)(3) + (-2)(-6) + (-2)(2)}{\sqrt{1^2 + 2^2 + 2^2}\sqrt{3^2 + 6^2 + 2^2}} \right|$$

$$= \left| \frac{3 + 12 - 4}{\sqrt{9}\ \sqrt{49}} \right| = \left| \frac{11}{3 \times 7} \right| = \frac{11}{21}$$

$\therefore$ Required angle, $\theta = \cos^{-1}\left(\frac{11}{21}\right)$

80. Equation of both the planes can be written as $2x - y + 2z = 5$ and $2x - y + 2z = 8$.

Distance between both the planes

$$= \left| \frac{8-5}{\sqrt{4+1+4}} \right| = \frac{3}{\sqrt{9}} = 1 \text{ unit}$$

81. The direction ratios of normal to the given plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 2$ are 1, 1, 1.

So, cartesian equation of plane passing through (a, b, c) and having d.r's of normal proportional to 1, 1, 1 is

$1 \cdot (x - a) + 1 \cdot (y - b) + 1 \cdot (z - c) = 0$
$\Rightarrow$ $x + y + z = a + b + c$
$\therefore$ Vector equation of the required plane is,
$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = a + b + c$

82. Let the eq. of the plane through (2, 1, –1) be

$a(x - 2) + b(y - 1) + c(z + 1) = 0$ ...(i)

Also, point (–1, 3, 4) lies on it

$\therefore$ $-3a + 2b + 5c = 0$ ...(ii)

Also (i) is $\perp$ to the plane $x - 2y + 4z = 10$

$\therefore$ $a \cdot 1 + b(-2) + c \cdot 4 = 0$
$\Rightarrow$ $a - 2b + 4c = 0$ ...(iii)

Solving (ii) and (iii), we get

$$\frac{a}{8+10} = \frac{b}{5+12} = \frac{c}{6-2} \quad \Rightarrow \quad \frac{a}{18} = \frac{b}{17} = \frac{c}{4}$$

$\therefore$ From (i), required eqn. of the plane is

$18(x - 2) + 17(y - 1) + 4(z + 1) = 0$
$\Rightarrow$ $18x + 17y + 4z = 49$

In vector form, eq. of this plane is

$\vec{r} \cdot (18\hat{i} + 17\hat{j} + 4\hat{k}) = 49.$

83. Cartesian equation of plane passing through points $A(2, 2, -1)$, $B(3, 4, 2)$ and $C(7, 0, 6)$ is given by

$$\begin{vmatrix} x-2 & y-2 & z+1 \\ 3-2 & 4-2 & 2-(-1) \\ 7-2 & 0-2 & 6-(-1) \end{vmatrix} = 0$$

www.examfree.in

$$\Rightarrow \begin{vmatrix} x-2 & y-2 & z+1 \\ 1 & 2 & 3 \\ 5 & -2 & 7 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(14+6)-(y-2)(7-15) + (z+1)(-2-10) = 0$$

$$\Rightarrow 20(x-2)+8(y-2)-12(z+1)=0$$

$$\Rightarrow 20x+8y-12z-68=0$$

$$\Rightarrow 5x+2y-3z-17=0 \quad \text{... (i)}$$

Now, let $\vec{r} = x\hat{i}+y\hat{j}+z\hat{k}$ be any vector on plane.

Then (i) becomes $\vec{r}\cdot(5\hat{i}+2\hat{j}-3\hat{k})=17$
which is the vector equation of plane.
Equation of plane parallel to plane (i) is given by
$5x + 2y - 3z = c$ ....(ii)
Since, (ii) passes through (4, 3, 1)
$\therefore \quad 5(4)+2(3)-3=c \Rightarrow c=23$
Substituting $c = 23$ in (ii), we get
$5x + 2y - 3z = 23$
or if $\vec{r} = x\hat{i}+y\hat{j}+z\hat{k}$ is any vector on plane, then its vector equation is given by
$\vec{r}\cdot(5\hat{i}+2\hat{j}-3\hat{k})=23$

84. The equation of any plane through the line of intersection of the given planes is

$$[\vec{r}\cdot(2\hat{i}-3\hat{j}+4\hat{k})-1]+\lambda[\vec{r}\cdot(\hat{i}-\hat{j})+4]=0$$

$$\Rightarrow \vec{r}\cdot[(2+\lambda)\hat{i}+(-3-\lambda)\hat{j}+4\hat{k}]=1-4\lambda \quad \text{...(i)}$$

If plane (i) is perpendicular to
$\vec{r}\cdot(2\hat{i}-\hat{j}+\hat{k})+8=0,$

then $[(2+\lambda)\hat{i}+(-3-\lambda)\hat{j}+4\hat{k}]\cdot[2\hat{i}-\hat{j}+\hat{k}]=0$

$\Rightarrow (2 + \lambda)2 + (-3 - \lambda)(-1) + 4(1) = 0$

$\Rightarrow 4 + 2\lambda + 3 + \lambda + 4 = 0$

$$\Rightarrow 3\lambda+11=0 \Rightarrow \lambda=\frac{-11}{3}$$

Putting $\lambda=\frac{-11}{3}$ in (i), we obtain the equation of required plane *i.e.*,

$$\vec{r}\cdot\left[\left(2-\frac{11}{3}\right)\hat{i}+\left(-3+\frac{11}{3}\right)\hat{j}+4\hat{k}\right]=1+\frac{44}{3}$$

$$\Rightarrow \vec{r}\cdot\left(-\frac{5}{3}\hat{i}+\frac{2}{3}\hat{j}+4\hat{k}\right)=\frac{47}{3}$$

$$\Rightarrow \vec{r}\cdot(-5\hat{i}+2\hat{j}+12\hat{k})=47$$

Now given line is $x - 1 = 2y - 4 = 3z - 12$

i.e., $\frac{x-1}{1}=\frac{y-2}{\frac{1}{2}}=\frac{z-4}{\frac{1}{3}}$ or $\frac{x-1}{6}=\frac{y-2}{3}=\frac{z-4}{2}$

Now, the line passes through the point (1, 2, 4) satisfies the equation of plane. So, the plane contains the line.

85. The equation of any plane through the line of intersection of the given planes is

$$[\vec{r}\cdot(\hat{i}+2\hat{j}+3\hat{k})-4]+\lambda[\vec{r}\cdot(2\hat{i}+\hat{j}-\hat{k})+5]=0$$

$$\Rightarrow \vec{r}\cdot[(1+2\lambda)\hat{i}+(2+\lambda)\hat{j}+(3-\lambda)\hat{k}]=4-5\lambda \quad \text{...(i)}$$

If plane (i) is perpendicular to
$\vec{r}\cdot(5\hat{i}+3\hat{j}-6\hat{k})+8=0$, then

$$[(1+2\lambda)\hat{i}+(2+\lambda)\hat{j}+(3-\lambda)\hat{k}]\cdot[5\hat{i}+3\hat{j}-6\hat{k}]=0$$

$\Rightarrow 5(1 + 2\lambda) + 3(2 + \lambda) - 6(3 - \lambda) = 0$

$\Rightarrow 5 + 10\lambda + 6 + 3\lambda - 18 + 6\lambda = 0$

$$\Rightarrow 19\lambda-7=0 \Rightarrow \lambda=\frac{7}{19}$$

Putting $\lambda=\frac{7}{19}$ in (i), we obtain the equation of required plane *i.e.*,

$$\vec{r}\cdot\left\{\left(1+2\times\frac{7}{19}\right)\hat{i}+\left(2+\frac{7}{19}\right)\hat{j}+\left(3-\frac{7}{19}\right)\hat{k}\right\} = 4-5\times\frac{7}{19}$$

$$\Rightarrow \vec{r}\cdot\left(\frac{33}{19}\hat{i}+\frac{45}{19}\hat{j}+\frac{50}{19}\hat{k}\right)=\frac{41}{19}$$

$$\Rightarrow \vec{r}\cdot\left(33\hat{i}+45\hat{j}+50\hat{k}\right)=41$$

86. (a) : Perpendicular distance of the origin (0, 0, 0) from the plane $-2x + 6y - 3z + 7 = 0$ is

$$\left|\frac{-2\times0+6\times0-3\times0+7}{\sqrt{(-2)^2+(6)^2+(-3)^2}}\right|=\left|\frac{7}{\sqrt{49}}\right|=\frac{7}{7}=1 \text{ unit}$$

87. (b) : The distance of the point (2, 3, 4) from the plane $\vec{r}\cdot(3\hat{i}-6\hat{j}+2\hat{k})=-11$ is given by

$$\frac{(2\hat{i}+3\hat{j}+4\hat{k})\cdot(3\hat{i}-6\hat{j}+2\hat{k})+11}{\sqrt{(3)^2+(-6)^2+(2)^2}}$$

$$=\frac{(6-18+8+11)}{\sqrt{9+36+4}}=\frac{7}{7}=1 \text{ unit}$$

88. Length of the $\perp$ from $O(0, 0, 0)$ to the plane $2x - 3y + 6z + 21 = 0$ is

www.exambits.in

$$\left|\frac{2\times0-3\times0+6\times0+21}{\sqrt{2^2+(-3)^2+6^2}}\right|=\frac{21}{7}=3 \text{ units}.$$

89. Perpendicular distance from the origin $(0,0,0)$ to the plane $3x - 4y + 12z - 3 = 0$ is

$$\left|\frac{3\times0-4\times0+12\times0-3}{\sqrt{3^2+(-4)^2+12^2}}\right|=\frac{3}{13} \text{ unit}.$$

90. The equation of plane passing through $(3, -1, 2)$ is

$$a(x-3)+b(y+1)+c(z-2)=0 \quad ...(i)$$

Since, plane (i) passes through the points $(5, 2, 4)$ and $(-1, -1, 6)$.

$\therefore \quad 2a + 3b + 2c = 0$ ...(ii)

and $-4a + 0b + 4c = 0$ ...(iii)

On solving (ii) and (iii), we get

$$\frac{a}{12-0}=\frac{b}{-8-8}=\frac{c}{0+12}$$

$$\Rightarrow \quad \frac{a}{12}=\frac{b}{-16}=\frac{c}{12}=\lambda \quad (\text{say})$$

$\Rightarrow \quad a = 12\lambda, b = -16\lambda, c = 12\lambda$

Substituting the values of a, b and c in (i), we get

$$12\lambda(x-3)+(-16\lambda)(y+1)+12\lambda(z-2)=0$$

$\Rightarrow \quad 12x - 16y + 12z = 76$

$\Rightarrow \quad 3x - 4y + 3z = 19$

Its vector equation is given by

$$\vec{r}\cdot(3\hat{i}-4\hat{j}+3\hat{k})=19$$

which is the equation of plane passing through three given points.

Now, distance of plane from origin is given by

$$d=\left|\frac{3(0)-4(0)+3(0)-19}{\sqrt{(3)^2+(-4)^2+(3)^2}}\right|=\frac{19}{\sqrt{34}} \text{ units}$$

91. Equation of given line is

$$\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}=k(\text{say})$$

$\Rightarrow \quad x = 3k + 2, y = 4k - 1, z = 12k + 2$

Since point $(3k + 2, 4k - 1, 12k + 2)$ lie on plane $x - y + z = 5$

$\therefore \quad 3k + 2 - 4k + 1 + 12k + 2 = 5$

$\Rightarrow \quad 11k = 0 \quad \Rightarrow k = 0$

$\therefore$ Point is $(2, -1, 2)$

Required distance

$$=\sqrt{(2+1)^2+(-1+5)^2+(2+10)^2}$$

$$=\sqrt{9+16+144}=\sqrt{169}=13 \text{ units}$$

92. *Refer to answer 91.*

93. Given that $-6, 3, 4$ are intercepts on x, y and z-axes respectively.

$\therefore$ Eq. of the plane is $\frac{x}{-6}+\frac{y}{3}+\frac{z}{4}=1$

$\Rightarrow \quad -2x + 4y + 3z - 12 = 0$

$\therefore$ Length of the perpendicular from origin to the plane $-2x + 4y + 3z - 12 = 0$ is

$$\left|\frac{-2\times0+4\times0+3\times0-12}{\sqrt{(-2)^2+4^2+3^2}}\right|=\frac{12}{\sqrt{29}} \text{ unit}$$

94. Let P be the point with position vector $\hat{i}+3\hat{j}+4\hat{k} \equiv (1, 3, 4)$ and the plane is $\vec{r}\cdot(2\hat{i}-\hat{j}+\hat{k})+3=0$

$\Rightarrow \quad 2x - y + z + 3 = 0$ ...(i)

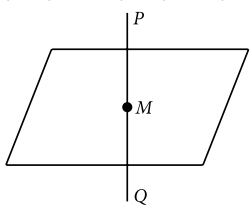
Any line $\perp$ to this plane passes through $P(1, 3, 4)$ is

$$\frac{x-1}{2}=\frac{y-3}{-1}=\frac{z-4}{1}=\lambda \text{ (say)}$$

Any point M on the line is $(2\lambda + 1, -\lambda + 3, \lambda + 4)$

This lies on the plane (i).

$\therefore \quad 2(2\lambda + 1) - (-\lambda + 3) + (\lambda + 4) + 3 = 0$



$\Rightarrow \quad 6\lambda + 6 = 0 \Rightarrow \lambda = -1.$

$\therefore$ Coordinates of point M are $(-1, 4, 3)$.

Let the image of P in the plane (i) be $Q(\alpha, \beta, \gamma)$, then M will be the midpoint of PQ.

$$\therefore \quad \frac{\alpha+1}{2}=-1, \frac{\beta+3}{2}=4, \frac{\gamma+4}{2}=3$$

$\Rightarrow \quad \alpha = -3, \beta = 5, \gamma = 2$

Hence, $Q(-3, 5, 2)$ is the image of $P(1, 3, 4)$ in plane (i).

95. Equation of the line joining the points $A(3, -4, -5)$ and $B(2, -3, 1)$ is

$$\frac{x-3}{2-3}=\frac{y+4}{-3+4}=\frac{z+5}{1+5}$$

$$\Rightarrow \quad \frac{x-3}{-1}=\frac{y+4}{1}=\frac{z+5}{6}=\lambda(\text{say})$$

Any point on it is $Q(-\lambda + 3, \lambda - 4, 6\lambda - 5)$

Let this point Q lies on the plane $2x + y + z = 7$

$\Rightarrow \quad 2(-\lambda + 3) + (\lambda - 4) + (6\lambda - 5) = 7$

$\Rightarrow \quad 5\lambda - 3 = 7 \Rightarrow \lambda = 2$

$\therefore \quad Q$ is $(1, -2, 7)$

www.examkrea.in

$\therefore$ Required distance,

$$PQ = \sqrt{(1-3)^2 + (-2-4)^2 + (7-4)^2}$$

$$= \sqrt{4+36+9} = 7 \text{ units}$$

96. Given line is $\vec{r} = (\hat{i}+\hat{j}) + \lambda(\hat{i}+2\hat{j}-\hat{k})$

The line passes through point having position vector $\vec{a} = \hat{i}+\hat{j}$ and parallel to vector $\vec{b_1} = \hat{i}+2\hat{j}-\hat{k}$.

The plane passes through point (–1, 3, –4).

$\therefore$ Plane is also parallel to vector

$$\vec{b_2} = (-1-1)\hat{i} + (3-2)\hat{j} + (-4+1)\hat{k}$$

$$i.e., \vec{b_2} = -2\hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{b_1} \times \vec{b_2} = \vec{n} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ -2 & 1 & -3 \end{bmatrix}$$

$$= \hat{i}(-6+1) - \hat{j}(-3-2) + \hat{k}(1+4) = -5\hat{i} + 5\hat{j} + 5\hat{k}$$

Now, plane passes through $\vec{a} = \hat{i}+\hat{j}$

and normal to vector $\vec{n} = -5\hat{i} + 5\hat{j} + 5\hat{k}$

Hence the equation of required plane is

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \Rightarrow \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$\Rightarrow \vec{r} \cdot (-5\hat{i} + 5\hat{j} + 5\hat{k}) = (\hat{i}+\hat{j}) \cdot (-5\hat{i} + 5\hat{j} + 5\hat{k})$$

$$\Rightarrow \vec{r} \cdot (-5\hat{i} + 5\hat{j} + 5\hat{k}) = 0 \Rightarrow -5x + 5y + 5z = 0$$

$\therefore$ Length of perpendicular from (2, 1, 4) to plane

$$= \frac{(-5)(2)+5(1)+5(4)}{\sqrt{25+25+25}} = \frac{-10+5+20}{\sqrt{75}} = \frac{15}{\sqrt{75}} = \frac{15}{5\sqrt{3}}$$

$$= \sqrt{3} \text{ units}$$

97. The given plane is $2x - y + z + 1 = 0$ ...(i)

Perpendicular distance from $P(3, 2, 1)$ to (i)

$$= \left| \frac{2 \times 3 - 2 + 1 + 1}{\sqrt{2^2 + (-1)^2 + 1^2}} \right| = \frac{6}{\sqrt{6}} = \sqrt{6} \text{ units}$$

Any line through $P(3, 2, 1)$ and $\perp$ to plane (i) is

$$\frac{x-3}{2} = \frac{y-2}{-1} = \frac{z-1}{1} = \lambda (\text{say}) \quad ...(ii)$$

Any point on (ii) is $Q(2\lambda + 3, -\lambda + 2, \lambda + 1)$.

Let it lie on (i)

$\therefore$ $2(2\lambda + 3) - (-\lambda + 2) + (\lambda + 1) + 1 = 0$

$\Rightarrow$ $6\lambda + 6 = 0 \Rightarrow \lambda = -1$

$\therefore$ $Q \equiv (1, 3, 0)$.

This is the foot of the perpendicular from P on (i).

Let $R(x_1, y_1, z_1)$ be the image of P in the plane (i).

Then, Q is the mid point of PR.

$$\therefore \quad \frac{3+x_1}{2} = 1, \frac{2+y_1}{2} = 3, \frac{1+z_1}{2} = 0$$

$\Rightarrow$ $x_1 = -1, y_1 = 4, z_1 = -1$

Thus $R(-1, 4, -1)$ is the image of P in (i).

98. Let the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad ...(i)$$

where, a, b, c are variables.

This meets X, Y and Z axes at $A(a, 0, 0)$, $B(0, b, 0)$ and $C(0, 0, c)$.

Let (α, β, γ) be the coordinates of the centroid of triangle ABC. Then,

$$\alpha = \frac{a+0+0}{3} = \frac{a}{3}, \beta = \frac{0+b+0}{3} = \frac{b}{3},$$

$$\gamma = \frac{0+0+c}{3} = \frac{c}{3} \quad ...(ii)$$

The plane (i) is at a distance $3p$ from the origin.

$\therefore$ $3p$ = Length of perpendicular from (0, 0, 0) to the plane (i)

$$\Rightarrow \quad 3p = \frac{\left| \frac{0}{a} + \frac{0}{b} + \frac{0}{c} - 1 \right|}{\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2 + \left(\frac{1}{c}\right)^2}}$$

$$\Rightarrow \quad 3p = \frac{1}{\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2 + \left(\frac{1}{c}\right)^2}}$$

$$\Rightarrow \quad \frac{1}{9p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad ...(iii)$$

From (ii), we have

$a = 3\alpha$, $b = 3\beta$ and $c = 3\gamma$

Substituting the values of a, b, c in (iii), we get

$$\frac{1}{9p^2} = \frac{1}{9\alpha^2} + \frac{1}{9\beta^2} + \frac{1}{9\gamma^2}$$

$$\Rightarrow \quad \frac{1}{p^2} = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$$

So, the locus of (α, β, γ) is $\frac{1}{p^2} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}$

www.examrites.in

99. Let M be the foot of the perpendicular drawn from the point $P(2, 3, 4)$ on the plane $\vec{r}\cdot(2\hat{i}+\hat{j}+3\hat{k})-26=0$

i.e., $2x + y + 3z - 26 = 0$

Then, PM is the normal to the plane. So, the direction ratios of PM are proportional to 2, 1, 3.

So, the equation of PM is

$$\frac{x-2}{2}=\frac{y-3}{1}=\frac{z-4}{3}=r \text{ (say)}$$

$\therefore$ The coordinates of $M \equiv (2r + 2, r + 3, 3r + 4)$.

Since M lies on the plane $2x + y + 3z - 26 = 0$

$\therefore$ $2(2r + 2) + r + 3 + 3(3r + 4) - 26 = 0$

$\Rightarrow$ $4r + 4 + r + 3 + 9r + 12 - 26 = 0$

$\Rightarrow$ $14r - 7 = 0 \Rightarrow r = \frac{1}{2}$

$\therefore$ The coordinates of M are $\left(3, \frac{7}{2}, \frac{11}{2}\right)$

Thus, the position vector of the foot of perpendicular is $3\hat{i}+\frac{7}{2}\hat{j}+\frac{11}{2}\hat{k}$

Now, length of the perpendicular from P on the given plane

$$=\left|\frac{2\times2+1\times3+3\times4-26}{\sqrt{4+1+9}}\right|=\frac{7}{\sqrt{14}}=\sqrt{\frac{7}{2}} \text{ units}$$

Let $Q(x_1, y_1, z_1)$ be the image of the point P in the given plane.

Then, the coordinates of M are

$$\left(\frac{x_1+2}{2}, \frac{y_1+3}{2}, \frac{z_1+4}{2}\right)$$

But, the coordinates of M are $\left(3, \frac{7}{2}, \frac{11}{2}\right)$

$$\therefore \left(\frac{x_1+2}{2}, \frac{y_1+3}{2}, \frac{z_1+4}{2}\right)=\left(3, \frac{7}{2}, \frac{11}{2}\right)$$

$$\Rightarrow \frac{x_1+2}{2}=3, \frac{y_1+3}{2}=\frac{7}{2}, \frac{z_1+4}{2}=\frac{11}{2}$$

$\Rightarrow$ $x_1 = 4, y_1 = 4, z_1 = 7$

Thus, the coordinates of the image of the point P in the given plane are (4, 4, 7).

100. Equation of line passing through (1, –2, 3) and having direction cosines proportional to 2, 3, –6 is

$$\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=\lambda \text{(say)} \quad \ldots(i)$$

Any point on (i) is $2\lambda + 1, 3\lambda - 2, -6\lambda + 3$

Let it lies on given plane $x - y + z = 5$

$\therefore$ $2\lambda + 1 - (3\lambda - 2) + (-6\lambda + 3) = 5$

$\Rightarrow$ $-7\lambda = -1 \Rightarrow \lambda = \frac{1}{7}$

$\therefore$ Point is $\left(\frac{2}{7}+1, \frac{3}{7}-2, \frac{-6}{7}+3\right)$ or $\left(\frac{9}{7}, \frac{-11}{7}, \frac{15}{7}\right)$

$\therefore$ Required distance

$$=\sqrt{\left(\frac{9}{7}-1\right)^2+\left(\frac{-11}{7}+2\right)^2+\left(\frac{15}{7}-3\right)^2}$$

$$=\frac{1}{7}\sqrt{4+9+36}=\frac{7}{7}=1 \text{ unit}$$

101. The given planes are

$2x + 3y + 4z = 5$ ...(i)

and $x + y + z = 1$ ...(ii)

Eq. of any plane through the line of intersection of these planes is given by

$2x + 3y + 4z - 5 + \lambda (x + y + z - 1) = 0$

$\Rightarrow$ $(2+\lambda)x+(3+\lambda)y+(4+\lambda)z-(5+\lambda)=0$ ...(iii)

$\because$ This plane is $\perp$ to the plane $x - y + z = 0$

$\therefore$ $(2 + \lambda)\cdot1 + (3 + \lambda)\cdot(-1) + (4 + \lambda)\cdot1 = 0$

$\Rightarrow$ $2 + \lambda - 3 - \lambda + 4 + \lambda = 0 \Rightarrow \lambda = -3$

$\therefore$ Eq. of the required plane is

$(2 - 3)x + (3 - 3)y + (4 - 3)z - (5 - 3) = 0$

$\Rightarrow$ $-x + z - 2 = 0 \Rightarrow x - z + 2 = 0$

Now distance of the point (1, 3, 6) from this plane

$$=\left|\frac{1-6+2}{\sqrt{1^2+(-1)^2}}\right|=\frac{3}{\sqrt{2}}=\frac{3\sqrt{2}}{2} \text{ units}$$

102. The equation of the plane passing through (2, 5, – 3) is

$a(x - 2) + b(y - 5) + c(z + 3) = 0$ ...(i)

If plane (i) passes through the points (–2, –3, 5) and (5, 3, – 3), then

$-4a - 8b + 8c = 0 \Rightarrow a + 2b - 2c = 0$ ...(ii)

and $3a - 2b + 0\cdot c = 0$ ...(iii)

Solving (ii) and (iii), we get

$$\frac{a}{-4}=\frac{b}{-6}=\frac{c}{-2-6} \Rightarrow \frac{a}{2}=\frac{b}{3}=\frac{c}{4}=\lambda \text{(say)}$$

$\Rightarrow$ $a = 2\lambda, b = 3\lambda, c = 4\lambda$

Substituting the values of a, b, c in (i), we get

$2\lambda(x - 2) + 3\lambda(y - 5) + 4\lambda(z + 3) = 0$

$\Rightarrow$ $2x + 3y + 4z = 7$ which is the equation of the plane passing through the given three points.

Now the distance of (7, 2, 4) from this plane is given by

$$\left|\frac{14+6+16-7}{\sqrt{4+9+16}}\right|=\frac{29}{\sqrt{29}}=\sqrt{29} \text{ units}$$

www.examrace.in

103. The given line is

$\vec{r} = (2\hat{i} - 4\hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$

$\Rightarrow \quad \frac{x-2}{3} = \frac{y+4}{4} = \frac{z-2}{2} = \lambda$ (say) ...(i)

Any point on it is $(3\lambda + 2, 4\lambda - 4, 2\lambda + 2)$

This lies on the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$

$\Rightarrow \quad x - 2y + z = 0$ ...(ii)

$\therefore \quad 3\lambda + 2 - 2(4\lambda - 4) + 2\lambda + 2 = 0$

$\Rightarrow \quad -3\lambda + 12 = 0 \Rightarrow \lambda = 4$

$\therefore$ The point of intersection of (i) and (ii) is

$(3 \times 4 + 2, 4 \times 4 - 4, 2 \times 4 + 2) = (14, 12, 10)$

Its distance from the point (2, 12, 5)

$= \sqrt{(14-2)^2 + (12-12)^2 + (10-5)^2}$

$= \sqrt{144 + 0 + 25} = \sqrt{169} = 13$ units.

104. *Refer to answer 101.*

We get the equation of plane is $x - z + 2 = 0$

Now length of $\perp$ from (0, 0, 0) on it

$= \left| \frac{0 - 0 + 2}{\sqrt{1^2 + (-1)^2}} \right| = \sqrt{2}$ units

105. *Refer to answer 91.*

106. Given planes are $2x + 3y - 2z = 5$ ...(i)

and $x + 2y - 3z = 8$ ...(ii)

Normal vectors of (i) and (ii) are respectively

$\vec{m}_1 = 2\hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{m}_2 = \hat{i} + 2\hat{j} - 3\hat{k}$

Since required plane is perpendicular to the planes (i) and (ii). So, normal to the required plane will be in the direction of $\vec{m}_1 \times \vec{m}_2$.

So, $\vec{m}_1 \times \vec{m}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -2 \\ 1 & 2 & -3 \end{vmatrix} = -5\hat{i} + 4\hat{j} + \hat{k} = \vec{m}.$

Also position vector of point (1, –1, 2) on the plane is, $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$.

So, equation of plane is $\vec{r} \cdot \vec{m} = \vec{a} \cdot \vec{m}$

$\Rightarrow \quad \vec{r} \cdot (-5\hat{i} + 4\hat{j} + \hat{k}) = (\hat{i} - \hat{j} + 2\hat{k}) \cdot (-5\hat{i} + 4\hat{j} + \hat{k})$

$\Rightarrow \quad 5x - 4y - z - 7 = 0$ ...(iii)

Distance of $P(-2, 5, 5)$ from (iii) is,

$d = \left| \frac{5(-2) - 4(5) - 5 - 7}{\sqrt{5^2 + (-4)^2 + (-1)^2}} \right| = \frac{42}{\sqrt{42}} = \sqrt{42}$ units.

107. *Refer to answer 102.*

108. Equation of plane passing through intersection of given planes is

$x + 3y + 6 + \lambda(3x - y - 4z) = 0$

$\Rightarrow \quad \vec{r} \cdot (\hat{i} + 3\hat{j}) + 6 + \lambda[\vec{r} \cdot (3\hat{i} - \hat{j} - 4\hat{k})] = 0$

$\Rightarrow \quad \vec{r} \cdot [(1 + 3\lambda)\hat{i} + (3 - \lambda)\hat{j} - 4\lambda\hat{k}] + 6 = 0$

$\because$ Distance of plane from origin is unity.

$\therefore \quad \left| \frac{6}{\sqrt{(1+3\lambda)^2 + (3-\lambda)^2 + 16\lambda^2}} \right| = 1$

$\Rightarrow \quad 26 = 26\lambda^2 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$

$\therefore$ Equation of the required plane is

$\vec{r} \cdot (4\hat{i} + 2\hat{j} - 4\hat{k}) + 6 = 0$

or $\vec{r} \cdot (-2\hat{i} + 4\hat{j} + 4\hat{k}) + 6 = 0$

109. *Refer to answer 100.*

110. *Refer to answer 108.*

111. The given plane is $2x + 4y - z = 2$ ...(i)

The d.r.s. of the normal to (i) are 2, 4, –1.

$\therefore$ Eq. of line $\perp$ to (i) through $P(7, 14, 5)$ is

$\frac{x-7}{2} = \frac{y-14}{4} = \frac{z-5}{-1} = k$ (say) ...(ii)

Any point on (ii) is $Q(2k + 7, 4k + 14, -k + 5)$

$\because$ It lies on the plane (i)

$\therefore \quad 2(2k + 7) + 4(4k + 14) - (-k + 5) = 2$

$\Rightarrow \quad 21k + 65 = 2 \Rightarrow 21k = -63 \Rightarrow k = -3.$

$\therefore \quad Q \equiv (2 \times (-3) + 7, 4 \times (-3) + 14, 3 + 5) \equiv (1, 2, 8)$

This is the foot of $\perp$ on the plane (i)

$\therefore \quad PQ = \sqrt{(7-1)^2 + (14-2)^2 + (5-8)^2}$

$= \sqrt{36 + 144 + 9} = 3\sqrt{21}$ units

which is the length of the $\perp$ from P on plane.

Again let $R(x_1, y_1, z_1)$ be the image of P in the plane (i). Then Q is the mid point of PR.

$\therefore \quad \frac{x_1 + 7}{2} = 1, \frac{y_1 + 14}{2} = 2, \frac{z_1 + 5}{2} = 8$

$\Rightarrow \quad x_1 = -5, y_1 = -10, z_1 = 11$

$\Rightarrow \quad R \equiv (-5, -10, 11)$

112. The line joining the given points $P(3, 4, -1)$ and $Q(2, -1, 5)$ has direction ratios $<3 - 2, 4 + 1, -1 - 5>$ *i.e.*, $<1, 5, -6>$

The plane through (3, –3, 1) and perpendicular to the line PQ is

$1(x - 3) + 5(y + 3) - 6(z - 1) = 0$

$\Rightarrow \quad x + 5y - 6z + 18 = 0$ ...(i)

www.examrace.in

Eq. of the line $\perp$ to (i) through the origin is

$$\frac{x-0}{1}=\frac{y-0}{5}=\frac{z-0}{-6}$$

$$\Rightarrow \quad \frac{x}{1}=\frac{y}{5}=\frac{z}{-6}=k\,(\text{say}) \qquad \ldots(ii)$$

Any point on line (ii) is $(k, 5k, -6k)$.
Let it lie on (i)

$$\Rightarrow \quad k+25k+36k+18=0 \Rightarrow k=-\frac{9}{31}.$$

$\therefore$ The foot of the perpendicular from origin on the plane (i) is $R \equiv \left(-\frac{9}{31}, \frac{-45}{31}, \frac{54}{31}\right)$

Also length of the perpendicular = OR

$$=\sqrt{\left(\frac{9}{31}\right)^2+\left(\frac{45}{31}\right)^2+\left(\frac{54}{31}\right)^2}$$

$$=\sqrt{\frac{5022}{31^2}}=\sqrt{\frac{162}{31}}=9\sqrt{\frac{2}{31}}\ \text{units}.$$

113. The given line is

$$2x=y=z \Rightarrow \frac{x}{1/2}=\frac{y}{1}=\frac{z}{1}\ i.e.,\ \frac{x}{1}=\frac{y}{2}=\frac{z}{2} \qquad \ldots(i)$$

The line through $P(3, 4, 5)$ and parallel to line (i) is

$$\frac{x-3}{1}=\frac{y-4}{2}=\frac{z-5}{2}=k\ (\text{say})$$

Any point on it is $Q(k + 3, 2k + 4, 2k + 5)$
Suppose Q lies on the given plane $x + y + z = 2$

$\Rightarrow \quad (k + 3) + (2k + 4) + (2k + 5) = 2$

$\Rightarrow \quad 5k = -10 \Rightarrow k = -2$

$\therefore$ Q has coordinates $(1, 0, 1)$

$\therefore$ The required distance PQ

$$=\sqrt{(3-1)^2+(4-0)^2+(5-1)^2}$$

$$=\sqrt{4+16+16}=6\ \text{units}$$

114. *Refer to answer 97.*

115. The given lines are

$\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and

$\vec{r}=(\hat{i}+\hat{j})+\mu(-\hat{i}+\hat{j}-2\hat{k})$

These lines pass through the point having position vector $\vec{a}_1=\hat{i}+\hat{j}$ and parallel to the vectors $\vec{b}_1=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}_2=-\hat{i}+\hat{j}-2\hat{k}$ respectively.

$\Rightarrow$ The plane containing them also passes through $\vec{a}=\hat{i}+\hat{j}$.

Since plane contains both the lines. So, the plane is normal to the vector $\vec{n}$.

$$i.e., \vec{n}=\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -1\\ -1 & 1 & -2\end{vmatrix}=-3\hat{i}+3\hat{j}+3\hat{k}$$

Hence the eqn. of the required plane is

$(\vec{r}-\vec{a})\cdot\vec{n}=0 \quad \Rightarrow \quad \vec{r}\cdot\vec{n}=\vec{a}\cdot\vec{n}$

$\Rightarrow \quad \vec{r}\cdot(-3\hat{i}+3\hat{j}+3\hat{k})=(\hat{i}+\hat{j})\cdot(-3\hat{i}+3\hat{j}+3\hat{k})$

$= -3 + 3 = 0$

$\Rightarrow \quad \vec{r}\cdot(-\hat{i}+\hat{j}+\hat{k})=0 \Rightarrow -x + y + z = 0$

$\therefore$ Perpendicular distance of the plane from $(2, 1, 4)$

$$=\left|\frac{-2+1+4}{\sqrt{(-1)^2+1^2+1^2}}\right|=\frac{3}{\sqrt{3}}=\sqrt{3}\ \text{units}.$$

116. *Refer to answer 112.*

117. (a) : The given plane is $z = 0$

$\therefore$ D.r's of line $\perp$ to this plane are $(1, 1, 0)$

So, the line has direction ratios proportional to 1, 1, 0.

$\therefore$ Equation of line through $(-1, 5, 4)$ and perpendicular to plane $z = 0$ is

$\vec{r}=(-\hat{i}+5\hat{j}+4\hat{k})+\lambda(\hat{i}+\hat{j})$.

118. The given plane is $\vec{r}\cdot(\hat{i}+2\hat{j}-5\hat{k})+9=0$

$\Rightarrow \quad \vec{n}=\hat{i}+2\hat{j}-5\hat{k}$

$\because$ D.r's of $\perp$ to this plane are 1, 2, –5.

So, the line has direction ratios proportional to 1, 2, –5.

$\therefore$ Eq. of line through $(1, 2, 3)$ and $\perp$ to the plane is

$\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(\hat{i}+2\hat{j}-5\hat{k})$.

119. The equation of line through $A(-1, 1, -8)$ and $B(5, -2, 10)$ is $\frac{x+1}{5+1}=\frac{y-1}{-2-1}=\frac{z+8}{10+8}$

$$i.e., \frac{x+1}{6}=\frac{y-1}{-3}=\frac{z+8}{18}=k(\text{say}) \qquad \ldots(i)$$

Any point on (i) is given by $(6k - 1, -3k + 1, 18k - 8)$.
We know that the coordinates of any point on the ZX-plane is given by $(x, 0, z)$.

$\therefore \quad -3k + 1 = 0 \Rightarrow k=\frac{1}{3}$

Thus the coordinates of the point where the line joining A and B crosses the ZX-plane are

$$\left(6\times\frac{1}{3}-1, 0, 18\times\frac{1}{3}-8\right)=(1, 0, -2)$$

www.examkits.in

120. The equation of the line through $A(3, 4, 1)$ and $B(5, 1, 6)$ is $\frac{x-3}{5-3} = \frac{y-4}{1-4} = \frac{z-1}{6-1}$

i.e., $\frac{x-3}{2} = \frac{y-4}{-3} = \frac{z-1}{5} = k$ (say) ...(i)

Any point on (i) is given by $(2k + 3, -3k + 4, 5k + 1)$

We know that the coordinates of any point on the XZ plane are $(x_1, 0, z_1)$.

$\because$ $(2k + 3, -3k + 4, 5k + 1)$ lies on the XZ plane

$\therefore$ $-3k + 4 = 0 \Rightarrow k = \frac{4}{3}$

Thus the coordinates of the point where the line joining A and B crosses the XZ plane are

$$\left(2 \times \frac{4}{3} + 3, 0, 5 \times \frac{4}{3} + 1\right) = \left(\frac{17}{3}, 0, \frac{23}{3}\right)$$

The vector equation of the XZ plane is $\vec{r} \cdot \hat{j} = 0$

The vector equation of the line AB is

$\vec{r} = (3\hat{i} + 4\hat{j} + \hat{k}) + \lambda(2\hat{i} - 3\hat{j} + 5\hat{k})$

Let θ be the angle between the XZ plane and the line AB, then

$$\sin\theta = \left|\frac{\vec{b} \cdot \vec{n}}{|\vec{b}||\vec{n}|}\right| = \left|\frac{(2\hat{i} - 3\hat{j} + 5\hat{k}) \cdot \hat{j}}{\sqrt{4 + 9 + 25}\sqrt{0 + 1 + 0}}\right|$$

$$\Rightarrow \sin\theta = \frac{|-3|}{\sqrt{38}} \Rightarrow \theta = \sin^{-1}\left(\frac{3}{\sqrt{38}}\right)$$

121. Eq. of any plane through $(-1, 2, 0)$ is

$a(x + 1) + b(y - 2) + cz = 0$ ...(i)

Also it passes through $(2, 2, -1)$

$\therefore$ $3a + 0 \cdot b - c = 0$ ...(ii)

Further plane (i) is parallel to the line

$$\frac{x-1}{1} = \frac{y+\frac{1}{2}}{1} = \frac{z+1}{-1}$$

$\therefore$ $a \cdot 1 + b \cdot 1 + c \cdot (-1) = 0$

$\Rightarrow$ $a + b - c = 0$ ...(iii)

Eliminating a, b and c from (ii) and (iii), we get

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{3} = \lambda(\text{say})$$

$\Rightarrow$ $a = \lambda$, $b = 2\lambda$ and $c = 3\lambda$

Put these values in (i), we get

$x + 2y + 3z = 3$

This is the eq. of the required plane.

122. Equation of plane passing through $(3, 2, 0)$ is

$a(x - 3) + b(y - 2) + c(z - 0) = 0$ ...(i)

Given line is $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$

Since plane contains the line, so

$a(3 - 3) + b(6 - 2) + c(4 - 0) = 0$

$\Rightarrow$ $0 \cdot a + 4b + 4c = 0$ ...(ii)

and $a(1) + b(5) + c(4) = 0$

$\Rightarrow$ $a + 5b + 4c = 0$ ...(iii)

Solving (ii) & (iii), we get

$$\frac{a}{16-20} = \frac{b}{4-0} = \frac{c}{0-4} = \lambda \text{ (say)}$$

$\Rightarrow$ $a = -4\lambda$, $b = 4\lambda$, $c = -4\lambda$

Putting values of a, b, c in (i), we get

$-4\lambda(x - 3) + 4\lambda(y - 2) - 4\lambda(z - 0) = 0$

$\Rightarrow$ $-4x + 12 + 4y - 8 - 4z = 0$

$\Rightarrow$ $x - y + z - 1 = 0$ is the required equation of plane.

123. The given equation of line is

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{2} = \lambda(\text{say})$$ ...(i)

and equation of plane is

$x - y + z - 5 = 0$ ...(ii)

From (i), we have

$x = 3\lambda + 2$, $y = 4\lambda - 1$, $z = 2\lambda + 2$

$\because$ Line (i) intersects the plane, so the point $(3\lambda + 2, 4\lambda - 1, 2\lambda + 2)$ will also lie on plane.

$\therefore$ $(3\lambda + 2) - (4\lambda - 1) + (2\lambda + 2) - 5 = 0$

$\Rightarrow$ $\lambda + 0 = 0 \Rightarrow \lambda = 0$

So, the required coordinates of the point are $(2, -1, 2)$. Now the direction ratios of line (i) are 3, 4, 2 and direction ratios of plane (ii) are 1, −1, 1. So, the angle between line (i) and plane (ii) is

$$\sin\theta = \left|\frac{3(1) + 4(-1) + 2(1)}{\sqrt{3^2 + 4^2 + 2^2}\sqrt{1^2 + (-1)^2 + 1^2}}\right| = \frac{1}{\sqrt{87}}$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{\sqrt{87}}\right)$$

124. The eq. of the line through $A(3, 4, 1)$ and $B(5, 1, 6)$ is $\frac{x-3}{5-3} = \frac{y-4}{1-4} = \frac{z-1}{6-1}$

$$\Rightarrow \frac{x-3}{2} = \frac{y-4}{-3} = \frac{z-1}{5}$$ ...(i)

Now (i) meets the XY-plane (whose eq. is $z = 0$)

$$\therefore \frac{x-3}{2} = \frac{y-4}{-3} = \frac{0-1}{5} = -\frac{1}{5}$$

$$\Rightarrow \frac{x-3}{2} = -\frac{1}{5}, \frac{y-4}{-3} = -\frac{1}{5}, z = 0$$

$$\Rightarrow x = 3 - \frac{2}{5}, y = 4 + \frac{3}{5}, z = 0$$

www.examrace.in

$\Rightarrow \quad x = \frac{13}{5},\ y = \frac{23}{5},\ z = 0$

$\Rightarrow \quad \left(\frac{13}{5}, \frac{23}{5}, 0\right)$ is the point where line (i) meets the *XY*-plane.

125. Eq. of the line through the points (3, –4, –5) and (2, –3, 1) is

$$\frac{x-2}{3-2} = \frac{y+3}{-4+3} = \frac{z-1}{-5-1}$$

$$\Rightarrow \quad \frac{x-2}{-1} = \frac{y+3}{1} = \frac{z-1}{6} = k \text{ (say)} \qquad ...(i)$$

Any point on (i) is $(-k + 2, k - 3, 6k + 1)$.
Suppose it lies on the plane $2x + y + z = 7$
$\Rightarrow \quad 2(-k + 2) + k - 3 + 6k + 1 = 7$
$\Rightarrow \quad 5k = 5 \Rightarrow k = 1$
$\therefore$ The required point is (1, –2, 7)

126. *Refer to answer 125.*

127. Plane is passing through $A(2, 1, -1)$ and is perpendicular to the line of intersection of plane
$P_1 : 2x + y - z = 3$;
$P_2 : x + 2y + z = 2$
Since, the required plane is $\perp$ to the line of intersection, the plane's normal is parallel to the line of intersection $\vec{l}$.
Now, $\vec{l}$ $\perp$ to normal to P_1, say n_1
and $\vec{l}$ $\perp$ to normal to P_2 say n_2

$$\text{Now, } n_1 \times n_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 3\hat{i} - 3\hat{j} + 3\hat{k}$$

$\therefore$ Equation of plane is $3x - 3y + 3z = C$
Since it passes through $A(2, 1, -1)$
$\therefore \quad 3(2) - 3(1) + 3(-1) = C$
$\Rightarrow \quad C = 0$
So, equation of required plane is $x - y + z = 0$

128. Given, line is $\vec{r} = (3\hat{i} - 2\hat{j} + 6\hat{k}) + \lambda(2\hat{i} - \hat{j} + 2\hat{k})$
and plane is $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 6$
For point of intersection,

$$((3\hat{i} - 2\hat{j} + 6\hat{k}) + \lambda(2\hat{i} - \hat{j} + 2\hat{k})) \cdot (\hat{i} - \hat{j} + \hat{k}) = 6$$

$\Rightarrow \quad (3 + 2\lambda) + (\lambda + 2) + (6 + 2\lambda) = 6$
$\Rightarrow \quad 5\lambda + 11 = 6 \Rightarrow \lambda = -1$
$\therefore \quad \vec{r} = (3\hat{i} - 2\hat{j} + 6\hat{k}) - (2\hat{i} - \hat{j} + 2\hat{k}) = \hat{i} - \hat{j} + 4\hat{k}$
And point of intersection is $Q(1, -1, 4)$
Also, given point is $P(-2, -4, 7)$

$$\therefore \quad \text{Required distance} = \sqrt{(3)^2 + (3)^2 + (-3)^2} = 3\sqrt{3}$$

Equation of PQ is

$$\vec{r} = (\hat{i} - \hat{j} + 4\hat{k}) + \alpha(-3\hat{i} - 3\hat{j} + 3\hat{k})$$

129. Here, the lines are

$$\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$$

$$\Rightarrow \quad \frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6} \qquad ...(i)$$

$$\text{and } \vec{r} = \hat{i} - 3\hat{j} + 5\hat{k} + \mu(\hat{i} + \hat{j} - \hat{k})$$

$$\Rightarrow \quad \frac{x-1}{1} = \frac{y+3}{1} = \frac{z-5}{-1} \qquad ...(ii)$$

Eq. of any plane through (1, 2, –4) is
$a(x - 1) + b(y - 2) + c(z + 4) = 0$ ...(iii)
For plane (iii) to be parallel to the lines (i) and (ii), we must have
$2a + 3b + 6c = 0$ and $a + b - c = 0$

$$\Rightarrow \frac{a}{-3-6} = \frac{b}{6+2} = \frac{c}{2-3} \Rightarrow \frac{a}{-9} = \frac{b}{8} = \frac{c}{-1} = \lambda \text{(say)}$$

$\Rightarrow \quad a = -9\lambda,\ b = 8\lambda,\ c = -\lambda$
Substituting values of a, b, c, in (iii), we get
$-9(x - 1) + 8(y - 2) - 1(z + 4) = 0$
$\Rightarrow \quad -9x + 8y - z - 11 = 0$
The vector equation of the plane is
$\vec{r} \cdot (-9\hat{i} + 8\hat{j} - \hat{k}) - 11 = 0$
Now, perpendicular distance from (9, –8, –10) to the plane

$$= \left| \frac{-9 \times 9 + 8(-8) - 1 \times (-10) - 11}{\sqrt{(-9)^2 + (8)^2 + (-1)^2}} \right|$$

$$= \left| \frac{-81 - 64 + 10 - 11}{\sqrt{81 + 64 + 1}} \right| = \frac{146}{\sqrt{146}} = \sqrt{146} \text{ units.}$$

130. Here, $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$
$\Rightarrow \quad x + y + z - 1 = 0$ ...(i)
and $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0$
$\Rightarrow \quad 2x + 3y - z + 4 = 0$ ...(ii)
Eq. of the plane passing through the line of intersection of the planes (i) and (ii) is given by
$x + y + z - 1 + \lambda(2x + 3y - z + 4) = 0$
$\Rightarrow \quad (1 + 2\lambda)x + (1 + 3\lambda)y + (1 - \lambda)z + (4\lambda - 1) = 0$ ...(iii)
$\because$ The plane is parallel to x-axis whose d.r's are 1, 0, 0.
$\therefore \quad (1 + 2\lambda) \cdot 1 + (1 + 3\lambda) \cdot 0 + (1 - \lambda) \cdot 0 = 0$

www.examrace.in

$\Rightarrow \quad 1+2\lambda=0 \Rightarrow \lambda=-\frac{1}{2}.$

Substituting value of λ in (iii), we get

$$\left[1+2\times\left(-\frac{1}{2}\right)\right]x+\left[1+3\left(-\frac{1}{2}\right)\right]y+\left(1+\frac{1}{2}\right)z+\left(4\times\left(-\frac{1}{2}\right)-1\right)=0$$

$\Rightarrow \quad -\frac{1}{2}y+\frac{3}{2}z-3=0$

$\Rightarrow \quad y-3z+6=0 \Rightarrow \vec{r}\cdot(\hat{j}-3\hat{k})+6=0$

This is the eq. of the required plane.

131. Let $A(2, 2, 1)$, $B(3, 0, 1)$ and $C(4, -1, 0)$.

Any plane passing through A is given by

$a(x-2)+b(y-2)+c(z-1)=0$ ...(i)

As B and C also lie on it, so

$a-2b+0\cdot c=0$

$2a-3b-c=0$

$\Rightarrow \quad \frac{a}{2-0}=\frac{b}{0+1}=\frac{c}{-3+4} \Rightarrow \frac{a}{2}=\frac{b}{1}=\frac{c}{1}$

$\therefore$ From (i), equation of plane is

$2(x-2)+(y-2)+(z-1)=0$

$\Rightarrow \quad 2x+y+z-7=0$ ...(iii)

Refer to answer 125.

132. Any line through (1, 2, 3) is

$\frac{x-1}{a}=\frac{y-2}{b}=\frac{z-3}{c}$ ...(i)

where a, b, c are the direction ratios of the line.

The given planes are

$\vec{r}\cdot(\hat{i}-\hat{j}+2\hat{k})=5 \Rightarrow x-y+2z=5$ ...(ii)

and $\vec{r}\cdot(3\hat{i}+\hat{j}+\hat{k})=6 \Rightarrow 3x+y+z=6$ ...(iii)

Since line (i) is parallel to both the planes.

$\therefore \quad a\cdot 1+b\cdot(-1)+c\cdot 2=0 \Rightarrow a-b+2c=0$

and $a\cdot 3+b\cdot 1+c\cdot 1=0 \Rightarrow 3a+b+c=0$

$\Rightarrow \quad \frac{a}{-1-2}=\frac{b}{6-1}=\frac{c}{1+3} \Rightarrow \frac{a}{-3}=\frac{b}{5}=\frac{c}{4}$

$\therefore$ From (i), the required line is

$\frac{x-1}{-3}=\frac{y-2}{5}=\frac{z-3}{4}.$

Its vector from is $\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(-3\hat{i}+5\hat{j}+4\hat{k})$.

133. The equation of the plane passing through the line of intersection of the planes $2x+y-z=3$ and $5x-3y+4z+9=0$ is

$(2x+y-z-3)+\lambda(5x-3y+4z+9)=0$

$\Rightarrow \quad x(2+5\lambda)+y(1-3\lambda)+z(4\lambda-1)+9\lambda-3=0$ ...(i)

Since, plane (i) is parallel to the line

$\frac{x-1}{2}=\frac{y-3}{4}=\frac{z-5}{5}$

$\therefore \quad 2(2+5\lambda)+4(1-3\lambda)+5(4\lambda-1)=0$

$\Rightarrow \quad 18\lambda+3=0 \Rightarrow \lambda=-\frac{1}{6}$

Putting the value of λ in (i), we obtain

$$x\left(2-\frac{5}{6}\right)+y\left(1+\frac{3}{6}\right)+z\left(-\frac{4}{6}-1\right)-\frac{9}{6}-3=0$$

$\Rightarrow \quad 7x+9y-10z-27=0$

This is the equation of the required plane.

134. The given lines are

$\vec{r}=2\hat{i}+\hat{j}-3\hat{k}+\lambda(\hat{i}+2\hat{j}+5\hat{k})$ ...(i)

and $\vec{r}=3\hat{i}+3\hat{j}+2\hat{k}+\mu(3\hat{i}-2\hat{j}+5\hat{k})$ ...(ii)

Here, $\vec{a}_1=2\hat{i}+\hat{j}-3\hat{k}, \vec{b}_1=\hat{i}+2\hat{j}+5\hat{k}$

$\vec{a}_2=3\hat{i}+3\hat{j}+2\hat{k}, \vec{b}_2=3\hat{i}-2\hat{j}+5\hat{k}$

The plane containing lines (i) and (ii) will pass through $\vec{a}_1=2\hat{i}+\hat{j}-3\hat{k}$.

Also the plane is parallel to two vectors $\vec{b}_1$ and $\vec{b}_2$.

$\therefore$ The plane is normal to the vector

$$\vec{n}=\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 5\\ 3 & -2 & 5\end{vmatrix}=20\hat{i}+10\hat{j}-8\hat{k}$$

$\therefore$ The vector eqn. of the required plane is

$(\vec{r}-\vec{a}_1)\cdot\vec{n}=0 \Rightarrow \vec{r}\cdot\vec{n}=\vec{a}_1\cdot\vec{n}$

$\Rightarrow \quad \vec{r}\cdot(20\hat{i}+10\hat{j}-8\hat{k})=(2\hat{i}+\hat{j}-3\hat{k})\cdot(20\hat{i}+10\hat{j}-8\hat{k})$

$= 40+10+24=74$

$\Rightarrow \quad \vec{r}\cdot(10\hat{i}+5\hat{j}-4\hat{k})=37$ ...(iii)

Its cartesian eqn. is $10x+5y-4z=37$

The given line is,

$\vec{r}=(2\hat{i}+5\hat{j}+2\hat{k})+p(3\hat{i}-2\hat{j}+5\hat{k})$ ...(iv)

The line (iv) will lie in the plane (iii) if the plane passes through the point $\vec{a}=2\hat{i}+5\hat{j}+2\hat{k}$ on line (iv) and is parallel to the line (iv).

Now, $\vec{a}\cdot(10\hat{i}+5\hat{j}-4\hat{k})$

$=(2\hat{i}+5\hat{j}+2\hat{k})\cdot(10\hat{i}+5\hat{j}-4\hat{k})=37$

$\therefore \quad \vec{a}$ lies in the plane (iii)

Also, $(10\hat{i}+5\hat{j}-4\hat{k})\cdot(3\hat{i}-2\hat{j}+5\hat{k})$

$= 10\times 3+5\times(-2)-4\times 5=0$

$\therefore$ Line (iv) is parallel to the plane (iii).

$\therefore$ Line (iv) lies in the plane (iii).

www.exambiter.in

CHAPTER
12

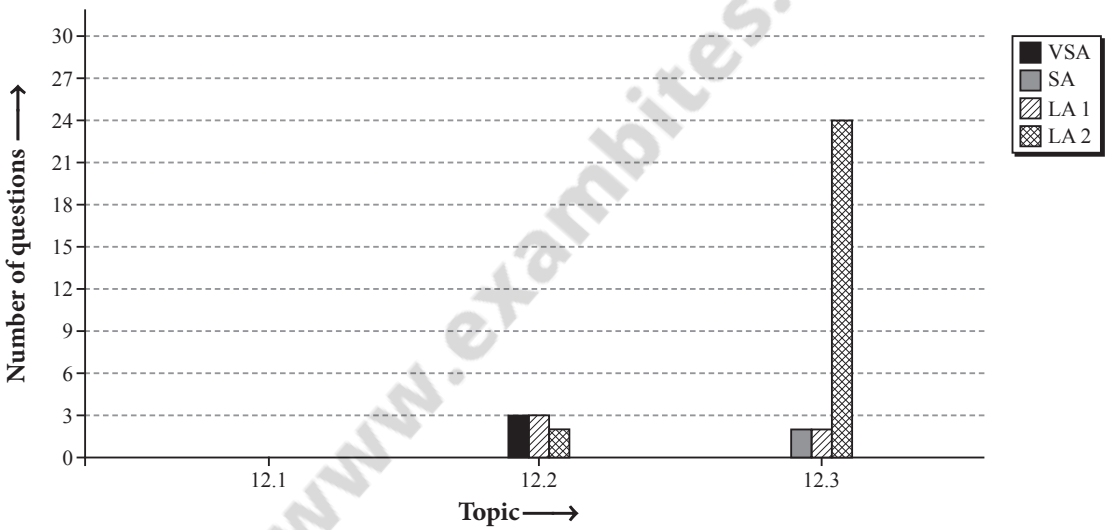
Linear Programming

12.1 Introduction

12.2 Linear Programming Problem and its Mathematical Formulation

12.3 Different Types of Linear Programming Problems

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Different Types of Linear Programming Problems*
- Maximum LA 2 type questions were asked from *Different Types of Linear Programming Problems*

QUICK RECAP

LINEAR PROGRAMMING

- Linear programming (LP) is an optimisation technique in which a linear function is optimised (*i.e.*, minimised or maximised) subject to certain restrictions which are in the form of linear inequalities.

LINEAR PROGRAMMING PROBLEM

- A linear programming problem (LPP) is a problem that is concerned with finding the optimal value of a linear function subject to given constraints.

WORKING RULE TO FORMULATE LPP

- The formulation of LPP as a mathematical model involves the following steps :

  Step 1 : Identify the aim or objective which is to be maximised or minimised and denote it by Z.

  Step 2 : Identify the decision variables and assign symbols $x, y \ldots$ or $x_1, x_2 \ldots$ to them.

  Step 3 : Identify all the restrictions or constraints in the problem and express them as linear inequalities or equations in terms of variables.

  Step 4 : Express the hidden conditions, generally involves non-negativity of variables.

- **Objective function :** The linear function $Z = ax + by$, which has to be optimised (maximised or minimised) is called the objective function.

- **Constraints :** The restrictions or inequalities in the linear programming problem.

- **Non-negativity constraints :** The assumption that negative values of variables are not possible in the solution. They are described as $x, y \geq 0$ or $x_1, x_2 \geq 0$.

OPTIMISATION PROBLEM

- A problem which maximise or minimise a linear function subject to the given constraints.

- **Feasible Region :** The common region determined by all the constraints of an LPP is called the feasible region.

  The feasible region may be either bounded or unbounded.

  (i) **Bounded feasible region :** If the feasible region is enclosed within a circle, then it is called bounded feasible region.

  (ii) **Unbounded feasible region :** If the feasible region is not bounded, then it is called unbounded feasible region.

- **Feasible Solution :** The set of points, within or on the boundary of the feasible region is said to be the feasible solution.

  Note :

  (i) The region other than feasible region is called infeasible region.

  (ii) Any point outside the feasible region is called an infeasible solution.

- **Optimal Value :** The maximum or minimum value of the objective function is called optimal value.

- **Optimal Solution :** Any point in the feasible region which gives the optimal value is called optimal solution.

- **Corner Point :** The intersection point of two boundary lines of the feasible region.

SOME IMPORTANT THEOREMS OF LPP

- **Theorem 1 :** Let R be the feasible region for a linear programming problem and let $Z = ax + by$ be the objective function. When Z has an optimal value (maximum or minimum), where the variables x and y are subject to constraints described by linear inequalities, the optimal value must occur at a corner point of the feasible region.

- **Theorem 2 :** Let R be the feasible region for a linear programming problem and let $Z = ax + by$ be the objective function. If R is bounded, then the objective function Z has both maximum and minimum value on R and each of these occurs at a corner point of R.

  Remark : If R is unbounded, then a maximum or a minimum value of the objective function may not exist. However, if it exists, it must occur at a corner point of R.

STEPS TO SOLVE LPP

- Here are the following steps to solve an LPP.

  Step 1 : Convert inequations into equations.

  Step 2 : Find the point of intersection.

  Step 3 : Draw the graph of inequations.

  Step 4 : Find the value of the objective function corresponding to each corner point.

www.examfees.in

Previous Years' CBSE Board Questions

12.2 Linear Programming Problem and its Mathematical Formulation

VSA (1 mark)

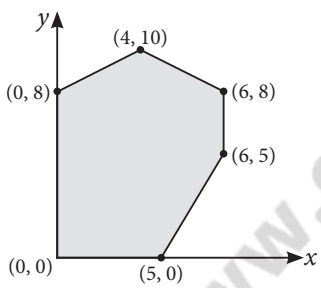
1. The corner points of the feasible region determined by the system of linear inequalities are (0, 0), (4, 0), (2, 4) and (0, 5). If the maximum value of $z = ax + by$, where $a, b > 0$ occurs at both (2, 4) and (4, 0), then

(a) $a = 2b$ (b) $2a = b$

(c) $a = b$ (d) $3a = b$ *(2020)*

2. The feasible region for an LPP is shown below :

Let $z = 3x - 4y$ be the objective function. Minimum of z occurs at



(a) (0, 0) (b) (0, 8)

(c) (5, 0) (d) (4, 10) *(2020)*

3. The graph of the inequality $2x + 3y > 6$ is

(a) half plane that contains the origin

(b) half plane that neither contains the origin nor the points of the line $2x + 3y = 6$.

(c) whole XOY-plane excluding the points on the line $2x + 3y = 6$.

(d) entire XOY-plane. *(2020)*

LA 1 (4 marks)

4. Solve the following LPP graphically :

Minimize $z = 5x + 7y$

subject to the constraints

$2x + y \geq 8$

$x + 2y \geq 10$

$x, y \geq 0$ *(2020)*

5. Solve the following LPP graphically :

Minimise $Z = 5x + 10y$

Subject to constraints $x + 2y \leq 120$, $x + y \geq 60$, $x - 2y \geq 0$ and $x, y \geq 0$ *(Delhi 2017)*

6. Maximise $Z = x + 2y$

subject to the constraints:

$x + 2y \geq 100$

$2x - y < 0$

$2x + y \leq 200$

$x, y \geq 0$

Solve the above LPP graphically. *(AI 2017)*

LA 2 (6 marks)

7. Find graphically, the maximum value of $z = 2x + 5y$, subject to constraints given below:

$2x + 4y \leq 8$, $3x + y \leq 6$, $x + y \leq 4$; $x \geq 0$, $y \geq 0$ *(Delhi 2015)*

8. Maximise $z = 8x + 9y$ subject to the constraints given below :

$2x + 3y \leq 6$, $3x - 2y \leq 6$, $y \leq 1$; $x, y \geq 0$ *(Foreign 2015)*

12.3 Different Types of Linear Programming Problems

SA (2 marks)

9. A small firm manufactures necklaces and bracelets. The total number of necklaces and bracelets that it can handle per day is at most 24. It takes one hour to make a bracelet and half an hour to make a necklace. The maximum number of hours available per day is 16. If the profit on a necklace is ₹ 100 and that on a bracelet is ₹ 300. Formulate an LPP for finding how many of each should be produced daily to maximize the profit ? It is being given that at least one of each must be produced. *(Delhi 2017)*

10. Two tailors, A and B, earn ₹ 300 and ₹ 400 per day respectively. A can stitch 6 shirts and 4 pairs of trousers while B can stitch 10 shirts and 4 pairs of trousers per day. To find how many days should each of them work and if it is desired to produce at least 60 shirts and 32

www.exambits.in

pairs of trousers at a minimum labour cost, formulate this as an LPP. *(AI 2017)*

LA 1 (4 marks)

11. A furniture trader deals in only two items—chairs and tables. He has ₹ 50,000 to invest and a space to store at most 35 items. A chair costs him ₹ 1,000 and a table costs him ₹ 2,000. The trader earns a profit of ₹ 150 and ₹ 250 on a chair and table, respectively. Formulate the above problem as an LPP to maximise the profit and solve it graphically. *(2020)*

12. A company manufactures two types of novelty souvenirs made of plywood. Souvenirs of type *A* requires 5 minutes each for cutting and 10 minutes each for assembling. Souvenirs of type *B* require 8 minutes each for cutting and 8 minutes each for assembling. Given that total time for cutting is 3 hours 20 minutes and for assembling 4 hours. The profit for type *A* souvenir is ₹ 100 each and for type *B* souvenir, profit is ₹ 120 each. How many souvenirs of each type should the company manufacture in order to maximize the profit? Formulate the problem as an LPP and solve it graphically. *(2020)*

LA 2 (6 marks)

13. A manufacturer has employed 5 skilled men and 10 semi-skilled men and makes two models *A* and *B* of an article. The making of one item of model *A* requires 2 hours work by a skilled man and 2 hours work by a semi-skilled man. One item of model *B* requires 1 hour by a skilled man and 3 hours by a semi-skilled man. No man is expected to work more than 8 hours per day. The manufacturer's profit on an item of model *A* is ₹ 15 and on an item of model *B* is ₹ 10. How many of items of each model should be made per day in order to maximize daily profit? Formulate the above LPP and solve it graphically and find the maximum profit. *(Delhi 2019)*

14. A dietician wishes to mix two types of food in such a way that the vitamin contents of the mixture contains atleast 8 units of vitamin A and 10 units of vitamin C. Food I contains 2 units/kg of vitamin A and 1 unit/kg of vitamin C. It costs ₹ 50 per kg to produce food I. Food II contains 1 unit/kg of vitamin A and 2 units/kg of vitamin C and it costs ₹ 70 per kg to produce food II. Formulate this problem as a LPP to minimise the cost of a mixture that will produce the required diet. Also find the minimum cost. *(AI 2019)*

15. A factory manufactures two types of screws *A* and *B*, each type requiring the use of two machines, an automatic and a hand-operated. It takes 4 minutes on the automatic and 6 minutes on the hand-operated machines to manufacture a packet of screws '*A*' while it takes 6 minutes on the automatic and 3 minutes on the hand-operated machine to manufacture a packet of screws '*B*'. Each machine is available for at most 4 hours on any day. The manufacturer can sell a packet of screws '*A*' at a profit of 70 paise and screws '*B*' at a profit of ₹ 1. Assuming that he can sell all the screws be manufactures, how many packets of each type should the factory owner produce in a day in order to maximize his profit? Formulate the above LPP and solve it graphically and find the maximum profit. *(2018)*

16. A manufacturer produces two products *A* and *B*. Both the products are processed on two different machines. The available capacity of first machine is 12 hours and that of second machine is 9 hours per day. Each unit of product *A* requires 3 hours on both machines and each unit of product *B* requires 2 hours on first machine and 1 hour on second machine. Each unit of product *A* is sold at ₹7 profit and that of *B* at a profit of ₹4. Find the production level per day for maximum profit graphically. *(Delhi 2016)*

17. There are two types of fertilisers '*A*' and '*B*'. '*A*' consists of 12% nitrogen and 5% phosphoric acid whereas '*B*' consists of 4% nitrogen and 5% phosphoric acid. After testing the soil conditions, farmer finds that he needs at least 12 kg of nitrogen and 12 kg of phosphoric acid for his crops. If '*A*' costs ₹ 10 per kg and '*B*' costs ₹ 8 per kg, then

www.examin.in

graphically determine how much of each type of fertiliser should be used so that the nutrient requirements are met at a minimum cost. *(AI 2016)*

18. In order to supplement daily diet, a person wishes to take *X* and *Y* tablets. The contents (in milligrams per tablet) of iron, calcium and vitamins in *X* and *Y* are given as below:

Tablets	Iron	Calcium	Vitamin
X	6	3	2
Y	2	3	4

The person needs to supplement at least 18 milligrams of iron, 21 milligrams of calcium and 16 milligrams of vitamins. The price of each tablet of *X* and *Y* is ₹ 2 and ₹ 1 respectively. How many tablets of each type should the person take in order to satisfy the above requirement at the minimum cost? Make an LPP and solve graphically.

(Foreign 2016)

19. A company manufactures three kinds of calculators : *A*, *B* and *C* in its two factories I and II. The company has got an order for manufacturing atleast 6400 calculators of kinds *A*, 4000 of kind *B* and 4800 of kind *C*. The daily output of factory I is of 50 calculators of kind *A*, 50 calculators of kind *B* and 30 calculators of kind *C*. The Daily output of factory II is of 40 calculators of kind *A*, 20 of kind *B* and 40 of kind *C*. The cost per day to run factory I is ₹ 12,000 and of factory II is ₹ 15,000. How many days do the two factories have to be in operation to produce the order with the minimum cost? Formulate this problem as an LPP and solve graphically. *(AI 2015)*

20. One kind of cake requires 200 g of flour and 25 g of fat, another kind of cake requires 100 g of flour and 50 g of fat. Find the maximum number of cakes which can be made from 5 kg of flour and 1 kg of fat, assuming that there is no shortage of the other ingredients used in making the cakes. Make it an LPP and solve it graphically.

(*Delhi 2015C, AI 2014C, 2011C*)

21. A manufacturer produces nuts and bolts. It take 2 hours work on machine *A* and 3 hours on machine *B* to produce a package of nuts. It takes 3 hours on machine *A* and 2 hours on machine *B* to produce a package of bolts. He earns a profit of ₹24 per package on nuts and ₹18 per package on bolts. How many package of each should be produced each day so as to maximise his profit, if he operates his machines for at the most 10 hours a day. Make an LPP from above and solve it graphically? *(AI 2015C)*

22. A dealer in rural area wishes to purchase a number of sewing machines. He has only ₹ 5760 to invest and has space for at most 20 items for storage. An electronic sewing machine costs him ₹ 360 and a manually operated sewing machine ₹ 240. He can sell an electronic sewing machine at a profit of ₹ 22 and a manually operated sewing machine at a profit of ₹ 18. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximise his profit? Make it as an LPP and solve graphically. *(Delhi 2014)*

23. A manufacturing company makes two types of teaching aids *A* and *B* of Mathematics for class XII. Each type of *A* requires 9 labour hours of fabricating and 1 labour hour for finishing. Each type of *B* requires 12 labour hours for fabricating and 3 labour hours for finishing. For fabricating and finishing, the maximum labour hours available per week are 180 and 30 respectively. The company makes a profit of ₹ 80 on each piece of type *A* and ₹ 120 on each piece of type *B*. How many pieces of type *A* and type *B* should be manufactured per week to get a maximum profit? Make it as an LPP and solve graphically. What is the maximum profit per week? (*AI 2014*)

24. A cottage industry manufactures pedestal lamps and wooden shades, each requiring the use of a grinding/cutting machine and a sprayer. It takes 2 hours on grinding/ cutting machine and 3 hours on the sprayer to manufacture a pedestal lamp. It takes

www.Examkites.in

1 hour on the grinding/cutting machine and 2 hours on the sprayer to manufacture a shade. On any day, the sprayer is available for at the most 20 hours and the grinding/cutting machine for at the most 12 hours. The profit from the sale of a lamp is ₹25 and that from a shade is ₹15. Assuming that the manufacturer can sell all the lamps and shades that he produces, how should he schedule his daily production in order to maximise his profit? Formulate an LPP and solve it graphically. *(Foreign 2014)*

25. A housewife wishes to mix together two kinds of food, *X* and *Y*, in such a way that the mixture contains at least 10 units of vitamin *A*, 12 units of vitamin *B* and 8 units of vitamin *C*. The vitamin contents of one kg of food is given below:

	Vitamin *A*	**Vitamin *B***	**Vitamin *C***
Food *X*	1	2	3
Food *Y*	2	2	1

One kg of food *X* costs ₹ 6 and one kg of food *Y* costs ₹ 10. Formulate the above problem as a linear programming problem and find the least cost of the mixture which will produce the diet graphically. What value will you like to attach with this problem? (*Delhi 2014C*)

26. If a young man rides his motorcycle at 25 km per hour, he had to spend ₹2 per km on petrol with very little pollution in the air. If he rides it at a faster speed of 40 km per hour, the petrol cost increases to ₹5 per km and rate of pollution also increases. He has ₹100 to spend on petrol and wishes to find what is the maximum distance he can travel within one hour. Express this problem as an LPP. Solve it graphically to find the distance to be covered with different speeds. What value is indicated in this question?

(Delhi 2014C, 2013C)

27. A cooperative society of farmers has 50 hectares of land to grow two crops *A* and *B*. The profits from crops *A* and *B* per hectare are estimated as ₹10,500 and ₹9,000 respectively. To control weeds a liquid herbicide has to be used for crops *A* and *B* at the rate of 20 litres and 10 litres per hectare, respectively. Further not more than 800 litres of herbicide should be used in order to protect fish and wildlife using a pond which collects drainage from this land. Keeping in mind that the protection of fish and other wildlife is more important than earning profit, how much land should be allocated to each crop so as to maximise the total profit? Form an LPP from the above information and solve it graphically. Do you agree with the message that the protection of wildlife is utmost necessary to preserve the balance in environment? *(Delhi 2013)*

28. A manufacturer considers that men and women workers are equally efficient and so he pays them at the same rate. He has 30 and 17 units of workers (male and female) and capital respectively; which he uses to produce two types of goods *A* and *B*. To produce one unit of *A*, 2 workers and 3 units of capital are required while 3 workers and 1 unit of capital is required to produce one unit of *B*. If *A* and *B* are priced at ₹100 and ₹120 per unit respectively, how should he use his resources to maximise the total revenue? Form the above as an LPP and solve graphically.

Do you agree with this view of the manufacturer that men and women workers are equally efficient and so should be paid at the same rate? (*AI 2013*)

29. A cottage industry manufactures pedestal lamps and wooden shades, each requiring the use of a cutting/grinding machine and a sprayer. It takes 2 hours on grinding/cutting machine and 3 hours on sprayer to manufacture a pedestal lamp. It takes 1 hour on grinding/cutting machine and 2 hours on sprayer to manufacture a shade. On any day, to keep the environment pollution under minimum level, sprayer can be used for at the most 20 hours while grinding/cutting machine can be used for at the most 12 hours. The profit from selling a

www.mtgexambites.in

pedestal lamp is ₹ 5 and for selling a shade is ₹ 3. Assuming that it can sell all that it produces, how should it schedule its daily production to maximize its profit? Make it as an LPP and solve it graphically. Which value is described in this question? (*AI 2013C*)

30. A manufacturer produces nuts and bolts. It takes 1 hour of work on machine *A* and 3 hours on machine *B* to produce a package of nuts. It takes 3 hours on machine *A* and 1 hour on machine *B* to produce a package of bolts. He earns a profit of ₹ 17.50 per package on nuts and ₹ 7 per package of bolts. How many packages of each should be produced each day so as to maximize his profits if he operates his machines for at the most 12 hours a day? Form the above as a linear programming problem and solve it graphically. *(Delhi 2012)*

31. A dietician wishes to mix two types of foods in such a way that the vitamin contents of the mixture contains at least 8 units of vitamin *A* and 10 units of vitamin *C*. Food I contains 2 units/kg of vitamin *A* and 1 unit/kg of vitamin *C* while food II contains 1 unit/kg of vitamin *A* and 2 units/kg of vitamin *C*. It costs ₹ 5 per kg to purchase food I and ₹ 7 per kg to purchase food II. Determine the minimum cost of such a mixture. Formulate the above as an LPP and solve it graphically. *(AI 2012)*

32. A decorative item dealer deals in two items *A* and *B*. He has ₹ 15,000 to invest and a space to store at the most 80 pieces. Item *A* costs him ₹ 300 and item *B* costs him ₹ 150. He can sell items *A* and *B* at respective profits of ₹ 50 and ₹ 28. Assuming he can sell all he buys, formulate the linear programming problem in order to maximise his profit and solve it graphically. *(Delhi 2012C)*

33. A manufacturer produces nuts and bolts. It takes 1 hour of work on machine *A* and 3 hours on machine *B* to produce a package of nuts while it takes 3 hours on machine *A* and 1 hour on machine *B* to produce a package of bolts. He earns a profit of ₹ 2.50 per package of nuts and ₹ 1.00 per package of bolts. How many packages of each type should he produce each day so as to maximise his profit, if he operates his machines for at most 12 hours a day? Formulate this problem as a linear programming problem and solve it graphically. *(AI 2012C)*

34. A factory makes tennis rackets and cricket bats. A tennis racket takes 1.5 hours of machine time and 3 hours of craftman's time in its making while a cricket bat takes 3 hours of machine time and 1 hour of craftman's time. In a day, the factory has the availability of not more than 42 hours of machine time and 24 hours of craftman's time. If the profit on a racket and on a bat are ₹ 20 and ₹ 10 respectively, then find the number of tennis rackets and cricket bats that the factory must manufacture to earn maximum profit. Form it as an LPP and solve it graphically. *(Delhi 2011)*

35. A merchant plans to sell two types of personal computers - a desktop model and a portable model that will cost ₹ 25,000 and ₹ 40,000 respectively. He estimates that the total monthly demand of computers will not exceed 250 units. Determine the number of units of each type of computers which the merchant should stock to get maximum profit, if he does not want to invest more than ₹ 70 lakhs and his profit on the desktop model is ₹ 4500 and on the portable model is ₹ 5,000. Form it as an LPP and solve it graphically. *(AI 2011)*

36. A dietician wishes to mix two types of foods in such a way that the vitamin contents of the mixture contain at least 8 units of vitamin *A* and 10 units of vitamin *C*. Food I contains 2 units per kg of vitamin *A* and 1 unit per kg of vitamin *C*. Food II contains 1 unit per kg of vitamin *A* and 2 units per kg of vitamin *C*. It costs ₹ 50 per kg to purchase Food I and ₹ 70 per kg to purchase Food II. Formulate the problem as a linear programming problem to minimise the cost of such mixture and find the minimum cost graphically. *(Delhi 2011C)*

www.examfear.in

Detailed Solutions

1. (a) : Since, maximum value of $z = ax + by$ occurs at both (2, 4) and (4, 0).

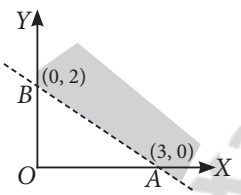
$\therefore \quad 2a + 4b = 4a + 0$

$\Rightarrow \quad 4b = 2a \Rightarrow 2b = a$

2. (b) : We know that minimum of objective function occurs at corner points.

Corner points	Value of $z = 3x - 4y$
(0, 0)	0
(5, 0)	15
(6, 5)	–2
(6, 8)	–14
(4, 10)	–28
(0, 8)	–32 ← Minimum

3. (b) : From the graph of inequality $2x + 3y > 6$. It is clear that it does not contain the origin nor the points of the line $2x + 3y = 6$.



4. We have, minimize $z = 5x + 7y$

subject to constraints,

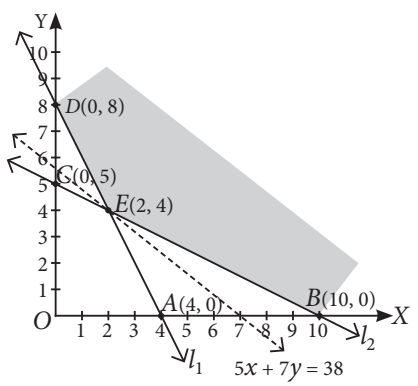
$2x + y \geq 8, x + 2y \geq 10, x, y \geq 0$

To solve LPP graphically, we convert inequations into equations.

Now, $l_1 : 2x + y = 8$, $l_2 : x + 2y = 10$ and $x = 0, y = 0$

l_1 and l_2 intersect at $E(2, 4)$

Let us draw the graph of these equations as shown below.



The corner points of the feasible region are $D(0, 8)$, $B(10, 0)$ and $E(2, 4)$.

Corner points	Value of $z = 5x + 7y$
D (0, 8)	56
B (10, 0)	50
E (2, 4)	38 (Minimum)

From the table, we find that 38 is the minimum value of z at $E(2, 4)$. Since the region is unbounded, so we draw the graph of inequality $5x + 7y < 38$ to check whether the resulting open half plane has any point common with the feasible region. Since it has no point in common. So, the minimum value of z is obtained at $E(2, 4)$ and the minimum value of $z = 38$.

5. We have, Minimise $Z = 5x + 10y$

subject to constraints :

$x + 2y \leq 120$

$x + y \geq 60$

$x - 2y \geq 0$

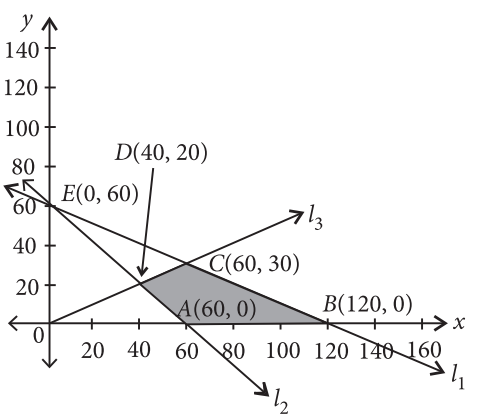
and $x, y \geq 0$

To solve L.P.P graphically, we convert inequations into equations.

$l_1 : x + 2y = 120$, $l_2 : x + y = 60$, $l_3 : x - 2y = 0$ and $x = 0, y = 0$

l_1 and l_2 intersect at $E(0, 60)$, l_1 and l_3 intersect at $C(60, 30)$, l_2 and l_3 intersect at $D(40, 20)$.

The shaded region $ABCD$ is the feasible region and is bounded. The corner points of the feasible region are $A(60, 0)$, $B(120, 0)$, $C(60, 30)$ and $D(40, 20)$.



Corner points	Value of $Z = 5x + 10y$
A(60, 0)	300 ← (Minimum)

www.exambees.in

$B(120, 0)$	600
$C(60, 30)$	600
$D(40, 20)$	400

Hence, Z is minimum at $A(60, 0)$ *i.e.*, 300.

6. Maximise $Z = x + 2y$

Subject to constraints : $x + 2y \geq 100$, $2x - y < 0$, $2x + y \leq 200$ and $x, y \geq 0$.

Converting the inequations into equations, we obtain the lines

$l_1 : x + 2y = 100$ ...(i)

$l_2 : 2x - y = 0$ ...(ii)

$l_3 : 2x + y = 200$ ...(iii)

$l_4 : x = 0$ ...(iv)

and $l_5 : y = 0$ ...(v)

By intercept form, we get

$$l_1 : \frac{x}{100} + \frac{y}{50} = 1$$

$\Rightarrow$ The line l_1 meets the coordinate axes at $(100, 0)$ and $(0, 50)$.

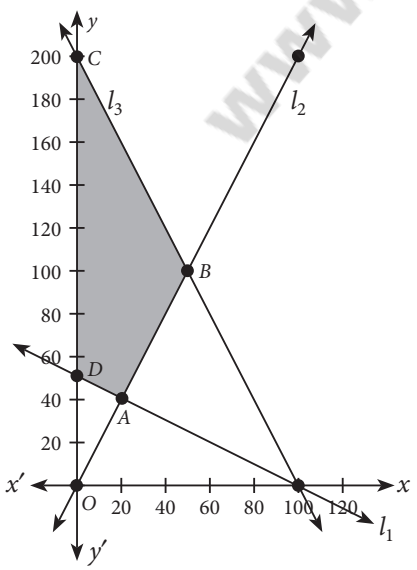
$l_2 : 2x = y$

$\Rightarrow$ The line l_2 passes through origin and $(50, 100)$, $(100, 200)$.

$$l_3 : \frac{x}{100} + \frac{y}{200} = 1$$

$\Rightarrow$ The line l_3 meets the coordinates axes at $(100, 0)$ and $(0, 200)$.

$l_4 : x = 0$ is the y-axis, $l_5 : y = 0$ is the x-axis



Now, plotting the above points on the graph, we get the feasible region of the LPP as shaded region $ABCD$. The coordinates of the corner points of the feasible region $ABCD$ are $A(20, 40)$, $B(50, 100)$, $C(0, 200)$, $D(0, 50)$.

Now, $Z_A = 20 + 2 \times 40 = 100$

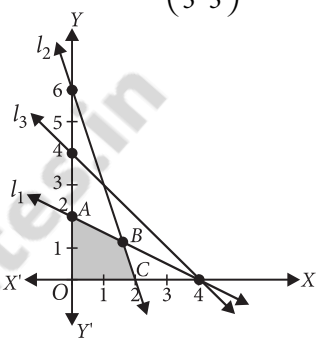
$Z_B = 50 + 2 \times 100 = 250$, $Z_C = 0 + 2 \times 200 = 400$

$Z_D = 0 + 2 \times 50 = 100$

$\therefore$ Z is maximum at $C(0, 200)$ and having value 400.

7. Let $l_1 : 2x + 4y = 8$, $l_2 : 3x + y = 6$, $l_3 : x + y = 4$; $x = 0$, $y = 0$

Solving l_1 and l_2 we get $B\left(\frac{8}{5}, \frac{6}{5}\right)$



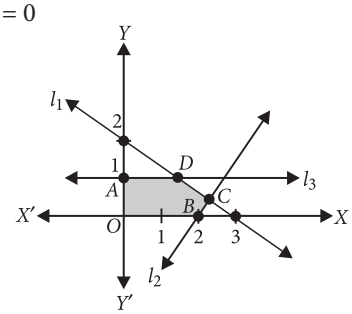
Shaded portion $OABC$ is the feasible region, where coordinates of the corner points are $O(0, 0)$, $A(0, 2)$, $B\left(\frac{8}{5}, \frac{6}{5}\right)$, $C(2, 0)$

The value of objective function at these points are :

Corner Points	Value of the objective function $z = 2x + 5y$
$O(0, 0)$	$2 \times 0 + 5 \times 0 = 0$
$A(0, 2)$	$2 \times 0 + 5 \times 2 = 10$ (Maximum)
$B\left(\frac{8}{5}, \frac{6}{5}\right)$	$2 \times \frac{8}{5} + 5 \times \frac{6}{5} = 9.2$
$C(2, 0)$	$2 \times 2 + 5 \times 0 = 4$

$\therefore$ The maximum value of z is 10, which is at $A(0, 2)$.

8. Let $l_1 : 2x + 3y = 6$, $l_2 : 3x - 2y = 6$, $l_3 : y = 1$; $x = 0$, $y = 0$



www.exambites.in

Solving l_1 and l_3, we get D (1.5, 1)

Solving l_1 and l_2, we get $C\left(\frac{30}{13}, \frac{6}{13}\right)$

Shaded portion $OADCB$ is the feasible region, where coordinates of the corner points are $O(0, 0)$, $A(0, 1)$, $D(1.5, 1)$, $C\left(\frac{30}{13}, \frac{6}{13}\right)$, $B(2, 0)$.

The value of the objective function at these points are :

Corner Points	Value of the objective function $z = 8x + 9y$
O (0, 0)	$8 \times 0 + 9 \times 0 = 0$
A (0, 1)	$8 \times 0 + 9 \times 1 = 9$
D (1.5, 1)	$8 \times 1.5 + 9 \times 1 = 21$
$C\left(\frac{30}{13}, \frac{6}{13}\right)$	$8\times\frac{30}{13}+9\times\frac{6}{13}=22.6$ (Maximum)
B (2, 0)	$8 \times 2 + 9 \times 0 = 16$

The maximum value of z is 22.6, which is at $C\left(\frac{30}{13}, \frac{6}{13}\right)$

9. Let x necklaces and y bracelets be manufactured per day to maximize the profit.

$\therefore$ Maximize $Z = 100x + 300y$

Subject to the constraints : $x + y \le 24$,

$$(1)x + \left(\frac{1}{2}\right)y \le 16 \Rightarrow 2x + y \le 32$$

and $x \ge 1$, $y \ge 1 \Rightarrow x - 1 \ge 0$ and $y - 1 \ge 0$

10. Suppose tailors A and B work for x and y days respectively.

Tailors / Product	$A(x)$	$B(y)$	Requirements
Shirts	6	10	60
Trousers	4	4	32

Then the L.P.P is

Minimise $Z = 300x + 400y$

Subject to constraints : $6x + 10y \ge 60$ or

$3x + 5y \ge 30$, $4x + 4y \ge 32$ or $x + y \ge 8$, $x \ge 0$, $y \ge 0$.

11. Let the number of chairs be x and the number of tables be y.

$\therefore$ $x + y \le 35$

Cost of a chair is ₹ 1,000 and cost of a table is ₹ 2,000.

$\therefore$ $1000x + 2000y \le 50000$

$\Rightarrow$ $x + 2y \le 50$

Now, total profit $Z = 150x + 250y$.

Above LPP can be stated mathematically as :

Maximize , $Z = 150x + 250y$

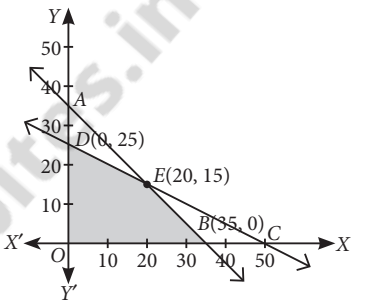
Subject to the constraints

$x + y \le 35$, $x + 2y \le 50$ and $x, y \ge 0$

To solve graphically, we convert the inequations into equations to obtain the following lines :

$x + y = 35$, $x + 2y = 50$, $x = 0$, $y = 0$

Let us draw the graph of these lines as shown below.



The feasible region is shown in the graph. We observe that the region is bounded.

The corner points of the feasible region $OBED$ are $O(0, 0)$, $B(35, 0)$, $E(20, 15)$ and $D(0, 25)$.

The value of the objective function at corner points of the feasible region are :

Corner points	**Value of $Z = 150x + 250y$**
$O(0, 0)$	0
$B(35, 0)$	5250
$E(20, 15)$	6750 (Maximum)
$D(0, 25)$	6250

$\therefore$ Z is maximum at $x = 20$, $y = 15$.

12. Let the number of souvenirs of type A and type B be x and y respectively. Then by the given data, the required LPP is

Maximize $Z = 100x + 120y$

Subject to constraints :

$5x + 8y \le 200$, $10x + 8y \le 240$, $x \ge 0$, $y \ge 0$

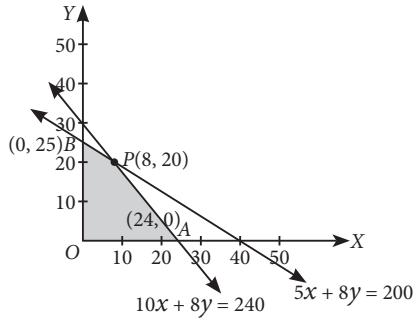
We have the lines

$l_1 : 5x + 8y = 200$ and $l_2 : 10x + 8y = 240$

These lines intersect at $P(8, 20)$.

Let us draw the graph of the lines.

www.exambits.in



The feasible region is $OAPB$ and the corner points are $O(0, 0)$, $A(24, 0)$, $P(8, 20)$ and $B(0, 25)$.

Corner points	Value of $Z = 100x + 120y$
$O(0, 0)$	0
$A(24, 0)$	2400
$P(8, 20)$	3200 (Maximum)
$B(0, 25)$	3000

$\therefore$ Profit, Z = ₹ 3200 will be maximum, when 8 souvenirs of type A and 20 souvenirs of type B are manufactured.

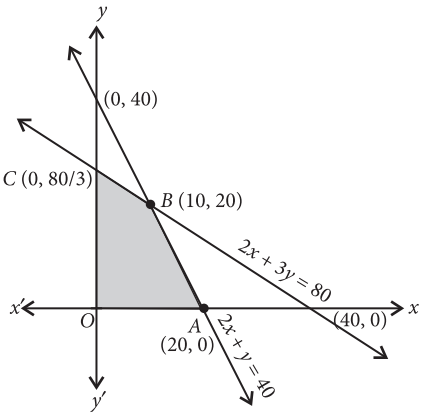
13. Let x items of model A and y items of model B be produced.

	Skilled men (5)	Semi-skilled men (10)	Cost (in ₹)
Model A (x)	2	2	15
Model B (y)	1	3	10
Capacity	8	8	

The inequations thus formed based on the given problem will be as follows :

$2x + y \le 40$ and $2x + 3y \le 80$

Let z be the total profit of the models.
Then $z = 15x + 10y$
Mathematical formulation of LPP is :
Maximize $z = 15x + 10y$ subject to the constraints :
$2x + y \le 40$ $2x + 3y \le 80$, $x \ge 0$, $y \ge 0$
To solve the LPP graphically, we convert the inequations into equations to obtain the following lines :
$2x + y = 40$, $2x + 3y = 80$, $x = 0$ and $y = 0$
The point of intersection of lines $2x + y = 40$ and $2x + 3y = 80$ is B (10, 20).



Shaded region $OABC$ is the feasible region, where $O(0, 0)$, $A(20, 0)$, B (10, 20) and $C(0, 80/3)$.
The value of objective function at these points are as follows :

Corner points	Value of objective function $z = 15x + 10y$
$O(0, 0)$	15(0) + 10(0) = 0
$A(20, 0)$	15(20) + 10(0) = 300
$B(10, 20)$	15(10) + 10(20) = 350 (maximum)
$C\left(0, \frac{80}{3}\right)$	$15(0)+10\left(\frac{80}{3}\right)=\frac{800}{3}$

$\therefore$ The maximum value of z = 350 which is at $B(10, 20)$.
Hence, manufacturer should produce 10 items of model A and 20 items of model B.

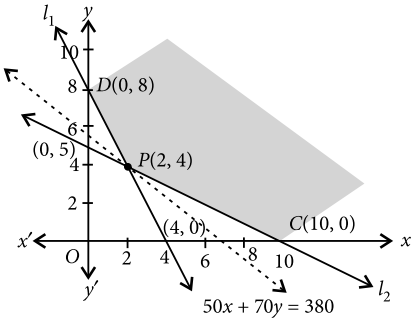
14. Let x kg of food I and y kg of food II be produced, then the given data can be represented in the tabular form as follows:

Nutrient	Food I	Food II	Requirements
Vitamin A	2	1	8
Vitamin C	1	2	10
Cost (in ₹)	50	70	

The given LPP is as follows:
Minimise $z = 50x + 70y$
Subject to the constraints:
$2x + y \ge 8$, $x + 2y \ge 10$, $x \ge 0$, $y \ge 0$
Now, we convert inequations into equations,
$l_1 : 2x + y = 8$, $l_2 : x + 2y = 10$

www.examrites.in

These lines intersect at $P(2, 4)$.



From the graph, the corner points of the feasible region are $C(10, 0)$, $P(2, 4)$ and $D(0, 8)$.

Corner points	Value of $z = 50x + 70y$
$C(10, 0)$	$50(10) + 70(0) = 500$
$P(2, 4)$	$50(2) + 70(4) = 380$ (minimum)
$D(0, 8)$	$50(0) + 70(8) = 560$

From the table, we find that 380 is the minimum value of z at $P(2, 4)$. Since, the region is unbounded, we have to check the inequality $50x + 70y < 380$ in open half planes has any point in common or not. Since, it has no point in common, so, the minimum cost is ₹ 380 and this is attained at $P(2, 4)$ *i.e.*, when 2 units of food I and 4 units of food II are produced.

15. Let the manufacturer produce x packages of screws A and y packages of screws B. Clearly, $x \geq 0$, $y \geq 0$. We make the following table from the given data.

Machines	Time required to produce product		Maximum machine mins. available
	A	B	
Automatic	4	6	$= 4 \times 60 = 240$
Hand operated	6	3	$= 4 \times 60 = 240$
Profit (in ₹)	7	10	

Since the machine is available for at the most 4 hours a day, we have the constraints

$4x + 6y \leq 240$

$6x + 3y \leq 240$

Total profit $Z = 7x + 10y$

Hence the mathematical formulation of the problem is Maximise $Z = 7x + 10y$

subject to the constraints

$2x + 3y \leq 120$ ...(1)

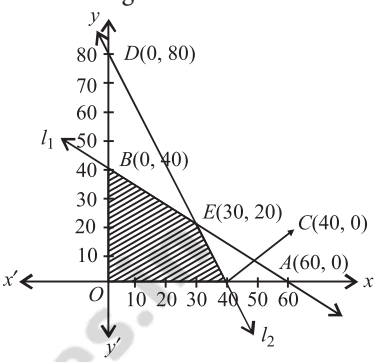
$2x + y \leq 80$ ...(2)

$x, y \geq 0$ ...(3)

Convert the inequations into equations, we get

$l_1 : 2x + 3y = 120$, $l_2 : 2x + y = 80$

The feasible region is shown in the graph. We observe that the region is bounded.



The corner points of the feasible region $OCEB$ are $O(0, 0)$, $C(40, 0)$, $E(30, 20)$ and $B(0, 40)$.

The value of the objective function at these points is given as :

Corner Points	Value of $Z = 7x + 10y$
O (0, 0)	0
C (40, 0)	280
E (30, 20)	410 (Maximum)
B (0, 40)	400

We find that the maximum value of Z is 410 at $E(30, 20)$. Hence the factory must sell 30 packages of screw A and 20 packages of screw B to realise maximum profit and maximum profit is ₹ 410.

16. The given information can be represented in the tabular form as below:

Machines	Time required to produce product		Maximum machine hours available
	A	B	
First machine	3	2	12
Second machine	3	1	9
Profit (in ₹)	7	4	

Let the manufacturer produces x units of product A and y units of product B per day.

$\therefore \quad 3x + 2y \leq 12$ and $3x + y \leq 9$

Let Z denote the total profit.

$\therefore \quad Z = 7x + 4y$

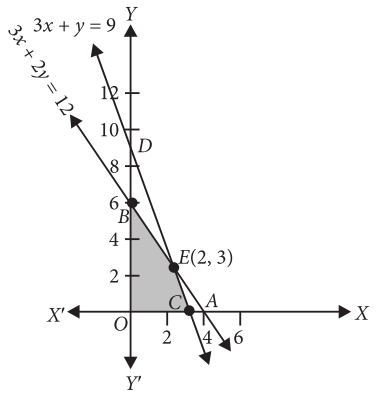
Clearly $x \geq 0$ and $y \geq 0$.

Above LPP can be stated mathematically as:

Maximise $Z = 7x + 4y$

www.examfree.com

subject to the constraints
$3x + 2y \le 12$, $3x + y \le 9$ and $x, y \ge 0$
To solve graphically, we convert the inequations into equations to obtain the following lines:
$3x + 2y = 12$, $3x + y = 9$, $x = 0$, $y = 0$
The line $3x + 2y = 12$ meets the coordinate axes at $A(4, 0)$ and $B(0, 6)$. Similarly $3x + y = 9$ meets the coordinate axes at $C(3, 0)$ and $D(0, 9)$
The point of intersection of lines $3x + 2y = 12$ and $3x + y = 9$ is $E(2, 3)$.



Coordinates of the corner points of the feasible region $OCEB$ are $O(0, 0)$, $C(3, 0)$, $E(2, 3)$, $B(0, 6)$
Values of the objective function at corner points of the feasible region are

Corner Points	Value of $Z = 7x + 4y$
$O(0, 0)$	0
$C(3, 0)$	21 + 0 = 21
$E(2, 3)$	14 + 12 = 26 (Maximum)
$B(0, 6)$	0 + 24 = 24

$\therefore$ Z is maximum at $x = 2$, $y = 3$
So, for maximum profit the manufacturer should manufacture 2 units of product A and 3 units of product B.

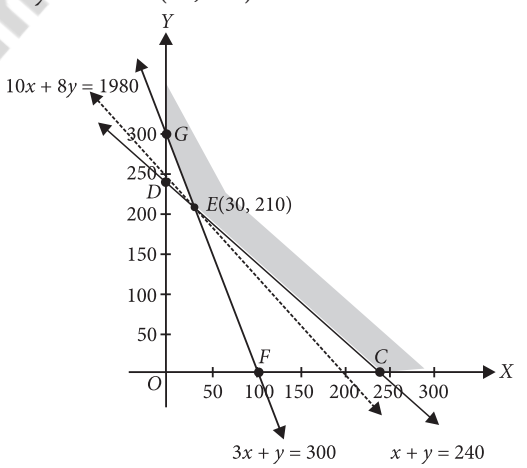
17. Let the requirement of fertiliser A by the farmer be x kg and that of B be y kg.

	Fertiliser A	Fertiliser B	Minimum requirement (in kg)
Nitrogen (in %)	12	4	12
Phosphoric acid/(in %)	5	5	12
Cost (in ₹ kg)	10	8	

The inequations thus formed based on the given problem will be as follows:

$$\frac{12x}{100} + \frac{4y}{100} \ge 12 \Rightarrow 12x + 4y \ge 1200$$
$$\Rightarrow 3x + y \ge 300$$
$$\text{Also, } \frac{5x}{100} + \frac{5y}{100} \ge 12$$
$$\Rightarrow 5x + 5y \ge 1200 \Rightarrow x + y \ge 240 \text{ and } x \ge 0, y \ge 0.$$

Let Z be the total cost of the fertilisers. Then
$Z = 10x + 8y$
The LPP can be stated mathematically as
Minimise $Z = 10x + 8y$
subject to constraints $3x + y \ge 300$, $x + y \ge 240$, $x \ge 0$, $y \ge 0$.
To solve the LPP graphically, we convert the inequations into equations to obtain the following lines:
$3x + y = 300$, $x + y = 240$, $x = 0$ and $y = 0$
Equation $3x + y = 300$ meets the coordinate axes at points $F(100, 0)$ and $G(0, 300)$
Equation $x + y = 240$ meets the coordinate axes at points $C(240, 0)$ and $D(0, 240)$.
The point of intersection of lines $3x + y = 300$ and $x + y = 240$ is $E(30, 210)$



The shaded region GEC represents the feasible region of given LPP and it is unbounded.

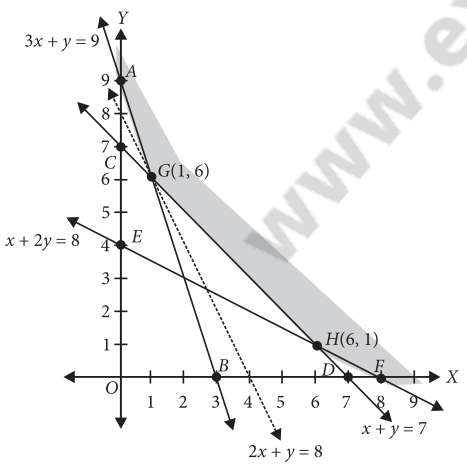
Corner points	Value of $Z = 10x + 8y$
$G(0, 300)$	2400
$C(240, 0)$	2400
$E(30, 210)$	1980 (Minimum)

From the table, we find that 1980 is the minimum value of Z at $E(30, 210)$. Since the region is unbounded, we have check that the inequality $10x + 8y < 1980$ in open half plane has any point in common or not. Since, it has no point in common.

www.examrite.in

So, the minimum value of Z is obtained at $E(30, 210)$ and the minimum value of Z is 1980.
So, the minimum requirement of fertiliser of type A will be 30 kg and that of type B will be 210 kg.

18. Let the person takes x tablets of type X and y tablets of type Y.
According to the given conditions, we have
$6x + 2y \geq 18 \Rightarrow 3x + y \geq 9$,
$3x + 3y \geq 21 \Rightarrow x + y \geq 7$
$2x + 4y \geq 16 \Rightarrow x + 2y \geq 8$
Let z be the total cost of tablets.
$\therefore \quad z = 2x + y$
Hence, the given LPP is
Minimise $Z = 2x + y$ subject to the constraints
$3x + y \geq 9, x + y \geq 7, x + 2y \geq 8$ and $x, y \geq 0$
To solve graphically, we convert the inequations into equations.
$3x + y = 9, x + y = 7, x + 2y = 8, x = 0, y = 0$
The line $3x + y = 9$ meets the coordinate axes at $A(0, 9)$ and $B(3, 0)$. Similarly, $x + y = 7$ meets the coordinate axes at $C(0, 7)$ and $D(7, 0)$. Also, line $x + 2y = 8$ meets the coordinate axes at $E(0, 4), F(8, 0)$
The point of intersection of the lines $3x + y = 9$ and $x + y = 7$ is $G(1, 6)$. Also, the point of intersection of the lines $x + y = 7$ and $x + 2y = 8$ is $H(6, 1)$.



The shaded region $AGHF$ represents the feasible region of the given LPP. The corner points of the feasible region are $A(0, 9)$, $G(1, 6)$, $H(6, 1)$ and $F(8, 0)$.
The values of the objective function at these points are given in the following table :

Corner Points	Value of $Z = 2x + y$
$A(0, 9)$	$2 \times 0 + 9 = 9$
$G(1, 6)$	$2 \times 1 + 6 = 8$ (Minimum)
$H(6, 1)$	$2 \times 6 + 1 = 13$
$F(8, 0)$	$2 \times 8 + 0 = 16$

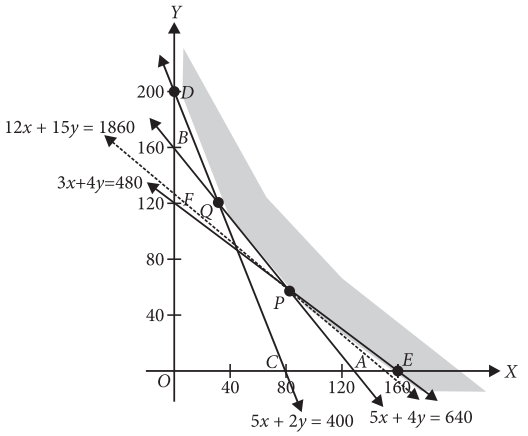
From the table, we find that 8 is the minimum value of Z at $G(1, 6)$. Since the region is unbounded we have to check that the inequality $2x + y < 8$ in open half plane has any point in common or not.
Since, it has no point in common. So, Z is minimum at $G(1, 6)$ and the minimum value of Z is 8.
Hence, the person should take 1 tablet of type X and 6 tablets of type Y in order to meet the requirements at the minimum cost.

19. Let x and y be the number of days, factory I and factory II have to be in operation to produce the order, respectively.

Calculator	Factory I	Factory II	Requirement
A	50	40	6400
B	50	20	4000
C	30	40	4800
Cost (in ₹)	12000	15000	

The inequations thus formed based on the given problem are as follows :
$50x + 40y \geq 6400 \Rightarrow 5x + 4y \geq 640$,
$50x + 20y \geq 4000 \Rightarrow 5x + 2y \geq 400$,
$30x + 40y \geq 4800 \Rightarrow 3x + 4y \geq 480$; $x, y \geq 0$
Let Z be the total cost of production.
$\therefore \quad Z = 12000x + 15000y$
So, the given LPP can be mathematically stated as
Minimise $Z = 12000x + 15000y$
Subject to constraints
$5x + 4y \geq 640, 5x + 2y \geq 400, 3x + 4y \geq 480; x, y \geq 0$
To solve the LPP graphically, we convert inequations into equations to obtain the following lines :
$5x + 4y = 640, 5x + 2y = 400, 3x + 4y = 480, x = 0$ and $y = 0$
The line $5x + 4y = 640$ meets the coordinate axes at $A(128, 0)$ and $B(0, 160)$
The line $5x + 2y = 400$ meets the coordinate axes at $C(80, 0)$ and $D(0, 200)$
The line $3x + 4y = 480$ meets the coordinate axes at $E(160, 0)$ and $F(0, 120)$
The point of intersection of lines $5x + 4y = 640$ and $5x + 2y = 400$ is $Q(32, 120)$.
The point of intersection of lines $5x + 4y = 640$ and $3x + 4y = 480$ is $P(80, 60)$.

www.exambites.in



The shaded region $DQPE$ represents the feasible region of the given LPP.

The corner points of the feasible region are $D(0, 200)$, $Q(32, 120)$, $P(80, 60)$ and $E(160, 0)$

The values of the objective function at these points are given as follows:

Corner Points	Value of $Z = 12000x + 15000y$
$D(0, 200)$	30,00,000
$Q(32, 120)$	21,84,000
$P(80, 60)$	18,60,000 (Minimum)
$E(160, 0)$	19,20,000

From the table, we find that 1860000 is the minimum value of Z at $P(80, 60)$. Since the region is unbounded we have to check that the inequality $12000x + 15000y < 1860000$ in open half plane has any point in common or not. Since it has no point in common. So, minimum value of Z is at $P(80, 60)$ and the minimum value of Z is 18,60,000.

20. Let x be the number of cakes of I kind and y be the number of cakes of II kind.

	Flour	Fat
Cake I	200 g	25 g
Cake II	100 g	50 g
Availability	5 kg	1 kg

The required LPP is

Maximise $Z = x + y$ subject to constraints

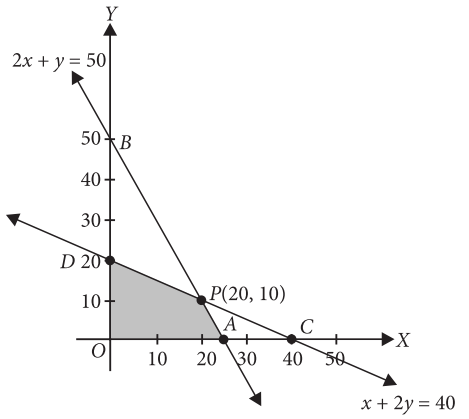
$200x + 100y \le 5{,}000 \Rightarrow 2x + y \le 50$

$25x + 50y \le 1{,}000 \Rightarrow x + 2y \le 40; x \ge 0, y \ge 0$

To solve LPP graphically, we convert inequations into equations.

$2x + y = 50$ ...(i), $x + 2y = 40$ ...(ii)

Lines (i) and (ii) intersect at $P(20, 10)$.



Shaded region is the feasible region *i.e.* $OAPD$. The corner points of the feasible region are $O(0, 0)$, $A(25, 0)$, $P(20, 10)$, $D(0, 20)$.

Corner Points	Value of $Z = x + y$
$O(0, 0)$	0
$A(25, 0)$	25
$P(20, 10)$	30 (Maximum)
$D(0, 20)$	20

Clearly, the number of cakes is maximum at $P(20, 10)$ *i.e.,* when 20 cakes of I kind and 10 cakes of II kind are made.

21. Let x and y be the number of packages of nuts and bolts manufactured respectively by the manufacturer. Then by the given data, the required LPP is

Maximize $Z = 24x + 18y$

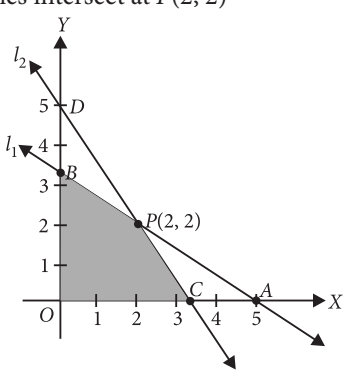
Subject to constraints

$2x + 3y \le 10$, $3x + 2y \le 10$, $x \ge 0$, $y \ge 0$.

We have the lines

$l_1 : 2x + 3y = 10$ and $l_2 : 3x + 2y = 10$

These lines intersect at $P(2, 2)$



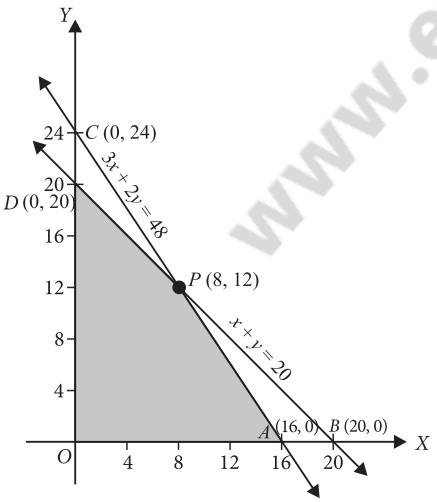
The feasible region is $OCPB$ and the corner points are $O(0, 0)$, $B\left(0, \frac{10}{3}\right), C\left(\frac{10}{3}, 0\right)$, $P(2, 2)$.

www.exambin.in

Corner points	Value of $Z = 24x + 18y$
$O(0, 0)$	0
$C\left(\frac{10}{3}, 0\right)$	80
$P(2, 2)$	84 (Maximum)
$B\left(0, \frac{10}{3}\right)$	60

$\therefore$ Profit, Z = ₹84 will be maximum, when 2 packages nuts and 2 packages of bolts are manufactured.

22. Let x be the number of electronic sewing machines and y be the number of manually operated sewing machines, the dealer sells. The given problem can be formulated as
Maximise $Z = 22x + 18y$
Subject to constraints
$x + y \le 20$, $360x + 240y \le 5760$
$\Rightarrow 3x + 2y \le 48$ and $x, y \ge 0$
To solve LPP graphically, we convert the inequations into equations *i.e.*, $x + y = 20$, $3x + 2y = 48$, $x = y = 0$

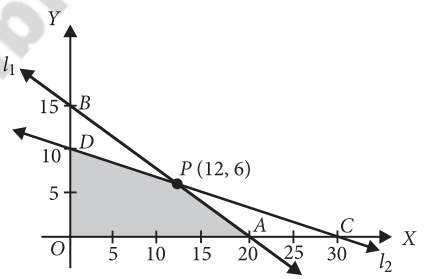


The shaded region $APDO$ is the feasible region.
The corner points of the feasible region are $A(16, 0)$, $P(8, 12)$, $D(0, 20)$ and $O(0, 0)$.

Corner Points	Value of $Z = 22x + 18y$
$A(16, 0)$	352
$P(8, 12)$	392 (Maximum)
$D(0, 20)$	360
$O(0, 0)$	0

We see that the point $P(8, 12)$ is giving the maximum value of Z.
Hence, the dealer should purchase 8 electronic sewing machines and 12 manually operated sewing machines to obtain the maximum profit under the given conditions.

23. Let x and y be the number of teaching aids of type A and B respectively to be manufactured per week. Then the LPP can be stated mathematically as
Maximise $Z = 80x + 120y$
subject to constraints
$9x + 12y \le 180$, $x + 3y \le 30$; $x \ge 0$, $y \ge 0$
To solve LPP graphically, we convert inequations into equations
$l_1 : 9x + 12y = 180 \Rightarrow 3x + 4y = 60$
and $l_2 : x + 3y = 30$
Both the lines intersect at $P(12, 6)$.



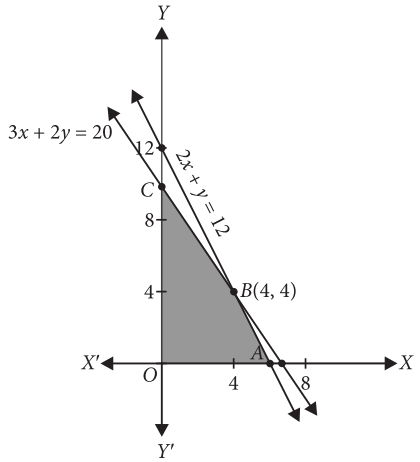
The feasible region is $OAPDO$.

Corner Points	Value of $Z = 80x + 120y$
$O(0, 0)$	0
$A(20, 0)$	1600
$P(12, 6)$	1680 (Maximum)
$D(0, 10)$	1200

$\therefore$ The profit is maximum at $P(12, 6)$ *i.e.*, when the teaching aids of types A and B are 12 and 6 respectively.
Also, maximum profit = ₹ 1680 per week

24. Let the cottage industry manufactures x pedestal lamps and y wooden shades. Then the LPP can be stated mathematically as
Maximize $Z = 25x + 15y$
Subject to constraints :
$2x + y \le 12$, $3x + 2y \le 20$, $x \ge 0$, $y \ge 0$.
Both the lines intersect at $B(4, 4)$.

www.examples.in



The corner points of feasible region are :
$A(6, 0)$, $B(4, 4)$, $C(0, 10)$ & $O(0, 0)$

Corner Points	Value of $Z = 25x + 15y$
$O(0, 0)$	0
$A(6, 0)$	150
$B(4, 4)$	160 (Maximum)
$C(0, 10)$	150

Clearly, Z is maximum at $B(4, 4)$
So, maximum profit of ₹160 is obtained when 4 pedestal lamps and 4 wooden shades are manufactured.

25. *Refer to answer 18.*

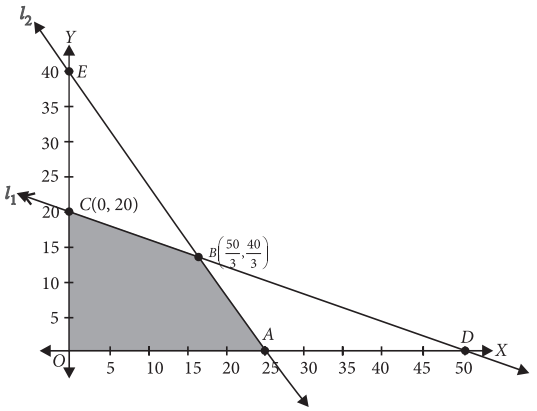
26. Suppose that the young man rides x km at 25 km per hour and y km at 40 km per hour. Then, the given problem can be formulated as
Maximize $Z = x + y$.
Subject to the constraints,
$x \geq 0, y \geq 0, 2x + 5y \leq 100,$

$$\frac{x}{25} + \frac{y}{40} \leq 1$$

$\Rightarrow \quad 8x + 5y \leq 200$
Now, we convert the system of the inequations into equations.
$l_1 : 2x + 5y = 100$ and $l_2 : 8x + 5y = 200$

Both the lines intersect at $B\left(\frac{50}{3}, \frac{40}{3}\right)$

The solution set of the given system is the shaded region $OABC$.



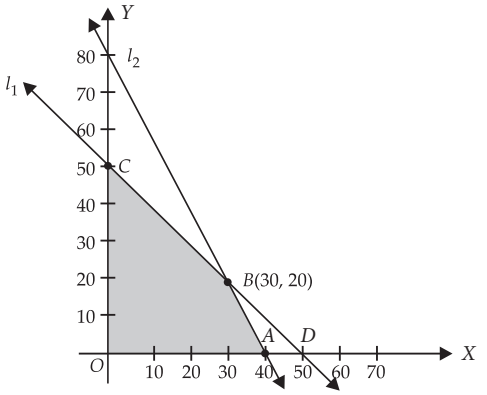
The coordinates of corner points O, A, B, C are (0, 0), (25, 0), $\left(\frac{50}{3}, \frac{40}{3}\right)$ and (0, 20) respectively.

Corner Points	Value of $Z = x + y$
$O(0, 0)$	0
$A(25, 0)$	25
$B\left(\frac{50}{3}, \frac{40}{3}\right)$	30 (Maximum)
$C(0, 20)$	20

So, $Z = x + y$ is maximum when $x = \frac{50}{3}$ and $y = \frac{40}{3}$.
Thus, the student can cover the maximum distance of 30 km, if he rides $\frac{50}{3}$ km at 25 km/hr and $\frac{40}{3}$ km at 40 km/hr.
The value indicated in this question is that maximum distance is covered in one hour with less pollution.

27. Let x hectare of land to be allocated to crop A and y hectare to B.
Thus, the LPP can be formulated as
Maximise $Z = 10{,}500x + 9{,}000y$
subject to the constraints
$x + y \leq 50, 20x + 10y \leq 800 \Rightarrow 2x + y \leq 80, x \geq 0, y \geq 0$
To solve LPP graphically, we convert inequations into equations.
$l_1 : x + y = 50, l_2 : 2x + y = 80$
$x = 0, y = 0$
Both the lines intersect at $B(30, 20)$

www.exambites.in



OABC is the feasible region which is bounded.
The corner points are $O(0, 0)$, $A(40, 0)$, $B(30, 20)$, $C(0, 50)$.

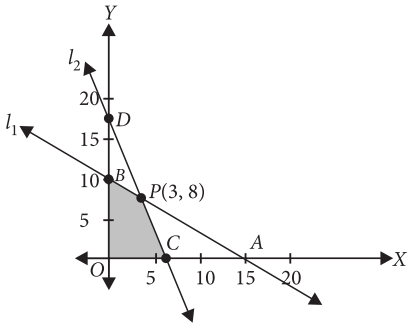
Corner Points	Value of $Z = 10{,}500x + 9{,}000y$
$O(0, 0)$	0
$A(40, 0)$	420000
$B(30, 20)$	495000 (Maximum)
$C(0, 50)$	450000

Hence society will allocate 30 hectares of land to crop A and 20 hectares of land to crop B to maximise the total profit.
Yes, protection of wildlife is necessary to preserve balance in environment because it will be a loss of biodiversity. The wild animals would get extinct ultimately if we would not provide protection to them. Wild life animals are being killed for their valuable ivory, skin, fur etc. to make products such as leather, meat etc.

28. Let x units of the goods A and y units of goods B be produced to maximise the total revenue.

	Workers	Capital (in units)	Revenue per unit (in ₹)
Goods A	2	3	100
Goods B	3	1	120
Total units	30	17	

The LPP is given by
Maximise $Z = 100x + 120y$
Subject to the constraints
$2x + 3y \le 30$, $3x + y \le 17$, $x \ge 0$, $y \ge 0$
To solve LPP graphically, we convert the inequations into equations.
$l_1 : 2x + 3y = 30$, $l_2 : 3x + y = 17$ and $x = 0$, $y = 0$
These lines meet at $P(3, 8)$.



The feasible region *OCPB* has been shaded.
The corner points of the feasible region are $O(0, 0)$, $C(5.6, 0)$, $P(3, 8)$, $B(0, 10)$

Corner Points	Value of $Z = 100x + 120y$
$O(0, 0)$	0
$C(5.6, 0)$	560
$P(3, 8)$	1260 (Maximum)
$B(0, 10)$	1200

Clearly, the maximum revenue is obtained at $P(3, 8)$, *i.e.*, when 3 units of good A and 8 units of good B are produced.
Yes, I agree with the view of the manufacturer. Men and women workers should be equally paid so that they can do their work efficiently and accurately.

29. *Refer to answer 24.*

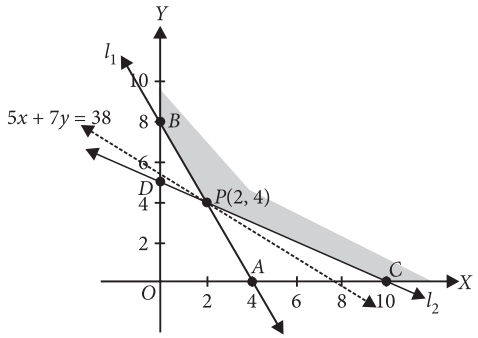
30. *Refer to answer 21.*

31. Let x kg of food I and y kg of food II be purchased, then the given data can be represented in the tabular form as follows :

Nutrients	Food I	Food II	Requirements
Vitamin A	2	1	8
Vitamin C	1	2	10
Cost (in ₹)	5	7	

$\therefore$ The given LPP is as follows.
Minimise $Z = 5x + 7y$
Subject to the constraints
$2x + y \ge 8$, $x + 2y \ge 10$, $x \ge 0$, $y \ge 0$
To solve LPP graphically, we convert inequations into equations.
$l_1 : 2x + y = 8$, $l_2 : x + 2y = 10$
These lines intersect at $P(2, 4)$

www.exambites.in



From the graph, the corner points of the feasible region are $C(10, 0)$, $P(2, 4)$ and $B(0, 8)$.

Corner Points	Value of $Z = 5x + 7y$
$C(10, 0)$	50
$P(2, 4)$	38 (Minimum)
$B(0, 8)$	56

From the table, we find that 38 is the minimum value of Z at $P(2, 4)$. Since the region is unbounded, we have to check that the inequality $5x + 7y < 38$ in open half plane has any point in common or not. Since, it has no point in common. So the minimum cost is ₹ 38 and this is attained at $P(2, 4)$, *i.e.*, when 2 units of food I and 4 units of food II are purchased.

32. Let the number of items of the type A and B be x and y respectively. Then the LPP is

Maximise $Z = 50x + 28y$

Subject to the constraints,

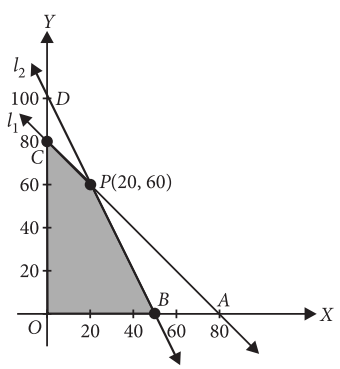
$x + y \leq 80$,

$300x + 150y \leq 15000 \Rightarrow 2x + y \leq 100$,

$x \geq 0, y \geq 0$

To solve LPP graphically, we convert inequations into equations.

$l_1 : x + y = 80$, $l_2 : 2x + y = 100$



Their point of intersection is $P(20, 60)$. The feasible region $OBPC$ is shown shaded in the figure with corner points $O(0, 0)$, $B(50, 0)$, $P(20, 60)$ and $C(0, 80)$

Corner Points	Value of $Z = 50x + 28y$
$O(0, 0)$	0
$C(0, 80)$	2240
$P(20, 60)$	2680 (Maximum)
$B(50, 0)$	2500

Thus, the maximum profit of ₹ 2680 is at P (20, 60) *i.e.*, when 20 items of type A and 60 items of type B are purchased and sold.

33. *Refer to answer 21.*

34. Let the factory makes 'x' tennis rackets and 'y' cricket bats.

We make the following table from the given data

	Tennis Rackets	Cricket Bats	Availability
Machine time (in hrs)	1.5	3	42
Craftsman's time (in hrs)	3	1	24
Profit (in ₹)	20	10	

Hence, the mathematical formulation of the problem is

Maximize $Z = 20x + 10y$

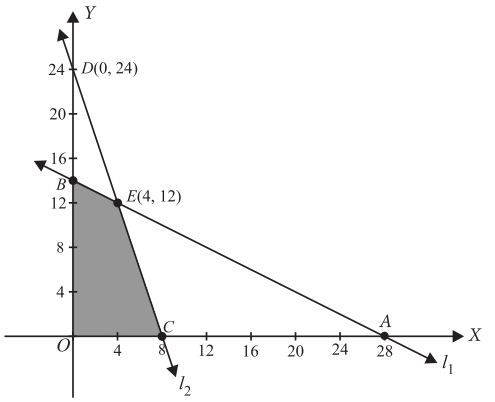
subject to the constraints

$1.5x + 3y \leq 42 \Rightarrow x + 2y \leq 28$, $3x + y \leq 24$; $x, y \geq 0$

Convert the given inequations into equations, we have

$l_1 : x + 2y = 28$, $l_2 : 3x + y = 24$ and $x = 0$, $y = 0$

These lines intersect at $E(4, 12)$.



www.exambits.in

The feasible region $OCEB$ is shown in the graph and the corner points are $O(0, 0)$, $C(8, 0)$, $E(4, 12)$, $B(0, 14)$

Corner Points	Value of $Z = 20x + 10y$
O (0, 0)	0
C (8, 0)	160
E (4, 12)	200 (Maximum)
B (0, 14)	140

Hence, the profit is maximum *i.e.*, ₹ 200 when 4 tennis rackets and 12 cricket bats are manufactured.

35. Let the merchant stocks x desktop model computers and y portable model computers.

Hence, the mathematical formulation of the problem is

Maximize $Z = 4500x + 5000y$

subject to the constraints

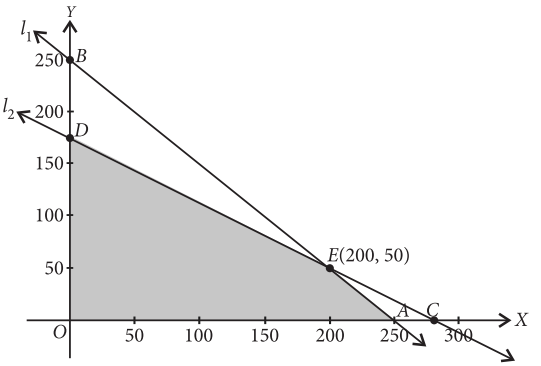
$x + y \leq 250$, $25000x + 40000y \leq 7000000$,

$5x + 8y \leq 1400$; $x, y \geq 0$

To solve LPP graphically we convert inequations into equations

$l_1 : x + y = 250$, $l_2 : 5x + 8y = 1400$ and $x = 0$, $y = 0$

These lines intersect at $E(200, 50)$



The feasible region $OAED$ is shown in the graph. Here we observe that the feasible region is bounded. The corner points are $O(0, 0)$, $A(250, 0)$, $E(200, 50)$ and $D(0, 175)$.

Corner Points	Value of $Z = 4500x + 5000y$
O (0, 0)	0
A (250, 0)	1125000
E (200, 50)	1150000 (Maximum)
D (0, 175)	875000

We find that maximum value of Z is 1150000 at $E(200, 50)$. Hence the merchant should stock 200 units of desktop model and 50 units of portable model to realise maximum profit and maximum profit is ₹ 1150000.

36. *Refer to answer 31.*

www.examfiesta.in

CHAPTER
13

Probability

13.1 Introduction

13.2 Conditional Probability

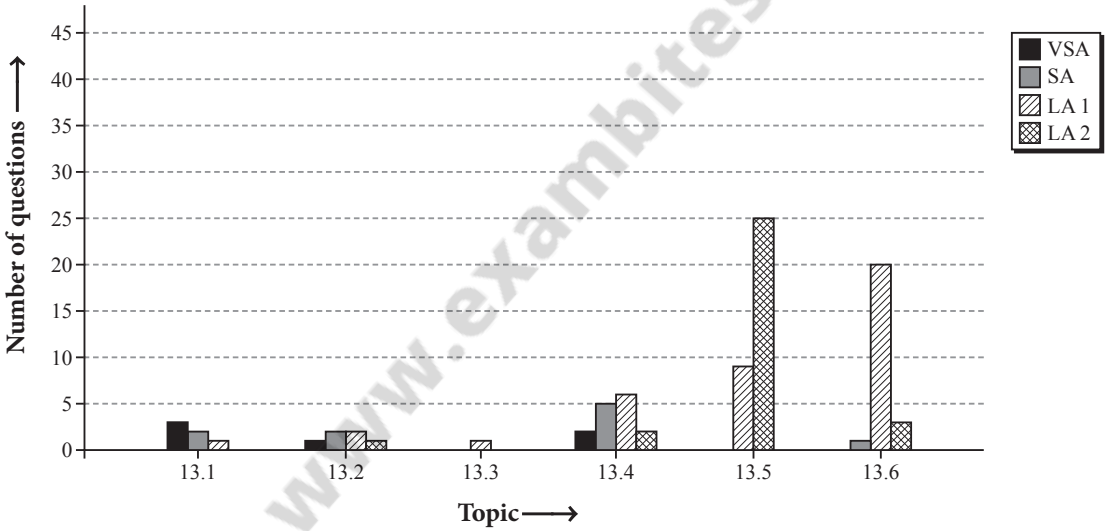
13.3 Multiplication Theorem on Probability

13.4 Independent Events

13.5 Bayes' Theorem

13.6 Random Variables and its Probability Distributions

Topicwise Analysis of Last 10 Years' CBSE Board Questions (2020-2011)



www.exambites.in

- Maximum weightage is of *Bayes' Theorem*
- Maximum LA 1 type questions were asked from *Random Variables and its Probability Distributions*
- Maximum LA 2 type questions were asked from *Bayes' Theorem*

QUICK RECAP

CONDITIONAL PROBABILITY

Let E and F be two events of a random experiment, then the conditional probability of occurrence of E under the condition that F has already occurred, *i.e.*, $P(E/F)$ is given by

$$P(E/F)=\frac{P(E \cap F)}{P(F)}, \text{ where } P(F) \neq 0$$

- **Properties of Conditional Probability**
  - (i) For an event A and sample space S, $P(S/A) = P(A/A) = 1$
  - (ii) If A and B are any two events of a sample

space S and F is an event of S such that $P(F) \neq 0$, then
$P((A \cup B)/F) = P(A/F) + P(B/F) - P((A \cap B)/F)$
In particular, if A and B are disjoint events, then $P((A \cup B)/F) = P(A/F) + P(B/F)$

(iii) $P(A'/B) = 1 - P(A/B)$

(iv) $0 \leq P(A/B) \leq 1$

MULTIPLICATION THEOREM ON PROBABILITY

Let A and B are two events associated with a sample space S. Then $A \cap B$ denotes the event that both A and B have occurred. The event $A \cap B$ is also written as AB.
Also, $P(A \cap B) = P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$, provided $P(A) \neq 0$, $P(B) \neq 0$
Note : If A, B and C are three events associated with a random experiment, then
$P(A \cap B \cap C) = P(A) \cdot P(B/A) \cdot P(C/(A \cap B))$
$= P(A) \cdot P(B/A) \cdot P(C/AB)$
Similarly, multiplication rule of probability can be extended to four or more events.

INDEPENDENT EVENTS

Events are said to be independent, if the occurrence or non-occurrence of one does not affect the probability of the occurrence or non-occurrence of the other.

Two events A and B associated with a random experiment are said to be independent if

(i) $P(A/B) = P(A)$ provided $P(B) \neq 0$

(ii) $P(B/A) = P(B)$ provided $P(A) \neq 0$

Note :

(i) Two events A and B associated with a random experiment are independent if $P(A \cap B) = P(A) \cdot P(B)$

(ii) Two events A and B are said to be dependent if they are not independent, *i.e.*, if $P(A \cap B) \neq P(A) \cdot P(B)$

(iii) Events $A_1, A_2, ..., A_n$ are mutually independent if the probability of the simultaneous occurrence of (any) finite number of events is equal to the product of their separate probabilities. While these events are pairwise independent if $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$.

(iv) If A and B are independent events, then so are the events A and B'; A' and B; A' and B'.

THEOREM OF TOTAL PROBABILITY

If $E_1, E_2, E_3, ..., E_n$ are mutually exclusive and exhaustive events associated with a sample space S of a random experiment and A is any event associated with S, then
$P(A) = P(E_1)\, P(A/E_1) + P(E_2)\, P(A/E_2) + ... + P(E_n)\, P(A/E_n)$

BAYES' THEOREM

If $E_1, E_2, E_3, ..., E_n$ are mutually exclusive and exhaustive events associated with a random experiment and A is any event associated with the experiment, then

$$P(E_i/A) = \frac{P(E_i)P(A/E_i)}{\sum_i P(E_i)P(A/E_i)},$$

where $i = 1, 2, 3, ..., n$

RANDOM VARIABLE

A random variable is a real valued function, whose domain is the sample space of a random experiment. Generally, it is denoted by X.

PROBABILITY DISTRIBUTIONS

Let real numbers $x_1, x_2, ..., x_n$ are the possible values of a random variable X and $p_1, p_2, ..., p_n$ are the corresponding probabilities to each value of the random variable X. Then the probability distribution is

X :	x_1	x_2	...	x_n
$P(X)$:	p_1	p_2	...	p_n.

Note :

(i) $1 > p_i > 0$

(ii) Sum of probabilities $(p_1 + p_2 + ... + p_n) = 1$

▶ **Mean :** If X is a random variable, then mean $(\bar{X})$ of X is defined as

$$\bar{X} = \mu = p_1x_1 + p_2x_2 + ... + p_nx_n = \sum_{i=1}^{n} p_ix_i$$

Mean of a random variable is also called the expectation of X, denoted by $E(X)$.

▶ **Variance :** If X is a random variable, then variance of X is defined as

$$\text{Var}\,(X) = \sigma^2 = \sum_{i=1}^{n} p_ix_i^2 - \left(\sum_{i=1}^{n} p_ix_i\right)^2$$

or $\text{Var}(X) = E(X^2) - \{E(X)\}^2$

Note : Standard deviation $(\sigma) = \sqrt{\text{Var}(X)}$

www.exambites.in

Previous Years' CBSE Board Questions

13.1 Introduction

VSA (1 mark)

1. Two cards are drawn at random and one-by-one without replacement from a well-shuffled pack of 52 playing cards. Find the probability that one card is red and the other is black. *(2020)*

2. A bag contains 3 black, 4 red and 2 green balls. If three balls are drawn simultaneously at random, then the probability that the balls are of different colours is _____. *(2020)*

3. A die is thrown once. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$

(c) 0 (d) 1 *(2020)*

SA (2 marks)

4. If A and B are two events such that $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cup B) = 0.6$, then find $P(B' \cap A)$. *(2020)*

5. Out of 8 outstanding students of a school, in which there are 3 boys and 5 girls, a team of 4 students is to be selected for a quiz competition. Find the probability that 2 boys and 2 girls are selected. *(AI 2019)*

LA 1 (4 marks)

6. A bag A contains 4 black and 6 red balls and bag B contains 7 black and 3 red balls. A die is thrown. If 1 or 2 appears on it, then bag A is chosen, otherwise bag B. If two balls are drawn at random (without replacement) from the selected bag, find the probability of one of them being red and another black. *(Delhi 2015)*

13.2 Conditional Probability

VSA (1 mark)

7. A card is picked at random from a pack of 52 playing cards. Given that the picked card is a queen, the probability of this card to be a card of spade is

(a) $\frac{1}{3}$ (b) $\frac{4}{13}$

(c) $\frac{1}{4}$ (d) $\frac{1}{2}$ *(2020)*

SA (2 marks)

8. 12 cards numbered 1 to 12 (one number on one card), are placed in a box and mixed up thoroughly. Then a card is drawn at random from the box. If it is known that the number on the drawn card is greater than 5, find the probability that the card bears an odd number. *(AI 2019)*

9. A black and a red die are rolled together. Find the conditional probability of obtaining the sum 8, given that the red die resulted in a number less than 4. *(2018)*

LA 1 (4 marks)

10. Assume that each born child is equally likely to be a boy or a girl. If a family has two children, what is the conditional probability that both are girls? Given that

(i) the youngest is a girl.

(ii) atleast one is a girl. *(Delhi 2014)*

11. A couple has 2 children. Find the probability that both are boys, if it is known that

(i) one of them is a boy,

(ii) the older child is a boy. *(Delhi 2014C)*

LA 2 (6 marks)

12. Consider the experiment of tossing a coin. If the coin shows head, toss it again, but if it shows tail, then throw a die. Find the conditional probability of the event that 'the die shows a number greater than 4' given that 'there is at least one tail'. *(Delhi 2014C)*

www.exambites.in

13.3 Multiplication Theorem on Probability

LA 1 (4 marks)

13. A bag contains 3 red and 7 black balls. Two balls are selected at random one-by-one without replacement. If the second selected ball happens to be red, what is the probability that the first selected ball is also red? *(Delhi 2014C)*

13.4 Independent Events

VSA (1 mark)

14. If A and B are two independent events with $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$, then $P(B'|A)$ is equal to

(a) $\frac{1}{4}$ (b) $\frac{1}{8}$

(c) $\frac{3}{4}$ (d) 1 *(2020)*

15. A problem is given to three students whose probabilities of solving it are $\frac{1}{3}, \frac{1}{4}$ and $\frac{1}{6}$ respectively. If the events of solving the problem are independent, find the probability that at least one of them solves it. *(2020)*

SA (2 marks)

16. The probability of finding a green signal on a busy crossing X is 30%. What is the probability of finding a green signal on X on two consecutive days out of three? *(2020)*

17. Given two independent events A and B such that $P(A) = 0.3$ and $P(B) = 0.6$, find $P(A' \cap B')$. *(2020)*

18. A die marked 1, 2, 3 in red and 4, 5, 6 in green is tossed. Let A be the event "number is even" and B be the event "number is marked red". Find whether the events A and B are independent or not. *(Delhi 2019)*

19. Prove that if E and F are independent events, then the events E' and F' are also independent. *(Delhi 2017)*

20. A die, whose faces are marked 1, 2, 3 in red and 4, 5, 6 in green, is tossed. Let A be the event "number obtained is even" and B be the event "number obtained is red". Find if A and B are independent events. *(AI 2017)*

LA 1 (4 marks)

21. A and B throw a pair of dice alternately. A wins the game if he gets a total of 7 and B wins the game if he gets a total of 10. If A starts the game, then find the probability that B wins. *(Delhi 2016)*

22. Probability of solving specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that

(i) The problem is solved

(ii) exactly one of them solved the problem. *(Delhi 2015C, 2011)*

23. P speaks truth in 70% of the cases and Q in 80% of the cases. In what percent of cases are they likely to agree in stating the same fact? Do you think, when they agree, means both are speaking truth? *(Delhi 2013)*

24. A speaks truth in 75% of the cases, while B in 90% of the cases. In what per cent of cases are they likely to contradict each other in stating the same fact?
Do you think that statement of B is true? *(Delhi 2013)*

25. A speaks truth in 60% of the cases, while B in 90% of the cases. In what per cent of cases are they likely to contradict each other in stating the same fact? In the cases of contradiction do you think, the statement of B will carry more weight as he speaks truth in more number of cases than A? *(Delhi 2013)*

26. The probabilities of two students A and B coming to the school in time are $\frac{3}{7}$ and $\frac{5}{7}$ respectively. Assuming that the events, 'A coming in time' and 'B coming in time' are independent, find the probability of only one of them coming to the school in time.
Write at least one advantage of coming to school in time. *(AI 2013)*

www.examhits.in

LA 2 (6 marks)

27. If A and B are two independent events such that $P(\bar{A} \cap B) = \frac{2}{15}$ and $P(A \cap \bar{B}) = \frac{1}{6}$, then find $P(A)$ and $P(B)$. *(Delhi 2015)*

28. In a hockey match, both teams A and B scored same number of goals up to the end of the game, so as to decide the winner, the referee asked both the captains to throw a die alternately and decided that the team, whose captain gets a six first, will be declared the winner. If the captain of team A was asked to start, find their respective probabilities of winning the match and state whether the decision of the referee was fair or not. *(AI 2013)*

13.5 Bayes' Theorem

LA 1 (4 marks)

29. There are two bags, I and II. Bag I contains 3 red and 5 black balls and Bag II contains 4 red and 3 black balls. One ball is transferred randomly from Bag I to Bag II and then a ball is drawn randomly from Bag II. If the ball so drawn is found to be black in colour, then find the probability that the transferred ball is also black. *(2020)*

30. A bag contains two coins, one biased and the other unbiased. When tossed, the biased coin has a 60% chance of showing heads. One of the coins is selected at random and on tossing it shows tails. What is the probability it was an unbiased coin? *(2020)*

31. In a shop X, 30 tins of ghee of type A and 40 tins of ghee of type B which look alike, are kept for sale. While in shop Y, similar 50 tins of ghee of type A and 60 tins of ghee of type B are there. One tin of ghee is purchased from one of the randomly selected shop and is found to be of type B. Find the probability that it is purchased from shop Y. *(2020)*

32. Suppose a girl throws a die. If she gets 1 or 2, she tosses a coin three times and notes the number of tails. If she gets 3, 4, 5 or 6, she tosses a coin once and notes whether a 'head' or 'tail' is obtained. If she obtained exactly one 'tail', what is the probability that she threw 3, 4, 5 or 6 with the die? *(2018)*

33. Often it is taken that a truthful person commands, more respect in the society. A man is known to speak the truth 4 out of 5 times. He throws a die and reports that it is a six. Find the probability that it is actually a six. Do you also agree that the value of truthfulness leads to more respect in the society? *(Delhi 2017)*

34. Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous year results report that 70% of all students who have 100% attendance attain A grade and 10% irregular students attain A grade in their annual examination. At the end of the year, one student is chosen at random from the school and he was found to have an A grade. What is the probability that the student has 100% attendance ? Is regularity required only in school ? Justify your answer. *(AI 2017)*

35. Three persons A, B and C apply for a job of Manager in a Private Company. Chances of their selection (A, B and C) are in the ratio 1 : 2 : 4. The probabilities that A, B and C can introduce changes to improve profits of the company are 0.8, 0.5 and 0.3 respectively. If the change does not take place, find the probability that it is due to the appointment of C. *(Delhi 2016)*

36. A bag contains 4 balls. Two balls are drawn at random (without replacement) and are found to be white. What is the probability that all balls in the bag are white ? *(AI 2016)*

37. Three machines E_1, E_2 and E_3 in a certain factory producing electric bulbs, produce 50%, 25% and 25% respectively, of the total daily output of electric bulbs. It is known that 4% of the bulbs produced by each of machines E_1 and E_2 are defective and that 5% of those produced by machine E_3 are defective. If one bulb is picked up at random from a day's production, calculate the probability that it is defective. *(Foreign 2015)*

www.examtest.in

LA 2 (6 marks)

38. A manufacturer has three machine operators *A*, *B* and *C*. The first operator *A* produces 1% of defective items, whereas the other two operators *B* and *C* produces 5% and 7% defective items respectively. *A* is on the job for 50% of the time. *B* on the job 30% of the time and *C* on the job for 20% of the time. All the items are put into one stockpile and then one item is chosen at random from this and is found to be defective. What is the probability that it was produced by *A*?
(Delhi 2019)

39. An insurance company insured 3000 cyclists, 6000 scooter drivers and 9000 car drivers. The probability of an accident involving a cyclist, a scooter driver and a car driver are 0.3, 0.05 and 0.02 respectively. One of the insured persons meets with an accident. What is the probability that he is a cyclist?
(AI 2019)

40. Bag *A* contains 3 red and 5 black balls, while bag *B* contains 4 red and 4 black balls. Two balls are transferred at random from bag *A* to bag *B* and then a ball is drawn from bag *B* at random. If the ball drawn from bag *B* is found to be red find the probability that two red balls were transferred from *A* to *B*.
(Foreign 2016)

41. In a factory which manufactures bolts, machines *A*, *B* and *C* manufacture respectively 30%, 50% and 20% of the bolts. Of their outputs 3, 4 and 1 percent respectively are defective bolts. *A* bolt is drawn at random from the product and is found to be defective. Find the probability that this is not manufactured by machine *B*. *(AI 2015)*

42. A bag contains 4 red and 4 black balls, another bag contains 2 red and 6 black balls. One of the two bags is selected at random and two balls are drawn at random without replacement from the bag and are found to be both red. Find the probability that the balls are drawn from the first bag.
(Delhi 2015C)

43. In answering a question on a multiple choice test, a student either knows the answer or guesses. Let $\frac{3}{5}$ be the probability that he knows the answer and $\frac{2}{5}$ be the probability that he guesses. Assuming that a student who guesses the answer will be correct with probability $\frac{1}{3}$, what is the probability that the student knows the answer given that he answered it correctly? *(AI 2015C)*

44. A card from a pack of 52 playing cards is lost. From the remaining cards of the pack three cards are drawn at random (without replacement) and are found to be all spades. Find the probability of the lost card being a spade. *(Delhi 2014)*

45. There are three coins. One is a two-headed coin (having head on both faces), another is a biased coin that comes up heads 75% of the times and third is also a biased coin that comes up tails 40% of the times. One of the three coins is chosen at random and tossed, and it shows head. What is the probability that it was the two-headed coin?
(AI 2014)

46. An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident for them are 0.01, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a scooter driver or a car driver? *(Foreign 2014)*

47. A man is known to speak the truth 3 out of 5 times. He throws a die and reports that it is '1'. Find the probability that it is actually 1.
(Delhi 2014C)

48. An urn contains 4 balls. Two balls are drawn at random from the urn (without replacement) and are found to be white. What is the probability that all the four balls in the urn are white? *(AI 2014C)*

49. Assume that the chances of a patient having a heart attack is 40%. Assuming that a meditation and yoga course reduces the risk of heart attack by 30% and prescription of certain drug reduces its chance by 25%. At

www.exambees.in

a time a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options, the patient selected at random suffers a heart attack. Find the probability that the patient followed a course of meditation and yoga. Interpret the result and state which of the above stated methods is more beneficial for the patient. *(Delhi 2013)*

50. In a group of 400 people, 160 are smokers and non-vegetarian, 100 are smokers and vegetarian and the remaining are non-smokers and vegetarian. The probabilities of getting a special chest disease are 35%, 20% and 10% respectively. A person is chosen from the group at random and is found to be suffering from the disease. What is the probability that the selected person is a smoker and non-vegetarian? What value is reflected in this question? *(Delhi 2013C)*

51. Assume that the chances of a patient having a heart attack are 50%. Assume that a meditation and yoga course reduces the risk of heart attack by 30% and the prescription of certain drugs and certain restrictions reduces the risk by 25%. At a time a patient chooses only one of the two options with equal probabilities. After going through one of the two options, the patient selected at random suffers a heart attack. Find the probability that the patient followed a course of meditation and yoga. What values are reflected in this question? *(AI 2013C)*

52. Of the students in a college, it is known that 60% reside in hostel and 40% are day scholars (not residing in hostel). Previous year results report that 30% of all students who reside in hostel attain 'A' grade and 20% of day scholars attain 'A' grade in their annual examination. At the end of the year, one student is chosen at random from the college and he has an 'A' grade, what is the probability that the student is a hosteler? *(Delhi 2012, 2011C)*

53. A girl throws a die. If she gets a 5 or 6, she tosses a coin three times and notes the number of heads. If she gets 1, 2, 3 or 4, she tosses a coin two times and notes the number of heads obtained. If she obtained exactly two heads, what is the probability that she threw 1, 2, 3 or 4 with the die? *(Delhi 2012)*

54. Bag I contains 3 red and 4 black balls and bag II contains 4 red and 5 black balls. Two balls are transferred at random from bag I to bag II and then a ball is drawn from bag II. The ball so drawn is found to be red in colour. Find the probability that the transferred balls were both black. *(Delhi 2012)*

55. Suppose a girl throws a die. If she gets a 5 or 6, she tosses a coin 3 times and notes the number of heads. If she gets 1, 2, 3 or 4, she tosses a coin once and notes whether a head or tail is obtained. If she obtained exactly one head, what is the probability that she threw 1, 2, 3 or 4 with the die? *(AI 2012)*

56. A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn at random and are found to be hearts. Find the probability of the missing card to be a heart. *(Delhi 2012C)*

57. An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers, the probabilities of their meeting an accident respectively are 0.01, 0.03 and 0.15. One of the insured persons meets with an accident. What is the probability that he is a car driver? *(AI 2012C)*

58. Suppose 5% of men and 0.25% of women have grey hair. A grey haired person is selected at random. What is the probability of this person being male ? Assume that there are equal number of males and females. *(Delhi 2011)*

59. Bag I contains 3 red and 4 black balls and bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag II. *(Delhi 2011)*

60. A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six. *(Delhi 2011)*

www.exambits.in

61. Given three identical boxes I, II and III each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold? *(AI 2011)*

62. There are three coins. One is a two-tailed coin (having tail on both faces), another is a biased coin that comes up heads 60% of the times and third is an unbiased coin. One of the three coins is chosen at random and tossed, and it shows tails. What is the probability that it was a two-tailed coin? *(AI 2011C)*

13.6 Random Variables and its Probability Distributions

SA (2 marks)

63. The random variable X has a probability distribution $P(X)$ of the following form, where 'k' is some number,

$$P(X=x)=\begin{cases} k, & \text{if } x=0 \\ 2k, & \text{if } x=1 \\ 3k, & \text{if } x=2 \\ 0, & \text{otherwise} \end{cases}$$

Determine the value of 'k'. *(Delhi 2019)*

LA 1 (4 marks)

64. An urn contains 5 red, 2 white and 3 black balls. Three balls are drawn, one-by-one at random without replacement. Find the probability distribution of the number of white balls. Also, find the mean and the variance of the number of white balls drawn. *(2020)*

65. The probability distribution of a random variable X, where k is a constant is given below :

$$P(X=x)=\begin{cases} 0.1, & \text{if } x=0 \\ kx^2, & \text{if } x=1 \\ kx, & \text{if } x=2 \text{ or } 3 \\ 0, & \text{otherwise} \end{cases}$$

Determine
(a) the value of k
(b) $P(x \le 2)$
(c) Mean of the variable X *(2020)*

66. Three rotten apples are mixed with seven fresh apples. Find the probability distribution of the number of rotten apples, if three apples are drawn one by one with replacement. Find the mean of the number of rotten apples. *(2020)*

67. Two numbers are selected at random (without replacement) from the first five positive integers. Let X denote the larger of the two numbers obtained. Find the mean and variance of X. *(2018)*

68. The random variable X can take only the values 0, 1, 2, 3. Given that $P(X=0)=P(X=1)=p$ and $P(X=2)=P(X=3)$ such that $\Sigma p_i x_i^2 = 2\Sigma p_i x_i$, find the value of p. *(Delhi 2017)*

69. There are 4 cards numbered 1, 3, 5 and 7, one number on one card. Two cards are drawn at random without replacement. Let X denote the sum of the numbers on the two drawn cards. Find the mean and variance of X. *(AI 2017)*

70. In a game, a man wins ₹ 5 for getting a number greater than 4 and loses ₹ 1 otherwise, when a fair die is thrown. The man decided to throw a die thrice but to quit as and when he gets a number greater than 4. Find the expected value of the amount he wins/loses. *(AI 2016)*

71. Let X denote the number of colleges where you will apply after your results and $P(X=x)$ denotes your probability of getting admission in x number of colleges. It is given that

$$P(X=x)=\begin{cases} kx & \text{, if } x=0 \text{ or } 1 \\ 2kx & \text{, if } x=2 \\ k(5-x) & \text{, if } x=3 \text{ or } 4 \\ 0 & \text{, if } x>4 \end{cases}$$

where k is a positive constant. Find the value of k. Also find the probability that you will get admission in
(i) exactly one college
(ii) atmost 2 colleges
(iii) atleast 2 colleges. *(Foreign 2016)*

www.examprep.in

72. Three cards are drawn successively with replacement from a well shuffled pack of 52 cards. Find the probability distribution of the number of spades. Hence find the mean of the distribution. *(AI 2015)*

73. Two numbers are selected at random (without replacement) from positive integers 2, 3, 4, 5, 6 and 7. Let X denote the larger of the two numbers obtained. Find the mean and variance of the probability distribution of X. *(Foreign 2015)*

74. From a lot of 15 bulbs which include 5 defectives, a sample of 2 bulbs is drawn at random (without replacement). Find the probability distribution of the number of defective bulbs. *(Delhi 2015C)*

75. Three cards are drawn at random (without replacement) from a well shuffled pack of 52 playing cards. Find the probability distribution of number of red cards. Hence find the mean of the distribution. *(Foreign 2014)*

76. A class has 15 students whose ages are 14, 17, 15, 14, 21, 17, 19, 20, 16, 18, 20, 17, 16, 19 and 20 years. One student is selected in such a manner that each has the same chance of being chosen and the age X of the selected student is recorded. What is the probability distribution of the random variable X? Find the mean of X. *(AI 2014C)*

77. Out of a group of 30 honest people, 20 always speak the truth. Two persons are selected at random from the group. Find the probability distribution of the number of selected persons who speak the truth. Also find the mean of the distribution. What values are described in this question? *(Delhi 2013C)*

78. In a group of 30 scientists working on an experiment, 20 never commit error in their work and are reporting results elaborately. Two scientists are selected at random from the group. Find the probability distribution of the number of selected scientists who never commit error in work and reporting. Also find the mean of the distribution. What values are described in this question? *(AI 2013C)*

79. Two cards are drawn simultaneously (without replacement) from a well-shuffled pack of 52 cards. Find the mean and variance of the number of red cards. *(AI 2012)*

80. An urn contains 4 white and 6 red balls. Four balls are drawn at random (without replacement) from the urn. Find the probability distribution of the number of white balls. *(Delhi 2012C)*

81. Two cards are drawn successively with replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of kings and hence find the mean of the distribution. *(AI 2012C)*

82. A random variable X has the following probability distribution:

X	0	1	2	3	4	5	6	7
$P(X)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Determine:

(i) K (ii) $P(X < 3)$

(iii) $P(X > 6)$ (iv) $P(0 < X < 3)$

(AI 2011)

83. Two cards are drawn successively with replacement from a pack of 52 cards. Find the probability distribution of the number of aces drawn. *(Delhi 2011C)*

LA 2 (6 marks)

84. An urn contains 3 white and 6 red balls. Four balls are drawn one by one with replacement from the urn. Find the probability distribution of the number of red balls drawn. Also find mean and variance of the distribution. *(Delhi 2016)*

85. Two numbers are selected at random (without replacement) from the first six positive integers. Let X denote the larger of the two numbers obtained. Find the probability distribution of the random variable X, and hence find the mean of the distribution. *(AI 2014)*

86. In a game, a man wins rupees five for a six and loses rupee one for any other number, when a fair die is thrown. The man decided to throw a die thrice but to quit as and when he gets a six. Find the expected value of the amount he wins/loses. *(AI 2014C)*

www.exambits.in

Detailed Solutions

1. Required probability $= \frac{26}{52}\times\frac{26}{51}+\frac{26}{52}\times\frac{26}{51}$

$= 2\times\frac{26}{52}\times\frac{26}{51}=\frac{26}{51}$

2. Required probability $= \frac{{}^4C_1\times{}^3C_1\times{}^2C_1}{{}^9C_3}$

$= \frac{4\times3\times2}{\frac{9\times7\times8}{3\times2}}=\frac{2}{7}$

3. **(d) :** Here, $A = \{4, 5, 6\}$, $B = \{1, 2, 3, 4\}$

$A \cap B = \{4\}$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{3}{6}+\frac{4}{6}-\frac{1}{6}=1$

4. We have, $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cup B) = 0.6$

So, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= 0.4 + 0.3 - 0.6 = 0.1$

Now, $P(B' \cap A) = P(A - B) = P(A) - P(A \cap B)$

$= 0.4 - 0.1 = 0.3$

5. Total number of students = 8

The number of ways to select 4 students out of 8 students $= {}^8C_4 = \frac{8!}{4!\,4!}=70$

The number of ways to select 2 boys and 2 girls

$= {}^3C_2\times{}^5C_2=\frac{3!}{2!1!}\times\frac{5!}{2!3!}=3\times10=30$

$\therefore$ Required probability $=\frac{30}{70}=\frac{3}{7}$.

6. Probability of choosing bag $A = P(A)= \frac{2}{6}=\frac{1}{3}$

Probability of choosing bag $B = P(B) = \frac{4}{6}=\frac{2}{3}$

Let E_1 and E_2 be the events of drawing a red and a black ball from bag A and B respectively.

$\therefore$ $P(E_1) = \frac{6\times4}{{}^{10}C_2}$ and $P(E_2) = \frac{7\times3}{{}^{10}C_2}$

$\therefore$ Required probability
$= P(A) \times P(E_1) + P(B) \times P(E_2)$

$= \frac{1}{3}\times\frac{6\times4}{{}^{10}C_2}+\frac{2}{3}\times\frac{7\times3}{{}^{10}C_2}=\frac{8}{45}+\frac{14}{45}=\frac{22}{45}$

7. **(c) :** Let A be the event that the card is a spade and B be the event that the picked card is a queen.

We have a total of 13 spades and 4 queen cards.

Also only one queen is from spade.

$\therefore$ $P(A|B) = \frac{P(A\cap B)}{P(B)}=\frac{\frac{1}{52}}{\frac{4}{52}}=\frac{1}{4}$

8. The sample space, S is given by

$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Let A be the event that number on the drawn card is odd, and B be the event that number on the drawn card is greater than 5.

$\therefore$ $A = \{1, 3, 5, 7, 9, 11\}$

$B = \{6, 7, 8, 9, 10, 11, 12\}$

and, $A \cap B = \{7, 9, 11\}$

Now, $P(A)=\frac{n(A)}{n(S)}=\frac{6}{12}$, $P(B)=\frac{n(B)}{n(S)}=\frac{7}{12}$

$P(A\cap B)=\frac{n(A\cap B)}{n(S)}=\frac{3}{12}$

Now, $P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=\frac{3/12}{7/12}=\frac{3}{7}$

Hence, required probability is $\frac{3}{7}$.

9. E : 'a total of 8' and F : 'red die resulted in a number less than 4'

i.e., $E = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$

and $F = \{(x, y) : x \in \{1, 2, 3, 4, 5, 6\}, y \in \{1, 2, 3\}\}$

i.e., $F = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3), (6, 1), (6, 2), (6, 3)\}$

Hence, $E \cap F = \{(5, 3), (6, 2)\}$, $P(E) = 5/36$, $P(F) = 18/36$, $P(E \cap F) = 2/36$

$\therefore$ Required probability $= P(E/F)$

$= \frac{P(E\cap F)}{P(F)}=\frac{2/36}{18/36}=\frac{2}{18}=\frac{1}{9}$.

10. Let G_i $(i = 1, 2)$ and B_i $(i = 1, 2)$ denote the ith child is a girl or a boy respectively.

Then sample space is,

$S = \{G_1G_2, G_1B_2, B_1G_2, B_1B_2\}$

Let A be the event that both children are girls, B be the event that the youngest child is a girl and C be

www.exambits.in

the event that at least one of the children is a girl.
Then $A = \{G_1G_2\}$, $B = \{G_1G_2, B_1G_2\}$
and $C = \{B_1G_2, G_1G_2, G_1B_2\}$
$\Rightarrow A \cap B = \{G_1G_2\}$ and $A \cap C = \{G_1G_2\}$

(i) Required probability = $P(A/B)$

$$= \frac{P(A \cap B)}{P(B)} = \frac{1/4}{2/4} = \frac{1}{2}$$

(ii) Required probability = $P(A/C)$

$$= \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}$$

11. Let $B_i (i = 1, 2)$ and $G_i (i = 1, 2)$ denote the ith child is a boy or a girl respectively.
Then sample space is,

$S = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$

Let A be the event that both are boys, B be the event that one of them is a boy and C be the event that the older child is a boy.
$A = \{B_1B_2\}$, $B = \{G_1B_2, B_1G_2, B_1B_2\}$
$C = \{B_1B_2, B_1G_2\} \Rightarrow A \cap B = \{B_1B_2\}$ and $A \cap C = \{B_1B_2\}$

(i) Required probability = $P(A/B)$

$$= \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/4} = \frac{1}{3}$$

(ii) Required probability = $P(A/C)$

$$= \frac{P(A \cap C)}{P(C)} = \frac{1/4}{2/4} = \frac{1}{2}$$

12. The sample space S of the given random experiment is
$S = \{(H, H), (H, T), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$
Let A be the event that the die shows a number greater than 4 and B be the event that there is at least one tail.

$\therefore$ $A = \{(T, 5), (T, 6)\}$
and $B = \{(T,1),(T,2),(T,3),(T,4),(T,5),(T,6),(H,T)\}$
$A \cap B = \{(T, 5), (T, 6)\}$

$\therefore$ $P(B) = P\{(T, 1)\} + P\{(T, 2)\} + P\{(T, 3)\}$
$+ P\{(T, 4)\} + P\{(T, 5)\} + P\{(T, 6)\} + P\{(H, T)\}$

$$= \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{4} = \frac{3}{4}$$

$$P(A \cap B) = P\{(T,5)\} + P\{(T,6)\} = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$\therefore$ Required probability $= P(A/B) = \dfrac{P(A \cap B)}{P(B)}$

$$= \frac{1/6}{3/4} = \frac{2}{9}$$

13. Let A be the event of drawing a red ball in first draw and B be the event of drawing a red ball in second draw.

$$\therefore \quad P(A) = \frac{{}^3C_1}{{}^{10}C_1} = \frac{3}{10}$$

Now, $P(B/A)$ = Probability of drawing a red ball in the second draw, when a red ball already has been drawn in the first draw $= \dfrac{{}^2C_1}{{}^9C_1} = \dfrac{2}{9}$

$\therefore$ The required probability = $P(A \cap B)$

$$= P(A) \cdot P(B/A) = \frac{3}{10} \times \frac{2}{9} = \frac{1}{15}$$

14. **(c) :** Given, A and B are independent events.

Also, $P(A) = \dfrac{1}{3}$ and $P(B) = \dfrac{1}{4}$

Now, $P(B'|A) = \dfrac{P(B' \cap A)}{P(A)}$

$$= \frac{P(B')P(A)}{P(A)} \qquad [\because A, B \text{ are independent events}]$$

$$= P(B') = 1 - P(B) = 1 - \frac{1}{4} = \frac{3}{4}$$

15. Let A, B, C be respectively the events of solving problem by three students and $P(A)$, $P(B)$, $P(C)$ be their probability of solving the problem respectively.

$$\therefore \quad P(A) = \frac{1}{3}, P(B) = \frac{1}{4} \text{ and } P(C) = \frac{1}{6}$$

Required probability = $1 - P(\overline{A} \cap \overline{B} \cap \overline{C})$
$= 1 - P(\overline{A})\, P(\overline{B})\, P(\overline{C})$
($\because$ A, B, C are independent $\therefore$ $\overline{A}$, $\overline{B}$, $\overline{C}$ are also independent)
$= 1 - [1 - P(A)]\,[1 - P(B)]\,[1 - P(C)]$

$$= 1 - \frac{2}{3} \times \frac{3}{4} \times \frac{5}{6} = 1 - \frac{5}{12} = \frac{7}{12}$$

16. Let G be the event of a green signal.
Required probability = $P(GGG') + P(G'GG)$

$$= \left(\frac{3}{10}\right)^2 \frac{7}{10} + \frac{7}{10} \cdot \left(\frac{3}{10}\right)^2 = \frac{9}{100} \cdot \frac{7}{10} + \frac{7}{10} \cdot \frac{9}{100}$$

$$= \frac{63}{1000} + \frac{63}{1000} = \frac{126}{1000}$$

17. Given, A and B are independent events. So, A' and B' are also independent events.
Now, $P(A' \cap B') = P(A') \times P(B')$
$= [1 - P(A)][1 - P(B)] = [1 - 0.3][1 - 0.6]$
[Given, $P(A) = 0.3$ and $P(B) = 0.6$]
$= 0.7 \times 0.4 = 0.28$

www.exambres.in

18. We have, $S = \{1, 2, 3, 4, 5, 6\}$ and A be the event that number is even $= \{2, 4, 6\}$

$$\Rightarrow \quad P(A) = \frac{3}{6} = \frac{1}{2}$$

B be the event that number is red $= \{1, 2, 3\}$

$$\Rightarrow \quad P(B) = \frac{3}{6} = \frac{1}{2}$$

and $A \cap B = \{2\}$

$$\Rightarrow \quad P(A \cap B) = \frac{1}{6} \qquad ...(i)$$

$$\text{Also, } P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \qquad ...(ii)$$

From (i) and (ii),

$P(A) \cdot P(B) \neq P(A \cap B)$

So, A and B are not independent.

19. Since, E and F are independent events.

$\therefore \quad P(E \cap F) = P(E)\, P(F) \qquad ...(i)$

Now, $P(E' \cap F') = 1 - P(E \cup F)$

$[\because P(E' \cap F') = P((E \cup F)')]$

$= 1 - [P(E) + P(F) - P(E \cap F)]$

$= 1 - P(E) - P(F) + P(E)\, P(F)$ [Using (i)]

$= (1 - P(E))\,(1 - P(F)) = P(E')\, P(F')$

Hence, E' and F' are independent events.

20. *Refer to answer 18.*

21. Total outcomes = 36

Favourable outcomes for A to win

$= \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$

$\therefore$ Probability of A to win, $P(A) = \frac{6}{36} = \frac{1}{6}$

Probability of A to lose, $P(\bar{A}) = 1 - \frac{1}{6} = \frac{5}{6}$

Favourable outcomes for B to win

$= \{(4, 6), (6, 4), (5, 5)\}$

$\therefore$ Probability of B to win, $P(B) = \frac{3}{36} = \frac{1}{12}$

Probability of B to lose, $P(\bar{B}) = 1 - \frac{1}{12} = \frac{11}{12}$

$\therefore$ Required probability

$$= P(\bar{A})P(B) + P(\bar{A})P(\bar{B})P(\bar{A})P(B) + P(\bar{A})P(\bar{B})P(\bar{A})P(\bar{B})P(\bar{A})P(B) + ...$$

$$= \frac{5}{6} \times \frac{1}{12} + \frac{5}{6} \times \frac{11}{12} \times \frac{5}{6} \times \frac{1}{12} + \frac{5}{6} \times \frac{11}{12} \times \frac{5}{6} \times \frac{11}{12} \times \frac{5}{6} \times \frac{1}{12} + ...$$

$$= \frac{5/72}{1 - \frac{5}{6} \times \frac{11}{12}} = \frac{5}{17}$$

22. Let X and Y denote the respective events of solving the given specific problem by A and B,

then $P(X) = \frac{1}{2}$ and $P(Y) = \frac{1}{3}$

(i) P (problem is solved)

$$= P(X \cup Y) = 1 - P(\bar{X})P(\bar{Y})$$

$$= 1 - \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 1 - \frac{1}{2} \times \frac{2}{3} = \frac{2}{3}.$$

(ii) P (Exactly one of A and B solves the problem)

$$P(X) \cdot P(\bar{Y}) + P(\bar{X}) \cdot P(Y)$$

$$= \frac{1}{2}\left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{2}\right) \cdot \frac{1}{3} = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3}$$

$$= \frac{1}{2}\left(\frac{2}{3} + \frac{1}{3}\right) = \frac{1}{2}$$

23. Let E be the event that P speaks truth and F be the event that Q speaks truth. Then, E and F are independent events such that $P(E) = \frac{70}{100} = \frac{7}{10}$ and $P(F) = \frac{80}{100} = \frac{4}{5}$

P and Q will agree to each other in stating the same fact in the following mutually exclusive ways:

(I) P speaks truth and Q speaks truth *i.e.* $E \cap F$

(II) P tells a lie and Q tells a lie *i.e.* $\bar{E} \cap \bar{F}$.

$\therefore$ $P(P$ and Q agree to each other)

$$= P(E)\, P(F) + P(\bar{E})\, P(\bar{F}) = \frac{7}{10} \times \frac{4}{5} + \left(1 - \frac{7}{10}\right)\left(1 - \frac{4}{5}\right)$$

$$= \frac{28}{50} + \frac{3}{50} = \frac{31}{50} = \frac{62}{100}.$$

Hence, in 62% of the cases P and Q are likely to agree in stating the same fact.

P and Q agree means either both P and Q are speaking truth or both are telling lie.

24. Probability of A speaking the truth is,

$$P(A) = \frac{75}{100} = \frac{3}{4} \Rightarrow P(\bar{A}) = 1 - \frac{3}{4} = \frac{1}{4}$$

Probability of B speaking the truth is,

$$P(B) = \frac{90}{100} = \frac{9}{10} \Rightarrow P(\bar{B}) = 1 - \frac{9}{10} = \frac{1}{10}$$

Now A and B will contradict each other in the following mutually exclusive cases:

(i) A speaks the truth and B does not.

(ii) B speaks the truth and A does not.

www.examples.in

$\therefore$ Probability that A and B will contradict each other $= P(A)P(\bar{B}) + P(B)P(\bar{A})$

$$= \frac{3}{4} \times \frac{1}{10} + \frac{9}{10} \times \frac{1}{4} = \frac{12}{40} = \frac{3}{10} = \frac{30}{100}$$

i.e., they will contradict each other in 30% of the cases. We think that statement of B may be false.

25. *Refer to answer 23.*

26. Let E be the event that A is coming in time; $P(E) = \frac{3}{7}$

and F be the event that B is coming in time, $P(F) = \frac{5}{7}$

Also E and F are given to be independent events.

$\therefore$ Probability of only one of them coming to the school in time $= P(E) \cdot P(\bar{F}) + P(\bar{E}) \cdot P(F)$

$$= \frac{3}{7} \cdot \left(1 - \frac{5}{7}\right) + \left(1 - \frac{3}{7}\right) \cdot \frac{5}{7}$$

$$= \frac{3}{7} \cdot \frac{2}{7} + \frac{4}{7} \cdot \frac{5}{7} = \frac{26}{49}$$

One advantage of coming to school in time is vital to the education process and encourages for a good pattern of work.

27. It is given that A and B are independent events and $P(\bar{A} \cap B) = \frac{2}{15}$

$$\Rightarrow \quad P(\bar{A})P(B) = \frac{2}{15} \qquad \text{...(i)}$$

Also, $P(A \cap \bar{B}) = \frac{1}{6} \Rightarrow P(A)P(\bar{B}) = \frac{1}{6}$ ...(ii)

Let $p = P(A) \Rightarrow P(\bar{A}) = 1 - P(A) = 1 - p$

and $q = P(B) \Rightarrow P(\bar{B}) = 1 - P(B) = 1 - q$

Now, from (i) and (ii), we get

$$(1 - p)q = \frac{2}{15} \qquad \text{...(iii)}$$

and $p(1 - q) = \frac{1}{6}$ ...(iv)

Subtracting (iii) from (iv), we get

$$p - q = \frac{1}{6} - \frac{2}{15} = \frac{1}{30} \Rightarrow p = q + \frac{1}{30}$$

Putting this value of p in (iii), we get

$$\left(1 - q - \frac{1}{30}\right)q = \frac{2}{15} \Rightarrow \frac{29}{30}q - q^2 = \frac{2}{15}$$

$$\Rightarrow \quad 30q^2 - 29q + 4 = 0$$

$$\Rightarrow \quad 30q^2 - 24q - 5q + 4 = 0$$

$$\Rightarrow \quad 6q(5q - 4) - 1(5q - 4) = 0$$

$$\Rightarrow \quad (5q - 4)(6q - 1) = 0$$

$$\Rightarrow \quad q = \frac{4}{5} \text{ or } \frac{1}{6}$$

For $q = \frac{4}{5}$, using (iv), we have

$$p\left(1 - \frac{4}{5}\right) = \frac{1}{6} \Rightarrow p\left(\frac{1}{5}\right) = \frac{1}{6} \Rightarrow p = \frac{5}{6}$$

For $q = \frac{1}{6}$, using (iv), we have

$$p\left(1 - \frac{1}{6}\right) = \frac{1}{6} \Rightarrow p\left(\frac{5}{6}\right) = \frac{1}{6} \Rightarrow p = \frac{1}{5}$$

$$\therefore \quad P(A) = \frac{5}{6}, P(B) = \frac{4}{5} \text{ or } P(A) = \frac{1}{5}, P(B) = \frac{1}{6}$$

28. Probability of getting a six by the captains of both the teams A and B is

$$P(A) = \frac{1}{6} = P(B)$$

$$\therefore \quad P(\bar{A}) = P(\bar{B}) = 1 - \frac{1}{6} = \frac{5}{6}$$

Since A starts the game, he can throw a six in the following mutually exclusive ways :

$(A), (\bar{A}\bar{B}A), (\bar{A}\bar{B}\bar{A}\bar{B}A), \ldots$

Probability that A wins

$$= P(A) + P(\bar{A}\bar{B}A) + P(\bar{A}\bar{B}\bar{A}\bar{B}A) + \ldots$$

$$= P(A) + P(\bar{A})P(\bar{B})P(A) + P(\bar{A})P(\bar{B})P(\bar{A})P(\bar{B})P(A) + \ldots$$

$$= \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \ldots$$

$$= \frac{1}{6} + \frac{1}{6} \cdot \left(\frac{5}{6}\right)^2 + \frac{1}{6} \cdot \left(\frac{5}{6}\right)^4 + \ldots$$

This is an infinite G.P., with $a = \frac{1}{6}$ and $r = \left(\frac{5}{6}\right)^2$

Hence the probability of the team A winning the match

$$= \frac{\frac{1}{6}}{1 - \left(\frac{5}{6}\right)^2} = \frac{6}{11}$$

Since the total probability is unity, the probability of team B winning the match $= 1 - \frac{6}{11} = \frac{5}{11}$. The decision of the referee was not fair as whosoever starts throwing the die gets an upper hand.

www.exambites.in

29. Let E_1 be the event that ball transferred from bag I to bag II is red and E_2 be the event that ball transferred from bag I to bag II is black. B be the event that ball drawn from bag II is black.

So, $P(E_1)=\frac{3}{8},\ P(E_2)=\frac{5}{8}$

$P(B\,|\,E_1)=\frac{3}{8},\ P(B\,|\,E_2)=\frac{4}{8}$

So, required probability = $P(E_2\,|\,B)$

$$=\frac{P(E_2)\times P(B\,|\,E_2)}{P(E_1)\times P(B\,|\,E_1)+P(E_2)\times P(B\,|\,E_2)}$$

$$=\frac{\frac{5}{8}\times\frac{4}{8}}{\frac{3}{8}\times\frac{3}{8}+\frac{5}{8}\times\frac{4}{8}}=\frac{20}{9+20}=\frac{20}{29}$$

30. Let E_1 be the event of choosing a biased coin and E_2 be the event of choosing an unbiased coin.

$\Rightarrow \quad P(E_1)=P(E_2)=\frac{1}{2}$

Given, probability of biased coin has the chance of showing heads is 60%

$\therefore$ Probability of biased coin has the chance of showing tail is 40%

Let A be the event of showing tails.

$\therefore \quad P(A\,|\,E_1)=\frac{40}{100}=\frac{2}{5}$

$P(A\,|\,E_2)=\frac{1}{2}$

Using Bayes' theorem, we get

$$P(E_2\,|\,A)=\frac{P(E_2)P(A\,|\,E_2)}{P(E_1)(A\,|\,E_1)+P(E_2)(A\,|\,E_2)}$$

$$=\frac{\frac{1}{2}\times\frac{1}{2}}{\frac{1}{2}\times\frac{2}{5}+\frac{1}{2}\times\frac{1}{2}}=\frac{\frac{1}{4}}{\frac{1}{5}+\frac{1}{4}}=\frac{\frac{1}{4}}{\frac{9}{20}}=\frac{5}{9}$$

31. Let E_1 be the event of getting ghee from shop X, E_2 be the event of getting ghee from shop Y and A be the event of getting ghee of type B.

$\therefore \quad P(E_1)=\frac{1}{2},\ P(E_2)=\frac{1}{2},\ P(A|E_1)=\frac{40}{70}=\frac{4}{7},$

$P(A|E_2)=\frac{60}{110}=\frac{6}{11}$

Using Bayes' Theorem, we have

$$P(E_2|A)=\frac{P(E_2)P(A\,|\,E_2)}{P(E_1)P(A\,|\,E_1)+P(E_2)P(A\,|\,E_2)}$$

$$\Rightarrow \quad P(E_2|A)=\frac{\frac{1}{2}\times\frac{6}{11}}{\frac{1}{2}\times\frac{4}{7}+\frac{1}{2}\times\frac{6}{11}}$$

$$=\frac{\frac{6}{11}}{\frac{4}{7}+\frac{6}{11}}=\frac{42}{44+42}=\frac{42}{86}=\frac{21}{43}$$

32. Let E_1 be the event that the outcome on the die is 1 or 2, E_2 be the event that the outcome on the die is 3, 4, 5, 6.

$\therefore \quad P(E_1)=\frac{2}{6}=\frac{1}{3}$ and $P(E_2)=\frac{4}{6}=\frac{2}{3}$

Let A be the event of getting exactly one tail.

Now, $P(A/E_1)$ be the probability of getting exactly one tail by tossing the coin three times if she gets 1 or 2 = $\frac{3}{8}$ and

$P(A/E_2)$ be the probability of getting exactly one tail in a single throw of coin if she gets 3, 4, 5, 6 = $\frac{1}{2}$

The probability that the girl threw 3, 4, 5, 6 with the die, if she obtained exactly one tail is given by $P(E_2/A)$.

$$\therefore\ P(E_2/A)=\frac{P(E_2)\cdot P(A/E_2)}{P(E_1)\cdot P(A/E_1)+P(E_2)\cdot P(A/E_2)}$$

$$=\frac{\frac{2}{3}\cdot\frac{1}{2}}{\frac{1}{3}\cdot\frac{3}{8}+\frac{2}{3}\cdot\frac{1}{2}}=\frac{8}{11}$$

33. Let E_1 be the event that '6' occurs, E_2 be the event that '6' does not occur and A be the event that the man reports that it is '6'.

$\therefore \quad P(E_1)=\frac{1}{6},\ P(E_2)=\frac{5}{6}$

Now, $P(A/E_1)$ be the probability that the man reports that there is '6' on the die and '6' actually occurs

= Probability that the man speaks the truth $=\frac{4}{5}$

And $P(A/E_2)$ be the probability that the man reports that there is '6' when actually '6' does not occurs

= Probability that man does not speaks the truth

$=1-\frac{4}{5}=\frac{1}{5}$

www.examplites.in

$\therefore$ Required probability $= P(E_1/A)$

$$= \frac{P(E_1)\cdot P(A/E_1)}{P(E_1)\cdot P(A/E_1)+P(E_2)\cdot P(A/E_2)}$$

$$= \frac{\frac{1}{6}\times\frac{4}{5}}{\frac{1}{6}\times\frac{4}{5}+\frac{5}{6}\times\frac{1}{5}} = \frac{4}{4+5} = \frac{4}{9}$$

Yes, we are agree that the value of truthfulness leads to more respect in the society.

34. Let E_1 is event of students which have 100% attendance, E_2 is event of students which are irregular and A is the event of students which have an A grade.

Then, $P(E_1) = 0.3$, $P(E_2) = 0.7$, $P(A/E_1) = 0.7$ and $P(A/E_2) = 0.1$

So, P(Probability that student has 100% attendance given that he has A grade)

$$= P(E_1/A) = \frac{P(E_1)\cdot P(A/E_1)}{P(E_1)\cdot P(A/E_1)+P(E_2)\cdot P(A/E_2)}$$

[Using Baye's theorem]

$$= \frac{0.3\times 0.7}{0.3\times 0.7+0.7\times 0.1}$$

$$= \frac{0.3\times 0.7}{0.7(0.3+0.1)} = \frac{0.3}{0.4} = \frac{3}{4} = 0.75$$

As per answer, the probability of regular students having grade A is more than 50%. So, the regularity is required. No, regularity is required everywhere as it maintains our respect in society.

35. Let I be the event that changes take place to improve profits.

Probability of selection of A, $P(A) = \frac{1}{7}$

Probability of selection of B, $P(B) = \frac{2}{7}$

Probability of selection of C, $P(C) = \frac{4}{7}$

Probability that A does not introduce changes, $P(\bar{I}/A) = 1-0.8 = 0.2$

Probability that B does not introduce changes, $P(\bar{I}/B) = 1-0.5 = 0.5$

Probability that C does not introduce changes, $P(\bar{I}/C) = 1-0.3 = 0.7$

So, required probability $= P(C/\bar{I})$

$$= \frac{P(C)P(\bar{I}/C)}{P(A)P(\bar{I}/A)+P(B)P(\bar{I}/B)+P(C)P(\bar{I}/C)}$$

$$= \frac{\frac{4}{7}\times 0.7}{\frac{1}{7}\times 0.2+\frac{2}{7}\times 0.5+\frac{4}{7}\times 0.7} = 0.7$$

36. Consider the following events.

E : Two balls drawn are white

A : There are 2 white balls in the bag

B : There are 3 white balls in the bag

C : There are 4 white balls in the bag

$$P(A) = P(B) = P(C) = \frac{1}{3}$$

$$P(E/A) = \frac{{}^2C_2}{{}^4C_2} = \frac{1}{6},\ P(E/B) = \frac{{}^3C_2}{{}^4C_2} = \frac{3}{6} = \frac{1}{2}$$

$$P(E/C) = \frac{{}^4C_2}{{}^4C_2} = 1$$

$$\therefore\quad P(C/E) = \frac{P(C)\cdot P(E/C)}{P(A)\cdot P(E/A)+P(B)\cdot P(E/B)+P(C)\cdot P(E/C)}$$

$$= \frac{\frac{1}{3}\times 1}{\frac{1}{3}\times\frac{1}{6}+\frac{1}{3}\times\frac{1}{2}+\frac{1}{3}\times 1} = \frac{3}{5}$$

37. Let A be the event that the bulb is defective.

$$\therefore\quad P(E_1) = \frac{50}{100},\ P(E_2) = \frac{25}{100},\ P(E_3) = \frac{25}{100}$$

$$P(A/E_1) = \frac{4}{100},\ P(A/E_2) = \frac{4}{100},\ P(A/E_3) = \frac{5}{100}$$

$\therefore$ Required probability,

$$P(A) = P(E_1)P(A/E_1)+P(E_2)P(A/E_2)+P(E_3)P(A/E_3)$$

$$= \frac{50}{100}\times\frac{4}{100}+\frac{25}{100}\times\frac{4}{100}+\frac{25}{100}\times\frac{5}{100}$$

$$= \frac{200+100+125}{10000} = \frac{425}{10000} = \frac{17}{400}$$

38. Let E_1, E_2, E_3 and E be the events defined as follows:

E_1 : The item is manufactured by operator A

E_2 : The item is manufactured by operator B

E_3 : The item is manufactured by operator C

E : The item is defective.

$$\therefore\quad P(E_1) = \frac{50}{100} = \frac{5}{10},\ P(E_2) = \frac{30}{100} = \frac{3}{10},$$

$$P(E_3) = \frac{20}{100} = \frac{2}{10}$$

$$P(E/E_1) = \frac{1}{100}; P(E/E_2) = \frac{5}{100}; P(E/E_3) = \frac{7}{100}$$

www.exambites.in

Now, we have, to find $P(E_1/E)$ (*i.e.*, item is defective and it is produced by operator A)

$$\therefore P(E_1/E)=\frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1)+P(E_2)P(E/E_2)+P(E_3)P(E/E_3)}$$

$$=\frac{\frac{5}{10}\times\frac{1}{100}}{\frac{5}{10}\times\frac{1}{100}+\frac{3}{10}\times\frac{5}{100}+\frac{2}{10}\times\frac{7}{100}}=\frac{5}{5+15+14}$$

$$=\frac{5}{34}$$

39. Total number of persons insured
$= 3000 + 6000 + 9000 = 18000$

Let E_1, E_2 and E_3 be the event that the person is a cyclist, a scooter driver and a car driver respectively.

$$\therefore\ P(E_1)=\frac{3000}{18000}=\frac{1}{6},\quad P(E_2)=\frac{6000}{18000}=\frac{1}{3}$$

and $P(E_3)=\frac{9000}{18000}=\frac{1}{2}$

Let E be the event that insured person meets with an accident.

$\therefore$ $P(E/E_1) = 0.3$, $P(E/E_2) = 0.05$, $P(E/E_3) = 0.02$

By Bayes' theorem,

$\therefore$ Required probability

$$=\frac{P(E/E_1)\cdot P(E_1)}{P(E/E_1)\cdot P(E_1)+P(E/E_2)\cdot P(E_2)+P(E/E_3)P(E_3)}$$

$$=\frac{0.3\times\frac{1}{6}}{0.3\times\frac{1}{6}+0.05\times\frac{1}{3}+0.02\times\frac{1}{2}}$$

$$=\frac{\frac{0.3}{6}}{\frac{0.3+0.1+0.06}{6}}=\frac{0.3}{0.46}=\frac{15}{23}.$$

40. Let E_1, E_2, E_3 and C be the events as defined below :

E_1 : Two red balls are transferred from bag A to bag B.

E_2 : One red ball and one black ball is transferred from bag A to bag B.

E_3 : Two black balls are transferred from bag A to bag B.

C : Ball drawn from bag B is red.

So, $P(E_1)=\frac{{}^3C_2}{{}^8C_2}=\frac{3}{28}$, $P(E_2)=\frac{{}^3C_1\times{}^5C_1}{{}^8C_2}=\frac{15}{28}$

$P(E_3)=\frac{{}^5C_2}{{}^8C_2}=\frac{10}{28}$

Also, $P(C/E_1)=\frac{6}{10}$, $P(C/E_2)=\frac{5}{10}$, $P(C/E_3)=\frac{4}{10}$

$\therefore$ Required probability, $P(E_1/C)$

$$=\frac{P(E_1)P(C/E_1)}{P(E_1)P(C/E_1)+P(E_2)P(C/E_2)+P(E_3)P(C/E_3)}$$

$$=\frac{\frac{3}{28}\times\frac{6}{10}}{\frac{3}{28}\times\frac{6}{10}+\frac{15}{28}\times\frac{5}{10}+\frac{10}{28}\times\frac{4}{10}}=\frac{18}{18+75+40}=\frac{18}{133}$$

41. Let E_1, E_2, E_3 and E be the events as follows:

E_1 : The bolt is manufactured by the machine A

E_2 : The bolt is manufactured by the machine B

E_3 : The bolt is manufactured by the machine C

E : The bolt is defective.

$$\therefore\ P(E_1)=\frac{30}{100}=\frac{3}{10};P(E_2)=\frac{50}{100}=\frac{5}{10};$$

$$P(E_3)=\frac{20}{100}=\frac{2}{10}$$

$$P(E/E_1)=\frac{3}{100};P(E/E_2)=\frac{4}{100};\ P(E/E_3)=\frac{1}{100}$$

We find : $P(E_2/E)$ (*i.e.*, Bolt is defective and it is manufactured by machine B)

$$P(E_2/E)=\frac{P(E_2)\cdot P(E/E_2)}{\sum_{i=1}^{3}P(E_i)\cdot P(E/E_i)}$$

$$=\frac{\frac{5}{10}.\frac{4}{100}}{\frac{3}{10}.\frac{3}{100}+\frac{5}{10}.\frac{4}{100}+\frac{2}{10}.\frac{1}{100}}=\frac{20}{9+20+2}=\frac{20}{31}$$

$\therefore$ Required probability

= The probability that bolt is defective and not manufactured by machine B

$$=1-P(E_2/E)=1-\frac{20}{31}=\frac{11}{31}.$$

42. Let E_1 and E_2 denote the events of selection of first bag and second bag respectively. Let A be the event that 2 balls drawn are both red.

$$\therefore\ P(E_1)=\frac{1}{2}=P(E_2)$$

Now, $P(A/E_1)=\frac{{}^4C_2}{{}^8C_2}=\frac{4\cdot3}{8\cdot7}=\frac{3}{14}$

www.exambrain.in

$$P(A/E_2)=\frac{{}^2C_2}{{}^8C_2}=\frac{1\times 2}{8\cdot 7}=\frac{1}{28}$$

The required probability = $P(E_1/A)$

$$=\frac{P(E_1)\cdot P(A/E_1)}{P(E_1).P(A/E_1)+P(E_2).P(A/E_2)}$$

$$=\frac{\frac{1}{2}.\frac{3}{14}}{\frac{1}{2}.\frac{3}{14}+\frac{1}{2}.\frac{1}{28}}=\frac{3\times 2}{3\times 2+1}=\frac{6}{7}.$$

43. Let E_1, E_2 and A be the events defined as follows:

E_1 : The student knows the answer

E_2 : The student guesses the answer

A : The student answers correctly

We have, $P(E_1)=\frac{3}{5}, P(E_2)=\frac{2}{5}$

Also, $P(A/E_2)=\frac{1}{3}$ and $P(A/E_1)=1$

$\therefore$ Required probability

$$=P(E_1/A)=\frac{P(E_1)\cdot P(A/E_1)}{P(E_1)\cdot P(A/E_1)+P(E_2)\cdot P(A/E_2)}$$

$$=\frac{\frac{3}{5}\cdot 1}{\frac{3}{5}\cdot 1+\frac{2}{5}\cdot\frac{1}{3}}=\frac{3\times 3}{3\times 3+2}=\frac{9}{11}.$$

44. Let E_1, E_2, E_3, E_4 and A be the events defined as below :

E_1 : Missing card is a card of heart.

E_2 : Missing card is a card of spade.

E_3 : Missing card is a card of club.

E_4 : Missing card is a card of diamond.

A : Drawing three spade cards from the remaining cards.

Now, $P(E_1)=P(E_2)=P(E_3)=P(E_4)=\frac{13}{52}=\frac{1}{4}$

$$P(A/E_2)=\frac{{}^{12}C_3}{{}^{51}C_3}$$

$$P(A/E_1)=P(A/E_3)=P(A/E_4)=\frac{{}^{13}C_3}{{}^{51}C_3}$$

$\therefore$ Required probability = $P(E_2/A)$

$$=\frac{P(E_2)P(A/E_2)}{P(A/E_1)P(E_1)+P(A/E_2)P(E_2)+P(A/E_3)P(E_3)+P(A/E_4)P(E_4)}$$

$$=\frac{\frac{1}{4}\times\frac{{}^{12}C_3}{{}^{51}C_3}}{\frac{1}{4}\times\frac{{}^{13}C_3}{{}^{51}C_3}+\frac{1}{4}\times\frac{{}^{12}C_3}{{}^{51}C_3}+\frac{1}{4}\times\frac{{}^{13}C_3}{{}^{51}C_3}+\frac{1}{4}\times\frac{{}^{13}C_3}{{}^{51}C_3}}$$

$$=\frac{220}{286+220+286+286}=\frac{220}{1078}=\frac{10}{49}$$

45. Let A be the two-headed coin, B be the biased coin showing up heads 75% of the times and C be the biased coin showing up tails 40% (*i.e.*, showing up heads 60%) of the times.

Let E_1, E_2 and E_3 be the events of choosing coins of the type A, B and C respectively. Let S be the event of getting a head. Then

$$P(E_1)=\frac{1}{3}, P(E_2)=\frac{1}{3}, P(E_3)=\frac{1}{3}$$

$$P(S/E_1)=1, P(S/E_2)=\frac{75}{100}=\frac{3}{4},$$

$$P(S/E_3)=\frac{60}{100}=\frac{3}{5}$$

$\therefore$ Required probability

$$=P(E_1/S)=\frac{P(E_1)\cdot P(S/E_1)}{\sum_{i=1}^{3}P(E_i)\cdot P(S/E_i)}$$

$$=\frac{\frac{1}{3}\cdot 1}{\frac{1}{3}\cdot 1+\frac{1}{3}\cdot\frac{3}{4}+\frac{1}{3}\cdot\frac{3}{5}}=\frac{20}{20+15+12}=\frac{20}{47}$$

46. Let the events are defined as

E_1 : Person is a scooter driver

E_2 : Person is a car driver

E_3 : Person is a truck driver

A : Person meets with an accident.

Then, $P(E_1)=\frac{2000}{12000}=\frac{1}{6}, P(E_2)=\frac{4000}{12000}=\frac{2}{6},$

$$P(E_3)=\frac{6000}{12000}=\frac{3}{6}.$$

Also, $P(A/E_1)=0.01=\frac{1}{100}, P(A/E_2)=0.03=\frac{3}{100},$

$$P(A/E_3)=0.15=\frac{15}{100}.$$

$\therefore$ Required probability = 1 – P(the person who meets with accident is a truck driver)

i.e., Required probability = 1 – $P(E_3/A)$

www.examkites.in

$$=1-\frac{P(A/E_3)P(E_3)}{P(A/E_1)P(E_1)+P(A/E_2)P(E_2)+P(A/E_3)P(E_3)}$$

$$=1-\frac{\frac{15}{100}\times\frac{3}{6}}{\frac{1}{100}\times\frac{1}{6}+\frac{3}{100}\times\frac{2}{6}+\frac{15}{100}\times\frac{3}{6}}=1-\frac{45}{1+6+45}$$

$$=1-\frac{45}{52}=\frac{7}{52}.$$

47. *Refer to answer 33.*

48. *Refer to answer 36.*

49. Let A, E_1 and E_2 respectively be the events that a person has a heart attack, the selected person followed the course of yoga and meditation and the person adopted the drug prescription.

$$P(A)=\frac{40}{100}=0.40,\ P(E_1)=P(E_2)=\frac{1}{2}$$

$P(A/E_1) = 0.40 \times 0.70 = 0.28$,

$P(A/E_2) = 0.40 \times 0.75 = 0.30$

Probability that the patient suffering from heart attack followed the course of meditation and yoga is

$$P(E_1/A)=\frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)}$$

$$=\frac{\frac{1}{2}\times 0.28}{\frac{1}{2}\times 0.28+\frac{1}{2}\times 0.30}=\frac{0.14}{0.14+0.15}=\frac{14}{29}$$

Now, $P(E_2/A)=\frac{P(E_2)P(A/E_2)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)}$

$$=\frac{\frac{1}{2}\times 0.30}{\frac{1}{2}\times 0.28+\frac{1}{2}\times 0.30}=\frac{0.15}{0.14+0.15}=\frac{15}{29}$$

Since $P(E_1/A) < P(E_2/A)$. So, the course of yoga and meditation is more beneficial for a person having heart attack.

50. Let A, B, C and E are respectively the events that a person is smoker and non-vegetarian, smoker and vegetarian, non-smoker and vegetarian, and the selected person is suffering from the disease.

Here, $n(A) = 160$, $n(B) = 100$,

$n(C) = 400 - (160 + 100) = 140$.

Also, $P(A)=\frac{160}{400}, P(B)=\frac{100}{400}, P(C)=\frac{140}{400}$

and $P(E/A)=\frac{35}{100}, P(E/B)=\frac{20}{100}, P(E/C)=\frac{10}{100}$

$\therefore$ Required probability

$$=P(A/E)=\frac{P(A)\cdot P(E/A)}{P(A)\cdot P(E/A)+P(B)\cdot P(E/B)+P(C)\cdot P(E/C)}$$

$$=\frac{\frac{160}{400}\times\frac{35}{100}}{\frac{160}{400}\times\frac{35}{100}+\frac{100}{400}\times\frac{20}{100}+\frac{140}{400}\times\frac{10}{100}}$$

$$=\frac{5600}{5600+2000+1400}=\frac{5600}{9000}=\frac{28}{45}$$

The value reflected in this question is that being non-vegetarian and smokers has more chances of suffering from disease.

51. *Refer to answer 49.*

52. Let E_1, E_2 and S be the following events :

E_1 : The student resides in hostel

E_2 : The student is a day-scholar

S : The student attains A grade.

$$P(E_1)=\frac{60}{100};\ P(E_2)=\frac{40}{100}$$

$$P(S/E_1)=\frac{30}{100};\ P(S/E_2)=\frac{20}{100}$$

$\therefore$ Required probability

$$=P(E_1/S)=\frac{P(E_1)\cdot P(S/E_1)}{P(E_1)\cdot P(S/E_1)+P(E_2)\cdot P(S/E_2)}$$

$$=\frac{\frac{60}{100}\times\frac{30}{100}}{\frac{60}{100}\times\frac{30}{100}+\frac{40}{100}\times\frac{20}{100}}$$

$$=\frac{6\times 3}{6\times 3+4\times 2}=\frac{18}{18+8}=\frac{9}{13}$$

53. Let A_1, A_2 and A be the following events :

A_1 : The girl gets 5 or 6 and hence tosses a coin 3-times.

A_2 : The girl gets 1, 2, 3 or 4 and hence tosses a coin 2-times.

A : The girl gets exactly two heads.

Now, $P(A_1) = P(5 \text{ or } 6) = P(5) + P(6) = \frac{1}{6}+\frac{1}{6}=\frac{1}{3}$

www.exambites.in

$P(A_2) = P(1) + P(2) + P(3) + P(4)$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{2}{3}$$

$$P(A/A_1) = \frac{3}{8},\ P(A/A_2) = \frac{1}{4}$$

$\therefore$ Required probability

$$= P(A_2/A) = \frac{P(A_2)\cdot P(A/A_2)}{P(A_1)\cdot P(A/A_1) + P(A_2)\cdot P(A/A_2)}$$

$$= \frac{\frac{2}{3}\times\frac{1}{4}}{\frac{1}{3}\times\frac{3}{8} + \frac{2}{3}\times\frac{1}{4}} = \frac{\frac{1}{6}}{\frac{1}{8}+\frac{1}{6}} = \frac{\frac{1}{6}}{\frac{3+4}{24}} = \frac{4}{7}$$

54. *Refer to answer 40.*

55. *Refer to answer 53.*

56. Let E_1, E_2, E_3, E_4 be the events of the missing card to be of hearts, spade, club and diamond respectively and A be the event of drawing 2 heart cards from the remaining cards. Then

$$P(E_1) = P(E_2) = P(E_{3)} = P(E_4) = \frac{13}{52} = \frac{1}{4}$$

$$P(A/E_1) = \frac{{}^{12}C_2}{{}^{51}C_2},\ P(A/E_2) = \frac{{}^{13}C_2}{{}^{51}C_2}$$

Similarly, $P(A/E_3) = P(A/E_4) = \frac{{}^{13}C_2}{{}^{51}C_2}$

$\therefore$ Required probability

$$= P(E_1/A) = \frac{P(E_1)P(A/E_1)}{\sum_{i=1}^{4} P(E_i)P(A/E_i)}$$

$$= \frac{\frac{1}{4}\cdot\frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{1}{4}\cdot\frac{{}^{12}C_2}{{}^{51}C_2} + 3\cdot\frac{1}{4}\cdot\frac{{}^{13}C_2}{{}^{51}C_2}} = \frac{12\times11}{12\times11+3\times13\times12}$$

$$= \frac{11}{11+39} = \frac{11}{50}$$

57. *Refer to answer 46.*

$\therefore$ Required probability

$$= P(E_2/A) = \frac{P(E_2)\cdot P(A/E_2)}{\sum_{i=1}^{3} P(E_i)\cdot P(A/E_i)}$$

$$= \frac{\frac{2}{6}\times\frac{3}{100}}{\frac{1}{6}\times\frac{1}{100} + \frac{2}{6}\times\frac{3}{100} + \frac{3}{6}\times\frac{15}{100}}$$

$$= \frac{6}{1+6+45} = \frac{6}{52} = \frac{3}{26}.$$

58. Let E_1 be the event of choosing a male, E_2 be the event of choosing a female and A be the event that a person has grey hair.

Then, $P(E_1) = P(E_2) = \frac{1}{2}$

and $P(A/E_1) = \frac{5}{100}, P(A/E_2) = \frac{0.25}{100}$

$\therefore$ Required probability

$$= P(E_1/A) = \frac{P(E_1)\,P(A/E_1)}{P(E_1)\,P(A/E_1) + P(E_2)\,P(A/E_2)}$$

$$= \frac{\frac{1}{2}\times\frac{5}{100}}{\frac{1}{2}\times\frac{5}{100} + \frac{1}{2}\times\frac{0.25}{100}} = \frac{5}{5+0.25} = \frac{5}{5.25} = \frac{500}{525} = \frac{20}{21}$$

59. Let E_1 be the event that bag I is chosen and E_2 be the event that bag II is chosen and A be the event that the chosen ball is red.

$\therefore$ $P(E_1) = \frac{1}{2} = P(E_2)$

Also $P(A/E_1) = \frac{3}{7}; P(A/E_2) = \frac{5}{11}$

$\therefore$ Required probability

$$= P(E_2/A) = \frac{P(E_2)\cdot P(A/E_2)}{P(E_1)\cdot P(A/E_1) + P(E_2)\cdot P(A/E_2)}$$

$$= \frac{\frac{1}{2}\times\frac{5}{11}}{\frac{1}{2}\times\frac{3}{7} + \frac{1}{2}\times\frac{5}{11}} = \frac{\frac{5}{11}}{\frac{3}{7}+\frac{5}{11}}$$

$$= \frac{5\times7}{3\times11+5\times7} = \frac{35}{68}$$

60. *Refer to answer 33.*

61. Consider the following events :

E_1 : Box I is chosen

E_2 : Box II is chosen

E_3 : Box III is chosen

A : The coin drawn is of gold.

www.exambites.in

We have,

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}.$$

$$P(A/E_1) = \frac{2}{2} = 1, P(A/E_2) = 0, P(A/E_3) = \frac{1}{2}$$

Probability that the other coin in the box is of gold = Probability that gold coin is drawn from the box I.
$= P(E_1/A)$

$$= \frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + P(E_3)P(A/E_3)}$$

$$= \frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times 0 + \frac{1}{3} \times \frac{1}{2}} = \frac{2}{3}.$$

62. Let C_1, C_2 and C_3 be the events of selecting a two-tailed coin, a biased coin and an unbiased coin. Let T be the event that the coin appears with a tail. Now, $P(C_1) = P(C_2) = P(C_3) = \frac{1}{3}$

$$P(T/C_1) = \frac{2}{2} = 1, \; P(T/C_2) = 1 - \frac{60}{100} = 1 - \frac{3}{5} = \frac{2}{5}$$

$$P(T/C_3) = \frac{1}{2}$$

$\therefore$ Required probability

$$= P(C_1/T) = \frac{P(C_1) \cdot P(T/C_1)}{\sum_{i=1}^{3} P(C_i) \cdot P(T/C_i)}$$

$$= \frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{1}{2}} = \frac{1}{1 + \frac{2}{5} + \frac{1}{2}} = \frac{10}{19}$$

63. We have, $P(X = x) = \begin{cases} k, & \text{if } x = 0 \\ 2k, & \text{if } x = 1 \\ 3k, & \text{if } x = 2 \\ 0, & \text{otherwise} \end{cases}$

Since, $\Sigma P(x_i) = 1 \Rightarrow k + 2k + 3k = 1$

$$\Rightarrow 6k = 1 \Rightarrow k = \frac{1}{6}$$

64. Let X be the event that white balls are drawn. So, possible values of X are 0,1 and 2.
Here, total number of balls = 5 + 2 + 3 = 10

(i) $P(X = 0) = \frac{{}^8C_3}{{}^{10}C_3} = \frac{8!}{3! \times 5!} \times \frac{3! \times 7!}{10!} = \frac{7}{15}$

(ii) $P(X = 1) = \frac{{}^8C_2 \times {}^2C_1}{{}^{10}C_3} = \frac{7}{15}$

(iii) $P(X = 2) = \frac{{}^8C_1 \times {}^2C_2}{{}^{10}C_3} = \frac{1}{15}$

So, probability distribution table is

X	0	1	2
$P(X)$	$\frac{7}{15}$	$\frac{7}{15}$	$\frac{1}{15}$

$$\text{Mean} = \sum X P(X) = 0 \times \frac{7}{15} + 1 \times \frac{7}{15} + 2 \times \frac{1}{15}$$

$$= 0 + \frac{7}{15} + \frac{2}{15} = \frac{9}{15} = \frac{3}{5}$$

$$\text{Variance} = \sum X^2 P(X) - [E(X)]^2$$

$$= 0^2 \times \frac{7}{15} + 1^2 \times \frac{7}{15} + 2^2 \times \frac{1}{15} - \frac{9}{25}$$

$$= 0 + \frac{7}{15} + \frac{4}{15} - \frac{9}{25} = \frac{28}{75}$$

65. The probability distribution of x is

$X = x$	0	1	2	3
$P(X = x)$	0.1	k	$2k$	$3k$

(a) $\because \sum P(X) = 1$

$\Rightarrow$ $0.1 + k + 2k + 3k = 1$

$\Rightarrow$ $6k = 1 - 0.1 \Rightarrow 6k = 0.9$

$$\Rightarrow k = \frac{0.9}{6} = 0.15$$

(b) $P(x \le 2) = P(0) + P(1) + P(2)$
$= 0.1 + 0.15 + 0.3 = 0.55$

(c) Mean, $\bar{X} = \sum X \cdot P(X)$
$= 0.15 \times 1 + 2 \times 0.3 + 3 \times 0.45 = 2.1$

66. We have, number of rotten apples = 3 and number of good apples = 7
$\therefore$ Total number of apples = 10
Let X be number of rotten apples.
So, X can take values 0, 1, 2, 3
Let E be the event of getting a rotten apple.

$$P(E) = \frac{3}{10}, \; P(\bar{E}) = \frac{7}{10}$$

Now, $P(X = 0) = {}^3C_0 \cdot \frac{7}{10} \cdot \frac{7}{10} \cdot \frac{7}{10} = \frac{343}{1000}$

$$P(X = 1) = {}^3C_1 \cdot \frac{3}{10} \cdot \frac{7}{10} \cdot \frac{7}{10} = \frac{441}{1000}$$

www.exambites.in

$P(X=2) = {}^3C_2 \cdot \frac{3}{10}\cdot\frac{3}{10}\cdot\frac{7}{10} = \frac{189}{1000}$

$P(X=3) = {}^3C_3 \cdot \frac{3}{10}\cdot\frac{3}{10}\cdot\frac{3}{10} = \frac{27}{1000}$

So, probability distribution table is given by

X	0	1	2	3
$P(X)$	$\frac{343}{1000}$	$\frac{441}{1000}$	$\frac{189}{1000}$	$\frac{27}{1000}$

Now, mean $(\bar{X}) = \sum X\cdot P(X)$

$= 0\times\frac{343}{1000} + 1\times\frac{441}{1000} + 2\times\frac{189}{1000} + 3\times\frac{27}{1000}$

$= \frac{441}{1000} + \frac{378}{1000} + \frac{81}{1000} = \frac{900}{1000} = \frac{9}{10}$

67. Let X denote the random variable, which represents the larger of the two numbers drawn from first five positive integers.

$\therefore$ X can take values 2, 3, 4 or 5

Total number of ways = $5 \times 4 = 20$

$\therefore$ $P(X=2) = P(\{1, 2\}, \{2, 1\})$

$= \frac{2}{20} = \frac{1}{10}$

Similarly $P(X=3) = P\{(1, 3), (3, 1), (2, 3), (3, 2)\}$

$= \frac{4}{20} = \frac{2}{10}$

$P(X=4) = P\{(1, 4), (4, 1), (2, 4), (4, 2), (3, 4), (4, 3)\}$

$= \frac{6}{20} = \frac{3}{10}$

$P(X=5) = P\{(1, 5), (5, 1), (2, 5), (5, 2), (3, 5), (5, 3), (4, 5), (5, 4)\}$

$= \frac{8}{20} = \frac{4}{10}$

Hence, probability distribution is

X	$P(X)$	X^2	$X^2P(X)$
2	1/10	4	4/10
3	2/10	9	18/10
4	3/10	16	48/10
5	4/10	25	100/10

$\therefore$ $E(X) = \sum XP(X)$

$= \left(2\times\frac{1}{10}\right) + \left(3\times\frac{2}{10}\right) + \left(4\times\frac{3}{10}\right) + \left(5\times\frac{4}{10}\right)$

$= \frac{2}{10} + \frac{6}{10} + \frac{12}{10} + \frac{20}{10} = \frac{40}{10} = 4$

Variance $= \sum X^2P(X) - \left(\sum XP(X)\right)^2$

$= \left(\frac{4}{10} + \frac{18}{10} + \frac{48}{10} + \frac{100}{10}\right) - 4^2$

$= \frac{170}{10} - 16 = 1$

68. We have, $P(X=0) = P(X=1) = p$

Let $P(X=2) = P(X=3) = k$

Since, X is a random variable taking values 0, 1, 2, 3

$\therefore$ $P(X=0) + P(X=1) + P(X=2) + P(X=3) = 1$

$\Rightarrow p + p + k + k = 1 \Rightarrow 2p + 2k = 1 \Rightarrow p + k = \frac{1}{2}$ ...(i)

Now, $\Sigma p_i x_i^2 = 2\Sigma p_i x_i$

$\Rightarrow p(0) + p(1) + k(4) + k(9) = 2[p(0) + p(1) + k(2) + k(3)]$

$\Rightarrow p + 13k = 2p + 10k$

$\Rightarrow p - 3k = 0$ ...(ii)

Subtracting (ii) from (i), we get

$4k = \frac{1}{2} \Rightarrow k = \frac{1}{8}$

$\therefore$ From (i), we get $p = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$

69. Since, X denotes the sum of the numbers on the two drawn cards.

$\therefore$ X can take values 4, 6, 8, 10, 12.

$P(X=4) = \{1, 3\}$ or $\{3, 1\}$

$= \left(\frac{1}{4}\times\frac{1}{3}\right) + \left(\frac{1}{4}\times\frac{1}{3}\right) = \frac{2}{12} = \frac{1}{6}$

$P(X=6) = \{1, 5\}$ or $\{5, 1\}$ $= \frac{1}{4}\times\frac{1}{3} + \frac{1}{4}\times\frac{1}{3} = \frac{1}{6}$

$P(X=8) = \{3, 5\}$ or $\{5, 3\}$ or $\{1, 7\}$ or $\{7, 1\}$

$= \frac{1}{4}\times\frac{1}{3} + \frac{1}{4}\times\frac{1}{3} + \frac{1}{4}\times\frac{1}{3} + \frac{1}{4}\times\frac{1}{3} = \frac{4}{12} = \frac{1}{3}$

$P(X=10) = \{3, 7\}$ or $\{7, 3\}$

$= \frac{1}{4}\times\frac{1}{3} + \frac{1}{4}\times\frac{1}{3} = \frac{1}{6}$

$P(X=12) = \{5, 7\}$ or $\{7, 5\}$ $= \frac{1}{4}\times\frac{1}{3} + \frac{1}{4}\times\frac{1}{3} = \frac{1}{6}$

X	$P(X)$	$XP(X)$	X^2	$X^2P(X)$
4	1/6	4/6	16	16/6
6	1/6	1	36	6
8	1/3	8/3	64	64/3
10	1/6	10/6	100	100/6
12	1/6	2	144	144/6

www.exambites.in

$\therefore$ Mean $= E(X) = \sum XP(X)$

$= \frac{4}{6} + 1 + \frac{8}{3} + \frac{10}{6} + 2 = \frac{2}{3} + 1 + \frac{8}{3} + \frac{5}{3} + 2 = \frac{24}{3} = 8$

Variance $= E(X^2) - [E(X)]^2$

$= \sum X^2 P(X) - (8)^2$

$= \frac{16}{6} + 6 + \frac{64}{3} + \frac{100}{6} + \frac{144}{6} - (8)^2$

$= \frac{8}{3} + 6 + \frac{64}{3} + \frac{50}{3} + \frac{72}{3} - 64 = \frac{20}{3} = 6.67$

70. Let X be the amount he wins/loses.

Then, X can take values –3, 3, 4, 5.

$P(X = 5) = P$(Getting a number greater than 4 in the first throw) $= \frac{2}{6} = \frac{1}{3}$

$P(X = 4) = P$(Getting a number less than or equal to 4 in the first throw and a number greater than 4 in the second throw) $= \frac{4}{6} \times \frac{2}{6} = \frac{2}{9}$

$P(X = 3) = P$(Getting a number less than or equal to 4 in the first two throws and a number greater than 4 in the third throw) $= \frac{4}{6} \times \frac{4}{6} \times \frac{2}{6} = \frac{4}{27}$

$P(X = -3) = P$(Getting a number less than or equal to 4 in all three throws) $= \frac{4}{6} \times \frac{4}{6} \times \frac{4}{6} = \frac{8}{27}$

$\therefore$ The probability distribution is

X	5	4	3	–3
$P(X)$	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{4}{27}$	$\frac{8}{27}$

$\therefore E(X) = \sum XP(X) = 5\left(\frac{1}{3}\right) + 4\left(\frac{2}{9}\right) + 3\left(\frac{4}{27}\right) - 3\left(\frac{8}{27}\right)$

$= \frac{57}{27} = \frac{19}{9}$

Hence, expected value of the amount he wins/loses is $\frac{19}{9}$.

71. The probability distribution of X is

X	0	1	2	3	4
$P(X)$	0	k	$4k$	$2k$	k

The given distribution is a probability distribution.

$\therefore \quad \sum_{i=0}^{4} p_i = 1$

$\Rightarrow \quad 0 + k + 4k + 2k + k = 1 \Rightarrow 8k = 1$

$\Rightarrow \quad k = \frac{1}{8} = 0.125$

(i) P (getting admission in exactly one college) $= P(X = 1) = k = 0.125$

(ii) P (getting admission in atmost 2 colleges) $= P(X \le 2) = 0 + k + 4k = 5k = 0.625$

(iii) P (getting admission in atleast 2 colleges) $= P(X \ge 2) = 4k + 2k + k = 7k = 0.875$

72. Let X denote the number of spade cards in a sample of 3 cards drawn from a well-shuffled pack of 52 cards.

Since there are 13 spade cards in the pack, so in a sample of 3 cards drawn, either there is no spade card or one spade card or two spade cards or 3 spade cards. Thus $X = 0, 1, 2$ and 3.

Now, $P(X = 0)$ = Probability of getting no spade card

$= \frac{39}{52} \cdot \frac{39}{52} \cdot \frac{39}{52} = \frac{27}{64}$

$P(X = 1)$ = Probability of getting one spade card

$= \frac{13}{52} \cdot \frac{39}{52} \cdot \frac{39}{52} + \frac{39}{52} \cdot \frac{13}{52} \cdot \frac{39}{52} + \frac{39}{52} \cdot \frac{39}{52} \cdot \frac{13}{52} = \frac{27}{64}$

$P(X = 2)$ = Probability of getting 2 spade cards

$= \frac{13}{52} \cdot \frac{13}{52} \cdot \frac{39}{52} + \frac{13}{52} \cdot \frac{39}{52} \cdot \frac{13}{52} + \frac{39}{52} \cdot \frac{13}{52} \cdot \frac{13}{52} = \frac{9}{64}$

$P(X = 3)$ = Probability of getting 3 spade cards

$= \frac{13}{52} \cdot \frac{13}{52} \cdot \frac{13}{52} = \frac{1}{64}$

Hence, the probability distribution of X is

X	0	1	2	3
$P(X)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$

Now, mean of this distribution is given by

$\bar{X} = 0 \times \frac{27}{64} + 1 \times \frac{27}{64} + 2 \times \frac{9}{64} + 3 \times \frac{1}{64} = \frac{48}{64} = \frac{3}{4}$

73. Let X be the random variable.

$\therefore$ X can take values 3, 4, 5, 6 or 7.

Total number of ways $= {}^6C_2 = 15$

The probability distribution of a random variable X is given by

www.examprep.in

X	3	4	5	6	7
$P(X)$	1/15	2/15	3/15	4/15	5/15

$\therefore$ Mean $= \sum XP(X)$

$= 3\times\frac{1}{15}+4\times\frac{2}{15}+5\times\frac{3}{15}+6\times\frac{4}{15}+7\times\frac{5}{15}$

$= \frac{3}{15}+\frac{8}{15}+\frac{15}{15}+\frac{24}{15}+\frac{35}{15}=\frac{85}{15}=\frac{17}{3}$

and variance $= \sum X^2P(X)-\left(\sum XP(X)\right)^2$

$= \frac{9}{15}+\frac{32}{15}+\frac{75}{15}+\frac{144}{15}+\frac{245}{15}-\left(\frac{17}{3}\right)^2$

$= \frac{505}{15}-\frac{289}{9} = \frac{210}{135}=\frac{14}{9}$

74. Let X denote the number of defective bulbs in a sample of 2 bulbs which are to be drawn.
Here number of defective bulbs = 5
Number of non-defective bulbs = 15 – 5 = 10
$\therefore$ X can take values 0, 1, 2.
Now, $P(X = 0)$ = Probability of getting no defective bulb
= Probability of getting 2 non-defective bulbs.

$= \frac{{}^{10}C_2}{{}^{15}C_2}=\frac{10\times9}{15\times14}=\frac{3}{7}=\frac{9}{21}$

$P(X = 1)$ = Probability of getting 1 defective bulb

$= \frac{{}^{5}C_1\times{}^{10}C_1}{{}^{15}C_2}=\frac{5\times10\times2}{15\times14}=\frac{10}{21}$

$P(X = 2)$ = Probability of getting 2 defective bulbs

$= \frac{{}^{5}C_2}{{}^{15}C_2}=\frac{5\times4}{15\times14}=\frac{2}{21}$

Thus the probability distribution of X is given by

X	0	1	2
$P(X)$	$\frac{9}{21}$	$\frac{10}{21}$	$\frac{2}{21}$

75. Let X denote the number of red cards. So X can take values 0, 1, 2, 3.
Total number of cards = 52
Number of red cards = 26.

Now, $P(X = 0) = \frac{{}^{26}C_3}{{}^{52}C_3}=\frac{26\times25\times24}{52\times51\times50}=\frac{4}{34}$

$P(X = 1) = \frac{{}^{26}C_1\times{}^{26}C_2}{{}^{52}C_3}=\frac{26\times26\times25\times6}{2\times52\times51\times50}=\frac{13}{34}$

$P(X = 2) = \frac{{}^{26}C_2\times{}^{26}C_1}{{}^{52}C_3}=\frac{26\times25\times26\times6}{2\times52\times51\times50}=\frac{13}{34}$

$P(X = 3) = \frac{{}^{26}C_3}{{}^{52}C_3}=\frac{26\times25\times24}{52\times51\times50}=\frac{4}{34}$

$\therefore$ Probability distribution of X is given by

X	0	1	2	3
$P(X)$	$\frac{4}{34}$	$\frac{13}{34}$	$\frac{13}{34}$	$\frac{4}{34}$

Mean $= \sum XP(X)$

$= 0\left(\frac{4}{34}\right)+1\left(\frac{13}{34}\right)+2\left(\frac{13}{34}\right)+3\left(\frac{4}{34}\right)=\frac{3}{2}$

76. Here the ages of the given 15 students are 14, 17, 15, 14, 21, 17, 19, 20, 16, 18, 20, 17, 16, 19 and 20 years.
$\therefore$ The required probability distribution of X is given by

X	14	15	16	17	18	19	20	21
$P(X)$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{1}{15}$

$\therefore$ Mean, $\bar{X} = \sum XP(X)$

$$=14\times\frac{2}{15}+15\times\frac{1}{15}+16\times\frac{2}{15}+17\times\frac{3}{15}$$
$$+18\times\frac{1}{15}+19\times\frac{2}{15}+20\times\frac{3}{15}+21\times\frac{1}{15}$$
$$=\frac{1}{15}(28+15+32+51+18+38+60+21)=\frac{263}{15}$$

77. Total number of honest people = 30
The number of people who speak truth = 20
The number of people who do not speak truth
= 30 – 20 = 10
Number of selected persons = 2
Let X denote the number of people who speak truth.
$\therefore$ X can take values 0, 1, 2

Now, $P(X=0)=\frac{{}^{20}C_0\times{}^{10}C_2}{{}^{30}C_2}=\frac{10\times9}{30\times29}=\frac{9}{87}$

www.examfear.in

$$P(X=1)=\frac{{}^{20}C_1\times{}^{10}C_1}{{}^{30}C_2}=\frac{20\times10\times2}{30\times29}=\frac{40}{87}$$

$$P(X=2)=\frac{{}^{20}C_2\times{}^{10}C_0}{{}^{30}C_2}=\frac{20\times19}{30\times29}=\frac{38}{87}$$

Hence, the probability distribution of X is

X	0	1	2
$P(X)$	$\frac{9}{87}$	$\frac{40}{87}$	$\frac{38}{87}$

Mean of the distribution

$$\bar{X}=0\times\frac{9}{87}+1\times\frac{40}{87}+2\times\frac{38}{87}=\frac{40+76}{87}=\frac{116}{87}=\frac{4}{3}$$

Since, out of 30 honest people, 20 always speak truth. So, the value of truthfulness and morality is described here.

78. *Refer to answer 77.*

79. Let X denote the number of red cards. Clearly X can take values 0, 1 or 2.

Here we have 26 red cards in the pack.

$\therefore$ $P(X = 0)$ = probability of getting no red card

$$=\frac{{}^{26}C_2}{{}^{52}C_2}=\frac{26\times25}{52\times51}=\frac{25}{102}$$

$P(X = 1)$ = Probability of getting 1 red card and the other non-red card

$$=\frac{{}^{26}C_1\times{}^{26}C_1}{{}^{52}C_2}=\frac{26\times26\times2}{52\times51}=\frac{52}{102}$$

$P(X = 2)$ = Probability of getting two red cards

$$=\frac{{}^{26}C_2}{{}^{52}C_2}=\frac{26\times25}{52\times51}=\frac{25}{102}$$

Hence, the probability distribution of X is,

X	0	1	2
$P(X)$	$\frac{25}{102}$	$\frac{52}{102}$	$\frac{25}{102}$

$\therefore$ Mean, $\bar{X}=\sum XP(X)$

$$=0\times\frac{25}{102}+1\times\frac{52}{102}+2\times\frac{25}{102}=\frac{102}{102}=1$$

and variance $=\sum X^2P(X)-(\text{Mean})^2$

$$=0^2\times\frac{25}{102}+1^2\times\frac{52}{102}+2^2\times\frac{25}{102}-1^2$$

$$=\frac{152}{102}-1=\frac{50}{102}=\frac{25}{51}$$

80. Let X denote the number of white balls drawn from the urn. Then $X = 0, 1, 2, 3, 4$.

Now $P(X = 0)$ = Probability of getting no white ball

$$=\frac{{}^4C_0\cdot{}^6C_4}{{}^{10}C_4}=\frac{15}{210}$$

$P(X = 1)$ = Probability of getting 1 white ball

$$=\frac{{}^4C_1\cdot{}^6C_3}{{}^{10}C_4}=\frac{80}{210}$$

$P(X = 2)$ = Probability of getting 2 white balls

$$=\frac{{}^4C_2\cdot{}^6C_2}{{}^{10}C_4}=\frac{90}{210}$$

$P(X = 3)$ = Probability of getting 3 white balls

$$=\frac{{}^4C_3\cdot{}^6C_1}{{}^{10}C_4}=\frac{24}{210}$$

$P(X = 4)$ = Probability of getting 4 white balls

$$=\frac{{}^4C_4\cdot{}^6C_0}{{}^{10}C_4}=\frac{1}{210}$$

Thus the probability distribution of X is given by

X	0	1	2	3	4
$P(X)$	$\frac{15}{210}$	$\frac{80}{210}$	$\frac{90}{210}$	$\frac{24}{210}$	$\frac{1}{210}$

81. Let X denote the number of kings drawn. Then X can take the values 0, 1 or 2.

$P(X = 0)$ = Probability of not getting a king in two draws $=\frac{48}{52}\cdot\frac{48}{52}=\frac{144}{169}$

$P(X = 1)$ = Probability of getting one king in two draws $=\frac{4}{52}\cdot\frac{48}{52}+\frac{48}{52}\cdot\frac{4}{52}=\frac{12}{169}+\frac{12}{169}=\frac{24}{169}$

$P(X = 2)$ = Probability of getting two kings in two draws $=\frac{4}{52}\cdot\frac{4}{52}=\frac{1}{169}$

Thus the probability distribution of X is

X	0	1	2
$P(X)$	$\frac{144}{169}$	$\frac{24}{169}$	$\frac{1}{169}$

Now, Mean $=\sum XP(X)$

$$=0\times\frac{144}{169}+1\times\frac{24}{169}+2\times\frac{1}{169}=\frac{26}{169}=\frac{2}{13}$$

www.exambites.in

82. (i) Since $\Sigma P(X) = 1$

$\therefore \quad 0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$

$\Rightarrow \quad 10K^2 + 9K - 1 = 0$

$\Rightarrow \quad K = \dfrac{-9 \pm \sqrt{81+40}}{20} = \dfrac{-9 \pm 11}{20} = \dfrac{1}{10}, -1$

Since the probability of the event lies between 0 and 1, therefore, rejecting $K = -1$ Hence, $K = \dfrac{1}{10}$

(ii) $P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$

$= 0 + K + 2K = 3K = \dfrac{3}{10} \qquad \left(\because K = \dfrac{1}{10}\right)$

(iii) $P(X > 6) = P(7)$

$= 7K^2 + K = \dfrac{7}{100} + \dfrac{1}{10} = \dfrac{17}{100} \qquad \left(\because K = \dfrac{1}{10}\right)$

(iv) $P(0 < X < 3) = P(X = 1) + P(X = 2)$

$= K + 2K = 3K = \dfrac{3}{10} \qquad \left(\because K = \dfrac{1}{10}\right)$

83. *Refer to answer 81.*

84. Number of white balls = 3

Number of red balls = 6

Total number of balls = 9

Let X be the random variable denoting the number of red balls drawn.

$\therefore$ X can take values 0, 1, 2, 3 or 4

$P(X = 0)$ = Probability of getting no red ball in four draws

$= \dfrac{1}{3} \times \dfrac{1}{3} \times \dfrac{1}{3} \times \dfrac{1}{3} = \dfrac{1}{81}$

$P(X = 1)$ = Probability of getting one red ball in four draws

$= 4 \times \left[\dfrac{1}{3} \times \dfrac{1}{3} \times \dfrac{1}{3} \times \dfrac{2}{3}\right] = \dfrac{8}{81}$

$P(X = 2)$ = Probability of getting two red balls in four draws

$= 6 \times \left(\dfrac{1}{3} \times \dfrac{1}{3} \times \dfrac{2}{3} \times \dfrac{2}{3}\right) = \dfrac{24}{81}$

$P(X = 3)$ = Probability of getting three red balls in four draws

$= 4 \times \left(\dfrac{1}{3} \times \dfrac{2}{3} \times \dfrac{2}{3} \times \dfrac{2}{3}\right) = \dfrac{32}{81}$

$P(X = 4)$ = Probability of getting four red balls in four draws

$= \dfrac{2}{3} \times \dfrac{2}{3} \times \dfrac{2}{3} \times \dfrac{2}{3} = \dfrac{16}{81}$

The probability distribution of X is

X	0	1	2	3	4
$P(X)$	1/81	8/81	24/81	32/81	16/81

Mean $= \sum XP(X)$

$= 0 \times \dfrac{1}{81} + 1 \times \dfrac{8}{81} + 2 \times \dfrac{24}{81} + 3 \times \dfrac{32}{81} + 4 \times \dfrac{16}{81}$

$= \dfrac{216}{81} = \dfrac{8}{3}$

Variance $= \sum X^2 P(X) - (\text{Mean})^2$

$= 0 \times \dfrac{1}{81} + 1 \times \dfrac{8}{81} + 4 \times \dfrac{24}{81} + 9 \times \dfrac{32}{81} + 16 \times \dfrac{16}{81} - \left(\dfrac{8}{3}\right)^2$

$= \dfrac{648}{81} - \dfrac{64}{9} = \dfrac{8}{9}$

85. *Refer to answer 73.*

86. *Refer to answer 70.*

www.exambites.in

PRACTICE PAPER 1
(Solved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. If $A=\begin{bmatrix}5 & x\\ y & 0\end{bmatrix}$ and $A = A^T$, then

(a) $x = y$ (b) $x = -y$

(c) $x = 2y$ (d) $x = 3y$

2. If $y=\frac{\ln x}{x}$, then the value of $y''(e)$ is

(a) 1 (b) $-\frac{1}{e}$

(c) $-\frac{1}{e^2}$ (d) $-\frac{1}{e^3}$

3. The function $f(x) = x^2 - 2x$ is strictly decreasing in the interval

(a) $(-\infty, 1)$ (b) $[1, \infty)$

(c) R (d) None of these

4. Evaluate : $\int[\sec^2(7-4x)]dx$

(a) $-\frac{1}{4}\tan(7-4x)+C$

(b) $\frac{1}{4}\tan(7-4x)+C$

(c) $\frac{1}{4}\tan(7+4x)+C$

(d) $-\frac{1}{4}\tan(7x-4)+C$

5. $\int\frac{a}{(1+x^2)\tan^{-1}x}dx =$

(a) $a\log|\tan^{-1}x|+C$ (b) $\frac{a}{2}(\tan^{-1}x)^2+C$

(c) $a\log(1+x^2)+C$ (d) None of these

6. Integrating factor of $\frac{xdy}{dx}-y=x^4-3x$ is

(a) x (b) $\log x$

(c) $\frac{1}{x}$ (d) $-x$

7. If direction cosines of two lines are proportional to (2, 3, –6) and (3, –4, 5), then find the acute angle between them.

(a) $\sin^{-1}\left(\frac{18\sqrt{2}}{35}\right)$ (b) $\cos^{-1}\left(\frac{18\sqrt{2}}{35}\right)$

(c) $\sin^{-1}\left(\frac{9\sqrt{2}}{35}\right)$ (d) $\cos^{-1}\left(\frac{9\sqrt{2}}{35}\right)$

8. The equation of a line which passes through the point (1, 2, 3) and is parallel to the vector $3\hat{i}+2\hat{j}-2\hat{k}$, is

(a) $\vec{r}=(3\hat{i}+2\hat{j}-2\hat{k})+\lambda(\hat{i}+2\hat{j}+3\hat{k})$

(b) $\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(3\hat{i}+2\hat{j}-2\hat{k})$

(c) $\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(2\hat{i}-5\hat{k})$

(d) $\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(4\hat{i}+4\hat{j}+\hat{k})$

9. If $A(0, 4, 1)$, $B(2, 3, -1)$, $C(4, 5, 0)$, then find the angle between the lines AB and BC.

(a) 45° (b) 90° (c) 30° (d) 60°

10. The region represented by the inequalities $x \geq 6, y \geq 2, 2x + y \leq 10, x \geq 0, y \geq 0$ is

(a) unbounded (b) a polygon
(c) a triangle (d) None of these

(Q.11 - Q.15) Fill in the blanks.

11. If $\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = k \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, then k_____ .

12. If $y = 3x^2 + b$, then $\frac{dy}{dx}$ at $x = 2$ is_____ .

13. The order of the differential equation given by $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{\frac{1}{4}} + x^{\frac{1}{5}} = 0$ is_____ .

14. The principal value of $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$ is_____.

OR

The value of $\text{cosec}^{-1}(2/\sqrt{3})$ is _____.

15. If the points with position vectors $60\hat{i} + 3\hat{j}$, $40\hat{i} - 8\hat{j}$ and $a\hat{i} - 52\hat{j}$ are collinear, then the value of a = _____ .

OR

The magnitude of the vector $\vec{a} = 2\hat{i} - 6\hat{j} - 3\hat{k}$ is _____ .

(Q.16 – Q.20) Answer the following questions.

16. Find the principal value of $\tan^{-1} 1 + \cos^{-1}\left(\frac{-1}{2}\right) + \sin^{-1}\left(\frac{-1}{2}\right)$.

17. The sides of an equilateral triangle are increasing at the rate of 2 cm/s. Find the rate at which the area increases, when the side is $20\sqrt{3}$ cm.

OR

Show that tangents to the curve $y = x^3 + 3x$ at $x = -1$ and $x = 1$ are parallel.

18. Evaluate : $\int_{-\pi/2}^{\pi/2} \sin x \cos^2 x (\sin^2 x + \cos x)\, dx$

19. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors, such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$, then find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.

20. If a line makes an angle θ_1, θ_2, θ_3 with the axes respectively, then find the value of $\cos 2\theta_1 + \cos 2\theta_2 + \cos 2\theta_3$.

SECTION - B

21. Find the value of $\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right)$ (where $x > y > 0$).

OR

If $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$, then find the value of x.

22. Find cofactors of a_{21} and a_{31} of the matrix $A = [a_{ij}] = \begin{bmatrix} 1 & 3 & -2 \\ 4 & -5 & 6 \\ 3 & 5 & 2 \end{bmatrix}$.

23. Evaluate the definite integral :

$$\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x}\, dx$$

24. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$, then show that $\vec{a} = \vec{0}$ or $\vec{b} = \vec{c}$ or $\vec{a}$ is perpendicular to $\vec{b} - \vec{c}$.

25. Write the vector equation of the following line : $\frac{x-5}{3} = \frac{y+4}{7} = \frac{6-z}{2}$

26. The probability that a person will get an electric contract is $\frac{2}{5}$ and the probability that he will not get plumbing contract is $\frac{4}{7}$. If the probability of getting at least one contract is $\frac{2}{3}$, what is the probability that he will get both?

OR

A random variable X has the following distribution.

X	1	2	3	4	5	6	7	8
$P(X)$	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the event $E = \{X \text{ is prime number}\}$ and $F = \{X < 4\}$, find $P(E \cup F)$.

www.exambites.in

SECTION - C

27. Let $f: N \to R$ be a function defined as $f(x) = 4x^2 + 12x + 15$.
Show that $f: N \to S$ is invertible, where S is the range of f. Also, find inverse of f.

28. Determine the values of a, b and c for which the function

$$f(x) = \begin{cases} \dfrac{\sin(a+1)x + \sin x}{x}, & x < 0 \\ c, & x = 0 \\ \dfrac{\sqrt{x+bx^2} - \sqrt{x}}{b\sqrt{x^3}}, & x > 0 \end{cases}$$

may be continuous at $x = 0$.

OR

It is given that for the function $f(x) = x^3 + bx^2 + ax + 5$ on [1, 3], Rolle's theorem holds with $c = 2 + \frac{1}{\sqrt{3}}$. Find the values of a and b.

29. Find $\int \frac{x^4}{(x-1)(x^2+1)} dx$.

30. Solve the differential equation :

$$\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$$

31. Show that the lines $\frac{x}{1} = \frac{y-2}{2} = \frac{z+3}{3}$ and $\frac{x-2}{2} = \frac{y-6}{3} = \frac{z-3}{4}$ are coplanar. Also, find the equation of the plane containing these lines.

OR

Find the shortest distance between the lines whose vector equations are

$\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$, and

$\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}$.

32. A factory owner wants to purchase two types of machines A and B for his factory. The machine A requires an area of 1000 m^2 and 12 skilled men for running it and its daily output is 50 units, whereas the machine B requires area of 1200 m^2 and 8 skilled men, and its daily output is 40 units. If an area of 7600 m^2 and 72 skilled men be available to operate the machines, then how many machines of each type should be bought to maximize the daily output?

SECTION - D

33. If $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix}$, then find A^{-1}, using elementary row operations.

OR

Three shopkeepers A, B, C are using polythene, handmade bag and newspaper envelopes as carry bags. It is found that the shopkeepers A, B, C are using (20, 30, 40), (30, 40, 20), (40, 20, 30) polythene, handmade bags and newspapers envelopes, respectively. The shopkeepers A, B, C spend ₹ 250, ₹ 270 and ₹ 200 on these carry bags respectively.
(i) Represent the above situation by a matrix equation and form linear equations using matrix multiplication.
(ii) Is it possible to find the cost of each carry bag using matrices?

34. A point on the hypotenuse of a triangle is at distance a and b from the sides of the triangle. Show that the minimum length of the hypotenuse is $(a^{\frac{2}{3}} + b^{\frac{2}{3}})^{\frac{3}{2}}$.

OR

Show that the triangle of maximum area that can be inscribed in a given circle is an equilateral triangle.

35. Find the area of the region

$\{(x, y) : x^2 + y^2 \le 1 \le x + y\}$.

36. Assume that the chances of a patient having a heart attack is 40%. Assuming that a meditation and yoga course reduces the risk of heart attack by 30% and prescription of certain drug reduces its chance by 25%. At a time, a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options, the patient selected at random suffers a heart attack. Find the probability
(i) the patient followed a course of meditation and yoga.
(ii) the patient followed the prescription of certain drug.

www.examtrain.in

Detailed Solutions

1. (a) : We have, $A=\begin{bmatrix}5 & x\\ y & 0\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}5 & y\\ x & 0\end{bmatrix}$

Now, $A=A^T \Rightarrow \begin{bmatrix}5 & x\\ y & 0\end{bmatrix}=\begin{bmatrix}5 & y\\ x & 0\end{bmatrix} \Rightarrow \quad x=y$

2. (d) : We have, $y=\dfrac{\ln x}{x}$

On differentiating w.r.t. x, we get

$$y'=\frac{(1-\ln x)}{x^2} \Rightarrow y''=\frac{x^2\left(-\frac{1}{x}\right)-(1-\ln x)2x}{x^4}$$

$$\therefore \quad y''(e)=\frac{-e-0}{e^4}=-\frac{1}{e^3}$$

3. (a) : $f'(x) = 2x - 2 = 2(x - 1) < 0$ if $x < 1$ *i.e.*, $x \in (-\infty, 1)$. Hence f is strictly decreasing in $(-\infty, 1)$.

4. (a) : $\int \sec^2(7-4x)dx=-\dfrac{1}{4}\tan(7-4x)+C$

5. (a) : Let $I=a\int\dfrac{dx}{(1+x^2)\tan^{-1}x}$

Put $\tan^{-1}x=t \Rightarrow \dfrac{1}{1+x^2}dx=dt$

$$\Rightarrow I=a\int\frac{dt}{t}=a\log|t|+C=a\log|\tan^{-1}x|+C$$

6. (c) : $\dfrac{xdy}{dx}-y=x^4-3x$

$$\Rightarrow \quad \frac{dy}{dx}-\frac{y}{x}=\frac{x^4-3x}{x} \Rightarrow \frac{dy}{dx}-\frac{1}{x}\cdot y=x^3-3$$

I.F. $= e^{-\int\frac{1}{x}dx} = e^{-\log x}=e^{\log x^{-1}}=\dfrac{1}{x}$

7. (b) : We know that,

$$\cos\theta=\frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

$$=\frac{|6-12-30|}{\sqrt{4+9+36}\sqrt{9+16+25}}=\frac{36}{7\cdot5\sqrt{2}}=\frac{18\sqrt{2}}{35}$$

$$\Rightarrow \quad \theta=\cos^{-1}\left(\frac{18\sqrt{2}}{35}\right)$$

8. (b) : It is given that the line passes through the point $A(1, 2, 3)$ and parallel to the vector $\vec{b}=3\hat{i}+2\hat{j}-2\hat{k}$. . Therefore, the position vector of A is $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$.

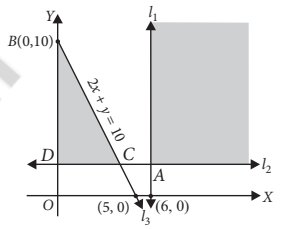
$\therefore \quad \vec{r}=\hat{i}+2\hat{j}+3\hat{k}+\lambda(3\hat{i}+2\hat{j}-2\hat{k})$ is the required equation of the line.

9. (b) : d.r.'s of line AB are $< 2 - 0, 3 - 4, -1 - 1 >$ *i.e.*, $< 2, -1, - 2 >$.

d.r.'s of line BC are $<4 - 2, 5 - 3, 0 + 1 >$ *i.e.*, $< 2, 2, 1 >$.

$$\therefore \quad \cos\theta=\frac{(2)(2)+(-1)(2)+(-2)(1)}{\sqrt{4+1+4}\sqrt{4+4+1}}=0\Rightarrow\theta=90°$$

10. (d) : Let $l_1 : x = 6$, $l_2 : y = 2$, $l_3 : 2x + y = 10$, $l_4 : x = 0$ and $l_5 : y = 0$



The graph of the inequalities $2x + y \le 10$, $y > 2$, $x \ge 0$, $y \ge 0$ is the region bounded by ΔBCD. This region has no point in common with the region $x \ge 6$, $y \ge 2$ as is clear from the figure. Hence, the region of the given inequalities is an empty set.

11. We have

$$\begin{vmatrix}b+c & c+a & a+b\\ c+a & a+b & b+c\\ a+b & b+c & c+a\end{vmatrix}=\begin{vmatrix}b & c & a\\ c & a & b\\ a & b & c\end{vmatrix}+\begin{vmatrix}c & a & b\\ a & b & c\\ b & c & a\end{vmatrix}$$

$$=\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix}+\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix}=2\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix}$$

$\therefore \quad k = 2$

12. We have, $y = 3x^2 + b \Rightarrow \dfrac{dy}{dx}=6x$

$$\left[\frac{dy}{dx}\right]_{x=2}=6\times2=12$$

13. We have, $\dfrac{d^2y}{dx^2}+\left(\dfrac{dy}{dx}\right)^{1/4}+(x)^{1/5}=0$

Clearly, order of given differential equation is 2.

14. Principal value of $\cos^{-1}\dfrac{1}{2}$ is $\dfrac{\pi}{3}$.

Principal value of $\sin^{-1}\dfrac{1}{2}$ is $\dfrac{\pi}{6}$.

So, the value of $\cos^{-1}\left(\dfrac{1}{2}\right)+2\sin^{-1}\left(\dfrac{1}{2}\right)=\dfrac{\pi}{3}+2\times\dfrac{\pi}{6}$

$$=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

www.examrace.in

OR

Let $\text{cosec}^{-1}\left(\frac{2}{\sqrt{3}}\right)=\theta \Rightarrow \text{cosec}\theta=\frac{2}{\sqrt{3}} \Rightarrow \theta=\frac{\pi}{3}$

15. Let $\overrightarrow{OA}=60\hat{i}+3\hat{j}$, $\overrightarrow{OB}=40\hat{i}-8\hat{j}$
and $\overrightarrow{OC}=a\hat{i}-52\hat{j}$
$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=-20\hat{i}-11\hat{j}$
$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=(a-60)\hat{i}-55\hat{j}$
As A, B, C are collinear.
i.e. $(-20\hat{i}-11\hat{j})=\lambda\{(a-60)\hat{i}-55\hat{j}\}$
On comparing, we get
$-20=(a-60)\lambda$ and $-11=-55\lambda \Rightarrow \lambda=1/5$
$\therefore \quad -20=(a-60)(1/5)$
$\Rightarrow \quad a-60=-100 \Rightarrow a=-100+60 \Rightarrow a=-40.$

OR

Given, $\vec{a}=2\hat{i}-6\hat{j}-3\hat{k}$
$\therefore \quad |\vec{a}|=\sqrt{2^2+(-6)^2+(-3)^2}=\sqrt{4+36+9}=7$

16. Principal value of $\tan^{-1}(1)$ is $\frac{\pi}{4}$.

Principal value of $\cos^{-1}\left(\frac{-1}{2}\right)$ is $\frac{2\pi}{3}$.

Also, principal value of $\sin^{-1}\left(\frac{-1}{2}\right)$ is $\left(\frac{-\pi}{6}\right)$.

$\therefore$ Principal value of $\tan^{-1}(1)+\cos^{-1}\left(\frac{-1}{2}\right)+\sin^{-1}\left(\frac{-1}{2}\right)$

$$=\frac{\pi}{4}+\frac{2\pi}{3}-\frac{\pi}{6}=\frac{3\pi}{4}$$

17. If x is the side of an equilateral triangle and A is its area, then

$$A=\frac{\sqrt{3}}{4}x^2 \Rightarrow \frac{dA}{dt}=\frac{\sqrt{3}}{4}2x\frac{dx}{dt}$$

Here, $x=20\sqrt{3}$ cm and $\frac{dx}{dt}=2$ cm/s

$$\therefore \quad \frac{dA}{dt}=\frac{\sqrt{3}}{4}2(20\sqrt{3})2=60 \text{ cm}^2/\text{s}$$

OR

Given, $y=x^3+3x \Rightarrow \frac{dy}{dx}=3x^2+3$

Since, slopes of tangents at $x=1$ and $z=-1$ are equal.
$\therefore$ The two tangents are parallel.

18. $\int_{-\pi/2}^{\pi/2} \sin x\cos^2 x(\sin^2 x+\cos x)\,dx$

$$=\int_{-\pi/2}^{\pi/2}\sin^3 x\cos^2 x\,dx+\int_{-\pi/2}^{\pi/2}\sin x\cos^3 x\,dx=0$$

($\because$ $\sin^3 x\cos^2 x$ and $\sin x\cos^3 x$ are odd functions)

19. $(\vec{a}+\vec{b}+\vec{c})^2=|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$
$\Rightarrow \quad 0=1^2+2^2+3^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$
$\Rightarrow \quad 2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=-14$
$\Rightarrow \quad \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-7$

20. $\cos 2\theta_1+\cos 2\theta_2+\cos 2\theta_3$
$=2(\cos^2\theta_1+\cos^2\theta_2+\cos^2\theta_3)-3=2(1)-3=-1$

21. $\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left(\frac{x-y}{x+y}\right)$

$$=\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left[\frac{1-\frac{y}{x}}{1+\frac{y}{x}}\right]$$

$$=\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}1+\tan^{-1}\left(\frac{y}{x}\right)$$

$$=\tan^{-1}\left(\frac{x}{y}\right)+\cot^{-1}\left(\frac{x}{y}\right)-\tan^{-1}1=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{4}$$

OR

We have, $\tan^{-1}2x+\tan^{-1}3x=\frac{\pi}{4}$

$$\Rightarrow \quad \frac{3x+2x}{1-6x^2}=\tan\frac{\pi}{4} \Rightarrow 5x=1-6x^2$$

$$\Rightarrow \quad 6x^2+5x-1=0 \Rightarrow x=-1,\frac{1}{6}$$

But when $x=-1$, $\tan^{-1}2x=\tan^{-1}(-2)<0$ and $\tan^{-1}3x=\tan^{-1}(-3)<0$
This value will not satisfy the given equation.
Hence, $x=\frac{1}{6}$.

22. Let M_{ij} and c_{ij} respectively denote the minor and cofactor of element a_{ij} in A. Then,
$M_{21}=6+10=16 \Rightarrow c_{21}=-M_{21}=-16$
$M_{31}=18-10=8 \Rightarrow c_{31}=M_{31}=8.$

23. Let $I=\int_0^{\pi/2}\frac{\sin x-\cos x}{1+\sin x\cos x}dx$

$$I=\int_0^{\pi/2}\frac{\sin\left(\frac{\pi}{2}-x\right)-\cos\left(\frac{\pi}{2}-x\right)}{1+\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)}dx$$

www.exambites.in

$$= \int_0^{\pi/2} \frac{\cos x - \sin x}{1+\cos x \sin x}dx = -\int_0^{\pi/2} \frac{\sin x - \cos x}{1+\sin x \cos x}dx = -I$$

$\Rightarrow \quad 2I = 0 \Rightarrow I = 0.$

24. Given $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c} \Rightarrow \vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{c} = 0$

$\Rightarrow \quad \vec{a}\cdot(\vec{b}-\vec{c}) = 0$

$\Rightarrow \quad \vec{a} = \vec{0}$ or $\vec{b}-\vec{c} = \vec{0}$ or $\vec{a}$ is perpendicular to $\vec{b}-\vec{c}$

$\Rightarrow \quad \vec{a} = \vec{0}$ or $\vec{b} = \vec{c}$ or $\vec{a}$ is perpendicular to $\vec{b}-\vec{c}$.

25. Line is $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{-2}$

It passes through the point (5, –4, 6) and direction ratios are 3, 7, –2.

$\therefore$ Vector equation of line is

$\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} - 2\hat{k})$

26. Consider the following events :

A = Person gets an electric contract,

B = Person gets plumbing contract

We have, $P(A) = \frac{2}{5}, P(\bar{B}) = \frac{4}{7}$ and $P(A \cup B) = \frac{2}{3}$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \quad \frac{2}{3} = \frac{2}{5} + \left(1 - \frac{4}{7}\right) - P(A \cap B)$

$\Rightarrow \quad P(A \cap B) = \frac{2}{5} + \frac{3}{7} - \frac{2}{3} = \frac{17}{105}$

OR

$P(E) = P(X = 2) + P(X = 3) + P(X = 5) + P(X = 7)$
$= 0.23 + 0.12 + 0.20 + 0.07 = 0.62$

$P(F) = P(X = 1) + P(X = 2) + P(X = 3)$
$= 0.15 + 0.23 + 0.12 = 0.50$

$P(E \cap F) = P(X = 2) + P(X = 3) = 0.23 + 0.12 = 0.35$

$\therefore \quad P(E \cup F) = P(E) + P(F) - P(E \cap F)$
$= 0.62 + 0.50 - 0.35 = 0.77.$

27. Given, $f(x) = 4x^2 + 12x + 15$

Let $f(x_1) = f(x_2)$ where $x_1, x_2 \in N$

$\Rightarrow \quad 4x_1^2 + 12x_1 + 15 = 4x_2^2 + 12x_2 + 15$

$\Rightarrow \quad 4x_1^2 + 12x_1 = 4x_2^2 + 12x_2$

$\Rightarrow \quad x_1^2 + 3x_1 = x_2^2 + 3x_2 \Rightarrow (x_1^2 - x_2^2) + 3(x_1 - x_2) = 0$

$\Rightarrow \quad (x_1 - x_2)(x_1 + x_2 + 3) = 0$

$\Rightarrow \quad x_1 = x_2 \ (\because x_1 + x_2 + 3 \neq 0$ for $x_1, x_2 \in N)$

Hence, f is one-one.

$\because$ Codomain is equal to the range. $\therefore$ f is onto.

Since, f is both one-one and onto.

Hence, $f : N \to S$ is invertible.

Let $y = f(x) \quad \Rightarrow 4x^2 + 12x + 15 = y$

$\Rightarrow \quad 4x^2 + 12x + 9 - 9 + 15 = y$

$\Rightarrow (2x + 3)^2 + 6 = y \quad \Rightarrow (2x + 3)^2 = y - 6$

$\Rightarrow \quad 2x + 3 = \sqrt{y-6} \quad \Rightarrow 2x = \sqrt{y-6} - 3$

$\Rightarrow \quad x = \frac{\sqrt{y-6} - 3}{2} \quad \Rightarrow f^{-1}(y) = \frac{\sqrt{y-6} - 3}{2}$

$\therefore \quad f^{-1}(x) = \frac{\sqrt{x-6} - 3}{2}$

28. Here, $f(0) = c$;

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0} \frac{\sin(a+1)x + \sin x}{x}$$

$$= \lim_{x\to 0}\left(\frac{\sin(a+1)x}{x} + \frac{\sin x}{x}\right)$$

$$= \lim_{x\to 0}\left\{\frac{\sin(a+1)x}{(a+1)x}\right\}\cdot(a+1) + \lim_{x\to 0}\frac{\sin x}{x}$$

$= (a + 1) + 1 = a + 2$

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0} \frac{\sqrt{x + bx^2} - \sqrt{x}}{b\sqrt{x^3}}$$

$$= \lim_{x\to 0} \frac{\sqrt{x}\{\sqrt{1+bx} - 1\}}{bx\sqrt{x}}$$

$$= \lim_{x\to 0}\left(\frac{\sqrt{1+bx} - 1}{bx}\right) \times \left(\frac{\sqrt{1+bx} + 1}{\sqrt{1+bx} + 1}\right)$$

$$= \lim_{x\to 0} \frac{1 + bx - 1}{bx(\sqrt{1+bx} + 1)}$$

$$= \lim_{x\to 0} \frac{1}{\sqrt{1+bx} + 1} = \frac{1}{\sqrt{1} + 1} = \frac{1}{2}$$

Now, f is continuous at $x = 0$ if

$$\lim_{x\to 0^-} f(x) = f(0) = \lim_{x\to 0^+} f(x)$$

i.e., if $a + 2 = c = \frac{1}{2} \Rightarrow a = -\frac{3}{2}$ and $c = \frac{1}{2}$

Hence, for f to be continuous at $x = 0$, we must have $a = -\frac{3}{2}$, $c = \frac{1}{2}$; b may have any real value.

OR

We have, $f(x) = x^3 + bx^2 + ax + 5$ satisfy Rolle's theorem on [1, 3] with $c = 2 + \frac{1}{\sqrt{3}}$

$\therefore \quad f(1) = f(3)$ and $f'(c) = 0$

$f(1) = 1 + b + a + 5 = a + b + 6$

$f(3) = 27 + 9b + 3a + 5 = 3a + 9b + 32$

Now, $f(1) = f(3)$

$\Rightarrow \quad a + b + 6 = 3a + 9b + 32 \Rightarrow 2a + 8b + 26 = 0$

$\Rightarrow \quad a + 4b + 13 = 0$ ...(i)

Now, $f'(x) = 3x^2 + 2bx + a$

$f'(c) = 0$ with $c = 2 + \frac{1}{\sqrt{3}}$

www.examfiles.in

$$\Rightarrow\ 3\left(2+\frac{1}{\sqrt{3}}\right)^2+2b\left(2+\frac{1}{\sqrt{3}}\right)+a=0$$

$$\Rightarrow\ 3\times\frac{(2\sqrt{3}+1)^2}{3}+2b\left(\frac{2\sqrt{3}+1}{\sqrt{3}}\right)+a=0$$

$$\Rightarrow\ 12+1+4\sqrt{3}+4b+\frac{2b}{\sqrt{3}}+a=0$$

$$\Rightarrow\ 13+4\sqrt{3}+\frac{2b}{\sqrt{3}}+4b+a=0$$

$$\Rightarrow\ 13+4\sqrt{3}+\frac{2b}{\sqrt{3}}+4b-4b-13=0 \qquad \text{[Using (i)]}$$

$$\Rightarrow\ 4\sqrt{3}+\frac{2b}{\sqrt{3}}=0$$

$\Rightarrow$ $12+2b=0 \Rightarrow b=-6$

$\therefore$ From (i), $a=24-13=11$.

29. Let $I=\int\frac{x^4}{(x-1)(x^2+1)}dx$

Let $\frac{x^4}{(x-1)(x^2+1)}=\frac{A}{x-1}+\frac{Bx+C}{x^2+1}$

$\Rightarrow\ x^4=A(x^2+1)+(Bx+C)(x-1)$

Putting $x=0$, we get $A=C$

Putting $x=1$, we get $A=\frac{1}{2}$

$\therefore\ C=\frac{1}{2}$

Putting $x=-1$, we get $B=\frac{1}{2}$

$$\therefore\ I=\frac{1}{2}\int\frac{dx}{x-1}+\frac{1}{2}\int\frac{x}{x^2+1}dx+\frac{1}{2}\int\frac{dx}{x^2+1}$$

$$=\frac{1}{2}\left[\log(x-1)+\int\frac{x}{x^2+1}dx+\tan^{-1}x\right]+c$$

Put $x^2+1=t\Rightarrow 2x\,dx=dt$

$$\therefore\ I=\frac{1}{2}\left[\log(x-1)+\frac{1}{2}\int\frac{dt}{t}+\tan^{-1}x\right]+c$$

$$=\frac{1}{2}\left[\log(x-1)+\frac{1}{2}\log t+\tan^{-1}x\right]+c$$

$$=\frac{1}{2}\left[\log(x-1)+\frac{1}{2}\log(x^2+1)+\tan^{-1}x\right]+c$$

30. We have, $\sec^2 y\frac{dy}{dx}+2x\tan y=x^3$

Put $\tan y=t\ \Rightarrow\ \sec^2 y\frac{dy}{dx}=\frac{dt}{dx}$

$$\Rightarrow\ \frac{dt}{dx}+2xt=x^3 \qquad ...(i)$$

(i) is a linear differential equation of the form $\frac{dt}{dx}+Pt=Q$, where $P=2x$, $Q=x^3$

I.F. $=e^{\int P\,dx}=e^{x^2}$

The solution of the equation (i) is given by

$$t(e^{x^2})=\int e^{x^2}\times x^3\,dx \qquad ...(ii)$$

Put $x^2=z$, $2x\,dx=dz$ $\therefore$ (ii) becomes

$$t(e^z)=\frac{1}{2}\int e^z z\,dz=\frac{1}{2}\left[ze^z-\int e^z dz\right]$$

$$\Rightarrow\ t(e^z)=\frac{1}{2}(ze^z-e^z)$$

$$\Rightarrow\ t(e^{x^2})=\frac{1}{2}\left(x^2e^{x^2}-e^{x^2}\right)$$

$$\Rightarrow\ \tan y\,(e^{x^2})=\frac{1}{2}\left(x^2e^{x^2}-e^{x^2}\right)$$

$$\Rightarrow\ \tan y=\frac{1}{2}(x^2-1)$$

31. Here, $x_1=0, y_1=2, z_1=-3; x_2=2, y_2=6, z_2=3; a_1=1,$ $b_1=2,\ c_1=3;\ a_2=2,\ b_2=3,\ c_2=4.$

$$\therefore\ \begin{vmatrix}x_2-x_1 & y_2-y_1 & z_2-z_1\\ a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\end{vmatrix}=\begin{vmatrix}2 & 4 & 6\\ 1 & 2 & 3\\ 2 & 3 & 4\end{vmatrix}=0.$$

Hence, the two given lines are coplanar.

The equation of the plane containing these lines is

$$\begin{vmatrix}x-0 & y-2 & z+3\\ 1 & 2 & 3\\ 2 & 3 & 4\end{vmatrix}=0$$

$\Rightarrow x(8-9)-(y-2)(4-6)+(z+3)(3-4)=0$

$\Rightarrow\ -x+2(y-2)-(z+3)=0\Rightarrow x-2y+z+7=0$

OR

Given equations of lines are

$\vec{r}=(1-t)\hat{i}+(t-2)\hat{j}+(3-2t)\hat{k}$

i.e., $\vec{r}=\hat{i}-2\hat{j}+3\hat{k}+t(-\hat{i}+\hat{j}-2\hat{k})$

and $\vec{r}=(s+1)\hat{i}+(2s-1)\hat{j}-(2s+1)\hat{k}$

i.e., $\vec{r}=\hat{i}-\hat{j}-\hat{k}+s(\hat{i}+2\hat{j}-2\hat{k})$

Here, $\vec{a}_1=\hat{i}-2\hat{j}+3\hat{k}, \vec{b}_1=-\hat{i}+\hat{j}-2\hat{k}$

$\vec{a}_2=\hat{i}-\hat{j}-\hat{k}$, $\vec{b}_2=\hat{i}+2\hat{j}-2\hat{k}$

$\Rightarrow\ \vec{a}_2-\vec{a}_1=(\hat{i}-\hat{j}-\hat{k})-(\hat{i}-2\hat{j}+3\hat{k})=\hat{j}-4\hat{k}$

and $\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -1 & 1 & -2\\ 1 & 2 & -2\end{vmatrix}$

www.examrites.in

$= \hat{i}(-2+4) - \hat{j}(2+2) + \hat{k}(-2-1) = 2\hat{i} - 4\hat{j} - 3\hat{k}$

$\therefore$ Shortest distance, $d = \left|\dfrac{(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}\right|$

$= \left|\dfrac{(\hat{j} - 4\hat{k})\cdot(2\hat{i} - 4\hat{j} - 3\hat{k})}{\sqrt{(2)^2 + (-4)^2 + (-3)^2}}\right| = \left|\dfrac{0 - 4 + 12}{\sqrt{4 + 16 + 9}}\right|$

$= \dfrac{8}{\sqrt{29}}$ units

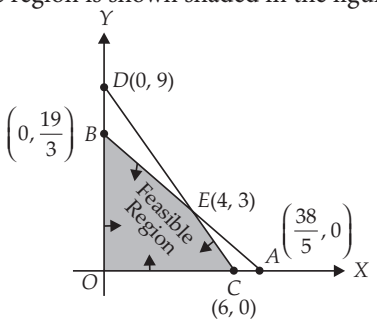
32. If x machines of type A and y machines of type B are bought, then our problem is to maximize $Z = 50x + 40y$ subject to constraints

$x \geq 0, y \geq 0$

$1000x + 1200y \leq 7600 \Rightarrow 5x + 6y \leq 38$

$12x + 8y \leq 72 \Rightarrow 3x + 2y \leq 18$

Feasible region is shown shaded in the figure.



Points	$Z = 50x + 40y$	Value
$O(0, 0)$	$50 \times 0 + 40 \times 0$	0
$B\left(0, \frac{19}{3}\right)$	$50 \times 0 + 40 \times \frac{19}{3}$	253.3
$E(4, 3)$	$50 \times 4 + 40 \times 3$	320
$C(6, 0)$	$50 \times 6 + 40 \times 0$	300

i.e., Daily output should be maximized when 4 machines of type A and 3 machines of type B are bought.

33. We have, $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix}$

We know that, $A = IA$

$\Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 2 & 5 & 3 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$

Applying $R_2 \to R_2 - 2R_1$

$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 7 & 3 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$

Applying $R_3 \to 4R_3 - R_2$

$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 7 & 3 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -1 & 4 \end{bmatrix} A$

Applying $R_2 \leftrightarrow R_3$

$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 7 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 4 \\ -2 & 1 & 0 \end{bmatrix} A$

Applying $R_1 \to R_1 + R_2$ and $R_3 \to R_3 - 7R_2$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -4 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 4 \\ 2 & -1 & 4 \\ -16 & 8 & -28 \end{bmatrix} A$

Applying $R_3 \to -\dfrac{1}{4}R_3$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 4 \\ 2 & -1 & 4 \\ 4 & -2 & 7 \end{bmatrix} A$

Applying $R_1 \to R_1 - R_3$ and $R_2 \to R_2 - R_3$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & -3 \\ -2 & 1 & -3 \\ 4 & -2 & 7 \end{bmatrix} A$

$\therefore \quad A^{-1} = \begin{bmatrix} -1 & 1 & -3 \\ -2 & 1 & -3 \\ 4 & -2 & 7 \end{bmatrix}$

OR

(i) Let x, y, z be the cost of polythene, handmade bags and newspaper envelopes respectively used by shopkeepers A, B and C.

$\therefore \quad \begin{bmatrix} 20 & 30 & 40 \\ 30 & 40 & 20 \\ 40 & 20 & 30 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 250 \\ 270 \\ 200 \end{bmatrix}$

$20x + 30y + 40z = 250$

$30x + 40y + 20z = 270$

$40x + 20y + 30z = 200$

(ii) Let $P = 1000 \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

$\Rightarrow \quad |P| = 1000[2(12 - 4) - 3(9 - 8) + 4(6 - 16)]$

$\Rightarrow \quad |P| = 1000[16 - 3 - 40] = 1000[-27] = -27000$

$\because \quad |P| \neq 0$ or P^{-1} exists.

www.exambites.in

$\therefore$ Equations have a unique solution.

$\therefore$ We can find the cost of carry bag using matrices.

34. Let AB $(= l)$ be the hypotenuse of a triangle AOB and let P be a point on the hypotenuse AB such that $PL \perp OA$ and $PM \perp OB$ and $PL = a$, $PM = b$.

Let $\angle OAB = \theta$, then $\angle MPB = \angle OAB = \theta$

In ΔPLA, $\sin\theta = \frac{PL}{PA} = \frac{a}{PA}$

$$\Rightarrow \quad PA = \frac{a}{\sin\theta} = a\operatorname{cosec}\theta \qquad \text{...(i)}$$

In ΔMPB, $\cos\theta = \frac{PM}{PB} = \frac{b}{PB}$

$$\Rightarrow \quad PB = \frac{b}{\cos\theta} = b\sec\theta \qquad \text{...(ii)}$$

Now, $l = AB = PA + PB$

$\Rightarrow$ $l = a\operatorname{cosec}\theta + b\sec\theta$ [Using (i) and (ii)]

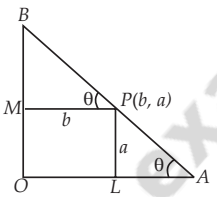
$$\therefore \quad \frac{dl}{d\theta} = -a\operatorname{cosec}\theta\cot\theta + b\sec\theta\tan\theta$$

Put $\frac{dl}{d\theta} = 0 \quad \Rightarrow \quad -a\operatorname{cosec}\theta\cot\theta + b\sec\theta\tan\theta = 0$

$$\Rightarrow \quad \frac{a\cos\theta}{\sin^2\theta} = \frac{b\sin\theta}{\cos^2\theta}$$

$$\Rightarrow \quad \frac{a}{b} = \frac{\sin^3\theta}{\cos^3\theta} = \tan^3\theta$$

$$\Rightarrow \quad \tan\theta = \left(\frac{a}{b}\right)^{\frac{1}{3}}$$



Further, $\frac{d^2l}{d\theta^2} = -a[-\operatorname{cosec}\theta\cdot\cot\theta]\cot\theta -$

$a\operatorname{cosec}\theta(-\operatorname{cosec}^2\theta)] + b[\sec\theta\tan\theta]\tan\theta$

$+ b[\sec\theta(\sec^2\theta)]$

$= a\operatorname{cosec}\theta\cot^2\theta + a\operatorname{cosec}^3\theta + b\sec\theta\tan^2\theta$

$+ b\sec^3\theta$

$= a\operatorname{cosec}\theta(\cot^2\theta + \operatorname{cosec}^2\theta) + b\sec\theta$

$(\tan^2\theta + \sec^2\theta)$

$$= a\sqrt{1+\cot^2\theta}(\cot^2\theta + 1 + \cot^2\theta)$$

$$+ b\sqrt{1+\tan^2\theta}(\tan^2\theta + 1 + \tan^2\theta)$$

$$= a\sqrt{1+\cot^2\theta}(1+2\cot^2\theta) + b\sqrt{1+\tan^2\theta}(1+2\tan^2\theta)$$

Now, $\left[\frac{d^2l}{d\theta^2}\right]_{\tan\theta=\left(\frac{a}{b}\right)^{1/3}}$

$$= a\sqrt{1+\left(\frac{b}{a}\right)^{2/3}}\left[1+2\left(\frac{b}{a}\right)^{2/3}\right] + b\sqrt{1+\left(\frac{a}{b}\right)^{2/3}}\left[1+2\left(\frac{a}{b}\right)^{2/3}\right]$$

$$= \sqrt{a^{2/3}+b^{2/3}}\left[\frac{a^{1/3}}{\sqrt{a^{2/3}}}(a^{2/3}+2b^{2/3}) + \frac{b^{1/3}}{\sqrt{b^{2/3}}}(b^{2/3}+2a^{2/3})\right]$$

$$\sqrt{a^{2/3}+b^{2/3}} = \left[\left(a^{2/3}+2b^{2/3}\right) + \left(b^{2/3}+2a^{2/3}\right)\right]$$

$$= \sqrt{a^{2/3}+b^{2/3}}\left[3a^{2/3}+3b^{2/3}\right] > 0$$

$$\left[\because \quad a^{2/3}+b^{2/3} > 0, a > 0, b > 0\right]$$

Length of the hypotenuse is minimum when

$$\tan\theta = \left(\frac{a}{b}\right)^{1/3}$$

Length of the hypotenuse $= l$

$= a\operatorname{cosec}\theta + b\sec\theta$

$$= \left[a\sqrt{1+\cot^2\theta} + b\sqrt{1+\tan^2\theta}\right]$$

$$= \left[a\sqrt{1+\frac{b^{2/3}}{a^{2/3}}} + b\sqrt{1+\frac{a^{2/3}}{b^{2/3}}}\right]$$

$$= \left[a\sqrt{\frac{a^{2/3}+b^{2/3}}{a^{2/3}}} + b\sqrt{\frac{b^{2/3}+a^{2/3}}{b^{2/3}}}\right]$$

$$= \sqrt{a^{2/3}+b^{2/3}}\left[\frac{a}{a^{1/3}} + \frac{b}{b^{1/3}}\right]$$

$$= \sqrt{a^{2/3}+b^{2/3}}\left[a^{2/3}+b^{2/3}\right] = \left(a^{2/3}+b^{2/3}\right)^{3/2}$$

Hence, the minimum length of the hypotenuse is $(a^{2/3}+b^{2/3})^{3/2}$.

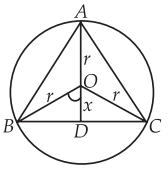
OR

Let ABC be a triangle inscribed in a given circle with centre O and radius r. The area of the triangle will be maximum if its vertex A opposite to the base BC is at a maximum distance from the base BC. This is possible only when A lies on the diameter perpendicular to BC. Thus, $AD \perp BC$. So, triangle ABC must be an isosceles triangle.

Let $OD = x$.

In right triangle ODB, we have $OB^2 = OD^2 + BD^2$

$\Rightarrow$ $r^2 = x^2 + BD^2$



$$\Rightarrow \quad BD = \sqrt{r^2 - x^2}$$

$$\Rightarrow \quad BC = 2BD = 2\sqrt{r^2 - x^2}.$$

Also, $AD = AO + OD = r + x$.

Let A denote the area of ΔABC.

Then, $A = \frac{1}{2}(BC \times AD)$

$$\Rightarrow \quad A = \frac{1}{2} \times 2\sqrt{r^2 - x^2} \times (r + x)$$

$$\Rightarrow \quad A = (r + x)\sqrt{r^2 - x^2}$$

www.examblues.in

$$\Rightarrow \quad \frac{dA}{dx} = \sqrt{r^2 - x^2} - \frac{x(r+x)}{\sqrt{r^2 - x^2}}$$

$$\Rightarrow \quad \frac{dA}{dx} = \frac{r^2 - rx - 2x^2}{\sqrt{r^2 - x^2}}$$

For maximum or minimum values of A, we must have $\frac{dA}{dx} = 0 \Rightarrow \frac{r^2 - rx - 2x^2}{\sqrt{r^2 - x^2}} = 0$

$\Rightarrow (r - 2x)(r + x) = 0 \Rightarrow r - 2x = 0 \quad [\because \quad r + x \neq 0]$

$$\Rightarrow \quad x = \frac{r}{2}$$

Now, $\frac{dA}{dx} = \frac{r^2 - rx - 2x^2}{\sqrt{r^2 - x^2}}$

$$\Rightarrow \quad \frac{d^2A}{dx^2} = \frac{(-r - 4x)}{\sqrt{r^2 - x^2}} + \frac{(r^2 - rx - 2x^2)x}{(r^2 - x^2)^{3/2}}$$

$$\Rightarrow \quad \left(\frac{d^2A}{dx^2}\right)_{x = r/2} = -2\sqrt{3} < 0$$

Thus, A is maximum when $x = \frac{r}{2}$

$$\therefore \quad BD = \sqrt{r^2 - x^2} = \frac{\sqrt{3}r}{2}$$

In ΔODB, let $\angle BOD = \theta$

$$\tan\theta = \frac{BD}{OD} \Rightarrow \tan\theta = \frac{\frac{\sqrt{3}r}{2}}{\frac{r}{2}} = \sqrt{3} \Rightarrow \theta = 60°$$

$\therefore \quad \angle BAC = \theta = 60°$

But $AB = AC \quad \therefore \quad \angle B = \angle C$

$\Rightarrow \quad \angle A = \angle B = \angle C = 60°$

$\Rightarrow$ A is maximum when ΔABC is equilateral.

35. Let $R = \{(x, y) : x^2 + y^2 \leq 1 \leq x + y\}$. Then

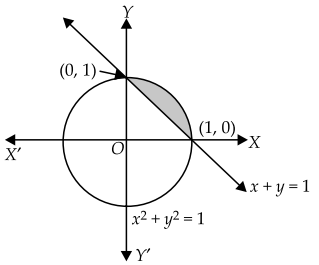
$\Rightarrow \quad R = R_1 \cap R_2$, where

$R_1 = \{(x, y) : x^2 + y^2 \leq 1\}$ and $R_2 = \{(x, y) : 1 \leq x + y\}$

R_1 : Clearly $x^2 + y^2 = 1$ represents a circle with centre at (0, 0) and radius 1.

Since $x^2 + y^2 \leq 1$, so region R_1 represents the interior of circle $x^2 + y^2 = 1$.

R_2 : Since $x + y = 1$ is the equation of a straight line cutting x and y-axes at (1, 0) and (0, 1) respectively. This line divides the xy-plane in two parts represented by $x + y \leq 1$ and $x + y \geq 1$. Since (0, 0) does not satisfy the inequality $x + y \geq 1$. So, R_2 is the region lying above the line $x + y = 1$.



Hence, region R is the shaded region shown in figure.

So, required area is given by

$$A = \int_0^1 \left\{\sqrt{1 - x^2} - (1 - x)\right\} dx$$

$$\Rightarrow \quad A = \left[\frac{1}{2}x\sqrt{1 - x^2} + \frac{1}{2}\sin^{-1}\left(\frac{x}{1}\right) - x + \frac{x^2}{2}\right]_0^1$$

$$\Rightarrow \quad A = \left\{0 + \frac{1}{2}\sin^{-1}1 - 1 + \frac{1}{2}\right\} - 0 = \left(\frac{\pi}{4} - \frac{1}{2}\right) \text{sq.units}$$

36. Let A, E_1 and E_2 respectively denotes the event that a person has a heart attack, the selected person followed the course of yoga and meditation and the person adopted the drug prescription.

$$\therefore \quad P(A) = \frac{40}{100} = 0.40 \quad ; \quad P(E_1) = P(E_2) = \frac{1}{2}$$

$P(A/E_1) = 0.40 \times 0.70 = 0.28$

$P(A/E_2) = 0.40 \times 0.75 = 0.30$

Probability that the patient suffering heartattack followed course of meditation and yoga is

$$P(E_1 / A) = \frac{P(E_1)P(A / E_1)}{P(E_1)P(A / E_1) + P(E_2)P(A / E_2)}$$

$$= \frac{\frac{1}{2} \times 0.28}{\frac{1}{2} \times 0.28 + \frac{1}{2} \times 0.30} = \frac{0.14}{0.14 + 0.15} = \frac{14}{29}$$

$$\text{Now, } P(E_2 / A) = \frac{P(E_2)P(A / E_2)}{P(E_1)P(A / E_1) + P(E_2)P(A / E_2)}$$

$$= \frac{\frac{1}{2} \times 0.30}{\frac{1}{2} \times 0.28 + \frac{1}{2} \times 0.30} = \frac{0.15}{0.14 + 0.15} = \frac{15}{29}$$

www.exambites.in

PRACTICE PAPER 2
(Solved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.examples.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. The value of $\cot\left(\text{cosec}^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is

(a) $\frac{5}{17}$ (b) $\frac{6}{17}$ (c) $\frac{3}{17}$ (d) $\frac{4}{17}$

2. The equation of the line joining the points (–3, 4, 11) and (1, –2, 7) is

(a) $\frac{x+3}{2}=\frac{y-4}{3}=\frac{z-11}{4}$

(b) $\frac{x+3}{-2}=\frac{y-4}{3}=\frac{z-11}{2}$

(c) $\frac{x+3}{-2}=\frac{y+4}{3}=\frac{z+11}{4}$

(d) $\frac{x+3}{2}=\frac{y+4}{-3}=\frac{z+11}{2}$

3. The equation of the tangent to the curve $y = e^{2x}$ at the point (0, 1) is

(a) $y + 1 = 2x$ (b) $1 - y = 2x$
(c) $y - 1 = 2x$ (d) none of these

4. $\int\frac{10x^9+10^x\log_e 10}{10^x+x^{10}}dx$ is equal to

(a) $10^x - x^{10} + C$ (b) $10^x + x^{10} + C$
(c) $(10^x - x^{10})^{-1} + C$ (d) $\log_e(10^x + x^{10}) + C$

5. The optimal value of the objective function is attained at the points

(a) on X-axis
(b) on Y-axis
(c) which are corner points of the feasible region
(d) none of these

6. The solution of the differential equation $\frac{dy}{dx}=\frac{1+y^2}{1+x^2}$ is

(a) $y = \tan^{-1}x$ (b) $y - x = k(1 + xy)$
(c) $x = \tan^{-1}y$ (d) $\tan(xy) = k$

7. A line makes angles α, β and γ with the co-ordinate axes. If $\alpha + \beta = 90°$, then γ is equal to

(a) 0 (b) 90°
(c) 180° (d) None of these

8. If $\sin\left(\sin^{-1}\frac{1}{5}+\cos^{-1}x\right)=1$, then the value of x is

(a) –1 (b) $\frac{2}{5}$ (c) $\frac{1}{3}$ (d) $\frac{1}{5}$

9. The shortest distance between the lines $x = y = z$ and $x=1-y=\frac{z}{0}$ is

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{6}}$

10. $\int xe^{x^2}\,dx$ is equal to

(a) $-\frac{e^{x^2}}{2}+C$ (b) $\frac{e^{x^2}}{2}+C$

(c) $\frac{e^{x}}{2}+C$ (d) $-\frac{e^{x}}{2}+C$

(Q.11 - Q.15) Fill in the blanks.

11. If A is a square matrix of order 3×3 such that $|\text{adj } A| = 324$, then $|A|$ = _____ .

12. The value of $\int_0^{2\pi} \frac{dx}{e^{\sin x}+1}$ = _____ .

OR

The value of $\int_0^1 \frac{1}{2x-3}\,dx$ = _____ .

13. If $\vec{a}=2\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=5\hat{i}-3\hat{j}+\hat{k}$, then the projection of $\vec{b}$ on $\vec{a}$ is_____ .

14. The sum of order and degree of the differential equation

$\left(\frac{d^3y}{dx^3}\right)^2-3\frac{d^2y}{dx^2}+2\left(\frac{dy}{dx}\right)^4=y^4$ is_____ .

OR

The integrating factor for solving the differential equation $(1+y^2)\,dx=(\tan^{-1}y-x)\,dy$ is _____ .

15. If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined to x-axis at angles 30° and 120° respectively, then $|\vec{a}+\vec{b}|$ = _____ .

(Q.16 – Q.20) Answer the following questions.

16. Write the value of $3xyz$, if

$$\begin{bmatrix}2&0&0\\0&3&0\\0&0&4\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}4\\-6\\8\end{bmatrix}.$$

17. Find the value of c in Mean value theorem for the function $f(x) = x(x - 2)$, $x \in [1, 2]$.

18. If $y = e^{3x+7}$, then find the value of $\frac{dy}{dx}$ at $x = 0$.

OR

Is the function defined by $|x^2 + 2|$, a continuous function?

19. The total revenue received from the sale of x units of a product is given by $R(x) = 15x^2 + 4x + 30$. Find the marginal revenue when $x = 5$, where by marginal revenue we mean the rate of $R(x)$ with respect to the number of items sold at an instant.

20. Find the value of λ so that the vectors $2\hat{i}-4\hat{j}+\hat{k}$ and $4\hat{i}-8\hat{j}+\lambda\hat{k}$ are perpendicular.

SECTION - B

21. Find the value of $\sin^{-1}\left[\cos\left(\sin^{-1}\frac{\sqrt{3}}{2}\right)\right]$.

OR

Prove that, $\tan^{-1}\left(\frac{1-x}{1+x}\right)-\tan^{-1}\left(\frac{1-y}{1+y}\right)$

$=\sin^{-1}\left(\frac{y-x}{\sqrt{1+x^2}\sqrt{1+y^2}}\right)$.

22. For what values of x will the given matrix

$$\begin{bmatrix}-x&x&2\\2&x&-x\\x&-2&-x\end{bmatrix}$$ be singular?

23. Evaluate : $\int_0^{\pi/4}\sqrt{1+\sin 2x}\,dx$

24. Find the area of a parallelogram whose adjacent sides are given by the vectors $\vec{a}=\hat{i}+3\hat{j}+4\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$.

25. If the line through the points (4, 1, 2) and (5, λ, 0) is parallel to the line through the points (2, 1, 1) and (3, 3, –1), then find λ.

26. A four digit number is formed using the digits 1, 2, 3, 5 with no repetitions. Write the probability that the number is divisible by 5.

OR

A bag contains 12 white pearls and 18 black pearls. Two pearls are drawn in succession without replacement. Find the probability that the first pearl is white and the second is black.

SECTION - C

27. Let S and T be two sets. Show that $f: S \times T \to T \times S$ defined as $f(s, t) = (t, s)$ is one-one and onto.

www.exambits.in

28. Let $f(x)=\begin{cases}\dfrac{1-\cos 4x}{x^2}, & \text{if } x<0\\ a, & \text{if } x=0\\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}, & \text{if } x>0\end{cases}$

Determine the value of a so that $f(x)$ is continuous at $x = 0$.

OR

If $y=\dfrac{\sqrt{1-x^2}(2x-3)^{1/2}}{(x^2+2)^{2/3}}$, find $\dfrac{dy}{dx}$.

29. Prove that $\int_0^{\pi}\dfrac{x}{1-\cos\alpha\sin x}dx=\dfrac{\pi(\pi-\alpha)}{\sin\alpha}$.

30. Solve the differential equation :
$(1 + y^2)\,(1 + \log x)\,dx + x\,dy = 0$.

31. Show that the lines

$$\frac{x-a+d}{\alpha-\delta}=\frac{y-a}{\alpha}=\frac{z-a-d}{\alpha+\delta}$$

and $\dfrac{x-b+c}{\beta-\gamma}=\dfrac{y-b}{\beta}=\dfrac{z-b-c}{\beta+\gamma}$ are coplanar.

OR

Find the length and the foot of the perpendicular from the point (7, 14, 5) to the plane $2x + 4y - z = 2$.

32. A village has 500 hectares of land to grow two types of plants, X and Y. The contribution of total amount of oxygen produced by plant X and plant Y are 60% and 40% per hectare respectively. To control weeds, a liquid herbicide has to be used for X and Y at rates of 20 litres and 10 litres per hectare, respectively. Further no more than 8000 litres of herbicides should be used in order to protect aquatic animals in a pond which collects drainage from this land. How much land should be allocated to each plant so as to maximise the total production of oxygen?

SECTION - D

33. Using elementary row transformations find the inverse of the matrix $A=\begin{bmatrix}3 & -1 & -2\\ 2 & 0 & -1\\ 3 & -5 & 0\end{bmatrix}$.

OR

If $A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}$, $6A^{-1} = A^2 + cA + dI$, then find the values of c and d.

34. Find the points on the curve $9y^2 = x^3$, where normal to the curve makes equal intercepts with the axes.

OR

A square piece of tin of side 24 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. What should be the side of the square to be cut off so that the volume of the box is maximum?

35. Find the area bounded by the lines $y = 4x + 5$, $y = 5 - x$ and $4y = x + 5$.

36. Three cards are drawn successively with replacement from a well-shuffled deck of 52 cards. A random variable X denotes the number of hearts in the three cards drawn. Find the mean and variance of X.

Detailed Solutions

1. **(b)** : $\cot\left(\text{cosec}^{-1}\dfrac{5}{3}+\tan^{-1}\dfrac{2}{3}\right)$

Let $\text{cosec}^{-1}\dfrac{5}{3}=\theta,\ \tan^{-1}\dfrac{2}{3}=\phi$

$\Rightarrow \quad \text{cosec}\,\theta=\dfrac{5}{3},\ \tan\phi=\dfrac{2}{3}$

$\Rightarrow \quad \cot\theta=\dfrac{4}{3},\ \cot\phi=\dfrac{3}{2}$

Now, $\cot(\theta+\phi)=\dfrac{\cot\theta\cot\phi-1}{\cot\theta+\cot\phi}=\dfrac{\left(\frac{4}{3}\times\frac{3}{2}\right)-1}{\frac{4}{3}+\frac{3}{2}}=\dfrac{6}{17}$

www.exambites.in

2. (b) : D.r.'s of the line joining the given points are (1 – (– 3), – 2 – 4, 7 – 11)
i.e., (4, – 6, – 4) or (– 2, 3, 2)

$\therefore$ Equation of line is $\frac{x+3}{-2}=\frac{y-4}{3}=\frac{z-11}{2}$

3. (c) : $\left(\frac{dy}{dx}\right)_{(0,1)}=2e^0=2.$ Hence equation of tangent to the given curve at (0, 1) is

$y - 1 = 2(x - 0)$ *i.e.*, $y - 1 = 2x$.

4. (d) : Put $10^x + x^{10} = t$

$\Rightarrow \quad (10^x \log_e 10 + 10x^9)dx = dt$

$$\therefore \quad \int \frac{10x^9 + 10^x \log_e 10}{10^x + x^{10}}\,dx = \int \frac{dt}{t}$$

$= \log_e t + c = \log_e(10^x + x^{10}) + C$

5. (c) : When we solve an L.P.P. graphically, the optimal (or optimum) value of the objective function is attained at corner points of the feasible region.

6. (b) : $\frac{dy}{dx}=\frac{1+y^2}{1+x^2} \Rightarrow \frac{dy}{1+y^2}=\frac{dx}{1+x^2}$

On integrating, we get

$\tan^{-1}y = \tan^{-1}x + c \quad \Rightarrow \tan^{-1}y - \tan^{-1}x = c$

$$\Rightarrow \quad \tan^{-1}\left(\frac{y-x}{1+xy}\right)=c$$

$$\Rightarrow \quad \frac{y-x}{1+xy} = \tan c = k(\text{say})$$

$\Rightarrow \quad y - x = k(1 + xy)$

7. (b) : We know that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
$\Rightarrow \quad \cos^2\alpha + \cos^2(90° - \alpha) + \cos^2\gamma = 1$
$\Rightarrow \quad \cos^2\alpha + \sin^2\alpha + \cos^2\gamma = 1 \Rightarrow 1 + \cos^2\gamma = 1$
$\Rightarrow \quad \cos^2\gamma = 0$
$\Rightarrow \quad \cos\gamma = 0 \Rightarrow \gamma = 90°.$

8. (d) : We have, $\sin\left(\sin^{-1}\frac{1}{5}+\cos^{-1}x\right)=1$

$$\Rightarrow \quad \sin^{-1}\frac{1}{5}+\cos^{-1}x=\sin^{-1}(1)$$

$$\Rightarrow \quad \sin^{-1}\frac{1}{5}+\frac{\pi}{2}-\sin^{-1}x=\frac{\pi}{2}$$

$$\Rightarrow \quad \sin^{-1}\frac{1}{5}-\sin^{-1}x=0 \Rightarrow \sin^{-1}\frac{1}{5}=\sin^{-1}x$$

$$\Rightarrow \quad x=\sin\left(\sin^{-1}\frac{1}{5}\right)\Rightarrow x=\frac{1}{5}$$

9. (d) : The lines are $\frac{x}{1}=\frac{y}{1}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y-1}{-1}=\frac{z}{0}$.

Now, $\vec{a}_1 = 0, \vec{b}_1 = \hat{i}+\hat{j}+\hat{k}$, $\vec{a}_2 = \hat{j}$, $\vec{b}_2 = \hat{i}-\hat{j}$

$\vec{b}_1 \times \vec{b}_2 = (\hat{i}+\hat{j}+\hat{k})\times(\hat{i}-\hat{j}) = \hat{i}+\hat{j}-2\hat{k}$

$$\text{S.D.} = \frac{|\hat{j}\cdot(\hat{i}+\hat{j}-2\hat{k})|}{|\hat{i}+\hat{j}-2\hat{k}|}=\frac{1}{\sqrt{6}}.$$

10. (b) : Let $I = \int xe^{x^2}\,dx$

Put $x^2 = t \Rightarrow 2xdx = dt \Rightarrow x\,dx = \frac{dt}{2}$

$$\therefore \quad I=\frac{1}{2}\int e^t\,dt=\frac{e^t}{2}+C=\frac{e^{x^2}}{2}+C$$

11. If A is a non-singular square matrix of order $n \times n$, then $|\text{adj } A| = |A|^{n-1}$.
Here, $|\text{adj } A| = 324$
$\therefore \quad |A|^2 = 324 \Rightarrow |A| = 18$

12. Let $I=\int_0^{2\pi}\frac{dx}{e^{\sin x}+1}$ ...(i)

$$\Rightarrow \quad I=\int_0^{2\pi}\frac{dx}{e^{\sin(2\pi-x)}+1} \qquad \text{(By property)}$$

$$\Rightarrow \quad I=\int_0^{2\pi}\frac{dx}{e^{-\sin x}+1} \Rightarrow I=\int_0^{2\pi}\frac{e^{\sin x}}{e^{\sin x}+1}\,dx \quad \text{...(ii)}$$

Adding (i) and (ii), we get

$$2I=\int_0^{2\pi}1\cdot dx=2\pi \quad \therefore \quad I=\pi$$

OR

$$\int_0^1\frac{1}{2x-3}\,dx = \frac{1}{2}\Big[\log|2x-3|\Big]_0^1$$

$$= \frac{1}{2}\big[\log|-1| - \log|-3|\big]$$

$$= \frac{1}{2}[0 - \log 3] = -\frac{1}{2}\log 3$$

13. Given that, $\vec{a} = 2\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b} = 5\hat{i}-3\hat{j}+\hat{k}$

The projection of $\vec{b}$ on $\vec{a}$ $=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$

$$=\frac{(2\hat{i}+\hat{j}+2\hat{k})\cdot(5\hat{i}-3\hat{j}+\hat{k})}{\sqrt{(2)^2+(1)^2+(2)^2}}=\frac{10-3+2}{\sqrt{9}}=\frac{9}{3}=3$$

www.examrace.in

14. We have, $\left(\frac{d^3y}{dx^3}\right)^2 - 3\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^4 = y^4$

Clearly, it is a third order differential equation with degree 2.

$\therefore$ Required sum = 3 + 2 = 5

OR

We have, $(1 + y^2)dx = (\tan^{-1}y - x)dy$

$\Rightarrow \quad \frac{dx}{dy} + \frac{1}{1+y^2}\cdot x = \frac{\tan^{-1}y}{1+y^2}$

$\therefore$ Integrating factor $= e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$

15. Clearly, angle between $\vec{a}$ and $\vec{b} = \frac{\pi}{2}$

$\Rightarrow \quad \vec{a}\cdot\vec{b} = 0$

$\therefore \quad |\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b} = 1 + 1 + 0 = 2$

$\Rightarrow \quad |\vec{a}+\vec{b}| = \sqrt{2}$

16. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x \\ 3y \\ 4z \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \\ 8 \end{bmatrix}$

$\Rightarrow \quad 2x = 4, 3y = -6, 4z = 8 \Rightarrow x = 2, y = -2, z = 2$

$\Rightarrow \quad 3xyz = 3 \times 2 \times (-2) \times 2 = -24$

17. Since Mean value theorem is satisfied.

$\therefore \quad f'(c) = \frac{f(2) - f(1)}{2-1} \Rightarrow 2c - 2 = 1 \Rightarrow c = \frac{3}{2}$

18. $\because \quad y = e^{3x+7} \Rightarrow \frac{dy}{dx} = 3e^{3x+7}$

$\therefore \quad \left.\frac{dy}{dx}\right|_{x=0} = 3e^{3\times 0+7} = 3e^7$

OR

Let $f(x) = |x^2 + 2|$ and a be a real number.

$\lim\limits_{x\to a^+} f(x) = \lim\limits_{h\to 0} |(a+h)^2 + 2| = |a^2 + 2| = a^2 + 2$

$\lim\limits_{x\to a^-} f(x) = \lim\limits_{h\to 0} |(a-h)^2 + 2| = |a^2 + 2| = a^2 + 2$

Also, $f(a) = |a^2 + 2| = a^2 + 2$

$\therefore \quad \lim\limits_{x\to a^+} f(x) = \lim\limits_{x\to a^-} f(x) = f(a)$

Hence, $f(x) = |x^2 + 2|$ is a continuous function.

19. Marginal revenue $= \frac{dR}{dx}$

$= \frac{d}{dx}(15x^2 + 4x + 30) = 15 \times 2x + 4$

when $x = 5$, we have

Marginal revenue $= 15 \times 10 + 4 = 150 + 4 = 154$

Hence, the required marginal revenue is ₹ 154.

20. The given vectors are at right angles if their dot product vanishes.

$\therefore \quad (2\hat{i} - 4\hat{j} + \hat{k})\cdot(4\hat{i} - 8\hat{j} + \lambda\hat{k}) = 0$

$\Rightarrow \quad 8 + 32 + \lambda = 0 \Rightarrow \lambda = -40$

21. $\sin^{-1}\left[\cos\left(\sin^{-1}\frac{\sqrt{3}}{2}\right)\right] = \sin^{-1}\left[\cos\frac{\pi}{3}\right]$

$= \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

OR

We have,

L.H.S. $= \tan^{-1}\left(\frac{1-x}{1+x}\right) - \tan^{-1}\left(\frac{1-y}{1+y}\right)$

$= (\tan^{-1}1 - \tan^{-1}x) - (\tan^{-1}1 - \tan^{-1}y)$

$= \tan^{-1}y - \tan^{-1}x$

$= \tan^{-1}\left(\frac{y-x}{1+yx}\right)$

$= \sin^{-1}\left(\frac{y-x}{\sqrt{(1+yx)^2 + (y-x)^2}}\right)$

$= \sin^{-1}\left\{\frac{y-x}{\sqrt{(1+x^2)(1+y^2)}}\right\}$ = R.H.S.

22. Given matrix is singular if its determinant is zero.

$\therefore \quad \begin{vmatrix} -x & x & 2 \\ 2 & x & -x \\ x & -2 & -x \end{vmatrix} = 0$

$\Rightarrow \quad -x(-x^2 - 2x) - x(-2x + x^2) + 2(-4 - x^2) = 0$

$\Rightarrow \quad x^3 + 2x^2 + 2x^2 - x^3 - 8 - 2x^2 = 0$

$\Rightarrow \quad 2x^2 - 8 = 0 \Rightarrow x = \pm 2$

23. $\int_0^{\pi/4}\sqrt{1 + \sin 2x}\,dx$

$= \int_0^{\pi/4}\sqrt{\cos^2 x + \sin^2 x + 2\sin x\cos x}\,dx$

$= \int_0^{\pi/4}\sqrt{(\cos x + \sin x)^2}\,dx$

$= \int_0^{\pi/4}(\cos x + \sin x)\,dx = [\sin x - \cos x]_0^{\pi/4}$

$= \left(\sin\frac{\pi}{4} - \cos\frac{\pi}{4}\right) - (\sin 0 - \cos 0) = 1.$

24. We are given, adjacent sides of a parallelogram as $\vec{a} = \hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

www.examhive.in

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&3&4\\1&-1&1\end{vmatrix}$

$=(3+4)\hat{i}-(1-4)\hat{j}+(-1-3)\hat{k}$

$\Rightarrow\ \vec{a}\times\vec{b}=7\hat{i}+3\hat{j}-4\hat{k}$

Now, $|\vec{a}\times\vec{b}|=\sqrt{7^2+3^2+(-4)^2}$

$=\sqrt{49+9+16}=\sqrt{74}$

$\therefore$ Area of parallelogram

$=|\vec{a}\times\vec{b}|=\sqrt{74}$ sq. units.

25. Let the given points be A (4, 1, 2), B (5, λ, 0), C(2, 1, 1) and D(3, 3, –1).
Direction ratios of AB are (5 – 4, λ – 1, 0 – 2) *i.e.*, (1, λ – 1, –2)
Direction ratios of CD are (3 – 2, 3 –1, –1–1) *i.e.*, (1, 2, –2)
Now, given $AB \parallel CD$

$\therefore\quad \frac{1}{1}=\frac{\lambda-1}{2}=\frac{-2}{-2}$ or $\lambda-1=2\Rightarrow\lambda=3$

26. A four digit number formed using the digits 1, 2, 3 and 5 without repetitions will be divisible by 5 if and only if the unit's digit is 5.
So, the number of ways to fill the rest three places is 3!.
Further the total number of number that can be formed = 4!

$\therefore$ Required probability = $\frac{3!}{4!}=\frac{1}{4}$.

OR

Let A and B be the events of getting a white pearl in the first draw and a black pearl in the second draw.
Now, $P(A)$ = P(getting a white pearl in the first draw) $=\frac{12}{30}=\frac{2}{5}$

When second pearl is drawn without replacement, the probability that the second pearl is black is the conditional probability of the event B occurring when A has already occurred.

$\therefore\quad P(B\,|\,A)=\frac{18}{29}$

By multiplication rule of probability, we have

$P(A\cap B)=P(A).P(B\,|\,A)=\frac{2}{5}\times\frac{18}{29}=\frac{36}{145}$

27. One-One : Let (s_1, t_1) and $(s_2, t_2)\in S\times T$ such that $f(s_1, t_1)=f(s_2, t_2)$
$\Rightarrow\quad (t_1, s_1)=(t_2, s_2)\ \Rightarrow\ t_1=t_2$ and $s_1=s_2$
$\Rightarrow\quad (s_1, t_1)=(s_2, t_2)$
Thus, $f(s_1, t_1)=f(s_2, t_2)$
$\Rightarrow\quad (s_1, t_1)=(s_2, t_2)$ for all $(s_1, t_1), (s_2, t_2)\in S\times T$.
So, f is one-one.
Onto : Let (t, s) be an arbitrary element of $T\times S$.
Then, $t\in T$ and $s\in S$
$\Rightarrow\quad (s, t)\in S\times T$.
Thus, for all $(t, s)\in T\times S$ there exists $(s, t)\in S\times T$ such that $f(s, t)=(t, s)$
So, $f: S\times T\to T\times S$ is an onto function.
Hence, f is one-one and onto.

28. For $f(x)$ to be continuous at $x=0$, we must have $\lim\limits_{x\to0^-}f(x)=\lim\limits_{x\to0^+}f(x)=f(0)$

$\Rightarrow\quad \lim\limits_{x\to0^-}f(x)=\lim\limits_{x\to0^+}f(x)=a$ ...(i)

Now, $\lim\limits_{x\to0^-}f(x)=\lim\limits_{x\to0}\frac{1-\cos4x}{x^2}$

$\Rightarrow\quad \lim\limits_{x\to0^-}f(x)=\lim\limits_{x\to0}\frac{2\sin^2 2x}{x^2}$

$\Rightarrow\quad \lim\limits_{x\to0^-}f(x)=2\lim\limits_{x\to0}\left(\frac{\sin2x}{x}\right)^2$

$\Rightarrow\quad \lim\limits_{x\to0^-}f(x)=2\times4\cdot\lim\limits_{x\to0}\left(\frac{\sin2x}{2x}\right)^2=8(1)^2=8$ ...(ii)

Also, $\lim\limits_{x\to0^+}f(x)=\lim\limits_{x\to0}\frac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}$

$\Rightarrow\quad \lim\limits_{x\to0^+}f(x)=\lim\limits_{x\to0}\frac{\sqrt{x}}{16+\sqrt{x}-16}\cdot(\sqrt{16+\sqrt{x}}+4)$

$\Rightarrow\quad \lim\limits_{x\to0^+}f(x)=\lim\limits_{x\to0}\sqrt{16+\sqrt{x}}+4=4+4=8$ ...(iii)

From (i), (ii) and (iii), we get $a=8$

OR

Given, $y=\frac{\sqrt{1-x^2}\,(2x-3)^{1/2}}{(x^2+2)^{2/3}}$

Taking log on both sides, we get

$\log y=\frac{1}{2}\log(1-x^2)+\frac{1}{2}\log(2x-3)-\frac{2}{3}\log(x^2+2)$

On differentiating both sides w.r.t.x, we get

www.exambites.in

$$\frac{1}{y}\frac{dy}{dx}=\frac{1}{2(1-x^2)}(-2x)+\frac{1}{2(2x-3)}\times 2-\frac{2}{3}\times\frac{1}{x^2+2}\times 2x$$

$$\therefore\quad \frac{dy}{dx}=y\left\{-\frac{x}{1-x^2}+\frac{1}{2x-3}-\frac{4x}{3(x^2+2)}\right\}$$

$$\Rightarrow\quad \frac{dy}{dx}=\frac{\sqrt{1-x^2}\,(2x-3)^{1/2}}{(x^2+2)^{2/3}}\left\{-\frac{x}{1-x^2}+\frac{1}{2x-3}-\frac{4x}{3(x^2+2)}\right\}$$

29. Let $I=\int_0^{\pi}\frac{x}{1-\cos\alpha\sin x}dx$

$$\Rightarrow\quad I=\int_0^{\pi}\frac{(\pi-x)}{1-\cos\alpha\sin(\pi-x)}dx$$

$$\Rightarrow\quad I=\int_0^{\pi}\frac{\pi}{1-\cos\alpha\sin x}dx-\int_0^{\pi}\frac{x}{1-\cos\alpha\sin x}dx$$

$$\Rightarrow\quad I=\pi\int_0^{\pi}\frac{1}{1-\cos\alpha\sin x}dx-I$$

$$\Rightarrow\quad 2I=\pi\int_0^{\pi}\frac{1}{1-\cos\alpha\sin x}dx$$

$$\Rightarrow\quad 2I=\pi\int_0^{\pi}\frac{1+\tan^2(x/2)}{1+\tan^2(x/2)-2\cos\alpha\tan(x/2)}dx$$

$$\Rightarrow\quad I=\frac{\pi}{2}\int_0^{\pi}\frac{\sec^2(x/2)}{\tan^2(x/2)-2\cos\alpha\tan(x/2)+1}dx$$

Let $\tan\frac{x}{2}=t$

$$\therefore\quad d\left(\tan\frac{x}{2}\right)=dt\Rightarrow\sec^2\frac{x}{2}dx=2dt$$

Also, when $x=0$, $t=\tan 0=0$ and

when $x=\pi$, $t=\tan\frac{\pi}{2}=\infty$

$$\therefore\quad I=\frac{\pi}{2}\int_0^{\infty}\frac{2dt}{t^2-2t\cos\alpha+1}$$

$$\Rightarrow\quad I=\pi\int_0^{\infty}\frac{1}{(t-\cos\alpha)^2+(1-\cos^2\alpha)}dt$$

$$\Rightarrow\quad I=\pi\int_0^{\infty}\frac{1}{\sin^2\alpha+(t-\cos\alpha)^2}dt$$

$$\Rightarrow\quad I=\frac{\pi}{\sin\alpha}\left[\tan^{-1}\left(\frac{t-\cos\alpha}{\sin\alpha}\right)\right]_0^{\infty}$$

$$\Rightarrow\quad I=\frac{\pi}{\sin\alpha}\left[\tan^{-1}\infty-\tan^{-1}(-\cot\alpha)\right]$$

$$\Rightarrow\quad I=\frac{\pi}{\sin\alpha}\left[\frac{\pi}{2}+\tan^{-1}(\cot\alpha)\right]$$

$$\Rightarrow\quad I=\frac{\pi}{\sin\alpha}\left[\frac{\pi}{2}+\tan^{-1}\left\{\tan\left(\frac{\pi}{2}-\alpha\right)\right\}\right]$$

$$\Rightarrow\quad I=\frac{\pi}{\sin\alpha}\left[\frac{\pi}{2}+\frac{\pi}{2}-\alpha\right]=\frac{\pi(\pi-\alpha)}{\sin\alpha}$$

30. $(1+y^2)\,(1+\log x)\,dx+x\,dy=0$

$$\Rightarrow\quad (1+y^2)(1+\log x)dx=-x\,dy$$

$$\Rightarrow\quad \frac{(1+\log x)}{x}dx=\frac{-dy}{1+y^2}$$

Integrating both sides, we get

$$\int\left(\frac{1+\log x}{x}\right)dx=-\int\frac{dy}{1+y^2}$$

Put $1+\log x=t\Rightarrow\frac{1}{x}dx=dt$

$$\therefore\quad \int t\,dt=-\int\frac{dy}{1+y^2}$$

$$\Rightarrow\quad \frac{t^2}{2}=-\tan^{-1}y+c$$

$$\Rightarrow\quad (1+\log x)^2=-2\tan^{-1}y+2c$$

$$\Rightarrow\quad 2\tan^{-1}y=-(1+\log x)^2+2c$$

$$\Rightarrow\quad 2\tan^{-1}y=C-(1+\log x)^2\qquad[\text{Here, } C=2c]$$

31. We know that the lines

$$\frac{x-x_1}{l_1}=\frac{y-y_1}{m_1}=\frac{z-z_1}{n_1}$$

and $\frac{x-x_2}{l_2}=\frac{y-y_2}{m_2}=\frac{z-z_2}{n_2}$

are coplanar, if

$$\begin{vmatrix}x_2-x_1 & y_2-y_1 & z_2-z_1\\ l_1 & m_1 & n_1\\ l_2 & m_2 & n_2\end{vmatrix}=0$$

The equations of the given lines are

$$\frac{x-(a-d)}{\alpha-\delta}=\frac{y-a}{\alpha}=\frac{z-(a+d)}{\alpha+\delta}\text{ and}$$

$$\frac{x-(b-c)}{\beta-\gamma}=\frac{y-b}{\beta}=\frac{z-(b+c)}{\beta+\gamma}$$

Here, $x_1=a-d$, $y_1=a$, $z_1=a+d$

$x_2=b-c$, $y_2=b$, $z_2=b+c$

$l_1=\alpha-\delta$, $m_1=\alpha$, $n_1=\alpha+\delta$; $l_2=\beta-\gamma$, $m_2=\beta$, $n_2=\beta+\gamma$

www.exambites.in

$$\therefore \quad \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}$$

$$= \begin{vmatrix} b-c-a+d & b-a & b+c-a-d \\ \alpha-\delta & \alpha & \alpha+\delta \\ \beta-\gamma & \beta & \beta+\gamma \end{vmatrix}$$

Applying $C_1 \to C_1 + C_3$, we get

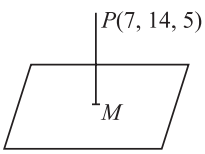
$$\begin{vmatrix} 2(b-a) & b-a & b+c-a-d \\ 2\alpha & \alpha & \alpha+\delta \\ 2\beta & \beta & \beta+\gamma \end{vmatrix} = 0$$

[$\because$ C_1 and C_2 are proportional]

Hence, given lines are coplanar.

OR

Let M be the foot of the perpendicular from the point $P(7, 14, 5)$ on the plane $2x + 4y - z = 2$. Then, PM is normal to the plane. So, its direction ratios are 2, 4, –1.



Since PM passes through $P(7, 14, 5)$. Therefore, its equation is

$$\frac{x-7}{2} = \frac{y-14}{4} = \frac{z-5}{-1} = r(\text{say})$$

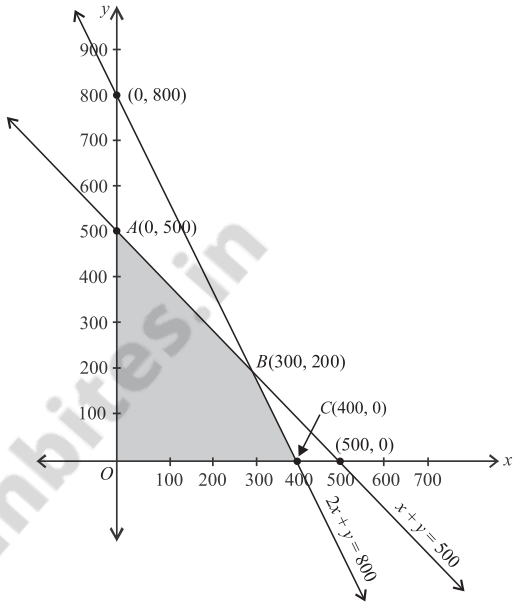
Let the coordinates of M be $(2r + 7, 4r + 14, -r + 5)$.
Since M lies on the plane $2x + 4y - z = 2$.
Therefore, $2(2r + 7) + 4(4r + 14) - (-r + 5) = 2$
$\Rightarrow \quad 21r + 63 = 0 \Rightarrow r = -3$
So, the coordinates of M are (1, 2, 8)
Now, PM = Length of the perpendicular from P

$$\Rightarrow \quad PM = \sqrt{(7-1)^2 + (14-2)^2 + (5-8)^2} = 3\sqrt{21}$$

32. Let plants X and Y be grown in x and y hectares.
So, $x \geq 0$ and $y \geq 0$
$x + y \leq 500$
Contribution of oxygen by the plants
= 60% of x + 40% of y

$$\therefore \quad Z = \frac{6x}{10} + \frac{4y}{10} = 0.6x + 0.4y$$

Also, Amount of liquid herbicides required = $(20x + 10y)$ litres
Given $20x + 10y \leq 8000$
$\Rightarrow \quad 2x + y \leq 800$
The L.P.P for given problem is
Maximise, $Z = 0.6x + 0.4y$
such that $x + y \leq 500$
and $2x + y \leq 800$
$x, y \geq 0$
Sketching a graph for the above L.P.P, we get the region shown in the figure.
Solving $x + y = 500$ and $2x + y = 800$, we get $B(300, 200)$



Corner points	$Z = 0.6x + 0.4y$
$O(0,0)$	$Z = 0 + 0 = 0$
$C(400,0)$	$Z = 0.6 \times 400 + 0.4 \times 0 = 240$
$B(300,200)$	$Z = 0.6 \times 300 + 0.4 \times 200 = 260$ (Maximum)
$A(0,500)$	$Z = 0.6 \times 0 + 0.4 \times 500 = 200$

Maximum production of oxygen will be achieved when plant X is planted in 300 hectares and plant Y is planted in 200 hectares of land.

33. We have, $A = IA$

$$\therefore \quad \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -1 \\ 3 & -5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_1 \to R_1 - R_2$, we get

$$\begin{bmatrix} 1 & -1 & -1 \\ 2 & 0 & -1 \\ 3 & -5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$, we get

www.exambites.in

$$\begin{bmatrix}1 & -1 & -1\\ 0 & 2 & 1\\ 0 & -2 & 3\end{bmatrix}=\begin{bmatrix}1 & -1 & 0\\ -2 & 3 & 0\\ -3 & 3 & 1\end{bmatrix}A$$

Applying $R_2 \to 1/2R_2$, we get

$$\begin{bmatrix}1 & -1 & -1\\ 0 & 1 & 1/2\\ 0 & -2 & 3\end{bmatrix}=\begin{bmatrix}1 & -1 & 0\\ -1 & 3/2 & 0\\ -3 & 3 & 1\end{bmatrix}A$$

Applying $R_1 \to R_1 + R_2$, $R_3 \to R_3 + 2R_2$, we get

$$\begin{bmatrix}1 & 0 & -1/2\\ 0 & 1 & 1/2\\ 0 & 0 & 4\end{bmatrix}=\begin{bmatrix}0 & 1/2 & 0\\ -1 & 3/2 & 0\\ -5 & 6 & 1\end{bmatrix}A$$

Applying $R_3 \to 1/4\ R_3$, we get

$$\begin{bmatrix}1 & 0 & -1/2\\ 0 & 1 & 1/2\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}0 & 1/2 & 0\\ -1 & 3/2 & 0\\ -5/4 & 3/2 & 1/4\end{bmatrix}A$$

Applying $R_1 \to R_1 + (1/2)\ R_3$, $R_2 \to R_2 + (-1/2)\ R_3$, we get

$$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}-5/8 & 5/4 & 1/8\\ -3/8 & 3/4 & -1/8\\ -5/4 & 3/2 & 1/4\end{bmatrix}A$$

Hence, $A^{-1}=\begin{bmatrix}-5/8 & 5/4 & 1/8\\ -3/8 & 3/4 & -1/8\\ -5/4 & 3/2 & 1/4\end{bmatrix}$.

OR

Given, $6A^{-1} = A^2 + cA + dI$

$\Rightarrow \quad 6A^{-1}\cdot A = A^2A + cA\cdot A + dI\cdot A$

$\Rightarrow \quad 6I = A^3 + cA^2 + dA$

$\Rightarrow \quad A^3 + cA^2 + dA - 6I = O$ ...(i)

Now, $A^2=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}$

$$=\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 5\\ 0 & -10 & 14\end{bmatrix}$$

$$A^3=\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 5\\ 0 & -10 & 14\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & -11 & 19\\ 0 & -38 & 46\end{bmatrix}$$

$\therefore$ From equation (i), we have

$$\begin{bmatrix}1 & 0 & 0\\ 0 & -11 & 19\\ 0 & -38 & 46\end{bmatrix}+c\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 5\\ 0 & -10 & 14\end{bmatrix}$$

$$+d\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & -2 & 4\end{bmatrix}-6\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=O$$

$$\Rightarrow \begin{bmatrix}1+c+d-6 & 0 & 0\\ 0 & -11-c+d-6 & 19+5c+d\\ 0 & -38-10c-2d & 46+14c+4d-6\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$\Rightarrow \quad 1 + c + d - 6 = 0, -11 - c + d - 6 = 0$

$\Rightarrow \quad c + d = 5, -c + d = 17 \Rightarrow c = -6, d = 11$

34. Let the required point be (x_1, y_1).

Equation of the curve is $9y^2 = x^3$.

Since (x_1, y_1) lies on the curve. Therefore,

$$9y_1^2 = x_1^3 \qquad ...(i)$$

Now, $9y^2 = x^3$

$$\Rightarrow \quad \frac{dy}{dx}=\frac{x^2}{6y} \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)}=\frac{x_1^2}{6y_1}$$

Since the normal to the curve at (x_1, y_1) makes equal intercepts with the coordinate axes.

Therefore, slope of the normal = ±1

$$\Rightarrow \quad -\frac{1}{\left(\frac{dy}{dx}\right)_{(x_1,y_1)}}=\pm 1 \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)}=\pm 1$$

$$\Rightarrow \quad \frac{x_1^2}{6y_1}=\pm 1 \Rightarrow x_1^2=\pm\, 6y_1 \Rightarrow x_1^4 = 36y_1^2$$

$$\Rightarrow \quad x_1^4 = 36\left(\frac{x_1^3}{9}\right) \qquad \text{[Using (i)]}$$

$$\Rightarrow \quad x_1^4 = 4x_1^3 \Rightarrow x_1^3(x_1 - 4) = 0 \Rightarrow x_1 = 0, 4$$

Putting $x_1 = 0$ in (i), we get $9y_1^2 = 0 \Rightarrow y_1 = 0$

Putting $x_1 = 4$ in (i), we get $9y_1^2 = 4^3 \Rightarrow y_1 = \pm\frac{8}{3}$

But, the line making equal intercepts with the coordinate axes cannot pass through the origin.

Hence, the required points are (4, 8/3) and (4, –8/3).

www.exambites.in

OR

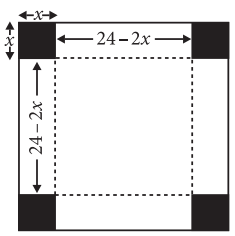
Let x cm be the length of a side of the square which is cut-off from each corner of the tin plate. Then, sides of the box as shown in figure are $24 - 2x$, $24 - 2x$ and x.

Let V be the volume of the box. Then,

$V = (24 - 2x)^2 x = 4x^3 - 96x^2 + 576x$

$$\Rightarrow \frac{dV}{dx} = 12x^2 - 192x + 576 \text{ and } \frac{d^2V}{dx^2} = 24x - 192$$

For maximum or minimum values of V, we must have $\frac{dV}{dx} = 0$



$\Rightarrow \quad 12x^2 - 192x + 576 = 0$

$\Rightarrow \quad x^2 - 16x + 48 = 0$

$\Rightarrow \quad (x - 12)(x - 4) = 0$

$\Rightarrow \quad x = 12, 4$

But, $x = 12$ is not possible.

Therefore, $x = 4$.

and $\left[\frac{d^2V}{dx^2}\right]_{x=4} = -96 < 0$

Hence, volume is maximum when $x = 4$.

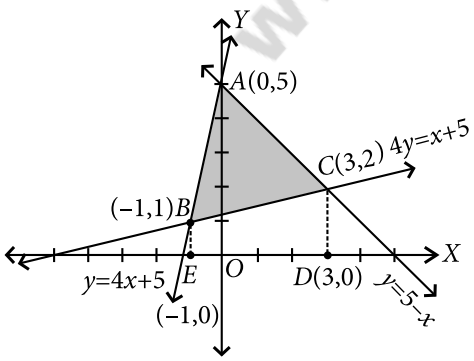
35. We have, $y = 4x + 5$, $y = 5 - x$ and $4y = x + 5$

On solving $y = 4x + 5$ and $y = 5 - x$, we get $x = 0$ and $y = 5$

On solving $y = 4x + 5$ and $4y = x + 5$, we get $x = -1$ and $y = 1$

On solving $y = 5 - x$ and $4y = x + 5$, we get $x = 3$ and $y = 2$

Let us draw a rough sketch of the line on the graph.



$\therefore$ Area of shaded region

$= \text{Area}(EBAOE) + \text{Area}(ACDOA) - \text{Area}(EBCDOE)$

$$= \int_{-1}^{0} (4x+5)\,dx + \int_{0}^{3} (5-x)\,dx - \frac{1}{4}\int_{-1}^{3} (x+5)\,dx$$

$$= \left[2x^2 + 5x\right]_{-1}^{0} + \left[5x - \frac{x^2}{2}\right]_{0}^{3} - \frac{1}{4}\left[\frac{x^2}{2} + 5x\right]_{-1}^{3}$$

$$= [0 - (2 - 5)] + \left[15 - \frac{9}{2}\right] - \frac{1}{4}\left[\left(\frac{9}{2} + 15\right) - \left(\frac{1}{2} - 5\right)\right]$$

$$= \frac{27}{2} - \frac{1}{4} \times \frac{48}{2} = \frac{27}{2} - 6 = \frac{15}{2} \text{ sq. units}$$

36. Let E = Event of drawing a heart.

Then, $P(E) = \frac{13}{52} = \frac{1}{4}$ and $P(\bar{E}) = \left(1 - \frac{1}{4}\right) = \frac{3}{4}$

Let X = number of hearts in a draw.

Then, $X = 0, 1, 2$ or 3

$X = x_i$	p_i
0	$\frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{27}{64}$
1	$\frac{1}{4} \times \frac{3}{4} \times \frac{3}{4} + \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} + \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} = \frac{27}{64}$
2	$\frac{1}{4} \times \frac{1}{4} \times \frac{3}{4} + \frac{1}{4} \times \frac{3}{4} \times \frac{1}{4} + \frac{3}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{9}{64}$
3	$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$

$\therefore$ Mean $(\mu) = \Sigma x_i p_i$

$$= \left(0 \times \frac{27}{64}\right) + \left(1 \times \frac{27}{64}\right) + \left(2 \times \frac{9}{64}\right) + \left(3 \times \frac{1}{64}\right) = \frac{3}{4}$$

Variance, $\sigma^2 = \Sigma x_i^2 p_i - \mu^2$

$$= \left(0 \times \frac{27}{64}\right) + \left(1^2 \times \frac{27}{64}\right) + \left(2^2 \times \frac{9}{64}\right) + \left(3^2 \times \frac{1}{64}\right) - \frac{9}{16} = \frac{9}{16}$$

www.examrace.in

PRACTICE PAPER 3
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. Find the principal value of

$$\cos^{-1}\left(\frac{-1}{2}\right)+2\sin^{-1}\left(\frac{-1}{2}\right).$$

(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{3\pi}{4}$ (d) $\frac{5\pi}{8}$

2. Solution set of the inequality $x \geq 0$ is

(a) half plane on the left of Y-axis

(b) half plane on the right of Y-axis excluding the points on Y-axis

(c) half plane on the right of Y-axis including the points on Y-axis

(d) none of these

3. The distance 's' metres covered by a body in t seconds, is given by $s = 3t^2 - 12t + 5$. The body will stop after

(a) 1 s (b) 3 s (c) 2 s (d) 4 s

4. If points (2, 0), (0, 5) and (x, y) are collinear, then

(a) $5x + 2y = 1$ (b) $5x + 2y = 10$

(c) $x + y = 10$ (d) $2x + 5y = 1$

5. Evaluate : $\int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}}\,dx$

(a) $\frac{x^2}{3}+C$ (b) $\frac{x^3}{3}+C$

(c) $\frac{x^2}{2}+C$ (d) $\frac{x^2}{4}+C$

6. If the direction ratios of a line are 1, –3, 2, then its direction cosines are

(a) $\frac{1}{\sqrt{14}}, \frac{-3}{\sqrt{14}}, \frac{2}{\sqrt{14}}$

(b) $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$

(c) $\frac{-1}{\sqrt{14}}, \frac{-3}{\sqrt{14}}, \frac{2}{\sqrt{14}}$

(d) $\frac{-1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{-3}{\sqrt{14}}$

7. Integrating factor of the differential equation $\frac{dy}{dx} + y\tan x - \sec x = 0$ is

(a) $\cos x$ (b) $\sec x$ (c) $e^{\cos x}$ (d) $e^{\sec x}$

8. $\int \frac{(x+3)e^x}{(x+4)^2}\,dx$ is equal to

(a) $\frac{1}{(x+4)^2}+C$ (b) $\frac{e^x}{(x+4)^2}+C$

(c) $\frac{e^x}{x+4}+C$ (d) $\frac{e^x}{x+3}+C$

9. What is order and degree of $y + \frac{dy}{dx} = \frac{1}{4}\int y\,dx$?

(a) Order = 2 and degree = 1

(b) Order = 1 and degree = 2

(c) Order = 2 and degree = 3
(d) Order = 3 and degree = 1

10. If there is an error of 2% in measuring the length of a simple pendulum, then percentage error in its period is
(a) 1% (b) 2% (c) 3% (d) 4%

(Q.11 - Q.15) Fill in the blanks.

11. The principal value of

$$\tan^{-1}(-\sqrt{3})+\sec^{-1}(-2)-\operatorname{cosec}^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

is_____ .

12. If $\begin{vmatrix} 2x+5 & 3 \\ 5x+2 & 9 \end{vmatrix} = 0$, then x = _____ .

OR

The cofactor of element a_{12} of the matrix $\begin{bmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{bmatrix}$ is_____ .

13. $\int_{\pi/4}^{\pi/2} \cos 2x\, dx =$ _____ .

14. If lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-5}{1}=\frac{z-6}{-5}$ are mutually perpendicular, then the value of k is_____.

OR

The direction cosines of the line joining the points (4, 3, –5) and (–2, 1, –8) are_____ .

15. The length of perpendicular from the origin to the plane $\vec{r}\cdot(3\hat{i}-4\hat{j}+12\hat{k})=5$ is_____.

(Q.16 – Q.20) Answer the following questions.

16. Find x, if $\begin{bmatrix} 8 & 6x \\ z & 2y \end{bmatrix}^T = \begin{bmatrix} 8 & 4 \\ 12 & 6 \end{bmatrix}$.

17. If $x = a\cos^4\theta$, $y = a\sin^4\theta$, then find $\frac{dy}{dx}$ at $\theta=\frac{3\pi}{4}$.

18. What is the cosine of the angle which the vector $\sqrt{2}\,\hat{i}+\hat{j}+\hat{k}$ makes with y-axis?

19. If $f(x)=-\sqrt{25-x^2}$, then find the value of $\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}$.

OR

Can Rolle's theorem be applied to the function $|x|$ in the interval [–2, 2] ?

20. If $|\vec{a}|=2, |\vec{b}|=5$ and $|\vec{a}\times\vec{b}|=8$, then find the value of $\vec{a}\cdot\vec{b}$.

SECTION - B

21. If $\tan^{-1}\left(\frac{x-1}{x+2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$, then find the value of x.

OR

Prove that : $\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}=\tan^{-1}\frac{2}{9}$

22. Find the value of $\begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix}$.

23. Evaluate : $\int_0^1 \frac{\tan^{-1}x}{1+x^2}dx$

24. Find the area of the parallelogram determined by the vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$.

25. Show that the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{5}=\frac{y-1}{2}=z$ intersect. Find their point of intersection.

26. A die is thrown three times, if the first throw is a four, find the chance of getting 15 as the sum.

OR

If $P(A)=\frac{3}{8}, P(B)=\frac{1}{2}$ and $P(A\cap B)=\frac{1}{4}$, find $P(\overline{A}/\overline{B})$ and $P(\overline{B}/\overline{A})$.

www.exambrief.in

SECTION - C

27. Let $A = \{1, 2, 3, ..., 9\}$ and R be the relation in $A \times A$ defined by $(a, b)\ R\ (c, d)$ if $a + d = b + c$ for $(a, b), (c, d)$ in $A \times A$. Prove that R is an equivalence relation.

28. If $f(x) = \begin{cases} 1 & , \text{ if } x \le 3 \\ ax + b & , \text{ if } 3 < x < 5 \\ 7 & , \text{ if } 5 \le x \end{cases}$

Determine the values of a and b so that $f(x)$ is continuous.

OR

Differentiate the function

$\sqrt{\log\left\{\sin\left(\frac{x^2}{3} - 1\right)\right\}}$ w.r.t. x.

29. Prove that : $\int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx = \pi^2$

30. Solve the following initial value problem.

$(xe^{y/x} + y)dx = xdy, y(1) = 1$

31. Show that the lines $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}$ and $\frac{x+2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$ do not intersect.

OR

A variable plane is at a constant distance p from the origin and meets the coordinate axes in A, B, C. Show that the locus of the centroid of the tetrahedron $OABC$ is $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$.

32. A company manufactures two types of chemicals A and B. Each chemical requires two types of raw material P and Q. The table below shows number of units of P and Q required to manufacture one unit of A and one unit of B and the total availability of P and Q.

Raw material \ Chemical	A	B	Availability
P	3	2	120
Q	2	5	160

The company gets profit of ₹ 350 and ₹ 400 by selling one unit of A and one unit of B respectively. How many units of the chemicals A and B respectively should be manufactured so that the company gets maximum profit? (Assume that the entire production of A and B can be sold.)

SECTION - D

33. Find the area bounded by the curves $y = -\sqrt{4 - x^2}$, $x^2 = -\sqrt{2}y$ and $x = y$.

34. If the tangent at $P(1, 1)$ on $y^2 = x(2 - x)^2$ meets the curve again at Q, then find the coordinates of Q.

OR

A figure consists of a semi-circle with a rectangle on its diameter. Given the perimeter of the figure as P units, find its dimensions in order that the area may be maximum.

35. Find the inverse of the following matrix by using elementary row transformations.

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

OR

Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then show that

$A^2 - 4A - 5I_3 = 0$. Hence find A^{-1}.

36. A chairman is biased so that he selects his relatives for a job 3 times as likely as others. If there are 3 posts for a job, find the probability distribution for selection of persons other than their relatives.

www.exambites.in

Hints & Solutions

1. (a) **2.** (c) **3.** (c) **4.** (b)
5. (b) **6.** (a) **7.** (b) **8.** (c)
9. (a) **10.** (a) **11.** 0 **12.** –13 **OR** 0

13. $\frac{-1}{2}$ **14.** $\frac{-10}{7}$ **OR** $\left(\frac{6}{7},\frac{2}{7},\frac{3}{7}\right)$

15. $\frac{5}{13}$ units **16.** 2 **17.** –1

18. $\frac{1}{2}$ **19.** $\frac{1}{\sqrt{24}}$ **OR** NO

20. 6 **21.** $\pm\sqrt{\frac{5}{2}}$ **22.** 0 **23.** $\frac{\pi^2}{32}$

24. $8\sqrt{3}$ sq. units

25. (–1, –1, –1) **26.** $\frac{1}{18}$ **OR** $\frac{3}{4}$ and $\frac{3}{5}$

28. $a = 3, b = -8$ **OR** $\dfrac{x\cot\left(\frac{x^2}{3}-1\right)}{3\sqrt{\log\left\{\sin\left(\frac{x^2}{3}-1\right)\right\}}}$

30. $\log x + e^{-y/x} = \frac{1}{e}$

32. $\frac{280}{11}$ units of chemical A and $\frac{240}{11}$ units of chemical B should be manufactured.

33. $\left(\pi+\frac{1}{3}\right)$ sq. units

34. $\left(\frac{9}{4}, \frac{3}{8}\right)$ **OR** $\frac{2P}{\pi+4}$ units, $\frac{P}{\pi+4}$ units

35. $\frac{1}{11}\begin{bmatrix}-1 & -3 & 5\\ -2 & 5 & -1\\ 7 & -1 & -2\end{bmatrix}$ **OR** $\frac{1}{5}\begin{bmatrix}-3 & 2 & 2\\ 2 & -3 & 2\\ 2 & 2 & -3\end{bmatrix}$

36.

x	0	1	2	3
$P(x)$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{3}{64}$	$\frac{1}{64}$

www.exambires.in

PRACTICE PAPER 4
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. Using principal values, write the value of $2\cos^{-1}\frac{1}{2} + 3\sin^{-1}\frac{1}{2}$.

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{7\pi}{6}$ (d) $\frac{5\pi}{6}$

2. If $\vec{a} = \hat{i} + \hat{j}; \vec{b} = \hat{j} + \hat{k}; \vec{c} = \hat{k} + \hat{i}$, then find a unit vector in the direction of $\vec{a} + \vec{b} + \vec{c}$.

(a) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{6}}$ (b) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{2}}$

(c) $\frac{\hat{i}+\hat{j}+\hat{k}}{6}$ (d) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

3. If A is a matrix of order 2×3 and B is a matrix of order 3×5, what is the order of matrix $(AB)'$?

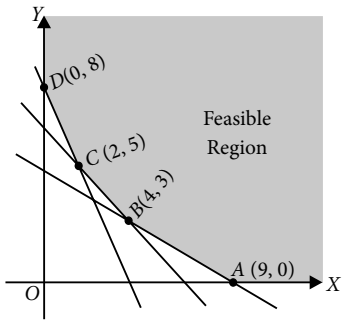
(a) 3×2 (b) 2×5
(c) 3×3 (d) 5×2

4. Find the second order derivative of $\tan^{-1}x$.

(a) $\frac{1}{1+x^2}$ (b) $\frac{2x}{1+x^2}$

(c) $\frac{-2x}{(1+x^2)^2}$ (d) $\frac{1}{(1+x^2)^2}$

5. Feasible region for an LPP is shown shaded in the following figure. Minimum of $Z = 4x + 3y$ occurs at the point



(a) (0, 8) (b) (2, 5)
(c) (4, 3) (d) (9, 0)

6. Point of intersection of the lines $\frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ and $\frac{x-4}{2} = \frac{y-0}{0} = \frac{z+1}{3}$ is

(a) (0, 1, –1) (b) (1, 1, –1)
(c) (4, 0, –1) (d) (3, 4, 2)

7. Find the order and the degree of the differential equation $x^2\frac{d^2y}{dx^2} = \left\{1+\left(\frac{dy}{dx}\right)^2\right\}^4$.

(a) order = 1 and degree = 2
(b) order = 2 and degree = 1
(c) order = 4 and degree = 2
(d) order = 2 and degree = 4

8. The equation of a line passing through the point (– 3, 2, –4) and equally inclined to the axes is

(a) $x - 3 = y + 2 = z - 4$

(b) $x + 3 = y - 2 = z + 4$

(c) $\frac{x+3}{1} = \frac{y-2}{2} = \frac{z+4}{3}$

(d) None of these

9. Evaluate :

$$\int (3\sin x - 2\cos x + 4\sec^2 x - 5\operatorname{cosec}^2 x)\, dx$$

(a) $-3\cos x - 2\sin x + 4\tan x + 5\cot x + C$

(b) $3\cos x + 2\sin x + 4\tan x + 5\cot x + C$

(c) $3\cos x + 2\sin x - 4\tan x - 5\cot x + C$

(d) $-3\cos x - 2\sin x - 4\tan x - 5\cot x + C$

10. A balloon, which always remains spherical, has a variable diameter $\frac{3}{2}(2x+3)$. Determine the rate of change of volume with respect to x.

(a) $\frac{27\pi}{8}(2x+3)^2$ (b) $\frac{9\pi}{10}(2x+3)^2$

(c) $\frac{27\pi}{8}(3x+2)$ (d) $\frac{9\pi}{16}(3x+2)$

(Q.11 - Q.15) Fill in the blanks.

11. If A and B are invertible matrices of order 3, $|A| = 2$ and $|(AB)^{-1}| = -\frac{1}{6}$, then the value of $|B|$ is_____.

12. The slope of the tangent to the curve $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at the point (1, 3) is_____.

OR

The interval on which the function $f(x) = 2x^3 + 9x^2 + 12x - 1$ is decreasing is _____ .

13. $\int_{-1/2}^{1/2} \cos x \log\left(\frac{1+x}{1-x}\right) dx$ _____ .

14. The angle between the planes $3x + 2y + z - 5 = 0$ and $x + y - 2z - 3 = 0$ is $\cos^{-1}\left(\frac{3}{2\sqrt{k}}\right)$.

Then k = _____ .

OR

Write the equation of the line parallel to the line $\frac{x-2}{-3} = \frac{y+3}{2} = \frac{z+5}{6}$ and passing through the point (1, 2, 3) is _____ .

15. If $\overrightarrow{AB} \times \overrightarrow{AC} = 2\hat{i} - 4\hat{j} + 4\hat{k}$, then the area of ΔABC is_____ .

(Q.16 - Q.20) Answer the following questions.

16. If $f(x) = \frac{\sqrt{4+x} - 2}{x}$, $x \neq 0$ be continuous at $x = 0$, then find the value of $f(0)$.

17. Evaluate : $\int_0^{1.5} [x]dx$ (where $[\cdot]$ is a greatest integer function)

18. Determine the sum of order and degree of the differential equation $\sqrt{x + \left(\frac{dy}{dx}\right)^2} = a\left(\frac{d^2y}{dx^2}\right)^{\frac{1}{3}}$.

19. If $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}$, find $(\vec{a} + 3\vec{b}) \cdot (2\vec{a} - \vec{b})$.

20. Let R be a relation in the set of natural numbers N, defined by $R = \{(a, b) \in N \times N : a < b\}$. Is relation R reflexive? Give reason.

OR

Show that the function $f : R \to R$ defined by $f(x) = 6^x + 6^{|x|}$ is many one and into.

SECTION - B

21. Solve the following equation :

$$\sin^{-1}\frac{3x}{5} + \sin^{-1}\frac{4x}{5} = \sin^{-1} x$$

OR

Solve the following equation :

$$\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$$

22. If A is a square matrix of order 3 such that $|\text{adj } A| = 64$. Find $|A|$.

23. Evaluate : $\int \frac{\cos x}{\sqrt{\sin^2 x - 2\sin x - 3}} dx$

www.exambits.in

24. If $\vec{a} = 3\hat{i} - 4\hat{j} - 4\hat{k}$, $\vec{b} = 2\hat{i} - 4\hat{j} - 3\hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, find $|3\vec{a} - 2\vec{b} + 4\vec{c}|$.

25. Find the distance of point $2\hat{i} + \hat{j} - \hat{k}$ from the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 4\hat{k}) = 9$.

26. Find the probability of drawing
(i) a diamond card (ii) an ace card in each of the two consecutive draws from a well shuffled pack of cards, if the card drawn is not replaced after the first draw.

OR

A and B are two events such that $P(A) = 0.3$ and $P(A \cup \bar{B}) = 0.8$. If A and B are independent events, find $P(B)$.

SECTION - C

27. Prove that the relation 'congruence modulo m' on the set Z of all integers is an equivalence relation.

28. If $f(x) = x \tan^{-1} x$, find $f'(\sqrt{3})$ by first principle.

OR

Show that the function

$$f(x) = \begin{cases} \left(\dfrac{e^{1/x} - 1}{e^{1/x} + 1}\right), & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

is discontinuous at $x = 0$.

29. Evaluate : $\int \dfrac{\sin x}{\sqrt{1 + \sin x}} dx$

30. Solve the following initial value problem:

$$2x^2 \frac{dy}{dx} - 2xy + y^2 = 0, y(e) = e$$

31. Find the shortest distance between lines

$\vec{r} = 6\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$

and $\vec{r} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$.

OR

Find the image of the point (1, 6, 3) in the line

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}.$$

32. If a young man rides his motorcycle at 25 km/hr, he had to spend ₹ 2 per km on petrol. If he rides at a faster speed of 40 km/hr. The petrol cost increases at ₹ 5 per km. He had ₹ 100 to spend on petrol and wishes to find what is the maximum distance he can travel in one hour. Express this as an LPP and solve it graphically.

SECTION - D

33. Write $A = \begin{bmatrix} 4 & 2 & -1 \\ 3 & 5 & 7 \\ 1 & -2 & 1 \end{bmatrix}$ as the sum of a symmetric matrix B and a skew-symmetric matrix C.

OR

Find the inverse of the matrix $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 7 & 2 \\ 2 & 4 & 1 \end{bmatrix}$ by using elementary row transformations.

34. Find the coordinates of the points on the curve $y = x^2 + 3x + 4$, the tangents at which pass through the origin.

OR

A wire of length 36 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a circle. What should be the lengths of the two pieces, so that the combined area of the square and the circle is minimum?

35. Find the area of the region bounded by the curve $y = \sqrt{16 - x^2}$ and x-axis.

36. Consider the experiment of tossing a coin. If the coin shows head toss it again but if it shows tail then throw a die. Find the conditional probability of the event, 'the die shows a number greater than 4, given that 'there is at least one tail'.

www.exambites.in

Hints & Solutions

1. (c) **2.** (d) **3.** (d) **4.** (c)
5. (b) **6.** (c) **7.** (b) **8.** (b)
9. (a) **10.** (a) **11.** -3
12. 2 **OR** $[-2, -1]$ **13.** 0
14. 21 **OR** $\frac{x-1}{-3}=\frac{y-2}{2}=\frac{z-3}{6}$
15. 3 sq. units **16.** 0.25 **17.** 0.5
18. 4 **19.** -15

20. NO, $(a, a)\notin R$ because a is not less than a.

21. 0, 1, -1 OR $\frac{1}{6}$ **22.** ± 8

23. $\log\left|(\sin x-1)+\sqrt{\sin^2 x-2\sin x-3}\right|+C$

24. $\sqrt{197}$

25. $\frac{13}{\sqrt{21}}$ units

26. (i) $\frac{1}{17}$ (ii) $\frac{1}{221}$ **OR** $\frac{2}{7}$

28. $\frac{3\sqrt{3}+4\pi}{12}$

29. $2\left(\sin\frac{x}{2}-\cos\frac{x}{2}\right)-\frac{1}{\sqrt{2}}\log\tan\left|\left(\frac{x}{4}+\frac{\pi}{8}\right)\right|+C$

30. $y=\frac{2x}{1+\log|x|}$

31. 9 units **OR** (1, 0, 7)

32. 30 km

33. $\begin{bmatrix} 4 & 5/2 & 0 \\ 5/2 & 5 & 5/2 \\ 0 & 5/2 & 1 \end{bmatrix}+\begin{bmatrix} 0 & -1/2 & -1 \\ 1/2 & 0 & 9/2 \\ 1 & -9/2 & 0 \end{bmatrix}$

OR

$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 2 & -5 \end{bmatrix}$

34. (2, 14) and (–2, 2

OR

$\frac{36\pi}{\pi+4}$ m and $\frac{144}{\pi+4}$ m

35. 8π sq. units

36. $\frac{2}{9}$

www.exambites.in

PRACTICE PAPER 5
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. Find the principal value of $\text{cosec}^{-1}\left(\frac{-2}{\sqrt{3}}\right)$.

(a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

2. Evaluate : $\int (e^x+1)^2 e^x dx$

(a) $e^{3x}+C$ (b) $\frac{(e^x+1)^3}{3}+C$

(c) $\frac{(e^x+1)^2}{2}+C$ (d) $\frac{e^x}{3}+C$

3. For what choice of a and b, the function $f(x)=\begin{cases} x^2 & , \ x \le c \\ ax+b & , \ x > c \end{cases}$ is differentiable at $x = c$?

(a) $a = c, b = c$ (b) $a = c, b = -c$

(c) $a = -c^2, b = 2c$ (d) $a = 2c, b = -c^2$

4. If $\angle A = 90°$ in the ΔABC, then $\tan^{-1}\left(\frac{c}{a+b}\right)+\tan^{-1}\left(\frac{b}{a+c}\right)$ is equal to

(a) 0 (b) 1

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

5. Evaluate : $\int \cos^3 x \, e^{\log \sin x} dx$

(a) $\frac{\cos^4 x}{4}+C$ (b) $-\frac{\cos^4 x}{4}+C$

(c) $\frac{\cos^4 x}{4x}+C$ (d) None of these

6. Form the differential equation corresponding to the function $y = c(x - c)^2$.

(a) $y'^2 - 4xyy' = 0$

(b) $y' - 4xyy' = 0$

(c) $y'^3 - 4xyy' + 8y^2 = 0$

(d) $y'^3 - 4xyy' - 8y^2 = 0$

7. The points $A(3, -5, 1)$, $B(-1, 0, 8)$ and $C(7, -10, -6)$ are

(a) collinear

(b) not collinear

(c) the mid-points of sides of a triangle

(d) None of these

8. Find the equation of the plane passing through (2, 3, –1) and is perpendicular to the vector $3\hat{i}-4\hat{j}+7\hat{k}$.

(a) $3x + 4y - 7z + 12 = 0$

(b) $3x - 4y + 7z + 13 = 0$

(c) $3x - 4y + 7z - 13 = 0$

(d) $3x - 4y - 7z - 13 = 0$

9. Diet of sick person must contain atleast 4000 units of vitamins, 50 units of minerals and 2500 calories. Two foods F_1 and F_2 cost ₹ 50 and ₹ 75 per unit respectively. Each unit of food F_1 contains 200 unit of vitamins, 2 units of minerals and 40 calories, whereas each unit of food F_2 contains 100 units of vitamins, 3 units of minerals and 35 calories. Formulate the above problem as LPP to fulfill sick person's requirements at minimum cost.

(a) Minimize $Z = 50x + 75y$
Subject to $200x + 100y \geq 4000$,
$2x + 3y \geq 50, 40x + 35y \geq 2500, x \geq 0, y \geq 0$

(b) Minimize $Z = 50x + 75y$
Subject to $200x + 100y \geq 4000$,
$2x + 3y \geq 50, 40x + 35y \leq 2500, x \geq 0, y \geq 0$

(c) Minimize $Z = 50x + 75y$
Subject to $200x + 100y \geq 4000$,
$2x + 3y \leq 50, 40x + 35y \geq 2500, x \geq 0, y \geq 0$

(d) Minimize $Z = 50x + 75y$
Subject to $200x + 100y \geq 4000$,
$2x + 3y \leq 50, 40x + 35y \leq 2500, x \geq 0, y \geq 0$

10. Find the direction cosines of the line $\frac{x-1}{2} = -y = \frac{z+1}{2}$.

(a) $\frac{2}{3}, \frac{-1}{3}, \frac{2}{3}$ (b) $\frac{2}{3}, \frac{1}{3}, \frac{2}{3}$

(c) $\frac{-2}{3}, \frac{1}{3}, \frac{2}{3}$ (d) $\frac{2}{3}, \frac{1}{3}, \frac{-2}{3}$

(Q.11 - Q.15) Fill in the blanks.

11. The acute angle which the line with direction cosines $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{6}}, n$ makes with positive direction of z-axis is _____.

12. If a matrix has 12 elements, then the number of possible orders is _____.

OR

If $A = \begin{bmatrix} 1 & 2 \\ -3 & 0 \\ 2 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{bmatrix}$, then the matrix C is _____ such that $A + B + C$ is zero matrix.

13. The order of the differential equation $\log\left(\frac{d^2y}{dx^2}\right) = \left(\frac{dy}{dx}\right)^3 + x$ is _____.

14. $\int_{-\pi/2}^{\pi/2} \sin x \cos^2 x (\sin^2 x + \cos x) dx$ = _____.

OR

$\int \frac{10x^9 + 10^x \log_e 10}{10^x + x^{10}} dx$ is equal to _____ .

15. The projection of $\vec{a} = \hat{i} - 3\hat{k}$ on $\vec{b} = 3\hat{i} + \hat{j} - 4\hat{k}$ is _____.

(Q.16 – Q.20) Answer the following questions.

16. Let $f(x) = \begin{cases} \frac{x^3 + x^2 - 16x + 20}{(x-2)^2}, & x \neq 2 \\ k, & x = 2 \end{cases}$

If $f(x)$ is continuous for all x, then find the value of k.

OR

If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in R$ and $f(0) = 0$, then find the value of $f(1)$.

17. A stone is dropped into a quiet lake and waves move in circles at a speed of 3 cm per second. At the instant when the radius of the circular wave is 8 cm, how fast is the enclosed area increasing ?

18. Find the value of $\begin{vmatrix} \sin 10° & -\cos 10° \\ \sin 80° & \cos 80° \end{vmatrix}$.

19. Find the slope of the tangent to the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at $t = 2$.

20. Find the area of the parallelogram determined by the vectors $2\hat{i} + 3\hat{j} + 4\hat{k}$ and $4\hat{i} - 3\hat{j} + 2\hat{k}$.

SECTION - B

21. Find the value of

$\cos^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right) + \text{cosec}^{-1}\left(\frac{1+\sqrt{x}}{1-\sqrt{x}}\right), 1 > x \geq 0.$

www.exambites.in

OR

Prove that

$$\cot^{-1}\left(\frac{ab+1}{a-b}\right)+\cot^{-1}\left(\frac{bc+1}{b-c}\right)+\cot^{-1}\left(\frac{ca+1}{c-a}\right)=0$$

22. If $x=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfies the equation

$x^2-(a+d)x+kI=0$ then find the value of k.

23. Evaluate : $\int_0^1 x(1-x)^{3/2}\,dx$

24. Find the projection of the vector $7\hat{i}+\hat{j}-4\hat{k}$ on $2\hat{i}+6\hat{j}+3\hat{k}$.

25. Find the angle between the lines whose direction ratios are (2, 3, 4) and (1, –3, –2).

26. One bag contains 3 red and 5 black balls. Another bag contains 6 red and 4 black balls. A ball is transferred from first bag to the second bag and then a ball is drawn from the second bag. Find the probability that the ball drawn is red.

OR

If $P(A)=0.6$, $P(B)=0.5$ and $P(A/B)=0.3$, then find $P(A\cup B)$.

SECTION - C

27. Let L be the set of all lines in a plane and R be the relation in L defined as $R=\{(L_1,L_2):L_1 \text{ is perpendicular to } L_2\}$. Show that R is symmetric but neither reflexive nor transitive.

28. Discuss the continuity of the function f, where f is defined by

$$f(x)=\begin{cases}2x, & \text{if } x<0\\ 0, & \text{if } 0\le x\le 1\\ 4x, & \text{if } x>1\end{cases}$$

OR

If $y=\tan x+\sec x$, prove that

$$\frac{d^2y}{dx^2}=\frac{\cos x}{(1-\sin x)^2}.$$

29. Evaluate $\int_{-1}^{1} e^x dx$ as limit of sums.

30. Form the differential equation representing the family of curves $y=A\cos(x+B)$, where A and B are parameters.

31. Show that the four points (0, –1, –1), (4, 5, 1), (3, 9, 4) and (–4, 4, 4) are coplanar. Also, find the equation of the plane containing them.

OR

Find the equation of perpendicular drawn from the point $P(2,4,-1)$ to the line $\frac{x+5}{1}=\frac{y+3}{4}=\frac{z-6}{-9}$.

32. Rajiv has two courses to prepare for TOEFL examination to go abroad. Each hour of study, he devotes to course A is expected to return ₹ 600 in terms of long range foreign job benefits. Each hour devoted to course B is expected to return ₹ 300 in terms of long range foreign job benefits. The stores are closed and Rajiv has only 15 chewing gums. He finds that he consumes one chewing gum every 20 minutes while studying course B and every 12 minutes while studying course A. Time is running short only four hours are left to prepare. Rajiv feels that he must devote atleast two hours to study. Using linear programming, determine an optimal policy for Rajiv that would maximise his return in terms of long range foreign job benefits.

SECTION - D

33. Using matrices, solve the following system of equations : $x+2y+z=7;\ x+3z=11;\ 2x-3y=1$

OR

Using elementary transformations, find the inverse of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}$.

34. Find the intervals in which the following function is strictly increasing or strictly decreasing : $f(x)=20-9x+6x^2-x^3$.

OR

For the curve $y=4x^3-2x^5$, find all the points at which the tangent passes through origin.

35. Find the area of the segment cut off from the parabola $y^2=4x$ by the line $y=x$.

36. There are two bags, one of which contains 3 black and 4 white balls, while the other contains 4 black and 3 white balls. A fair die is tossed, if the face 1 or 3 turns up, a ball is taken from the first bag, and if any other face turns up a ball is chosen from the second bag. Find the probability of choosing a black ball.

www.exambites.in

Hints & Solutions

1. (a) **2.** (b) **3.** (d) **4.** (c)
5. (b) **6.** (c) **7.** (a) **8.** (b)
9. (a) **10.** (a) **11.** $\frac{\pi}{4}$

12. 6 **OR** $C=\begin{bmatrix}-3 & -3\\ 4 & 0\\ -3 & -1\end{bmatrix}$ **13.** 2

14. 0 **OR** $\log_e(10^x + x^{10}) + C$

15. $\frac{15}{\sqrt{26}}$ **16.** 7 **OR** $f(1) = 0$

17. 48π cm^2/s **18.** 1 **19.** $\frac{6}{7}$

20. $6\sqrt{22}$ sq. units **21.** $\frac{\pi}{2}$

22. $k = ad - bc$ **23.** $\frac{4}{35}$ **24.** $\frac{8}{7}$

25. $\cos^{-1}\left[\frac{-15}{\sqrt{29}\sqrt{14}}\right]$

26. $\frac{51}{88}$ **OR** $P(A \cup B) = 0.95$

28. At $x = 0$, $f(x)$ is continuous but at $x = 1$, $f(x)$ is not continuous.

29. $e - \frac{1}{e}$ **30.** $\frac{d^2y}{dx^2} + y = 0$

31. $5x - 7y + 11z + 4 = 0$

OR

$\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

32. Maximize return is ₹1800 on devoting 3 hours of study to course *A* and no time to course *B*.

33. $x = 2$, $y = 1$ and $z = 3$

OR

$\begin{bmatrix}3 & -2 & -1\\ -4 & 1 & -1\\ 2 & 0 & 1\end{bmatrix}$

34. $f(x)$ is strictly increasing in (1, 3) and strictly decreasing in $(-\infty, 1) \cup (3, \infty)$.

OR

(0, 0), (1, 2) and (–1, –2)

35. $\frac{8}{3}$ sq. units **36.** $\frac{11}{21}$

www.exambites.in

PRACTICE PAPER 6
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. Find the principal value of $\cot^{-1}(-\sqrt{3})$.

(a) $\frac{5\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

2. The angle between the straight line $\frac{x-1}{2}=\frac{y+3}{-1}=\frac{z-5}{2}$ and the plane $4x - 2y + 4z = 9$ is

(a) 90° (b) 45° (c) 30° (d) 60°

3. Solve : $\frac{dy}{dx}+1=e^{x+y}$

(a) $x-e^{-2(x+y)}=c$ (b) $y+e^{-(x+y)}=c$

(c) $x+e^{-(x+y)}=c$ (d) $2y-e^{-(x+y)}=c$

4. Evaluate : $\int_0^{\pi/4}\sqrt{1-\sin 2x}\,dx$

(a) $\sqrt{2}+1$ (b) $2\sqrt{2}$

(c) $2\sqrt{2}-1$ (d) $\sqrt{2}-1$

5. The equation of the tangent to the curve $y = (4 - x^2)^{2/3}$ at $x = 2$ is

(a) $y = 2$ (b) $y = 4$

(c) $x = 2$ (d) $x = 0$

6. Find a unit vector perpendicular to each one of the vectors $\vec{a}=(4\hat{i}-\hat{j}+3\hat{k})$ and $\vec{b}=(2\hat{i}+2\hat{j}-\hat{k})$.

(a) $-\hat{i}+2\hat{j}+2\hat{k}$ (b) $\frac{1}{6}(-\hat{i}+2\hat{j}+2\hat{k})$

(c) $\frac{1}{2}(-\hat{i}+2\hat{j}+2\hat{k})$ (d) $\frac{1}{3}(-\hat{i}+2\hat{j}+2\hat{k})$

7. If an error k % is made in measuring the radius of a sphere, then percentage error in its volume is

(a) $3k$ % (b) k %

(c) $2k$ % (d) $\frac{k}{3}$%

8. Find the principal value of $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$.

(a) $\frac{5\pi}{6}$ (b) $\frac{\pi}{6}$ (c) $\frac{4\pi}{9}$ (d) $\frac{2\pi}{3}$

9. $\int e^x\left(\frac{1-x}{1+x^2}\right)^2 dx$ is equal to

(a) $\frac{e^x}{1+x^2}+C$ (b) $\frac{-e^x}{1+x^2}+C$

(c) $\frac{e^x}{(1+x^2)^2}+C$ (d) $\frac{-e^x}{(1+x^2)^2}+C$

10. A printing company prints two types of magazine A and B. The company earn ₹ 20 and ₹ 30 on each copy of magazines A and B respectively. The magazines are processed on three machines. Magazine A requires 2 hours

www.exambites.in

on machine I, 4 hours on machine II and 2 hours on machine III. Magazine *B* requires 3 hours on machine I, 5 hours on machine II and 3 hours on machine III. Machine I, II, III are available for 35, 50 and 70 hours per week respectively. Formulate the LPP to determine weekly production of magazines *A* and *B*, so that the total profit is maximum.

(a) Maximize $Z = 20x + 30y$
Subject to, $2x + 3y \le 35$, $4x + 5y \le 50$, $2x + 3y \ge 70$, $x \ge 0$, $y \ge 0$

(b) Maximize $Z = 20x + 30y$
Subject to, $2x + 3y \le 35$, $4x + 5y \le 50$, $2x + 3y \le 70$, $x \le 0$, $y \ge 0$

(c) Maximize $Z = 20x + 30y$
Subject to, $2x + 3y \le 35$, $4x + 5y \le 50$, $2x + 3y \ge 70$, $x \le 0$, $y \ge 0$

(d) Maximize $Z = 20x + 30y$
Subject to, $2x + 3y \le 35$, $4x + 5y \le 50$, $2x + 3y \le 70$, $x \ge 0$, $y \ge 0$

(Q.11 - Q.15) Fill in the blanks.

11. The distance of the plane $2x - 3y + 6z + 14 = 0$ from the origin is _____ .

OR

If *l*, *m* and *n* are the direction cosines of line l_1, then equation of the line (l_1) passing through (x_1, y_1, z_1) is_____ .

12. The value of $\int_1^2 \frac{dx}{x^2}$ = _____ .

13. The order of the differential equation of family $y = Ax + A^3$ of curves is _____ .

14. The area of the parallelogram whose diagonals are represented by the vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + 2\hat{k}$ is _____ .

15. $\begin{vmatrix} \frac{1}{a} & a & bc \\ \frac{1}{b} & b & ca \\ \frac{1}{c} & c & ab \end{vmatrix}$ = _____ .

OR

The minor of 6 and cofactor of 4 respectively in the determinant $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$ is_____ .

(Q.16 – Q.20) Answer the following questions.

16. If $x = 3\cos^4\theta$, $y = 3\sin^4\theta$, then find $\frac{dy}{dx}$ at $\theta = \frac{3\pi}{4}$.

17. Find the sum of *x*, *y*, *z* and *w* if $\begin{bmatrix} x-y & 2z+w \\ 2x-y & 2x+w \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 12 & 15 \end{bmatrix}$.

18. If $\vec{a}$ and $\vec{b}$ are non-collinear vectors and vectors $\vec{\alpha} = (x+4y)\vec{a} + (2x+y+1)\vec{b}$ and $\vec{\beta} = (-2x+y+2)\vec{a} + (2x-3y-1)\vec{b}$ are connected by the relation $3\vec{\alpha} = 2\vec{\beta}$, find $x - y$.

OR

If the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$, $-2\hat{i} + 3\hat{j} - 4\hat{k}$, $\lambda\hat{i} - \hat{j} + 2\hat{k}$ are coplanar, then find the value of λ.

19. Find the *x*-coordinate of the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+5}{4}$ meets the plane $2x + 4y - z = 3$.

20. If $f(2) = 4$ and $f'(2) = 1$, then find $\lim_{x \to 2} \frac{xf(2) - 2f(x)}{x-2}$.

SECTION - B

21. If $\cot^{-1} x + \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) = \frac{\pi}{4}$, then find the value of *x*.

22. Show that $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix} = 0$, where *a*, *b*, *c* are in A.P.

23. Evaluate : $\int_{-1}^{1} e^{|x|} dx$

24. Let $\vec{a}$ and $\vec{b}$ be two vectors of the same magnitude such that the angle between them is 60° and $\vec{a} \cdot \vec{b} = 8$. Find $|\vec{a}|$ and $|\vec{b}|$.

OR

If the points with vertices (p_1, q_1), (p_2, q_2) and $(p_1 + p_2, q_1 + q_2)$ are collinear, show that $p_1 q_2 = p_2 q_1$.

www.exambrics.in

25. Find the vector equation of the plane which is at a distance of $\frac{6}{\sqrt{29}}$ from the origin and its normal vector from the origin is $2\hat{i} - 3\hat{j} + 4\hat{k}$.

26. Probabilities of solving a specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that the problem is solved.

OR

Two cards are drawn without replacement from a well shuffled pack of 52 cards. Find the probability that one is a spade and other is a queen of red colour.

SECTION - C

27. Let $f : N \to R$ be a function defined as $f(x) = 4x^2 + 12x + 15$. Show that $f : N \to$ Range (f) is invertible. Find the inverse of f.

28. Find all the points of discontinuity of the function $f(x) = [x^2]$ on [1, 2], where $[\cdot]$ denotes the greatest integer function.

OR

If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x - y)$, prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.

29. Find the local maximum and local minimum values of $\frac{(x-1)(x-6)}{(x-10)}$.

30. Solve the initial value problem
$(x^2+1)y' - 2xy = (x^4+2x^2+1)\cos x, y(0) = 0.$

31. Find the equation of the plane passing through the point (1, 1, 1) and perpendicular to each of the following planes $x + 2y + 3z = 7$ and $2x - 3y + 4z = 0$.

OR

Show that the lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(3\hat{i} - \hat{j})$ and $\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + 3\hat{k})$ are coplanar. Also, find the plane containing these two lines.

32. An oil company requires 13000, 20000 and 15000 barrels of high grade, medium grade and low grade oil respectively. Refinery A produces 100, 300 and 200 barrels per day of high, medium and low grade oil respectively, whereas the Refinery B produces 200, 400 and 100 barrels per day respectively. If A costs ₹ 400 per day and B costs ₹ 300 per day to operate, how many days refinery A and B should be run respectively to minimize the cost of requirement?

SECTION - D

33. Find the inverse of the matrix $\begin{bmatrix} 0 & -1 & 1 \\ 2 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ by using elementary row transformations.

OR

Without expanding show that

$$\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix} = 0.$$

34. Evaluate :
$\int \tan^{-1}(\sec x + \tan x)dx; -\pi/2 < x < \pi/2$

OR

Evaluate :
$$\int \left\{(2x-3)^5 + \frac{1}{(7x-5)^3} + \frac{1}{\sqrt{5x-4}} + \frac{1}{2-3x} + \sqrt{3x+2}\right\}dx$$

35. Prove that the curves $y = x^2$ and $x = y^2$ divide the square bounded by $x = 0$, $y = 0$, $x = 1$ and $y = 1$ into three parts which are equal in area.

36. Coloured balls are distributed in three bags as shown in the following table :

Bag	Colour of ball		
	Green	Black	Blue
I	1	2	3
II	2	4	1
III	3	3	2

A bag is selected at random and then two balls are drawn from the selected bag. Find the probability that the balls drawn are black and blue.

www.exambites.in

Hints & Solutions

1. (a) **2.** (a) **3.** (c) **4.** (d)
5. (c) **6.** (d) **7.** (a) **8.** (a)
9. (a) **10.** (d)

11. 2 units **OR** $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$

12. $\frac{1}{2}$ **13.** 2 **14.** 3 sq. units

15. 0 **OR** – 6, 6 **16.** –1 **17.** 11
18. 3 **OR** $\lambda = 0$ **19.** 3 **20.** 2
21. 3 **23.** $2(e-1)$

24. $|\vec{a}|=|\vec{b}|=4$

25. $\vec{r}\cdot\left(\frac{2}{\sqrt{29}}\hat{i}+\frac{-3}{\sqrt{29}}\hat{j}+\frac{4}{\sqrt{29}}\hat{k}\right)=\frac{6}{\sqrt{29}}$

26. $\frac{2}{3}$ **OR** $\frac{1}{51}$

27. $f^{-1}(x)=\frac{-3+\sqrt{x-6}}{2}$

28. Points of discontinuity are $x=\sqrt{2}, x=\sqrt{3}$.

29. Local maximum at $x = 4$ is 1 and local minimum at $x = 16$ is 25.

30. $y = (x^2+1)\sin x$ defined for all $x \in R$.

31. $17x + 2y - 7z - 12 = 0$

OR

$3x + 9y - 2z - 14 = 0$

32. Refinery A should run for $\frac{170}{3}$ days and refinery B should run for $\frac{110}{3}$ days.

33. $\begin{bmatrix} 0 & -1 & 1 \\ -4 & 3 & -2 \\ -3 & 3 & -2 \end{bmatrix}$

34. $\frac{\pi}{4}x+\frac{x^2}{4}+C$

OR

$$\frac{1}{12}(2x-3)^6-\frac{1}{14}(7x-5)^{-2}+\frac{2}{5}\sqrt{5x-4}$$
$$-\frac{1}{3}\log|2-3x|+\frac{2}{9}(3x+2)^{3/2}+C$$

36. $\frac{169}{630}$

www.exambites.in

PRACTICE PAPER 7
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. Let $A = \{1, 2, 3\}$. Then the number of equivalence relations containing (1, 2) is
(a) 1 (b) 2 (c) 3 (d) 4

2. If $f(x) = 3x^2 + 15x + 5$, then the approximate value of $f(3.02)$ is
(a) 47.66 (b) 57.66
(c) 67.66 (d) 77.66

3. Shamali wants to invest ₹ 50,000 in saving certificates and PPF. She wants to invest atleast ₹ 15, 000 in saving certificates and at least ₹ 20,000 in PPF. The rate of interest on saving certificates is 8% p.a. and that on PPF is 9% p.a. Formulate the above problem as LPP to determine maximum yearly income.
(a) Maximize $Z = 0.08x + 0.09y$
Subject to, $x + y \le 50000$, $x \ge 15000$, $y \ge 20000$
(b) Maximize $Z = 0.08x + 0.09y$
Subject to, $x + y \le 50000$, $x \ge 15000$, $y \le 20000$
(c) Maximize $Z = 0.08x + 0.09y$
Subject to, $x + y \le 50000$, $x \le 15000$, $y \ge 20000$
(d) Maximize $Z = 0.08x + 0.09y$
Subject to, $x + y \le 50000$, $x \le 15000$, $y \le 20000$

4. Distance between the two planes $2x + 3y + 4z = 4$ and $4x + 6y + 8z = 12$ is
(a) 2 units (b) 4 units
(c) 8 units (d) $\frac{2}{\sqrt{29}}$ unit

5. The normal at the point (1, 1) on the curve $2y + x^2 = 3$ is
(a) $x + y = 0$ (b) $x - y = 0$
(c) $x + y + 1 = 0$ (d) $-x + y + 2 = 0$

6. Let $A = \{1, 2, 3\}$. Then the number of relations containing (1, 2) and (1, 3), which are reflexive and symmetric but not transitive is
(a) 1 (b) 2
(c) 3 (d) 4

7. The vector and the cartesian equations of the line that passes through the origin and (5, –2, 3) are A and B respectively. Here, A and B refer to
(a) $\vec{r} = 5\hat{i} - 2\hat{j} + 3\hat{k}$; $5x = -2y = 3z$
(b) $\vec{r} = \lambda(5\hat{i} - 2\hat{j} + 3\hat{k})$; $5x = -4y = 3z$
(c) $\vec{r} = \lambda(5\hat{i} - 2\hat{j} + 3\hat{k})$; $\frac{x}{5} = \frac{y}{-2} = \frac{z}{3}$
(d) None of these

8. If the matrix A is both symmetric and skew symmetric, then
(a) A is a diagonal matrix
(b) A is a zero matrix
(c) A is a square matrix
(d) None of these

9. If $|\vec{a}\cdot\vec{b}| = |\vec{a}\times\vec{b}|$ and θ is the angle between vectors $\vec{a}$ and $\vec{b}$, then θ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

10. The planes $2x - y + 4z = 5$ and $5x - 2.5y + 10z = 6$ are

(a) Perpendicular (b) Parallel

(c) Intersect y-axis

(d) Pass through $\left(0,0,\frac{5}{4}\right)$

(Q.11 - Q.15) Fill in the blanks.

11. The value of x, if $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$ is _____ .

12. The value of $\hat{i}\cdot(\hat{j}\times\hat{k}) + \hat{j}\cdot(\hat{i}\times\hat{k}) + \hat{k}\cdot(\hat{i}\times\hat{j})$ is _____ .

13. $\int_0^1 |2x-1|\,dx =$ _____ .

OR

$\int xe^{x^2}\,dx$ is equal to_____ .

14. $\int \frac{a}{(1+x^2)\tan^{-1}x}\,dx =$ _____ .

OR

$\int \frac{1}{a^x b^x}\,dx =$ _____ .

15. The order of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$ is _____ .

(Q.16 – Q.20) Answer the following questions.

16. Evaluate : $\int_{-\pi/4}^{\pi/4} \sin^5 x\,dx$

17. For what value of k, the function

$$f(x) = \begin{cases} kx^2, & \text{if } x \le 2 \\ 3, & \text{if } x > 2 \end{cases}$$

is continuous at $x = 2$?

18. If $\vec{a}$ is unit vector, then find $|\vec{x}|$, for $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a}) = 8$.

19. If $y = A\sin x + B\cos x$, then find the value of $\frac{d^2y}{dx^2} + y$.

OR

If $ax^2 + 2hxy + by^2 = 1$, then find $\frac{dy}{dx}$.

20. Evaluate : $\int_0^3 [x]\,dx$, where [·] denotes the greatest integer function.

SECTION - B

21. Prove that $\frac{1}{2}\tan^{-1}x = \cos^{-1}\left(\sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}\right)$.

22. Find the adjoint of matrix $A = \begin{bmatrix} 3 & 1 \\ -5 & 4 \end{bmatrix}$.

23. Find the value of $\int \frac{(x^4 - x)^{\frac{1}{4}}}{x^5}\,dx$.

24. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}$, then show that $\vec{a} = 0$ or $\vec{b} = \vec{c}$ or $\vec{a} \perp (\vec{b} - \vec{c})$.

OR

Prove that the vectors $3\hat{i} + 5\hat{j} + 2\hat{k}$, $2\hat{i} - 3\hat{j} - 5\hat{k}$ and $5\hat{i} + 2\hat{j} - 3\hat{k}$ forms the sides of an equilateral triangle.

25. Find the angle between the lines

$\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$

and $\vec{r} = (5\hat{j} - 2\hat{k}) + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$.

26. The probability distribution of x is

x	0	1	2	3
$P(x)$	0.2	k	k	$2k$

Write the value of k.

OR

If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$, then evaluate $P(A/B)$.

www.exambites.in

SECTION - C

27. Let $f: [-1, 1] \to Y$ such that $f(x) = \frac{x}{(x+2)}$, $x \neq -2$ and Y = range (f). Show that f is invertible and find f^{-1}.

28. Show that the function $f(x)$ given by

$$f(x) = \begin{cases} \frac{\sin x}{x} + \cos x, & x \neq 0 \\ 2, & x = 0 \end{cases}$$ is continuous at $x = 0$.

OR

If $(x - a)^2 + (y - b)^2 = c^2$, then prove that $\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ is a constant independent of a and b.

29. Show that semi-vertical angle of a right circular cone of given surface area and maximum volume is $\sin^{-1}\frac{1}{3}$.

30. Find the particular solution of the differential equation $\frac{dy}{dx} + y\cot x = 2x + x^2\cot x, x \neq 0$ given that $y = 0$, when $x = \frac{\pi}{2}$.

31. Find the equation of the plane which contains the two parallel lines $\frac{x-3}{3} = \frac{y+4}{2} = \frac{z-1}{1}$ and $\frac{x+1}{3} = \frac{y-2}{2} = \frac{z}{1}$.

OR

A plane meets the coordinates axes in A, B, C, such that the centroid of the triangle ABC is the point (α, β, γ). Show that the equation of the plane is $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 3$.

32. A house wife wishes to mix together two kinds of food, X and Y, in such a way that the mixture contains at least 10 units of vitamin A, 12 units of vitamin B and 8 units of vitamin C.

The vitamin contents of one kg of food is given below :

	Vitamin A	Vitamin B	Vitamin C
Food X	1	2	3
Food Y	2	2	1

One kg of food X costs ₹ 6 and one kg of food Y costs ₹ 10. Find the least cost of the mixture which will produce the diet.

SECTION - D

33. Using elementary row transformations, find the inverse of the matrix $A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$.

OR

Prove that

$$\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2(a+b+c)(ab+bc+ca-a^2-b^2-c^2)$$

34. Evaluate : $\int \frac{1}{\sin x(5-4\cos x)}dx$

OR

Evaluate : $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$

35. A cuboidal shaped godown with square base is to be constructed. Three times as much cost per square meter is incurred for constructing the roof as compared to the walls. Find the dimensions of the godown if it is to enclose a given volume and minimize the cost of constructing the roof and the walls.

36. In a multiple choice test, there are four options for a question of which exactly one is correct. A student either knows the answer or he guesses or he copies. The probability that he guesses is $\frac{3}{8}$ and that he copies is $\frac{1}{2}$. The probability of answer being correct when he copies is $\frac{1}{3}$. If the student got marks for the question, find the probability that he knew the answer.

www.exambites.in

Hints & Solutions

1. (b) **2.** (d) **3.** (a) **4.** (d)
5. (b) **6.** (a) **7.** (c) **8.** (b)
9. (b) **10.** (b) **11.** 2 **12.** 1

13. $\frac{1}{2}$ **OR** $\frac{e^{x^2}}{2}+C$

14. $a \log|\tan^{-1}x| + C$ **OR** $\frac{-(ab)^{-x}}{\log(ab)}+C$

15. 2 **16.** 0 **17.** $\frac{3}{4}$ **18.** 3

19. 0 **OR** $-\left(\frac{ax+hy}{hx+by}\right)$ **20.** 3

22. adj $A=\begin{bmatrix}4 & -1\\ 5 & 3\end{bmatrix}$

23. $\frac{4}{15}\left(1-\frac{1}{x^3}\right)^{5/4}+C$

25. $\theta=\cos^{-1}\left(\frac{19}{21}\right)$

26. $k = 0.2$ **OR** $\frac{4}{9}$

27. $f^{-1}(x)=\frac{2x}{1-x}$

30. $y=x^2-\frac{\pi^2}{4}\text{cosec } x$

31. $8x + y - 26z + 6 = 0$

32. Least cost is ₹ 52.

33. $A^{-1}=\begin{bmatrix}3 & -1 & 1\\ -15 & 6 & -5\\ 5 & -2 & 2\end{bmatrix}$

34. $\frac{1}{2}\log|1-\cos x| - \frac{1}{18}\log|1+\cos x| - \frac{4}{9}\log|5-4\cos x| + C$

OR

$-2\sqrt{1-x}-\sin^{-1}\sqrt{x}+\sqrt{x-x^2}+C$

35. Cost is minimum, when $x=\left(\frac{2V}{3}\right)^{1/3}$ and $y=\frac{(18V)^{1/3}}{2}$, where x is length and breadth and y is height.

36. $\frac{12}{37}$

www.examrites.in

PRACTICE PAPER 8
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. The tangent to the parabola $x^2 = 2y$ at the point $\left(1, \frac{1}{2}\right)$ makes with the x-axis an angle of

(a) 0° (b) 45° (c) 30° (d) 60°

2. Let R be the relation on the set of all real numbers defined by aRb iff $|a - b| \le 1$. Then, R is

(a) Reflexive and symmetric
(b) Symmetric only (c) Transitive only
(d) Anti-symmetric only

3. The anti derivative of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ equals

(a) $\frac{1}{3}x^{1/3} + 2x^{1/2} + C$ (b) $\frac{2}{3}x^{2/3} + \frac{1}{2}x^2 + C$

(c) $\frac{2}{3}x^{3/2} + 2x^{1/2} + C$ (d) $\frac{3}{2}x^{3/2} + \frac{1}{2}x^{1/2} + C$

4. If there is an error of a% in measuring the edge of a cube, then percentage error in its surface area is

(a) $2a$% (b) $(a/2)$% (c) $3a$% (d) $(a/3)$%

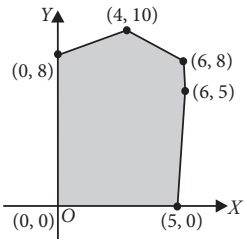
5. Let $A = \{1, 2, 3\}$ and $B = \{2, 4, 6, 8\}$. Consider the rule $f: A \to B, f(x) = 2x \ \forall \ x \in A$. The domain, co-domain and range of f respectively are

(a) {1, 2, 3}, {2, 4, 6}, {2, 4, 6, 8}
(b) {1, 2, 3}, {2, 4, 6, 8}, {2, 4, 6}
(c) {2, 4, 6, 8}, {2, 4, 6, 7}, {1, 2, 3}
(d) {2, 4, 6}, {2, 4, 6, 8}, {1, 2, 3}

6. Evaluate : $\int_0^{\pi} \sqrt{\frac{\cos 2x + 1}{2}}\, dx$

(a) 0 (b) 2 (c) –2 (d) 4

7. The feasible region for an LPP is shown shaded in the figure. Let $Z = 3x - 4y$ be the objective function. Minimum of Z occurs at



(a) (0, 0)
(b) (0, 8) (c) (5, 0) (d) (4, 10)

8. Find the angle between the line $\vec{r} = \hat{i} + 2\hat{j} - \hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k})$ and the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 4$.

(a) $\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ (b) $\sin^{-1}\left(\frac{2}{3}\right)$

(c) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (d) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

9. If $\frac{d}{dx} f(x) = 4x^3 - \frac{3}{x^4}$, such that $f(2) = 0$. Then, $f(x)$ is equal to

(a) $x^4 + \frac{1}{x^3} - \frac{129}{8}$ (b) $x^3 + \frac{1}{x^4} + \frac{129}{8}$

(c) $x^4 + \frac{1}{x^3} + \frac{129}{8}$ (d) $x^3 + \frac{1}{x^4} - \frac{129}{8}$

10. The lines $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ and $\frac{x-4}{1}$

$= \frac{y+3}{k} = \frac{z+1}{7}$ are coplanar if k =

(a) 4 (b) –4 (c) 2 (d) –2

(Q.11 - Q.15) Fill in the blanks.

11. The number of solutions of $\frac{dy}{dx} = \frac{y+1}{x-1}$ when $y(1) = 2$ is _____ .

OR

The solution of the differential equation $\cos x \sin y\, dx + \sin x \cos y\, dy = 0$ is_____ .

12. If $(2\hat{i} + 6\hat{j} + 14\hat{k}) \times (\hat{i} - \lambda\hat{j} + 7\hat{k}) = 0$, then λ = _____ .

OR

If $\left(\frac{1}{2}, \frac{1}{3}, n\right)$ are the direction cosines of a line, then the value of n is_____ ,

13. The minor of a_{22} of the matrix $\begin{bmatrix} 1 & 6 & 1 \\ 5 & 3 & 0 \\ 2 & 2 & 9 \end{bmatrix}$ is _____ .

14. The degree of the differential equation $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x$ is _____ .

15. The distance of the point $(-\alpha, -\beta, -\gamma)$ from z-axis is _____ units.

(Q.16 – Q.20) Answer the following questions.

16. Find the value of k for which the function $f(x) = \begin{cases} \frac{\sin 3x}{7x}, x \neq 0 \\ k, x = 0 \end{cases}$ is continuous at $x = 0$.

17. If the matrix $A = \begin{bmatrix} 5 & 2 & x \\ y & 2 & -3 \\ 4 & t & -7 \end{bmatrix}$ is a symmetric matrix, then find the value of $x + y + t$.

OR

If A and B are two matrices of the order $3 \times m$ and $3 \times n$, respectively, and $m = n$, then find the order of matrix $(5A - 2B)$.

18. Can Rolle's theorem be applied to the function $|x|$ in the interval $[-2, 2]$?

19. Find the magnitude of the resultant of vectors $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

20. Find λ, if the vectors $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ are perpendicular to each other.

SECTION - B

21. Find the value of $\tan^{-1}\left\{2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right\}$.

22. If $A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then find k so that $A^2 = 8A + kI$.

23. Evaluate : $\int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}}\, dx$

24. Write the direction ratios of the vector $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$ and hence calculate its direction cosines.

OR

Find the angle between the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

25. Find the angle between the planes $x + y + 2z = 9$ and $2x - y + z = 15$.

26. A bag contains 2 white and 4 black balls while another bag contains 6 white and 4 black balls. A bag is selected at random and a ball is drawn. Find the probability that the ball drawn is of white colour.

OR

If A and B are two independent events, prove that A' and B are also independent.

SECTION - C

27. If R_1 and R_2 be two equivalence relations on a set A, prove that $R_1 \cap R_2$ is also an equivalence relation on A.

28. Show that the function defined by $g(x) = x - [x]$ is discontinuous at all integral points. Here $[x]$ denotes greatest integer less than or equal to x.

OR

If $y = \sqrt{\frac{1+e^x}{1-e^x}}$, show that $\frac{dy}{dx} = \frac{e^x}{(1-e^x)\sqrt{1-e^{2x}}}$, $x < 0$.

29. Find the area of the largest isosceles triangle whose perimeter is 36 units.

30. Solve the differential equation $(1 + e^{2x})dy + e^x(1 + y^2)dx = 0$, given that $y = 1$ when $x = 0$.

www.examples.in

31. Find the equation of plane containing the line of intersection of the planes $x + y + z - 6 = 0$ and $2x + 3y + 4z + 5 = 0$ and passing through (1, 1, 1).

OR

Show that the lines $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}$ and $\frac{x+2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$ do not intersect.

32. A catering agency has two kitchens to prepare food at two places *A* and *B*. From these places `Mid-day Meal' is to be supplied to three different schools situated at *P, Q, R*. The monthly requirements of the schools are respectively 40, 40 and 50 food packets. A packet contains lunch for 1000 students. Preparing capacity of kitchens *A* and *B* are 60 and 70 packets per month respectively. The transportation cost per packet from the kitchens to schools is given below :

Transportation cost per packet (in rupees)		
To	From	
	A	*B*
P	5	4
Q	4	2
R	3	5

How many packets from each kitchen should be transported to school so that the cost of transportation is minimum? Also find the minimum cost.

SECTION - D

33. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$. Find A^{-1} and use it to solve the system of equations :
$x + y + 2z = 0, x + 2y - z = 9, x - 3y + 3z = -14$

OR

For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers *a* and *b* such that $A^2 + aA + bI = 0$. Hence, find A^{-1}.

34. Evaluate : $\int \frac{\tan\theta + \tan^3\theta}{1 + \tan^3\theta}\, d\theta$.

OR

Evaluate : $\int_0^{\pi} \frac{x}{a^2\cos^2 x + b^2 \sin^2 x}\, dx$.

35. Find the area of the region enclosed between the two circles $x^2 + y^2 = 1$ and $(x - 1)^2 + y^2 = 1$.

36. A company has two plants to manufacture bicycles. The first plant manufactures 60% of the bicycles and the second plant, 40%. Also, 80% of the bicycles are rated of standard quality at the first plant and 90% of standard quality at the second plant. A bicycle is picked up at random and found to be of standard quality. Find the probability that it comes from the second plant.

www.examfear.in

Hints & Solutions

1. (b) **2.** (a) **3.** (c) **4.** (a)
5. (b) **6.** (b) **7.** (b) **8.** (a)
9. (a) **10.** (b) **11.** 1 **OR** $\sin x \sin y = c$

12. –3 **OR** $n = \frac{\sqrt{23}}{6}$ **13.** 7

14. 2 **15.** $\sqrt{\alpha^2 + \beta^2}$ **16.** 3/7

17. 3 **OR** $(5A - 2B)$ is of order $3 \times m$ or $3 \times n$.

18. No **19.** $\sqrt{34}$ **20.** –9 **21.** $\pi/4$

22. $k = -7$ **23.** $\frac{x^3}{3} + C$

24. $(1, 1, -2)$ and $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}\right)$. **OR** $\cos^{-1}\left(\frac{-1}{3}\right)$

25. $\frac{\pi}{3}$ **26.** $\frac{7}{15}$ **29.** 62.352 sq. units.

30. $\tan^{-1} y + \tan^{-1} e^x = \frac{\pi}{2}$

31. $20x + 23y + 26z - 69 = 0$

32. 10 packets to *P*, 0 packets to *Q*, 50 packets to *R* transported from kitchen *A* and 30 packets to *P*, 40 packets to *Q*, 0 packets to *R* transported from kitchen *B*. Minimum cost = ₹ 400

33. $x = 1, y = 3, z = -2$

OR

$a = -4, b = 1, A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

34. $\frac{-1}{3}\log|1 + \tan\theta| + \frac{1}{6}\log|\tan^2\theta - \tan\theta + 1| + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2\tan\theta - 1}{\sqrt{3}}\right) + C$ **OR** $\frac{\pi^2}{2ab}$

35. $\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$ sq. units. **36.** $\frac{3}{7}$

PRACTICE PAPER 9
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.exambites.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. $\int_{-\pi/4}^{\pi/4} \frac{e^x \cdot \sec^2 x \, dx}{e^{2x}-1}$ is equal to

(a) 0 (b) 2
(c) e (d) None of these

2. The function $f(x) = 1 - x^3 - x^5$ is decreasing for

(a) $1 \le x \le 5$ (b) $x \le 1$
(c) $x \ge 1$ (d) all values of x

3. Find the magnitude of two vectors $\vec{a}$ and $\vec{b}$, having same magnitude and such that the angle between them is 60° and their scalar product is 8.

(a) 3 (b) 4 (c) 6 (d) 8

4. If $y = \log_5 (\log x)$, then $\frac{dy}{dx} =$

(a) $\frac{1}{x \log 5 \log x}$ (b) $\frac{1}{x \log 5 (\log x)^2}$

(c) $\frac{1}{x^2 (\log 5) \log x}$ (d) $\frac{1}{(x \log 5)^2 \log x}$

5. Solution of differential equation $xdy - ydx = 0$ represents

(a) a rectangular hyperbola
(b) parabola whose vertex is at origin
(c) straight line passing through origin
(d) a circle whose centre is at origin

6. Let $f(x) = \frac{\ln(1+ax) - \ln(1-bx)}{x}$, $x \neq 0$.

If $f(x)$ is continuous at $x = 0$, then find the value of $f(0)$.

(a) $a - b$ (b) $a + b$
(c) $b - a$ (d) $\ln a + \ln b$

7. The component of $\hat{i}$ in the direction of the vector $\hat{i} + \hat{j} + 2\hat{k}$ is

(a) $\sqrt{6}$ (b) 6
(c) $6\sqrt{6}$ (d) $\frac{\sqrt{6}}{6}$

8. Tangents to the curve $y = x^3 + 3x$ at $x = -1$ and $x = 1$ are

(a) parallel
(b) intersecting obliquely but not at an angle of 45°
(c) intersecting at right angles
(d) intersecting at an angle of 45°

9. $\int \frac{dx}{\sin^2 x \cos^2 x}$ equals

(a) $\tan x + \cot x + C$ (b) $\tan x - \cot x + C$
(c) $\tan x \cot x + C$ (d) $\tan x - \cot 2x + C$

10. Write the vector equation of the plane, passing through the point (a, b, c) and parallel to the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 2$.

(a) $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = a + b + c$
(b) $\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = a + b - c$
(c) $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = a - b + c$
(d) $\vec{r} \cdot (\hat{i} - \hat{j} - \hat{k}) = a - b - c$

(Q.11 - Q.15) Fill in the blanks.

11. $\int \frac{e^x(1+x)}{\cos^2(e^x x)}dx$ equals _____ .

12. The value of $\tan\left(2\tan^{-1}\frac{1}{5}\right)$ _____ .

13. If the line joining (2, 3, –1) and (3, 5, –3) is perpendicular to the line joining (1, 2, 3) and (3, 5, λ), then the value of λ is _____ .

14. $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(\frac{-1}{2}\right)\right]$ = _____ .

OR

If $\tan^{-1} 3 + \tan^{-1} x = \tan^{-1} 8$, then x = _____ .

15. If for a unit vector $\vec{a}$, $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a}) = 15$, then $|\vec{x}|$ = _____ .

OR

If $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$, where $(\vec{a}\neq 0, \vec{b}\neq 0)$, then angle between $\vec{a}$ and $\vec{b}$ is _____.

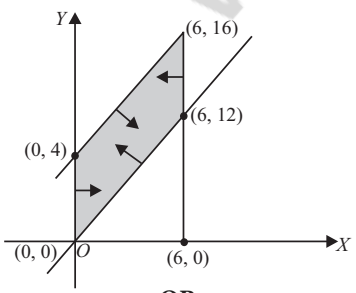
(Q.16 – Q.20) Answer the following questions.

16. Find the order of the differential equation whose general solution is given by $y = (C_1 + C_2)\cos(x + C_3) - C_4 e^{x+C_5}$, where C_1, C_2, C_3, C_4, C_5 are arbitrary constants.

17. Find the intercept cut off by the plane $2x + y - z = 5$ on x-axis.

18. A matrix A, of order 3 × 3, has determinant 4. Find the value of $|3A|$.

19. The feasible region for an LPP is shown shaded in the figure. Let $F = 3x - 4y$ be the objective function. Find the maximum value of F.



OR

Corner points of the feasible region for an LPP are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). Let $F = 4x + 6y$ be the objective function. At which point minimum value of F occurs

20. Find the number of all possible matrices of order 3 × 3 with each entry 0 or 1.

SECTION - B

21. Give an example to show that the relation R in the set of natural numbers, defined by $R = \{(x, y), x, y \in N, x \leq y^2\}$ is not transitive.

22. Evaluate, $\begin{vmatrix} 9 & 9 & 12 \\ 1 & 3 & -4 \\ 1 & 9 & 12 \end{vmatrix}$ using the properties of determinants.

23. Evaluate (i) $\int \tan^3 x\,dx$ (ii) $\int \cot^4 x\,dx$

24. Find $|\vec{a}-\vec{b}|$, if two vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a}\cdot\vec{b} = 4$.

OR

Show that the vectors $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form right angled triangle.

25. Find the coordinates of the point where the line through (5, 1, 6) and (3, 4, 1) crosses the yz-plane.

26. Two balls are drawn at random from a bag containing 5 white and 7 black balls.

(i) What is the probability that both balls are white?

(ii) What is the probability of drawing one white and one black ball?

OR

A problem in mathematics is given to 3 students whose chance of solving it are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. What is the probability that the problem is solved?

SECTION - C

27. Consider $f : R^+ \to [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$. Show that f is invertible with $f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$.

28. If $y = \cos(m\cos^{-1}x)$, then show that $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + m^2 y = 0$.

OR

Show that $f(x) = |x - 2|$ is continuous but not differentiable at $x = 2$.

29. Given the sum of the perimeter of a square and a circle, show that the sum of their areas is least when the side of the square is equal to diameter of the circle.

www.exambee.in

30. Solve the differential equation $\frac{d^2y}{dx^2} = \frac{1}{x} + 6$ given that $\frac{dy}{dx} = 0$ and $y = 0$ when $x = 1$.

31. Find the equation of the plane through the points (1, 0, –1), (3, 2, 2) and parallel to the line $\frac{x-1}{1} = \frac{y-1}{-2} = \frac{z-2}{3}$.

OR

Show that the line $\vec{r} = (2\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ is parallel to the plane $\vec{r} \cdot (\hat{i} + 5\hat{j} + \hat{k}) = 5$. Also, find the distance between them.

32. A chemical industry produces two fertilizers, *A* and *B*. The following table gives the units of ingredients *C* and *D* (per kg) of fertilizers *A* and *B* as well as minimum requirements of *C* and *D*, and costs per kg of *A* and *B*.

	Fertilizers (in units)		Minimum requirement (in units)
	A	*B*	
Ingredient *C* (per kg)	1	2	80
Ingredient *D* (per kg)	3	1	75
Cost per kg (in ₹)	4	6	

Find the quantities of *A* and *B* which would minimize the cost.

SECTION - D

33. Show that : $\begin{vmatrix} \sin^2 A & \sin A & \cos^2 A \\ \sin^2 B & \sin B & \cos^2 B \\ \sin^2 C & \sin C & \cos^2 C \end{vmatrix}$

$= -(\sin A - \sin B)(\sin B - \sin C)(\sin C - \sin A)$.

OR

If $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1} A^{-1}$.

34. Evaluate : $\int \frac{1}{\sqrt[4]{(x-1)^3 (x+2)^5}} dx$

OR

Evaluate the integral $\int_1^2 (2x^2 + 5x) dx$ using limit of sums.

35. Find the area of the region $\{(x, y) : x^2 + y^2 \le 8, x^2 \le 2y\}$.

36. Three cards are drawn successively with replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of kings drawn. Also find the mean and variance of the distribution.

Hints & Solutions

1. (a) **2.** (d) **3.** (b) **4.** (a)
5. (c) **6.** (b) **7.** (d) **8.** (a)
9. (b) **10.** (a) **11.** $\tan(xe^x) + C$ **12.** 5/12
13. 7 **14.** 1 **OR** $\frac{1}{5}$ **15.** 4 **OR** $\frac{\pi}{2}$.
16. 3 **17.** 5/2 **18.** 108
19. 0

OR

The minimum value of *F* occurs at every point of the segment joining the points (0, 2) and (3, 0).

20. 512
21. $x = 14, y = 4, z = 2$ **22.** 576
23. (i) $\frac{1}{2}\tan^2 x + \log|\cos x| + C$

(ii) $\frac{-1}{3}\cot^3 x + \cot x + x + C$

24. $\sqrt{5}$
25. $\left(0, \frac{17}{2}, \frac{-13}{2}\right)$ **26.** (i) $\frac{5}{33}$ (ii) $\frac{35}{66}$ **OR** $\frac{3}{4}$
30. $y = x \log|x| - 7x + 3x^2 + 4$
31. $4x - y - 2z - 6 = 0$ **OR** $\frac{10}{3\sqrt{3}}$ units
32. For minimum cost, 14 kg of *A* and 33 kg of *B* must be taken.
34. $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + C$ **OR** $\frac{73}{6}$
35. $\left(2\pi + \frac{4}{3}\right)$ sq. units
36.

X	0	1	2	3
P(*X*)	$\frac{1728}{2197}$	$\frac{432}{2197}$	$\frac{36}{2197}$	$\frac{1}{2197}$

Mean $= \frac{3}{13}$, Variance $= \frac{36}{169}$

www.exambites.in

PRACTICE PAPER 10
(Unsolved)

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

Time allowed : 3 hours **Maximum marks :** 80

www.examrules.in

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is

(a) 30° (b) 45° (c) 90° (d) 0°

2. Integrating factor of the differential equation $(1-x^2)\frac{dy}{dx} - xy = 1$ is

(a) $-x$ (b) $\frac{x}{1+x^2}$

(c) $\sqrt{1-x^2}$ (d) $\frac{1}{2}\log(1-x^2)$

3. Projection vector of $\vec{a}$ on $\vec{b}$ is

(a) $\left(\frac{\vec{a}\times\vec{b}}{|\vec{b}|^2}\right)\vec{b}$ (b) $\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

(c) $\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$ (d) $\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|^2}\right)\hat{b}$

4. The corner points of the feasible region determined by the system of linear constraints are (0, 0), (0, 40), (20, 40), (60, 20), (60, 0). The objective function is $Z = 4x + 3y$.

Compare the quantity in Column A and Column B.

Column A	Column B
Maximum of Z	325

(a) The quantity in column A is greater
(b) The quantity in column B is greater
(c) The two quantities are equal
(d) The relationship cannot be determined on the basis of the information supplied.

5. If $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$, then angle between $\vec{a}$ and $\vec{b}$ is $(\vec{a}\neq 0, \vec{b}\neq 0)$

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

6. $\int_0^{2/3} \frac{dx}{4+9x^2}$ equals

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{12}$ (c) $\frac{\pi}{24}$ (d) $\frac{\pi}{4}$

7. The angle between the line $\frac{x+1}{3} = \frac{y-1}{2} = \frac{z-2}{4}$ and the plane $2x + y - 3z + 4 = 0$ is

(a) $\cos^{-1}\left(\frac{4}{\sqrt{406}}\right)$ (b) $\sin^{-1}\left(\frac{4}{\sqrt{406}}\right)$

(c) 30° (d) None of these

8. $\int x^2 e^{x^3} dx$ equals

(a) $\frac{1}{3}e^{x^3} + C$ (b) $\frac{1}{3}e^{x^2} + C$

(c) $\frac{1}{2}e^{x^3} + C$ (d) $\frac{1}{2}e^{x^2} + C$

9. The direction cosines of the unit vector perpendicular to the plane $\vec{r}\cdot(6\hat{i}-3\hat{j}-2\hat{k})+1=0$ passing through the origin are

(a) $\frac{6}{7},\frac{3}{7},\frac{2}{7}$ (b) 6, 3, 2

(c) $-\frac{6}{7},\frac{3}{7},\frac{2}{7}$ (d) –6, 3, 2

10. The curve $y = x^{1/5}$ at (0, 0) has

(a) a vertical tangent (parallel to y-axis)
(b) a horizontal tangent (parallel to x-axis)
(c) an oblique tangent (d) no tangent

(Q.11 - Q.15) Fill in the blanks.

11. Let relation $R=\{(x,y)\in W\times W : y=2x-4\}$. If $(a, 2)$ and $(4, b^2)$ belongs to the relation R, then $|ab|$ = _____ .

OR

Let $f : R \to R$ be given by $f(x) = \tan x$. Then $f^{-1}(1)$ is _____ .

12. If the points with position vectors $60\hat{i}+3\hat{j}$, $40\hat{i}-8\hat{j}$ and $a\hat{i}-52\hat{j}$ are collinear, then the value of a is _____ .

13. The value of $2\int_0^{\pi/4}\sin 2x\,dx$ is _____ .

14. The value of $\sec^2(\tan^{-1}2)+\text{cosec}^2(\cot^{-1}3)$ is ____ .

15. An edge of a variable cube is increasing at the rate of 3 cm/s. Then, volume of the cube is increasing at the rate of _____ ,when the edge is 10 cm long .

OR

The points on the curve $y^2 = x + \sin x$ at which the tangent is parallel to x-axis lies on a ____ .

(Q.16 – Q.20) Answer the following questions.

16. Determine the value of k for which the following function is continuous at $x = 3$.

$$f(x)=\begin{cases}\dfrac{x^2-9}{x-3}, & x\neq 3\\ k, & x=3\end{cases}$$

OR

If $y=\log_{10}x+\log_e x$, then find $\frac{dy}{dx}$.

17. If the matrix has 8 elements, what are the number of possible orders it can have?

18. Find the sum of order and degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]=\frac{d^2y}{dx^2}$.

19. Find the number of points of non-differentiability of function $y = f(x) = |x - 3|$.

20. If ω is a complex cube root of unity, then find the value of $\begin{vmatrix}1 & \omega & \omega^2\\ \omega & \omega^2 & 1\\ \omega^2 & 1 & \omega\end{vmatrix}$.

SECTION - B

21. Prove that,

$$\cos^{-1}\left(\frac{\sin x+\cos x}{\sqrt{2}}\right)=x-\frac{\pi}{4},\ \frac{\pi}{4}\le x\le\frac{5\pi}{4}.$$

22. If $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$, then find the values of θ satisfying the equation $A^T + A = I_2$.

23. Evaluate : $\int\frac{1+\tan x}{x+\log\sec x}dx$

24. If $\vec{a}=\hat{i}+2\hat{j}$, $\vec{b}=\hat{j}+2\hat{k}$, then write a unit vector along the vector $3\vec{a}-2\vec{b}$.

OR

Let $\vec{a}=5\hat{i}-\hat{j}+7\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\lambda\hat{k}$. Find λ, such that $\vec{a}+\vec{b}$ is orthogonal to $\vec{a}-\vec{b}$.

25. Find the point on the line $\frac{x+2}{3}=\frac{y+1}{2}=\frac{z-3}{2}$ at a distance of $3\sqrt{2}$ from the point (1, 2, 3).

26. The probability that student entering a university will graduate is 0.4. Find the probability that out of 3 students of the university : (i) none will graduate, (ii) only one will graduate.

OR

Ten cards numbered 1 to 10 are placed in a box, mixed up thoroughly and then one card is drawn randomly. If it is known that the number on the drawn card is more than 3, then what is the probability that it is an even number ?

SECTION - C

27. Prove that the relation R in the set $\mathbb{R}$ of real numbers, defined by $R = \{(a, b) : 1 + ab > 0\}$, is reflexive, symmetric but not transitive.

28. Verify Rolle's Theorem for $f(x) = \sin x + \cos x$ in $\left[0,\frac{\pi}{2}\right]$.

OR

If $x = \sin t$, $y = \sin pt$, prove that

$$(1-x^2)\frac{d^2y}{dx^2}-x\frac{dy}{dx}+p^2y=0.$$

www.exambtec.in

29. Find the intervals in which the function $f(x) = (x + 1)^3(x - 3)^3$ is strictly increasing or decreasing.

30. Form the differential equation of family of parabolas having vertex at the origin and axis along positive y-axis.

31. Prove that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{4}=\frac{z-6}{7}$ are coplanar. Also, prove that the plane containing these two lines is $x - 2y + z = 0$.

OR

A plane meets the co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (α, β, γ). Show that the equation of the plane is $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$.

32. A furniture firm manufactures chairs and tables, each requiring the use of three machines A, B and C. Production of one chair requires 2 hours on machine A, 1 hour on machine B and 1 hour on machine C. Each table requires 1 hour each on machine A and B and 3 hours on machine C. The profit obtained by selling one chair is ₹ 30 while by selling one table the profit is ₹ 60. The total time available per week on machine A is 70 hours, on machine B is 40 hours and on machine C is 90 hours. How many chairs and tables should be made per week so as to maximize profit ?

SECTION - D

33. If $A=\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$, prove that $A^n=\begin{bmatrix}3^{n-1}&3^{n-1}&3^{n-1}\\3^{n-1}&3^{n-1}&3^{n-1}\\3^{n-1}&3^{n-1}&3^{n-1}\end{bmatrix}$ for all values of $n \in N$.

OR

Show that $\begin{vmatrix}1&1+p&1+p+q\\2&3+2p&1+3p+2q\\3&6+3p&1+6p+3q\end{vmatrix} = 1$.

34. Evaluate : $\int\frac{(3x+1)dx}{2x^2-2x+3}$

OR

Prove that $\int_0^{\pi} x\sin^3 x\,dx=\frac{2\pi}{3}$.

35. Find the area of the region bounded by the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$.

36. In a bolt factory machines, A, B and C manufacture respectively 25%, 30% and 45% of the total bolts. Of their output 5, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product.
(i) What is the probability that the bolt drawn is defective? (ii) If the bolt is found to be defective, find the probability that it is a product of machine B.

www.examdrives.in

Hints & Solutions

1. (c) **2.** (c) **3.** (b) **4.** (b)

5. (d) **6.** (c) **7.** (b) **8.** (a)

9. (c) **10.** (b)

11. 6 **OR** $f^{-1}(1) = \left\{n\pi+\frac{\pi}{4}; n\in Z\right\}$

12. –40 **13.** 1 **14.** 15

15. 900 cm^3/s **OR** Straight line

16. 3 **OR** $\frac{dy}{dx}=\frac{1}{x}\log_{10}e+\frac{1}{x}$

17. 4 **18.** 3 **19.** 1 **20.** 0

22. $\theta = 2n\pi \pm \frac{\pi}{3}, n \in Z$

23. $\log|x+\log\sec x|+C$

24. $\frac{1}{\sqrt{41}}(3\hat{i}+4\hat{j}-4\hat{k})$ **OR** $\pm\sqrt{73}$

25. (–2, –1, 3) and $\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$

26. (i) 0.216 (ii) 0.432 **OR** $\frac{4}{7}$

29. $f(x)$ is strictly increasing in (1, 3) and (3, ∞) strictly decreasing in (– ∞, – 1) and (–1, 1).

30. $\frac{dy}{dx}=\frac{2y}{x}$

32. To maximize the profit 15 chairs and 25 tables should be made per week.

34. $\frac{3}{4}\log|2x^2-2x+3|+\frac{\sqrt{5}}{2}\tan^{-1}\left(\frac{2x-1}{\sqrt{5}}\right)+C$

35. 20 π sq. units

36. (i) 0.0335 (ii) $\frac{24}{67}$

CBSE SAMPLE PAPER 2019-20
(Solved)

BLUE PRINT

Time Allowed : 3 hours **Maximum Marks :** 80

S. No.	Chapter	VSA-Objective Type (1 mark)	SA (2 marks)	LA-I (4 marks)	LA-II (6 marks)	Total
1.	Relations and Functions	1(1)	–	4(1)	–	5(2)
2.	Inverse Trigonometric Functions	1(1)	2(1)*	–	–	3(2)
3.	Matrices	2(2)	–	–	–	2(2)
4.	Determinants	2(2)	–	–	6(1)*	8(3)
5.	Continuity and Differentiability	1(1)	2(1)	4(1)*	–	7(3)
6.	Application of Derivatives	1(1)*	2(1)	–	6(1)*	9(3)
7.	Integrals	4(4)#	–	4(1)	–	8(5)
8.	Application of Integrals	–	–	–	6(1)	6(1)
9.	Differential Equations	1(1)	–	4(1)	–	5(2)
10.	Vector Algebra	2(2)#	2(1)*	–	–	4(3)
11.	Three Dimensional Geometry	2(2)	2(1)	–	6(1)	10(4)
12.	Linear Programming	1(1)	–	4(1)	–	5(2)
13.	Probability	2(2)	2(1)	4(1)*	–	8(4)
	Total	**20(20)**	**12(6)**	**24(6)**	**24(4)**	**80(36)**

*It is a choice based question.
#Out of the two or more questions only one question is choice based.

www.exambites.in

General Instructions :

(i) All the questions are compulsory.

(ii) This question paper contains 36 questions.

(iii) Section A comprises of 20 questions of 1 mark each. Section B comprises of 6 questions of 2 marks each. Section C comprises of 6 questions of 4 marks each. Section D comprises of 4 questions of 6 marks each.

(iv) There is no overall choice. However, an internal choice has been provided in three questions of 1 mark each, two questions of 2 marks each, two questions of 4 marks each, and two questions of 6 marks each. You have to attempt only one of the alternatives in all such questions.

(v) Use of calculators is not permitted.

SECTION - A

Q.1 - Q.10 are multiple choice type questions. Select the correct option.

1. If A is any square matrix of order 3×3 such that $|A| = 3$, then the value of $|\text{adj } A|$ is
(a) 3 (b) 1/3 (c) 9 (d) 27

2. Suppose P and Q are two different matrices of order $3 \times n$ and $n \times p$, then the order of the matrix $P \times Q$ is
(a) $3 \times p$ (b) $p \times 3$
(c) $n \times n$ (d) 3×3

3. If $(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + p\hat{j} + q\hat{k}) = \vec{0}$, then the values of p and q are
(a) $p = 6, q = 27$ (b) $p = 3, q = \frac{27}{2}$
(c) $p = 6, q = \frac{27}{2}$ (d) $p = 3, q = 27$

4. If A and B are two events such that $P(A) = 0.2$, $P(B) = 0.4$ and $P(A \cup B) = 0.5$, then value of $P(A/B)$ is
(a) 0.1 (b) 0.25
(c) 0.5 (d) 0.08

5. The point which does not lie in the half plane $2x + 3y - 12 \le 0$ is
(a) (1, 2) (b) (2, 1)
(c) (2, 3) (d) (−3, 2)

6. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then the value of $\cos^{-1}x + \cos^{-1}y$ is
(a) $2\pi/3$ (b) $\pi/3$
(c) $\pi/2$ (d) π

7. An urn contains 6 balls of which two are red and four are black. Two balls are drawn at random. Probability that they are of the different colours is
(a) $\frac{2}{5}$ (b) $\frac{1}{15}$ (c) $\frac{8}{15}$ (d) $\frac{4}{15}$

8. $\int \frac{dx}{\sqrt{9 - 25x^2}} =$
(a) $\sin^{-1}\left(\frac{5x}{3}\right) + c$
(b) $\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right) + c$
(c) $\frac{1}{6}\log\left(\frac{3+5x}{3-5x}\right) + c$
(d) $\frac{1}{30}\log\left(\frac{3+5x}{3-5x}\right) + c$

9. What is the distance (in units) between the two planes $3x + 5y + 7z = 3$ and $9x + 15y + 21z = 9$?
(a) 0 (b) 3 (c) $\frac{6}{\sqrt{83}}$ (d) 6

10. The equation of the line in vector form passing through the point$(-1, 3, 5)$ and parallel to line $\frac{x-3}{2} = \frac{y-4}{3}, z = 2$ is
(a) $\vec{r} = (-\hat{i} + 3\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + \hat{k})$
(b) $\vec{r} = (-\hat{i} + 3\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 3\hat{j})$
(c) $\vec{r} = (2\hat{i} + 3\hat{j} - 2\hat{k}) + \lambda(-\hat{i} + 3\hat{j} + 5\hat{k})$
(d) $\vec{r} = (2\hat{i} + 3\hat{j}) + \lambda(-\hat{i} + 3\hat{j} + 5\hat{k})$

(Q.11 - Q.15) Fill in the blanks.

11. If f be the greatest integer function defined as $f(x) = [x]$ and g be the modulus function defined as $g(x) = |x|$, then the value of gof $(-5/4)$ is _____.

12. If the function $f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & \text{when } x \neq 1 \\ k, & \text{when } x = 1 \end{cases}$ is given to be continuous at $x = 1$, then the value of k is _____.

13. If $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$, then the value of y is _____.

14. If tangent to the curve $y^2 + 3x - 7 = 0$ at the point (h, k) is parallel to line $x - y = 4$, then value of k is _____.

OR

For the curve $y = 5x - 2x^3$, if x increases at the rate of 2 units/sec, then at $x = 3$ the slope of the curve is changing at _____.

www.exambites.in

15. The magnitude of projection of $(2\hat{i}-\hat{j}+\hat{k})$ on $(\hat{i}-2\hat{j}+2\hat{k})$ is _____.

OR

Vector of magnitude 5 units and in the direction opposite to $2\hat{i}+3\hat{j}-6\hat{k}$ is _____.

(Q.16-Q.20) Answer the following questions.

16. Check whether $(l + m + n)$ is a factor of the determinant $\begin{vmatrix} l+m & m+n & n+l \\ n & l & m \\ 2 & 2 & 2 \end{vmatrix}$ or not. Give reason.

17. Evaluate : $\int_{-2}^{2}(x^3+1)\,dx$

18. Find : $\int\frac{3+3\cos x}{x+\sin x}dx$

OR

Find : $\int(\cos^2 2x - \sin^2 2x)dx$

19. Find : $\int xe^{(1+x^2)}\,dx$

20. Write the general solution of differential equation $\frac{dy}{dx} = e^{x+y}$.

SECTION - B

21. Express $\sin^{-1}\left(\frac{\sin x+\cos x}{\sqrt{2}}\right)$, where $-\frac{\pi}{4}<x<\frac{\pi}{4}$, in the simplest form.

OR

Let R be the relation in the set Z of integers given by $R = \{(a, b) : 2 \text{ divides } a - b\}$. Show that the relation R is transitive? Write the equivalence class of 0.

22. If $y = ae^{2x} + be^{-x}$, then show that $\frac{d^2y}{dx^2}-\frac{dy}{dx}-2y=0$.

23. A particle moves along the curve $x^2 = 2y$. At what point, ordinate increases at the same rate as abscissa increases?

24. For three non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$, prove that $[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}]=0$.

OR

If $\vec{a}+\vec{b}+\vec{c}=0$ and $|\vec{a}|=3, |\vec{b}|=5, |\vec{c}|=7$, then find the value of $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$.

25. Find the acute angle between the lines $\frac{x-4}{3}=\frac{y+3}{4}=\frac{z+1}{5}$ and $\frac{x-1}{4}=\frac{y+1}{-3}=\frac{z+10}{5}$.

26. *A* speaks truth in 80% cases and *B* speaks truth in 90% cases. In what percentage of cases are they likely to agree with each other in stating the same fact?

SECTION - C

27. Let $f : A \to B$ be a function defined as $f(x)=\frac{2x+3}{x-3}$, where $A = R - \{3\}$ and $B = R - \{2\}$. Is the function f one–one and onto? Is f invertible? If yes, then find its inverse.

28. If $\sqrt{1-x^2}+\sqrt{1-y^2}=a(x-y)$, then prove that $\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$.

OR

If $x = a(\cos 2\theta + 2\theta \sin 2\theta)$ and $y = a(\sin 2\theta - 2\theta \cos 2\theta)$, find $\frac{d^2y}{dx^2}$ at $\theta=\frac{\pi}{8}$.

29. Solve the differential equation :

$$xdy - ydx = \sqrt{x^2+y^2}\,dx$$

30. Evaluate : $\int_1^3|x^2-2x|\,dx$

31. Two numbers are selected at random (without replacement) from first 7 natural numbers. If X denotes the smaller of the two numbers obtained, find the probability distribution of X. Also, find mean of the distribution.

OR

There are three coins, one is a two headed coin (having head on both the faces), another is a biased coin that comes up heads 75% of the time and the third is an unbiased coin.

www.examkites.in

One of the three coins is chosen at random and tossed. If it shows head, then what is the probability that it was the two headed coin ?

32. Two tailors A and B earn ₹ 150 and ₹ 200 per day respectively. A can stitch 6 shirts and 4 pants per day, while B can stitch 10 shirts and 4 pants per day. Form a L.P.P. to minimize the labour cost to produce (stitch) at least 60 shirts and 32 pants and solve it graphically.

SECTION - D

33. Using the properties of determinants, prove that

$$\begin{vmatrix} (y+z)^2 & x^2 & x^2 \\ y^2 & (z+x)^2 & y^2 \\ z^2 & z^2 & (x+y)^2 \end{vmatrix} = 2xyz\,(x+y+z)^3.$$

OR

If $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$, find A^{-1}. Hence, solve the system of equations

$2x + 3y + 4z = 17$; $x - y = 3$; $y + 2z = 7$.

34. Using integration, find the area of the region, $\{(x, y) : x^2 + y^2 \le 1, x + y \ge 1, x \ge 0, y \ge 0\}$.

35. A given quantity of metal is to be cast into a solid half circular cylinder with a rectangular base and semi-circular ends. Show that in order that total surface area is minimum, the ratio of length of cylinder to the diameter of semi-circular ends is $\pi : \pi + 2$.

OR

Show that the triangle of maximum area that can be inscribed in a given circle is an equilateral triangle.

36. Find the equation of a plane passing through the points $A(2, 1, 2)$ and $B(4, -2, 1)$ and perpendicular to plane $\vec{r} \cdot (\hat{i} - 2\hat{k}) = 5$. Also, find the coordinates of the point, where the line passing through the points (3, 4, 1) and (5, 1, 6) crosses the plane thus obtained.

Detailed Solutions

1. (c) : Given, A is a square matrix of order 3×3 and $|A| = 3$.
Now, $|\text{adj } A| = |A|^{n-1}$, where n is order of A.
$\Rightarrow$ $|\text{adj } A| = (3)^2 = 9$

2. (a) : Given, P is a $3 \times n$ matrix and Q is a $n \times p$ matrix. So, order of $P \times Q$ is $3 \times p$.

3. (b) : Given, $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 27 \\ 1 & p & q \end{vmatrix} = 0$

$\Rightarrow$ $\hat{i}\,(6q - 27p) - \hat{j}\,(2q - 27) + \hat{k}\,(2p - 6) = 0$

$\Rightarrow$ $6q - 27p = 0$, $2q - 27 = 0$ and $2p - 6 = 0$

$\Rightarrow$ $q = \dfrac{27}{2}$ and $p = 3$.

4. (b) : We have, $P(A) = 0.2$, $P(B) = 0.4$ and $P(A \cup B) = 0.5$

$\because$ $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
$= 0.2 + 0.4 - 0.5 = 0.1$

Now, $P(A/B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.1}{0.4} = \dfrac{1}{4} = 0.25$

5. (c) : We have, $2x + 3y - 12 \le 0$ ...(i)
Putting (1, 2) in (i), we get $-4 \le 0$
Putting (2, 1) in (i), we get $-5 \le 0$
Putting (2, 3) in (i), we get $1 \le 0$, which is not true.
So, (2, 3) does not lie in the half plane.

6. (b) : We have, $\sin^{-1}x + \sin^{-1}y = \dfrac{2\pi}{3}$

$\Rightarrow$ $\dfrac{\pi}{2} - \cos^{-1}x + \dfrac{\pi}{2} - \cos^{-1}y = \dfrac{2\pi}{3}$

$\Rightarrow$ $\pi - (\cos^{-1}x + \cos^{-1}y) = \dfrac{2\pi}{3}$

$\Rightarrow$ $\cos^{-1}x + \cos^{-1}y = \pi - \dfrac{2\pi}{3} = \dfrac{\pi}{3}$

www.exambites.in

7. (c) : Total number of possible outcomes = ${}^6C_2 = 15$
Possible outcomes = ${}^2C_1\ {}^4C_1 = 2 \times 4 = 8$

$\therefore$ Required Probability = $\frac{8}{15}$

8. (b): Let $I = \int \frac{dx}{\sqrt{9-25x^2}} = \frac{1}{5}\int \frac{dx}{\sqrt{\frac{9}{25}-x^2}}$

$= \frac{1}{5}\int \frac{dx}{\sqrt{\left(\frac{3}{5}\right)^2 - x^2}}$

$= \frac{1}{5}\sin^{-1}\left(\frac{x}{3/5}\right) + c = \frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right) + c$

9. (a): Given planes are $3x + 5y + 7z = 3$ and $9x + 15y + 21z = 9$
$\Rightarrow$ $3x + 5y + 7z = 3$
Both the planes are coincident planes, so distance between them is zero.

10. (b): Given line is $\frac{x-3}{2} = \frac{y-4}{3}$, $z - 2 = 0$

Vector in the direction of this line, $\vec{b} = 2\hat{i} + 3\hat{j}$

$\therefore$ Equation of line passing through the point (–1, 3, 5) and parallel to $\vec{b} = 2\hat{i} + 3\hat{j}$ is given by $\vec{r} = (-\hat{i} + 3\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 3\hat{j})$

11. Given, $f(x) = [x]$ and $g(x) = |x|$

$\therefore$ $gof\left(\frac{-5}{4}\right) = g\left(f\left(\frac{-5}{4}\right)\right) = g\left(\left[\frac{-5}{4}\right]\right) = g(-2) = 2$

12. We have, $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & \text{when } x \neq 1 \\ k, & \text{when } x = 1 \end{cases}$

$\therefore$ $\lim\limits_{x\to 1} f(x) = \lim\limits_{x\to 1} \frac{x^2-1}{x-1} = \lim\limits_{x\to 1}(x+1) = 2$ and $f(1) = k$
Since, the function is continuous at $x = 1$
$\therefore$ $\lim\limits_{x\to 1} f(x) = f(1) \Rightarrow k = 2$

13. Given, $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} x+2y \\ 2x+y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$

Comparing, we get $x + 2y = 5$ and $2x + y = 4$
Solving both the equations, we get $y = 2$.

14. Given, $y^2 + 3x = 7$

$\Rightarrow$ $2y\frac{dy}{dx} + 3 = 0 \Rightarrow \frac{dy}{dx} = \frac{-3}{2y}$

Slope of tangent to the curve $y^2 + 3x - 7 = 0$ at (h, k) is given by $\frac{-3}{2k}$.

Since tangent to the given curve is parallel to $x - y = 4$.

$\therefore$ $\frac{-3}{2k} = 1 \Rightarrow k = \frac{-3}{2}$

OR

Given curve is $y = 5x - 2x^3$

Slope of this curve is $\frac{dy}{dx} = 5 - 6x^2 = m$(say)

As given, $\frac{dx}{dt}$ = 2 units/sec

Now, $\frac{dm}{dt} = -12 \cdot x \cdot \frac{dx}{dt}$

$\therefore$ $\left(\frac{dm}{dt}\right)_{x=3} = -12 \cdot 3 \cdot 2 = -72$

$\therefore$ Slope of the curve is decreasing at the rate of 72 units/sec.

15. Projection of $\left(2\hat{i} - \hat{j} + \hat{k}\right)$ on $\left(\hat{i} - 2\hat{j} + 2\hat{k}\right)$

$= \left|\frac{2\cdot 1 + 1\cdot 2 + 2\cdot 1}{\sqrt{1+4+4}}\right| = \left|\frac{2+2+2}{\sqrt{9}}\right| = \left|\frac{6}{\pm 3}\right| = 2$

OR

A vector in the direction opposite to $2\hat{i} + 3\hat{j} - 6\hat{k}$ is $-2\hat{i} - 3\hat{j} + 6\hat{k}$.

Its magnitude is $\left|\sqrt{4+9+36}\right| = \left|\sqrt{49}\right| = 7$

So, a vector in the direction opposite to $2\hat{i} + 3\hat{j} - 6\hat{k}$ of magnitude 5 units is $\frac{5}{7}(-2\hat{i} - 3\hat{j} + 6\hat{k})$.

16. We have, $\begin{vmatrix} l+m & m+n & n+l \\ n & l & m \\ 2 & 2 & 2 \end{vmatrix}$

$= \begin{vmatrix} l+m+n & m+n+l & n+l+m \\ n & l & m \\ 2 & 2 & 2 \end{vmatrix}$ [Apply $R_1 \to R_1 + R_2$]

www.examples.in

$= 2(l + m + n)\begin{vmatrix} 1 & 1 & 1 \\ n & l & m \\ 1 & 1 & 1 \end{vmatrix}$; yes, $(l + m + n)$ is a factor.

17. $\int_{-2}^{2}(x^3+1)\,dx = \int_{-2}^{2}(x^3)\,dx + \int_{-2}^{2}1\,dx = I_1 + I_2$

$= 0 + [x]_{-2}^{2}$ ($\because$ I_1 is odd function)

$= 2 + 2 = 4$

18. Let $I = \int \frac{3+3\cos x}{x+\sin x}dx$

Putting $x + \sin x = t$, we get $(1 + \cos x)dx = dt$

$\therefore \quad I = 3\int \frac{dt}{t} = 3\log|t| + C = 3\log|(x+\sin x)| + C$

OR

We have, $\int(\cos^2 2x - \sin^2 2x)dx$

$= \int \cos 4x dx = \frac{\sin 4x}{4} + C$

19. Let $I = \int xe^{(1+x^2)}dx$

Putting $(1 + x^2) = t$, we get, $2xdx = dt$

$\therefore \quad I = \frac{1}{2}\int e^t\,dt = \frac{1}{2}e^t + C = \frac{1}{2}e^{(1+x^2)} + C$

20. We have, $\frac{dy}{dx} = e^{x+y}$

$\Rightarrow \quad \frac{dy}{dx} = e^x e^y \quad \Rightarrow \quad \frac{dy}{e^y} = e^x dx$

Integrating both sides, we get

$-e^{-y} + c = e^x \Rightarrow e^x + e^{-y} = c$

21. We have, $\sin^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right)$ if $\frac{-\pi}{4} < x < \frac{\pi}{4}$

$= \sin^{-1}\left(\frac{\sin x}{\sqrt{2}} + \frac{\cos x}{\sqrt{2}}\right)$ if $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$= \sin^{-1}\left(\sin x\cos\frac{\pi}{4} + \cos x\sin\frac{\pi}{4}\right)$

if $\frac{-\pi}{4} + \frac{\pi}{4} < x + \frac{\pi}{4} < \frac{\pi}{4} + \frac{\pi}{4}$

$= \sin^{-1}\left(\sin\left(x + \frac{\pi}{4}\right)\right)$ if $0 < \left(x + \frac{\pi}{4}\right) < \frac{\pi}{2}$

i.e., principal values $= \left(x + \frac{\pi}{4}\right)$

OR

Let 2 divides $(a - b)$ and 2 divides $(b - c)$: where $a, b, c \in Z$

So, 2 divides $[(a - b) + (b - c)]$

$\Rightarrow$ 2 divides $(a - c)$: Yes relation R is transitive.

Equivalence class of 0 = {0, ± 2, ± 4, ± 6, ...}

22. We have, $y = ae^{2x} + be^{-x}$ ...(i)

$\Rightarrow \quad \frac{dy}{dx} = 2ae^{2x} - be^{-x}$ ...(ii)

$\Rightarrow \quad \frac{d^2y}{dx^2} = 4ae^{2x} + be^{-x}$ ...(iii)

L.H.S. $= \frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y$

$= (4ae^{2x} + be^{-x}) - (2ae^{2x} - be^{-x}) - 2(ae^{2x} + be^{-x})$

[From (i), (ii) and (iii)

$= 4ae^{2x} + be^{-x} - 2ae^{2x} + be^{-x} - 2ae^{2x} - 2be^{-x} = 0$

= R.H.S.

23. We have, $x^2 = 2y$ ...(i)

$\Rightarrow \quad 2x\frac{dx}{dt} = 2\frac{dy}{dt}$

$\Rightarrow \quad 2x\frac{dx}{dt} = 2\frac{dx}{dt} \Rightarrow x = 1 \qquad \left(\text{Given, } \frac{dy}{dt} = \frac{dx}{dt}\right)$

From (i), $y = \frac{1}{2}$

So, point is $\left(1, \frac{1}{2}\right)$.

24. L.H.S. $= \left[\vec{a} - \vec{b}\ \vec{b} - \vec{c}\ \vec{c} - \vec{a}\right]$

$= (\vec{a} - \vec{b}) \cdot \{(\vec{b} - \vec{c}) \times (\vec{c} - \vec{a})\}$

$= (\vec{a} - \vec{b}) \cdot \{\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{c} \times \vec{c} + \vec{c} \times \vec{a}\}$

$= (\vec{a} - \vec{b}) \cdot \{\vec{b} \times \vec{c} - \vec{b} \times \vec{a} + \vec{c} \times \vec{a}\}$

$= (\vec{a} - \vec{b}) \cdot \{\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}\}$

$= \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{a} \times \vec{b}) + \vec{a} \cdot (\vec{c} \times \vec{a}) - \vec{b} \cdot (\vec{b} \times \vec{c})$

$- \vec{b} \cdot (\vec{a} \times \vec{b}) - \vec{b} \cdot (\vec{c} \times \vec{a})$

$= \vec{a} \cdot (\vec{b} \times \vec{c}) + 0 + 0 - 0 - 0 - \vec{b} \cdot (\vec{c} \times \vec{a})$

$= \vec{a} \cdot (\vec{b} \times \vec{c}) - \vec{b} \cdot (\vec{c} \times \vec{a})$

$= 0$ = R.H.S. ($\because [\vec{a}\,\vec{b}\,\vec{c}] = [\vec{b}\,\vec{c}\,\vec{a}]$)

www.exambites.in

OR

We have, $(\vec{a}+\vec{b}+\vec{c})\cdot(\vec{a}+\vec{b}+\vec{c})=0$ $[\because\ \vec{a}+\vec{b}+\vec{c}=0]$

$\Rightarrow\ \vec{a}\cdot\vec{a}+\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$
$+\vec{c}\cdot\vec{b}+\vec{c}\cdot\vec{c}=0$

$\Rightarrow\ |\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0$

$\Rightarrow\ 3^2+5^2+7^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0$

$\Rightarrow\ 2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})\ =-(9+25+49)$

$\Rightarrow\ (\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=-\dfrac{83}{2}$

25. Vector in the direction of first line, $\vec{b}$
$=(3\hat{i}+4\hat{j}+5\hat{k})$

Vector in the direction of second line, $\vec{d}$
$=(4\hat{i}-3\hat{j}+5\hat{k})$

Angle θ between two lines is given by cos θ
$=\dfrac{\vec{b}.\vec{d}}{|\vec{b}||\vec{d}|}$

$\Rightarrow\ \cos\theta=\dfrac{(3\hat{i}+4\hat{j}+5\hat{k}).(4\hat{i}-3\hat{j}+5\hat{k})}{|(3\hat{i}+4\hat{j}+5\hat{k})||(4\hat{i}-3\hat{j}+5\hat{k})|}$

$\Rightarrow\ \cos\theta=\dfrac{12-12+25}{\sqrt{9+16+25}\sqrt{16+9+25}}$

$\Rightarrow\ \cos\theta=\dfrac{25}{\sqrt{50}\sqrt{50}}\ \Rightarrow\ \cos\theta=\dfrac{1}{2}\ \Rightarrow\ \theta=\dfrac{\pi}{3}$

26. We have, $P(A)=\dfrac{80}{100}=\dfrac{4}{5}$ and

$P(B)=\dfrac{90}{100}=\dfrac{9}{10}$

P(Agree) = P(Both speaking truth or both telling lie)
$= P(AB$ or $\bar{A}\bar{B})$
$= P(A)P(B)$ or $P(\bar{A})\,P(\bar{B})$
$=\left(\dfrac{4}{5}\right)\left(\dfrac{9}{10}\right)+\left(\dfrac{1}{5}\right)\left(\dfrac{1}{10}\right)=\dfrac{36+1}{50}=\dfrac{37}{50}$
$=\dfrac{74}{100}=74\%$

27. Let $y=f(x)=\dfrac{2x+3}{x-3}$ ...(i)

Let $x_1, x_2 \in A = R-\{3\}$
Let $f(x_1)=f(x_2)$

$\Rightarrow\ \dfrac{2x_1+3}{x_1-3}=\dfrac{2x_2+3}{x_2-3}$

$\Rightarrow\ (2x_1+3)(x_2-3)=(2x_2+3)(x_1-3)$
$\Rightarrow\ (2x_1x_2-6x_1+3x_2-9)=(2x_1x_2-6x_2+3x_1-9)$
$\Rightarrow\ -6x_1+3x_2=-6x_2+3x_1$
$\Rightarrow\ 9x_1=9x_2\ \Rightarrow\ x_1=x_2$
Now, $f(x_1)=f(x_2)\Rightarrow x_1=x_2$
So $f(x)$ is one-one.
For onto:

$\because\ y=\dfrac{2x+3}{x-3}\ \Rightarrow\ xy-3y=2x+3$

$\Rightarrow\ xy-2x=3y+3\ \Rightarrow\ x(y-2)=3(y+1)$

$\Rightarrow\ x=\dfrac{3(y+1)}{(y-2)}$ ...(ii)

Equation (ii) is defined for all real values of y except 2 *i.e.*, $y \in R-\{2\}$, which is same as given set $B = R-\{2\}$
(co-domain = range)
Also, $y=f(x)$

$f(x)=f\left(\dfrac{3(y+1)}{y-2}\right)$ [From (ii)]

$=\dfrac{2\left[\dfrac{3(y+1)}{(y-2)}\right]+3}{\dfrac{3(y+1)}{(y-2)}-3}\left(\text{since } f(x)=\dfrac{2x+3}{x-3}\right)$

$=\dfrac{2(3y+3)+3(y-2)}{3y+3-3y+6}=\dfrac{9y}{9}=y$

Thus, for every $y \in B$, there exist $x \in A$ such that $f(x) = y$
Thus function is onto.
Since, $f(x)$ is one-one and onto. So, $f(x)$ is invertible.
Inverse is given by $x=f^{-1}(y)=\dfrac{3(y+1)}{(y-2)}$

28. We have, $\sqrt{1-x^2}+\sqrt{1-y^2}=a(x-y)$
Let $x = \sin A$, $y = \sin B$

$\therefore\ \sqrt{1-\sin^2 A}+\sqrt{1-\sin^2 B}=a(\sin A-\sin B)$

$\Rightarrow\ \cos A+\cos B=a(\sin A-\sin B)$

$\Rightarrow\ 2\cos\left(\dfrac{A+B}{2}\right)\cos\left(\dfrac{A-B}{2}\right)$

$=2a\cos\left(\dfrac{A+B}{2}\right)\sin\left(\dfrac{A-B}{2}\right)$

$\Rightarrow\ \cos\left(\dfrac{A-B}{2}\right)=a\sin\left(\dfrac{A-B}{2}\right)$

www.exambites.in

$$= \cot\left(\frac{A-B}{2}\right) = a \Rightarrow \frac{A-B}{2} = \cot^{-1} a$$

$\Rightarrow$ $A - B = 2 \cot^{-1} a$

$\Rightarrow$ $\sin^{-1} x - \sin^{-1} y = 2 \cot^{-1} a$

Differentiating w.r.t. x, we get

$$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

OR

Given, $x = a(\cos 2\theta + 2\theta \sin 2\theta)$

$$\Rightarrow \frac{dx}{d\theta} = a(-2 \sin 2\theta + 2 \sin 2\theta + 4\theta \cos 2\theta)$$

$$\Rightarrow \frac{dx}{d\theta} = a(4\theta \cos 2\theta) \quad ...(i)$$

Again, $y = a(\sin 2\theta - 2\theta \cos 2\theta)$

$$\Rightarrow \frac{dy}{d\theta} = a(2\cos 2\theta + 4\theta \sin 2\theta - 2 \cos 2\theta)$$

$$\Rightarrow \frac{dy}{d\theta} = a(4\theta \sin 2\theta) \quad ...(ii)$$

From (i) and (ii), we get

$$\frac{dy}{dx} = \frac{a(4\theta \sin 2\theta)}{a(4\theta \cos 2\theta)} = \tan 2\theta$$

Differentiating again w.r.t. x, we get

$$\frac{d^2y}{dx^2} = 2\sec^2 2\theta . \frac{d\theta}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = 2\sec^2 2\theta \cdot \frac{1}{a(4\theta \cos 2\theta)}$$

$$\left[\frac{d^2y}{dx^2}\right]_{\theta=\frac{\pi}{8}} = 2\sec^2\frac{\pi}{4} \cdot \frac{1}{a\left(4\frac{\pi}{8}\cos\frac{\pi}{4}\right)} = \frac{8\sqrt{2}}{\pi a}$$

29. We have, $xdy - ydx = \sqrt{x^2 + y^2}\, dx$

$$\Rightarrow x\frac{dy}{dx} - y = \sqrt{x^2 + y^2}$$

$$\Rightarrow x\frac{dy}{dx} = y + \sqrt{x^2 + y^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \quad ...(i)$$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Putting in (i), we get

$$v + x\frac{dv}{dx} = \frac{vx + \sqrt{x^2 + v^2x^2}}{x}$$

$$\Rightarrow v + x\frac{dv}{dx} = \frac{x\left(v + \sqrt{1+v^2}\right)}{x}$$

$$\Rightarrow x\frac{dv}{dx} = v + \sqrt{1+v^2} - v$$

$$\Rightarrow x\frac{dv}{dx} = \sqrt{1+v^2} \Rightarrow \frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$$

Integrating both sides, we get

$$\int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \log\left(v + \sqrt{1+v^2}\right) = \log x + \log c$$

$$\Rightarrow \log\left(v + \sqrt{1+v^2}\right) = \log cx$$

$$\Rightarrow \left(v + \sqrt{1+v^2}\right) = cx$$

$$\Rightarrow \left(\frac{y}{x} + \sqrt{1+\left(\frac{y}{x}\right)^2}\right) = cx$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = cx^2$$

30. Let $I = \int_1^3 |x^2 - 2x|\, dx$

$$\because \quad |x^2 - 2x| = \begin{cases} -(x^2 - 2x), & \text{when } 1 \le x < 2 \\ (x^2 - 2x), & \text{when } 2 \le x \le 3 \end{cases}$$

$$\Rightarrow I = \int_1^2 |x^2 - 2x|\, dx + \int_2^3 |x^2 - 2x|\, dx$$

$$= -\int_1^2 (x^2 - 2x)\, dx + \int_2^3 (x^2 - 2x)\, dx$$

$$= -\left[\frac{x^3}{3} - x^2\right]_1^2 + \left[\frac{x^3}{3} - x^2\right]_2^3$$

$$= -\left(-\frac{4}{3} + \frac{2}{3}\right) + \left(\frac{4}{3}\right) = \frac{6}{3} = 2$$

31. Total cases when two numbers can be selected from first 7 natural numbers = ${}^7C_2$

Let X denotes the smaller of the two numbers obtained.

www.exambites.in

So, X can take values 1, 2, 3, 4, 5, 6.

$\therefore \quad P(X=1)=\frac{6}{{}^{7}C_2}=\frac{6}{21}=\frac{2}{7}, P(X=2)=\frac{5}{{}^{7}C_2}=\frac{5}{21},$

$P(X=3)=\frac{4}{{}^{7}C_2}=\frac{4}{21}, P(X=4)=\frac{3}{{}^{7}C_2}=\frac{3}{21}=\frac{1}{7},$

$P(X=5)=\frac{2}{{}^{7}C_2}=\frac{2}{21}, P(X=6)=\frac{1}{{}^{7}C_2}=\frac{1}{21}$

X_i	1	2	3	4	5	6
P_i	$\frac{6}{21}$	$\frac{5}{21}$	$\frac{4}{21}$	$\frac{3}{21}$	$\frac{2}{21}$	$\frac{1}{21}$
P_iX_i	$\frac{6}{21}$	$\frac{10}{21}$	$\frac{12}{21}$	$\frac{12}{21}$	$\frac{10}{21}$	$\frac{6}{21}$

$\therefore \quad \text{Mean} = \sum P_iX_i = \frac{6}{21}+\frac{10}{21}+\frac{12}{21}+\frac{12}{21}+\frac{10}{21}+\frac{6}{21}$

$=\frac{56}{21}=\frac{8}{3}$

OR

Let E_1 = event of selecting a two headed coin.
E_2 = event of selecting a biased coin, which shows 75% times head.
E_3 = event of selecting an unbiased coin.
A = event that tossed coin shows head.

$\therefore \quad P(E_1)=P(E_2)=P(E_3)=\frac{1}{3}$

$P\left(\frac{A}{E_1}\right)$ = P(coin showing head given that it is two headed coin) = 1

$P\left(\frac{A}{E_2}\right)$ = P(coin showing head given that it is biased coin) = $\frac{75}{100}=\frac{3}{4}$

$P\left(\frac{A}{E_3}\right)$ = P(coin showing head given that it is an unbiased coin) =1/2

By Bayes' theorem

P(getting two headed coin when it is known that it shows head)

$$P\left(\frac{E_1}{A}\right)=\frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right)+P(E_2)P\left(\frac{A}{E_2}\right)+P(E_3)P\left(\frac{A}{E_3}\right)}$$

$$=\frac{\frac{1}{3}\times 1}{\frac{1}{3}\times 1+\frac{1}{3}\times\frac{3}{4}+\frac{1}{3}\times\frac{1}{2}}=\frac{\frac{1}{3}\times 1}{\frac{1}{3}\left(1+\frac{3}{4}+\frac{1}{2}\right)}=\frac{\frac{1}{3}}{\frac{1}{3}\times\frac{9}{4}}=\frac{4}{9}$$

$\therefore$ Required probability = 4/9

32. Let tailor A works for x days and tailor B works for y days.

Objective function:

To minimize labour cost $Z = 150x + 200y$ (in ₹)

Subject to constraints

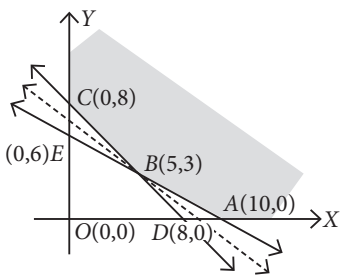
$6x + 10y \geq 60$ *i.e.*, $3x + 5y \geq 30$,

$4x + 4y \geq 32$ *i.e.*, $x + y \geq 8$, $x \geq 0$, $y \geq 0$

Consider equations to draw the graph and then we will shade feasible region

$3x + 5y = 30$

$x + y = 8$



Corner points of feasible region are $A(10, 0)$, $B(5, 3)$ and $C(0, 8)$

Value of Z at these corner points

Point	$Z = 150x + 200y$ (in ₹)
$A(10, 0)$	1500 + 0 = 1500
$B(5, 3)$	750 + 600 = 1350 (minimum)
$C(0, 8)$	0 + 1600 = 1600

So, minimum value of Z is ₹ 1350 when tailor A works for 5 days and tailor B works for 3 days.

To check draw $150x + 200y < 1350$ *i.e.*, $3x + 4y < 27$

As, there is no region common with feasible region. So minimum value is ₹ 1350.

www.examrites.in

33. L.H.S. $= \begin{vmatrix} (y+z)^2 & x^2 & x^2 \\ y^2 & (z+x)^2 & y^2 \\ z^2 & z^2 & (x+y)^2 \end{vmatrix}$

Applying $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$, we get

$$\begin{vmatrix} (y+z)^2 & x^2-(y+z)^2 & x^2-(y+z)^2 \\ y^2 & (z+x)^2-y^2 & 0 \\ z^2 & 0 & (x+y)^2-z^2 \end{vmatrix}$$

$$= \begin{vmatrix} (y+z)^2 & (x+y+z)(x-y-z) & (x+y+z)(x-y-z) \\ y^2 & (z+x+y)(z+x-y) & 0 \\ z^2 & 0 & (x+y+z)(x+y-z) \end{vmatrix}$$

Taking $(x + y + z)$ common from C_2 as well as C_3, we get

$$(x+y+z)^2 \begin{vmatrix} (y+z)^2 & (x-y-z) & (x-y-z) \\ y^2 & (z+x-y) & 0 \\ z^2 & 0 & (x+y-z) \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2 - R_3$, we get

$$(x+y+z)^2 \begin{vmatrix} 2yz & -2z & -2y \\ y^2 & (z+x-y) & 0 \\ z^2 & 0 & (x+y-z) \end{vmatrix}$$

Taking $\frac{1}{y}$ and $\frac{1}{z}$ common from C_2 and C_3 respectively, we get

$$\frac{(x+y+z)^2}{yz} \begin{vmatrix} 2yz & -2yz & -2yz \\ y^2 & (yz+yx-y^2) & 0 \\ z^2 & 0 & (zx+zy-z^2) \end{vmatrix}$$

Applying $C_2 \to C_2 + C_1$ and $C_3 \to C_3 + C_1$, we get

$$\frac{(x+y+z)^2}{yz} \begin{vmatrix} 2yz & 0 & 0 \\ y^2 & (yz+yx) & y^2 \\ z^2 & z^2 & (zx+zy) \end{vmatrix}$$

Expanding along R_1, we get

$$\left(\frac{(x+y+z)^2}{yz}\right) 2yz[(yz+yx)(zx+zy)-y^2z^2]$$

$= 2(x + y + z)^2[xyz^2 + x^2yz + xy^2z + y^2z^2 - y^2z^2]$

$= 2xyz(x + y + z)^2(x + y + z)$

$= 2xyz(x + y + z)^3$

OR

We have, $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

$\therefore \quad |A| = 2(-2 - 0) - 3(2 - 0) + 4(1 - 0) = -6 \neq 0$

$\therefore \quad A^{-1}$ exists.

Cofactors are

$A_{11} = -2, A_{12} = -2, A_{13} = 1$

$A_{21} = -2, A_{22} = 4, A_{23} = -2$

$A_{31} = 4, A_{32} = 4, A_{33} = -5$

$$\therefore \quad \text{adj}\,A = \begin{bmatrix} -2 & -2 & 1 \\ -2 & 4 & -2 \\ 4 & 4 & -5 \end{bmatrix}'$$

$$\Rightarrow \quad \text{adj}\,A = \begin{bmatrix} -2 & -2 & 4 \\ -2 & 4 & 4 \\ 1 & -2 & -5 \end{bmatrix}$$

$$\therefore \quad A^{-1} = \frac{\text{adj}\,A}{|A|} = \frac{1}{-6}\begin{bmatrix} -2 & -2 & 4 \\ -2 & 4 & 4 \\ 1 & -2 & -5 \end{bmatrix} \qquad \text{...(i)}$$

System of equations can be written as $AX = B$

where $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 17 \\ 3 \\ 7 \end{bmatrix}$

Now, $AX = B \Rightarrow X = A^{-1}B$

$$\Rightarrow \quad X = \frac{1}{-6}\begin{bmatrix} -2 & -2 & 4 \\ -2 & 4 & 4 \\ 1 & -2 & -5 \end{bmatrix}\begin{bmatrix} 17 \\ 3 \\ 7 \end{bmatrix} \qquad \text{(From (i))}$$

$$\Rightarrow \quad X = -\frac{1}{6}\begin{bmatrix} -34-6+28 \\ -34+12+28 \\ 17-6-35 \end{bmatrix} = \frac{-1}{6}\begin{bmatrix} -12 \\ 6 \\ -24 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$$\Rightarrow \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

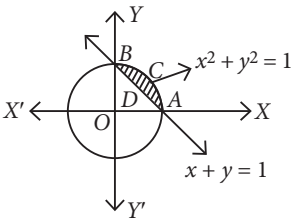
$\Rightarrow \quad x = 2, y = -1, z = 4$

www.exambites.in

34. $x^2 + y^2 = 1$ ...(i)

$x + y = 1$ ...(ii)

Solving (i) and (ii), we get



$x^2 + (1 - x)^2 = 1$

$\Rightarrow\ x^2 + x^2 - 2x + 1 = 1$

$\Rightarrow\ 2x^2 - 2x = 0$

$\Rightarrow\ 2x(x - 1) = 0$

$\Rightarrow\ x = 0$ or $x = 1$

Required area = shaded area $ACBDA$

= area ($OACBO$) – area ($OADBO$)

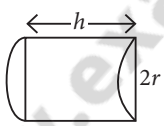
$$= \int_0^1 (y_{\text{circle}} - y_{\text{line}})\,dx = \int_0^1 \sqrt{1-x^2}\,dx - \int_0^1 (1-x)\,dx$$

$$= \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2}\sin^{-1}x\right]_0^1 - \left[x - \frac{x^2}{2}\right]_0^1$$

$$= \left[\left(0 + \frac{1}{2}\cdot\frac{\pi}{2}\right) - 0\right] - \left[\left(1 - \frac{1}{2}\right) - 0\right]$$

$$= \left(\frac{\pi}{4} - \frac{1}{2}\right) \text{ sq. units.}$$

35. Let r be the radius and h be the height of half cylinder.



Volume $= \frac{1}{2}\pi r^2 h = V$(constant) ...(i)

Total surface area of half cylinder is

$$S = 2\left(\frac{1}{2}\pi r^2\right) + \pi rh + 2rh \quad \text{...(ii)}$$

From (i) putting the value of h in (ii), we get

$$S = (\pi r^2) + \pi r\left(\frac{2V}{\pi r^2}\right) + 2r\left(\frac{2V}{\pi r^2}\right)$$

$$\Rightarrow\ S = (\pi r^2) + \left(\frac{1}{r}\right)\left[\frac{4V}{\pi} + 2V\right]$$

$$\therefore\quad \frac{dS}{dr} = (2\pi r) + \left(\frac{-1}{r^2}\right)\left[\frac{4V}{\pi} + 2V\right] \quad \text{...(iii)}$$

For maxima or minima, $\frac{dS}{dr} = 0$

$$\Rightarrow\ (2\pi r) + \left(\frac{-1}{r^2}\right)\left[\frac{4V}{\pi} + 2V\right] = 0$$

$$\Rightarrow\ (2\pi r) = \left(\frac{1}{r^2}\right)\left[\frac{4V + 2V\pi}{\pi}\right]$$

$$\Rightarrow\ \pi r^3 = V\left[\frac{2+\pi}{\pi}\right]$$

$$\Rightarrow\ V = \frac{\pi^2 r^3}{\pi + 2} \quad \text{...(iv)}$$

From (i) and (iv), we get

$$\frac{1}{2}\pi r^2 h = \frac{\pi^2 r^3}{\pi+2} \quad \Rightarrow \quad \frac{h}{2r} = \frac{\pi}{\pi+2}$$

$\Rightarrow$ height : diameter = $\pi : \pi + 2$

Differentiating (iii) w.r.t., r we get

$$\frac{d^2S}{dr^2} = (2\pi) + \left(\frac{2}{r^3}\right)\left[\frac{4V}{\pi} + 2V\right], \text{ which is positive}$$

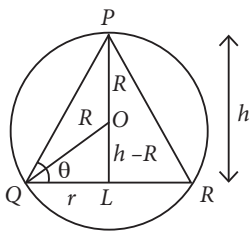
(as all quantities are $+ve$)

So, S is minimum when height : diameter = $\pi : \pi + 2$

OR

Let $2r$ be the base and h be the height of triangle PQR, which is inscribed in a circle of radius R.

$\therefore$ Area of triangle PQR, $A = \frac{1}{2}(2r)(h) = rh$ ...(i)



Area being positive quantity, A will be maximum or minimum if A^2 is maximum or minimum.

Let $Z = A^2 = r^2h^2$ ...(ii)

Now, in triangle OLQ, $QL^2 = OQ^2 - OL^2$

$\Rightarrow\ r^2 = R^2 - (h - R)^2 \Rightarrow r^2 = 2hR - h^2$

From (ii), $Z = h^2(2hR - h^2) \Rightarrow Z = (2h^3R - h^4)$

$$\Rightarrow\ \frac{dZ}{dh} = 6h^2R - 4h^3 \quad \text{...(iii)}$$

For maxima or minima, $\frac{dZ}{dh} = 0$

$$\Rightarrow\ 6h^2R - 4h^3 = 0 \ \Rightarrow\ 6R = 4h(h \neq 0) \Rightarrow h = \frac{3R}{2}$$

Differentiating (iii) w.r.t. h, we get

$$\frac{d^2Z}{dh^2} = 12hR - 12h^2$$

$$\Rightarrow\ \left[\frac{d^2Z}{dh^2}\right]_{h=\frac{3R}{2}} = 12\left(\frac{3R}{2}\right)R - 12\left(\frac{3R}{2}\right)^2$$

www.exambits.in

$= 18R^2 - 27R^2 = -9R^2 < 0$

So, $Z = A^2$ is maximum when $h = \frac{3R}{2}$

$\Rightarrow$ A is maximum when $h = \frac{3R}{2}$

When $h = \frac{3R}{2}$, $r^2 = 2hR - h^2$

$= 2R.\frac{3R}{2} - \left(\frac{3R}{2}\right)^2 = \frac{3R^2}{4} \Rightarrow r = \frac{\sqrt{3}R}{2}$

Now, in ΔPQL, $\tan\theta = \frac{h}{r} = \frac{\frac{3R}{2}}{\frac{\sqrt{3}R}{2}} = \sqrt{3}$

$= \tan\frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{3}$

Similarly, $\angle P = \angle R = \frac{\pi}{3}$

$\therefore$ Triangle ABC is an equilateral triangle.

36. Let $P(x, y, z)$ be any point on the plane in which $A(2, 1, 2)$ and $B(4, -2, 1)$ lie.

$\therefore$ $\overrightarrow{AP}$ and $\overrightarrow{AB}$ lie on required plane.

Also required plane is perpendicular to given plane $\vec{r}.(\hat{i} - 2\hat{k}) = 5$

$\therefore$ Normal to given plane $\overrightarrow{n_1} = (\hat{i} - 2\hat{k})$ lie on required plane.

$\Rightarrow$ $\overrightarrow{AP}, \overrightarrow{AB}$ and $\overrightarrow{n_1}$ are coplanar.

Where $\overrightarrow{AP} = (x-2)\hat{i} + (y-1)\hat{j} + (z-2)\hat{k}$

and $\overrightarrow{AB} = 2\hat{i} - 3\hat{j} - \hat{k}$

$\therefore$ $\left[\overrightarrow{AP}\ \overrightarrow{AB}\ \overrightarrow{n_1}\right] = 0$

$\Rightarrow$ $\begin{vmatrix} x-2 & y-1 & z-2 \\ 2 & -3 & -1 \\ 1 & 0 & -2 \end{vmatrix} = 0$

$\Rightarrow$ $(x-2)(6-0) - (y-1)(-4+1) + (z-2)(0+3) = 0$

$\Rightarrow$ $6x - 12 + 3y - 3 + 3z - 6 = 0$

$\Rightarrow$ $2x + y + z = 7$ ...(i)

Line passing through points $L(3, 4, 1)$ and $M(5, 1, 6)$ is

$\frac{x-3}{2} = \frac{y-4}{-3} = \frac{z-1}{5} = \lambda$ ...(ii)

$\Rightarrow$ General point on the line is $Q(2\lambda + 3, -3\lambda + 4, 5\lambda + 1)$

As line (ii) crosses plane (i) so point Q should satisfy equation (i).

$\therefore$ $2(2\lambda + 3) + (-3\lambda + 4) + (5\lambda + 1) = 7$

$\Rightarrow$ $4\lambda + 6 - 3\lambda + 4 + 5\lambda + 1 = 7$

$\Rightarrow$ $6\lambda = -4 \Rightarrow \lambda = -\frac{2}{3}$

$\therefore$ $Q\left(-\frac{4}{3} + 3, 2 + 4, -\frac{10}{3} + 1\right) i.e., Q\left(\frac{5}{3}, 6, -\frac{7}{3}\right).$

www.exambites.in

EXAM BITES

This Pdf Is
Downloaded From
www.exambites.in

Visit www.exambites.in for
More Premium Stuffs,Latest
Books,Test Papers,Lectures etc.


jeeneetadda


jeeneetadda_official


jeeneetadda

VISIT NOW !!